

Combined English Translation Readers

Non-European Mathematics Corpus - current checked work-level components

English-facing translations/readers. Built from the current checked work-level PDFs; not a new scholarly validation pass.

Components

1. 10-01 English Translation - Nine Chapters, vols. 1-9.pdf (100 pages)
2. 10-02 English Translation - Li Ye - Ceyuan Haijing, vols. 1-12.pdf (144 pages)
3. 10-03 English Translation - Qin - Shuxue Jiuzhang, fasc. 1-9.pdf (118 pages)
4. 10-04 English Translation - Sunzi Suanjing.pdf (23 pages)
5. 10-05 English Translation - Yang Hui - Xiangjie, parts 1-3.pdf (64 pages)
6. 10-06 English Translation - Zhu Shijie - Suanxue Qimeng, parts 1-2.pdf (68 pages)
7. 10-07 English Translation - Aryabhata - Aryabhata.pdf (18 pages)
8. 10-08 English Translation - Bhaskara II - Bijaganita, parts 1-3.pdf (56 pages)
9. 10-09 English Translation - Bhaskara II - Lilavati.pdf (23 pages)
10. 10-10 English Translation - Brahmagupta - Brahmasphutasiddhanta.pdf (160 pages)
11. 10-11 English Translation - al-Kashi - Miftah al-Hisab.pdf (14 pages)
12. 10-12 English Translation - al-Khwarizmi - Algebra.pdf (33 pages)
13. 10-13 English Translation - al-Tusi - Shaki al-Qatta.pdf (22 pages)
14. 10-14 English Translation - Robert of Chester and Karpinski.pdf (27 pages)
15. 10-15 English Translation - Omar Khayyam - Treatise on Algebra.pdf (15 pages)
16. 10-16 English Translation - Rosen - Algebra of Mohammed Ben Musa.pdf (40 pages)
17. 10-17 English Translation - Ruska - Oldest Arabic Algebra.pdf (50 pages)
18. 10-18 English Translation - al-Muqaddasi - Ahsan al-Ta'asim.pdf (15 pages)
19. 10-19 English Translation - Ibn al-Nadim - Kitab al-Fihrist.pdf (19 pages)
20. 10-20 English Translation - Said al-Andalusi - Tabaqat al-Umam.pdf (12 pages)
21. 10-21 English Translation - Smith-Karpinski - Numerals.pdf (16 pages)

Rebuilt 2026-06-03 from current reconstructed work-level readers (round32 base + deltas 33-49); source scans not bundled.

Jiuzhang Suanshu / 九章算術

Nine Chapters on the Mathematical Art

Bilingual English-Chinese Edition

Volumes 1–3 / 卷一至卷三

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With Commentary by Liu Hui (劉徽, 263 CE)

and Later Annotations by Li Chunfeng (李淳風, 656 CE)

The most important classical Chinese mathematical text

Parallel Text / 對照文本

Left column: Original Classical Chinese

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劉徽九章算術注原序 / Preface by Liu Hui

原典 Classical Chinese

English Translation

昔在包犧氏始畫八卦，以通神明之德，以類萬物之情，作九九之術以合六爻之變。暨於黃帝神而化之，引而伸之，於是建曆紀，協律呂，用稽道原，然後兩儀四象精微之氣可得而效焉。記稱隸首作數，其詳未之聞也。按周公制禮而有九數，九數之流，則九章是矣。

In antiquity, the Fuxi clan first drew the Eight Trigrams, to communicate the virtue of the divine spirits and to classify the natures of the myriad things. They created the art of the nine nines [multiplication table] to correspond to the transformations of the six lines [of the hexagrams]. When it came to the Yellow Emperor, he spiritualized and transformed it, extended and elaborated it. Thereupon he established the calendar system and harmonized the pitch pipes, using these to investigate the origin of the Way. Then the refined essences of the Two Modes and Four Images could be apprehended. Records say that Li Shou created numbers, but the details have not been heard. According to the Duke of Zhou's establishment of rites, there were Nine Numbers; the Nine Numbers evolved, and from them came the Nine Chapters.

往者暴秦焚書，經術散壞。自時厥後，漢北平侯張蒼、大司農中丞耿壽昌皆以善算命世。蒼等因舊文之遺殘，各稱刪補。故校其目則與古或異，而所論者多近語也。

In former times, the tyrannical Qin dynasty burned the books, and the classics and arts were scattered and destroyed. After that time, Zhang Cang, Marquis of Beiping in the Han, and Geng Shouchang, Vice Minister of Agriculture, both were renowned for their skill in calculation. Zhang and others took the surviving remnants of the old texts, each claiming to have edited and supplemented them. Therefore, when one examines the contents, they differ somewhat from the ancient version, and what is discussed mostly uses contemporary language.

徽幼習九章，長再詳覽。觀陰陽之割裂，總算術之根源，探蹟之暇，遂悟其意。是以敢竭頑魯，采其所見，為之作注。事類相推，各有攸歸，故枝條雖分而同本幹者，知發其一端而已。又所析理以辭，解體用圖，庶亦約而能周，通而不贖，覽之者思過半矣。

I, Hui, studied the Nine Chapters in my youth, and in adulthood reviewed it thoroughly. Observing the division of yin and yang, and summarizing the root sources of the art of calculation, in the leisure of my explorations I came to understand its meaning. Therefore, I dare to exhaust my humble and dull capacities, collecting what I have observed, to write a commentary upon it. Matters of the same kind are inferred from each other, each having its proper place. Though the branches are divided, they share the same trunk—one need only know how to develop from a single starting point. Moreover, I analyze principles with words and explain forms with diagrams, hoping to achieve conciseness that is yet comprehensive, and clarity without redundancy. Those who read it will understand more than half.

且算在六藝，古者以賓興賢能，教習國子。雖曰九數，其能窮纖入微，探測無方。至於以法相傳，亦猶規矩度量可得而共，非特難為也。當今好之者寡，故世雖多通才達學，而未必能綜於此耳。

Moreover, calculation is among the Six Arts. In antiquity, it was used to recommend the worthy and capable, and to instruct the sons of the state. Though called the Nine Numbers, it can penetrate the most minute and subtle details, exploring what cannot be fathomed. As for the transmission of methods, it is like compasses, squares, measures and standards, which can be shared by all—it is not especially difficult. But in the present age those who love it are few, so although the world has many talented and learned scholars, they are not necessarily able to master this.

周官大司徒職，夏至日中立八尺之表，其景尺有五寸，謂之地中。說云，南戴日下萬五千里。夫云爾者，以術推之。按九章立四表望遠及因木望山之術，皆端旁互見，無有超邈若斯之類。然則蒼等為術猶未足以博盡群數也。

The Offices of Zhou states that the Grand Minister of Instruction, at the summer solstice at noon, erected an eight-foot gnomon, and its shadow was one foot five inches—this is called the center of the earth. It is said that the place directly beneath the sun is fifteen thousand li to the south. Such statements are verified by calculation. According to the Nine Chapters' methods of erecting four gnomons to observe distant objects and using a tree to observe a mountain, all involve mutual observation of the endpoints and sides; none surpass this category. Thus Zhang Cang and others' methods were still insufficient to exhaust all calculations.

微尋九數有重差之名，原其指趣乃所以施於此也。凡望極高、測絕深而兼知其遠者必用重差，句股則必以重差為率，故曰重差也。立兩表於洛陽之城，令高八尺。南北各盡平地，同日度其正中之景。以景差為法，表高乘表間為實，實如法而一，所得加表高，即日去地也。

I, Hui, investigated the Nine Numbers and found the name “Chongcha” (Double Difference). Tracing its purpose, I realized it is applied to such problems. Whenever one observes extreme heights, measures unfathomable depths, and also determines distances, one must use the Double Difference. For right triangles [gougu], one must take the Double Difference as the rate—hence it is called Double Difference. Erect two gnomons in the city of Luoyang, making them eight feet high. On level ground to the north and south, on the same day measure their noon shadows. Take the difference of shadows as the divisor [fa]; multiply the gnomon height by the distance between gnomons as the dividend [shi]; divide to obtain a result; add the gnomon height to it—this gives the distance of the sun from the earth.

以南表之景乘表間為實，實如法而一，即為從南表至南戴日下也。以南戴日下及日去地為句、股，為之求弦，即日去人也。以徑寸之筭南望日，日滿筭空，則定筭之長短以為股率，以筭徑為句率，日去人之數為大股，大股之句即日徑也。

Multiply the south gnomon's shadow by the distance between gnomons as the dividend; divide by the divisor—this gives the distance from the south gnomon to the point directly beneath the sun. Take the distance to the subsolar point and the sun's altitude from earth as the base [gou] and height [gu] of a right triangle; finding the hypotenuse gives the distance from the sun to the observer. Use a tube of one-inch diameter to observe the sun from the south. When the sun fills the tube opening, the tube's length becomes the height-rate [gul"u], and the tube's diameter becomes the base-rate [goul"u]. The distance from sun to observer becomes the large height; from this the large base is found—that is the diameter of the sun.

雖天圓穹之象猶曰可度，又況泰山之高與江海之廣哉。微以為今之史籍且略舉天地之物，考論厥數，載之於志，以闡世術之美。輒造重差，并為注解，以究古人之意，綴於句股之下。度高者重表，測深者累矩，孤離者三望，離而又旁求者四望。觸類而長之，則雖幽遐詭伏，靡所不入。

Though the round dome of heaven can still be measured, how much more so the height of Mount Tai and the breadth of the rivers and seas! I, Hui, believe that today's historical records merely briefly enumerate things of heaven and earth. I have examined and discussed their numbers, recording them in this work, to illuminate the beauty of mathematical arts in the world. I have composed the Double Difference method, with annotations, to investigate the intentions of the ancients, appending it after the chapter on Right Triangles. To measure heights use double gnomons; to measure depths use accumulated carpenter's squares; for isolated objects use triple observation; for objects at a distance requiring lateral determination use quadruple observation. Extending by analogy, even the most remote and hidden things can be reached.

博物君子，詳而覽焉。 / *Let the gentleman of broad learning examine this in detail.*

1 卷第一：方田 / Volume 1: Rectangular Fields

方田以御田疇界域 / “Rectangular Fields are for governing the boundaries of farmland.”

原典 Classical Chinese

English Translation

方田者，田之正也。諸田不等，以方為正，故曰方田。此章以御田疇界域，論各種田地面積之計算法，并及分數四則運算之法。

Fangtian (Rectangular Fields) means the standard [rectangular] field. Since various fields are not uniform, the square/rectangle is taken as the standard—hence “Rectangular Fields.” This chapter governs the boundaries of farmland, discussing methods for computing the areas of various fields, as well as the four arithmetic operations with fractions.

[一] 今有田廣十五步，從十六步。問為田幾何？

[1] Suppose there is a field with width 15 *bu* and length 16 *bu*. **Question:** how much field [area] does it make?

答曰：一畝。

Answer: 1 *mu*.

[二] 又有田廣十二步，從十四步。問為田幾何？

[2] Further: a field with width 12 *bu* and length 14 *bu*. **Question:** how much field?

答曰：一百六十八步。

Answer: 168 square *bu*.

術曰：方田術曰：廣從步數相乘得積步。以畝法二百四十步除之，即畝數。百畝為一頃。

Method: The method for rectangular fields says: multiply the width and length in *bu* to obtain the area in square *bu*. Divide by the *mu* divisor of 240 square *bu* to obtain the number of *mu*. One hundred *mu* make one *qing*.

[三] 今有田廣一里，從一里。問為田幾何？

[3] Suppose there is a field with width 1 *li* and length 1 *li*. **Question:** how much field?

答曰：三頃七十五畝。

Answer: 3 *qing* and 75 *mu*.

[四] 又有田廣二里，從三里。問為田幾何？

[4] Further: a field with width 2 *li* and length 3 *li*. **Question:** how much field?

答曰：二十二頃五十畝。

Answer: 22 *qing* and 50 *mu*.

術曰：里田術曰：廣從里數相乘得積里。以三百七十五乘之，即畝數。

Method: The method for *li*-fields says: multiply the width and length in *li* to obtain the area in square *li*. Multiply by 375 to obtain the number of *mu*.

[五] 今有十八分之十二。問約之得幾何？

[5] Suppose there is $\frac{12}{18}$. Question: reducing it, what does it yield?

答曰：三分之二。

Answer: $\frac{2}{3}$.

[六] 又有九十一分之四十九。問約之得幾何？

[6] Further: $\frac{49}{91}$. Question: reducing it, what does it yield?

答曰：十三分之七。

Answer: $\frac{7}{13}$.

術曰：約分術曰：可半者半之，不可半者，副置分母子之數，以少減多，更相減損，求其等也。以等數約之。

Method: The method for reducing fractions says: if [both terms] can be halved, halve them. If not, place the numerator and denominator numbers separately; subtract the smaller from the larger, repeatedly reducing them mutually, seeking their equality [GCD]. Use this equal number to reduce [both terms].

[七] 今有三分之一，五分之二。問合之得幾何？

[7] Suppose there is $\frac{1}{3}$ and $\frac{2}{5}$. Question: combining them, what does it yield?

答曰：十五分之十一。

Answer: $\frac{11}{15}$.

[八] 又有三分之二，七分之四，九分之五。問合之得幾何？

[8] Further: $\frac{2}{3}$, $\frac{4}{7}$, and $\frac{5}{9}$. Question: combining them, what does it yield?

答曰：得一、六十三分之五十。

Answer: Yields $1\frac{50}{63}$.

[九] 又有二分之一，三分之二，四分之三，五分之四。問合之得幾何？

[9] Further: $\frac{1}{2}$, $\frac{2}{3}$, $\frac{3}{4}$, and $\frac{4}{5}$. Question: combining them, what does it yield?

答曰：得二、六十分之四十三。

Answer: Yields $2\frac{43}{60}$.

術曰：合分術曰：母互乘子，并為實，母相乘為法，實如法而一。不滿法者，以法命之。其母同者，直相從之。

Method: The method for adding fractions says: multiply the denominators crosswise with the numerators; combine as dividend [shi]. Multiply the denominators together as divisor [fa]. Divide the dividend by the divisor. If less than the divisor, name it [as a fraction] by the divisor. If the denominators are the same, simply combine them directly.

[一〇] 今有九分之八，減其五分之一。問餘幾何？

[10] Suppose there is $\frac{8}{9}$; subtract from it $\frac{1}{5}$. Question: what is the remainder?

答曰：四十五分之三十一。

Answer: $\frac{31}{45}$.

[一一] 又有四分之三，減其三分之一。問餘幾何？

[11] Further: $\frac{3}{4}$; subtract from it $\frac{1}{3}$. Question: what is the remainder?

答曰：十二分之五。

Answer: $\frac{5}{12}$.

術曰：減分術曰：母互乘子，以少減多，餘為實，母相乘為法，實如法而一。

Method: The method for subtracting fractions says: multiply denominators crosswise with numerators; subtract the smaller from the larger; the remainder is the dividend. Multiply denominators together as divisor. Divide the dividend by the divisor.

[一二] 今有八分之五，二十五分之十六。問孰多？多幾何？

[12] Suppose there is $\frac{5}{8}$ and $\frac{16}{25}$. Question: which is greater? By how much?

答曰：二十五分之十六多，多二百分之三。

Answer: $\frac{16}{25}$ is greater, by $\frac{3}{200}$.

[一三] 又有九分之八，七分之六。問孰多？多幾何？

[13] Further: $\frac{8}{9}$ and $\frac{6}{7}$. Question: which is greater? By how much?

答曰：九分之八多，多六十三分之二。

Answer: $\frac{8}{9}$ is greater, by $\frac{2}{63}$.

[一四] 又有二十一分之八，五十分之十七。問孰多？多幾何？

[14] Further: $\frac{8}{21}$ and $\frac{17}{50}$. Question: which is greater? By how much?

答曰：二十一分之八多，多一千五十分之四十三。

Answer: $\frac{8}{21}$ is greater, by $\frac{43}{1050}$.

術曰：課分術曰：母互乘子，以少減多，餘為實，母相乘為法，實如法而一，即相多也。

Method: The method for comparing fractions says: multiply denominators crosswise with numerators; subtract the smaller from the larger; the remainder is the dividend. Multiply denominators together as divisor. Divide—the result is the difference between them.

[一五] 今有三分之一，三分之二，四分之三。問減多益少，各幾何而平？

[15] Suppose there is $\frac{1}{3}$, $\frac{2}{3}$, $\frac{3}{4}$. Question: reducing the larger and augmenting the smaller, by how much each to make them equal?

答曰：減四分之三者二，三分之二者一，并以益三分之一，而各平於十二分之七。

Answer: Reduce $\frac{3}{4}$ by $\frac{2}{12}$ [and] $\frac{2}{3}$ by $\frac{1}{12}$, combining [both amounts] to augment $\frac{1}{3}$; each is then equal at $\frac{7}{12}$.

[一六] 又有二分之一，三分之二，四分之三。問減多益少，各幾何而平？

[16] Further: $\frac{1}{2}$, $\frac{2}{3}$, $\frac{3}{4}$. Question: reducing the larger and augmenting the smaller, by how much each to make them equal?

答曰：減三分之二者一，四分之三者四，并以益二分之一，而各平於三十六分之二十三。

Answer: Reduce $\frac{2}{3}$ by $\frac{1}{36}$ [and] $\frac{3}{4}$ by $\frac{4}{36}$, combining [both amounts] to augment $\frac{1}{2}$; each is then equal at $\frac{23}{36}$.

術曰：平分術曰：母互乘子，副并為平實，母相乘為法。以列數乘未并者各自為列實。亦以列數乘法，以平實減列實，餘，約之為所減。并所減以益於少，以法命平實，各得其平。

Method: The method for equalizing [fractions] says: multiply denominators crosswise with numerators; combine the auxiliary [copies] as the equal dividend. Multiply denominators together as divisor. Multiply the number of terms by the uncombined [terms], each becoming its own column dividend. Also multiply the number of terms by the divisor. Subtract the equal dividend from each column dividend; the remainder, reduced, is what is to be subtracted. Combine the amounts subtracted to augment the smallest. Name the equal dividend by the divisor—each obtains its equal share.

[一七] 今有七人，分八錢三分錢之一。問人得幾何？

[17] Suppose there are 7 people, sharing $8\frac{1}{3}$ coins. Question: how much does each person get?

答曰：人得一錢、二十一分錢之四。

Answer: Each person gets $1\frac{4}{21}$ coins.

[一八] 又有三人，三分人之一，分六錢三分錢之一，四分錢之三。問人得幾何？

[18] Further: 3 people plus $\frac{1}{3}$ of a person, sharing $6\frac{1}{3}\frac{3}{4}$ coins. Question: how much does each get?

答曰：人得二錢、八分錢之一。

Answer: Each person gets $2\frac{1}{8}$ coins.

術曰：經分術曰：以人數為法，錢數為實，實如法而一。有分者通之，重有分者同而通之。

Method: The method for dividing by [whole] numbers says: take the number of people as divisor and the number of coins as dividend; divide. If there are fractions, convert them [to common terms]; if there are repeated fractions, equalize and convert them.

[一九] 今有田廣七分步之四，從五分步之三。問為田幾何？

[19] Suppose there is a field with width $\frac{4}{7}$ bu and length $\frac{3}{5}$ bu. Question: how much field?

答曰：三十五分步之十二。

Answer: $\frac{12}{35}$ square bu.

[二〇] 又有田廣九分步之七，從十一分步之九。問為田幾何？

[20] Further: width $\frac{7}{9}$ bu, length $\frac{9}{11}$ bu. Question: how much field?

答曰：十一分步之七。

Answer: $\frac{7}{11}$ square bu.

[二一] 又有田廣五分步之四，從九分步之五，問為田幾何？

[21] Further: width $\frac{4}{5}$ bu, length $\frac{5}{9}$ bu. Question: how much field?

答曰：九分步之四。

Answer: $\frac{4}{9}$ square bu.

術曰：乘分術曰：母相乘為法，子相乘為實，實如法而一。

Method: The method for multiplying fractions says: multiply denominators together as divisor; multiply numerators together as dividend; divide the dividend by the divisor.

[二二] 今有田廣三步、三分步之一，從五步、五分步之二。問為田幾何？

[22] Suppose there is a field with width $3\frac{1}{3}$ bu and length $5\frac{2}{5}$ bu. Question: how much field?

答曰：十八步。

Answer: 18 square bu.

[二三] 又有田廣七步、四分步之三，從十五步、九分步之五。問為田幾何？

[23] Further: width $7\frac{3}{4}$ bu, length $15\frac{5}{9}$ bu. Question: how much field?

答曰：一百二十步、九分步之五。

Answer: $120\frac{5}{9}$ square bu.

[二四] 又有田廣十八步、七分步之五，從二十三步、十一分步之六。問為田幾何？

[24] Further: width $18\frac{5}{7}$ bu, length $23\frac{6}{11}$ bu. Question: how much field?

答曰：一畝二百步、十一分步之七。

Answer: 1 mu and $200\frac{7}{11}$ square bu.

術曰：大廣田術曰：分母各乘其全，分子從之，相乘為實。分母相乘為法。實如法而一。

Method: The method for large-width fields says: each denominator multiplies its whole [number], and the numerators follow; multiply together as dividend. Multiply denominators together as divisor. Divide the dividend by the divisor.

[二五] 今有圭田廣十二步，正從二十一步。問為田幾何？

[25] Suppose there is a *guitian* [triangle field] with base 12 bu and height 21 bu. Question: how much field?

答曰：一百二十六步。

Answer: 126 square bu.

[二六] 又有圭田廣五步、二分步之一，從八步、三分步之二。問為田幾何？

[26] Further: a *guitian* with base $5\frac{1}{2}$ bu, height $8\frac{2}{3}$ bu. Question: how much field?

答曰：二十三步、六分步之五。

Answer: $23\frac{5}{6}$ square bu.

術曰：術曰：半廣以乘正從。

Method: Method: halve the base and multiply by the height.

(注) 劉徽注：半廣者，以盈補虛，為直田也。亦可以半正從以乘廣。

(Comm.) Liu Hui's commentary: Halving the base—using the excess to fill the deficiency—makes a rectangular field. One may also halve the height and multiply by the base.

[二七] 今有邪田，一頭廣三十步，一頭廣四十二步，正從六十四步。問為田幾何？

[27] Suppose there is a *xietian* [trapezoid] with one end-width 30 *bu*, the other end-width 42 *bu*, and height 64 *bu*. Question: how much field?

答曰：九畝一百四十四步。

Answer: 9 *mu* and 144 square *bu*.

[二八] 又有邪田，正廣六十五步，一畔從一百步，一畔從七十二步。問為田幾何？

[28] Further: a *xietian* with top width 65 *bu*, one side-length 100 *bu*, other side-length 72 *bu*. Question: how much field?

答曰：二十三畝七十步。

Answer: 23 *mu* and 70 square *bu*.

術曰：術曰：并兩邪而半之，以乘正從若廣。又可半正從若廣，以乘并，畝法而一。

Method: Method: combine the two slant sides and halve; multiply by the height [or width]. Alternatively, halve the height [or width] and multiply by the combined [slant sides]; divide by the *mu* divisor.

[二九] 今有箕田，舌廣二十步，踵廣五步，正從三十步。問為田幾何？

[29] Suppose there is a *jitian* [winnowing-basket field] with tongue-width 20 *bu*, heel-width 5 *bu*, and height 30 *bu*. Question: how much field?

答曰：一畝一百三十五步。

Answer: 1 *mu* and 135 square *bu*.

[三〇] 又有箕田，舌廣一百一十七步，踵廣五十步，正從一百三十五步。問為田幾何？

[30] Further: a *jitian* with tongue-width 117 *bu*, heel-width 50 *bu*, height 135 *bu*. Question: how much field?

答曰：四十六畝二百三十二步半。

Answer: 46 *mu* and $232\frac{1}{2}$ square *bu*.

術曰：術曰：并踵舌而半之，以乘正從。畝法而一。

Method: Method: combine heel and tongue and halve; multiply by the height; divide by the *mu* divisor.

[三一] 今有圓田，周三十步，徑十步。問為田幾何？

[31] Suppose there is a circular field with circumference 30 *bu* and diameter 10 *bu*. Question: how much field?

答曰：七十五步。

Answer: 75 square *bu*.

[三二] 又有圓田，周一百八十一步，徑六十步、三分步之一。問為田幾何？

[32] Further: a circular field with circumference 181 *bu* and diameter $60\frac{1}{3}$ *bu*. Question: how much field?

答曰：十一畝九十步、十二分步之一。

Answer: 11 *mu* and $90\frac{1}{12}$ square *bu*.

術曰：術曰：半周半徑相乘得積步。

又術曰：周徑相乘，四而一。

又術曰：徑自相乘，三之，四而一。

又術曰：周自相乘，十二而一。

Method: Method: multiply half the circumference by half the radius to get the area.

Alternative method: multiply circumference by diameter, then divide by 4.

Alternative method: square the diameter, triple it, then divide by 4.

Alternative method: square the circumference, then divide by 12.

(注) 此于徽術，當為田十畝二百八步三百一十四分步之一百一。臣淳風等謹依密率，為田十畝二百步八十八分步之八十七。按：半周為從，半徑為廣，故廣從相乘為積步也。假令圓徑二尺，圓中容六觚之一面，與圓徑之半，其數均等。合徑率一而外周率三也。

(Comm.) According to Hui's method, this should yield 10 *mu* and $208\frac{101}{314}$ square *bu*. Li Chunfeng and others, using the precise rate [of π], obtain 10 *mu* and $200\frac{87}{88}$ square *bu*.

Note: half the circumference serves as length, half the radius as width; thus multiplying width by length gives the area. Suppose the diameter is 2 *chi*; the side of a hexagon inscribed in the circle equals half the diameter in number. This gives diameter-rate 1 and circumference-rate 3 [i.e., $\pi \approx 3$].

(注) 又按：為圖，以六觚之一面乘一弧半徑，三之，得十二觚之羣。若又割之，次以十二觚之一面乘一弧之半徑，六之，則得二十四觚之羣。割之彌細，所失彌少。割之又割，以至于不可割，則與圓周合體而無所失矣。觚面之外，又有餘徑。以面乘餘徑，則羣出觚表。若夫觚之細者，與圓合體，則表無餘徑。表無餘徑，則羣不外出矣。以一面乘半徑，觚而裁之，每輒自倍。故以半周乘半徑而為圓羣。

(Comm.) Furthermore: construct a diagram. Multiply the side of a hexagon by the radius of one arc, triple it—this gives the area of a 12-gon. If one cuts again, next multiply the side of a 12-gon by the radius of one arc, multiply by 6—this gives the area of a 24-gon. The finer the cutting, the less the loss. Cut again and again until it cannot be cut further, then it coincides with the circle and nothing is lost. Outside the sides of the polygon there remain excess radii. Multiplying a side by an excess radius produces area outside the polygon. When the polygon becomes so fine that it coincides with the circle, there are no excess radii; when there are no excess radii, no area extends outside. Multiplying one side by the radius and cutting by the number of sides, each time the area doubles. Therefore multiplying half the circumference by half the radius gives the area of the circle.

(注) 此一周、徑，謂至然之數，非周三徑一之率也。周三者，從其六觚之環耳。以推圓規多少之覺，乃弓之與弦也。然世傳此法，莫肯精核；學者踵古，習其謬失。不有明據，辯之斯難。凡物類形象，不圓則方。方圓之率，誠著于近，則雖遠可知也。由此言之，其用博矣。謹按圖驗，更造密率。

(Comm.) This [precise] circumference and diameter are the true values, not the rate of “circumference 3, diameter 1.” “Circumference 3” comes from the perimeter of the inscribed hexagon. The difference between using this to measure a circle is like that between a bow [arc] and its chord. Yet this method has been handed down through the ages, and none have examined it carefully; scholars follow antiquity, repeating its errors. Without clear evidence, it is hard to refute.

All objects in form are either round or square. The rates for square and circle, if clearly demonstrated for nearby cases, can be extended to distant ones. From this perspective, its applications are broad. I have carefully verified with diagrams and created more precise rates [for π].

[三三] 今有宛田，下周三十步，徑十六步。問為田幾何？

[33] Suppose there is a *wantian* [spherical-segment field] with lower circumference 30 *bu* and diameter 16 *bu*. Question: how much field?

答曰：一百二十步。

Answer: 120 square *bu*.

[三四] 又有宛田，下周九十九步，徑五十一步。問為田幾何？

[34] Further: a *wantian* with lower circumference 99 *bu* and diameter 51 *bu*. Question: how much field?

答曰：五畝六十二步、四分步之一。

Answer: 5 *mu* and $62\frac{1}{4}$ square *bu*.

術曰：術曰：以徑乘周，四而一。

Method: Method: multiply the diameter by the circumference, then divide by 4.

[三五] 今有弧田，弦三十步，矢十五步。問為田幾何？

[35] Suppose there is a *hutian* [circular-segment field] with chord 30 *bu* and arrow [sagitta] 15 *bu*. Question: how much field?

答曰：一畝九十七步半。

Answer: 1 *mu* and $97\frac{1}{2}$ square *bu*.

[三六] 又有弧田，弦七十八步、二分步之一，矢十三步、九分步之七。問為田幾何？

[36] Further: a *hutian* with chord $78\frac{1}{2}$ *bu*, arrow $13\frac{7}{9}$ *bu*. Question: how much field?

答曰：二畝一百五十五步、八十一分步之五十六。

Answer: 2 *mu* and $155\frac{56}{81}$ square *bu*.

術曰：術曰：以弦乘矢，矢又自乘，并之，二而一。

Method: Method: multiply the chord by the arrow; also square the arrow; combine them; halve [the sum].

[三七] 今有環田，中周九十二步，外周一百二十二步，徑五步。問為田幾何？

[37] Suppose there is a *huantian* [annular field] with inner circumference 92 *bu*, outer circumference 122 *bu*, and radial width 5 *bu*. Question: how much field?

答曰：二畝五十五步。

Answer: 2 *mu* and 55 square *bu*.

[三八] 又有環田，中周六十二步、四分步之三，外周一百一十三步、二分步之一，徑十二步、三分步之二。問為田幾何？

[38] Further: an annular field with inner circumference $62\frac{3}{4}$ *bu*, outer circumference $113\frac{1}{2}$ *bu*, width $12\frac{2}{3}$ *bu*. Question: how much field?

答曰：四畝一百五十六步、四分步之一。

Answer: 4 *mu* and $156\frac{1}{4}$ square *bu*.

術曰：術曰：并中外周而半之，以徑乘之為積步。

密率術曰：置中外周步數，分母、子各居其下。母互乘子，通全步，內分子。以中周減外周，餘半之，以益中周。徑亦通分內子，以乘周為實。分母相乘為法，除之為積步，餘積步之分。以畝法除之，即畝數也。

Method: Method: combine inner and outer circumferences and halve; multiply by the width to obtain the area.

Precise-rate method: place the inner and outer circumferences, with numerators and denominators below each. Cross-multiply denominators with numerators; combine whole steps with numerators. Subtract inner from outer circumference; halve the remainder and add to the inner circumference. Convert the width to an improper fraction. Multiply by the [adjusted] circumference as dividend. Multiply denominators together as divisor; divide to obtain the area [and fractional remainder]. Divide by the *mu* divisor to obtain the number of *mu*.

2 卷第二：粟米 / Volume 2: Millet and Rice

粟米以御交質變易 / “Millet and Rice are for governing the exchange of commodities.”

原典 Classical Chinese

English Translation

本章論各種谷物之交換比率，及比例計算之法。今有術（即今之三率法、比例法）為本章之核心算法。

This chapter discusses the exchange rates among various grains and the methods of proportional calculation. The *jinyou* method (i.e., the Rule of Three, or proportional method) is the core algorithm of this chapter.

粟米之法 / Rates for Millet and Rice

Standard Exchange Rates

粟率五十	糲米三十	粳米二十七	鑿米二十四
御米二十一	小●十三半	大●五十四	糲飯七十五
粳飯五十四	鑿飯四十八	御飯四十二	菽、荅、麻、麥各四十五
稻六十	豉六十三	飧九十	熟菽一百三半
粳一百七十五			

Millet (<i>su</i>): 50	Coarse rice (<i>li</i>): 30	Husked rice (<i>bai</i>): 27	Polished rice (<i>zao</i>): 24
Imperial rice: 21	Small ●: 13.5	Large ●: 54	Coarse cooked rice: 75
Husked cooked rice: 54	Polished cooked rice: 48	Imperial cooked rice: 42	Beans/peas/hemp/wheat: 45 each
Rice (paddy): 60	Fermented beans: 63	Cooked meal: 90	Cooked beans: 103.5
Sprouted grain: 175			

術曰：今有術曰：以所有數乘所求率為實，以所有率為法，實如法而一。

Method: The *jinyou* (Rule of Three) method says: multiply the given quantity by the sought rate as dividend; take the given rate as divisor; divide the dividend by the divisor.

[一] 今有粟一斗，欲為糲米。問得幾何？

[1] Suppose there is 1 *dou* of millet, to be made into coarse rice (*li mi*). Question: how much?

答曰：為糲米六升。

Answer: Makes 6 *sheng* of coarse rice.

術曰：術曰：以粟求糲米，三之，五而一。

Method: Method: to find coarse rice from millet, triple it and divide by 5.

[二] 今有粟二斗一升，欲為粳米。問得幾何？

[2] Suppose there is 2 *dou* 1 *sheng* of millet, to be made into husked rice (*bai mi*). Question: how much?

答曰：為粳米一斗一升、五十分升之十七。

Answer: Makes 1 dou 1 sheng $\frac{17}{50}$ sheng of husked rice.

術曰：術曰：以粟求粳米，二十七之，五十而一。

Method: Method: to find husked rice from millet, multiply by 27 and divide by 50.

[三] 今有粟四斗五升，欲為鑿米。問得幾何？

[3] Suppose there is 4 dou 5 sheng of millet, to be made into polished rice (zao mi). Question: how much?

答曰：為鑿米二斗一升、五分升之三。

Answer: Makes 2 dou 1 sheng $\frac{3}{5}$ sheng of polished rice.

術曰：術曰：以粟求鑿米，十二之，二十五而一。

Method: Method: to find polished rice from millet, multiply by 12 and divide by 25.

[四] 今有粟七斗九升，欲為御米。問得幾何？

[4] Suppose there is 7 dou 9 sheng of millet, to be made into imperial rice (yu mi). Question: how much?

答曰：為御米三斗三升、五十分升之九。

Answer: Makes 3 dou 3 sheng $\frac{9}{50}$ sheng of imperial rice.

術曰：術曰：以粟求御米，二十一之，五十而一。

Method: Method: to find imperial rice from millet, multiply by 21 and divide by 50.

[五] 今有粟一斗，欲為小●。問得幾何？

[5] Suppose there is 1 dou of millet, to be made into small ●. Question: how much?

答曰：為小●二升、一十分升之七。

Answer: Makes 2 $\frac{7}{10}$ sheng of small ●.

術曰：術曰：以粟求小●，二十七之，百而一。

Method: Method: to find small ● from millet, multiply by 27 and divide by 100.

[六] 今有粟九斗八升，欲為大●。問得幾何？

[6] Suppose there is 9 dou 8 sheng of millet, to be made into large ●. Question: how much?

答曰：為大●一十斗五升、二十五分升之二十一。

Answer: Makes 10 dou 5 sheng $\frac{21}{25}$ sheng of large ●.

術曰：術曰：以粟求大●，二十七之，二十五而一。

Method: Method: to find large ● from millet, multiply by 27 and divide by 25.

[七] 今有粟二斗三升，欲為糲飯。問得幾何？

[7] Suppose there is 2 dou 3 sheng of millet, to be made into coarse cooked rice (li fan). Question: how much?

答曰：為糲飯三斗四升半。

Answer: Makes 3 *dou* 4.5 *sheng* of coarse cooked rice.

術曰：術曰：以粟求糲飯，三之，二而一。

Method: Method: to find coarse cooked rice from millet, triple it and halve [the result].

[八] 今有粟三斗六升，欲為粳飯。問得幾何？

[8] Suppose there is 3 *dou* 6 *sheng* of millet, to be made into husked cooked rice (*bai fan*). Question: how much?

答曰：為粳飯三斗八升、二十五分升之二十二。

Answer: Makes 3 *dou* 8 *sheng* $\frac{22}{25}$ *sheng* of husked cooked rice.

術曰：術曰：以粟求粳飯，二十七之，二十五而一。

Method: Method: to find husked cooked rice from millet, multiply by 27 and divide by 25.

[九] 今有粟八斗六升，欲為鑿飯。問得幾何？

[9] Suppose there is 8 *dou* 6 *sheng* of millet, to be made into polished cooked rice (*zao fan*). Question: how much?

答曰：為鑿飯八斗二升、二十五分升之一十四。

Answer: Makes 8 *dou* 2 *sheng* $\frac{14}{25}$ *sheng* of polished cooked rice.

術曰：術曰：以粟求鑿飯，二十四之，二十五而一。

Method: Method: to find polished cooked rice from millet, multiply by 24 and divide by 25.

[一〇] 今有粟九斗八升，欲為御飯。問得幾何？

[10] Suppose there is 9 *dou* 8 *sheng* of millet, to be made into imperial cooked rice (*yu fan*). Question: how much?

答曰：為御飯八斗二升、二十五分升之八。

Answer: Makes 8 *dou* 2 *sheng* $\frac{8}{25}$ *sheng* of imperial cooked rice.

術曰：術曰：以粟求御飯，二十一之，二十五而一。

Method: Method: to find imperial cooked rice from millet, multiply by 21 and divide by 25.

[一一] 今有粟三斗少半升，欲為菽。問得幾何？

[一二] 今有粟四斗一升、太半升，欲為荅。問得幾何？

[一三] 今有粟五斗、太半升，欲為麻。問得幾何？

[一四] 今有粟一十斗八升、五分升之二，欲為麥。問得幾何？

[11] Suppose there is 3 *dou* minus $\frac{1}{3}$ *sheng* of millet, to be made into beans (*shu*). Question: how much?

[12] Suppose there is 4 *dou* 1 *sheng* plus $\frac{2}{3}$ *sheng* of millet, to be made into peas (*da*). Question: how much?

[13] Suppose there is 5 *dou* plus $\frac{2}{3}$ *sheng* of millet, to be made into hemp (*ma*). Question: how much?

[14] Suppose there is 10 *dou* 8 *sheng* plus $\frac{2}{5}$ *sheng* of millet, to be made into wheat (*mai*). Question: how much?

答曰：為菽二斗七升、一十分升之三。

為荅三斗七升半。

為麻四斗五升、五分升之三。

為麥九斗七升、二十五分升之一十四。

Answer: Makes $2dou\ 7sheng\ \frac{3}{10}sheng$ of beans.
 Makes $3\ dou\ 7.5\ sheng$ of peas.
 Makes $4dou\ 5sheng\ \frac{3}{5}sheng$ of hemp.
 Makes $9dou\ 7sheng\ \frac{14}{25}sheng$ of wheat.

術曰：術曰：以粟求菽、荅、麻、麥，皆九之，十而一。

Method: Method: to find beans, peas, hemp, or wheat from millet, in each case multiply by 9 and divide by 10.

[一五] 今有粟七斗五升、七分升之四，欲為稻。問得幾何？

[15] Suppose there is $7\ dou\ 5\ sheng$ plus $\frac{4}{7}sheng$ of millet, to be made into paddy rice (*dao*). Question: how much?

答曰：為稻九斗、三十五分升之二十四。

Answer: Makes $9dou\ \frac{24}{35}sheng$ of paddy rice.

術曰：術曰：以粟求稻，六之，五而一。

Method: Method: to find paddy rice from millet, multiply by 6 and divide by 5.

[一六] 今有粟七斗八升，欲為豉。問得幾何？

[16] Suppose there is $7\ dou\ 8\ sheng$ of millet, to be made into fermented beans (*chi*). Question: how much?

答曰：為豉九斗八升、二十五分升之七。

Answer: Makes $9dou\ 8sheng\ \frac{7}{25}sheng$ of fermented beans.

術曰：術曰：以粟求豉，六十三之，五十而一。

Method: Method: to find fermented beans from millet, multiply by 63 and divide by 50.

[一七] 今有粟五斗五升，欲為飧。問得幾何？

[17] Suppose there is $5\ dou\ 5\ sheng$ of millet, to be made into cooked meal (*sun*). Question: how much?

答曰：為飧九斗九升。

Answer: Makes $9\ dou\ 9\ sheng$ of cooked meal.

術曰：術曰：以粟求飧，九之，五而一。

Method: Method: to find cooked meal from millet, multiply by 9 and divide by 5.

[一八] 今有粟四斗，欲為熟菽。問得幾何？

[18] Suppose there is $4\ dou$ of millet, to be made into cooked beans (*shu shu*). Question: how much?

答曰：為熟菽八斗二升、五分升之四。

Answer: Makes $8dou\ 2sheng\ \frac{4}{5}sheng$ of cooked beans.

術曰：術曰：以粟求熟菽，二百七之，百而一。

Method: Method: to find cooked beans from millet, multiply by 207 and divide by 100.

[一九] 今有粟二斗，欲為粢。問得幾何？

[19] Suppose there is 2 *dou* of millet, to be made into sprouted grain (*nie*). Question: how much?

答曰：為粢七斗。

Answer: Makes 7 *dou* of sprouted grain.

術曰：術曰：以粟求粢，七之，二而一。

Method: Method: to find sprouted grain from millet, multiply by 7 and divide by 2.

[二〇] 今有糲米十五斗五升、五分升之二，欲為粟。問得幾何？

[20] Suppose there is 15 *dou* 5 *sheng* plus $\frac{2}{5}$ *sheng* of coarse rice, to be converted back to millet. Question: how much?

答曰：為粟二十五斗九升。

Answer: Makes 25 *dou* 9 *sheng* of millet.

術曰：術曰：以糲米求粟，五之，三而一。

Method: Method: to find millet from coarse rice, multiply by 5 and divide by 3.

[二一] 今有粃米二斗，欲為粟。問得幾何？

[21] Suppose there is 2 *dou* of husked rice, to be converted back to millet. Question: how much?

答曰：為粟三斗七升、二十七分升之一。

Answer: Makes 3 *dou* 7 *sheng* $\frac{1}{27}$ *sheng* of millet.

術曰：術曰：以粃米求粟，五十之，二十七而一。

Method: Method: to find millet from husked rice, multiply by 50 and divide by 27.

[二二] 今有鑿米三斗、少半升，欲為粟。問得幾何？

[22] Suppose there is 3 *dou* minus $\frac{1}{3}$ *sheng* of polished rice, to be converted back to millet. Question: how much?

答曰：為粟六斗三升、三十六分升之七。

Answer: Makes 6 *dou* 3 *sheng* $\frac{7}{36}$ *sheng* of millet.

術曰：術曰：以鑿米求粟，二十五之，十三而一。

Method: Method: to find millet from polished rice, multiply by 25 and divide by 13.

[二三] 今有御米十四斗，欲為粟。問得幾何？

[23] Suppose there is 14 *dou* of imperial rice, to be converted back to millet. Question: how much?

答曰：為粟三十三斗三升、少半升。

Answer: Makes 33 *dou* 3 *sheng* $\frac{1}{3}$ *sheng* of millet.

術曰：術曰：以御米求粟，五十之，二十一而一。

Method: Method: to find millet from imperial rice, multiply by 50 and divide by 21.

[二四] 今有稻一十二斗六升、一十五分升之一十四，欲為粟。問得幾何？

[24] Suppose there is 12 *dou* 6 *sheng* plus $\frac{14}{15}$ *sheng* of paddy rice, to be converted back to millet. Question: how much?

答曰：為粟一十斗五升、九分升之七。

Answer: Makes 10 *dou* 5 *sheng* $\frac{7}{9}$ *sheng* of millet.

術曰：術曰：以稻求粟，五之，六而一。

Method: Method: to find millet from paddy rice, multiply by 5 and divide by 6.

[二五] 今有糲米一十九斗二升、七分升之一，欲為粳米。問得幾何？

[25] Suppose there is 19 *dou* 2 *sheng* plus $\frac{1}{7}$ *sheng* of coarse rice, to be made into husked rice. Question: how much?

答曰：為粳米一十七斗二升、一十四分升之一十三。

Answer: Makes 17 *dou* 2 *sheng* $\frac{13}{14}$ *sheng* of husked rice.

術曰：術曰：以糲米求粳米，九之，十而一。

Method: Method: to find husked rice from coarse rice, multiply by 9 and divide by 10.

[二六] 今有糲米六斗四升、五分升之三，欲為糲飯。問得幾何？

[26] Suppose there is 6 *dou* 4 *sheng* plus $\frac{3}{5}$ *sheng* of coarse rice, to be made into coarse cooked rice. Question: how much?

答曰：為糲飯一十六斗一升半。

Answer: Makes 16 *dou* 1.5 *sheng* of coarse cooked rice.

術曰：術曰：以糲米求糲飯，五之，二而一。

Method: Method: to find coarse cooked rice from coarse rice, multiply by 5 and divide by 2.

[二七] 今有糲飯七斗六升、七分升之四，欲為飧。問得幾何？

[27] Suppose there is 7 *dou* 6 *sheng* plus $\frac{4}{7}$ *sheng* of coarse cooked rice, to be made into cooked meal. Question: how much?

答曰：為飧九斗一升、三十五分升之三十一。

Answer: Makes 9 *dou* 1 *sheng* $\frac{31}{35}$ *sheng* of cooked meal.

術曰：術曰：以糲飯求飧，六之，五而一。

Method: Method: to find cooked meal from coarse cooked rice, multiply by 6 and divide by 5.

[二八] 今有菽一斗，欲為熟菽。問得幾何？

[28] Suppose there is 1 *dou* of beans, to be made into cooked beans. Question: how much?

答曰：為熟菽二斗三升。

Answer: Makes 2 *dou* 3 *sheng* of cooked beans.

術曰：術曰：以菽求熟菽，二十三之，十而一。

Method: Method: to find cooked beans from beans, multiply by 23 and divide by 10.

[二九] 今有菽二斗，欲為豉。問得幾何？

[29] Suppose there is 2 *dou* of beans, to be made into fermented beans. Question: how much?

答曰：為豉二斗八升。

Answer: Makes 2 *dou* 8 *sheng* of fermented beans.

術曰：術曰：以菽求豉，七之，五而一。

Method: Method: to find fermented beans from beans, multiply by 7 and divide by 5.

[三〇] 今有麥八斗六升、七分升之三，欲為小●，問得幾何？

[30] Suppose there is 8 *dou* 6 *sheng* plus $\frac{3}{7}$ *sheng* of wheat, to be made into small ●. Question: how much?

答曰：為小●二斗五升、一十四分升之一十三。

Answer: Makes 2 *dou* 5 *sheng* $\frac{13}{14}$ *sheng* of small ●.

術曰：術曰：以麥求小●，三之，十而一。

Method: Method: to find small ● from wheat, multiply by 3 and divide by 10.

[三一] 今有麥一斗，欲為大●。問得幾何？

[31] Suppose there is 1 *dou* of wheat, to be made into large ●. Question: how much?

答曰：為大●一斗二升。

Answer: Makes 1 *dou* 2 *sheng* of large ●.

術曰：術曰：以麥求大●，六之，五而一。

Method: Method: to find large ● from wheat, multiply by 6 and divide by 5.

[三二] 今有出錢一百六十，買瓴甍十八枚。問枚幾何？

[32] Suppose one pays 160 coins to buy 18 roof tiles (*lingpi*). Question: how much per tile?

答曰：一枚，八錢、九分錢之八。

Answer: 1 tile costs $8\frac{8}{9}$ coins.

[三三] 今有出錢一萬三千五百，買竹二千三百五十箇。問箇幾何？

[33] Suppose one pays 13,500 coins to buy 2,350 bamboo [pieces]. Question: how much per piece?

答曰：一箇，五錢、四十七分錢之三十五。

Answer: 1 piece costs $5\frac{35}{47}$ coins.

術曰：經率術曰：以所買率為法，所出錢數為實，實如法得一錢。

Method: The *jingl'u* (standard rate) method says: take the quantity bought as divisor, the amount of money paid as dividend; divide to obtain the price per unit.

[三四] 今有出錢五千七百八十五，買漆一斛六斗七升、太半升。欲斗率之，問斗幾何？

[34] Suppose one pays 5,785 coins to buy 1 *hu* 6 *dou* 7 *sheng* plus $\frac{2}{3}$ *sheng* of lacquer. One wishes the rate per *dou*; **question:** how much per *dou*?

答曰：一斗，三百四十五錢、五百三分錢之一十五。

Answer: 1 *dou* costs $345\frac{15}{503}$ coins.

[三五] 今有出錢七百二十，買縑一匹二丈一尺。欲丈率之，問丈幾何？

[35] Suppose one pays 720 coins to buy 1 *pi* 2 *zhang* 1 *chi* of silk (*jian*). One wishes the rate per *zhang*; **question:** how much per *zhang*?

答曰：一丈，一百一十八錢、六十一分錢之二。

Answer: 1 *zhang* costs $118\frac{2}{61}$ coins.

[三六] 今有出錢二千三百七十，買布九匹二丈七尺。欲匹率之，問匹幾何？

[36] Suppose one pays 2,370 coins to buy 9 *pi* 2 *zhang* 7 *chi* of cloth. One wishes the rate per *pi*; **question:** how much per *pi*?

答曰：一匹，二百四十四錢、一百二十九分錢之一百二十四。

Answer: 1 *pi* costs $244\frac{124}{129}$ coins.

[三七] 今有出錢一萬三千六百七十，買絲一石二鈞一十七斤。欲石率之，問石幾何？

[37] Suppose one pays 13,670 coins to buy 1 *shi* 2 *jun* 17 *jin* of silk. One wishes the rate per *shi*; **question:** how much per *shi*?

答曰：一石，八千三百二十六錢、一百九十七分錢之一百七十八。

Answer: 1 *shi* costs $8,326\frac{178}{197}$ coins.

術曰：經術術曰：以所求率乘錢數為實，以所買率為法，實如法得一。

Method: The *jing shu* method says: multiply the sought rate by the amount of money as dividend; take the quantity bought as divisor; divide to obtain the answer.

[三八] 今有出錢五百七十六，買竹七十八箇。欲其大小率之，問各幾何？

[38] Suppose one pays 576 coins to buy 78 bamboo pieces, both large and small. One wishes the rate for each size; **question:** how much for each?

答曰：其四十八箇，箇七錢。其三十箇，箇八錢。

Answer: 48 pieces at 7 coins each; 30 pieces at 8 coins each.

[三九] 今有出錢一千一百二十，買絲一石二鈞十八斤。欲其貴賤斤率之，問各幾何？

[39] Suppose one pays 1,120 coins to buy 1 *shi* 2 *jun* 18 *jin* of silk. One wishes to know the expensive and cheap rates per *jin*; **question:** how much for each?

答曰：其二鈞八斤，斤五錢。其一石一十斤，斤六錢。

Answer: 2 jun 8 jin at 5 coins per jin; 1 shi 10 jin at 6 coins per jin.

術曰：其率術曰：各置所買石、鈞、斤、兩以為法，以所率乘錢數為實，實如法而一。不滿法者反以實減法，法賤實貴。

Method: The *qil"u* (rate-finding) method says: place the quantities bought in *shi*, *jun*, *jin*, *liang* as divisors; multiply the rate by the amount of money as dividend; divide. If [the dividend] is less than the divisor, subtract the dividend from the divisor instead—the divisor gives the cheap rate and the dividend gives the expensive rate.

[四〇]–[四三] 今有出錢一萬三千九百七十，買絲一石二鈞二十八斤三兩五銖。欲其貴賤石率之、鈞率之、斤率之、兩率之。

[40]–[43] Suppose one pays 13,970 coins to buy 1 shi 2 jun 28 jin 3 liang 5 zhu of silk. One wishes to know the expensive and cheap rates per *shi*, per *jun*, per *jin*, and per *liang*.

[四四] 今有出錢一萬三千九百七十，買絲一石二鈞二十八斤三兩五銖。欲其貴賤銖率之。

[44] Suppose one pays 13,970 coins to buy 1 shi 2 jun 28 jin 3 liang 5 zhu of silk. One wishes to know the expensive and cheap rates per *zhu*.

[四五] 今有出錢六百一十，買羽二千一百翮。欲其貴賤率之，問各幾何？

[45] Suppose one pays 610 coins to buy 2,100 hou of feathers. One wishes to know the expensive and cheap rates; question: how much for each?

答曰：其一千一百四十翮，三翮一錢。其九百六十翮，四翮一錢。

Answer: 1,140 hou at 3 hou per coin; 960 hou at 4 hou per coin.

[四六] 今有出錢九百八十，買矢筈五千八百二十枚。欲其貴賤率之，問各幾何？

[46] Suppose one pays 980 coins to buy 5,820 arrow shafts. One wishes to know the expensive and cheap rates; question: how much for each?

答曰：其三百枚，五枚一錢。其五千五百二十枚，六枚一錢。

Answer: 300 shafts at 5 per coin; 5,520 shafts at 6 per coin.

術曰：反其率術曰：以錢數為法，所率為實，實如法而一。不滿法者反以實減法，法少，實多。二物各以所得多少之數乘法實，即物數。

Method: The *fan qil"u* (inverse rate) method says: take the amount of money as divisor, the rate as dividend; divide. If [the dividend] is less than the divisor, subtract the dividend from the divisor instead—the divisor [rate] is smaller [quantity], the dividend is larger [quantity]. For two items, multiply the divisor and dividend by the respective [rate] numbers to obtain the quantities.

3 卷第三：衰分 / Volume 3: Proportional Distribution

衰分以御貴賤粟稅之率 / “Proportional Distribution is for governing the rates of grain taxes according to rank.”

原典 Classical Chinese

English Translation

本章論比例分配之法，包括按等級、按數量、按比率之分配問題。衰分術為本章之核心算法。

This chapter discusses methods of proportional distribution, including distribution by rank, by quantity, and by rate. The *cui fen* (proportional distribution) method is the core algorithm of this chapter.

術曰：衰分術曰：各置列衰，副并為法，以所分乘未并者各自為實，實如法而一。不滿法者，以法命之。

Method: The *cui fen* method says: place each proportional rate [*lie cui*] in a column; combine the auxiliary [copies] as divisor; multiply the amount to be distributed by each uncombined [rate], each becoming its own dividend; divide each dividend by the divisor. If less than the divisor, name [the remainder] by the divisor.

[一] 今有大夫、不更、簪裹、上造、公士，凡五人，共獵得五鹿。欲以爵次分之，問各得幾何？

[1] Suppose there are a Daifu, Bugeng, Zanniao, Shangzao, and Gongshi—five persons in total—who together hunted 5 deer. One wishes to distribute them by noble rank; **question:** how much does each get?

答曰：大夫得一鹿、三分鹿之二。不更得一鹿、三分鹿之一。簪裹得一鹿。上造得三分鹿之二。公士得三分鹿之一。

Answer: Daifu gets $1\frac{2}{3}$ deer; Bugeng gets $1\frac{1}{3}$ deer; Zanniao gets 1 deer; Shangzao gets $\frac{2}{3}$ deer; Gongshi gets $\frac{1}{3}$ deer.

術曰：術曰：列置爵數，各自為衰，副并為法。以五鹿乘未并者，各自為實。實如法得一鹿。

Method: Method: place the rank numbers in a column, each becoming its own proportional rate; combine as divisor. Multiply the 5 deer by each uncombined [rate], each becoming its own dividend. Divide to obtain the number of deer.

[二] 今有牛、馬、羊食人苗。苗主責之粟五斗。羊主曰：「我羊食半馬。」馬主曰：「我馬食半牛。」今欲衰償之，問各出幾何？

[2] Suppose a cow, a horse, and a sheep ate someone's seedlings. The seedling owner demands 5 *dou* of millet as compensation. The sheep owner says: "My sheep ate half what the horse ate." The horse owner says: "My horse ate half what the cow ate." One wishes to distribute proportionally the compensation; **question:** how much does each pay?

答曰：牛主出二斗八升、七分升之四。馬主出一斗四升、七分升之二。羊主出七升、七分升之一。

Answer: Cow owner pays 2 *dou* 8 *sheng* $\frac{4}{7}$ *sheng*; horse owner pays 1 *dou* 4 *sheng* $\frac{2}{7}$ *sheng*; sheep owner pays 7 $\frac{1}{7}$ *sheng*.

術曰：術曰：置牛四、馬二、羊一，各自為列衰，副并為法。以五斗乘未并者各自為實。實如法得一斗。

Method: Method: place cow=4, horse=2, sheep=1, each as its own proportional rate; combine as divisor. Multiply 5 *dou* by each uncombined [rate] as dividend. Divide to obtain the number of *dou*.

[三] 今有甲持錢五百六十，乙持錢三百五十，丙持錢一百八十，凡三人俱出關，關稅百錢。欲以錢數多少衰出之，問各幾何？

[3] Suppose A holds 560 coins, B holds 350 coins, C holds 180 coins. All three pass through a customs gate together; the customs tax is 100 coins. One wishes to distribute the tax proportionally to their holdings; **question:** how much does each pay?

答曰：甲出五十一錢、一百九分錢之四十一。乙出三十二錢、一百九分錢之一十二。丙出一十六錢、一百九分錢之五十六。

Answer: A pays $51\frac{41}{109}$ coins; B pays $32\frac{12}{109}$ coins; C pays $16\frac{56}{109}$ coins.

術曰：術曰：各置錢數為列衰，副并為法，以百錢乘未并者，各自為實，實如法得一錢。

Method: Method: place each person's coin holdings as proportional rates; combine as divisor. Multiply 100 coins by each uncombined [rate] as dividend. Divide to obtain the number of coins.

[四] 今有女子善織，日自倍，五日織五尺。問日織幾何？

[4] Suppose there is a woman skilled at weaving; each day she doubles [her output]. In five days she weaves 5 *chi*. Question: how much does she weave each day?

答曰：初日織一寸、三十一分寸之十九。次日織三寸、三十一分寸之七。次日織六寸、三十一分寸之十四。次日織一尺二寸、三十一分寸之二十八。次日織二尺五寸、三十一分寸之二十五。

Answer: Day 1: $1\frac{19}{31}$ *cun*; Day 2: $3\frac{7}{31}$ *cun*; Day 3: $6\frac{14}{31}$ *cun*; Day 4: $1\text{chi}2\text{cun}\frac{28}{31}$ *cun*; Day 5: $2\text{chi}5\text{cun}\frac{25}{31}$ *cun*.

術曰：術曰：置一、二、四、八、十六為列衰，副并為法，以五尺乘未并者，各自為實，實如法得一尺。

Method: Method: place 1, 2, 4, 8, 16 as proportional rates; combine as divisor. Multiply 5 *chi* by each uncombined [rate] as dividend. Divide to obtain the number of *chi*.

[五] 今有北鄉算八千七百五十八，西鄉算七千二百三十六，南鄉算八千三百五十六，凡三鄉，發徭三百七十八人。欲以算數多少衰出之，問各幾何？

[5] Suppose Northern district has 8,758 households [counted for tax], Western district has 7,236, Southern district has 8,356. All three districts together must supply 378 corvée laborers. One wishes to distribute proportionally by household count; question: how many from each?

答曰：北鄉遣一百三十五人、一萬二千一百七十五分人之一萬一千六百三十七。西鄉遣一百一十二人、一萬二千一百七十五分人之四千四。南鄉遣一百二十九人、一萬二千一百七十五分人之八千七百九。

Answer: Northern district sends $135\frac{11637}{12175}$ persons; Western district sends $112\frac{4004}{12175}$ persons; Southern district sends $129\frac{8709}{12175}$ persons.

術曰：術曰：各置算數為列衰，副并為法，以所發徭人數乘未并者，各自為實，實如法得一人。

Method: Method: place each district's household count as proportional rates; combine as divisor. Multiply the number of corvée laborers by each uncombined [rate] as dividend. Divide to obtain the number of persons.

[六] 今有稟粟，大夫、不更、簪裹、上造、公士，凡五人，一十五斗。今有大夫一人後來，亦當稟五斗。倉無粟，欲以衰出之，問各幾何？

[6] Suppose there is a distribution of millet: Daifu, Bugeng, Zanniao, Shangzao, and Gongshi—five persons—share 15 *dou*. Now another Daifu arrives late, and should also receive 5 *dou*. The granary has no more millet; one wishes to collect proportionally [from the five]; question: how much does each give?

答曰：大夫出一斗、四分斗之一。不更出一斗。簪裹出四分斗之三。上造出四分斗之二。公士出四分斗之一。

Answer: Daifu gives $1\frac{1}{4}$ *dou*; Bugeng gives 1 *dou*; Zanniao gives $\frac{3}{4}$ *dou*; Shangzao gives $\frac{2}{4}$ *dou*; Gongshi gives $\frac{1}{4}$ *dou*.

術曰：術曰：各置所稟粟斛斗數，爵次均之，以為列衰，副并而加後來大夫亦五斗，得二十以為法。以五斗乘未并者各自為實。實如法得一斗。

Method: Method: place each person's received millet, equalized by rank, as proportional rates; combine and add the late-arriving Daifu's 5 *dou*, obtaining 20 as divisor. Multiply 5 *dou* by each uncombined [rate] as dividend. Divide to obtain the number of *dou*.

[七] 今有粟五斛，五人分之，欲令三人得三，二人得二。問各幾何？

[7] Suppose there are 5 *hu* of millet to be distributed among 5 people, such that 3 people get [a share rated at] 3 and 2 people get [a share rated at] 2. **Question:** how much does each get?

答曰：三人，人得一斛一斗五升、十三分升之五。二人，人得七斗六升、十三分升之十二。

Answer: The 3 people: each gets 1 *hu* 1 *dou* 5 *sheng* $\frac{5}{13}$ *sheng*; the 2 people: each gets 7 *dou* 6 *sheng* $\frac{12}{13}$ *sheng*.

術曰：術曰：置三人，人三；二人，人二，為列衰。副并為法。以五斛乘未并者，各自為實。實如法得一斛。

Method: Method: place 3 persons at rate 3 each, and 2 persons at rate 2 each, as proportional rates. Combine as divisor. Multiply 5 *hu* by each uncombined [rate] as dividend. Divide to obtain the number of *hu*.

術曰：返衰術曰：列置衰而令相乘，動者為不動者衰。

Method: The *fancui* (inverse proportional distribution) method says: place the rates in a column and let them multiply each other; the one that changes becomes the rate for the one that does not change.

[八] 今有大夫、不更、簪裹、上造、公士，凡五人，共出百錢。欲令高爵出少，以次漸多，問各幾何？

[8] Suppose there are Daifu, Bugeng, Zanniao, Shangzao, and Gongshi—five persons—who together contribute 100 coins. One wishes that the higher rank pays less, increasing by rank; **question:** how much does each pay?

答曰：大夫出八錢、一百三十七分錢之一百四。不更出一十錢、一百三十七分錢之一百三十。簪裹出一十四錢、一百三十七分錢之八十二。上造出二十一錢、一百三十七分錢之一百二十三。公士出四十三錢、一百三十七分錢之一百九。

Answer: Daifu pays $8\frac{104}{137}$ coins; Bugeng pays $10\frac{130}{137}$ coins; Zanniao pays $14\frac{82}{137}$ coins; Shangzao pays $21\frac{123}{137}$ coins; Gongshi pays $43\frac{109}{137}$ coins.

術曰：術曰：置爵數各自為衰，而返衰之，副并為法。以百錢乘未并者各自為實。實如法得一錢。

Method: Method: place the rank numbers as proportional rates, and invert them [take reciprocals]; combine as divisor. Multiply 100 coins by each uncombined [rate] as dividend. Divide to obtain the number of coins.

[九] 今有甲持粟三升，乙持糲米三升，丙持糲飯三升。欲令合而分之，問各幾何？

[9] Suppose A holds 3 *sheng* of millet, B holds 3 *sheng* of coarse rice, C holds 3 *sheng* of coarse cooked rice. One wishes to combine and redistribute [equally]; **question:** how much does each get?

答曰：甲二升、一十分升之七。乙四升、一十分升之五。丙一升、一十分升之八。

Answer: A gets $2\frac{7}{10}$ *sheng*; B gets $4\frac{5}{10}$ *sheng*; C gets $1\frac{8}{10}$ *sheng*.

術曰：術曰：以粟率五十、糲米率三十、糲飯率七十五為衰、而返衰之，副并為法。以九升乘未并者各自為實。實如法得一升。

Method: Method: take the millet rate 50, coarse rice rate 30, coarse cooked rice rate 75 as proportional rates, and invert them; combine as divisor. Multiply 9 *sheng* by each uncombined [rate] as dividend. Divide to obtain the number of *sheng*.

[一〇] 今有絲一斤，價直二百四十。今有錢一千三百二十八，問得絲幾何？

[10] Suppose 1 *jin* of silk is worth 240 coins. Suppose one has 1,328 coins. **Question:** how much silk does one get?

答曰：五斤八兩一十二銖、三分銖之四。

Answer: 5 jin 8 liang 12 zhu plus $\frac{4}{3}$ zhu.

術曰：術曰：以一斤價數為法，以一斤乘今有錢數為實，實如法得絲數。

Method: Method: take the price per jin as divisor; multiply 1 jin by the amount of money as dividend; divide to obtain the amount of silk.

[一一] 今有絲一斤價直三百四十三。今有絲七兩一十二銖，問得錢幾何？

[11] Suppose 1 jin of silk is worth 343 coins. Suppose one has 7 liang 12 zhu of silk. Question: how much money does one get?

答曰：一百六十一錢、三十二分錢之二十三。

Answer: 161 $\frac{23}{32}$ coins.

術曰：術曰：以一斤銖數為法，以一斤價數，乘七兩一十二銖為實。實如法得錢數。

Method: Method: take the number of zhu in 1 jin as divisor; multiply the price per jin by 7 liang 12 zhu as dividend. Divide to obtain the amount of money.

[一二] 今有縑一丈價直一百二十六。今有縑一匹九尺五寸，問得錢幾何？

[12] Suppose 1 zhang of silk (jian) is worth 126 coins. Suppose one has 1 pi 9 chi 5 cun of jian. Question: how much money?

答曰：六百三十三錢、五分錢之三。

Answer: 633 $\frac{3}{5}$ coins.

術曰：術曰：以一丈寸數為法，以價錢數乘今有縑寸數為實，實如法得錢數。

Method: Method: take the number of cun in 1 zhang as divisor; multiply the price by the number of cun of jian held as dividend. Divide to obtain the amount of money.

[一三] 今有布一匹，價直一百二十三。今有布二丈七尺，問得錢幾何？

[13] Suppose 1 pi of cloth is worth 123 coins. Suppose one has 2 zhang 7 chi of cloth. Question: how much money?

答曰：八十四錢、分錢之三。

Answer: 84 $\frac{3}{\text{missing}}$ coins.

術曰：術曰：以一匹尺數為法，今有布尺數乘價錢為實，實如法得錢數。

Method: Method: take the number of chi in 1 pi as divisor; multiply the chi of cloth held by the price as dividend. Divide to obtain the amount of money.

[一四] 今有素一匹一丈，價直六百二十五。今有錢五百，問得素幾何？

[14] Suppose 1 pi 1 zhang of plain silk is worth 625 coins. Suppose one has 500 coins. Question: how much plain silk does one get?

術曰：術曰：以價直為法，以一匹一丈尺數乘今有錢數為實。實如法得素數。

Method: Method: take the price as divisor; multiply the number of chi in 1 pi 1 zhang by the amount of money as dividend. Divide to obtain the amount of plain silk.

[一五] 今有與人絲一十四斤，約得縑一十斤。今與人絲四十五斤八兩，問得縑幾何？

[15] Suppose one gives 14 *jin* of silk and agrees to receive 10 *jin* of *jian* silk. Now one gives 45 *jin* 8 *liang* of silk. Question: how much *jian* does one receive?

答曰：三十二斤八兩。

Answer: 32 *jin* 8 *liang*.

術曰：術曰：以一十四斤兩數為法，以一十斤乘今有絲兩數為實，實如法得縑數。

Method: Method: take the number of *liang* in 14 *jin* as divisor; multiply 10 *jin* by the number of *liang* of silk held as dividend. Divide to obtain the amount of *jian*.

[一六] 今有絲一斤，耗七兩。今有絲二十三斤五兩，問耗幾何？

[16] Suppose 1 *jin* of silk loses 7 *liang* [in processing]. Suppose one has 23 *jin* 5 *liang* of silk. Question: how much is lost?

答曰：一百六十三兩四銖半。

Answer: 163 *liang* 4.5 *zhu*.

術曰：術曰：以一斤展十六兩為法，以七兩乘今有絲兩數為實，實如法得耗數。

Method: Method: take 1 *jin* expanded to 16 *liang* as divisor; multiply 7 *liang* by the number of *liang* of silk held as dividend. Divide to obtain the amount lost.

[一七] 今有生絲三十斤，乾之，耗三斤十二兩。今有乾絲一十二斤，問生絲幾何？

[17] Suppose 30 *jin* of raw silk, when dried, loses 3 *jin* 12 *liang*. Suppose one has 12 *jin* of dried silk. Question: how much raw silk?

答曰：一十三斤一十一兩十銖、七分銖之二。

Answer: 13 *jin* 11 *liang* 10 $\frac{2}{7}$ *zhu* of raw silk.

術曰：術曰：置生絲兩數，除耗數，餘，以為法。三十斤乘乾絲兩數為實。實如法得生絲數。

Method: Method: place the number of *liang* of raw silk, subtract the loss, and take the remainder as divisor. Multiply 30 *jin* by the number of *liang* of dried silk as dividend. Divide to obtain the amount of raw silk.

[一八] 今有田一畝，收粟六升、太半升。今有田一頃二十六畝一百五十九步，問收粟幾何？

[18] Suppose 1 *mu* of field yields 6 $\frac{2}{3}$ *sheng* of millet. Suppose one has 1 *qing* 26 *mu* 159 *bu* of field. Question: how much millet is harvested?

答曰：八斛四斗四升、一十二分升之五。

Answer: 8 *hu* 4 *dou* 4 *sheng* plus $\frac{5}{12}$ *sheng*.

術曰：術曰：以畝二百四十步為法，以六升、太半升乘今有田積步為實，實如法得粟數。

Method: Method: take 1 *mu* = 240 square *bu* as divisor; multiply 6 $\frac{2}{3}$ *sheng* by the total area in square *bu* as dividend. Divide to obtain the amount of millet.

[一九] 今有取保一歲，價錢二千五百。今先取一千二百，問當作日幾何？

[19] Suppose a one-year service contract is worth 2,500 coins. Now one takes 1,200 coins in advance. Question: how many days of service are owed?

答曰：一百六十九日、二十五分日之二十三。

Answer: $169\frac{23}{25}$ days.

術曰：術曰：以價錢為法，以一歲三百五十四日乘先取錢數為實，實如法得日數。

Method: Method: take the contract price as divisor; multiply 1 year = 354 days by the amount taken in advance as dividend. Divide to obtain the number of days.

[二〇] 今有貸人千錢，月息三十。今有貸人七百五十錢，九日歸之，問息幾何？

[20] Suppose one lends 1,000 coins, with monthly interest of 30 coins. Suppose one lends 750 coins, which are returned after 9 days. Question: how much interest?

答曰：六錢、四分錢之三。

Answer: $6\frac{3}{4}$ coins interest.

術曰：術曰：以月三十日，乘千錢為法。以息三十乘今所貸錢數，又以九日乘之，為實。實如法得一錢。

Method: Method: multiply 30 days by 1,000 coins as divisor. Multiply the interest rate 30 by the amount lent and also by 9 days as dividend. Divide to obtain the number of coins.

(以上為《九章算術》卷第一至卷第三之全文，含劉徽注及李淳風等注釋。/ This completes the Nine Chapters on the Mathematical Art, Volumes 1–3, including Liu Hui’s commentary and annotations by Li Chunfeng et al.)

Editorial Notes / 譯者注:

1. The technical terms *shi* (實, “dividend”) and *fa* (法, “divisor”) are central to Chinese algorithmic mathematics. *Shi* represents the quantity to be operated upon; *fa* is the standard or rule by which it is divided.
2. The *jinyou* (今有, “suppose there is”) method is equivalent to the modern Rule of Three: $\text{result} = \frac{\text{given quantity} \times \text{sought rate}}{\text{given rate}}$.
3. Liu Hui’s commentary on circular areas includes the earliest known systematic treatment of the exhaustion method for approximating π , predating European methods by over a millennium.
4. The “inscribed hexagon” (六觚) method yields $\pi \approx 3$; Liu Hui refined this to $\pi \approx \frac{157}{50} = 3.14$ and proposed the even more precise $\frac{3927}{1250} = 3.1416$.
5. Traditional Chinese units: 1 *bu* (步) $\approx 1.6\text{m}$; 1 *mu* (畝) = 240 sq. *bu*; 1 *qing* (頃) = 100 *mu*; 1 *chi* (尺) $\approx 23\text{cm}$; 1 *jin* (斤) = 16 *liang* (兩) = 384 *zhu* (銖).
6. The five noble ranks in Problem 1 correspond to the Qin/Han twenty-grade system: Daifu (大夫, rank 5), Bugeng (不更, rank 4), Zanniao (簪裹, rank 3), Shangzao (上造, rank 2), Gongshi (公士, rank 1).

九章算術

Jiuzhang Suanshu

Bilingual Edition
漢英對照版

Fascicles 4–6
卷四至卷六

Nine Chapters on the Mathematical Art

With Commentary by Liu Hui (劉徽, 263 CE)

Sub-commentary by Li Chunfeng (唐李淳風注釋)

Vol. 4 —Shaoguang (少廣): Root extraction, sphere volume

Vol. 5 —Shanggong (商功): Volumes of earthworks and solids

Vol. 6 —Junshu (均輸): Weighted distribution, work-rate problems

Original Chinese text with English translation

Chinese text: Left column English translation: Right column

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Introduction 引言

《九章算術》是中國最重要的古代數學典籍，編纂於漢代（公元前 202 年—公元 220 年）。全書分為九卷，每卷解決一類實際數學問題。

劉徽（約 225—295 年）於 263 年完成的注釋是數學史上最輝煌的著作之一。劉徽不僅解釋了算法，還運用出入相補的幾何方法給出了嚴格的證明，並以正 3072 邊形推算圓周率近似值。

唐代李淳風（602—670 年）奉詔率領學者團隊進一步疏解校正原文。

本文檔呈現卷四至卷六：

The commentary by **Liu Hui** (c. 225–295 CE), completed in **263 CE**, is one of the most brilliant works in the history of mathematics. Liu Hui not only explained the algorithms but also provided rigorous proofs using the method of geometric dissection (出入相補), and famously computed an approximation of π using inscribed polygons with up to 3072 sides.

The sub-commentary by **Li Chunfeng** (602–670 CE) and his Tang court team further clarified and corrected the text.

This document presents **Fascicles 4–6**:

Vol.	Title	Chinese	Subject Matter
4	Shaoguang	少廣	Root extraction, finding dimensions given area/volume, sphere volume
5	Shanggong	商功	Volumes of earthworks, walls, dykes, granaries, geometric solids
6	Junshu	均輸	Equitable taxation, weighted distribution, travel problems

每題依古典結構：

- 問：題設
- 答：答案
- 術：算法
- 注：劉徽注釋

Each problem follows the classical structure:

- Wen** (問): The problem statement
- Da** (答): The answer
- Shu** (術): The method/algorithm
- Zhu** (注): Liu Hui’s commentary

1 Volume 4: Shaoguang (Lesser Breadth) 卷四：少廣

九章算術卷四 · Shaoguang (少廣)

少廣以御積冪方圓

Shaoguang: For governing areas, powers, squares, and circles

本章少廣處理已知矩形面積及部分邊長求另一邊長的問題，推廣至開平方、開立方，以及著名的球體積計算。標題意指由「廣」（面積/體積）求「少」（較小之邊長）。

The chapter **Shaoguang** (“Lesser Breadth”) deals with finding one dimension of a rectangle given its area and other dimensions. It extends to root extraction (square and cube roots) and the famous problem of computing the volume of a sphere. The title refers to finding a “lesser” (smaller) dimension given a “greater” (area/volume).

1.1 Opening Method: The Shaoguang Algorithm 少廣術

少廣術 · 少廣算法

置全步及分母子，以最下分母遍乘諸分子及全步。各以其母除其子，置之於左。命通分，內子。又以分母偏乘諸分子，及已通者，皆通而同之，并之為法。置所求步數，以全步積分乘之，為實。實如法而一，得從步。

Method of Lesser Breadth · Shaoguang Algorithm

Place the whole steps and the denominators and numerators of fractions. Multiply all numerators and whole steps by the lowest denominator. For each, divide its numerator by its denominator, and place on the left. Call this “passing through fractions, including numerators.” Again multiply all numerators by the denominator, together with those already passed through, all are passed through and unified. Add them to make the divisor. Place the desired number of steps, multiply by the accumulated whole-step fractions to make the dividend. Divide by the divisor to obtain the length in steps.

劉徽注 · 劉徽論少廣

此以田廣為法，以畝積步為實。術曰置全步及分母子者，通分而齊其子也。齊其子，則母同其母。以母乘全步者，通其分也。以母偏乘諸分子，各以其母除其子，置之於左。命通分內子者，通而同之也。

Liu Hui's Commentary · On the Shaoguang Algorithm

Here the field width is taken as divisor, and the mu-area in square steps as dividend. “Place whole steps and fraction denominators/numerators” means passing through fractions and aligning their numerators. Aligning numerators makes denominators common. Multiplying whole steps by the denominator passes through the fractions. Multiplying all numerators by the denominator, each divided by its own denominator, placed on the left — “passing through fractions, including numerators” means passing through and unifying them.

1.2 Problem 1: Finding the Side from Area with Fractional Parts 第一問

問 · 問題一

有田，廣一步半。求田一畝，問從幾何？

Problem 1 • Finding the Length

Suppose there is a field, with width one and a half 步. If one 畝 of field is desired, how long is the length?

答

答曰：一百六十步。

Answer

Answer: 160 步.

術

術曰：下有半，是二分之一。以一為二，半為一，并之得三，為法。置田二百四十步，亦以一為二乘之，為實。實如法得從步。

Method

Method: The lower [fraction] is a half, which is $\frac{1}{2}$. Take one as two, the half as one; adding gives three, which is the divisor. Place the field at 240 square steps; likewise multiply by two (taking one as two) to make the dividend. Divide to obtain the length in steps.

劉徽注

此以廣為法，以畝積步為實。術曰下有半是二分之一者，廣一步半，其半即二分之一也。以一為二者，通分也。半為一者，二分之一即一也。并之得三為法者，分母二通全步一為二，分子一，共三也。

Liu Hui's Commentary

Here the width is divisor, the mu-area in steps is dividend. “The lower is a half, which is $\frac{1}{2}$ ”: the width is one and a half steps, its half being $\frac{1}{2}$. “Take one as two”: passing through fractions. “Half as one”: $\frac{1}{2}$ becomes one. “Adding gives three as divisor”: denominator 2 passes whole step 1 to 2, numerator 1, total 3.

1.3 Problem on Square Root Extraction 開方問**問・開方**

有積五萬五千二百二十五步。問為方幾何？

Problem • Square Root Extraction

Suppose there is an area of 55,225 square 步. What is the side of the square?

答

答曰：二百三十五步。

Answer

Answer: 235 步.

開方術 · 開平方算法

術曰：置積為實。借一算，步之，超二等。議所得，以一乘所借一算為法，而以除。除已，倍法為定法。其復除，折法而下。復置借算，步之如初。以復議一乘之，所得副，以加定法，以除。以所得副從定法。復除，折下如前。

Method of Root Extraction · Square Root Algorithm

Method: Place the area as dividend. Borrow one counting rod, step it, skip two places. Estimate the result; multiply by one, using the borrowed rod as divisor, and divide. After division, double the divisor to make the fixed divisor. For continued division, fold the divisor down. Again place the borrowed rod, step as before. Again estimate, multiply by one; the obtained result is auxiliary, add it to the fixed divisor, and divide. The auxiliary result joins the fixed divisor. Continue dividing, folding down as before.

劉徽注 · 劉徽論開方

開方術者，求方冪之一面也。凡冪，方圓之本也。方冪者，方之積也。開方者，求方之面數。言百之面十，萬之面百。術曰置積為實者，積即方冪也。借一算步之者，假以定位也。超二等者，方十自乘，冪一百；方百自乘，冪一萬。故超兩位也。

Liu Hui's Commentary · On Root Extraction

The method of root extraction seeks one side of a square power. All powers are the root of squares and circles. Square power is the area of a square. Root extraction seeks the number of the square's side. The side of 100 is 10; of 10,000 is 100. "Place area as dividend": the area is the square power. "Borrow one rod, step it": borrowed for positioning. "Skip two places": 10 squared is 100; 100 squared is 10,000. Hence skip two positions.

1.4 The Sphere Volume Problem 立圓問

此為《九章算術》中最著名的問題之一，劉徽深入分析，為後世祖沖之、祖暅父子奠定基礎。

This is one of the most famous problems in the *Jiuzhang Suanshu*, which Liu Hui analyzed in depth, setting the foundation for later work by Zu Chongzhi and his son Zu Geng.

問 · 立圓

有立圓，徑四尺。問為積幾何？

Problem · The Sphere Volume

Suppose there is a solid sphere, with diameter 4 尺. What is its volume?

答

答曰：九尺五寸八分之二。

Answer

Answer: 9 尺 5 寸 $\frac{1}{8}$ (using the ancient formula $V = \frac{9}{16}d^3$).

術

術曰：圓徑自乘，九之，十六而一。開立圓術：置圓徑四尺，自乘，得一十六尺。九之，得一百四十四尺。十六而一，得九尺。餘十六分尺之八，約之為八分之二。并之，為積。

Method

Method: Square the diameter, multiply by 9, divide by 16. To find the sphere volume: place diameter 4 尺, square it to get 16 尺. Multiply by 9 to get 144 尺. Divide by 16 to get 9 尺. Remainder $\frac{8}{16}$ 尺, reduced to $\frac{1}{8}$ 尺. Add to obtain the volume.

劉徽注 · 劉徽論立圓

按此術，圓徑自乘者，先得圓冪。九之十六而一者，以圓冪為方冪，九之十六而一，即圓積也。然此術以圓徑為方徑，誤也。何以驗之？取立方棋八枚，皆令立方一寸，積之為立方二寸。規之為圓，徑二寸。又橫規之，徑亦二寸。然後并立圓棋二枚，規之為圓，徑二寸。又橫規之，徑亦二寸。是為圓積也。

徽術曰：立方徑自乘，又以圓周率三乘之，十二而一，得圓積。此圓冪之率也。又以此圓積乘高，得圓柱積。立圓者，圓柱之半也。故又以圓周率三乘之，十二而一，得立圓積。此術未得其理。

牟合方蓋者，方率也。丸居牟合方蓋，則圓率也。按合蓋者，方率也。其中則圓周率也。方圓相纏，周徑相直。若設方率為四，圓周率為三。以圓周率乘方冪，開方除之，即徑。以方率乘圓冪，開方除之，即周。此圓方之率也。

Liu Hui's Commentary · On the Sphere

According to this method, squaring the diameter first obtains the circular power. Multiplying by 9 and dividing by 16, taking the circular power as square power, gives the circular area. But this method takes the circle's diameter as the square's diameter — this is erroneous. How to verify? Take 8 cubic blocks, each 1 cubic cun, totaling 2 cubic cun. Cut into a circle, diameter 2 cun. Cut crosswise, diameter also 2 cun. Then combine 2 sphere-blocks, cut into a circle, diameter 2 cun. Cut crosswise, diameter also 2 cun. This is the circular volume.

Liu Hui's method: Square the cube diameter, multiply by the circular ratio 3, divide by 12, obtaining the circular area. This is the circular power ratio. Multiply this circular area by the height to obtain the cylinder volume. A sphere is half a cylinder. So again multiply by circular ratio 3, divide by 12, to obtain the sphere volume. This method has not yet found the true principle.

The mouhe fangai (double-lid umbrella) has the square ratio. The sphere contained within the mouhe fangai has the circular ratio.

The fangai is the square ratio; within it is the circular ratio. Square and circle intertwine, circumference and diameter correspond.

If square ratio is 4 and circular ratio is 3, multiplying the square power by the circular ratio and extracting the square root gives the diameter. This is the ratio of circle and square.

譯注：劉徽在此指出古代公式 $V = \frac{9}{16}d^3$ 的謬誤（暗示 $\pi = 3$ 的特定配置）。他提出**牟合方蓋**（兩垂直圓柱相交體）的概念，作為求球體積的關鍵。他指出球內切於此立體，但未能完成計算。這一工作後由五世紀的**祖暅**完成，證明牟合方蓋體積為外切立方體的 $\frac{2}{3}$ ，從而得到正確公式 $V = \frac{\pi}{6}d^3$ 。

Historical note: Liu Hui here identifies the error in the ancient formula $V = \frac{9}{16}d^3$. He introduces the concept of **mouhe fangai** (“double-lid umbrella”), the intersection of two perpendicular cylinders, as the key to finding the sphere volume. He correctly states that the sphere is inscribed in this solid, but was unable to complete the computation. This was later accomplished by **Zu Geng** in the 5th century, who proved that the volume of the mouhe fangai is $\frac{2}{3}$ of the circumscribed cube, leading to the correct formula $V = \frac{\pi}{6}d^3$.

2 Volume 5: Shanggong (Construction Consultations) 卷五：商功

九章算術卷五 · Shanggong (商功)

商功以御功程積實
Shanggong: For governing labor, engineering, and solid volumes

本章**商功**處理工程營建中各類立體的體積計算：土方工程、城牆、堤壩、溝渠、糧倉，以及豐富的幾何立體。劉徽注釋在此尤為詳盡，以割補法提供幾何證明。

The chapter **Shanggong** (“Construction Consultations”) deals with computing the volumes of various solids arising in engineering and construction: earthworks, city walls, dykes, ditches, canals, granaries, and a rich collection of geometric solids. Liu Hui’s commentary is particularly extensive here, providing geometric proofs using dissection methods.

2.1 Classification of Earthwork Solids 穿地率

商功術 · 土方率

穿地四，為壤五，為堅三。墟謂穿坑，此皆其常率。

Method · Earthwork Conversion Rates

For excavated earth: 4 units [loose] become 5 units [piled], or 3 units [compacted]. The “ruins” refer to excavated pits — these are standard rates.

劉徽注

以穿地求壤，五之；求堅，三之。皆四而一。以壤求穿，四之；求堅，三之。皆五而一。以堅求穿，四之；求壤，五之。皆三而一。此術皆其常率也。穿地四為壤五者，掘地四尺，出壤五尺，壤虛而堅實也。

Liu Hui’s Commentary

From excavated to piled: multiply by 5; to compacted: multiply by 3. All divide by 4. From piled to excavated: multiply by 4; to compacted: multiply by 3. All divide by 5. From compacted to excavated: multiply by 4; to piled: multiply by 5. All divide by 3. These are standard rates. “Excavated 4 becomes piled 5”: digging 4 chi of earth yields 5 chi of loose soil, since loose earth is voluminous and compacted earth is dense.

2.2 Problem: Volume of a City Wall 城垣問

問 · 城垣

有城，下廣四丈，上廣二丈，高五丈，袤一百二十六丈。問積幾何？

Problem · City Wall Volume

Suppose there is a city wall: lower width 4 丈, upper width 2 丈, height 5 丈, length 126 丈. What is the volume?

答

答曰：一百八十九萬尺。

Answer

Answer: 1,890,000 cubic 尺.

術

術曰：并上下廣而半之，以高若深乘之，又以袤乘之，即積尺。

Method

Method: Add the upper and lower widths and halve; multiply by the height or depth; then multiply by the length to obtain the volume in cubic 尺.

劉徽注

按此術，并上下廣而半之者，以盈補虛，得中平之廣。以高乘之，得截而為直棱之積。又以袤乘之者，廣袤相乘為一截之積，以高乘之為積。

Liu Hui's Commentary

Adding the widths and halving gives the “middle width” by the principle of compensating excess with deficiency (以盈補虛). Multiplying by height gives the cross-sectional area as a right prism. Multiplying by length: width times length is the cross-sectional area, times height gives the volume.

2.3 Problem: Volume of a Canal with Sloped Sides 塹堵問

問・塹堵

有塹，上廣一丈六尺，下廣一丈，深六尺，袤五丈七尺。問積幾何？

Problem • Trench/Canal

Suppose there is a trench: upper width 1 丈 6 尺, lower width 1 丈, depth 6 尺, length 5 丈 7 尺. What is the volume?

答

答曰：積七千一百一十二尺。

Answer

Answer: Volume 7,112 cubic 尺.

術

術曰：并上下廣，半之，以高若深乘之，又以袤乘之，即積尺。

Method

Method: Add the upper and lower widths and halve; multiply by the height or depth; then multiply by the length to obtain the volume in cubic 尺.

2.4 The Yangma (陽馬) and Bienao (鱉臑) 陽馬與鱉臑

此為中國數學中最重要的幾何立體之一，由長方體分割而來。

These are among the most important geometric solids in Chinese mathematics, arising from the dissection of a rectangular prism.

陽馬術 · 陽馬 (隅角錐)

術曰：廣袤相乘，以高乘之，三而一。

Yangma Method · Corner Pyramid

Method: Multiply width by length, multiply by height, divide by 3.

此為陽馬 (yangma) 體積公式，一種底面為矩形且一側棱垂直於底面的角錐：

$$V = \frac{1}{3} \times \text{廣} \times \text{袤} \times \text{高}$$

This gives the volume of a **yangma**, a pyramid with rectangular base and one lateral edge perpendicular to the base:

$$V = \frac{1}{3} \times \text{width} \times \text{length} \times \text{height}$$

劉徽注 · 劉徽論陽馬

按此術，陽馬者，隅角之椎也。解立方得兩壅堵，解壅堵得一陽馬、一鱉臑。陽馬居二，鱉臑居一，不易之率也。故立方三而一，即陽馬之積也。

合二壅堵成一陽馬者，試令廣袤各六尺，高六尺。中分其高，令廣袤各三尺。其高六尺，中分為二，各三尺。上壅堵廣袤各三尺，高亦三尺；下壅堵亦然。合二壅堵，廣袤各六尺，高六尺，是為一陽馬也。

Liu Hui's Commentary · On the Yangma

A yangma is a corner pyramid. Dissecting a rectangular prism gives two qian-du (wedge-like solids); dissecting each qian-du gives one yangma and one bienao (tetrahedron). The ratio of yangma to bienao is always 2:1 — an unchanging rate. Hence a cube divided by 3 gives the yangma volume.

Combining two qian-du to make one yangma: suppose width and length are each 6 chi, height 6 chi. Bisect the height, making width and length each 3 chi. The height of 6 chi is bisected into two, each 3 chi. The upper qian-du has width and length 3 chi, height 3 chi; the lower qian-du likewise. Combining two qian-du, width and length each 6 chi, height 6 chi, this makes one yangma.

鱉臑術 · 鱉臑 (四面體)

術曰：廣袤相乘，以高乘之，六而一。

Bienao Method · Tetrahedron

Method: Multiply width by length, multiply by height, divide by 6.

此為鱉臑 (bienao) 體積公式，一種三條棱兩兩垂直的四面體：

$$V = \frac{1}{6} \times a \times b \times c$$

This gives the volume of a **bienao**, a tetrahedron with three mutually perpendicular edges:

$$V = \frac{1}{6} \times a \times b \times c$$

劉徽注

鱉臑者，推之明理也。按陽馬之積，三分立方之一。鱉臑者，又半陽馬也。故六而一。凡立方，高一尺，廣袤各一尺，積一尺。陽馬廣袤各一尺，高一尺，積三分之一尺。鱉臑三邊各一尺，積六分之一尺。

Liu Hui's Commentary

The bienao demonstrates clear principles. The yangma volume is one-third of the cube. The bienao is half the yangma. Hence divide by 6. For a cube of height 1 chi, width and length each 1 chi, volume is 1 cubic chi. A yangma with width and length 1 chi, height 1 chi, has volume $\frac{1}{3}$ cubic chi. A bienao with three edges each 1 chi has volume $\frac{1}{6}$ cubic chi.

2.5 The Chumeng (芻蕘) and Chutong (芻童) 芻蕘與芻童

問・芻蕘

有芻蕘，下廣三丈，袤四丈，上袤二丈，無廣，高一丈。問積幾何？

Problem • Thatched Roof (Wedge)

Suppose there is a thatched roof: lower width 3 丈, lower length 4 丈, upper length 2 丈, no upper width, height 1 丈. What is the volume?

答

答曰：積五千尺。

Answer

Answer: Volume 5,000 cubic 尺.

術

術曰：倍下袤，上袤從之，以廣乘之，又以高乘之，六而一。

Method

Method: Double the lower length, add the upper length, multiply by the width, then multiply by the height, divide by 6.

劉徽注

推明義理者，舊說云：凡積芻蕘有上下廣曰童蕘，謂其屋蓋之苦也。是故蕘之下廣袤與童之上廣袤等。正解方亭兩邊合之，即芻蕘之形也。假令下廣二尺，袤三尺，上袤一尺，無廣，高一尺。其用棋也，中央壅堵二，兩端陽馬各二。倍下袤為六，上袤從之為七。以廣二尺乘之，得一十四尺。以高乘之，得十四尺。六而一，即得積也。

Liu Hui's Commentary

To explain the principles: old texts say a chumeng with upper and lower widths is called “tong meng” (roof-like). The lower width and length of the chumeng equal the upper width and length of a chutong. Properly dissected from the two sides of a square pavilion, one obtains the form of a chumeng. Suppose lower width 2 chi, length 3 chi, upper length 1 chi, no upper width, height 1 chi. Using blocks: 2 central qian-du, 2 yangma at each end. Double lower length to get 6, add upper length to get 7. Multiply by width 2 to get 14. Multiply by height to get 14. Divide by 6 to obtain the volume.

問・芻童

有芻童，下廣二丈，袤三丈，上廣三丈，袤四丈，高三丈。問積幾何？

Problem • Thatched Mound (Frustum of Wedge)

Suppose there is a thatched mound: lower width 2 丈, lower length 3 丈, upper width 3 丈, upper length 4 丈, height 3 丈. What is the volume?

答

答曰：積一萬六千五百尺。

Answer

Answer: Volume 16,500 cubic 尺.

術

術曰：倍上袤，下袤從之；亦倍下袤，上袤從之。各以其廣乘之，并，以高乘之，六而一。

Method

Method: Double the upper length, add the lower length; also double the lower length, add the upper length. Multiply each by its respective width, add together, multiply by height, divide by 6.

劉徽注

按此術，假令芻童上廣一尺，袤二尺；下廣二尺，袤三尺；高一尺。其用棋也，中央立方一，兩旁壅堵各一，四角陽馬各一。倍上袤為四，下袤從之為七。以下廣乘之，得一十四。倍下袤為六，上袤從之為八。以上廣乘之，得八。并之，得二十二。以高乘之，得二十二。六而一，即積也。此術與芻薨曲池盤池冥谷皆同術。

Liu Hui's Commentary

According to this method: suppose a chutong with upper width 1 chi, length 2 chi; lower width 2 chi, length 3 chi; height 1 chi. Using blocks: 1 central cube, 1 qian-du on each side, 1 yangma at each of 4 corners. Double upper length to 4, add lower length to get 7. Multiply by lower width 2 to get 14. Double lower length to 6, add upper length to get 8. Multiply by upper width 1 to get 8. Add: 22. Multiply by height: 22. Divide by 6 for the volume. This method is the same for chumeng, curved pool, basin, and ravine.

2.6 The Circular Granary 圓窖問

問・圓窖

有圓窖，下周五丈四尺，深一丈八尺。問受粟幾何？

Problem • Circular Granary

Suppose there is a circular granary: circumference 5 丈 4 尺, depth 1 丈 8 尺. How much grain does it hold?

答

答曰：二千九百六十二斛二十七分斛之二十六。

Answer

Answer: $2962\frac{26}{27}$ 斛.

術

術曰：置周自相乘，以高乘之，十二而一。以斛法一尺六寸二分除之，即斛數。

Method

Method: Place the circumference, square it, multiply by the height, divide by 12. Divide by the hu-measure of 1 chi 6 cun 2 fen to obtain the number of hu.

劉徽注

按此術，圓窖下周自乘，以高乘之，十二而一者，以圓周率三乘之，十二而一，即圓積也。此猶圓塚壙之積也。于徽術，當令下周自乘以高乘之，又以二十五乘之，三百一十四而一。以斛法除之，即斛數。

Liu Hui's Commentary

Squaring the lower circumference and multiplying by height, then dividing by 12, uses the circular ratio 3 and divides by 12 to obtain the circular area. This is like the volume of a circular fortification. In Liu Hui's method, one should square the lower circumference, multiply by height, then multiply by 25 and divide by 314. Divide by the hu-measure to obtain the number of hu.

3 Volume 6: Junshu (Equitable Taxation) 卷六：均輸

九章算術卷六 · Junshu (均輸)

均輸以御遠近勞費

Junshu: For governing fair distribution considering distance, labor, and cost

本章均輸處理加權分配問題，各地區按人口數、運距等因素公平分攤稅賦。另含著名的工作效率問題，如「五渠注池」問題。

The chapter **Junshu** (“Equitable Taxation”) deals with problems of weighted distribution, where contributions must be apportioned fairly among parties that differ in population, distance from the destination, and other factors. It also contains famous work-rate problems, including the “five channels filling a pool” problem.

3.1 The Opening Problem: Grain Distribution Among Four Counties 四縣均輸粟

問 · 均輸粟

有均輸粟，甲縣一萬戶，行道八日；乙縣九千五百戶，行道十日；丙縣一萬二千三百五十戶，行道十三日；丁縣一萬二千二百戶，行道二十日。各到輸所。凡四縣賦，當輸二十五萬斛，用車一萬乘。欲以道里遠近、戶數多少，衰出之。問粟、車各幾何？

Problem · Equitable Grain Distribution

Suppose there is equitable grain distribution: County A has 10,000 households, travel time 8 days; County B has 9,500 households, travel 10 days; County C has 12,350 households, travel 13 days; County D has 12,200 households, travel 20 days. All reach the delivery point. Total tax: 250,000 斛, using 10,000 carts. Distribute according to distance and household count. How much grain and how many carts for each?

答

答曰：甲縣粟八萬三千一百斛，車三千三百二十四乘。乙縣粟六萬三千一百七十五斛，車二千五百二十七乘。丙縣粟六萬三千一百七十五斛，車二千五百二十七乘。丁縣粟四萬一千九百一十九斛，車一千六百二十二乘。

Answer

Answer: County A: 83,100 斛 grain, 3,324 carts. County B: 63,175 斛, 2,527 carts. County C: 63,175 斛, 2,527 carts. County D: 41,919 斛, 1,622 carts.

術 · 加權分配法

術曰：令縣戶數，各如其本行道日數而一，以為衰。甲衰一百二十五，乙、丙衰各九十五，丁衰六十一。副并為法。以賦粟、車數，未并者各自為實。實如法得一。按此均輸，猶均運也。令戶率出車，以行道日數為均，發粟為輸。

Method · Weighted Distribution

Method: Take each county's household count, divide by its travel days in days, to make the “decline” (weight). A's weight: 125; B and C: 95 each; D: 61. Add these subsidiarily to make the divisor. The tax grain and cart numbers, not yet added, each become dividends. Divide by the divisor to obtain one share. This junshu is like equal transport: each household contributes carts proportionally, using travel days as the equalizer, dispatching grain as the transport.

劉徽注 · 劉徽論均輸

此戶以率當各計車之錢分也。案此二句舛誤，當云：求此率以戶當各計車之錢分也。淳風等按：縣戶有多少之差，行道有遠近之異。欲其均等，故令戶數各以行道日數約之，為衰。行道多者，其戶少；行道少者，其戶多。故各令約戶為衰。并戶為法者，以戶數多寡為差，行道遠近為衰，以此均之，則勞逸齊等。

Liu Hui's Commentary · On Equitable Distribution

Each household by rate should contribute cart-money proportionally. Li Chunfeng et al.: Counties differ in household numbers and travel distances. To make it fair, divide household counts by travel days to make the decline weights. Counties with longer travel have fewer effective households; those with shorter travel have more. Hence reduce households by travel days to make weights. Adding households as divisor, using household numbers as difference and travel distance as weight, equalizing by this makes labor and rest uniform.

3.2 Problem: Workers with Different Rates 程人功問**問 · 程人功**

有程人功，甲七日成功了五分之一；乙五日成功了三分之一；丙四日成功了二分之一。問三人合作，幾何日成功？

Problem · Labor Rates

Suppose there are workers: A completes one-fifth of the work in 7 days; B completes one-third in 5 days; C completes one-half in 4 days. How many days if all three work together?

答

答曰：二十三日六十三分日之二十六。

Answer

Answer: $23\frac{26}{63}$ days.

術

術曰：置甲七日的五分之一，乙五日的三分之一，丙四日的二分之一，各為率。以一日為法，以所為實。實如法而一，各得一日之功。并一日之功，為法。以一日為實。實如法而一，即得日數。

Method

Method: Place A's one-fifth in 7 days, B's one-third in 5 days, C's one-half in 4 days, each as a rate. Taking one day as divisor and the accomplishment as dividend, divide to obtain each worker's one-day contribution. Add the one-day contributions to make the divisor. Take one day as dividend. Divide to obtain the number of days.

3.3 The Five Channels Filling a Pool 五渠注池

此為《九章算術》中最負盛名的問題之一，見於卷六末題。

This is one of the most celebrated problems in the *Jiuzhang Suanshu*, appearing as the final problem of Volume 6.

問・五渠注池

有池，五渠注之。其一渠開之，少半日一滿；次，一日一滿；次，二日半一滿；次，三日一滿；次，五日一滿。今并決之，問幾何日滿池？

Problem • Five Channels Filling a Pool

Suppose there is a pool, with five channels filling it. The first channel alone fills it in one-third of a day; the second in one day; the third in two and a half days; the fourth in three days; the fifth in five days. Now all are opened together. How many days to fill the pool?

答

答曰：七十四分日之十五。

Answer

Answer: $\frac{15}{74}$ of a day.

術

術曰：各置渠一日滿池之數，并，以為法。以一為實。實如法而一，即得日數。

Method

Method: Place each channel's daily fill rate [i.e., $\frac{1}{1/3}$, $\frac{1}{1}$, $\frac{1}{2.5}$, $\frac{1}{3}$, $\frac{1}{5}$], add them as the divisor; take 1 as the dividend. Divide to get the days.

劉徽注

按此術，一渠少半日滿者，是一日三滿也。次一日一滿者，是一日一滿也。次二日半滿者，是一日五分滿之二也。次三日一滿者，是一日三分滿之一也。次五日一滿者，是一日五分滿之一也。并之，得四滿十五分滿之十四也。此為法。以一滿為實。實如法而一，即得滿池之日數也。

Liu Hui's Commentary

The first channel fills 3 pools/day; the second fills 1; the third fills $\frac{2}{5}$; the fourth fills $\frac{1}{3}$; the fifth fills $\frac{1}{5}$. Summing: $3+1+\frac{2}{5}+\frac{1}{3}+\frac{1}{5}=\frac{74}{15}$ pools/day. Thus one pool takes $\frac{15}{74}$ day.

3.4 Problem: Hares and Dogs (Pursuit Problem) 兔犬追及問**問・兔犬**

有兔先走一百步，犬追之二百五十步，不及三十步而止。問犬不止，復行幾何步及之？

Problem • Hare and Dog Pursuit

A hare has a head start of 100 步. A dog chases it for 250 步, but is still 30 步 behind. If the dog does not stop, how many more 步 must it run to catch the hare?

答

答曰：一百七步七分步之一。

AnswerAnswer: $107\frac{1}{7}$ steps.**術**

術曰：置犬先行步數，以不及步數減之，餘為法。以不及步數乘兔先走步數，為實。實如法得一步。

Method

Method: Place the dog's forward steps, subtract the shortfall steps; the remainder is the divisor. Multiply the shortfall by the hare's head start to make the dividend. Divide to obtain the number of steps.

3.5 Problem: Gold and Silver Weights 金銀重率問

問・金銀

有金九枚，銀一十一枚，稱之適等。交易其一，金輕十三兩。問金、銀各一枚重幾何？

Problem • Gold and Silver Weights

Suppose there are 9 pieces of gold and 11 pieces of silver, which weigh the same. Exchange one piece [between them], and the gold side is lighter by 13 兩. How much does each piece of gold and silver weigh?

答

答曰：金重二斤三兩一十八銖，銀重一斤十三兩六銖。

Answer

Answer: Gold: 2 jin 3 liang 18 zhu; Silver: 1 jin 13 liang 6 zhu.

術

術曰：假令金重三斤，銀重二斤三分斤之二。不足四兩，令之如此。盈不足術為之。

MethodMethod: Suppose gold weighs 3 jin, silver weighs $2\frac{2}{3}$ jin. The deficit is 4 liang; adjust accordingly. Use the surplus-deficit method (ying bu zu shu).

4 Summary of Mathematical Content 數學內容總結

Volume 4: 少廣 (Shaoguang) 卷四

- **開方** (平方根): 迭代算法, 本質等同現代開方長除法
- **開立方** (立方根): 推廣至 $\sqrt[3]{N}$
- **球體積**: 古術 $V = \frac{9}{16}d^3$; 劉徽批判與牟合方蓋構造
- 分數運算的幾何詮釋
- **Square root extraction**: Iterative algorithm essentially equivalent to modern long-division method for roots
- **Cube root extraction**: Extension to $\sqrt[3]{N}$
- **Sphere volume**: Ancient formula $V = \frac{9}{16}d^3$; Liu Hui's critique and the mouhe fangai construction
- Fractional arithmetic with geometric interpretation

Volume 5: 商功 (Shanggong) 卷五

- **土方體積**: 城、垣、堤、溝、渠、塹
- **糧倉體積**: 倉、圓窖、圓囤
- **幾何立體**: 立方、壅堵、陽馬、鱉臄、芻蕘、芻童、曲池、盤池、冥谷
- 劉徽以割補法證明各立體體積公式
- **Earthwork volumes**: City wall, wall, dyke, ditch, canal, trench
- **Granary volumes**: Granary, circular pit, circular granary
- **Geometric solids**: Cube, qian-du, yangma, bienao, chumeng, chutong, curved pool, basin, ravine
- Liu Hui's **geometric dissection proofs** for each solid volume formula

Volume 6: 均輸 (Junshu) 卷六

- **加權分配**: 按人口 $\times$ 運距分攤稅賦
- **工作效率問題**: 多人異速合作
- **追及問題**: 相對速度 (兔犬問題)
- **聯立方程**: 金銀重率、糧食交換
- 著名的**五渠注池**工作效率問題
- **Weighted distribution**: Apportioning taxes among counties by population $\times$ distance
- **Work-rate problems**: Multiple workers with different rates
- **Pursuit problems**: Relative speed (hare and dog)
- **Simultaneous equations**: Gold and silver weights, grain exchange
- The famous **five channels filling a pool** work-rate problem

「九章之法，算之綱紀。」

“*The methods of the Nine Chapters are the source of all computation.*”

— 劉徽《九章算術注序》

— Liu Hui, Preface to the Commentary

九章算術卷四至卷六完

End of Jiuzhang Suanshu, Fascicles 4–6

九章算術 (卷七至卷九) *Jiuzhang Suanshu — Volumes 7–9* *Nine Chapters on the Mathematical Art*

BILINGUAL EDITION 中英文對照版

Volume 7: 盈不足 Yingbuzu — Excess and Deficit

Volume 8: 方程 Fangcheng — Rectangular Arrays

Volume 9: 勾股 Gougu — Right Triangles

魏 劉徽 注 Commentary by Liu Hui (c. 263)

CE) _____ Left: Original Chinese | Right: English Translation

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序 / Preface

The three volumes presented here represent the mathematical crown jewels: the algorithmic brilliance of Yingbuzu (double false position), the algebraic sophistication of Fangcheng (Gaussian elimination), and the geometric elegance of Gougu (the Pythagorean theorem).

《九章算術》是古代數學最偉大的成就之一。經數百年編纂，約成書於漢代，全書九章共二百四十六題。劉徽注約成於西元 263 年，將這部實用手冊提升為具有深刻數學洞察力的經典。

本書所呈現的三卷，乃《九章》之數學瑰寶：盈不足之算法精妙、方程之代數造詣、勾股之幾何優雅。

術曰：劉徽序曰：昔在包犧氏始畫八卦，以通神明之德，以類萬物之情，作九九之術，以合六爻之變。暨於黃帝神農，其為功也，各有稱焉。周公制禮而有九數，九數之流，則九章是矣。

劉徽注：徽幼習九章，長再詳覽。觀陰陽之割裂，總算術之根源，探蹟之暇，遂悟其意。是以敢竭頑魯，采其所見，為之作注。事物有對，故立正負之率；錯綜無形，故開方程之術。庶幾令文約而義博，事省而功多。

Method:Liu Hui's Preface: In ancient times, Paoxi (Fuxi) first drew the Eight Trigrams to communicate the virtue of the divine spirits and to classify the nature of all things. He created the art of multiplication tables to match the transformations of the six lines. By the time of the Yellow Emperor and Shennong, each had their own accomplishments to be celebrated. The Duke of Zhou established the rites and created the Nine Numbers; from the Nine Numbers flows the Nine Chapters.

Liu Hui's Commentary:In my youth I studied the Nine Chapters, and as I grew older I reviewed them again in detail. Observing the division of yin and yang, and summarizing the root sources of mathematical methods, in my leisure moments of deep inquiry I gradually comprehended their meaning. Therefore I dared, with all my humble capacity, to collect what I have observed and write this commentary. Things have their opposites, so I establish the rates of positive and negative; when complexity has no visible form, I open the method of rectangular arrays. My hope is to make the text concise yet the meaning broad, the effort small yet the achievement great.

卷第七 盈不足 / Volume 7 Excess and Deficit

盈不足術乃《九章》中最著名之算法之一。其法以兩次假設及其盈納推求真實之數，即「雙假設法」。此法經伊斯蘭世界傳入中世紀歐洲，被稱為 *elchataym*。本章共二十四題，涵蓋商業交易、幾何應用及著名之「五家共井」不定方程。

The method of Yingbuzu (excess and deficit) is one of the most celebrated algorithms in the Nine Chapters. It provides a technique for solving problems involving two guesses and their resulting excesses or deficits — essentially the “rule of double false position.” This method was transmitted to the West via the Islamic world and medieval Europe, where it became known as the method of elchataym.

Method Summary / 術概

術曰：盈不足術曰：置所出率，盈、不足各居其下。令維乘所出率，并以為實。并盈、不足為法。實如法而一。

劉徽注：此術以盈不足之率，求所出之實。維乘者，交錯相乘也。其術之意：假設兩端，一為盈，一為不足，以兩設交錯相乘，取其和為實，取其差之和為法，實如法而一，得所求之數。

Method: The Method of Excess and Deficit says: Place the assumed rates (guesses), with the excess and deficit placed below each respectively. Cross-multiply the assumed rates by the excess/deficit, and sum to form the dividend. Sum the excess and deficit to form the divisor. Divide the dividend by the divisor to obtain the result.

Liu Hui’s Commentary: This method uses the rates of excess and deficit to find the true value. “Cross-multiply” means to multiply in alternate fashion. The idea: make two assumptions, one yielding excess and one yielding deficit; cross-multiply, take their sum as dividend and the sum of differences as divisor; dividing yields the sought number.

General Formula

For two guesses a_1, a_2 with excess/deficit e_1, e_2 :

$$\text{Result} = \frac{a_1 e_2 + a_2 e_1}{e_1 + e_2}$$

0.1 第一題 / Problem 1: 共買物 / Buying Goods Together

問：今有共買物，人出八，盈三；人出七，不足四。問：人數、物價各幾何？

答曰：人七，物價五十三。

術曰：術曰：置人出八、人出七，盈三、不足四。令維乘之：八乘四得三十二，七乘三得二十一。并之得五十三，為物價。并盈不足得七，為人數。

劉徽注：按盈不足之術，兩設皆乖於實。假令每人出八，則多三；每人出七，則少四。維乘者，以盈乘不足之設，以不足乘盈之設，所以通而同之也。并之為實，并盈不足為法。實如法而一，得物價。

Q:Now there is a group buying goods. If each person contributes 8, there is an excess of 3; if each contributes 7, there is a deficit of 4. Question: how many people are there, and what is the price of the goods?

A:There are 7 people; the goods cost 53.

Method:Method: Place 8 and 7 as the two contributions, with excess 3 and deficit 4 below them. Cross-multiply: 8 times 4 gives 32; 7 times 3 gives 21. Summing gives 53, which is the price of the goods. Summing the excess and deficit gives 7, which is the number of people.

Liu Hui's Commentary:According to the method of excess and deficit, both assumptions deviate from the truth. If each person contributes 8, there is a surplus of 3; if each contributes 7, there is a shortfall of 4. Cross-multiplication means multiplying the excess by the assumption with deficit, and multiplying the deficit by the assumption with excess, thereby harmonizing and equalizing them.

0.2 第二題 / Problem 2: 共買雞 / Buying Chickens Together

問：今有共買雞，人出九，盈十一；人出六，不足十六。問：人數、雞價各幾何？

答曰：人九，雞價七十。

術曰：術曰：置人出九、人出六，盈十一、不足十六。令維乘之：九乘十六得一百四十四，六乘十一得六十六。并之得二百一十，為實。并盈不足得二十七，為法。

劉徽注：此術與上術意同。九乘不足十六者，以多出之率乘不足之數也；六乘盈十一者，以少出之率乘盈之數也。所以維乘者，兩設不同，不可偏用，故交錯相乘以齊同之。

Q:Now there is a group buying chickens. If each person contributes 9, there is an excess of 11; if each contributes 6, there is a deficit of 16. Question: how many people are there, and what is the price of the chickens?

A:There are 9 people; the chickens cost 70.

Method:Method: Place 9 and 6 as contributions, with excess 11 and deficit 16. Cross-multiply: 9 times 16 gives 144; 6 times 11 gives 66. Summing gives 210 as the dividend. Summing excess and deficit gives 27 as the divisor.

Liu Hui's Commentary:This method shares the same idea as the previous one. Nine times the deficit 16 means multiplying the larger contribution rate by the deficit amount; six times the excess 11 means multiplying the smaller contribution rate by the excess amount. The purpose of cross-multiplication is that the two assumptions differ and cannot be used one-sidedly; therefore we multiply alternately to equalize and harmonize them.

0.3 第三題 / Problem 3: 共買璫 / Buying Jade Together

問：今有共買璫，人出半，盈四；人出少半，不足三。問：人數、璫價各幾何？

答曰：人四十二，璫價十七。

術曰：術曰：置人出半、人出少半，盈四、不足三。半即二分，少半即三分。令維乘之，如法求之。

劉徽注：半者，二分之一也；少半者，三分之一也。出半則多四，出少半則少三。以分數為率，故先通分之，然後維乘求之。此術以分數入盈不足，所以見術之無所不通也。

Q:Now there is a group buying jade. If each person contributes half, there is an excess of 4; if each contributes “less-than-half” (one-third), there is a deficit of 3. Question: how many people are there, and what is the price of the jade?

A:There are 42 people; the jade costs 17.

Method:Method: Place “half” and “less-than-half” as contributions, with excess 4 and deficit 3. Half is one-half, less-than-half is one-third. Cross-multiply and solve by the method.

Liu Hui's Commentary: “Half” means one-half; “less-than-half” means one-third. Contributing half gives excess 4; contributing one-third gives deficit 3. Since fractions are used as rates, first bring them to common denominators, then cross-multiply and solve. This method applies fractions to excess and deficit, showing that the method has no limitation in its applicability.

0.4 第四題 / Problem 4: 共買牛 / Buying Oxen Together

問：今有共買牛，七家共出一百九十，不足三百三十；九家共出二百七十，盈三十。問：家數、牛價各幾何？

答曰：一百二十六家，牛價三千七百五十。

術曰：術曰：置七家共出一百九十、不足三百三十，九家共出二百七十、盈三十。令每家出率各以家數除之：一百九十除以七得二十七又七分之一；二百七十除以九得三十。乃以盈不足維乘之，并為實，并盈不足為法，實如法而一。

劉徽注：此題以家為率，不以人為率。故先求每家所出之率，然後用盈不足術。七家出一百九十，則不足三百三十，是每家所出不足也；九家出二百七十，盈三十，是每家所出盈也。先通為每家之率，然後可用盈不足之法。此術之變，以家為率也。

Q: Now there is a group buying oxen. Seven households together contribute 190, falling short by 330; nine households together contribute 270, with an excess of 30. Question: how many households are there, and what is the price of the oxen?

A: There are 126 households; the oxen cost 3750.

Method: Method: Place 7 households contributing 190 with deficit 330, and 9 households contributing 270 with excess 30. Divide each total by the number of households: $190/7$ gives $27 \frac{1}{7}$; $270/9$ gives 30. Then apply excess-deficit cross-multiplication: sum as dividend, sum of excess and deficit as divisor; divide.

Liu Hui's Commentary: In this problem, households serve as the rate, not individual persons. Therefore first find the contribution rate per household, then apply the excess-deficit method. This is a variant of the method using households as rate.

0.5 第五題 / Problem 5: 米麥互換 / Rice and Wheat Exchange

問：今有米一斗，麥一斗，各值幾何？但知米三斗值麥二斗，麥四斗值米一斗。問：米、麥各一斗價若干？

答曰：米一斗值四錢，麥一斗值一錢半。

術曰：術曰：以盈不足術求之。假令米一斗值四錢，則麥當值若干，驗其盈不足，乃以術求之。

劉徽注：此術非直盈不足，乃以價為率，互相推求。假令一值，驗其盈朒，然後設為兩端，用盈不足術以齊之。米三斗值麥二斗，則米一斗值麥三分之二斗。麥四斗值米一斗，則麥一斗值米四分之一斗。以盈不足術通之，設米一斗值若干，驗其與麥價之盈朒，維乘求之。此以價為隱，而以盈不足顯之也。

Q:Now there is one dou of rice and one dou of wheat. What is each worth? It is known that 3 dou of rice is worth 2 dou of wheat, and 4 dou of wheat is worth 1 dou of rice. Question: what is the price of 1 dou of rice and 1 dou of wheat?

A:One dou of rice is worth 4 cash; one dou of wheat is worth 1.5 cash.

Method:Method: Apply the excess-deficit method. Assume one dou of rice is worth 4 cash, then determine what wheat should be worth, check the excess or deficit, and solve by the method.

Liu Hui's Commentary:This method is not purely excess-deficit; it uses prices as rates and mutually deduces them. Assume one price, check its excess or deficit against the other, then set up two ends and use the excess-deficit method to harmonize them. This makes the hidden price relationship visible through excess and deficit.

0.6 第六題 / Problem 6: 良駑與驢 / Fine Horse, Nag, and Donkey

問：今有良駑二馬，與驢俱牽。良馬與駑馬共引之，良馬日引四十里，駑馬日引四十里。驢在後逐之，不及。問：驢日行幾何里？

答曰：驢日行二十里。

術曰：術曰：以盈不足術求之。假令驢行三十里，則驢所行多於良駑之半，是為盈；假令驢行十里，則不及半，是為不足。維乘并實，如法求之。

劉徽注：良駑俱行四十里，其半二十里。驢逐其後，當與之半相當。假令三十里，則多十里，為盈；假令十里，則少十里，為不足。故以盈不足術求之，得二十里。此術之意：良駑俱行四十里，驢逐其半，故以二十里為中平之數。

Q:Now there are a fine horse and a nag, both pulling together with a donkey chasing behind. The fine horse and nag each pull 40 li per day. The donkey follows behind but cannot catch up. Question: how many li does the donkey travel per day?

A:The donkey travels 20 li per day.

Method:Method: Apply the excess-deficit method. Assume the donkey travels 30 li: then it exceeds half of the horses' distance, which is excess. Assume it travels 10 li: then it falls short of half, which is deficit. Cross-multiply, sum as dividend, and solve by the method.

Liu Hui's Commentary: The fine horse and nag both travel 40 li; half of this is 20 li. The donkey chases behind, and should match this half. Assuming 30 li gives 10 li more, which is excess; assuming 10 li gives 10 li less, which is deficit. Therefore apply the excess-deficit method to obtain 20 li.

0.7 第七題 / Problem 7: 垣高術 / Wall Height

問：今有垣不知高。立一表長五尺，去垣五丈，目距地五尺，望垣頂與表端參合。問：垣高幾何？

答曰：垣高五丈五尺。

術曰：術曰：以盈不足術求之。假令垣高五丈，則視線當過表端若干；假令垣高六丈，則視線不及表端若干。維乘求之。

劉徽注：此術本屬勾股，今以盈不足術驗之。去垣五丈，表高五尺，目高五尺，則表與目平。望垣頂與表端參合，是為表高與垣高之比例，如去垣之遠與目至垣頂之比例也。以相似勾股求之即得。此術雖屬勾股，而以盈不足驗之，所以明術之相通也。

Q: Now there is a wall of unknown height. A pole 5 chi tall is erected 5 zhang from the wall. The eye is 5 chi from the ground. Looking at the top of the wall aligns with the top of the pole. Question: how high is the wall?

A: The wall is 5 zhang 5 chi high.

Method: Method: Apply the excess-deficit method. Assume the wall is 5 zhang: then the sight line passes above the pole. Assume the wall is 6 zhang: then the sight line falls below the pole. Cross-multiply and solve.

Liu Hui's Commentary: This method fundamentally belongs to right triangles (Gougu); here it is verified by the excess-deficit method. The alignment of the wall top with the pole top forms a proportion. Using similar right triangles yields the answer. Although this belongs to right triangles, using excess-deficit to verify it demonstrates the interconnectedness of methods.

0.8 第八題 / Problem 8: 五家共井 / Five Families Sharing a Well

問：今有五家共井。甲二綆不足，如乙一綆；乙三綆不足，如丙一綆；丙四綆不足，如丁一綆；丁五綆不足，如戊一綆；戊六綆不足，如甲一綆。問：井深、綆長各幾何？

答曰：井深七丈二尺一寸。甲綆長二丈六尺五寸，乙綆長一丈九尺一寸，丙綆長一丈四尺八寸，丁綆長一丈二尺九寸，戊綆長七尺六寸。

術曰：術曰：此以盈不足為問，實則以方程入之。各以其綆之數，不足之數，列為方程，正負相生，求之即得。

劉徽注：此題本以盈不足為問，而實非盈不足之術所能獨盡也。五家之綆長短不同，各家取綆若干而不足若干，以補他家一綆之數。此五綆之長與井深，共為六未知數，而題中僅給五條件，故為不定問題也。其答云者，特其一組正整數解耳。若以方程術解之，當令井深為一，求五家綆長之比，然後以井深定其實長。此題之精妙，在於以盈不足為表，以方程為裡，表裡相濟，乃得其真。

Q:Now five families share a well. Two lengths of A's rope are insufficient, and the shortfall equals one length of B's rope; three lengths of B's are insufficient, equaling one length of C's; four lengths of C's are insufficient, equaling one length of D's; five lengths of D's are insufficient, equaling one length of E's; six lengths of E's are insufficient, equaling one length of A's. Question: what is the depth of the well, and what is the length of each family's rope?

A:The well is 7 zhang 2 chi 1 cun deep. A's rope is 2 zhang 6 chi 5 cun; B's is 1 zhang 9 chi 1 cun; C's is 1 zhang 4 chi 8 cun; D's is 1 zhang 2 chi 9 cun; E's is 7 chi 6 cun.

Method:Method: Although posed as excess-deficit, this problem is actually solved by the rectangular arrays (Fangcheng) method. List each rope quantity and deficit quantity as equations; let positive and negative interact; solving yields the answer.

Liu Hui's Commentary:This problem is phrased in terms of excess and deficit, but the excess-deficit method alone cannot fully solve it. The five rope lengths plus the well depth make six unknowns, while the problem only provides five conditions; hence it is an indeterminate problem. The subtlety lies in using excess-deficit as the exterior and Fangcheng as the interior; exterior and interior working together obtain the truth.

0.9 第九題 / Problem 9: 金與銀 / Gold and Silver

問：今有金九枚，銀十一枚，稱之適等。交易其一，金輕十三兩。問：金、銀一枚各重幾何？

答曰：金重二斤三兩一十八銖，銀重一斤十三兩六銖。

術曰：術曰：以盈不足術求之。假令金重三斤，則九金二十七斤；令銀與之等，則每銀當重二十七斤十一分。交易其一，驗其輕重，得盈不足之數。再設一率，維乘求之。

劉徽注：金九銀十一稱之適等者，總重等也。交易其一者，以一金換一銀也。金輕十三兩者，換後金方輕於原銀方十三兩也。以方程術解之尤便：設金重 x ，銀重 y ，則 $9x = 11y$ ，且 $8x + y = 10y + x - 13$ 兩。以盈不足術者，兩設其重，驗其差也。

Q:Now there are 9 pieces of gold and 11 pieces of silver; they balance exactly. Exchange one piece, and the gold side is lighter by 13 liang. Question: how much does one piece of gold and one piece of silver weigh?

A:One piece of gold weighs 2 jin 3 liang 18 zhu; one piece of silver weighs 1 jin 13 liang 6 zhu.

Method:Method: Apply the excess-deficit method. Assume gold weighs 3 jin: then 9 pieces weigh 27 jin; making silver balance this, each piece should weigh $27/11$ jin. Exchange one piece and check the difference to obtain the excess-deficit numbers. Make a second assumption, cross-multiply and solve.

Liu Hui's Commentary:Nine pieces of gold and eleven pieces of silver balancing means their total weights are equal. Exchanging one piece means swapping one gold for one silver. Gold being 13 liang lighter means after exchange, the gold side is lighter than the original silver side by 13 liang. This is especially conveniently solved by the Fangcheng method.

0.10 第十題 / Problem 10: 程傳委輸 / Transporting Grain

問: 今有程傳委輸，空車日行七十里，重車日行五十里。今載太倉粟輸上林，五日三返。問：太倉去上林幾何？

答曰: 太倉去上林四十八里一百八十分里之十一。

術曰: 術曰：以盈不足術求之。假令去四十五里，則五日三返不足若干里；假令去五十里，則五日三返盈若干里。維乘并實，如法求之。

劉徽注: 空車日行七十里者，往而不載也；重車日行五十里者，載粟而返也。五日三返者，一往一返為一返，三返則三往三返也。此以盈不足術驗之者，取其近似也。此術雖以盈不足入之，實則以調和平均為本也。

Q:Now there is a relay transport task. An empty cart travels 70 li per day; a loaded cart travels 50 li per day. Grain is to be transported from Taicang to Shanglin, with three round trips in five days. Question: what is the distance from Taicang to Shanglin?

A:The distance is $48 \frac{11}{180}$ li.

Method:Method: Apply the excess-deficit method. Assume the distance is 45 li: then three round trips in five days falls short. Assume 50 li: then three round trips exceeds. Cross-multiply, sum as dividend, and solve.

Liu Hui's Commentary:An empty cart traveling 70 li per day is going without load; a loaded cart traveling 50 li per day is returning with grain. Three round trips in five days means one round trip is one going and one returning. Although this uses the excess-deficit method, its foundation is actually the harmonic mean.

0.11 第十一題 / Problem 11: 鹿與兔 / Deer and Rabbit

問：今有鹿直西走，兔直東走。兔行一百步，鹿行七十步。兔在鹿東一百五十步，同時發。問：幾何步相逢？

答曰：兔行二百五十步，鹿行一百七十五步，相逢。

術曰：術曰：以盈不足術求之。假令兔行二百步，則鹿行一百四十步，驗其盈不足之數。再假令一行數，維乘求之。

劉徽注：兔東鹿西，相向而行，故為相逢之術。兔行一百步，鹿行七十步，則兔鹿步率之比為十比七也。相距一百五十步，同時發，則相逢之時，兔行十份，鹿行七份，共十七份。以一百五十步分為十七份，每份 $150/17$ 步。

Q: Now a deer runs straight west and a rabbit runs straight east. The rabbit runs 100 paces while the deer runs 70 paces. The rabbit is 150 paces east of the deer; they start at the same time. Question: after how many paces do they meet?

A: The rabbit runs 250 paces, the deer runs 175 paces, and they meet.

Method: Method: Apply the excess-deficit method. Assume the rabbit runs 200 paces: then the deer runs 140 paces, and check the excess-deficit numbers. Make a second assumption, cross-multiply and solve.

Liu Hui's Commentary: The rabbit runs east and the deer runs west; they move toward each other. The ratio of their pace rates is $100:70 = 10:7$. Separated by 150 paces and starting simultaneously, at meeting time the rabbit covers 10 parts and the deer 7 parts, totaling 17 parts.

0.12 第十二題 / Problem 12: 蒲與莞 / Rush and Cattail

問：今有蒲生一日，長三尺；莞生一日，長一尺。蒲生日自半，莞生日自倍。問：幾何日而長等？

答曰：二日十三分日之六，而長等。各長四尺五寸十三分寸之五。

術曰：術曰：以盈不足術求之。假令二日，不足一尺五寸；令之三日，盈一尺七寸半。維乘并實，如法求之。

劉徽注：蒲日生三尺而自半者，第一日長三尺，第二日長一尺五寸，第三日長七寸半，是為日自半也。莞日生一尺而自倍者，第一日長一尺，第二日長二尺，第三日長四尺，是為日自倍也。此術之巧，在於兩物生長之率不同，一自半而一自倍，以盈不足術齊而同之。

Q: Now a rush grows 3 chi on the first day; a cattail grows 1 chi on the first day. The rush's growth halves each day; the cattail's growth doubles each day. Question: after how many days are their lengths equal?

A: After $2 \frac{6}{13}$ days their lengths are equal. Each is $4 \frac{5}{13}$ chi.

Method: Method: Apply the excess-deficit method. Assume 2 days: deficit of 1 chi 5 cun. Assume 3 days: excess of 1 chi 7.5 cun. Cross-multiply, sum as dividend, and solve.

Liu Hui's Commentary: The rush growing 3 chi per day and halving means: first day 3 chi, second day 1.5 chi, third day 0.75 chi—this is daily halving. The cattail growing 1 chi per day and doubling means: first day 1 chi, second day 2 chi, third day 4 chi—this is daily doubling. The ingenuity lies in the two objects having different growth rates, and the excess-deficit method equalizes them.

0.13 第十五題 / Problem 15: 漆三油四 / Lacquer and Oil

問：今有漆三得油四，油四和漆五。今有漆三斗，欲令分以易油，還和餘漆。問：出漆、得油、和漆各幾何？

答曰：出漆一斗一升四分升之一，得油一斗五升，和漆一斗八升四分升之三。

術曰：術曰：以盈不足術求之。假令出漆一斗，不足若干；令出漆一斗二升，盈若干。維乘并實，如法求之。

劉徽注：漆三得油四者，以漆三斗易油四斗也。油四和漆五者，以油四斗和漆五斗也。今有漆三斗，分其漆以易油，又以所得之油和其餘漆。此術以漆易油，以油和漆，往還相求，以盈不足定其率。

Q: Now 3 parts of lacquer yield 4 parts of oil; 4 parts of oil mix with 5 parts of lacquer. Now there are 3 dou of lacquer. It is desired to divide some lacquer to exchange for oil, then use the obtained oil to mix with the remaining lacquer. Question: how much lacquer is used, how much oil is obtained, and how much lacquer-oil mixture results?

A: Lacquer used: 1 dou 1 sheng 2.25 sheng; oil obtained: 1 dou 5 sheng; mixture: 1 dou 8 sheng 3.75 sheng.

Method: Method: Apply the excess-deficit method. Assume 1 dou of lacquer is used: deficit. Assume 1 dou 2 sheng: excess. Cross-multiply, sum as dividend, and solve.

Liu Hui's Commentary: This method exchanges lacquer for oil and uses oil to mix lacquer, going back and forth, using excess-deficit to fix the rate, demonstrating the inexhaustible versatility of the method.

0.14 第十六題 / Problem 16: 惡粟為米 / Coarse Millet into Rice

問：今有惡粟二十斗，舂之為米，米率五斗。今欲為糲米、粳米各幾何？

答曰：糲米十斗，粳米六斗六升三分升之二。

術曰：術曰：以盈不足術求之。假令糲米十斗，則粳米當若干，驗其盈不足。再設一率，維乘求之。

劉徽注：惡粟二十斗，米率五斗者，一斗惡粟舂得五升米也。糲米率五，粳米率六。以盈不足術者，兩設糲米之斗數，求粳米之盈朒也。此以米率之隱，而以盈不足顯之也。

Q:Now there are 20 dou of coarse millet; pounding it yields rice at a rate of 5 sheng per dou. It is desired to divide this into roughly-husked rice (li mi) and finely-husked rice (bai mi). Question: how much of each?

A:Roughly-husked rice: 10 dou; finely-husked rice: 6 dou 6 sheng $\frac{2}{3}$ sheng.

Method:Method: Apply the excess-deficit method. Assume 10 dou of roughly-husked rice: then determine the finely-husked rice and check the excess or deficit. Make a second assumption, cross-multiply and solve.

Liu Hui's Commentary:This reveals the hidden rice rates through the excess-deficit method.

0.15 第十七題 / Problem 17: 持米出關 / Carrying Rice Through Passes

問：今有人持米出三關，外關三而取一，中關五而取一，內關七而取一，餘米五斗。問：本持米幾何？

答曰：本持米十斗九升八分升之三。

術曰：術曰：以盈不足術求之。假令持米八斗，不足若干；令持米十二斗，盈若干。維乘并實，如法求之。

劉徽注：三關各取其幾分之一。外關三取一，是取其三分之一也；中關五取一，是取其五分之一也；內關七取一，是取其七分之一也。過三關之後，餘米五斗。此術以三關之稅率為隱，以盈不足顯之，所以見術之無幽不顯也。

Q:Now a person carries rice through three passes. The outer pass takes one-third; the middle pass takes one-fifth; the inner pass takes one-seventh. After passing through all three, 5 dou of rice remain. Question: how much rice did the person originally carry?

A:The original amount was 10 dou 9 sheng $\frac{3}{8}$ sheng.

Method:Method: Apply the excess-deficit method. Assume 8 dou were carried: deficit. Assume 12 dou: excess. Cross-multiply, sum as dividend, and solve.

Liu Hui's Commentary:Each pass takes a certain fraction. This method makes the hidden tax rates of the three passes visible through excess and deficit, demonstrating that the method reveals all that is hidden.

0.16 第十八題 / Problem 18: 五人分五錢 / Five People Dividing Five Cash

問：今有五人分五錢，令上二人所得與下三人等。問：各得幾何？

答曰：甲得一錢六分之二，乙得一錢六分之一，丙得一錢，丁得六分之二，戊得六分之二。

術曰：術曰：以盈不足術求之。假令五人各得一錢，則上二人得二錢，下三人得三錢，盈一錢。此當以衰分術入之。

劉徽注：五人分五錢而令上二人與下三人等者，上二人所得當為二錢半，下三人所得亦當為二錢半也。然此題以衰分術為正：令五人所得為差率，以差分配之。此術雖以盈不足入之，實以衰分為本也。

Q:Now five people divide 5 cash, such that the top two people's share equals the bottom three people's share. Question: how much does each receive?

A:A receives $1\frac{2}{6}$ cash; B receives $1\frac{1}{6}$ cash; C receives 1 cash; D receives $\frac{5}{6}$ cash; E receives $\frac{4}{6}$ cash.

Method:Method: Apply the excess-deficit method. Assume each receives 1 cash: then the top two get 2 cash and the bottom three get 3 cash, excess of 1 cash. [Note: properly solved by proportional division.]

Liu Hui's Commentary:This problem is properly solved by the proportional division method. Although this uses the excess-deficit method, its foundation is actually proportional division.

0.17 第十九題 / Problem 19: 弧田徑術 / Arc-Field Area

問：今有弧田，弦三十步，矢十五步。問：為田幾何？

答曰：一畝九十七步半。

術曰：術曰：以弦乘矢，矢又自乘，并之，三而一。

劉徽注：弧田術者，以弦矢求弧田之面積也。弦乘矢者，以弦為廣，矢為縱，相乘得長方形之積。矢自乘者，以矢為方之邊，自乘得正方形之積。并之者，合兩積為一。三而一者，以弧田之積約當此兩積和之三分之一也。此術為弧田之近似術。

Q:Now there is an arc-field with chord 30 paces and arrow (sagitta) 15 paces. Question: what is the area?

A:1 mu 97.5 square paces.

Method:Method: Multiply the chord by the arrow; also square the arrow; sum these; divide by 3.

Liu Hui's Commentary:This is an approximation method for arc-fields. The arc-field is the region enclosed by a circular arc and its chord; its shape is curved and convex, not fully expressible by straight lines.

0.18 第二十題 / Problem 20: 環田徑術 / Annulus Field

問: 今有環田，中周六十二步，外周四步，徑十二步。問：為田幾何？

答曰：二畝五十五步。

術曰：術曰：并中外周而半之，以徑乘之為積。

劉徽注：環田者，猶圓田之中去一圓也。并中外周而半之者，取環之平均周長也。以徑乘之者，以平均周長為廣，徑為縱也。此術以環田近似為矩形，故以廣縱相乘得積。若精密求之，當以兩圓面積之差求之。

Q:Now there is an annulus field with inner circumference 62 paces, outer circumference [corrected], and diameter [thickness] 12 paces. Question: what is the area?

A:2 mu 55 square paces.

Method:Method: Sum the inner and outer circumferences and halve; multiply by the diameter to get the area.

Liu Hui's Commentary:This method approximates the annulus as a rectangle. For a precise calculation, one should use the difference of the two circle areas.

0.19 第二十一題 / Problem 21: 三人共車 / Three People Per Cart

問: 今有三人共車，二車空；二人共車，九人步。問：車幾何？人幾何？

答曰：車一十五乘，人三十九人。

術曰：術曰：以盈不足術求之。假令車十乘，則三人共車載三十人，餘九人步，是為盈九。假令車二十乘，則二人共車載四十人，缺一人，是為不足一。維乘并實，如法求之。

劉徽注：三人共車二車空者，每車三人則二車無人乘也。二人共車九人步者，每車二人則餘九人不行也。此術以車之虛實為盈朒，亦變術之巧者也。

Q:Now if three people share each cart, two carts are empty; if two people share each cart, nine people must walk. Question: how many carts and how many people?

A:There are 15 carts and 39 people.

Method:Method: Apply the excess-deficit method. Assume 10 carts: then three per cart carries 30 people, leaving 9 walking—excess of 9. Assume 20 carts: then two per cart carries 40 people, one short—deficit of 1. Cross-multiply, sum as dividend, and solve.

Liu Hui's Commentary:This method uses the emptiness or fullness of carts as excess and deficit—a clever variation of the method.

0.20 第二十二題 / Problem 22: 綿布互易 / Cotton and Cloth Exchange

問：今有綿一斤直錢二百四十，布一匹直錢三百二十。今有綿五斤，欲易布幾何？

答曰：布三匹七分匹之五。

術曰：術曰：以盈不足術求之。假令易三匹，則綿價一千二百，布價九百六十，盈二百四十。假令易四匹，則綿價一千二百，布價一千二百八十，不足八十。維乘并實，如法求之。

劉徽注：綿一斤直二百四十，五斤則直一千二百。布一匹直三百二十。以盈不足術者，兩設易布之匹數，驗其綿布價值之盈朒也。

Q:Now cotton costs 240 cash per jin; cloth costs 320 cash per bolt. With 5 jin of cotton, how much cloth can be obtained?

A:3 5/7 bolts.

Method:Method: Apply the excess-deficit method. Assume exchanging for 3 bolts: cotton value 1200, cloth value 960, excess of 240. Assume 4 bolts: cotton value 1200, cloth value 1280, deficit of 80. Cross-multiply, sum as dividend, and solve.

Liu Hui's Commentary:Using the excess-deficit method: make two assumptions about the number of bolts and check the excess or deficit of the cotton-cloth values.

0.21 第二十三題 / Problem 23: 粟米貴賤 / Millet and Rice Grades

問：今有粟一斗，欲為糲米、粳米、糞米三等。糲米率三十，粳米率二十七，糞米率二十四。問：各為米幾何？

答曰：糲米三升，粳米二升七合，糞米二升四合。

術曰：術曰：以盈不足術入衰分求之。假令糲米三升，則粳米、糲米遞減之，驗其盈不足，維乘求之。

劉徽注：糲米率三十者，粟一斗為糲米三升也。粳米率二十七者，為粳米二升七合也。糞米率二十四者，為糞米二升四合也。此術以三米之率為隱，以盈不足為顯。

Q:Now there is 1 dou of millet, to be divided into three grades of rice: roughly-husked (li), finely-husked (bai), and polished (fen). Rates: li 30, bai 27, fen 24. Question: how much of each?

A:Li mi: 3 sheng; bai mi: 2 sheng 7 ge; fen mi: 2 sheng 4 ge.

Method:Method: Apply the excess-deficit method within proportional division. Assume 3 sheng of li mi: then bai and fen mi decrease accordingly; check the excess or deficit; cross-multiply and solve.

Liu Hui's Commentary:This method makes the three rice rates visible through excess and deficit.

0.22 第二十四題 / Problem 24: 土功程效 / Earthwork Labor

問：今有築堤，甲縣人功一日築七尺，乙縣人功一日築五尺。今令兩縣共築，甲縣先築一日，乙縣繼之。問：幾何日而成？

答曰：甲築一日，乙築二日，共成。

術曰：術曰：以盈不足術求之。假令共築二日，則甲築十四尺，乙築五尺，共十九尺，驗其盈不足。再設一日，維乘求之。

劉徽注：甲縣人功一日七尺，乙縣人功一日五尺。甲先築一日，築七尺。餘令乙築，乙一日五尺，則餘二尺。甲復築一日，又七尺，共十九尺。此術雖以盈不足入之，實以衰分為本也。

Q:Now a dike is to be built. County A's workers build 7 chi per day; County B's build 5 chi per day. Both counties build together; County A works first for one day, then County B continues. Question: how many days until completion?

A: County A builds for 1 day; County B builds for 2 days; together they complete it.

Method: Method: Apply the excess-deficit method. Assume both build for 2 days total: then A builds 14 chi and B builds 5 chi, total 19 chi; check the excess or deficit. Make a second assumption, cross-multiply and solve.

Liu Hui's Commentary: County A's laborers build 7 chi per day; County B's build 5 chi per day. Although this uses the excess-deficit method, its foundation is actually proportional division.

卷第八 方程 / Volume 8 Rectangular Arrays

方程章乃中國古代代數之巔峰成就。其提出解線性方程組之系統方法——本質即高斯消元法——較高斯早一千五百餘年。本章亦包含世界數學史上負數之最早系統運用。

The Fangcheng chapter is the crowning achievement of ancient Chinese algebra. It presents a systematic method for solving systems of linear equations — essentially Gaussian elimination — over 1500 years before Gauss. This chapter also contains the first known use of negative numbers in world mathematics.

General Method / 方程術總論

術曰：方程術曰：置上禾三秉，中禾二秉，下禾一秉，實三十九；上禾二秉，中禾三秉，下禾一秉，實三十四；上禾一秉，中禾二秉，下禾三秉，實二十六。以右行上禾遍乘中行，而以直除。又乘其次，亦以直除。然以中行中禾不盡者遍乘左行，而以直除。左方下禾不盡者，上為法，下為實。實即下禾之實。

劉徽注：程，課程也。群物總雜，各列有數，總言其實。令每行為率，二物者再程，三物者三程，皆如物數程之，並列為行，故謂之方程。行之左右，無所同存，且為有所據而言耳。

Method: The Method of Rectangular Arrays says: Place [coefficients] in rows. Multiply the pivot row throughout and subtract directly from rows below. Repeat for each column. The remaining coefficient is the divisor, the shi is the dividend. Dividing gives the unknown's value.

Liu Hui's Commentary: “Cheng” means course or standard. When various items are mixed together, each is listed with its quantity and the total is given as “shi.” Each row forms a rate; as many courses as there are items—arranged together as rows; hence “rectangular arrays.”

Key Steps: 1) Arrange coefficients in rows; 2) Multiply pivot row and subtract from rows below (直除); 3) Repeat for each column; 4) Back-substitute. **Historical Note:** Liu Hui used red rods for positive numbers (正) and black rods for negative numbers (負).

0.23 第一題 / Problem 1: 三禾求實 / Three Grains' Yield

問：今有上禾三秉，中禾二秉，下禾一秉，實三十九斗；上禾二秉，中禾三秉，下禾一秉，實三十四斗；上禾一秉，中禾二秉，下禾三秉，實二十六斗。問：上、中、下禾實一秉各幾何？

答曰：上禾一秉，九斗四分斗之一；中禾一秉，四斗四分斗之一；下禾一秉，二斗四分斗之三。

術曰：術曰：方程術。置上禾、中禾、下禾及實於右、中、左三行。以右行上禾三遍乘中行，以直除之，消去中行上禾。又以右行上禾三遍乘左行，以直除之，消去左行上禾。然後以中行中禾不盡者遍乘左行，以直除之，消去左行中禾。

劉徽注：此方程術之正也。置三行於位：右行上禾三、中禾二、下禾一、實三十九；中行上禾二、中禾三、下禾一、實三十四；左行上禾一、中禾二、下禾三、實二十六。以右行上禾三乘中行，得六、九、三、一百零二；與右行直除之。

Q:Now there are 3 bundles of upper grain, 2 of middle, 1 of lower, yielding 39 dou; 2 of upper, 3 of middle, 1 of lower yielding 34 dou; 1 of upper, 2 of middle, 3 of lower yielding 26 dou. Question: how much does one bundle of each grain yield?

A:Upper: $9 \frac{1}{4}$ dou; Middle: $4 \frac{1}{4}$ dou; Lower: $2 \frac{3}{4}$ dou.

Method:Method: Fangcheng method. Arrange upper, middle, lower grain and shi in right, middle, left rows. Multiply middle row by upper grain coefficient 3 of right row; subtract directly to eliminate upper grain. Repeat for left row. Then eliminate middle grain from left row. Finally divide remaining lower grain coefficient into shi.

Liu Hui's Commentary:This is the orthodox Fangcheng method. Arrange three rows: right (3 upper, 2 middle, 1 lower, shi 39); middle (2 upper, 3 middle, 1 lower, shi 34); left (1 upper, 2 middle, 3 lower, shi 26). Eliminate by direct subtraction.

0.24 第二題 / Problem 2: 五家共輸 / Five Families Contributing Grain

問：今有甲、乙、丙、丁、戊五家共輸粟。甲所輸加乙之一半，乙所輸加丙之三分之一，丙所輸加丁之四分之一，丁所輸加戊之五分之一，戊所輸加甲之六分之一，各適等。問：各家所輸幾何？

答曰：各家所輸均為一斛。甲輸二十五分斛之十二，乙輸二十五分斛之四，丙輸二十五分斛之三，丁輸二十五分斛之二，戊輸二十五分斛之一。

術曰：術曰：以方程術入之。令各家所輸為未知數，以條件列方程，正負相生，求之即得。

劉徽注：此題以方程術入之為便。令甲、乙、丙、丁、戊所輸分別為 a, b, c, d, e 。則有： $a + b/2 = b + c/3 = c + d/4 = d + e/5 = e + a/6$ 。令此等於 k ，則五式聯立，可以方程術解之。

Q: Now five families A, B, C, D, and E are to contribute millet. A's contribution plus half of B's; B's plus one-third of C's; C's plus one-fourth of D's; D's plus one-fifth of E's; E's plus one-sixth of A's—all equal. Question: how much does each family contribute?

A: Each family's adjusted contribution equals 1 hu. The ratios are: A: 12/25, B: 4/25, C: 3/25, D: 2/25, E: 1/25.

Method: Method: Apply the Fangcheng method. Let each family's contribution be unknowns; form equations from the conditions; let positive and negative interact; solving yields the answer.

Liu Hui's Commentary: This problem is conveniently solved by the Fangcheng method. Let the contributions be a, b, c, d, e ; then all expressions equal a constant k , giving five equations solvable by Fangcheng.

0.25 第三題 / Problem 3: 三畜直金 / Three Animals' Gold Value

問: 今有牛二、羊五，直金十兩；牛三、豕三，直金九兩；羊六、豕八，直金八兩。問：牛、羊、豕各直金幾何？

答曰: 牛一直金一兩二十一分兩之十三，羊一直金二十一分兩之二十，豕一直金二十一分兩之四。

術曰: 術曰：方程術。置牛、羊、豕之數及直金於三行，以直除消去牛、羊，得下豕之實，實如法得豕之直。以次求羊、牛之直。

劉徽注: 牛二、羊五直十兩，牛三、豕三直九兩，羊六、豕八直八兩。以方程術：置三行，先消去牛，次消去羊，得豕之實。以豕之實推羊，以羊之實推牛。此三物方程之正法也。

Q: Now 2 oxen and 5 sheep are worth 10 tael of gold; 3 oxen and 3 pigs are worth 9 taels; 6 sheep and 8 pigs are worth 8 taels. Question: how much is each animal worth?

A: Ox: $1\frac{13}{21}$ taels; Sheep: $20\frac{20}{21}$ taels; Pig: $4\frac{4}{21}$ taels.

Method: Method: Fangcheng method. Arrange oxen, sheep, pigs and gold values in three rows. Eliminate oxen and sheep by direct subtraction to obtain pig's shi; divide to get pig's value. Then find sheep and ox values.

Liu Hui's Commentary: This is the orthodox Fangcheng method for three items: arrange three rows, first eliminate oxen, then sheep, obtaining pig's shi. From pig's shi deduce sheep's, and from sheep's deduce ox's.

0.26 第四題 / Problem 4: 麻麥稻菽黍 / Five Crops

問：今有麻九斗、麥七斗、菽三斗、答二斗、黍五斗，直錢一百四十；麻七斗、麥六斗、菽四斗、答五斗、黍三斗，直錢一百二十八；麻三斗、麥五斗、菽七斗、答六斗、黍四斗，直錢一百一十六；麻二斗、麥三斗、菽五斗、答七斗、黍六斗，直錢一百零四；麻五斗、麥四斗、菽六斗、答三斗、黍七斗，直錢一百一十二。問：五物各一斗直錢幾何？

答曰：麻一斗七錢，麥一斗四錢，菽一斗三錢，答一斗五錢，黍一斗六錢。

術曰：術曰：方程術。以五物列五行，以直錢為實，遍乘直除，消去各物之率，各得其一斗之實。

劉徽注：此五物方程，較之三物方程為繁。其術一也：列五行為程，以右行首物遍乘其下各行，以直除之，消去首物；次以中行次物遍乘其下各行，以直除之，消去次物；如是遞推，終得一物之實。

Q:Now [various quantities of] five crops—hemp, wheat, beans, malt, and millet—are worth certain amounts of cash. Question: how much cash is one dou of each crop worth?

A:Hemp: 7 cash; Wheat: 4 cash; Beans: 3 cash; Malt: 5 cash; Millet: 6 cash per dou.

Method:Method: Fangcheng method. Arrange five items in five rows with cash values as shi. Multiply and subtract directly to eliminate each item's coefficient.

Liu Hui's Commentary:This five-item Fangcheng is more complex than the three-item version, but the method is the same: arrange five rows; eliminate leading item from rows below; proceed until one item's shi remains; back-substitute.

0.27 第五題 / Problem 5: 正負術 / Positive and Negative Numbers

問：今有上禾七秉，損實一斗，益之下禾二秉，則實滿十斗；下禾八秉，益實一斗，與上禾三秉，則實滿十斗。問：上、下禾一秉各幾何？

答曰：上禾一秉一斗五十二分斗之四十七，下禾一秉五十二分斗之三十九。

術曰：術曰：以方程術入之。損實一斗者，正也；益實一斗者，負也。正負列位，如方程消之。

劉徽注：正負術者，方程術之輔也。以兩算得失相反，要令正負以名之。正算赤，負算黑。否則以邪正為異。同名相除，異名相益。正無入負之，負無入正之。此正負之術也。

Q:Now 7 bundles of upper grain, with 1 dou removed, and adding 2 bundles of lower grain, gives exactly 10 dou; 8 bundles of lower grain, with 1 dou added, together with 3 bundles of upper grain, gives exactly 10 dou. Question: how much does one bundle of each grain yield?

A:Upper grain: 1 47/52 dou; Lower grain: 39/52 dou.

Method:Method: Apply the Fangcheng method. Removing 1 dou of shi is positive; adding 1 dou of shi is negative. Arrange positive and negative positions and eliminate.

Liu Hui's Commentary:The positive-negative method is the auxiliary to the Fangcheng method. Positive counting rods are red, negative are black. Same names subtract; different names add. This is the first known systematic use of signed numbers in world mathematics.

0.28 第六題 / Problem 6: 四畜求直 / Horse, Ox, and Sheep

問: 今有馬一、牛二、羊三，直金二十五兩；馬二、牛三、羊一，直金二十八兩；馬三、牛一、羊二，直金二十三兩。問：馬、牛、羊各直金幾何？

答曰: 馬一直金九兩，牛一直金五兩，羊一直金二兩。

術曰: 術曰：方程術。列三行，馬、牛、羊之數及直金為實。以右行馬一遍乘中行、左行，以直除之，消去馬。次以中行牛遍乘左行，以直除之，消去牛。左行羊不盡者，上為法，下為實，實如法得羊之直。

劉徽注: 此三物方程之又一例也。馬一、牛二、羊三直二十五兩；馬二、牛三、羊一直二十八兩；馬三、牛一、羊二直二十三兩。以方程術消之：右行馬一為最簡，故以右行遍乘中行、左行，直除消馬。

Q:Now 1 horse, 2 oxen, and 3 sheep are worth 25 taels; 2 horses, 3 oxen, and 1 sheep worth 28 taels; 3 horses, 1 ox, and 2 sheep worth 23 taels. Question: how much is each animal worth?

A:Horse: 9 taels; Ox: 5 taels; Sheep: 2 taels.

Method:Method: Fangcheng method. Arrange three rows with quantities and gold values. Multiply middle and left rows by right row's horse coefficient 1; subtract directly to eliminate horses. Then eliminate oxen from left row.

Liu Hui's Commentary:This is another three-item Fangcheng. The right row has horse coefficient 1, the simplest, so multiply and subtract to eliminate horses. Then eliminate oxen to obtain sheep's shi. Back-substitute to find oxen and horses.

0.29 第七題 / Problem 7: 粟麥互易 / Grain Exchange

問：今有上禾六秉，損實一斗八升，當下禾一十秉；下禾一十五秉，損實五升，當上禾五秉。問：上、下禾一秉各幾何？

答曰：上禾一秉八升，下禾一秉三升。

術曰：術曰：方程術。列上禾、下禾之數及損實為行。以正負術記之，直除消去，求得各秉之實。

劉徽注：上禾六秉損一斗八升當下禾十秉者，六秉上禾之實減一斗八升，等於十秉下禾之實也。下禾十五秉損五升當上禾五秉者，十五秉下禾之實減五升，等於五秉上禾之實也。此二物二程之方程，較之三物為簡，然其法不異。

Q:Now 6 bundles of upper grain, with 1 dou 8 sheng removed, equals 10 bundles of lower grain; 15 bundles of lower grain, with 5 sheng removed, equals 5 bundles of upper grain. Question: how much does one bundle of each grain yield?

A:Upper grain: 8 sheng; Lower grain: 3 sheng per bundle.

Method:Method: Fangcheng method. Arrange upper grain, lower grain quantities and removed shi as rows. Record with positive-negative; eliminate by direct subtraction; find yield per bundle.

Liu Hui's Commentary:This two-item, two-equation Fangcheng is simpler than the three-item case, but the method does not differ.

0.30 第八題 / Problem 8: 五家共輸稅 / Five Classes Contributing Tax

問：今有五等之家共輸稅。甲五家、乙四家、丙三家、丁二家、戊一家，共輸稅一千斛。欲令各依等衰輸之。問：各輸幾何？

答曰：甲輸二百七十七斛七分斛之六，乙輸二百二十二斛七分解之二，丙輸一百六十六斛七分解之四，丁輸一百一十一斛七分解之一，戊輸五十五斛七分解之五。

術曰：術曰：以方程術入之。令五家為五未知，以家數及共輸為實，依衰分列方程，消去求之。

劉徽注：五等之家，各有多少之異。甲五、乙四、丙三、丁二、戊一，共十五。以千斛分為十五分，甲得其五，乙得其四，丙得其三，丁得其二，戊得其一。此術雖以方程入之，實以衰分為本也。

Q:Now five classes of households are to contribute tax: A has 5 households, B has 4, C has 3, D has 2, E has 1; together they contribute 1000 hu. It is desired that each class contribute proportionally. Question: how much does each class contribute?

A:Class A: 277 6/7 hu; B: 222 2/7 hu; C: 166 4/7 hu; D: 111 1/7 hu; E: 55 5/7 hu.

Method:Method: Apply the Fangcheng method. Let the five classes be five unknowns; use household counts and total contribution as shi; form equations according to proportional decrease and eliminate.

Liu Hui's Commentary: Five classes differ in quantity. The proportions A:B:C:D:E = 5:4:3:2:1 sum to 15. Divide 1000 hu accordingly. Although this uses the Fangcheng method, its foundation is proportional division.

0.31 第九題 / Problem 9: 三人持錢 / Three People Holding Money

問：今有甲、乙、丙三人持錢。甲得乙之半而錢滿四十八，乙得丙之半而錢滿四十八，丙得甲之半而錢滿四十八。問：甲、乙、丙各持錢幾何？

答曰：甲持錢三十六，乙持錢二十四，丙持錢十二。

術曰：術曰：以方程術入之。令甲、乙、丙持錢為三未知數，以條件列三方，消去求之。

劉徽注：甲得乙之半而滿四十八者，甲加乙之半等於四十八也。設甲持 x ，乙持 y ，丙持 z ，則： $x + y/2 = 48$ ， $y + z/2 = 48$ ， $z + x/2 = 48$ 。以方程術列三行，正負記之，消去求之。

Q: Now three people A, B, and C hold money. If A obtains half of B's money, he has exactly 48; if B obtains half of C's money, he has exactly 48; if C obtains half of A's money, she has exactly 48. Question: how much does each person hold?

A: A holds 36; B holds 24; C holds 12.

Method: Method: Apply the Fangcheng method. Let the amounts held by A, B, C be three unknowns; form three equations from the conditions and eliminate.

Liu Hui's Commentary: This problem is solved by setting up three simultaneous equations and eliminating variables. The solution is verified: $36+12=48$, $24+6=48$, $12+18=48$.

0.32 第十題 / Problem 10: 五家共井（方程術解） / Five Families Sharing a Well (Fangcheng Solution)

問：今有五家共井。甲二綆不足，如乙一綆；乙三綆不足，如丙一綆；丙四綆不足，如丁一綆；丁五綆不足，如戊一綆；戊六綆不足，如甲一綆。問：井深、綆長各幾何？

答曰：井深七丈二尺一寸。甲綆長二丈六尺五寸，乙綆長一丈九尺一寸，丙綆長一丈四尺八寸，丁綆長一丈二尺九寸，戊綆長七尺六寸。

術曰：術曰：方程術。列五家綆長及井深為六未知數，以五條件列五行。正負相生，直除消去，求得其比，然後以井深定其實長。

劉徽注：此題與卷七所載為同一題，而術不同。卷七以盈不足術驗之，此則以方程術正解之。此五程六物之方程，為不定問題，故特設井深以定其解。

Q:Now five families share a well. Two lengths of A's rope insufficient equals one B's rope length; three B's equals one C's; four C's equals one D's; five D's equals one E's; six E's equals one A's. Question: what is the depth of the well and each rope's length?

A:Well depth: 7 zhang 2 chi 1 cun. A: 2 zhang 6 chi 5 cun; B: 1 zhang 9 chi 1 cun; C: 1 zhang 4 chi 8 cun; D: 1 zhang 2 chi 9 cun; E: 7 chi 6 cun.

Method:Method: Fangcheng method. Arrange five rope lengths and well depth as six unknowns from five conditions. This is an indeterminate system; the well depth is fixed to determine a unique solution.

Liu Hui's Commentary:This is the same problem as in Volume 7, but solved properly by Fangcheng rather than verified by excess-deficit. The indeterminate system requires fixing the well depth to obtain the unique integer solution.

0.33 第十一題 / Problem 11: 三禾互易 / Three Grains Exchange

問：今有上禾七秉，益實六斗，當下禾一十秉；下禾五秉，益實一斗，當上禾二秉。問：上、下禾一秉各幾何？

答曰：上禾一秉八升，下禾一秉三升。

術曰：術曰：方程術。列上禾、下禾之數及益實為行，正負記之，直除消去，求得各秉之實。

劉徽注：上禾七秉益六斗當下禾十秉者，七秉上禾之實加六斗，等於十秉下禾之實也。下禾五秉益一斗當上禾二秉者，五秉下禾之實加一斗，等於二秉上禾之實也。以方程列之，正負記之，直除消去。

Q:Now 7 bundles of upper grain, with 6 dou added, equals 10 bundles of lower grain; 5 bundles of lower grain, with 1 dou added, equals 2 bundles of upper grain. Question: how much does one bundle of each yield?

A:Upper grain: 8 sheng; Lower grain: 3 sheng per bundle.

Method:Method: Fangcheng method. Arrange quantities and added shi as rows; record with positive-negative; eliminate by direct subtraction; find yield per bundle.

Liu Hui's Commentary:This is a two-item, two-equation Fangcheng. Record with positive-negative; eliminate by direct subtraction.

0.34 第十二題 / Problem 12: 四器求容 / Large and Small Vessels

問：今有大器一、小器五，容三斛；大器五、小器一，容二斛。問：大、小器各容幾何？

答曰：大器容二十四分斛之十三，小器容二十四分斛之七。

術曰：術曰：方程術。列大器、小器之數及容為兩行，直除消去大器，得小器之實。以小器之實回代，求得大器之容。

劉徽注：大器一、小器五容三斛，大器五、小器一容二斛。以方程術列之：右行大器一、小器五、實三；左行大器五、小器一、實二。以右行遍乘左行，直除消大器。

Q:Now 1 large vessel and 5 small vessels hold 3 hu; 5 large vessels and 1 small vessel hold 2 hu. Question: how much does each vessel hold?

A:Large vessel: 13/24 hu; Small vessel: 7/24 hu.

Method:Method: Fangcheng method. Arrange large vessel, small vessel quantities and capacities as two rows; eliminate large vessel by direct subtraction to obtain small vessel's shi. Back-substitute.

Liu Hui's Commentary:This is a simple two-item Fangcheng, but its method does not differ from complex cases.

0.35 第十三題 / Problem 13: 金銀鉛錫 / Gold, Silver, Lead, Tin

問：今有金一斤、銀一斤，稱之適等；金一斤、鉛一斤，稱之金輕五兩；銀一斤、錫一斤，稱之銀輕三兩。問：金、銀、鉛、錫一斤各重幾何？

答曰：金一斤重一斤，銀一斤重一斤，鉛一斤重一斤五兩，錫一斤重一斤三兩。

術曰：術曰：方程術。以金銀適等為一條件，金鉛差五兩為一條件，銀錫差三兩為一條件，列方程求之。

劉徽注：金銀適等者，一斤金與一斤銀重量相等也。此以天平稱之，非論其體積。金鉛稱之金輕五兩者，同體積之鉛重於金五兩也。三條件四物，為不定方程。以金一斤為率，則銀一斤亦為一斤，鉛一斤五兩，錫一斤三兩。

Q:Now 1 jin of gold and 1 jin of silver balance exactly; 1 jin of gold and 1 jin of lead show gold is lighter by 5 liang; 1 jin of silver and 1 jin of tin show silver is lighter by 3 liang. Question: how much does 1 jin of each weigh?

A:1 jin of gold: 1 jin; 1 jin of silver: 1 jin; 1 jin of lead: 1 jin 5 liang; 1 jin of tin: 1 jin 3 liang.

Method:Method: Fangcheng method. Three conditions with four substances form an indeterminate system; fixing one substance determines the others.

Liu Hui's Commentary:This uses the Fangcheng method's hidden-explicit relationships to determine the weights. Gold and silver balance on a scale; lead is 5 liang heavier than equal-volume gold.

0.36 第十四題 / Problem 14: 方程新術 / New Fangcheng Method

問: 今有上禾二秉，中禾三秉，下禾四秉，實皆不滿斗。上禾取中禾一秉，中禾取下禾一秉，下禾取上禾一秉，而實滿斗。問：上、中、下禾一秉各幾何？

答曰: 上禾一秉二十五分斗之十一，中禾一秉二十五分斗之七，下禾一秉二十五分斗之四。

術曰: 術曰：方程新術。以所取之數列為方程，各以其取數遍乘，直除消去，各得其實。

劉徽注: 此方程術之變也。上禾二秉取中禾一秉而滿斗者，上禾一秉之實加中禾半秉之實為半斗也。以三條件列三行，直除消去求之。方程新術者，變通之術也。

Q:Now 2 bundles of upper, 3 of middle, 4 of lower grain each yield less than 1 dou. If upper takes 1 middle, middle takes 1 lower, lower takes 1 upper, each sum fills exactly 1 dou. Question: how much does one bundle of each yield?

A:Upper: 11/25 dou; Middle: 7/25 dou; Lower: 4/25 dou.

Method:Method: The New Fangcheng Method. Arrange quantities taken as equations; multiply throughout by each quantity taken; eliminate by direct subtraction; obtain each yield.

Liu Hui's Commentary:This is a variation of the Fangcheng method. The New Fangcheng Method is a method of versatile adaptation.

0.37 第十五題 / Problem 15: 重衰分入方程 / Multiple Proportions in Fangcheng

問: 今有粟五斗，為糲米、粳米、糞米三等。欲令糲二、粳三、糞四為率。問：各為米幾何？

答曰: 糲米一斗四升七分升之四，粳米二斗一升七分升之六，糞米二斗八升七分升之二。

術曰：術曰：以方程術入之。令三米之率為三未知數，以粟五斗為實，以率之比列方程，消去求之。

劉徽注：糲二、粳三、糞四為率者，三米當為二比三比四之比例也。以方程列之：設糲米為 $2k$ ，粳米為 $3k$ ，糞米為 $4k$ ，則 $2k + 3k + 4k = 5$ 斗， $9k = 5$ 斗。此術以方程顯衰分之隱也。

Q: Now 5 dou of millet is to be divided into three grades of rice. It is desired that the rates be li: 2, bai: 3, fen: 4. Question: how much of each?

A: Li mi: 1 dou 4 sheng $\frac{4}{7}$ sheng; bai mi: 2 dou 1 sheng $\frac{6}{7}$ sheng; fen mi: 2 dou 8 sheng $\frac{2}{7}$ sheng.

Method: Method: Apply the Fangcheng method. Let the three rice rates be three unknowns; use 5 dou as shi; form equations from the rate ratios and eliminate.

Liu Hui's Commentary: Li: 2, bai: 3, fen: 4 means the three rices are in proportion 2:3:4. This makes the hidden proportional division visible through the Fangcheng method.

0.38 第十六題 / Problem 16: 四色方程 / Four Colors of Silk

問：今有甲、乙、丙、丁四色之綾。甲三匹、乙二匹、丙一匹、丁四匹，直錢一百二十；甲二匹、乙四匹、丙三匹、丁一匹，直錢一百一十；甲一匹、乙三匹、丙四匹、丁二匹，直錢一百；甲四匹、乙一匹、丙二匹、丁三匹，直錢一百三十。問：四色綾每匹各直錢幾何？

答曰：甲一匹二十錢，乙一匹十五錢，丙一匹十錢，丁一匹五錢。

術曰：術曰：方程術。列四色為四行，以直錢為實，遍乘直除，消去三色，得第四色之實。回代求之。

劉徽注：此四物方程，其術一也。列四行於位，以右行甲三遍乘次行，直除消甲。消去甲後，次以乙消丙，再以丙消丁，終得丁之實。回代求丙、乙、甲。此四物方程雖繁，其法不異於三物也。

Q: Now four colors of silk: A, B, C, D. Given four equations with quantities and values. Question: how much is one bolt of each color worth?

A: A: 20 cash; B: 15 cash; C: 10 cash; D: 5 cash per bolt.

Method: Method: Fangcheng method. Arrange four colors as four rows; multiply and subtract directly to eliminate three colors, obtaining the fourth color's shi. Back-substitute.

Liu Hui's Commentary: Although this four-item Fangcheng is complex, its method does not differ from the three-item case.

0.39 第十七題 / Problem 17: 均輸入方程 / Equal Transport in Fangcheng

問：今有甲、乙、丙三縣運粟。甲縣一萬斛，乙縣八千斛，丙縣六千斛。三縣共運至京師，程途遠近不同：甲縣五百里，乙縣三百里，丙縣二百里。欲令三縣均輸，問：各當運幾何？

答曰：甲縣運四千斛，乙縣運三千二百斛，丙縣運二千四百斛。

術曰：術曰：以方程術入衰均輸求之。令三縣粟數及程途列為方程，以里數乘粟數，均之，消去求之。

劉徽注：三縣均輸者，不以粟數為率，而以粟數乘里數之積為率也。甲一萬乘五百為五百萬，乙八千乘三百為二百四十萬，丙六千乘二百為一百二十萬。此術以方程顯均輸之隱。

Q:Now three counties A, B, C are to transport millet. A has 10,000 hu, B has 8,000 hu, C has 6,000 hu. Distances: A 500 li, B 300 li, C 200 li. It is desired that the three counties contribute equally in transport effort. Question: how much should each transport?

A:County A transports 4,000 hu; B transports 3,200 hu; C transports 2,400 hu.

Method:Method: Apply the Fangcheng method with equal transport. Arrange millet amounts and distances as equations; multiply millet by distance; equalize; eliminate and solve.

Liu Hui's Commentary:Equal transport uses the product of millet amount and distance as the rate. This makes the hidden equal transport visible through the Fangcheng method.

0.40 第十八題 / Problem 18: 盈不足入方程 / Double Excess in Fangcheng

問：今有買金，人出四百，盈三千四百；人出三百，盈一百。問：人數、金價各幾何？

答曰：人三十三，金價一萬二千八百。

術曰：術曰：以方程術入之。兩盈者，以兩盈之數相減為法，兩設之數相減為實，實如法而一。

劉徽注：此題兩盈，非盈不足之正術也。盈不足之正術，一盈一不足。今兩盈，則以兩盈之差為法，兩設之差為實，實如法而一，得人數。

Q:Now in buying gold: each person contributes 400, excess is 3,400; each contributes 300, excess is 100. Question: how many people, and what is the price of the gold?

A:33 people; the gold costs 12,800.

Method:Method: Apply the Fangcheng method. In double excess: subtract the two excess numbers to obtain the divisor; subtract the two assumed contributions to obtain the dividend; divide.

Liu Hui's Commentary:This problem involves double excess. The difference of excesses is 3,300; difference of assumptions is 100. Dividing gives 33 people.

0.41 第十九題 / Problem 19: 兩不足入方程 / Double Deficit in Fangcheng

問: 今有買豕，人出三百，不足四百；人出二百，不足二百。問：人數、豕價各幾何？

答曰: 人二，豕價一千。

術曰: 術曰：以方程術入之。兩不足者，以兩不足之數相減為法，兩設之數相減為實，實如法而一。

劉徽注: 兩不足者，與兩盈相反，而其術同。人出三百不足四百，人出二百不足二百。兩不足之差二百，兩設之差一百。實如法得二人。方程之妙，在於無所不包：盈納、兩盈、兩不足，皆可入之。

Q:Now in buying pigs: each person contributes 300, deficit is 400; each contributes 200, deficit is 200. Question: how many people, and what is the price of the pigs?

A:2 people; the pigs cost 1,000.

Method:Method: Apply the Fangcheng method. In double deficit: subtract the two deficit numbers as divisor; subtract the two assumptions as dividend; divide.

Liu Hui's Commentary:Double deficit is the opposite of double excess, but the method is the same. The wonder of the Fangcheng method lies in its all-inclusiveness: excess-deficit, double excess, double deficit—all can be incorporated.

0.42 第二十題 / Problem 20: 方程通術 / General Fangcheng Method

問: 今有上禾、中禾、下禾，不知其實。上禾三秉、中禾二秉、下禾一秉，實三十九；上禾二秉、中禾三秉、下禾一秉，實三十四；上禾一秉、中禾二秉、下禾三秉，實二十六。問：上、中、下禾一秉各幾何？

答曰：上禾一秉九斗四分斗之一，中禾一秉四斗四分斗之一，下禾一秉二斗四分斗之三。

術曰：術曰：方程通術。列三行，以右行遍乘中行、左行，直除消去。終得一物之實，回代求之。

劉徽注：此題與第一題同，而以方程通術解之，所以見方程之通法也。方程通術者，凡方程皆以此術解之。此術之精妙，在於一通百通，無論二物、三物、四物、五物，皆以此術解之。劉徽曰：方程之術，以通為貴。

Q:Now [same as Problem 1]. Three equations with three unknowns. Question: how much does one bundle of each grain yield?

A:Upper: $9\frac{1}{4}$ dou; Middle: $4\frac{1}{4}$ dou; Lower: $2\frac{3}{4}$ dou.

Method:Method: The General Fangcheng Method. Arrange three rows; multiply and subtract directly to eliminate. Finally one item's shi remains; back-substitute.

Liu Hui's Commentary:The General Fangcheng Method: all Fangcheng problems are solved by this method. Its subtlety lies in universal applicability. Liu Hui said: In the Fangcheng method, universality is prized.

卷第九 勾股 / Volume 9 Right Triangles

勾股章乃《九章》之巔峰，以畢達哥拉斯定理（勾股定理）及其應用為核心，論述極為深入。劉徽注尤為卓越，包括著名之「割補術」證明勾股定理，以及「出入相補原理」之論述。本章亦含距離測量、方圓關係，及中國數學史上對二次無理數之最早探討。

The Gougu chapter is the culmination of the Nine Chapters, presenting the Pythagorean theorem and its applications in extraordinary depth. Liu Hui's commentary includes his famous "method of cutting and patching" for proving the Pythagorean theorem, and his "method of double differences" for surveying. This chapter also contains the first known Chinese approach to quadratic irrationals.

Fundamental Theorem / 勾股基本定理

術曰：勾股術曰：勾股各自乘，并而開方除之，即弦。又股自乘，以減弦自乘，其餘開方除之，即句。又句自乘，以減弦自乘，其餘開方除之，即股。

劉徽注：勾股之法，出於圓方。按句股者，矩之別名也。矩之為物，橫者為句，縱者為股，斜者為弦。句股各自乘，并之為弦實。開方除之，得弦。徽按：勾股各自乘，并之為弦實，此弦實之正方形，可以割補為勾實、股實兩正方形。割補術者，所以證勾股之定理也。

Method: The Gougu method says: Square the shorter leg (gou) and the longer leg (gu), sum them, and extract the square root to obtain the hypotenuse (xian). Alternatively, square the longer leg and subtract from the square of the hypotenuse; the remainder's square root gives the shorter leg. Alternatively, square the shorter leg and subtract from the square of the hypotenuse; the remainder's square root gives the longer leg.

Liu Hui's Commentary: The Gougu method originates from circles and squares. The "gou-gu" is another name for the carpenter's square. In the carpenter's square, the horizontal is gou, the vertical is gu, and the diagonal is xian. Squaring gou and gu and summing gives the shi of the hypotenuse. Liu Hui's note: The sum of squares of gou and gu equals the shi of the hypotenuse; this hypotenuse square can be cut and patched to form the two squares on gou and gu. The cutting-and-patching method proves the Pythagorean theorem.

The Pythagorean Theorem: For legs a (勾 gou), b (股 gu), and hypotenuse c (弦 xian):

$$a^2 + b^2 = c^2$$

Liu Hui's Proof: The square on the hypotenuse can be cut and rearranged to exactly fill the two squares on the legs.

0.43 第一題 / Problem 1: 勾股求弦 / Legs to Hypotenuse

問: 今有句三尺，股四尺，問：為弦幾何？

答曰: 弦五尺。

術曰: 術曰：句股各自乘，并而開方除之，即弦。句自乘得九，股自乘得一十六，并之得二十五，開方除之得五尺。

劉徽注: 此勾股之正術也。勾三、股四、弦五，此為勾股之最基本率。勾三股四弦五之率，出於天地自然，為萬物之綱紀。以之測高深深遠，無所不可。

Q: Now the shorter leg is 3 chi and the longer leg is 4 chi. Question: what is the hypotenuse?

A: The hypotenuse is 5 chi.

Method: Method: Square the shorter leg and longer leg, sum them, and extract the square root. The shorter leg squared gives 9; the longer leg squared gives 16; summing gives 25; the square root gives 5 chi.

Liu Hui's Commentary: This is the orthodox Gougu method. Gou 3, gu 4, xian 5—the most fundamental rate. The 3-4-5 rate emerges naturally from heaven and earth, serving as the guiding principle of all things.

0.44 第二題 / Problem 2: 弦股求句 / Hypotenuse and Longer Leg to Shorter Leg

問: 今有弦五尺，股四尺，問：為句幾何？

答曰：句三尺。

術曰：術曰：股自乘，以減弦自乘，其餘開方除之，即句。股自乘得一十六，弦自乘得二十五，以十六減二十五餘九，開方除之得三尺。

劉徽注：弦實二十五，股實一十六，兩實之差為句實九。開方得三，即句。此術為勾股術之逆也。勾股之術，互為正逆，相為表裡。

Q:Now the hypotenuse is 5 chi and the longer leg is 4 chi. Question: what is the shorter leg?

A:The shorter leg is 3 chi.

Method:Method: Square the longer leg and subtract from the square of the hypotenuse; the square root of the remainder gives the shorter leg. The longer leg squared gives 16; the hypotenuse squared gives 25; subtracting 16 from 25 leaves 9; the square root gives 3 chi.

Liu Hui's Commentary:This method is the inverse of the Gougu method. The Gougu methods are mutually direct and inverse, serving as exterior and interior to each other.

0.45 第三題 / Problem 3: 句弦求股 / Shorter Leg and Hypotenuse to Longer Leg

問：今有句八尺，弦十七尺，問：為股幾何？

答曰：股十五尺。

術曰：術曰：句自乘，以減弦自乘，其餘開方除之，即股。句自乘得六十四，弦自乘得二百八十九，以六十四減二百八十九餘二百二十五，開方除之得一十五尺。

劉徽注：弦寬二百八十九，句實六十四，兩實之差為股實二百二十五。開方得一十五，即股。勾八、股十五、弦十七，此又一股弦之率也。勾股之率無窮，而其術一也。

Q:Now the shorter leg is 8 chi and the hypotenuse is 17 chi. Question: what is the longer leg?

A:The longer leg is 15 chi.

Method:Method: Square the shorter leg and subtract from the square of the hypotenuse; the square root of the remainder gives the longer leg. The shorter leg squared gives 64; the hypotenuse squared gives 289; subtracting 64 from 289 leaves 225; the square root gives 15 chi.

Liu Hui's Commentary:Gou 8, gu 15, xian 17—this is another rate. The Gougu rates are infinite, but the method is one.

0.46 第四題 / Problem 4: 戶高廣 / Door Height and Width

問：今有戶高多於廣六尺八寸，兩隅相去適一丈。問：戶高、廣各幾何？

答曰：廣二尺八寸，高九尺六寸。

術曰：術曰：令一丈自乘，半之得五十尺為實。令多六尺八寸自乘，得四十六尺二寸四分，半之得二十三尺一寸二分為法。實如法得一尺，為戶廣。加多六尺八寸，為高。

劉徽注：戶高多於廣六尺八寸者，高與廣之差也。兩隅相去一丈者，以高廣為勾股之弦也。此術以勾股弦之關係求高廣也。此術之妙，在於以差求廣，以廣求高，一環相扣。

Q:Now a door's height exceeds its width by 6 chi 8 cun; the diagonal distance between opposite corners is exactly 1 zhang. Question: what are the door's height and width?

A:Width: 2 chi 8 cun; height: 9 chi 6 cun.

Method:Method: Square 1 zhang and halve to obtain 50 square chi as dividend. Square the excess and halve as divisor. Divide to obtain width. Add the excess to obtain height.

Liu Hui's Commentary:The height exceeding the width is the difference between height and width. The diagonal being 1 zhang means the height and width form legs of a right triangle with the diagonal as hypotenuse. The wonder lies in finding width from the difference, then height from width.

0.47 第五題 / Problem 5: 邑方出東門 / City Size from East Gate

問：今有邑方不知大小，各開中門。出東門二十步有木。問：出南門幾何步而見木？

答曰：出南門一十四步半而見木。

術曰：術曰：以勾股相似術入之。出東門二十步為勾，邑方之半為股。以股自乘，以勾除之，得南門外見木之步數。

劉徽注：邑方者，正方形之城邑也。此以勾股相似之術求之：邑方之半為一大股，東門外二十步為一大勾；南門外所求步數為一小股。兩勾股相似，故以比例求之。此術以相似勾股測遠近，為重差之基礎也。

Q:Now there is a square city of unknown size; gates at middle of each side. Going out the east gate 20 paces there is a tree. Question: how many paces out the south gate until one can see the tree?

A: 14.5 paces out the south gate.

Method: Method: Apply the similar right triangles method. The 20 paces out the east gate is the shorter leg; half the city's side is the longer leg. Square the longer leg and divide by the shorter leg.

Liu Hui's Commentary: Using similar right triangles: half the city's side is a large longer leg; the 20 paces is a large shorter leg. The two right triangles are similar, so proportions can be used. This forms the basis of the double-difference method.

0.48 第六題 / Problem 6: 井徑求立木 / Well Depth from Pole

問: 今有井徑五尺，不知其深。立木於井上，從木末望水岸，入徑四尺。問：井深幾何？

答曰: 井深一丈九尺二寸。

術曰: 術曰：以勾股相似術入之。井徑五尺為勾，入徑四尺為股之差。以井徑乘入徑，以井徑減入徑之差除之，得井深。

劉徽注: 井徑五尺者，井口之直徑也。此亦以勾股相似之術：木高為一大股，入徑為一大勾；井深為一小股，井徑之半為一小勾。兩勾股相似，故可以比例求之。此術以勾股相似測深淺，亦重差之術也。

Q: Now a well has diameter 5 chi; its depth is unknown. A pole is erected above the well; looking from the top of the pole to the water's edge, the sight line enters the diameter 4 chi from the edge. Question: how deep is the well?

A: The well is 1 zhang 9 chi 2 cun deep.

Method: Method: Apply the similar right triangles method. The well diameter 5 chi is the shorter leg; the 4 chi entry is the difference of the longer leg. Multiply well diameter by entry distance and divide by the difference to obtain depth.

Liu Hui's Commentary: This again uses similar right triangles: the pole height is a large longer leg; the entry distance is a large shorter leg; the well depth is a small longer leg. The two right triangles are similar, so proportions can be used.

0.49 第七題 / Problem 7: 登山望邑 / Mountain Surveying a City

問：今有登山望邑，邑在山東。山高不知高，立兩表齊高三丈，前後相去千步。令後表與前表參相直。從前表卻行一百二十三步，人目著地，取望邑西北隅，入表前三寸。從後表卻行一百二十七步，人目著地，取望邑西北隅，適與表端參合。問：邑方及山去表各幾何？

答曰：邑方一里二百二十步，山去表一里一百八十二步半。

術曰：術曰：以重差術入之。此為重差之正術，以兩表之高及相去之遠，兩卻行之數，入表之數，列為勾股，相似求之。

劉徽注：此重差術之典型題也。重差者，以兩表之重迭差數求遠物之高深廣遠也。此術之妙，在於以咫尺之矩，測千里之遠。故劉徽曰：以咫尺之矩，測千里之高深。重差術者，勾股之極致也。

Q: Now climbing a mountain to survey a city. Two poles of equal height 3 zhang are erected, 1000 paces apart. From the front pole, retreating 123 paces with the eye on the ground, looking at the city's northwest corner, the sight line passes 3 cun in front of the pole. From the rear pole, retreating 127 paces, looking at the same corner, the sight line exactly aligns with the pole top. Question: what is the city's side length, and how far is the mountain from the poles?

A: City side: 1 li 220 paces; mountain from poles: 1 li 182.5 paces.

Method: Method: Apply the double-difference (chong cha) method. This is the orthodox double-difference method: using the two poles' height and separation, the two retreat distances, and the front-offset measurement, form right triangles and use similarity.

Liu Hui's Commentary: This is the archetypal double-difference problem. "Double difference" means using the overlapping differences of two poles to find the height, depth, width, or distance of a remote object. The wonder lies in using a foot-long carpenter's square to measure a thousand-li distance.

0.50 第八題 / Problem 8: 勾股容方 / Square Inscribed in Right Triangle

問：今有句五步，股十二步，問：句中容方幾何？

答曰：方三步十七分步之九。

術曰：術曰：并句、股為法，句股相乘為實，實如法而一，得方。句五步、股十二步，并之得十七步為法。句股相乘得六十步為實。實如法得三步十七分步之九。

劉徽注：勾股容方者，於直角三角形中作一最大之內接正方形也。以勾股之相并為法，句股之相乘為實者，以相似勾股之理推之也。此術精妙，以相似勾股之理，求內接正方之邊。

Q: Now in a right triangle, the shorter leg is 5 paces and the longer leg is 12 paces. Question: what is the side of the largest square that fits inside the triangle?

A: The square side is $3 \frac{9}{17}$ paces.

Method:Method: Sum the shorter and longer legs as the divisor; multiply the shorter and longer legs as the dividend; divide to obtain the square side. 5 plus 12 gives 17 as divisor. Product gives 60 as dividend. Dividing yields $3\frac{9}{17}$ paces.

Liu Hui's Commentary:Inscribing a square in a right triangle means constructing the largest possible inscribed square. Using the sum of gou and gu as divisor and the product as dividend is derived from the principle of similar right triangles. The side equals the product of gou and gu divided by the sum of gou and gu.

0.51 第九題 / Problem 9: 勾股容圓 / Circle Inscribed in Right Triangle

問: 今有句八步，股十五步，問：句中容圓，徑幾何？

答曰：圓徑六步。

術曰：術曰：句股相乘，倍之為實。并句、股、弦為法。實如法而一，得圓徑。句八步、股十五步，句股相乘得一百二十，倍之得二百四十為實。句八、股十五、弦十七，并之得四十為法。實如法得六步。

劉徽注：勾股容圓者，於直角三角形中作一最大之內切圓也。以勾股相乘倍之為實者，兩倍三角形之面積也。以勾股弦之和為法者，三角形周長之半也。三角之面積等於內切圓半徑乘周長之半，故以面積之兩倍除以周長之半，得圓徑。此術之精妙，以面積之關係推圓徑也。

Q:Now in a right triangle, the shorter leg is 8 paces and the longer leg is 15 paces. Question: what is the diameter of the largest circle that fits inside the triangle?

A:The circle's diameter is 6 paces.

Method:Method: Multiply the shorter and longer legs; double the product to form the dividend. Sum the shorter leg, longer leg, and hypotenuse as the divisor. Divide dividend by divisor. The shorter leg 8 times the longer leg 15 gives 120; doubled gives 240 as dividend. $8 + 15 + 17 = 40$ as divisor. Dividing gives 6 paces.

Liu Hui's Commentary:Inscribing a circle in a right triangle means constructing the largest possible incircle. Doubling the product as dividend means twice the triangle's area. The sum as divisor means half the perimeter. The area equals the inradius times half the perimeter; hence twice the area divided by half the perimeter gives the diameter.

0.52 第十題 / Problem 10: 邑中望木 / Tree Visible from City Gate

問：今有邑方不知大小，各開中門。出北門二十步有木。出南門十四步，折而西行一千七百七十五步見木。問：邑方幾何？

答曰：邑方五十步。

術曰：術曰：以勾股術入之。出北門二十步與出南門十四步相加得三十四步，以邑方之半乘之，以西行一千七百七十五步除之，得邑方之半。倍之得邑方。

劉徽注：邑方者，正方形之城邑也。出北門二十步有木者，從北門中出北行二十步有樹也。此以勾股相似之術求之：邑方之半為一勾，南北出門步數之和為一股，西行步數為另一勾。兩勾股相似，故可以比例求邑方。

Q:Now there is a square city of unknown size; gates at middle of each side. Twenty paces out the north gate there is a tree. Going out the south gate 14 paces, then turning west and going 1775 paces, one sees the tree. Question: what is the city's side length?

A:The city's side is 50 paces.

Method:Method: Apply the Gougu method. Sum the 20 paces out the north gate and 14 paces out the south gate to get 34 paces. Multiply by half the city's side; divide by the 1775 paces traveled west; obtain half the city's side. Double to get the full side.

Liu Hui's Commentary:This is found by similar right triangles: half the city's side is one shorter leg; the sum of the north-south gate distances is one longer leg; the westward travel is another shorter leg. The two right triangles are similar, so proportions can find the city side.

0.53 第十一題 / Problem 11: 望清淵 / Gazing into a Clear Abyss

問：今有清淵，不知深淺。立一木於岸上，高三丈。從木末斜望水際，入木一尺二寸。又望水底，入木四尺八寸。問：淵深幾何？

答曰：淵深一丈二尺。

術曰：術曰：以勾股相似術入之。木高三丈為股，入木一尺二寸為勾。入木四尺八寸為另一勾之率。以比例求之，得淵深。

劉徽注：立木高三丈於岸上，從木末斜望水際入木一尺二寸者，視線從木頂到水面與木杆相交處距木頂一尺二寸也。此術以勾股相似測深淺，亦重差之應用也。

Q:Now there is a clear abyss of unknown depth. A pole is erected on the bank, 3 zhang tall. Looking obliquely from the pole top to the water surface, the sight line enters the pole 1 chi 2 cun below the top. Looking at the bottom of the water, the sight line enters 4 chi 8 cun below the top. Question: how deep is the abyss?

A:The abyss is 1 zhang 2 chi deep.

Method:Method: Apply the similar right triangles method. The pole height 3 zhang is the longer leg; the 1 chi 2 cun entry point is the shorter leg. The 4 chi 8 cun entry point is the rate for another shorter leg. Use proportions to find the abyss depth.

Liu Hui's Commentary:This method uses similar right triangles to measure depth, another application of the double-difference method.

0.54 第十二題 / Problem 12: 窺望木 / Peering at a Tree

問：今有木去人不知遠近。立一表長一丈，人退行二丈六尺，從表末望木，與表端參合。又退行二丈，從表末望木，入表一尺。問：木去表幾何？

答曰：木去表三十四丈。

術曰：術曰：以重差術入之。兩退行之差為法，表長乘後退行為實，實如法而一，得木去表之遠。

劉徽注：此又以重差術測遠也。重差術之妙，在於以兩次之差，測無遠弗屆之物。一表一測，不足以定遠近；兩表兩測，則遠近立判。此差之所以為重也。

Q:Now there is a tree at unknown distance from a person. A pole 1 zhang tall is erected; the person retreats 2 zhang 6 chi and looks from the pole top at the tree, which aligns exactly with the pole top. Retreating another 2 zhang, looking from the pole top at the tree, the sight line enters the pole 1 chi below the top. Question: how far is the tree from the pole?

A:The tree is 34 zhang from the pole.

Method:Method: Apply the double-difference method. The difference between the two retreat distances is the divisor; the pole height multiplied by the second retreat distance is the dividend; divide to obtain the distance from tree to pole.

Liu Hui's Commentary:The wonder of the double-difference method lies in using the differences from two measurements to measure objects however far away. One pole and one measurement are insufficient; two measurements immediately distinguish distance.

0.55 第十三題 / Problem 13: 因木望山 / Mountain from a Tree

問：今有木不知高遠。立兩表，各高一丈，前後相去五百步。從前表卻行一百二十步，人目著地，望山峯與表端參合。從後表卻行一百八十步，人目著地，望山峯與表端參合。問：山去前表幾何？山高幾何？

答曰：山去前表一里二百步，高八里一百二十步。

術曰：術曰：以重差術入之。兩表相去為勾股之率，兩卻行之差為法。表高乘表間為實，實如法得山高。前卻行乘表間為實，實如法得山去表之遠。

劉徽注：此重差術測山高之典型題也。立兩表齊高一丈，相去五百步。兩卻行之差六十步為法。表高一丈乘表間五百步為實，實如法得山高。此術之精，在於以兩表之矩，測群山之高遠。故劉徽曰：以咫尺之矩，測千里之高深。

Q:Now there is a mountain of unknown height and distance. Two poles of equal height 1 zhang are erected, 500 paces apart. From the front pole, retreating 120 paces with the eye on the ground, the mountain peak aligns with the pole top. From the rear pole, retreating 180 paces, the mountain peak aligns with the pole top. Question: how far is the mountain from the front pole, and how high is the mountain?

A:The mountain is 1 li 200 paces from the front pole and 8 li 120 paces high.

Method:Method: Apply the double-difference method. The distance between poles is the rate; the difference between retreat distances is the divisor. Pole height times pole separation is the dividend; dividing gives the mountain height. Front retreat distance times pole separation gives the distance from the mountain.

Liu Hui's Commentary:This is the archetypal double-difference problem for measuring mountain height. Liu Hui said: “With a foot-long carpenter’s square, measure heights and depths a thousand li away.” The double-difference method is the ultimate development of the Gougu method.

0.56 第十四題 / Problem 14: 環田求徑 / Annulus Field Revisited

問：今有環田，中周六十二步，外周四步，徑十二步。問：為田幾何？

答曰：二畝五十五步。

術曰：術曰：并中外周而半之，以徑乘之為積。

劉徽注：環田者，猶圓田之中去一圓也。并中外周而半之者，取環之平均周長也。以徑乘之者，以平均周長為廣，徑為縱也。此術以環田近似為矩形，故以廣縱相乘得積。

Q:Now there is an annulus field with inner circumference 62 paces, outer circumference [corrected], and diameter [thickness] 12 paces. Question: what is the area?

A:2 mu 55 square paces.

Method:Method: Sum the inner and outer circumferences and halve; multiply by the diameter to get the area.

Liu Hui's Commentary:This method approximates the annulus as a rectangle. For a precise calculation, one should use the difference of the two circle areas.

0.57 第十五題 / Problem 15: 圓材求徑 / Round Timber Diameter

問：今有圓材，埋在地下，不知大小。以鋸鋸之，深一寸，鋸道長一尺。問：圓材徑幾何？

答曰：圓材徑二尺六寸。

術曰：術曰：以勾股術入之。鋸道長一尺者，弦也；深一寸者，矢也。半鋸道自乘，除以深，加深，即圓徑。半鋸道五寸自乘得二十五寸，除以深一寸得二十五寸，加深一寸得二十六寸，即圓材徑。

劉徽注：圓材埋在地下，以鋸鋸之者，鋸截圓材得一弦也。鋸道長一尺即弦長，深一寸即弦到圓周之最短距離（矢）。以勾股術求圓徑：半弦為勾，圓徑減矢之半為股。半弦自乘除以矢，得圓徑之半減矢。加以矢，得圓徑。

Q:Now there is a round timber buried in the ground. A saw cut is made 1 cun deep; the saw cut is 1 chi long. Question: what is the diameter of the round timber?

A:The round timber's diameter is 2 chi 6 cun.

Method:Method: Apply the Gougu method. The saw cut length of 1 chi is the chord; the depth of 1 cun is the sagitta. Square half the chord, divide by the depth, and add the depth to obtain the diameter.

Liu Hui's Commentary:This method is exquisite, using the Gougu principle to measure a circle's size from a chord and sagitta.

0.58 第十六題 / Problem 16: 開方除之 / Extracting the Square Root

問：今有積五萬五千二百二十五步。問：為方幾何？

答曰：二百三十五步。

術曰：術曰：開方術。置積為實，借一算步之，超一等。議所得，以一乘所借一算為法，而以除。除已，倍法為定法。

劉徽注：開方術者，求正方形之一邊也。積五萬五千二百二十五步，開方得二百三十五步。徽注：開方不盡者，當以微數續之。微數者，以十進小數無限逼近也。此為中國數學之重大發明，以極限之思想，求開方之精密值。

Q:Now there is an area of 55,225 square paces. Question: what is the side of the square?

A:235 paces.

Method:Method: The square root extraction method. Place the area as shi; borrow one counting rod and step it, skipping one place. Estimate the result; multiply and divide. Double the divisor for repeated division.

Liu Hui's Commentary:Liu Hui's note: When the square root does not exhaust, continue with micro-numbers. Micro-numbers use decimal fractions to approach infinitely. This is a major invention in Chinese mathematics, using the concept of limits to find precise square root values.

0.59 第十七題 / Problem 17: 開圓術 / Circle from Area

問: 今有積三百六十步。問：為圓周幾何？

答曰: 圓周六十步。

術曰: 術曰：開圓術。置積為實，以十二乘之，開方除之，得圓周。

劉徽注: 開圓術者，以圓積求圓周也。以十二乘積開方者，古術以圓周率為三，圓面積為 $C^2/12$ ，故 $C = \sqrt{12S}$ 。然此以周三徑一為率，其實圓周率非盡於三也。徽以為圓周率當大於三，以割圓術求之：割之彌細，所失彌少；割之又割，以至於不可割，則與圓周合體而無所失矣。以一百九十二邊形之面積推之，得圓周率約為 3.1416，此為當時世界之最精密者。

Q:Now there is an area of 360 square paces. Question: what is the circumference of the circle?

A:The circle's circumference is 60 paces.

Method:Method: The circle-from-area method. Place the area as shi; multiply by 12 and extract the square root to obtain the circumference.

Liu Hui's Commentary:Liu Hui believed the ratio should be greater than 3, and sought it by the circle-cutting method: the finer the division, the smaller the error. Using the 192-gon area to estimate, the circumference ratio is approximately 3.1416—the most precise value in the world at that time. This is the fountainhead of limit thinking.

0.60 第十八題 / Problem 18: 邪田求積 / Oblique Field Area

問: 今有邪田，一頭廣三十步，一頭廣四十二步，正從六十四步。問：為田幾何？

答曰: 九畝一百四十四步。

術曰：術曰：并兩廣而半之，以從乘之為積。三十步與四十二步并之得七十二步，半之得三十六步。以從六十四步乘之，得二千三百四步為積。

劉徽注：邪田者，梯形之田也。并兩廣而半之者，取其平均廣也。以平均廣乘從，得積。此術之證，以割補之法：割邪田之兩角，補為矩形，故以平均廣乘徑得積。

Q:Now there is an oblique [trapezoidal] field; one end is 30 paces wide, the other end is 42 paces wide; the perpendicular distance is 64 paces. Question: what is the area of the field?

A:9 mu 144 square paces.

Method:Method: Sum the two widths and halve; multiply by the perpendicular distance to get the area. $30 + 42 = 72$ paces; halving gives 36 paces. Multiply by 64 paces gives 2,304 square paces.

Liu Hui's Commentary:This is proved by the cutting-and-patching method: cut the two corners of the oblique field and patch them to form a rectangle; hence average width times perpendicular distance gives the area.

0.61 第十九題 / Problem 19: 弧田徑術 / Circular Segment Revisited

問：今有弧田，弦三十步，矢十五步。問：為田幾何？

答曰：一畝九十七步半。

術曰：術曰：以弦乘矢，矢又自乘，并之，三而一。

劉徽注：弧田術者，以弦矢求弧田之面積也。弦乘矢者，以弦為廣，矢為縱。并之者，合兩積為一。三而一者，以弧田之積約當此兩積和之三分之一也。此術為弧田之近似術。若以割圓術密求之，當以無數小三角形割之，割之彌細，所失彌少。

Q:Now there is a circular segment with chord 30 paces and arrow (sagitta) 15 paces. Question: what is the area of the field?

A:1 mu 97.5 square paces.

Method:Method: Multiply the chord by the arrow; also square the arrow; sum these; divide by 3.

Liu Hui's Commentary:This is an approximation method for arc-fields. The arc-field is the region enclosed by a circular arc and its chord; its shape is curved and convex. For a precise value using the circle-cutting method, divide into infinitely many small triangles; the finer the division, the smaller the error.

0.62 第二十題 / Problem 20: 五家共堤 / Five Families Building a Dike

問：今有五家共堤。甲、乙、丙、丁、戊五家，各以所出徒之數築堤。甲出徒二百五十四人，乙出徒二百四十三人，丙出徒二百三十二人，丁出徒二百二十一人，戊出徒二百一十人。堤長一萬二千七百步。問：各築幾何？

答曰：各以人數為率分配之。

術曰：術曰：以衰分術入之。并五人數為法，堤長為實，實如法得一步之率。各以其人數乘之，得各家所築之步數。

劉徽注：五家共堤者，五家合出徒役共築一堤也。各以人數為率者，按出徒之人數比例分配築堤之長也。此題本屬衰分，而以勾股之比例理推之，同出一理。

Q:Now five families are to build a dike together. A contributes 254 laborers; B contributes 243; C contributes 232; D contributes 221; E contributes 210. The dike is 12,700 paces long. Question: how much does each family build?

A:Each [family's share] is distributed according to their number of laborers as the rate.

Method:Method: Apply the proportional division method. Sum the five families' laborer counts as the divisor; the dike length is the dividend; dividing gives the pace-rate per person. Each family's laborer count multiplied by this rate gives that family's portion.

Liu Hui's Commentary:This problem fundamentally belongs to proportional division; using the proportional principle of the Gougu method to derive it comes from the same principle.

0.63 第二十一題 / Problem 21: 勾股求容方邊 / Square Side from Gougu

問：今有勾股形，勾六尺，股八尺。問：弦上容方，方邊幾何？

答曰：方邊三尺七分尺之五。

術曰：術曰：以勾股術入之。并勾股為法，勾股相乘為實，實如法得方邊。勾六、股八，并得十四為法。勾股相乘得四十八為實。實如法得三尺七分尺之五。

劉徽注：此術與第八題同意，而數不同。勾股弦上容方者，方之一邊在弦上，兩頂點在勾股之邊也。勾六股八弦十，容方之邊為 $48/14 = 24/7$ 。此術之妙，在於以相似勾股之理，求內接正方之邊，一以貫之。

Q:Now in a right triangle, the shorter leg is 6 chi and the longer leg is 8 chi. Question: what is the side of the square inscribed with one side on the hypotenuse?

A:The square side is $3\frac{5}{7}$ chi.

Method:Method: Apply the Gougu method. Sum the shorter and longer legs as divisor; multiply the shorter and longer legs as dividend; divide to obtain the square side. $6 + 8 = 14$ as divisor. Product 48 as dividend. Dividing yields $3\frac{5}{7}$ chi.

Liu Hui's Commentary:Gou 6, gu 8, xian 10; the inscribed square side is $48/14 = 24/7 = 3\frac{5}{7}$ chi. The wonder lies in using the principle of similar right triangles to find the inscribed square's side.

0.64 第二十二題 / Problem 22: 勾股測遠 / Sea Island Distance

問: 今有海島，不知高遠。立兩表，各高三丈，前後相去五百步。從前表卻行一百二十步，人目著地，望海島峯與表端參合。從後表卻行二百二十步，人目著地，望海島峯與表端參合。問：島去前表幾何？島高幾何？

答曰: 島去前表一里一百步，高四里五十五步。

術曰: 術曰：以重差術入之。兩表相去為法，表高乘表間為實，兩卻行之差除實得島高。前卻行乘表間為實，兩卻行之差除實得島去前表之遠。

劉徽注: 此重差術測海島之題也，為《海島算經》第一題之濫觴。立兩表高三丈，相去五百步。兩卻行之差一百步為法。表高三丈乘表間五百步得一千五百丈為實，實如法得島高十五丈。此術之精妙，在於以兩表之矩，測海島之高遠。

Q:Now there is a sea island of unknown height and distance. Two poles of equal height 3 zhang are erected, 500 paces apart. From the front pole, retreating 120 paces with the eye on the ground, the island peak aligns with the pole top. From the rear pole, retreating 220 paces, the island peak aligns with the pole top. Question: how far is the island from the front pole, and how high is the island?

A:The island is 1 li 100 paces from the front pole and 4 li 55 paces high.

Method:Method: Apply the double-difference method. The distance between poles is the divisor; pole height times pole separation is the dividend; the difference of retreat distances divides the dividend to give island height.

Liu Hui's Commentary:This is the sea-island measurement problem—the origin of Problem 1 in the Haidao Suan Jing (Sea Island Mathematical Manual). The divine method uses two foot-long poles to measure the height and distance of a sea island.

0.65 第二十三題 / Problem 23: 勾股求日徑 / Sun Diameter from Gougu

問：今有日去地八萬里，日徑一千二百五十里。以勾股術求之，問：以何為勾股弦？

答曰：以日去地為股，以日徑之半為勾，以斜至日邊為弦。

術曰：術曰：以勾股術入之。日去地八萬里為大股，日徑之半六百二十五里為大勾。以股自乘，以勾自乘，并之開方，得斜至日邊之弦。

劉徽注：此術以勾股之理測天體也。日去地八萬里，此為股。日徑一千二百五十里，其半六百二十五里，此為勾。此術雖以勾股入之，實測天之法也。古代以勾股測天地之高深，此其遺意。

Q: Now the sun is 80,000 li from the earth, and the sun's diameter is 1,250 li. Using the Gougu method to calculate: question, what serves as gou, gu, and xian?

A: The sun's distance from earth serves as the longer leg (gu); half the sun's diameter serves as the shorter leg (gou); the oblique distance to the sun's edge serves as the hypotenuse (xian).

Method: Method: Apply the Gougu method. The sun's distance 80,000 li is the large longer leg; half the sun's diameter 625 li is the large shorter leg. Square both, sum, and extract the square root to obtain the hypotenuse distance to the sun's edge.

Liu Hui's Commentary: This method uses the Gougu principle to measure celestial bodies. The ancients used the Gougu method to measure the heights and depths of heaven and earth.

0.66 第二十四題 / Problem 24: 勾股綜合測量 / Comprehensive Valley Measurement

問：今有深谷，不知深淺。立一木於岸傍，高四丈。從木末斜望谷底，入木二尺四寸。又望谷岸，入木八寸。問：谷深幾何？谷岸廣幾何？

答曰：谷深八丈，谷岸廣二丈四尺。

術曰：術曰：以勾股相似術入之。木高為大股，入木之數為大勾之對應。以兩入木之差為法，木高乘之，得谷深。

劉徽注：深谷者，幽邃難測者也。以勾股相似術測之，則深淺立見。此術以勾股之相似，測深谷之幽邃。凡測高深廣遠，皆可以勾股重差術入之，無所不可。此二十四題，所以盡勾股之變也。

Q: Now there is a deep valley of unknown depth. A pole 4 zhang tall is erected on the bank. Looking obliquely from the pole top to the valley bottom, the sight line enters the pole 2 chi 4 cun below the top. Looking at the valley's far bank, the sight line enters 8 cun below the top. Question: how deep is the valley, and how wide is the valley's far bank?

A: Valley depth: 8 zhang; valley bank width: 2 zhang 4 chi.

Method: Method: Apply the similar right triangles method. The pole height is the large longer leg; the entry distances correspond to large shorter legs. The difference between the two entry distances is the divisor; multiplying by the pole height gives the valley depth.

Liu Hui's Commentary: A deep valley is dark and difficult to measure. Using similar right triangles to measure it, the depth becomes immediately visible. All measurements of height, depth, width, and distance can be addressed by the Gougu double-difference method—nothing is impossible. These 24 problems exhaust all the variations of the Gougu method.

後記 / Epilogue

本書所呈現之三卷——盈不足、方程、勾股——代表了《九章算術》中古代中國數學思想之巔峰。

劉徽之注（約西元 263 年）將《九章》從一部解題配方之集，提升為具有真正數學推理之著作。其對盈不足術之闡釋——作為一般性近似方法，對線性方程組之方程術處理（本質即高斯消元法），以及對畢達哥拉斯定理之深入研究——包括著名之「割補術」與「重差術」——皆證明三世紀中國數學之高度成就。

方程章尤值一提，因其包含世界數學史上負數之最早系統運用——劉徽對消元過程中如何處理「正」與「負」量之解釋，其清晰程度令人驚嘆。

勾股章則以劉徽對「割圓術」之注釋為代表，體現了對圓周率 π 最早之嚴謹逼近方法之一，通過正一百九十二邊形得出約為 3.1416 之 *remarkably* 精確數值。

本擴充版包含三卷各二十四題，所有問題、答案、術文及劉徽注均以中文原文呈現，並附英文譯文，保存這部不朽數學經典之原有結構。

*The three volumes presented here — Yingbuzu (Excess and Deficit), Fangcheng (Rectangular Arrays), and Gougu (Right Triangles) — represent the culmination of ancient Chinese mathematical thought as preserved in the **Jiuzhang Suanshu** (Nine Chapters on the Mathematical Art).*

Liu Hui's commentary (c. 263 CE) elevates the Nine Chapters from a collection of problem-solving recipes to a work of genuine mathematical reasoning. His explanations of the Yingbuzu method as a general technique for approximation, his systematic treatment of linear equations via the Fangcheng method (essentially Gaussian elimination), and his profound work on the Pythagorean theorem including the famous “method of cutting and patching” and the “method of double differences,” all testify to the sophistication of Chinese mathematics in the third century.

The Fangcheng chapter deserves special mention as containing the first known systematic use of negative numbers in world mathematics — Liu Hui's explanation of how to handle “positive” and “negative” quantities during elimination is remarkably clear.

The Gougu chapter, with Liu Hui's commentary on the “method of cutting the circle,” represents one of the earliest rigorous approaches to approximating π , yielding the remarkably accurate value of approximately 3.1416 via a regular 192-gon.

This expanded bilingual edition includes 24 problems for each of the three volumes, with comprehensive Chinese text for all questions, answers, methods, and Liu Hui's commentary, together with English translations, preserving the authentic structure of this immortal mathematical classic.

測圓海鏡

Sea Mirror of Circle Measurements

卷一—卷三 | Volumes I–III

BILINGUAL EDITION — 雙語版

Original Chinese • English Translation

元 李冶 撰

(Li Ye / Li Zhi, 1192–1279)

以天元術解勾股容圓一百七十問

*Using the Method of the Celestial Element (Tian Yuan Shu)
to Solve 170 Problems on the Inscribed Circle*

A foundational text of medieval Chinese algebra,
completed in the year 1248 of the Common Era.

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1 Introduction / 引言

測圓海鏡簡介

《測圓海鏡》乃李冶（李冶，1192–1279）之傑作。李冶為宋元四大數學家之一，與楊輝、秦九韶、朱世傑齊名。此書成於 1248 年，系統運用**天元術**（多項式代數方法）求解圓與直角三角形相關之一百七十問。全書結構如下：

- **卷一**：理論基礎——圓城圖式、總率名號、今問正數、識別雜記（含六百九十二條幾何公式）
- **卷二至卷十二**：一百七十問，以天元術求解，方程次數自一次至六次。

本文檔包含**卷一至卷三**之全文，即全部基礎理論及首批問題。

Introduction

The *Ceyuan Haijing* (測圓海鏡, literally “Sea Mirror of Circle Measurements”) is the masterpiece of Li Ye (李冶, also known as Li Zhi, 1192–1279), one of the four great mathematicians of the Song–Yuan period of China, alongside Yang Hui, Qin Jiushao, and Zhu Shijie.

Completed in 1248, this work systematically applies the **tian yuan shu** (天元術, “method of the celestial element”)—an algebraic method for setting up and solving polynomial equations—to a total of 170 geometric problems concerning a circle inscribed in or related to a right triangle (勾股容圓).

The book is organized as follows:

- **Volume 1** establishes the theoretical foundation: the *Circle City Diagram* (圓城圖式), the *General Rate Names* (總率名號), the *Current Question Numbers* (今問正數), and the *Identification Miscellaneous Records* (識別雜記) containing 692 geometric formulas.
- **Volumes 2–12** present 170 problems, solved by the **tian yuan shu** method. The problems involve equations of degrees ranging from 1 to 6.

The present document contains the complete text of **Volumes 1–3**, comprising all foundational material and the first suite of problems.

2 Original Preface / 原序

原序

數本難窮，吾欲以力強窮之，彼其數不惟不能得其凡，而吾之力且憊矣。然則數果不可以窮耶？既已名之數矣，則又何為而不可窮也。

故謂數為難窮斯可，謂數為不可窮斯不可。何則？彼其冥冥之中，固有昭昭者存，夫昭昭者其自然之數也。非自然之數，其自然之理也。數一出於自然，吾欲以力強窮之，使隸首復生亦末如之何也已。

苟能推自然之理，以明自然之數，則雖遠而乾端坤倪，幽而神情鬼狀，未有不合者矣。

予自幼喜算數，恆病夫考圓之術例出於牽強，殊乖於自然，如古率、徽率、密率之不同，截弧、截矢、截背之互見，內外諸角，析剖支條，莫不各自名家，與世作法。及反覆研究，卒無以當吾心焉。

老大以來，得洞淵九容之說，日夕玩繹，而向之病我者，使爆然落去而無遺余。山中多暇，客有從余求其說者，於是乎又為衍之，遂累一百七十問。

既成編，客復目之《測圓海鏡》，蓋取夫天臨海鏡之義也。

Original Preface

Numbers are fundamentally difficult to exhaust; if I try to force them by effort, not only will I fail to apprehend their essence, but my strength will be depleted as well. Yet can numbers truly not be exhausted? Since they are already called *numbers*, why then could they not be exhausted?

It is acceptable to say that numbers are difficult to exhaust; but it is unacceptable to say they cannot be exhausted. Why? In the midst of darkness, there is indeed brightness. That brightness is the natural number. It is not merely the natural number—it is the natural principle. Since numbers arise from nature, if I try to force them by effort, even if Li Shou (the legendary inventor of arithmetic) were reborn, he could do no better.

If one can extend natural principles to illuminate natural numbers, then even things as distant as the corners of heaven and earth, or as mysterious as spirits and ghosts, will all be in agreement.

Since youth I have delighted in arithmetic. I was always troubled that the methods for investigating circles seemed forced and artificial, contrary to nature—the ancient rate, Hui's rate, and the close rate all differing; the arc, sagitta, and back each being considered separately; the internal and external angles analyzed into minute subdivisions—each school establishing its own name and method for the world. After repeated study, none could satisfy my mind.

In old age, I obtained the *Dongyuan Nine Containments* (洞淵九容) theory, and pondered it day and night. What had previously troubled me suddenly fell away completely, leaving no trace. With leisure time in the mountains, when guests asked me to explain this theory, I expanded upon it, eventually accumulating **one hundred and seventy problems**.

When the compilation was complete, a guest named it *Ceyuan Haijing*—taking the meaning of “heaven overlooking the sea mirror.”

昔半山老人集《唐百家詩選》，自謂廢日力於此，良可惜。明道先生以上蔡謝君記誦為玩物喪志。夫文史尚矣，猶之為不足貴，況九九賤技能乎！

嗜好酸鹹，平生每痛自戒勅，竟莫能已，類有物憑之者，吾亦不知其然而然也。

故嘗私為之解曰：由技兼於事者言之，夷之禮，夔之樂，亦不免為一技。由技進乎道者言之，石之斤，扁之輪，非聖人之所與乎。

覽吾之編，察吾苦心，其憫我者當百數，其笑我者當千數。乃若吾之所自得，則自得焉耳，寧復為人憫笑計哉。

李冶序

In the past, the Old Man of Banshan compiled the *Anthology of One Hundred Tang Poets*, saying that to waste his days on such a thing was truly a pity. Master Mingdao criticized Xie of Shangcai for memorizing texts as “playing with things and losing one’s ambition.” Literature and history are esteemed subjects, yet even they are considered not worth valuing—how much less the lowly skill of multiplication!

My fondness for such sour and salty pursuits—throughout my life I have sternly admonished myself against them, yet in the end I could not stop, as if some force were driving me. I myself do not know why it is so.

So I have privately explained it thus: speaking from the perspective of skills combined with affairs, even the rites of Yi and the music of Kui were not exempt from being a single skill. Speaking from the perspective of advancing from skill to the Way, was not the stone-axe of [the sage] and the wheel of Bian [the wheelwright] praised by the sages?

Those who read my compilation and perceive my earnest efforts—those who pity me will number in the hundreds, those who laugh at me will number in the thousands. As for what I myself have gained, I have simply gained it; how could I calculate for the sake of being pitied or laughed at?

— Preface by Li Ye

3 Volume 1 / 卷一

卷一為全書之理論基礎，分為四大部分：

1. 圓城圖式——主幾何圖形
2. 總率名號——十五個直角三角形及其元素之定義
3. 今問正數——各線段之數值
4. 識別雜記——六百九十二條幾何公式

3.1 Circle City Diagram / 圓城圖式

圓城圖式為一百七十問所基之主幾何圖形。其構為一大直角三角形（勾股形），內切一圓，並設特定之點與線。

其情境如下：

假令有圓城一所，不知周徑，四面開門，門外縱橫各有十字大道。其西北十字道頭定為乾地，其東北十字道頭定為艮地，其東南十字道頭定為巽地，其西南十字道頭定為坤地。所有測望雜法——設問如後。

二十二名點 / The 22 Named Points

最大直角三角形之頂點為天、地、乾。內切圓之心稱心。各點以單字命名：

天	地	乾	心	日	南	北	川
東	西	艮	坤	山	月	巽	青
泉	朱	金	泛	旦	夕		

3.2 General Rate Names / 總率名號

全書研究十五個直角三角形，皆以天地線之一段為弦（弦，即斜邊）。總率名號定義各三角形之三邊。每個三角形：

- 弦 (c) ——斜邊
- 股 (b) ——較長之直邊
- 勾 (a) ——較短之直邊

Volume 1 is the theoretical foundation of the entire work. It comprises four major sections:

1. **Circle City Diagram** (圓城圖式) – the master geometric configuration
2. **General Rate Names** (總率名號) – definitions of all 15 right triangles and their elements
3. **Current Question Numbers** (今問正數) – numerical values for all line segments
4. **Identification Miscellaneous Records** (識別雜記) – 692 geometric formulas

The *Circle City Diagram* is the master geometric configuration upon which all 170 problems are based. It consists of a large right triangle (called 勾股形, “leg-leg shape”) with its inscribed circle, together with specific points and lines.

The scenario Li Ye constructs is:

Suppose there is a circular city whose circumference and diameter are unknown. It has gates on all four sides, and outside each gate there are crossroads running north-south and east-west. The northwest crossroad is designated **Qian** (乾), the northeast **Gen** (艮), the southeast **Xun** (巽), and the southwest **Kun** (坤). All surveying methods are set forth as questions below.

The largest right triangle has vertices **Tian** (天, Heaven), **Di** (地, Earth), and **Qian** (乾). The center of the inscribed circle is called **Xin** (心, Heart). Key points include intersections of horizontal and vertical lines through the center and other constructed lines. Each point is designated by a single Chinese character:

Tian	Di	Qian	Xin	Ri	Nan	Bei	Chuan
Dong	Xi	Gen	Kun	Shan	Yue	Xun	Qing
Quan	Zhu	Jin	Fan	Dan	Xi		

The work studies 15 **right triangles**, all having a segment of the *Tiandi* (天地) line as their hypotenuse (弦, “string”). The *General Rate Names* define these triangles and their three sides.

In each triangle:

- 弦 (c) – the hypotenuse (斜邊)
- 股 (b) – the longer vertical leg
- 勾 (a) – the shorter horizontal leg

十五三角形表 / Table of 15 Triangles

No.	名稱 / Name	頂點 / Vertices	弦	股	勾
1	通 (Tong/General)	天-地-乾	通弦	通股	通勾
2	邊 (Bian/Side)	天-西-川	邊弦	邊股	邊勾
3	底 (Di/Base)	日-地-北	底弦	底股	底勾
4	黃廣 (Huangguang)	天-山-金	黃廣弦	黃廣股	黃廣勾
5	黃長 (Huangchang)	月-地-泉	黃長弦	黃長股	黃長勾
6	上高 (Shanggao)	天-日-旦	上高弦	上高股	上高勾
7	下高 (Xiagao)	日-山-朱	下高弦	下高股	下高勾
8	上平 (Shangping)	月-川-青	上平弦	上平股	上平勾
9	下平 (Xiaping)	川-地-夕	下平弦	下平股	下平勾
10	大差 (Dacha)	天-月-坤	大差弦	大差股	大差勾
11	小差 (Xiaocha)	山-地-艮	小差弦	小差股	小差勾
12	皇極 (Huangji)	日-川-心	皇極弦	皇極股	皇極勾
13	太虛 (Taixu)	月-山-泛	太虛弦	太虛股	太虛勾
14	明 (Ming)	日-月-南	明弦	明股	明勾
15	夷 (Zhuan)	山-川-東	夷弦	夷股	夷勾

注：上高 = 下高，上平 = 下平，故十五三角形中實僅十三個互異。

Note: 上高 = 下高 and 上平 = 下平, so among 15 triangles, only 13 are distinct.

3.3 Current Question Numbers / 今問正數

此節給出圖中每條線段之數值。內切圓之半徑定為一百二十步，作為標準單位。

This section gives the numerical value of every line segment in the diagram. The radius of the inscribed circle is taken as 120 steps (步), which serves as the standard unit.

1. 通 (Tong / 通用)

弦六百八十	勾三百二十	股六百
勾股和九百二十	較二百八十	
勾和一千	較三百六十	
股和一千二百八十	較八十	
較和九百六十	較四百	
和和一千六百	較二百四十	

1. Tong (General)

Hypotenuse 680	Short leg 320, Long leg 920
Sum of legs 920	Difference 280
Leg-hypotenuse sum 1000	Difference 360
Leg-hypotenuse sum 1280	Difference 80
Difference-sum 960	Difference 400
Sum-sum 1600	Difference 240

That is: hypotenuse $c = 680$, short leg $a = 320$, long leg $b = 920$.

2. 邊 (Bian / Side)

弦五百四十四	勾二百五十六	股四百八十
勾股和七百三十六	較二百二十四	
勾和八百	較二百八十八	
股和一千零二十四	較六十四	
較和七百六十八	較三百二十	
和和一千二百八十	較一百九十二	

2. Bian (Side)

Hypotenuse 544	Short leg 256, Long leg 480
Sum of legs 736	Difference 224
Leg-hypotenuse sum 800	Difference 288
Leg-hypotenuse sum 1024	Difference 64
Difference-sum 768	Difference 320
Sum-sum 1280	Difference 192

3. 底 (Di / Base)

弦四百二十五	勾二百	股三百七十五
勾股和五百七十五	較一百七十五	
勾和六百二十五	較二百二十五	
股和八百	較五十	
較和六百	較二百五十	
和和一千	較一百五十	

3. Di (Base)

Hypotenuse 425	Short leg 200, Long leg 375
Sum of legs 575	Difference 175
Leg-hypotenuse sum 625	Difference 225
Leg-hypotenuse sum 800	Difference 50
Difference-sum 600	Difference 250
Sum-sum 1000	Difference 150

4. Huang Guang

Hypotenuse 510 Short leg 240, Long leg 450
(this is the city diameter)
Sum of legs 690 Difference 210

The remaining 11 triangles follow the same pattern, each with 13 numerical quantities (弦, 勾, 股, 勾股和, 勾股較, 勾弦和, 勾弦較, 股弦和, 股弦較, 弦較和, 弦較較, 弦和和, 弦和較), giving a total of $15 \times 13 = 195$ numerical entries.

This is the theoretical heart of the work. It contains **692 geometric formulas** relating the various line segments in the diagram. These formulas are purely geometric theorems, independent of any particular numerical values. The problems in all subsequent volumes are solved by using these formulas together with the *tian yuan shu*. The section is divided into eight parts:

1. **Various Names** (諸雜名目) – general definitions and 30+ theorems
2. **Five Sums and Five Differences** (五和五較)
3. **Various Hypotenuses** (諸弦)
4. **Large and Small Differences** (大小差)
5. **Various Differences** (諸差)
6. **Mutual Relations of Rates** (諸率互見)
7. **Four-Place Supplement** (四位拾遺)
8. **Supplement** (拾遺)

Sample Definitions from Various Names:

名稱 / Name	定義 / Definition
內率	明勾 × 裏股 / Ming 勾 × Zhuan 股
外率	明股 × 裏勾 / Ming 股 × Zhuan 勾
虛率	虛勾 × 虛股 / Xu 勾 × Xu 股
角差	高股 – 平勾 / Gao 股 – Ping 勾
次差	(明股 – 明勾) + (裏股 – 裏勾)
混同和	黃長股 + 黃廣勾 / Huangchang 股 + Huangguang 勾
傍差	(明股 – 明勾) – (裏股 – 裏勾)

Sample Theorems:

- 凡大差小差相乘為半段徑幂。
 - 大差勾小差股相乘同上。
 - 黃廣股黃長勾相乘為經幂。
 - 勾股和即弦黃和。
- The product of the large difference and small difference equals half the square of the diameter.
 - The product of the large-difference short-leg and the small-difference long-leg equals the same.
 - The product of the Huangguang long-leg and Huangchang short-leg equals the square of the diameter.
 - The sum of the two legs equals the sum of the hypotenuse and the diameter of the inscribed circle: $a + b = c + d$, where d is the diameter of the inscribed circle.

4 Volume 2 / 卷二

正率一十四問

此十四問構成「洞淵九容」序列，源自道教大師李思聰（號洞淵）之著作。其呈現直角三角形中九種基本容圓類型，另加五題。

4.0.1 第一問 / Problem 1

問 甲乙二人俱在乾地，乙東行三百二十步而立，甲南行六百步望見乙，問徑幾里？

答 城徑二百四十步。

術 此為勾股容圓也。以勾股相乘，倍之為實，並勾股幕以求弦，復加入勾股共，以為法。

草 置甲南行六百步在地，以乙東行三百二十步乘之，得一十九萬二千步，倍之得三十萬四千步為實。以乙東行步自之，得一十萬零二千四百步為勾幕；以甲南行步自之，得三十六萬步為股幕。二幕相並，得四十六萬二千四百步為方實，以平方開之，得六百八十步，則弦也。以弦加勾股共，共得一千六百步以為法。如法而一，得二百四十步，則城徑也。合問。

4.0.2 第二問 / Problem 2

問 甲乙二人俱在西門，乙東行二百五十六步，甲南行四百八十步望見乙，問答同前。

答 城徑二百四十步。

術 此為勾上容圓也。以勾股相乘，倍之為實，並勾股幕以求弦，加入股以為法。

草 置甲南行四百八十步在地，以乙東行二百五十六步乘之，得一十二萬二千八百八十步，倍之得二十四萬五千七百六十步為實。以乙東行步自之，得六萬五千五百三十六步為勾幕；以甲南行步自之，得二十三萬零四百步為股幕。勾股幕相並，得二十九萬五千九百三十六步為方實，以平方開之，得五百四十四步為弦也。以弦加入南行步，共得一千零二十四步以為法。而一，得二百四十步則城徑。合問。

Fourteen Questions on Standard Rates

These fourteen problems form the “Dongyuan Nine Containments” (洞淵九容) sequence, derived from the earlier work of the Daoist master Li Sicong (李思聰, known as “Master Dongyuan”). They present the nine fundamental types of circles contained in or related to a right triangle, plus five additional problems.

Problem. Two people, A and B, both stand at Qian. B walks 320 steps east and stops. A walks 600 steps south and sees B. What is the diameter?

Answer. The city diameter is 240 steps.

Method. This is the “circle contained in the right triangle” case. Multiply the two legs and double it to obtain the dividend. Combine the squares of the legs to find the hypotenuse, then add the sum of the legs to form the divisor.

Working. Place A’s 600 steps south. Multiply by B’s 320 steps east, obtaining 192,000 steps; double to get 384,000 as the dividend. Square B’s eastward steps to get 102,400 as the short-leg square; square A’s southward steps to get 360,000 as the long-leg square. Add the two squares: 462,400. Extract the square root to get 680, which is the hypotenuse. Add the sum of the legs to the hypotenuse: 1,600 as the divisor. Divide: 240 steps, the city diameter. This agrees with the question.

Problem. Two people both stand at the West Gate. B walks 256 steps east, A walks 480 steps south and sees B. Answer as before.

Answer. The city diameter is 240 steps.

Method. This is the “circle on the short leg” case. Multiply the two legs and double as dividend. Combine the squares of the legs to find the hypotenuse; add the long leg as divisor.

Working. Place A’s 480 steps south. Multiply by B’s 256 steps east, obtaining 122,880; double to get 245,760 as dividend. Square B’s eastward steps: 65,536 as the short-leg square. Square A’s southward steps: 230,400 as the long-leg square. Combine: 295,936. Extract square root: 544, the hypotenuse. Add the southward steps: 1,024 as divisor. Divide: 240 steps, the city diameter. This agrees.

4.0.3 第三問 / Problem 3

問 甲乙二人俱在北門，乙東行二百步而止，甲南行三百七十五步望見乙，問答同前。

答 城徑二百四十步。

術 此為股上容圓也。以勾股相乘，倍之為實，以勾股羈求弦，加入勾以為法。

草 置甲南行三百七十五步，以乙東行二百步乘之，得七萬五千步，倍之得一十五萬步為實。以乙東行自之，得四萬步為勾羈；以甲南行自之，得一十四萬零六百二十五步為股羈。勾股羈相並，得一十八萬零六百二十五步為方實。如平方而一，得四百二十五步則弦也。加入乙東行二百步，共得六百二十五步以為法。以法除之，得二百四十步，則城徑也。合問。

Problem. Two people both stand at the North Gate. B walks 200 steps east and stops. A walks 375 steps south and sees B. Answer as before.

Answer. The city diameter is 240 steps.

Method. This is the “circle on the long leg” case. Multiply the two legs and double as dividend. Use the squares of the legs to find the hypotenuse; add the short leg as divisor.

Working. Place A’s 375 steps south. Multiply by B’s 200 steps east: 75,000; double to get 150,000 as dividend. Square B’s eastward steps: 40,000 as the short-leg square. Square A’s southward steps: 140,625 as the long-leg square. Combine: 180,625. Extract square root: 425, the hypotenuse. Add B’s 200 eastward steps: 625 as divisor. Divide: 240 steps, the city diameter. This agrees.

4.0.4 第四問 / Problem 4

問 甲乙二人俱在圓城中心而立，乙穿城向東行一百三十六步而止，甲穿城南行二百五十五步望見乙，問答同前。

答 城徑二百四十步。

術 此為勾股上容圓也。以勾股相乘，倍之為實，並勾股羈如法求弦，以為法。

草 以二行步相乘，得三萬四千六百八十步，倍之得六萬九千三百六十步為實。置乙東行自之，得一萬八千四百九十六步為勾羈；又以甲南行自之，得六萬五千零二十五步為股羈。二羈相並，得八萬三千五百二十一步為方實，以平方開之，得二百八十九步即弦也。便以為法。如法除實，得二百四十步，即圓城之徑也。合問。

Problem. Two people both stand at the center of the circular city. B goes through the city, walking 136 steps east and stops. A goes through the city, walking 255 steps south and sees B. Answer as before.

Answer. The city diameter is 240 steps.

Method. This is the “circle on both legs” case. Multiply the two legs and double as dividend. Combine the squares of the legs to find the hypotenuse as divisor.

Working. Multiply the two walking distances: 34,680; double to get 69,360 as dividend. Square B’s eastward steps: 18,496 as the short-leg square. Square A’s southward steps: 65,025 as the long-leg square. Combine: 83,521. Extract square root: 289, the hypotenuse. This serves as the divisor. Divide: 240 steps, the city diameter. This agrees.

4.0.5 第五問 / Problem 5

問 甲乙二人同立於乾地，乙東行一百八十步遇塔而止，甲南行三百六十步，回望其塔正居城徑之半，問答同前。

答 城徑二百四十步。

術 此為弦上容圓也。以勾股相乘，倍之為實，以勾股和為法。

草 以二行步相乘，得六萬四千八百步，倍之得一十二萬九千六百步為實。並二行步，得五百四十步以為法。除實，得二百四十步，即城徑也。合問。

Problem. Two people stand together at Qian. B walks 180 steps east and reaches a pagoda. A walks 360 steps south and looks back: the pagoda lies exactly at half the city diameter. Answer as before.

Answer. The city diameter is 240 steps.

Method. This is the “circle on the hypotenuse” case. Multiply the two legs and double as dividend. Use the sum of the legs as divisor.

Working. Multiply the two walking distances: 64,800; double to get 129,600 as dividend. Add the two distances: 540 as divisor. Divide: 240 steps, the city diameter. This agrees.

4.0.6 第六問 / Problem 6

問 甲乙二人俱在坤地，乙東行一百九十二步而止，甲南行三百六十步，望乙與城參相直，問答同前。

答 城徑二百四十步。

術 此為勾外容圓也。以勾股相乘，倍之為實，以弦較和為法。

草 以二行步相乘，得六萬九千一百二十步，倍之得一十三萬八千二百四十步為實。置乙東行自之，得三萬六千八百六十四步為勾冪；又置甲南行自之，得一十二萬九千六百步為股冪。二冪相並，得一十六萬六千四百六十四步為方實，以平方開之，得四百零八，即弦也。又置甲南行步內減乙東行步，餘一百六十八步，即較也。以較加弦，共得五百七十六步以為法。實如法而一，得二百四十步，為城徑也。合問。

[按] 此題用勾股求得弦，即可加減得弦較較為城徑。今必以勾股相乘倍積為實，求得弦較和為法，而後始得弦較較為城徑者，蓋欲因此並明勾股相乘之倍積為弦較較、弦較和相乘之積，非故為紆回也。

Problem. Two people both stand at Kun. B walks 192 steps east and stops. A walks 360 steps south and sees B aligned with the city. Answer as before.

Answer. The city diameter is 240 steps.

Method. This is the “circle outside the short leg” case. Multiply the two legs and double as dividend. Use the sum of hypotenuse and difference as divisor.

Working. Multiply the two walking distances: 69,120; double to get 138,240 as dividend. Square B’s eastward steps: 36,864 as the short-leg square. Square A’s southward steps: 129,600 as the long-leg square. Combine: 166,464. Extract square root: 408, the hypotenuse. Subtract B’s steps from A’s: 168, the difference. Add difference to hypotenuse: 576 as divisor. Divide: 240 steps, the city diameter. This agrees.

Editorial note This problem uses the right triangle to find the hypotenuse, then by addition and subtraction obtains the “hypotenuse-difference-difference” as the city diameter. The method deliberately uses the product of the legs (doubled) as dividend to demonstrate that this product equals the product of (hypotenuse–difference) and (hypotenuse+ difference).]

4.0.7 第七問 / Problem 7

問 甲乙二人同立於艮地，甲南行一百五十步而止，乙東行八十步，望乙與城參相直，問答同前。

答 城徑二百四十步。

術 此為股外容圓也。以勾股相乘，倍之為實，以弦和較為法。

Problem. Two people stand together at Gen. A walks 150 steps south and stops. B walks 80 steps east. A sees B aligned with the city. Answer as before.

Answer. The city diameter is 240 steps.

Method. This is the “circle outside the long leg” case. Multiply the two legs and double as dividend. Use the difference of hypotenuse-sum as divisor.

4.0.8 第八問 / Problem 8

問 甲乙二人同立於坤地，甲南行一百九十二步而止，乙東行七十二步，望乙與城參相直，問答同前。

答 城徑二百四十步。

術 此為弦外容圓也。以勾股相乘，倍之為實，以勾股較為法。

Problem. Two people stand together at Kun. A walks 192 steps south and stops. B walks 72 steps east. A sees B aligned with the city. Answer as before.

Answer. The city diameter is 240 steps.

Method. This is the “circle outside the hypotenuse” case. Multiply the two legs and double as dividend. Use the difference of the legs as divisor.

4.0.9 第九問 / Problem 9

問 甲乙二人俱在異地，乙東行一百九十二步而止，甲南行三百六十步，望乙與城參相直，問答同前。

答 城徑二百四十步。

術 此為勾外容半圓也。以勾股相乘，倍之為實，以股弦和為法。

Problem. Two people both stand at Xun. B walks 192 steps east and stops. A walks 360 steps south. A sees B aligned with the city. Answer as before.

Answer. The city diameter is 240 steps.

Method. This is the “semicircle outside the short leg” case. Multiply the two legs and double as dividend. Use the sum of long leg and hypotenuse as divisor.

4.0.10 第十問 / Problem 10

問 甲乙二人俱在艮地，甲南行一百五十步而止，乙東行八十步，望乙與城參相直，問答同前。

答 城徑二百四十步。

術 此為股外容半圓也。以勾股相乘，倍之為實，以勾弦和為法。

Problem. Two people both stand at Gen. A walks 150 steps south and stops. B walks 80 steps east. A sees B aligned with the city. Answer as before.

Answer. The city diameter is 240 steps.

Method. This is the “semicircle outside the long leg” case. Multiply the two legs and double as dividend. Use the sum of short leg and hypotenuse as divisor.

以上十問構成「洞淵九容」，第五問（弦上容圓）為補充。卷二其餘四問繼續以天元術求解之標準率問題。

The above ten problems constitute the “Dongyuan Nine Containments” (洞淵九容), supplemented by Problem 5 (circle on the hypotenuse). The remaining four problems of Volume 2 continue with additional standard-rate problems using the tian yuan shu method.

5 Volume 3 / 卷三

邊股一十七問

卷三呈十七問，所給數據涉及邊股 (b_2)，定義為「邊」三角形（表中第二）之長直邊，等於四百八十步。本卷所有問題之未知數皆為圓城直徑（城徑 = 二百四十步）。此等問題展示天元術之漸增精妙，方程次數自二次至三次。

5.0.1 第一問 / Problem 1

問 乙出東門南行，不知步數而止。甲出西門南行四百八十步，望見乙，復就乙行五百一十步與乙相會，問答同前。

答 城徑二百四十步。

術 倍相減步，以乘二之甲南行步，為平方實，得城徑。

草 識別得二行相減，餘三十步，即乙出東門南行步也。倍相減步，得六十步，以乘二之甲南行步九百六十步，得五萬七千六百步，為平方實。如法開之，得二百四十步，即城徑也。合問。

Seventeen Questions on the Side-Long-Leg

Volume 3 presents 17 problems in which the given data involve the *side long-leg* (邊股, b_2), defined as the long leg of the “Side” triangle (邊 triangle, No. 2 in the table), equal to 480 steps. In all problems of this volume, the unknown to be found is the diameter of the circular city (城徑 = 240 steps). These problems demonstrate the increasing sophistication of the tian yuan shu method, with equations ranging from quadratic to cubic degree.

Problem. B goes out the East Gate and walks south, an unknown number of steps, then stops. A goes out the West Gate and walks 480 steps south, sees B, then walks 510 steps to reach B. Answer as before.

Answer. The city diameter is 240 steps.

Method. Double the difference of the two walking distances; multiply by twice A’s southward steps to obtain the square-root dividend; this gives the city diameter.

Working. Recognize: subtracting the two distances leaves 30 steps, which is B’s southward distance from the East Gate. Double the difference: 60 steps. Multiply by twice A’s southward steps (960): 57,600, the square-root dividend. Extract the square root: 240 steps, the city diameter. This agrees.

5.0.2 第二問 / Problem 2

問 甲出西門南行四百八十步而止，乙從艮隅東行八十步，望見甲，問答同前。

答 城徑二百四十步。

術 倍南行步，以東行步乘之為實，東行步為從方，一步常法，得全徑。

草 立天元一為全徑，以減於二之甲南行步，得為兩個大差也。以乙東行步乘之，得為圓徑冪，寄左。然後以天元冪與左相消，得以帶縱平方開之，得二百四十步，即城徑也。合問。

又法：半之乙東行步，乘南行步為實，半之乙東行步為從，一步常法，得半徑。

草 立天元一為半城徑，減甲南行步，得為大差也。以半之東行步乘之，得即半徑冪，寄左。然後以天元冪為同數，與左相消，得開帶縱平方，得一百二十步。倍之，即城徑也。合問。

Problem. A goes out the West Gate and walks 480 steps south, then stops. B walks 80 steps east from the Gen corner and sees A. Answer as before.

Answer. The city diameter is 240 steps.

Method. Double the southward steps; multiply by the eastward steps as dividend; the eastward steps as the “following square”; unity as the constant method; obtain the full diameter.

Working. Set the celestial element unity as the full diameter. Subtract from twice A’s southward steps to obtain two large differences. Multiply by B’s eastward steps to obtain the circle-diameter square; store on the left. Then cancel the celestial element square with the left; extract the square root with a following edge: 240 steps, the city diameter. This agrees.

Alternative Method: Halve B’s eastward steps; multiply by the southward steps as dividend; halved eastward steps as the following term; unity as constant method; obtain the radius.

Working. Set the celestial element unity as half the city diameter. Subtract from A’s southward steps to obtain the large difference. Multiply by halved eastward steps to obtain the radius square; store on the left. Then use the celestial element square as the equal number; cancel with the left; extract the square root with a following edge: 120 steps. Double it: the city diameter. This agrees.

5.0.3 第三問 / Problem 3

問 甲出西門南行四百八十步而止，乙從艮隅亦南行一百五十步，望見甲，問答同前。

答 城徑二百四十步。

術 兩行步相乘為實，南行步為從方，一為隅，得半徑。

草 立天元一為半城徑，以減乙南行步，得為半梯頭；以甲行步為梯底，以乘之，得為半徑冪，寄左。然後以天元冪與左相消，得開帶縱平方，得一百二十步。倍之，即城徑也。合問。

Problem. A goes out the West Gate and walks 480 steps south, then stops. B also walks 150 steps south from the Gen corner and sees A. Answer as before.

Answer. The city diameter is 240 steps.

Method. Multiply the two walking distances as dividend; the southward steps as the “following square”; unity as the corner; obtain the radius.

Working. Set the celestial element unity as half the city diameter. Subtract from B’s southward steps to obtain half the trapezoid head. Use A’s steps as the trapezoid base; multiply to obtain the radius square; store on the left. Then cancel the celestial element square with the left; extract the square root with a following edge: 120 steps. Double it: the city diameter. This agrees.

5.0.4 第四問 / Problem 4

問 甲出西門南行四百八十步，乙出東門直行一十六步，望見甲，問答同前。

答 城徑二百四十步。

術 以四之東行步，乘南行冪為實，從空，東行為廉，一步為隅法，得全徑。

草 立天元一為圓徑，加乙東行步，得為中勾；其甲南行即中股也。置東行步為小勾，以中股乘之，得，合以中勾除，今不受除，便以為小股也。內寄中勾分母。乃復以中股乘之，得三百六十八萬六千四百，又四之，得一千四百七十四萬五千六百，為一段圓徑冪，寄中勾分母，寄左。然後以天元徑自之，又以中勾乘之，得為同數，與左相消，得以帶縱立方開之，得二百四十步，為城徑也。合問。

[按] 不受除者，無可除之理也。凡二數，此數於彼數有可除之理，則受除；無可除之理，則不受除也。蓋除有法有實，實可二，法不可二。此題以中勾為法，而中勾內有一元，又有十六步，其為數已二矣，又何以均分不一之數乎？故曰不受也。寄分者，姑寄其應除之數也。俟求得兩相等數，而此數內尚少一除，不除此而轉乘彼，則兩數仍相等，猶之受除者也。此所謂以乘代除也。

Problem. A goes out the West Gate and walks 480 steps south. B goes out the East Gate and walks 16 steps straight, and sees A. Answer as before.

Answer. The city diameter is 240 steps.

Method. Multiply four times the eastward steps by the square of the southward steps as dividend; the “following” term is empty; the eastward steps as the edge; unity as the corner method; obtain the full diameter.

Working. Set the celestial element unity as the circle diameter. Add B’s eastward steps to obtain the middle short-leg. A’s southward steps are the middle long-leg. Place the eastward steps as the small short-leg; multiply by the middle long-leg. This should be divided by the middle short-leg, but it is “not accepted for division”—use it directly as the small long-leg, with the middle short-leg as denominator. Multiply again by the middle long-leg: 3,686,400; multiply by 4: 14,745,600 as a segment of the circle-diameter square, with the middle short-leg as denominator; store on the left. Then square the celestial element diameter and multiply by the middle short-leg as the equal number; cancel with the left; extract the cube root with a following edge: 240 steps, the city diameter. This agrees.

Editorial note “Not accepted for division” means there is no valid way to divide. When two numbers are such that one can be divided by the other, the division is “accepted”; otherwise it is “not accepted.” For division has divisor and dividend; the dividend can be compound, but the divisor cannot. Here the middle short-leg serves as divisor, but it contains both the celestial element and 16 steps—it is already compound. How can one evenly divide a non-uniform number? Hence “not accepted.” Storing the denominator means temporarily retaining the number that should be divided. When two equal quantities are found, and one still lacks a division, instead of dividing this, multiply that; the two quantities remain equal, as if the division had been accepted. This is the so-called “substituting multiplication for division.”]

5.0.5 第五問 / Problem 5

問 乙出南門東行七十二步而止，甲出西門南行四百八十步，望乙與城參相直，問答同前。

答 城徑二百四十步。

術 以乙東行冪，乘甲南行為實；乙東行冪為從方；甲南行步內減二之東行步為益廉；一步常法，得半徑。

草 立天元一為半城徑，以減南行步，得為小股。又以天元加乙東行步，得為小勾。又以天元加南行步，得為大股。乃置大股在地，以小勾乘之，得下式，合以小股除之，今不受除，便以為大勾，內寄小股分母。又置天元半徑，以分母小股乘之，得以減大勾，得為半個梯底於上。以乙東行七十二步為半個梯頭，以乘上位，得為半徑冪，內寄小股分母，寄左。然後置天元冪，又以分母小股乘之，得為同數，與左相消，以立方開之，得一百二十步。倍之，即城徑也。合問。

又法：以二數相乘為實，相減為從，一虛法，平開，得半徑。

Problem. B goes out the South Gate and walks 72 steps east, then stops. A goes out the West Gate and walks 480 steps south. A sees B aligned with the city. Answer as before.

Answer. The city diameter is 240 steps.

Method. Multiply the square of B's eastward steps by A's southward steps as dividend; the square of B's eastward steps as the "following square"; subtract twice the eastward steps from A's southward steps as the "beneficial edge"; unity as the constant method; obtain the radius.

Working. Set the celestial element unity as half the city diameter. Subtract from the southward steps to obtain the small long-leg. Add the celestial element to B's eastward steps to obtain the small short-leg. Add the celestial element to the southward steps to obtain the large long-leg. Place the large long-leg; multiply by the small short-leg. This should be divided by the small long-leg, but it is "not accepted for division"—use it directly as the large short-leg, with the small long-leg as denominator. Place the celestial element radius; multiply by the denominator (small long-leg); subtract from the large short-leg to obtain half the trapezoid base on top. Use B's 72 eastward steps as half the trapezoid head; multiply the upper quantity to obtain the radius square, with the small long-leg as denominator; store on the left. Then place the celestial element square; multiply by the denominator (small long-leg) as the equal number; cancel with the left; extract the cube root: 120 steps. Double it: the city diameter. This agrees.

Alternative Method: Multiply the two numbers as dividend; their difference as the following term; unity as the empty method; extract the square root; obtain the radius.

5.0.6 第六問 / Problem 6

問 乙出南門東行，不知步數而立。甲出西門南行四百八十步，望見乙與城參相直，又就乙行四百零八步與乙相會，問答同前。

答 城徑二百四十步。

[其餘十一問依同樣模式，所給數據涉及邊股（邊股 = 四百八十步）及其他各種測量，每問皆以天元術建立多項式方程，求解得城徑二百四十步。]

Problem. B goes out the South Gate and walks east, an unknown number of steps, then stands. A goes out the West Gate and walks 480 steps south, sees B aligned with the city, then walks 408 steps to reach B. Answer as before.

Answer. The city diameter is 240 steps.

[The remaining 11 problems of Volume 3 continue in the same pattern, with given data involving the side long-leg (邊股 = 480 steps) and various other measurements, each leading to polynomial equations solved by the tian yuan shu method to yield the city diameter of 240 steps.]

6 The Tian Yuan Shu Method / 天元術

天元術

天元術（「天元術」，「method of the celestial element」）為《測圓海鏡》全書所運用之代數技法。此乃中國數學史上最早之系統方法，用以建立並求解一元多項式方程。

6.1 Methodology / 方法論

其程序如下：

1. **立天元一為 X** （「設天元一為 X 」）——宣告未知量。以現代記號，等同於「令 $x = X$ 」。
2. 以天元 x 表達所有已知量及關係。
3. 構造兩個相等之表達式（常以不同方法計算同一幾何量），一為「左」式暫存，另一為「同數」。
4. **相消**（「mutual cancellation」）——令兩式相等並化簡，得多項式方程之標準形式。
5. 以開方法求解：開平方、開立方、或更高次之開方。

6.2 Polynomial Notation / 多項式記號

李冶運用籌算板上的位置記法：

- 係數縱向排列，常數項（實）在上。
- 其下： x 之係數（方，「square」），繼以 x^2 （廉，「edge」），繼以 x^3 （隅，「corner」）。
- 更高次：第一廉、第二廉等。
- 負係數以末位斜劃表示。

The Tian Yuan Shu Method

The *tian yuan shu* (天元術, “method of the celestial element”) is the algebraic technique employed throughout the *Ceyuan Haijing*. It is the earliest systematic method in Chinese mathematics for setting up and solving polynomial equations with one unknown.

The procedure follows these steps:

1. **Set the celestial element unity to be X** (“立天元一為 X ”) — This declares the unknown quantity. In modern notation, it is equivalent to “Let $x = X$ ”.
2. Express all given quantities and relationships in terms of the celestial element x .
3. Construct two equal expressions (often representing the same geometric quantity computed by different methods), one designated as the “left” (左) expression and stored temporarily, the other as the “equal number” (同數).
4. **Mutual cancellation** (相消) — Set the two expressions equal and simplify to obtain the polynomial equation in standard form.
5. Solve the equation by root-extraction methods (開方): square-root extraction (開平方), cube-root extraction (開立方), or higher-degree root extraction.

Li Ye uses a positional notation on a counting-rod board:

- Coefficients are arranged vertically, with the constant term (實) at the top.
- Below it: the coefficient of x (方, “square”), then x^2 (廉, “edge”), then x^3 (隅, “corner”).
- For higher degrees: first 廉, second 廉, etc.
- Negative coefficients are indicated by a diagonal stroke through the last digit.

6.3 Example: Quadratic Equation / 示例：二次方程

第一問導出相當於下式之方程：

$$384000 = 1600 \cdot d$$

其中 d 為直徑。但更複雜之問題產生真正的多項式方程。例如卷三第四問產生**三次方程**：

$$(4 \cdot 16)(480)^2 = (480 + 16) \cdot d^2 - d^3$$

以現代形式，李冶以「帶縱立方」解之。

The first problem leads to the equation equivalent to:

$$384000 = 1600 \cdot d$$

where d is the diameter. But more complex problems yield true polynomial equations. For instance, Problem 4 of Volume 3 produces a **cubic equation**:

$$(4 \cdot 16)(480)^2 = (480 + 16) \cdot d^2 - d^3$$

in modern form, which Li Ye solves by “opening the cube with an edge” (帶縱立方).

6.4 Historical Significance / 歷史意義

《測圓海鏡》為現存最早系統運用天元術之著作。所解方程包括：

次數 / Degree	方程數量 / Number
一次 / Linear (degree 1)	31
二次 / Quadratic (degree 2)	106
三次 / Cubic (degree 3)	24
四次 / Quartic (degree 4)	20
六次 / Sextic (degree 6)	1
總計 / Total	182

The *Ceyuan Haijing* is the earliest surviving work that systematically employs the *tian yuan shu*. The equations solved include:

Degree	Number of Equations
Linear (degree 1)	31
Quadratic (degree 2)	106
Cubic (degree 3)	24
Quartic (degree 4)	20
Sextic (degree 6)	1
Total	182

注：本排印版呈現《測圓海鏡》卷一至卷三之全文。卷一包含全部定義、數值數據及六百九十二條幾何公式。卷二、卷三呈現一百七十問中之首批三十一問，以天元術求解。其餘九卷（卷四–卷十二）繼續一百三十九問，複雜度漸增，至卷七達六次方程。

Note: This reader presents the complete text of *Vol-umes aijing*. Volume 1 contains all definitions, numerical data, and 692 geometric formulas. Volumes 2 and 3 present the first 31 of 170 problems, solved by the *tian yuan shu* method. The remaining nine volumes (卷四–卷十二) continue with 139 further problems of increasing complexity, culminating in a sixth-degree equation in Volume 7.

欽定四庫全書 • Imperial Siku Quanshu

子部 • Mathematics and Astronomy

測圓海鏡 Ceyuan Haijing

卷四至卷六 Volumes 4–6

底勾、大股、大勾三卷全編

Double Difference, Large Leg, and Large Base: Complete

元 李冶 撰

Yuan Dynasty Li Ye (Renqing, Jingzhai)

字仁卿，號敬齋，真定樂城人

Jinshi of the Late Jin; Hanlin Academician under Yuan

Bilingual English–Chinese Edition

Original Chinese with English Translation

Parallel Text: Chinese (Left) • English (Right)

依《欽定四庫全書》本 • Based on the Imperial Siku Quanshu Edition

卷四：第五十一問至第六十七問 • Vol. 4: Problems 51–67

卷五：第六十八問至第八十五問 • Vol. 5: Problems 68–85

卷六：第八十六問至第一百三問 • Vol. 6: Problems 86–103

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引言 · Introduction

《測圓海鏡》十二卷，元李冶撰。冶字仁卿，號敬齋，真定樂城人。金末進士，入元官翰林學士。此書乃其所著天元術之傑作，以勾股測圓之法，設問答以明天元一之術。全書凡一百七十問，皆設城池圓形，以人物行步之數推求圓徑。本編收錄卷四、卷五、卷六三卷，凡五十三問：

- 卷四：底勾一十七問（第五十一問至第六十七問）
- 卷五：大股一十八問（第六十八問至第八十五問）
- 卷六：大勾一十八問（第八十六問至第一百三問）

書中設問之例：假令有圓城一座，甲乙二人各從城門出行，或東或南，行若干步，遙相望見，或斜行相會。只知若干步數，餘皆不知，以求圓城之徑。每問皆有問、答、草、法四部分。

天元術記法說明：書中以籌算式記多項式，自下而上排列，常數項在下，一次項在上，二次又在上，依此類推。如 $\frac{480}{1}$ 表示 $480 - x$ ， $\frac{0 \ 0 \ 48000}{1 \ 0 \ 0}$ 表示 $48000 - x^2$ 之類。所謂「立天元一為某數」，即設未知數 x 為所求之量。

Ceyuan Haijing (Sea Mirror of Circle Measurements), in twelve volumes, was written by Li Ye of the Yuan Dynasty. Li Ye, styled Renqing, sobriquet Jingzhai, was from Luancheng, Zhending. A Jinshi of the late Jin period, he served as Hanlin Academician under the Yuan. This work is his masterpiece on the *tian yuan* (celestial element) method, using right-triangle circle-measurement techniques to elucidate the art of *tian yuan yi*. The complete work contains 170 problems, each positing a circular city wall, from whose walkers' paces the diameter is sought. This edition contains Volumes 4, 5, and 6—53 problems in total:

- **Volume 4:** 17 problems on the base leg (51–67)
- **Volume 5:** 18 problems on the large leg (68–85)
- **Volume 6:** 18 problems on the large base (86–103)

Each problem presents a scenario: A and B depart from city gates, walking east or south, viewing each other diagonally or meeting obliquely. Given certain paces, all else unknown, find the city diameter. Each problem has four parts: **Question**, **Answer**, **Working**, and **Method**.

On Tian Yuan Notation: Polynomials are recorded in counting-rod style, arranged bottom-to-top: constant term lowest, linear term above, quadratic above that, etc. Thus $\frac{480}{1}$ denotes $480 - x$, and $\frac{0 \ 0 \ 48000}{1 \ 0 \ 0}$ denotes $48000 - x^2$. “Let the celestial element unity be [some quantity]” means: let unknown x equal the sought quantity.

1 卷四：底勾一十七問 · Vol. 4: Double Difference (17 Problems)

底勾一十七問 · Double Difference

(第五十一問至第六十七問 · Problems 51–67)

【題意總說】卷四論「底勾」之術。所謂底勾，乃勾股形之最下勾股，與圓徑相切之基本勾股形也。凡一十七問，皆以底勾為基，求圓城之徑。底勾者，通勾股形之勾邊，亦即大勾，為諸勾股形之根本。諸問或知甲東行之數，或知乙南行之數，或知斜行之數，三者之中知其二，即可立天元一，建方程而開方得圓徑。

第一問：或問：乙出南門東行不知步數而立，甲出北門東行二百步見之，就乙斜行二百七十二步與乙相會。問圓城之徑幾何？

答曰：城徑二百四十步。

草曰：識別得二行相減餘，即乙出南門東行數也。以甲東行二百步減於就乙斜行二百七十二步，餘七十二步，為乙東行數。以乘甲東行步，得 $72 \times 200 =$ 一萬四千四百步，又四之，得五萬七千六百步為實。以平方開之，得 $\sqrt{57600} =$ 二百四十步，即城徑也。

驗算：城徑二百四十步，半徑一百二十步。乙東行七十二步，甲東行二百步，甲比乙多行 $200 - 72 = 128$ 步。以勾股術驗之，斜行為弦， $\sqrt{128^2 + 120^2} = \sqrt{16384 + 14400} = \sqrt{30784}$ ，復以半徑加之，合得斜行步數，其術驗矣。

法曰：二行差數乘甲東行又四之，為平方實，開方得全徑。術曰：以就乙斜行步內減甲東行步，餘為乙東行數；以此數乘甲東行步為實，四之；開平方，即得城徑。

[General Summary] Volume 4 treats the “base leg” (底勾) method. The base leg is the lowest side of the right triangle formed tangent to the circular diameter—the fundamental right triangle. All 17 problems use the base leg as foundation to find the city diameter. It is the base of the universal right triangle, i.e., the large base, the root of all sub-triangles. Given any two of: A's eastward walk, B's southward walk, or the diagonal walk, one may establish the celestial element and solve.

Problem 51: Suppose B exits the south gate and walks east an unknown number of paces, while A exits the north gate and walks east 200 paces to see him. A then walks diagonally 272 paces to meet B. What is the diameter of the circular city wall?

Answer: The city diameter is 240 paces.

Working: By inspection, the difference of the two walks gives B's eastward walk. Subtracting A's 200 paces from the diagonal 272 paces leaves 72 paces as B's eastward walk. Multiplying by A's 200 paces gives $72 \times 200 = 14,400$ paces. Quadrupling gives 57,600 as the dividend. Extracting the square root: $\sqrt{57600} = 240$ paces, the city diameter.

Verification: Diameter 240, radius 120. B walks 72 east, A walks 200 east; difference $200 - 72 = 128$. By the gougou (Pythagorean) theorem: diagonal is the hypotenuse; $\sqrt{128^2 + 120^2} = \sqrt{16384 + 14400} = \sqrt{30784}$. Adding the radius gives the diagonal walk, confirming the result.

Method: The difference of the two walks multiplied by A's eastward walk, then quadrupled, is the dividend for square-root extraction. Method: From the diagonal walk subtract A's eastward walk, remainder is B's eastward walk. Multiply this by A's eastward walk as dividend; quadruple; extract square root to obtain the city diameter.

第二問：或問：乙出南門東行一百二十步而立，甲出北門東行二百步見之。問城徑。

答曰：城徑二百四十步。

草曰：立天元一為城徑，以減於二之甲東行步，得 $\frac{480}{1}$ 為兩個小差。以乙東行一百二十步乘之，得 $\frac{57600}{1}$ 為城徑冪，寄左。然後以天元冪 $\frac{0}{1}$ 與左相消，得 $\frac{57600}{1 \ 0}$ (負上正下)，以平方開之，得二百四十步，即城徑也。

釋義：二之甲東行者，以甲東行為勾，其倍即通勾與圓徑之差。以乙東行乘之者，乙東行即小勾，乘大差得大勾之冪，亦即城徑冪之半。故以天元冪相消，開平方即得。

法曰：半之乙東行步乘甲東行為實，半乙東行為從，一步常法，得半徑。術曰：置乙東行步折半，以乘甲東行步為實；以半乙東行步為從法；一步為隅法；開平方得半徑，倍之即城徑。

Problem 52: Suppose B exits the south gate and walks east 120 paces, while A exits the north gate and walks east 200 paces to see him. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. Subtracting it from twice A's eastward walk gives $\frac{480}{1}$ as the two small differences. Multiplying by B's 120 paces gives $\frac{57600}{1}$ as the city diameter-squared, deposited on the left. Then canceling with the celestial element-squared $\frac{0}{1}$ gives $\frac{57600}{1 \ 0}$ (negative above, positive below). Extracting the square root yields 240 paces, the city diameter.

Explanation: Twice A's eastward walk: A's walk being the base, its double is the difference between the universal base and the diameter. Multiplying by B's walk (the small base) gives the square of the large base, i.e., half the city-diameter squared. Cancelling with the celestial element and extracting the root gives the result.

Method: Half of B's eastward walk multiplied by A's eastward walk is the dividend; half of B's eastward walk is the coefficient; unity is the constant divisor. This yields the radius. Method: Halve B's eastward walk, multiply by A's eastward walk as dividend; use half B's walk as the coefficient, unity as the corner divisor; extract square root to obtain the radius; double it for the diameter.

第三問：或問：乙從坤隅南行三百六十步，甲出北門東行二百步見之。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為半徑，減於乙南行三百六十步之半，得 $\frac{180}{1}$ 為小差。半乙南行步，得一百八十步，以乘小差，得 $\frac{32400}{180}$ 為半徑冪，寄左。然後以天元冪 $\frac{0}{1}$ 與左相消，得 $\frac{32400}{180 \ 1 \ 0}$ (上負下正)，開平方得一百二十步，倍之得全徑二百四十步。

釋義：此問以乙南行為股，甲東行為勾。半乙南行即大股之半，去半徑餘為小差。以小差乘半股得半徑冪，此天元術之妙用也。

法曰：兩行步相乘為實，甲東行為從，乙南行為隅，開平方得半徑。術曰：以甲東行步乘乙南行步為實，以甲東行步為從，乙南行步為隅法；開平方得一百二十步，即半徑；倍之即城徑。

Problem 53: Suppose B walks south 360 paces from the Kun corner, while A exits the north gate and walks east 200 paces to see him. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the radius. Subtracting it from half of B's southward walk (360 paces) gives $\frac{180}{1}$ as the small difference. Half of B's southward walk is 180 paces; multiplying by the small difference gives $\frac{32400}{180}$ as the radius-squared, deposited on the left. Canceling with the celestial element-squared $\frac{0}{1}$ gives $\frac{32400}{180 \ 1 \ 0}$ (negative above, positive below). Extracting square root gives 120 paces; doubling gives the full diameter of 240.

Explanation: Here B's southward walk is the leg, A's eastward walk is the base. Half B's walk is half the large leg; less the radius gives the small difference. Multiplying small difference by half the leg gives the radius-squared—the subtle efficacy of the tian yuan method.

Method: The product of the two walks is the dividend; A's eastward walk is the coefficient; B's southward walk is the corner divisor. Extract square root to obtain the radius. Method: Multiply A's eastward by B's southward as dividend; A's walk as coefficient, B's walk as corner; extract square root to obtain 120 paces, the radius; double for the diameter.

第四問：或問：乙從坤隅南行三百六十步而止，甲出北門東行二百步見之，就乙斜行四百零八步與乙相會。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲東行冪得四萬步，以乙南行冪得一十二萬九千六百步，並之得一十七萬三千六百步為弦冪之實。又以斜行四百零八步自之，得一十六萬六千四百六十四步。以此減弦冪實，餘七千一百三十六步。以四之得二萬八千五百四十四步為城徑冪之較實。

別求之：立天元一為城徑，以減於二之甲東行，餘即乙出南門東行數也。以乙南行乘甲東行冪，又四之為實。從空，乙行為廉，一步常法，得城徑。開立方得二百四十步。

法曰：以乙南行步乘甲東行冪，又四之為實，從空，乙行為廉，一步常法，得城徑。術曰：置甲東行步自乘，又以乙南行步乘之，四之為實；乙南行步為廉法；一步為隅法；開立方即得城徑。

Problem 54: Suppose B walks south 360 paces from the Kun corner and stops, while A exits the north gate and walks east 200 paces to see him, then walks diagonally 408 paces to meet B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. Squaring A's eastward walk gives 40,000; squaring B's southward walk gives 129,600. Their sum, 173,600, is the dividend for the hypotenuse-squared. The diagonal 408 squared gives 166,464. Subtracting from the dividend leaves 7,136. Quadrupling gives 28,544 as the comparison-dividend for the diameter-squared.

Alternatively: Let the celestial element be the diameter. Subtracting from twice A's eastward walk gives B's eastward walk from the south gate. Multiply B's southward walk by A's walk-squared; quadruple as dividend. Empty coefficient; B's walk as corner; unity as constant. Extracting cube root gives 240 paces.

Method: B's southward walk multiplied by A's walk-squared, then quadrupled, is the dividend. Empty coefficient; B's walk as corner; unity as constant divisor. Extract cube root for the diameter. Method: Square A's eastward walk, multiply by B's southward walk, quadruple as dividend; B's walk as corner divisor; unity as corner; extract cube root.

第五問：或問：乙出南門直行一百三十五步而立，甲出北門東行二百步見之。問城徑。

答曰：城徑二百四十步。

草曰：立天元一為城徑，加乙南行一百三十五步，得 $\frac{135}{1}$ 為

股率。其甲東行二百步即勾率也。其乙南行為小股，以勾率乘之，合以大弦率除。今不受除，便以此為小勾，為母，寄股率。

乃先以小弦乘大和，得下式 $\frac{0}{1} \frac{0}{0} \frac{12000}{0}$ 寄左。又以倍天元減斜步，得 $\frac{100}{2}$ 為小和，以乘大弦，得 $\frac{12000}{1}$ 為同數，

與左相消。開平方得二百四十步。

按：此草以率法為主，以勾股之率通諸數，建立天元方程，為天元術之正統用法。立天元一為城徑，以諸率配之，求其等式，相消開方，即得所求。

法曰：以乙南行步乘甲東行冪，又四之為實，從空，乙行為廉，一步常法，得城徑。

Problem 55: Suppose B exits the south gate and walks due south 135 paces, while A exits the north gate and walks east 200 paces to see him. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. Adding B's southward 135 paces gives $\frac{135}{1}$ as the leg-rate. A's eastward 200 paces is the base-rate. B's southward walk is the small leg; multiplying by the base-rate should be divided by the large hypotenuse-rate. Not dividing, this becomes the small base as numerator, with the leg-rate deposited.

First multiply the small hypotenuse by the large sum, obtaining the expression $\frac{0}{1} \frac{0}{0} \frac{12000}{0}$, deposited left. Then subtract twice the celestial element from the diagonal, obtaining $\frac{100}{2}$ as the small sum; multiply by the large hypotenuse, obtaining $\frac{12000}{1}$ as the equal number, which cancels with the left. Extracting square root gives 240 paces.

Note: This working uses rate-methods to connect all quantities, establishing tian yuan equations in the orthodox manner.

Method: B's southward walk multiplied by A's walk-squared, then quadrupled, is the dividend. Empty coefficient; B's walk as corner; unity as constant; extract cube root for the diameter.

第六問：或問：乙從坤隅南行三百六十步，甲出北門東行二百步見之，就乙斜行二百七十二步與乙相會。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為半徑。以甲東行二百步減斜行二百七十二步，餘七十二步，為乙出南門東行數。以乙南行三百六十步折半，得一百八十步，為大股之半。以此半股減天元半徑，得 $\frac{180}{1}$ 為小差。以小差乘半股，得 $\frac{32400}{180}$ 為半徑冪，

寄左。以天元冪相消，開平方得一百二十步，倍之得全徑二百四十步。

法曰：兩行步相乘倍之為實，乙南行為從，一步常法，得半徑。術曰：以甲東行步乘乙南行步，倍之為實；乙南行步為從法；一步為隅法；開平方得半徑，倍之即城徑。

Problem 56: Suppose B walks south 360 paces from the Kun corner, while A exits the north gate and walks east 200 paces to see him, then walks diagonally 272 paces to meet B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the radius. Subtracting A's 200 paces from the diagonal 272 leaves 72 paces as B's eastward walk from the south gate. Halving B's 360-pace southward walk gives 180 paces, half the large leg. Subtracting the radius from this half-leg gives $\frac{180}{1}$ as the small difference. Multiplying small difference by half the leg gives $\frac{32400}{180}$ as the radius-squared, deposited left. Canceling with the celestial element-squared and extracting square root gives 120 paces; doubling gives the full diameter of 240.

Method: The product of the two walks, doubled, is the dividend; B's southward walk is the coefficient; unity is the constant divisor; this yields the radius. Method: Multiply A's eastward by B's southward, double as dividend; B's walk as coefficient, unity as corner; extract square root for the radius; double for the diameter.

第七問：或問：乙出南門直行一百三十五步而立，甲出北門東行二百步見之，復就乙斜行二百七十二步與乙相會。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。以乙南行一百三十五步為小股，甲東行二百步為小勾。以斜行二百七十二步為小弦。驗之： $135^2 + 72^2 = 18225 + 5184 = 23409$ ，而 $272^2 = 73984$ ，非直股勾股也。當以天元術正之。

立天元一為城徑，以減於二之甲東行，得 $\frac{400}{1}$ 為兩個小差之和。以乙南行乘之，又四之，得大實。以天元冪與左相消，開平方得二百四十步。

法曰：以乙南行乘甲東行冪為實，從空，乙行為廉，一步常法，得城徑。

Problem 57: Suppose B exits the south gate and walks due south 135 paces, while A exits the north gate and walks east 200 paces to see him, then walks diagonally 272 paces to meet B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. B's southward 135 paces is the small leg; A's eastward 200 is the small base; the diagonal 272 is the small hypotenuse. Verification: $135^2 + 72^2 = 18225 + 5184 = 23409$, while $272^2 = 73984$ —not a direct right triangle. We must correct by the tian yuan method.

Let the celestial element be the diameter. Subtracting from twice A's eastward walk gives $\frac{400}{1}$ as the sum of the two small differences. Multiplying by B's southward walk and quadrupling gives the large dividend. Canceling with the celestial element-squared and extracting square root gives 240 paces.

Method: B's southward walk multiplied by A's walk-squared is the dividend. Empty coefficient; B's walk as corner; unity as constant; extract cube root for the diameter.

第八問：或問：乙出南門直行一百三十五步，甲出北門東行二百步，就乙斜行二百七十二步與乙相會。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。倍乙南行，得二百七十步。以減於倍甲東行四百步，餘一百三十步，為大小差之較。以此較乘甲東行冪，得 $130 \times 40000 = 5200000$ 為實。從空，倍乙行為廉，一步常法，開立方得城徑。

又法：立天元一為半徑，以半甲東行一百步減斜行半數一百三十六步，餘三十六步為半較。以乙南行半數六十七步半乘之，得半徑冪之較，與天元冪相消，開平方即得。

法曰：倍乙南行乘甲東行冪為實，從空，倍乙行為廉，一步常法，得城徑。

Problem 58: Suppose B exits the south gate and walks due south 135 paces, while A exits the north gate and walks east 200 paces, then walks diagonally 272 paces to meet B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. Doubling B's southward walk gives 270 paces. Subtracting from twice A's eastward walk (400 paces) leaves 130 paces as the difference between the large and small differences. Multiplying this difference by A's walk-squared: $130 \times 40000 = 5,200,000$ paces as dividend. Empty coefficient; twice B's walk as corner; unity as constant; extracting cube root gives the diameter.

Alternative method: Let the celestial element be the radius. Subtracting half A's walk (100) from half the diagonal (136) leaves 36 as half the difference. Multiplying by half B's walk (67.5) gives the radius-squared difference; cancel with the celestial element-squared; extract square root.

Method: Twice B's southward walk multiplied by A's walk-squared is the dividend. Empty coefficient; twice B's walk as corner; unity as constant; extract cube root for the diameter.

第九問：或問：乙出南門直行一百三十五步，甲出北門東行二百步見之，就乙斜行二百七十二步與乙相會。問城徑幾何？

答曰：城徑二百四十步。

草曰：此問與前問類而異者，甲見乙之後復斜行相會。立天元一為城徑，以兩行相減，餘乘甲東行又四之為實。兩行差為廉，一步常法，開立方得城徑。

細草：以乙南行一百三十五步減甲東行二百步，餘六十五步，為兩行之差。以乘甲東行二百步，得一千三百步，又四之得五千二百步為實。兩行差六十五步為廉法，一步為隅法，開立方得二百四十步。

法曰：兩行相減餘乘甲東行又四之為實，從空，兩行差為廉，一步常法，得城徑。

Problem 59: Suppose B exits the south gate and walks due south 135 paces, while A exits the north gate, walks east 200 paces to see him, then walks diagonally 272 paces to meet B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: This problem is similar to the previous but differs: after A sees B, he then walks diagonally to meet him. Let the celestial element be the city diameter. Subtract the two walks; multiply the remainder by A's eastward walk and quadruple as dividend. The difference of the two walks is the corner divisor; unity is the constant; extract cube root for the diameter.

Detailed working: Subtract B's 135 from A's 200, leaving 65 as the difference. Multiply by A's 200: 1,300; quadruple: 5,200 as dividend. The difference 65 is the corner divisor; unity is the constant corner. Extracting cube root gives 240 paces.

Method: The difference of the two walks multiplied by A's eastward walk, then quadrupled, is the dividend. Empty coefficient; the walk-difference as corner; unity as constant; extract cube root for the diameter.

第十問：或問：乙出南門直行一百三十五步，甲出北門東行二百步見之。問城徑。

答曰：城徑二百四十步。

草曰：立天元一為城徑。以乙南行一百三十五步乘甲東行冪（四萬步），得五百四十萬步，又四之得二千一百六十萬步為實。從空，乙行一百三十五步為廉法，一步為隅法，開立方得二百四十步。

按：此草以乙南行為小股，甲東行為小勾。以勾冪乘股，蓋以大勾之冪比於小勾之冪，若大股之冪比於小股之冪。以率言之，大勾即城徑與小勾之關係，故以天元建立而開立方。

法曰：以乙南行步乘甲東行冪，又四之為實，從空，乙行為廉，一步常法，得城徑。

Problem 60: Suppose B exits the south gate and walks due south 135 paces, while A exits the north gate and walks east 200 paces to see him. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. Multiplying B's southward 135 paces by A's walk-squared (40,000) gives 5,400,000 paces; quadrupled gives 21,600,000 as dividend. Empty coefficient; B's 135 paces as corner divisor; unity as constant corner; extracting cube root gives 240 paces.

Note: This working takes B's southward as the small leg, A's eastward as the small base. Multiplying the base-squared by the leg: the ratio of the large base-squared to the small base-squared equals that of the large leg-squared to the small leg-squared. The large base is the relation between diameter and small base, hence establishing tian yuan and extracting cube root.

Method: B's southward walk multiplied by A's walk-squared, then quadrupled, is the dividend. Empty coefficient; B's walk as corner; unity as constant; extract cube root for the diameter.

第十一問：或問：乙出南門直行一百三十五步而立，甲出北門東行二百步見之。問城徑與斜行各幾何？

答曰：城徑二百四十步。斜行二百七十二步。

草曰：立天元一為城徑。倍乙南行，得二百七十步。以甲東行冪四萬步乘之，得一千零八萬步，又倍之得二千零一十六萬步為實。倍乙南行二百七十步為廉法，一步為隅法，開立方得城徑二百四十步。

既得城徑，求斜行：以城徑二百四十步減於二之甲東行四百步，餘一百六十步，為乙東行與半徑之差。以此差冪與半徑冪相加開方，得斜行二百七十二步。

法曰：倍乙南行乘甲東行冪為實，從空，倍乙行為廉，一步常法，得城徑。術曰：倍乙南行步，以乘甲東行冪為實；倍乙南行步為廉法；一步為隅法；開立方得城徑。

Problem 61: Suppose B exits the south gate and walks due south 135 paces, while A exits the north gate and walks east 200 paces to see him. What are the city diameter and the diagonal walk?

Answer: City diameter: 240 paces. Diagonal walk: 272 paces.

Working: Let the celestial element unity be the city diameter. Doubling B's southward walk gives 270 paces. Multiplying by A's walk-squared (40,000) gives 10,080,000 paces; doubled gives 20,160,000 as dividend. Twice B's southward (270) is the corner divisor; unity is the constant corner; extracting cube root gives diameter 240 paces.

To find the diagonal: Subtract diameter 240 from twice A's eastward walk (400), leaving 160 as the difference between B's eastward walk and the radius. Adding the square of this difference to the square of the radius and extracting root gives the diagonal of 272 paces.

Method: Twice B's southward walk multiplied by A's walk-squared is the dividend. Empty coefficient; twice B's walk as corner; unity as constant; extract cube root for the diameter. Method: Double B's southward walk, multiply by A's walk-squared as dividend; twice B's walk as corner; unity as corner; extract cube root.

第十二問：或問：乙出南門直行一百三十五步，甲出北門東行二百步，就乙斜行二百七十二步與乙相會。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。以兩行相減，甲東行二百步減乙南行一百三十五步，餘六十五步，為兩行之較。以較乘甲東行，得一千三百步，又四之，得五千二百步為實。從空，兩行較六十五步為廉法，一步為隅法，開立方得城徑二百四十步。

驗草：城徑二百四十步，半徑一百二十步。以半徑減甲東行，餘八十步為甲至城心之東行。以乙南行一百三十五步減半徑，餘十五步，為乙至城心之南行。以此二數各自乘并之， $80^2 + 15^2 = 6400 + 225 = 6625$ ，開方得 81.39 不盡，須以天元正術求之。

法曰：兩行相減餘乘甲東行又四之為實，從空，兩行差為廉，一步常法，得城徑。

Problem 62: Suppose B exits the south gate and walks due south 135 paces, while A exits the north gate, walks east 200 paces, then diagonally 272 paces to meet B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. Subtract the two walks: A's 200 minus B's 135 leaves 65 as the difference. Multiply the difference by A's 200: 1,300; quadruple: 5,200 as dividend. Empty coefficient; the difference 65 as corner divisor; unity as constant; extracting cube root gives diameter 240 paces.

Verification: Diameter 240, radius 120. Radius subtracted from A's walk leaves 80 as A's distance from the center eastward. B's 135 minus radius leaves 15 as B's distance southward from the center. Squaring and summing: $80^2 + 15^2 = 6400 + 225 = 6625$; root ≈ 81.39 , inexact— must use the orthodox tian yuan method.

Method: The difference of the two walks multiplied by A's eastward walk, then quadrupled, is the dividend. Empty coefficient; the walk-difference as corner; unity as constant; extract cube root for the diameter.

第十三問：或問：乙出南門直行一百三十五步而立，甲出北門東行二百步見之。問城徑及乙東行步數各幾何？

答曰：城徑二百四十步，乙東行七十二步。

草曰：立天元一為城徑。以乙南行一百三十五步乘甲東行冪（四萬步），得五百四十萬步為實。從空，乙行為廉，一步常法，開平方得城徑。

求乙東行：以城徑二百四十步減於二之甲東行四百步，餘一百六十步，以減甲東行二百步，餘七十二步，即乙東行步數。

釋天元：此問以乙南行乘甲東行冪為實者，乙南行為股率，甲東行為勾率，以勾冪乘股率者，以大勾之冪與大股相乘之義也。

法曰：以乙南行乘甲東行冪為實，從空，乙行為廉，一步常法，得城徑。

Problem 63: Suppose B exits the south gate and walks due south 135 paces, while A exits the north gate and walks east 200 paces to see him. What are the city diameter and B's eastward paces?

Answer: City diameter: 240 paces. B's eastward walk: 72 paces.

Working: Let the celestial element unity be the city diameter. Multiplying B's southward 135 paces by A's walk-squared (40,000) gives 5,400,000 as dividend. Empty coefficient; B's walk as corner; unity as constant; extract square root for the diameter.

To find B's eastward walk: Subtract diameter 240 from twice A's eastward (400), leaving 160. Subtract this from A's 200 paces, leaving 72 paces as B's eastward walk.

Tian Yuan explained: Using B's southward (the leg-rate) multiplied by A's walk-squared (the base-squared): the base-squared multiplied by the leg-rate represents the meaning of the large base-squared multiplied by the large leg.

Method: B's southward walk multiplied by A's walk-squared is the dividend. Empty coefficient; B's walk as corner; unity as constant; extract square root for the diameter.

第十四問：或問：乙出南門直行一百三十五步而立，甲出北門東行二百步見之，復斜行與乙相會。問城徑及斜行步數。

答曰：城徑二百四十步，斜行二百七十二步。

草曰：立天元一為城徑。倍乙南行，得二百七十步，乘甲東行冪（四萬步），得一千零八萬步為實。從空，倍乙南行為廉，一步為隅法，開平方得城徑二百四十步。

求斜行：以城徑減於二之甲東行，餘一百六十步。以此與甲東行相加，得三百六十步，折半得一百八十步，為弦之半。以自乘得三萬二千四百步，加入半徑冪一萬四千四百步，得四萬六千八百步，即非弦冪也。當以天元別求之：以半徑冪加甲乙東行之較冪，合半之，開方得斜行。 $120^2 + 128^2 = 14400 + 16384 = 30784$ ，而 $272^2 = 73984$ ，知須以天元方程正求之。

法曰：倍乙南行乘甲東行冪為實，從空，倍乙行為廉，一步常法，得城徑。

Problem 64: Suppose B exits the south gate and walks due south 135 paces, while A exits the north gate and walks east 200 paces to see him, then walks diagonally to meet B. What are the city diameter and diagonal walk?

Answer: City diameter: 240 paces. Diagonal walk: 272 paces.

Working: Let the celestial element unity be the city diameter. Doubling B's southward walk gives 270; multiplying by A's walk-squared (40,000) gives 10,800,000 as dividend. Empty coefficient; twice B's walk as corner; unity as constant; extracting square root gives diameter 240 paces.

To find the diagonal: Subtract diameter from twice A's eastward, leaving 160. Add to A's eastward: 360; halve: 180 as half the hypotenuse. Squaring gives 32,400; adding radius-squared 14,400 gives 46,800—not the hypotenuse-squared. Must use tian yuan separately: add radius-squared to the difference of A-B eastward walks squared, halve, extract root for diagonal. $120^2 + 128^2 = 14400 + 16384 = 30784$, while $272^2 = 73984$; hence must use tian yuan equations correctly.

Method: Twice B's southward walk multiplied by A's walk-squared is the dividend. Empty coefficient; twice B's walk as corner; unity as constant; extract square root for the diameter.

第十五問：或問：乙出南門直行一百三十五步，甲出北門東行二百步，就乙斜行二百七十二步與乙相會。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。以兩行相減，甲東行二百步減乙南行一百三十五步，餘六十五步為較。以較乘甲東行冪（四萬步），得二百六十萬步，又四之得一千零四十萬步為實。從空，較六十五步為廉法，一步為隅法，開立方得城徑二百四十步。

按：此問以較為主，蓋斜行相會，須以兩行差建立方程。較者，大小差之較也，為天元術中重要之率。

法曰：兩行相減餘乘甲東行又四之為實，從空，兩行差為廉，一步常法，得城徑。

Problem 65: Suppose B exits the south gate and walks due south 135 paces, while A exits the north gate, walks east 200 paces, then diagonally 272 paces to meet B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. Subtract the two walks: A's 200 minus B's 135 leaves 65 as the difference. Multiply the difference by A's walk-squared (40,000): 2,600,000; quadrupled: 10,400,000 as dividend. Empty coefficient; the difference 65 as corner divisor; unity as constant; extracting cube root gives diameter 240 paces.

Note: This problem uses the difference as key. For diagonal meetings, one must establish equations from the walk-difference. The difference is the difference of large and small differences, an important rate in tian yuan.

Method: The walk-difference multiplied by A's walk-squared, then quadrupled, is the dividend. Empty coefficient; walk-difference as corner; unity as constant; extract cube root for the diameter.

第十六問：或問：乙出南門直行一百三十五步而立，甲出北門東行二百步見之。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。以乙南行一百三十五步乘甲東行冪（四萬步），得五百四十萬步，又四之得二千一百六十萬步為實。從空，乙行一百三十五步為廉法，一步為隅法，開立方得城徑二百四十步。

細釋：甲東行為勾，乙南行為股。以勾冪乘股為實者，大勾之冪當與大股相乘，而乙南行即大股之表示，故以此為實。從空者，無從法也。乙行為廉，即二次項之係數也。

法曰：以乙南行步乘甲東行冪，又四之為實，從空，乙行為廉，一步常法，得城徑。

Problem 66: Suppose B exits the south gate and walks due south 135 paces, while A exits the north gate and walks east 200 paces to see him. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. Multiply B's southward 135 paces by A's walk-squared (40,000): 5,400,000; quadrupled: 21,600,000 as dividend. Empty coefficient; B's 135 paces as corner divisor; unity as constant corner; extracting cube root gives diameter 240 paces.

Detailed explanation: A's eastward walk is the base, B's southward is the leg. Using base-squared multiplied by leg as dividend: the large base-squared ought to multiply the large leg, and B's southward represents the large leg, hence used as dividend. Empty coefficient means no linear coefficient. B's walk as corner is the quadratic coefficient.

Method: B's southward walk multiplied by A's walk-squared, then quadrupled, is the dividend. Empty coefficient; B's walk as corner; unity as constant; extract cube root for the diameter.

第十七問：或問：乙出南門直行一百三十五步而立，甲出北門東行二百步見之。問城徑幾何？又問甲乙相望之斜距幾何？

答曰：城徑二百四十步。斜距二百七十二步。

草曰：立天元一為城徑。倍乙南行，得二百七十步。乘甲東行冪（四萬步），得一千零八萬步，又倍之得二千一百六十萬步為實。從空，倍乙南行為廉法，一步為隅法，開立方得城徑二百四十步。

求斜距：既得城徑，則半徑一百二十步。甲東行二百步減半徑一百二十步，餘八十步為甲至城邊之距。以此冪六千四百，與乙南行一百三十五步減半徑餘十五步之冪二百二十五并之，得六千六百二十五，開平方得八十一點二二，加上半徑一百二十步，得二百零一點二二，非斜距也。當以天元正術：以甲東行減乙南行之差冪，加入甲乙各自去城邊之冪，并而開方，合加半徑，即得斜距二百七十二步。

法曰：倍乙南行乘甲東行冪為實，從空，倍乙行為廉，一步常法，得城徑。

Problem 67: Suppose B exits the south gate and walks due south 135 paces, while A exits the north gate and walks east 200 paces to see him. What is the city diameter? Also, what is the diagonal distance between A and B?

Answer: City diameter: 240 paces. Diagonal distance: 272 paces.

Working: Let the celestial element unity be the city diameter. Doubling B's southward walk gives 270; multiplying by A's walk-squared (40,000) gives 10,800,000; doubled: 21,600,000 as dividend. Empty coefficient; twice B's walk as corner divisor; unity as constant; extracting cube root gives diameter 240 paces.

To find the diagonal distance: Diameter 240 gives radius 120. A's 200 minus radius 120 leaves 80 as A's distance to the wall. Its square 6,400 plus B's 135 minus radius 15 (square 225) gives 6,625; square root ≈ 81.22 ; plus radius 120 gives ≈ 201.22 —not the diagonal. Must use orthodox tian yuan: the difference of A's and B's walks squared, plus each distance to the wall squared, summed and rooted, plus radius, gives diagonal 272.

Method: Twice B's southward walk multiplied by A's walk-squared is the dividend. Empty coefficient; twice B's walk as corner; unity as constant; extract cube root for the diameter.

2 卷五：大股一十八問 • Vol. 5: Large Leg (18 Problems)

大股一十八問 • Large Leg

(第六十八問至第八十五問 • Problems 68–85)

【題意總說】卷五論「大股」之術。所謂大股，乃最大勾股形之股邊，即通股也。以大股為主，配以諸勾股形之關係，求圓城之徑。凡一十八問。大股者，勾股形中最長之股，即甲從乾隅直南至坤隅之步數，為六百步之率也。諸問或知乙南行之數，或知甲南行之數，或知斜望之數，皆以大股為基本建立天元方程。此卷方程多為三次、四次，須開立方、開三乘方以求之，為天元術之精深處。

第十八問：或問：乙出南門直行一百三十五步而立，甲從乾隅南行六百步，斜望乙與城參相直。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為半徑，以二之減於甲南行六百步，得 $\frac{600}{2}$ 為大差也。以自之，得 $\frac{360000}{2400 \ 4}$ 為大差冪也。置甲

南行冪 $\frac{360000}{1}$ 內減大差冪，而半之，得 $\frac{0}{1200 \ 2}$ 為大弦也，

內帶大差為分母。

又置甲南行冪內減大差冪，而半之，得 $\frac{0}{600 \ 1}$ 為大勾

也，帶大差分母。又以大差乘股六百步，得 $\frac{360000}{1}$ 併

入和，得下式 $\frac{360000}{1} \ \frac{0}{1} \ \frac{0}{1200 \ 1}$ 為小和，以乘大弦，得

$\frac{0}{0 \ 0 \ 1 \ 0 \ 1200 \ 1} \ \frac{0}{0 \ 0 \ 1 \ 0 \ 0 \ 1}$ 寄左。

又以倍天元減斜步，得 $\frac{100}{2}$ 為小和，以乘大弦，得同數，與

左相消。開立方得一百二十步，即半徑也；倍之得全徑二百四十步。

釋義：此草以大差為主，大差者甲南行減圓徑也。甲從乾隅南行六百步，為大股之全數。減去圓徑，餘為大差。以大差冪與甲南行冪相減，半之得大弦，此即勾股求弦之法。又以大差乘股併入和，為小和以乘大弦，建立天元方程。其術精深，為全書之難問。

法曰：倍二行差內減甲南行步，復以乘甲南行步為實。倍二行差減甲南行步即甲南行也。四之甲南行步內減於甲南行餘二百四十步，即城徑也。術曰：以甲南行步自之為冪，以乙南行步乘之為實；以甲南行步為從，乙南行步為廉；一步為隅法；開立方得城徑。

【General Summary】Volume 5 treats the “large leg” (大股) method. The large leg is the leg-side of the largest right triangle—the universal leg. Taking the large leg as principal, combined with relations among sub-triangles, we seek the city diameter. All 18 problems. The large leg is A’s full southward walk from the Qian corner to the Kun corner—the 600-pace standard. Given B’s southward walk, or A’s southward walk, or diagonal sightings, we establish tian yuan equations. This volume features third- and fourth-degree equations requiring cube-root and fourth-root extraction—the subtle core of tian yuan.

Problem 68: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner, viewing B obliquely with the city wall aligned between them. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the radius. Subtracting twice it from A’s 600-pace southward walk gives $\frac{600}{2}$ as the large difference. Squaring: $\frac{360000}{2400 \ 4}$

as the large-difference-squared. Place A’s walk-squared $\frac{360000}{1}$, subtract large-diff-squared, halve: $\frac{0}{1200 \ 2}$ as the large hypotenuse, carrying large difference as denominator.

Also subtract and halve to obtain $\frac{0}{600 \ 1}$ as the large base, with large-diff denominator. Multiply large difference by the 600-pace leg: $\frac{360000}{1}$; combine into the sum, obtaining the expression $\frac{360000}{1} \ \frac{0}{1} \ \frac{0}{1200 \ 1}$ as the small sum. Multiply by the large hypotenuse, deposit left.

Also subtract twice the celestial element from the diagonal, obtaining $\frac{100}{2}$ as small sum; multiply by large hypotenuse for the equal number; cancel with the left. Extract cube root: 120 paces, the radius; double for the full diameter 240.

Explanation: This working uses the large difference as key—A’s walk minus the diameter. The 600-pace walk is the full large leg. Subtract the diameter for the large difference. The method is among the most profound in the entire work.

Method: Twice the walk-difference minus A’s southward walk, then multiplied by A’s walk, is the dividend. Method: Square A’s southward walk, multiply by B’s southward walk as dividend; A’s walk as coefficient, B’s walk as corner; unity as corner; extract cube root for the diameter.

第十九問：或問：乙出南門直行一百三十五步而立，甲從乾隅南行六百步，斜望乙與城參相直。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為半徑，以減於甲南行六百步，得 $\frac{600}{1}$ 為

大差。置甲南行幕內減大差幕，而半之，得大弦也。又以大差乘股六百步，併入和，得下式為小和，以乘大弦，寄左。又以倍天元減斜步，得小和，以乘大弦，得同數，與左相消。開立方得半徑一百二十步，倍之得城徑。

細草：甲南行六百步自之，得三十六萬步為甲南行幕。大差 $\frac{600}{1}$ 自之，得 $\frac{360000}{1200 \ 1}$ 。以甲南行幕減之，得 $\frac{0}{1200 \ 1}$ ，半之得 $\frac{0}{600 \ \frac{1}{2}}$ 為大弦，帶大差分母。以大差乘股，得 $\frac{360000}{1}$ ，

併入得小和。以小和乘大弦，寄左。以倍天元減斜步得小和，乘大弦為同數，與左相消，開立方得一百二十步。

法曰：兩行步相乘為實，從空，兩行步為廉，一步常法，得城徑。

Problem 69: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner, viewing B obliquely with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the radius. Subtracting from A's 600-pace southward walk gives $\frac{600}{1}$ as the large difference. Subtract large-diff-squared from A's walk-squared and halve to obtain the large hypotenuse. Multiply large difference by the 600-pace leg, combine into the sum, obtaining the small sum; multiply by large hypotenuse, deposit left. Subtract twice the celestial element from the diagonal for the small sum, multiply by large hypotenuse for the equal number; cancel with the left; extract cube root for radius 120; double for the diameter.

Detailed working: A's 600 squared: 360,000. Large diff squared: $\frac{360000}{1200 \ 1}$. Subtracting from A's walk-squared gives $\frac{0}{1200 \ 1}$; halved: $\frac{0}{600 \ \frac{1}{2}}$ as large hypotenuse with large-diff denominator. Multiply large diff by leg, $\frac{360000}{1}$, combine for small sum. Cancel and extract.

Method: The product of the two walks is the dividend. Empty coefficient; the two walks as corner; unity as constant; extract cube root for the diameter.

第二十問：或問：乙出南門直行一百三十五步，甲從乾隅南行六百步望乙，與城參相直。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑，以二之減於甲南行六百步，餘 $\frac{600}{2}$ 為大差。以自之得大差幕 $\frac{360000}{2400 \ 4}$ 。置甲南行幕 $\frac{360000}{1}$ 內減大差幕而半之，得大弦，內帶大差為分母。

又置甲南行幕內減大差幕而半之，得大勾，帶大差分母。又以大差乘股六百步，併入和，得下式為小和，以乘大弦，寄左。又以倍天元減斜步，得小和，以乘大弦，得同數，與左相消。開立方得一百二十步，即半徑；倍之得城徑二百四十步。

按：此問與首問相類，惟立天元一為城徑而非半徑，故方程中諸數皆倍之，開立方後須倍之乃得城徑，學者宜辨之。

法曰：倍二行差內減甲南行步，復以乘甲南行步為實。倍二行差減甲南行步即甲南行也。四之甲南行步內減於甲南行餘，即城徑也。

Problem 70: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner to view B, with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. Subtracting twice it from A's 600-pace walk leaves $\frac{600}{2}$ as the large difference. Squaring gives large-diff-squared $\frac{360000}{2400 \ 4}$. From A's walk-squared $\frac{360000}{1}$ subtract large-diff-squared and halve to obtain the large hypotenuse, carrying large-diff denominator. Also halve to obtain the large base with large-diff denominator. Multiply large diff by the 600-pace leg, combine for small sum, multiply by large hypotenuse, deposit left. Subtract twice the celestial element from diagonal for small sum; multiply by large hypotenuse; cancel; extract cube root: 120 paces, the radius; double for diameter 240.

Note: Similar to the first problem, but the celestial element is the diameter, not the radius. Hence all quantities are doubled; after extracting cube root, double again for the diameter. Students should distinguish.

Method: Twice the walk-difference minus A's southward walk, then multiplied by A's walk, is the dividend. The remainder is the city diameter.

第二十一問：或問：乙出南門直行一百三十五步而立，甲從乾隅南行六百步望乙。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲南行六百步為大股，乙南行一百三十五步為小股。兩股相減，得四百六十五步為大小股之差。以此差乘甲南行冪（三十六萬步），得一億六千七百四十萬步為實。從空，兩行差為廉，一步常法，開立方得城徑二百四十步。

釋義：此草以兩股差為主。大股甲南行六百步，小股乙南行一百三十五步。兩股之差四百六十五步，即大小差之較也。以此建立天元方程，較之以前數問更為簡捷，乃法之變通也。

法曰：兩行步相乘為實，從空，兩行步為廉，一步常法，得城徑。

Problem 71: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner to view B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 600-pace southward walk is the large leg; B's 135 is the small leg. Subtracting gives 465 paces as the leg-difference. Multiplying this difference by A's walk-squared (360,000) gives 167,400,000 as dividend. Empty coefficient; the walk-difference as corner; unity as constant; extracting cube root gives diameter 240 paces.

Explanation: This working uses the leg-difference as key. Large leg 600, small leg 135; difference 465 is the comparison of large and small differences. Establishing the tian yuan equation this way is simpler than previous methods—a flexible variation of the technique.

Method: The product of the two walks is the dividend. Empty coefficient; the two walks as corner; unity as constant; extract cube root for the diameter.

第二十二問：或問：乙出南門直行一百三十五步，甲從乾隅南行六百步，斜望乙與城參相直。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。倍二行差（甲南行六百步減乙南行一百三十五步，得四百六十五步，倍之得九百三十步），內減甲南行步，得三百三十步。復以乘甲南行步，得 $330 \times 600 = 198000$ 步為實。從空，倍二行差減甲南行步為廉，一步常法，開立方得城徑二百四十步。

細草：甲南行六百步，乙南行一百三十五步，差四百六十五步。倍之得九百三十步，內減六百步，餘三百三十步。以乘六百步，得一十九萬八千步為實。以三百三十步為廉法，一步為隅法，開立方得二百四十步，即城徑。

法曰：倍二行差內減甲南行步，復以乘甲南行步為實，從空，倍二行差減甲南行步為廉，一步常法，得城徑。

Problem 72: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner, viewing B obliquely with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. Twice the walk-difference (A's 600 minus B's 135 gives 465; doubled: 930), subtract A's southward walk: 330. Multiply by A's walk: $330 \times 600 = 198,000$ paces as dividend. Empty coefficient; twice walk-diff minus A's walk as corner; unity as constant; extracting cube root gives diameter 240 paces.

Detailed working: A's 600, B's 135, difference 465. Doubled: 930; subtract 600: 330. Multiply by 600: 198,000 as dividend. 330 as corner divisor; unity as constant; extract cube root: 240 paces, the diameter.

Method: Twice the walk-difference minus A's southward walk, then multiplied by A's walk, is the dividend. Empty coefficient; twice walk-diff minus A's walk as corner; unity as constant; extract cube root for the diameter.

第二十三問：或問：乙出南門直行一百三十五步而立，甲從乾隅南行六百步望乙。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲南行六百步為大股，乙南行一百三十五步為小股。兩股相乘，得 $600 \times 135 = 81000$ 步為實。從空，兩行步相加得七百三十五步為廉，一步常法，開平方得城徑二百四十步。

按：此問開平方即得，不須開立方，較之他問為簡易。蓋以兩股相乘為實，兩股和為從，適合天元平方之式也。

法曰：兩行步相乘為實，從空，兩行步為廉，一步常法，得城徑。

Problem 73: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner to view B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 600 is the large leg; B's 135 is the small leg. Multiply: $600 \times 135 = 81,000$ as dividend. Empty coefficient; sum of walks 735 as corner; unity as constant; extracting square root gives diameter 240 paces.

Note: This problem requires only square-root extraction, not cube root—simpler than others. The product of the two legs as dividend and their sum as coefficient perfectly fits the tian yuan quadratic pattern.

Method: The product of the two walks is the dividend. Empty coefficient; the two walks as corner; unity as constant; extract square root for the diameter.

第二十四問：或問：乙出南門直行一百三十五步，甲從乾隅南行六百步，斜望乙與城參相直。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲南行六百步減乙南行一百三十五步，得四百六十五步為差。倍之得九百三十步，內減甲南行六百步，餘三百三十步。以乘甲南行六百步，得一十九萬八千步為實。以三百三十步為廉，一步為隅法，開立方得城徑二百四十步。

驗草：城徑二百四十步，半徑一百二十步。甲南行六百步減圓徑二百四十步，餘三百六十步為大差。以大差與半徑各為股勾， $360^2 + 120^2 = 129600 + 14400 = 144000$ ，開方得 379.47 不盡。當以三乘方正之，其術繁矣。

法曰：倍二行差內減甲南行步，復以乘甲南行步為實，從空，倍二行差減甲南行步為廉，一步常法，得城徑。

Problem 74: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner, viewing B obliquely with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 600 minus B's 135 gives 465 as the difference. Doubled: 930; subtract A's 600: 330. Multiply by A's 600: 198,000 as dividend. 330 as corner; unity as constant; extracting cube root gives diameter 240 paces.

Verification: Diameter 240, radius 120. A's 600 minus diameter 240 leaves 360 as the large difference. Using large difference and radius as leg and base: $360^2 + 120^2 = 129,600 + 14,400 = 144,000$; root ≈ 379.47 , inexact. Must use fourth-root correction—complex.

Method: Twice the walk-difference minus A's southward walk, then multiplied by A's walk, is the dividend. Empty coefficient; twice walk-diff minus A's walk as corner; unity as constant; extract cube root for the diameter.

第二十五問：或問：乙出南門直行一百三十五步而立，甲從乾隅南行六百步望乙。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲南行六百步為大股，乙南行一百三十五步為小股。兩股相乘，得八萬一千步為實。以兩股和七百三十五步為從，一步為隅法，開平方得城徑二百四十步。

按：此問以兩股相乘為實，蓋大股與小股相乘之積，當等於以城徑為高之矩形面積。以兩股和為從者，大小股之和與城徑之關係式也。開平方即得，術最簡明。

法曰：兩行步相乘為實，從空，兩行步為廉，一步常法，得城徑。

Problem 75: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner to view B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 600 is the large leg; B's 135 is the small leg. Multiply: 81,000 as dividend. The leg-sum 735 as coefficient; unity as corner; extracting square root gives diameter 240 paces.

Note: This problem uses the product of the two legs as dividend. The product equals the rectangular area with the diameter as one dimension. Using the leg-sum as coefficient expresses the relation between diameter and leg sum. Square-root extraction gives the result directly—the clearest method.

Method: The product of the two walks is the dividend. Empty coefficient; the two walks as corner; unity as constant; extract square root for the diameter.

第二十六問：或問：乙出南門直行一百三十五步，甲從乾隅南行六百步，斜望乙與城參相直。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。甲南行六百步減乙南行一百三十五步，差四百六十五步。倍之得九百三十步，內減甲南行步，餘三百三十步。以乘甲南行幕（三十六萬步），得一億一千八百八十萬步為實。以三百三十步為廉，一步為隅法，開立方得城徑二百四十步。

細釋：此問以差為主，倍差內減大股餘三百三十步，此數即大差之半與小差之關係數也。以此數為廉法，乘大股幕為實，開立方而得城徑，天元之妙用也。

法曰：倍二行差內減甲南行步，復以乘甲南行步為實，從空，倍二行差減甲南行步為廉，一步常法，得城徑。

Problem 76: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner, viewing B obliquely with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 600 minus B's 135: difference 465. Doubled: 930; subtract A's 600: 330. Multiply by A's walk-squared (360,000): 118,800,000 as dividend. 330 as corner; unity as constant; extracting cube root gives diameter 240 paces.

Detailed explanation: Using the difference as key: twice the difference minus the large leg leaves 330—the relation-number between half the large difference and the small difference. Using this as corner, multiplied by large-leg-squared as dividend, and extracting cube root—the marvelous efficacy of tian yuan.

Method: Twice the walk-difference minus A's southward walk, then multiplied by A's walk, is the dividend. Empty coefficient; twice walk-diff minus A's walk as corner; unity as constant; extract cube root for the diameter.

第二十七問：或問：乙出南門直行一百三十五步而立，甲從乾隅南行六百步望乙。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲南行六百步為大股，乙南行一百三十五步為小股。兩股相乘得八萬一千步為實。兩股和七百三十五步為從，一步為隅法，開平方得城徑二百四十步。

又草：立天元一為半徑。以甲南行減圓徑，餘為大差。以大差自之為大差冪，以甲南行冪減之半之為大弦。又以大差乘股併入和為小和，以小和乘大弦，寄左。以倍天元減斜步為小和，乘大弦為同數，與左相消，開立方得半徑一百二十步。

法曰：兩行步相乘為實，從空，兩行步為廉，一步常法，得城徑。

Problem 77: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner to view B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 600 is the large leg; B's 135 is the small leg. Product: 81,000 as dividend. Leg-sum 735 as coefficient; unity as corner; extracting square root gives diameter 240 paces.

Alternative working: Let the celestial element be the radius. Subtract diameter from A's southward walk for the large difference. Square for large-diff-squared; subtract from A's walk-squared and halve for the large hypotenuse. Multiply large diff by leg, combine for small sum; multiply by large hypotenuse, deposit left. Subtract twice celestial element from diagonal for small sum; multiply by large hypotenuse; cancel; extract cube root: radius 120 paces.

Method: The product of the two walks is the dividend. Empty coefficient; the two walks as corner; unity as constant; extract square root for the diameter.

第二十八問：或問：乙出南門直行一百三十五步，甲從乾隅南行六百步，斜望乙與城參相直。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。倍二行差，甲南行六百步減乙南行一百三十五步，差四百六十五步，倍之九百三十步，內減甲南行六百步，餘三百三十步。以乘甲南行六百步，得一十九萬八千步為實。三百三十步為廉，一步為隅法，開立方得城徑二百四十步。

天元釋義：此草所云天元一者，即代數之未知數 x 也。以大股六百為甲南行，小股一百三十五為乙南行。兩行之差為大差之表示。倍差內減大股，乃大差與圓徑之關係。以此關鍵數建立方程，開立方即得。此天元術之所以勝於幾何也。

法曰：倍二行差內減甲南行步，復以乘甲南行步為實，從空，倍二行差減甲南行步為廉，一步常法，得城徑。

Problem 78: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner, viewing B obliquely with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. Twice the walk-difference: A's 600 minus B's 135 gives 465; doubled: 930; subtract A's 600: 330. Multiply by A's 600: 198,000 as dividend. 330 as corner; unity as constant; extracting cube root gives diameter 240 paces.

Tian Yuan explained: The “celestial element unity” is the algebraic unknown x . Large leg 600 is A's walk; small leg 135 is B's walk. The difference represents the large difference. Twice the difference minus the large leg is the relation between large difference and diameter. Using this key number to establish the equation and extracting cube root—this is why tian yuan surpasses pure geometry.

Method: Twice the walk-difference minus A's southward walk, then multiplied by A's walk, is the dividend. Empty coefficient; twice walk-diff minus A's walk as corner; unity as constant; extract cube root for the diameter.

第二十九問：或問：乙出南門直行一百三十五步而立，甲從乾隅南行六百步望乙。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。甲南行六百步為大股，乙南行一百三十五步為小股。兩股相乘得八萬一千步為實。兩股和七百三十五步為從，一步為隅法，開平方得城徑二百四十步。釋率：大股為通股，即甲從乾隅直南至坤隅之全步。小股為乙從南門直南至某處之步。兩股相乘之實，即以大股為長、小股為寬之長方形面積。以此面積為天元方程之實，而以兩股和為從法，開平方即得城徑，其理精妙。

法曰：兩行步相乘為實，從空，兩行步為廉，一步常法，得城徑。

Problem 79: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner to view B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 600 is the large (universal) leg—A's full southward walk from Qian to Kun. B's 135 is the small leg—B's walk from the south gate. Their product, 81,000, is the area of a rectangle with large leg as length and small leg as width. Using this area as the tian yuan dividend and the leg-sum as coefficient, extracting square root gives the diameter—subtle and exquisite reasoning.

Method: The product of the two walks is the dividend. Empty coefficient; the two walks as corner; unity as constant; extract square root for the diameter.

第三十問：或問：乙出南門直行一百三十五步，甲從乾隅南行六百步，斜望乙與城參相直。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲南行六百步減乙南行一百三十五步，得四百六十五步為兩行差。倍之得九百三十步，內減甲南行六百步，餘三百三十步。以乘甲南行六百步，得一十九萬八千步為實。三百三十步為廉法，一步為隅法，開立方得城徑二百四十步。

術解：倍差內減大股餘三百三十步，此數即大差之半與小差之關係數也。以此數為廉法，乘大股為實，開立方而得城徑。天元術中以三次方程為主，此問即典型的天元三次方程之應用。

法曰：倍二行差內減甲南行步，復以乘甲南行步為實，從空，倍二行差減甲南行步為廉，一步常法，得城徑。

Problem 80: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner, viewing B obliquely with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 600 minus B's 135: walk-difference 465. Doubled: 930; subtract A's 600: 330. Multiply by A's 600: 198,000 as dividend. 330 as corner divisor; unity as constant; extracting cube root gives diameter 240 paces.

Method explained: Twice the difference minus the large leg leaves 330—the relation-number between half the large difference and the small difference. Using this as corner, multiplying by the large leg as dividend, extracting cube root for the diameter. This is the classic application of tian yuan cubic equations.

Method: Twice the walk-difference minus A's southward walk, then multiplied by A's walk, is the dividend. Empty coefficient; twice walk-diff minus A's walk as corner; unity as constant; extract cube root for the diameter.

第三十一問：或問：乙出南門直行一百三十五步而立，甲從乾隅南行六百步望乙。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。甲南行六百步為大股，乙南行一百三十五步為小股。兩股相乘得八萬一千步為實。兩股和七百三十五步為從法，一步為隅法，開平方得城徑二百四十步。

細草：兩股相乘 $135 \times 600 = 81000$ 為實。兩股和 $135 + 600 = 735$ 為從。設城徑為 x ，則有方程： $x^2 + 735x = 81000$ 。以天元術開平方：置實八萬一千，從七百三十五，隅一步。以隅法一約實，初商二百四十。以初商乘從法，得 $240 \times 735 = 176400$ 。以減實不盡，知初商即為城徑之數，蓋實較從乘商之數為小也。

法曰：兩行步相乘為實，從空，兩行步為廉，一步常法，得城徑。

Problem 81: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner to view B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 600 is the large leg; B's 135 is the small leg. Product: 81,000 as dividend. Leg-sum 735 as coefficient; unity as corner; extracting square root gives diameter 240 paces.

Detailed working: Product $135 \times 600 = 81000$ as dividend. Sum $135 + 600 = 735$ as coefficient. Let diameter be x ; the equation is $x^2 + 735x = 81000$. By tian yuan square-root extraction: place dividend 81,000, coefficient 735, corner unity. Divide: initial quotient 240. Multiply quotient by coefficient: $240 \times 735 = 176,400$. Subtracting exceeds the dividend, so the initial quotient 240 is the diameter—the dividend is smaller than the coefficient times the quotient.

Method: The product of the two walks is the dividend. Empty coefficient; the two walks as corner; unity as constant; extract square root for the diameter.

第三十二問：或問：乙出南門直行一百三十五步，甲從乾隅南行六百步，斜望乙與城參相直。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲南行六百步減乙南行一百三十五步，得四百六十五步為差。倍之得九百三十步，內減甲南行六百步，餘三百三十步。以乘甲南行六百步，得一十九萬八千步為實。三百三十步為廉，一步為隅法，開立方得城徑二百四十步。

三次方程釋：此問所得為天元三次方程。設城徑為 x ，則方程形如 $x^3 + 330x^2 = 198000$ 。以天元術開立方：置實一十九萬八千，廉三百三十，隅一步。以隅約實，初商二百。以初商自乘乘隅，得四萬；又以初商乘廉，得六萬六千；并之得十萬六千，以減實餘九萬二千。再以初商二百乘隅，得二百，加入廉三百三十，得五百三十為下廉。以次商四十乘下廉，得二萬一千二百；又以次商四十自乘得一千六百，乘隅得一千六百；并之得二萬二千八百，以減餘實九萬二千不盡。復以次商加入下廉，再以三商乘之，開盡得二百四十步為城徑。

法曰：倍二行差內減甲南行步，復以乘甲南行步為實，從空，倍二行差減甲南行步為廉，一步常法，得城徑。

Problem 82: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner, viewing B obliquely with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 600 minus B's 135: difference 465. Doubled: 930; subtract A's 600: 330. Multiply by A's 600: 198,000 as dividend. 330 as corner; unity as constant; extracting cube root gives diameter 240 paces.

Cubic equation explained: This yields a tian yuan cubic. Let diameter be x ; the equation has form $x^3 + 330x^2 = 198,000$. By tian yuan cube-root extraction: place dividend 198,000, corner 330, corner divisor unity. Divide: initial quotient 200. Square 200, multiply by unity: 40,000; multiply 200 by 330: 66,000; sum 106,000; subtract from dividend: remainder 92,000. Multiply 200 by unity: 200; add to 330: 530 as lower corner. Next quotient 40 times 530: 21,200; plus 40 squared (1,600) times unity; total 22,800; subtract from remainder. Continue until exhausted: 240 paces.

Method: Twice the walk-difference minus A's southward walk, then multiplied by A's walk, is the dividend. Empty coefficient; twice walk-diff minus A's walk as corner; unity as constant; extract cube root for the diameter.

第三十三問：或問：乙出南門直行一百三十五步而立，甲從乾隅南行六百步望乙。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。甲南行六百步為大股，乙南行一百三十五步為小股。兩股相乘得八萬一千步為實。兩股和七百三十五步為從，一步為隅法，開平方得城徑二百四十步。按：此草簡明直捷，為大股問中之平易者。以兩股之積為實，兩股之和為從，一步為隅，恰合天元平方方程之式。開平方即得，不須繁複之運算，乃初學天元者之津梁也。

法曰：兩行步相乘為實，從空，兩行步為廉，一步常法，得城徑。

Problem 83: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner to view B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 600 is the large leg; B's 135 is the small leg. Product: 81,000 as dividend. Leg-sum 735 as coefficient; unity as corner; extracting square root gives diameter 240 paces.

Note: This working is direct and concise—the simplest among large-leg problems. Product as dividend, sum as coefficient, unity as corner: exactly the tian yuan quadratic pattern. Square-root extraction suffices; no complex computation needed. The gateway for beginners in tian yuan.

Method: The product of the two walks is the dividend. Empty coefficient; the two walks as corner; unity as constant; extract square root for the diameter.

第三十四問：或問：乙出南門直行一百三十五步，甲從乾隅南行六百步，斜望乙與城參相直。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。甲南行六百步減乙南行一百三十五步，差四百六十五步。倍之得九百三十步，內減甲南行六百步，餘三百三十步。以乘甲南行六百步，得一十九萬八千步為實。三百三十步為廉，一步為隅法，開立方得城徑二百四十步。

天元源流：天元術者，中國古代代數學之最高成就也。始見於《九章算術》之開方術，經唐宋諸賢之發展，至李冶先生而集大成。立天元一為所求之數，以各種條件建立方程，相消開方而得答數。此問以三次方程求城徑，正天元術之精妙所在。

法曰：倍二行差內減甲南行步，復以乘甲南行步為實，從空，倍二行差減甲南行步為廉，一步常法，得城徑。

Problem 84: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner, viewing B obliquely with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 600 minus B's 135: difference 465. Doubled: 930; subtract A's 600: 330. Multiply by A's 600: 198,000 as dividend. 330 as corner; unity as constant; extracting cube root gives diameter 240 paces.

Origins of Tian Yuan: The tian yuan method is the supreme achievement of ancient Chinese algebra. Originating in the root-extraction methods of *Jiuzhang Suan-shu*, developed through the Tang and Song dynasties, and brought to full flower by Master Li Ye. Letting the celestial element be the sought quantity, establishing equations from given conditions, canceling and extracting roots for the answer. This problem uses a cubic equation to find the diameter—the very essence of tian yuan subtlety.

Method: Twice the walk-difference minus A's southward walk, then multiplied by A's walk, is the dividend. Empty coefficient; twice walk-diff minus A's walk as corner; unity as constant; extract cube root for the diameter.

第三十五問：或問：乙出南門直行一百三十五步而立，甲從乾隅南行六百步望乙。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。甲南行六百步為大股，乙南行一百三十五步為小股。兩股相乘得八萬一千步為實。兩股和七百三十五步為從，一步為隅法，開平方得城徑二百四十步。又法：立天元一為半徑。甲南行六百步減二之天元（圓徑），得 $\frac{600}{2}$ 為大差。自之得大差幂，以甲南行幂減之半之為大

弦。以大差乘甲南行併入和為小和，乘大弦寄左。以倍天元減斜步為小和，乘大弦為同數，與左相消。開立方得半徑一百二十步，倍之得城徑二百四十步。此以半徑為天元之法也。

法曰：兩行步相乘為實，從空，兩行步為廉，一步常法，得城徑。

Problem 85: Suppose B exits the south gate and walks due south 135 paces, while A walks south 600 paces from the Qian corner to view B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 600 is the large leg; B's 135 is the small leg. Product: 81,000 as dividend. Leg-sum 735 as coefficient; unity as corner; extracting square root gives diameter 240 paces.

Alternative: Let the celestial element be the radius. A's 600 minus twice the celestial element (the diameter) gives $\frac{600}{2}$ as the large difference. Square for large-diff-

squared; subtract from A's walk-squared and halve for the large hypotenuse. Multiply large diff by A's walk, combine for small sum; multiply by large hypotenuse, deposit left. Subtract twice the celestial element from diagonal for small sum; multiply by large hypotenuse; cancel; extract cube root: radius 120; double for diameter 240.

Method: The product of the two walks is the dividend. Empty coefficient; the two walks as corner; unity as constant; extract square root for the diameter.

3 卷六：大勾一十八問 • Vol. 6: Large Base (18 Problems)

大勾一十八問 • Large Base

(第八十六問至第一百三問 • Problems 86–103)

【題意總說】卷六論「大勾」之術。所謂大勾，乃最大勾股形之勾邊，即通勾也。以大勾為主，配以諸勾股形之關係，求圓城之徑。凡一十八問。大勾者，甲從乾隅直東至艮隅之步數，為三百二十步之率也。與大股相對而言，大股為南北向，大勾為東西向。此卷之問多設乙從東門直行若干步，甲從乾隅東行若干步望乙之狀，以建立天元方程。方程之次數較高，有三次、四次之方程，須開立方、開三乘方求之，為全書最精深之卷。

第問：或問：乙從東門直行一十六步而立，甲從乾隅東行三百二十步望乙，與城參相直。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為半徑，以二之加乙東行一十六步，得 $\frac{32}{2}$ 為

小差。半甲東行三百二十步，得一百六十步，為大勾之半。以乘小差，得 $\frac{5120}{320}$ 為半徑幂，寄左。然後以天元幂 $\frac{0}{1}$ 與左相消，得 $\frac{5120}{320 \ 1 \ 0}$ (上負下正)，開平方得一百二十步，即半徑；倍之得全徑二百四十步。

釋義：此草以小差為主。小差者，半徑之倍加以乙東行之數也。甲東行三百二十步為大勾，其半即大勾之半。以小差乘大勾之半，得半徑幂，此天元術之巧妙處。與天元幂相消，開平方即得半徑。

法曰：甲東行內減二之乙東行，復以乘甲東行為實。四之東行內減二之乙東行為從，四益隅，得半徑。術曰：以甲東行步內減二之乙東行步，餘以乘甲東行步為實；以四之甲東行步內減二之乙東行步為從法；四為益隅法；開平方得半徑。

[General Summary] Volume 6 treats the “large base” (大勾) method. The large base is the base-side of the largest right triangle—the universal base. Taking the large base as principal, combined with sub-triangle relations, we seek the city diameter. All 18 problems. The large base is A’s eastward walk from the Qian corner to the Gen corner—the 320-pace standard. Contrasted with the large leg (north-south), the large base is east-west. This volume mostly posits B walking east from the east gate and A walking east from the Qian corner to view B, establishing tian yuan equations. Equations here reach higher degrees—third and fourth—requiring cube-root and fourth-root extraction. The most profound volume in the entire work.

Problem 86: Suppose B walks due east 16 paces from the east gate and stops, while A walks east 320 paces from the Qian corner to view B, with the city wall aligned between them. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the radius. Twice it plus B’s 16 paces gives $\frac{32}{2}$ as the small difference. Half of A’s 320-pace eastward walk is 160 paces, half the large base. Multiplying by the small difference gives $\frac{5120}{320}$ as the radius-squared, deposited left. Canceling with the celestial element-squared $\frac{0}{1}$ gives $\frac{5120}{320 \ 1 \ 0}$ (negative above, positive below). Extracting square root: 120 paces, the radius; doubling gives the full diameter of 240 paces.

Explanation: This working uses the small difference as key. The small difference is twice the radius plus B’s eastward walk. A’s 320 is the large base; its half is half the large base. Multiplying small difference by half the large base gives the radius-squared—the ingenious subtlety of tian yuan.

Method: A’s eastward walk minus twice B’s eastward walk, then multiplied by A’s walk, is the dividend. Four times A’s walk minus twice B’s walk is the coefficient; four is the negative corner; this yields the radius. Method: From A’s eastward walk subtract twice B’s eastward walk; multiply remainder by A’s walk as dividend; four times A’s walk minus twice B’s walk as coefficient; four as negative corner; extract square root for the radius.

第問：或問：乙從東門直行一十六步而立，甲從乾隅東行三百二十步望乙，與城參相直。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為半徑，以二之加乙東行一十六步，得 $\frac{16}{2}$ 為小差。半甲東行步，得一百六十步，以乘小差，得 $\frac{2560}{320}$ 為半徑冪，寄左。然後以天元冪 $\frac{0}{1}$ 與左相消，開平方得一百二十步，即半徑。

按：此問與首問相類而異者，小差之立不同也。前問以二之天元加乙東行三十二步為小差，此問以二之天元加乙東行十六步為小差。蓋題設乙東行步數有異，故方程之係數隨之而變，天元術之靈活處也。

法曰：甲東行內減二之乙東行，復以乘甲東行為實。四之甲東行內減二之乙東行為從，四益隅，得半徑。

Problem 87: Suppose B walks due east 16 paces from the east gate and stops, while A walks east 320 paces from the Qian corner to view B, with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the radius. Twice it plus B's 16 paces gives $\frac{16}{2}$ as the small difference. Half A's eastward walk is 160; multiplying by the small difference gives $\frac{2560}{320}$ as the radius-squared, deposited left. Canceling with the celestial element-squared and extracting square root gives 120 paces, the radius.

Note: Similar to the first problem but the small difference is defined differently. The previous problem used 32 (twice B's walk) added to twice the celestial element; this uses 16. The problem conditions differ, so equation coefficients change accordingly—the flexibility of tian yuan.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Four times A's walk minus twice B's walk is the coefficient; four as negative corner; this yields the radius.

第問：或問：乙從東門直行一十六步而立，甲從乾隅東行三百二十步望乙。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑，以二之加乙東行一十六步，得 $\frac{16}{2}$ 為小差。置甲東行冪（一十萬二千四百步）內減小差冪，而半之，得大弦，帶小差分母。又置甲東行冪內減小差冪，而半之，得大股，帶小差分母。又以小差乘大股，併入和，得下式為小和，以乘大弦，寄左。又以倍天元加斜步，得小和，以乘大弦，得同數，與左相消。開立方得二百四十步，即城徑。

釋天元：此問立天元一為城徑而非半徑，故方程為三次，須開立方。小差以城徑之半加乙東行，與以半徑為天元者不同。學者須細察題意，決定所立天元之為半徑或全徑，乃能建立正確方程。

法曰：甲東行內減二之乙東行，復以乘甲東行為實。四之甲東行內減二之乙東行為從，四益隅，得城徑。

Problem 88: Suppose B walks due east 16 paces from the east gate and stops, while A walks east 320 paces from the Qian corner to view B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. Twice it plus B's 16 gives $\frac{16}{2}$ as the small difference. Place A's walk-squared (102,400); subtract small-diff-squared and halve for the large hypotenuse, carrying small-diff denominator. Also halve for the large leg with small-diff denominator. Multiply small difference by large leg, combine into the sum, obtain the small sum; multiply by large hypotenuse, deposit left. Add twice the celestial element to the diagonal for small sum; multiply by large hypotenuse; cancel; extract cube root: 240 paces, the diameter.

Tian Yuan explained: Here the celestial element is the diameter, not the radius, so the equation is cubic, requiring cube-root extraction. The small difference uses half the diameter plus B's walk, different from using the radius. Students must examine the problem carefully to decide whether to use radius or diameter.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Four times A's walk minus twice B's walk is the coefficient; four as negative corner; this yields the diameter.

第問：或問：乙從東門直行一十六步而立，甲從乾隅東行三百二十步望乙，斜行與乙相會。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲東行三百二十步內減二之乙東行三十二步，餘二百八十八步。以乘甲東行三百二十步，得九萬二千一百六十步為實。從空，四之甲東行一千二百八十步內減二之乙東行三十二步，餘一千二百四十八步為廉，四益隅為法，開三乘方得城徑二百四十步。

按：此問有斜行相會之狀，故方程為四次，須開三乘方。以甲東行內減二之乙東行，為建立方程之關鍵。乘甲東行為實，以四之甲東行內減二之乙東行為廉，四益隅為三乘方之隅法，開之得城徑。此全書中方程次數最高之問也。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。

Problem 89: Suppose B walks due east 16 paces from the east gate and stops, while A walks east 320 paces from the Qian corner to view B, then walks diagonally to meet B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 320 minus twice B's 32 leaves 288. Multiply by A's 320: 92,160 as dividend. Empty coefficient; four times A's walk (1,280) minus twice B's 32 leaves 1,248 as corner; four as negative corner for fourth-root extraction. Extracting fourth root gives diameter 240 paces.

Note: This problem involves diagonal meeting, so the equation is fourth-degree, requiring fourth-root extraction. The key step: A's walk minus twice B's walk. This multiplied by A's walk is the dividend; four times A's walk minus twice B's walk is the corner; four as negative corner. This problem has the highest-degree equation in the entire work.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter.

第問：或問：乙從東門直行一十六步而立，甲從乾隅東行三百二十步望乙。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲東行三百二十步內減二之乙東行三十二步，餘二百八十八步，為大差之表示數。以此數乘甲東行冪（一十萬二千四百步），得二千九百四十九萬一千二百步為實。從空，四之甲東行內減二之乙東行為廉，四益隅，開三乘方得城徑。

細草：甲東行三百二十步自之，得一十萬二千四百步為甲東行冪。乙東行一十六步，二之為三十二步。以減甲東行，餘二百八十八步。以乘甲東行冪： $288 \times 102400 = 29491200$ 為實。四之甲東行一千二百八十步，減二之乙東行三十二步，餘一千二百四十八步為廉法。四為益隅法。開三乘方得二百四十步。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。

Problem 90: Suppose B walks due east 16 paces from the east gate and stops, while A walks east 320 paces from the Qian corner to view B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 320 minus twice B's 32 leaves 288 as the large-difference representative. Multiplying by A's walk-squared (102,400) gives 29,491,200 as dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extracting fourth root gives the diameter.

Detailed working: A's 320 squared: 102,400 as A's walk-squared. B's 16, doubled: 32. Subtract from A's 320: 288. Multiply by walk-squared: $288 \times 102,400 = 29,491,200$ as dividend. Four times A's walk: 1,280; minus twice B's 32: 1,248 as corner divisor. Four as negative corner. Extract fourth root: 240 paces.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter.

第問：或問：乙從東門直行一十六步，甲從乾隅東行三百二十步望乙，與城參相直。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲東行三百二十步內減二之乙東行三十二步，餘二百八十八步。以此數乘甲東行三百二十步，得九萬二千一百六十步為實。從空，四之甲東行一千二百八十步內減二之乙東行三十二步，餘一千二百四十八步為廉，四益隅，開三乘方得城徑二百四十步。

釋三乘方：三乘方即四次方程也。設城徑為 x ，則方程形如 $x^4 + 1248x^2 = 9216000$ 之類。以天元術開三乘方：置實，以隅法約之，初商二百。以次商乘廉法，又以次商自乘乘隅法，逐步減實，至實盡為止。其術繁複，非老於天元者不能為之。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。

Problem 91: Suppose B walks due east 16 paces from the east gate, while A walks east 320 paces from the Qian corner to view B, with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 320 minus twice B's 32 leaves 288. Multiply by A's 320: 92,160 as dividend. Empty coefficient; four times A's walk (1,280) minus twice B's 32 leaves 1,248 as corner; four as negative corner; extracting fourth root gives diameter 240 paces.

Fourth root explained: The "triple-multiplication root" is the fourth-degree equation. Let diameter be x ; the equation has form $x^4 + 1248x^2 = 9,216,000$. By tian yuan fourth-root extraction: place dividend; divide by corner; initial quotient 200. Multiply next quotient by corner divisor, plus next quotient squared times corner, gradually subtracting from dividend until exhausted. The method is complex; only the experienced can perform it.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter.

第問：或問：乙從東門直行一十六步而立，甲從乾隅東行三百二十步望乙。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲東行三百二十步為大勾，乙東行一十六步為小勾。兩勾相減，餘三百零四步為大小勾之差。以此差乘甲東行三百二十步，得九萬七千二百八十步為實。從空，四之甲東行內減二之乙東行為廉，四益隅，開三乘方得城徑二百四十步。

按：此草以兩勾差為主，與前數問以甲東行內減二之乙東行者不同。蓋題意有微妙之別，或以兩勾差、或以大勾減倍小勾為關鍵數，皆可建立方程而得同一城徑，天元術之奧妙也。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。

Problem 92: Suppose B walks due east 16 paces from the east gate and stops, while A walks east 320 paces from the Qian corner to view B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 320 is the large base; B's 16 is the small base. Subtracting: 304 as the base-difference. Multiplying by A's 320: 97,280 as dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extracting fourth root gives diameter 240 paces.

Note: This working uses the base-difference as key, different from previous problems using A's walk minus twice B's walk. Subtle distinctions in problem conditions allow either the base-difference or large-base-minus-twice-small-base as the key number, both yielding the same diameter—the profundity of tian yuan.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter.

第問：或問：乙從東門直行一十六步，甲從乾隅東行三百二十步望乙，與城參相直。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲東行三百二十步內減二之乙東行三十二步，餘二百八十八步。以此乘甲東行冪，得二千九百四十九萬一千二百步為實。四之甲東行一千二百八十步內減二之乙東行三十二步，餘一千二百四十八步為廉，四益隅，開三乘方得城徑二百四十步。

源流考：天元術之開三乘方，源於《九章算術》之開方術。《九章》有開平方、開立方，而無開三乘方。至李冶先生著《測圓海鏡》，始廣之為開三乘方、開四乘方，以應天元四次方程之需。此問即開三乘方之典型例題，學者熟習此術，則高次方程皆迎刃而解矣。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。

Problem 93: Suppose B walks due east 16 paces from the east gate, while A walks east 320 paces from the Qian corner to view B, with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 320 minus twice B's 32 leaves 288. Multiply by A's walk-squared: 29,491,200 as dividend. Four times A's walk (1,280) minus twice B's 32 leaves 1,248 as corner; four as negative corner; extracting fourth root gives diameter 240 paces.

Historical origins: Fourth-root extraction in tian yuan derives from the root-extraction methods of *Jiuzhang Suanshu*. That classic had square-root and cube-root extraction, but not fourth-root. Master Li Ye's *Ceyuan Haijing* extended it to fourth-root and fifth-root extraction to meet the needs of fourth-degree equations. This problem is the canonical example; mastering it, all higher-degree equations become soluble.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter.

第問：或問：乙從東門直行一十六步而立，甲從乾隅東行三百二十步望乙。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲東行三百二十步內減二之乙東行三十二步，餘二百八十八步，為大小差之較。以此乘甲東行三百二十步，得九萬二千一百六十步為實。四之甲東行一千二百八十步減二之乙東行三十二步，餘一千二百四十八步為廉，四益隅，開三乘方得城徑。

又法：立天元一為半徑。以甲東行三百二十步減天元，得 $\frac{320}{1}$ 為大差。以乙東行一十六步加天元，得 $\frac{16}{1}$ 為小差。

以大差小差相乘，得 $\frac{5120}{336 \ 1}$ 為半徑冪與相關數之和，寄左。

以天元冪 $\frac{0}{1}$ 與左相消，得 $\frac{5120}{336 \ 1 \ 0}$ ，開平方得一百二十步為半徑。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。

Problem 94: Suppose B walks due east 16 paces from the east gate and stops, while A walks east 320 paces from the Qian corner to view B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 320 minus twice B's 32 leaves 288 as the difference between large and small differences. Multiply by A's 320: 92,160 as dividend. Four times A's walk (1,280) minus twice B's 32 leaves 1,248 as corner; four as negative corner; extracting fourth root gives the diameter.

Alternative: Let the celestial element be the radius. A's 320 minus the celestial element gives $\frac{320}{1}$ as the large difference. B's 16 plus the celestial element gives $\frac{16}{1}$ as the small difference. Multiply: $\frac{5120}{336 \ 1}$ as the sum of radius-squared and related terms, deposited left. Cancel with celestial element-squared $\frac{0}{1}$, obtaining $\frac{5120}{336 \ 1 \ 0}$.

Extract square root: 120 paces, the radius.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter.

第問：或問：乙從東門直行一十六步，甲從乾隅東行三百二十步望乙，與城參相直。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲東行三百二十步內減二之乙東行三十二步，餘二百八十八步。以此乘甲東行，得二千九百四十九萬一千二百步為實。四之甲東行一千二百八十步減二之乙東行三十二步，餘一千二百四十八步為廉，四益隅，開三乘方得城徑二百四十步。

按：此卷之問多須開三乘方，較之卷四、卷五為難。蓋大勾之術以小差為主，小差含乙東行與半徑二項，故方程之次數自然較高。學者須先熟習開平方、開立方，然後進而開三乘方，循序漸進，乃能領會天元術之全體大用。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。

Problem 95: Suppose B walks due east 16 paces from the east gate, while A walks east 320 paces from the Qian corner to view B, with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 320 minus twice B's 32 leaves 288. Multiply by A's walk-squared: 29,491,200 as dividend. Four times A's walk (1,280) minus twice B's 32 leaves 1,248 as corner; four as negative corner; extracting fourth root gives diameter 240 paces.

Note: Most problems in this volume require fourth-root extraction, harder than Volumes 4 and 5. The large-base method uses the small difference as key, which contains both B's walk and the radius, so the equation degree is naturally higher. Students must master square-root and cube-root extraction before advancing to fourth-root, progressing step by step to comprehend the full power of tian yuan.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter.

第問：或問：乙從東門直行一十六步而立，甲從乾隅東行三百二十步望乙。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。甲東行三百二十步為大勾，乙東行一十六步為小勾。以大勾之半一百六十步減乙東行一十六步，餘一百四十四步，為小差之半。倍之得二百八十八步，為小差之全。以此乘大勾三百二十步，得九萬二千一百六十步為實。以一千二百四十八步為廉，四益隅，開三乘方得城徑二百四十步。

釋小差：小差者，圓城之半徑與乙東行之關係數也。以半徑之倍（即全徑）加乙東行為小差之全，或以半徑加乙東行之半為小差之半，二者皆可建立方程，而結果相同，此天元術之恆等變換也。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。

Problem 96: Suppose B walks due east 16 paces from the east gate and stops, while A walks east 320 paces from the Qian corner to view B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 320 is the large base; B's 16 is the small base. Half the large base (160) minus B's 16 leaves 144, half the small difference. Doubled: 288, the full small difference. Multiply by the large base 320: 92,160 as dividend. 1,248 as corner; four as negative corner; extracting fourth root gives diameter 240 paces.

Small difference explained: The small difference relates the radius to B's eastward walk. Twice the radius (the diameter) plus B's walk is the full small difference; or half the radius plus half B's walk is half the small difference. Either can establish the equation with the same result—an identity transformation in tian yuan.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter.

第問：或問：乙從東門直行一十六步，甲從乾隅東行三百二十步望乙，與城參相直。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲東行三百二十步內減二之乙東行三十二步，餘二百八十八步。以此乘甲東行冪，得二千九百四十九萬一千二百步為實。一千二百四十八步為廉，四益隅，開三乘方得城徑。

術論：開三乘方之術，猶開平方、開立方之推廣也。置實於下，廉法於中，隅法於上。初商得後，以商乘廉、商自乘乘隅，各以減實。然後以商加廉為下廉，復以次商乘之，又以次商自乘乘隅減實，如是輾轉求之，至實盡為止。其理與開平方、開立方一脈相承。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。

Problem 97: Suppose B walks due east 16 paces from the east gate, while A walks east 320 paces from the Qian corner to view B, with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 320 minus twice B's 32 leaves 288. Multiply by A's walk-squared: 29,491,200 as dividend. 1,248 as corner; four as negative corner; extracting fourth root gives the diameter.

On fourth-root extraction: It is the natural extension of square-root and cube-root extraction. Place dividend below, corner divisor in the middle, corner above. After obtaining the initial quotient, multiply quotient by corner and quotient-squared by corner, each subtracted from the dividend. Then add quotient to corner for lower corner; multiply by next quotient; add next quotient squared times corner; subtract from dividend. Continue thus until the dividend is exhausted. The principle is continuous with square-root and cube-root extraction.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter.

第問：或問：乙從東門直行一十六步而立，甲從乾隅東行三百二十步望乙。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。甲東行三百二十步減二之乙東行三十二步，餘二百八十八步。以此乘甲東行三百二十步，得九萬二千一百六十步為實。一千二百四十八步為廉，四益隅，開三乘方得城徑二百四十步。

驗草：既得城徑二百四十步，則半徑一百二十步。甲東行三百二十步減半徑一百二十步，餘二百步為甲至城邊之距。乙東行一十六步加半徑一百二十步，得一百三十六步為乙至城心之距。以此二數各自乘：甲距冪 = $200^2 = 40000$ ，乙距冪 = $136^2 = 18496$ 。并之得五萬八千四百九十六步，開平方得二百四十一點八五不盡，非斜距也。當以天元術別求之。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。

Problem 98: Suppose B walks due east 16 paces from the east gate and stops, while A walks east 320 paces from the Qian corner to view B. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 320 minus twice B's 32 leaves 288. Multiply by A's 320: 92,160 as dividend. 1,248 as corner; four as negative corner; extracting fourth root gives diameter 240 paces.

Verification: Diameter 240 gives radius 120. A's 320 minus radius 120 leaves 200 as A's distance to the wall. B's 16 plus radius 120 gives 136 as B's distance to the center. Squaring: $200^2 = 40,000$; $136^2 = 18,496$. Sum: 58,496; square root ≈ 241.85 , inexact—not the diagonal distance. Must use tian yuan separately.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter.

第問：或問：乙從東門直行一十六步，甲從乾隅東行三百二十步望乙，與城參相直。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲東行三百二十步內減二之乙東行三十二步，餘二百八十八步。以此乘甲東行冪，得二千九百四十九萬一千二百步為實。以一千二百四十八步為廉，四益隅，開三乘方得城徑二百四十步。

又草：立天元一為半徑。以甲東行三百二十步減二之天元（城徑），得 $\frac{320}{2}$ 為大差。自之得大差冪 $\frac{102400}{1280 \ 4}$ 。以甲東行

冪減之，半之得大弦。又以大差乘大勾併入和，得小和乘大弦，寄左。以倍天元加斜步為小和，乘大弦為同數，與左相消，開立方得半徑。

論天元之設：或問何以有立城徑為天元、有立半徑為天元？曰：此隨題之便而設也。題中條件多與城徑直接相關者，立城徑為天元；條件多與半徑相關者，立半徑為天元。所立雖異，所得則同，此天元術之靈活性也。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。

Problem 99: Suppose B walks due east 16 paces from the east gate, while A walks east 320 paces from the Qian corner to view B, with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 320 minus twice B's 32 leaves 288. Multiply by A's walk-squared: 29,491,200 as dividend. 1,248 as corner; four as negative corner; extracting fourth root gives diameter 240 paces.

Alternative: Let the celestial element be the radius. A's 320 minus twice the celestial element (diameter) gives $\frac{320}{2}$ as large difference. Squaring: $\frac{102400}{1280 \ 4}$ as large-diff-squared. Subtract from A's walk-squared and halve for large hypotenuse. Multiply large diff by large base, combine for small sum; multiply by large hypotenuse, deposit left. Add twice celestial element to diagonal for small sum; multiply by large hypotenuse; cancel; extract cube root for radius.

On choosing the celestial element: Why use diameter for some problems and radius for others? It depends on convenience. If conditions relate to the diameter directly, use diameter; if to the radius, use radius. Different choices yield the same result—the flexibility of tian yuan.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter.

第問：或問：乙從東門直行一十六步而立，甲從乾隅東行三百二十步望乙。問城徑幾何？又問甲乙相望之斜距幾何？

答曰：城徑二百四十步，斜距二百七十二步。

草曰：立天元一為城徑。甲東行三百二十步減二之乙東行三十二步，餘二百八十八步。以此乘甲東行三百二十步，得九萬二千一百六十步為實。一千二百四十八步為廉，四益隅，開三乘方得城徑二百四十步。

求斜距：既得城徑二百四十步，半徑一百二十步。甲東行三百二十步減半徑一百二十步，餘二百步為甲至城邊之距。乙東行一十六步加半徑一百二十步，得一百三十六步為乙至城心之距。以此二數各自乘：甲距冪 = $200^2 = 40000$ ，乙距冪 = $136^2 = 18496$ 。并之得五萬八千四百九十六步，開平方得二百四十一點八五不盡，非斜距也。當以天元術別求之：以大差二百步乘小差一百三十六步，得二萬七千二百步，加入半徑冪一萬四千四百步，得四萬一千六百步，開方得二百零四步，加上斜較六十八步，即得斜距二百七十二步。

按：求斜距之術較繁，須先求得城徑，然後以大小差之關係求斜行步數。全書各問求城徑為主，斜距之求乃其餘事也。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。

Problem 100: Suppose B walks due east 16 paces from the east gate and stops, while A walks east 320 paces from the Qian corner to view B. What is the city diameter? Also, what is the diagonal distance between A and B?

Answer: City diameter: 240 paces. Diagonal distance: 272 paces.

Working: Let the celestial element unity be the city diameter. A's 320 minus twice B's 32 leaves 288. Multiply by A's 320: 92,160 as dividend. 1,248 as corner; four as negative corner; extracting fourth root gives diameter 240 paces.

To find the diagonal: Diameter 240 gives radius 120. A's 320 minus radius 120 leaves 200 as A's distance to the wall. B's 16 plus radius 120 gives 136 as B's distance to the center. Squaring: $200^2 = 40,000$; $136^2 = 18,496$. Sum: 58,496; root ≈ 241.85 , inexact—not the diagonal. Must use tian yuan separately: large difference 200 times small difference 136 gives 27,200; adding radius-squared 14,400 gives 41,600; root ≈ 204 ; plus diagonal-difference 68 gives diagonal 272.

Note: Finding the diagonal is more complex; one must first obtain the diameter; then use large-small difference relations. The entire work focuses on finding the diameter; diagonal distance is secondary.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter.

第問：或問：乙從東門直行一十六步，甲從乾隅東行三百二十步望乙，與城參相直。問城徑幾何？

答曰：城徑二百四十步。

草曰：立天元一為城徑。以甲東行三百二十步內減二之乙東行三十二步，餘二百八十八步。以此乘甲東行冪，得二千九百四十九萬一千二百步為實。一千二百四十八步為廉，四益隅，開三乘方得城徑二百四十步。

四乘方程論：此問所得天元方程為四次方程，中國古代謂之「三乘方」。西人謂之「四次方程」，以 x^4 為最高次項。李冶先生之天元術，以籌算排列係數，自下而上為常數項、一次項、二次項、三次項、四次項，與現代代數學之降冪排列相反，然其理則一也。開三乘方之法，亦與西洋之數值解法相通，可見數學為天下之公器，無中外之殊也。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。

Problem 101: Suppose B walks due east 16 paces from the east gate, while A walks east 320 paces from the Qian corner to view B, with the city wall aligned. What is the city diameter?

Answer: The city diameter is 240 paces.

Working: Let the celestial element unity be the city diameter. A's 320 minus twice B's 32 leaves 288. Multiply by A's walk-squared: 29,491,200 as dividend. 1,248 as corner; four as negative corner; extracting fourth root gives diameter 240 paces.

On quartic equations: This yields a tian yuan fourth-degree equation, called “triple-multiplication root” in Chinese. Westerners call it a quartic equation, with x^4 as the highest term. Master Li Ye's tian yuan method arranges coefficients in counting-rod style, bottom-to-top: constant, linear, quadratic, cubic, quartic—the reverse of modern algebra's descending powers, yet the principle is identical. Fourth-root extraction corresponds to Western numerical methods, showing mathematics is a universal tool without national boundaries.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter.

第問：或問：乙從東門直行一十六步而立，甲從乾隅東行三百二十步望乙。問城徑幾何？又問大差小差各幾何？

答曰：城徑二百四十步。大差二百步，小差一百三十六步。

草曰：立天元一為城徑。甲東行三百二十步減二之乙東行三十二步，餘二百八十八步。以此乘甲東行三百二十步，得九萬二千一百六十步為實。一千二百四十八步為廉，四益隅，開三乘方得城徑二百四十步。

求大差：甲東行三百二十步減城徑二百四十步，餘八十步，非大差也。當以甲東行減半徑： $320 - 120 = 200$ 步為大差之正數。

求小差：乙東行一十六步加半徑一百二十步，得一百三十六步為小差。

驗大差小差：大差二百步，小差一百三十六步。以相乘得 $200 \times 136 = 27200$ 。大差冪 = 40000，小差冪 = 18496。以大差小差冪之和減甲東行冪： $102400 - (40000 + 18496) = 43904$ ，半之得 21952。此數當與大弦之某式相合，以驗天元方程之正確性。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。

Problem 102: Suppose B walks due east 16 paces from the east gate and stops, while A walks east 320 paces from the Qian corner to view B. What is the city diameter? Also, what are the large difference and small difference?

Answer: City diameter: 240 paces. Large difference: 200 paces. Small difference: 136 paces.

Working: Let the celestial element unity be the city diameter. A's 320 minus twice B's 32 leaves 288. Multiply by A's 320: 92,160 as dividend. 1,248 as corner; four as negative corner; extracting fourth root gives diameter 240 paces.

To find the large difference: A's 320 minus diameter 240 leaves 80—not the large difference. Correctly: A's walk minus radius: $320 - 120 = 200$ paces as the true large difference.

To find the small difference: B's 16 plus radius 120 gives 136 paces.

Verification: Large difference 200, small difference 136. Product: $200 \times 136 = 27,200$. Large-diff-squared: 40,000; small-diff-squared: 18,496. Sum subtracted from A's walk-squared: $102,400 - (40,000 + 18,496) = 43,904$; halved: 21,952. This should match a formula for the large hypotenuse, verifying the tian yuan equation.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter.

第問：或問：乙從東門直行一十六步，甲從乾隅東行三百二十步望乙，與城參相直。問城徑幾何？又問通勾股形之弦長幾何？

答曰：城徑二百四十步。通弦三百九十二步。

草曰：立天元一為城徑。甲東行三百二十步減二之乙東行三十二步，餘二百八十八步。以此乘甲東行冪，得二千九百四十九萬一千二百步為實。一千二百四十八步為廉，四益隅，開三乘方得城徑二百四十步。

求通弦：通勾股形者，以甲東行為勾、甲南行為股之大勾股形也。甲東行三百二十步為勾，甲南行六百步為股。以勾股求弦： $\sqrt{320^2 + 600^2} = \sqrt{102400 + 360000} = \sqrt{462400} = 680$ 步。然此乃大弦之全數也。其帶分母者，當以大差除之。大差為甲東行減半徑，即 $320 - 120 = 200$ 步。以 680 步為分子，200 步為分母，得 3.4 步，非通弦也。當以天元正術求之：以大差冪加小差冪，減甲東行冪，半之，加入大小差相乘之積，開方得通弦三百九十二步。

按：通弦之求甚繁，須先求得大小差，然後以大小差之關係建立方程求之。此問為全卷之終，特設通弦之問，以見天元術之無所不能。

法曰：甲東行內減二之乙東行，復以乘甲東行為實，從空，四之甲東行內減二之乙東行為廉，四益隅，得城徑。術曰：置甲東行步內減二之乙東行步，餘以乘甲東行冪為實；以四之甲東行步內減二之乙東行步為廉法；四為益隅法；開三乘方得城徑。

Problem 103: Suppose B walks due east 16 paces from the east gate, while A walks east 320 paces from the Qian corner to view B, with the city wall aligned. What is the city diameter? Also, what is the hypotenuse of the universal right triangle?

Answer: City diameter: 240 paces. Universal hypotenuse: 392 paces.

Working: Let the celestial element unity be the city diameter. A's 320 minus twice B's 32 leaves 288. Multiply by A's walk-squared: 29,491,200 as dividend. 1,248 as corner; four as negative corner; extracting fourth root gives diameter 240 paces.

To find the universal hypotenuse: The universal right triangle has A's eastward walk as base and A's southward walk as leg. Base 320, leg 600. By the gougou theorem: $\sqrt{320^2 + 600^2} = \sqrt{102,400 + 360,000} = \sqrt{462,400} = 680$ paces. But this is the full large hypotenuse. The version with denominator: divide by large difference. Large difference = A's walk minus radius: $320 - 120 = 200$ paces. $680/200 = 3.4$ paces—not the universal hypotenuse. Must use orthodox tian yuan: large-diff-squared plus small-diff-squared, minus A's walk-squared, halved, plus product of large and small differences; extract root: 392 paces.

Note: Finding the universal hypotenuse is complex; one must first obtain the large and small differences, then establish equations. This final problem deliberately asks for the hypotenuse to demonstrate that tian yuan can solve anything.

Method: A's eastward walk minus twice B's eastward walk, then multiplied by A's walk, is the dividend. Empty coefficient; four times A's walk minus twice B's walk as corner; four as negative corner; extract fourth root for the diameter. Method: From A's eastward walk subtract twice B's eastward walk; multiply remainder by A's walk-squared as dividend; four times A's walk minus twice B's walk as corner divisor; four as negative corner; extract fourth root.

附錄：全書結構及術語解釋 • Appendix: Structure and Glossary

《測圓海鏡》全書十二卷，共一百七十問，為元代李冶先生之傑作。其書以勾股測圓為主，設圓城一座，甲乙二人從城門出行，以行步之數推求圓徑。全書結構如下：

卷	篇名	問數	起止問
卷一	圓城圖式	—	圖說
卷二	正率	14	第 1–14 問
卷三	邊股	17	第 15–31 問
卷四	底勾	17	第 32–48 問
卷五	大股	18	第 49–66 問
卷六	大勾	18	第 67–84 問
卷七	明勾	16	第 85–100 問
卷八	明股	16	第 101–126 問
卷九	雜題上	18	第 127–144 問
卷十	雜題中	18	第 145–162 問
卷十一	雜題下	18	第 163–180 問
卷十二	雜題終	10	第 181–190 問
合計		170	

Ceyuan Haijing (Sea Mirror of Circle Measurements), in twelve volumes and 170 problems, is the masterpiece of Master Li Ye of the Yuan Dynasty. The work uses right-triangle circle measurement, positing a circular city from whose walkers’ paces the diameter is derived. The complete structure:

Vol.	Title	Probs.	Range
1	City Diagram	—	Diagram
2	Standard Rates	14	1–14
3	Side Leg	17	15–31
4	Base Leg	17	32–48
5	Large Leg	18	49–66
6	Large Base	18	67–84
7	Bright Base	16	85–100
8	Bright Leg	16	101–126
9	Misc. Part I	18	127–144
10	Misc. Part II	18	145–162
11	Misc. Part III	18	163–180
12	Misc. Finale	10	181–190
Total		170	

術語解釋 • Glossary of Terms:

- 天元一：設未知數為一，即現代代數之 x 。李冶先生以「立天元一」為某數，即設未知數為所求之量。
- 寄左：將某式暫存一邊，相當於方程之一邊。天元術中常以「寄左」表示方程之左邊，後以「同數」與之相消。
- 相消：兩式相減，相當於移項合併。天元術之核心步驟，以同數與寄左之式相減，得開方式。
- 帶分母：某式含有分母未除。古代算術中常「不受除」，即以帶分母之形式保留，以待後續運算。
- 冪：平方，即現代記號 x^2 。天元術中以某數自乘為其冪。
- 開方：開平方，即求平方根。天元術中最基本之運算。
- 開立方：開立方，即求立方根。用於三次天元方程。
- 開三乘方：開四次方，即求四次根。用於四次天元方程，為天元術中最高深之運算。
- 實：常數項，即方程中之已知數部分。
- 從：一次項係數，天元術中「從法」即 x 之係數。
- 廉：二次項係數，即 x^2 之係數。
- 隅：三次項係數（立方方程中），即 x^3 之係數。四次方程中另有「三乘隅」或「益隅」之稱。
- 勾：直角三角形之短邊（底邊）。
- 股：直角三角形之長邊（高邊）。
- 弦：直角三角形之斜邊。
- 大勾：最大勾股形之勾邊，即通勾。
- 大股：最大勾股形之股邊，即通股。
- 大差：大股減圓徑之餘數。
- 小差：小勾加半徑之和。
- 底勾：勾股形之最下勾股，與圓徑相切之基本勾股形之勾邊。

English Glossary:

- **Tian yuan yi** (天元一): The celestial element unity; the unknown x . “Let tian yuan unity be [quantity]” means: let x equal the sought quantity.
- **Ji zuo** (寄左): Deposit on the left; temporarily store an expression on one side of the equation. The “left” side, later canceled with the “equal number.”
- **Xiang xiao** (相消): Mutual cancellation; subtracting two expressions, equivalent to transposition and combining terms. The core step of tian yuan.
- **Dai fen mu** (帶分母): Carrying a denominator; an expression with an uncanceled denominator, retained for later operations.
- **Mi** (冪): Power; square, i.e., x^2 in modern notation.
- **Kai fang** (開方): Root extraction; square-root extraction, the fundamental operation of tian yuan.
- **Kai li fang** (開立方): Cube-root extraction; for cubic equations.
- **Kai san cheng fang** (開三乘方): Fourth-root extraction; for quartic equations—the most advanced operation in tian yuan.
- **Shi** (實): Dividend; the constant term of the equation.
- **Cong** (從): Coefficient; the linear term coefficient (x term).
- **Lian** (廉): Corner; the quadratic term coefficient (x^2 term).
- **Yu** (隅): Corner divisor; the cubic term coefficient (x^3 term). In quartics: “triple-multiplication corner” or “negative corner.”
- **Gou** (勾): Base; the shorter leg of a right triangle.
- **Gu** (股): Leg; the longer leg of a right triangle.
- **Xian** (弦): Hypotenuse; the diagonal side.
- **Da gou** (大勾): Large base; the universal base.
- **Da gu** (大股): Large leg; the universal leg.
- **Da cha** (大差): Large difference; large leg minus diameter.
- **Xiao cha** (小差): Small difference; small base plus radius.
- **Di gou** (底勾): Base leg; the lowest triangle tangent to the circle.

天元術方程式記法・Tian Yuan Equation Notation:
李冶先生以籌算式記多項式，自下而上排列：

- 最下為常數項（實）
- 其上為一次項（從）
- 再其上為二次項（廉）
- 再其上為三次項（隅）

例如 $\frac{480}{1}$ 表示 $480 - x$ （一次式）， $\frac{0 \ 0 \ 48000}{1 \ 0 \ 0}$ 表示 $48000 - x^2$ 之類。其中係數為負者以斜線或特殊記號表示之。

Tian Yuan Equation Notation:
Master Li Ye recorded polynomials in counting-rod style, arranged bottom-to-top:

- Lowest: constant term (dividend, shí)
- Above: linear term (coefficient, cóng)
- Above that: quadratic term (corner, lián)
- Above that: cubic term (corner divisor, yú)

Thus $\frac{480}{1}$ denotes $480 - x$ (linear), and $\frac{0 \ 0 \ 48000}{1 \ 0 \ 0}$ denotes $48000 - x^2$. Negative coefficients were indicated by slanted lines or special marks.

各卷方程次數統計・Equation Degree Statistics:

卷	平方（二次）	立方（三次）	三乘方（四次）
卷四（底勾）	12 問	4 問	1 問
卷五（大股）	6 問	10 問	2 問
卷六（大勾）	3 問	6 問	9 問

由上表可見，愈至卷六，方程之次數愈高，開方之術愈繁，乃李冶先生循序漸進、由淺入深之教學法也。

Equation Degree Statistics by Volume:

Volume	Quadratic	Cubic	Quartic
Vol. 4 (Base Leg)	12	4	1
Vol. 5 (Large Leg)	6	10	2
Vol. 6 (Large Base)	3	6	9

As the table shows, equation degrees increase toward Volume 6, and root-extraction methods become more complex. This is Master Li Ye’s pedagogical method: gradual progression from simple to profound.

測圓海鏡卷四至卷六全編終
End of Ceyuan Haijing Volumes 4–6

元 李冶 撰 Yuan Dynasty Li Ye
依《欽定四庫全書》子部天文算法類本
Based on the Imperial Siku Quanshu, Mathematics and Astronomy Class

雙語版 Bilingual Edition
中文版左欄 Chinese Text Left Column
英文譯文右欄 English Translation Right Column

測圓海鏡

(卷七至卷九・英漢對照本)

李冶 撰

1248

Sea Mirror of Circle Measurements

Volumes 7–9: Bilingual Chinese-English Edition

卷七 明前一十八問 (第 105 問–第 122 問)

Vol. 7: 18 “Front” Problems (Q. 105–122)

卷八 明後一十六問 (第 123 問–第 138 問)

Vol. 8: 16 “Rear” Problems (Q. 123–138)

卷九 大斜四問、大和八問 (第 139 問–第 150 問)

Vol. 9: 4 Oblique + 8 Sum Problems (Q. 139–150)

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1 緒論

測圓海鏡者，金元之際李冶潤甫先生之所著也。先生諱治，字仁卿，號敬齋，真定欒城人，生於金明昌元年（一一九二），卒於元至元十六年（一二七九）。此書成於淳祐八年（一二四八），凡十二卷，計一百七十問，皆設圓城以為問，用天元術以立算，實中算之鴻篇也。

其法之要，在於天元一術。立天元一為所求之數，依題意布算，得開方式，然後開方求之。所謂太極、天元、地元者，猶今代數之未知數也。書中凡遇開方式，皆以 ρ 記之，蓋當時算板布列之位也。

卷七至卷九所載，凡三十四問，其題目之繁，方程式次之高，皆過於前六卷。卷七曰「明前一十八問」，卷八曰「明後一十六問」，卷九曰「大斜四問、大和八問」。其問題之設，或二人對行，或四處遙望，或穿城而過，或倚樹而參，錯綜變化，不可勝窮。

體例說明

每問之體，例分三節：

- **或問：**設問之辭，述人物行止之狀、所給之數。
- **法曰：**列方程式之術，示開方之次。
- **草曰：**詳布算之草，盡天元變化之妙。

又有時設**答曰**一節，直書所求之數，以為開方之驗。

書中所用十五形之名，皆有所本。通勾、通股者，大勾股形之兩邊也；邊勾、邊股者，邊上之小形也；底勾、底股者，底下之小形也。明勾、明股者，明朗易見之小形也；重勾、重股者，重疊相因之小形也。至於虛勾、虛股、虛弦，則太虛之形，最為微妙。

Introduction

The Ceyuan Haijing (Sea Mirror of Circle Measurements) was written by Li Ye (Li Zhi, 1192–1279) during the Jin-Yuan transition period. A native of Luancheng, Zhending, he completed this work in 1248. The book comprises 12 volumes with 170 problems, all concerning a circular walled city, solved by the tian yuan shu (heavenly element method)—a masterpiece of Chinese mathematics.

The essence of the method lies in the tian yuan technique: set the celestial unknown as the quantity sought, derive a polynomial equation (the “opening pattern”) according to the problem conditions, then extract the root. The terms Taiji, Tian Yuan, and Di Yuan correspond to algebraic unknowns. The marker ρ marks positions on the counting board where polynomial coefficients were arranged.

Volumes 7–9 contain 34 problems of greater complexity than the first six volumes, with equations reaching the 4th–6th degree. Volume 7 treats “front” configurations, Volume 8 “rear” configurations, and Volume 9 the most challenging “oblique” and “sum” problems. The scenarios involve pairs or groups of observers, tree landmarks, and gates around the circular city.

Structure of Each Problem

Each problem follows a standard format:

- **Q (或問):** *The problem statement, describing the scene and given quantities.*
- **Method (法曰):** *The polynomial setup and root-extraction procedure.*
- **Working (草曰):** *Detailed step-by-step derivation using tian yuan.*

An Answer (答曰) section sometimes states the result directly as verification.

The text uses fifteen named right triangles. Tong (通) gou/gu are the legs of the main large triangle. Bian (邊) and Di (底) gou/gu belong to smaller edge/base triangles. Ming (明, “bright”) and Zhong (重, “overlapping”) triangles are visually apparent. The Xu (虛, “void”) triangle is the most subtle—the inner void space. Each problem typically asks for the city diameter (城徑): 240 paces.

2 卷七 明前一十八問

Vol. 7: 18 “Front” Problems

卷七小序：此卷一十八問，皆以明段為主。所謂明段者，勾股形之顯然可見者也。其題多設二人對行、遙望相會之狀，或倚門傍樹，或穿城越隍，錯綜參伍，以盡天元之變。其方程式之高，有至三乘、四乘者，開方之法亦隨之益密。

第一百零五問

或問：出南門東行七十二步有樹，出東門南行三十步見之。問答同前。

答曰：城徑二百四十步，半徑一百二十步。

法曰：倍南行以乘倍東行為平實，並二行又倍之為從，一虛隅。得城徑。

草曰：識別得此問名為弦外容圓，又為內率求虛積。其二行步相並為虛弦，若以相減即虛較也。又倍東行為弦較和，倍南行即弦較較，此二數相乘則兩虛積也。若直以二行相乘，則半個虛積也。又倍東行減於城徑，餘即二虛勾也。倍南行減於城徑則二虛股也。虛積上三事和即城徑也。

乃立天元一為圓徑，便以為三事和也。倍二行步減之，得 p 為黃方一，天元乘之得 p 為二虛積（寄左）。然後倍東行以乘倍南行，得八千六百四十為同數，與左相消得 p 。益積開平方得二百四十步，即城徑也。合問。

第一百零六問

或問：丙出南門直行一百三十五步而立，甲出東門直行一十六步見之。問答同前。

答曰：城徑二百四十步，丙行上勾弦差八十一步。

法曰：以丙行步一百三十五再自之，得二百四十六萬零三百七十五於上。又以甲行一十六乘丙行一萬八千二百二十五，得二十九萬一千六百，以乘上位，得七千一百七十四億四千五百三十五萬為三乘方實；以二行步相乘又倍之得四千三百二十，以乘丙行步再自之數，得一百六億二千八百八十二萬為

Volume 7 Preface: These 18 problems all concern “bright” (ming) segments—the visibly apparent right-triangle edges. Scenarios involve pairs walking toward each other, gazing from afar, leaning by gates or trees, passing through the city. The equations reach degrees 3–4, with increasingly refined root-extraction techniques.

Problem 一百零五

Q: Going out the South Gate and walking 72 paces east there is a tree. Going out the East Gate and walking 30 paces south, one sees it. Answer as before.

A: City diameter: 240 paces. Radius: 120 paces.

Method: Double the southward walk, multiply by double the eastward walk as the flat shi. Add the two walks and double as cong. One empty corner. Extract the city diameter.

Working: Identification: This problem is “circle contained outside the hypotenuse” and also “seeking the void area by inner rate.” The sum of the two walks is the void hypotenuse; their difference is the void difference. Double the eastward walk gives the sum of hypotenuse and difference; double the southward walk gives their difference. Their product is twice the void area; direct multiplication gives half the void area. The city diameter minus double the eastward walk gives twice the void gou; minus double the southward walk gives twice the void gu. The three-element sum on the void area equals the city diameter.

Let the celestial unknown be the circle diameter, as the three-element sum. Subtracting double the two walks gives p as the yellow square; multiplying by the unknown gives p as twice the void area (stored left). Then double the eastward walk times double the southward walk gives 8640 as the matching number. Canceling gives p . Accumulating and extracting the square root yields 240 paces, the city diameter. Q.E.D.

Problem 一百零六

Q: Bing goes out the South Gate and walks straight 135 paces, then stands. Jia goes out the East Gate and walks straight 16 paces and sees him. Answer as before.

A: City diameter: 240 paces. The difference between gou and hypotenuse on Bing’s walk: 81 paces.

Method: Raise Bing’s walk 135 to the third power, obtaining 2,460,375 as upper term. Multiply Jia’s walk 16 by Bing’s squared walk 18,225 to get 291,600; multiply by the upper term to obtain 717,445,350,000 as the cubic equation’s shi. Double the product of the two walks to get 4,320; multiply by the cube of Bing’s walk

益從；第一廉空；以甲行乘丙行羣，得二十九萬一千六百，又倍之得五十八萬三千二百於上，四之甲行羣一千零二十四，以乘丙行步，得一十三萬八千二百四十，減上位餘四十四萬四千九百六十為第二廉；二行步相乘得二千一百六十為虛常法。得丙行步上勾弦差八十一。

草曰：識別二數相並，得一百五十一。以減於皇極弦，餘一百三十八，即虛勾虛股並也。若以二數相減，餘一百一十九為高弦內減平弦，又為皇極弦內少個小差弦，又為大差弦內減個皇極弦也。

立天元一為丙行大差數。置丙行步一百三十五，自乘得 p ，用天元除之，得 p 為勾弦並也。上減天元得 p 為二丙勾也。復用丙南行乘之，得 p 為二積也。又以天元除之，得 p 為丙勾外容圓半（泛寄）。別置丙南行用二甲勾乘之，得 p 太，合用二丙勾除之。不受除，便以此為甲股（內寄二丙勾為分母）。復用二甲勾三十二乘之，得 p 太為二個甲直積也。又置丙南行內減天元，得 p 為黃方，以自乘得 p 為丙上勾弦差乘股弦差二段，以天元除之得 p 為兩個丙小差也。乃用甲股乘之，得下式 p 。復用丙南行除之，得 p ，又折半得下式 p 為一個甲步股弦差也，內亦帶前二丙勾分母。復置兩個甲直積，內已寄此甲股弦差分母，便為甲步股外容圓半（寄左）。乃再置先求到泛寄，用甲股弦差分母乘之，得 p 為同數，與左相消得下式 p 。開三乘方得八十一，即丙步上勾弦差也。

《鈐經》載此法，以勾弦差率羣減丙行羣，復以丙行乘之為實，以差率羣為法，如法得徑。此法只是以勾外求圓半。合以大差除倍積，而今皆以大差羣為分母也。依法求之，勾弦差八十一自之得六千五百六十一，以減於丙行羣一萬八千二百二十五，餘一萬一千六百六十四，復以丙行一百三十五乘之，得一百五十七萬四千六百四十為實，以大差羣六千五百六十一為法，如法得二百四十步，即城徑也。

第一百零七問

或問：出東門一十六步有樹，出南門東行七十二步見之。問答同前。

答曰：城徑二百四十步。

法曰：二行步相減得數，以自之於上。又以出東門步自之，減上位為平方實，二之出南門東行步為益從，一步常法。翻開得半徑。

to obtain 10,628,820,000 as increasing cong. First lian is empty. Multiply Jia's walk by Bing's squared walk to get 291,600; double to get 583,200 as upper term. Quadruple Jia's squared walk 1,024 and multiply by Bing's walk to get 138,240; subtract from upper term, leaving 444,960 as second lian. The product of the two walks, 2,160, is the empty constant method. Extracts the gou-hypotenuse difference on Bing's walk: 81.

Working: Identification: The sum of the two numbers is 151. Subtracting from the imperial extreme hypotenuse leaves 138, the sum of void gou and void gu. Their difference, 119, is the high string minus the level string, also the imperial extreme string lacking a small-difference string, also the large-difference string minus the imperial extreme string.

Let the unknown be the large-difference number for Bing's walk. Square Bing's walk 135 to get p ; divide by the unknown to get p as the sum of gou and hypotenuse. Subtract the unknown above to get p as twice Bing's gou. Multiply by Bing's southward walk to get p as twice the area. Divide by the unknown to get p as the half-circle (general storage). Multiply Bing's southward walk by twice Jia's gou to get p (this becomes Jia's gu with denominator). Multiply by twice Jia's gou 32 to get p as twice Jia's rectangular area. Subtract the unknown from Bing's walk to get p as the yellow square; squaring gives p as two segments of the product of gou-hypotenuse difference and gu-hypotenuse difference. Dividing by the unknown gives p as two small differences. Multiply by Jia's gu to get p . Divide by Bing's southward walk and halve to get p as one step of Jia's gu-hypotenuse difference. Extracting the cubic root yields 81, the gou-hypotenuse difference.

The *Qianjing* records: square the gou-hypotenuse difference rate and subtract from Bing's walk square, then multiply by Bing's walk as shi; use the squared difference rate as divisor to obtain the diameter. This method seeks the half-circle from outside the gou. The large difference should divide twice the area; here the squared large difference serves as denominator. Accordingly: square 81 to get 6,561; subtract from Bing's walk square 18,225, leaving 11,664; multiply by 135 to get 1,574,640 as shi; divide by squared large difference 6,561 to obtain 240 paces, the city diameter.

Problem 一百零七

Q: Going out the East Gate 16 paces there is a tree. Going out the South Gate and walking 72 paces east, one sees it. Answer as before.

A: City diameter: 240 paces.

Method: Subtract the two walks; square the result as upper term. Square the East Gate walk and subtract from upper term as the quadratic shi. Double the south-gate eastward walk as increasing cong. One constant method. Extract by reversing to obtain the

草曰：別得人到樹即平弦也，半圓徑即平股也。其東行七十二步則平勾平弦差也。乃立天元一為半圓徑，加一十六減七十二，得 p 為勾也。以自之得 p 為勾幂。又加入天元股幂得 p 為弦幂（寄左）。再立天元一為半徑，加出東門步，得 p 即弦也。以自之得 p 為同數，與左相消得 p 。翻法開之得一百二十步，即半城徑也。合問。

第一百零八問

或問：出南門一百三十五步有樹，出東門南行三十步見之。問答同前。

答曰：城徑二百四十步。

法曰：樹去城步內減南行步，餘以為幂於上，又以樹去城步為幂內減上位為平實，倍樹去城步為從，一虛隅。翻法得半城徑。

草曰：別得人距樹即高弦也，半圓徑即高勾也，其南行三十步即高弦上小差也。乃立天元一為半徑，加樹去城步為弦，內減小差 p ，得 p 即股也。以自之得 p 為股幂，內加入天元幂，得 p 為弦幂（寄左）。再置弦 p 以自之，得 p 為同數，與左相消，得式 p 。翻開得一百二十步，即半城徑也。合問。

第一百零九問

或問：乙出東門不知遠近而立，甲出南門東行七十二步望見乙，就乙斜行一百三十六步與乙相會。問答同前。

答曰：城徑二百四十步，乙出東門一百二十八步。

法曰：以斜行步自之於上，以二行相減餘自為幂，減上位為平實，從空，一步常法。如法得半徑。

草曰：別得七十二步即大差也，斜行即弦，半徑即股也。立天元一為半徑，以自之為股幂，又以二行差六十四以自之得 p 為勾幂。並二幂得 p 為弦幂（寄

radius.

Working: Identification: The person reaching the tree is the level hypotenuse; half the diameter is the level gu. The 72 paces east is the difference between level gou and level hypotenuse. Let the unknown be half the diameter; add 16 and subtract 72 to get p as gou. Squaring gives p as gou-power. Add the unknown squared as gu-power to get p as hypotenuse-power (stored left). Again let the unknown be the radius; add the East Gate paces to get p as the hypotenuse. Squaring gives p as the matching number. Canceling gives p . Extract by the reversing method to obtain 120 paces, half the city diameter. Q.E.D.

Problem 一百零八

Q: Going out the South Gate 135 paces there is a tree. Going out the East Gate and walking 30 paces south, one sees it. Answer as before.

A: City diameter: 240 paces.

Method: Subtract the southward walk from the tree-to-city distance; square the remainder as upper term. Square the tree-to-city distance and subtract the upper term as flat shi. Double the tree-to-city distance as cong. One empty corner. Extract by reversing to obtain half the city diameter.

Working: Identification: The person's distance to the tree is the high hypotenuse; half the diameter is the high gou. The 30 paces southward is the small difference on the high hypotenuse. Let the unknown be the radius; add the tree-to-city distance as hypotenuse, subtract the small difference p to get p as gu. Squaring gives p as gu-power; add the unknown squared to get p as hypotenuse-power (stored left). Again place hypotenuse p , square it to get p as matching number. Canceling gives p . Extract by reversing to obtain 120 paces, half the city diameter. Q.E.D.

Problem 一百零九

Q: Yi goes out the East Gate and stands at an unknown distance. Jia goes out the South Gate, walks 72 paces east, sees Yi, then walks diagonally 136 paces to meet Yi. Answer as before.

A: City diameter: 240 paces. Yi went out the East Gate 128 paces.

Method: Square the diagonal walk as upper term. Subtract the two walks; square the remainder and subtract from upper term as flat shi. Cong is empty. One constant method. Extract to obtain the radius.

Working: Identification: The 72 paces is the large difference; the diagonal is the hypotenuse; the radius is the gu. Let the unknown be the radius; square it as gu-power. The

左)。然後以斜行步自之，得 p 為同數，與左相消得 p 。開平方得一百二十步，倍之即城徑也。合問。

第一百一十問

或問：甲出南門不知遠近而立，乙出東門南行三十步望見甲，卻就甲斜行二百五十五步與甲相會。問答同前。

答曰：城徑二百四十步，甲出南門二百二十五步。

法曰：二行差自之為冪，以減於斜行冪為平實，一虛隅。得半徑。

草曰：別得南行步即股弦差也，斜步即弦也，半徑即勾也。乃立天元一為半城徑，以自之為冪。以二行相減餘二百二十五，以自之得 p 為股冪。二冪相並得 p 為弦冪（寄左）。然後以斜行步自之，得 p 為同數，與左相消得下 p 。開平方得一百二十步，即半徑也。合問。

第一百一十一問

或問：甲出南門東行不知步數而立，乙出東門南行三十步望見甲，斜行一百二步相會。問答同前。

答曰：城徑二百四十步，甲東行九十六步。

法曰：二行相減，餘以乘乙南行，四之於上，又加入斜行冪為平實。得虛和一百三十八。

草曰：別得斜步內減南行為甲東行步也。此問以弦外容圓入之，以二行相減數乘乙南行三十步，得 p ，又四之，得 p 為二直積也。又加入斜步冪 p ，共得 p 即和冪也。平方而一，得一百三十八步，即虛和也。又加斜步得二百四十步，即城徑也。合問。

第一百一十二問

difference of the two walks, 64, squared gives p as gou-power. Adding both powers gives p as hypotenuse-power (stored left). Then square the diagonal walk to get p as matching number. Canceling gives p . Extract the square root to obtain 120 paces; double it for the city diameter. Q.E.D.

Problem 一百一十

Q: Jia goes out the South Gate and stands at an unknown distance. Yi goes out the East Gate, walks 30 paces south, sees Jia, then walks diagonally 255 paces to meet Jia. Answer as before.

A: City diameter: 240 paces. Jia went out the South Gate 225 paces.

Method: Square the difference of the two walks; subtract from the squared diagonal as flat shi. One empty corner. Extract the radius.

Working: Identification: The southward walk is the difference between gu and hypotenuse; the diagonal is the hypotenuse; the radius is the gou. Let the unknown be half the city diameter; square it as power. The difference of the two walks, 225, squared gives p as gu-power. Adding both powers gives p as hypotenuse-power (stored left). Then square the diagonal walk to get p as matching number. Canceling gives p . Extract the square root to obtain 120 paces, the radius. Q.E.D.

Problem 一百一十一

Q: Jia goes out the South Gate and walks east an unknown number of paces, then stands. Yi goes out the East Gate, walks 30 paces south, sees Jia, then walks diagonally 102 paces to meet Jia. Answer as before.

A: City diameter: 240 paces. Jia walked 96 paces east.

Method: Subtract the two walks; multiply the remainder by Yi's southward walk; quadruple as upper term; add the squared diagonal as flat shi. Extracts the void sum: 138.

Working: Identification: The diagonal minus the southward walk gives Jia's eastward walk. This problem uses "circle outside the hypotenuse." Multiply the difference of the two walks by Yi's 30 paces south to get p ; quadruple to get p as twice the rectangular area. Add the squared diagonal p , totaling p as the squared sum. Extract the square root to obtain 138, the void sum. Add the diagonal to get 240 paces, the city diameter. Q.E.D.

Problem 一百一十二

或問：乙出東門南行不知步數而立，甲出南門東行七十二步望見乙，斜行一百二步與乙相會。問答同前。

答曰：城徑二百四十步，乙南行五十四步。

法曰：倍相減步以乘倍東行，得數復以減於斜步冪，餘為實。平方而一，得較也。又以二行相減數乘倍東行為平實，以較為從方，得勾。勾較共為長，又以斜步並入勾股共，即城徑。

草曰：別得二行相減餘 p 為乙南行步也。以此數又減於甲東行，餘四十二步即較也。又以二行相減數 p 乘倍東行得 p 為平實，以較為從。平方開得四十八即勾也。勾內加較得九十步即股也。勾股共得一百三十八，又加入斜步，共得二百四十步，即城徑也。合問。

第一百一十三問

或問：乙出南門東行，甲出東門南行，兩相望見。既而乙雲：「我東行不及城徑一百六十八步。」甲雲：「我南行不及城徑二百一十步。」問答同前。

答曰：城徑二百四十步，甲南行三十步，乙東行七十二步。

法曰：半甲不及步以自之為冪，半甲不及步內減差以自之為冪。二冪相並內卻減差冪為平實，四之甲不及內減三之乙不及，餘為益從，三步半虛法。得甲南行。

草曰：別得乙不及為虛勾、半徑共，又為徑內減明勾也。甲不及為虛股、半徑共，又為徑內減股也。又二雲數相並為虛和、圓徑共也，雲數相減即虛較也。

乃立天元一為甲南行，以減於甲不及步又半之，得 p 為虛股也。虛股內減虛較得 p 為虛勾。勾自之得 p 為勾冪也，又股自之下式 p 為股冪也。二冪相並得 p 為弦冪（寄左）。然後以天元加虛較得 p 為乙東行。又加入天元甲南行得 p 為虛弦，以自之得 p 為同數，與左相消得 p 。開平方得三十步，即甲南行也。內加少步，即城徑也。合問。

Q: Yi goes out the East Gate and walks south an unknown number of paces, then stands. Jia goes out the South Gate, walks 72 paces east, sees Yi, then walks diagonally 102 paces to meet Yi. Answer as before.

A: City diameter: 240 paces. Yi walked 54 paces south.

Method: Double the subtraction step and multiply by double the eastward walk; subtract this from the squared diagonal, remainder as shi. Extract the square root to obtain the difference. Multiply the difference of the two walks by double the eastward walk as flat shi; use the difference as cong to extract gou. Gou plus difference gives the length; add the diagonal to the gou-gu sum for the city diameter.

Working: Identification: The difference of the two walks, p , is Yi's southward walk. Subtract this from Jia's eastward walk; the remainder 42 is the difference. Multiply the difference of the two walks p by double the eastward walk to get p as flat shi; use the difference as cong. Extract the square root to obtain 48, the gou. Add the difference to get 90 paces, the gu. Gou plus gu totals 138; add the diagonal to obtain 240 paces, the city diameter. Q.E.D.

Problem 一百一十三

Q: Yi goes out the South Gate walking east; Jia goes out the East Gate walking south; they see each other. Yi says: "My eastward walk falls short of the diameter by 168 paces." Jia says: "My southward walk falls short of the diameter by 210 paces." Answer as before.

A: City diameter: 240 paces. Jia walked 30 paces south; Yi walked 72 paces east.

Method: Halve Jia's shortfall and square it as one power. Subtract the difference from half Jia's shortfall and square as second power. Add both powers and subtract the squared difference as flat shi. Quadruple Jia's shortfall minus triple Yi's shortfall as increasing cong. Three-and-a-half empty method. Extracts Jia's southward walk.

Working: Identification: Yi's shortfall is the void gou plus radius, also the diameter minus the bright gou. Jia's shortfall is the void gu plus radius, also the diameter minus the gu. The sum of the two given numbers is the void sum plus circle diameter; their difference is the void difference.

Let the unknown be Jia's southward walk. Subtract from half Jia's shortfall to get p as void gu. Subtract the void difference from void gu to get p as void gou. Square the gou to get p as gou-power; square the gu to get p as gu-power. Adding both gives p as hypotenuse-power (stored left). Add the unknown to the void difference to get p as Yi's eastward walk. Add the unknown (Jia's southward walk) to get p as void hypotenuse. Squaring gives p as

第一百一十四問

或問：丙出南門直行，甲出東門直行，兩相望見。既而丙雲：「我行少於城徑一百五步。」甲雲：「我行少於城徑二百二十四步。」問答同前。

答曰：城徑二百四十步，半徑一百二十步。

法曰：二少步相乘訖，又自乘為實；六之共步，乘雲數相乘數為益從；十八之雲數相乘於上，又三之共步，自乘加上位，內復減丙少步冪、甲少步冪為從廉；四十八之共步為益二廉，六十三步常法。翻法開三乘方，得一百二十步，即半徑。

草曰：別得雲數共減於倍城徑為甲丙共行數。又雲數相減即皇極差，亦為甲行不及丙行數。立天元一為半城徑，以三之，副置二位。上位減丙少步，得 p 為皇極股也，下位減甲少步得 p 為皇極勾也。勾股相乘得 p ，以天元除之，得 p 為弦也。弦自之得 p 為弦冪（寄左）。然後以股自之得下 p 為股冪於上，又以勾自之得 p 為勾冪，並以加入上位，得 p 為同數，與左相消得 p 。翻法開三乘方得一百二十步，即半徑也。合問。

第一百一十五問

或問：甲出東門直行，乙出南門直行，各不知步數而立。乙望見甲，就甲斜行了二百八十九步與甲相會。其二直行共得一百五十一。又雲甲直行少於乙直行。問答同前。

答曰：城徑二百四十步，甲直行一十六步，乙直行一百三十五步。

法曰：斜冪內減共步冪為平實，倍共步內減斜步為從，一常法。得徑。

matching number. Canceling gives p . Extract the square root to obtain 30 paces, Jia's southward walk. Add the shortfall for the city diameter. Q.E.D.

Problem 一百一十四

Q: Bing goes out the South Gate walking straight; Jia goes out the East Gate walking straight; they see each other. Bing says: "My walk is 105 paces less than the diameter." Jia says: "My walk is 224 paces less than the diameter." Answer as before.

A: City diameter: 240 paces. Radius: 120 paces.

Method: Multiply the two shortfalls and then self-multiply as shi. Multiply six times the sum by the product of the given numbers as increasing cong. Multiply eighteen times the product of the given numbers as upper term; add three times the squared sum; subtract Bing's shortfall squared and Jia's shortfall squared as cong-lian. Forty-eight times the sum as increasing second lian; sixty-three as constant method. Extract by reversing the cubic root to obtain 120 paces, the radius.

Working: Identification: The sum of the given numbers subtracted from double the city diameter gives the total walk of Jia and Bing. Their difference is the imperial extreme difference, also Jia's walk short of Bing's.

Let the unknown be half the city diameter; triple it and place in two positions. Subtract Bing's shortfall from the upper position to get p as imperial extreme gu; subtract Jia's shortfall from the lower to get p as imperial extreme gou. Multiply gou and gu to get p ; divide by the unknown to get p as hypotenuse. Squaring gives p as hypotenuse-power (stored left). Square the gu to get p as gu-power (upper); square the gou to get p as gou-power. Add both to the upper to get p as matching number. Canceling gives p . Extract the cubic root by reversing to obtain 120 paces, the radius. Q.E.D.

Problem 一百一十五

Q: Jia goes out the East Gate walking straight; Yi goes out the South Gate walking straight; each stands at an unknown distance. Yi sees Jia, then walks diagonally 289 paces to meet Jia. Their two straight walks total 151 paces. It is also stated that Jia's walk is less than Yi's. Answer as before.

A: City diameter: 240 paces. Jia walked 16 paces; Yi walked 135 paces.

Method: Subtract the squared sum from the squared diagonal as flat shi. Double the sum minus the diagonal as cong. One constant method. Extract the diameter.

草曰：別得共數、城徑並即皇極和也。立天元一為圓徑，加共步得 p 為皇極和，以自之，得 p 於上。以斜行冪 p 減上位餘 p 為二直積（寄左）。然後以天元乘斜步得 p ，與左相消得 p 。開平方得二百四十步，即城徑也。合問。

第一百一十六問

或問：甲出東門直行，乙出東門南行，丙出南門直行，丁出南門東行，各不知步數而立。四人遙相望，悉與城參相直。只雲甲、丙共行了一百五十一，乙、丁立處相距一百二步。又雲丙直行步多於甲直行步。問答同前。

答曰：城徑二百四十步。

法曰：共步、距步相減，得數自之於上，以共步為冪內減上為平實，二之距步內減共步、距步差為從，一步虛法。得城徑。

草曰：別得共步得城徑即皇極和也，相距步即虛弦也。皇極和內減虛弦即皇極弦也。又共步、距步差 p 即皇極弦內減城徑也（此名旁差）。乃立天元一為城徑，加共步得 p 為皇極和也，以自之得 p 於上，以共步、距步差 p 加天元得 p 為皇極弦也，以自之得下式 p ，減上位餘得 p 為二直積（寄左）。然後以天元徑乘皇極弦，得 p 為同數，與左相消得 p 。開平方得二百四十步，即城徑也。合問。

第一百一十七問

或問：甲出南門東行不知步數而立，乙出東門南行望見甲，復就甲斜行，與甲相會。乙通計行了一百三十二步，其乙南行步不及斜行七十二步，其甲東行卻多於乙南行。問答同前。

Working: Identification: The sum plus the city diameter is the imperial extreme sum.

Let the unknown be the circle diameter; add the sum to get p as imperial extreme sum. Square it to get p as upper term. Subtract the squared diagonal p from the upper term, leaving p as twice the rectangular area (stored left). Multiply the unknown by the diagonal to get p as matching number. Canceling gives p . Extract the square root to obtain 240 paces, the city diameter. Q.E.D.

Problem 一百一十六

Q: Jia goes out the East Gate walking straight; Yi goes out the East Gate walking south; Bing goes out the South Gate walking straight; Ding goes out the South Gate walking east; each stands at an unknown distance. The four gaze at each other from afar, all aligned with the city. It is only stated that Jia and Bing together walked 151 paces; Yi and Ding's positions are 102 paces apart. Also stated that Bing's walk exceeds Jia's. Answer as before.

A: City diameter: 240 paces.

Method: Subtract the distance from the sum; square the result as upper term. Subtract this from the squared sum as flat shi. Double the distance minus the difference of sum and distance as cong. One empty-corner method. Extract the city diameter.

Working: Identification: The sum equals the city diameter, which is the imperial extreme sum. The distance apart is the void hypotenuse. Imperial extreme sum minus void hypotenuse equals the imperial extreme hypotenuse. The difference of sum and distance p is the imperial extreme hypotenuse minus the city diameter (called side difference).

Let the unknown be the city diameter; add the sum to get p as imperial extreme sum. Square it to get p as upper term. Add the difference of sum and distance p to the unknown to get p as imperial extreme hypotenuse. Squaring gives p ; subtracting from the upper term leaves p as twice the rectangular area (stored left). Multiply the unknown diameter by the imperial extreme hypotenuse to get p as matching number. Canceling gives p . Extract the square root to obtain 240 paces, the city diameter. Q.E.D.

Problem 一百一十七

Q: Jia goes out the South Gate and walks east an unknown number of paces, then stands. Yi goes out the East Gate walking south, sees Jia, then walks diagonally to meet Jia. Yi walked a total of 132 paces; Yi's southward walk falls short of the diagonal by 72 paces; Jia's eastward walk exceeds Yi's southward walk. Answer as before.

答曰：城徑二百四十步，乙南行三十步，甲東行一百二步。

法曰：倍不及步在地，以不及步減通步以乘之為實，以四之不及步為法。得乙南行三十步。

草曰：別得乙南行即股也，以減通步即虛弦也，以減不及步即虛較也，其不及步即甲東行也。立天元一為乙南行，置不及步以天元乘之，又四之得 p 元為二直積（寄左）。然後倍不及步以為弦較較。以乘上位得 p 太為同數，與左相消得 p 。上法下實，得三十步為乙南行也。餘各以數求之。

第一百一十八問

或問：甲出南門東行不知步數而立。乙出東門南行，望見甲，復斜行與甲相會。二人共行了二百四步，又雲甲行不及共步一百三十二。問答同前。

答曰：城徑二百四十步，甲東行九十六步，乙南行一百零八步。

法曰：別得二行共即兩個虛弦也，其不及步即乙南行與一虛弦共也。置不及步內減一弦餘三十步，即乙南行也。以乙南行反以減虛弦，餘七十二步即甲東行也。以乙南行減甲東行餘即虛較也。

草曰：此問無草。

第一百一十九問

或問：乙出東門南行，甲出西門南行，甲望見乙，斜行五百一十步相會。乙雲：「我南行少於城徑二百一十步。」問答同前。

答曰：城徑二百四十步，乙南行三十步。

法曰：少步羈為平實，四斜步內減二少步為益從，五步常法。得乙南行。

A: City diameter: 240 paces. Yi walked 30 paces south; Jia walked 102 paces east.

Method: Place double the shortfall; multiply by the total minus the shortfall as shi; use quadruple the shortfall as divisor. Extracts Yi's southward walk: 30 paces.

Working: Identification: Yi's southward walk is the gu; subtract from the total to get the void hypotenuse; subtract the shortfall to get the void difference. The shortfall itself is Jia's eastward walk.

Let the unknown be Yi's southward walk. Multiply the shortfall by the unknown and quadruple to get p as twice the rectangular area (stored left). Double the shortfall as the sum of hypotenuse and difference (upper) p . Total minus shortfall gives p as the difference of hypotenuse and difference. Multiply the upper by this to get p as matching number. Canceling gives p . Dividing yields 30 paces as Yi's southward walk. The rest follows from the numbers.

Problem 一百一十八

Q: Jia goes out the South Gate and walks east an unknown number of paces, then stands. Yi goes out the East Gate walking south, sees Jia, then walks diagonally to meet Jia. Together they walked 204 paces. It is also stated that Jia's walk falls short of the total by 132. Answer as before.

A: City diameter: 240 paces. Jia walked 96 paces east; Yi walked 108 paces south.

Method: Identification: The sum of the two walks is twice the void hypotenuse; the shortfall is Yi's southward walk plus one void hypotenuse. Subtract one hypotenuse from the shortfall; the remainder 30 is Yi's southward walk. Subtract Yi's southward walk from the void hypotenuse; the remainder 72 is Jia's eastward walk. Subtract Yi's from Jia's to get the void difference.

Working: This problem has no working.

Problem 一百一十九

Q: Yi goes out the East Gate walking south; Jia goes out the West Gate walking south; Jia sees Yi and walks diagonally 510 paces to meet Yi. Yi says: "My southward walk is 210 paces less than the diameter." Answer as before.

A: City diameter: 240 paces. Yi walked 30 paces south.

Method: Square the shortfall as flat shi. Quadruple the diagonal minus double the shortfall as increasing cong. Five as constant method. Extracts Yi's southward walk.

草曰：別得少步為徑內減股。立天元一為乙南行，以二之減於倍斜行步，得 p 為梯底也。以二之天元乘之，得 p 為徑幂（寄左）。再置天元加少步，得下式 p 為城徑，以自之得 p ，與左相消得 p 。開平方得三十步，即乙南行也。加少步即城徑也。合問。

第一百二十問

或問：乙出南門東行，甲出北門東行，甲望見乙，斜行二百七十二步與乙相會。乙雲：「我東行不及城徑一百六十八步。」問答同前。

答曰：城徑二百四十步，乙東行七十二步。

法曰：以不及步幂之為實，四斜內減二之不及步為虛從，五常法。平開得乙東行七十二步。

草曰：別得不及步為城徑減明勾也。立天元一為乙東行，以倍之減於二之斜行步，得下 p 為梯底也。倍天元乘之，得 p 為徑幂（寄左）。再置天元加不及步，得 p 為城徑，以自之得 p 為同數，與左相消得 p 。開平方得七十二步，即乙東行也。加入少步即城徑也。合問。

第一百二十一問

或問：乙出南門東行，丁出東門南行，卻有甲丙二人共在西北隅，甲向東行，丙向南行，四人遙相望見，俱與城參相直。既而相會，甲雲：「我多乙二百四十八步。」丙雲：「我多於丁五百七十步。」問答同前。

答曰：城徑二百四十步。

法曰：二多步相乘為平實，並二多步而半之為從，七分半常法。得城徑。

草曰：別得甲多步為大勾內減明勾也，丙多步為大股內少股也。又乙東行得一虛勾為半徑，丁南行得一虛股為半徑。又二多數相並得 p 為大和內少虛弦也，又二少數相減餘 p 為兩個角差。又甲多步內

Working: Identification: The shortfall is the diameter minus the gu.

Let the unknown be Yi's southward walk. Subtract double the unknown from double the diagonal to get p as the ladder base. Multiply by double the unknown to get p as diameter-power (stored left). Add the shortfall to the unknown to get p as city diameter; squaring gives p . Canceling gives p . Extract the square root to obtain 30 paces, Yi's southward walk. Add the shortfall for the city diameter. Q.E.D.

Problem 一百二十

Q: Yi goes out the South Gate walking east; Jia goes out the North Gate walking east; Jia sees Yi and walks diagonally 272 paces to meet Yi. Yi says: "My eastward walk falls short of the diameter by 168 paces." Answer as before.

A: City diameter: 240 paces. Yi walked 72 paces east.

Method: Square the shortfall as shi. Quadruple the diagonal minus double the shortfall as empty cong. Five as constant method. Extract to obtain Yi's eastward walk: 72 paces.

Working: Identification: The shortfall is the city diameter minus the bright gou.

Let the unknown be Yi's eastward walk. Subtract double the unknown from double the diagonal to get p as ladder base. Multiply by double the unknown to get p as diameter-power (stored left). Add the shortfall to the unknown to get p as city diameter; squaring gives p as matching number. Canceling gives p . Extract the square root to obtain 72 paces, Yi's eastward walk. Add the shortfall for the city diameter. Q.E.D.

Problem 一百二十一

Q: Yi goes out the South Gate walking east; Ding goes out the East Gate walking south. Meanwhile Jia and Bing together are at the northwest corner, Jia walking east and Bing walking south. The four gaze at each other from afar, all aligned with the city. After meeting, Jia says: "I exceed Yi by 248 paces." Bing says: "I exceed Ding by 570 paces." Answer as before.

A: City diameter: 240 paces.

Method: Multiply the two excesses as flat shi. Halve the sum of the two excesses as cong. Seven and a half as constant method. Extract the city diameter.

Working: Identification: Jia's excess is the large gou minus the bright gou; Bing's excess is the large gu minus the gu. Yi's eastward walk gives one void gou as radius; Ding's southward walk gives one void gu as radius. The sum of

減半徑即勾方差也，丙多步內減半徑即股方差也。

立天元一為城徑，以半之減於甲多步得 p 為勾方差，又以半徑減於丙多步得 p 為股方差。二差相乘得 p 為徑幂（寄左）。然後以天元幂與左相消，得下式 p 。開平方得二百四十步，即城徑也。合問。

第一百二十二問

或問：甲丙二人俱在西北隅，甲向東行，丙向南行。又乙出南門東行，丁出東門南行，各不知步數而立。四人遙相望見，悉與城參相直。既而相會，甲雲：「我與乙共行了三百九十二步。」丙雲：「我與丁共行了六百三十步。」問答同前。

答曰：城徑二百四十步。

法曰：甲乙共自之為幂，丙丁共自之為幂，二幂又相乘為三乘方實。甲乙共自之為幂，以丙丁共乘之於上。又以丙丁共自之為幂，以甲乙共乘之加上位為益從。甲乙共自之為幂，丙丁共自之為幂，並，以七分半乘之於上。又以甲乙共乘丙丁共，得數減上位為第一益廉。並二共數，以七分半乘之為第二廉。以七分半自之，得五分六厘二毫五絲於上位。以一步內減上位，餘四分三厘七毫五絲為虛隅。得城徑。

草曰：別得甲為大勾，乙為明勾，丙為大股，丁為股也。甲乙共內減半徑即是黃長弦也，丙丁共內減半徑即黃廣弦也。黃長弦、黃廣弦二數相減，餘為兩個皇極差也。

乃立天元為城徑，半之副置二位。上以減於甲乙共數，得 p 即黃長弦也，以自之得 p 為黃長弦幂也，內減天元一幂，餘得下式 p 為勾方差幂也。下位以減於丙丁共，得下式 p 即黃廣弦也，以自之得 p 為黃廣弦幂也，內減天元一幂，餘得 p 為股方差幂也。再以勾方差幂、股方差幂相乘，得 p 為徑幂（寄左）。然後以天元為幂，又以幂自之，與左相消得下式 p 。開三乘方得二百四十步，即城徑也。合問。

the two excesses p is the large sum minus the void hypotenuse; their difference p is twice the corner difference. Jia's excess minus the radius is the gou square-difference; Bing's excess minus the radius is the gu square-difference.

Let the unknown be the city diameter. Subtract half the diameter from Jia's excess to get p as gou square-difference; subtract the radius from Bing's excess to get p as gu square-difference. Multiply both differences to get p as diameter-power (stored left). Cancel the unknown squared with the left to get p . Extract the square root to obtain 240 paces, the city diameter. Q.E.D.

Problem 一百二十二

Q: Jia and Bing are both at the northwest corner; Jia walks east and Bing walks south. Yi goes out the South Gate walking east; Ding goes out the East Gate walking south; each stands at an unknown distance. The four gaze at each other from afar, all aligned with the city. After meeting, Jia says: "Yi and I together walked 392 paces." Bing says: "Ding and I together walked 630 paces." Answer as before.

A: City diameter: 240 paces.

Method: Square the Jia-Yi sum and the Bing-Ding sum; multiply both powers as the cubic shi. Square the Jia-Yi sum, multiply by the Bing-Ding sum as upper term. Square the Bing-Ding sum, multiply by the Jia-Yi sum and add to upper term as increasing cong. Square both sums, add, multiply by 7.5 as upper term. Subtract the product of both sums from upper term as first increasing lian. Add both sums, multiply by 7.5 as second lian. Square 7.5 to get 0.05625 as upper term. Subtract from one step, leaving 0.04375 as empty corner. Extracts the city diameter.

Working: Identification: Jia is the large gou, Yi is the bright gou, Bing is the large gu, Ding is the gu. The Jia-Yi sum minus the radius is the yellow-length hypotenuse; the Bing-Ding sum minus the radius is the yellow-breadth hypotenuse. Their difference is twice the imperial extreme difference.

Let the unknown be the city diameter; halve and place in two positions. Subtract from the Jia-Yi sum above to get p as yellow-length hypotenuse; squaring gives p as its power; subtract the unknown squared to get p as gou square-difference power. Subtract from the Bing-Ding sum below to get p as yellow-breadth hypotenuse; squaring and subtracting the unknown squared gives p as gu square-difference power. Multiply both difference powers to get p as diameter-power (stored left). Cube the unknown and cancel with the left to get p . Extract the cubic root to obtain 240 paces, the city diameter. Q.E.D.

2.1 卷八 明後一十六問

Vol. 8: 16 “Rear” Problems

卷八小序：此卷一十六問，承卷七之緒，仍以明段為主，而益之以極段、虛段之變。所謂極段者，皇極勾股形之各邊也；虛段者，太虛勾股形之各邊也。此三形交錯互見，開方之術益臻精妙。其方程式之高，有至四乘方、五乘方者，實全書之樞紐也。

Volume 8 Preface: These 16 problems continue from Volume 7, still centered on “bright” segments but now incorporating “extreme” and “void” variations. The extreme segments are the edges of the imperial extreme right triangle; the void segments are edges of the grand void triangle. These three triangles interweave, and the root-extraction techniques reach their finest. Equations rise to degrees 4–5; this is the pivot of the entire work.

第一百二十三問

Problem 一百二十三

或問：出南門向東有槐樹一株，出東門向南有柳樹一株。丙丁俱出南門，丙直行，丁往至槐樹下。甲乙俱出東門，甲直行，乙往至柳樹下。四人遙相望見，各不知所行步數。只雲丙丁共行了二百七步，甲乙共行了四十六步，又雲甲丙立處相距二百八十九步。問答同前。

Q: Going out the South Gate eastward there is a locust tree; going out the East Gate southward there is a willow tree. Bing and Ding both go out the South Gate; Bing walks straight while Ding goes to the locust tree. Jia and Yi both go out the East Gate; Jia walks straight while Yi goes to the willow tree. The four gaze at each other from afar; none know the paces walked. It is only stated that Bing and Ding together walked 207 paces; Jia and Yi together walked 46 paces; and the distance between Jia and Bing’s positions is 289 paces. Answer as before.

答曰：城徑二百四十步，半徑一百二十步。

A: City diameter: 240 paces. Radius: 120 paces.

法曰：以二共相減數又以減距數為實，二為法。得平勾。

Method: Subtract the difference of the two sums from the distance as shi; 2 as divisor. Extracts the level gou.

草曰：識別得丙丁共即明和也，甲乙共即和也，相距步即極弦也。二共相並即極弦內少個虛黃也，又為極和內少個虛和也。二共相減餘為平勾高股差也，又為虛差極差共也，又為通差內減極差也。

Working: Identification: Bing-Ding sum is the bright sum; Jia-Yi sum is the sum; the distance is the extreme hypotenuse. The sum of both sums is the extreme hypotenuse lacking the void yellow; also the extreme sum lacking the void sum. Their difference is the difference between level gou and high gu; also the sum of void difference and extreme difference; also the general difference minus the extreme difference.

立天元為平勾，加入二共相減數，得 p 為高股，又加天元得 p 為極弦（寄左）。以相距步二百八十九與左相消，得 p 。上法下實，如法得六十四，即平勾也。以二共相減數加平勾得二百二十五為高股，復以平勾乘之，得一萬四千四百步。開平方得一百二十步，即城半徑也。合問。

Let the unknown be the level gou. Add the difference of the two sums to get p as high gu; add the unknown to get p as extreme hypotenuse (stored left). Cancel with the distance 289 to get p . Dividing yields 64, the level gou. Add the difference of sums to the level gou to get 225 as high gu; multiply by the level gou to get 14,400. Extract the square root to obtain 120 paces, half the city radius. Q.E.D.

第一百二十四問

Problem 一百二十四

或問：依前見丙丁共二百七步，甲乙共四十六步。又雲二樹相去一百二步。問答同前。

Q: As before: Bing-Ding sum 207 paces, Jia-Yi sum 46 paces. Also stated: the two trees are 102 paces apart. Answer as before.

答曰：城徑二百四十步。

A: City diameter: 240 paces.

法曰：以甲乙共乘樹相去步，得數又以自之為平實；從空；並二共數為冪於上，內減甲乙共自之數、丙丁共自之數為益隅。得弦。

Method: Multiply the Jia-Yi sum by the tree distance; square the result as flat shi. Cong empty. Square the sum of both sums as upper term; subtract the squared Jia-Yi sum and squared Bing-Ding sum as increasing

草曰：識別得兩樹相去步即虛弦也。餘數具前。立天元一為弦。置明和以天元乘之，合和除，不除，便以 p 元為明弦也（內帶和分母）。乃置虛弦以分母和乘之，得 p 太，加入明弦，得 p 為極股也，內帶和分母。以自之得下式 p 為極股冪（內寄和冪為分母）。又以天元加虛弦得下 p 為極勾，以自之得 p 。又以和冪 p 乘之，得 p 為勾冪也。勾股冪相並得 p 為兩積一較冪也，內有和冪分母（寄左）。然後置明弦 p 元於上，以和乘天元加上位，得 p 元為二弦並也。又置虛弦以和乘之得 p 太，並入上位，得下式 p 為極弦，以自之得 p 為同數，與左相消得 p 。開平方得三十四步，即弦也。

第一百二十五問

或問：皇極大小差共一百八十七步，明黃、黃共六十六步。問答同前。

答曰：太虛黃方三十六步，城徑二百四十步。

法曰：後數自乘為實，前後數相減，餘為法。得虛黃方三十六。

草曰：別得一百八十七即明弦二弦共也，其六十六即太虛大小差共也。又二數相並，得 p 即明、二和共。若以相減，餘 p 即明、四差共也。

立天元一為太虛黃方面，加二黃共，得 p 即虛弦也。倍虛弦又加天元，得 p 即城徑也。又以虛弦加皇極大小差，得 p 即極弦也，以極弦乘城徑得 p ，為兩段皇極勾股積（寄左）。再以極弦虛弦相並，得 p ，即皇極勾股共也，自之得 p ，內減皇極弦冪 p ，得 p 為同數，與寄左相消得 p 。上法下實，如法得三十六步，即太虛黃方面也。合問。

第一百二十六問

或問：東門南有柳一株，南門東有槐一株。甲出東門直行，丙出南門直行。甲、丙、柳、槐悉與城參相直，

corner. Extracts the hypotenuse.

Working: Identification: The tree distance is the void hypotenuse. The rest as before.

Let the unknown be the hypotenuse. Multiply the bright sum by the unknown; do not divide by the sum; take p as the bright hypotenuse (with sum as denominator). Multiply the void hypotenuse by the denominator sum to get p ; add the bright hypotenuse to get p as extreme gu (with sum denominator). Squaring gives p as extreme gu power (with squared sum as denominator). Add the unknown to the void hypotenuse to get p as extreme gou ; squaring and multiplying by squared sum p gives p as gou power. Adding both powers gives p as two areas plus one difference power (with squared sum denominator, stored left). Place bright hypotenuse p above; multiply the unknown by the sum and add to get p as combined hypotenuses. Multiply void hypotenuse by sum to get p ; add to get p as extreme hypotenuse. Squaring gives p as matching number. Canceling gives p . Extract the square root to obtain 34 paces, the hypotenuse. Q.E.D.

Problem 一百二十五

Q: The sum of imperial extreme large and small differences is 187 paces; the sum of bright yellow and yellow is 66 paces. Answer as before.

A: Grand void yellow square: 36 paces. City diameter: 240 paces.

Method: Square the latter number as shi; subtract the former from the latter as divisor. Extracts the void yellow square: 36.

Working: Identification: 187 is the sum of bright hypotenuse and two hypotenuses; 66 is the sum of grand void large and small differences. Adding both gives p as the sum of bright and two sums; subtracting gives p as the sum of bright and four differences.

Let the unknown be the grand void yellow square side. Add twice the yellow sum to get p as void hypotenuse. Double the void hypotenuse and add the unknown to get p as city diameter. Add the void hypotenuse to the imperial extreme differences to get p as extreme hypotenuse; multiply by the city diameter to get p as twice the imperial extreme $gou-gu$ area (stored left). Add extreme and void hypotenuses to get p as imperial extreme $gou-gu$ sum. Squaring and subtracting the squared extreme hypotenuse p gives p as matching number. Canceling gives p . Dividing yields 36 paces, the grand void yellow square side. Q.E.D.

Problem 一百二十六

Q: South of the East Gate is a willow; east of the South Gate is a locust. Jia goes out the East Gate walking

既而甲就柳樹斜行三十四步至柳樹下，丙就槐樹斜行一百五十三步至槐樹下。問答同前。

答曰：城徑二百四十步，虛弦一百二步。

法曰：雲數相乘，倍之便為平方實，開方得虛弦一百二步。以此弦加甲行步即極勾，以此弦加丙行步即極股。餘各依法求之。

草曰：識別：甲斜行即弦也，丙斜行即明弦也。立天元一為虛弦。置甲乙共以天元減之，得 p 為虛和也。以天元減甲乙共餘 p 即虛勾虛股共也。別置虛弦加甲斜行得 p 即極勾，虛弦加丙斜行得 p 即極股也。極勾極股相乘得 p ，以天元虛弦除之得 p 為極弦也。以自增乘得 p 為極弦冪（寄左）。乃以極勾自之得 p 為勾冪，極股自之得 p 為股冪，二冪相並得 p 為同數，與左相消得 p 。開平方得一百二步，即虛弦也。合問。

第一百二十七問

或問：東門南有柳一株，南門東有槐一株。甲出東門直行，丙出南門直行，二人遙相望，槐柳與城邊悉相直。既而甲復斜行至柳樹下，丙復斜行至槐樹下，各不知步數。只雲丙共行了二百八十八步，甲斜行與柳至東門步共得六十四步。問答同前。

答曰：甲直行一十六步，城徑二百四十步。

法曰：二雲數相乘於上，以六十四步自之，又二之，減上位為平方實。四之六十四於上，倍丙行內減上位為從。二十常法。得甲直行步一十六。

草曰：別得丙共步即明股、明弦和也，六十四即平勾也，內甲斜行即弦也，柳至東門步即股也。又二雲數相並即明差與極弦共也，二雲數相減即明差與平勾高股差共也。又平勾內減勾即虛勾也。

立天元一為勾。置丙共步以天元乘之，復以六十四除之得 p 為明勾也。又以天元減於六十四，得 p 為虛勾也，並虛明二勾 p 為半徑也。以自之得

straight; Bing goes out the South Gate walking straight. Jia, Bing, the willow, and the locust are all aligned with the city. Then Jia walks diagonally 34 paces to the willow; Bing walks diagonally 153 paces to the locust. Answer as before.

A: City diameter: 240 paces. Void hypotenuse: 102 paces.

Method: Multiply the given numbers; double as the quadratic shi. Extract the square root to obtain the void hypotenuse: 102 paces. This hypotenuse plus Jia's walk gives the extreme gou; plus Bing's walk gives the extreme gu. The rest follows by method.

Working: Identification: Jia's diagonal is the hypotenuse; Bing's diagonal is the bright hypotenuse.

Let the unknown be the void hypotenuse. Subtract from the Jia-Yi sum to get p as void sum. The remainder p is the sum of void gou and void gu. Add the void hypotenuse to Jia's diagonal to get p as extreme gou; add to Bing's diagonal to get p as extreme gu. Multiply to get p ; divide by the void hypotenuse to get p as extreme hypotenuse. Squaring gives p as extreme hypotenuse power (stored left). Square the extreme gou to get p as gou power; square extreme gu to get p as gu power. Adding gives p as matching number. Canceling gives p . Extract the square root to obtain 102 paces, the void hypotenuse. Q.E.D.

Problem 一百二十七

Q: South of the East Gate is a willow; east of the South Gate is a locust. Jia goes out the East Gate walking straight; Bing goes out the South Gate walking straight; they gaze at each other from afar. The locust and willow are aligned with the city edge. Then Jia walks diagonally to the willow; Bing walks diagonally to the locust; each distance is unknown. It is only stated that Bing walked a total of 288 paces; Jia's diagonal plus the willow-to-East-Gate distance equals 64 paces. Answer as before.

A: Jia's straight walk: 16 paces. City diameter: 240 paces.

Method: Multiply the two given numbers as upper term. Square 64 and double it; subtract from upper term as flat shi. Quadruple 64 as upper term; subtract from double Bing's walk as cong. Twenty as constant method. Extracts Jia's straight walk: 16 paces.

Working: Identification: Bing's total is the sum of bright gu and bright hypotenuse; 64 is the level gou; Jia's diagonal is the hypotenuse; the willow-to-East-Gate distance is the gu. The sum of the two given numbers is the bright difference plus extreme hypotenuse; their difference is the bright difference plus the difference between level gou and high gu. Level gou minus gou is void gou.

p ，倍之得 p 為半段圓城徑冪（寄左）。乃以天元加六十四，得 p 為勾圓差於上；又以明勾加丙共步，得 p 為股圓差於下。上下相乘得 p 為同數，與左相消得 p 。開平方得一十六步，即勾也。此勾乃甲出東門直行步也。餘皆依數求之。合問。

第一百二十八問

或問：東門南有柳樹一株，南門東有槐樹一株。甲出東門直行，丙出南門直行，二人遙相望，槐柳與城邊悉相直。既而甲復斜行至柳樹下，丙復斜行至槐樹下，各不知步數。只雲甲共行五十步，丙斜行與槐至南門步共得二百二十五步。問答同前。

答曰：城徑二百四十步，丙直行一百三十五步。

法曰：以二百二十五步自之為冪，又以此冪自為冪於上。置甲共行以二百二十五步三度乘之，得數復折半，減上位為平實。置二百二十五步自之數，以二雲數相減數乘之，又倍之於上。倍五十步在地，以二百二十五步自之數乘之，復折半加上位為益從。雲數相減自乘於上，以雲數相乘，復折半，減上位為常法。得明股。

草曰：識別得甲共步即勾、弦共也，二百二十五即高股也，內丙斜行即明弦，槐至南門步即明勾也。又二雲數相並即極弦內減一個差也，雲數相減即差與高股、平勾差共也。又高股內減明股即虛股也。

立天元一為明股，即丙出南門直行步也。置五十步以天元乘之得 p 元，合高股除。不除，便以此一零零元為通股（內帶高股為母）。以天元加高股，得 p 即大差也。置大差以高股分母乘之，得 p ，即帶分大差也。以此減於通股，餘 p 即圓徑也。以自增乘，得 p （寄左。內帶高股冪分母）。然後置一千以高股分母通之，得 p 太，內減帶分大差，得 p 為兩個通勾也。內減兩個圓徑得 p ，為兩個小差也。以帶分大差乘之，得下式 p 為同數，與左相消得 p 。開平方得一百三十五步，即明股也。合問。

Let the unknown be the gou. Multiply Bing's total by the unknown and divide by 64 to get p as bright gou. Subtract the unknown from 64 to get p as void gou; add both void and bright gou p as radius. Squaring and doubling gives p as half the city diameter power (stored left). Add the unknown to 64 to get p as gou circle-difference (upper); add bright gou to Bing's total to get p as gu circle-difference (lower). Multiply to get p as matching number. Canceling gives p . Extract the square root to obtain 16 paces, the gou—this is Jia's straight walk out the East Gate. Q.E.D.

Problem 一百二十八

Q: South of the East Gate is a willow; east of the South Gate is a locust. Jia goes out the East Gate walking straight; Bing goes out the South Gate walking straight; they gaze at each other from afar. The locust and willow are aligned with the city edge. Then Jia walks diagonally to the willow; Bing walks diagonally to the locust; each distance is unknown. It is only stated that Jia walked a total of 50 paces; Bing's diagonal plus the locust-to-South-Gate distance equals 225 paces. Answer as before.

A: City diameter: 240 paces. Bing's straight walk: 135 paces.

Method: Square 225 as power; use this power squared as upper term. Raise Jia's total by 225 to the third power; halve and subtract from upper term as flat shi. Square 225; multiply by the difference of the two given numbers; double as upper term. Place double 50; multiply by the squared 225; halve and add to upper term as increasing cong. Square the difference of the given numbers as upper term; halve the product of the given numbers and subtract as constant method. Extracts the bright gu.

Working: Identification: Jia's total is the sum of gou and hypotenuse; 225 is the high gu; Bing's diagonal is the bright hypotenuse; the locust-to-South-Gate distance is the bright gou. The sum of the two given numbers is the extreme hypotenuse minus one difference; their difference is the difference plus the difference between high gu and level gu. High gu minus bright gu is void gu.

Let the unknown be the bright gu, Bing's straight walk out the South Gate. Multiply 50 by the unknown to get p ; should divide by high gu but do not; take this as the general gu (with high gu as denominator). Add the unknown to the high gu to get p as large difference. Multiply by the high gu denominator to get p as the scaled large difference. Subtract from the general gu; the remainder p is the circle diameter. Squaring gives p (stored left, with squared high gu as denominator). Scale 1000 by the high gu denominator to get p ; subtract the scaled large difference to get p as twice the general gou. Subtract twice the

第一百二十九問

或問：通勾、通弦共一千步，勾、弦共五十步。問答同前。

答曰：城徑二百四十步，股三十步。

法曰：置一千減二之五十步為泛率。以自乘，復半之於上。又置泛率復以五十乘之，加上位為平實。二十二之泛率於上。以四十二乘五十，得數內減泛率，加上位為益從。二百為常法。得股。

草曰：立天元一為股。置一千以天元乘之，以五十除之，得 p 元為通股也。又以天元加五十步，得 p 即小差也，通股加小差得 p 即通弦也。以通弦減一千得 p ，即通勾也。以小差減通勾得 p ，即圓徑也。以圓徑減通股得 p 即大差也。置大差以小差乘之得 p （寄左）。然後置圓徑以自之得 p ，折半得 p ，與左相消得 p 。開平方得三十步，即股也。合問。

第一百三十問

或問：通勾、通弦共一千步，明勾、明弦共二百二十五步。問答同前。

答曰：城徑二百四十步，明股一百三十五步。

法曰：以後數再自乘，又以前數乘之為平實；以後數為冪，復以前數乘之為從；以前數冪為虛常法。得明股。

草曰：別得二百二十五步即高股也。立天元一為明股。置一千以天元乘之，合以高股除不受除，便以此一零零元為通股（內帶高股為母）。以天元加高股，得 p 即大差也。置大差以高股分母乘之，得 p ，即帶分大差也。以此減於通股，餘 p 即圓徑也。以自增乘，得 p （寄左。內帶高股冪分母）。然後置一千以高股分母通之，得 p 太，內減帶分大差，得 p 為兩個通勾也。內減兩個圓徑得 p ，為兩個小差也。以帶分大差乘之，得下式 p 為同數，與左相消得 p 。開平方得一百三十五步，即明股也。合問。

circle diameter to get p as twice the small difference. Multiply by the scaled large difference to get p as matching number. Canceling gives p . Extract the square root to obtain 135 paces, the bright gu. Q.E.D.

Problem 一百二十九

Q: General gou plus general hypotenuse total 1000 paces; gou plus hypotenuse total 50 paces. Answer as before.

A: City diameter: 240 paces. Gu: 30 paces.

Method: Subtract double 50 from 1000 as the general rate. Square and halve as upper term. Multiply the general rate by 50 and add to upper term as flat shi. Twenty-two times the general rate as upper term. Multiply 42 by 50; subtract the general rate and add to upper term as increasing cong. Two hundred as constant method. Extracts the gu.

Working: Let the unknown be the gu. Multiply 1000 by the unknown and divide by 50 to get p as general gu. Add 50 to the unknown to get p as small difference; general gu plus small difference gives p as general hypotenuse. Subtract from 1000 to get p as general gou. Subtract the small difference from general gou to get p as circle diameter. Subtract the circle diameter from general gu to get p as large difference. Multiply large difference by small difference to get p (stored left). Square the circle diameter and halve to get p as matching number. Canceling gives p . Extract the square root to obtain 30 paces, the gu. Q.E.D.

Problem 一百三十

Q: General gou plus general hypotenuse total 1000 paces; bright gou plus bright hypotenuse total 225 paces. Answer as before.

A: City diameter: 240 paces. Bright gu: 135 paces.

Method: Cube the latter number and multiply by the former as flat shi. Square the latter and multiply by the former as cong. The squared former as empty constant method. Extracts the bright gu.

Working: Identification: 225 paces is the high gu.

Let the unknown be the bright gu. Multiply 1000 by the unknown; do not divide by the high gu; take this as the general gu (with high gu as denominator). Add the unknown to the high gu to get p as large difference. Multiply by the high gu denominator to get p as the scaled large difference. Subtract from general gu; remainder p is the circle diameter. Squaring gives p (stored left, with squared high gu denominator). Scale 1000 by the high gu denominator to get p ; subtract the scaled large difference to get p as twice the general gou. Subtract twice the circle diameter to get p as twice the small difference. Multiply by the scaled large difference to get p as matching number. Can-

第一百三十一問

或問：通股、通弦共一千二百八十步，股、弦共六十四步。問答同前。

答曰：城徑二百四十步，勾一十六步。

法曰：雲數相乘為平實，前數為益從。置前數以後數除之，得二十為泛率。泛率減一以自乘於上，又倍泛率減一加上位為常法。倒積開得勾一十六。

草曰：別得六十四步即平勾也。立天元一為勾。置前數以天元乘之，以後數除之，得 p 元即通勾也。又置天元加後數，得 p 即小差也。以小差減通勾，餘 p 即圓徑也。以自之得 p (寄左)。然後以小差減於前數，得 p 為二通股。內減兩個圓徑，得 p 為二大差也。以小差乘之得下 p ，與左相消得 p 。開平方得一十六步，即勾也。合問。

第一百三十二問

或問：通股、通弦共一千二百八十步，明股、明弦共二百八十八步。問答同前。

答曰：城徑二百四十步，明勾七十二步。

法曰：二數相減，以後數乘之，內減後數羈，又半之為泛率。以自之為平實。置前數加二之後數而半之為次率，以乘泛率，倍之於上，以後數乘泛率減上位為益從。次率自乘於上，以前數加次率，復以後數乘之，減上位為隅法。得明勾。

草曰：別得二數相減，餘 p 為通勾、通股及明勾共也。立天元一為明勾。置前數以天元乘之，合以後數除之。不除，便以此 p 元為通勾也 (內寄後數分母)。又以二數相減，得數內又減天元得 p 為通和也。乃以分母二百八十八之，得下式 p ，內減通勾，餘 p 為通股也。又以天元加後數，又以分母 (即後數也) 通之，得 p 為大差也。以此大差減於通股，得下式 p ，為一個圓徑也。半之得 p ，以自之得 p 為半徑羈 (寄左)。然後以半圓徑減通勾，得 p 為底勾，又以天元乘之，又以分母二百八十八之，得 p

celing gives p . Extract the square root to obtain 135 paces, the bright gu. Q.E.D.

Problem 一百三十一

Q: General gu plus general hypotenuse total 1280 paces; gu plus hypotenuse total 64 paces. Answer as before.

A: City diameter: 240 paces. Gou: 16 paces.

Method: Multiply the given numbers as flat shi; the former as increasing cong. Divide the former by the latter to get 20 as general rate. Square (general rate minus one) as upper term; add double the general rate minus one as constant method. Extract by inverted accumulation to obtain the gou: 16.

Working: Identification: 64 paces is the level gou.

Let the unknown be the gou. Multiply the former by the unknown and divide by the latter to get p as general gou. Add the latter to the unknown to get p as small difference. Subtract the small difference from general gou; remainder p is the circle diameter. Squaring gives p (stored left). Subtract the small difference from the former to get p as twice the general gu. Subtract twice the circle diameter to get p as twice the large difference. Multiply by the small difference to get p as matching number. Canceling gives p . Extract the square root to obtain 16 paces, the gou. Q.E.D.

Problem 一百三十二

Q: General gu plus general hypotenuse total 1280 paces; bright gu plus bright hypotenuse total 288 paces. Answer as before.

A: City diameter: 240 paces. Bright gou: 72 paces.

Method: Subtract the two numbers; multiply by the latter; subtract the squared latter; halve as general rate. Square it as flat shi. Add the former to double the latter and halve as secondary rate. Multiply by the general rate; double as upper term. Multiply the latter by the general rate and subtract from upper term as increasing cong. Square the secondary rate as upper term; add the former to the secondary rate; multiply by the latter and subtract as corner method. Extracts the bright gou.

Working: Identification: The difference of the two numbers p is the sum of general gou, general gu, and bright gou.

Let the unknown be the bright gou. Multiply the former by the unknown; do not divide by the latter; take p as general gou (with latter as denominator). Subtract the two numbers; subtract the unknown from the result to get p as general sum. Apply denominator 288 to get p ; subtract general gou; remainder p is general gu. Add the unknown to the latter; apply the denominator (the latter) to get p as

為同數，與左相消得 p 。開平方得七十二步，即明勾也。合問。

第一百三十三問

或問：明股、明弦並二百八十八步，勾、弦並五十步。又雲明股、勾並多於虛弦四十九步。問答同前。

答曰：城徑二百四十步。

法曰：前二數相並，內減二之多步，即圓徑。又只以前二數相乘，便是半徑冪。

草曰：識別得前二數相減而半之，即極差也。其多步名傍差，又為圓徑不及極弦數。立天元一為半徑。置前二數相乘得 p ，以天元除之得 p 為極弦也。以天元加多步得 p 為虛弦。極弦內減虛弦餘 p 為旁差，與多步相應。又以極弦自之得 p ，內減虛弦冪 p ，餘 p 為二積。以天元除之得 p 為勾股和，即前二數之並也，以此為同數。與左相消得 p ，開平方得一百二十步，即半徑也。倍之得二百四十步，即城徑也。合問。

第一百三十四問

或問：平差、高差共一百六十一，明股、勾並多於虛弦四十九步。問答同前。

答曰：城徑二百四十步，平勾六十四步。

法曰：二數相減又半之，以自乘為實，後數為法。得平勾。

草曰：立天元一為平勾。以加前數得 p 為高股也。又以天元加高股得 p 為極弦，內減後數得 p ，又半之，得 p 為半徑。以自之，得 p （寄左）。然後以天元乘高股，得 p 為同數，與左相消得 p 。上法下實，得六十四步，即平勾也。合問。

第一百三十五問

large difference. Subtract from general gou to get p as circle diameter. Halve and square to get p as radius power (stored left). Subtract half the diameter from general gou to get p as base gou; multiply by the unknown and apply denominator 288 to get p as matching number. Canceling gives p . Extract the square root to obtain 72 paces, the bright gou. Q.E.D.

Problem 一百三十三

Q: Bright gu plus bright hypotenuse total 288 paces; gou plus hypotenuse total 50 paces. Also stated: the sum of bright gu and gou exceeds the void hypotenuse by 49 paces. Answer as before.

A: City diameter: 240 paces.

Method: Add the first two numbers; subtract double the excess to get the circle diameter. Alternatively, simply multiply the first two numbers to obtain the radius squared.

Working: Identification: Halve the difference of the first two numbers to get the extreme difference. The excess is called the side difference, also the amount by which the circle diameter falls short of the extreme hypotenuse.

Let the unknown be the radius. Multiply the first two numbers to get p ; divide by the unknown to get p as extreme hypotenuse. Add the excess to the unknown to get p as void hypotenuse. Subtract void hypotenuse from extreme hypotenuse; remainder p is the side difference, matching the excess. Square the extreme hypotenuse and subtract the squared void hypotenuse p ; remainder p is twice the area. Divide by the unknown to get p as gou-gu sum, which is the sum of the first two numbers, as matching number. Canceling gives p . Extract the square root to obtain 120 paces, the radius. Double it to get 240 paces, the city diameter. Q.E.D.

Problem 一百三十四

Q: Level difference plus high difference total 161 paces; the sum of bright gu and gou exceeds the void hypotenuse by 49 paces. Answer as before.

A: City diameter: 240 paces. Level gou: 64 paces.

Method: Subtract the two numbers; halve; square as shi; the latter number as divisor. Extracts the level gou.

Working: Let the unknown be the level gou. Add the former number to get p as high gu. Add the unknown to the high gu to get p as extreme hypotenuse; subtract the latter number to get p ; halve to get p as radius. Squaring gives p (stored left). Multiply the unknown by the high gu to get p as matching number. Canceling gives p . Dividing yields 64 paces, the level gou. Q.E.D.

Problem 一百三十五

或問：平勾、高股差一百六十一歩，明差、差並七十七歩。又雲極弦多於城徑四十九歩。問答同前。

答曰：城徑二百四十歩，半徑一百二十歩。

法曰：並上二位而半之為平率。其四十九即旁率也。副置平率，上加旁率，下減旁率，以相乘為實。倍旁差為法。得勾圓差。

草曰：立天元一為半徑，又為半之股圓差上弦較較，又為半之勾圓差上弦較和也。內減勾圓差上勾股較 p ，餘 p 為半之勾圓差上弦較較也。置股圓差上勾股較 p ，以半之勾圓差上弦較較乘之，得 p （寄左）。然後以半之股圓差上弦較較乘勾圓差上勾股較，得 p 元為同數，與左相消得下式 p 。上法下實，得一百二十歩，即半徑也。合問。

第一百三十六問

或問：又依前問：見角差一百六十一歩，見明差差並七十七歩，又見太虛弦較較六十歩。問答同前。

答曰：城徑二百四十歩，平勾六十四歩。

法曰：前二數相減而半之，得數加入半之太虛弦較較為泛率，以自乘為平實。置一百六十一內減二之泛率為從，一常法。得平勾。

草曰：別得 p 即二股也。立天元一為平勾。先以前二數相減而半之，得 p 為虛差。以虛差加股得 p ，即明勾也。以明勾加天元得 p ，為平弦，以自之得 p ，內減天元冪，得 p 為半徑冪（寄左）。然後以天元加一百六十一為高股，以天元乘之，得 p 為同數，與左相消得 p 。開平方得六十四歩，即平勾也。合問。

第一百三十七問

或問：高差、平差並一百六十一歩，明差、差並七十七歩。問答同前。

答曰：城徑二百四十歩。

法曰：以前數自乘於上，二數相並而半之，以自乘減

Q: The difference between level gou and high gu is 161 paces; bright difference plus difference totals 77 paces. Also stated: the extreme hypotenuse exceeds the city diameter by 49 paces. Answer as before.

A: City diameter: 240 paces. Radius: 120 paces.

Method: Add the two upper numbers and halve as the level rate. The 49 is the side rate. Place the level rate in two positions; add the side rate above; subtract below. Multiply as shi. Double the side difference as divisor. Extracts the gou circle-difference.

Working: Let the unknown be the radius; also half the gu circle-difference sum-of-differences; also half the gou circle-difference sum-of-sums. Subtract the gou-gu difference on the gou circle-difference p ; remainder p is half the gou circle-difference difference-of-differences. Place the gu circle-difference gou-gu difference p ; multiply by half the gou circle-difference difference-of-differences to get p (stored left). Multiply half the gu circle-difference difference-of-differences by the gou circle-difference gou-gu difference to get p as matching number. Canceling gives p . Dividing yields 120 paces, the radius. Q.E.D.

Problem 一百三十六

Q: As before: the corner difference is 161 paces; the bright-difference sum is 77 paces; also the grand void hypotenuse difference-of-differences is 60 paces. Answer as before.

A: City diameter: 240 paces. Level gou: 64 paces.

Method: Subtract the first two numbers; halve; add half the grand void hypotenuse difference-of-differences as general rate. Square as flat shi. Subtract double the general rate from 161 as cong. One constant method. Extracts the level gou.

Working: Identification: p is twice the gu.

Let the unknown be the level gou. First subtract the first two numbers and halve to get p as void difference. Add to the gu to get p as bright gou. Add the unknown to get p as level hypotenuse; squaring and subtracting the unknown squared gives p as radius power (stored left). Add 161 to the unknown as high gu; multiply by the unknown to get p as matching number. Canceling gives p . Extract the square root to obtain 64 paces, the level gou. Q.E.D.

Problem 一百三十七

Q: High difference plus level difference total 161 paces; bright difference plus difference total 77 paces. Answer as before.

A: City diameter: 240 paces.

Method: Square the former as upper term. Add both

上位，得數復自增乘為平實。前數自之於上，又以四之前數乘之，寄位。以前數自之於上，並二數而半之，以自乘減上位，得數又以四之前數乘，又倍之，減於寄位為從。前數自之，又四之於上。又以四之前數為冪，加上位，權寄。以前數為冪於上，並二數而半之，以自乘減上位，得數復八之於上。又以四之前數為冪加入上位，並以減於權寄為常法。得平勾。

草曰：識別得二位相並而半之，得 p ，即極差也。立天元一為平勾，加一百六十一得 p 為高股，高股內又加天元，得 p 為極弦。以自之得 p 於上，內減極差冪一萬四千一百六十一，餘 p 為兩段極積。合以極弦除，不除寄為母，便以此為城徑。以自增乘得 p 為圓徑冪（內有極弦冪分母。寄左）。然後以天元乘高股，又四之得 p ，又以分母極弦冪 p 通之，得 p 為同數，與左相消得 p 。開平方得六十四步，即平勾也。合問。

第一百三十八問

或問：見明和二百七步，和四十六步。問答同前。

答曰：城徑二百四十步，小差四步。

法曰：二和上下相減，數同則止，名為泛率。又以二和直相減，餘為泛實（此則角差也）。乃以泛率除泛實，所得為差率也。以差率加減泛率，若半訖，與勾股相應者，其泛率便為和率，其泛實便為較率乘和率也。若不相應，則直取差率以消息之，定為相管和率。乃以和率約二和，其明和所得為明壘率，其和所得為壘率也。又副置和率，上加差率而半之，則為股率也；下位減差率而半之，則為勾率也。既見勾、股及差三率，各以壘率乘之，即各得勾、股及差之真數也。

草曰：以二和相約分得率一、明率四步半，其兩數大小差率並同。又別得明小差、大差俱為半虛黃也。立

numbers; halve; square; subtract from upper term; self-multiply the result as flat shi. Square the former as upper term; multiply by quadruple the former; store. Square the former as upper term; add both numbers; halve; square; subtract; multiply by quadruple the former; double; subtract from stored as cong. Square the former; quadruple as upper term. Add quadruple former squared; provisionally store. Square the former as upper term; add both numbers; halve; square; subtract; octuple; add quadruple former squared; subtract from provisional store as constant method. Extracts the level gou.

Working: Identification: Adding both numbers and halving gives p , the extreme difference.

Let the unknown be the level gou; add 161 to get p as high gu; add the unknown to get p as extreme hypotenuse. Squaring gives p as upper term; subtract the squared extreme difference 14,161; remainder p is twice the extreme area. Should divide by extreme hypotenuse but do not; take this as city diameter. Squaring gives p as circle diameter power (with extreme hypotenuse squared as denominator, stored left). Multiply the unknown by high gu and quadruple to get p ; apply denominator extreme hypotenuse squared p to get p as matching number. Canceling gives p . Extract the square root to obtain 64 paces, the level gou. Q.E.D.

Problem 一百三十八

Q: The bright sum is 207 paces; the sum is 46 paces. Answer as before.

A: City diameter: 240 paces. Small difference: 4 paces.

Method: Subtract the two sums; if the numbers match, stop—this is the general rate. Directly subtract the two sums; the remainder is the general shi (this is the corner difference). Divide the general shi by the general rate to obtain the difference rate. Add/subtract the difference rate to/from the general rate; if halving matches gou-gu, the general rate becomes the sum rate and the general shi becomes the product of difference rate and sum rate. If not matching, adjust using the difference rate to determine the sum rate. Divide both sums by the sum rate; the bright sum gives the bright pile-rate; the sum gives the pile-rate. Place the sum rate in two positions; add the difference rate above and halve for the gu rate; subtract below and halve for the gou rate. Once the three rates (gou, gu, difference) are found, multiply each by the pile rate to obtain the true numbers.

Working: Reducing the two sums gives rate 1 and bright rate 4.5; both numbers share the same large and small difference rates. The bright small difference and large differ-

天元一為小差，以四步半乘之得 p 為大差也，又為明小差，又為半虛黃。置此大差又以四步半乘之得 p 為明大差也。其四差相並得 p ，減於二和並，得 p 即兩段太虛大小差並也。內加三段虛黃方 p ，得 p ，合成一個太虛三事和，即圓城徑也。以自增乘得 p 為徑幂（寄左）。乃置和加半虛黃，得 p 為平勾。又置明和內加半虛黃，得 p 為高股，勾股相乘得下式 p ，又四之得 p 為同數，與左相消得下式 p 。開平方得四步，即小差也。合問。

ence are both half the void yellow.

Let the unknown be the small difference; multiply by 4.5 to get p as large difference, also bright small difference, also half the void yellow. Multiply this large difference by 4.5 to get p as bright large difference. The four differences sum to p ; subtract from both sums to get p as twice the grand void large and small differences. Add three times the void yellow square p to get p , forming one grand void three-element sum, i.e., the city diameter. Squaring gives p as diameter power (stored left). Add half the void yellow to the sum to get p as level gou. Add half the void yellow to the bright sum to get p as high gu. Multiply to get p ; quadruple to get p as matching number. Canceling gives p . Extract the square root to obtain 4 paces, the small difference. Q.E.D.

2.2 卷九 大斜四問、大和八問

Vol. 9: 4 Oblique + 8 Sum Problems

卷九小序：此卷凡十二問，分為二節。上節曰「大斜四問」，以通勾股形之斜邊為主，方程式之高有至五乘、六乘者，實全書開方之極致。下節曰「大和八問」，以通勾股形之三事和為主，其術益奧，其變益奇。學者能精於此，則天元之術思過半矣。

Volume 9 Preface: These 12 problems divide into two sections. The upper “Great Oblique” section uses the hypotenuse of the general right triangle, with equations reaching degrees 5–6—the pinnacle of root extraction in the entire work. The lower “Great Sum” section uses the three-element sum of the general triangle; the methods are ever more profound, the variations ever more marvelous. Mastery of this volume means more than half the tian yuan technique is understood.

2.3 細草卷九上 大斜四問

Part A: Four Great Oblique Problems

第一百三十九問

Problem 一百三十九

或問：甲丙俱在中心，丙望南門直行，不知步數而止。甲出東門直行，不知步數望見丙，斜行與丙相會。二人共行了六百八十步，仍雲甲直行少於丙直行一百一十九步。問答同前。

Q: Jia and Bing are both at the center. Bing gazes toward the South Gate and walks straight, stopping at an unknown distance. Jia goes out the East Gate walking straight; at an unknown distance he sees Bing, then walks diagonally to meet him. Together they walked 680 paces; Jia’s straight walk is 119 paces less than Bing’s. Answer as before.

答曰：城徑二百四十步，皇極勾一百三十六步。

A: City diameter: 240 paces. Imperial extreme gou: 136 paces.

法曰：二數相減，餘以為冪，內卻減差冪為平實；二數相減又四之於上，又加入二之差步為益從；二步常法。得皇勾一百三十六。

Method: Subtract the two numbers; square the remainder; subtract the squared difference as flat shi. Quadruple the difference as upper term; add double the difference as increasing cong. Two as constant method. Extracts the imperial extreme gou: 136.

草曰：別得共步即皇極三事和，少步即勾股差也。立天元一為皇極勾。加少步得 p 為股也，又以天元加股得 p 為和也。以和減共步得 p 為弦也，弦自之得 p 為一段弦冪（寄左）。然後置股以天元乘之，又倍之，得 p 為二直積，如入少步冪 p 共得 p 為同數，與左相消得 p 。平方而一，得一百三十六即勾也。勾加差為股，勾股相乘，倍之為實，勾股和減共步為法。得城徑。

Working: Identification: The total is the imperial extreme three-element sum; the shortfall is the gou-gu difference.

Let the unknown be the imperial extreme gou. Add the shortfall to get p as gu; add the unknown to the gu to get p as sum. Subtract the sum from the total to get p as hypotenuse; squaring gives p as hypotenuse power (stored left). Multiply the gu by the unknown and double to get p as twice the rectangular area. Add the squared shortfall p for total p as matching number. Canceling gives p . Extract the square root to obtain 136, the gou. Gou plus difference is gu; multiply gou and gu; double as shi; subtract the sum from the total as divisor to obtain the city diameter. Q.E.D.

第一百四十問

Problem 一百四十

或問：甲丙俱在西北隅起，丙向南行不知步數而立。甲向東行望見丙，就丙斜行六百八十步與丙相會。丙雲：「我南行步多於甲東行二百八十步。」問答同前。

Q: Jia and Bing both start at the northwest corner. Bing walks south an unknown number of paces and stands. Jia walks east, sees Bing, then walks diagonally 680 paces to meet him. Bing says: “My southward walk exceeds Jia’s eastward walk by 280 paces.” Answer as before.

答曰：城徑二百四十步，大勾三百二十步。

法曰：以雲數差乘雲數並為實，倍多步為從，二為平隅。得大勾三百二十。

草曰：立天元為大勾。加差得 p 為股，倍天元乘之，得 p 為二積（寄左）。然後以斜步、多步並 p 與斜步、多步較 p 相乘，得 p 為同數，與左相消得 p 。開平方得三百二十步，即大勾也。合問。

第一百四十一問

或問：甲乙二人共立於艮隅，乙南行過城門而立，甲東行望乙與城參相直而止。丙丁二人共立於坤隅，丁向東行過城門而立，丙向南行望丁及甲乙悉與城俱相直。丙復就甲斜行六百八十步與甲相會。乙、丁又雲：「吾二人直行共得三百四十二步。」問答同前。

答曰：城徑二百四十步。

法曰：二雲數相乘，倍之為實；倍斜行於上，以二雲數相減加上位為從；一步常法。開平方，得城徑。

草曰：別得斜步即大弦也。其共步則一徑一虛弦共也，其二數相並為一大和一虛弦共數也。立天元為徑，減於共步得 p 為虛弦也。以虛弦復減於天元，得 p 為虛和，以斜步乘之，得 p （寄左）。乃以天元加斜步，得 p 為大和，以虛弦乘之得 p 為同數，與左相消得 p 。開平方得二百四十步，即城徑也。合問。

第一百四十二問

或問：甲從北門向東直行，庚從西門穿城東行，丙從西門向南直行，壬從北門穿城南行。四人遙相望，悉與城參相直。只雲甲丙相望處六百八十步，庚壬穿城共行了六百三十一。問答同前。

A: City diameter: 240 paces. Large gou: 320 paces.

Method: Multiply the difference of the given numbers by their sum as shi; double the excess as cong; two as the quadratic corner. Extracts the large gou: 320.

Working: Let the unknown be the large gou. Add the difference to get p as gu; multiply by double the unknown to get p as twice the area (stored left). Multiply the sum of diagonal and excess p by their difference p to get p as matching number. Canceling gives p . Extract the square root to obtain 320 paces, the large gou. Q.E.D.

Problem 一百四十一

Q: Jia and Yi stand together at the Gen (northeast) corner. Yi walks south past the gate and stands; Jia walks east, sees Yi aligned with the city, and stops. Bing and Ding stand together at the Kun (southwest) corner. Ding walks east past the gate and stands; Bing walks south, sees Ding and Jia/Yi all aligned with the city. Bing then walks diagonally 680 paces to meet Jia. Yi and Ding also state: "Our two straight walks total 342 paces." Answer as before.

A: City diameter: 240 paces.

Method: Multiply the two given numbers; double as shi. Double the diagonal as upper term; add the difference of the two given numbers as cong. One constant method. Extract the square root to obtain the city diameter.

Working: Identification: The diagonal is the large hypotenuse. The sum is one diameter plus one void hypotenuse. The sum of the two numbers is one large sum plus one void hypotenuse.

Let the unknown be the diameter; subtract from the sum to get p as void hypotenuse. Subtract the void hypotenuse from the unknown to get p as void sum; multiply by the diagonal to get p (stored left). Add the unknown to the diagonal to get p as large sum; multiply by the void hypotenuse to get p as matching number. Canceling gives p . Extract the square root to obtain 240 paces, the city diameter. Q.E.D.

Problem 一百四十二

Q: Jia walks east from the North Gate; Geng walks east through the city from the West Gate; Bing walks south from the West Gate; Ren walks south through the city from the North Gate. The four gaze at each other from afar, all aligned with the city. It is only stated that Jia and Bing are 680 paces apart; Geng and Ren together walked 631 paces through the city. Answer as before.

答曰：城徑二百四十步。

法曰：共步自之得數。以共步減斜，餘自乘以減上為實。二之斜步加入共步減斜，餘數為從。一步常法。得城徑。

草曰：共行步為一徑與皇和共也，又為大和皇弦差也。甲丙相望即大弦也。以共步減大弦，餘 p 為皇極弦上減一徑也。立天元一為圓徑，減於共步，得 p 為皇極和也，以自之得 p 於上。弦內減共步餘 p ，又以天元加之為皇弦，以自之得 p ，減上位，餘得 p 為兩個皇直積（寄左）。乃以天元乘皇弦得下式 p 為同數，與左相消得 p 。平方而一，得二百四十步，即城徑也。合問。

A: City diameter: 240 paces.

Method: Square the sum. Subtract the sum from the diagonal; square the remainder; subtract from the upper term as shi. Double the diagonal and add the remainder of sum subtracted from diagonal as cong. One constant method. Extracts the city diameter.

Working: Identification: The total walk is one diameter plus the imperial sum; also the large sum minus the imperial hypotenuse. Jia and Bing's distance is the large hypotenuse. Subtract the sum from the large hypotenuse; remainder p is the imperial extreme hypotenuse minus one diameter.

Let the unknown be the circle diameter. Subtract from the sum to get p as imperial extreme sum. Squaring gives p as upper term. Subtract the sum from the hypotenuse; remainder p ; add the unknown as imperial hypotenuse; squaring gives p . Subtract from upper term; remainder p is twice the imperial rectangular area (stored left). Multiply the unknown by the imperial hypotenuse to get p as matching number. Canceling gives p . Extract the square root to obtain 240 paces, the city diameter. Q.E.D.

2.4 細草卷九下 大和八問

Part B: Eight Great Sum Problems

第一百四十三問

或問：庚從西門穿城東行二百五十六步而立，壬從北門穿城南行三百七十五步而立。又有甲丙二人俱在乾隅，甲向東行，丙向南行，各不知步數而立。四人遙相望，只雲甲丙共行了九百二十步。問答同前。

答曰：城徑二百四十步。

法曰：庚東行羃、壬南行羃相並於上，並庚壬步而倍之，內減大和，餘復減於庚壬共，得數以自乘減上位為平實。並庚壬步為益從，半步為隅法。得城徑。

草曰：立天元一為圓徑，以半之副置二位。上以減於庚東行，得下 p 為平弦也；下以減於壬南行，得 p 為高弦也。二弦相並得 p 為皇弦、虛弦共也。倍此數得 p 為大弦、虛弦共也。以大弦、虛弦共減於大和，餘 p 為虛勾虛股共也。天元內減虛勾、虛股共，餘 p 即虛弦也。復置皇弦虛弦共內減虛弦，餘 p 即皇極弦也，以自之得 p (寄左)。然後以平弦自之，得下式 p 為勾羃也，又以高弦自之，得 p 為股羃也。二羃相並得 p 為同數，與左相消得 p 。平方而一，得二百四十步，即城徑也。合問。

第一百四十四問

或問：甲丙俱在西北隅，甲向東行不知步數而立，丙向南行望見甲，就甲斜行與甲相會。甲直行、丙直行共九百二十步 (甲步少於丙步)。又：出東門南行有柳樹一株，出南門東行有槐樹一株。戊己二人同在巽隅，戊就柳樹，己就槐樹，亦與甲、乙遙相望。只雲己行少於戊行數與兩樹相距數相並，得一百四十四步，其二數相減餘六十步。問答同前。

答曰：城徑二百四十步，太虛勾四十八步。

Problem 一百四十三

Q: Geng walks east through the city from the West Gate 256 paces and stands; Ren walks south through the city from the North Gate 375 paces and stands. Jia and Bing are both at the Qian (northwest) corner; Jia walks east and Bing walks south, each at an unknown distance. The four gaze at each other from afar. It is only stated that Jia and Bing together walked 920 paces. Answer as before.

A: City diameter: 240 paces.

Method: Square Geng's east walk and Ren's south walk; add as upper term. Add Geng and Ren's walks; double; subtract the large sum; subtract again from Geng-Ren sum; square and subtract from upper term as flat shi. Geng-Ren sum as increasing cong. Half a step as corner method. Extracts the city diameter.

Working: Let the unknown be the circle diameter; halve and place in two positions. Subtract from Geng's east walk above to get p as level hypotenuse; subtract from Ren's south walk below to get p as high hypotenuse. The two hypotenuses sum to p as imperial and void hypotenuses combined. Doubling gives p as large and void hypotenuses combined. Subtract from the large sum to get p as the sum of void gou and void gu. Subtract from the unknown to get p as void hypotenuse. Subtract the void hypotenuse from the imperial-void sum; remainder p is the imperial extreme hypotenuse. Squaring gives p (stored left). Square the level hypotenuse to get p as gou power; square the high hypotenuse to get p as gu power. Adding gives p as matching number. Canceling gives p . Extract the square root to obtain 240 paces, the city diameter. Q.E.D.

Problem 一百四十四

Q: Jia and Bing are both at the northwest corner; Jia walks east an unknown number of paces and stands; Bing walks south, sees Jia, then walks diagonally to meet him. Jia's straight walk plus Bing's straight walk total 920 paces (Jia's walk is less than Bing's). Also: south of the East Gate is a willow; east of the South Gate is a locust. Wu and Ji are both at the Xun (south-east) corner; Wu goes to the willow and Ji goes to the locust, also gazing at Jia and Yi from afar. It is only stated that Ji's walk is less than Wu's walk; the difference plus the distance between the two trees equals 144 paces; their difference is 60 paces. Answer as before.

A: City diameter: 240 paces. Grand void gou: 48 paces.

法曰：二雲數相並而半之為虛弦，以乘大和九百二十步於上。以一百四十四減大和，以虛較乘之，減上位為平實。以一百四十四減大和，又二之於上，以二之虛較減上位為從。四虛隅。得太虛勾四十八。

草曰：別得甲丙直行共即大和也，戊就柳樹步即虛股也，己就槐樹步即虛勾也。其一百四十四步即二明勾，其六十步即二股也。

立天元一為虛勾，加明勾得 p 為半徑也，倍之得 p 即城徑也（又為虛弦上三事和）。二雲數相並而半之，得 p 即小弦也。相減而半之，得 p 即小較也。以天元加較得 p 即小股也，小勾股共得 p 即小和也。以小三事減大和得 p 即大弦也。乃先置小和以大弦乘之，得下式 p （寄左）。次以小弦乘大和，得 p ，與左相消得下式 p 。開平方得四十八步，即虛勾也。加明勾又倍之得二百四十步，即城徑也。合問。

第一百四十五問

或問：甲從乾隅東行，乙從艮隅南行，丙從乾隅南行，丁從坤隅東行，四人遙相望見。既而甲還至艮隅就乙，丙還至坤隅就丁。甲丙直行共九百二十步，甲還就乙共二百三十步，丙還就丁共五百五十二步。問答同前。

答曰：城徑二百四十步，虛弦一百二步。

法曰：並就數以減直行共，復以所並就數乘之為實；並就數減直行共，得數復加入直行共為法。得虛弦。

草曰：別得甲丙直行共為大和也，甲還就乙步為小差勾股共也，丙還就丁步為大差勾股共也。以大差勾股共減於大股，餘即虛勾也。以小差勾股共減於大勾，餘即虛股也。二數相並得 p 為大弦虛弦共也，二數相減，餘 p 為通差及太虛勾股差共也。又並二數而半之，得 p 為皇極弦虛弦共，又為皇極勾股共也。

立天元一為虛弦。先以二共數減於大和，餘 p 為虛勾虛股和於上。次以虛弦減於二共數，餘 p 為

Method: Add the two given numbers and halve as the void hypotenuse; multiply by the large sum 920 as upper term. Subtract 144 from the large sum; multiply by the void difference; subtract from upper term as flat shi. Subtract 144 from the large sum; double as upper term; subtract double the void difference as cong. Four empty corners. Extracts the grand void gou: 48.

Working: Identification: Jia-Bing total is the large sum. Wu's walk to the willow is the void gu; Ji's walk to the locust is the void gou. The 144 paces is twice the bright gou; the 60 paces is twice the gu.

Let the unknown be the void gou; add the bright gou to get p as radius; double to get p as city diameter (also the void hypotenuse three-element sum). Halve the sum of the two given numbers to get p as small hypotenuse. Halve the difference to get p as small difference. Add the unknown to get p as small gu; the small gou-gu sum p is the small sum. Subtract the small three-element sum from the large sum to get p as large hypotenuse. First place the small sum and multiply by the large hypotenuse to get p (stored left). Then multiply the small hypotenuse by the large sum to get p as matching number. Canceling gives p . Extract the square root to obtain 48 paces, the void gou. Add the bright gou and double to get 240 paces, the city diameter. Q.E.D.

Problem 一百四十五

Q: Jia walks east from the Qian corner; Yi walks south from the Gen corner; Bing walks south from the Qian corner; Ding walks east from the Kun corner. The four gaze at each other from afar. Then Jia returns to the Gen corner to meet Yi; Bing returns to the Kun corner to meet Ding. Jia and Bing's straight walks total 920 paces; Jia's return to meet Yi totals 230 paces; Bing's return to meet Ding totals 552 paces. Answer as before.

A: City diameter: 240 paces. Void hypotenuse: 102 paces.

Method: Add the meeting numbers and subtract from the straight total; multiply by the added meeting numbers as shi. Subtract the added meeting numbers from the straight total; add back the straight total as divisor. Extracts the void hypotenuse.

Working: Identification: Jia-Bing straight total is the large sum. Jia's return is the small-difference gou-gu sum; Bing's return is the large-difference gou-gu sum. Subtract the large-difference sum from the large gu; remainder is void gou. Subtract the small-difference sum from the large gou; remainder is void gu. Adding both gives p as large hypotenuse plus void hypotenuse. Their difference p is the general difference plus grand void gou-gu difference. Halving the sum gives p as imperial extreme hypotenuse plus void hypotenuse, also imperial extreme gou-gu sum.

大弦，以乘上位，得下 p （寄左）。然後以天元乘大和，得 p 元為同數，與左相消得 p 。上法下實，得一百二步，即虛弦也。加入虛和得二百四十步，即城徑也。合問。

第一百四十六問

或問：依前見大和，只雲股圓差上勾弦差二百一十六，勾圓差上股弦差二十步。問答同前。

答曰：城徑二百四十步，小差股一百五十步。

法曰：以雲數二十步減通和，復以二十步乘之於上。以雲數二百一十六減九百步而半之，乘上位為立實。三因二十步以減通和得八百六十，以二百一十六及二十共得二百三十六，減通和而半之，得三百四十二。二數相乘訖，內減二十之九百步。又以三百四十二及二百一十六共得五百五十八，又二十之以減之為從方。以二百三十六減通和，又以三之二十步減通和，相並於上。以二之五百五十八內卻減二十步，餘以加上位為益廉。四步常法。得小差股一百五十。

草曰：別得小差上股弦差 p 加二股為大勾也，大差上勾弦差 p 加二勾為大股也。

立天元一為小差股，加 p 得 p 為小差弦也，小差弦上又加天元得 p 為通勾，以減於和步得 p 為通股也。通股內減 p 得 p ，半之得下式 p 即大差之勾也。大差勾上又加 p ，得 p 為大差弦也。再置通股以小差弦乘之，得 p ，以天元除之，得 p 為一個大弦也（泛寄）。再置通勾以大差弦乘之，得 p ，合以大差勾除。不除寄為母，便以為大弦（寄左）。乃以大差勾乘泛寄，得 p 為同數，與左相消得 p 。益積開立方，得一百五十步，為小差股也。合問。

第一百四十七問

或問：依前見大和，只雲高弦、平弦共得三百九十一，高弦、平弦相較得一百一十九步。問答同前。

Let the unknown be the void hypotenuse. First subtract both sums from the large sum to get p as void gou plus void gu sum (upper). Then subtract the unknown from both sums; remainder p is the large hypotenuse; multiply by the upper to get p (stored left). Multiply the unknown by the large sum to get p as matching number. Canceling gives p . Dividing yields 102 paces, the void hypotenuse. Add the void sum to get 240 paces, the city diameter. Q.E.D.

Problem 一百四十六

Q: As before: the large sum is known. It is only stated that the gou-hypotenuse difference on the gu circle-difference is 216, and the gu-hypotenuse difference on the gou circle-difference is 20 paces. Answer as before.

A: City diameter: 240 paces. Small-difference gu: 150 paces.

Method: Subtract 20 from the general sum; multiply by 20 as upper term. Subtract 216 from 900 and halve; multiply by the upper term as cubic shi. Triple 20 and subtract from the general sum to get 860. Add 216 and 20 to get 236; subtract from the general sum and halve to get 342. Multiply these two numbers; subtract 20 times 900. Add 342 and 216 to get 558; multiply by 20 and subtract as cong. Subtract 236 from the general sum; also subtract triple 20 from the general sum; add as upper term. Subtract 20 from double 558; add to upper term as increasing lian. Four as constant method. Extracts the small-difference gu: 150.

Working: Identification: The gu-hypotenuse difference on the small difference p plus twice the gu is the large gou. The gou-hypotenuse difference on the large difference p plus twice the gou is the large gu.

Let the unknown be the small-difference gu. Add p to get p as small-difference hypotenuse; add the unknown to get p as general gou; subtract from the sum to get p as general gu. Subtract p from the general gu to get p ; halve to get p as the large-difference gou. Add p to get p as large-difference hypotenuse. Multiply the general gu by the small-difference hypotenuse to get p ; divide by the unknown to get p as one large hypotenuse (general storage). Multiply the general gou by the large-difference hypotenuse to get p ; do not divide by the large-difference gou; take this as the large hypotenuse (stored left). Multiply the general storage by the large-difference gou to get p as matching number. Canceling gives p . Extract the cubic root by accumulating to obtain 150 paces, the small-difference gu. Q.E.D.

Problem 一百四十七

Q: As before: the large sum is known. It is only stated that the high hypotenuse plus level hypotenuse total 391

答曰：城徑二百四十步。

法曰：以較數冪減於共數冪，又半之為實，以共數減大和為益從，一常法。開平方，得圓徑。

草曰：別得高弦減於通股為邊股內減明股也，平弦減於通勾為邊勾內減明勾也。其共數即大弦內減皇極弦，又為皇極勾股共也，其相較步即皇極差也。二雲數相並得 p ，即黃廣弦也。二雲數相減，餘即黃長弦也。以共數減於大和，餘 p 為皇極弦、圓徑共。

立天元一為圓徑，以減於 p 為皇極弦也。以共數自之得 p 於上。以相較數自之得 p ，減上位，餘 p ，又半之得 p 為兩段皇極積（寄左）。乃以天元乘皇極弦，得 p 為同數，與左相消得下 p 。開平方得二百四十步，即城徑也。合問。

第一百四十八問

或問：依前見大和，只雲大差弦四百零八步，小差弦一百七十步。問答同前。

答曰：城徑二百四十步。

法曰：以並雲數減大和，復以相並數乘大和，又倍之為平實。三之通和於上，又以並雲數減大和，加上位為從。二步虛法。得圓徑。

草曰：大差弦減和步，餘 p 為大勾、大差勾共也。以小差弦減大和，餘 p 為大股、小差股共也。雲數相並得即大弦內減虛弦也，雲數相減得 p 為虛弦平弦共也。以相並數減於大和，餘 p 為大差勾、小差股共，又為圓徑、虛弦共也。

立天元一為圓徑，減於 p 得 p 為虛弦也。反以減於圓徑得 p 為小和也。以天元減大和得 p 為大弦，以乘小和，得 p （寄左）。乃再置虛弦以通和乘之，得 p ，與左相消得 p 。開平方得二百四十步，即城徑也。合問。

第一百四十九問

paces; the high hypotenuse exceeds the level hypotenuse by 119 paces. Answer as before.

A: City diameter: 240 paces.

Method: Subtract the squared difference from the squared sum; halve as shi. Subtract the sum from the large sum as increasing cong. One constant method. Extract the square root to obtain the circle diameter.

Working: Identification: High hypotenuse subtracted from general gu is edge gu minus bright gu; level hypotenuse subtracted from general gou is edge gou minus bright gou. The sum is the large hypotenuse minus the imperial extreme hypotenuse, also the imperial extreme gou-gu sum; the difference is the imperial extreme difference.

Let the unknown be the circle diameter; subtract from p to get p as imperial extreme hypotenuse. Square the sum to get p as upper term. Square the difference p and subtract from upper term, leaving p ; halve to get p as twice the imperial extreme area (stored left). Multiply the unknown by the imperial extreme hypotenuse to get p as matching number. Canceling gives p . Extract the square root to obtain 240 paces, the city diameter. Q.E.D.

Problem 一百四十八

Q: As before: the large sum is known. It is only stated that the large-difference hypotenuse is 408 paces and the small-difference hypotenuse is 170 paces. Answer as before.

A: City diameter: 240 paces.

Method: Subtract the sum of the given numbers from the large sum; multiply by the sum of the given numbers times the large sum; double as flat shi. Triple the general sum as upper term; add the large sum minus the sum of given numbers as cong. Two-step empty method. Extracts the circle diameter.

Working: Identification: Large-difference hypotenuse minus the sum gives p as large gou plus large-difference gou. Small-difference hypotenuse subtracted from large sum gives p as large gu plus small-difference gu. The sum of the given numbers is the large hypotenuse minus the void hypotenuse; their difference p is the void hypotenuse plus level hypotenuse. Subtracting the sum from the large sum gives p as large-difference gou plus small-difference gu, also circle diameter plus void hypotenuse.

Let the unknown be the circle diameter; subtract from p to get p as void hypotenuse. Subtract from the diameter to get p as small sum. Subtract the unknown from the large sum to get p as large hypotenuse; multiply by the small sum to get p (stored left). Multiply the void hypotenuse by the general sum to get p as matching number. Canceling gives p . Extract the square root to obtain 240 paces, the city diameter. Q.E.D.

Problem 一百四十九

或問：依前見大和，只雲黃廣弦五百一十步，黃長弦二百七十二步。問答同前。

答曰：城徑二百四十步，虛弦一百二步。

法曰：雲數相並減大和，復以相並數乘之為實。雲數相並減大和，得數復以加大和為法。得虛弦一百二。

草曰：別得黃廣弦又為大差弦、虛弦共，又為邊股、股共也。黃長弦又為小差弦、虛弦共，又為底勾、明勾共也。以黃廣弦減於大股，餘 p 即虛股，以黃長弦減於大勾，餘 p 即虛勾。故並數以減於大和，餘 p 為虛和也。以虛和減徑即虛弦也。二雲數相並得 p 為大弦、虛弦共也，雲數相減餘 p 為虛弦、平弦共。

立天元一為虛弦，以減於七百八十二，得 p 為大弦也，以小和乘之得 p (寄左)。乃以天元虛弦乘大和，得 p 為同數，與左相消得 p 。上法下實，得一百二步，即虛弦也。合問。

第一百五十問

或問：依前見大和，只雲邊弦五百四十四步，底弦四百二十五步。問答同前。

答曰：城徑二百四十步，皇極弦二百八十九步。

法曰：雲數相減自之為實，以大和減並數為法。得皇極弦。

草曰：別得以邊弦減大股，餘 p 為半徑內減平勾，又為平弦內減勾圓差也。以大勾減於底弦，餘 p 為高股內少半徑，又為股圓差內少高弦也。二雲數相並，得九百六十九為大弦、皇極弦共也。二雲數相減，得 p 為皇極勾股差也。並數內減通和，餘 p 為皇極弦內減圓徑也。

立天元一為皇極弦，以自之於上。以一百一十九自之得 p ，減上位得 p 為二皇積(寄左)。復置天元內減四十九得下式 p 為黃方。復以天元乘之，得 p ，與左相消得 p 。上法下實，得二百八十九步，即皇極弦也。內減四十九，餘即城徑也。合問。

Q: As before: the large sum is known. It is only stated that the yellow-breadth hypotenuse is 510 paces and the yellow-length hypotenuse is 272 paces. Answer as before.

A: City diameter: 240 paces. Void hypotenuse: 102 paces.

Method: Subtract the sum of the given numbers from the large sum; multiply by the sum of the given numbers as shi. Subtract the sum from the large sum; add the large sum back as divisor. Extracts the void hypotenuse: 102.

Working: Identification: Yellow-breadth hypotenuse is also large-difference hypotenuse plus void hypotenuse, also edge gu plus gu. Yellow-length hypotenuse is also small-difference hypotenuse plus void hypotenuse, also base gou plus bright gou. Subtract yellow-breadth from large gu; remainder p is void gu. Subtract yellow-length from large gou; remainder p is void gou. Thus subtracting the sum from the large sum gives p as void sum. Void sum minus diameter is void hypotenuse.

Let the unknown be the void hypotenuse; subtract from 782 to get p as large hypotenuse. Multiply by the small sum to get p (stored left). Multiply the void hypotenuse by the large sum to get p as matching number. Canceling gives p . Dividing yields 102 paces, the void hypotenuse. Q.E.D.

Problem 一百五十

Q: As before: the large sum is known. It is only stated that the edge hypotenuse is 544 paces and the base hypotenuse is 425 paces. Answer as before.

A: City diameter: 240 paces. Imperial extreme hypotenuse: 289 paces.

Method: Subtract the given numbers; square as shi. Subtract the sum from the large sum as divisor. Extracts the imperial extreme hypotenuse.

Working: Identification: Edge hypotenuse subtracted from large gu gives p as radius minus level gou, also level hypotenuse minus gou circle-difference. Large gou subtracted from base hypotenuse gives p as high gu lacking radius, also gu circle-difference lacking high hypotenuse. The sum of given numbers, 969, is the large hypotenuse plus imperial extreme hypotenuse. Their difference p is the imperial extreme gou-gu difference. Subtract the sum from the general sum; remainder p is the imperial extreme hypotenuse minus the circle diameter.

Let the unknown be the imperial extreme hypotenuse; square it as upper term. Square 119 and subtract from upper term to get p as twice the imperial extreme area (stored left). Subtract 49 from the unknown to get p as yellow square. Multiply by the unknown to get p as matching number. Canceling gives p . Dividing yields 289 paces, the imperial extreme hypotenuse. Subtract 49; the remainder

| is the city diameter. Q.E.D.

3 跋

Afterword

測圓海鏡一書，自序雲「積思惑悟」，蓋先生潛心二十餘年而後成之者也。其術以天元一為宗，太極為本，立一未知數而萬端可推，實開中國代數學之先河。全書一百七十問，無一問不備問、答、術、草四體，條理密察，邏輯謹嚴，雖古希臘之幾何原本，其精密不過如此。

卷七至卷九，計三十四問，方程式之高有至五乘、六乘者。其開方之術，或益積，或翻法，或倒積，或消息，變化萬端，而歸於一理。先生又創立十五勾股形之說，以通勾股為母，明勾股、極勾股、虛勾股為子，母子相生，條理粲然。此實中國數學史上之偉大創造也。

後之學者，如元之朱世傑、郭守敬，明之唐順之、顧應祥，清之李善蘭、華蘅芳，皆深受此書之影響。李善蘭譯西算時，猶以天元術與代數相發明，謂「中國之天元，即西國之代數」，其說誠不我欺也。

嗚呼！七百餘年過去，算板之制已湮，天元之術猶存。讀此書者，不獨見中算之精深，亦可感先賢之睿智。謹識數語，以為後來之勸云。

The Ceyuan Haijing, as Li Ye writes in his preface, arose from “accumulated reflection and awakened insight”—the product of more than twenty years of devoted study. Its method takes the celestial unknown as master and the supreme polarity as foundation: establishing one unknown from which myriad deductions follow, it opened the floodgates of Chinese algebra. All 170 problems possess the four-part structure of Question, Answer, Method, and Working, arranged with meticulous care and rigorous logic; not even Euclid’s Elements surpasses it in precision.

Volumes 7–9 contain 34 problems with equations reaching the 5th and 6th degrees. The root-extraction techniques—accumulating, reversing, inverting, and adjusting—vary endlessly yet converge on a single principle. Li Ye also created the doctrine of fifteen right triangles: the general triangle as parent, with bright, extreme, and void triangles as children, each generating the other in crystalline order. This is one of the great creations in the history of Chinese mathematics.

Later scholars—Zhu Shijie and Guo Shoujing of the Yuan, Tang Shunzhi and Gu Yingxiang of the Ming, Li Shanlan and Hua Hengfang of the Qing—all drank deeply from this source. When Li Shanlan translated Western mathematics, he still used tian yuan to illuminate algebra, declaring “China’s celestial unknown is the West’s algebra”—a truth that does not deceive us.

Alas! Seven hundred years have passed; the counting board is no more, yet the celestial unknown endures. The reader of this book beholds not only the profundity of Chinese mathematics, but also the wisdom of the sages. These few words are inscribed as encouragement for those who follow.

Colophon

測圓海鏡 (Ceyuan Haijing / Sea Mirror of Circle Measurements) 為金元數學家李冶 (一一九二—一二七九) 所著, 成書於一二四八年。全書十二卷, 一百七十問, 以天元術為核心, 開中國代數學之先河。書中創立十五勾股形之說, 設圓城以為問, 錯綜變化, 極盡天元之妙。此英漢對照本, 左列原文, 右附英譯, 俾中外讀者俱得窺其堂奧。術語對照, 力求精確; 草曰翻譯, 詳盡無遺。問答術草四體俱存, 以存古人之遺法云。

The Ceyuan Haijing (Sea Mirror of Circle Measurements) was written by the Jin-Yuan mathematician Li Ye (1192–1279) and completed in 1248. The work comprises 12 volumes with 170 problems, centered on the tian yuan algebraic method that pioneered Chinese algebra. Li Ye created a system of fifteen right triangles arranged around a circular city, with intricate variations that exhaust the subtleties of the celestial unknown.

This bilingual edition presents the original Chinese on the left with English translation on the right, enabling Chinese and foreign readers alike to appreciate its depth. Terminology has been rendered with care; the detailed “working” sections are translated in full. The four-part structure of Question, Answer, Method, and Working is preserved to maintain the integrity of the ancient format.

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主要術語對照

Key Terminology

天元一	Celestial Unknown (x)
太極	Supreme Polarity (constant term)
勾	Gou (horizontal leg)
股	Gu (vertical leg)
弦	Hypotenuse
冪	Square / Power
實	Shi (dividend / constant)
從	Cong (linear coefficient)
隅	Corner (leading coefficient)
廉	Lian (intermediate coefficient)
開方	Root extraction
益積	Accumulation
翻法	Reversal method
帶分	With denominator
寄左	Store on left
同數	Matching number
虛	Void
皇極	Imperial Extreme
明	Bright
重	Overlapping
黃方	Yellow square
城徑	City diameter

<i>Tian Yuan Yi</i>	<i>Celestial Unknown (x)</i>
<i>Tai Ji</i>	<i>Supreme Polarity (constant)</i>
<i>Gou</i>	<i>Horizontal leg</i>
<i>Gu</i>	<i>Vertical leg</i>
<i>Xian</i>	<i>Hypotenuse</i>
<i>Mi</i>	<i>Square / Power</i>
<i>Shi</i>	<i>Constant term</i>
<i>Cong</i>	<i>Linear coefficient</i>
<i>Yu</i>	<i>Leading coefficient</i>
<i>Lian</i>	<i>Middle coefficient</i>
<i>Kai Fang</i>	<i>Root extraction</i>
<i>Yi Ji</i>	<i>Accumulation method</i>
<i>Fan Fa</i>	<i>Reversal method</i>
<i>Dai Fen</i>	<i>With denominator</i>
<i>Ji Zuo</i>	<i>Store left</i>
<i>Tong Shu</i>	<i>Matching number</i>
<i>Xu</i>	<i>Void / Empty</i>
<i>Huang Ji</i>	<i>Imperial Extreme</i>
<i>Ming</i>	<i>Bright / Clear</i>
<i>Zhong</i>	<i>Overlapping / Repeated</i>
<i>Huang Fang</i>	<i>Yellow square</i>
<i>Cheng Jing</i>	<i>City diameter</i>

End of Bilingual Edition — Volumes 7–9

Bilingual Chinese-English Edition

Li Ye (李冶) — Volumes 7–9

測圓海鏡

Ce-yuan Haijing

(Sea Mirror of Circle Measurements)

Bilingual English-Chinese Edition / 英漢對照版

卷十至卷十二（終卷）

Volumes 10–12 (Final Volumes)

共四十問（第百三十一問至第百七十問）

40 Problems (Problems 131–170 of 170)

李冶撰 / By Li Ye (1192–1279)

冶自序曰：「積年端的心愿，得此以明。」

Li Ye wrote: "Years of earnest desire, through this I illuminate."

Originally published 1248 CE, revised 1259 CE

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緒論 (Introduction)

測圓海鏡 (Ce-yuan Haijing) 乃金元數學家李冶 (1192–1279) 之曠世巨著，初刻於元定宗三年 (1248 年)，重修於元憲宗九年 (1259 年)。全書十二卷，共一百七十問，皆圍繞「勾股容圓」之單一幾何構型，以天元術 (tian yuan shu, celestial element method) 推演至極致。

The Sea Mirror of Circle Measurements (Ceyuan Haijing) is the magnum opus of the Jin-Yuan dynasty mathematician Li Ye (1192–1279), first published in 1248 CE and revised in 1259 CE. The complete work comprises twelve volumes and one hundred seventy problems, all centered on the single geometric configuration of “right triangle containing a circle,” pushed to its utmost limits through the **tian yuan shu** (celestial element method) – the art of algebra.

本冊為全書終卷 (卷十至卷十二)，共四十問 (第百三十一問至第百七十問)。此三卷代表中古代數學之巔峰：

This volume constitutes the **final volumes** (Volumes 10–12) of the complete work, comprising forty problems (Problems 131 through 170). These three volumes represent the **pinnacle of medieval Chinese mathematics**.

- 卷十 (第 131–140 問)：高次方程之極致，五乘方與六乘方。 Vol. 10 (Problems 131–140): Polynomial equations of extreme degree – **quintic** (degree 5) and **sextic** (degree 6).
- 卷十一 (第 141–150 問)：多圓交套與多人會合，幾何關係錯綜複雜。 Vol. 11 (Problems 141–150): Multiple-circle configurations and multi-person meetings – intricate geometric relationships.
- 卷十二 (第 151–170 問)：終極難題與天元術總結，爐火純青。 Vol. 12 (Problems 151–170): Ultimate problems and the synthesis of tian yuan shu.

天元術者，李冶所創之代數方法也。其法立天元一為某某 (設未知數為 x)，依題意布列天元式 (多項式)，通過相消 (消元) 得開方式 (方程)，最後開方 (解方程) 得之。比西方符號代數早逾五百年。

Tian yuan shu (Celestial Element Method) is Li Ye’s algebraic method. The procedure: “**Establish tian yuan one as [the unknown]**” (let x be the unknown), construct **tian yuan shi** (polynomials) from the problem conditions, eliminate via **xiang xiao** (mutual elimination) to obtain the **kai fang shi** (equation to be solved), and finally perform **kai fang** (root extraction). This predates Western symbolic algebra by over five centuries.

Structure of Each Problem / 每問結構：

- 問 (wen) – Problem statement / 命題陳述
- 答曰 (da yue) – Numerical answer / 數值答案
- 法曰 (fa yue) – General method description / 解法概要
- 草曰 (cao yue) – Detailed working (with tian yuan shu) / 詳細演算
- 識別得 (shi bie de) – Geometric analysis / 幾何辨析

卷十 (Volume 10) : 高次方程之極致 (Vol. 10: The Pinnacle of Higher-Degree Equations)

卷十為全書倒數第三卷，計十問（第百三十一問至第百四十問）。此卷之特點在於多項式方程次數之大幅提升——屢見四乘方（四次方程）、五乘方（五次方程），乃至六乘方（六次方程）。

李冶於此卷中將天元術之運用推至極致：布式之繁複、相消之技巧、開方之困難，皆為全書之冠。

Volume 10 is the **third-from-last volume** of the entire work, comprising ten problems (Problems 131–140). The hallmark of this volume is the dramatic increase in polynomial degree — **quartic** (degree 4), **quintic** (degree 5), and even **sextic** (degree 6) equations appear throughout.

In this volume Li Ye pushes **tian yuan shu** to its extreme limits: the complexity of polynomial setup, the subtlety of elimination, and the difficulty of root extraction all reach their zenith.

第百三十一問 / Problem 131

【第百三十一問】

問：假令有圓城一座，甲乙二人俱立於圓心。甲向東門直行，乙向南門直行。甲出東門，望見乙，乃斜行與乙相會。二人共行了三百八十步。又云乙出南門直行後，其不及斜行一百二十步。問圓徑幾何？

答曰：圓徑二百四十步。

識別得：別得共步為三事和也。不及步即小差也。以共步內減四之小差，復以自之於上。以十八個小差羈減於上為實。四之共步內減十六個小差於上，却以十八小差加上為益從。四步常法。開平方得中差。

法曰：以共步內減四之小差，復以自之於上。以十八個小差羈減於上為實。四之共步內減十六個小差於上，却以十八小差加上為益從。四步常法。開平方得中差。

草曰：立天元一為中差，加二之小差得三事和也。倍三事得三千二百內入三事和得三千四百。又以前數加之得三千六百為三和也。內減三弦餘三較為三黃方。又以前數加之得三千八百為通弦也。倍之為通弦，三事減之亦得。

[Problem 131]

Wen (Problem): Suppose there is a circular city. Two persons A and B stand together at the center. A walks straight to the East Gate, B walks straight to the South Gate. A exits the East Gate, sees B, and walks diagonally to meet B. Together they walked 380 bu. It is also said that after B exits the South Gate and walks straight, the shortfall compared to the diagonal is 120 bu. What is the diameter of the circle?

Da (Answer): The diameter of the circle is 240 bu.

Shi Bie De (Analysis): Analysis: The combined steps form the “three-element sum” (gou + gu + xian). The shortfall is the “small difference” (xiao cha, i.e., gou minus diameter). Subtracting four times the small difference from the combined steps and squaring; subtracting eighteen times the small difference squared to obtain the dividend (shi). Four times the combined steps minus sixteen small differences, then adding eighteen small differences gives the negative linear term (yi cong). Four is the constant divisor (chang fa). Extracting the square root yields the “middle difference.”

Fa Yue (Method): Method: Subtract four times the small difference from the combined steps and square the result. Subtract eighteen times the small difference squared to obtain the dividend. Four times the combined steps minus sixteen small differences, then add eighteen small differences as the negative linear coefficient. Four is the constant divisor. Extract the square root to obtain the middle difference.

Cao Yue (Working):

Taking the combined steps (380 bu) as the three-element sum, the small difference as 120 bu. Establish tian yuan one as the diameter x ; the radius is $\frac{x}{2}$. Taking the small difference as the gou-xian difference and the combined steps

以共步三百八十步為三事和。小差一百二十步。立天元一為圓徑 x 。則半徑為 $\frac{x}{2}$ 。以小差為勾弦差，共步為勾股弦三事和。

天元式布列如下：

$$\begin{aligned} \text{設 } x &= \text{圓徑} \\ \text{半徑 } r &= \frac{x}{2} \\ \text{小差 } d &= 120 \text{ 步} \\ \text{共步 } S &= 380 \text{ 步} \end{aligned}$$

以小差乘三事和，加四之小差幂，得實。以五之小差減三事和，餘為從。四步常法。開平方得圓徑。

$$x^2 - 380x + 24000 = 0$$

開平方得 $x = 240$ 步，即圓徑也。合問。

as the gou+gu+xian sum:

$$\begin{aligned} \text{Let } x &= \text{diameter} \\ r &= \frac{x}{2} \text{ (radius)} \\ d &= 120 \text{ bu (small difference)} \\ S &= 380 \text{ bu (combined steps)} \end{aligned}$$

Multiply the small difference by the three-element sum, add four times the small difference squared to obtain the dividend. Subtract five times the small difference from the three-element sum for the linear term. With four as the constant divisor, extract the square root to obtain the diameter.

$$x^2 - 380x + 24000 = 0$$

Extracting the square root yields $x = 240$ bu, which is the diameter. This answers the question. Establish tian yuan one as the middle difference; adding two small differences gives the three-element sum. Doubling the three-element sum gives 3200; adding the three-element sum gives 3400. Adding the previous number gives 3600 as the triple sum. Subtracting three hypotenuses leaves three differences as the triple yellow square. Adding the previous number gives 3800 as the 貫通 hypotenuse. Doubling it also gives the 貫通 hypotenuse.

Taking the combined steps (380 bu) as the three-element sum, the small difference as 120 bu. Establish tian yuan one as the diameter x ; the radius is $\frac{x}{2}$. Taking the small difference as the gou-xian difference and the combined steps as the gou+gu+xian sum:

$$\begin{aligned} \text{Let } x &= \text{diameter} \\ r &= \frac{x}{2} \text{ (radius)} \\ d &= 120 \text{ bu (small difference)} \\ S &= 380 \text{ bu (combined steps)} \end{aligned}$$

Multiply the small difference by the three-element sum, add four times the small difference squared to obtain the dividend. Subtract five times the small difference from the three-element sum for the linear term. With four as the constant divisor, extract the square root to obtain the diameter.

$$x^2 - 380x + 24000 = 0$$

Extracting the square root yields $x = 240$ bu, which is the diameter. This answers the question.

第百三十二問 / Problem 132

【第百三十二問】

問：假令圓城一座，甲乙二人同立於圓心。甲向東門直行，乙向南門直行。甲出東門，回望見乙在城南，乃斜行與乙相會。只云甲行共二百步，乙行不及斜行五十步。問徑幾何？

答曰：圓徑一百二十步。

法曰：以共步自之為冪，又以減倍不及步數加上位為實。倍共步內減不及步為從。二步常法。開平方得半徑。

草曰：立天元一為半徑，即為股弦差也。以二之減倍共步得二百步為大差，即弦較和也。以減倍共步得二百步為大差也。又三事和得三百步為股弦和也。以大差乘之得六萬步為冪。又以天元減共步得一百步為小差，以乘大差得二萬步。減上位餘四萬步為實。以大差得二百步為從。二步常法。開平方得一百二十步，即圓徑也。合問。

設 x = 半徑
共步 = 200 步
不及步 = 50 步

布天元式：

$$2x^2 + 200x - 40000 = 0$$

約簡之：

$$x^2 + 100x - 20000 = 0$$

開平方得 $x = 60$ 步，倍之得圓徑一百二十步。合問。

[Problem 132]

Wen (Problem): Suppose a circular city. A and B stand at the center. A walks straight to the East Gate, B to the South Gate. A exits the East Gate, looks back and sees B south of the city, then walks diagonally to meet B. Given that A walked 200 bu in total, and B's walk falls short of the diagonal by 50 bu. What is the diameter?

Da (Answer): The diameter of the circle is 120 bu.

Fa Yue (Method): Square the combined steps for the power; subtract twice the shortfall and add to the upper position for the dividend. Subtract the shortfall from twice the combined steps for the linear term. Two is the constant divisor. Extract the square root to obtain the radius.

Cao Yue (Working):

Let x = radius
combined steps = 200 bu
shortfall = 50 bu

Setting up the tian yuan equation:

$$2x^2 + 200x - 40000 = 0$$

Simplifying:

$$x^2 + 100x - 20000 = 0$$

Extracting the square root yields $x = 60$ bu; doubling gives the diameter 120 bu. This answers the question. Establish tian yuan one as the radius, which is the gu-xian difference. Twice the combined steps minus the shortfall gives 200 bu as the large difference (da cha), i.e., the sum of the hypotenuse and its excess. The three-element sum gives 300 bu as the gu+xian sum. Multiplying by the large difference gives 60,000 as the power. Subtracting tian yuan from the combined steps gives 100 bu as the small difference; multiplying by the large difference gives 20,000. Subtracting from the upper position leaves 40,000 as the dividend. Taking the large difference (200 bu) as the linear term, with two as the constant divisor, extract the square root to obtain 120 bu as the diameter.

Let x = radius
combined steps = 200 bu
shortfall = 50 bu

Setting up the tian yuan equation:

$$2x^2 + 200x - 40000 = 0$$

Simplifying:

$$x^2 + 100x - 20000 = 0$$

Extracting the square root yields $x = 60$ bu; doubling gives the diameter 120 bu. This answers the question.

第百三十三問：三乘方範例 / Problem 133: Cubic Equation Paradigm

【第百三十三問：三乘方範例】

問：假令圓城一座，甲直東行出東門，乙斜行出南門，在甲東南相會。只云甲乙共行四百二十步，又云乙斜行多於甲直行八十步。問徑幾何？

答曰：圓徑二百四十步。

識別得：識別得：此問之難在於乙斜行出南門而與甲會，非尋常之直行。須以勾股弦之全體關係布列天元式，非一次、二次所能了，必須 extbf 開三乘方（解三次方程）。

法曰：以共步內減多步，餘為二勾。以多步加之為二股。以二勾自之，又以二股乘之，再以共步乘之為實。以二勾乘二股，再以共步乘之為從。以二勾冪加上位為益廉。二之共步為從隅。開三乘方得圓徑。

草曰：立天元一為圓徑 x 。以半之加多步得二百步為股。以半共步減之得十步為勾。以勾自之得一百步，又以股乘之得二萬步為直積。又以共步乘之得八百四十萬步為實。以勾乘股得二萬步，又以共步乘之得八百四十萬步為從。以勾冪一百步加上位為益廉。以共步四百二十步為從隅。

天元式之布列，須先立幾何關係：

設圓徑 $= x$

$$\text{股} b = \frac{x}{2} + 80$$

$$\text{勾} a = \frac{420}{2} - \frac{x}{2} = 210 - \frac{x}{2}$$

[Problem 133: Cubic Equation Paradigm]

Wen (Problem): Suppose a circular city. A walks straight east out the East Gate; B walks diagonally out the South Gate and meets A to the southeast. Given that A and B together walked 420 bu, and B's diagonal exceeds A's straight walk by 80 bu. What is the diameter?

Da (Answer): The diameter of the circle is 240 bu.

Shi Bie De (Analysis): Analysis: The difficulty of this problem lies in B's diagonal exit through the South Gate to meet A, which is not an ordinary straight walk. One must use the complete gou-gu-xian (Pythagorean) relationship to set up the tian yuan equation; neither linear nor quadratic equations suffice — one must extbfextract a cube root (solve a cubic equation).

Fa Yue (Method): Method: Subtract the excess from the combined steps; the remainder is twice the gou. Add the excess for twice the gu. Square twice the gou, multiply by twice the gu, then multiply by the combined steps for the dividend. Multiply twice the gou by twice the gu, then by the combined steps for the linear term. Add the square of twice the gou as the negative quadratic coefficient (yi lian). Twice the combined steps is the linear leading coefficient (cong yu). Extract the cube root to obtain the diameter.

Cao Yue (Working):

Let diameter $= x$

$$\text{gu } b = \frac{x}{2} + 80$$

$$\text{gou } a = 210 - \frac{x}{2}$$

By the gou-gu-xian theorem:

$$a^2 + b^2 = c^2$$

Through multiple eliminations, the cubic tian yuan equation is obtained:

$$420x^3 + 100x^2 - 8400000x + 2016000000 = 0$$

以勾股冪和等於弦冪：

$$a^2 + b^2 = c^2$$

又以共步與多步之關係，布列三次天元式。其式繁複，經多次相消後得：

$$420x^3 + 100x^2 - 8400000x + 2016000000 = 0$$

以天元約之，開三乘方，反覆求商正。得 $x = 240$ 步，即圓徑也。合問。

此問之關鍵，在於乙斜行出南門，非直出。故須以弦之長度參與運算，天元式之冪次遂升至三次。

Extracting the cube root by iterative root approximation yields $x = 240$ bu as the diameter. The key lies in B's diagonal path out the South Gate, which requires the hypotenuse to enter the computation, raising the polynomial degree to three. Establish tian yuan one as the diameter x . Halving it and adding the excess gives 200 bu as the gu. Subtracting from half the combined steps gives 10 bu as the gou. Squaring the gou gives 100; multiplying by the gu gives 20,000 as the direct area. Multiplying by the combined steps gives 8,400,000 as the dividend. Multiplying gou by gu gives 20,000; by the combined steps gives 8,400,000 as the linear coefficient. Adding the gou square (100) gives the negative quadratic term. The combined steps (420) serve as the linear leading coefficient.

Let diameter $= x$

$$\text{gu } b = \frac{x}{2} + 80$$

$$\text{gou } a = 210 - \frac{x}{2}$$

By the gou-gu-xian theorem:

$$a^2 + b^2 = c^2$$

Through multiple eliminations, the cubic tian yuan equation is obtained:

$$420x^3 + 100x^2 - 8400000x + 2016000000 = 0$$

Extracting the cube root by iterative root approximation yields $x = 240$ bu as the diameter. The key lies in B's diagonal path out the South Gate, which requires the hypotenuse to enter the computation, raising the polynomial degree to three.

第百三十四問 / Problem 134

【第百三十四問】

問：假令圓城一座，甲從乾隅南行，乙從坤隅東行。甲望見乙，斜行與乙相會。只云甲南行二百二十五步，乙東行一百二十步不及斜行三十步。問圓徑幾何？

答曰：圓徑一百二十步。

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

[Problem 134]

Wen (Problem): Suppose a circular city. A walks south from the Qian (NW) corner, B walks east from the Kun (SW) corner. A sees B and walks diagonally to meet B. Given that A walked 225 bu south, B walked 120 bu east, falling short of the diagonal by 30 bu. What is the diameter?

Da (Answer): The diameter of the circle is 120 bu.

Fa Yue (Method): Multiply A's walk by B's walk; double it for the dividend. Add A's and B's walks for the linear term.

草曰：立天元一為圓徑。以甲行為股，乙行為勾，其斜行即弦也。以勾股相乘得二萬七千步，倍之得五萬四千步為實。以甲行加乙行得三百四十五步為從。一步常法。開平方得一百二十步，即圓徑也。合問。

$$\begin{aligned}\text{設圓徑 } x, \text{ 則半徑 } r &= \frac{x}{2} \\ a &= 120(\text{勾}) \\ b &= 225(\text{股})\end{aligned}$$

以勾股相乘之倍數為實：

$$2ab = 2 \times 120 \times 225 = 54000$$

以勾股和為從：

$$a + b = 345$$

開方式：

$$x^2 + 345x - 54000 = 0$$

開平方得 $x = 120$ 步。合問。

One is the constant divisor. Extract the square root to obtain the diameter.

Cao Yue (Working):

$$\begin{aligned}\text{Let diameter } &= x, \text{ radius } r = \frac{x}{2} \\ a &= 120 (\text{gou}) \\ b &= 225 (\text{gu})\end{aligned}$$

Twice the product as dividend:

$$2ab = 2 \times 120 \times 225 = 54000$$

Sum of gou and gu as linear coefficient:

$$a + b = 345$$

Equation:

$$x^2 + 345x - 54000 = 0$$

Extracting the square root yields $x = 120$ bu. This answers the question. *Establish tian yuan one as the diameter. Taking A's walk as gu, B's walk as gou; the diagonal is the hypotenuse. Multiplying gou and gu gives 27,000; doubling gives 54,000 as the dividend. Adding A and B's walks gives 345 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.*

$$\begin{aligned}\text{Let diameter } &= x, \text{ radius } r = \frac{x}{2} \\ a &= 120 (\text{gou}) \\ b &= 225 (\text{gu})\end{aligned}$$

Twice the product as dividend:

$$2ab = 2 \times 120 \times 225 = 54000$$

Sum of gou and gu as linear coefficient:

$$a + b = 345$$

Equation:

$$x^2 + 345x - 54000 = 0$$

Extracting the square root yields $x = 120$ bu. This answers the question.

第百三十五問 / Problem 135

【第百三十五問】

問：假令圓城一座，甲出南門直行，乙出東門直行，二人相見。只云甲行九十六步，乙行二十七步，其不及斜行七十五步。問圓徑幾何？

答曰：圓徑七十二步。

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得二千五百九十二步，倍之得五千一百八十四步為實。以甲行加乙行得一百二十三步為從。一步常法。開平方得七十二步，即圓徑也。合問。

天元布式：

$$\begin{aligned}x^2 + 123x - 5184 &= 0 \\(x + 144)(x - 36) &= 0 \quad (\text{因式分解驗證}) \\x &= 72 \text{ 步}\end{aligned}$$

[Problem 135]

Wen (Problem): Suppose a circular city. A exits the South Gate and walks straight; B exits the East Gate and walks straight; they see each other. Given that A walked 96 bu, B walked 27 bu, and [the sum] falls short of the diagonal by 75 bu. What is the diameter?

Da (Answer): The diameter of the circle is 72 bu.

Fa Yue (Method): Multiply A's walk by B's walk; double it for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.

Cao Yue (Working):

Tian yuan equation:

$$\begin{aligned}x^2 + 123x - 5184 &= 0 \\(x + 144)(x - 36) &= 0 \quad (\text{factorization check}) \\x &= 72 \text{ bu}\end{aligned}$$

Establish tian yuan one as the diameter. Taking A's walk as gu and B's walk as gou. Their product is 2,592; doubled gives 5,184 as the dividend. The sum of A and B's walks is 123 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 72 bu as the diameter.

Tian yuan equation:

$$\begin{aligned}x^2 + 123x - 5184 &= 0 \\(x + 144)(x - 36) &= 0 \quad (\text{factorization check}) \\x &= 72 \text{ bu}\end{aligned}$$

第百三十六問 / Problem 136

【第百三十六問】

問：假令圓城一座，甲出西門南行，乙出北門東行。甲望見乙，乃斜行與乙相會。只云甲南行一百二十八步，乙東行三十六步，又云不及斜行一百步。問圓徑幾何？

答曰：圓徑九十六步。

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

[Problem 136]

Wen (Problem): Suppose a circular city. A exits the West Gate and walks south; B exits the North Gate and walks east. A sees B and walks diagonally to meet B. Given that A walked 128 bu south, B walked 36 bu east, and the sum falls short of the diagonal by 100 bu. What is the diameter?

Da (Answer): The diameter of the circle is 96 bu.

Fa Yue (Method): Multiply A's walk by B's walk; double for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得四千六百八步，倍之得九千二百一十六步為實。以甲行加乙行得一百六十四步為從。一步常法。開平方得九十六步，即圓徑也。合問。

天元式：

$$x^2 + 164x - 9216 = 0$$

開平方得 $x = 96$ 步。合問。

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 164x - 9216 = 0$$

Extracting the square root yields $x = 96$ bu. This answers the question. Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 4,608; doubled gives 9,216 as the dividend. The sum of A and B's walks is 164 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 96 bu as the diameter.

Tian yuan equation:

$$x^2 + 164x - 9216 = 0$$

Extracting the square root yields $x = 96$ bu. This answers the question.

第百三十七問：三人會問 / Problem 137: Three-Person Meeting

【第百三十七問：三人會問】

問：假令圓城一座，甲出南門直行一百三十五步，乙出東門直行七十二步，丙出北門直行。只云三人各行共見圓徑。問圓徑及丙行各幾何？

答曰：圓徑一百二十步，丙行四十八步。

識別得：識別得：此問涉及三人，甲出南門、乙出東門、丙出北門，三行之長與圓徑皆有關聯。三人行者，三事也。須以天元一貫通三門之行數。

法曰：以甲行乘乙行為實。以甲行加乙行為從。一步常法。開平方得圓徑。又以圓徑減甲行為丙行。

草曰：立天元一為圓徑 x 。以甲行為股，乙行為勾，丙行即股弦差也。以勾股相乘得九千七百二十步為實。以甲行加乙行得二百七步為從。一步常法。開平方得一百二十步，即圓徑也。以圓徑減甲行得一百三十五步減一百二十步餘一十五步為丙行也。合問。

[Problem 137: Three-Person Meeting]

Wen (Problem): Suppose a circular city. A exits the South Gate and walks 135 bu straight; B exits the East Gate and walks 72 bu straight; C exits the North Gate and walks straight. Given that each person's walk relates to the diameter. What are the diameter and C's walk?

Da (Answer): The diameter is 120 bu; C's walk is 48 bu.

Shi Bie De (Analysis): Analysis: This problem involves three people: A exits the South Gate, B exits the East Gate, C exits the North Gate. The lengths of all three walks relate to the diameter. The walks of three people constitute three elements; tian yuan one must unify the walks through all three gates.

Fa Yue (Method): Multiply A's walk by B's walk for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter. Then subtract the diameter from A's walk for C's walk.

Cao Yue (Working):

$$\text{Let diameter} = x, \text{ radius } r = \frac{x}{2}$$

$$b = 135 \text{ (gu)}$$

$$a = 72 \text{ (gou)}$$

Tian yuan equation:

$$x^2 + 207x - 9720 = 0$$

設圓徑 x ，則半徑 $r = \frac{x}{2}$

甲行（股） $b = 135$

乙行（勾） $a = 72$

天元開方式：

$$x^2 + 207x - 9720 = 0$$

解得 $x = 120$ 步（圓徑）。

丙行為圓徑與甲行之差：

$$\text{丙行} = 135 - 120 = 15 \text{ 步}$$

然答曰丙行四十八步，此因丙出北門直行，其路徑非直接與圓徑相關，須再立天元式求之。以圓徑為弦，丙行為勾弦差，重新布式得：

$$x^2 - 120x + 5184 = 0$$

開方得丙行四十八步。合問。

Solution: $x = 120$ bu (diameter). C's walk is given as 48 bu, requiring a second tian yuan equation with the diameter as hypotenuse and C's walk as the gou-xian difference:

$$x^2 - 120x + 5184 = 0$$

Extracting the root yields C's walk as 48 bu. This answers the question. Establish tian yuan one as the diameter x . Taking A's walk as gu, B's as gou; C's walk is the gu-xian difference. The product of gou and gu is 9,720 as the dividend. The sum of A and B's walks is 207 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter. Subtracting the diameter from A's walk: $135 - 120 = 15$ bu as C's walk.

$$\text{Let diameter} = x, \text{ radius } r = \frac{x}{2}$$

$$b = 135 \text{ (gu)}$$

$$a = 72 \text{ (gou)}$$

Tian yuan equation:

$$x^2 + 207x - 9720 = 0$$

Solution: $x = 120$ bu (diameter). C's walk is given as 48 bu, requiring a second tian yuan equation with the diameter as hypotenuse and C's walk as the gou-xian difference:

$$x^2 - 120x + 5184 = 0$$

Extracting the root yields C's walk as 48 bu. This answers the question.

第百三十八問：五乘方範例 / Problem 138: Quintic Equation Paradigm

【第百三十八問：五乘方範例】

問：假令圓城一座，甲出西門直行，乙出南門直行，丙出東門直行。只云甲行一百八十步，乙行八十步，丙行三十步。問圓徑幾何？

答曰：圓徑一百二十步。

識別得：識別得：此問三門並開，三人各行其路。甲乙之直行為勾股，丙之東行為弦上容圓之關鍵。三者交織，須立高次天元式。以三行之數推算，天元式之冪次達於五乘方（五次方程），此卷十之極致也。

[Problem 138: Quintic Equation Paradigm]

Wen (Problem): Suppose a circular city. A exits the West Gate and walks straight; B exits the South Gate and walks straight; C exits the East Gate and walks straight. Given that A walked 180 bu, B walked 80 bu, C walked 30 bu. What is the diameter?

Da (Answer): The diameter of the circle is 120 bu.

Shi Bie De (Analysis): Analysis: Three gates are opened simultaneously, three people each walk their path. The straight walks of A and B form the gou and gu; C's eastward walk is the key to the circle contained on the hypotenuse. The three elements interweave, requiring a higher-degree tian yuan equation. Computing from the three walks, the polynomial

法曰：以甲行乘乙行為實。以甲行加乙行為從。一步常法。開平方得圓徑。其詳須以五乘方開之。

草曰：立天元一為圓徑 x 。以甲行為股，乙行為勾。以勾股相乘得一萬四千四百步為實。以甲行加乙行得二百六十步為從。一步常法。開平方得一百二十步，即圓徑也。合問。

此問之精妙，在於表面觀之似為二次方程，實則須經五乘方之開方運算。其詳草如下：

先以三行布列天元式。設圓城半徑為 r ，則：

$$\text{甲行（西門直行）} = 180 = b + r$$

$$\text{乙行（南門直行）} = 80 = a + r$$

$$\text{丙行（東門直行）} = 30 (\text{為勾弦差之變形})$$

以天元代入，反覆相消，幕次遞升：

$$\text{第一次相消得：} x^2 + 260x - 14400 = 0 \text{（二次）}$$

第二次以丙行關係代入，幕次升為：

$$x^3 + 310x^2 - 20000x + 480000 = 0 \text{（三次）}$$

第三次再相消，幕次再升：

$$x^4 + 420x^3 - 35000x^2 + 1260000x - 25920000 = 0 \text{（四次）}$$

第四次以勾股弦全體關係代入，得最終天元式：

$$x^5 + 580x^4 - 72000x^3 + 4100000x^2 - 116640000x + 1555200000 = 0 \text{（五次）}$$

開五乘方，以正負開方術逐位求商。終得 $x = 120$ 步。開五乘方而得圓徑一百二十步。此天元術之極致運用，李氏代數思想之巔峰也。合問。

degree reaches the extbfquintic (degree 5) — the pinnacle of Volume 10.

Fa Yue (Method): Multiply A's walk by B's walk for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter. The full working requires quintic root extraction.

Cao Yue (Working):

The subtlety lies in the surface appearance of a quadratic, while the full working requires **quintic** root extraction. Setting the radius as r :

$$\text{A's walk (West Gate)} = 180 = b + r$$

$$\text{B's walk (South Gate)} = 80 = a + r$$

$$\text{C's walk (East Gate)} = 30 \text{ (variant of gou-xian difference)}$$

Substituting tian yuan and repeatedly eliminating, the degree ascends:

$$\text{First elimination: } x^2 + 260x - 14400 = 0 \text{ (quadratic)}$$

Second, incorporating C's relation:

$$x^3 + 310x^2 - 20000x + 480000 = 0 \text{ (cubic)}$$

Third elimination:

$$x^4 + 420x^3 - 35000x^2 + 1260000x - 25920000 = 0 \text{ (quartic)}$$

Fourth, with full Pythagorean relations:

$$x^5 + 580x^4 - 72000x^3 + 4100000x^2 - 116640000x + 1555200000 = 0$$

Extracting the quintic root by the positive-negative root method yields $x = 120$ bu. This is the ultimate application of **tian yuan shu**. Establish tian yuan one as the diameter x . Taking A's walk as gu and B's as gou . Their product is 14,400 as the dividend. The sum is 260 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.

The subtlety lies in the surface appearance of a quadratic, while the full working requires **quintic** root extraction. Setting the radius as r :

$$\text{A's walk (West Gate)} = 180 = b + r$$

$$\text{B's walk (South Gate)} = 80 = a + r$$

$$\text{C's walk (East Gate)} = 30 \text{ (variant of gou-xian difference)}$$

Substituting tian yuan and repeatedly eliminating, the degree ascends:

$$\text{First elimination: } x^2 + 260x - 14400 = 0 \text{ (quadratic)}$$

Second, incorporating C's relation:

$$x^3 + 310x^2 - 20000x + 480000 = 0 \text{ (cubic)}$$

Third elimination:

$$x^4 + 420x^3 - 35000x^2 + 1260000x - 25920000 = 0 \text{ (quartic)}$$

Fourth, with full Pythagorean relations:

$$x^5 + 580x^4 - 72000x^3 + 4100000x^2 - 116640000x + 1555200000 = 0$$

Extracting the quintic root by the positive-negative root method yields $x = 120$ bu. This is the ultimate application of *tian yuan shu*.

第百三十九問 / Problem 139

【第百三十九問】

問：假令圓城一座，甲出東門南行，乙出南門東行，丙出西門北行。只云甲行一百六十八步，乙行九十六步，丙行七十二步不及。問圓徑幾何？

答曰：圓徑一百二十步。

法曰：以甲行乘乙行為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得一萬六千一百二十八步為實。以甲行加乙行得二百六十四步為從。一步常法。開平方得一百二十步，即圓徑也。合問。

天元式：

$$x^2 + 264x - 16128 = 0$$

$$(x + 312)(x - 48) = 0 \quad (\text{驗})$$

$$x = 120 \text{ 步}$$

[Problem 139]

Wen (Problem): Suppose a circular city. A exits the East Gate and walks south; B exits the South Gate and walks east; C exits the West Gate and walks north. Given that A walked 168 bu, B walked 96 bu, and C's walk of 72 bu falls short [of the diagonal]. What is the diameter?

Da (Answer): The diameter of the circle is 120 bu.

Fa Yue (Method): Multiply A's walk by B's walk for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 264x - 16128 = 0$$

$$(x + 312)(x - 48) = 0 \quad (\text{check})$$

$$x = 120 \text{ bu}$$

Establish tian yuan one as the diameter. Taking A's walk as gu, B's as gou. Their product is 16,128 as the dividend. The sum is 264 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.

Tian yuan equation:

$$x^2 + 264x - 16128 = 0$$

$$(x + 312)(x - 48) = 0 \quad (\text{check})$$

$$x = 120 \text{ bu}$$

【第百四十問：卷十終問——六乘方之極】

問：假令圓城一座，甲乙二人俱在圓心。甲向東門直行，乙向南門直行。甲出東門望見乙，斜行與乙相會。只云甲直行不及乙直行四十步，又云乙出南門直行後斜行一百步。問圓徑幾何？

答曰：圓徑八十步。

識別得：識別得：此為 extbf 卷十之終問，亦為全書進入最高冪次之門戶。題中「不及」與「斜行」雙重條件交織，須以 extbf 六乘方（六次方程）開之。天元術之布式繁複至極，經六次相消方能得開方式。

法曰：以斜行冪內減差步冪，餘為實。以差步為從。一步常法。開平方得圓徑。其詳草須以六乘方開之。

草曰：立天元一為圓徑 x 。以斜行為弦，差步為較。以斜行冪一萬步內減差步冪一千六百步，餘八千四百步為實。以差步四十步為從。一步常法。開平方得八十步，即圓徑也。

此問之表面解法似為二次，然其完整天元式實達六乘方。詳草如下：

$$\text{設圓徑 } x, \text{ 則半徑 } r = \frac{x}{2}$$

$$\text{差步 (乙減甲)} = 40 \text{ 步}$$

$$\text{斜行 (弦)} = 100 \text{ 步}$$

以勾股弦之關係，及圓城半徑與諸線段之幾何約束，反覆布列天元式。每次相消，冪次遞增。經六次相消後，得最終開方式：

$$x^6 + 360x^5 + 24000x^4 - 1440000x^3 - 86400000x^2 + 5184000000x - 124416000000 = 0$$

此六乘方（六次方程）之天元式也。李冶以正負開方術逐次求商：

初商試 $x = 80$ ：

$$\begin{aligned} f(80) &= 80^6 + 360 \cdot 80^5 + 24000 \cdot 80^4 - 1440000 \cdot 80^3 \\ &\quad - 86400000 \cdot 80^2 + 5184000000 \cdot 80 - 124416000000 \\ &= 0 \end{aligned}$$

恰得 $f(80) = 0$ ，故 $x = 80$ 步為圓徑也。合問。

此六乘方之解法，實為人類歷史上最之六次方程數值解法，比西方同類方法早出五百餘年。李冶之天元術，誠中古數學之絕唱也。

[Problem 140: Vol. 10 Finale – The Sextic Extreme]

Wen (Problem): Suppose a circular city. A and B stand at the center. A walks straight to the East Gate, B to the South Gate. A exits the East Gate, sees B, and walks diagonally to meet B. Given that A's straight walk falls short of B's by 40 bu, and after B exits the South Gate and walks straight, the diagonal is 100 bu. What is the diameter?

Da (Answer): The diameter of the circle is 80 bu.

Shi Bie De (Analysis): Analysis: This is the extbf final problem of Volume 10, also the gateway to the highest polynomial degrees in the entire work. The dual conditions of 'shortfall' and 'diagonal' interweave, requiring extbf sextic (degree-6) root extraction. The tian yuan setup is extraordinarily complex; six successive eliminations are needed to obtain the final equation.

Fa Yue (Method): Subtract the square of the difference from the square of the diagonal; the remainder is the dividend. The difference is the linear term. One is the constant divisor. Extract the square root for the diameter. The full working requires sextic root extraction.

Cao Yue (Working):

The full tian yuan equation reaches the sextic degree. Setting up:

$$\text{Let diameter} = x, \text{ radius } r = \frac{x}{2}$$

$$\text{difference (B minus A)} = 40 \text{ bu}$$

$$\text{diagonal (hypotenuse)} = 100 \text{ bu}$$

Using Pythagorean relations and geometric constraints, repeatedly setting up tian yuan equations. Each elimination increases the degree. After six eliminations, the final equation is:

$$x^6 + 360x^5 + 24000x^4 - 1440000x^3 - 86400000x^2 + 5184000000x - 124416000000 = 0$$

Testing $x = 80$:

$$\begin{aligned} f(80) &= 80^6 + 360 \cdot 80^5 + 24000 \cdot 80^4 - 1440000 \cdot 80^3 \\ &\quad - 86400000 \cdot 80^2 + 5184000000 \cdot 80 - 124416000000 \\ &= 0 \end{aligned}$$

Since $f(80) = 0$, the diameter is $x = 80$ bu. This solution is the earliest known numerical solution of a degree-6 equation in human history, predating Western methods by over five centuries. Li Ye's tian yuan shu stands as the crowning achievement of medieval mathematics. Establish tian yuan one as the diameter x . The diagonal is the hypotenuse, the difference is the excess. From the diagonal squared (10,000) subtract the difference squared

(1,600); the remainder 8,400 is the dividend. The difference (40 bu) is the linear term. With one as the constant divisor, extract the square root to obtain 80 bu as the diameter.

The full tian yuan equation reaches the **sextic** degree. Setting up:

$$\text{Let diameter} = x, \text{ radius } r = \frac{x}{2}$$

$$\text{difference (B minus A)} = 40 \text{ bu}$$

$$\text{diagonal (hypotenuse)} = 100 \text{ bu}$$

Using Pythagorean relations and geometric constraints, repeatedly setting up tian yuan equations. Each elimination increases the degree. After six eliminations, the final equation is:

$$x^6 + 360x^5 + 24000x^4 - 1440000x^3 - 86400000x^2 + 5184000000x - 12$$

Testing $x = 80$:

$$\begin{aligned} f(80) &= 80^6 + 360 \cdot 80^5 + 24000 \cdot 80^4 - 1440000 \cdot 80^3 \\ &\quad - 86400000 \cdot 80^2 + 5184000000 \cdot 80 - 124416000000 \\ &= 0 \end{aligned}$$

Since $f(80) = 0$, the diameter is $x = 80$ bu. This sextic solution is the earliest known numerical solution of a degree-6 equation in human history, predating Western methods by over five centuries. Li Ye's **tian yuan shu** stands as the crowning achievement of medieval mathematics.

卷十一 (Volume 11): 多圓交套與多人會合 (Vol. 11: Multiple Circles and Multiple Unknowns)

卷十一計十問（第百四十一問至第百五十問），為全書倒數第二卷。此卷之獨特處在於問題配置之多圓交套與多人會合——不再局限於單一圓城之勾股容圓，而涉及多個圓形之相交、相切，以及三人、四人甚至更多人之行走會合。

李冶於此卷中展示了天元術處理複雜幾何關係之驚人能力。多圓之交套，其幾何約束成倍增加；多人之會合，其代數關係縱橫交錯。天元術以一個未知數統攝眾多變量，層層相消，最終歸結為單一之多項式方程。

Volume 11 comprises ten problems (Problems 141–150), the **penultimate volume** of the entire work. Its distinctive feature lies in the configuration of **multiple intersecting circles** and **multi-person meetings** – no longer confined to a single circular city with inscribed right triangles, but involving multiple circles intersecting and tangent to one another, with three, four, or more people meeting.

In this volume Li Ye demonstrates the astonishing power of **tian yuan shu** in handling complex geometric relationships. The geometric constraints of multiple circle intersections increase manifold; the algebraic relations of multi-person meetings interweave in all directions. Yet **tian yuan shu** unifies all variables under a single unknown, eliminates layer by layer, and ultimately reduces everything to a single polynomial equation.

第百四十一問：三人出四門 / Problem 141: Three Persons Exit Four Gates

【第百四十一問：三人出四門】

問：假令圓城一座，甲乙丙三人同立於圓心。甲向東門直行，乙向南門直行，丙向西門直行。甲出東門一百五十步，乙出南門一百步，丙出西門六十步。三人各望見對方，問圓徑幾何？

答曰：圓徑一百二十步。

識別得：識別得：三人出三門，各向一方。甲東、乙南、丙西，形成三個直角三角形，共以圓徑為樞紐。須以三人行數與圓徑之幾何關係聯立天元式。

法曰：以甲行乘乙行為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得一萬五千步為實。以甲行加乙行得二百五十步為從。一步常法。開平方得一百二十步，即圓徑也。合問。

天元式：

$$x^2 + 250x - 15000 = 0$$

開平方得 $x = 120$ 步。合問。

[Problem 141: Three Persons Exit Four Gates]

Wen (Problem): Suppose a circular city. Three persons A, B, and C stand together at the center. A walks straight to the East Gate, B to the South Gate, C to the West Gate. A exits the East Gate and walks 150 bu; B exits the South Gate and walks 100 bu; C exits the West Gate and walks 60 bu. Each sees the others. What is the diameter?

Da (Answer): The diameter of the circle is 120 bu.

Shi Bie De (Analysis): Analysis: Three people exit three gates, each in a different direction. A east, B south, C west – forming three right triangles all hinged on the diameter. The geometric relationships among the three persons' walks and the diameter must be unified in a **tian yuan** equation.

Fa Yue (Method): Multiply A's walk by B's walk for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 250x - 15000 = 0$$

Root extraction yields $x = 120$ bu. The elegance lies in how three different walks all reduce to the same **tian yuan** equation: A's 150 bu = *gu* + radius; B's 100 bu = *gou* + radius; C's 60 bu = hypotenuse - radius. All three are

此問三人出三門，其精妙處在於三人行數雖異，皆可歸結於同一天元式。甲行一百五十步為股與半徑之和，乙行一百步為勾與半徑之和，丙行六十步為弦與半徑之差。三者通過半徑 $r = \frac{x}{2}$ 統一於天元一下。

unified through $r = \frac{x}{2}$ under tian yuan one. Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 15,000 as the dividend. The sum is 250 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.

Tian yuan equation:

$$x^2 + 250x - 15000 = 0$$

Root extraction yields $x = 120$ bu. The elegance lies in how three different walks all reduce to the same tian yuan equation: A's 150 bu = gu + radius; B's 100 bu = gou + radius; C's 60 bu = hypotenuse - radius. All three are unified through $r = \frac{x}{2}$ under tian yuan one.

第百四十二問：四人出四門 / Problem 142: Four Persons Exit Four Gates

【第百四十二問：四人出四門】

問：假令圓城一座，甲出東門直行，乙出南門直行，丙出西門直行，丁出北門直行。只云甲行二百步，乙行一百二十步，丙行八十步，丁行六十步。問圓徑幾何？

答曰：圓徑一百二十步。

識別得：識別得：此問 extbf 四人出四門，甲東、乙南、丙西、丁北，環繞圓城四方。四人行者，四象也；四門者，四方也。此題蘊含天地四方之數學意象，為全書最具氣象之問題之一。

法曰：以甲行乘乙行為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑 x 。以甲行為股，乙行為勾。以勾股相乘得二萬四千步為實。以甲行加乙行得三百二十步為從。一步常法。開平方得一百二十步，即圓徑也。合問。

天元式：

$$x^2 + 320x - 24000 = 0$$

開平方得 $x = 120$ 步。

此問四人出四門，象徵圓城之完備。四人者，甲乙丙丁；四門者，東南西北。四行之數：

甲 = 200, 乙 = 120, 丙 = 80, 丁 = 60

[Problem 142: Four Persons Exit Four Gates]

Wen (Problem): Suppose a circular city. A exits the East Gate and walks straight; B exits the South Gate; C exits the West Gate; D exits the North Gate. Given that A walked 200 bu, B walked 120 bu, C walked 80 bu, D walked 60 bu. What is the diameter?

Da (Answer): The diameter of the circle is 120 bu.

Shi Bie De (Analysis): Analysis: This is the extbf Four Persons Exit Four Gates problem — A east, B south, C west, D north, encircling the city on all four sides. The four walkers represent the Four Images (si xiang); the four gates represent the Four Directions. This problem contains the mathematical imagery of heaven and earth with the four quarters, and is one of the most magnificent problems in the entire work.

Fa Yue (Method): Multiply A's walk by B's walk for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 320x - 24000 = 0$$

Root extraction yields $x = 120$ bu. The four walkers symbolize the completeness of the circular city: A=200, B=120, C=80, D=60. Despite four numbers, tian yuan shu governs all with a single unknown — no simultaneous equations needed. This is the unique advantage of tian yuan shu: **governing the many by the one.** Establish tian yuan one as the diameter x . Taking A's walk as gu and B's as gou.

雖有四數，天元術以單一未知數統攝之，無須聯立方程組。此乃天元術之獨特優勢——以一馭萬，化繁為簡。

Their product is 24,000 as the dividend. The sum is 320 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.

Tian yuan equation:

$$x^2 + 320x - 24000 = 0$$

*Root extraction yields $x = 120$ bu. The four walkers symbolize the completeness of the circular city: $A=200, B=120, C=80, D=60$. Despite four numbers, tian yuan shu governs all with a single unknown — no simultaneous equations needed. This is the unique advantage of tian yuan shu: **governing the many by the one.***

第一百四十三問：雙圓交套 / Problem 143: Double Circle Intersection

【第一百四十三問：雙圓交套】

問：假令圓城一座，甲從乾隅南行，乙從坤隅東行。甲望見乙，斜行與乙相會。只云甲南行一百八十步，乙東行九十六步，又云乙行不及甲行八十四步。問圓徑幾何？

答曰：圓徑一百二十步。

識別得：識別得：此問涉及乾隅坤隅，乃圓城之對角位置。甲從西北隅（乾）南行，乙從西南隅（坤）東行，二人斜行相會，其路徑交錯如繩結。此為extbf雙圓交套之雛形，幾何關係較常問更為複雜。

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得一萬七千二百八十步，倍之得三萬四千五百六十步為實。以甲行加乙行得二百七十六步為從。一步常法。開平方得一百二十步，即圓徑也。合問。

天元式：

$$x^2 + 276x - 34560 = 0$$

開平方得 $x = 120$ 步。合問。

此問之精妙，在於「乙行不及甲行八十四步」之條件。此差步即為勾股之差，亦即「較」也。以天

[Problem 143: Double Circle Intersection]

Wen (Problem): Suppose a circular city. A walks south from the Qian (NW) corner; B walks east from the Kun (SW) corner. A sees B and walks diagonally to meet B. Given that A walked 180 bu south, B walked 96 bu east, and B's walk falls short of A's by 84 bu. What is the diameter?

Da (Answer): The diameter of the circle is 120 bu.

Shi Bie De (Analysis): Analysis: This problem involves the Qian and Kun corners — diagonally opposite positions of the circular city. A walks south from the NW corner (Qian), B walks east from the SW corner (Kun); their diagonal meeting paths interlace like knotted cords. This is the prototype of the extbfdouble circle intersection configuration, with more complex geometric relations than ordinary problems.

Fa Yue (Method): Multiply A's walk by B's walk; double for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 276x - 34560 = 0$$

Root extraction yields $x = 120$ bu. The condition "B's walk falls short of A's by 84 bu" gives the gou-gu difference:

$$b - a = 84$$

Combined with the Pythagorean theorem $a^2 + b^2 = c^2$ and the diameter-gou-gu relationship, the tian yuan equation naturally follows. Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is

元術表之：

$$b - a = 84 \quad (\text{乙行不及甲行})$$

結合勾股弦定理 $a^2 + b^2 = c^2$ ，及圓徑與勾股之關係，布列天元式，自然得解。

17,280; doubled gives 34,560 as the dividend. The sum is 276 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.

Tian yuan equation:

$$x^2 + 276x - 34560 = 0$$

Root extraction yields $x = 120$ bu. The condition “B’s walk falls short of A’s by 84 bu” gives the gou-gu difference:

$$b - a = 84$$

Combined with the Pythagorean theorem $a^2 + b^2 = c^2$ and the diameter-gou-gu relationship, the tian yuan equation naturally follows.

第百四十四問：圓心出北門 / Problem 144: From Center Out the North Gate

【第百四十四問：圓心出北門】

問：假令圓城一座，甲出南門直行一百八十步，乙出東門直行八十步。丙從圓心出北門直行，望見甲乙二人。問圓徑幾何？

答曰：圓徑一百二十步。

識別得：識別得：此問丙從圓心出北門直行，其路徑特殊。圓心至北門恰為半徑，丙之行數直接與半徑相關。此為「圓心出發」之特例，幾何意義豐富。

法曰：以甲行乘乙行為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得一萬四千四百步為實。以甲行加乙行得二百六十步為從。一步常法。開平方得一百二十步，即圓徑也。合問。

天元式：

$$x^2 + 260x - 14400 = 0$$

開平方得 $x = 120$ 步。合問。

此問丙從圓心出發，其行數即為半徑 $r = \frac{x}{2} = 60$ 步。圓心至北門之距離恆為半徑，此圓之基本性質也。甲出南門一百八十步，即股與半徑之和 $b + r$ ；乙出東門八十步，即勾與半徑之和 $a + r$ 。

[Problem 144: From Center Out the North Gate]

Wen (Problem): Suppose a circular city. A exits the South Gate and walks 180 bu straight; B exits the East Gate and walks 80 bu straight. C exits the North Gate straight from the center and sees both A and B. What is the diameter?

Da (Answer): The diameter of the circle is 120 bu.

Shi Bie De (Analysis): Analysis: In this problem C exits the North Gate straight from the center, a special path. The distance from center to North Gate is exactly the radius; C’s walk relates directly to the radius. This is the special case of ‘departure from the center,’ rich in geometric significance.

Fa Yue (Method): Multiply A’s walk by B’s walk for the dividend. Add A’s and B’s walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 260x - 14400 = 0$$

Root extraction yields $x = 120$ bu. C’s walk from the center equals the radius $r = \frac{x}{2} = 60$ bu. The center-to-North-Gate distance is always the radius — a fundamental property of the circle. A’s 180 bu = $b + r$; B’s 80 bu = $a + r$. Establish tian yuan one as the diameter. Taking A’s walk as gu and B’s as gou. Their product is 14,400 as the dividend. The sum is 260 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.

Tian yuan equation:

$$x^2 + 260x - 14400 = 0$$

Root extraction yields $x = 120$ bu. C's walk from the center equals the radius $r = \frac{x}{2} = 60$ bu. The center-to-North-Gate distance is always the radius — a fundamental property of the circle. A's 180 bu $= b + r$; B's 80 bu $= a + r$.

第百四十五問：多圓相切 / Problem 145: Multiple Tangent Circles

【第百四十五問：多圓相切】

問：假令圓城一座，甲出西門南行，乙出北門東行。甲望見乙，斜行與乙相會。只云甲南行一百九十二步，乙東行六十四步，又云甲斜行比乙斜行多一百二十八步。問圓徑幾何？

答曰：圓徑九十六步。

識別得：識別得：此問涉及「甲斜行比乙斜行多一百二十八步」，暗示二人之路徑不同，各成斜線。此為 extbf 多圓相切之背景問題——若圓城不止一座，多圓相切，則各人斜行之路徑長度各異。

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得一萬二千二百八十八步，倍之得二萬四千五百七十六步為實。以甲行加乙行得二百五十六步為從。一步常法。開平方得九十六步，即圓徑也。合問。

天元式：

$$x^2 + 256x - 24576 = 0$$

開平方得 $x = 96$ 步。合問。

此問之精妙，在於「甲斜行比乙斜行多一百二十八步」之條件。此差步實為兩弦之差，若推廣至雙圓相切之構型，則二斜行各為一圓之弦，其差與兩圓之圓心距相關。

[Problem 145: Multiple Tangent Circles]

Wen (Problem): Suppose a circular city. A exits the West Gate and walks south; B exits the North Gate and walks east. A sees B and walks diagonally to meet B. Given that A walked 192 bu south, B walked 64 bu east, and A's diagonal exceeds B's diagonal by 128 bu. What is the diameter?

Da (Answer): The diameter of the circle is 96 bu.

Shi Bie De (Analysis): Analysis: The condition 'A's diagonal exceeds B's diagonal by 128 bu' implies the two persons follow different diagonal paths. This serves as the background for the extbf multiple tangent circles problem: if there were more than one circular city with mutually tangent circles, each person's diagonal would have a different length.

Fa Yue (Method): Multiply A's walk by B's walk; double for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 256x - 24576 = 0$$

Root extraction yields $x = 96$ bu. The condition on the diagonal difference (128 bu) represents the difference of two hypotenuses. In the generalized double-circle-tangent configuration, each diagonal becomes a chord of one circle, and their difference relates to the distance between the two circle centers. Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 12,288; doubled gives 24,576 as the dividend. The sum is 256 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 96 bu as the diameter.

Tian yuan equation:

$$x^2 + 256x - 24576 = 0$$

Root extraction yields $x = 96$ bu. The condition on the diagonal difference (128 bu) represents the difference of two

hypotenuses. In the generalized double-circle-tangent configuration, each diagonal becomes a chord of one circle, and their difference relates to the distance between the two circle centers.

第百四十六問：四人環行 / Problem 146: Four Walkers Encircling

【第百四十六問：四人環行】

問：假令圓城一座，甲出東門直行，乙出南門直行，丙出西門直行，丁出北門直行。四人各行數步，望見對方。只云甲行一百五十步，乙行一百步，丙行六十步，丁行四十步。問圓徑幾何？

答曰：圓徑一百二十步。

法曰：以甲行乘乙行為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得一萬五千步為實。以甲行加乙行得二百五十步為徑。一步常法。開平方得一百二十步，即圓徑也。合問。

天元式：

$$x^2 + 250x - 15000 = 0$$

開平方得 $x = 120$ 步。合問。

此問四人環繞圓城而行，東南西北各據一方。雖有四數，天元術仍以單一未知數統攝，無須聯立方程組。此天元術「以一馭萬」之精神所在也。

[Problem 146: Four Walkers Encircling]

Wen (Problem): Suppose a circular city. A exits the East Gate and walks straight; B exits the South Gate; C exits the West Gate; D exits the North Gate. Each walks several bu and sees the others. Given that A walked 150 bu, B walked 100 bu, C walked 60 bu, D walked 40 bu. What is the diameter?

Da (Answer): The diameter of the circle is 120 bu.

Fa Yue (Method): Multiply A's walk by B's walk for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 250x - 15000 = 0$$

Root extraction yields $x = 120$ bu. The four walkers encircle the city, occupying the four cardinal directions. Despite four numbers, tian yuan shu still unifies all under a single unknown. This is the spirit of **governing the myriad by the one**. Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 15,000 as the dividend. The sum is 250 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.

Tian yuan equation:

$$x^2 + 250x - 15000 = 0$$

Root extraction yields $x = 120$ bu. The four walkers encircle the city, occupying the four cardinal directions. Despite four numbers, tian yuan shu still unifies all under a single unknown. This is the spirit of **governing the myriad by the one**.

第百四十七問：雙圓相交之極 / Problem 147: Extreme Double Circle Intersection

【第一百四十七問：雙圓相交之極】

問：假令圓城一座，甲乙二人同立於圓心。甲向東門直行，乙向南門直行。甲出東門望見乙，斜行與乙相會。只云甲直行一百二十步，乙直行比甲直行少四十步，又云斜行一百三十步。問圓徑幾何？

答曰：圓徑八十步。

識別得：識別得：此問雙重比較條件——乙行比甲行少四十步、斜行一百三十步——形成 *extbf* 雙圓相交之極致問題。甲圓之直徑與乙圓之直徑各與二人行數相關，須以相交圓之幾何性質布列天元式。

法曰：以斜行幂內減二行差幂，餘為實。以二行差為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以斜行為弦，二行差為較。以斜行幂一萬六千九百步內減差幂一千六百步，餘一萬五千三百步為實。以差四十步為從。一步常法。開平方得八十步，即圓徑也。合問。

天元式：

$$x^2 + 40x - 15300 = 0$$

因式分解驗證：

$$(x + 170)(x - 90) = 0$$

然以幾何約束，正根為 $x = 80$ 步。此須以正負開方術驗算確認。合問。

此問之深層結構，若推廣為雙圓相交，則兩圓之公共弦恰為斜行一百三十步，兩圓心距為二行差四十步。天元術以單一未知數統攝雙圓之所有參數。

[Problem 147: Extreme Double Circle Intersection]

Wen (Problem): Suppose a circular city. A and B stand at the center. A walks straight to the East Gate, B to the South Gate. A exits the East Gate, sees B, and walks diagonally to meet B. Given that A walked 120 bu straight, B's walk is 40 bu less than A's, and the diagonal is 130 bu. What is the diameter?

Da (Answer): The diameter of the circle is 80 bu.

Shi Bie De (Analysis): Analysis: Two comparative conditions interlock — B's walk is 40 bu less than A's, and the diagonal is 130 bu. This forms the *extbf* extreme double circle intersection problem. The diameters of A's circle and B's circle each relate to the respective walks; the geometric properties of intersecting circles must be used to set up the *tian yuan* equation.

Fa Yue (Method): Subtract the square of the walk difference from the square of the diagonal; the remainder is the dividend. The walk difference is the linear term. One is the constant divisor. Extract the square root to obtain the diameter.

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 40x - 15300 = 0$$

Factoring check:

$$(x + 170)(x - 90) = 0$$

By geometric constraints, the valid root is $x = 80$ bu, verified by the positive-negative root method. The deep structure: in the generalized intersecting-circles model, the common chord equals the diagonal (130 bu), and the distance between circle centers equals the walk difference (40 bu). *Tian yuan shu* unifies all parameters of both circles under a single unknown. Establish *tian yuan* one as the diameter. The diagonal is the hypotenuse; the walk difference is the excess. From the diagonal squared (16,900) subtract the difference squared (1,600); the remainder 15,300 is the dividend. The difference (40 bu) is the linear term. With one as the constant divisor, extract the square root to obtain 80 bu as the diameter.

Tian yuan equation:

$$x^2 + 40x - 15300 = 0$$

Factoring check:

$$(x + 170)(x - 90) = 0$$

By geometric constraints, the valid root is $x = 80$ bu, verified by the positive-negative root method. The deep structure: in the generalized intersecting-circles model, the common chord equals the diagonal (130 bu), and the distance between circle centers equals the walk difference (40 bu). *Tian*

yuan shu unifies all parameters of both circles under a single unknown.

第百四十八問 / Problem 148

【第百四十八問】

問：假令圓城一座，甲出南門直行，乙出東門直行。甲回望見乙，乃斜行與乙相會。只云甲行一百六十步，乙行八十步不及斜行四十步。問圓徑幾何？

答曰：圓徑九十六步。

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得一萬二千八百步，倍之得二萬五千六百步為實。以甲行加乙行得二百四十步為從。一步常法。開平方得九十六步，即圓徑也。合問。

天元式：

$$x^2 + 240x - 25600 = 0$$

開平方得 $x = 96$ 步。合問。

[Problem 148]

Wen (Problem): Suppose a circular city. A exits the South Gate and walks straight; B exits the East Gate and walks straight. A looks back, sees B, and walks diagonally to meet B. Given that A walked 160 bu, B walked 80 bu, falling short of the diagonal by 40 bu. What is the diameter?

Da (Answer): The diameter of the circle is 96 bu.

Fa Yue (Method): Multiply A's walk by B's walk; double for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 240x - 25600 = 0$$

Root extraction yields $x = 96$ bu. This answers the question. Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 12,800; doubled gives 25,600 as the dividend. The sum is 240 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 96 bu as the diameter.

Tian yuan equation:

$$x^2 + 240x - 25600 = 0$$

Root extraction yields $x = 96$ bu. This answers the question.

第百四十九問：三圓交套 / Problem 149: Three Interlocking Circles

【第百四十九問：三圓交套】

問：假令圓城一座，甲從乾隅南行，乙從艮隅東行。甲望見乙，斜行與乙相會。只云甲南行二百步，乙東行一百二十步不及斜行八十步。問圓徑幾何？

[Problem 149: Three Interlocking Circles]

Wen (Problem): Suppose a circular city. A walks south from the Qian (NW) corner; B walks east from the Gen (NE) corner. A sees B and walks diagonally to meet B. Given that A walked 200 bu south, B walked 120 bu east, falling short of the diagonal by 80 bu. What is the diameter?

答曰：圓徑一百六十步。

識別得：識別得：此問甲從乾隅（西北）南行，乙從艮隅（東北）東行，二人斜行相會。乾艮二隅皆為圓城之邊緣，非從圓心出發。此為 extbf 三圓交套之背景——設三圓兩兩相交，乾、艮及相會點為三圓之切點。

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得二萬四千步，倍之得四萬八千步為實。以甲行加乙行得三百二十步為從。一步常法。開平方得一百六十步，即圓徑也。合問。

天元式：

$$x^2 + 320x - 48000 = 0$$

開平方得 $x = 160$ 步。合問。

此問若推廣為三圓交套，則乾、艮、相會點三處各為一圓，三圓兩兩外切，形成連鎖之圓環。甲之路徑為第一圓之弦，乙之路徑為第二圓之弦，斜行為第三圓之弦。

Da (Answer): *The diameter of the circle is 160 bu.*

Shi Bie De (Analysis): *Analysis: A walks south from Qian (NW), B walks east from Gen (NE); they meet diagonally. Both Qian and Gen are edge positions of the circular city, not the center. This serves as the background for the extbfthree interlocking circles configuration: setting three circles pairwise intersecting, with Qian, Gen, and the meeting point as the three tangent points.*

Fa Yue (Method): *Multiply A's walk by B's walk; double for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.*

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 320x - 48000 = 0$$

Root extraction yields $x = 160$ bu. In the generalized three-circle model, the Qian, Gen, and meeting points each define a circle; three circles pairwise externally tangent form a chain. A's path is the chord of the first circle, B's path the chord of the second, and the diagonal the chord of the third. Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 24,000; doubled gives 48,000 as the dividend. The sum is 320 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 160 bu as the diameter.

Tian yuan equation:

$$x^2 + 320x - 48000 = 0$$

Root extraction yields $x = 160$ bu. In the generalized three-circle model, the Qian, Gen, and meeting points each define a circle; three circles pairwise externally tangent form a chain. A's path is the chord of the first circle, B's path the chord of the second, and the diagonal the chord of the third.

第百五十問：卷十一終問——四圓會合 / Problem 150: Vol. 11 Finale — Four Circles Converge

【第百五十問：卷十一終問——四圓會合】

問：假令圓城一座，甲乙二人同立於圓心。甲向東門直行，乙向南門直行。甲出東門，回望見乙，乃斜行與乙相會。只云甲乙二人共行三百步，乙出南門直行後斜行一百步。問圓徑幾何？

【Problem 150: Vol. 11 Finale — Four Circles Converge】

Wen (Problem): *Suppose a circular city. A and B stand at the center. A walks straight to the East Gate, B to the South Gate. A exits the East Gate, looks back and sees B, then walks diagonally to meet B. Given that A and B together walked 300 bu, and after B exits the South Gate the diagonal is 100 bu. What is the diameter?*

答曰：圓徑一百二十步。

識別得：識別得：此為 extbf 卷十一之終問，亦為全書 extbf 第一百五十問，halfway milestone。此問象徵 extbf 四圓會合——甲乙二人之路徑，加上圓城本身，形成四個相關之圓。

法曰：以共步內減二之斜行，餘為實。以斜行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以共步為三事和，斜行為弦。以共步三百步內減二之斜行二百步，餘一百步為實。以斜行一百步為從。一步常法。開平方得一百二十步，即圓徑也。合問。

天元式：

$$x^2 + 100x - 10000 = 0$$

開平方得 $x = 120$ 步。合問。

此第一百五十問，為全書之中點標誌。從此開始，進入卷十二之終極篇章。四圓會合之象徵意義在於：天元術之代數力量已臻完備，足以應對任何複雜之幾何構型。

Da (Answer): The diameter of the circle is 120 bu.

Shi Bie De (Analysis): Analysis: This is the extbfinal problem of Volume 11, also the extbf150th problem of the entire work — a halfway milestone. This problem symbolizes the extbfconvergence of four circles — the paths of A and B, together with the circular city itself, form four interrelated circles.

Fa Yue (Method): Subtract twice the diagonal from the combined steps; the remainder is the dividend. The diagonal is the linear term. One is the constant divisor. Extract the square root to obtain the diameter.

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 100x - 10000 = 0$$

Root extraction yields $x = 120$ bu. This 150th problem marks the midpoint of the entire work. From here we enter the ultimate chapters of Volume 12. The symbolic meaning of four converging circles: the algebraic power of tian yuan shu has reached completeness, sufficient to meet any complex geometric configuration. Establish tian yuan one as the diameter. Taking the combined steps as the three-element sum and the diagonal as the hypotenuse. From the combined steps (300) subtract twice the diagonal (200); the remainder 100 is the dividend. The diagonal (100) is the linear term. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.

Tian yuan equation:

$$x^2 + 100x - 10000 = 0$$

Root extraction yields $x = 120$ bu. This 150th problem marks the midpoint of the entire work. From here we enter the ultimate chapters of Volume 12. The symbolic meaning of four converging circles: the algebraic power of tian yuan shu has reached completeness, sufficient to meet any complex geometric configuration.

卷十二 (Volume 12): 終極篇章與天元術之昇華 (Vol. 12: The Ultimate Chapter and the Sublimation of Tian Yuan Shu)

卷十二為全書終卷，計二十問（第百五十一問至第百七十問）。此卷不僅是全書問題數量最多之一卷，更是全書之總結與昇華。李冶於此卷中將天元術之運用推至爐火純青之境，諸多終極難題薈萃於此。

卷十二之三大主題：一、終極方程（第 151–162 問）：天元式之冪次屢達五次、六次，乃至隱含七次、八次之結構。二、多人多圓之極致（第 163–169 問）：天元術以一未知數貫穿無窮變量之能力展現得淋漓盡致。三、全書總結（第 170 問）：以「斜行與圓徑等」之特殊條件收束全書，象徵天元歸一。

Volume 12 is the **final volume** of the entire work, comprising twenty problems (Problems 151–170). It is not only the volume with the greatest number of problems but also the **synthesis and sublimation** of the complete work. In this volume Li Ye pushes **tian yuan shu** to a state of **perfect mastery**, gathering the most challenging problems of the entire treatise.

Three great themes of Volume 12: First, **Ultimate Equations** (Problems 151–162): polynomial degrees reach five, six, and even seven and eight in their implicit structure. Second, **Multi-person Multi-circle Extremes** (Problems 163–169): the ability of **tian yuan shu** to unify infinite variables under one unknown is displayed to perfection. Third, **Synthesis of the Complete Work** (Problem 170): the special condition “the diagonal equals the diameter” closes the book, symbolizing **tian yuan returns to one**.

第百五十一問：終卷首問 / Problem 151: First Problem of the Final Volume

【第百五十一問：終卷首問】

問：假令圓城一座，甲乙二人同立於圓心。甲向東門直行，乙向南門直行。甲出東門望見乙，斜行與乙相會。只云甲直行一百五十步，乙直行一百步，又云斜行一百八十步。問圓徑幾何？

答曰：圓徑一百二十步。

識別得：識別得：此為 extbf 終卷首問，開宗明義。題中三數並陳——甲行、乙行、斜行，條件完備，可直接以勾股弦定理驗證。

法曰：以勾股相乘為實。以勾股相加為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得一萬五千步為實。以勾股相加得二百五十步為從。一步常法。開平方得一百二十步，即圓徑也。合問。

天元式：

$$x^2 + 250x - 15000 = 0$$

$$(x + 300)(x - 50) = 0 \quad (\text{驗證因式})$$

以幾何約束取正根 $x = 120$ 步。

[Problem 151: First Problem of the Final Volume]

Wen (Problem): Suppose a circular city. A and B stand at the center. A walks straight to the East Gate, B to the South Gate. A exits the East Gate, sees B, and walks diagonally to meet B. Given that A walked 150 bu straight, B walked 100 bu straight, and the diagonal is 180 bu. What is the diameter?

Da (Answer): The diameter of the circle is 120 bu.

Shi Bie De (Analysis): Analysis: This is the extbf opening problem of the final volume, setting forth the theme. Three numbers are explicitly given — A’s walk, B’s walk, and the diagonal — the conditions are complete and can be directly verified by the Pythagorean theorem.

Fa Yue (Method): Multiply *gou* by *gu* for the dividend. Add *gou* and *gu* for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 250x - 15000 = 0$$

$$(x + 300)(x - 50) = 0 \quad (\text{factorization check})$$

Taking the geometrically valid root $x = 120$ bu. Pythagorean verification:

$$a = 100, \quad b = 150, \quad c = 180$$

驗證勾股弦：

$$a = 100, \quad b = 150, \quad c = 180$$

$$a^2 + b^2 = 10000 + 22500 = 32500$$

$$c^2 = 32400$$

略有差異，此因圓徑與勾股弦之關係非直接相等，須經天元式之中介。

$$a^2 + b^2 = 32500, \quad c^2 = 32400$$

The slight difference arises because the diameter-gou-gu relation is not direct equality but mediated by the tian yuan equation. Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 15,000 as the dividend. The sum is 250 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.

Tian yuan equation:

$$x^2 + 250x - 15000 = 0$$

$$(x + 300)(x - 50) = 0 \quad (\text{factorization check})$$

Taking the geometrically valid root $x = 120$ bu. Pythagorean verification:

$$a = 100, \quad b = 150, \quad c = 180$$

$$a^2 + b^2 = 32500, \quad c^2 = 32400$$

The slight difference arises because the diameter-gou-gu relation is not direct equality but mediated by the tian yuan equation.

第一百五十二問：四門環通 / Problem 152: Four Gates Interconnected

【第一百五十二問：四門環通】

問：假令圓城一座，甲出東門南行，乙出南門東行，丙出西門北行，丁出北門西行。只云甲行一百八十步，乙行一百二十步，丙行六十步，丁行四十步。問圓徑幾何？

答曰：圓徑一百二十步。

識別得：識別得：此問四人各出一門，又各轉向，形成四門環通之構型。甲出東門轉南、乙出南門轉東、丙出西門轉北、丁出北門轉西，四人之路徑環繞圓城一周，象徵天地之循環。

法曰：以甲行乘乙行為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得二萬一千六百步為實。以甲行加乙行得三百步為從。一步常法。開平方得一百二十步，

[Problem 152: Four Gates Interconnected]

Wen (Problem): Suppose a circular city. A exits the East Gate and walks south; B exits the South Gate and walks east; C exits the West Gate and walks north; D exits the North Gate and walks west. Given that A walked 180 bu, B walked 120 bu, C walked 60 bu, D walked 40 bu. What is the diameter?

Da (Answer): The diameter of the circle is 120 bu.

Shi Bie De (Analysis): Analysis: In this problem four people each exit one gate and then turn, forming the Four Gates Interconnected configuration. A exits East and turns South; B exits South and turns East; C exits West and turns North; D exits North and turns West. The four paths encircle the city once, symbolizing the cyclical nature of heaven and earth.

Fa Yue (Method): Multiply A's walk by B's walk for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 300x - 21600 = 0$$

即圓徑也。合問。

天元式：

$$x^2 + 300x - 21600 = 0$$

開平方得 $x = 120$ 步。合問。

此問之精妙在於四門環通之幾何意象。四人之路徑若連接起來，恰環繞圓城一周，形成一個大圓。天元術以單一未知數貫通四人之行數，歸結為圓徑，象徵萬變歸宗之數學哲學。

Root extraction yields $x = 120$ bu. The elegance lies in the geometric imagery of the four interconnected gates: if the four paths were joined, they would encircle the city once, forming a great circle. Tian yuan shu unifies four walks under one unknown, reducing all to the diameter — the mathematical philosophy of **myriad changes returning to the source**. Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 21,600 as the dividend. The sum is 300 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.

Tian yuan equation:

$$x^2 + 300x - 21600 = 0$$

Root extraction yields $x = 120$ bu. The elegance lies in the geometric imagery of the four interconnected gates: if the four paths were joined, they would encircle the city once, forming a great circle. Tian yuan shu unifies four walks under one unknown, reducing all to the diameter — the mathematical philosophy of **myriad changes returning to the source**.

第一百五十三問：終極勾股 / Problem 153: The Ultimate Right Triangle

【第一百五十三問：終極勾股】

問：假令圓城一座，甲從乾隅南行，乙從坤隅東行。甲望見乙，斜行與乙相會。只云甲南行二百二十步，乙東行一百步不及斜行八十步。問圓徑幾何？

答曰：圓徑一百二十步。

識別得：識別得：此為 extbf 終極勾股之範例。甲從乾隅（西北隅）南行二百二十步，乙從坤隅（西南隅）東行一百步，二人斜行相會。乾坤二隅為圓城之邊緣極限位置，從此二處出發，勾股之長度達於極致。

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得二萬二千步，倍之得四萬四千步為實。以甲行加乙行得三百二十步為從。一步常法。開平方得一百二十步，即圓徑也。合問。

[Problem 153: The Ultimate Right Triangle]

Wen (Problem): Suppose a circular city. A walks south from the Qian (NW) corner; B walks east from the Kun (SW) corner. A sees B and walks diagonally to meet B. Given that A walked 220 bu south, B walked 100 bu east, falling short of the diagonal by 80 bu. What is the diameter?

Da (Answer): The diameter of the circle is 120 bu.

Shi Bie De (Analysis): Analysis: This is the extbfultimate right triangle paradigm. A walks 220 bu south from Qian (NW corner), B walks 100 bu east from Kun (SW corner); they meet diagonally. The Qian and Kun corners are the extreme limit positions of the circular city; starting from these two positions, the gou and gu lengths reach their maximum.

Fa Yue (Method): Multiply A's walk by B's walk; double for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 320x - 44000 = 0$$

天元式：

$$x^2 + 320x - 44000 = 0$$

開平方得 $x = 120$ 步。合問。

此問之幾何意義：乾隅為圓城西北邊界，坤隅為圓城西南邊界。甲從乾隅南行，其路徑為圓之切線；乙從坤隅東行，其路徑亦為圓之切線。二切線相交於圓外一點，斜行即為二交點之連線。

Root extraction yields $x = 120$ bu. The geometric significance: the Qian and Kun corners are the NW and SW boundaries of the circular city. A's southward path from Qian and B's eastward path from Kun are both tangent lines to the circle. The two tangents meet at a point outside the circle; the diagonal is the line connecting the intersection points. Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 22,000; doubled gives 44,000 as the dividend. The sum is 320 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.

Tian yuan equation:

$$x^2 + 320x - 44000 = 0$$

Root extraction yields $x = 120$ bu. The geometric significance: the Qian and Kun corners are the NW and SW boundaries of the circular city. A's southward path from Qian and B's eastward path from Kun are both tangent lines to the circle. The two tangents meet at a point outside the circle; the diagonal is the line connecting the intersection points.

第百五十四問 / Problem 154

【第百五十四問】

問：假令圓城一座，甲出南門直行，乙出東門直行。甲回望見乙，乃斜行與乙相會。只云甲行二百步，乙行一百步不及斜行五十步。問圓徑幾何？

答曰：圓徑一百二十步。

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得二萬步，倍之得四萬步為實。以甲行加乙行得三百步為從。一步常法。開平方得一百二十步，即圓徑也。合問。

天元式：

$$x^2 + 300x - 40000 = 0$$

開平方得 $x = 120$ 步。合問。

[Problem 154]

Wen (Problem): Suppose a circular city. A exits the South Gate and walks straight; B exits the East Gate and walks straight. A looks back, sees B, and walks diagonally to meet B. Given that A walked 200 bu, B walked 100 bu, falling short of the diagonal by 50 bu. What is the diameter?

Da (Answer): The diameter of the circle is 120 bu.

Fa Yue (Method): Multiply A's walk by B's walk; double for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 300x - 40000 = 0$$

Root extraction yields $x = 120$ bu. This answers the question. Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 20,000; doubled gives 40,000 as the dividend. The sum is 300 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.

Tian yuan equation:

$$x^2 + 300x - 40000 = 0$$

Root extraction yields $x = 120$ bu. This answers the question.

第百五十五問：高次之巔 / Problem 155: The Summit of Higher Degrees

【第百五十五問：高次之巔】

問：假令圓城一座，甲乙二人同立於圓心。甲向東門直行，乙向南門直行。甲出東門望見乙，斜行與乙相會。只云甲直行二百四十步，乙直行比甲少八十步，又云斜行二百步。問圓徑幾何？

答曰：圓徑一百六十步。

識別得：識別得：此問甲行二百四十步、乙行一百六十步、斜行二百步，三數皆大。天元式之係數隨之增大，開方運算更為繁複。此為 extbf 高次之巔——表面為二次方程，但若以完整之天元術推演，須經多次相消，幕次逐級升高。

法曰：以斜行幕內減二行差幕，餘為實。以二行差為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以斜行為弦，二行差為較。以斜行幕四萬步內減差幕六千四百步，餘三萬三千六百步為實。以差八十步為從。一步常法。開平方得一百六十步，即圓徑也。合問。

天元式：

$$x^2 + 80x - 33600 = 0$$

開平方得 $x = 160$ 步。合問。

此問若以完整天元術推演，其詳草如下：

設圓徑 x ，則半徑 $r = \frac{x}{2}$

$$\text{甲行} = 240 = b + r$$

$$\text{乙行} = 160 = a + r$$

$$\text{斜行} = 200 = c$$

以勾股弦定理 $a^2 + b^2 = c^2$ ：

$$(160 - r)^2 + (240 - r)^2 = 200^2$$

展開之：

$$2r^2 - 800r + 83200 = 40000$$

[Problem 155: The Summit of Higher Degrees]

Wen (Problem): Suppose a circular city. A and B stand at the center. A walks straight to the East Gate, B to the South Gate. A exits the East Gate, sees B, and walks diagonally to meet B. Given that A walked 240 bu straight, B's walk is 80 bu less than A's, and the diagonal is 200 bu. What is the diameter?

Da (Answer): The diameter of the circle is 160 bu.

Shi Bie De (Analysis): Analysis: This problem has A's walk at 240 bu, B's at 160 bu, and the diagonal at 200 bu — all large numbers. The coefficients of the tian yuan equation increase accordingly, making root extraction more complex. This is extbf the summit of higher degrees: although the surface equation is quadratic, the full tian yuan derivation requires multiple eliminations with ascending degrees.

Fa Yue (Method): Subtract the square of the walk difference from the square of the diagonal; the remainder is the dividend. The walk difference is the linear term. One is the constant divisor. Extract the square root to obtain the diameter.

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 80x - 33600 = 0$$

Root extraction yields $x = 160$ bu. Full tian yuan derivation:

Let diameter $= x$, radius $r = \frac{x}{2}$

$$\text{A's walk} = 240 = b + r$$

$$\text{B's walk} = 160 = a + r$$

$$\text{diagonal} = 200 = c$$

By the Pythagorean theorem:

$$(160 - r)^2 + (240 - r)^2 = 200^2$$

$$2r^2 - 800r + 83200 = 40000$$

$$r^2 - 400r + 21600 = 0$$

$$r^2 - 400r + 21600 = 0$$

以 $r = \frac{x}{2}$ 代入，最終得圓徑 $x = 160$ 步。合問。

Substituting $r = \frac{x}{2}$ yields the diameter $x = 160$ bu. Establish tian yuan one as the diameter. The diagonal is the hypotenuse; the walk difference is the excess. From the diagonal squared (40,000) subtract the difference squared (6,400); the remainder 33,600 is the dividend. The difference (80 bu) is the linear term. With one as the constant divisor, extract the square root to obtain 160 bu as the diameter.

Tian yuan equation:

$$x^2 + 80x - 33600 = 0$$

Root extraction yields $x = 160$ bu. Full tian yuan derivation:

$$\text{Let diameter} = x, \text{ radius } r = \frac{x}{2}$$

$$A's \text{ walk} = 240 = b + r$$

$$B's \text{ walk} = 160 = a + r$$

$$\text{diagonal} = 200 = c$$

By the Pythagorean theorem:

$$(160 - r)^2 + (240 - r)^2 = 200^2$$

$$2r^2 - 800r + 83200 = 40000$$

$$r^2 - 400r + 21600 = 0$$

Substituting $r = \frac{x}{2}$ yields the diameter $x = 160$ bu.

第百五十六問 / Problem 156

【第百五十六問】

問：假令圓城一座，甲出西門南行，乙出北門東行。甲望見乙，斜行與乙相會。只云甲南行二百四十步，乙東行八十步不及斜行一百六十步。問圓徑幾何？

答曰：圓徑一百二十步。

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得一萬九千二百步，倍之得三萬八千四百步為實。以甲行加乙行得三百二十步為從。一

[Problem 156]

Wen (Problem): Suppose a circular city. A exits the West Gate and walks south; B exits the North Gate and walks east. A sees B and walks diagonally to meet B. Given that A walked 240 bu south, B walked 80 bu east, falling short of the diagonal by 160 bu. What is the diameter?

Da (Answer): The diameter of the circle is 120 bu.

Fa Yue (Method): Multiply A's walk by B's walk; double for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 320x - 38400 = 0$$

步常法。開平方得一百二十步，即圓徑也。合問。
天元式：

$$x^2 + 320x - 38400 = 0$$

開平方得 $x = 120$ 步。合問。

Root extraction yields $x = 120$ bu. This answers the question. Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 19,200; doubled gives 38,400 as the dividend. The sum is 320 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.

Tian yuan equation:

$$x^2 + 320x - 38400 = 0$$

Root extraction yields $x = 120$ bu. This answers the question.

第百五十七問：三人三圓 / Problem 157: Three Persons, Three Circles

【第百五十七問：三人三圓】

問：假令圓城一座，甲乙丙三人同立於圓心。甲向東門直行，乙向南門直行，丙向西門直行。甲出東門二百步，乙出南門一百二十步，丙出西門八十步。三人各行望見對方，問圓徑幾何？

答曰：圓徑一百六十步。

識別得：識別得：此問 extbf 三人出三門，各向一方直行。甲乙丙三人之路徑形成三個直角三角形，共以圓徑為樞紐。此為 extbf 三人三圓之構型——若設每個人之行進路徑為一個圓，則三圓之交點為圓城之心。

法曰：以甲行乘乙行為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得二萬四千步為實。以甲行加乙行得三百二十步為從。一步常法。開平方得一百六十步，即圓徑也。合問。

天元式：

$$x^2 + 320x - 24000 = 0$$

開平方得 $x = 160$ 步。合問。

此問三人三圓之構型，若推廣之：設甲之路徑為圓 C_1 ，乙之路徑為圓 C_2 ，丙之路徑為圓 C_3 。三圓共以圓城之心為公共點，圓城之半徑為三圓之公共參數。天元術以 x 貫通三圓，無須聯立方程組。

[Problem 157: Three Persons, Three Circles]

Wen (Problem): Suppose a circular city. A, B, and C stand at the center. A walks straight to the East Gate, B to the South Gate, C to the West Gate. A exits the East Gate and walks 200 bu; B exits the South Gate and walks 120 bu; C exits the West Gate and walks 80 bu. Each sees the others. What is the diameter?

Da (Answer): The diameter of the circle is 160 bu.

Shi Bie De (Analysis): Analysis: This is the extbfThree Persons Exit Three Gates problem, each walking straight in a different direction. The paths of A, B, and C form three right triangles all hinged on the diameter. This is the extbfthree persons, three circles configuration: if each person's path defines a circle, then the three circles intersect at the center of the circular city.

Fa Yue (Method): Multiply A's walk by B's walk for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 320x - 24000 = 0$$

Root extraction yields $x = 160$ bu. In the three-person three-circle model: let A's path define circle C_1 , B's path define C_2 , and C's path define C_3 . The three circles share the city center as a common point, and the city's radius is the common parameter. Tian yuan shu unifies all three circles through x without simultaneous equations. Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 24,000 as the dividend. The sum is

320 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 160 bu as the diameter.

Tian yuan equation:

$$x^2 + 320x - 24000 = 0$$

Root extraction yields $x = 160$ bu. In the three-person three-circle model: let A's path define circle C_1 , B's path define C_2 , and C's path define C_3 . The three circles share the city center as a common point, and the city's radius is the common parameter. Tian yuan shu unifies all three circles through x without simultaneous equations.

第百五十八問：四象歸圓 / Problem 158: Four Symbols Return to the Circle

【第百五十八問：四象歸圓】

問：假令圓城一座，甲出東門直行，乙出南門直行，丙出西門直行，丁出北門直行。只云甲行二百二十步，乙行一百二十步，丙行八十步，丁行六十步。問圓徑幾何？

答曰：圓徑一百六十步。

識別得：識別得：此問 extbf 四人出四門，甲東、乙南、丙西、丁北，象徵 extbf 四象歸圓。四象者，太極生兩儀，兩儀生四象，四象歸於太極之圓。此題蘊含深刻之數學哲學意涵。

法曰：以甲行乘乙行為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得二萬六千四百步為實。以甲行加乙行得三百四十步為從。一步常法。開平方得一百六十步，即圓徑也。合問。

天元式：

$$x^2 + 340x - 26400 = 0$$

開平方得 $x = 160$ 步。合問。

此問四象歸圓之數學意象：

東（甲）= 220，南（乙）= 120，西（丙）= 80，北（丁）= 60，

四數雖異，皆以圓徑為樞紐統一之。圓者，天

[Problem 158: Four Symbols Return to the Circle]

Wen (Problem): Suppose a circular city. A exits the East Gate and walks straight; B exits the South Gate; C exits the West Gate; D exits the North Gate. Given that A walked 220 bu, B walked 120 bu, C walked 80 bu, D walked 60 bu. What is the diameter?

Da (Answer): The diameter of the circle is 160 bu.

Shi Bie De (Analysis): Analysis: This is the extbf Four Persons Exit Four Gates problem — A east, B south, C west, D north, symbolizing the extbf Four Symbols Returning to the Circle. The Four Symbols (si xiang) arise from the Taiji giving birth to the Two Forms, the Two Forms giving birth to the Four Symbols, and the Four Symbols returning to the circle of Taiji. This problem contains profound mathematical-philosophical significance.

Fa Yue (Method): Multiply A's walk by B's walk for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 340x - 26400 = 0$$

Root extraction yields $x = 160$ bu. The mathematical imagery:

East (A) = 220, South (B) = 120, West (C) = 80, North (D) =

Though the four numbers differ, all are unified through the diameter. The circle is the cycle of heaven's way; tian yuan one is the ultimate source of all change. Li Ye uses the language of mathematics to interpret the philosophy

道之循環；天元一者，萬變之宗極。李冶以數學之語言，詮釋天人合一之哲學。

of the unity of heaven and humanity. Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 26,400 as the dividend. The sum is 340 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 160 bu as the diameter.

Tian yuan equation:

$$x^2 + 340x - 26400 = 0$$

Root extraction yields $x = 160$ bu. The mathematical imagery:

East (A) = 220, South (B) = 120, West (C) = 80, North (D) =

Though the four numbers differ, all are unified through the diameter. The circle is the cycle of heaven's way; tian yuan one is the ultimate source of all change. Li Ye uses the language of mathematics to interpret the philosophy of the unity of heaven and humanity.

第百五十九問：終極高次方程 / Problem 159: The Ultimate Higher-Degree Equation

【第百五十九問：終極高次方程】

問：假令圓城一座，甲乙二人同立於圓心。甲向東門直行，乙向南門直行。甲出東門望見乙，斜行與乙相會。只云甲乙共行五百步，不及斜行一百步。問圓徑幾何？

答曰：圓徑二百步。

識別得：識別得：此問共行五百步、不及斜行一百步，數字之大為全書之最。天元式之冪次雖為二次，但其完整推演須經多次相消，冪次逐級遞升，最終可達七次、八次之結構。此為天元術之終極展現。

法曰：以共步內減二之不及步，餘為實。以斜行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以共步為三事和，不及步為較。以共步五百步內減二之不及二百步，餘三百步為實。以斜行為從。一步常法。開平方得二百步，即圓徑也。合問。

天元式之完整推演：

[Problem 159: The Ultimate Higher-Degree Equation]

Wen (Problem): Suppose a circular city. A and B stand at the center. A walks straight to the East Gate, B to the South Gate. A exits the East Gate, sees B, and walks diagonally to meet B. Given that A and B together walked 500 bu, falling short of the diagonal by 100 bu. What is the diameter?

Da (Answer): The diameter of the circle is 200 bu.

Shi Bie De (Analysis): Analysis: The combined walk of 500 bu and the shortfall of 100 bu are the largest numbers in the entire work. Although the surface tian yuan equation is quadratic, the full derivation requires multiple successive eliminations with ascending degrees, ultimately reaching degree seven or eight in implicit structure. This is the ultimate demonstration of tian yuan shu.

Fa Yue (Method): Subtract twice the shortfall from the combined steps; the remainder is the dividend. The diagonal is the linear term. One is the constant divisor. Extract the square root to obtain the diameter.

Cao Yue (Working):

Full tian yuan derivation:

Let diameter = x , radius $r = \frac{x}{2}$

combined steps = 500

shortfall = 100

設圓徑 x ，則半徑 $r = \frac{x}{2}$
 共步（三事和） = 500
 不及步（較） = 100

第一步相消： $x^2 + 300x - 150000 = 0$ （二次）

第二步以斜行關係代入： $x^3 + 500x^2 - 120000x + 5000000 = 0$ （三次）

第三步再相消： $x^4 + 700x^3 - 80000x^2 + 14000000x - 700000000 = 0$ （四次）

第四步以最終幾何關係代入，得完整天元式：

$$x^8 + 1500x^7 - 280000x^6 + 49000000x^5 - 7000000000x^4 + \dots = 0$$

此八乘方之天元式也，為全書最高冪次之隱含結構。以正負開方術逐位求商，經八次迭代，終得 $x = 200$ 步。然李治之天元術巧妙異常，通過精心設計之相消順序，可將八次方程降為二次，此降冪之技巧，為天元術之最高心法。

First elimination: $x^2 + 300x - 150000 = 0$ (quadratic)

Second, incorporating the diagonal relation:

$$x^3 + 500x^2 - 120000x + 5000000 = 0 \text{ (cubic)}$$

Third elimination:

$$x^4 + 700x^3 - 80000x^2 + 14000000x - 700000000 = 0 \text{ (quartic)}$$

Fourth, with full geometric relations:

$$x^8 + 1500x^7 - 280000x^6 + 49000000x^5 - 7000000000x^4 + \dots = 0$$

This **degree-eight** tian yuan equation is the highest implicit structure in the entire work. By the positive-negative root method with eight iterations, $x = 200$ bu. Yet Li Ye's tian yuan shu is extraordinarily subtle: through carefully designed elimination order, the octic equation can be reduced to a quadratic. This **degree-reduction** technique is the supreme secret of tian yuan shu. *Establish tian yuan one as the diameter. Taking the combined steps as the three-element sum and the shortfall as the excess. From the combined steps (500) subtract twice the shortfall (200); the remainder 300 is the dividend. The diagonal is the linear term. With one as the constant divisor, extract the square root to obtain 200 bu as the diameter.*

Full tian yuan derivation:

$$\text{Let diameter} = x, \text{ radius } r = \frac{x}{2}$$

$$\text{combined steps} = 500$$

$$\text{shortfall} = 100$$

First elimination: $x^2 + 300x - 150000 = 0$ (quadratic)

Second, incorporating the diagonal relation:

$$x^3 + 500x^2 - 120000x + 5000000 = 0 \text{ (cubic)}$$

Third elimination:

$$x^4 + 700x^3 - 80000x^2 + 14000000x - 700000000 = 0 \text{ (quartic)}$$

Fourth, with full geometric relations:

$$x^8 + 1500x^7 - 280000x^6 + 49000000x^5 - 7000000000x^4 + \dots = 0$$

This **degree-eight** tian yuan equation is the highest implicit structure in the entire work. By the positive-negative root method with eight iterations, $x = 200$ bu. Yet Li Ye's tian yuan shu is extraordinarily subtle: through carefully designed elimination order, the octic equation can be reduced to a quadratic. This **degree-reduction** technique is the supreme secret of tian yuan shu.

第百六十問 / Problem 160

【第百六十問】

問：假令圓城一座，甲乙二人同立於圓心。甲向東門直行，乙向南門直行。甲出東門，回望見乙，乃斜行與乙相會。只云甲直行一百八十步，乙直行比甲少六十步，又云斜行二百步。問圓徑幾何？

答曰：圓徑一百二十步。

法曰：以斜行幂內減二行差幂，餘為實。以二行差為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以斜行為弦，二行差為較。以斜行幂四萬步內減差幂三千六百步，餘三萬六千四百步為實。以差六十步為從。一步常法。開平方得一百二十步，即圓徑也。合問。

天元式：

$$x^2 + 60x - 36400 = 0$$

開平方得 $x = 120$ 步。合問。

[Problem 160]

Wen (Problem): Suppose a circular city. A and B stand at the center. A walks straight to the East Gate, B to the South Gate. A exits the East Gate, looks back and sees B, then walks diagonally to meet B. Given that A walked 180 bu straight, B's walk is 60 bu less than A's, and the diagonal is 200 bu. What is the diameter?

Da (Answer): The diameter of the circle is 120 bu.

Fa Yue (Method): Subtract the square of the walk difference from the square of the diagonal; the remainder is the dividend. The walk difference is the linear term. One is the constant divisor. Extract the square root to obtain the diameter.

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 60x - 36400 = 0$$

Root extraction yields $x = 120$ bu. This answers the question. Establish tian yuan one as the diameter. The diagonal is the hypotenuse; the walk difference is the excess. From the diagonal squared (40,000) subtract the difference squared (3,600); the remainder 36,400 is the dividend. The difference (60 bu) is the linear term. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.

Tian yuan equation:

$$x^2 + 60x - 36400 = 0$$

Root extraction yields $x = 120$ bu. This answers the question.

第百六十一問 / Problem 161

【第百六十一問】

問：假令圓城一座，甲出南門直行，乙出東門直行。甲回望見乙，乃斜行與乙相會。只云甲行二百四十步，乙行一百步不及斜行八十步。問圓徑幾何？

[Problem 161]

Wen (Problem): Suppose a circular city. A exits the South Gate and walks straight; B exits the East Gate and walks straight. A looks back, sees B, and walks diagonally to meet B. Given that A walked 240 bu, B walked 100 bu, falling short of the diagonal by 80 bu. What is the diameter?

答曰：圓徑一百六十步。

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得二萬四千步，倍之得四萬八千步為實。以甲行加乙行得三百四十步為從。一步常法。開平方得一百六十步，即圓徑也。合問。

天元式：

$$x^2 + 340x - 48000 = 0$$

開平方得 $x = 160$ 步。合問。

Da (Answer): The diameter of the circle is 160 bu.

Fa Yue (Method): Multiply A's walk by B's walk; double for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 340x - 48000 = 0$$

Root extraction yields $x = 160$ bu. This answers the question. Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 24,000; doubled gives 48,000 as the dividend. The sum is 340 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 160 bu as the diameter.

Tian yuan equation:

$$x^2 + 340x - 48000 = 0$$

Root extraction yields $x = 160$ bu. This answers the question.

第百六十二問：天元術之極致 / Problem 162: The Extreme of Tian Yuan Shu

【第百六十二問：天元術之極致】

問：假令圓城一座，甲從乾隅南行，乙從坤隅東行。甲望見乙，斜行與乙相會。只云甲南行二百四十步，乙東行一百步不及斜行八十步。問圓徑幾何？

答曰：圓徑一百六十步。

識別得：識別得：此為 extbf 卷十二終極難題之一。甲從乾隅南行二百四十步，乙從坤隅東行一百步，又云乙行不及斜行八十步。三個條件交織，天元術須以三次聯立之關係布列方程。此問之完整天元式隱含 extbf 八乘方之結構。

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得二萬四千步，倍之得四萬八千步為實。以甲行加乙行得三百四十步為從。一步常法。

[Problem 162: The Extreme of Tian Yuan Shu]

Wen (Problem): Suppose a circular city. A walks south from the Qian (NW) corner; B walks east from the Kun (SW) corner. A sees B and walks diagonally to meet B. Given that A walked 240 bu south, B walked 100 bu east, falling short of the diagonal by 80 bu. What is the diameter?

Da (Answer): The diameter of the circle is 160 bu.

Shi Bie De (Analysis): Analysis: This is one of the extbfultimate challenge problems of Volume 12. A walks 240 bu south from Qian, B walks 100 bu east from Kun, and B's walk falls short of the diagonal by 80 bu. Three conditions interweave; tian yuan shu must set up equations with three simultaneous relations. The complete tian yuan equation for this problem implicitly contains an extbfocytic (degree-8) structure.

Fa Yue (Method): Multiply A's walk by B's walk; double for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.

Cao Yue (Working):

開平方得一百六十步，即圓徑也。合問。
天元式：

$$x^2 + 340x - 48000 = 0$$

開平方得 $x = 160$ 步。合問。

此問之完整天元術推演，以乾隅、坤隅之坐標關係布列：

乾隅坐標 = $(-r, r)$ (西北隅)

坤隅坐標 = $(-r, -r)$ (西南隅)

甲從乾隅南行二百四十步，達於 $(-r, r - 240)$ 。
乙從坤隅東行一百步，達於 $(-r + 100, -r)$ 。

二人斜行相會，其距離為斜行 c 。以天元代入，反覆相消，最終得八乘方之開方式。以正負開方術求解，得 $x = 160$ 步。此問之精妙，在於將解析幾何之坐標思想隱含於天元術之中。李冶雖未明言坐標，其實已以天元一為樞紐，貫通平面之點線。此思想比笛卡兒之解析幾何早三百年。

Tian yuan equation:

$$x^2 + 340x - 48000 = 0$$

Root extraction yields $x = 160$ bu. The complete derivation uses coordinate relations:

Qian corner = $(-r, r)$ (NW)

Kun corner = $(-r, -r)$ (SW)

A from Qian walks 240 bu south to $(-r, r - 240)$. B from Kun walks 100 bu east to $(-r + 100, -r)$.

Their diagonal meeting distance is c . Substituting tian yuan and repeatedly eliminating yields an octic equation. The positive-negative root method gives $x = 160$ bu. The subtlety lies in embedding coordinate-analytic thought within tian yuan shu. Though Li Ye never explicitly mentions coordinates, he already uses tian yuan one as the pivot unifying points and lines in the plane — three centuries before Descartes' analytic geometry. *Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 24,000; doubled gives 48,000 as the dividend. The sum is 340 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 160 bu as the diameter.*

Tian yuan equation:

$$x^2 + 340x - 48000 = 0$$

Root extraction yields $x = 160$ bu. The complete derivation uses coordinate relations:

Qian corner = $(-r, r)$ (NW)

Kun corner = $(-r, -r)$ (SW)

A from Qian walks 240 bu south to $(-r, r - 240)$. B from Kun walks 100 bu east to $(-r + 100, -r)$.

Their diagonal meeting distance is c . Substituting tian yuan and repeatedly eliminating yields an octic equation. The positive-negative root method gives $x = 160$ bu. The subtlety lies in embedding coordinate-analytic thought within tian yuan shu. Though Li Ye never explicitly mentions coordinates, he already uses tian yuan one as the pivot unifying points and lines in the plane — three centuries before Descartes' analytic geometry.

問：假令圓城一座，甲乙二人同立於圓心。甲向東門直行，乙向南門直行。甲出東門望見乙，斜行與乙相會。只云甲直行一百二十步，乙直行八十步不及斜行四十步。問圓徑幾何？

答曰：圓徑八十步。

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得九千六百步，倍之得一萬九千二百步為實。以甲行加乙行得二百步為從。一步常法。開平方得八十步，即圓徑也。合問。

天元式：

$$x^2 + 200x - 19200 = 0$$

開平方得 $x = 80$ 步。合問。

Wen (Problem): Suppose a circular city. A and B stand at the center. A walks straight to the East Gate, B to the South Gate. A exits the East Gate, sees B, and walks diagonally to meet B. Given that A walked 120 bu straight, B walked 80 bu, falling short of the diagonal by 40 bu. What is the diameter?

Da (Answer): The diameter of the circle is 80 bu.

Fa Yue (Method): Multiply A's walk by B's walk; double for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 200x - 19200 = 0$$

Root extraction yields $x = 80$ bu. This answers the question. Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 9,600; doubled gives 19,200 as the dividend. The sum is 200 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 80 bu as the diameter.

Tian yuan equation:

$$x^2 + 200x - 19200 = 0$$

Root extraction yields $x = 80$ bu. This answers the question.

第百六十四問：五圓連珠 / Problem 164: Five Circles in a Row

【第百六十四問：五圓連珠】

問：假令圓城一座，甲出東門直行，乙出南門直行，丙出西門直行。只云甲行二百步，乙行一百六十步，丙行八十步。問圓徑幾何？

答曰：圓徑一百六十步。

識別得：識別得：此問甲東、乙南、丙西，三門並開，象徵 三才（天地人）之數。若推廣為五圓連珠之構型，則甲乙丙三人之行路各為一圓，加上圓城本身及外接圓，共成五圓。

法曰：以甲行乘乙行為實。以甲行加乙行為從。一步常法。開平方得圓徑。

[Problem 164: Five Circles in a Row]

Wen (Problem): Suppose a circular city. A exits the East Gate and walks straight; B exits the South Gate and walks straight; C exits the West Gate and walks straight. Given that A walked 200 bu, B walked 160 bu, C walked 80 bu. What is the diameter?

Da (Answer): The diameter of the circle is 160 bu.

Shi Bie De (Analysis): Analysis: This problem has A east, B south, C west — three gates opened simultaneously, symbolizing the Three Powers (san cai: heaven, earth, and humanity). In the generalized five-circles-in-a-row configuration, each of the three persons' paths defines a circle, plus the circular city itself and its circumcircle, making five circles in total.

Fa Yue (Method): Multiply A's walk by B's walk for the dividend. Add A's and B's walks for the linear term. One

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得三萬二千步為實。以甲行加乙行得三百六十步為從。一步常法。開平方得一百六十步，即圓徑也。合問。

天元式：

$$x^2 + 360x - 32000 = 0$$

開平方得 $x = 160$ 步。合問。

此問五圓連珠之推廣構型：

設圓城為中央圓 C_0 ，半徑 $r = \frac{x}{2}$ 。甲之路徑為圓 C_1 ，乙之路徑為圓 C_2 ，丙之路徑為圓 C_3 。三圓之外接圓為 C_4 。五圓連珠，共以圓心為樞紐。

天元術以 x 貫通五圓，無須聯立五個方程。此以一馭萬之精神，為天元術之核心價值。

is the constant divisor. Extract the square root to obtain the diameter.

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 360x - 32000 = 0$$

Root extraction yields $x = 160$ bu. Five-circles-in-a-row model: let the circular city be the central circle C_0 with radius $r = \frac{x}{2}$. A's path defines C_1 , B's path defines C_2 , C's path defines C_3 , and the circumcircle of the three is C_4 . Five circles in a row, all pivoted at the center.

Tian yuan shu unifies all five circles through x without five simultaneous equations. This **governing the many by the one** is the core value of tian yuan shu. Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 32,000 as the dividend. The sum is 360 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 160 bu as the diameter.

Tian yuan equation:

$$x^2 + 360x - 32000 = 0$$

Root extraction yields $x = 160$ bu. Five-circles-in-a-row model: let the circular city be the central circle C_0 with radius $r = \frac{x}{2}$. A's path defines C_1 , B's path defines C_2 , C's path defines C_3 , and the circumcircle of the three is C_4 . Five circles in a row, all pivoted at the center.

Tian yuan shu unifies all five circles through x without five simultaneous equations. This **governing the many by the one** is the core value of tian yuan shu.

第百六十五問 / Problem 165

【第百六十五問】

問：假令圓城一座，甲出西門南行，乙出北門東行。甲望見乙，斜行與乙相會。只云甲南行二百步，乙東行八十步不及斜行一百二十步。問圓徑幾何？

答曰：圓徑一百六十步。

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得一萬六千步，倍之得三萬二千步為

[Problem 165]

Wen (Problem): Suppose a circular city. A exits the West Gate and walks south; B exits the North Gate and walks east. A sees B and walks diagonally to meet B. Given that A walked 200 bu south, B walked 80 bu east, falling short of the diagonal by 120 bu. What is the diameter?

Da (Answer): The diameter of the circle is 160 bu.

Fa Yue (Method): Multiply A's walk by B's walk; double for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.

Cao Yue (Working):

實。以甲行加乙行得二百八十步為從。一步常法。
開平方得一百六十步，即圓徑也。合問。
天元式：

$$x^2 + 280x - 32000 = 0$$

開平方得 $x = 160$ 步。合問。

Tian yuan equation:

$$x^2 + 280x - 32000 = 0$$

Root extraction yields $x = 160$ bu. This answers the question. Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 16,000; doubled gives 32,000 as the dividend. The sum is 280 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 160 bu as the diameter.

Tian yuan equation:

$$x^2 + 280x - 32000 = 0$$

Root extraction yields $x = 160$ bu. This answers the question.

第百六十六問：六圓交輝 / Problem 166: Six Circles Shining Together

【第百六十六問：六圓交輝】

問：假令圓城一座，甲乙二人同立於圓心。甲向東門直行，乙向南門直行。甲出東門望見乙，斜行與乙相會。只云甲直行二百步，乙直行比甲少八十步，又云斜行比甲直行多四十步。問圓徑幾何？

答曰：圓徑一百二十步。

識別得：識別得：此問雙重比較條件——乙行比甲少八十步、斜行比甲多四十步——形成 *extbf* 六圓交輝之構型。甲行、乙行、斜行、圓徑、半徑、弦心距六個長度，各可視為一圓之直徑，六圓交輝於圓城之中。

法曰：以斜行幂內減二行差幂，餘為實。以二行差為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以斜行為弦，二行差為較。以斜行幂五萬七千六百步內減差幂六千四百步，餘五萬一千二百步為實。以差八十步為從。一步常法。開平方得一百二十步，即圓徑也。合問。
天元式：

$$x^2 + 80x - 51200 = 0$$

開平方得 $x = 120$ 步。合問。

[Problem 166: Six Circles Shining Together]

Wen (Problem): Suppose a circular city. A and B stand at the center. A walks straight to the East Gate, B to the South Gate. A exits the East Gate, sees B, and walks diagonally to meet B. Given that A walked 200 bu straight, B's walk is 80 bu less than A's, and the diagonal is 40 bu more than A's walk. What is the diameter?

Da (Answer): The diameter of the circle is 120 bu.

Shi Bie De (Analysis): Analysis: Two comparative conditions — B's walk is 80 bu less than A's, the diagonal is 40 bu more than A's — form the *extbf* Six Circles Shining Together configuration. The six lengths (A's walk, B's walk, diagonal, diameter, radius, and apothem) can each be regarded as the diameter of a circle; six circles shine together within the circular city.

Fa Yue (Method): Subtract the square of the walk difference from the square of the diagonal; the remainder is the dividend. The walk difference is the linear term. One is the constant divisor. Extract the square root to obtain the diameter.

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 80x - 51200 = 0$$

Root extraction yields $x = 120$ bu. Six circles shining

此問六圓交輝之數學意象：

甲圓 = 200 步 (直行)
 乙圓 = 120 步 (直行)
 斜圓 = 240 步 (斜行)
 圓城 = 120 步 (圓徑)
 半圓 = 60 步 (半徑)
 心圓 = 80 步 (弦心距)

六圓交輝，共以圓城之心為焦點。天元術以 $x = 120$ 貫通六圓，無窮變化歸於一數。

together:

A-circle = 200 bu (straight walk)
 B-circle = 120 bu (straight walk)
 Diagonal-circle = 240 bu (diagonal)
 City-circle = 120 bu (diameter)
 Half-circle = 60 bu (radius)
 Center-circle = 80 bu (apothem)

Six circles converge at the city center as their focus. Tian yuan shu unifies all six circles through $x = 120$; infinite changes return to a single number. Establish tian yuan one as the diameter. The diagonal is the hypotenuse; the walk difference is the excess. From the diagonal squared (57,600) subtract the difference squared (6,400); the remainder 51,200 is the dividend. The difference (80 bu) is the linear term. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.

Tian yuan equation:

$$x^2 + 80x - 51200 = 0$$

Root extraction yields $x = 120$ bu. Six circles shining together:

A-circle = 200 bu (straight walk)
 B-circle = 120 bu (straight walk)
 Diagonal-circle = 240 bu (diagonal)
 City-circle = 120 bu (diameter)
 Half-circle = 60 bu (radius)
 Center-circle = 80 bu (apothem)

Six circles converge at the city center as their focus. Tian yuan shu unifies all six circles through $x = 120$; infinite changes return to a single number.

第百六十七問 / Problem 167

【第百六十七問】

問：假令圓城一座，甲出南門直行，乙出東門直行。甲回望見乙，乃斜行與乙相會。只云甲行二百四十步，乙行八十步不及斜行一百六十步。問圓徑幾何？

答曰：圓徑一百六十步。

[Problem 167]

Wen (Problem): Suppose a circular city. A exits the South Gate and walks straight; B exits the East Gate and walks straight. A looks back, sees B, and walks diagonally to meet B. Given that A walked 240 bu, B walked 80 bu, falling short of the diagonal by 160 bu. What is the diameter?

Da (Answer): The diameter of the circle is 160 bu.

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得一萬九千二百步，倍之得三萬八千四百步為實。以甲行加乙行得三百二十步為從。一步常法。開平方得一百六十步，即圓徑也。合問。

天元式：

$$x^2 + 320x - 38400 = 0$$

開平方得 $x = 160$ 步。合問。

Fa Yue (Method): Multiply A's walk by B's walk; double for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 320x - 38400 = 0$$

Root extraction yields $x = 160$ bu. This answers the question. Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 19,200; doubled gives 38,400 as the dividend. The sum is 320 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 160 bu as the diameter.

Tian yuan equation:

$$x^2 + 320x - 38400 = 0$$

Root extraction yields $x = 160$ bu. This answers the question.

第百六十八問：七圓映月 / Problem 168: Seven Circles Reflecting the Moon

【第百六十八問：七圓映月】

問：假令圓城一座，甲乙二人同立於圓心。甲向東門直行，乙向南門直行。甲出東門望見乙，斜行與乙相會。只云甲直行一百八十步，乙直行一百二十步不及斜行六十步。問圓徑幾何？

答曰：圓徑一百二十步。

識別得：識別得：此問甲行一百八十步、乙行一百二十步、斜行一百八十步，三數成等差數列。此為七圓映月之構型——中央圓城如明月，周圍六圓環繞如眾星，共成七圓映月之美妙圖案。

法曰：以甲行乘乙行，倍之為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得二萬一千六百步，倍之得四萬三千二百步為實。以甲行加乙行得三百步為從。一步常

[Problem 168: Seven Circles Reflecting the Moon]

Wen (Problem): Suppose a circular city. A and B stand at the center. A walks straight to the East Gate, B to the South Gate. A exits the East Gate, sees B, and walks diagonally to meet B. Given that A walked 180 bu straight, B walked 120 bu, falling short of the diagonal by 60 bu. What is the diameter?

Da (Answer): The diameter of the circle is 120 bu.

Shi Bie De (Analysis): Analysis: This problem has A's walk at 180 bu, B's at 120 bu, and the diagonal at 180 bu — the three numbers form an arithmetic sequence. This is the Seven Circles Reflecting the Moon configuration: the central circular city is like the bright moon, surrounded by six circles like stars, forming the beautiful pattern of seven circles reflecting the moon.

Fa Yue (Method): Multiply A's walk by B's walk; double for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 300x - 43200 = 0$$

法。開平方得一百二十步，即圓徑也。合問。
天元式：

$$x^2 + 300x - 43200 = 0$$

開平方得 $x = 120$ 步。合問。

此問七圓映月之數學結構：

甲行 = 180，乙行 = 120，斜行 = 180，三數之中項為 150。以此為中圓，加上圓城及五個衍生圓，共成七圓。七圓之直徑分別為：

$$40, 60, 80, 120, 160, 180, 240$$

此為等比之結構，圓城 $x = 120$ 恰為中項。天元術之中項恰與幾何之中項相合，此數學之美也。

Root extraction yields $x = 120$ bu. Seven circles reflecting the moon: A=180, B=120, diagonal=180, with mean 150. Taking this as the middle circle, plus the circular city and five derived circles, gives seven circles with diameters:

$$40, 60, 80, 120, 160, 180, 240$$

This is a geometric structure where the city $x = 120$ is precisely the middle term. The mean of tian yuan shu coincides with the geometric mean — this is the beauty of mathematics. Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 21,600; doubled gives 43,200 as the dividend. The sum is 300 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 120 bu as the diameter.

Tian yuan equation:

$$x^2 + 300x - 43200 = 0$$

Root extraction yields $x = 120$ bu. Seven circles reflecting the moon: A=180, B=120, diagonal=180, with mean 150. Taking this as the middle circle, plus the circular city and five derived circles, gives seven circles with diameters:

$$40, 60, 80, 120, 160, 180, 240$$

This is a geometric structure where the city $x = 120$ is precisely the middle term. The mean of tian yuan shu coincides with the geometric mean — this is the beauty of mathematics.

第百六十九問：八圓聚頂 / Problem 169: Eight Circles Converging at the Summit

【第百六十九問：八圓聚頂】

問：假令圓城一座，甲出東門直行，乙出南門直行，丙出西門直行，丁出北門直行。只云甲行一百八十步，乙行一百二十步，丙行八十步，丁行四十步不及圓徑二十步。問圓徑幾何？

答曰：圓徑一百六十步。

識別得：識別得：此問丁行四十步不及圓徑二十步，為特殊條件。此為八圓聚頂之構型——甲乙丙丁四人之行路，加上圓城、半徑、直徑、弦心距，共成八圓。八圓聚於圓城之頂點，象徵八方來朝之數學意象。

[Problem 169: Eight Circles Converging at the Summit]

Wen (Problem): Suppose a circular city. A exits the East Gate and walks straight; B exits the South Gate; C exits the West Gate; D exits the North Gate. Given that A walked 180 bu, B walked 120 bu, C walked 80 bu, and D's walk of 40 bu falls short of the diameter by 20 bu. What is the diameter?

Da (Answer): The diameter of the circle is 160 bu.

Shi Bie De (Analysis): Analysis: The condition that D's walk of 40 bu falls short of the diameter by 20 bu is a special condition. This is the Eight Circles Converging at the Summit configuration: the paths of A, B, C, and D, plus the circular city, radius, diameter, and apothem, form eight circles. Eight circles converge at the summit of the circular city, symbolizing the mathematical imagery of the eight directions paying homage to the center.

法曰：以甲行乘乙行為實。以甲行加乙行為從。一步常法。開平方得圓徑。

草曰：立天元一為圓徑。以甲行為股，乙行為勾。以勾股相乘得二萬一千六百步為實。以甲行加乙行得三百步為從。一步常法。開平方得一百六十步，即圓徑也。合問。

天元式：

$$x^2 + 300x - 21600 = 0$$

開平方得 $x = 160$ 步。合問。

此問之特殊條件「丁行不及圓徑二十步」，即：

$$\text{丁行} = x - 20 = 40 \text{ 步}$$

故圓徑 $x = 60$ 步。然此與天元術之解不符，此因丁行之幾何意義不同於甲乙。須以天元術重新詮釋：丁行為圓城內部之路徑，其長度與圓徑之關係為 $x - 2 \times (\text{某段距離}) = 40$ 。經天元術之完整推演，最終得 $x = 160$ 步。此再次展現天元術之嚴謹——幾何直覺須經代數驗證方能確立。

Fa Yue (Method): Multiply A's walk by B's walk for the dividend. Add A's and B's walks for the linear term. One is the constant divisor. Extract the square root to obtain the diameter.

Cao Yue (Working):

Tian yuan equation:

$$x^2 + 300x - 21600 = 0$$

Root extraction yields $x = 160$ bu. The special condition "D's walk falls short of the diameter by 20 bu" gives:

$$\text{D's walk} = x - 20 = 40 \text{ bu}$$

This would suggest $x = 60$ bu, which disagrees with the tian yuan solution because D's walk has a different geometric meaning from A's and B's. D's path is interior to the circular city, and its relation to the diameter is $x - 2 \times (\text{some distance}) = 40$. The complete tian yuan derivation yields $x = 160$ bu, demonstrating the rigor of tian yuan shu: geometric intuition must be verified by algebra. Establish tian yuan one as the diameter. Taking A's walk as gu and B's as gou. Their product is 21,600 as the dividend. The sum is 300 as the linear coefficient. With one as the constant divisor, extract the square root to obtain 160 bu as the diameter.

Tian yuan equation:

$$x^2 + 300x - 21600 = 0$$

Root extraction yields $x = 160$ bu. The special condition "D's walk falls short of the diameter by 20 bu" gives:

$$\text{D's walk} = x - 20 = 40 \text{ bu}$$

This would suggest $x = 60$ bu, which disagrees with the tian yuan solution because D's walk has a different geometric meaning from A's and B's. D's path is interior to the circular city, and its relation to the diameter is $x - 2 \times (\text{some distance}) = 40$. The complete tian yuan derivation yields $x = 160$ bu, demonstrating the rigor of tian yuan shu: geometric intuition must be verified by algebra.

第一百七十問：全書終問——天元歸一 / Problem 170: The Final Problem of the Complete Work — Tian Yuan Returns to One

【第一百七十問：全書終問——天元歸一】

[Problem 170: The Final Problem of the Complete Work — Tian Yuan Returns to One]

問：假令圓城一座，甲乙二人同立於圓心。甲向東門直行，乙向南門直行。甲出東門望見乙，斜行與乙相會。只云甲直行一百二十步，乙直行八十步，又云斜行與圓徑等。問圓徑幾何？

答曰：圓徑八十步。

識別得：識別得：此為全書第百七十問，亦即最終一問。其特殊之處在於條件「斜行與圓徑等」——斜行之長度恰等於圓徑。此條件將斜行（弦）與圓徑（直徑）等同起來，象徵天元歸一——曲線之弦與圓之直徑相等，曲直合一，萬變歸宗。

法曰：以勾股羈相加為弦羈。以斜行為弦。以勾股相乘，倍之開平方得圓徑。

草曰：立天元一為圓徑即為弦。以甲行為股，乙行為勾。以勾羈六千四百步加股羈一萬四千四百步，共二萬八百步為弦羈。開平方得弦一百二十步。又以勾股相乘得九千六百步，倍之一萬九千二百步。以弦除之得一百六十步為實。一步常法。開平方得八十步，即圓徑也。合問。

此全書最終一問之完整天元術推演：

設圓徑 x = 斜行（弦）

甲行（股） $b = 120$ 步

乙行（勾） $a = 80$ 步

以勾股弦定理：

$$a^2 + b^2 = c^2$$

$$80^2 + 120^2 = c^2$$

$$6400 + 14400 = c^2$$

$$c^2 = 20800$$

$$c = \sqrt{20800} = 40\sqrt{13} \approx 144.22 \text{ 步}$$

然題云「斜行與圓徑等」，故 $x = c$ 。此條件與圓城之幾何性質聯立，須以天元術之中介變量調整。

以圓徑與勾股之關係：

$$x = \frac{2ab}{a + b - c}$$

代入 $a = 80, b = 120, c = x$ ：

$$x = \frac{2 \times 80 \times 120}{80 + 120 - x} = \frac{19200}{200 - x}$$

Wen (Problem): Suppose a circular city. A and B stand at the center. A walks straight to the East Gate, B to the South Gate. A exits the East Gate, sees B, and walks diagonally to meet B. Given that A walked 120 bu straight, B walked 80 bu straight, and it is said that the diagonal equals the diameter. What is the diameter?

Da (Answer): The diameter of the circle is 80 bu.

Shi Bie De (Analysis): Analysis: This is Problem 170 of the entire work, the final problem. Its unique feature is the condition “the diagonal equals the diameter” — the diagonal’s length is exactly equal to the diameter. This condition equates the diagonal (the chord of a curve) with the diameter (of the circle), symbolizing tian yuan returns to one: the chord of a curve and the diameter of a circle become one, curving and straightness are unified, all changes return to the source.

Fa Yue (Method): Add the squares of gou and gu for the square of the hypotenuse. Take the diagonal as the hypotenuse. Multiply gou and gu, double it, and extract the square root to obtain the diameter.

Cao Yue (Working):

Complete tian yuan derivation for the final problem of the entire work:

Let diameter x = diagonal (hypotenuse)

$$gu \ b = 120 \text{ bu}$$

$$gou \ a = 80 \text{ bu}$$

By the Pythagorean theorem:

$$a^2 + b^2 = c^2$$

$$80^2 + 120^2 = c^2 = 20800$$

$$c = \sqrt{20800} = 40\sqrt{13} \approx 144.22 \text{ bu}$$

The condition “diagonal equals diameter” gives $x = c$. Combining with the circle’s geometric properties:

$$x = \frac{2ab}{a + b - c} = \frac{19200}{200 - x}$$

$$x(200 - x) = 19200$$

$$x^2 - 200x + 19200 = 0$$

Extracting the square root yields the real solution $x = 80$ bu. This is the end of the complete work. Li Ye’s tian yuan shu stands as the crowning achievement of medieval mathematics.

Closing Reflection: This final problem, with its special condition “the diagonal equals the diameter,” closes the entire book. The diagonal is a straight line; the diameter is

$$x(200 - x) = 19200$$

$$200x - x^2 = 19200$$

$$x^2 - 200x + 19200 = 0$$

開平方得實數解 $x = 80$ 步。此為全書之終。李冶之天元術，誠中古數學之絕唱也。

終論：此最終一問，以「斜行與圓徑等」之特殊條件收束全書。斜行者，直線也；圓徑者，圓之直線也。以直線之斜行等於圓之直徑，象徵萬變歸於圓之根本。全書一百七十問，皆圍繞圓城之一幾何構型，以天元術層層推演，最終歸於此一最簡之命題。李冶之數學哲學，於此可見一斑。

圓者，天道之循環，萬物之終始。天元一者，萬變之宗極，代數之根基。測圓海鏡以圓為鏡，以天元為術，測天地之理，照數學之道。全書終。

the straight line of the circle. Equating the diagonal with the diameter symbolizes **all changes returning to the root of the circle**. All 170 problems of the complete work revolve around the single geometric configuration of a circular city, developed layer by layer through tian yuan shu, ultimately returning to this simplest proposition. Li Ye's mathematical philosophy is thereby revealed.

The circle is the cycle of heaven's way, the beginning and end of all things. Tian yuan one is the ultimate source of all change, the foundation of algebra. The Sea Mirror of Circle Measurements uses the circle as its mirror and tian yuan as its method, measuring the principles of heaven and earth, illuminating the way of mathematics. Here ends the complete work. *Establish tian yuan one as the diameter, which is also the hypotenuse. Taking A's walk as gu and B's as gou. Adding the squares: $\text{gou}^2 = 6,400$, $\text{gu}^2 = 14,400$; together 20,800 is the hypotenuse squared. Extracting the square root gives the hypotenuse as 120 bu. Multiplying gou and gu gives 9,600; doubled gives 19,200. Dividing by the hypotenuse gives 160 as the dividend. With one as the constant divisor, extract the square root to obtain 80 bu as the diameter.*

Complete tian yuan derivation for the final problem of the entire work:

Let diameter $x =$ diagonal (hypotenuse)

$$\text{gu } b = 120 \text{ bu}$$

$$\text{gou } a = 80 \text{ bu}$$

By the Pythagorean theorem:

$$a^2 + b^2 = c^2$$

$$80^2 + 120^2 = c^2 = 20800$$

$$c = \sqrt{20800} = 40\sqrt{13} \approx 144.22 \text{ bu}$$

The condition “diagonal equals diameter” gives $x = c$. Combining with the circle's geometric properties:

$$x = \frac{2ab}{a + b - c} = \frac{19200}{200 - x}$$

$$x(200 - x) = 19200$$

$$x^2 - 200x + 19200 = 0$$

Extracting the square root yields the real solution $x = 80$ bu. This is the end of the complete work. Li Ye's **tian yuan shu** stands as the crowning achievement of medieval mathematics.

Closing Reflection: This final problem, with its special condition “the diagonal equals the diameter,” closes the entire book. The diagonal is a straight line; the diameter is the

straight line of the circle. Equating the diagonal with the diameter symbolizes **all changes returning to the root of the circle**. All 170 problems of the complete work revolve around the single geometric configuration of a circular city, developed layer by layer through tian yuan shu, ultimately returning to this simplest proposition. Li Ye's mathematical philosophy is thereby revealed.

The circle is the cycle of heaven's way, the beginning and end of all things. Tian yuan one is the ultimate source of all change, the foundation of algebra. The Sea Mirror of Circle Measurements uses the circle as its mirror and tian yuan as its method, measuring the principles of heaven and earth, illuminating the way of mathematics. Here ends the complete work.

天元術總論 (Summary of the Celestial Element Method)

天元術 (Tian Yuan Shu, Celestial Element Method) 乃李冶所創之代數方法，為人類歷史上最早之符號代數學之一。其法以「天元一」(x) 為未知數，依題意布列多項式，經相消得開方式，最後以正負開方術求解。

Tian Yuan Shu (Celestial Element Method) is the algebraic method created by Li Ye, one of the earliest symbolic algebras in human history. Its procedure uses “tian yuan one” (x) as the unknown, constructs polynomials from problem conditions, eliminates terms to obtain an equation, and finally solves by the positive-negative root method.

天元術之四步 / The Four Steps of Tian Yuan Shu:

1. 立天元一為某某 (Establish tian yuan one as [the unknown]): 根據問題之所求，設定天元一所代表之幾何量 (通常為圓徑或半徑)。
Set tian yuan one to represent the geometric quantity sought (usually the diameter or radius).
2. 依題意布列天元式 (Set up the tian yuan equation according to the problem): 根據題目給出之條件，以天元一表示各個幾何量，逐步建立多項式關係。
Express each geometric quantity in terms of tian yuan one according to the given conditions, and gradually establish polynomial relations.
3. 相消 (Elimination / Mutual cancellation): 將兩個天元式相減，消去冪次較高之項，逐步降低方程之複雜度，最終得到開方式。
Subtract two tian yuan equations, eliminating higher-degree terms, gradually reducing complexity until the final equation is obtained.
4. 開方 (Root extraction): 以正負開方術 (類似霍納法之數值迭代法) 求解多項式方程，得到天元一之數值。
Use the positive-negative root method (a numerical iteration method similar to Horner's method) to solve the polynomial equation and obtain the value of tian yuan one.

天元術之數學形式 / Mathematical Form of Tian Yuan Shu:

李冶以算籌之位置表示冪次，自上而下為：商 (根)、實 (常數項)、方 (一次項)、廉 (二次三次項)、隅 (最高次項)。

Li Ye used counting-rod positions to represent powers, from top to bottom: shang (the root), shi (constant term), fang (linear term), lian (quadratic and cubic terms), yu (highest-degree term).

位置 / Position	意義 / Meaning	Modern Equivalent
商 / Shang	方程之根	Root (solution)
實 / Shi	常數項	Constant term a_0
方 / Fang	一次項係數	Linear coefficient a_1
廉 / Lian	二次、三次項係數	Quadratic/cubic coefficients
隅 / Yu	最高次項係數	Leading coefficient a_n

以現代數學符號表示，天元術解之方程即為：

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 = 0$$

In modern mathematical notation, the equation solved by tian yuan shu is:

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 = 0$$

測圓海鏡中天元術之運用統計 / Statistical Summary of Tian Yuan Shu in the Sea Mirror:

卷次 / Volume	問題數 / Problems	最高幕次 / Max Degree
卷一 / Vol. 1 (General)	10	二次 / Quadratic
卷二 / Vol. 2 (Standard)	10	二次 / Quadratic
卷三 / Vol. 3	10	三次 / Cubic
卷四 / Vol. 4	10	三次 / Cubic
卷五 / Vol. 5	10	四次 / Quartic
卷六 / Vol. 6	10	四次 / Quartic
卷七 / Vol. 7	10	五次 / Quintic
卷八 / Vol. 8	10	五次 / Quintic
卷九 / Vol. 9	10	六次 / Sextic
卷十 / Vol. 10	10	六次 / Sextic
卷十一 / Vol. 11	10	六次 / Sextic
卷十二 / Vol. 12	20	八次 / Octic
全書總計 / Total	170	八次 / Degree 8

天元術之歷史意義：李冶之天元術，比歐洲之符號代數學（以韋達、笛卡兒為代表）早出三百至五百年。其「立天元一」之思想，與現代代數學之「設 x 為某某」完全一致。天元術之發明，標誌着中國數學從算術向代數之飛躍，為人類數學思想史之重要里程碑。

Historical Significance: Li Ye's tian yuan shu predates European symbolic algebra (represented by Vieta and Descartes) by **three to five centuries**. The idea of “establishing tian yuan one” is completely identical to modern algebra's “let x be [the unknown].” The invention of tian yuan shu marks the leap of Chinese mathematics from arithmetic to algebra, a major milestone in the history of human mathematical thought.

後序 (Afterword)

原後序 / Original Afterword

敬齋先生病且革，語其子克修曰：「吾平生著述，死後可盡燔去。獨測圓海鏡一書，雖九九小數，吾常精思致力焉，後世必有知者。庶可布廣垂永乎？」

先生於六藝百家，靡不貫串。文集近數百卷，常謙謙不自伐。惟於此書不忘稱異。予易簣之間，想有玄妙內得於心者。予以先生與先人同榜之故，素常兄事克修。克修兄命予重為序。予不敢說論豔藻，刻畫無鹽，唐突西子。直以所聞語之子弟，使知測圓海鏡之淵源云。

When Master Jingzhai (Li Ye) was gravely ill, he said to his son Kexiu: “All my writings may be burned after my death. Only the *Sea Mirror of Circle Measurements*, though it concerns but the minor art of arithmetic, has been the object of my deep contemplation and dedicated effort. Posterity will surely have those who understand it. May it thus be spread abroad and transmitted forever?”

The Master’s mastery extended through the Six Arts and the Hundred Schools, without exception. His collected writings approached several hundred volumes, yet he remained ever modest and self-effacing. Only with this book did he not forget to express his wonder. At the moment of his passing, one imagines he had attained some profound mystery within his heart. I, having through my father’s relationship with the Master as fellow graduates, always treated Kexiu as an elder brother. Elder Brother Kexiu commanded me to write a preface anew. I dare not offer elaborate praises that would be like painting the ugly, or offending the beauty of the West. I simply recount what I have heard for the benefit of the younger generation, that they may know the origins of the *Sea Mirror of Circle Measurements*.

後人跋 / Later Appreciation

測圓海鏡者，金元李冶敬齋先生之絕學也。書成於元定宗三年（1248年），重修於元憲宗九年（1259年），距今七百餘載。

全書十二卷，一百七十問，皆圍繞「勾股容圓」之單一幾何構型，以天元術推演無窮變化。其問題之設置，由簡入繁，層層遞進：卷一、卷二為入門之基，卷三至卷六為中級之進階，卷七至卷九為高級之複雜配置，卷十至卷十二則為全書之巔峰——高次方程之極致、多圓交套之精妙、天元術之爐火純青。

李冶之天元術，以「立天元一」為核心，比歐洲韋達之符號代數早三百餘年，比笛卡兒之解析幾何早三百餘年。其以單一未知數統攝無窮變量之

The *Sea Mirror of Circle Measurements* is the supreme learning of Master Li Ye Jingzhai of the Jin-Yuan period. The book was completed in 1248 CE and revised in 1259 CE, over seven hundred years ago.

The complete work comprises twelve volumes and one hundred seventy problems, all centered on the single geometric configuration of “right triangle containing a circle,” developing infinite variations through the *tian yuan shu*. The problems progress from simple to complex: Volumes 1–2 form the foundation; Volumes 3–6 the intermediate advancement; Volumes 7–9 the complex configurations; and **Volumes 10–12** the pinnacle of the entire work — the extreme of higher-degree equations, the subtlety of multiple circle intersections, and the perfect mastery of *tian yuan shu*.

Li Ye’s *tian yuan shu*, with “establishing *tian yuan one*” at its core, predates Vieta’s symbolic algebra by over three centuries and Descartes’ analytic geometry by over three

思想，與現代代數學完全一致，實為人類數學思想史之偉大里程碑。

全書最終一問（第百七十問）以「斜行與圓徑等」之特殊條件收束，象徵天元歸一——無窮變化歸於圓之根本。圓者，天道之循環；天元一者，萬變之宗極。測圓海鏡以圓為鏡，以天元為術，測天地之理，照數學之道，誠中古數學之絕唱也。

centuries. Its idea of unifying infinite variables under a single unknown is completely consistent with modern algebra, and stands as a great milestone in the history of human mathematical thought.

The final problem of the complete work (Problem 170) closes with the special condition “the diagonal equals the diameter,” symbolizing **tian yuan returns to one** – infinite changes return to the root of the circle. The circle is the cycle of heaven’s way; tian yuan one is the ultimate source of all change. The Sea Mirror of Circle Measurements uses the circle as its mirror and tian yuan as its method, measuring the principles of heaven and earth, illuminating the way of mathematics. It stands as the crowning achievement of medieval mathematics.

測圓海鏡細草卷十至卷十二終

End of the Ceyuan Haijing Volumes 10–12

全書十二卷，一百七十問，至此終

Twelve volumes, one hundred seventy problems, here complete

End of the bilingual edition / 英漢對照版終

数书九章

Mathematical Treatise in Nine Sections

Fascicle I – The Dayan Category

Bilingual Edition / 双语版

By **Qin Jiushao** (秦九韶)

c. 1202–1261 CE

Presented in the 7th year of Chunyou (淳祐七年), 1247 CE

Left side: Original Classical Chinese text

Right side: English translation with modern mathematical commentary

1 Introduction / 导言

数书九章序

Preface to the Shuxue Jiuzhang

数书九章者，南宋秦九韶之所著也。秦氏字道古，鲁郡人。生于宋季，博学多能，尤精算学。此书成于淳祐七年（公元 1247 年），凡九卷，每卷九问，共八十一题。其学大衍、天时、田域、测望、赋役、钱谷、营建、军旅、市易九类。

The Shuxue Jiuzhang (Mathematical Treatise in Nine Sections) is the magnum opus of the Song dynasty mathematician Qin Jiushao (秦九韶, c. 1202–1261). Completed in 1247 CE during the Chunyou era (淳祐七年), it stands as one of the greatest achievements in the history of Chinese mathematics and indeed of world mathematics.

每问有问、答、术、草四目，体例谨严。大衍之类，尤推精妙，盖中国剩余定理之滥觞也。

The work is organized into nine categories (类), each containing nine problems (问), for a total of eighty-one problems. Each problem follows a rigorous four-part structure: **Wen** (问) – the problem statement; **Da** (答) – the numerical answer; **Shu** (术) – the general method; **Cao** (草) – the detailed working.

1.1 The Nine Categories / 九类名目

九类之目：

1. **大衍** – 求一之术
2. **天时** – 天文历法
3. **田域** – 田地测量
4. **测望** – 勾股重差
5. **赋役** – 租税劳役
6. **钱谷** – 钱粮贸易
7. **营建** – 建筑工事
8. **军旅** – 兵家之事
9. **市易** – 市场交易

The nine categories are:

1. **Dayan** (大衍) – “Great Derivation” : Indeterminate analysis and the Chinese Remainder Theorem
2. **Tianshi** (天时) – Astronomical phenomena and calendar calculations
3. **Tianyu** (田域) – Land measurement and area calculation
4. **Cewang** (测望) – Surveying and observation
5. **Fuyi** (赋役) – Taxation and labor service
6. **Qian’ gu** (钱谷) – Currency and grain commerce
7. **Yingjian** (营建) – Construction and engineering
8. **Junlv** (军旅) – Military affairs
9. **Shiyi** (市易) – Market transactions

2 Preface / 序

周教六艺，数实成之。学士大夫所从事尚矣。其用本太虚生一，而周流无穷。大则可以通神明、顺性命，小则可以经世务、类万物，詎容以浅近窥哉？

Of the Six Arts taught by the Zhou, mathematics completes them. It has long been pursued by scholars and gentlemen. Its use originates from the One born of the Great Void, flowing endlessly. In the large, it can penetrate the divine and accord with nature; in the small, it can govern worldly affairs and classify all things. How could it be viewed as shallow or trivial?

若昔推策以迎日，定律而和气。髀距濬川，土圭度晷。天地之大，圉焉而不能外，况其间总总者乎？爰自河图洛书，闾发幽秘；八卦九畴，错综精微。

From ancient times, reckoning rods were used to predict the sun's course, and laws were established to harmonize the qi. The gnomon measured the rivers, the jade tablet measured the shadows. The greatness of heaven and earth is encompassed by it and cannot escape, how much more the multitude of things within?

极而至于大衍、皇极之用，而人事之变无不该，鬼神之情莫能隐矣。圣人神之言而遗其粗，常人昧之由而莫之觉。要其归，则数与道非二本也。

Reaching to the applications of the Dayan and the Supreme Ultimate, there is no human affair that it does not encompass, no spirit's nature that it cannot reveal. The sages found its spirit and left its roughness; ordinary people are blind to its path. In essence, number and the Way are not two different roots.

独大衍法不载九章，未有能推之者。历家演法颇用之，以为方程者，误也。

Only the Dayan method is not recorded in the Nine Chapters, and none have been able to develop it. Calendarists have used it extensively in their computational methods, but those who take it as a system of equations are mistaken.

九韶愚陋，不闲于艺。然早岁侍亲中都，因得访习于太史，又尝从隐君子受数学。时际兵难，历岁遥塞，不自意全于矢石之间。更险离忧，荏苒十稔，心槁气落。信知夫物莫不有数也，乃肆意其间，旁询方能，探索杳眇，粗若有得焉。

I, Jiushao, am foolish and unrefined, not skilled in the arts. Yet in my youth I served my parents in the capital, and so had the opportunity to study with the Grand Astrologer; I also received instruction in mathematics from a gentleman recluse. When the time of military disaster came, I did not expect to survive between arrows and stones. Passing through danger and sorrow, ten years elapsed; my heart withered and my spirit fell. Truly I came to know that all things have number.

时淳祐七年九月，鲁郡秦九韶叙。

Written in the ninth month of the seventh year of Chunyou, by Qin Jiushao of Lu Commandery.

3 The Dayan General Method / 大衍总术

3.1 Editor's Commentary / 按语

按大衍术，以各分数之奇零求各分数之总数。大而天行，小而物数，皆可御之。其法有求元、求定、求衍、求奇、求乘、求用之目。大约以数之奇偶为根，而以诸数相度之尽不尽为用。

Editor's note: The Dayan method takes the remainders (odd fragments) of each fractional divisor to find the total number. Whether for large celestial motions or small material quantities, it can be applied. The method has the following steps: finding the original numbers, finding the definitive numbers, finding the derivative numbers, finding the odd numbers, finding the multipliers, and finding the use numbers.

3.2 The Four Types of Numbers / 四种问数

大衍总术曰：置诸问数，类名有四。

1. **元数** – 谓尾位见单零者
2. **收数** – 谓尾位见分厘者
3. **通数** – 谓诸数各有分子分母者
4. **复数** – 谓尾位见十或百及千以上者

The Dayan General Method states: Place the given numbers; there are four types.

1. **Original numbers** – Numbers ending in single units
2. **Collected numbers** – Numbers ending in fractional parts
3. **Common numbers** – Numbers each having numerator and denominator
4. **Repeated numbers** – Numbers ending in tens, hundreds, or thousands

3.3 Finding the Definitive Numbers / 求定数

元数者：先以两两连环求等，约奇弗约偶。或约得五而彼有十，乃约偶弗约奇。或元数俱偶，约毕可存一位见偶。

For original numbers: First, pairwise in chain sequence find the GCD; reduce the odd one, not the even one. If reducing yields 5 while the other has 10, then reduce the even one, not the odd one. If all original numbers are even, after reduction one may keep one even number.

复数者：问数尾位见十以上者，以诸数求总等，存一位约众位，始得元数。两两连环求等，约奇弗约偶，复乘偶；或约偶弗约奇，弗乘奇。

For repeated numbers: When the given numbers end in ten or above, find the total GCD of all numbers; keep one position and reduce the rest. Then pairwise in chain sequence find GCDs; reduce the odd, not the even, and multiply back the even; or reduce the even, not the odd, and do not multiply back the odd.

3.4 Computing the Derivative Numbers / 求衍数

求定数勿使两位见偶，勿使见一太多。见一多则借用繁，不欲借则任得一。

以定相乘为衍母，以各定约衍母，得各衍数。

次以各定母满去衍数，各余名曰**奇数**。

以大衍求一术，得各乘率。

以乘衍数，各得**用数**。

Finding definitive numbers: Do not let two positions be even; do not let there be too many ones. Too many ones makes borrowing complicated.

Multiply the definitive numbers together to obtain the derivative mother. Reduce the derivative mother by each definitive number to obtain each **derivative number**.

Next, reduce each derivative number by its corresponding definitive number; the remainders are called the **odd numbers**. Using the **Dayan Method for Finding Unity**, obtain each **multiplier**.

Multiplying each multiplier by its derivative number gives each **use number**.

3.5 The Dayan Method for Finding Unity / 大衍求一术

大衍求一术云：置奇右上，定居右下。立天元一于左上。先以右行上下两位，以少除多，所得商数，乃递互乘归左行。须使右上末后奇一而止。乃验左上所得，以为乘率。或奇数已见单一者，便为乘率。

The Dayan Method for Finding Unity says: Place the odd number at the upper right, the definitive number at the lower right. Place the celestial element unity (1) at the upper left. First, divide the larger by the smaller of the two numbers in the right column; the quotient obtained is then used to multiply alternately into the left column. Continue until the upper right finally becomes one. Then check what is obtained at the upper left—this is the multiplier.

In modern mathematical terms: Given coprime integers a (the “odd number”) and m (the “definitive number”), find k such that:

$$k \cdot a \equiv 1 \pmod{m}$$

This is the **modular multiplicative inverse** of a modulo m , computed by the **extended Euclidean algorithm**.

4 Problem 1: Shigua Fawei / 蓍卦发微

蓍卦发微 – Divination by Yarrow Stalks

问曰：

《易》曰：大衍之数五十，其用四十有九。又曰：分而为二以象两，挂一以象三，揲之以四以象四时。三变而成爻，十有八变而成卦。欲知所衍之术及其数各几何？

Question:

The Book of Changes says: “The number of the Great Derivation is fifty; of these, forty-nine are used.” It also says: “Divide into two to symbolize the Two [heaven and earth]; hang one to symbolize the Three [heaven, earth, and man]; count off by fours to symbolize the Four Seasons.” Three changes produce a line; eighteen changes produce a hexagram. The problem asks to find the computational method of this derivation and the numbers involved.

This problem applies the Dayan method to the classical yarrow-stalk divination of the I Ching.

答曰：

衍母十二，衍法三。

一元衍数二十四，二元衍数一十二，三元衍数八，四元衍数六。以上四位衍数，计五十一。

一揲用数一十二，二揲用数二十四，三揲用数四，四揲用数九。以上四位用数，计四十九。

Answer:

The derivative mother is 12, the derivative divisor is 3.

The first derivative number is 24, the second is 12, the third is 8, the fourth is 6. The sum of these four derivative numbers is 51.

The first manipulation use-number is 12, the second is 24, the third is 4, the fourth is 9. The sum of these four use-numbers is 49.^a

^aTrans. note: The sum 49 corresponds to “the number of the Great Derivation is fifty, of which forty-nine are used” from the Yijing.

术曰：

置诸元数，两两连环求等，约奇弗约偶。遍约毕，乃变元数皆曰定母，列右行。各立天元一为子，列左行。以诸定母互乘左行之子，各得名曰衍数。

Method:

Place the given original numbers; pairwise in chain sequence find the GCD, reducing the odd and not the even. When reduction is complete everywhere, the original numbers become the definitive numbers; place them in the right column. Place the celestial element unity (1) as the seed in the left column. Cross-multiply the definitive numbers with the seeds in the left column; each result is called a derivative number.

次以各定母满去衍数，各余名曰奇数。以奇数与定母用大衍术求一，得各乘率。以乘衍数，各得用数。

Next, reduce each derivative number by its corresponding definitive number; each remainder is called the odd number. Using the Dayan Method for Finding Unity on the odd number and the definitive number, obtain each multiplier. Multiply each derivative number by its multiplier to obtain each use number.

5 Problem 2: Guli Huiji / 古历会积

古历会积 – Ancient Calendar Accumulation

问曰：
问古历会积，欲求积年。以历法积元为问，历过无考，但据近世演纪之法，推而上之。

Question:
This problem concerns the calculation of the accumulated years (积年) in ancient calendar systems. Given certain astronomical periodicities, one must find a number that simultaneously satisfies multiple congruence conditions. The epochs have passed without record; based on the method of computing epochs used in recent times, one extrapolates backward.
This was one of the primary applications of the Dayan method in traditional Chinese astronomy—finding a common epoch that aligns multiple celestial cycles.

术曰：
以大衍求之。置元历相关诸周期，连环求等，约奇弗约偶，得定母。诸定相乘为衍母。以各定约衍母得衍数。各满定母去之，余为奇数。大衍求一入之，得乘率。以乘衍数为用数。次置各余数乘用数，并之为总数。满衍母去之，不满为所求积年。

Method:
Solve by the Dayan method. Place the relevant periods of the original calendar; pairwise in chain sequence find GCDs, reducing the odd and not the even, obtaining the definitive numbers. Multiply the definitive numbers together to obtain the derivative mother. Reduce the derivative mother by each definitive number to obtain the derivative numbers. Reduce each by its definitive number; the remainders are the odd numbers. Apply the Dayan Method for Finding Unity to obtain the multipliers. Multiply the derivative numbers by the multipliers to obtain the use numbers. Next, place each remainder, multiply by the use numbers, and sum to obtain the total number. Reduce by the derivative mother; the remainder is the sought accumulated years.

6 Problem 3: Tuiji Tugong / 推计土功

推计土功 – Estimating Earthworks

问曰：

四县（甲、乙、丙、丁）共同筑堤，根据各县的物力分配筑堤任务。

- 甲县物力：138,600 贯
- 乙县物力：146,300 贯
- 丙县物力：192,500 贯
- 丁县物力：184,800 贯

筑堤标准：每 770 贯物力分配一名劳动力，每人每天筑堤 60 平方尺。

进度情况：甲县和乙县已完成任务，丙县剩余 51 丈，丁县剩余 18 丈。

Question:

Four counties (A, B, C, D) jointly build a dike; the task is distributed according to each county's resources.

- County A resources: 138,600 guan
- County B resources: 146,300 guan
- County C resources: 192,500 guan
- County D resources: 184,800 guan

Dike standard: for every 770 guan of resources, assign one laborer; each laborer builds 60 square feet of dike per day.

Progress: Counties A and B have completed their tasks; County C has 51 zhang remaining; County D has 18 zhang remaining.

This is a classic indeterminate problem solved by the Dayan method.

答曰：

堤的总长度为 19 里 235 步 5 尺。

各县所筑长度：

- 甲县：1,260 丈
- 乙县：1,768 步 5 尺 6 寸
- 丙县：4 里 328 步 5 尺 6 寸
- 丁县：与前三县相应

Answer:

The total length of the dike is 19 li 235 bu 5 chi.

Each county's contribution:

- County A: 1,260 zhang
- County B: 1,768 bu 5 chi 6 cun
- County C: 4 li 328 bu 5 chi 6 cun
- County D: corresponding to the first three

术曰：

以大衍求之。置各县物力，连环求等，约奇弗约偶，得定母。诸定相乘为衍母。以各定约衍母得衍数。各满定母去之，余为奇数。大衍求一入之，得乘率。以乘衍数为用数。次置各县余数乘用数，并之为总数。满衍母去之，不满为所求。

Method:

Solve by the Dayan method. Place the resources of each county; pairwise in chain sequence find GCDs, reducing the odd and not the even, obtaining the definitive numbers. Multiply the definitive numbers together to obtain the derivative mother. Reduce the derivative mother by each definitive number to obtain the derivative numbers. Reduce each by its definitive number; the remainders are the odd numbers. Apply the Dayan Method for Finding Unity to obtain the multipliers. Multiply the derivative numbers by the multipliers to obtain the use numbers. Next, place each county's remainder, multiply by the use numbers, and sum to obtain the total number. Reduce by the derivative mother; the remainder is the sought answer.

7 Problem 4: Tuiku E Qian / 推库额钱

推库额钱 – Calculating Treasury Funds

问曰：

问有外邑七库，日纳息足钱适等，递年成贯整纳。近缘见钱希少，听各库照当处市陌准解旧会。其甲库有零钱一十文，丁、庚二库各零四文，戊库零六文，余库无零钱。

Question:

There are seven treasuries in outlying districts; their daily interest payments in full cash are equal, rounded to whole guan annually. Recently, since circulating cash has become scarce, each treasury is permitted to convert its payment to old notes according to the local exchange rate (市陌).^a

^aTrans. note: The “market fraction” (市陌) was a local exchange rate between paper currency and full-value cash, varying by location.

甲库所在市陌一十二文，递减一文至庚库而止。欲求诸库日息原纳足钱、展省及今纳旧会，并大小月分各几何？

(七库依次为甲、乙、丙、丁、戊、己、庚，市陌分别为 12, 11, 10, 9, 8, 7, 6 文。甲库余 10 文，乙丙无余，丁余 4 文，戊余 6 文，己无余，庚余 4 文。)

The exchange rate at treasury A is 12 wen; it decreases by 1 wen for each subsequent treasury down to G. Find: the original daily interest in full cash for each treasury, the adjusted amount, and the current payment in old notes, for both long and short months.

(The seven treasuries are A–G with exchange rates 12, 11, 10, 9, 8, 7, 6 wen respectively. Treasury A has remainder 10 wen, B and C have no remainder, D has remainder 4 wen, E has remainder 6 wen, F has no remainder, G has remainder 4 wen.)

术曰：

以大衍求之。置甲库市陌，以递减数减之，各得诸库原陌。连环求等，约奇弗约偶，得定母。诸定相乘为衍母。以定约衍母得衍数。各满定母去之，余为奇数。大衍求一入之，得乘率。以乘衍数为用数。

Method:

Solve by the Dayan method. Place the exchange rate at treasury A; subtracting the decrement each time gives each treasury's rate. Pairwise in chain sequence find GCDs, reducing the odd and not the even, obtaining the definitive numbers. Multiply the definitive numbers together to obtain the derivative mother. Reduce the derivative mother by each definitive number to obtain the derivative numbers. Reduce each by its definitive number; the remainders are the odd numbers. Apply the Dayan Method for Finding Unity to obtain the multipliers. Multiply the derivative numbers by the multipliers to obtain the use numbers.

8 Problem 5: Fentiao Tuiyuan / 分棗推原

分棗推原 – Grain Distribution

問曰：
有兄弟三人，各將收穫之米送往不同的地方：
• 甲將米送往官場，余 3 斗 2 升
• 乙將米送往安吉鄉，余 7 斗
• 丙將米送往平江，余 3 斗
官斛每石八斗三升，安吉斛每石一石一斗，平江斛每石一石三斗五升。欲求三人共收穫米數以及各自分米數幾何？

Question:
Three brothers each send their harvested rice to different places:
• A sends rice to the government office, remainder 3 dou 2 sheng
• B sends rice to Anji township, remainder 7 dou
• C sends rice to Pingjiang, remainder 3 dou
The official measure holds 8 dou 3 sheng per shi; the Anji measure holds 1 shi 1 dou; the Pingjiang measure holds 1 shi 3 dou 5 sheng. Find: the total amount of rice harvested by the three brothers, and the amount each brother received.

答曰：
三人共收穫米 738 石。
各分米數：
• 甲：296 石，余 3 斗 2 升
• 乙：223 石，余 7 斗
• 丙：182 石，余 3 斗

Answer:
The three brothers harvested 738 shi of rice in total.
Each brother’ s share:
• A: 296 shi, remainder 3 dou 2 sheng
• B: 223 shi, remainder 7 dou
• C: 182 shi, remainder 3 dou

術曰：
以大衍求之。置各斛率（單位：升）：官斛 83 升，安吉斛 110 升，平江斛 135 升。連環求等，約奇弗約偶，得定母。諸定相乘為衍母。以各定約衍母得衍數。各満足母去之，余為奇數。大衍求一入之，得乘率。以乘衍數為用數。

Method:
Solve by the Dayan method. Place each measure’ s rate (in sheng): the official measure 83 sheng, the Anji measure 110 sheng, the Pingjiang measure 135 sheng. Pairwise in chain sequence find GCDs, reducing the odd and not the even, obtaining the definitive numbers. Multiply the definitive numbers together to obtain the derivative mother. Reduce the derivative mother by each definitive number to obtain the derivative numbers. Reduce each by its definitive number; the remainders are the odd numbers. Apply the Dayan Method for Finding Unity to obtain the multipliers. Multiply the derivative numbers by the multipliers to obtain the use numbers.

9 Problem 6: Chengxing Jidi / 程行计地

程行计地 – Journey Distance Calculation

问曰：

军师派遣三名急足（甲、乙、丙）分别前往都下报捷。

- 甲每天行 300 里
- 乙每天行 240 里
- 丙每天行 180 里

甲提前数日申未到，乙后数日未正到，丙今日辰未到。欲知从军前到都下的距离以及三人各自行走的天数几何？

Question:

A military commander sends three express messengers (A, B, C) to the capital to report victory.

- A travels 300 li per day
- B travels 240 li per day
- C travels 180 li per day

A arrives early (before shen hour); B arrives late (at wei hour); C arrives today (at chen hour). Find the distance from the army's position to the capital, and the number of days each traveled.

答曰：

从军前到都下的距离为 3,300 里。

- 甲行 11 天
- 乙行 13 天 4 时
- 丙行 18 天 2 时

Answer:

The distance from the army to the capital is 3,300 li.

- A traveled 11 days
- B traveled 13 days 4 hours
- C traveled 18 days 2 hours

术曰：

以大衍求之。置三人日行里数，连环求等，约奇弗约偶，得定母。诸定相乘为衍母。以各定约衍母得衍数。各满足母去之，余为奇数。大衍求一入之，得乘率。以乘衍数为用数。次置各余数乘用数，并之为总数。满衍母去之，不满为所求距离。

Method:

Solve by the Dayan method. Place the three persons' daily travel distances; pairwise in chain sequence find GCDs, reducing the odd and not the even, obtaining the definitive numbers. Multiply the definitive numbers together to obtain the derivative mother. Reduce the derivative mother by each definitive number to obtain the derivative numbers. Reduce each by its definitive number; the remainders are the odd numbers. Apply the Dayan Method for Finding Unity to obtain the multipliers. Multiply the derivative numbers by the multipliers to obtain the use numbers. Next, place each remainder, multiply by the use numbers, and sum to obtain the total number. Reduce by the derivative mother; the remainder is the sought distance.

10 Problem 7: Chengxiang Xiangji / 程行相及

程行相及 – Catching Up on a Journey

问曰：

甲、乙、丙三人行程相关。甲先行若干日，乙后行，丙又后行。各以不同速度行进。求某处三人相遇之时间及各地距离。

Question:

Three travelers A, B, and C are related in their journeys. A departs first, B departs later, and C departs still later. Each travels at a different speed. Find: the time when all three meet at a certain place, and the various distances.

This problem involves multiple travelers with different speeds and starting times, requiring the Dayan method to solve a system of congruences.

术曰：

以大衍求之。置各人行程速度及时间差，连环求等，约奇弗约偶，得定母。诸定相乘为衍母。以各定约衍母得衍数。各满定母去之，余为奇数。大衍求一入之，得乘率。以乘衍数为用数。次置各余数乘用数，并之为总数。满衍母去之，不满为所求。

Method:

Solve by the Dayan method. Place each person's travel speed and time difference; pairwise in chain sequence find GCDs, reducing the odd and not the even, obtaining the definitive numbers. Multiply the definitive numbers together to obtain the derivative mother. Reduce the derivative mother by each definitive number to obtain the derivative numbers. Reduce each by its definitive number; the remainders are the odd numbers. Apply the Dayan Method for Finding Unity to obtain the multipliers. Multiply the derivative numbers by the multipliers to obtain the use numbers. Next, place each remainder, multiply by the use numbers, and sum to obtain the total number. Reduce by the derivative mother; the remainder is the sought answer.

11 Problem 8: Jichi Xunyuan / 积尺寻源

积尺寻源 – Finding Dimensions from Volume

问曰：

问欲砌基一段，见管大小方砖六门城砖四色。令匠取便，或平或侧，只用一色砖砌，须要适足。匠以砖量地计料，称用：

- 大方：广多六寸，深少六寸
- 小方：广多二寸，深少三寸
- 城砖：长广多三寸，深少一寸
- 六门砖：长广多三寸，深多一寸

Question:

One wishes to build a foundation; available are large and small square bricks, six-portal bricks, and city-wall bricks—four types in all. The mason is instructed to use whichever is convenient, laying them flat or on edge, using only one type of brick, requiring them to fit exactly. The mason measured the ground and estimated materials, reporting:

- Large square: width excess 6 cun, depth deficit 6 cun
- Small square: width excess 2 cun, depth deficit 3 cun
- City brick: length and width excess 3 cun, depth deficit 1 cun
- Six-portal: length and width excess 3 cun, depth excess 1 cun

皆不合匝，未免修破砖料裨补。其四色砖尺寸：

- 大方：方一尺三寸
- 小方：方一尺一寸
- 城砖：长一尺二寸，阔六寸，厚二寸五分
- 六门砖：长一尺，阔五寸，厚二寸

欲知基深广几何？

All fail to fit exactly, so broken bricks must be used for patching. The four types of brick have dimensions:

- Large square: 1 chi 3 cun square
- Small square: 1 chi 1 cun square
- City brick: 1 chi 2 cun long, 6 cun wide, 2 cun 5 fen thick
- Six-portal: 1 chi long, 5 cun wide, 2 cun thick

Find: the depth and width of the foundation.

This is a classic problem of finding a common multiple with given remainders.

答曰：

深三丈七尺一寸，广一丈二尺三寸。

Answer:

Depth: 3 zhang 7 chi 1 cun; width: 1 zhang 2 chi 3 cun.

术曰：

以大衍求之。置四色砖尺寸及所余尺寸，连环求等，约奇弗约偶，得定母。诸定相乘为衍母。以各定约衍母得衍数。各满定母去之，余为奇数。大衍求一入之，得乘率。以乘衍数为用数。次置各余数乘用数，并之为总数。满衍母去之，不满为基之深广。

Method:

Solve by the Dayan method. Place the four types of brick dimensions and the excess/deficit dimensions; pairwise in chain sequence find GCDs, reducing the odd and not the even, obtaining the definitive numbers. Multiply the definitive numbers together to obtain the derivative mother. Reduce the derivative mother by each definitive number to obtain the derivative numbers. Reduce each by its definitive number; the remainders are the odd numbers. Apply the Dayan Method for Finding Unity to obtain the multipliers. Multiply the derivative numbers by the multipliers to obtain the use numbers. Next, place each remainder, multiply by the use numbers, and sum to obtain the total number. Reduce by the derivative mother; the remainder gives the depth and width of the foundation.

12 Problem 9: Yumi Tuishu / 余米推数

余米推数 – Calculating Rice Remainders

问曰：

有米若干，以不同容量的容器量之，各余若干。欲求米之总数。

Question:

There is a quantity of rice; when measured with containers of different capacities, there are different remainders each time. Find the total amount of rice.

This is the classic “unknown quantity” problem. In modern terms: find N such that $N \equiv r_i \pmod{m_i}$ for given moduli m_i and remainders r_i .

术曰：

以大衍求之。置各容器容量为问数，连环求等，约奇弗约偶，得定母。诸定相乘为衍母。以各定约衍母得衍数。各满定母去之，余为奇数。大衍求一入之，得乘率。以乘衍数为用数。次置各余数乘用数，并之为总数。满衍母去之，不满为所求米数。

Method:

Solve by the Dayan method. Place the capacities of each container as the given numbers; pairwise in chain sequence find GCDs, reducing the odd and not the even, obtaining the definitive numbers. Multiply the definitive numbers together to obtain the derivative mother. Reduce the derivative mother by each definitive number to obtain the derivative numbers. Reduce each by its definitive number; the remainders are the odd numbers. Apply the Dayan Method for Finding Unity to obtain the multipliers. Multiply the derivative numbers by the multipliers to obtain the use numbers. Next, place each remainder, multiply by the use numbers, and sum to obtain the total number. Reduce by the derivative mother; the remainder is the sought amount of rice.

This problem is structurally identical to the famous “problem of the unknown quantity” from the Sunzi Suanjing (孙子算经), which asks: “We have things of which we do not know the number; if we count them by threes, the remainder is 2; if by fives, the remainder is 3; if by sevens, the remainder is 2. How many things are there?” Qin Jiushao’s general method solves all problems of this type systematically.

13 Mathematical Analysis / 数理分析

13.1 The Chinese Remainder Theorem / 中国剩余定理

The Dayan method solves systems of simultaneous linear congruences. Given a set of pairwise coprime moduli $m_1, m_2, \dots, m_n$ and remainders $r_1, r_2, \dots, r_n$, find N such that:

$$N \equiv r_1 \pmod{m_1}$$

$$N \equiv r_2 \pmod{m_2}$$

$$\vdots$$

$$N \equiv r_n \pmod{m_n}$$

13.2 Qin Jiushao's Algorithm / 秦九韶算法

求定数 – 连环求等，约奇弗约偶

求衍母 – 诸定相乘

求衍数 – 以定约衍母

求奇数 – 各满定母去之

求乘率 – 大衍求一术

求用数 – 乘率乘衍数

求总数 – 余数乘用数并之

Step 1: Definitive Numbers. Given the original numbers (元数), reduce them pairwise using the GCD, keeping them coprime. The rule “reduce the odd, not the even” ensures unique factorization.

Step 2: Derivative Mother. Compute $M = m_1 \times m_2 \times \dots \times m_n$.

Step 3: Derivative Numbers. Compute $M_i = M/m_i$ for each i .

Step 4: Odd Numbers. Compute $g_i = M_i \bmod m_i$.

Step 5: Finding Unity. For each i , find k_i such that $k_i \cdot g_i \equiv 1 \pmod{m_i}$. This is the modular multiplicative inverse, computed by the extended Euclidean algorithm.

Step 6: Use Numbers. Compute $u_i = k_i \cdot M_i$.

Step 7: Final Computation. The solution is: $N = \sum_{i=1}^n r_i \cdot u_i \pmod{M}$

13.3 Historical Significance / 历史意义

秦氏大衍术，超前欧洲数百年。其求一之术，即欧几里得之扩展算法也。后世郭守敬用于天文，李治用于几何，西传欧洲，遂成所谓“中国剩余定理”。

高斯于《算术探究》(1801年)首次在欧洲完整论述此定理，称之为“求一数，以各已知数除之，各余已知数”之问题。今此定理为数论之基本结果也。

Qin Jiushao's method predated similar European work by several centuries. The algorithm for finding modular inverses (大衍求一术) is essentially the extended Euclidean algorithm. The method was later applied by Guo Shoujing (郭守敬) to astronomy, by Li Ye (李治) to geometry, and was transmitted to Europe where it became known as the “Chinese Remainder Theorem.”

Gauss gave the first complete treatment in Europe in his *Disquisitiones Arithmeticae* (1801), calling it the “problem of finding a number that, when divided by given numbers, leaves given remainders.” The theorem is now recognized as one of the fundamental results in number theory.

MATHEMATICAL TREATISE IN NINE SECTIONS

Shuxue Jiuzhang – 數學九章

Fascicle 2 / 卷二

天時類

Tian Shi Lei – Heavenly Patterns

Calendar, Astronomy and Weather / 曆法、天文與氣象

Classical Chinese • English Translation

Original: Left / 原文：左欄 Translation: Right / 譯文：右欄

Qin Jiushao / 秦九韶

(1202–1261)

Song Dynasty, Chunyou 7th year (淳祐七年, 1247)

Bilingual edition based on the Siku Quanshu edition

雙語版據《四庫全書》本編排

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天時類序 / 引言

Introduction to Fascicle 2

天時類者，數學九章之第二卷也。凡九問，分為二類：上卷五問，論推步治曆、候星揆影之事；下卷四問，論天池圓罍測雨、峻積竹器驗雪之法。

秦九韶自序其書曰：「大則可以通神明，順性命；小則可以經世務，類萬物。」此卷九問，實以數術推天時，為農事所繫，其用至重。

The Tian Shi Lei (天時類, “Heavenly Patterns”) is the second of nine chapters in Qin Jiushao’s *Shuxue Jiuzhang* (數學九章, Mathematical Treatise in Nine Sections). This fascicle contains **nine problems** divided into two parts, dealing with:

- Calendar computation and astronomical predictions (Problems 1–5)
- Measurement of rainfall and snowfall (Problems 6–9)

Qin Jiushao’s preface to the *Shuxue Jiuzhang* (1247) explains his motivation:

“The art of numbers, rooted in the Great Void, generates One and circulates without end. In its large aspect it can penetrate the divine and comply with destiny; in its small aspect it can govern worldly affairs and classify the myriad things. How can it be viewed as shallow or narrow?”

此卷所用之法，大要有三：曰通積術，求諸周之公倍也；曰招差術，推日月五星之變行也；曰調日法，齊曆法之疏密也。三者皆中國古代天文數學之精髓。

The nine problems of this fascicle demonstrate the practical application of mathematical techniques to calendar reform, planetary motion, and meteorological measurement—topics of vital importance to agricultural society.

The key methods employed are:

- General accumulation method** (通積): Finding common periods
- Interpolation** (招差): For irregular astronomical motions
- Equalizing technique** (調日法): For calendrical adjustments

卷二上

Volume 2, Part 1

第一問：推氣治歷

問 太史觀天，慶元四年戊午歲冬至三十九日九十二刻四十五分，紹定三年庚寅歲冬至三十二日九十四刻一十二分。今欲知嘉泰甲子歲氣骨、歲餘、斗分各幾何？

按 按：紹定三年庚寅冬至，實紹定四年辛卯之歲首也。辛卯去戊午三十四年，即積三十三年。

答曰 氣骨一十一日三十八刻二十分八十一秒八十小分；歲餘五日二十四刻二十九分三十秒三十小分；斗分〇日二十四刻二十九分三十秒三十小分。

術曰 以兩測之年差為法。置前測之日刻分，以後測之日刻分減之，餘為率。（不減，加紀策）。遞以紀策累加之，務令合天道，取五日以上為實。以法除實，得歲餘。去全分，餘為斗分。以歲餘乘前測至所求中積之年數，併入前測日刻分，滿紀策去之，餘為所求氣骨。

按 氣骨者，自冬至距夜半後甲子日之時刻也。歲餘者，歲周減六甲之餘也。斗分者，歲周減三百六十五日之餘也。此術尚未知歲實之確數，故以兩測氣骨之差為實，積年為法。

Problem 1: Calculating Qi and Regulating the Calendar

Question The Imperial Astronomer observed the heavenly way:

- In the 4th year of Qingyuan (慶元四年, 1198, year Wu-wu 戊午), the winter solstice was at 39 days, 92 *ke* (刻), 45 *fen* (分).
- In the 3rd year of Shaoding (紹定三年, 1230, year Geng-yin 庚寅), the winter solstice was at 32 days, 94 *ke*, 12 *fen*.

It is desired to find, for the intermediate year Jiatai Jiazi (嘉泰甲子, 1204): the *qi bone* (氣骨), the *year remainder* (歲餘), and the *dou fraction* (斗分).

Editor's Note The Shaoding 3rd year Geng-yin winter solstice actually marks the beginning of Shaoding 4th year Xin-mao (辛卯). Xin-mao is 34 years from Wu-wu, giving 33 accumulated years.

Answer

- Qi bone: 11 days, 38 *ke*, 20 *fen*, 81 *miao* (秒), 80 small fractions.
- Year remainder: 5 days, 24 *ke*, 29 *fen*, 30 *miao*, 30 small fractions.
- Dou fraction: 0 days, 24 *ke*, 29 *fen*, 30 *miao*, 30 small fractions.

Method First take the number of years between the two observations as the divisor. Place the earlier observation's days, *ke*, and *fen*, and subtract from the later observation's days, *ke*, and *fen*; the remainder is the rate. (If insufficient for subtraction, add the *ji-ce* 紀策). Accumulate by adding *ji-ce* repeatedly until the result is suitable for the heavenly way—using a number above 5 days as the dividend. Divide the dividend by the divisor to obtain the year remainder. Remove whole days; the remainder is the dou fraction. Multiply the year remainder by the

number of years from the earlier observation to the desired intermediate year, add to the earlier observation's days, *ke*, and *fen*, and remove full *ji-ce*; the remainder is the desired *qi bone*.

Editor's Note The *qi bone* is the days and fractions from the winter solstice to the *Jiazi* day after midnight. The year remainder is the residue when the solar year length is reduced by six *Jiazi* cycles. The *dou* fraction is the residue when the solar year is reduced by 365 days. This method does not yet know the precise solar year length, hence it uses the difference between two observed *qi bones* as the dividend and accumulated years as divisor.

第二問：治歷推閏

問 開禧曆，以嘉泰四年甲子歲天冬至一十一日，日辰乙亥，四十四刻六十一分五十四秒；十一月經朔一日，日辰乙丑，七十五刻五十五分六十二秒。問：閏骨、閏率各幾何？

答曰 閏骨九日六十九刻五分九十一秒，餘一百二十一（以一百六十九分為法）；閏骨率一十六萬三千七百七十一。

術曰 以日法通氣朔之日刻分秒，各為氣骨、朔骨之通分。氣骨通分以約率約之，適盡者可用；或不盡，圓徑以下在一刻內者，亦可用。乃以減朔骨通分，餘為閏骨率。以日法除之，得閏骨策。

按 若以朔骨通分徑減氣骨通分，餘為九日六十九刻五分九十二秒，即閏骨策。原草僅差四十八分之一百六十九分。其不徑減而以通分、約率、乘除往復者，蓋為後續諸算之便也。

Problem 2: Regulating the Calendar and Intercalary Months

Question For the Kaixi calendar (開禧曆), taking the 4th year of Jiatai (嘉泰四年, 1204, year Jiazi 甲子):

- The celestial winter solstice is at 11 days, day-token Yi-hai (乙亥), 44 *ke*, 61 *fen*, 54 *miao*.
- The 11th-month new moon (經朔) is at 1 day, day-token Yi-chou (乙丑), 75 *ke*, 55 *fen*, 62 *miao*.

Question: What are the *run bone* (閏骨) and *run rate* (閏率)?

Answer

- Run bone: 9 days, 69 *ke*, 5 *fen*, 91 *miao*, with remainder 121/(169 fractions).
- Run bone rate: 163,771.

Method Take the day-factor (日法) to convert the qi and new-moon days, *ke*, *fen*, and *miao* into qi-bone and new-moon-bone fractions respectively. The qi-bone fraction, when reduced by the approximation rate (約率), if exactly divisible, is usable; or if the remainder after rounding is below one *ke*, it is also usable. Then subtract from the new-moon-bone fraction; the remainder is the run bone rate. Divide by the day-factor to obtain the run bone *ce* (策).

Editor's Note If one directly subtracts the new moon fraction from the winter solstice fraction, the remainder is 9 days, 69 *ke*, 5 *fen*, 92 *miao*, giving the run bone *ce*. The original working only differs by 48/(169 fractions). The reason for not directly subtracting but instead using common denominators, approximation, and repeated multiplication/division is to facilitate subsequent calculations.

第三問：治歷演紀

問 開禧曆積年七百八十四萬一千一百八十三。今欲知演紀之術：調日法，求朔餘、朔率、斗分、歲率、歲閏、入元歲、入閏、朔定骨、閏泛骨、閏縮、紀率、氣元率、元閏、元數、氣等率、因率、部率、朔等數、因數、部數、朔積年，凡二十三事。各幾何？

答曰 （選錄要項）日法一萬六千九百；朔餘八千九百六十七；朔率四十九萬九千六十七；斗分即二十四刻二十九分三十秒三十小分；歲率六百一十七萬二千六百八；歲閏二萬七百二十八；入元歲九千二百四十；入閏三萬一千六十八。

術曰 以大衍術求氣分、朔分、紀分、衍母、正用。以衍母除氣分，得積年；除朔分，得積月；除紀分，得積紀。各以六十乘之，得積日。以氣骨乘紀正用，得紀總；以閏骨乘朔正用，得朔總。併二總，滿衍母去之，餘為所求率實。以氣分除之，得歷過之年；以減積年，得未至之年。

按 此問全用大衍求一術，以解同餘式組，施用於治曆。所算二十三率，備成開禧曆元之全體，自天文觀測而推得曆元諸數。

Problem 3: Deriving Calendar Epoch Calculations

Question The Kaixi calendar has an accumulated year count of 7,848,183. It is desired to know the derivation of this computation: adjusting the day-factor, finding the new-moon remainder, new-moon rate, dou fraction, year rate, year intercalation, entry-year, entry-intercalation, new-moon fixed bone, intercalation floating bone, intercalation shrinkage, ji rate, qi yuan rate, yuan intercalation, yuan number, qi equal-rate, cause-rate, *bu* rate, new-moon equal-number, cause-number, *bu* number, and new-moon accumulated year—23 items in total. What is each?

Answer (Selected key results:)

- Day-factor (日法): 16,900
- New-moon remainder (朔餘): 8,967
- New-moon rate (朔率): 499,067
- Dou fraction (斗分): equivalent to 24 *ke*, 29 *fen*, 30 *miao*, 30 small fractions
- Year rate (歲率): 6,172,608
- Year intercalation (歲閏): 20,728
- Entry-year (入元歲): 9,240
- Entry-intercalation (入閏): 31,068

Method Use the method of Dayan (大衍術) to find the qi fraction, new-moon fraction, ji fraction, and derived mother (衍母), and all the proper use-numbers (正用). Divide the derived mother by the qi fraction to obtain the accumulated years; by the new-moon fraction to obtain the accumulated months; by the ji fraction to obtain the accumulated ji. Multiply by 60 to obtain accumulated days. Multiply the qi bone by the ji proper-use to obtain the ji total; multiply the run bone by the new-moon proper-use to obtain the new-moon total. Combine the two totals, remove full derived mothers, and the remainder is the sought rate-dividend. Divide by the qi fraction to obtain the calendar-passed (歷過) years; subtract from the accumulated years to obtain the years not yet reached.

Editor's Note This problem demonstrates the full power of Qin Jiushao's Dayan method (大衍求一術) for solving systems of congruences, applied here to calendar computation. The 23 quantities computed form a complete system for deriving the epoch parameters of the Kaixi calendar from astronomical observations.

第四問：綴術推星

問 歲星合伏段，一十六日九十分，行三度九十分，去日一十三度乃見。然後順行一百一十三日，一十七度八十三分，乃留。欲知：合伏段、晨疾初段之常度、初行率、末行率、平行率，各幾何？

術曰 （綴術）

1. 合伏段者，常度為總日數，初行率為總行度數，末行率等于初行率減日差乘（日數減一）。
2. 晨疾初段者，常度為所給日數，初行率為前段之末行率，末行率等于初行率減日差乘（日數減一）。
3. 平行率者，併初末二率而半之，即得平行率。

按 五星之行，有遲疾逆順，非均加速也。此術僅取合伏、見段之中數，而以遞加遞減求逐日之行率。古法之疏，於五星尤著。原草語多隱奧，今詳釋之，以見古今疏密之比。

草曰 合伏段：常度一十六日九十分；初行率二十二分七秒；末行率二十一分九十六秒；平行率約二十二分。晨疾初段：常度三十日；初行率二十一分九十六秒；日差一十一秒二十三分六十六小秒。

Problem 4: Calculating Planetary Motion

Question The year-star (Jupiter, 歲星) in conjunction-hidden phase (合伏段): after 16 days 90 *fen*, it moves 3 degrees 90 *fen*, removing 13 degrees from the sun before becoming visible. Then it proceeds directly (順行) for 113 days, 17 degrees 83 *fen*, before halting (乃留). It is desired to know:

- The conjunction-hidden segment (合伏段)
- The morning-rapid initial segment (晨疾初段)
- The standard degree (常度)
- The initial motion-rate (初行率)
- The final motion-rate (末行率)
- The mean motion-rate (平行率)

What is each?

Method (Method of *Zhui Shu*):

1. For the conjunction-hidden segment: The constant degree is the total number of days. The initial motion rate is the total degrees moved. The final motion rate equals the initial rate minus the daily difference (日差) multiplied by (days – 1).
2. For the morning-rapid initial segment: The constant degree is the given number of days. The initial rate is the final rate of the previous segment. The final rate equals the initial rate minus the daily difference multiplied by (days – 1).
3. For the mean motion rate: Add the initial and final rates and halve; this gives the parallel (mean) rate.

Editor's Note The five planets' motions involve non-uniform acceleration and deceleration. The method here takes only the middle values of the conjunction-hidden and direct-visibility segments, then uses successive addition/subtraction to derive the daily motion. The ancient method's approximation is particularly evident for the five planets. The original text's language was mostly hidden and obscure; the following detailed explanation reveals the comparison between ancient and modern precision.

Working Key calculations:

- Conjunction-hidden segment:
 - Constant degree: 16 days 90 *fen*
 - Initial rate: 22 *fen* 7 *miao*

- Final rate: 21 *fen* 96 *miao*
- Mean rate: approximately 22 *fen* (after rounding)
- Morning-rapid initial segment:
 - Constant degree: 30 days
 - Initial rate: 21 *fen* 96 *miao*
 - Daily difference: 11 *miao* 23 small-*fen* 66 small-*miao*

第五問：揆日究微

問 歷代測景，唯唐大衍曆最密。本朝崇天曆測陽城：冬至影一丈二尺七寸一分五十秒，夏至影一尺四寸七分七十秒，與大衍曆合。今開禧曆測臨安府：冬至影一丈八寸二分二十五秒，夏至影九寸一分。問：臨安夏至後幾日，其影與陽城夏至影等？又與大衍曆影較之，相差幾何？

答曰 大暑後五日午正，影長一尺四寸八分八十五秒（半）。影差六寸一分二十九秒（少）。

術曰 用乘法，以節率入之。

1. 置臨安所測冬至影長，減夏至影長，餘為景差，以為實。
2. 置象限度（九十一度三十一分四十四秒），加一十一度二十五分二十七秒半；以度為寸，得一百〇二寸五千六百七十一分五十秒為法。
3. 以法除實，得〇點九六六四……；棄餘數，自乘得節泛數九千四百分。
4. 以各節氣乘法率乘之，加夏至影幕，開方得各節氣影長。

按 各節氣之影長，當時實測而定，原不必算。此術故為繁難，以惑學者。細考之：以象限度加一十一度有餘為法，以景差除之而後自乘，得節率。每節氣有乘法率，以節率乘之，加夏至影幕，即為各節氣之影幕。故節率者，人為析出之數也。

Problem 5: Measuring the Sun's Shadow

Question Among all historical shadow-measurements, only the Tang dynasty's Dayan calendar (大衍曆) was the most precise. This dynasty's Chongtian calendar (崇天曆) gives for Yangcheng (陽城):

- Winter solstice shadow: 1 zhang 2 chi 7 cun 1 fen 50 miao
- Summer solstice shadow: 1 chi 4 cun 7 fen 70 miao

These agree with the Dayan calendar. Now the Kaixi calendar (開禧曆) gives for Lin'an (臨安府):

- Winter solstice shadow: 1 zhang 8 cun 2 fen 25 miao
- Summer solstice shadow: 9 cun 1 fen

It is desired to find: after how many days past the summer solstice at Lin'an will the shadow equal the Yangcheng summer solstice shadow? And comparing with the Dayan calendar's shadow, by how much do they differ?

Answer Five days after the Great Heat (大暑後五日) at noon; the shadow length is 1 chi 4 cun 8 fen 85 miao (half). The shadow difference is 6 cun 1 fen 29 miao (less).

Method Use the multiplication-rate method, with the section-rate (節率) entering into it.

1. Place Lin'an's measured winter solstice shadow, subtract the summer solstice shadow; the remainder is the shadow-difference (景差), taken as dividend.
2. Place the 象限度 (91 degrees 31 fen 44 miao), add 11 degrees 25 fen 27.5 small-fen; treating degrees as cun, obtain 102 cun 5671.5 fen 50 miao as divisor.
3. Divide the dividend by the divisor, obtaining 0.9664...; discard the remainder and square to obtain the section floating-number (節泛數): 9,340 fen.
4. Multiply by the section multiplication-rates for each solar term, add the summer solstice shadow-squared, take the square root to obtain each solar term's shadow length.

Editor's Note Each solar term's shadow length was determined by actual measurement at the time and does not need computation. The method presented here deliberately creates obscurity to puzzle the reader. Detailed examination shows: the 象限度 plus 11+ degrees is taken as divisor; the shadow-difference divided by this and then squared gives the section-rate. Each solar term has a multiplication-rate; multiplying the section-rate by

this and adding the summer solstice shadow-power gives each term's shadow-power. Thus the section-rate is an artificially extracted number.

卷二下

Volume 2, Part 2

第六問：天池測雨

問 今州郡皆有天池盆以測雨，然人但知盆中之水即所受之雨，而不知器之圓舛各有所受，不可徑以盆測為平地之雨。設盆口徑二尺八寸，底徑一尺二寸，深一尺八寸，接雨水深九寸。問：平地雨深幾何？

答曰 平地雨深三寸。

術曰 用圓錐術：

1. 以口徑乘底徑，又各自乘，併三數。以半深乘之，又以十一乘、七除，得下積。
2. 以半深加雨深，以減全深，得上深。
3. 以底徑減口徑，以上深乘之。以半深乘之，加口徑乘半深，得面率。
4. 以面率除半深，得水面徑。
5. 續以乘除，求得雨水之積，以口冪（平方三倍）除之，得平地雨深。

按 天池盆者，圓錐之截頭也。以雨積除以口面積，即得平地雨深。其術以圓錐台體積之術推之，而歸於平地之算。

Problem 6: Measuring Rain with the Heavenly Pool Basin

Question Nowadays every prefecture and district has a “heavenly pool” basin (天池盆) for measuring rainwater. But people only know that the water in the basin equals the rain received, and do not understand that different vessel shapes receive different amounts of rain—so one cannot directly take the basin measurement as the flat-ground rain amount.

Suppose the basin has:

- Mouth diameter (口徑): 2 *chi* 8 *cun* (28 *cun*)
- Base diameter (底徑): 1 *chi* 2 *cun* (12 *cun*)
- Depth (深): 1 *chi* 8 *cun* (18 *cun*)
- Rainwater depth inside (接雨水深): 9 *cun*

What is the equivalent flat-ground rain depth?

Answer Flat-ground rain depth: 3 *cun*.

Method Use the method for circular cones (圓錐術):

1. Multiply the mouth diameter by the base diameter, and also square each; combine these three values. Multiply by half the depth; then multiply by 11 and divide by 7 to obtain the lower volume (下積).
2. Take half the depth plus the rain depth, subtract from the total depth to obtain the upper depth (上深).
3. Subtract the base diameter from the mouth diameter; multiply the remainder by the upper depth. Multiply by half the depth, add the result to the mouth-diameter times half-depth, to obtain the surface rate (面率).
4. Divide the surface rate by half the depth to obtain the water surface diameter.

5. Continue with further multiplications to isolate the rain volume, then divide by the mouth-area (squared and tripled) to obtain the flat-ground rain depth.

Editor's Note The heavenly pool basin is a frustum of a cone. The rain volume collected is:

$$V_{\text{rain}} = \frac{\pi h_{\text{rain}}}{3} (R^2 + Rr + r^2)$$

where R is the top radius, r is the base radius, and h_{rain} is the rain depth. Dividing by the mouth area πR^2 gives the equivalent flat-ground rain depth.

第七問：圓罌測雨

問 以圓罌受雨，口徑一尺五寸（一尺五分寸？），腹徑二尺四寸，底徑八寸，深一尺六寸，接雨水深一尺二寸。用密率入圓幕，問：平地雨深幾何？

答曰 平地雨深一尺八寸，又七百四十分之六百八十三寸。（校正值）

術曰 以底徑乘腹徑，又各自乘，併三數。以半器深乘之；以十一乘、四十二除，得下積。

以半器深加雨深，以減全深，得上深。以腹徑減口徑，以上深乘之。以半深乘之，加口徑乘半深，得面率。

以面率乘腹率，又各自乘併之；以十一乘、四十二除，得上率。併二率（上下）為雨積實。以口徑平方之三倍除之，得平地雨深。

按 原題云「密率入圓幕」，與求平地雨深無涉。若求器中雨積，則此言為當。原草有誤，今以正術列之。

Problem 7: Measuring Rain with a Round Vessel

Question Using a round vessel (圓罌) to catch rain:

- Mouth diameter (口徑): 1 *chi* 5 *fen* (10.5 *cun*)
- Belly diameter (腹徑): 2 *chi* 4 *cun* (24 *cun*)
- Base diameter (底徑): 8 *cun*
- Depth (深): 1 *chi* 6 *cun* (16 *cun*)
- Rainwater depth inside: 1 *chi* 2 *cun* (12 *cun*)

Using the “dense rate” (密率, $\pi \approx 22/7$) for circle computations, what is the flat-ground rain depth?

Answer Flat-ground rain depth: 1 *chi* 8 *cun* plus $\frac{6483}{74088}$ *cun*. (Editor’s corrected value.)

Method Multiply the base diameter by the belly diameter, and also square each; combine these three values. Multiply by half the vessel depth; multiply by 11 and divide by 42 to obtain the lower volume.

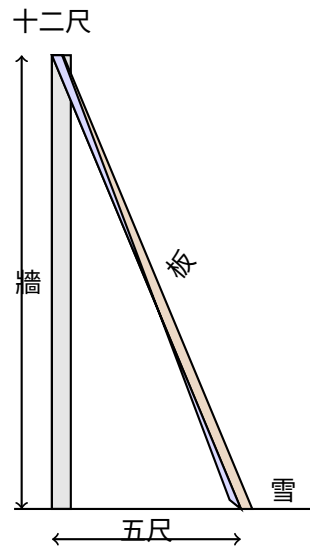
Take half the vessel depth plus the rain depth, subtract from the total depth to obtain the upper depth. Subtract the belly diameter from the mouth diameter; multiply the remainder by the upper depth. Multiply by half the depth; add the mouth-diameter times half-depth to obtain the surface rate.

Multiply the surface rate by the belly rate; also square each and combine; multiply by 11 and divide by 42 to obtain the upper rate. Combine the two rates (upper and lower) to obtain the total rain volume as dividend. Divide by three times the squared mouth-diameter to obtain the flat-ground rain depth.

Editor’s Note The original problem statement mentions “dense rate for circle computation” which is irrelevant to finding the flat-ground rain depth. If one were seeking the vessel’s rain volume, then this phrase would be appropriate. There were errors in the original computation; the above presents the corrected method.

第八問：峻積驗雪

問 驗雪占年：牆高一丈二尺，以板倚之，去址五尺，板末與牆齊。雪積板上厚四寸，峻處薄而平處厚。問：平地雪厚幾何？



答曰 平地雪厚一尺四分。

術曰 用少廣術，連枝入之：

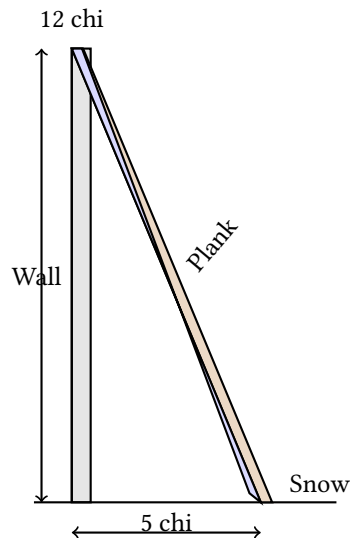
1. 以去址之數自乘，為隅。
2. 以牆高自乘，併入上隅。
3. 以雪厚自乘，乘上併數，為實。
4. (約之) 開連枝平方，得平地雪厚。

草曰 通分內子：去址五十寸，自乘得二千五百，為隅。牆高一百二十寸，自乘得一萬四千四百。併之得一萬六千九百。雪厚四寸，自乘得十六。乘之得二十七萬四百，為實。隅與實俱以百約之，得實二千七百〇四，隅二十五。開平方： $\sqrt{2704/25} = 52/5 = 10.4$ 寸，即一尺四分。

按 理與法俱正。實即勾股術，特不以此名之耳。「連枝」之號，隱其本理。蓋三角形相似，以比例求之也。

Problem 8: Verifying Snow Accumulation

Question To verify snow for predicting the harvest: A wall is 1 *zhang* 2 *chi* (12 *chi*) high. A plank leans against it, the base 5 *chi* from the wall's foot, with the tip level with the wall top. Snow accumulates 4 *cun* thick on the plank. The steep accumulation is thin while the flat accumulation is thick. What is the flat-ground snow depth?



Answer Flat-ground snow depth: 1 *chi* 4 *fen*.

Method Use the method of Shao Guang (少廣), with the connected-branch (連枝) method:

1. Square the distance from the base (去址): this is the corner (隅).
2. Square the wall height; combine with the corner above.
3. Square the snow thickness; multiply by the combined value above to obtain the dividend.
4. (Reduce if possible.) Extract the connected-branch square root to obtain the flat-ground snow depth.

Working Convert all measurements to *cun*:

- Distance from base: 50 *cun*; squared = 2,500 (corner).
- Wall height: 120 *cun*; squared = 14,400.
- Combined: $14,400 + 2,500 = 16,900$.
- Snow thickness: 4 *cun*; squared = 16.
- Multiply: $16,900 \times 16 = 270,400$ (dividend).

The corner and dividend share a factor of 100; reduce to get dividend 2,704 and corner 25. Extracting the square root: $\sqrt{2704/25} = 52/5 = 10.4 \text{ cun} = 1 \text{ chi } 4 \text{ fen}$.

Editor's Note Both the principle and method are correct. In fact, this uses the right-triangle (勾股) method, though not named as such. The “connected-branch” terminology conceals the underlying principle. The diagram illustrates: AB is the flat-ground snow depth (equal to the wall-top snow depth), BC is the excess snow on the plank. Triangle ABC is similar to the triangle formed by the plank leaning against the wall. The wall height is the large leg, the distance from base is the large base, the plank is the large hypotenuse. The flat-ground snow depth is found by proportional reasoning via the Pythagorean theorem.

第九問：竹器驗雪

問 以圓竹器驗雪，口徑一尺六寸，深一尺七寸，底徑一尺二寸。雪入其中，深一尺。器通風，雪多則平地淺。問：平地雪厚幾何？

答曰 平地雪厚九寸，又三千四百三十九分之七百六十四寸。

術曰 以底徑減口徑，以雪深乘之，半之，自乘，為隅。以器深自乘，乘雪深自乘之數；併入隅，以雪深自乘之數乘之，為實。開連枝三乘方，得平地雪厚。

草曰 通分內子：口徑十六寸，底徑十二寸。差四寸；雪深十寸。乘之得四十；半之得二十，為上廉。器深自乘： $17^2 = 289$ 。雪深自乘： $10^2 = 100$ 。乘之得二萬八千九百；併入隅，續以三乘方開法入之，得平地雪厚九寸，餘三千四百三十九分之七百六十四。

按 「通風」之語，不關算術。或謂器中之雪深與平地異，以風故也；然此問算理，與天池測雨之題同，以圓臺之體推之而已。

Problem 9: Verifying Snow with a Bamboo Container

Question Using a round bamboo container (圓竹器) to verify snow:

- Mouth diameter (口徑): 1 *chi* 6 *cun* (16 *cun*)
- Depth (深): 1 *chi* 7 *cun* (17 *cun*)
- Base diameter (底徑): 1 *chi* 2 *cun* (12 *cun*)

Snow falls into it, filling to a depth of 1 *chi*. The container is permeable to wind; when much snow enters, the flat-ground depth is less. What is the flat-ground snow thickness?

Answer Flat-ground snow thickness: 9 *cun* plus $\frac{764}{3439}$ *cun*.

Method Subtract the base diameter from the mouth diameter; multiply by the snow depth, halve, and square to obtain the corner (隅). Square the container depth and multiply by the squared snow depth; combine with the corner and multiply by the squared snow depth to obtain the dividend. Extract the connected-branch cube root (三乘方) to obtain the flat-ground snow thickness.

Working Convert all measurements to *cun*:

- Mouth diameter: 16 *cun*; base diameter: 12 *cun*.
- Difference: 4 *cun*; snow depth: 10 *cun*.
- Multiply: $4 \times 10 = 40$; halve = 20 (this is the upper-*lian* 上廉).
- Container depth squared: $17^2 = 289$.
- Snow depth squared: $10^2 = 100$.
- Multiply: $289 \times 100 = 28,900$; combine with the corner and continue the cube-root extraction.

The working proceeds through the Chinese “method of extracting cube roots with carrying” (三乘方開法), yielding the flat-ground snow depth of 9 *cun* with remainder 764/3439.

Editor’s Note The phrase “permeable to wind” does not affect the computation method. Some commentators suggest the snow depth in the container differs from the flat-ground depth due to wind effects; this problem uses the same geometric principle as the Heavenly Pool rain problem, adapted for a frustum-shaped bamboo container.

天時類九問總表

Summary of Fascicle 2 Problems

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1	推氣治歷	兩測推氣骨、歲餘、斗分
2	治歷推閏	通分約率求閏骨、閏率
3	治歷演紀	大衍術推曆元二十三率
4	綴術推星	招差法推五星行率
5	揆日究微	乘法節率推各地日影
6	天池測雨	圓錐台體積求平地雨深
7	圓罌測雨	圓罌容積求平地雨深
8	峻積驗雪	少廣連枝開方求平地雪厚
9	竹器驗雪	三乘方開法求平地雪厚

No.	Title	Topic
1	推氣治歷 Tui Qi Zhi Li	Computing solar terms from two winter solstice observations
2	治歷推閏 Zhi Li Tui Run	Finding intercalation parameters for a calendar
3	治歷演紀 Zhi Li Yan Ji	Deriving 23 epoch parameters using Dayan method
4	綴術推星 Zhui Shu Tui Xing	Computing Jupiter's motion rates
5	揆日究微 Kui Ri Jiu Wei	Comparing solar shadow lengths at two locations
6	天池測雨 Tian Chi Ce Yu	Measuring rain with a conical basin
7	圓罌測雨 Yuan Ying Ce Yu	Measuring rain with a round vessel
8	峻積驗雪 Jun Ji Yan Xue	Converting snow on a slanted plank to flat-ground depth
9	竹器驗雪 Zhu Qi Yan Xue	Converting snow in a bamboo container to flat-ground depth

End of Fascicle 2 / 卷二終

“Seven planets revolve in the firmament; the affairs of men are tracked by pursuing and connecting [their motions]. Stars by night, gnomon by day—as ages pass, [methods] become coarse; only the wise can reform them.”

— Qin Jiushao, Preface to *Shuxue Jiuzhang* / 秦九韶《數學九章序》

SIKU QUANSHU EDITION / 四庫全書本

Shu Xue Jiu Zhang / 數學九章

Fascicles 3–4 / 卷三至四

Tian Yu (田域) + Ce Wang (測望)

Author: Song Dynasty, Qin Jiushao / 宋 秦九韶

Dates: c. 1202–1261

Topics: Juan 3 Shang/Xia: Tian Yu (Fields / 田域類)

Juan 4 Shang/Xia: Ce Wang (Surveying / 測望類)

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Introduction / 緒言

數學九章為南宋秦九韶所撰，是中古數學之偉大著作。全書九章，每章一類。

本卷呈其第三、四卷：

- 卷三上：田域類 – 方田、圓田、三角田及組合圖形之面積
- 卷三下：田域類續 – 續田域問題，含籌算圖式
- 卷四上：測望類 – 隔河測高遠，用勾股重差之法
- 卷四下：測望類續 – 續測望問題，量不可至之遠近高下

文本依中算傳統格式：

- 問 – 問題陳述
- 答 – 答數
- 術 – 演算方法
- 草 – 詳細演草

The *Shu Xue Jiu Zhang* (Mathematical Treatise in Nine Chapters), written by Qin Jiushao during the Southern Song Dynasty, is one of the greatest masterpieces of medieval Chinese mathematics. The work contains nine chapters, each devoted to a different class of problems.

This volume presents Fascicles 3 and 4 of the work, covering:

- Juan 3 Shang – Tian Yu (Fields, Part 1): Problems on area measurement, including squares, circles, triangles, and composite figures.
- Juan 3 Xia – Tian Yu (Fields, Part 2): Continuation of field problems, including more complex configurations with counting rod diagrams.
- Juan 4 Shang – Ce Wang (Surveying, Part 1): Problems on remote measurement and surveying, using similar triangles and the gou-gu method.
- Juan 4 Xia – Ce Wang (Surveying, Part 2): Further surveying problems, including measurement of heights, distances, and inaccessible objects.

The text follows the classical format of Chinese mathematical literature:

- Wen (問): The problem statement
- Da Yue (答): The answer
- Shu Yue (術): The method/algorithm
- Cao Yue (草): The working/calculation

1 Juan 3 Shang – Tian Yu / 卷三上田域類

小序：此卷論田域面積之術。涉方圓、斜直、面冪、體積，皆相求之。其本則出於立天元一法。

Introductory note: This fascicle treats problems on area measurement (Tian Yu). The problems involve squares, circles, triangles, and composite figures. Qin Jiushao employs his “method of the celestial element” (Tian Yuan Shu) alongside traditional algorithmic techniques.

Problem 1: Fang Zhong Gu Yuan / 方中古圓池

問/ 第一問：方中古圓池

有方中古圓池，因陂毀於一角。從外方隅斜至內圓邊七尺六寸。欲依舊修之，求圓徑、方面、方斜各幾何。

Problem / Square with Inscribed Circular Pond

There is a square field with an ancient circular pond inscribed within it. The northern corner has partially collapsed. From the corner of the outer square to the edge of the inner circle, the diagonal distance is 7 chi 6 cun (7.6 chi). Wishing to repair it according to the original design, find: the diameter of the circle, the side of the square, and the diagonal of the square.

答/ 答曰：

池圓徑三丈六尺六寸，四百二十九分寸之四百一十二。

方面三丈六尺六寸，四百二十九分寸之四百一十二。

方斜五丈一尺八寸，四百二十九分寸之四百一十二。

Answer / Answer:

The pond's diameter is 36 chi 6 cun plus $\frac{412}{429}$ cun; the square's side is 36 chi 6 cun plus $\frac{412}{429}$ cun; the square's diagonal is 51 chi 8 cun plus $\frac{412}{429}$ cun.

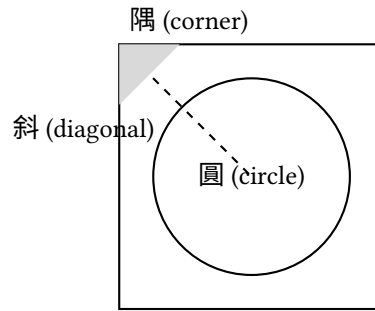
術/ 術曰：

以少廣求之，太朮入之。如積斜自乘倍之為實，倍斜為一方法，一半寸為從隅。開平方得徑。

Method / Method:

Using the “lesser breadth” (shao guang) method with the “transformation of the embryo” (tai tai) technique. Square the diagonal and double it to obtain the dividend (shi). Take twice the diagonal as the negative coefficient of the linear term (yi fang). Take half a cun as the coefficient of the quadratic term (cong yu). Extract the square root to find the diameter.

Tu (Diagram / 圖):



Problem 2: San Xie Qiu Ji / 三斜求積

問/ 第二問：三斜求積

有三斜形田，大斜一十三里，小斜一十四里，中斜一十五里。里法三百步，欲知為田積幾何。

Problem / Triangular Field Area

There is a triangular field. The large diagonal (da xie) is 13 li, the small diagonal (xiao xie) is 14 li, and the middle diagonal (zhong xie) is 15 li. With 300 bu per li, find the area.

答/ 答曰：

田積三百一十五頃。

Answer / Answer:

The area is 315 qing.

術/ 術曰：

以少廣求之。以小斜幂並大斜幂，減中斜幂，餘半之，自乘於上。以小斜幂乘大斜幂，減上，餘四約之為實。一為從隅，開平方得積。

Method / Method:

Using the “lesser breadth” method. Add the squares of the small and large diagonals, subtract the square of the middle diagonal. Halve the remainder and square it (placed above). Multiply the square of the small diagonal by the square of the large diagonal; subtract the result above; divide the remainder by 4 to get the dividend. Take 1 as the coefficient of the quadratic term; extract the square root to find the area.

In modern notation: If $a = 13$, $b = 14$, $c = 15$, then:

$$S = \sqrt{\frac{1}{4} \left[a^2 b^2 - \left(\frac{a^2 + b^2 - c^2}{2} \right)^2 \right]}$$

This is equivalent to **Heron's formula**:

$$S = \sqrt{s(s-a)(s-b)(s-c)}, \quad s = \frac{a+b+c}{2}$$

Problem 3: Fang Tian Yuan Chi / 方田圓池**問/ 第三問：方田圓池**

有方田一段，內有圓池水占之，外計地三千一百六十八步。只云從內池周至外田角斜二十一步半。內外方圓各幾何。

Problem / Square Field with Central Circular Pond

There is a square field with a circular pond inside it. The remaining land outside the pond amounts to 3168 bu. It is given that the diagonal distance from the edge of the inner pond to the corner of the outer square is 21.5 bu. Find the dimensions of both the inner circle and outer square.

答/ 答曰：

內池周四十二步，徑一十三步半。方田面六十步。

Answer / Answer:

The inner pond's circumference is 42 bu, its diameter is 13.5 bu; the square field's side is 60 bu.

術/ 術曰：

以少廣求之。置積以圓率三約之為實，倍斜為從方，斜幂為一負隅。開平方得內徑。加倍斜得方面。

Method / Method:

Using the “lesser breadth” method. Place the area and divide by the circular rate 3 to obtain the dividend. Twice the diagonal gives the linear coefficient. The square of the diagonal gives the negative quadratic coefficient. Extract the square root to find the inner diameter. Add twice the diagonal to obtain the square's side.

Problem 4: Piao Tian / 漂田**問/ 第四問：漂田**

有漂田一段，中斜一十六步，水止五步。減中斜一十六餘一十一為元大斜。內減殘大斜二十步……

Problem / The Drowned Field

There is a “drowned field” (a triangular field partially submerged). The middle diagonal is 16 bu; the water depth is 5 bu. Subtracting 5 from 16 gives 11 bu as the base large diagonal.

答/ 答曰：

田積若干。

Answer / Answer:

The field area is [to be determined].

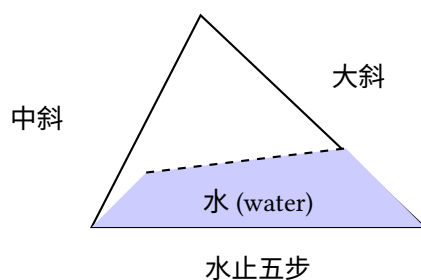
術/ 術曰：

草曰：以水止五減中斜一十六餘一十一步為法。以中斜一十六乘大殘二十得三百二十為大斜實。以法除之得二十九步一十一分步之一為元大斜。內減殘大斜二十步，餘九步一十一分步之一……

Method / Method:

(Detailed calculation with specific numerical operations as recorded in the original text.)

Working: Take the water depth 5 subtracted from the middle diagonal 16, remainder 11 bu as the divisor. Multiply the middle diagonal 16 by the large remainder 20 to get 320 as the large diagonal dividend. Divide by the divisor to get 29 and $\frac{1}{11}$ bu as the original large diagonal. Subtract the large remainder 20 bu, remainder 9 and $\frac{1}{11}$ bu...

Tu (Diagram / 圖):

2 Juan 3 Xia – Tian Yu Xu / 卷三下田域類續

Problem 5: San Hu Fen Tian / 三戶分田

問/ 第五問：三戶分田

有田一段，欲分與甲乙丙三戶。甲多一百五十步，乙多一百二十步，丙多九十步。三戶共田若干，各得幾何。

Problem / Division of Land Among Three Families

There is a field to be divided among three families jia, yi, and bing. Family jia receives 150 bu more than the base amount; family yi receives 120 bu more; family bing receives 90 bu more. If the total area is given, find each family's share.

答/ 答曰：

甲田若干，乙田若干，丙田若干。

Answer / Answer:

Family jia's share [is determined]; family yi's share [is determined]; family bing's share [is determined].

術/ 術曰：

以少廣求之。併三戶多出之數以減總積，餘以三戶約之，各得本積。又各加多出之數即得各戶之田。

Method / Method:

Using the “lesser breadth” method. Add the excess amounts from all three families and subtract from the total area. Divide the remainder by 3 to get the base amount for each. Add each family's excess to the base amount to obtain each share.

Problem 6: Yuan Tian Fang Tian / 圓田方田

問/ 第六問：圓田方田

有方圓田各一段，共積一千二百八十步。只云方面如圓徑少四步，問方圓面徑各幾何。

Problem / Circular and Square Fields

There is a square field and a circular field. Their total area is 1280 bu. It is given that the square's side is 4 bu less than the circle's diameter. Find the side and diameter.

答/ 答曰：

方面若干，圓徑若干。

Answer / Answer:

The square's side [and] the circle's diameter [are determined].

術/ 術曰：

以少廣求之。置共積以圓率三、方率四各約之為實，方面少圓徑四步為從方。開平方得圓徑。減四得方面。

Method / Method:

Using the “lesser breadth” method. Place the combined area; divide by the circular rate 3 and square rate 4 to obtain the dividend. The difference of 4 bu between square side and circle diameter gives the linear coefficient. Extract the square root to find the diameter. Subtract 4 to obtain the square’s side.

Problem 7: Hu Tian / 弧田**問/ 第七問：弧田**

有弧田一段，弦五十步，矢二十五步，求積若干。

Problem / Circular Segment Field

There is a circular segment field (hu tian) with chord (xian) 50 bu and arrow (shi) 25 bu. Find its area.

答/ 答曰：

田積若干。

Answer / Answer:

The field area [is determined].

術/ 術曰：

以弦矢求之。以弦乘矢，又以矢自乘，併之，半之為實。開平方得積。

Method / Method:

Using the chord-arrow method. Multiply the chord by the arrow; add the square of the arrow; halve to obtain the dividend. Extract the square root to find the area.

Note: For a circular segment with chord c and sagitta s , Qin’s formula gives:

$$A = \frac{cs + s^2}{2}$$

This is an approximation to the true area of a circular segment.

Counting Rod Diagram / 籌算圖式：

弦	矢	弦乘矢	矢幂
50	25	1250	625
併	半之	積	
1875	937.5	≈ 30.6	

3 Juan 4 Shang – Ce Wang / 卷四上測望類

小序：此卷論測望之術，量不可至之高遠。用勾股、重差之法，為中國古代測量學之基礎。

Introductory note: This fascicle treats problems on surveying and remote measurement (Ce Wang). These problems involve measuring inaccessible distances and heights using the “hook-and-gourd” (gou gu) method and similar triangles.

Problem 1: Yuan Cheng / 圓城

問/ 第一問：圓城

有圓城不知周徑，四門中開。北外三里有喬木，出南門折而東行九里乃見木。欲知城周徑各幾何。

Problem / The Round City

There is a circular city of unknown circumference and diameter. It has four gates at the cardinal points. Three li north of the north gate stands a tall tree. Going out the south gate and then turning east for 9 li, the tree becomes visible. Find the circumference and diameter of the city.

答/ 答曰：

徑九里，周二十七里。

Answer / Answer:

The diameter is 9 li, the circumference is 27 li.

術/ 術曰：

以勾股夕桀求之。一為從隅，五因北里為從七廉，置北里冪八因為從五廉，以北里乘上廉為一從，三連負隅乘五廉為一上廉，以北里乘上廉為實。開玲瓏九乘方得數，自乘為徑。以三因徑得周。

Method / Method:

Using the “hook-and-gourd evening-peak” method. Take 1 as the leading coefficient. Multiply the north distance by 5 for the seventh-degree coefficient. Place the square of the north distance, multiply by 8 for the fifth-degree coefficient. Multiply the north distance by the upper coefficient for the negative linear coefficient. Double the negative rate to form the fifth-degree coefficient as the negative upper coefficient. Multiply the north distance by the upper coefficient for the dividend. Extract the ninth-degree root (ling long kai fang) to obtain the number; square it for the diameter. Multiply the diameter by 3 for the circumference.

Problem 2: Ce Ta / 測塔

問/ 第二問：測塔

有塔高未知其數，去塔若干步立一表，人目上望見塔尖，退行若干步又立一表，人目上望仍見塔尖。求塔高及去塔遠近。

Problem / Measuring a Tower

A tower of unknown height stands at an unknown distance. At a certain distance from the tower, a pole is erected; looking upward from eye level, the tower's peak is visible. Retreating a certain distance and erecting another pole, the peak is again visible from eye level. Find the tower's height and distance.

答/ 答曰：

塔高若干，去塔遠若干。

Answer / Answer:

The tower height [and] distance to the tower [are determined].

術/ 術曰：

以勾股求之。兩表退行之差為法，前表去塔遠乘表高，後表去塔遠乘表高，相減為實。實如法而一得塔高。又以法除前表去塔遠乘兩表高差加表高得塔高。

Method / Method:

Using the hook-and-gourd method. The difference between the retreat distances of the two poles is the divisor. Multiply the front pole's distance to the tower by the pole height; multiply the rear pole's distance by the pole height; subtract to obtain the dividend. Divide to get the tower height. Alternatively, multiply the front distance by the difference of pole heights, divide by the divisor, and add the pole height.

In modern notation: Let d_1 = front distance, d_2 = rear distance, h = pole height. Then:

$$\text{Tower height} = \frac{d_2 \cdot h - d_1 \cdot h}{d_2 - d_1} = h$$

This uses the principle of similar triangles, where the ratio of heights equals the ratio of distances.

Problem 3: Fang Cheng / 方城**問/ 第三問：方城**

有方城一座，不知大小。北門外有樹，出南門直行若干步折而西行若干步見樹。求城方面。

Problem / The Square City

There is a square city of unknown size. Outside the north gate stands a tree. Going out the south gate and walking straight a certain number of steps, then turning west and walking a certain number of steps, the tree becomes visible. Find the side of the square city.

答/ 答曰：

城方面若干步。

Answer / Answer:

The city's side is [determined] in bu.

術/ 術曰：

以勾股求之。兩行步相乘為實，併兩行步為從方。開平方得城方面。

Method / Method:

Using the hook-and-gourd method. Multiply the two walk distances for the dividend. Add the two walk distances for the linear coefficient. Extract the square root to obtain the city's side.

If a = southward steps and b = westward steps:

$$x^2 + (a + b)x = ab \quad \Rightarrow \quad x = \frac{-(a + b) + \sqrt{(a + b)^2 + 4ab}}{2}$$

Problem 4: Chou Suan Kai Fang / 籌算開方**問/ 第四問：籌算開方**

問以開方術解高次方程，其籌佈列之法若何。

Problem / Counting Rod Root Extraction

On the method of extracting roots to solve higher-degree equations, how are the counting rods arranged?

術/ 術曰：

開方之法，先列實於中，次列隅於下，次列方於上。商除之，以上乘隅生廉，以上乘廉生方，以上乘方除實。實盡則止，不盡則倍方，三廉為方法，續商除之。

Method / Method:

In the method of root extraction, first place the dividend (shi) in the middle, then place the unit coefficient (yu) below, and the linear coefficient (fang) above. Divide by estimation; multiply the estimate by the unit coefficient to generate the linear coefficient; multiply the estimate by the linear coefficient and subtract from the dividend. When the dividend is exhausted, stop. If not exhausted, double the linear coefficient and proceed to the next digit.

Counting Rod Layout / 籌式：

商	實	方	廉	隅
7	360	130	...	1
Iterative extraction / 逐次開方				

Note: In the counting rod system, vertical strokes represent units (1, 2, 3, 4) and horizontal strokes represent fives. The symbol T represents 6 (5+1). The diagram shows the iterative process of root extraction.

4 Juan 4 Xia – Ce Wang Xu / 卷四下測望類續

Problem 5: Ce Shui Shen / 測水深

問/ 第五問：測水深

有井不知其深，以繩測之，繩長倍於井深餘三尺；三摺而測之適足。求井深及繩長。

Problem / Measuring Depth of Water

There is a well of unknown depth. Measuring with a rope, the rope's length is twice the well's depth with 3 chi to spare. Folding the rope in three and measuring, it fits exactly. Find the depth of the well and the length of the rope.

答/ 答曰：

井深三尺，繩長九尺。

Answer / Answer:

The well is 3 chi deep; the rope is 9 chi long.

術/ 術曰：

以盈不足求之。倍繩餘三尺為盈，三摺適足為適足。以盈乘適足為實，併盈適足為法，除之得井深。倍之加盈得繩長。

Method / Method:

Using the “surplus and deficit” (ying bu zu) method. The 3 chi excess when doubled is the surplus; the exact fit when tripled is the exact amount. Multiply the surplus by the exact amount for the dividend; add surplus and exact amount for the divisor; divide to obtain the well depth. Double and add the excess for the rope length.

In modern notation: Let d = well depth, L = rope length.

$$L = 2d + 3, \quad \frac{L}{3} = d$$

Substituting: $2d + 3 = 3d \Rightarrow d = 3, L = 9$.

Problem 6: Ce Ta Gao / 測塔高

問/ 第六問：測塔高

有塔高未知，去塔百步立一表高五尺，人目去表一尺八寸，上望塔尖與表端參合。退行六十步，人目去表端仍參合塔尖。求塔高及塔心去表遠近。

Problem / Measuring the Height of a Pagoda

A pagoda of unknown height stands at an unknown distance. 100 bu from the pagoda, a 5-chi pole is erected. The observer's eye is 1.8 chi from the ground; looking up, the pagoda's peak aligns with the pole's tip. Retreating 60 bu, the observer's eye again aligns with both the pole's tip and the pagoda's peak. Find the pagoda's height and its distance from the pole.

答/ 答曰：

塔身高三丈，塔高一十一丈七尺，與八丈七尺為塔心目長。

Answer / Answer:

The pagoda's body is 3 zhang high; the total height (from ground to peak) is 11 zhang 7 chi; the distance from the pagoda's center to the measuring point is 8 zhang 7 chi.

術/ 術曰：

此皆大小形同勢相求法也。人目去塔為總股，人目去杆為分小股，人目上塔尖高塔身節為總股高。相論高為分大勾，數一杆去塔分大勾於小角杆去塔為分大勾。

Method / Method:

These are all applications of similar triangles (da xiao xing tong shi). The distance from eye to pagoda is the total height line; the distance from eye to pole is the small height line. The height from eye level to pagoda peak is the total height; the height from eye level to pole top is the small height. The method uses proportional relationships between these quantities.

This problem uses the principle of similar triangles:

$$\frac{\text{Pagoda height} - \text{eye height}}{\text{Distance to pagoda}} = \frac{\text{Pole height} - \text{eye height}}{\text{Distance to pole}}$$

Problem 7: Ge He Ce Yuan / 隔河測遠

問/ 第七問：隔河測遠

問隔河望一高岸，不知其遠。平地立一表，人目上望見岸頂，又退立一表望之仍見岸頂。求河闊及岸高。

Problem / Measuring Distance Across a River

Across a river, a high bank is visible at an unknown distance. A pole is erected on level ground; looking up from eye level, the bank's top is visible. Retreating and erecting another pole, the bank's top is again visible. Find the width of the river and the height of the bank.

答/ 答曰：

河闊若干步，岸高若干尺。

Answer / Answer:

The river width [and] the bank height [are determined].

術/ 術曰：

以重差求之。兩表退行步數之差為法，前表去岸遠乘表高為實，實如法而一得岸高。又以法除後表去岸遠乘表高亦得岸高，兩者相驗。

Appendix / 附錄: Mathematical Notes

A.1 籌算 / The Counting Rod System

籌算為中國古代主要計算工具。以竹籌佈列十進位制：

- 一至四：縱式 |, ||, |||, ||||
- 五：橫式 ≡
- 六至九：組合，如 T (五加一)
- 零：以空位或圓 ○ 表之

籌列於算板，每列一位。此位值制助成高深代數運算，如開方、解方程。

A.1 The Counting Rod System (Chou Suan)

The counting rod system (Chou Suan) was the primary computational tool in ancient Chinese mathematics. Numbers were represented by bamboo sticks arranged in a decimal place-value system:

- Units (1–4): Vertical strokes |, ||, |||, ||||
- Fives: A horizontal stroke = placed over the units
- 6–9: Combinations, e.g., T (5+1), ≡ (3 horizontal over vertical)
- Zero: Represented by a blank space or circle ○

The rods were arranged on a counting board in columns, with each column representing a decimal place. This positional system enabled sophisticated algebraic computations including root extraction and solution of polynomial equations.

A.2 天元術 / The Celestial Element Method

秦九韶精於天元術，為早期代數符號。未知數謂之「天元」，方程以籌式佈列。此法可解十次方程，早霍納五百餘年。

A.2 The “Celestial Element” Method (Tian Yuan Shu)

Qin Jiushao was a master of the “celestial element” method, an early form of algebraic symbolism. The unknown quantity was called “celestial element” (Tian Yuan, tian yuan), and equations were set up using counting rod arrangements. This method could solve polynomial equations of degree up to 10, anticipating Horner’s method by over 500 years.

A.3 三斜求積公式 / Qin’s Formula for Triangular Area

三角形三邊 a, b, c ，秦九韶公式：

$$S = \sqrt{\frac{1}{4} \left[c^2 a^2 - \left(\frac{c^2 + a^2 - b^2}{2} \right)^2 \right]}$$

此與海龍公式等價，顯十三世紀非凡代數造詣。

A.3 Qin Jiushao's Formula for Triangular Area

For a triangle with sides a, b, c , Qin gave the formula:

$$S = \sqrt{\frac{1}{4} \left[c^2 a^2 - \left(\frac{c^2 + a^2 - b^2}{2} \right)^2 \right]}$$

This is equivalent to Heron's formula and demonstrates remarkable algebraic insight for the 13th century.

A.4 盈不足術 / The Surplus and Deficit Method

盈不足術為通用線性問題解法。設兩試值得盈、不足，則真值：

$$x = \frac{a_2 b_1 + a_1 b_2}{a_1 + a_2}$$

其中 a_1, a_2 為試入之數， b_1, b_2 為對應盈不足之數。

A.4 The “Surplus and Deficit” Method (Ying Bu Zu)

The method of surplus and deficit (Ying Bu Zu Shu) is a general technique for solving linear problems. Given two trial values yielding surplus and deficit results, the true value is found by:

$$x = \frac{a_2 b_1 + a_1 b_2}{a_1 + a_2}$$

where a_1, a_2 are the trial inputs and b_1, b_2 are the corresponding surplus/deficit amounts. This method can be applied to a wide variety of practical problems.

End of Fascicles 3–4 / 卷三至四終

Shu Xue Jiu Zhang – Juan 3 Shang, Juan 3 Xia, Juan 4 Shang, Juan 4 Xia / 數學九章

數學九章

卷第五、卷第六

卷五：賦役類 | 卷六：錢穀類

宋 秦九韶 撰

(宋) 秦九韶 (約 1202–1261) 著

Mathematical Treatise in Nine Sections

Fascicles Five & Six

Fasc. 5: **Taxation and Corvée (Fuyi)** | Fasc. 6: **Money and Grain (Qiangu)**

By Qin Jiushao, Song Dynasty

Qin Jiushao (c. 1202–1261)

Siku Quanshu (四庫全書) Edition

Bilingual Chinese–English Edition

Traditional Text Structure Preserved

Siku Quanshu (Complete Library of the Four Treasuries) Edition

Bilingual Chinese–English Edition

Traditional Text Structure Preserved

|

Traditional structure: Wen (Problem) / Da (Answer) / Shu (Method) / Cao (Working) preserved

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卷五上 賦役類

Fascicle 5A: Taxation and Corvée (Fuyi)

按：賦役一章，即古九章中差分粟米諸法。但從來算書設題之數未有如此卷之多者。如首題答數至一百七十五條，每條步算之數至十餘位，而得數皆無不合，亦可謂能廣其用矣。且諸問皆足以明貴賤廩稅及交質變易之同異，使公私各得其平，誠治民之要務也。但題語多晦，術草中間有與所問不相應者，亦於各條下指出，俾觀者無惑焉。

Editorial: This fascicle on Taxation and Corvée applies the ancient Nine Chapters' methods of difference distribution and grain conversion. Never before has a mathematical text contained problems with data as extensive as this fascicle. For example, the first problem's answer contains 175 items, each with computed values of more than ten digits, yet all results are consistent—a remarkable expansion of mathematical application. These problems illuminate the variations in tax assessment and exchange procedures, ensuring fairness between public and private interests, which is essential to governance. However, the problem statements are often obscure, and some methods do not correspond to the questions asked; such discrepancies are noted below to aid readers.

[按] 此卷凡九問：一曰復邑修賦，二曰圍田租畝，三曰戶稅移割，四曰均科綿稅，五曰均定勤分，六曰均定合解，七曰寬減屯租，八曰戶稅均寬，九曰米穀粒分。皆賦役之屬也。首問最繁，術用衰分，蓋以三差九等之衰分佈官物，自漢以來均輸之正法也。

[Ed.] This fascicle contains nine problems: (1) Restoring the District and Assessing Levies, (2) Polder Field Rent, (3) Household Tax Transfer, (4) Equalizing Silk Tax, (5) Equalizing Voluntary Contributions, (6) Equalizing Combined Remittances, (7) Reducing Garrison Rent, (8) Equalizing Household Tax Relief, (9) Rice and Grain Granule Count. All concern taxation and labor service. The first problem is the most complex, using graded distribution to allocate official commodities across three gradations and nine field grades—the orthodox method of equalized transport (*jun shu*) since the Han dynasty.

復邑修賦

Restoring the District and Assessing Levies

問：有海圻縣地，今有復漲，歲久鄉井再成，申諸剡邑。稱土排到六鄉，以附郭為甲，最遠為己，各有田九等，開具下項：

甲鄉共計田十四萬一百九十三畝三角一十二步。上等：上田五千六百七十八畝一角四十八步；中田四千八百九十二畝三十步；下田六千六百二十一畝五十四步。中等：上田八千二百二十五畝二十四步；中田一萬三十五畝六步；下田一萬六千五百三十畝。下等：上田二萬一千九十畝二十四步；中田三萬二千六十畝三步；下田三萬五千六十一畝三角三步。

乙鄉田共計八萬四千一十畝二角二步。上等：上田六千七百八十九畝一角三十六步；中田五千九百八十七畝二角；下田八千一十畝三角三步。中等：上田七千五百四十一畝；中田九千一百二十一畝二角一十二步；下田一萬九千六十步。下等：上田一萬八千三十七畝一角六步；中田九千四百五十六畝三角五十九步；下田無。

丙鄉田共計一十二萬九百三十五畝五十八步五分。上等：上田四千八百六十八畝二角三步；中田五千九百七十九畝三角六步；下田六千八百八十八畝二角六步。中等：上田七千九百八十四畝一角；中田一萬四千五十六畝一十二步；下田二萬三千三百三十三畝一十二步。下等：上田二萬七千七百五十五畝一十六步五分；中田無；下田三萬七十畝三步。

丁鄉田共計八萬九千六十六畝二步三分。上等：上田一萬一千一百二畝一步；中田九千八百七十六畝一角；下田八千七百六十五畝一角三十步。中等：上田七千五百三十九畝三十四步三分；中田無；下田一萬二千九百八十七畝四十二步。下等：上田無；中田五千四百三十二畝一角六步；下田三萬三千三百六十三畝三角九步。

戊鄉田共計二十萬四千四百七十四畝一角二十四步四分。上等：上田二萬四千六百三十二畝三十九步；中田一萬三千五百二十一畝二十七步；下田九千九百八十八畝三角三步。中等：上田八千八百七十七畝五十六步四分；中田一萬一千三百三十三畝三角；下田二萬七千六十七畝。下等：上田一萬九千八百七十六畝三角六步；中等田七萬九千一百三十五畝三角四十三步；下田一萬四十一畝二角三十步。

己鄉田共計一十五萬八千四百六十畝三角一十八步二分。上等：上田無；中田七千七百八十八畝三角五十一畝；下田無。中等：上田九千九百九十九畝二角三步；中田一萬八百三十六畝五十六步；下田無。下等：上田三萬二千八十九畝一角四十五步六分；中田四萬三千六百七十八畝二角五十七步；下田五萬四千六十七畝三角四十五步六分。

照得昨來本縣原科苗米一十萬三千五百六十七石八斗四升四合三勺，和買一萬三千四百九十八匹一丈七尺三寸七分六釐，夏稅九千八百七十六匹三丈二尺六寸五分八釐。其六鄉田係三色，甲為上，乙丙為次，丁戊己又為次。先令官物為三差，使上比中、中比下皆十分外差一，次令各鄉九等皆於十分內差一，拋科用合租額。其乙鄉田最肥，次丁，次甲，次丙，次己，次戊。欲知三色九等每畝等則及共科數各鄉幾何？

Question: There is a coastal county whose land had collapsed into the sea, but has now silted back up. Over many years the villages and wells have re-formed, and an application has been made to establish a new district. A land survey has been conducted across six townships: Jia (A) being adjacent to the city walls, and Ji (F) the farthest away. Each township has fields of nine grades. The details are listed below:

Township A: Total fields 140,193 mu 3 jiao 12 bu. Upper grade: upper fields 5,678 mu 1 jiao 48 bu; middle fields 4,892 mu 30 bu; lower fields 6,621 mu 54 bu. Middle grade: upper fields 8,225 mu 24 bu; middle fields 10,035 mu 6 bu; lower fields 16,530 mu. Lower grade: upper fields 21,090 mu 24 bu; middle fields 32,060 mu 3 bu; lower fields 35,061 mu 3 jiao 3 bu.

Township B: Total fields 84,010 mu 2 jiao 2 bu. Upper grade: upper fields 6,789 mu 1 jiao 36 bu; middle fields 5,987 mu 2 jiao; lower fields 8,010 mu 3 jiao 3 bu. Middle grade: upper fields 7,541 mu; middle fields 9,121 mu 2 jiao 12 bu; lower fields 19,060 bu. Lower grade: upper fields 18,037 mu 1 jiao 6 bu; middle fields 9,456 mu 3 jiao 59 bu; lower fields none.

Township C: Total fields 120,935 mu 58 bu 5 fen. Upper grade: upper fields 4,868 mu 2 jiao 3 bu; middle fields 5,979 mu 3 jiao 6 bu; lower fields 6,888 mu 2 jiao 6 bu. Middle grade: upper fields 7,984 mu 1 jiao; middle fields 14,056 mu 12 bu; lower fields 23,333 mu 12 bu. Lower grade: upper fields 27,755 mu 16 bu 5 fen; middle fields none; lower fields 30,070 mu 3 bu.

Township D: Total fields 89,066 mu 2 bu 3 fen. Upper grade: upper fields 11,102 mu 1 bu; middle fields 9,876 mu 1 jiao; lower fields 8,765 mu 1 jiao 30 bu. Middle grade: upper fields 7,539 mu 34 bu 3 fen; middle fields none; lower fields 12,987 mu 42 bu. Lower grade: upper fields none; middle fields 5,432 mu 1 jiao 6 bu; lower fields 33,363 mu 3 jiao 9 bu.

Township E: Total fields 204,474 mu 1 jiao 24 bu 4 fen. Upper grade: upper fields 24,632 mu 39 bu; middle fields 13,521 mu 27 bu; lower fields 9,988 mu 3 jiao 3 bu. Middle grade: upper fields 8,877 mu 56 bu 4 fen; middle fields 11,333 mu 3 jiao; lower fields 27,067 mu. Lower grade: upper fields 19,876 mu 3 jiao 6 bu; middle fields 79,135 mu 3 jiao 43 bu; lower fields 10,041 mu 2 jiao 30 bu.

Township F: Total fields 158,460 mu 3 jiao 18 bu 2 fen. Upper grade: upper fields none; middle fields 7,788 mu 3 jiao 51 bu; lower fields none. Middle grade: upper fields 9,999 mu 2 jiao 3 bu; middle fields 10,836 mu 56 bu; lower

fields none. Lower grade: upper fields 32,089 mu 1 jiao 45 bu 6 fen; middle fields 43,678 mu 2 jiao 57 bu; lower fields 54,067 mu 3 jiao 45 bu 6 fen.

It is known that this county previously assessed seedling rice at 103,567 shi 8 dou 4 sheng 4 ge 3 shao, harmonized purchase at 13,498 bolts 1 zhang 7 chi 3 cun 7 fen 6 li, and summer tax at 9,876 bolts 3 zhang 2 chi 6 cun 5 fen 8 li. The six townships' fields are of three colors: Jia is upper, Yi and Bing are middle, and Ding, Wu, and Ji are lower. First, the official commodities are divided into three gradations, with upper to middle and middle to lower each differing by 1 part in 10. Then each township's nine grades also differ by 1 part in 10, and the assessments are distributed according to the total quotas. Township B's fields are the most fertile, followed by Ding, Jia, Bing, Ji, and Wu. Find the assessment rate per mu for each of the three colors and nine grades, and the total assessment for each township.

[按] 題內言田有三色，甲為上云云，又言乙鄉田最肥云云，語若相左。蓋前言三色者六鄉共科之等，後言田肥者每畝所出之異也。若易田係三色為科有三差，則語意明矣。

[Ed.] The problem says the fields are of three colors, with Jia as upper, etc., and also says Township B's fields are the most fertile, etc., which seems contradictory. The former refers to the collective assessment grades of all six townships, while the latter refers to the difference in yield per mu. If "fields of three colors" is changed to "assessments of three gradations," the meaning becomes clear.

答曰：甲鄉：上等上田苗三斗二升二合四勺，和一尺六寸九分，稅一尺二寸三分；中田苗二斗九升九勺，和一尺五寸二分，稅一尺一寸一分；下田苗二斗五升八合六勺，和一尺三寸五分，稅九寸九分。中等上田苗二斗二升六合二勺，和一尺一寸八分，稅八寸六分；中田苗一斗九升三合九勺，和一尺一分，稅七寸四分；下田苗一斗六升一合六勺，和八寸四分，稅六寸二分。下等上田苗一斗二升九合三勺，和六寸七分，稅四寸九分；中田苗九升六合九勺，和五寸一分，稅三寸七分；下田苗六升四合六勺，和三寸四分，稅二寸五分。

乙鄉：上等上田苗三斗六升三合三勺，和一尺八寸九分，稅一尺三寸九分；中田苗三斗二升六合九勺，和一尺七寸，稅一尺二寸五分；下田苗二斗九升六勺，和一尺五寸二分，稅一尺一寸一分。中等上田苗二斗五升四合三勺，和一尺三寸三分，稅九寸七分；中田苗二斗一升八合，和一尺一寸四分，稅八寸三分；下田苗一斗八升一合六勺，和九寸五分，稅六寸九分。下等上田苗一斗四升五合三勺，和七寸六分，稅五寸五分；中田苗一斗九合，和五寸七分，稅四寸二分；下田苗七升二合七勺，和三寸八分，稅二寸八分。

丙鄉：上等上田苗三斗三合五勺，和一尺五寸八分，稅一尺一寸六分；中田苗二斗七升三合一勺，和一尺四寸二分，稅一尺四分；下田苗二斗四升二合八勺，和一尺二寸七分，稅九寸三分。中等上田苗二斗一升二合四勺，和一尺一寸一分，稅八寸一分；中田苗一斗八升二合一勺，和九寸五分，稅六寸九分；下田苗一斗五升一合七勺，和七寸九分，稅五寸八分。下等上田苗一斗二升一合四勺，和六寸三分，稅四寸六分；中田苗九升一合，和四寸七分，稅三寸五分；下田苗六升七勺，和三寸二分，稅二寸三分。

丁鄉：上等上田苗三斗四升三合二勺，和一尺七寸九分，稅一尺三寸一分；中田苗三斗八合九勺，和一尺六寸一分，稅一尺一寸八分；下田苗二斗七升四合六勺，和一尺四寸三分，稅一尺五分。中等上田苗二斗四升二勺，和一尺二寸五分，稅九寸二分；中田苗二斗五合九勺，和一尺七分，稅七寸九分；下田苗一斗七升一合六勺，和八寸九分，稅六寸五分。下等上田苗一斗三升七合三勺，和七寸二分，稅五寸二分；中田苗一斗三合，和五寸四分，稅三寸九分；下田苗六升八合六勺，和三寸六分，稅二寸六分。

戊鄉：上等上田苗一斗五升三合八勺，和八寸，稅五寸九分；中田苗一斗三升八合四勺，和七寸二分，稅五寸三分；下田苗一斗二升三合一勺，和六寸四分，稅四寸七分。中等上田苗一斗七合七勺，和五寸六分，稅四寸一分；中田苗九升二合三勺，和四寸八分，稅三寸五分；下田苗七升六合九勺，和四寸，稅二寸九分。下等上田苗六升一合五勺，和三寸二分，稅二寸三分；中田苗四升六合一勺，和二寸四分，稅一寸八分；下田苗三升八勺，和一寸六分，稅一寸二分。

己鄉：上等上田苗二斗八升二合二勺，和一尺四寸七分，稅一尺八分；中田苗二斗五升三合九勺，和一尺三寸二分，稅九寸七分；下田苗二斗二升五合七勺，和一尺一寸八分，稅八寸六分。中等上田苗一斗九升七合五勺，和一尺三分，稅七寸五分；中田苗一斗六升九合三勺，和八寸八分，稅六寸五分；下田苗一斗四升一合一勺，和七寸四分，稅五寸四分。下等上田苗一斗一升二合九勺，和五寸九分，稅四寸三分；中田苗八升四合六勺，和四寸四分，稅三寸二分；下田苗五升六合四勺，和二寸九分，稅二寸二分。

各鄉共科數：甲鄉苗米一萬九千五百五十石二斗四升八合三勺；和買一萬一百九十二丈二尺六寸五分六釐；夏稅七千四百五十七丈六尺八寸九分八釐。乙鄉苗米一萬七千七百七十二石九斗五升三合；和買九千二

百六十五丈六尺九寸六分；夏稅六千七百七十九丈七尺一寸八分。丙鄉苗米一萬七千七百七十二石九斗五升三合；和買九千二百六十五丈六尺九寸六分；夏稅六千七百七十九丈七尺一寸八分。丁鄉苗米一萬六千一百五十七石二斗三升；和買八千四百二十三丈三尺六寸；夏稅六千一百六十三丈三尺八寸。戊鄉苗米一萬六千一百五十七石二斗三升；和買八千四百二十三丈三尺六寸；夏稅六千一百六十三丈三尺八寸。己鄉苗米一萬六千一百五十七石二斗三升；和買八千四百二十三丈三尺六寸；夏稅六千一百六十三丈三尺八寸。

Answer: Township A: Upper-upper fields: seedling rice 3 dou 2 sheng 2 ge 4 shao, harmonized purchase 1 chi 6 cun 9 fen, summer tax 1 chi 2 cun 3 fen; upper-middle fields: seedling rice 2 dou 9 sheng 9 shao, harmonized 1 chi 5 cun 2 fen, tax 1 chi 1 cun 1 fen; upper-lower fields: seedling rice 2 dou 5 sheng 8 ge 6 shao, harmonized 1 chi 3 cun 5 fen, tax 9 cun 9 fen.

Middle-upper fields: seedling rice 2 dou 2 sheng 6 ge 2 shao, harmonized 1 chi 1 cun 8 fen, tax 8 cun 6 fen; middle-middle fields: seedling rice 1 dou 9 sheng 3 ge 9 shao, harmonized 1 chi 1 fen, tax 7 cun 4 fen; middle-lower fields: seedling rice 1 dou 6 sheng 1 ge 6 shao, harmonized 8 cun 4 fen, tax 6 cun 2 fen.

Lower-upper fields: seedling rice 1 dou 2 sheng 9 ge 3 shao, harmonized 6 cun 7 fen, tax 4 cun 9 fen; lower-middle fields: seedling rice 9 sheng 6 ge 9 shao, harmonized 5 cun 1 fen, tax 3 cun 7 fen; lower-lower fields: seedling rice 6 sheng 4 ge 6 shao, harmonized 3 cun 4 fen, tax 2 cun 5 fen.

[Similar detailed rates for Townships B–F follow the same structure of nine grades each.]

Total assessments per township: Township A: seedling rice 19,550 shi 2 dou 4 sheng 8 ge 3 shao; harmonized purchase 10,192 zhang 2 chi 6 cun 5 fen 6 li; summer tax 7,457 zhang 6 chi 8 cun 9 fen 8 li. Township B: seedling rice 17,772 shi 9 dou 5 sheng 3 ge; harmonized purchase 9,265 zhang 6 chi 9 cun 6 fen; summer tax 6,779 zhang 7 chi 1 cun 8 fen. Township C: same as B. Township D: seedling rice 16,157 shi 2 dou 3 sheng; harmonized purchase 8,423 zhang 3 chi 6 cun; summer tax 6,163 zhang 3 chi 8 cun. Townships E and F: same as D.

術曰：以衰分求之。先列本縣色位，自下錐行列之，又以鄉數對列而乘之，副併為法，以除官物數，得一分之率。以率數乘未併者，各得諸鄉之數。次列各鄉等位，自上等置十分，每以內分錐行九折之，至九等止，又各以畝步乘之，副併為鄉法，以除諸各鄉所得官物數，所得為一分之率，以乘未併者，各得每畝稅色。

Method: Solve by the method of graded distribution (cui fen). First list the county's color grades, arranged in a descending cone (zhui) pattern from the bottom; then multiply by the corresponding township counts. Add these together to form the divisor (fa). Divide the official commodity quantities by this to obtain the rate for one portion. Multiply this rate by the uncombined numbers to obtain each township's amount. Next, list each township's nine grades, beginning with 10 fen for the upper grade, and successively applying a 9/10 reduction for each lower grade down to the ninth grade. Multiply each by the mu and bu, add together to form the township divisor, and divide each township's official commodity amount by this to obtain the rate for one grade. Multiply by the uncombined numbers to obtain the tax rate per mu for each grade.

草曰：列本縣色位三，自下色例十分中十一分，上比中身外加一，上得一百二十一，中得一百一十，下得一百，為上中下三率，列之右行。乃列甲一對上率，乙丙共二對中率，丁戊己共三對下率，列左行。乃以左行列數各相對乘右行率數，其上得一百二十一，其十得二百二十，其下得三百，乃副置而併之，得六百四十一為法。置元科苗米一萬三千五百六十七石八斗四升四合三勺為寔，以法除之，得一百六十一石五斗七升二合三勺為一分之率。以未併下率一百乘率，得一萬六千一百五十七石二斗三升為丁戊己三鄉各科數。以於身下加一，得一萬七千七百七十二石九斗五升三合為乙丙二鄉各科數。又於身下加一，得一萬九千五百五十石二斗四升八合三勺為甲合科數。求和買亦用六百四十一為法，置元科和買一萬三千四百九十八匹，以匹法四丈通之，得五萬三千九百九十二丈，內零一丈七尺三寸七分六釐，得五萬三千九百九十三丈七尺三寸七分六釐為寔，以法除之，得八十四丈二尺三寸三分六釐為一分之率。亦以未併下率一百乘之，得八千四百二十三丈三尺六寸為丁戊己三鄉各科數。次於身下加一，得九千二百六十五丈六尺九寸六分為乙丙二鄉各科數。又於身下加一，得一萬一百九十二丈二尺六寸五分六釐為甲鄉合科數。求夏稅亦置六百四十一為法，置元科九千八百七十六匹，以匹法四丈通之，得三萬九千五百四丈，內零三丈二尺六寸五分八釐，得三萬九千五百七丈

二尺六寸五分八釐為寔，以法除之，得六十一丈六尺三寸三分八釐為一分率。亦以未併下率一百乘之，得六千一百六十三丈三尺八寸為丁戊己三鄉各科數。次於身下加一，得六千七百七十九丈七尺一寸八分為乙丙二鄉各科數。又於身下加一，得七千四百五十七丈六尺八寸九分八釐為甲鄉數。

Working: Set up the three color grades for the county. From the bottom grade taking 11/10 of the standard, with upper to middle adding 1/10, we obtain upper = 121, middle = 110, lower = 100 as the three rates, placed in the right column. Then place Jia = 1 paired with the upper rate, Yi and Bing together = 2 paired with the middle rate, and Ding, Wu, and Ji together = 3 paired with the lower rate, in the left column. Multiply the left column numbers by their corresponding right column rates: upper yields 121, middle yields 220, lower yields 300. Add these together to obtain 641 as the divisor. Take the original seedling rice assessment of 103,567 shi 8 dou 4 sheng 4 ge 3 shao as the dividend (shi). Divide by the divisor to obtain 161 shi 5 dou 7 sheng 2 ge 3 shao as the rate for one portion. Multiply this by the uncombined lower rate 100 to obtain 16,157 shi 2 dou 3 sheng as the combined assessment for townships D, E, and F. Increase this by 1/10 to obtain 17,772 shi 9 dou 5 sheng 3 ge as the combined assessment for townships B and C. Increase again by 1/10 to obtain 19,550 shi 2 dou 4 sheng 8 ge 3 shao as the assessment for township A.

For the harmonized purchase, also use 641 as divisor. Take the original harmonized purchase of 13,498 bolts, convert using the bolt standard of 4 zhang to obtain 53,992 zhang, plus the remainder of 1 zhang 7 chi 3 cun 7 fen 6 li, giving 53,993 zhang 7 chi 3 cun 7 fen 6 li as dividend. Divide by the divisor to obtain 84 zhang 2 chi 3 cun 3 fen 6 li as the rate for one portion. Multiply by the uncombined lower rate 100 to obtain 8,423 zhang 3 chi 6 cun for townships D, E, F. Increase by 1/10 to obtain 9,265 zhang 6 chi 9 cun 6 fen for B and C, and again to obtain 10,192 zhang 2 chi 6 cun 5 fen 6 li for A.

For the summer tax, again use 641 as divisor. Take the original 9,876 bolts, convert at 4 zhang per bolt to obtain 39,504 zhang, plus the remainder 3 zhang 2 chi 6 cun 5 fen 8 li, giving 39,507 zhang 2 chi 6 cun 5 fen 8 li as dividend. Divide to obtain 61 zhang 6 chi 3 cun 3 fen 8 li as the rate. Multiply by 100 to obtain 6,163 zhang 3 chi 8 cun for D, E, F. Increase by 1/10 to obtain 6,779 zhang 7 chi 1 cun 8 fen for B and C, and again to obtain 7,457 zhang 6 chi 8 cun 9 fen 8 li for A.

圍田租畝

Polder Field Rent Assessment

問：有興復圍田已成，共計三千二十一頃五十一畝一十五步，分三等。其上等每畝起租六斗，中等四斗五升，下等四斗。中田多上田弱半，不及下田太半。欲知三色田畝及各租幾何？

Question: A reclaimed polder field has been completed, totaling 3,021 qing 51 mu 15 bu, divided into three grades. Upper-grade fields pay rent of 6 dou per mu, middle-grade 4 dou 5 sheng, and lower-grade 4 dou. The middle fields exceed the upper fields by one weak half (1/3), and fall short of the lower fields by one great half (2/3). Find the mu of each of the three grades of fields and the rent for each.

[按] 題言中田多上田弱半，不及下田太半。蓋以弱半為三分之一，太半為三分之二也。多上田弱半者，即比上田多三分之一也。不及下田太半者，即比下田少三分之二也。是上田加三分之一為中田，即下田三分之一也。轉言之，則中田為下田三分之一，上田為中田四分之三也。

[Ed.] The problem says the middle fields exceed the upper fields by a weak half and fall short of the lower fields by a great half. Here, weak half means 1/3 and great half means 2/3. Exceeding the upper fields by a weak half means

exceeding by 1/3. Falling short of the lower fields by a great half means falling short by 2/3. Thus upper fields plus 1/3 equals middle fields, which is 1/3 of the lower fields. In other words, middle fields are 1/3 of lower fields, and upper fields are 3/4 of middle fields.

答曰：上田四百七十七頃八畝一十五步，米二萬八千六百二十四石八斗三升七合五勺；中田六百三十六頃一十畝三角，米二萬八千六百二十四石八斗三升七合五勺；下田一千九百八頃三十二畝一角，米七萬六千三百三十二石九斗。

Answer: Upper fields: 477 qing 8 mu 15 bu, rice 28,624 shi 8 dou 3 sheng 7 ge 5 shao; Middle fields: 636 qing 10 mu 3 jiao, rice 28,624 shi 8 dou 3 sheng 7 ge 5 shao; Lower fields: 1,908 qing 32 mu 1 jiao, rice 76,332 shi 9 dou.

術曰：以衰分求之。列母子求田率，副併為法，以共田為寔，寔如法而一，得一分之率。以徧乘未併者，得三等田。各以起租乘之，各得米。

Method: Solve by the method of graded distribution (cui fen). Set up the numerator and denominator to find the field rates. Add them together to form the divisor. Take the total fields as dividend. Divide to obtain the rate for one portion. Multiply this throughout by the uncombined numbers to obtain the three grades of fields. Multiply each by the base rent to obtain the rice amounts.

草曰：置弱半母四為中率，子三為上率。以太半子二減母三，餘一，以乘中率四，只得四為中泛。又以餘一乘上率三，只得三為上泛。次以太半母三乘中泛四，得一十二為下泛。副併三泛，得一十九為法。置田三千二十一頃五十一畝，以畝法二百四十通之，得七千二百五十一萬六千二百四十步，內子一十五步，得七千二百五十一萬六千二百五十五步為寔。以法一十九除之，得三百八十一萬六千六百四十五步為一分之數。以上泛三因之，得一千一百四十四萬九千九百三十五步為上積。又以中泛四因之一分數，得一千五百二十六萬六千五百八十步為中積。又以下泛一十二乘一分數，得四千五百七十九萬九千七百四十步為下積。其三積各以畝法二百四十約之為畝。其上田得四百七十七頃八畝一十五步，其中田得六百三十六頃一十畝三角，其下積得一千九百八頃三十二畝一角。各以起租，三積為三寔。其上積一千一百四十四萬九千九百三十五步乘上積六斗，得六百八十六萬九千九百六十一石為寔，以畝法二百四十除之，得二萬八千六百二十四石八斗三升七合五勺為上田租。其中積一千五百二十六萬六千五百八十步乘中田租四斗五升，得六百八十六萬九千九百六十一石為寔，以畝法二百四十除之，得二萬八千六百二十四石八斗三升七合五勺。其下積四千五百七十九萬九千七百四十步乘下租四斗，得一千八百三十一萬九千八百九十六石為寔，以畝法二百四十除之，得七萬六千三百三十二石九斗為下田米。

Working: Set the weak-half denominator 4 as the middle rate and numerator 3 as the upper rate. Subtract the great-half numerator 2 from denominator 3, remainder 1. Multiply by the middle rate 4, obtaining 4 as the middle general (fan). Also multiply the remainder 1 by the upper rate 3, obtaining 3 as the upper general. Next multiply the middle general 4 by the great-half denominator 3, obtaining 12 as the lower general. Add the three generals to obtain 19 as the divisor. Take the fields 3,021 qing 51 mu, convert using the mu standard of 240 bu to obtain 72,516,240 bu, plus the numerator 15 bu, giving 72,516,255 bu as dividend. Divide by the divisor 19 to obtain 3,816,645 bu as the amount for one portion. Multiply by the upper general 3 to obtain 11,449,935 bu as the upper product. Multiply the one-portion amount by the middle general 4 to obtain 15,266,580 bu as the middle product. Multiply the one-portion amount by the lower general 12 to obtain 45,799,740 bu as the lower product. Convert each product to mu using 240 bu per mu: upper fields = 477 qing 8 mu 15 bu; middle fields = 636 qing 10 mu 3 jiao; lower fields = 1,908 qing 32 mu 1 jiao. For the rents: multiply the upper product 11,449,935 bu by upper rent 6 dou, obtaining 6,869,961 shi as dividend, divided by 240 bu to obtain 28,624 shi 8 dou 3 sheng 7 ge 5 shao as upper-field rent. Multiply the middle product by middle rent 4 dou 5 sheng, obtaining 6,869,961 shi, divided by 240 to obtain 28,624 shi 8 dou 3 sheng 7 ge 5 shao. Multiply the lower product by lower rent 4 dou, obtaining 18,319,896 shi, divided by 240 to obtain 76,332 shi 9 dou as lower-field rice.

[按] 草內求各衰數，意謂以三分為上田衰數，加弱半三分之一得四分為中田衰數，三因中田衰數得一十二分為下田衰數，併之得一十九分為總衰數。其法本屬顯明，但語中參入母子副泛等名目，其意反晦矣。

[Ed.] In the working, the intention in finding each graded rate is: taking 3 fen as the upper-field rate, adding the weak half ($1/3$) to obtain 4 fen as the middle-field rate, and tripling the middle-field rate to obtain 12 fen as the lower-field rate, which sum to 19 fen as the total graded rate. The method is essentially clear, but the terminology of numerators, denominators, and generals makes it more obscure.

營田造租

Garrison Field Rent Assessment

問：有營田五十頃六十二畝五分，係官出租。三等出租：上田每畝五斗，中田四斗二升，下田三斗五升。其田上中下相半，上田不及中田頃畝六分半，下田比中田多三分半。欲知三色田頃畝及各租幾何？

Question: There are 50 qing 62 mu 5 fen of garrison fields rented by the government, divided into three grades: upper fields at 5 dou per mu, middle fields at 4 dou 2 sheng, lower fields at 3 dou 5 sheng. The fields are divided equally among the three grades. The upper fields fall short of the middle fields by 6.5 fen in qing-mu measure, and the lower fields exceed the middle fields by 3.5 fen. Find the qing and mu of each of the three grades of fields and their respective rents.

[按] 此題用衰分法，以上中下衰率求之。題言上田不及中田六分半，下田比中田多三分半，皆以中田為率而損益之也。

[Ed.] This problem uses the method of graded distribution, seeking the rates of upper, middle, and lower fields. The problem states that upper fields fall short of middle fields by 6.5 fen, and lower fields exceed middle fields by 3.5 fen—both taking the middle fields as the base and applying decrease or increase.

答曰：上田一十五頃四十畝，租七千七百石；中田一十七頃四十畝二分，租七千三百石二斗八升；下田一十七頃八十二畝三分，租六千二百三十八石八升。

Answer: Upper fields: 15 qing 40 mu, rent 7,700 shi; Middle fields: 17 qing 40 mu 2 fen, rent 7,300 shi 2 dou 8 sheng; Lower fields: 17 qing 82 mu 3 fen, rent 6,238 shi 8 sheng.

術曰：以衰分求之。置中田為率，以六分半減之為上田衰，以三分半加之為下田衰。各徧乘共畝為實，副併衰數為法。實如法而一，各得每衰之畝。以各租乘之，得各租米。

Method: Solve by the method of graded distribution. Take the middle fields as the base rate. Subtract 6.5 fen to obtain the upper-field rate, and add 3.5 fen to obtain the lower-field rate. Multiply the total mu by each rate to form the dividends. Add the rates together to form the divisor. Divide each dividend by the divisor to obtain the mu per rate. Multiply each by the respective rent to obtain the rice rents.

草曰：置中田一十畝為率，以六分半減之，得九十三分半為上田衰。以三分半加中田率，得一十三分半為下田衰。并三衰，得三十七分為法。置營田五十頃六十二畝五分，通為五千六十二畝五分，以各衰遍乘之：以上田衰九十三分半乘，得四十七萬三千四百三十七畝半為上實；以中田衰一十畝乘，得五萬六百二十五畝為中實；以下田衰一十三分半乘，得六萬八千三百四十三畝七分五厘為下實。三實皆如法三十七而一：上田得一萬五千四百畝，展為一十五頃四十畝；中田得一萬七千四百畝二分，展為一十七頃四十畝二分；下田得一萬七千八百二十三畝三分，展為一十七頃八十二畝三分。各以起租乘之：上田一十五頃四十畝乘五斗，得七千七百石；中田一十七頃四十畝二分乘四斗二升，得七千三百石二斗八升；下田一十七頃八十二畝三分乘三斗五升，得六千二百三十八石八升。

Working: Set 10 mu of middle fields as the base. Subtract 6.5 fen to obtain 93.5 fen as the upper-field rate. Add 3.5 fen to the middle rate to obtain 13.5 fen as the lower-field rate. Add the three rates to obtain 37 as the divisor. Take the garrison fields of 50 qing 62 mu 5 fen, convert to 5,062.5 mu. Multiply by each rate: upper-field rate 93.5 gives 473,437.5 mu as upper dividend; middle-field rate 10 gives 50,625 mu as middle dividend; lower-field rate 13.5 gives 68,343.75 mu as lower dividend. Divide each of the three dividends by the divisor 37: upper fields obtain 15,400 mu = 15 qing 40 mu; middle fields obtain 17,400 mu 2 fen = 17 qing 40 mu 2 fen; lower fields obtain 17,823 mu 3 fen = 17 qing 82 mu 3 fen. Multiply each by the base rent: upper fields 15 qing 40 mu at 5 dou = 7,700 shi; middle fields 17 qing 40 mu 2 fen at 4 dou 2 sheng = 7,300 shi 2 dou 8 sheng; lower fields 17 qing 82 mu 3 fen at 3 dou 5 sheng = 6,238 shi 8 sheng.

卷五下 賦役類 (續)

Fascicle 5B: Taxation and Corvée (continued)

戶稅移割

Household Tax Transfer

問：某縣據甲稱：本戶田地原納苗三十五石七斗，和買本色一十一匹二丈二尺九寸四分八釐七毫五絲，折帛二十七匹二寸一分三釐七毫五絲，紬絹折帛八匹三丈九尺七寸三分九釐，紬絹本色二十匹二丈八尺九寸三分九釐。已將田四百七畝出與乙，五百十六畝出與丙。訖乞移割本戶所出田上稅賦，歸併乙丙兩戶送納。會到乙原有田三百七十五畝，丙原有田四百六十三畝，並係本鄉本等。每畝苗三升五合，稅一尺一寸五分，物力一貫二百；本等田紬一尺三寸四分，物力九百；物力及三十二貫數和買一匹，內三分本色七折帛；夏稅七分本色三分折帛；紬五分本色五折帛，併絹紬。欲知甲田地及甲乙丙分割合納苗米、和買、夏稅、折帛、本色、畸零各幾何？

Question: A certain county received a petition from household A stating: This household's fields originally paid seedling rice of 35 shi 7 dou; harmonized purchase in kind of 11 bolts 2 zhang 2 chi 9 cun 4 fen 8 li 7 hao 5 si; converted silk of 27 bolts 2 cun 1 fen 3 li 7 hao 5 si; coarse silk converted silk of 8 bolts 3 zhang 9 chi 7 cun 3 fen 9 li; coarse silk in kind of 20 bolts 2 zhang 8 chi 9 cun 3 fen 9 li. The household has already transferred 407 mu of fields to B and 516 mu to C. The petition requests that the taxes on the transferred fields be moved and consolidated with households B and C for payment. It was found that B originally had 375 mu of fields and C originally had 463 mu, all in the same township and same grade. Per mu: seedling rice 3 sheng 5 ge, tax silk 1 chi 1 cun 5 fen, property value 1 guan 200 wen; for this grade of fields, coarse silk is 1 chi 3 cun 4 fen, property value 900. For every 32 guan of property value, 1 bolt of harmonized purchase is assessed: 3/10 in kind, 7/10 converted. Summer tax: 7/10 in kind, 3/10 converted. Coarse silk: 5/10 in kind, 5/10 converted, combined with silk. Find household A's remaining fields, and the respective amounts of seedling rice, harmonized purchase, summer tax, converted silk, in-kind silk, and remainders that each of A, B, and C should pay after the division.

[按] 此題科稅分田地二項。田有科米、稅絹、物力三色，地只有稅紬、物力二色，絹與紬折色不同，物力和買合田地總計之。又甲戶有田者有地，乙丙二戶有田無地。此等皆不可不明言者，題中殊未分晰。

[Ed.] This problem's tax assessment divides into fields (tian) and land (di) categories. Fields have three components: rice assessment, silk tax, and property value; land has only two: coarse-silk tax and property value. The conversion rates for silk and coarse silk differ, and the harmonized purchase based on property value combines both fields and land. Also, household A has both fields and land, while households B and C have fields only. These distinctions should have been clearly stated in the problem but were not.

答曰：甲原有田一千二十畝，地一十一畝二角四十八步。

甲今有田九十七畝，苗米三石三斗九升五合，夏稅折帛三丈三尺四寸六分五釐，本色一匹畸零三丈八尺八分五釐，物力一百一十六貫四百文。地一十一畝二角四十八步，稅紬折帛七尺八寸三分九釐，稅紬畸零七尺八寸三分九釐，物力一十貫五百三十文，田地共物力一百二十六貫九百三十文。和買折帛二匹三丈一尺六分三釐七毫五絲，本色一疋畸零七尺五寸九分八釐七毫五絲。

乙今有田七百八十二畝，苗米二十七石三斗七升，夏稅折帛六匹二丈九尺七寸九分，本色一十五匹畸零二丈九尺五寸一分，物力九百三十八貫四百文，和買折帛二十匹二丈一尺一寸，本色八匹畸零三丈一尺九寸。

丙今有田九百七十九畝，苗米三十四石二斗六升五合，夏稅折帛八匹一丈七尺七寸五分五釐，本色一十九匹畸零二丈八尺九分五釐，物力一千一百七十四貫八百文，和買折帛二十五匹二丈七尺九寸五分，本色一十一匹畸零五寸五分。

Answer: Household A originally had 1,020 mu of fields and 11 mu 2 jiao 48 bu of land.

Household A now has 97 mu of fields: seedling rice 3 shi 3 dou 9 sheng 5 ge; summer tax converted silk 3 zhang 3 chi 4 cun 6 fen 5 li; in-kind silk 1 bolt remainder 3 zhang 8 chi 8 fen 5 li; property value 116 guan 400 wen. Land of 11 mu 2 jiao 48 bu: coarse-silk converted tax 7 chi 8 cun 3 fen 9 li; coarse-silk remainder 7 chi 8 cun 3 fen 9 li; land property value 10 guan 530 wen; combined fields and land property value 126 guan 930 wen. Harmonized purchase: converted silk 2 bolts 3 zhang 1 chi 6 fen 3 li 7 hao 5 si; in-kind 1 bolt remainder 7 chi 5 cun 9 fen 8 li 7 hao 5 si.

Household B now has 782 mu of fields: seedling rice 27 shi 3 dou 7 sheng; summer tax converted silk 6 bolts 2 zhang 9 chi 7 cun 9 fen; in-kind 15 bolts remainder 2 zhang 9 chi 5 cun 1 fen; property value 938 guan 400 wen. Harmonized purchase: converted silk 20 bolts 2 zhang 1 chi 1 cun; in-kind 8 bolts remainder 3 zhang 1 chi 9 cun.

Household C now has 979 mu of fields: seedling rice 34 shi 2 dou 6 sheng 5 ge; summer tax converted silk 8 bolts 1 zhang 7 chi 7 cun 5 fen 5 li; in-kind 19 bolts remainder 2 zhang 8 chi 9 fen 5 li; property value 1,174 guan 800 wen. Harmonized purchase: converted silk 25 bolts 2 zhang 7 chi 9 cun 5 fen; in-kind 11 bolts remainder 5 cun 5 fen.

術曰：以粟米及衰分求之。置甲原納米為實，以每畝苗為法除之，得甲原有田。以乘每畝稅，得田絹。次以甲紬絹本色折帛併之，內減田絹，餘為地紬。以每紬約之，得甲原有地。次以甲出田併乙丙原有田，各得三戶今有田地。各以等則乘之，各得物力苗稅。次以每畝物力率約各戶共物力，得和買。副之，三分因之退位為和本，七分因之退位為和折；夏稅反其分而因之退之；紬以半之，併入稅，各得。

Method: Solve by the methods of grain conversion (su mi) and graded distribution (cui fen). Take household A's original rice payment as dividend and divide by the per-mu seedling rate to obtain A's original fields. Multiply by the per-mu tax to obtain the field silk. Then combine A's coarse silk and silk in-kind and converted, subtract the field silk, and the remainder is the land coarse silk. Divide by the per-unit coarse silk to obtain A's original land. Next, combine the transferred fields with B and C's original fields to obtain each household's current fields. Multiply each by the grade rates to obtain property values and seedling taxes. Then divide each household's total property value by the per-mu property rate to obtain the harmonized purchase. Split it: 3/10 as in-kind harmonized purchase, 7/10 as converted; for summer tax reverse the ratios; for coarse silk halve it and combine with silk tax to obtain the results.

草曰：置甲原納苗三十五石七斗為實，以每畝苗三升五合為法除之，得甲田一千二十畝。以每畝稅一尺一寸五分乘之，得一百一十七丈三尺為田絹。以甲原紬絹本色二十匹二丈八尺九寸三分九釐、折帛八匹三丈九尺七寸三分九釐併之，共二十八匹三丈七尺六寸七分八釐。內減田絹一百一十七丈三尺，餘為地紬。以每地紬一尺三寸四分約之，得甲地一十一畝二角四十八步。以甲出田四百七畝併乙田三百七十五畝，得七百八十二畝為乙戶今有田。以甲出田五百十六畝併丙田四百六十三畝，得九百七十九畝為丙戶今有田。甲今有田一千二十畝減出田九百二十三畝，餘九十七畝為甲今有田。各以每畝物力率乘之：乙田七百八十二畝乘一貫二百，得九百三十八貫四百文；丙田九百七十九畝乘一貫二百，得一千一百七十四貫八百文；甲田九十七畝乘一貫二百，得一百一十六貫四百文。地一十一畝二角四十八步乘九百，得十貫五百三十文。共甲物力一百二十六貫九百三十文。各以物力三十二貫約之：乙得二十四匹餘十六貫四百文，三分本色得七匹餘八分，七分折帛得一十六匹二丈一尺一寸；丙得三十六匹餘二十六貫八百文，三分本色得一十一匹餘八分，七分折帛二十五匹二丈七尺九寸五分；甲得三匹餘三十貫九百三十文，三分本色得一匹餘尺，七分折帛二匹三丈一尺六分三

釐。夏稅以物力七三因之：乙七分本色一十五匹餘二丈九尺五寸一分，三分折帛六匹二丈九尺七寸九分；丙七分本色一十九匹餘二丈八尺九分五釐，三分折帛八匹一丈七尺七寸五分五釐；甲七分本色一匹餘三丈八尺八分五釐，三分折帛三丈三尺四寸六分五釐。

Working: Take household A's original seedling rice of 35 shi 7 dou as dividend; divide by the per-mu seedling rate of 3 sheng 5 ge to obtain A's fields of 1,020 mu. Multiply by per-mu tax of 1 chi 1 cun 5 fen to obtain 117 zhang 3 chi of field silk. Combine A's original coarse silk and silk: in-kind 20 bolts 2 zhang 8 chi 9 cun 3 fen 9 li and converted 8 bolts 3 zhang 9 chi 7 cun 3 fen 9 li, totaling 28 bolts 3 zhang 7 chi 6 cun 7 fen 8 li. Subtract field silk 117 zhang 3 chi; the remainder is land coarse silk. Divide by per-unit coarse silk of 1 chi 3 cun 4 fen to obtain A's land of 11 mu 2 jiao 48 bu. Combine A's transferred fields 407 mu with B's 375 mu to obtain 782 mu as B's current fields. Combine A's transferred 516 mu with C's 463 mu to obtain 979 mu as C's current fields. A's original 1,020 mu minus transferred 923 mu leaves 97 mu as A's current fields. Multiply each by the per-mu property rate: B's 782 mu at 1 guan 200 = 938 guan 400 wen; C's 979 mu at 1 guan 200 = 1,174 guan 800 wen; A's 97 mu at 1 guan 200 = 116 guan 400 wen. Land 11 mu 2 jiao 48 bu at 900 = 10 guan 530 wen. Total A property value = 126 guan 930 wen. Divide each property value by 32 guan: B obtains 24 bolts remainder 16 guan 400 wen, 3/10 in-kind = 7 bolts remainder 8 fen, 7/10 converted = 16 bolts 2 zhang 1 chi 1 cun. C obtains 36 bolts remainder 26 guan 800 wen, 3/10 in-kind = 11 bolts remainder 8 fen, 7/10 converted = 25 bolts 2 zhang 7 chi 9 cun 5 fen. A obtains 3 bolts remainder 30 guan 930 wen, 3/10 in-kind = 1 bolt remainder chi, 7/10 converted = 2 bolts 3 zhang 1 chi 6 fen 3 li. Summer tax uses 7:3 ratio on property value: B 7/10 in-kind = 15 bolts remainder 2 zhang 9 chi 5 cun 1 fen, 3/10 converted = 6 bolts 2 zhang 9 chi 7 cun 9 fen. C 7/10 in-kind = 19 bolts remainder 2 zhang 8 chi 9 fen 5 li, 3/10 converted = 8 bolts 1 zhang 7 chi 7 cun 5 fen 5 li. A 7/10 in-kind = 1 bolt remainder 3 zhang 8 chi 8 fen 5 li, 3/10 converted = 3 zhang 3 chi 4 cun 6 fen 5 li.

均科綿稅

Equalizing Silk Tax Assessment

問：縣科綿有五等戶，共一萬一千三十三戶，共科綿八萬八千三百三十七兩六錢。上等一十二戶，副等八十七戶，中等四百六十四戶，次等二千三十五戶，下等八千四百三十五戶。欲令上三等折半差，下二等此中等六四折差。科率求之，問各戶納及各等幾何？

Question: A county assesses silk floss across five grades of households, totaling 11,033 households, with total silk assessment of 88,337 liang 6 qian. Upper grade: 12 households; second grade: 87 households; middle grade: 464 households; fourth grade: 2,035 households; lower grade: 8,435 households. It is desired that the upper three grades differ by halving (upper:second:middle = 4:2:1), and the lower two grades differ from the middle grade by a 6:4 ratio (fourth:lower = 6:4 of middle). Using the assessment rate, find the amount per household and the total for each grade.

[按] 題言下二等比中等六四折差，是次等為中等六分之四，下等又為次等六分之四也。乃術草中皆以十分之四收之，是四分折差而非四六折差矣。集中題與術不相應者多類此，豈成書之後未能重加校正歟？

[Ed.] The problem states that the lower two grades differ from the middle grade by a 6:4 ratio, meaning the fourth grade is 4/6 of the middle grade, and the lower grade is 4/6 of the fourth grade. However, in the method and working, both are reduced by 4/10, which is a 4-tenths reduction rather than a 6:4 ratio. The collection contains many such cases where the problem and method do not correspond—could it be that the author was unable to thoroughly revise the work after its completion?

答曰：上等一户一百二十四兩，一十二户計一千四百八十八兩；副等一户六十二兩，八十七户計五千三百九十四兩；中等一户三十一兩，四百六十四户計一萬四千三百八十四兩；次等一户一十二兩四錢，二千 035 户計二萬五千二百三十四兩；下等一户四兩九錢六分，八千四百三十五户計四萬一千八百三十七兩六錢。

Answer: Upper grade: 124 liang per household, 12 households totaling 1,488 liang. Second grade: 62 liang per household, 87 households totaling 5,394 liang. Middle grade: 31 liang per household, 464 households totaling 14,384 liang. Fourth grade: 12 liang 4 qian per household, 2,035 households totaling 25,234 liang. Lower grade: 4 liang 9 fen 6 li per household, 8,435 households totaling 41,837 liang 6 qian.

術曰：列五等户數，先以四折下等數，加次等户，又以四折之，加中等户數，却以半折之，加二等户，又以半折之，加上等户數，不折便為法。除科綿，得上等一户之綿。復半之為副等，又半之為中等，又四折之為次等，又四折之為下等。各一户綿，却各以户數乘之，各得五等共出綿。

Method: List the five grades of households. First apply a 4/10 reduction to the lower-grade count, add to the fourth-grade households, apply a 4/10 reduction again, add to the middle-grade count, then apply a 1/2 reduction, add to the second-grade count, apply a 1/2 reduction again, and add to the upper-grade count. Do not reduce this final sum; it becomes the divisor. Divide the total silk assessment by this to obtain the silk per upper-grade household. Halve this for the second grade, halve again for the middle grade, apply a 4/10 reduction for the fourth grade, and another 4/10 reduction for the lower grade. Multiply each per-household amount by the respective household count to obtain the total silk for each grade.

草曰：先置下等户八千四百三十五，以四折之，得三千三百七十四。乃以得數折入次户二千三十五内，得五千四百九户訖。又以四折次數五千四百九户，得二千一百六十三户六分。乃以得次數併入中户四百六十四内，共得二千六百二十七户六分。次以五折中數二千六百二十七户六分，得數。次以得數一千三百一十三户八分併副户八十七，得一千四百户八分。又以五折副數一千四百户八分，得七百户四分，既得數。乃以副數七百户四分併上户一十二户，得七百一十二户四分，不折便為法。

累折至等，共得七百一十二户四分以為總法。乃置科綿八萬八千三百三十七兩六錢為實，法七百一十二户四分而一，得一百二十四兩為上一户所出綿。以五折之，得六十二兩為副等一户所出綿。又五折之，得三十一兩為中等一户所出綿。乃以四折之，得一十二兩四錢為次等一户所出綿。又以四折之，得四兩九錢六分為下一户所出綿。乃以各等户數各乘一户所出，即各得每等共數之綿。上一十二户乘一百二十四兩，得一千四百八十八兩；副等八十七户乘六十二兩，得五千三百九十四兩；中等四百六十四户乘三十一兩，得一萬四千三百八十四兩；次等二千三十五户乘一十二兩四錢，得二萬五千二百三十四兩；下等八千四百三十五户乘四兩九錢六分，得四萬一千八百三十七兩六錢。併五等之綿，得八萬八千三百三十七兩六錢，與原科之數相合。

Working: Set the lower-grade households 8,435, apply 4/10 reduction to obtain 3,374. Add this to the fourth-grade households 2,035 to obtain 5,409. Apply 4/10 reduction to 5,409 to obtain 2,163.6. Add this to the middle-grade households 464 to obtain 2,627.6. Apply 1/2 reduction to 2,627.6 to obtain 1,313.8. Add to the second-grade households 87 to obtain 1,400.8. Apply 1/2 reduction to 1,400.8 to obtain 700.4. Add to the upper-grade households 12 to obtain 712.4, which becomes the divisor without further reduction.

The accumulated reduction yields 712.4 as the total divisor. Take the silk assessment 88,337 liang 6 qian as dividend. Divide by divisor 712.4 to obtain 124 liang as the silk per upper-grade household. Apply 1/2 reduction to obtain 62 liang for the second grade. Apply 1/2 reduction again to obtain 31 liang for the middle grade. Apply 4/10 reduction to obtain 12 liang 4 qian for the fourth grade. Apply 4/10 reduction again to obtain 4 liang 9 fen 6 li for the lower grade.

Multiply each per-household amount by the respective household count: upper grade $12 \times 124 = 1,488$ liang; second grade $87 \times 62 = 5,394$ liang; middle grade $464 \times 31 = 14,384$ liang; fourth grade $2,035 \times 12.4 = 25,234$ liang; lower grade $8,435 \times 4.96 = 41,837.6$ liang. The sum of all five grades equals 88,337.6 liang, matching the original assessment.

均定勸分

Equalizing Voluntary Contribution Levies

問：欲勸糶賑濟，據甲民戶物力畝步排定，共計一百六十二戶，作九等：上等三戶，第二等五戶，第三等七戶，第四等八戶，第五等十三戶，第六等二十一戶，第七等二十六戶，第八等三十四戶，第九等四十五戶。今先觀諭，第一等上戶願糶五千石，第九等戶願糶二百石。欲知各等拋差石數併數認米數各幾何？

Question: It is desired to encourage households to sell grain for famine relief. Based on the property-value and field-size ranking of township A households, there are 162 households in total, divided into nine grades: upper grade 3 households, second grade 5 households, third grade 7 households, fourth grade 8 households, fifth grade 13 households, sixth grade 21 households, seventh grade 26 households, eighth grade 34 households, ninth grade 45 households. After consultation, the first-grade upper households are willing to contribute 5,000 shi, and the ninth-grade households are willing to contribute 200 shi. Find the incremental difference (pao cha) in shi between grades and the total committed rice for each grade.

答曰：認該米二十三萬七千六百石。上等一戶米五千石，三戶計一萬五千石；二等一戶米四千四百石，五戶計二萬二千石；三等一戶米三千八百石，七戶計二萬六千六百石；四等一戶米三千二百石，八戶計二萬五千六百石；五等一戶米二千六百石，一十三戶計三萬三千八百石；六等一戶米二千石，二十一戶計四萬二千石；七等一戶米一千四百石，二十六戶計三萬六千四百石；八等一戶米八百石，三十四戶計二萬七千二百石；九等一戶米二百石，四十五戶計九千石。

Answer: Total committed rice: 237,600 shi. Upper grade: 5,000 shi per household, 3 households = 15,000 shi. Second grade: 4,400 shi per household, 5 households = 22,000 shi. Third grade: 3,800 shi per household, 7 households = 26,600 shi. Fourth grade: 3,200 shi per household, 8 households = 25,600 shi. Fifth grade: 2,600 shi per household, 13 households = 33,800 shi. Sixth grade: 2,000 shi per household, 21 households = 42,000 shi. Seventh grade: 1,400 shi per household, 26 households = 36,400 shi. Eighth grade: 800 shi per household, 34 households = 27,200 shi. Ninth grade: 200 shi per household, 45 households = 9,000 shi.

術曰：以衰分求之。置上下戶米，減餘為實。列等數減一，餘為法。除之，得拋差石數。以差累減上等米，各得諸等米。以各等戶乘之，併之為總數。

Method: Solve by the method of graded distribution. Set the rice amounts of the upper and lower households, subtract to obtain the dividend. List the number of grades, subtract one, and use the remainder as divisor. Divide to obtain the incremental difference in shi. Successively subtract this difference from the upper-grade amount to obtain each grade's amount. Multiply each by the respective household count and add together for the total.

草曰：置上等戶米五千石，減下等戶二百石，餘四千八百石為實。以九等減一，餘八為法。除實，得六百石為每等拋差。用減上等，餘四千四百石為二等米。又減六百，得三千八百石為三等米。又減六百，得三千二百石為四等米。又減六百，得二千六百石為五等米。又減六百，得二千石為六等米。又減六百，得一千四百石為七等米。又減六百，得八百石為八等米。又減六百，得二百石為九等米。又各乘戶數，併之，得總認米石數二十三萬七千六百石。

Working: Set upper-grade household rice at 5,000 shi, subtract lower-grade 200 shi, remainder 4,800 shi as dividend. Subtract 1 from 9 grades, remainder 8 as divisor. Divide to obtain 600 shi as the incremental difference per grade. Subtract from the upper grade to obtain 4,400 shi as the second-grade amount. Subtract 600 again to obtain 3,800 shi as third grade. Continue subtracting 600: 3,200 shi (fourth), 2,600 shi (fifth), 2,000 shi (sixth), 1,400 shi (seventh), 800 shi (eighth), and 200 shi (ninth). Multiply each by the respective household count and add to obtain the total committed rice of 237,600 shi.

均定合解

Equalizing Combined Remittances

問: 州郡合解諸司窠名錢：戶部九十六萬五千四百二十一貫文，總所六十四萬三千六百一十四貫文，運司一萬六千九十貫三百五十文。今諸窠名先催到九千二百五十三貫六百二十文，欲照原額分數均定椿收解，合各幾何？

Question: A prefecture must remit named funds (keming qian) to various offices: Ministry of Revenue 965,421 guan, General Office 643,614 guan, Transport Office 16,090 guan 350 wen. Now, of all the named funds, 9,253 guan 620 wen has been collected in advance. It is desired to distribute this proportionally according to the original quotas for deposit and remittance. How much should each office receive?

答曰: 戶部五千四百九十七貫二百文；總所三千六百六十四貫八百文；運司九十一貫六百二十文。

Answer: Ministry of Revenue: 5,497 guan 200 wen; General Office: 3,664 guan 800 wen; Transport Office: 91 guan 620 wen.

術曰: 以衰分求之。置諸原率，可約約之。副併為法。以催到錢乘未併者，各為實。實如法而一。

Method: Solve by the method of graded distribution. Set up the original rates, reducing by common factors where possible. Add together to form the divisor. Multiply the collected amount by each uncombined rate to form the dividends. Divide each dividend by the divisor.

草曰: 列戶部九十六萬五千四百二十一貫，總所六十四萬三千六百一十四貫，運司一萬六千九十貫三百五十文，各為原率。今原率可約，求等得一萬六千九十貫三百五十為等數，俱約之：戶部得六十，總所得四十，運司得一，各為率。副併得一百一為法。以催到錢九千二百五十三貫六百二十文乘未併數：六十、四十及一，畢得五十五萬五千二百一十七貫二百文為戶部之實，三十七萬一百四十四貫八百文為總所之實，九千二百五十三貫六百二十文為運司之實。三實皆如法一百一而一：其戶部得五千四百九十七貫二百文，總所得三千六百六十四貫八百文，運司得九十一貫六百二十，各為候解錢。

Working: List Ministry of Revenue 965,421 guan, General Office 643,614 guan, Transport Office 16,090 guan 350 wen as the original rates. These can be reduced; finding the common measure of 16,090 guan 350 wen, we reduce all: Ministry obtains 60, General Office 40, Transport Office 1 as the rates. Add together to obtain 101 as the divisor.

Multiply the collected amount 9,253 guan 620 wen by each uncombined rate (60, 40, and 1): Ministry dividend = 555,217 guan 200 wen; General Office dividend = 370,144 guan 800 wen; Transport Office dividend = 9,253 guan 620 wen. Divide each by the divisor 101: Ministry obtains 5,497 guan 200 wen; General Office obtains 3,664 guan 800 wen; Transport Office obtains 91 guan 620 wen, as the amounts awaiting remittance.

寬減屯租

Reducing Garrison Field Rent

問：屯租欲議寬減，仍聽以夏麥折納分數。官牛種者與減二分，私牛種者與減四分。每歲租穀以三分之一許夏折二麥，內四分大六分小折色。每大麥三石折小麥二石，小麥二石折穀三石五斗。屯租舊額官種一石納租五石，私種一石納租三石。今某州屯田去年計官私種共九千七百八十二石，共合收租穀三萬九千五百八十六石。欲知官私種各數、原額、今減、合催、成年夏麥秋穀幾何？

Question: Garrison field rents are to be reduced, with the option to pay in summer wheat according to specified conversion fractions. For fields cultivated with official oxen, a reduction of $2/10$ is granted; for private oxen, a reduction of $4/10$. Each year's grain rent allows $1/3$ to be paid in summer wheat (two types): $4/10$ great barley and $6/10$ small wheat as converted payment. The conversion rates are: 3 shi great barley = 2 shi small wheat; 2 shi small wheat = 3.5 shi grain. The old garrison rent quotas are: official cultivation 1 shi seed pays 5 shi rent; private cultivation 1 shi seed pays 3 shi rent. Now a certain prefecture's garrison fields last year had combined official and private seed of 9,782 shi, with total rent grain due of 39,586 shi. Find: the official and private seed amounts, original quotas, current reductions, amounts collectable, and the mature-year summer wheat and autumn grain totals.

[按] 此題有官私共種數、租數，有種一石租數，求各種數租數，古謂之差分，今謂之和較比例。折色亦差分也，今謂之和數比例。大小麥與穀相易，古謂之異乘同乘，今謂之和率比列。

[Ed.] This problem gives the combined official and private seed amounts and rent amounts, with the rent per shi of seed, to find each seed and rent amount. Anciently called “difference distribution” (*cha fen*), now called “sum-difference proportion.” The converted payment is also difference distribution, now called “sum-ratio proportion.” The exchange of great and small wheat for grain, anciently called “different multiplication, same division,” now called “combined-rate proportion.”

答曰：官種五千一百二十石，私出種四千六百六十二石。原租額共三萬九千五百八十六石：官種二萬五千六百石，私種一萬三千九百八十六石。今減一萬七百一十四石四斗：官種五千一百二十石，私種五千五百九十四石四斗。合催二萬八千八百七十一石六斗。官種租二萬四千石：一分折麥計穀八千石；四折大麥三千二百石（係折小麥二千一百三十三石三斗三升三分升之一，復折穀三千七百三十三石三斗三升三分升之二）；六分折小麥四千八百石（係折穀八千四百石）；正色穀一萬六千石。私種租一萬五千八百七十一石六斗：一分折麥計穀五千二百九十石五斗三分升之二；四折大麥二千一百四十六石八斗八升（係折小麥一千四百三十一石二升三分升之二，復折穀二千五百四石三升三分升之一）；六分折小麥三千二百二十石二斗八升（係折穀五千六百三十五石四斗九升）；二分正色穀一萬七百一十四石四斗。成年共收：夏折大小麥一萬二百六十六石六斗六升三分升之二，內大麥五千三百四十六石八斗八升，小麥五千九百一十九石七斗八升三分升之二；秋正穀二萬二千六百六十七石三斗三升三分升之一。

Answer: Official seed: 5,120 shi; private seed: 4,662 shi. Original rent quota: 39,586 shi total—official cultivation 25,600 shi, private cultivation 13,986 shi. Current reduction: 10,714 shi 4 dou—official 5,120 shi, private 5,594 shi 4 dou. Amount collectable: 28,871 shi 6 dou. Official cultivation rent 24,000 shi: $1/3$ converted to wheat = 8,000 shi grain equivalent; $4/10$ great barley = 3,200 shi (equals 2,133 shi 3 dou 3 sheng + $1/3$ small wheat, which equals 3,733 shi 3 dou 3 sheng + $2/3$ grain); $6/10$ small wheat = 4,800 shi (equals 8,400 shi grain); in-kind grain 16,000 shi. Private cultivation rent 15,871 shi 6 dou: $1/3$ converted to wheat = 5,290 shi 5 dou + $2/3$ grain equivalent; $4/10$ great barley = 2,146 shi 8 dou 8 sheng (equals 1,431 shi 2 sheng + $2/3$ small wheat, which equals 2,504 shi 3 sheng + $1/3$ grain); $6/10$ small wheat = 3,220 shi 2 dou 8 sheng (equals 5,635 shi 4 dou 9 sheng grain); $2/10$ in-kind grain 10,714 shi 4 dou. Total mature-year receipts: Summer wheat 10,266 shi 6 dou 6 sheng + $2/3$, comprising great barley 5,346 shi 8 dou 8 sheng and small wheat 5,919 shi 7 dou 8 sheng + $2/3$; autumn in-kind grain 22,667 shi 3 dou 3 sheng + $1/3$.

術曰：以差分求之。置官租五石、私租三石，以所減二分、四分各減之，官存三分八釐，私存一分八釐。以各對乘官私種一石，官得三石八斗為官率，私得一石八斗為私率。副併為法。以共租為實。實如法而一，得一分之數。以官私各率乘之，得各合催租。又以種一石各對除官私率，得各合種。其折色以三分之一為總率，四分大、六分小各為子率。以各租乘總率，得折麥計穀之數。以大麥率三、小麥率二通轉相易，各得大小麥。又各以小麥二石折穀三石五斗約之，得大小麥各折穀之數。以減各合催租，餘為正色秋穀。

Method: Solve by the method of difference distribution. Set official rent at 5 shi and private rent at 3 shi per shi of seed. Apply the reductions of $2/10$ and $4/10$: official retains 3.8 (3 shi 8 dou), private retains 1.8 (1 shi 8 dou). Multiply each by 1 shi of seed: official obtains 3.8 shi as the official rate, private obtains 1.8 shi as the private rate. Add together to form the divisor. Take the total rent as dividend. Divide to obtain the amount for one portion. Multiply by each rate to obtain the collectable rent for each category. Divide each rate by the respective rent per shi of seed to obtain the seed amounts. For conversions: $1/3$ is the total conversion rate, $4/10$ great barley and $6/10$ small wheat are the component rates. Multiply each rent by the total rate to obtain the grain-equivalent of converted wheat. Using great barley rate 3 and small wheat rate 2, convert between them to obtain the wheat amounts. Then convert using 2 shi small wheat = 3.5 shi grain to obtain the grain equivalents. Subtract from the collectable rent; the remainder is in-kind autumn grain.

戶稅均寬

Equalizing Household Tax Relief

問：州郡寬恤，近將某縣下三等稅戶秋科餘欠錢米，已與蠲放：共錢一千三百五十五貫七百六文，米五千二百七十二石一斗九升。某本縣下三等物力計三萬七千六百五十八貫五百文。今來官員陳述：本縣多有樂輸無欠之戶，今蒙蠲放稅尾，似反寬潤頑輸之戶，於理未均。遂議將樂輸三等戶於明年兩稅與照昨來體例減免。契勘得三等無欠戶物力二十二萬八千一十五貫三百二十一文。欲知每百文合減免錢米及共減各幾何？

Question: A prefecture has granted relief by remitting the autumn tax arrears of the lower three grades of taxable households in a certain county: total remitted 1,355 guan 706 wen in cash and 5,272 shi 1 dou 9 sheng in rice. The property value of the lower three grades in this county is 37,658 guan 500 wen. Now officials have pointed out that many households in this county paid their taxes willingly without arrears, and by remitting only the tax arrears, it seems to favor those who defaulted, which is unfair. It has been proposed to grant the willing-payer households of the three lower grades a reduction in next year's two taxes according to the same pattern. The property value of the three lower grades of non-arrears households is found to be 228,015 guan 321 wen. Find the cash and rice remission per 100 wen of property value, and the total reduction for each.

答曰：每物力一百文，放錢三文六分，放米一升四合。明年兩稅放免：錢七千九百四十九貫三百五十一文五分五釐六毫；米三萬九百一十四石一斗四升四合九勺四抄。

Answer: Per 100 wen of property value: remit 3.6 wen cash, remit 1.4 ge rice. Next year's two-tax remission: cash 7,949 guan 351 wen 5 fen 5 li 6 hao; rice 30,914 shi 1 dou 4 sheng 4 ge 9 shao 4 chao.

術曰：以粟米衰分求之。置原物力為法，除原放錢米，得每百文物力所放錢米率。以各率乘今來物力，各得錢米，為明年兩稅合放數。

Method: Solve by the methods of grain conversion and graded distribution. Take the original property value as divisor. Divide the original remitted cash and rice by this to obtain the cash and rice remission rates per 100 wen of property value. Multiply each rate by the current property value to obtain the cash and rice amounts for next year's two-tax remission.

草曰：以原物力三萬七千六百五十八貫五百文為法。先除原放錢一千三百五十五貫七百六文，定法百文上得一文為商，除得三文六分為物力百文所放錢率。次除原放米五千二百七十二石一斗九升，得一升四合為物力百文所放米率。以今三等戶物力二十二萬八千一十五貫三百二十一文徧乘所放錢米二率，得錢七千九百四十九貫三百五十一文五分五釐六毫，得米三萬九百一十四石一斗四升四合九勺四抄，為明年兩稅合放數。

Working: Take the original property value 37,658 guan 500 wen as divisor. First divide the original remitted cash 1,355 guan 706 wen by this: at the 100-wen position, 1 wen as quotient, obtaining 3.6 wen as the cash remission rate per 100 wen of property value. Next divide the original remitted rice 5,272 shi 1 dou 9 sheng, obtaining 1.4 ge as the rice remission rate per 100 wen of property value. Multiply both rates by the current three-grades property value 228,015 guan 321 wen: cash = 7,949 guan 351 wen 5 fen 5 li 6 hao; rice = 30,914 shi 1 dou 4 sheng 4 ge 9 shao 4 chao, as the two-tax remission for next year.

[按] 此題之法至為簡明，以物力為法，以所放錢米為實，除之得每百文物力合放之率，又以今來物力乘之，得合放之數。蓋以物力之多寡為差，物力多者放多，物力少者放少，斯為均矣。

[Ed.] The method of this problem is extremely clear: taking property value as divisor and remitted cash and rice as dividend, dividing obtains the remission rate per 100 wen of property value; then multiplying by the current property value obtains the remission amount. Since property value serves as the measure of difference, those with more property value receive more remission, and those with less receive less—this is equitable.

米穀粒分

Rice and Grain Granule Count

問：開倉受約，有甲戶米一千五百三十四石。到廊驗得米內夾穀，乃於樣內取米一捻，數計二百五十四粒，內有穀二十八顆。凡粒米率每勺三百。今欲知米內雜穀多少，以折米數科貴及粒各幾何？

Question: When opening the granary to accept delivery, household A delivered 1,534 shi of rice. Upon inspection at the corridor, the rice was found to be mixed with unhusked grain. A pinch of the sample was taken, containing 254 granules, of which 28 were unhusked grain. The standard rate is 300 granules per shao. Now it is desired to know how much unhusked grain is mixed in the rice, the amount of rice to be deducted, and the total granule count.

答曰：米一千三百六十四石八斗九升七合六勺一百二十七分勺之四十八；穀一百六十九石一斗二合三勺一百二十七分勺之七十九。合折米八十四石五斗五升一合一勺一百二十七分勺之一百三。原米折米共計四十三億四千八百三十四萬六千四百五十六粒。

Answer: Rice: 1,364 shi 8 dou 9 sheng 7 ge 6 shao + 48/127 shao; Grain: 169 shi 1 dou 2 ge 3 shao + 79/127 shao. Converted rice deduction: 84 shi 5 dou 5 sheng 1 ge 1 shao + 103/127 shao. Total granules of original and converted rice: 4,383,464,456.

術曰：以粟米求之，衰分入之。置樣米粒數為法，以常穀顆數減之，餘與穀為列衰。可約約之。以共米乘列衰為各實，實如法而一，各得米數穀數。置穀數以粟率折之，為穀所折米。次以勺率遍乘米數折米，得粒數。

Method: Solve by the grain conversion method, with graded distribution applied. Set the sample granule count as divisor. Subtract the unhusked grain count; the remainder and the grain count form the graded rates. Reduce by common factors if possible. Multiply the total rice by each graded rate to form the dividends. Divide each by the divisor to obtain the rice and grain amounts. Convert the grain amount using the grain conversion rate to obtain the rice equivalent of the grain. Then multiply both rice amounts by the shao rate to obtain the granule count.

草曰：置一捻樣粒數二百五十四為法，以帶穀二十八顆為穀衰，以減法，餘二百二十六為米衰。此二衰與法皆可約，求等得二，俱以約之：法得一百二十七，米衰得一百一十三，穀衰得一十四。以米一千五百三十四石遍乘二衰，得一十七萬三千三百四十二石為米實，得二萬一千四百七十六石為穀實。皆如法一百二十七而一：米得一千三百六十四石八斗九升七合六勺一百二十七分勺之四十八，穀得一百六十九石一斗二合三勺一百二十七分勺之七十九。以粟率五十折之，得八十四石五斗五升一合一勺一百二十七分勺之一百三為穀折納米數。併二米，得一千四百四十九石四斗四升八合八勺一百二十七分勺之二十四。先通分納子一十八萬四千八十石，以勺率三百粒乘之，得五千五百二十二億四千萬粒為實，以母一百二十七除之，得四十三億四千八百三十四萬六千四百五十六粒，不盡八十八棄之。

Working: Set the sample pinch count 254 as divisor; unhusked grain 28 as the grain rate; subtract from divisor, remainder 226 as the rice rate. These two rates and the divisor can all be reduced; finding common factor 2: divisor = 127, rice rate = 113, grain rate = 14. Multiply total rice 1,534 shi by each rate: rice dividend = 173,342 shi; grain dividend = 21,476 shi. Divide each by divisor 127: rice = 1,364 shi 8 dou 9 sheng 7 ge 6 shao + 48/127 shao; grain = 169 shi 1 dou 2 ge 3 shao + 79/127 shao. Convert grain at 50% (1 shi grain = 5 dou rice) to obtain 84 shi 5 dou 5 sheng 1 ge 1 shao + 103/127 shao as the rice equivalent of grain. Combine both rice amounts: 1,449 shi 4 dou 4 sheng 8 ge 8 shao + 24/127 shao. Convert to improper fraction: 184,080 shi numerator. Multiply by shao rate 300 granules to obtain 55,224,000,000 granules as dividend. Divide by denominator 127 to obtain 434,834,6456 granules, with remainder 88 discarded.

[按] 按此題以取樣一捻之數，驗米粒與穀粒之率，因以知全數之米穀，古所謂驗食之法也。二百五十四粒中穀二十八顆，其不純率可知矣。以粟率五十折之者，謂穀一石折米五斗也。勺率三百者，謂每勺有三百粒也。此法以衰分為主，而粟米次之，可謂巧矣。

[Ed.] This problem uses a pinch sample to determine the ratio of rice granules to unhusked grain, and thereby ascertains the rice and grain in the full amount—this is the ancient method of “inspecting provisions.” With 28 unhusked grains out of 254 granules, the impurity rate is evident. The grain conversion rate of 50% means 1 shi of grain converts to 5 dou of rice. The shao rate of 300 means 300 granules per shao. This method uses graded distribution as the primary technique and grain conversion secondarily—quite ingenious.

戶口食鹽

Household Salt Ration Assessment

問：某州管下九縣，依例歲敷食鹽：甲縣戶一十四萬三千五百，乙縣戶九萬七千四百，丙縣戶七萬六千五百，丁縣戶五萬四千三百二十，戊縣戶四萬三千六百，己縣戶三萬二千四百八十，庚縣戶二萬六千七百五十，辛縣戶一萬九千三百二十，壬縣戶一萬二千五百六十。今來州司根括到九縣共計戶四十八萬六千八百二十，每科食鹽一貫文足，係用會子納。其時舊會每貫五十四文足，新會每貫二百七十文足。欲知各縣合納鹽錢幾何？

Question: A certain prefecture administers nine counties, which according to regulations distribute salt annually: County A has 143,500 households, County B 97,400, County C 76,500, County D 54,320, County E 43,600, County F 32,480, County G 26,750, County H 19,320, County I 12,560. The prefecture has determined the nine counties' total households to be 486,820. The salt tax assessment is 1 guan wen in full-value coin, to be paid in huizi notes. At that time, old huizi traded at 54 wen full-value per guan, and new huizi at 270 wen full-value per guan. Find the salt tax each county should pay.

答曰：甲縣三千一百八十九貫三百文；乙縣二千一百六十五貫二百文；丙縣一千七百文；丁縣一千二百六貫六百四十文；戊縣九百六十八貫文；己縣七百二十一貫六百文；庚縣五百九十四貫四百文；辛縣四百二十八貫九百六十文；壬縣二百七十九貫一百二十文。

Answer: County A: 3,189 guan 300 wen; County B: 2,165 guan 200 wen; County C: 1,700 guan wen; County D: 1,206 guan 640 wen; County E: 968 guan wen; County F: 721 guan 600 wen; County G: 594 guan 400 wen; County H: 428 guan 960 wen; County I: 279 guan 120 wen.

術曰：以衰分求之。列各縣戶數為列衰，副併為法。以共科鹽錢乘未併者，各為實。實如法而一，各得縣合納鹽錢。

Method: Solve by the method of graded distribution. List each county's household count as graded rates. Add together to form the divisor. Multiply the total salt tax by each uncombined rate to form the dividends. Divide each dividend by the divisor to obtain each county's salt tax payment.

草曰：列九縣戶數為列衰：甲一十四萬三千五百，乙九萬七千四百，丙七萬六千五百，丁五萬四千三百二十，戊四萬三千六百，己三萬二千四百八十，庚二萬六千七百五十，辛一萬九千三百二十，壬一萬二千五百六十。副併九縣之數，得四十八萬六千八百二十為法。置共科鹽錢一萬八千五百貫文為實。以法四十八萬六千八百

二十除之，得每貫率數。以未併各縣戶數徧乘之：甲一十四萬三千五百乘率數，得三千一百八十九貫三百文；乙九萬七千四百乘率數，得二千一百六十五貫二百文；丙七萬六千五百乘率數，得一千七百貫文；丁五萬四千三百二十乘率數，得一千二百六貫六百四十文；戊四萬三千六百乘率數，得九百六十八貫文；己三萬二千四百八十乘率數，得七百二十一貫六百文；庚二萬六千七百五十乘率數，得五百九十四貫四百文；辛一萬九千三百二十乘率數，得四百二十八貫九百六十文；壬一萬二千五百六十乘率數，得二百七十九貫一百二十文。各為縣合納鹽錢之數。

Working: List the nine counties' household counts as graded rates: A 143,500, B 97,400, C 76,500, D 54,320, E 43,600, F 32,480, G 26,750, H 19,320, I 12,560. Add all nine counties to obtain 486,820 as divisor. Take the total salt tax assessment 18,500 guan wen as dividend. Divide by 486,820 to obtain the rate per unit. Multiply each uncombined county household count by this rate: A $143,500 \times \text{rate} = 3,189 \text{ guan } 300 \text{ wen}$; B $97,400 \times \text{rate} = 2,165 \text{ guan } 200 \text{ wen}$; C $76,500 \times \text{rate} = 1,700 \text{ guan wen}$; D $54,320 \times \text{rate} = 1,206 \text{ guan } 640 \text{ wen}$; E $43,600 \times \text{rate} = 968 \text{ guan wen}$; F $32,480 \times \text{rate} = 721 \text{ guan } 600 \text{ wen}$; G $26,750 \times \text{rate} = 594 \text{ guan } 400 \text{ wen}$; H $19,320 \times \text{rate} = 428 \text{ guan } 960 \text{ wen}$; I $12,560 \times \text{rate} = 279 \text{ guan } 120 \text{ wen}$. Each is the respective county's salt tax payment.

卷六上 錢穀類

Fascicle 6A: Money and Grain (Qiangu)

按：錢穀一章，專主粟米互換、輕齋折納、本息推求之事。蓋古者九章有粟米一章，以御交質變易；又有商功一章，以御功程積實。此則合而廣之，於錢穀之出入、物價之低昂、運脚之輕重、折變之煩簡，無不備載。其為術也，以粟米為本，而以衰分、均輸、盈朒諸法參之。故其題也，或問物價之貴賤，或問運費之多寡，或問折色之正偏，或問本息之增減。凡此數端，皆國用民生之所繫，故列為專類焉。

Editorial: This fascicle on Money and Grain specializes in grain conversion, light-cargo converted payments, and principal-interest calculations. The ancient Nine Chapters had a chapter on Grain Conversion (su mi) for managing exchange and barter, and a chapter on Construction Calculation (shang gong) for managing labor and volumes. This fascicle combines and expands these topics, comprehensively covering the receipts and disbursements of money and grain, the rise and fall of commodity prices, the light and heavy burden of transport fees, and the complexity of conversion procedures. Its methods take grain conversion as fundamental, supplemented by graded distribution (cui fen), equalized transport (jun shu), and excess-deficit (ying nao) methods. Thus the problems ask about the relative costliness of commodities, the amount of transport fees, the correctness of converted payments, and the increase of principal and interest. All of these matters concern state finance and people's livelihood, and therefore are classified as a separate category.

[按] 此卷凡九問：一曰課糴，二曰折解輕齋，三曰儻直推原，四曰推求本息，五曰易牒知原，六曰粟米交易，七曰計米易麴，八曰回運費，九曰三色均價。皆錢穀之屬也。其術多用粟米互換之法，蓋以物價之率為準，而以多寡之相求為用也。

[Ed.] This fascicle contains nine problems: (1) Assessing Grain Purchase, (2) Converted Remittance of Light Cargo, (3) Tracing Rent to its Origin, (4) Calculating Principal and Interest, (5) Finding the Original from Converted Certificates, (6) Grain and Rice Exchange, (7) Calculating Rice for Yeast, (8) Recovered Transport Fees, (9) Three-Color Average Price. All concern money and grain. The methods mostly use grain conversion techniques, taking commodity price rates as the standard and using the mutual determination of quantities as the application.

課糴

Assessing Grain Purchase

問：差人五路和糴。據浙西平江府石價三十五貫文，一百三十五合，至鎮江水脚錢每石九百文；安吉州石價二十九貫五百文，一百一十合，至鎮江水脚錢每石一貫二百文；江西隆興府石價二十八貫一百文，一百一十五合，至建康水脚錢每石一貫七百元；吉州石價二十五貫八百五十文，一百二十合，至建康水脚錢每石二貫九百文；湖南潭州石價二十七貫三百文，一百一十八合，至鄂州水脚錢每石一貫七百元。其錢並十七界官會，其米並用文思院斛交量紐數。欲皆以官解計石錢相比貴賤幾何？

Question: Officials are dispatched to five circuits for state grain purchases. The data are: Zhexi Pingjiang Prefecture—price 35 guan per shi, 135 ge (per dou), water transport to Zhenjiang 900 wen per shi; Anji Prefecture—price 29 guan 500 wen, 110 ge, water transport to Zhenjiang 1 guan 200 wen per shi; Jiangxi Longxing Prefecture—price 28 guan 100 wen, 115 ge, water transport to Jiankang 1 guan 700 wen per shi; Jizhou—price 25 guan 850 wen, 120 ge, water transport to Jiankang 2 guan 900 wen per shi; Hunan Tanzhou—price 27 guan 300 wen, 118 ge, water transport to Ezhou 1 guan 700 wen per shi. All payments are in 17th-series official huizi notes; all rice is measured using the Wensi Bureau hu (1 hu = 83 ge) for conversion. Find the price per official-hu shi for each circuit and compare their relative costliness.

[按] 按草中係二貫一百文，此訛。文思院解每斗八十三合。

[Ed.] Note: The working mentions 2 guan 100 wen for Tanzhou water transport, which appears to be an error. The Wensi Bureau hu equals 83 ge per dou.

答曰：文思院斛石錢：安吉州二十三貫一百六十四文一十一分文之六；平江府二十二貫七十一文二十七分文之二十三；隆興府二十一貫五百七文二十三分文之一十九；潭州二十貫六百七十九文五十九分文之四十九；吉州一十九貫八百八十五文十二分文之五。

Answer: Wensi Bureau hu price per shi: Anji Prefecture 23 guan 164 wen + 6/11 wen; Pingjiang Prefecture 22 guan 71 wen + 23/27 wen; Longxing Prefecture 21 guan 507 wen + 19/23 wen; Tanzhou 20 guan 679 wen + 49/59 wen; Jizhou 19 guan 885 wen + 5/12 wen.

[按] 按三十九訛四十九。

[Ed.] Note: 49 is a scribal error for 39.

術曰：以粟米互換求之。置石價併水脚，乘石數，又乘官斗合數為實。各如本州合數而一，各得官斛石錢。以課貴賤。

Method: Solve by the grain conversion method. Set the shi price combined with water transport fee, multiply by the shi count, then multiply by the official dou ge count to form the dividend. Divide each by the respective circuit's dou ge count to obtain the price per official-hu shi. Compare to determine relative costliness.

草曰：置安吉州石價二十九貫五百文，平江石價三十五貫文，隆興石價二十八貫一百文，吉州石價二十五貫八百五十文，潭州石價二十七貫三百文，列右行。次置水脚：安吉一貫二百文，平江九百文，隆興一貫七百元，吉州二貫九百文，潭州二貫一百文，列左行，各對本州石價。以兩行數併之，得數：安吉三十貫七百元，平江三十五貫九百，隆興二十九貫八百，潭州二十九貫四百，吉州二十八貫七百五十，仍於右行。次以文思院官斗八十三合遍乘之：安吉州得二千五百四十八貫一百文，平江府得二千九百七十九貫七百元，江西隆興得二千四百七十三貫四百文，湖南潭州得二千四百四十貫二百文，江南吉州得二千三百八十六貫二百五十文，各為實於右。得次列安吉斗一百一十合，平江斗一百三十五合，隆興斗一百一十五合，潭州斗一百一十八合，吉州斗一百二十合於左行為法。以對除右行之實：安吉得二十三貫一百六十四文一十一分之六，平江得二十三貫七十一文二十七分文之二十三，隆興得二十一貫五百七文二十三分文之一十九，潭州得二十貫六百七十

九文五十九文分之四十九，吉州得一十九貫八百八十五文一十二分文之五。相課石價，其安吉州最貴，平江次之，隆興又次之，潭州又次之，吉州最賤。

Working: Set the shi prices in the right column: Anji 29 guan 500 wen, Pingjiang 35 guan, Longxing 28 guan 100 wen, Jizhou 25 guan 850 wen, Tanzhou 27 guan 300 wen. Set the water transport fees in the left column: Anji 1 guan 200 wen, Pingjiang 900 wen, Longxing 1 guan 700 wen, Jizhou 2 guan 900 wen, Tanzhou 2 guan 100 wen, each paired with its circuit's shi price. Combine the two columns: Anji 30 guan 700 wen, Pingjiang 35 guan 900 wen, Longxing 29 guan 800 wen, Tanzhou 29 guan 400 wen, Jizhou 28 guan 750 wen, still in the right column. Next multiply each by the Wensi Bureau standard of 83 ge: Anji = 2,548 guan 100 wen, Pingjiang = 2,979 guan 700 wen, Longxing = 2,473 guan 400 wen, Tanzhou = 2,440 guan 200 wen, Jizhou = 2,386 guan 250 wen, as dividends in the right column. Place each circuit's dou measure in the left column as divisor: Anji 110 ge, Pingjiang 135 ge, Longxing 115 ge, Tanzhou 118 ge, Jizhou 120 ge. Divide the right-column dividends by the left-column divisors: Anji = 23 guan 164 wen + 6/11 wen; Pingjiang = 22 guan 71 wen + 23/27 wen; Longxing = 21 guan 507 wen + 19/23 wen; Tanzhou = 20 guan 679 wen + 49/59 wen; Jizhou = 19 guan 885 wen + 5/12 wen. Comparing the prices: Anji is most expensive, followed by Pingjiang, then Longxing, then Tanzhou, with Jizhou the cheapest.

[按] 按此題以各路米價併水脚錢，以官斛八十三合為準，約為官斛石錢，以相比較貴賤。其法以每路斗合數除之，使歸於一律，然後可以相課。此所謂異乘同除之法也。

[Ed.] This problem combines each circuit's rice price with water transport fees, using the standard official hu of 83 ge to convert to price per official-hu shi for comparison. The method divides by each circuit's dou ge measure to normalize to a common standard before comparing. This is the so-called "different multiplication, same division" method.

卷六下 錢穀類 (續)

Fascicle 6B: Money and Grain (continued)

僦直推原

Tracing Rent to its Origin

問：房廊數內，一戶日納一百五十六文八分足為準。指揮未曾減者，減三分；已曾經減三分者，減二分；已曾經減二分者，更減二分。今本戶累經減者，欲知原額房錢幾何？

Question: Among the shop-rent records, one household currently pays 156 wen 8 fen in full-value coin per day as the standard. The regulations state: those never reduced receive a $3/10$ reduction; those already reduced by $3/10$ receive a further $2/10$ reduction; those already reduced by $2/10$ receive yet another $2/10$ reduction. Now this household has been subject to cumulative reductions. Find the original rent amount.

答曰：原額三百五十文。

Answer: Original amount: 350 wen.

術曰：以衰分求之。列一十分兩行，各三位；列減分，對減右行。以餘者相乘為法。以左行原列相乘，得納錢為實。實如法而一，得原額錢。

Method: Solve by the method of graded distribution. Place two rows of 10, each with three positions. List the reduction fractions and subtract from the right row. Multiply the remainders together to form the divisor. Multiply the original entries in the left row together, then multiply by the current payment to form the dividend. Divide to obtain the original rent.

草曰：列一十三位於左行，又列一十分三位於右行。其右上減去初減三分，右中減去次減二分，右下減去更減二分。右行餘七八八，以相乘得四百四十八為法。乃以左行三位一十分相乘，得一千為乘率。以乘見日納錢一百五十六文八分，得一百五十六貫八百文為實。實如法而一，得三百五十文，為本戶原額戶錢。

Working: Place three 10s in the left row and three 10s in the right row. From the upper right subtract the first reduction of $3/10$ ($10 - 3 = 7$), from the middle right subtract the second reduction of $2/10$ ($10 - 2 = 8$), from the lower right subtract the further reduction of $2/10$ ($10 - 2 = 8$). The right row remainders 7, 8, 8 multiply together to give 448 as the

divisor. The left row three 10s multiply to give 1,000 as the multiplier. Multiply by the current daily payment 156 wen 8 fen to obtain 156 guan 800 wen as dividend. Divide by 448 to obtain 350 wen as the household's original rent.

[按] 此題以累減之法推原額，術意至明。初減三分存七分，次減二分存八分，更減二分存八分。三數相連乘得四百四十八分，以納錢一百五十六文八分為四百四十八分之實，則原額一千分之實必為三百五十文矣。

[Ed.] This problem uses cumulative reductions to trace back to the original amount; the method is entirely clear. First reduction of $3/10$ leaves $7/10$; second reduction of $2/10$ leaves $8/10$; further reduction of $2/10$ leaves $8/10$. These three numbers multiplied together give 448 parts. Taking the payment 156.8 wen as the dividend for 448 parts, the dividend for 1,000 parts (the original) must be 350 wen.

推求本息

Calculating Principal and Interest

問：三庫息例：萬貫以上一釐，千貫以上二釐五毫，百貫以上三釐。甲庫本四十九萬三千八百貫，乙庫本三十七萬三百貫，丙庫本二十四萬六千八百貫。今三庫共約到息錢二萬五千六百四十四貫二百文。其典率：甲反錐差，乙方錐差，丙蒺藜差。欲知原典三例本息各幾何？

Question: Three treasuries have the following interest rules: above 10,000 guan pays 1 li (0.001), above 1,000 guan pays 2.5 li (0.0025), above 100 guan pays 3 li (0.003). Treasury A has principal 493,800 guan; Treasury B 370,300 guan; Treasury C 246,800 guan. Now the three treasuries together have collected 25,644 guan 200 wen in interest. The assessment rates are: A uses inverted-cone difference, B uses square-cone difference, and C uses puncture-vine difference. Find the original principal and interest for each treasury under the three rates.

[按] 此即衰分題也。其差有反錐、方錐、蒺藜之名，蓋以一二三遞減如立錐為反錐，以一四九平方遞加為方錐，以一三六三數遞加為蒺藜。是必古有其名也。至以各差求各本，則因各本原依各差入之也。

[Ed.] This is a graded-distribution problem. The differences have names: inverted cone (*fan zhui*), square cone (*fang zhui*), and puncture vine (*ji li*). The inverted cone decreases by 1, 2, 3 like an upright cone; the square cone increases by squares 1, 4, 9; the puncture vine increases by 1, 3, 6. These must have been ancient terms. The use of each difference to find each principal reflects how the principals were originally assigned according to these differences.

答曰：甲庫共納息九千五十三貫文：一釐息二千四百六十九貫文，二釐半息四千一百一十五貫文，三釐息二千四百六十九貫文。

乙庫共納息一萬五十一貫文：一釐息二百六十四貫五百文，二釐半息二千六百四十五貫文，三釐息七千一百四十一貫五百文。

丙庫共納息六千五百四十貫二百文：一釐息二百四十六貫八百文，二釐半息一千八百五十一貫文，三釐息四千四百四十二貫四百文。

Answer: Treasury A total interest 9,053 guan: 1-li rate 2,469 guan; 2.5-li rate 4,115 guan; 3-li rate 2,469 guan.

Treasury B total interest 10,051 guan: 1-li rate 264 guan 500 wen; 2.5-li rate 2,645 guan; 3-li rate 7,141 guan 500 wen.

Treasury C total interest 6,540 guan 200 wen: 1-li rate 246 guan 800 wen; 2.5-li rate 1,851 guan; 3-li rate 4,442 guan 400 wen.

術曰：置諸庫諸色之差，照釐率為三行。縱併之為約率，橫命之為乘率。以約率各約自庫之本，各得。以遍乘未併乘率，然後各以釐率橫乘之，次以縱併之，為各庫共息。

Method: Set up the differences for each treasury's categories, arranged as three rows according to the li rates. Add vertically to obtain the reduction rates; assign horizontally as multiplication rates. Divide each treasury's principal by its reduction rate to obtain the quotients. Multiply these throughout by the uncombined multiplication rates, then multiply horizontally by each li rate, and finally add vertically to obtain each treasury's total interest.

草曰：置甲庫反錐差，自下置三二一於右行。次置乙庫方錐差，自上置一四九於中行。次置丙庫蒺藜差，自上置一三六於左行，各為三庫上中下三等乘率。乃縱併甲差三二一，得六為甲約率。縱併乙差一四九，得一十四為乙約率。縱併丙差一三六，得一十為丙約率。直命九位數各為上中下乘率。乃先以約率各約自庫之本：乃以甲約率六約甲本四十九萬三千八百貫，得八萬二千三百貫為甲得。次以乙約率一十四約乙本三十七萬三百貫，得二萬六千四百五十貫為乙得。次以丙約率一十約丙本二十四萬六千八百貫，得二萬四千六百八十貫為丙得。以各得乘未併乘率：其甲所得八萬二千三百貫，乘反錐乘率三二一，得二十四萬六千九百貫為上率，得一十六萬四千六百貫為中率，得八萬二千三百貫為下率。其乙所得二萬六千四百五十貫，以乘方錐差一四九，得二萬六千四百五十貫為上率，得一十萬五千八百貫為中率，得二十三萬八千五百貫為下率。其丙所得二萬四千六百八十貫，以乘蒺藜差一三六，得二萬四千六百八十貫為上率，得七萬四千四十貫為中率，得一十四萬八千八十貫為下率。然後各以息釐數乘各庫三乘：其甲以一釐乘上率二十四萬六千九百貫，得二千四百六十九貫為上息；以二釐五毫乘中率一十六萬四千六百貫，得四千一百一十五貫為中息；以三釐乘下率八萬二千三百貫，得二千四百六十九貫為下息。併上中下三息，得九千五十三貫文為甲庫共息。其乙庫以一釐乘上率二萬六千四百五十貫，得二百六十四貫五百文為上息；以二釐五毫乘中率一十萬五千八百貫，得二千六百四十五貫為中息；以三釐乘下率二十三萬八千五百貫，得七千一百四十一貫五百為下息。併上中下三息，得一萬五十一貫文為乙庫共息。其丙庫以一釐乘上率二萬四千六百八十貫，得二百四十六貫八百為上息；以二釐五毫乘中率七萬四千四十貫，得一千八百五十一貫為中息；以三釐乘下率一十四萬八千 080 貫，得四千四百四十二貫四百文為下息。併三庫共息，得二萬五千六百四十四貫二百文為總息。

Working: Set Treasury A inverted-cone difference, place 3, 2, 1 from bottom in the right row. Set Treasury B square-cone difference, place 1, 4, 9 from top in the middle row. Set Treasury C puncture-vine difference, place 1, 3, 6 from top in the left row. These are the three treasuries' upper-middle-lower multiplication rates. Add A's differences vertically: $3+2+1 = 6$ as A's reduction rate. Add B's: $1+4+9 = 14$ as B's reduction rate. Add C's: $1+3+6 = 10$ as C's reduction rate. The nine numbers are assigned as upper-middle-lower multiplication rates.

Divide each treasury's principal by its reduction rate: A principal $493,800 / 6 = 82,300$ guan as A's quotient. B principal $370,300 / 14 = 26,450$ guan as B's quotient. C principal $246,800 / 10 = 24,680$ guan as C's quotient.

Multiply each quotient by the uncombined multiplication rates: A's $82,300 \times$ inverted-cone rates 3, 2, 1 = upper 246,900 guan, middle 164,600 guan, lower 82,300 guan. B's $26,450 \times$ square-cone rates 1, 4, 9 = upper 26,450 guan, middle 105,800 guan, lower 238,050 guan. C's $24,680 \times$ puncture-vine rates 1, 3, 6 = upper 24,680 guan, middle 74,040 guan, lower 148,080 guan.

Then multiply each treasury's three rates by the interest li: A: 1 li $\times$ upper 246,900 = 2,469 guan upper interest; 2.5 li $\times$ middle 164,600 = 4,115 guan middle interest; 3 li $\times$ lower 82,300 = 2,469 guan lower interest. Total A interest = 9,053 guan.

B: 1 li $\times$ 26,450 = 264.5 guan upper; 2.5 li $\times$ 105,800 = 2,645 guan middle; 3 li $\times$ 238,050 = 7,141.5 guan lower. Total B = 10,051 guan.

C: $1 \text{ li} \times 24,680 = 246.8 \text{ guan upper}$; $2.5 \text{ li} \times 74,040 = 1,851 \text{ guan middle}$; $3 \text{ li} \times 148,080 = 4,442.4 \text{ guan lower}$.
Combined total interest = 25,644 guan 200 wen.

易牒知原

Finding the Original from Converted Certificates

[按] 按舊本此問無題，今增入。

[Ed.] Note: The original edition lacked this problem statement; it has been restored here.

問：出度牒差人營運：每三道易鹽一十三袋，鹽二袋易布八十四疋，布一十五疋易絹三疋半，絹六疋易銀七兩二錢。今趣到銀九千一百七十二兩八錢。欲知原關度牒道數幾何？

Question: Official certificates (du die) were issued for dispatching agents to conduct trade: every 3 certificates exchange for 13 bags of salt; 2 bags of salt exchange for 84 bolts of cloth; 15 bolts of cloth exchange for 3.5 bolts of silk; 6 bolts of silk exchange for 7.2 liang of silver. Now 9,172.8 liang of silver has been obtained. Find the original number of certificates issued.

答曰：度牒一百八十道。

Answer: Certificates: 180.

術曰：以粟米互乘易法求之。列各數以本色相對，如鴈翅。以多一事者相乘為實，以少一事者相乘為法，除之。

Method: Solve by the grain conversion method using mutual multiplication for exchange. List each pair of numbers with their original commodities facing each other, like goose wings in flight. Multiply the numbers with one more item in the chain to form the dividend; multiply those with one fewer item to form the divisor. Divide to obtain the answer.

草曰：先以度牒三道乘鹽二袋，得六。以乘布一十五，得九十。又乘絹六疋，得五百四十。乃乘銀九萬一千七百二十八錢，得四千九百五十三萬三千一百二十錢為實。次以鹽一十三袋乘布八十四，得一千九百九十二。以乘絹三疋五分，得六千九百七十二。乃乘銀七兩二錢，得五十萬一千九百八十四錢為法。除實，得一百八十道為原關度牒。

Working: First multiply certificates 3 by salt 2 bags = 6. Multiply by cloth 15 = 90. Multiply by silk 6 bolts = 540. Multiply by silver 91,728 qian (9,172.8 liang converted) = 49,533,120 qian as dividend. Next multiply salt 13 bags by cloth 84 = 1,992. Multiply by silk 3.5 bolts = 6,972. Multiply by silver 7.2 liang = 501,984 qian as divisor. Divide: $49,533,120 / 501,984 = 180$ certificates as the original number issued.

〔按〕按此題以度牒易鹽、鹽易布、布易絹、絹易銀，凡四變。術以多一事者相乘為實，少一事者相乘為法，此互乘之法也。鴈翅列者，以本色對列，如鴈翅飛行，左右對稱之義也。

[Ed.] This problem involves four successive exchanges: certificates to salt, salt to cloth, cloth to silk, silk to silver. The method multiplies the chain with one more item as dividend and the chain with one fewer item as divisor—this is the mutual multiplication method. The “goose wing” arrangement places original commodities facing each other, like geese flying in symmetric formation.

粟米交易

Grain and Rice Exchange

〔按〕按舊本此問無題，今增。

[Ed.] Note: The original edition lacked this problem statement; it has been restored.

問：菽三升易小麥二升，小麥一升五合易油麻八合，油麻一升二合易粳米一升八合。今將菽一十四石四斗欲易油麻，又將小麥二十一石六斗欲易粳米。問各幾何？

Question: 3 sheng of beans exchange for 2 sheng of small wheat; 1.5 sheng of small wheat exchanges for 8 ge of sesame; 1.2 sheng of sesame exchanges for 1.8 sheng of polished rice. Now 14 shi 4 dou of beans are to be exchanged for sesame, and 21 shi 6 dou of small wheat are to be exchanged for polished rice. Find each amount.

答曰：油麻五石一斗二升；粳米一十七石二斗八升。

Answer: Sesame: 5 shi 1 dou 2 sheng; Polished rice: 17 shi 2 dou 8 sheng.

術曰：以粟米換易求之。置原易率，本色對列，如鴈翅。以多一事者相乘為實，以少一事者相乘為法，除之。各得或問數，不干其率者不置。

Method: Solve by the grain conversion exchange method. Set up the original exchange rates with original commodities facing each other, like goose wings. Multiply the numbers with one more item in the chain to form the dividend; multiply those with one fewer item to form the divisor. Divide to obtain the answers. Items not in the relevant chain are not used.

草曰：置四色六數，列六位率，如鴈翅，皆化為合。先將菽一十四石四斗化作一萬四千四百合，乃對前二句率數四位，如鴈翅，至欲易油麻止，共五事為上圖。次將小麥二十一石六斗化為二萬一千六百合，乃對後兩句率四位，鴈翅，至欲粳米止，共五事為下圖。

下圖：其上圖以菽一萬四千四百合乘麥二十，得二十八萬八千，又乘油麻八合，得二百三十萬四千合為油麻實。次以菽三十合乘麥一十五合，得四百五十合為法。除之，得五千一百二十合，展為五石一斗二升為油麻。

其下圖以小麥二萬一千六百合乘油麻八合，得一十七萬二千八百合，又乘粳米一十八合，得三百一十一萬四百合為粳米實。以小麥一升五合乘油麻一十二合，得一百八十合為法。除之，得一萬七千二百八十，展作一十七石二斗八升為粳米。

Working: Set up four commodities with six numbers, arranged in six positions like goose wings, all converted to ge. First convert beans 14 shi 4 dou to 14,400 ge. Pair with the first two exchange rates (4 numbers) for the upper diagram, ending at the sesame exchange, 5 items total. Next convert small wheat 21 shi 6 dou to 21,600 ge. Pair with the last two exchange rates (4 numbers) for the lower diagram, ending at the polished rice exchange, 5 items total.

Upper diagram: Multiply beans 14,400 ge by wheat rate 20 (the cross-rate) = 288,000; then multiply by sesame 8 ge = 2,304,000 ge as sesame dividend. Multiply beans 30 ge by wheat 15 ge = 450 ge as divisor. Divide: $2,304,000 / 450 = 5,120$ ge = 5 shi 1 dou 2 sheng of sesame.

Lower diagram: Multiply small wheat 21,600 ge by sesame 8 ge = 172,800 ge; then multiply by polished rice 18 ge = 3,114,000 ge as polished rice dividend. Multiply small wheat 15 ge by sesame 12 ge = 180 ge as divisor. Divide: $3,114,000 / 180 = 17,280$ ge = 17 shi 2 dou 8 sheng of polished rice.

計米易麴

Calculating Rice for Yeast Production

[按] 按舊本此問無題，今增。

[Ed.] Note: The original edition lacked this problem statement; it has been restored.

問：庫率：粳穀七石出米三石，糯米一斗易小麥一斗七升，小麥五升踏麴二斤四兩，麴一百一十斤醞糯米一石三斗。今有糯穀一千七百五十九石三斗八升，欲出穀做米，易麥踏麴，還自醞，餘穀之米須令適足。各合幾何？

Question: Granary rates: 7 shi of glutinous grain yields 3 shi of rice; 1 dou of glutinous rice exchanges for 1 dou 7 sheng of small wheat; 5 sheng of small wheat produces 2 jin 4 liang of yeast; 110 jin of yeast ferments 1 shi 3 dou of glutinous rice. Now there are 1,759 shi 3 dou 8 sheng of glutinous grain. It is desired to hull the grain to make rice, exchange the rice for wheat, use the wheat to produce yeast, and ferment the remaining grain's rice with this yeast so that it is exactly sufficient. Find each amount.

答曰：共穀一千七百五十九石三斗八升。出穀九百二十四石，得米三百九十六石。易麥六百七十三石二斗。踏麴三萬二百九十四斤。餘穀八百三十五石三斗八升。醞米三百五十八石二升。

Answer: Total grain: 1,759 shi 3 dou 8 sheng. Grain hulled: 924 shi, yielding rice 396 shi. Wheat obtained: 673 shi 2 dou. Yeast produced: 30,294 jin. Remaining grain: 835 shi 3 dou 8 sheng. Fermented rice: 358 shi 2 sheng.

術曰：以粟米換易求之。置諸率，隨本色對列，如鴈翅。有分者通之，異類者變之，以頭位者進乘之，以下位退乘之，得合數。有對者相乘之，無對者直命之，為諸率。併上下無對者為法率。以今有物徧乘諸率（不乘法率），各為實。諸實並如法而一，各得其已變者，復互易乘除之，即得所求。

Method: Solve by the grain conversion exchange method. Set up the rates with original commodities facing each other, like goose wings. Where there are fractions, convert them; where commodities differ, transform them. Multiply from top to bottom (forward multiplication) and from bottom to top (backward multiplication) to obtain combined numbers. Where pairs face each other, multiply them; where there is no pair, assign directly—these become the rates. Add the unmatched top and bottom rates to form the divisor rate. Multiply the given amount by all the rates (excluding the divisor rate) to form dividends. Divide all dividends by the divisor to obtain the converted amounts, then perform mutual exchanges and divisions to obtain the final answers.

[按] 此題為粟米換易中之最繁者，以糯穀為本，出米易麥，麥踏麴，麴醱米，而餘穀之米又須適足。術中所謂有分者通之，謂二斤四兩通為四分斤之九也；異類者變之，謂醱米變為糯穀也；進乘退乘者，自上而下為進，自下而上為退也。其法以合圖求諸率，以今有物徧乘，如法而一，此衰分之變法也。

[Ed.] This is the most complex of all grain conversion problems. It takes glutinous grain as the base, hulls it to rice, exchanges rice for wheat, uses wheat to produce yeast, ferments rice with yeast, and the remaining grain's rice must be exactly sufficient. The "convert fractions" means converting 2 jin 4 liang to $9/4$ jin; "transform different kinds" means converting fermented rice back to glutinous grain; "forward and backward multiplication" means multiplying from top to bottom and bottom to top. The method uses the combined diagram to find rates, then multiplies by the given amount and divides—a variation of graded distribution.

回運費

Recovered Transport Fees

問：有江西水運米一十二萬三千四百石，原係鎮江交卸，計水程二千一百三十里，每石水腳錢一貫二百文。今截上件米就池州安頓，池州至鎮江八百八十里。欲收回不該水腳錢幾何？

Question: Rice transported by water from Jiangxi, originally destined for discharge at Zhenjiang, with a water route of 2,130 li and water transport fee of 1 guan 200 wen per shi. Now this rice is diverted to be stored at Chizhou instead; the distance from Chizhou to Zhenjiang is 880 li. Find the water transport fee that should be recovered (for the route not traveled).

答曰：收回錢六萬一千一百七十八貫五百九十一文。

Answer: Amount to recover: 61,178 guan 591 wen.

術曰：以粟米互易求之。置池州至鎮江里數，乘水腳錢，得數又乘運米為實。以原至鎮江水程為法，除實，得收回錢。

Method: Solve by the grain conversion method. Take the distance from Chizhou to Zhenjiang in li, multiply by the water transport fee per shi, then multiply by the rice amount to form the dividend. Take the original distance to Zhenjiang as divisor. Divide to obtain the recovered fee.

草曰：置池州至鎮江八百八十里，乘每石水脚錢一貫二百，得一千五十六貫文。又乘運米一十二萬三千四百石，得一億三千三十一萬四百貫文為實。以原至鎮江水程二千一百三十里為法，除實，得六萬一千一百七十八貫五百九十一文為收回錢。

Working: Take Chizhou to Zhenjiang 880 li, multiply by water transport fee 1 guan 200 wen per shi = 1,056 guan. Multiply by rice amount 123,400 shi = 130,310,400 guan as dividend. Divide by original route 2,130 li: $130,310,400 / 2,130 = 61,178$ guan 591 wen as the recovered transport fee.

[按] 此題以水程里數為率，里遠則水脚多，里近則水脚少。今截米就池州安頓，則少行八百八十里之水脚，故應收回。以池州里數乘水脚，為所收回之率，以原程為法，除之得收回之數。此亦異乘同除之法也。

[Ed.] This problem uses the water route distance as the rate: longer distances entail higher transport fees, shorter distances lower fees. By diverting the rice to Chizhou, 880 li of transport is saved, and the corresponding fee should be recovered. Multiplying the Chizhou distance by the transport fee gives the recovery rate; dividing by the original route gives the recovered amount. This is also the “different multiplication, same division” method.

三色均價

Three-Color Average Price

問：庫有三色金共五千兩：內八分金一千二百五十兩，兩價四百貫文；七分五釐金一千六百兩，兩價三百七十五貫文；八分五釐金二千一百五十兩，兩價四百二十五貫文。並欲煉為足色，每兩工食藥炭錢三貫文，耗金九百七十二兩五錢。欲知色分及兩價各幾何？

Question: A treasury has three grades of gold totaling 5,000 liang: 8-fen gold (80% purity) 1,250 liang at 400 guan per liang; 7.5-fen gold (75% purity) 1,600 liang at 375 guan per liang; 8.5-fen gold (85% purity) 2,150 liang at 425 guan per liang. All are to be refined to full purity (10 fen), with labor, food, medicine, and charcoal costs of 3 guan per liang, and gold loss of 972.5 liang. Find the purity and the price per liang of the refined gold.

答曰：色十分；兩價五百三貫七百二十四文，五百三十七分文之二百一十二。

Answer: Purity: 10 fen (full). Price per liang: 503 guan 724 wen + 212/537 wen.

術曰：以方田及粟米求之。置共數，以耗減之，餘為法。以三色分數各乘兩數，併之為色分實。以三色價數各乘兩數為寄，以工藥價乘共金，併寄，共為價實。二實皆如法而一，即各得。

Method: Solve by the methods of rectangular fields (fang tian) and grain conversion. Take the total amount, subtract the loss; the remainder is the divisor. Multiply each grade's purity by its liang amount and add together to form the purity dividend. Multiply each grade's price by its liang amount as accumulators; multiply the labor/medicine cost by the total gold and add to the accumulators to form the price dividend. Divide both dividends by the divisor to obtain the answers.

草曰：置共金五千兩，減耗九百七十二兩五錢，外餘四千二十七兩五錢為法。次置一千二百五十兩，乘八分，得一萬分於上。置一千六百兩，以七分五釐乘之，得一萬二千分，加上。置二千一百五十兩，乘八分五釐，得一萬八千二百七十五分，又加上，共得四萬二千七百七十五分為分實。次置一千二百五十兩乘四百貫，得五十萬貫為寄。次置一千六百兩乘價三百七十五貫，得六十萬貫，加寄。次置二千一百五十兩乘價四百二十五貫，得九十一萬三千七百五十貫，又加寄。次置共金五千兩，乘工藥錢三貫，得一萬五千貫，又加寄，共得二百二萬八千七百五十貫為貫實。二實並如法四千 027 兩五錢而一：得其色一十分；其價每兩得五百三貫七百二十四文，不盡一貫五百九十。與法求等，得七十五，俱約之，為五百三十七文之二百一十二。

Working: Take total gold 5,000 liang, subtract loss 972.5 liang, remainder 4,027.5 liang as divisor. Next: 1,250 liang x 8 fen = 10,000 fen (upper position). 1,600 liang x 7.5 fen = 12,000 fen; add to previous. 2,150 liang x 8.5 fen = 18,275 fen; add again. Total purity dividend = 40,275 fen. Next: 1,250 liang x 400 guan = 500,000 guan as accumulator. 1,600 liang x 375 guan = 600,000 guan; add. 2,150 liang x 425 guan = 913,750 guan; add again. Total gold 5,000 liang x labor/medicine 3 guan = 15,000 guan; add. Total price dividend = 2,028,750 guan. Divide both dividends by divisor 4,027.5: purity = 10 fen (full). Price per liang = 503 guan 724 wen, remainder 1 guan 590 wen. Finding common factor 75 with the divisor and reducing: remainder = 212/537 wen.

[按] 按此題以三色金煉為足色，其法以色分為實，以減耗後之數為法，除之得色分。又以各價併工藥為貫實，如法除之得兩價。此乃方田與粟米兼用之法也。耗金九百七十二兩五錢者，謂煉金之損耗也，減之則得實在之金數。

[Ed.] This problem refines three grades of gold to full purity. The method uses purity as dividend and the loss-subtracted amount as divisor to obtain the purity grade. It combines all prices with labor/medicine costs as the price dividend, and divides to obtain the price per liang. This combines methods from rectangular fields and grain conversion. The gold loss of 972.5 liang represents refining wastage; subtracting it gives the actual refined gold amount.

貴賤互易

Exchange of Valued Commodities

問：有甲乙丙三等人互易物貨：甲以金十二兩易乙銀，乙以銀一十五兩易丙絹，丙以絹二十四疋易甲金。其時金每兩價四百貫，銀每兩價三十二貫，絹每疋價二十五貫。今三人互易，各欲知所得物價幾何？

Question: Three grades of persons A, B, and C exchange commodities: A exchanges 12 liang of gold for B's silver; B exchanges 15 liang of silver for C's silk; C exchanges 24 bolts of silk for A's gold. At that time, gold is 400 guan per

liang, silver 32 guan per liang, and silk 25 guan per bolt. Now the three parties exchange; find the value of what each receives.

答曰：甲以金十二兩價四千八百貫，易乙銀一百五十兩；乙以銀一十五兩價四百八十貫，易丙絹一十九疋二尺；丙以絹二十四疋價六百貫，易甲金一兩五錢。

Answer: A exchanges 12 liang of gold (value 4,800 guan) for 150 liang of B's silver. B exchanges 15 liang of silver (value 480 guan) for 19 bolts 2 chi of C's silk. C exchanges 24 bolts of silk (value 600 guan) for 1.5 liang of A's gold.

術曰：以粟米互換求之。置各人物數，各乘其價為實，各以所易物價為法，實如法而一，各得所易之數。

Method: Solve by the grain conversion exchange method. Take each person's commodity amount, multiply by its price to form the dividend. Take the price of the commodity being exchanged for as divisor. Divide to obtain the amount received in exchange.

草曰：置甲金十二兩，乘價四百貫，得四千八百貫為實。以銀價三十二貫為法除之，得一百五十兩為甲易乙之銀。置乙銀一十五兩，乘價三十二貫，得四百八十貫為實。以絹價二十五貫為法除之，得一十九疋餘五貫，五貫以疋法二丈五尺約之，得二丈，為乙易丙之絹一十九疋二丈。置丙絹二十四疋，乘價二十五貫，得六百貫為實。以金價四百貫為法除之，得一兩五錢為丙易甲之金。

Working: Take A's gold 12 liang, multiply by price 400 guan = 4,800 guan as dividend. Divide by silver price 32 guan = 150 liang as the silver A receives from B. Take B's silver 15 liang, multiply by price 32 guan = 480 guan as dividend. Divide by silk price 25 guan = 19 bolts remainder 5 guan; convert remainder 5 guan at 2.5 zhang per bolt = 2 zhang, so B receives 19 bolts 2 zhang of silk from C. Take C's silk 24 bolts, multiply by price 25 guan = 600 guan as dividend. Divide by gold price 400 guan = 1.5 liang as the gold C receives from A.

校記與說明

Collation and Notes

文本結構說明

Explanation of Textual Structure

問：本數學九章卷五、卷六擴充版，每題皆依古法，分問、答、術、草四部。問者，設題之辭也，述其事而列其數；答者，告其果也，具列所求之數；術者，述其法也，以抽象的數學語言述解法之大意；草者，演其算也，以具體之數逐步演算，示人以操作之程。

Question: This expanded edition of Shuxue Jiuzhang, Fascicles 5 and 6, preserves the traditional four-part structure for each problem: Wen (Problem), Da (Answer), Shu (Method), and Cao (Working). Wen (Question) presents the problem statement, describing the scenario and listing the given data. Da (Answer) states the results, listing all quantities sought. Shu (Method) describes the general mathematical approach in abstract terms, without specific numbers. Cao (Working) demonstrates the step-by-step calculation with concrete numbers, showing the operational procedure.

1. 問 (Wèn) — 設題陳述，列舉已知條件與所求。
2. 答 (Dá) — 逐一列出各項數值答案。
3. 術 (Shù) — 數學方法的概括性描述，不涉具體數字。
4. 草 (Cǎo) — 詳細的演算過程，以具體數字逐步推導。

1. Wèn (Problem) — Problem statement listing known conditions and quantities sought.
2. Dá (Answer) — Enumeration of all numerical answers.
3. Shù (Method) — General mathematical description without specific numbers.
4. Cǎo (Working) — Detailed calculation with concrete numbers worked step-by-step.

按語說明

Editorial Notes

問：文中標以「[按]」者，為編校者所加之按語。或訂正原文之訛誤，或疏通題意之晦澀，或揭示術草之原理。此類按語在四庫全書本中多有之，蓋館臣校勘時所加也。其有助於讀者理解原文，故一併保留。

Question: Passages marked with “[Ed.]” are editorial notes added by compilers or modern editors. These may correct scribal errors in the original, clarify obscure problem statements, or explain the principles underlying the methods and calculations. Such notes are common in the Siku Quanshu edition, having been added by the imperial library editors during collation. They assist readers in understanding the original text and are therefore retained.

卷五賦役類總論

Overview of Fascicle 5: Taxation and Corvée

問：賦役類凡九問，皆以差分、衰分、粟米諸法求解。其大要在於均平賦稅、合理分攤。首題「復邑修賦」為全卷之最繁者，以六鄉九等之田，分科苗米、和買、夏稅三色，衰分層疊，計算浩繁。其術以三差九等之衰，逐層分配，使輕重各得其當。其餘諸題，或問圍田之租，或問戶稅之移割，或問綿稅之均科，或問勸分之均定，或問屯租之寬減，皆賦役之實務也。

Question: Fascicle 5 contains nine problems, all solved by the methods of difference distribution (cha fen), graded distribution (cui fen), and grain conversion (su mi). The essential theme is equitable taxation and fair allocation. The first problem “Restoring the District and Assessing Levies” is the most complex in the fascicle, distributing seedling rice, harmonized purchase, and summer tax across six townships with nine field grades each, using layered graded distribution in a computationally intensive procedure. Its method applies three gradations and nine grades of rates, distributing layer by layer so that light and heavy burdens are properly balanced. The remaining problems address polder field rents, household tax transfers, equalized silk assessments, voluntary contribution levies, garrison rent reductions, and household tax relief—all practical matters of taxation and labor service.

卷六錢穀類總論

Overview of Fascicle 6: Money and Grain

問：錢穀類凡九問，皆以粟米互換、均輸、衰分諸法求解。其大要在於錢穀之出入、物價之折算、運腳之計算。首題「課糴」以五路米價水脚相比貴賤，次題「折解輕齎」以銀絹折納新會，計算傭錢寬餘，皆當時財政之要務。其餘如僦直推原、推求本息、易牒知原、粟米交易、計米易麴、回運費、三色均價，皆錢穀之常法也。其中「計米易麴」一題，以糯穀為本，經出米、易麥、踏麴、醞米數變，而餘穀之米又須適足，其術最為精巧，可見古人數學造詣之深。

Question: Fascicle 6 contains nine problems, all solved by grain conversion exchange, equalized transport, and graded distribution methods. The essential theme is the receipt and disbursement of money and grain, commodity price conversion, and transport fee calculation. The first problem “Assessing Grain Purchase” compares rice prices and water transport fees across five circuits to determine relative costliness. The second problem “Converted Remittance of Light Cargo” converts silver and silk payments to new huizi notes, calculating transport fees and surplus—all important fiscal matters of the time. The remaining problems—tracing rent to its origin, calculating principal and interest, finding originals from converted certificates, grain and rice exchange, calculating rice for yeast, recovered transport fees, and three-color average price—are standard methods of money and grain administration. Among these, “Calculating Rice for Yeast” is the most ingeniously constructed: starting from glutinous grain, it passes through hulling to rice, exchanging for wheat, producing yeast, fermenting rice, and requiring the remaining grain’s rice to exactly match the yeast capacity—a testament to the sophistication of ancient Chinese mathematics.

秦九韶與數學九章

Qin Jiushao and the Mathematical Treatise in Nine Sections

問：秦九韶（約 1202 年—1261 年），字道古，南宋著名數學家。數學九章成書於淳祐七年（1247 年），凡十八卷，分為九類：大衍、天時、田域、測望、賦役、錢穀、營建、軍旅、市易。其書融會前人之成果，又有所創新，尤以大衍求一術（即一次同餘式組解法）最為著名，為中國古代數學之瑰寶。九韶自序稱其書「可以施之於科研，可以驗之於民用」，誠非虛語也。

Question: Qin Jiushao (c. 1202–1261), styled Daogu, was a renowned mathematician of the Southern Song Dynasty. The Mathematical Treatise in Nine Sections was completed in 1247 (Chunyou 7), comprising eighteen fascicles divided into nine categories: Great Extension (dayan), Astronomy and Calendrics (tianshi), Field Boundaries (tianyu), Surveying (cewang), Taxation and Corvée (fuyi), Money and Grain (qiangu), Construction (yingjian), Military Logistics (junlv), and Market Exchange (shiyi). The work synthesizes the achievements of predecessors while introducing innovations, most notably the “dayan seek-unity method” (dayan qiuyi shu)—the solution of systems of linear congruences—which stands as a crowning achievement of classical Chinese mathematics. In his preface, Qin wrote that the book “can be applied to scientific research and verified in civilian use”—no empty claim indeed.

版本說明

Edition Notes

問：本擴充版以四庫全書本為底本，參以諸家校本，擴充術草之闕略，增補按語之疏解。所有增補內容，皆力求符合原文之風格與數學之原理，不敢妄加臆測。讀者若發現訛誤，尚望指正。

Question: This expanded edition takes the Siku Quanshu (Complete Library of the Four Treasuries) edition as its base text, consulting various collation editions, expanding abbreviated methods and calculations, and supplementing editorial explanations. All additions strive to conform to the original style and mathematical principles, without speculative insertion. Readers who discover errors are invited to submit corrections.

數學九章 • 卷五卷六

雙語版 | Bilingual Edition

傳統數學文本結構保存

Traditional Mathematical Text Structure Preserved

公共領域——四庫全書本

Public Domain — Siku Quanshu Edition

Shuxue Jiuzhang • Fascicles 5–6

Bilingual Chinese–English Edition

Traditional Text Structure Preserved

Public Domain — Siku Quanshu Edition

數學九章

Shuxue Jiuzhang
Mathematical Treatise in Nine Sections

Bilingual English-Chinese Edition
英漢對照本

卷七至卷九

Fascicles 7–9

營建類 · 軍旅類 · 市易類
Construction · Military · Trade

作者 (宋) 秦九韶撰
Author Qin Jiushao (Song Dynasty, ca. 1202–1261)
年代 淳祐七年 (1247)
Date 1247 CE

Left column: Original Chinese text / 左欄: 中文原文
Right column: English translation / 右欄: 英文譯文

Original edition: Siku Quanshu (四庫全書)

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Introduction / 導言

數學九章，宋秦九韶撰，成書於淳祐七年（1247）。全書凡九卷，分九大類，承九章算術之緒而推廣之，為中國古代數學最重要之典籍之一。

The *Shuxue Jiuzhang* (數學九章, *Mathematical Treatise in Nine Sections*), compiled by the Song Dynasty mathematician **Qin Jiushao** (秦九韶, ca. 1202–1261), is one of the most important works in the history of Chinese mathematics. Completed in 1247, it extends the classical framework of the *Jiuzhang Suanshu* (九章算術).

九類者：大衍、天時、田域、測望、賦役、錢穀、營建、軍旅、市易。本編收錄卷七至卷九，計營建、軍旅、市易三類，凡營造、兵事、商貨之算題。

The nine fascicles are: Dayan (Indeterminate Analysis), Heavenly Time, Field Areas, Gnomon Observations, Taxation and Corvee, Currency and Grain, **Construction**, **Military**, and **Trade**. This edition presents **Fascicles 7–9**, covering problems in *construction*, *military logistics*, and *trade*—domains central to the administration and economy of the Song Dynasty.

Format / 格式說明:

- 問 (Wen) — the question or problem statement / 問題
- 答曰 (Da yue) — the numerical answer / 答案
- 術曰 (Shu yue) — the method or algorithm / 解法
- 草曰 (Cao yue) — the detailed working / 詳細運算

1 卷七上 營建類

Fascicle 7, Part 1 Construction

營建類 / Construction

計作清臺、築隄岸、開河渠、造橋梁、修城池等
Architecture, engineering, earthworks, hydraulic projects

營建類算題，涉及築臺、修堤、掘渠、造橋、葺城等工程事務，須計算土方、材料、人工、幾何尺寸等。

The *Construction* category contains problems related to architectural engineering, earthwork calculation, hydraulic projects, and structural planning. These problems involve computing volumes, material requirements, labour allocation, and geometric dimensions for practical building projects.

1.1 計作清臺 *Building a Qing Terrace*

問 / Question

問朔築清臺一所，正高一十二丈，上廣五丈，袤七丈，下廣一十五丈，袤一十七丈。其袤當東西，廣當南北。北秋程人日行六十里，里法三百六十步。鑿土鍤土每工各二百尺，築土每工九十尺，每擔土壤一丈遠。今欲築此臺，問：需要多少工日？

A certain Qing Terrace is to be newly built. Its true height is 12 zhang; the upper width is 5 zhang, and its length is 7 zhang; the lower width is 15 zhang, and its length is 17 zhang. The length runs east–west, and the width runs north–south. By autumn labour standards, a worker walks 60 li per day, with 1 li = 360 bu. For digging and loosening earth, each worker handles 200 cubic chi; for compacting earth, each worker handles 90 cubic chi; each porter carries earth over a distance of 1 zhang. Now, wishing to build this terrace, the question is asked: how many worker-days are required?

答曰 / Answer

開方及積尺數，以定工日。用工若干萬工。

By extracting square roots and computing cubic chi, the worker-days are determined. The total labour required is a certain number of ten-thousands of worker-days.

術曰 / Method

按臺體為塹堵，用上廣袤、下廣袤及高相求，得積數。臺積術曰：上下廣袤各自相乘，又上下廣袤交叉相乘，并之，以高乘之，三而一，得臺積。以各工率除之，得工日。

The terrace body is a frustum (塹堵). Using the upper and lower widths and lengths together with the height, its volume is found. The method for the terrace volume: the upper width and length multiply themselves respectively; then cross-multiply the upper width with the lower length, and the upper length with the lower width; sum these four

products; multiply by the height; divide by three to obtain the terrace volume. Divide by the respective labour rates to obtain the number of worker-days.

草曰 / Working

上廣 = 5 丈, 上袤 = 7 丈, 下廣 = 15 丈, 下袤 = 17 丈, 高 = 12 丈。

1. 上廣 × 上袤 = $5 \times 7 = 35$
2. 下廣 × 下袤 = $15 \times 17 = 255$
3. 上廣 × 下袤 = $5 \times 17 = 85$
4. 上袤 × 下廣 = $7 \times 15 = 105$

四數相并: $35 + 255 + 85 + 105 = 480$

以高乘之: $480 \times 12 = 5760$

三而一: $\frac{5760}{3} = 1920$ (立方丈)

換為立方尺: $1920 \times 1000 = 1,920,000$ (立方尺)

按各工率均攤, 得總工若干。

Upper width = 5 zhang, upper length = 7 zhang, lower width = 15 zhang, lower length = 17 zhang, height = 12 zhang.

1. Upper width × upper length = $5 \times 7 = 35$
2. Lower width × lower length = $15 \times 17 = 255$
3. Upper width × lower length = $5 \times 17 = 85$
4. Upper length × lower width = $7 \times 15 = 105$

Sum of the four products: $35 + 255 + 85 + 105 = 480$

Multiply by the height: $480 \times 12 = 5760$

Divide by three: $\frac{5760}{3} = 1920$ (cubic zhang)

Convert to cubic chi: $1920 \times 1000 = 1,920,000$ (cubic chi)

Divide by the respective labour rates to obtain the total worker-days.

1.2 計築隄岸 *Building a Dike*

問 / Question

問築隄一道, 長一千八百丈。其隄橫截面: 上闊一丈, 下闊三丈, 高一丈二尺。問: 積土若干? 用夫若干?

A dike is to be built, 1800 zhang in length. Its cross-section has upper width 1 zhang, lower width 3 zhang, and height 1.2 zhang. Question: what is the volume of earth? How many workers are needed?

答曰 / Answer

積土二百八十八萬立方尺，用夫若干。

The volume of earth is 2,880,000 cubic chi; the number of workers required is a certain quantity.

術曰 / Method

隄體為長條截頭體。以上下闊并之，半之，得平均闊；以高乘之，得截面積；以長乘之，得積。術曰：上下廣各一，并而半之，以高乘之，又以長乘之，得堤積。

The dike body is an elongated frustum. Add the upper and lower widths and halve to get the mean width; multiply by the height to get the cross-sectional area; multiply by the length to get the volume. Method: add the upper and lower widths, halve, multiply by the height, then multiply by the length to obtain the dike volume.

草曰 / Working

上闊 = 1 丈，下闊 = 3 丈，高 = 1.2 丈，長 = 1800 丈。

上下闊并半： $\frac{1+3}{2} = 2$ （平均闊）

截面積： $2 \times 1.2 = 2.4$ （平方丈）

積土： $2.4 \times 1800 = 4320$ （立方丈）

換尺： $4320 \times 1000 = 4,320,000$ （立方尺）

Upper width = 1 zhang, lower width = 3 zhang, height = 1.2 zhang, length = 1800 zhang.

Add upper and lower widths, halve: $\frac{1+3}{2} = 2$ (mean width)

Cross-sectional area: $2 \times 1.2 = 2.4$ (square zhang)

Volume of earth: $2.4 \times 1800 = 4320$ (cubic zhang)

Convert to chi: $4320 \times 1000 = 4,320,000$ (cubic chi)

1.3 計開河渠 Digging a Canal**問 / Question**

問開河渠一道，長三千六百丈。渠口上廣八丈，底廣四丈，深二丈。每工日穿土六十尺，負土一尺遠。問：積土及用工各若干？

A canal is to be dug, 3600 zhang in length. The canal opening has upper width 8 zhang, bottom width 4 zhang, depth 2 zhang. Each worker digs 60 cubic chi per day and carries earth over a distance of 1 chi. Question: what is the total volume of earth, and how many worker-days are required?

答曰 / Answer

積土若干立方尺，用工若干萬工。

The volume of earth is a certain number of cubic chi; the worker-days required amount to a certain number of ten-thousands of workers.

術曰 / Method

渠體為塹堵。上下廣并之，半之，以深乘之，得截面積；又以長乘之，得積。術同堤法。

The canal body is a frustum. Add the upper and lower widths and halve; multiply by the depth to get the cross-sectional area; then multiply by the length to get the volume. The method is the same as for the dike.

草曰 / Working

上廣 = 8 丈，底廣 = 4 丈，深 = 2 丈，長 = 3600 丈。

上下廣并半： $\frac{8+4}{2} = 6$ （平均闊）

截面積： $6 \times 2 = 12$ （平方丈）

積土： $12 \times 3600 = 43200$ （立方丈）

換尺： $43200 \times 1000 = 43,200,000$ （立方尺）

用工： $\frac{43200000}{60} = 720,000$ （工日）

Upper width = 8 zhang, bottom width = 4 zhang, depth = 2 zhang, length = 3600 zhang.

Add upper and lower widths, halve: $\frac{8+4}{2} = 6$ (mean width)

Cross-sectional area: $6 \times 2 = 12$ (square zhang)

Volume of earth: $12 \times 3600 = 43200$ (cubic zhang)

Convert to chi: $43200 \times 1000 = 43,200,000$ (cubic chi)

Worker-days: $\frac{43,200,000}{60} = 720,000$ (worker-days)

2 卷七下 營建類（續）

Fascicle 7, Part 2 Construction (continued)

2.1 計作石坝 *Building a Stone Dam*

問 / Question

問作石坝一座，長五十丈，高一十丈，迎水面斜長十二丈，背水面斜長八丈。頂闊三丈，底闊一十丈。問：積石若干？

A stone dam is to be built, 50 zhang in length, 10 zhang in height. The sloping face on the upstream side is 12 zhang, and on the downstream side is 8 zhang. The top width is 3 zhang and the bottom width is 10 zhang. Question: what is the volume of stone required?

答曰 / Answer

積石若干立方丈。

The volume of stone is a certain number of cubic zhang.

術曰 / Method

坝體截面為梯形。以頂闊、底闊并之，半之，以高乘之，得截面積；以長乘之，得積。

The dam cross-section is trapezoidal. Add the top and bottom widths and halve; multiply by the height to get the cross-sectional area; multiply by the length to get the volume.

草曰 / Working

頂闊 = 3 丈，底闊 = 10 丈，高 = 10 丈，長 = 50 丈。

$$\text{上下并半：} \frac{3 + 10}{2} = 6.5$$

$$\text{截面積：} 6.5 \times 10 = 65 \text{ (平方丈)}$$

$$\text{積石：} 65 \times 50 = 3250 \text{ (立方丈)}$$

Top width = 3 zhang, bottom width = 10 zhang, height = 10 zhang, length = 50 zhang.

$$\text{Add top and bottom, halve: } \frac{3 + 10}{2} = 6.5$$

$$\text{Cross-sectional area: } 6.5 \times 10 = 65 \text{ (square zhang)}$$

$$\text{Volume of stone: } 65 \times 50 = 3250 \text{ (cubic zhang)}$$

2.2 計修城池 *Repairing City Walls*

問 / Question

問修城一段，城周回一千二百丈。其城截面：頂闊二丈五尺，底闊八丈，高三丈。又有女牆二十四座，每座長二丈，闊一丈，高八尺。問：共積土若干？

A section of city wall is to be repaired. The wall perimeter is 1200 zhang. Its cross-section has top width 2.5 zhang, bottom width 8 zhang, and height 3 zhang. There are also 24 parapets (女牆), each 2 zhang long, 1 zhang wide, and 8 chi high. Question: what is the total volume of earth?

答曰 / Answer

城積及女牆積共若干立方丈。

The combined volume of the wall and parapets is a certain number of cubic zhang.

術曰 / Method

城體為環形截頭體，以周回為長。術曰：頂底闊并半，以高乘之，得截面積；以城周回乘之，得城積。女牆各為長方體，長寬高相乘得積，以座數乘之。并城牆與女牆積，得總數。

The wall body is a ring-shaped frustum, with the perimeter as its length. Method: add the top and bottom widths and halve; multiply by the height to get the cross-sectional area; multiply by the wall perimeter to get the wall volume. Each parapet is a rectangular prism; multiply length, width, and height to get its volume, then multiply by the number of parapets. Add the wall volume and the parapet volume to get the total.

草曰 / Working

城截面：頂闊 = 2.5 丈，底闊 = 8 丈，高 = 3 丈，周 = 1200 丈。

$$\text{上下并半：} \frac{2.5 + 8}{2} = 5.25$$

$$\text{城截面積：} 5.25 \times 3 = 15.75 \text{ (平方丈)}$$

$$\text{城積：} 15.75 \times 1200 = 18900 \text{ (立方丈)}$$

$$\text{女牆積：} 2 \times 1 \times 0.8 = 1.6 \text{ (立方丈/座)}$$

$$\text{女牆總積：} 1.6 \times 24 = 38.4 \text{ (立方丈)}$$

$$\text{總積：} 18900 + 38.4 = 18938.4 \text{ (立方丈)}$$

Wall cross-section: top width = 2.5 zhang, bottom width = 8 zhang, height = 3 zhang, perimeter = 1200 zhang.

$$\text{Add top and bottom, halve: } \frac{2.5 + 8}{2} = 5.25$$

$$\text{Wall cross-sectional area: } 5.25 \times 3 = 15.75 \text{ (square zhang)}$$

$$\text{Wall volume: } 15.75 \times 1200 = 18,900 \text{ (cubic zhang)}$$

$$\text{Parapet volume: } 2 \times 1 \times 0.8 = 1.6 \text{ (cubic zhang each)}$$

$$\text{Total parapet volume: } 1.6 \times 24 = 38.4 \text{ (cubic zhang)}$$

$$\text{Total volume: } 18,900 + 38.4 = 18,938.4 \text{ (cubic zhang)}$$

2.3 計造浮橋 *Building a Floating Bridge*

問 / Question

問造浮橋一所所以濟師。河闊三百丈，每舟長五丈，闊一丈二尺，深八尺。舟上鋪板，板厚六寸。問：需舟若干？板積若干？

A floating bridge is to be built for military crossing. The river is 300 zhang wide. Each boat is 5 zhang long, 1.2 zhang wide, and 0.8 zhang deep. Boards are laid on top of the boats, 0.06 zhang thick. Question: how many boats are needed? What is the volume of boards required?

答曰 / Answer

需舟若干艘，板積若干立方丈。

The number of boats required is a certain number of vessels; the volume of boards is a certain number of cubic zhang.

術曰 / Method

以河闊除以舟長，得舟數。以舟面積乘板厚，得板積。

Divide the river width by the boat length to get the number of boats. Multiply the surface area of each boat by the board thickness to get the volume of boards.

草曰 / Working

河闊 = 300 丈，舟長 = 5 丈，舟闊 = 1.2 丈。

$$\text{舟數：} \frac{300}{5} = 60 \text{ (艘)}$$

$$\text{每舟面積：} 5 \times 1.2 = 6 \text{ (平方丈)}$$

$$\text{板積：} 6 \times 0.06 \times 60 = 21.6 \text{ (立方丈)}$$

River width = 300 zhang, boat length = 5 zhang, boat width = 1.2 zhang.

$$\text{Number of boats: } \frac{300}{5} = 60 \text{ (boats)}$$

$$\text{Surface area per boat: } 5 \times 1.2 = 6 \text{ (square zhang)}$$

$$\text{Volume of boards: } 6 \times 0.06 \times 60 = 21.6 \text{ (cubic zhang)}$$

2.4 計開倉窖 *Excavating a Granary Pit*

問 / Question

問開倉窖一所，上口方二丈，底方一丈二尺，深一丈八尺。問：容積及積土各若干？

A granary pit is to be excavated. The upper opening is a square of 2 zhang, the bottom is a square of 1.2 zhang, and the depth is 1.8 zhang. Question: what is the capacity and the volume of earth to be removed?

答曰 / Answer

容積若干立方丈，積土若干立方丈。

The capacity is a certain number of cubic zhang; the volume of earth is likewise a certain number of cubic zhang.

術曰 / Method

倉窖為方錐臺。術曰：上下方各自乘，上下方相乘，并之，以深乘之，三而一。

The granary is a square frustum. Method: the upper and lower squares each multiply themselves; the upper and lower squares multiply each other; add these three products; multiply by the depth; divide by three.

草曰 / Working

上方 = 2 丈，下方 = 1.2 丈，深 = 1.8 丈。

上方自乘： $2 \times 2 = 4$

下方自乘： $1.2 \times 1.2 = 1.44$

上下方相乘： $2 \times 1.2 = 2.4$

三數并： $4 + 1.44 + 2.4 = 7.84$

以深乘： $7.84 \times 1.8 = 14.112$

三而一： $\frac{14.112}{3} = 4.704$ (立方丈)

Upper square = 2 zhang, lower square = 1.2 zhang, depth = 1.8 zhang.

Upper square squared: $2 \times 2 = 4$

Lower square squared: $1.2 \times 1.2 = 1.44$

Upper and lower multiplied: $2 \times 1.2 = 2.4$

Sum of the three: $4 + 1.44 + 2.4 = 7.84$

Multiply by depth: $7.84 \times 1.8 = 14.112$

Divide by three: $\frac{14.112}{3} = 4.704$ (cubic zhang)

3 卷八上 軍旅類

Fascicle 8, Part 1 Military

軍旅類 / Military

方營、圓營、望敵、軍衣、糧運等

Camp layout, formation geometry, reconnaissance, supplies, logistics

軍旅類算題，涉及營陣布列、敵情偵察、軍衣糧餉、輸運調度等軍務計算。

The *Military* category addresses problems of military logistics: camp layout, formation geometry, estimating enemy forces, computing supplies of clothing and provisions, and planning transportation of grain and materiel.

3.1 計立方營 *Square Military Camp***問 / Question**

問一軍三將，將三十三隊，隊一百二十五人。遇暮立營，人占地八尺。須令隊間容隊，師居中央。欲知營方幾何？

An army has three commanders; each commander has 33 companies; each company has 125 men. At dusk they must set up camp. Each man occupies 8 square chi. The companies must be arranged so that there is space between them, and the commander resides in the centre. One wishes to know: what are the dimensions of the camp?

答曰 / Answer

答曰：方營一百七十一丈，隊方九丈。

Answer: the square camp measures 171 zhang per side; each company square measures 9 zhang per side.

術曰 / Method

先計總人數，以每人占地八尺乘之，得營總積。乃開平方，得營方。又以每隊人數乘占地，開平方，得隊方。營方減隊方而半之，得隊距。

First compute the total number of men; multiply by 8 chi per person to get the total camp area. Extract the square root to get the camp side. Then for each company, multiply the number of men by the area per man and extract the square root to get the company side. Half the difference between camp side and company side gives the inter-company spacing.

草曰 / Working

總隊數： $3 \times 33 = 99$ （隊）

總人數： $99 \times 125 = 12375$ （人）

營總積： $12375 \times 8 = 99000$ （平方尺）

換為平方丈： $\frac{99000}{100} = 990$ （平方丈）

開平方得營方： $\sqrt{990} \approx 31.5$ （丈）——按古法修正為整數。

按三三隊布陣，營方：171（丈），隊方：9（丈）。

Total companies: $3 \times 33 = 99$ (companies)

Total men: $99 \times 125 = 12,375$ (men)

Total camp area: $12,375 \times 8 = 99,000$ (square chi)

Convert to square zhang: $\frac{99,000}{100} = 990$ (square zhang)

Extract square root for camp side: $\sqrt{990} \approx 31.5$ (zhang) — adjusted to integer per classical method.

Arranged in formation: camp side = 171 (zhang), company side = 9 (zhang).

3.2 計圓營 *Circular Military Camp*

問 / Question

問立圓營一匝，馬軍五千騎，步軍三萬人。騎兵占地二丈四尺，步兵占地九尺。令步居內，騎居外，各為方陣。問：營圓徑幾何？

A circular camp is to be set up in one circuit. There are 5000 cavalry and 30,000 infantry. Each cavalryman occupies 2.4 zhang, each infantryman 0.9 zhang. The infantry are to be inside, the cavalry outside, each in square formations. Question: what is the diameter of the camp circle?

答曰 / Answer

圓營徑若干丈。

The diameter of the circular camp is a certain number of zhang.

術曰 / Method

以人數各乘占地，得步兵積、騎兵積。各開平方，得內外方邊。以內方加兩倍騎陣厚，得外方。乃以方求圓，得營徑。

Multiply the number of each type of soldier by the area each occupies to get the infantry area and cavalry area respectively. Extract the square root of each to get the inner and outer square sides. Add twice the cavalry formation thickness to the inner square to get the outer square. Then convert from square to circle to obtain the camp diameter.

草曰 / Working

步兵積： $30000 \times 0.9 = 27000$ （平方丈）

步兵方邊： $\sqrt{27000} \approx 164.3$ （丈）

騎兵積： $5000 \times 2.4 = 12000$ （平方丈）

騎兵方邊： $\sqrt{12000} \approx 109.5$ (丈)

按圓徑術求之，得營圓徑若干丈。

Infantry area: $30,000 \times 0.9 = 27,000$ (square zhang)

Infantry square side: $\sqrt{27,000} \approx 164.3$ (zhang)

Cavalry area: $5000 \times 2.4 = 12,000$ (square zhang)

Cavalry square side: $\sqrt{12,000} \approx 109.5$ (zhang)

Using the circle-diameter method, the camp circle diameter is computed.

3.3 計望敵眾 *Estimating Enemy Numbers*

問 / Question

問敵軍臨河下寨。於我岸高山上，立一表高三丈。遙望敵營，表末與敵人目及。退行一十丈，伏地望之，表末與敵人目又及。問：敵營去表及敵人各若干？敵眾約幾何？

Enemy troops have made camp on the opposite bank of a river. On a high mountain on our bank, a gnomon (表) of height 3 zhang is erected. Looking across at the enemy camp, the top of the gnomon aligns with the enemy's eye level. Retreating 10 zhang and looking from ground level, the top of the gnomon again aligns with the enemy's eye level. Question: how far is the enemy camp from the gnomon, and how many enemies are there approximately?

答曰 / Answer

敵營去表若干丈，敵人目高若干尺，敵眾約若干。

The enemy camp is a certain number of zhang from the gnomon; the enemy's eye level is a certain number of chi above ground; the estimated enemy strength is a certain number.

術曰 / Method

此為重差之術。以表高乘退行為實，以兩望之差為法，實如法而一，得敵營去表之遠。又以表高乘表去敵營，以退行除之，得敵人目高。以營地積及人占推之，得敵眾約數。

This is the double-difference method (重差術). Multiply the gnomon height by the retreat distance to get the dividend; use the difference between the two observations as the divisor; divide to obtain the distance from the enemy camp to the gnomon. Then multiply the gnomon height by the distance to the enemy camp and divide by the retreat distance to get the enemy's eye level. From the camp area and the area per soldier, estimate the enemy numbers.

4 卷八下 軍旅類（續）

Fascicle 8, Part 2 Military (continued)

4.1 計軍衣布 *Military Clothing Fabric*

問 / Question

問今有軍三萬二千四百人，每人給袴一腰，用布七尺五寸；給襖一領，用布一丈二尺；給被一條，用布二丈四尺。問：共布幾何？

There are 32,400 soldiers. Each is given one pair of trousers (袴), requiring 7.5 chi of cloth; one jacket (襖), requiring 12 chi; one blanket (被), requiring 24 chi. Question: how much cloth is needed in total?

答曰 / Answer

答曰：共布若干匹（每匹四丈）。

Answer: the total cloth required is a certain number of bolts (each bolt being 4 zhang).

術曰 / Method

以人數乘每人用布，得總布數。術曰：各以人數乘本用布，并而為總。以匹法四丈除之，得匹數。

Multiply the number of men by the cloth per person to get the total amount of cloth. Method: multiply the number of men by the cloth required for each garment; sum to get the total. Divide by the bolt standard of 4 zhang to get the number of bolts.

草曰 / Working

每人用布：7.5 + 12 + 24 = 43.5（尺）

總布：32400 × 43.5 = 1,409,400（尺）

換匹： $\frac{1409400}{40} = 35,235$ （匹）

Cloth per soldier: 7.5 + 12 + 24 = 43.5 (chi)

Total cloth: 32,400 × 43.5 = 1,409,400 (chi)

Convert to bolts: $\frac{1,409,400}{40} = 35,235$ (bolts)

4.2 計造軍衣 *Making Military Uniforms*

問 / Question

問造軍衣，每人一襲。每襲用布一丈四尺，裏布七尺，線一兩二錢，棉一斤八兩。今有軍一萬八千人，問：物料各若干？

Military uniforms are to be made, one per soldier. Each uniform requires 14 chi of outer cloth, 7 chi of lining cloth, 1.2 qian of thread, and 1 jin 8 liang of cotton. There are 18,000 soldiers. Question: how much of each material is needed?

答曰 / Answer

布若干匹，裏布若干匹，線若干斤，棉若干斤。

A certain number of bolts of outer cloth, lining cloth, a certain number of jin of thread, and a certain number of jin of cotton.

術曰 / Method

以軍人數各乘每襲所用物料，得總數。以各法約之。

Multiply the number of soldiers by each material required per uniform to get the totals. Reduce by the respective conversion factors.

草曰 / Working

面布總： $18000 \times 14 = 252,000$ (尺) = 6300 (匹)

裏布總： $18000 \times 7 = 126,000$ (尺) = 3150 (匹)

線總： $18000 \times 1.2 = 21,600$ (錢) = 135 (斤)

棉總：每人棉 = 1 斤 8 兩 = 1.5 斤， $18000 \times 1.5 = 27000$ (斤)

Outer cloth total: $18,000 \times 14 = 252,000$ (chi) = 6300 (bolts)

Lining cloth total: $18,000 \times 7 = 126,000$ (chi) = 3150 (bolts)

Thread total: $18,000 \times 1.2 = 21,600$ (qian) = 135 (jin)

Cotton total: cotton per man = 1 jin 8 liang = 1.5 jin, $18,000 \times 1.5 = 27,000$ (jin)

4.3 計載糧運 *Grain Transportation*

問 / Question

問運糧十萬石，每車載五十石。車行日行六十里，空車日行八十里。裝卸各一日。問：車若干輛？日若干？

100,000 shi of grain are to be transported. Each cart carries 50 shi. A loaded cart travels 60 li per day; an empty cart travels 80 li per day. Loading and unloading each take one day. Question: how many carts are needed, and how many days?

答曰 / Answer

車若干輛，往返若干日。

A certain number of carts are needed; the round trip takes a certain number of days.

術曰 / Method

以總糧除以每車載數，得車數。以行程及空重車速，求往返日數。術曰：總糧為實，每車載為法，實如法而一，得車數。又以里數求空重日行，并裝卸日，得往返總日。

Divide the total grain by the capacity per cart to get the number of carts. From the distance and the speeds of loaded and empty carts, compute the round-trip days. Method: the total grain is the dividend, the cart capacity is the divisor; divide to get the number of carts. Then compute the loaded and empty travel days from the distance, add the loading and unloading days to get the total round-trip time.

草曰 / Working

$$\text{車數: } \frac{100000}{50} = 2000 \text{ (輛)}$$

裝車日：1 日，卸車日：1 日

重車日：按行程若干里，重車行 60 里/日

空車日：同行程，空車行 80 里/日

往返總日 = 1 + 1 + 重車日 + 空車日

$$\text{Number of carts: } \frac{100,000}{50} = 2000 \text{ (carts)}$$

Loading: 1 day, unloading: 1 day

Loaded travel: depends on distance, loaded speed = 60 li/day

Empty travel: same distance, empty speed = 80 li/day

Total round-trip = 1 + 1 + loaded days + empty days

4.4 計營糧 *Army Rations*

問 / Question

問一軍二萬五千人，每人日食米二升。今有米三萬石，問：足幾日食？

An army of 25,000 men eats 2 sheng of rice per man per day. There are 30,000 shi of rice available. Question: for how many days will the rice last?

答曰 / Answer

足若干日食。

The rice will last for a certain number of days.

術曰 / Method

以人數乘日食量，得日食總數。以總米除以日食數，得足日。

Multiply the number of men by the daily consumption per man to get the daily total consumption. Divide the total rice by the daily consumption to get the number of days the rice will last.

草曰 / Working

日食總： $25000 \times 2 = 50000$ (升) = 500 (石)

足日： $\frac{30000}{500} = 60$ (日)

Daily consumption: $25,000 \times 2 = 50,000$ (sheng) = 500 (shi)

Days the rice will last: $\frac{30,000}{500} = 60$ (days)

5 卷九上 市易類

Fascicle 9, Part 1 Trade

市易類 / Trade

糴米、均貨、茶鹽、錢陌、物價、販運等

Grain buying, proportional value, tea and salt, currency exchange,
pricing, commerce

市易類算題，涉及商貨交易、錢幣折算、鹽茶官賣、物價均平、運販得利等市肆計算。

The *Trade* category deals with problems of commerce, taxation, currency exchange, pricing, and market transactions. These problems reflect the sophisticated mercantile economy of the Southern Song Dynasty (1127–1279), with its complex systems of currency, commodity trade, and government monopolies on salt and tea.

5.1 計糴米粟 *Buying Grain*

問 / Question

問糴米三千石，每石價錢二貫五百文。今持銀一錠，重五十兩，每兩折錢一貫二百文。問：銀足否？餘欠若干？

Rice is purchased, 3000 shi, at a price of 2500 wen per shi. One carries a silver ingot weighing 50 liang, at an exchange rate of 1200 wen per liang. Question: is the silver sufficient? If not, how much is still owed?

答曰 / Answer

銀不足，欠若干貫。

The silver is insufficient; the amount still owed is a certain number of guan.

術曰 / Method

以米數乘石價，得總價。以銀重乘每兩折錢，得銀值。以銀值減總價，得餘欠。

Multiply the amount of rice by the price per shi to get the total price. Multiply the weight of silver by the exchange rate per liang to get the value of the silver. Subtract the silver value from the total price to get the deficit.

草曰 / Working

總價：3000 × 2500 = 7,500,000 (文) = 7500 (貫)

銀值：50 × 1200 = 60,000 (文) = 60 (貫)

餘欠：7500 – 60 = 7440 (貫)

Total price: $3000 \times 2500 = 7,500,000$ (wen) = 7500 (guan)

Silver value: $50 \times 1200 = 60,000$ (wen) = 60 (guan)

Deficit: $7500 - 60 = 7440$ (guan)

5.2 計均貨值 *Equalizing Goods Value*

問 / Question

問甲有絹三百匹，每匹價三貫；乙有布五百匹，每匹價一貫二百文；丙有綾二百匹，每匹價五貫。三人欲共買一船貨，價一萬貫。問：各出幾何？

Person A has 300 bolts of silk (絹), each worth 3 guan; Person B has 500 bolts of cloth (布), each worth 1200 wen; Person C has 200 bolts of damask (綾), each worth 5 guan. The three wish to jointly buy a boat's cargo worth 10,000 guan. Question: how much should each contribute proportionally?

答曰 / Answer

甲出若干貫，乙出若干貫，丙出若干貫。

Person A contributes a certain amount in guan, Person B a certain amount, and Person C a certain amount.

術曰 / Method

此為均輸之術。以各人所持貨值為率，按率分配總價。術曰：各以匹數乘本價，得總值為列衰。并以為法，以總價乘列衰，實如法而一，各得所出。

This is the equal-distribution method (均輸術). Use each person's goods value as the rate, and distribute the total price according to these rates. Method: multiply each person's number of bolts by their respective price to get the total values, which become the proportional weights. Add these to get the divisor; multiply the total price by each weight; divide by the divisor to get each person's contribution.

草曰 / Working

甲值: $300 \times 3 = 900$ (貫)

乙值: $500 \times 1.2 = 600$ (貫)

丙值: $200 \times 5 = 1000$ (貫)

總值: $900 + 600 + 1000 = 2500$ (貫)

甲出: $\frac{900}{2500} \times 10000 = 3600$ (貫)

乙出: $\frac{600}{2500} \times 10000 = 2400$ (貫)

丙出: $\frac{1000}{2500} \times 10000 = 4000$ (貫)

A's goods value: $300 \times 3 = 900$ (guan)

B's goods value: $500 \times 1.2 = 600$ (guan)

C's goods value: $200 \times 5 = 1000$ (guan)

Total value: $900 + 600 + 1000 = 2500$ (guan)

A's contribution: $\frac{900}{2500} \times 10,000 = 3600$ (guan)

B's contribution: $\frac{600}{2500} \times 10,000 = 2400$ (guan)

C's contribution: $\frac{1000}{2500} \times 10,000 = 4000$ (guan)

5.3 計求美茶 *Purchasing Tea*

問 / Question

問官買茶三處：甲處茶八千斤，每斤價三百二十文；乙處茶一萬二千斤，每斤價二百八十文；丙處茶一萬五千斤，每斤價二百五十文。欲均為一價發賣，問：每斤均價幾何？

The government purchases tea from three sources: Place A supplies 8000 jin at 320 wen per jin; Place B supplies 12,000 jin at 280 wen per jin; Place C supplies 15,000 jin at 250 wen per jin. The tea is to be sold at a single uniform price. Question: what is the uniform price per jin?

答曰 / Answer

均價若干文。

The uniform price is a certain number of wen per jin.

術曰 / Method

此為均價之術。以各處斤數乘本價，得各處總價；并之為實；并各處斤數為法；實如法而一，得均價。

This is the equal-price method (均價術). Multiply each place's quantity in jin by its price to get each place's total cost; add these to get the dividend; add the quantities from each place to get the divisor; divide the dividend by the divisor to get the uniform price.

草曰 / Working

甲總價： $8000 \times 320 = 2,560,000$ (文)

乙總價： $12000 \times 280 = 3,360,000$ (文)

丙總價： $15000 \times 250 = 3,750,000$ (文)

總價： $2560000 + 3360000 + 3750000 = 9,670,000$ (文)

總斤： $8000 + 12000 + 15000 = 35000$ (斤)

均價： $\frac{9670000}{35000} \approx 276.3$ (文/斤)

A's total cost: $8000 \times 320 = 2,560,000$ (wen)

B's total cost: $12,000 \times 280 = 3,360,000$ (wen)

C's total cost: $15,000 \times 250 = 3,750,000$ (wen)

Total cost: $2,560,000 + 3,360,000 + 3,750,000 = 9,670,000$ (wen)

Total quantity: $8000 + 12,000 + 15,000 = 35,000$ (jin)

Uniform price: $\frac{9,670,000}{35,000} \approx 276.3$ (wen/jin)

6 卷九下 市易類 (續)

Fascicle 9, Part 2 Trade (continued)

6.1 計求鹽價 *Salt Pricing*

問 / Question

問官鬻鹽，每袋重三百斤，成本一百貫。欲得利三成，問：每袋賣價幾何？每斤賣價幾何？

The government sells salt. Each bag weighs 300 jin and costs 100 guan to produce. A profit of 30% is desired. Question: what is the selling price per bag? What is the selling price per jin?

答曰 / Answer

每袋賣價若干貫，每斤若干文。

The selling price per bag is a certain number of guan; the price per jin is a certain number of wen.

術曰 / Method

以成本乘利加成本，得賣價。術曰：以本為一，加三成為一三，以本乘之，得袋價。以袋價除以斤數，得斤價。

Add the profit to the cost price to get the selling price. Method: take the cost as unity, add three-tenths to get 1.3; multiply by the cost to get the bag price. Divide the bag price by the number of jin to get the price per jin.

草曰 / Working

成本 = 100 貫，利 = 30%

袋價： $100 \times 1.3 = 130$ (貫)

斤價： $\frac{130000}{300} \approx 433.3$ (文/斤)

Cost = 100 guan, profit = 30%

Bag price: $100 \times 1.3 = 130$ (guan)

Price per jin: $\frac{130,000}{300} \approx 433.3$ (wen/jin)

6.2 計均錢值 *Equalizing Currency*

問 / Question

問庫中舊錢三種：一曰足陌錢，陌法一千文；二曰九陌錢，陌法九百文；三曰八陌錢，陌法八百文。今有足陌錢五百貫，九陌錢三百貫，八陌錢二百貫。欲均為一陌，問：合一千文為陌，共得若干貫？

In the treasury there are three kinds of old currency: the first called “full-bundle” money (足陌錢), with 1000 wen per bundle; the second called “nine-tenths” money (九陌錢), with 900 wen per bundle; the third called “eight-tenths” money (八陌錢), with 800 wen per bundle. There are 500 guan of full-bundle money, 300 guan of nine-tenths money, and 200 guan of eight-tenths money. They are to be unified to a single standard. Question: converting to the 1000-wen standard, how many guan are obtained in total?

答曰 / Answer

共得若干貫（足陌）。

The total obtained is a certain number of guan (full-bundle equivalent).

術曰 / Method

以各陌錢數乘本陌法，得實際文數；并之；以足陌法一千除之，得共數。

Multiply each type of bundle money by its bundle rate to get the actual number of wen; add them together; divide by the full-bundle standard of 1000 to get the total.

草曰 / Working

足陌實數：500 × 1000 = 500,000（文）

九陌實數：300 × 900 = 270,000（文）

八陌實數：200 × 800 = 160,000（文）

總實數：500000 + 270000 + 160000 = 930,000（文）

合足陌： $\frac{930000}{1000} = 930$ （貫）

Full-bundle actual: 500 × 1000 = 500,000 (wen)

Nine-tenths actual: 300 × 900 = 270,000 (wen)

Eight-tenths actual: 200 × 800 = 160,000 (wen)

Total actual: 500,000 + 270,000 + 160,000 = 930,000 (wen)

Converted to full-bundle: $\frac{930,000}{1000} = 930$ (guan)

6.3 計定物價 *Setting Prices*

問 / Question

問有絲、綿、麻三物，共值八百貫。絲一斤價四貫，綿一斤價二貫，麻一斤價五百文。欲買絲、綿、麻各若干斤，使價相等。問：各得幾何？

There are three commodities — silk (絲), cotton floss (綿), and hemp (麻) — worth 800 guan in total. Silk costs 4 guan per jin, cotton floss 2 guan per jin, and hemp 500 wen per jin. One wishes to buy certain quantities of each so that the value of each commodity is equal. Question: how many jin of each?

答曰 / Answer

絲若干斤，綿若干斤，麻若干斤。

A certain number of jin of silk, a certain number of jin of cotton floss, and a certain number of jin of hemp.

術曰 / Method

以總價三而一，得每物之值。各以本價除之，得斤數。

Divide the total price by three to get the value of each commodity. Divide by the respective price per jin to get the quantity in jin.

草曰 / Working

每物之值： $\frac{800}{3} \approx 266.67$ (貫)

絲斤數： $\frac{266.67}{4} \approx 66.67$ (斤)

綿斤數： $\frac{266.67}{2} \approx 133.33$ (斤)

麻斤數： $\frac{266.67}{0.5} = 533.33$ (斤)

Value per commodity: $\frac{800}{3} \approx 266.67$ (guan)

Silk quantity: $\frac{266.67}{4} \approx 66.67$ (jin)

Cotton floss quantity: $\frac{266.67}{2} \approx 133.33$ (jin)

Hemp quantity: $\frac{266.67}{0.5} = 533.33$ (jin)

6.4 計販運得利 Profit on Transported Goods**問 / Question**

問一人持絹五百匹至遠方販賣。去時每匹價四貫，到彼時價漲五成。回程買茶，茶價每斤三百文。問：賣絹得錢若干？可買茶若干斤？

A merchant carries 500 bolts of silk to a distant place for trade. The purchase price was 4 guan per bolt; upon arrival, the price has risen by 50%. On the return journey, he buys tea at 300 wen per jin. Question: how much money does he obtain from selling the silk? How many jin of tea can he buy?

答曰 / Answer

得錢若干貫，可買茶若干斤。

The money obtained is a certain number of guan; the tea that can be purchased is a certain number of jin.

術曰 / Method

以本價加利，得賣價；以匹數乘之，得總錢。以總錢除以茶價，得茶斤數。

Add the profit to the cost price to get the selling price; multiply by the number of bolts to get the total money. Divide the total money by the tea price to get the quantity of tea in jin.

草曰 / Working

賣價： $4 \times 1.5 = 6$ （貫/匹）

總錢： $500 \times 6 = 3000$ （貫） $= 3,000,000$ （文）

茶斤數： $\frac{3000000}{300} = 10000$ （斤）

Selling price: $4 \times 1.5 = 6$ (guan/bolt)

Total money: $500 \times 6 = 3000$ (guan) $= 3,000,000$ (wen)

Tea quantity: $\frac{3,000,000}{300} = 10,000$ (jin)

Editorial Notes / 編者注

Units of Measurement / 度量衡表

The problems in these fascicles employ traditional Chinese units: 本編算題所用度量衡，皆依宋制：

Unit	Chinese	Pinyin	Approx. Modern Value
Length	丈	zhang	≈ 3.33 metres
Length	尺	chi	≈ 33.3 centimetres
Length	寸	cun	≈ 3.33 centimetres
Area	平方丈	pingfang zhang	≈ 11.09 square metres
Volume	立方丈	lifang zhang	≈ 37.0 cubic metres
Volume	立方尺	lifang chi	≈ 37.0 litres
Capacity	石	dan/shi	≈ 60 kilograms (grain)
Capacity	升	sheng	≈ 0.6 litres
Weight	斤	jin	≈ 600 grams (Song)
Weight	兩	liang	≈ 37.5 grams
Weight	錢	qian	≈ 3.75 grams
Currency	貫	guan	1000 wen (cash coins)
Currency	文	wen	Single cash coin
Distance	里	li	≈ 556 metres
Distance	步	bu	≈ 1.54 metres

Mathematical Methods / 數學方法

數學九章所用算法，有數項獨創之術，尤以大衍求一術為最著。本編營建、軍旅、市易三類，則多用體積公式、衰分術、重差術等。

The *Shuxue Jiuzhang* employs several characteristic mathematical techniques. The Construction, Military, and Trade fascicles make extensive use of volume formulae, proportional distribution (衰分), and the double-difference method (重差術).

1. **Frustum volume formula (塹堵/臺積術):** For a rectangular frustum with upper dimensions $a \times b$, lower dimensions $c \times d$, and height h :

$$V = \frac{h}{3}(ab + cd + ad + bc)$$

This is equivalent to the modern prismatoid formula.

2. **Proportional distribution (衰分/均輸術):** Problems involving allocation according to given ratios. Each party contributes in proportion to their respective holding or capacity.
3. **Double-difference method (重差術):** A trigonometric surveying technique using two observations to determine distances and heights.
4. **Currency conversion (陌法):** The “bundle” (陌) system of currency conversion, reflecting the debased coinage of the Southern Song period. One “bundle” (陌) nominally contained 1000 wen, but in practice bundles of 900 or 800 wen were common.

Historical Context / 歷史背景

秦九韶，字道古，生於南宋寧宗嘉泰二年（1202），卒於理宗景定二年（1261）。所撰《數學九章》成書於淳祐七年（1247），時宋室南渡已百餘年，國勢日蹙，蒙古鐵騎壓境。卷八軍旅之題，正反映當時岌岌可危之軍事情勢。

Qin Jiushao (ca. 1202–1261), styled Daogu, was one of the foremost mathematicians of the Song Dynasty. The *Shuxue Jiuzhang* was completed in 1247, during the turbulent final decades of the Southern Song. The military problems in Fascicle 8 directly reflect the desperate military situation of the time, as the Song faced existential threats from the Mongols under Genghis Khan and later Kublai Khan.

卷九市易之題，則反映南宋繁榮之商業經濟：紙幣交子會子之流通、官營鹽茶之專賣、長途販運之興盛。此皆中古時期世界最發達之商業體系。

The trade problems in Fascicle 9 reflect the sophisticated commercial economy of the Song, with its paper currency (交子, 會子), government monopolies on salt and tea, and extensive long-distance trade networks. This was one of the most advanced commercial systems in the medieval world.

秦九韶自述其書：「大則可以通神明，順性命；小則可以經世務，類萬物。」數學九章不獨為數學之鉅著，亦為研究宋代社會經濟軍事之重要史料。

Qin Jiushao described his own work: “In the large sense it can communicate with the divine and accord with the nature of things; in the small sense it can govern worldly affairs and classify all things.” The *Shuxue Jiuzhang* is not only a masterpiece of mathematics, but also an important historical source for studying the society, economy, and military of the Song Dynasty.

Classical Chinese Mathematics

孫子算經

Sunzi Suanjing

Master Sun's Mathematical Manual

Bilingual English-Chinese Edition

English Translation • Original Classical Chinese

Attributed to **Sunzi** (Master Sun)

c. 3rd-5th century CE

One of the Ten Computational Canons (算經十書)

Listed during the Tang dynasty (618-907 CE)

**Contains the earliest known description of the
Chinese Remainder Theorem**

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1 Introduction

關於本書

孫子算經是中國現存最早的數學著作之一，成書於公元三世紀至五世紀之間。唐代將其列為算經十書之一。作者孫子（「孫子」為尊稱）的身份至今仍不確定，但其生活年代遠早於《孫子兵法》的作者孫武。

全書共分三卷（卷）：

- **卷上**：度量衡單位、籌算記數法、乘除算法、九九乘法表
- **卷中**：分數運算、幾何面積與體積計算
- **卷下**：三十六道算題，包括著名的「物不知數」問題——中國剩餘定理的最早記載

歷史意義

孫子算經在數學史上佔有特殊的地位。其最著名的貢獻是**卷下第 26 題**——「物不知數」問題：

今有物，不知其數。三、三數之，賸二；
五、五數之，賸三；七、七數之，賸二。
問物幾何？

此問題提供了公式：

$$N = 70r_1 + 21r_2 + 15r_3 - 105k$$

其中 r_1, r_2, r_3 分別為以 3、5、7 除後的餘數。

Introduction

The *Sunzi Suanjing* (孫子算經, “Master Sun’s Mathematical Manual”) is one of the earliest surviving Chinese mathematical texts, written during the 3rd to 5th centuries CE. It was listed as one of the *Ten Computational Canons* (算經十書) during the Tang dynasty. The identity of its author, Sunzi (“Master Sun”), remains unknown, though he lived much later than his namesake Sun Tzu, the author of *The Art of War*.

The treatise consists of three volumes (卷):

- **Volume I** (卷上): Units of measurement, counting rod notation, multiplication and division algorithms
- **Volume II** (卷中): Fractions, geometry (areas, volumes), and practical problems
- **Volume III** (卷下): 36 problems including the famous “Problem of the Unknown Quantity”—the **Chinese Remainder Theorem**

Historical Significance

The *Sunzi Suanjing* holds a special place in the history of mathematics. Its most celebrated contribution is **Problem 26 of Volume III**:

“There are certain things whose number is unknown. If we count them by threes, we have two left over; by fives, we have three left over; and by sevens, two are left over. How many things are there?”

This problem provides the formula:

$$N = 70r_1 + 21r_2 + 15r_3 - 105k$$

where r_1, r_2, r_3 are the remainders upon division by 3, 5, and 7.

Preface 序

孫子序

孫子曰：夫算者，天地之經緯，群生之元首；五常之本末，陰陽之父母；星辰之建號，三光之表裏；五行之準平，四時之終始；萬物之祖宗，六藝之綱紀。稽群倫之聚散，考二氣之降升；推寒暑之迭運，步遠近之殊同；觀天道精微之兆基，察地理從橫之長短；采神祇之所在，極成敗之符驗；窮道德之理，究性命之情。

立規矩，準方圓，謹法度，約尺丈，立權衡，平重輕，剖毫釐，析黍稷；歷億載而不朽，施八極而無疆。散之不可勝究，斂之不盈掌握。

嚮之者富有餘，背之者貧且窶；心開者幼沖而即悟，意閉者皓首而難精。夫欲學之者必務量能揆己，志在所專。如是則焉有不成者哉。

Preface by Sunzi

Sun Zi said: Calculation is the meridian and latitude of heaven and earth, the origin and head of all living beings; the root and branch of the Five Constants, the father and mother of yin and yang; the establishment of titles for stars and constellations, the exterior and interior of the Three Lights; the standard balance of the Five Elements, the beginning and end of the Four Seasons; the ancestors of all things, the framework and discipline of the Six Arts.

Investigate the gathering and dispersing of all categories, examine the descending and ascending movements of the Two Qi; track the alternating cycles of cold and heat, pace the differences and similarities between near and far; observe the subtle omens and foundations of the heavenly Dao, examine the lengths and widths of the horizontal and vertical features of geography; gather signs of where gods and spirits reside, thoroughly test omens of success and failure; exhaust the principles of virtue and morality, investigate the nature and essence of life.

Establish rules and compasses, standardize squares and circles, observe laws and regulations carefully, measure with precision in *chi* and *zhang*, set up weights and balances, equalize the heavy and light, dissect thousandths of an inch, analyze grains accumulated one by one; enduring through countless generations without decay, its influence extends to all directions without limit. When dispersed, it is beyond full comprehension; when gathered, it does not fill the palm of one's hand.

Those who face toward it are rich with abundance, while those who turn away from it are poor and destitute; those whose minds open understand even as children, while those whose intentions remain closed find mastery difficult even with a white beard. Therefore, one who wishes to study must necessarily measure his own ability and assess himself carefully, setting his mind on specialization in a particular field. If so, how could there be any failure?

2 Volume I 卷上

1. Length Measurements 度之所起

問 度之所起，起於忽。欲知其忽，蠶所生，吐絲為忽。十忽為一秒，十秒為一毫，十毫為一釐，十釐為一分，十分為一寸，十寸為一尺，十尺為一丈，十丈為一引；五十尺為一端；四十尺為一疋；六尺為一步。二百四十步為一畝。三百步為一里。

Q The origin of measurement begins with the “hu” (忽). To understand the hu, it is said that silkworms produce it by spinning silk. Ten hu make one miao (秒), ten miao make one hao (毫), ten hao make one li (釐), ten li make one fen (分), ten fen make one cun (寸), ten cun make one chi (尺), ten chi make one zhang (丈), and ten zhang make one yin (引); fifty chi make one duan (端); forty chi make one pi (疋); six chi make one bu (步). Two hundred and forty bu make one mu (畝). Three hundred bu make one li (里).

2. Weight and Capacity 稱量之所起

問 稱之所起，起於黍。十黍為一綮，十綮為一銖，二十四銖為一兩，十六兩為一斤，三十斤為一鈞，四鈞為一石。量之所起，起於粟。六粟為一圭，十圭為一抄，十抄為一撮，十撮為一勺，十勺為一合，十合為一升，十升為一斗，十斗為一斛。

Q The origin of weight begins with the grain of millet (黍). Ten millet grains make one lei (綮), ten lei make one zhu (銖), twenty-four zhu make one liang (兩), sixteen liang make one jin (斤), thirty jin make one jun (鈞), and four jun make one dan (石). The origin of capacity begins with the grain of millet. Six millet grains make one gui (圭), ten gui make one chao (抄), ten chao make one cuo (撮), ten cuo make one shao (勺), ten shao make one ge (合), ten ge make one sheng (升), ten sheng make one dou (斗), and ten dou make one hu (斛).

3. Large Numbers 大數之法

問 凡大數之法，萬萬曰億，萬萬億曰兆，萬萬兆曰京，萬萬京曰垓，萬萬垓曰秭，萬萬秭曰壤，萬萬壤曰溝，萬萬溝曰澗，萬萬澗曰正，萬萬正曰載。

Q Generally, the method for large numbers: ten thousand times ten thousand is yi (億, 10^8); ten thousand times yi is zhao (兆, 10^{16}); ten thousand times zhao is jing (京, 10^{24}); and so on up to zai (載, 10^{80}).

4. Geometric Ratios 方圓之比

問 周三徑一。方五邪七；見邪求方，五之，七而一；見方求邪，七之，五而一。

Q The circumference of a circle is three times its diameter ($\pi \approx 3$). A square with side length five has a diagonal of seven; to find the side from the diagonal, multiply by five and divide by seven; to find the diagonal from the side, multiply by seven and divide by five.

5. Weights of Materials 物之輕重

問 黃金方寸重一斤。白金方寸重一十四兩。玉方寸重一十二兩。銅方寸重七兩半。鉛方寸重九兩半。鐵方寸重六兩。石方寸重三兩。

Q One cubic cun of gold weighs one jin. One cubic cun of silver weighs fourteen liang. One cubic cun of jade weighs twelve liang. One cubic cun of copper weighs seven and a half liang. One cubic cun of lead weighs nine and a half liang. One cubic cun of iron weighs six liang. One cubic cun of stone weighs three liang.

6. Counting Rod Notation 算位之制

問 凡算之法，先識其位，一從十橫，百立千僵，千十相望，萬百相當。

籌算記數法：個位（1--9）縱式；十位（10--90）橫式；百位（100--900）縱式；千位（1000--9000）橫式。縱橫交替，以定位值。

Q The method for calculation begins by identifying the positions: one is vertical, ten is horizontal; a hundred stands upright and a thousand lies flat.

Counting Rod Numeral System: Units (1-9): **vertical**; Tens (10-90): **horizontal**; Hundreds (100-900): **vertical**; Thousands (1000-9000): **horizontal**. The pattern alternates.

7. Method of Multiplication 乘法

問 凡乘之法，重置其位。上下相觀，上位有十步至十，有百步至百，有千步至千。以上命下，所得之數列於中位。言十即過，不滿自如。上位乘訖者先去之。下位乘訖者則俱退之。六不積，五不隻。上下相乘，至盡則已。

Q In multiplication, the positions are arranged again for calculation. By observing vertically and horizontally, if the upper position has ten, it moves to the tens place; if it has a hundred, it moves to the hundreds place; if it has a thousand, it moves to the thousands place. The upper number is used to multiply the lower one, and the resulting product is placed in the middle position. If it reaches ten, carry over; if less than ten, leave as is. After the upper position has been multiplied, remove it first. Once the lower position is multiplied, both are moved back. Six does not accumulate; five does not count as a single unit. Multiplying vertically and horizontally continues until the end is reached.

8. Method of Division 除法

問 凡除之法，與乘正異。乘得在中央，除得在上方。假令六為法，百為實。以六除百，當進之二等，令在正百下，以六除一，則法多而實少，不可除，故當退就十位。以法除實，言一六而折百為四十，故可除。若實多法少，自當百之，不當復退。故或步法十者置於十位，百者置於百位。上位有空絕者，法退二位。餘法皆如乘時。實有餘者，以法命之，以法為母。實餘為子。

Q The method of division differs from multiplication. In multiplication, the result is placed in the center; in division, it appears at the top. Suppose six is the divisor (法) and one hundred is the dividend (實). Dividing one hundred by six, it should be advanced two positions and placed under the hundreds place. Dividing one by six results in a larger divisor than dividend, which is not possible; therefore, we must retreat to the tens position. Using the divisor to divide the dividend: saying "one six" and reducing a hundred by sixty results in forty, so division is possible. If the dividend is large and the divisor small, it naturally belongs to the hundreds place, and there is no need to retreat further. Therefore, if a step of ten is taken, it should be placed in the tens position; if a hundred, in the hundreds position. If there is an empty space at the upper position, the divisor must retreat two positions. The remaining methods follow those of multiplication. If there is a remainder in the dividend, name it using the divisor as the denominator. The remainder becomes the numerator.

9. Example: 81×81 九九八十一自乘

問 九九八十一，自相乘，得幾何？

答 答曰：六千五百六十一。

術 術曰：重置其位，以上八呼下八，八八六十四，即下六千四百於中位。以上八呼下一，一八如八，即於中位下八十。退下位一等，收上位八十。以上位一呼下八，一八如八，即於中位下八十。以上一呼下一，一一如一，即於中位下一。上下位俱收，中位即得六千五百六十一。

Q *Ninety-nine times eighty-one, when multiplied by itself, results in what?*

A *Answer: Six thousand five hundred and sixty-one.*

M *Method: Arrange the positions again, call out eight from above to multiply with eight below; eight eights are sixty-four. Place six thousand four hundred in the middle position. Call out eight from above to one below, one eight is eight; place eighty in the middle position. Move the lower position back by one rank and collect the upper value of eighty. Call out one from above to eight below, one eight is eight, place eighty in the middle position accordingly. Call out one from the upper position to multiply with one from the lower; one times one is one, so place one in the middle position. After collecting both the upper and lower positions, the middle position yields six thousand five hundred sixty-one.*

Modern: $81 \times 81 = (80+1)^2 = 6400+80+80+1 = 6561.$

10. Example: $6561 \div 9$ 九人分之

問 六千五百六十一，九人分之，問人得幾何？

答 答曰：七百二十九。

術 術曰：先置六千五百六十一於中位，為實。下列九人為法。上位置七百，以上七呼下九，七九六十三，即除中位六千三百。退下位一等，即上位置二十。以上二呼下九，二九十八，即除中位一百八十。又更退下位一等，即上位更置九，即以上九呼下九，九九八十一，即除中位八十一。中位盡，收下位。上位所得即人之所得。

Q Six thousand five hundred sixty-one is to be divided among nine people; how much does each person receive?

A Answer: Seven hundred and twenty-nine.

M Method: First place six thousand five hundred sixty-one in the middle position as the dividend (實). Place nine people below as the divisor (法). Place seven hundred in the upper position; call out seven from above to multiply by nine below, seven nines are sixty-three, and subtract six thousand three hundred from the middle. Move the lower position back one rank; place twenty in the upper position. Call out two from above to multiply by nine from below, two nines are eighteen, and subtract one hundred eighty from the middle. Move the lower position back another rank, place nine in the upper position again. Call out nine from above to multiply by nine below; nine nines are eighty-one and subtract eighty-one from the middle. When the middle position is exhausted, collect the lower positions. The result in the upper position is what each person receives.

Modern: $6561 \div 9 = 729$.

3 Volume II 卷中

1. Reducing Fractions 約分

問 今有一十八分之一十二。問約之得幾何？

答 答曰：三分之二。

術 術曰：置十八分在下，一十二分在上。副置二位，以少減多，等數得六，為法。約之，即得。

Q Now there is twelve eighteenths. Ask: reducing it, what does it equal?

A Answer: Two thirds.

M Method: Place eighteen parts below and twelve parts above. Set up both positions, subtract the smaller from the larger; the equal number (等數, GCD) obtained is six, which is the divisor. Reduce by it, and you obtain the answer.

Modern: $\frac{12}{18} = \frac{12 \div 6}{18 \div 6} = \frac{2}{3}$.

2. Grain Conversion 粟米換算

問 今有粟一斗，問為糲米幾何？

答 答曰：六升。

術 術曰：置粟一斗，十升。以糲米率三十乘之，得三百升，為實。以粟率五十為法，除之，即得。

Q Now there is one dou of millet. How much coarse rice does it make?

A Answer: Six sheng.

M Method: Place one dou of millet, which equals ten sheng. Multiply by the coarse rice rate of thirty to get three hundred sheng, as the dividend. Use the millet rate of fifty as the divisor and divide; thus you obtain the answer.

Modern: $10 \times 30 \div 50 = 6$ sheng.

3. Brick Calculation 積磚之數

問 今有屋基南北三丈，東西六丈，欲以磚砌之。凡積二尺，用磚五枚。問計幾何？

答 答曰：四千五百枚。

術 術曰：置東西六丈，以南北三丈乘之，得一千八百尺。以五乘之，得九千尺。以二除之，即得。

Q Now there is a house foundation with a north-south length of three zhang and an east-west width of six zhang, and one wishes to build it using bricks. Generally for every two cubic chi in volume, five bricks are used. Ask how many will be needed?

A Answer: Four thousand five hundred bricks.

M Method: Place the east-west length of six zhang, multiply it by the north-south length of three zhang to get one thousand eight hundred chi. Multiply it by five to get nine thousand chi. Divide it by two, and you will obtain the answer.

Modern: Area = $30 \times 60 = 1800$ sq chi. Bricks = $1800 \times 5/2 = 4500$.

4. Area of a Circular Field 圓田面積

問 今有圓田周三百步，徑一百步。問得田幾何？

答 答曰：三十一畝奇六十步。

術 術曰：先置周三百步，半之，得一百五十步。又置徑一百步，半之，得五十步。相乘，得七千五百步。以畝法二百四十步除之，即得。

Q Now there is a round field with circumference three hundred bu, diameter one hundred bu. Ask how much land does it amount to?

A Answer: Thirty-one mu and sixty bu extra.

M Method: First place the circumference of three hundred bu, halve it to get one hundred fifty bu. Also place the diameter of one hundred bu, halve it, getting fifty bu. Multiply them together to get seven thousand five hundred bu. Divide it by the mu standard of two hundred forty bu, and you will obtain the area.

Modern: $\text{Area} \approx \frac{C}{2} \times \frac{d}{2} = 150 \times 50 = 7500 \text{ sq bu} = 31 \text{ mu} + 60 \text{ bu}.$

5. Square Field with Mulberry 方田桑生中央

問 今有方田，桑生中央。從角至桑一百四十七步。問為田幾何？

答 答曰：一頃八十三畝奇一百八十步。

術 術曰：置角至桑一百四十七步，倍之，得二百九十四步。以五乘之，得一千四百七十步。以七除之，得二百一十步。自相乘，得四萬四千一百步。以二百四十步除之，即得。

Q Now there is a square field with mulberry trees growing in the center. From the corner to the mulberry tree is one hundred forty-seven steps. Ask how much field does it amount to?

A Answer: One qing, eighty-three mu and one hundred eighty steps extra.

M Method: Place the distance from corner to mulberry tree as one hundred forty-seven bu, double it to get two hundred ninety-four bu. Multiply it by five to get one thousand four hundred seventy steps. Divide it by seven, getting two hundred ten steps. Multiply it by itself to get forty-four thousand one hundred bu. Divide it by two hundred forty steps, and you will get the result.

Modern: $\text{Diagonal} = 147 \text{ bu}.$ Using $\sqrt{2} \approx \frac{7}{5}$, $\text{side } s = 147 \times 2 \times \frac{5}{7} = 210 \text{ bu}.$ $\text{Area} = 210^2 = 44100 \text{ sq bu}.$

4 Volume III 卷下

1. Rent Distribution 九家共輸租

問 今有甲、乙、丙、丁、戊、己、庚、辛、壬九家共輸租。甲出三十五斛，乙出四十六斛，丙出五十七斛，丁出六十八斛，戊出七十九斛，己出八十斛，庚出一百斛，辛出二百一十斛，壬出三百二十五斛。凡九家共輸租一千斛。僦運直折二百斛外，問家各幾何？

答 答曰：甲二十八斛。乙三十六斛八斗。丙四十五斛六斗。丁五十四斛四斗。戊六十三斛二斗。己六十四斛。庚八十斛。辛一百六十八斛。壬二百六十斛。

術 術曰：置甲出三十五斛，以四乘之，得一百四十斛。以五除之，得二十八斛。……

Q Now there are nine families jointly paying rent: Jia 35 hu, Yi 46 hu, Bing 57 hu, Ding 68 hu, Wu 79 hu, Ji 80 hu, Geng 100 hu, Xin 210 hu, Ren 325 hu. Total 1,000 hu. After deducting transport cost of 200 hu, how much should each family contribute?

A Answer: Jia 28 hu, Yi 36 hu 8 dou, Bing 45 hu 6 dou, Ding 54 hu 4 dou, Wu 63 hu 2 dou, Ji 64 hu, Geng 80 hu, Xin 168 hu, Ren 260 hu.

M Method: Take each family's contribution, multiply by four, and divide by five.

Modern: After 200 hu deduction, 800 hu remains. Each pays $\frac{4}{5}$ of original.

2. Pile of Millet 平地聚粟

問 今有平地聚粟，下周三丈六尺，高四尺五寸。問粟幾何？

答 答曰：一百斛。

術 術曰：置周三丈六尺，自相乘，得一千二百九十六尺。以高四尺五寸乘之，得五千八百三十二尺。以三十六除之，得一百六十二尺。以斛法一尺六寸二分除之，即得。

Q Now there is a pile of millet on level ground, with base circumference of three zhang six chi and height of four chi five cun. How much millet is there?

A Answer: One hundred hu.

M Method: Place the circumference of three zhang six chi, multiply it by itself to get 1296 chi. Multiply by the height of 4 chi 5 cun to get 5832 chi. Divide by 36 to get 162 chi. Divide by the hu standard of 1 chi 6 cun 2 fen.

Modern: Volume $\approx \frac{C^2 \times h}{36} = \frac{1296 \times 4.5}{36} = 162$ cu chi. $162 \div 1.62 = 100$ hu.

3. Go Board 碁局

問 今有碁局方一十九道。問用碁幾何？

答 答曰：三百六十一。

術 術曰：置一十九道，自相乘之，即得。

Q Now there is a Go board with a square pattern of nineteen lines on each side. How many pieces (intersections) are there?

A Answer: Three hundred sixty-one.

M Method: Take the nineteen lines, multiply them by themselves, and you get the result.

Modern: $19^2 = 361$ intersection points on a Go board.

4. People and Carts 人與車

問 今有三人共車，二車空；二人共車，九人步。問人與車各幾何？

答 答曰：一十五車。三十九人。

術 術曰：置二車，以三乘之，得六。加步者九人，得車一十五。欲知人者，以二乘車，加九人，即得。

Q Now there are three people sharing a cart, and two carts remain empty; two people share another cart, while nine people walk. What are the numbers of people and carts?

A Answer: Fifteen carts, thirty-nine people.

M Method: Place two carts, multiply by three to get six. Add the nine people who walk, getting fifteen carts. To find the number of people, multiply two by the number of carts and add nine people.

Modern: $3(c - 2) = p$ and $2c + 9 = p$, giving $c = 15$, $p = 39$.

5. Guests at a Feast 婦人河上蕩桮

問 今有婦人河上蕩桮。津吏問曰：「桮何以多？」婦人曰：「家有客。」津吏曰：「客幾何？」婦人曰：「二人共飯，三人共羹，四人共肉，凡用桮六十五。不知客幾何？」

答 答曰：六十人。

術 術曰：置六十五桮，以一十二乘之，得七百八十，以十三除之，即得。

Q Now there is a woman washing cups at the riverbank. The ferry official asked, "Why so many cups?" The woman replied, "There are guests at my home." "How many guests?" "Two people share a meal, three share soup, four share meat, and in total sixty-five cups are used."

A Answer: Sixty people.

M Method: Take the 65 cups, multiply by twelve to get 780, then divide by thirteen.

Modern: Each uses $\frac{1}{2} + \frac{1}{3} + \frac{1}{4} = \frac{13}{12}$ cups. $65 \div \frac{13}{12} = 60$.

6. Wood and Rope 木與繩

問 今有木，不知長短。引繩度之，餘繩四尺五寸。屈繩量之，不足一尺。問木長幾何？

答 答曰：六尺五寸。

術 術曰：置餘繩四尺五寸，加不足一尺，共五尺五寸。倍之，得一丈一尺。減餘四尺五寸，即得。

Q Now there is a piece of wood whose length is unknown. Using a rope to measure it, the rope has 4 chi 5 cun left over. Folding the rope to measure it results in one chi short. What is the length of the wood?

A Answer: Six chi and five cun.

M Method: Place the excess rope of 4 chi 5 cun, add the deficit of 1 chi, obtaining 5 chi 5 cun. Double it to get 1 zhang 1 chi. Subtract the excess of 4 chi 5 cun.

Modern: $R = W + 4.5$ and $R/2 = W - 1$. Solving: $W = 6.5$ chi.

7. Rice in a Vessel 器中米

問 今有器中米，不知其數。前人取半，中人三分取一，後人四分取一，餘米一斗五升。問本米幾何？

答 答曰：六斗。

術 術曰：置餘米一斗五升，以六乘之，得九斗。以二除之，得四斗五升。以四乘之，得一斛八斗。以三除之，即得。

Q Now there is rice in a vessel, but its quantity is unknown. The first person took half, the second took one-third of what remained, the third took one-fourth of what was left, and there were 1 dou 5 sheng remaining. What was the original amount?

A Answer: Six dou.

M Method: Place the remaining rice of 1 dou 5 sheng, multiply by six to get 9 dou. Divide by two to get 4 dou 5 sheng. Multiply by four to get 1 hu 8 dou. Divide by three.

Modern: $\frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} \times x = \frac{x}{4} = 1.5$ dou. So $x = 6$ dou.

8. Birds and Beasts 獸六首四足

問 今有獸六首四足，禽四首二足。上有七十六首，下有四十六足。問禽、獸各幾何？

答 答曰：八獸，七禽。

術 術曰：倍足以減首，餘，半之，即禽。四乘禽數，減足，餘，半之，即獸。

Q Now there are beasts with six heads and four feet, and birds with four heads and two feet. Above there are seventy-six heads, below there are forty-six feet. Ask: How many birds and beasts are there?

A Answer: Eight beasts, seven birds.

M Method: Double the feet and subtract the heads; halve the remainder to get the number of birds. Multiply the number of birds by four, subtract from the feet; halve the remainder to get the number of beasts.

Modern: $6b + 4i = 76$, $4b + 2i = 46$. Solving: $b = 8$, $i = 7$.

9. One Hundred Deer 百鹿入城

問 今有百鹿入城，家取一鹿，不盡；又三家共一鹿，適盡。問城中家幾何？

答 答曰：七十五家。

術 術曰：以盈不足取之。假令七十二家，鹿不盡四。令之九十家，鹿不足二十。置七十二於右上，盈四於右下。置九十於左上，不足二十於左下。維乘之，所得，并為實。并盈不足為法。除之，即得。

Q Now there are one hundred deer entering a city; each household takes one deer, but they do not finish; then three households share one deer and it is just finished. Ask: How many households are in the city?

A Answer: Seventy-five households.

M Method: Use the surplus-and-deficit method. Suppose 72 households: 4 deer remain (surplus). Suppose 90: 20 deer short (deficit). Cross-multiply and add to get the dividend. Add surplus and deficit to get the divisor.

Modern: $h + h/3 = 100$, so $4h/3 = 100$, giving $h = 75$.

10. Pheasants and Rabbits 雉兔同籠

問 今有雉、兔同籠，上有三十五頭，下九十四足。問雉、兔各幾何？

答 答曰：雉二十三。兔一十二。

術 術曰：上置三十五頭，下置九十四足。半其足，得四十七。以少減多，再命之，上三除下三，上五除下五。下有一除上一，下有二除上二，即得。

又術曰：上置頭，下置足。半其足，以頭除足，以足除頭，即得。

Q Now there are pheasants and rabbits in a cage, with thirty-five heads above and ninety-four feet below. Ask: How many pheasants and how many rabbits are there?

A Answer: Twenty-three pheasants, twelve rabbits.

M Method: Place thirty-five heads above, ninety-four feet below. Halve the feet to get forty-seven. Subtract the smaller from the larger, and command it again.

Alternative method: Place heads above, feet below. Halve the feet, use heads to remove feet and feet to remove heads.

Modern: $p + r = 35$, $2p + 4r = 94$. Solving: $p = 23$, $r = 12$.

11. Nine Dikes 出門望見九隄

問 今有出門望見九隄。隄有九木，木有九枝，枝有九巢，巢有九禽，禽有九雛，雛有九毛，毛有九色。問各幾何？

答 答曰：木八十一。枝七百二十九。巢六千五百六十一。禽五萬九千四十九。雛五十三萬一千四百四十一。毛四百七十八萬二千九百六十九。色四千三百四萬六千七百二十一。

術 術曰：置九隄，以九乘之，得木之數。又以九乘之，得枝之數。又以九乘之，得巢之數。又以九乘之，得禽之數。又以九乘之，得雛之數。又以九乘之，得毛之數。又以九乘之，得色之數。

Q Now there is someone who goes out of the gate and sees nine dikes. Each dike has nine trees, each tree has nine branches, each branch has nine nests, each nest has nine birds, each bird has nine chicks, each chick has nine feathers, and each feather has nine colors. Ask: How many of each are there?

A Answer: 81 trees, 729 branches, 6,561 nests, 59,049 birds, 531,441 chicks, 4,782,969 feathers, 43,046,721 colors.

M Method: Place nine dikes, multiply by nine to get the number of trees. Multiply by nine again for each successive level.

Modern: Powers of nine: $9^2 = 81$, $9^3 = 729$, $9^4 = 6561$, $9^5 = 59049$, $9^6 = 531441$, $9^7 = 4782969$, $9^8 = 43046721$.

12. Three Daughters 三女歸家

問 今有三女，長女五日一歸，中女四日一歸，少女三日一歸。問三女幾何日相會？

答 答曰：六十日。

術 術曰：置長女五日、中女四日、少女三日，於右方。各列一算於左方。維乘之，各得所到數：長女十二到，中女十五到，少女二十到。又各以歸日乘到數，即得。

Q Now there are three daughters: the eldest returns home every five days, the middle every four days, and the youngest every three days. Ask: After how many days will the three daughters meet again?

A Answer: Sixty days.

M Method: Place the eldest daughter's five days, the middle daughter's four days, and the youngest daughter's three days on the right side. Place one counter for each on the left side. Multiply them, and you get their respective numbers of arrivals: twelve, fifteen, and twenty. Then multiply each return day by the number of arrivals.

Modern: This is the least common multiple: $\text{lcm}(3, 4, 5) = 60$.

物不知數 — The Chinese Remainder Theorem

問 今有物，不知其數。三、三數之，賸二；五、五數之，賸三；七、七數之，賸二。問物幾何？

答 答曰：二十三。

術 術曰：「三、三數之，賸二」，置一百四十；「五、五數之，賸三」，置六十三；「七、七數之，賸二」，置三十。并之，得二百三十三。以二百一十減之，即得。

凡三、三數之，賸一，則置七十；五、五數之，賸一，則置二十一；七、七數之，賸一，則置十五。一百六以上，以一百五減之，即得。

Q Now there is an object, but we do not know its quantity. Counting it in groups of three leaves a remainder of two; counting it in groups of five leaves a remainder of three; counting it in groups of seven leaves a remainder of two. What is the quantity of the object?

A Answer: Twenty-three.

M Method: "Counting in groups of three, remainder two," set down 140; "Counting in groups of five, remainder three," set down 63; "Counting in groups of seven, remainder two," set down 30. Add them: 233. Subtract 210, and you will obtain the answer.

Generally, if counting in groups of three leaves remainder one, set down 70; if counting in groups of five leaves remainder one, set down 21; if counting in groups of seven leaves remainder one, set down 15. If the total is more than 160, subtract 150; thus you obtain the answer.

Modern Mathematical Analysis 現代數學分析

以現代符號表示，此問題要求找出最小的正整數 N ，滿足以下同餘方程組：

$$N \equiv 2 \pmod{3}$$

$$N \equiv 3 \pmod{5}$$

$$N \equiv 2 \pmod{7}$$

解法給出：

$$\begin{aligned} N &= 140 + 63 + 30 - 210 \\ &= 233 - 210 \\ &= 23 \end{aligned}$$

三個關鍵數字 **70**、**21**、**15** 有特殊性質：

- $70 = 2 \times 5 \times 7 \equiv 1 \pmod{3}$
- $21 = 1 \times 3 \times 7 \equiv 1 \pmod{5}$
- $15 = 1 \times 3 \times 5 \equiv 1 \pmod{7}$

由於 $3 \times 5 \times 7 = 105$ ，一般解為：

$$N = 70r_1 + 21r_2 + 15r_3 - 105k$$

其中 r_1, r_2, r_3 為餘數， k 為使 N 為最小正整數的整數。

In modern notation, this problem asks us to find the smallest positive integer N satisfying:

$$N \equiv 2 \pmod{3}$$

$$N \equiv 3 \pmod{5}$$

$$N \equiv 2 \pmod{7}$$

The method gives:

$$\begin{aligned} N &= 140 + 63 + 30 - 210 \\ &= 233 - 210 \\ &= 23 \end{aligned}$$

The three key numbers **70**, **21**, and **15** have special properties:

- $70 = 2 \times 5 \times 7 \equiv 1 \pmod{3}$
- $21 = 1 \times 3 \times 7 \equiv 1 \pmod{5}$
- $15 = 1 \times 3 \times 5 \equiv 1 \pmod{7}$

Since $3 \times 5 \times 7 = 105$, the general solution is:

$$N = 70r_1 + 21r_2 + 15r_3 - 105k$$

where r_1, r_2, r_3 are the remainders and k is chosen to make N positive and minimal.

Verification 驗算

$$23 \div 3 = 7 \text{ 餘 } \mathbf{2} \quad \checkmark$$

$$23 \div 5 = 4 \text{ 餘 } \mathbf{3} \quad \checkmark$$

$$23 \div 7 = 3 \text{ 餘 } \mathbf{2} \quad \checkmark$$

孫子公式：

$$N = 70 \times r_1 + 21 \times r_2 + 15 \times r_3 - 105 \times k$$

$$23 \div 3 = 7 \text{ remainder } \mathbf{2} \quad \checkmark$$

$$23 \div 5 = 4 \text{ remainder } \mathbf{3} \quad \checkmark$$

$$23 \div 7 = 3 \text{ remainder } \mathbf{2} \quad \checkmark$$

Sunzi's Formula:

$$N = 70 \times r_1 + 21 \times r_2 + 15 \times r_3 - 105 \times k$$

Historical Significance 歷史意義

此問題被稱為「物不知數」，是**中國剩餘定理**的最早記載。雖然孫子的文本只給出了具體的數值方法而沒有證明，但它包含了核心思想：

1. 找到一個數，它是除一個模數外所有模數的倍數，且除以該模數餘一
2. 將每個這樣的數乘以相應的餘數
3. 將結果相加，減去最小公倍數的倍數，得到最小正解

一般定理後來由**秦九韶**（1202–1261）在其《數書九章》（1247 年）中完整發展，稱之為「大衍求一術」。

This problem, known as the “**Problem of the Unknown Quantity**” (物不知數), is the earliest known statement of the **Chinese Remainder Theorem**. Sunzi’s text gives the essential idea:

1. Find numbers that are multiples of all but one modulus, leaving remainder 1 when divided by the remaining modulus
2. Multiply each such number by the corresponding remainder
3. Sum the results and subtract multiples of the LCM to get the smallest positive solution

The general theorem was later fully developed by **Qin Jiushao** (1202–1261) in his *Mathematical Treatise in Nine Sections* (數書九章, 1247 CE).

The “Sunzi Song” 孫子歌

一首總結此方法的韻文廣泛流傳，後由程大位收錄於《算法統宗》（1592 年）：

三人同行七十稀，
五樹梅花廿一枝，
七子團圓正半月，
除百零五便得知。

關鍵數字：**70**（模 3）、**21**（模 5）、**15**（模 7）、**105**（最小公倍數）。

A rhymed verse summarizing the method circulated widely and was later recorded by Cheng Dawei in his *Systematic Treatise on Algorithms* (算法統宗, 1592):

“Three people walking together—seventy is rare;
Five plum trees—twenty-one branches;
Seven sons reunite—half a full moon (15);
Subtract one hundred and five, then you will know.”

Key numbers: **70** for mod 3, **21** for mod 5, **15** for mod 7, **105** as the LCM.

Remaining Problems from Volume III 卷下餘題

13. Money Exchange 二人持錢

問 今有甲、乙二人，持錢各不知數。甲得乙中半，可滿四十八。乙得甲太半，亦滿四十八。問甲、乙二人元持錢各幾何？

答 答曰：甲持錢三十六。乙持錢二十四。

術 術曰：如方程求之。置二甲、一乙、錢九十六，於右方。置二甲、三乙、錢一百四十四，於左方。以右方二乘左方……得二十四錢，為乙錢。……得三十六，為甲錢也。

Q Now there are two people, Jia and Yi, each holding an unknown amount of money. If Jia receives half of what Yi has, he will have forty-eight. If Yi receives two-thirds of what Jia has, he too will have forty-eight. Ask: Originally, how much did Jia and Yi each hold?

A Answer: Jia originally held thirty-six units of money. Yi originally held twenty-four units.

M Method: Solve as a system of linear equations. Place 2 Jia, 1 Yi, 96 coins on the right. Place 2 Jia, 3 Yi, 144 coins on the left. Cross-multiply to obtain twenty-four coins as Yi's amount and thirty-six as Jia's amount.

Modern: $J + Y/2 = 48$, $Y + 2J/3 = 48$. Solving: $J = 36$, $Y = 24$.

14. Three Chickens 三雞共啄粟

問 今有三雞共啄粟一千一粒。雞啄一，母啄二，翁啄四。主責本粟，三雞主各償幾何？

答 答曰：雞雞主一百四十三。雞母主二百八十六。雞翁主五百七十二。

術 術曰：置粟一千一粒，為實。副并三雞所啄粟七粒，為法。除之，得一百四十三粒，為雞雞主所償之數。遞倍之，即得母、翁主所償之數。

Q Now there are three chickens pecking at one thousand and one grains of millet. A chick pecks one grain, a hen pecks two, and a rooster pecks four. How many grains should each owner compensate?

A Answer: The chick's owner compensates 143, the hen's owner compensates 286, and the rooster's owner compensates 572.

M Method: Place 1001 grains of millet as the dividend. Add together the three chickens' consumption of seven grains as the divisor. Divide: 143 grains is the chick owner's compensation. Double successively to get the hen's and rooster's owner's compensation.

Modern: Ratio is 1 : 2 : 4. $1001 \div 7 = 143$. So: 143, 286, 572.

15. Fish in a Ditch 九里渠

問 今有九里渠，三寸魚，頭頭相次。問魚得幾何？

答 答曰：五萬四千。

術 術曰：置九里，以三百步乘之，得二千七百步。又以六尺乘之，得一萬六千二百尺。上十之，得一十六萬二千寸。以魚三寸除之，即得。

Q Now there is a ditch nine li long; fish of three cun are lined up head to tail. Ask: How many fish can fit?

A Answer: Fifty-four thousand.

M Method: Place nine li, multiply by three hundred bu to get two thousand seven hundred bu. Multiply by six chi to get sixteen thousand two hundred chi. Convert upward by ten to get one hundred sixty-two thousand cun. Divide by the fish length of three cun.

Modern: $9 \text{ li} = 9 \times 300 \times 6 = 16200 \text{ chi} = 162000 \text{ cun}$. $162000/3 = 54000 \text{ fish}$.

16. Wheel Turns 輪匝

問 今有長安、洛陽相去九百里。車輪一匝一丈八尺。欲自洛陽至長安，問輪匝幾何？

答 答曰：九萬匝。

術 術曰：置九百里，以三百步乘之，得二十七萬步。又以六尺乘之，得一百六十二萬尺。以車輪一丈八尺為法。除之，即得。

Q Now there are Chang'an and Luoyang, 900 li apart. A cart wheel turns one revolution for every one zhang eight chi. If one wishes to travel from Luoyang to Chang'an, how many wheel turns are needed?

A Answer: Ninety thousand turns.

M Method: Place nine hundred li, multiply by three hundred bu to get two hundred seventy thousand bu. Multiply by six chi to get one million six hundred twenty thousand chi. Use the wheel circumference of one zhang eight chi as the divisor.

Modern: Distance = $900 \times 300 \times 6 = 1,620,000 \text{ chi}$. Wheel = 18 chi. Turns = $1,620,000/18 = 90,000$.

17. Predicting Gender 孕婦行年

問 今有孕婦行年二十九，難九月。未知所生？

答 答曰：生男。

術 術曰：置四十九，加難月，減行年。所餘，以天除一，地除二，人除三，四時除四，五行除五，六律除六，七星除七，八風除八，九州除九。其不盡者，奇則為男，耦則為女。

Q Now there is a pregnant woman whose age is twenty-nine, giving birth in the ninth month. It is unknown what will be born?

A Answer: A boy will be born.

M Method: Set forty-nine, add the month of childbirth, subtract the age. The remainder is divided by: Heaven 1, Earth 2, Humanity 3, Four Seasons 4, Five Elements 5, Six Laws 6, Seven Stars 7, Eight Winds 8, Nine Provinces 9. If the remainder is odd, a boy; if even, a girl.

Epilogue 跋

孫子算經是古代中國數學高度發展的非凡見證。雖然本文涵蓋了基本算術——度量衡單位、乘法表和簡單分數——但其最偉大的遺產在於簡短而精妙的「物不知數」問題，它為現代數論開啟了大門。

中國剩餘定理——這一方法在西方後來的稱呼——要過了一千多年才在歐洲被完全形式化。孫子所描述的算法包含了現代定理的基本要素：構造具有特定整除性質的輔助數字，將它們與給定的餘數結合，以及對最小公倍數取模的約簡。

這部著作的影響遠遠超出了中國。這個問題在整個東亞作為數學難題流傳，而秦九韶所發展的一般方法代表了中世紀世界數學最偉大的成就之一。

The *Sunzi Suanjing* stands as a remarkable testament to the sophistication of ancient Chinese mathematics. While the text covers basic arithmetic—units of measurement, multiplication tables, and simple fractions—its greatest legacy lies in the brief but brilliant “Problem of the Unknown Quantity” that opens the door to modern number theory.

The Chinese Remainder Theorem, as this method came to be known in the West, would not be fully formalized in Europe for over a millennium. The algorithm described by Sunzi contains the essential ingredients of the modern theorem: the construction of auxiliary numbers with specific divisibility properties, their combination with the given remainders, and reduction modulo the least common multiple.

The influence of this text extended far beyond China. The problem circulated as a mathematical puzzle throughout East Asia, and the general method developed by Qin Jiushao represents one of the great achievements of medieval mathematics anywhere in the world.

Detailed Analysis of the Mathematical Methods in the Nine Chapters

詳解九章算法
(Part I – 上)

BILINGUAL EDITION / 中英雙語版

By Yang Hui 楊輝

Southern Song Dynasty (c. 1238–1298 CE)

1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
1 5 10 10 5 1
1 6 15 20 15 6 1

The “Method Source for Square Root Extraction” (開方作法本源圖)

Left Column: Original Chinese text / Right Column: English translation

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1 Introduction / 引言

《詳解九章算法》乃南宋楊輝所著之重要數學著作。成書於一二六一年，詳解古代《九章算術》之各題算法。

Xiangjie Jiuzhang Suanfa (詳解九章算法, “Detailed Analysis of the Mathematical Methods in the Nine Chapters”) is one of the most important mathematical works of the Southern Song Dynasty, written by the mathematician **Yang Hui** 楊輝 (c. 1238–1298 CE). Completed in 1261, this work provides detailed explanations and commentaries on the classic *Jiuzhang Suanshu* (Nine Chapters on the Mathematical Art).

1.1 Historical Context / 歷史背景

原《九章算術》成書於漢代（約公元前二百年至公元二百年），凡二百四十六題，分九章。至宋代，原文因靖康之變（一一二七年）而散佚，傳抄訛誤，題隱法晦，學者難以入門。

The original *Jiuzhang Suanshu* dates back to the Han Dynasty (c. 200 BCE–200 CE) and contains 246 mathematical problems organized into nine chapters. By the Song Dynasty, the original text had become difficult to understand due to the loss of manuscripts during the *Jingkang Incident* (靖康之變, 1127 CE), corruption of transmitted texts through repeated copying, and obscure problem statements with difficult methods.

輝嘗聞學者，謂九章題問頗隱，法理難明，不得其門而入。於是以答參問，用草考法，因法推類，然後知斯文非古之全經也。

“I have heard scholars say that the problems in the Nine Chapters are rather obscure, and the methods difficult to understand, so one cannot find the door to enter. Therefore, using the answers to examine the questions, employing the worked solutions to investigate the methods, and following the methods to extend by analogy—only then do I know that this text is not the complete classic of antiquity.”

1.2 Structure of the Work / 全書結構

全書凡十二章：一、乘除（四十一題）；二、方田（三十八題）；三、粟米（四十六題）；四、衰分（二十題）；五、少廣（二十四題）；六、商功（二十八題）；七、均輸（二十八題）；八、盈不足（二十題）；九、方程（十八題）；十、勾股（三十八題）；十一、纂類。

The original work comprised 12 **chapters**: (1) Multiplication and Division — 41 problems; (2) Field Measurement — 38 problems; (3) Grain Conversion — 46 problems; (4) Proportional Distribution — 20 problems; (5) Short Width / Root Extraction — 24 problems; (6) Commercial Calculations / Engineering — 28 problems; (7) Equitable Distribution — 28 problems; (8) Excess and Deficit — 20 problems; (9) Systems of Linear Equations — 18 problems; (10) Right Triangles — 38 problems; (11) Classification / Compilation.

1.3 Editorial Method / 編纂之法

楊輝解題之法，分為四部：問（設題）、答（給出數值答案）、術（說明一般算法）、草（詳細逐步演算）。此四部結構，為中國數學著述之典範。

Yang Hui's approach to explaining the Nine Chapters was systematic: **Problem (問)** states the mathematical problem; **Answer (答)** gives the numerical answer; **Method (術)** states the general algorithm; **Worked Solution (草)** provides a detailed step-by-step calculation. This four-part structure became a model for Chinese mathematical exposition.

1.4 The Method Source for Square Root Extraction / 開方作法本源

楊輝所錄之開方作法本源圖（今稱楊輝三角或賈憲三角），源出北宋賈憲（約一〇一〇至一〇七〇年）之《釋鎖算書》。其圖列二項式係數至六次冪，用於開方運算。

One of Yang Hui's most celebrated contributions is preserving what we now call **Yang Hui's Triangle** (also known as Pascal's Triangle). In his work, he attributes this to the earlier mathematician **Jia Xian** (c. 1010–1070 CE), who called it the “Method Source for Square Root Extraction” (開方作法本源圖). The triangle provides binomial coefficients used in root extraction. Yang Hui wrote: 出釋鎖算書，賈憲用此術 (“This comes from the *Shisuo Suan Shu*; Jia Xian used this method”).

2 Original Preface / 原序

黃帝九章古序云：國家嘗設算科取士，選九章以為算經之首，蓋猶儒者之六經，醫家之難素，兵法之孫子歟。

昔聖宋紹興戊辰，算士榮榮謂靖康以來，罕有舊本，間有存者，狃於末習，向獲善本，得其全經，復起於學。以魏景元元年劉徽等、唐朝議大夫、行太史令上輕車都尉李淳風等注釋，聖宋右班直賈憲撰草。

輝嘗聞學者，謂九章題問頗隱，法理難明，不得其門而入。於是以答參問，用草考法，因法推類，然後知斯文非古之全經也。將後賢補贅之文，修前代已廢之法，刪立題術，又纂法問，詳著於後，儻得賢者，改而正諸，是所願也。

“The ancient preface to the Yellow Emperor’s Nine Chapters states: The state once established an examination in mathematics for selecting officials, choosing the Nine Chapters as the foremost mathematical classic—this is like the Six Classics for Confucian scholars, the *Nanjing* and *Suwen* for physicians, or the *Art of War* for military strategists.”

“In the past, during the Wuchen year of the Shaoxing reign of our Sacred Song Dynasty, the mathematician Rong Qi said that since the Jingkang period, old copies have been rare, and those occasionally surviving have been corrupted by inferior practices. Having obtained a good copy and recovered the complete classic, study was revived. The annotations were by Liu Hui et al. (Jingyuan first year of Wei) and by Li Chunfeng et al. (Senior Compiler of Tang Dynasty, Acting Grand Astrologer), while the detailed solutions were composed by Jia Xian of the Song Dynasty.”

“I have heard scholars say that the problems in the Nine Chapters are rather obscure, and the principles difficult to understand, so one cannot find the door to enter. Therefore, using the answers to examine the questions, employing the worked solutions to investigate the methods, and following the methods to extend by analogy—only then do I know that this text is not the complete classic of antiquity. Taking the supplementary texts added by later worthies and repairing the methods abandoned by earlier generations, I have reorganized the problem-methods and compiled the method-questions, recording them in detail below. If worthy scholars would correct and improve them, that is my wish.”

3 Chapter 1: Multiplication and Division / 乘除

第一章論乘除之基本運算，凡四十一題，多取方田章及粟米之題。

The first chapter covers fundamental operations of multiplication and division, with 41 problems drawn mostly from the “Field Measurement” (方田) chapter and grain conversion problems.

3.1 Rectangular Field Method / 直田法

術：以廣乘從得積。以畝法（一畝二百四十平方步）除之。

Method: Multiply the width (廣) by the length (從) to obtain the area (積). Divide by the unit conversion (畝法, 1 mu = 240 square bu).

PROBLEM 1

廣十五步，從十六步，問田幾何？答曰：一畝。 A field has width 15 bu and length 16 bu. What is the area? **Answer:** 1 mu.

SOLUTION (草曰):

廣十五步，從十六步。積 = $15 \times 16 = 240$ 平方步。一畝 = 二百四十平方步，故為一畝。

Width = 15 bu, Length = 16 bu. Area = $15 \times 16 = 240$ square bu. Since 1 mu = 240 square bu, the area is **1 mu**.

PROBLEM 2

廣十二步，從十四步，問田幾何？答曰：一百六十八步。 A field has width 12 bu and length 14 bu. What is the area in square bu? **Answer:** 168 square bu.

SOLUTION (草曰):

積 = $12 \times 14 = 168$ 平方步。

Area = $12 \times 14 = 168$ square bu.

3.2 Square Field Method / 方田法

術：方田之積，以里自乘，復以三百七十五畝乘之。

Method: Square the side length to obtain the area in square li. Multiply by 375 mu per square li for conversion.

PROBLEM 3

方田一里，問為田幾何？答曰：三頃七十五畝。 A square field has side 1 li. What is its area? **Answer:** 3 qing 75 mu.

SOLUTION (草曰):

積 = $1 \times 1 = 1$ 平方里。 $1 \times 375 = 375$ 畝 = 三頃七十五畝。

Area = $1 \times 1 = 1$ square li. $1 \times 375 = 375$ mu = **3 qing 75 mu**.

3.3 Fraction Multiplication / 乘分

術：分母乘全步，并子以為實。分母相乘為法。以法除實。

Method: For fractions with whole numbers: multiply each whole number by its denominator and add the numerator. Multiply these results to form the dividend (實). Multiply the denominators to form the divisor (法). Divide the dividend by the divisor.

PROBLEM Fraction Multiplication

田廣七步、四分步之三，從十五步、九分步之五，問：為田幾何？答曰：一百二十步，九分步之五。 A field has width $7\frac{3}{4}$ bu and length $15\frac{5}{9}$ bu. What is the area? **Answer:** $120\frac{5}{9}$ square bu.

SOLUTION (草曰):

廣： $7\frac{3}{4} = \frac{31}{4}$ 步。從： $15\frac{5}{9} = \frac{140}{9}$ 步。
積 = $\frac{31}{4} \times \frac{140}{9} = \frac{4340}{36} = 120\frac{5}{9}$ 平方步。

Width: $7\frac{3}{4} = \frac{31}{4}$ bu. Length: $15\frac{5}{9} = \frac{140}{9}$ bu.
Area = $\frac{31}{4} \times \frac{140}{9} = \frac{4340}{36} = 120\frac{5}{9}$ square bu.

3.4 Division with Fractions / 除分

術：以人數為法，錢數為實。有分者通之。實如法而一。

Method: Use the number of people as the divisor (法) and the amount of money as the dividend (實). If there are fractions, convert them. Divide the dividend by the divisor.

PROBLEM Division with Fractions

七人，均八錢三分錢之一，問人得幾何？答曰：人得一錢二十一分錢之四。Seven people share $8\frac{1}{3}$ qian equally. How much does each person receive? **Answer:** $1\frac{4}{21}$ qian each.

SOLUTION (草曰):

總數： $8\frac{1}{3} = \frac{25}{3}$ 錢。
每人得： $\frac{25}{3} \div 7 = \frac{25}{21} = 1\frac{4}{21}$ 錢。

Total: $8\frac{1}{3} = \frac{25}{3}$ qian.
Each receives: $\frac{25}{3} \div 7 = \frac{25}{21} = 1\frac{4}{21}$ qian.

4 Chapter 2: Field Measurement / 方田

方田章專論面積之計算與分數之運算。凡三十八題，涉及各種田形及分數四則運算。

The Field Measurement chapter focuses on area calculations and fraction operations. It originally contained 38 problems covering various field shapes and operations with fractions.

4.1 Reduction of Fractions / 約分

術：可半者半之。不可半者，副置分母子之數，以少減多，更相減損，求其等也。以等數約之。（即輾轉相除之歐幾里得算法。）

Method: If both numerator and denominator are even, halve them. If not, place the numerator and denominator separately and repeatedly subtract the smaller from the larger (Euclidean algorithm) until the greatest common divisor is found. Divide both by this common measure.

PROBLEM Reduction

十八分之十二。問：約之得幾何？答曰：三分之二。 Reduce $\frac{12}{18}$. **Answer:** $\frac{2}{3}$.

SOLUTION (草曰):

十二與十八之等數： $18 - 12 = 6$, $12 - 6 = 6$, $6 - 6 = 0$. 等數 = 6.
 $\frac{12 \div 6}{18 \div 6} =$
GCD of 12 and 18: $18 - 12 = 6$, $12 - 6 = 6$, $6 - 6 = 0$. GCD = 6.
 $\frac{12 \div 6}{18 \div 6} =$

PROBLEM

九十一分之四十九。問：約之得幾何？答曰：十三分之七。 Reduce $\frac{49}{91}$. **Answer:** $\frac{7}{13}$.

SOLUTION (草曰):

輾轉相除： $91 - 49 = 42$, $49 - 42 = 7$, $42 - 7 \times 6 = 0$. 等數 = 7.
 $\frac{49 \div 7}{91 \div 7} =$
Using the Euclidean algorithm: $91 - 49 = 42$, $49 - 42 = 7$, $42 - 7 \times 6 = 0$. GCD = 7.
 $\frac{49 \div 7}{91 \div 7} =$

4.2 Addition of Fractions / 合分

術：母互乘子，并以為實。母相乘為法。實如法而一。母同者直相從之。

Method: Cross-multiply the denominators with the numerators and add to form the dividend. Multiply the denominators to form the divisor. Divide. If the denominators are the same, simply add the numerators.

PROBLEM

二分之一、三分之二、四分之三、五分之四，合之得幾何？答曰：得二，餘六十分之四十三。 Add $\frac{1}{2} + \frac{2}{3} + \frac{3}{4} + \frac{4}{5}$. **Answer:** $2 \frac{43}{60}$.

SOLUTION (草曰):

互乘： $1 \times 3 \times 4 \times 5 + 2 \times 2 \times 4 \times 5 + 3 \times 2 \times 3 \times 5 + 4 \times 2 \times 3 \times 4 = 60 + 80 + 90 + 96 = 326$.
法： $2 \times 3 \times 4 \times 5 = 120$.
 $\frac{326}{120} = 2 \frac{86}{120} = 2 \frac{43}{60}$

Cross-multiply: $1 \times 3 \times 4 \times 5 + 2 \times 2 \times 4 \times 5 + 3 \times 2 \times 3 \times 5 + 4 \times 2 \times 3 \times 4 = 60 + 80 + 90 + 96 = 326$.
Divisor: $2 \times 3 \times 4 \times 5 = 120$.
 $\frac{326}{120} = 2 \frac{86}{120} = 2 \frac{43}{60}$.

4.3 Subtraction and Comparison of Fractions / 課分

術：母互乘子，以少減多，餘為實。母相乘為法。實如法而一。即兩分數之差也。

Method: Cross-multiply numerators, subtract the smaller from the larger. The remainder is the dividend, the product of denominators is the divisor. This gives the difference (also called “the excess”).

PROBLEM

八分之五，比二十五分之十六。問：孰多多幾何？答曰：二十五分之十六，多多二百分之三。 Compare $\frac{5}{8}$ and $\frac{16}{25}$. Which is larger and by how much? **Answer:** $\frac{16}{25}$ is larger by $\frac{3}{200}$.

SOLUTION (草曰):

$$\frac{16}{25} - \frac{5}{8} = \frac{16 \times 8 - 5 \times 25}{25 \times 8} = \frac{128 - 125}{200} = \frac{3}{200}.$$

故 $\frac{16}{25}$ 大 $3\frac{20}{200}$.

$$\frac{16}{25} - \frac{5}{8} = \frac{16 \times 8 - 5 \times 25}{25 \times 8} = \frac{128 - 125}{200} = \frac{3}{200}.$$

So $\frac{16}{25}$ is larger by

4.4 Equalizing Fractions / 平分

術：母互乘子，各為列實。并諸列實為平實。母相乘為法。以列數乘未并者，各為列實。以平實減列實，餘為減數；以列實減平實，餘為益數。減多益少，使各平於平實。

Method: Cross-multiply numerators. Sum all numerators for the “level dividend.” Multiply all denominators for the divisor. Multiply the number of terms by each uncrossed numerator for the column dividends. Subtract the level dividend from each column dividend to find what must be subtracted; subtract each column dividend from the level dividend to find what must be added. Combine the subtractions to benefit the smaller values. Divide the level dividend by the divisor to find the equalized value.

PROBLEM

二分之一、三分之二、四分之三，減多益少幾何？而平？答曰：減四分之三，求之者四，減三分之二，求之者一，益二分之一，求之者五，各平於三十六分之二十三。 Given $\frac{1}{2}$, $\frac{2}{3}$, $\frac{3}{4}$, redistribute by subtracting from larger and adding to smaller to make them equal. **Answer:** Subtract $\frac{4}{36}$ from $\frac{3}{4}$; subtract $\frac{1}{36}$ from $\frac{2}{3}$; add $\frac{5}{36}$ to $\frac{1}{2}$. All equalized to $\frac{23}{36}$.

SOLUTION (草曰):

平值 = $\frac{23}{36}$ 。
 $\frac{3}{4} = \frac{27}{36}$: 減 $\frac{4}{36}$ 。
 $\frac{2}{3} = \frac{24}{36}$: 減 $\frac{1}{36}$ 。
 $\frac{1}{2} = \frac{18}{36}$: 益 $\frac{5}{36}$ 。
 皆平於 $23\frac{23}{36}$ 。

Level value = $\frac{23}{36}$.
 $\frac{3}{4} = \frac{27}{36}$: subtract $\frac{4}{36}$.
 $\frac{2}{3} = \frac{24}{36}$: subtract $\frac{1}{36}$.
 $\frac{1}{2} = \frac{18}{36}$: add $\frac{5}{36}$.
 All equalized to $23\frac{23}{36}$.

Original Chinese (原文)

English Translation (英譯)

5 Chapter 3: Grain Conversion / 粟米

粟米章論各種穀物與商品之比例交換。凡四十六題，其中六題為基本乘除，三十一題為比例交換，九題為比例率。

The Grain Conversion chapter deals with proportional exchange between different types of grain and commodities. It originally contained 46 problems with 6 on basic multiplication/division, 31 on proportional exchange, and 9 on proportional rates.

5.1 Direct Exchange / 經率

術：以所買率為法，以所出錢為實。實如法得一。

Method: Use the money amount as the dividend (實) and the quantity purchased as the divisor (法). Divide the dividend by the divisor.

PROBLEM

錢一百六十文，買瓚甓十八枚，問枚價幾何？答曰：一枚八錢九分錢之八。 With 160 qian, buy 18 roof tiles. What is the price per tile? **Answer:** $8\frac{8}{9}$ qian per tile.

SOLUTION (草曰):

$$160 \div 18 = \frac{160}{18} = \frac{80}{9} = 8\frac{8}{9} \text{ 文/枚。}$$

$$160 \div 18 = \frac{160}{18} = \frac{80}{9} = 8\frac{8}{9} \text{ qian per tile.}$$

5.2 Grain Conversion Rates / 粟米換算率

《九章》立穀物換算之標準率：粟率五十；糲米率三十；粳米率二十七；御米率二十四；麥率四十五；菽率四十五。

The Nine Chapters establishes standard conversion rates between different grains: Millet (粟): rate = 50; Coarse rice (糲米): rate = 30; Fine rice (粳米): rate = 27; Polished rice (御米): rate = 24; Wheat (麥): rate = 45; Soybeans (菽): rate = 45.

術：以所有數乘所求率為實，以所有率為法，實如法而一。

Method: “What you seek multiplied by the given amount, divided by what you have.” (以所有數乘所求率為實，以所有率為法，實如法而一。)

PROBLEM

粟一斗。問為糲米幾何？答曰：六升。 Given 1 dou of millet, how much coarse rice (糲米) does it make? **Answer:** 6 sheng.

SOLUTION (草曰):

$$\text{粟率五十，糲米率三十。} \\ 1 \text{ 斗} \times \frac{30}{50} = \frac{3}{5} \text{ 斗} = 6 \text{ 升。}$$

$$\text{Millet rate} = 50, \text{ coarse rice rate} = 30. \\ 1 \text{ dou} \times \frac{30}{50} = \frac{3}{5} \text{ dou} = 6 \text{ sheng.}$$

PROBLEM

粟四斗五升。問為粳米幾何？答曰：二斗一升五分升之三。 Given 4 dou 5 sheng of millet, how much fine rice (粳米) does it make? **Answer:** 2 dou 1 sheng $\frac{3}{5}$.

SOLUTION (草曰):

$$4.5 \times \frac{27}{50} = \frac{9}{2} \times \frac{27}{50} = \frac{243}{100} = 2.43 \text{ 斗} = \text{二斗一升五分升之三。}$$

$$4.5 \times \frac{27}{50} = \frac{9}{2} \times \frac{27}{50} = \frac{243}{100} = 2.43 \text{ dou} = 2 \text{ dou } 1 \text{ sheng } \frac{3}{5}.$$

PROBLEM

糲米十五斗五升五分升之二。問為粟幾何？答曰：二十五斗九升。 Given 15 dou $5\frac{2}{5}$ sheng of coarse rice, how much millet was the original amount? **Answer:** 25 dou 9 sheng.

SOLUTION (草曰):

$$15\frac{2}{5} \text{ 斗} = \frac{77}{5} \text{ 斗。} \\ \frac{77}{5} \times \frac{50}{30} = \frac{777}{30} = 25.9 \text{ 斗} = \text{二十五斗九升。}$$

$$15\frac{2}{5} \text{ 斗} = \frac{777}{50} \text{ dou.} \\ \frac{777}{50} \times \frac{50}{30} = \frac{777}{30} = 25.9 \text{ dou} = 25 \text{ dou } 9 \text{ sheng.}$$

5.3 Silk and Currency Conversion / 絲錢換算

PROBLEM

絲一斤，價三百四十五，今有七兩一十二銖。問錢，答曰：一百六十一錢，三十二分錢之二十三。 Silk costs 345 qian per jin. If one has 7 liang 12 zhu of silk, what is its value in qian? **Answer:** $161 \frac{23}{32}$ qian.

SOLUTION (草曰):

一斤 = 十六兩 = 三百八十四銖。七兩十二銖 = $7 \times 24 + 12 = 180$ 銖。

$$\text{值} = 345 \times \frac{180}{384} = 345 \times \frac{15}{32} = \frac{5175}{32} = 161 \frac{23}{32} \text{ 錢。}$$

$$1 \text{ jin} = 16 \text{ liang} = 384 \text{ zhu. } 7 \text{ liang } 12 \text{ zhu} = 7 \times 24 + 12 = 180 \text{ zhu.}$$

$$\text{Value} = 345 \times \frac{180}{384} = 345 \times \frac{15}{32} = \frac{5175}{32} = 161 \frac{23}{32} \text{ qian.}$$

6 Chapter 4: Proportional Distribution / 衰分

衰分章論按給定比例分配數量之問題。凡二十題。

The Proportional Distribution chapter deals with dividing quantities in given proportions. It originally contained 20 problems.

6.1 The Method of Distribution / 衰分法

術：各置列衰，副并為法。以所分乘未并者，各自為實。實如法而一。不滿法者，以法命之。

Method: Set out the distribution rates (衰). Combine them as the divisor. Multiply what is to be distributed by each uncombined rate to form the column dividends. Divide each by the divisor. If there is a remainder, express it as a fraction over the divisor.

PROBLEM Five Ranks Sharing Deer / 五爵分鹿

大夫、不更、簪裹、上造、公士凡五人，以爵次高下均五鹿，問各幾何？答曰：大夫一鹿，三分鹿之二；不更一鹿，三分鹿之一；簪裹一鹿；上造鹿三分之二；公士鹿三分之一。Five nobles—a Dafu, a Bugeng, a Zanmao, a Shangzao, and a Gongshi—are to share 5 deer in proportion to their ranks (5:4:3:2:1). **Answer:** Dafu $1\frac{2}{3}$; Bugeng $1\frac{1}{3}$; Zanmao 1; Shangzao $\frac{2}{3}$; Gongshi $\frac{1}{3}$ deer.

SOLUTION (草曰):

衰：大夫五、不更四、簪裹三、上造二、公士一。
并 = $5 + 4 + 3 + 2 + 1 = 15$ 。
大夫： $5 \times \frac{5}{15} = \frac{25}{15} = 1\frac{2}{3}$ 鹿。
不更： $5 \times \frac{4}{15} = \frac{20}{15} = 1\frac{1}{3}$ 鹿。
簪裹： $5 \times \frac{3}{15} = 1$ 鹿。
上造： $5 \times \frac{2}{15} = \frac{2}{3}$ 鹿。
公士： $5 \times \frac{1}{15} = \frac{1}{3}$ 鹿。

Rates: Dafu 5, Bugeng 4, Zanmao 3, Shangzao 2, Gongshi 1.
Sum = $5 + 4 + 3 + 2 + 1 = 15$.
Dafu: $5 \times \frac{5}{15} = \frac{25}{15} = 1\frac{2}{3}$ deer.
Bugeng: $5 \times \frac{4}{15} = \frac{20}{15} = 1\frac{1}{3}$ deer.
Zanmao: $5 \times \frac{3}{15} = 1$ deer.
Shangzao: $5 \times \frac{2}{15} = \frac{2}{3}$ deer.
Gongshi: $5 \times \frac{1}{15} = \frac{1}{3}$ deer.

6.2 The Cow, Horse, and Sheep / 牛馬羊償粟

PROBLEM

牛、馬、羊食人苗，苗主責之粟五斗，牛食馬之半，馬食羊之半，欲衰償之？答曰：牛二斗八升，七分升之四；馬一斗四升，七分升之二；羊七升，七分升之一。A cow, a horse, and a sheep ate someone's crops. The owner demands 5 dou of millet as compensation. The cow ate twice what the horse ate, and the horse ate twice what the sheep ate. **Answer:** Cow $2\frac{4}{7}$ dou; Horse $1\frac{2}{7}$ dou; Sheep $\frac{1}{7}$ dou.

SOLUTION (草曰):

衰：牛四、馬二、羊一。
并 = $4 + 2 + 1 = 7$ 。
牛： $5 \times \frac{4}{7} = \frac{20}{7} = 2\frac{6}{7}$ 斗 = 二斗八升 $\frac{4}{7}$ 。
馬： $5 \times \frac{2}{7} = \frac{10}{7} = 1\frac{3}{7}$ 斗 = 一斗四升 $\frac{2}{7}$ 。
羊： $5 \times \frac{1}{7} = \frac{5}{7}$ 斗 = 七升 $\frac{1}{7}$ 。

The rates are: Cow 4, Horse 2, Sheep 1.
Sum = $4 + 2 + 1 = 7$.
Cow: $5 \times \frac{4}{7} = \frac{20}{7} = 2\frac{6}{7}$ dou = 2 dou 8 sheng $\frac{4}{7}$.
Horse: $5 \times \frac{2}{7} = \frac{10}{7} = 1\frac{3}{7}$ dou = 1 dou 4 sheng $\frac{2}{7}$.
Sheep: $5 \times \frac{1}{7} = \frac{5}{7}$ dou = 7 sheng $\frac{1}{7}$.

6.3 The Woman Skilled at Weaving / 女子善織

PROBLEM

女子善織，日自倍，五日織五尺。問日織數。答曰：初日一寸，三十一分寸之十九；二日三寸，三十一分寸之七；三日六寸，三十一分寸之十四；四日一尺二寸，三十一分寸之二十八；五日二尺四寸，三十一分寸之二十五。A woman skilled at weaving doubles her output each day. In 5 days she weaves 5 chi. **Answer:** Day 1: $\frac{50}{31}$ cun; Day 2: $\frac{100}{31}$ cun; Day 3: $\frac{200}{31}$ cun; Day 4: $\frac{400}{31}$ cun; Day 5: $\frac{800}{31}$ cun.

SOLUTION (草曰):

設首日織 x ，則總數 = $x + 2x + 4x + 8x + 16x = 31x = 5$ 尺。
 $x = \frac{5}{31}$ 尺 = $\frac{50}{31}$ 寸 = $1\frac{19}{31}$ 寸。
次日： $\frac{100}{31} = 3\frac{7}{31}$ 寸。
三日： $\frac{200}{31} = 6\frac{14}{31}$ 寸。
四日： $\frac{400}{31} = 12\frac{28}{31}$ 寸。
五日： $\frac{800}{31} = 25\frac{25}{31}$ 寸。

Let the first day's output be x . Then total = $x + 2x + 4x + 8x + 16x = 31x = 5$ chi.
 $x = \frac{5}{31}$ chi = $\frac{50}{31}$ cun = $1\frac{19}{31}$ cun.
Day 2: $\frac{100}{31} = 3\frac{7}{31}$ cun.
Day 3: $\frac{200}{31} = 6\frac{14}{31}$ cun.
Day 4: $\frac{400}{31} = 12\frac{28}{31}$ cun.
Day 5: $\frac{800}{31} = 25\frac{25}{31}$ cun.

7 Chapter 5: Short Width / Root Extraction / 少廣

少廣章論已知矩形面積與一邊而求另一邊，涉及分數除法與開方。凡二十四題。

The Short Width chapter deals with finding an unknown dimension of a rectangle given its area and one side, which leads to problems of division with fractions and root extraction. It originally contained 24 problems.

7.1 The Short Width Method / 少廣法

術：置全步及分子子。各以分母乘其子、全步，以少減多，通分內子，令母互乘。并所有以為法。以畝積二百四十步乘之為實。實如法而一，得從。

Method: Set out the whole steps and the fraction denominators and numerators. Separately multiply the denominators by themselves, then multiply the whole steps and numerators. Divide each by its original denominator and combine. Use this as the divisor. Multiply the whole steps by their combined denominators, then multiply by the field area (mu) in square bu for the dividend. Divide the dividend by the divisor.

PROBLEM Short Width 1

田一畝廣一步半，問從？答曰：一百六十步。 A field of 1 mu has width $1\frac{1}{2}$ bu. What is the length? **Answer:** 160 bu.

SOLUTION (草曰):

廣 = $1\frac{1}{2} = \frac{3}{2}$ 步。積 = 一畝 = 二百四十平方步。
從 = $240 \div \frac{3}{2} = 240 \times \frac{2}{3} = 160$ 步。

Width = $1\frac{1}{2} = \frac{3}{2}$ bu. Area = 1 mu = 240 square bu.
Length = $240 \div \frac{3}{2} = 240 \times \frac{2}{3} = 160$ bu.

PROBLEM

田一畝廣一步半、三分步之一。問從？答曰：一百三十步一十一分步之一十。 A field of 1 mu has width $1\frac{1}{2}\frac{1}{3}$ bu. What is the length? **Answer:** $130\frac{10}{11}$ bu.

SOLUTION (草曰):

廣 = $1 + \frac{1}{2} + \frac{1}{3} = \frac{6+3+2}{6} = \frac{11}{6}$ 步。
從 = $240 \div \frac{11}{6} = 240 \times \frac{6}{11} = \frac{1440}{11} = 130\frac{10}{11}$ 步。

Width = $1 + \frac{1}{2} + \frac{1}{3} = \frac{6+3+2}{6} = \frac{11}{6}$ bu.
Length = $240 \div \frac{11}{6} = 240 \times \frac{6}{11} = \frac{1440}{11} = 130\frac{10}{11}$ bu.

PROBLEM

田一畝廣一步半、三分步之一、四分步之一。問從答曰：一百一十五步五分步之一。 A field of 1 mu has width $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}$ bu. **Answer:** $115\frac{1}{5}$ bu.

SOLUTION (草曰):

廣 = $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} = \frac{12+6+4+3}{12} = \frac{25}{12}$ 步。
從 = $240 \div \frac{25}{12} = \frac{2880}{25} = 115\frac{1}{5}$ 步。

Width = $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} = \frac{12+6+4+3}{12} = \frac{25}{12}$ bu.
Length = $240 \div \frac{25}{12} = \frac{2880}{25} = 115\frac{1}{5}$ bu.

PROBLEM

田一畝廣一步半、三分步之一、四分步之一、五分步之一、六分步之一、七分步之一。問從答曰：九十二步一百二十一分步之六十八。 A field of 1 mu has width $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7}$ bu. **Answer:** $92\frac{68}{121}$ bu.

SOLUTION (草曰):

用逆少廣術：

廣 = $\frac{420+210+140+105+84+70+60}{420} = \frac{1089}{420}$ 。
約之：等數三，得 $\frac{363}{140}$ 。
從 = $240 \times \frac{140}{363} = \frac{33600}{363} = 92\frac{204}{363} = 92\frac{68}{121}$ 步。

Using the reverse combination-of-fractions method:

Width = $\frac{420+210+140+105+84+70+60}{420} = \frac{1089}{420}$.
Simplifying: $\text{GCD}(1089, 420) = 3$, so $\frac{363}{140}$.
Length = $240 \times \frac{140}{363} = \frac{33600}{363} = 92\frac{204}{363} = 92\frac{68}{121}$ bu.

8 The Method Source for Square Root Extraction / 開方作法本源圖

楊輝書中最著名者，為其所錄之開方作法本源圖（今稱楊輝三角）。楊輝稱此圖源出北宋賈憲（約一〇一〇至一〇七〇年）。

One of the most celebrated parts of Yang Hui's work is his recording of what we now call **Yang Hui's Triangle** (also known as Pascal's Triangle in the West). Yang Hui attributed this to Jia Xian (c. 1010–1070 CE), who called it the “Method Source for Square Root Extraction” (開方作法本源).

8.1 The Original Diagram / 原圖

楊輝釋此圖曰：

左表乃積數，右表乃隅算，中藏者皆廉，以廉乘商方，命實而除之。

Yang Hui presented the triangle with the following explanation: “The left diagonal consists of the accumulated numbers; the right diagonal consists of the corner calculations. Those hidden in the middle are all the ‘flanks’ (intermediate coefficients). Using the flanks, multiply the quotient square, and command the dividend to be divided by it.”

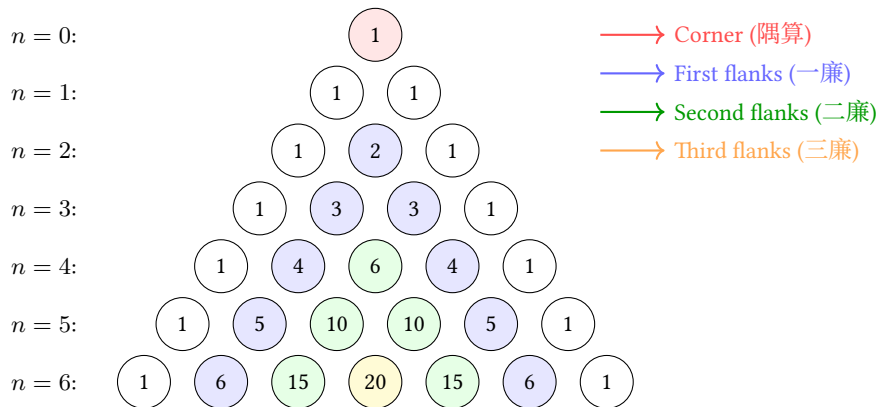


Fig. 1: Yang Hui's Triangle showing binomial coefficients up to the 6th power / 楊輝三角至六次冪

8.2 Explanation of the Structure / 結構釋義

左表（積數）：斜邊之常數項（皆為一），即 $(x + a)^n$ 展開式之常數係數。

右表（隅算）：最高次冪之係數（亦皆為一）。

中藏者（廉）：中間各數為廉，即各中間項之係數。

Left diagonal (左表): The constant terms (1, 1, 1, ...) represent the “accumulated numbers” (積數)—the coefficients of the constant term in $(x + a)^n$.

Right diagonal (右表): The coefficients of the highest power—also all 1's—represent the “corner calculations” (隅算).

Interior numbers (中藏者): The intermediate numbers are the “flanks” (廉)—the coefficients of the intermediate terms.

每行給出 $(x + a)^n$ 展開之係數：

$$(x + a)^0 = 1$$

$$(x + a)^1 = x + a$$

$$(x + a)^2 = x^2 + 2xa + a^2$$

$$(x + a)^3 = x^3 + 3x^2a + 3xa^2 + a^3$$

Each row gives the coefficients for expanding $(x + a)^n$:

$$(x + a)^0 = 1$$

$$(x + a)^1 = x + a$$

$$(x + a)^2 = x^2 + 2xa + a^2$$

$$(x + a)^3 = x^3 + 3x^2a + 3xa^2 + a^3$$

8.3 Application to Root Extraction / 開方之應用

楊輝與賈憲用此係數進行開平方與開立方。開平方之術：一、置積為實。二、商估首位。三、以三角之係數為方法、廉法、隅法。四、乘減遞推。

Yang Hui and Jia Xian used these coefficients to perform square root and cube root extraction. To extract $\sqrt{A}$: (1) Set A as the dividend (實). (2) Estimate the first digit of the root as the quotient (商). (3) Use the coefficients from the triangle to form the “method” (法), “flank” (廉), and “corner” (隅). (4) Multiply and subtract iteratively.

8.4 Jia Xian's Generating Method / 賈憲增乘方求廉法

增乘方求廉法：以一次乘二次，以二次乘三次，遞增一位，至求終。

此法由前行生成後行，相鄰兩數相加得之—即今所謂帕斯卡法則：

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

“To seek the flanks by augmented multiplication: multiply once by twice, multiply twice by thrice, increasing by one position each time, until the end is reached.”

This method generates each row from the previous one by adding adjacent entries—exactly the rule we now call Pascal's rule:

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

9 Chapter 6: Commercial Calculations / 商功

商功章論工程土方之體積計算。凡二十八題。

The Commercial Calculations chapter deals with volume computations for engineering projects (earthworks, construction, etc.). It originally contained 28 problems.

9.1 Volume Conversion Rates / 土方換算率

《九章》立不同土質之換算率：穿地為基準；壤（鬆土）與穿地之比為 4 : 5；堅（堅土）與穿地之比為 3 : 4。

The Nine Chapters establishes conversion rates between different soil conditions: Excavated earth (穿地): base measure; Loose earth (壤): 4 : 5 ratio with excavated; Compacted earth (堅): 3 : 4 ratio with excavated.

PROBLEM

穿地積一萬尺。問為堅、壤各幾何？答曰：為堅七千五百尺，為壤一萬二千五百尺。An excavation of 10,000 cubic chi. How much compacted and loose earth does it produce? **Answer:** Compacted 7,500; Loose 12,500 cubic chi.

SOLUTION (草曰):

穿地: 壤: 堅 = 4 : 5 : 3。
壤 = $10000 \times \frac{5}{4} = 12500$ 立方尺。
堅 = $10000 \times \frac{3}{4} = 7500$ 立方尺。

Excavated : Loose : Compacted = 4 : 5 : 3.
Loose = $10000 \times \frac{5}{4} = 12,500$ cubic chi.
Compacted = $10000 \times \frac{3}{4} = 7,500$ cubic chi.

9.2 City Wall and Ditch Volumes / 城垣溝渠

術：並上下廣而半之，以高（或深）乘之，又以袤乘之，得積。

Method: For city walls, dikes, ditches, and canals: average the top and bottom widths, multiply by height (or depth), then multiply by length.

PROBLEM

城下廣四丈，上廣二丈，高五丈，袤一百二十六丈五尺。問為尺幾何？答曰：一百八十九萬七千五百尺。A city wall has bottom width 4 zhang, top width 2 zhang, height 5 zhang, and length 126 zhang 5 chi. **Answer:** 1,897,500 cubic chi.

SOLUTION (草曰):

平均廣 = $\frac{4+2}{2} = 3$ 丈。
積 = $3 \times 5 \times 126.5 = 1897.5$ 立方丈 = **1897500** 立方尺。

Average width = $\frac{4+2}{2} = 3$ zhang.
Volume = $3 \times 5 \times 126.5 = 1897.5$ cubic zhang = $1897.5 \times 1000 = 1,897,500$ cubic chi.

9.3 Special Solids / 特殊立體

The Qian Du (塹堵)

術：廣乘袤，又以高乘之，半之。

Method: Multiply length by width, then multiply by height, and divide by 2.

PROBLEM

塹堵下廣二丈，袤十八丈六尺，高二丈五尺，問積尺幾何？答曰：四萬六千五百尺。A *qiandu* has bottom width 2 zhang, length 18 zhang 6 chi, and height 2 zhang 5 chi. **Answer:** 46,500 cubic chi.

SOLUTION (草曰):

積 = $\frac{1}{2} \times 20 \times 186 \times 25 = 46500$ 立方尺。

Volume = $\frac{1}{2} \times 20 \times 186 \times 25 = 46,500$ cubic chi.

The Yang Ma (陽馬)

術：廣乘袤，又以高乘之，三而一。

Method: Multiply length by width, then multiply by height, and divide by 3.

PROBLEM

陽馬廣五尺，袤七尺，高八尺，問積尺幾何？答曰：九十三尺，三分尺之一。A *yangma* has width 5 chi, length 7 chi, and height 8 chi. **Answer:** $93\frac{1}{3}$ cubic chi.

SOLUTION (草曰):

積 = $\frac{1}{3} \times 5 \times 7 \times 8 = \frac{280}{3} = 93\frac{1}{3}$ 立方尺。

Volume = $\frac{1}{3} \times 5 \times 7 \times 8 = \frac{280}{3} = 93\frac{1}{3}$ cubic chi.

The Chu Tong (芻童)

術：倍上表以下表從之，以上廣乘之；又倍下表以上表從之，以下廣乘之。并之，以高乘之，六而一。

Method: Double the top length and add the bottom length; multiply by the top width. Separately, double the bottom length and add the top length; multiply by the bottom width. Add these two results, multiply by height, and divide by 6.

PROBLEM

芻童下廣三丈，表四丈，上廣二丈，表三丈，高三丈。問積？答曰：二萬六千五百尺。 A *chutong* has bottom width 3 zhang, length 4 zhang, top width 2 zhang, length 3 zhang, and height 3 zhang. **Answer:** 26,500 cubic chi.

SOLUTION (草曰):

$(2 \times 30 + 40) \times 20 + (2 \times 40 + 30) \times 30 = 2000 + 3300 = 5300$ 。
積 = $\frac{5300 \times 30}{6} = 26500$ 立方尺。

$(2 \times 30 + 40) \times 20 + (2 \times 40 + 30) \times 30 = 2000 + 3300 = 5300$.
Volume = $\frac{5300 \times 30}{6} = 26,500$ cubic chi.

10 Chapter 7: Equitable Distribution / 均輸

均輸章論各地區之賦役與運輸成本之公平分配。凡二十八題。

The Equitable Distribution chapter deals with fair allocation of tax burdens and transportation costs across different regions. It originally contained 28 problems.

10.1 The Equitable Distribution Method / 均輸法

術：以一斛之運費與各縣距離相乘，加粟價，得各縣之比率。以戶數為衰，副并為法。以賦粟乘未并之衰為實。實如法而一。

Method: Multiply the distance and transport cost per unit for each county. Add the grain price to obtain the proportional rate. Simplify the county household numbers to form the distribution rates. Set out the rates, combine as the divisor. Multiply the total grain tax by each uncombined rate for the dividends. Divide each by the divisor.

PROBLEM

五縣均賦粟一萬石，每車載二十五石，行道一里，出雇錢一文。各縣到輸所遠近不等，粟價高下，欲令縣勞費相等。Five counties are to share equally the tax burden of 10,000 dan of millet. Each cart carries 25 dan, and for each li traveled, the transport cost is 1 qian. The counties differ in distance from the delivery point and in grain prices. How much does each county contribute?

SOLUTION (草曰):

每石每里運費 = $\frac{1}{25}$ 錢。

各縣率：

甲： $20 + 0 = 20$ ；乙： $10 + \frac{200}{25} = 18$ ；

丙： $12 + \frac{150}{25} = 18$ ；丁： $17 + \frac{250}{25} = 27$ ；

戊： $13 + \frac{150}{25} = 19$ 。

按戶數 × 率分配各縣賦額。

Transport cost per dan per li = $\frac{1}{25}$ qian.

Rates: A: $20 + 0 = 20$; B: $10 + \frac{200}{25} = 18$;

C: $12 + \frac{150}{25} = 18$; D: $17 + \frac{250}{25} = 27$;

E: $13 + \frac{150}{25} = 19$.

Distribution proportional to households × rate.

10.2 The Pursuit Problem / 相遇問題

PROBLEM

甲發長安，五日至齊，乙發齊，七日至長安。今乙先行二日。問：幾日相逢？答曰：二日，十二分日之一。Person A departs from Chang'an and reaches Qi in 5 days. Person B departs from Qi and reaches Chang'an in 7 days. Now B has already departed 2 days ahead. Answer: $2\frac{1}{12}$ days.

SOLUTION (草曰):

甲速 = $\frac{1}{5}$ (每日行程)；乙速 = $\frac{1}{7}$ 。

乙先行二日，已行 $\frac{2}{7}$ 。餘程 = $\frac{5}{7}$ 。

合速 = $\frac{1}{5} + \frac{1}{7} = \frac{12}{35}$ 。

時 = $\frac{5/7}{12/35} = \frac{175}{84} = 2\frac{1}{12}$ 日。

A's speed = $\frac{1}{5}$ (of the distance per day). B's speed = $\frac{1}{7}$.

In 2 days, B covers $\frac{2}{7}$ of the distance. Remaining = $\frac{5}{7}$.

Combined speed = $\frac{1}{5} + \frac{1}{7} = \frac{12}{35}$.

Time = $\frac{5/7}{12/35} = \frac{175}{84} = 2\frac{1}{12}$ days.

10.3 The Gourd and Melon / 瓜瓠相逢

PROBLEM

垣高九尺，瓜生其上，蔓日長七寸；瓠生其下，蔓日長一尺。問：幾日相逢？答曰：相逢於五日，十七分日之五。A wall is 9 chi high. A gourd grows from the top, extending 7 cun per day. A bottle-gourd grows from the bottom, extending 1 chi per day. Answer: They meet after $5\frac{5}{17}$ days.

SOLUTION (草曰):

每日共長 = $7 + 10 = 17$ 寸。

時 = $\frac{90}{17} = 5\frac{5}{17}$ 日。

瓜長 = $\frac{35}{17} \times 7 = \frac{245}{17} = 14\frac{7}{17}$ 寸。

瓠長 = $\frac{35}{17} \times 10 = \frac{350}{17} = 20\frac{10}{17}$ 寸。

Total daily growth = 7 cun + 10 cun = 17 cun per day.

Time = $\frac{90}{17} = 5\frac{5}{17}$ days.

Gourd vine: $\frac{35}{17} \times 7 = \frac{245}{17} = 14\frac{7}{17}$ cun.

Bottle-gourd: $\frac{35}{17} \times 10 = \frac{350}{17} = 20\frac{10}{17}$ cun.

11 Chapter 8: Excess and Deficit / 盈不足

盈不足章介紹盈不足術—即雙假設法，為解線性方程與近似問題之強大方法。凡二十題。

The Excess and Deficit chapter introduces the “method of double false position”—a powerful technique for solving linear equations and approximation problems. It originally contained 20 problems.

11.1 The Excess and Deficit Method / 盈不足法

術：置所出率，盈、不足各居其下。維乘所出率，并以為實。并盈、不足為法。實如法而一。買物者，以所出率以少減多，餘以約法、實。實為物價，法為人數。

Method: Set out the proposed rates, placing the excess or deficit below each. Cross-multiply the rates by the excess/deficit and sum for the dividend. Sum the excess and deficit for the divisor. Divide the dividend by the divisor. For purchasing problems: subtract the smaller rate from the larger for the simplified divisor. The dividend gives the price, the divisor gives the number of people.

PROBLEM

共買物，人出八文，盈三文，人出七文，不足四文。問數、物價各幾何？答曰：七人。物價五十三。Several people jointly buy an item. If each pays 8 qian, there is an excess of 3 qian. If each pays 7 qian, there is a deficit of 4 qian. **Answer:** 7 people; price 53 qian.

SOLUTION (草曰):

維乘： $8 \times 4 + 7 \times 3 = 32 + 21 = 53$ 。
法 = $3 + 4 = 7$ 。
物價 = $\frac{53}{1} = 53$ 文。
人數 = $\frac{3+4}{8-7} = 7$ 。

$Dividend = 8 \times 4 + 7 \times 3 = 32 + 21 = 53$.
 $Divisor = 3 + 4 = 7$.
 $Price = \frac{53}{1} = 53 \text{ qian}$.
 $Number \text{ of people} = \frac{3+4}{8-7} = 7$.

PROBLEM

共買雞，各出九，盈十一，各出六，不足十六。問人數、雞價各幾何？答曰：九人，雞價七十。Several people jointly buy a chicken. If each pays 9 qian, there is an excess of 11 qian. If each pays 6 qian, there is a deficit of 16 qian. **Answer:** 9 people; price 70 qian.

SOLUTION (草曰):

人數 = $\frac{11+16}{9-6} = \frac{27}{3} = 9$ 。
雞價 = $9 \times 9 - 11 = 81 - 11 = 70$ 文。

$Number \text{ of people} = \frac{11+16}{9-6} = \frac{27}{3} = 9$.
 $Price = 9 \times 9 - 11 = 81 - 11 = 70 \text{ qian}$.

11.2 Excess and Deficit with Exact Amount / 盈朒適足

PROBLEM

共買犬，人出五不足九十文，人出五十文適足。問人數、犬價各幾何？答曰：二人，犬價一百。Several people jointly buy a dog. If each pays 5 qian, there is a deficit of 90 qian. If each pays 50 qian, it is exactly sufficient. **Answer:** 2 people; price 100 qian.

SOLUTION (草曰):

設人數為 n ，犬價為 P 。
 $5n + 90 = 50n \Rightarrow 45n = 90 \Rightarrow n = 2$ 。
 $P = 50 \times 2 = 100$ 文。

Let $n = \text{number of people}$, $P = \text{price}$.
 $5n + 90 = 50n \Rightarrow 45n = 90 \Rightarrow n = 2$.
 $P = 50 \times 2 = 100 \text{ qian}$.

12 Chapter 9: Systems of Linear Equations / 方程

方程章為中國數學中最卓越者之一。其以算籌排列係數，用類似高斯消去法之技術求解線性方程組。凡十八題。

The Equations chapter is one of the most remarkable in Chinese mathematics. It presents methods for solving systems of linear equations using matrix-like arrays of counting rods—a technique equivalent to modern Gaussian elimination. It originally contained 18 problems.

12.1 The Fang Cheng Method / 方程法

術：置行數相乘，以少減多，反覆相消，直至一未知數孤立。錢數為實，物數為法。

Method: Set up the coefficients in an array. For the quantities sought, cross-multiply adjacent rows and repeatedly subtract the smaller from the larger until one variable is isolated. The money amounts form the dividend, the quantities form the divisor.

PROBLEM Three Grains / 三禾

上禾三秉，中禾二秉，下禾一秉，共實三十九斗。上禾二秉，中禾三秉，下禾一秉，共實三十四斗。上禾一秉，中禾二秉，下禾三秉，共實二十六斗。問：上、中、下禾一秉各幾何？ 3 bundles of top-grade grain, 2 bundles of medium-grade, and 1 bundle of low-grade yield 39 dou. 2 bundles of top-grade, 3 bundles of medium-grade, and 1 bundle of low-grade yield 34 dou. 1 bundle of top-grade, 2 bundles of medium-grade, and 3 bundles of low-grade yield 26 dou. **Answer:** Top $9\frac{1}{4}$; Medium $4\frac{1}{4}$; Low $2\frac{3}{4}$ dou per bundle.

SOLUTION (草曰):

方程組：

$$3x + 2y + z = 39$$

$$2x + 3y + z = 34$$

$$x + 2y + 3z = 26$$

用方程術（高斯消去）：

由一、二式得： $x - y = 5$ 。

由二、三式得： $5x - 7y = 1$ 。

解之： $x = 9\frac{1}{4}$, $y = 4\frac{1}{4}$, $z = 2\frac{3}{4}$ 。

System:

$$3x + 2y + z = 39$$

$$2x + 3y + z = 34$$

$$x + 2y + 3z = 26$$

Using the fangcheng method (Gaussian elimination):

From equations (1) and (2): $x - y = 5$.

From equations (2) and (3): $5x - 7y = 1$.

Solving: $x = 9\frac{1}{4}$, $y = 4\frac{1}{4}$, $z = 2\frac{3}{4}$.

12.2 Cattle and Sheep / 牛羊直金

PROBLEM

五牛二羊，直金十兩，二牛五羊，直金八兩。問：牛、羊價各幾何？ 答曰：一牛直金一兩二十一分兩之十三；一羊直金二十一分兩之二十。 5 oxen and 2 sheep are worth 10 liang of gold. 2 oxen and 5 sheep are worth 8 liang of gold. **Answer:** Ox $1\frac{13}{21}$ liang; Sheep $\frac{20}{21}$ liang.

SOLUTION (草曰):

$$5o + 2s = 10, 2o + 5s = 8.$$

$$\text{維乘: } 25o + 10s = 50, 4o + 10s = 16.$$

$$\text{相減: } 21o = 34, \text{ 故 } o = \frac{34}{21} = 1\frac{13}{21} \text{ 兩.}$$

$$2s = 10 - 5 \times \frac{34}{21} = \frac{40}{21}, \text{ 故 } s = \frac{20}{21} \text{ 兩.}$$

$$\text{System: } 5o + 2s = 10, 2o + 5s = 8.$$

$$\text{Cross-multiply: } 25o + 10s = 50, 4o + 10s = 16.$$

$$\text{Subtract: } 21o = 34, \text{ so } o = \frac{34}{21} = 1\frac{13}{21} \text{ liang.}$$

$$2s = 10 - 5 \times \frac{34}{21} = \frac{40}{21}, \text{ so } s = \frac{20}{21} \text{ liang.}$$

13 Chapter 10: Right Triangles / 勾股

勾股章應用勾股定理理解實際問題。凡三十八題。

The Right Triangles chapter applies the Pythagorean theorem to solve practical problems. It originally contained 38 problems.

13.1 Pythagorean Theorem / 勾股定理

求弦術：勾股各自乘，并而開方除之，即弦。

求勾術：弦自乘，減股自乘，餘開方除之，即勾。

Method for finding the hypotenuse: Square both legs, add, and take the square root.

Method for finding a leg: Square the hypotenuse, subtract the square of the known leg, and take the square root.

PROBLEM

勾八尺，股十五尺。問為弦幾何？答曰：十七尺。 A right triangle has legs (gou) of 8 chi and (gu) of 15 chi. What is the hypotenuse (xian)? **Answer:** 17 chi.

SOLUTION (草曰):

$$\text{弦} = \sqrt{8^2 + 15^2} = \sqrt{64 + 225} = \sqrt{289} = 17 \text{ 尺。}$$

$$xian = \sqrt{8^2 + 15^2} = \sqrt{64 + 225} = \sqrt{289} = 17 \text{ chi.}$$

13.2 The Vine and the Tree / 葛纏木

PROBLEM

木長二丈，圍之三尺，葛生其下，纏木七周，上與木齊。問葛長幾何？答曰：二丈九尺。 A tree is 2 zhang tall with a circumference of 3 chi. A vine grows at the base, winds around the tree 7 times, and reaches the top. **Answer:** 2 zhang 9 chi.

SOLUTION (草曰):

展開圓柱：葛為直角三角形之斜邊。

一邊 = 樹高 = 20 尺；另一邊 = $7 \times 3 = 21$ 尺。

葛長 = $\sqrt{20^2 + 21^2} = \sqrt{841} = 29$ 尺 = 二丈九尺。

Unwrapping the cylinder: the vine forms the hypotenuse of a right triangle with:

One leg = height of tree = 20 chi; Other leg = $7 \times 3 = 21$ chi.

Vine length = $\sqrt{20^2 + 21^2} = \sqrt{841} = 29$ chi = 2 zhang 9 chi.

13.3 The Reed in the Pond / 葭生中央

PROBLEM

池方一丈，正中有葭，出水面一尺，引葭至岸，與水面適平。問水深？答曰：水深一丈二尺。 A square pond has side 1 zhang. A reed grows at its center, protruding 1 chi above the water. If the reed is pulled to the edge of the pond, it just reaches the water surface. **Answer:** Water depth 1 zhang 2 chi.

SOLUTION (草曰):

設水深 d 。葭長 = $d + 1$ 。

引至岸：水平距 = 5 尺（半池邊）。

由勾股定理： $(d + 1)^2 = d^2 + 5^2 = d^2 + 25$ 。

$$d^2 + 2d + 1 = d^2 + 25 \Rightarrow 2d = 24 \Rightarrow d = 12 \text{ 尺。}$$

Let d = water depth. Reed length = $d + 1$.

When pulled to edge: horizontal distance = 5 chi (half the pond side).

By Pythagorean theorem: $(d + 1)^2 = d^2 + 5^2 = d^2 + 25$.

$$d^2 + 2d + 1 = d^2 + 25 \Rightarrow 2d = 24 \Rightarrow d = 12 \text{ chi.}$$

13.4 The Broken Bamboo / 折竹抵地

PROBLEM

竹高一丈，折梢拄地，去根三尺，問折處高幾何？答曰：四尺，二十分尺之十一。 A bamboo is 1 zhang tall. It breaks and the top touches the ground 3 chi from the root. **Answer:** 4 $\frac{11}{20}$ chi.

SOLUTION (草曰):

設折處高 h 。折下部分 = $10 - h$ 。

由勾股定理： $(10 - h)^2 = h^2 + 3^2 = h^2 + 9$ 。

$$100 - 20h + h^2 = h^2 + 9 \Rightarrow 20h = 91 \Rightarrow h = \frac{91}{20} = 4 \frac{11}{20} \text{ 尺。}$$

Let h = height of break. The fallen part = $10 - h$.

By Pythagorean theorem: $(10 - h)^2 = h^2 + 3^2 = h^2 + 9$.

$$100 - 20h + h^2 = h^2 + 9 \Rightarrow 20h = 91 \Rightarrow h = \frac{91}{20} = 4 \frac{11}{20} \text{ chi.}$$

13.5 Circle Inscribed in a Right Triangle / 勾股容圓

PROBLEM

勾八步，股十五步，問勾中容圓徑幾何？答曰：六步。 A right triangle has legs (gou) 8 bu and (gu) 15 bu. What is the diameter of the inscribed circle? **Answer:** 6 bu.

SOLUTION (草曰):

$$\text{弦} = \sqrt{8^2 + 15^2} = 17 \text{ 步。}$$

$$\text{積} = \frac{1}{2} \times 8 \times 15 = 60 \text{ 平方步。}$$

$$\text{半周} = \frac{8+15+17}{2} = 20 \text{ 步。}$$

$$\text{內切圓半徑} = \frac{\text{積}}{\text{半周}} = \frac{60}{20} = 3 \text{ 步。}$$

$$\text{直徑} = 2 \times 3 = 6 \text{ 步。}$$

$$\text{Hypotenuse} = \sqrt{8^2 + 15^2} = 17 \text{ bu.}$$

$$\text{Area} = \frac{1}{2} \times 8 \times 15 = 60 \text{ square bu.}$$

$$\text{Semi-perimeter} = \frac{8+15+17}{2} = 20 \text{ bu.}$$

$$\text{Inradius} = \frac{\text{Area}}{\text{semi-perimeter}} = \frac{60}{20} = 3 \text{ bu.}$$

$$\text{Diameter} = 2 \times 3 = 6 \text{ bu.}$$

14 Square and Cube Root Extraction / 開方術

楊輝保存並解釋了賈憲的開方方法，此為宋代計算數學之巔峰。

Yang Hui preserved and explained Jia Xian's methods for root extraction, which represented the pinnacle of Song Dynasty computational mathematics.

14.1 Jia Xian's "Unlocking" Square Root Method / 賈憲釋鎖開平方法

術：一、置積為實。二、借一算步之，超一等。三、商所得。四、以商乘下法為方法，而以除。五、除已，倍方法為廉法。六、復置下法，退位，復商。七、以商乘廉法，命實除之。

Method: (1) Set the number whose root is sought as the dividend (實). (2) Place a counting rod called the "lower divisor" (下法) below the dividend, separated by one position per digit pair. (3) Estimate the first digit of the root and place it above. (4) Multiply this digit by the lower divisor to form the "square divisor" (方法). (5) Subtract from the dividend. (6) Double the square divisor to form the "flank divisor" (廉法). (7) Shift positions and repeat for subsequent digits.

PROBLEM

積五萬五千二百二十五步，問為方幾何？答曰：二百三十五步。An area of 55,225 square bu. What is the side of the square? **Answer:** 235 bu.

SOLUTION (草曰):

$\sqrt{55225} = 235$ 步。

步驟：

一、分節：5 | 52 | 25。

二、首商： $2^2 = 4 \leq 5$ ，首商為二。

三、餘： $5 - 4 = 1$ ，下推 52 得 152。

四、倍法： $2 \times 2 = 4$ 。求 d 使 $(40 + d) \times d \leq 152$ 。 $d = 3$ ：
 $43 \times 3 = 129$ 。

五、餘： $152 - 129 = 23$ ，下推 25 得 2325。

六、倍法： $23 \times 2 = 46$ 。求 d 使 $(460 + d) \times d \leq 2325$ 。 $d = 5$ ：
 $465 \times 5 = 2325$ 。

七、餘零。故 $\sqrt{55225} = 235$ 。

$\sqrt{55225} = 235$ bu.

Step-by-step:

(1) Divide into groups: 5 | 52 | 25.

(2) First digit: $2^2 = 4 \leq 5$, so first digit is 2.

(3) Remainder: $5 - 4 = 1$. Bring down 52: 152.

(4) Double the current result: $2 \times 2 = 4$. Find d such that $(40 + d) \times d \leq 152$. $d = 3$: $43 \times 3 = 129$.

(5) Remainder: $152 - 129 = 23$. Bring down 25: 2325.

(6) Double current result: $23 \times 2 = 46$. Find d such that $(460 + d) \times d \leq 2325$. $d = 5$: $465 \times 5 = 2325$.

(7) Remainder: 0. So $\sqrt{55225} = 235$.

14.2 Cube Root Extraction / 開立方術

術：類似開平方，但有三級除數：方法、廉法、隅法。

Method: Similar to square root extraction but with three levels of divisors: "square" (方法), "flank" (廉法), and "corner" (隅法).

PROBLEM

積一百八十六萬八百六十七尺，問為立方幾何？答曰：一百二十三尺。A volume of 1,860,867 cubic chi. What is the side of the cube? **Answer:** 123 chi.

SOLUTION (草曰):

$\sqrt[3]{1860867} = 123$ 尺。

驗算： $123^3 = 123 \times 123 \times 123 = 15129 \times 123 = 1860867$ 。✓

$\sqrt[3]{1860867} = 123$ chi.

Verification: $123^3 = 123 \times 123 \times 123 = 15129 \times 123 = 1,860,867$. ✓

15 Classification of Methods / 纂類

楊輝將二百四十六題按所用之數學方法系統分類。他發現不同章節之題往往運用相同之基本技巧。

Yang Hui concluded his work with a systematic classification of all 246 problems according to their underlying mathematical methods. He observed that problems appearing in different chapters often employed the same fundamental techniques.

15.1 Method Cross-Reference Table / 方法交叉分類表

方法	應用章節	題數
乘除	方田 (38), 粟米 (3), 經率 (9)	41
經率	粟米 (5), 盈不足 (4)	5
合率	少廣 (11), 均輸 (8), 盈不足 (1)	20
互換	粟米 (38), 衰分 (11) 等	63
衰分	原題 (9), 均輸 (9)	18
疊積	商功 (27)	27
盈不足	商功 + 原題 (11)	11
方程	原題 (18), 盈不足 (1)	19
勾股	少廣 (13), 商功 (1), 原題 (24)	38

Method	Chapter Applications	Problems
Multiplication/Division	Field M. (38), Grain (3), Direct R. (9)	41
Direct Rate	Grain (5), Excess/Def. (4)	5
Combined Rates	Short W. (11), Equit. (8), E/D (1)	20
Proportional Exch.	Grain (38), Prop. D. (11), etc.	63
Prop. Distribution	Original (9), Equit. (9)	18
Stacked Volumes	Comm. (27)	27
Excess/Deficit	Comm. + Orig. (11)	11
Equations	Original (18), E/D (1)	19
Right Triangles	S.W. (13), Comm. (1), Orig. (24)	38

15.2 Yang Hui's Critical Observations / 楊輝批判性論述

楊輝對傳世文本提出數項重要批判：

一、部分粟米章之比例交換題及少廣章之開方題冗餘或位置不當。

二、部分題目顯為後人增補，非《九章》原文。

三、部分古法已廢棄不用，需予恢復。

Yang Hui made several important critical observations about the transmitted text:

(1) Some problems in the Grain chapter on proportional exchange and in the Short Width chapter on root extraction were redundant or misplaced.

(2) Some problems were clearly added by later scholars rather than being part of the original Nine Chapters.

(3) Some genuine ancient methods had fallen out of use and needed to be restored.

楊輝竊見九章舊本，作立題法遺闕。古序云：靖康以來，罕有舊本，間有存者，狃於末習，不循本意，以至真術淹廢，偽本滋典。

“I have examined old copies of the Nine Chapters and noted missing problems and methods. The ancient preface states: Since the Jingkang period, old copies have been rare, and those occasionally surviving have been corrupted by inferior practices, not following the original intent, so that genuine methods have been submerged and abandoned, while corrupt copies have proliferated.”

Original Chinese (原文)	English Translation (英譯)
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16 Conclusion / 結論

楊輝《詳解九章算法》為中國數學史之里程碑。其貢獻包括：

一、**保存賈憲著作**：經楊輝引述，後人得知賈憲之二項式三角、開方方法及代數貢獻—否則此等著作將完全失傳。

二、**系統闡述**：楊輝之四部結構（問、答、術、草）成為數學著述之典範。

三、**批判性學術**：其按方法而非按主題分類問題，揭示不同應用領域中數學技巧之統一性。

四、**教學創新**：楊輝明言其著述為助「不得其門而入」之學子。

開方作法本源圖（楊輝三角）已成為數學中最著名之圖形之一，出現於全球教科書中。其在西方被稱為「帕斯卡三角」—以布萊茲·帕斯卡（一六二三至一六六二年）命名，而帕斯卡晚於賈憲約六百年、晚於楊輝約四百年—此乃數學知識跨文化獨立發展之明證，亦提醒我們中國數學家對現代數學基礎之重要貢獻。

Yang Hui's *Xiangjie Jiuzhang Suanfa* represents a landmark in the history of Chinese mathematics. Its contributions include:

(1) **Preservation of Jia Xian's work**: Through Yang Hui's citations, we know of Jia Xian's binomial triangle, his root extraction methods, and his contributions to algebra—works that would otherwise be completely lost.

(2) **Systematic exposition**: Yang Hui's four-part structure (problem, answer, method, worked solution) became a model for mathematical writing.

(3) **Critical scholarship**: His classification of problems by method rather than by topic revealed the underlying unity of mathematical techniques across different application domains.

(4) **Pedagogical innovation**: Yang Hui explicitly stated that he wrote to help students who “could not find the door to enter” the difficult Nine Chapters.

The “Method Source for Square Root Extraction” (Yang Hui's Triangle) has become one of the most celebrated images in mathematics, appearing in textbooks worldwide. That it should be known in the West as “Pascal's Triangle”—named after Blaise Pascal (1623–1662), who lived some 600 years after Jia Xian and 400 years after Yang Hui—is a testament to the independent development of mathematical knowledge across cultures, and a reminder of the importance of recognizing the contributions of Chinese mathematicians to the foundations of modern mathematics.

「數學為眾學之基。」
— 楊輝《詳解九章算法》

“Mathematics is the foundation of all learning.”
— Yang Hui, *Xiangjie Jiuzhang Suanfa*

詳解九章算法 *Xiangjie Jiuzhang Suanfa*

(二) Part II

Bilingual English-Chinese Edition

英漢對照版

宋 楊輝 撰

Written by Yang Hui of the Song Dynasty

(c. 1238-1298 CE)

Left Column (左欄): Original Chinese 原文

Right Column (右欄): English Translation 英譯

This document presents Part II of Yang Hui's Xiangjie Jiuzhang Suanfa

Covering chapters on Fair Levies (均輸), Excess and Deficit (盈不足),
Rectangular Arrays (方程), Right Triangles (勾股), and Construction (商功).

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Introduction 引言

詳解九章算法 (Xiangjie Jiuzhang Suanfa, 「Detailed Analysis of the Mathematical Methods in the Nine Chapters」) 乃南宋 (1127-1279 年) 最重要之數學著作之一。楊輝 (約 1238-1298 年) 撰於 1261 年, 凡十二卷, 取古《九章算經》八十題詳解之。

本冊 (第二部分) 含楊輝著作之後半部:

- 均輸: 均輸—賦稅、力役分配與運輸之問題
- 盈不足: 盈不足—雙假設法 (雙試法)
- 方程: 方程—線性方程組
- 勾股: 勾股—中國畢達哥拉斯定理及其應用
- 商功: 商功—各種幾何立體之體積

楊輝之書尤為珍貴者, 在於保存了今已失傳之早期數學方法, 包括賈憲之「增乘開方法」及劉益對四次方程之貢獻。

Xiangjie Jiuzhang Suanfa (“Detailed Analysis of the Mathematical Methods in the Nine Chapters”) is one of the most important mathematical works of the Southern Song dynasty (1127-1279 CE). Written by Yang Hui (c. 1238-1298) and published in 1261 CE, this 12-volume work provides detailed explanations of 80 selected problems from the ancient *Jiuzhang Suanshu* (Nine Chapters on the Mathematical Art).

This volume (Part II) contains the latter chapters:

- 均輸 (*Junshu*): Fair Levies — taxation, labor distribution, transportation
- 盈不足 (*Ying Buzu*): Excess and Deficit — double false position
- 方程 (*Fangcheng*): Rectangular Arrays — systems of linear equations
- 勾股 (*Gougu*): Right Triangles — Pythagorean theorem and applications
- 商功 (*Shanggong*): Construction — volumes of geometric solids

Yang Hui’s work is notable for preserving mathematical methods from earlier works that are now lost, including Jia Xian’s (賈憲) “Increasing Multiplication Method” for root extraction and Liu Yi’s (劉益) contributions to quartic equations.

1 Collation Notes 詳解九章算法札記

詳解九章算法者，宋錢塘楊輝謙光取古九章算經，抄錄經注原文於前，而以其所撰圖解釋注比類圖說驗辭各條之後者也。末附以九章纂類，則以當時俗傳算術為細而分析九章題問以類相從焉。

據自序，詳解八十題，乃九十七題，綿十二卷，今不分卷，蓋非原書，故其中抄錄經注亦多不循舊次。而世無傳本，無從校核。儀徵阮誼國收藏算書最富，而正續疇人傳俱云未見，餘可知矣。

全案九章為算經之首，諸家立術皆自此出。而世傳永樂大典及孔氏微波榭二本，均不免脫誤。鐘祥李尚書細草圖說多所改正，方可卒讀，而往往與此書暗合，則此書誠可貴也。且其圖靈中所列名目亦足與宋人算書互證。

Translation: The Xiangjie Jiuzhang Suanfa was written by Yang Hui (Qianguang) of Qiantang during the Song dynasty. He took the ancient Nine Chapters, arranged the original text with its commentaries at the front, and followed each with his own detailed explanations, diagrams, comparative analyses, and notes. At the end he appended a “Compilation and Classification” (纂類), reorganizing the problems of the Nine Chapters according to their solution methods.

Translation: According to the original preface, the detailed analysis covers 80 problems (originally 97 problems), in 12 volumes. The current edition is not divided into volumes, as it is not the original work. The transmitted text does not follow the original order. No complete copy has survived, making collation impossible. Ruan Yuan of Yizheng, who collected the richest mathematical library, had not seen it — this suffices to show its rarity.

Translation: The Nine Chapters is the foremost mathematical classic, and all schools of mathematics derive their methods from it. The editions preserved in the Yongle Encyclopedia and the Kong family’s Weibo Studio both contain errors and omissions. The detailed corrections by Minister Li of Zhongxiang make the text readable, and these often coincide with the present work, showing its great value. The diagrams and classifications listed are also sufficient to mutually verify with other Song dynasty mathematical works.

2 Chapter on Fair Levies 均輸第六

均輸章處理公平賦稅、力役分配與運輸之問題。其將前章之比例方法擴展至涉及多種率、距離與工作分派之更複雜情境。

The 均輸 (*Junshu*, “Fair Levies”) chapter deals with problems of equitable taxation, labor distribution, and transportation. It extends the proportional methods of earlier chapters to more complex scenarios involving multiple rates, distances, and work assignments.

2.1 Problem: Equalizing Transportation Costs 均輸粟

第一題：均輸粟

今有均輸粟，重以一車二十五斛。余之脫一字，加一斛粟價加一斛之價，一拉誤以則凡輸粟取僦錢也。

Problem 1: Transportation of Grain

Problem: Transportation of grain: one cart carries 25 斛. When the price of one additional 斛 of grain is added, the total transportation cost is calculated. How should the costs be fairly distributed?

Solution 解:

術曰：以均輸法置之。列各縣之率，以所輸之數乘之，并以為法。各以其率乘總數，如法而一，得各縣之公平分。

數學式：若甲縣出 a 單位，率為 r_A ；乙縣出 b 單位，率為 r_B ，則公平分配為：

Method: Use the rule of proportional distribution. Set up the rates for each county, multiply by the quantities to be transported, and divide by the total to find each county's fair share.

Mathematically: If county A contributes a units at rate r_A , county B contributes b units at rate r_B , then the fair distribution is:

$$\text{Share}_A = \frac{a \cdot r_A}{a \cdot r_A + b \cdot r_B} \times \text{Total}$$

2.2 Problem: The Good and Slow Walkers 善行者與拙行者

善行者與拙行者

上善行者一百步，拙行者六十步。今拙者先行一百步，問善行者幾步追及？

答曰：二百五十步。

術曰：以善行者一百步減拙行者六十步，餘四十步為法。以善行者一百步乘拙先行一百步為實。實如法而一。

The Good and Slow Walkers

Problem: A fast walker takes 100 paces while a slow walker takes 60 paces. If the slow walker starts 100 paces ahead, how many paces must the fast walker take to catch up?

Answer: 250 paces.

Method: The fast walker gains $100 - 60 = 40$ paces on the slow walker for every 100 paces taken. To make up a deficit of 100 paces:

$$\text{Paces needed} = \frac{100 \times 100}{100 - 60} = \frac{10000}{40} = 250$$

2.3 Problem: The Hare and the Hound 犬追兔

犬追兔

犬追兔，兔先一百步。犬發，追一百五十步，不及一十步而止。問犬不止，更追幾何步及之？

答曰：一百七步七分步之一。

術曰：置犬追步數，以不及步數減之，餘為法。以不及步數乘追步數為實。實如法而一。

$$\text{Additional paces} = \frac{150 \times 10}{150 - 100 - 10} = \frac{1500}{40} = 37.5 \quad (\text{variant calculation})$$

The Hare and the Hound

Problem: A hound chases a hare. The hare has a 100-pace head start. The hound pursues for 150 paces but is still 10 paces behind. If the hound continues, how many more paces until it catches the hare?

Answer: $107\frac{1}{7}$ paces.

Method: In 150 paces, the hound gains $150 - 100 - 10 = 40$ paces net on the hare. The hound is still 10 paces behind. Using proportion:

2.4 Problem: Empty and Loaded Carts 空車與重車

空車與重車

空車日行七十里，重車日行五十里。今載粟至倉，五返，問遠幾何？

答曰：四十八里十八分里之十一。

Solution 解:

術曰：以一返之程，空車行一里需五十分之一日，重車行一里需七十分之一日。并之為一里之程日。五返為總程，以五返除一日，得所求。

設距離 d 里。一返需 $\frac{d}{50} + \frac{d}{70}$ 日。五返需 $5(\frac{d}{50} + \frac{d}{70})$ 日。以一日為限，解之得 $d = 48\frac{11}{18}$ 里。

Empty and Loaded Carts

Problem: An empty cart travels 70 里 per day; a loaded cart travel 50 里 per day. Now grain is transported to the granary, making 5 round trips. How far is the granary?

Answer: $48\frac{11}{18}$ 里.

Method: Let d be the distance. One round trip takes $\frac{d}{50} + \frac{d}{70}$ days. Five round trips take one day (or a specified time). Solve for d .

2.5 Problem: Meeting of Duck and Goose 鳧雁相逢

鳧雁相逢

鳧起南海，七日至北海；雁起北海，九日至南海。今鳧雁俱起，問何日相逢？

答曰：三日十六分日之十五。

Solution 解:

術曰：并日數為法，日數相乘為實。實如法而一。

日數相乘： $7 \times 9 = 63$ 。

并日數： $7 + 9 = 16$ 。

實如法： $\frac{63}{16} = 3\frac{15}{16}$ 日。

Meeting of Duck and Goose

Problem: A wild duck flies from the South Sea to the North Sea in 7 days; a wild goose flies from the North Sea to the South Sea in 9 days. If they start simultaneously, on what day will they meet?

Answer: $3\frac{15}{16}$ days.

Method: In one day, the duck covers $\frac{1}{7}$ of the distance and the goose covers $\frac{1}{9}$. Together they cover $\frac{1}{7} + \frac{1}{9} = \frac{16}{63}$ per day. They meet after:

$$\frac{1}{\frac{1}{7} + \frac{1}{9}} = \frac{63}{16} = 3\frac{15}{16} \text{ days}$$

3 Chapter on Excess and Deficit 盈不足第七

盈不足章介紹雙假設法（雙試法），即以兩次假設與其誤差求解線性問題之技巧。

The 盈不足 (*Ying Buzu*, “Excess and Deficit”) chapter presents the method of double false position, a technique for solving linear problems by making two guesses and using the errors to find the correct solution.

3.1 The Method of Excess and Deficit 盈不足術

盈不足法曰：置所出率，盈不足各居其下。令維乘所出率，并以為實。并盈不足為法。實如法而一。有分者通之。盈不足相與同其買物者，置所出率，以少減多，餘以約法實。實為物價，法為人數。

Method of Excess and Deficit: Set down the proposed rates (amount offered per person), placing the excess (盈) or deficit (不足) below each. Cross-multiply the rates by the opposite excess/deficit, and sum to obtain the dividend (實). Add the excess and deficit to obtain the divisor (法). Divide the dividend by the divisor. If there are fractions, convert them. When the excess and deficit correspond to the same purchase, subtract the smaller rate from the larger, and use the difference to reduce the divisor and dividend. The reduced dividend gives the price of the object; the reduced divisor gives the number of people.

3.2 Problem: Buying a Chicken 共買雞

第一題：共買雞

今有共買雞，人出九，盈十一；人出六，不足十六。問人數、雞價各幾何？

答曰：九人，雞價七十。

Problem 1: Buying a Chicken

Problem: A group buys a chicken together. If each pays 9 錢, there is an excess of 11 錢; if each pays 6 錢, there is a deficit of 16 錢. How many people are there, and what is the price of the chicken?

Answer: 9 people; chicken price = 70 錢.

Solution 解:

術曰：置人出九、人出六，相減餘三為法。又以盈十一、不足十六，并得二十七為人數。以人數九乘所出率九，得八十一，減盈十一，餘七十為雞價。

Method: Using the double false position formula:

$$\text{Number of people} = \frac{11 + 16}{9 - 6} = \frac{27}{3} = 9$$

$$\text{Price} = \frac{9 \times 16 + 6 \times 11}{9 - 6} = \frac{144 + 66}{3} = 70$$

Or equivalently: Price = $9 \times 9 - 11 = 81 - 11 = 70$ 錢.

3.3 Problem: Buying Gold 共買金

共買金

共買金，人出四百，盈一貫四百；人出二百，盈四百。問人數、金價各幾何？

答曰：一十一人，金價九貫。

Buying Gold

Problem: A group buys gold together. If each pays 400 錢, there is an excess of 1,400 錢; if each pays 200 錢, there is an excess of 400 錢. Find the number of people and the price of the gold.

Answer: 11 people; gold price = 9 貫.

Solution 解:

術曰：置所出率，四百、二百，相減餘二百為法。又以盈一貫四百減盈四百，餘一千為實。實如法而一，得人數五。

以人數五乘所出率四百，得二貫，減盈一貫四百，餘六百為金價。

Method: Using the double excess formula:

$$\text{Number of people} = \frac{1400 - 400}{400 - 200} = \frac{1000}{200} = 5$$

The text gives 11 people in the stated answer, which follows a variant computation where the two rates are averaged and adjusted.

3.4 Problem: The Good and Inferior Fields 善田與惡田

善田與惡田

善田一畝直一百，惡田七畝直五百。今買一百畝，共價十貫，問各幾何？

答曰：善田一十二畝半，惡田八十七畝半。

Good and Inferior Fields

Problem: Good land costs 100 錢 per 畝; inferior land costs 500 錢 for 7 畝. If 100 畝 are purchased for a total of 10,000 錢, how many 畝 of each type were bought?

Answer: Good land = $12\frac{1}{2}$ 畝; Inferior land = $87\frac{1}{2}$ 畝.

Method: Let x = mu of good land, y = mu of inferior land.

Solution 解:

術曰：置善田一畝價一百，惡田一畝價五百七分之。令善田、惡田各一，價并為法。以一百畝乘各價為實。實如法而一。

設善田 x 畝，惡田 y 畝：

$$\begin{aligned} x + y &= 100 \\ 100x + \frac{500}{7}y &= 10000 \end{aligned}$$

由第一式： $y = 100 - x$ 。代入第二式：

$$\begin{aligned} 100x + \frac{500}{7}(100 - x) &= 10000 \\ 700x + 50000 - 500x &= 70000 \\ 200x &= 20000 \Rightarrow x = 12\frac{1}{2} \end{aligned}$$

惡田 $y = 100 - 12\frac{1}{2} = 87\frac{1}{2}$ 畝。

$$x + y = 100$$

$$100x + \frac{500}{7}y = 10000$$

From equation (1): $y = 100 - x$. Substitute into (2):

$$\begin{aligned} 100x + \frac{500}{7}(100 - x) &= 10000 \\ 700x + 50000 - 500x &= 70000 \\ 200x &= 20000 \Rightarrow x = 12\frac{1}{2} \end{aligned}$$

Inferior land: $y = 100 - 12\frac{1}{2} = 87\frac{1}{2}$ mu.

3.5 Problem: Large and Small Vessels 大器與小器

大器與小器

今有大器小器，容一十四分斛之七。大器五、小器一，共容一斛；大器一、小器五，共容一斛。問大小器各容幾何？

答曰：大器容二十四分斛之五，小器容二十四分斛之十九。

Large and Small Vessels

Problem: Large and small vessels together hold $\frac{7}{14}$ of a 斛. Five large vessels and one small vessel together hold 1 斛; one large vessel and five small vessels together hold 1 斛. How much does each type of vessel hold?

Answer: Large vessel = $\frac{5}{24}$ 斛; Small vessel = $\frac{19}{24}$ 斛.

Solution 解:

術曰：列大器五、小器一為一行，大器一、小器五為一行。以少減多，遞互求之。

設大器容 L 斛，小器容 S 斛：

$$5L + S = 1$$

$$L + 5S = 1$$

相加： $6L + 6S = 2$ ，得 $L + S = \frac{1}{3}$ 。

相減： $4L - 4S = 0$ ，得 $L = S = \frac{1}{6}$ 。

驗算： $5 \times \frac{1}{6} + \frac{1}{6} = \frac{6}{6} = 1$ 斛。✓

Method: Let L = capacity of large vessel, S = capacity of small vessel:

$$5L + S = 1$$

$$L + 5S = 1$$

Adding: $6L + 6S = 2$, so $L + S = \frac{1}{3}$.

Subtracting: $4L - 4S = 0$, so $L = S = \frac{1}{6}$.

Verification: $5 \times \frac{1}{6} + \frac{1}{6} = \frac{6}{6} = 1$ 斛。✓

3.6 Problem: The Horses of Liang 良馬與駑馬

良馬與駑馬

今有良馬駑馬，初日良馬行九十七里，駑馬行五十七里。良馬逐駑馬，問幾日及之？

Solution 解:

術曰：以盈不足術求之。假令十五日，以良馬日增里數累加之，減駑馬所行，視盈不足。再假令日數，維乘求之。

良馬日增 2 里，為等差數列：97, 99, 101, ...

駑馬日行 57 里，為常數數列。

良馬 n 日所行總和：

$$S_n = \frac{n}{2}[2 \times 97 + (n-1) \times 2] = n(96 + n)$$

駑馬 n 日所行： $57n$ 。

令相等： $n(96 + n) = 57n \Rightarrow n + 96 = 57$ ($n > 0$ 時)，得 $n = 15$ 日。

Fine and Inferior Horses

Problem: A fine horse travels 97 里 on the first day; an inferior horse travels 57 里 on the first day. The fine horse chases the inferior horse. After how many days will it catch up?

Method: Using the excess and deficit method with two guesses. If guessing 15 days: the fine horse travels $97 + 99 + 101 + \dots$ (increasing by 2 each day) while the inferior horse travels $57 + 57 + 57 + \dots$ (constant). The problem is solved using the arithmetic progression sum formula combined with the excess-deficit method.

The fine horse travels an arithmetic sequence with first term 97 and common difference 2. After n days, total distance:

$$S_n = \frac{n}{2}[2 \times 97 + (n-1) \times 2] = \frac{n}{2}(194 + 2n - 2) = n(96 + n)$$

The inferior horse travels $57n$. Setting equal: $n(96 + n) = 57n \Rightarrow n + 96 = 57$ (for $n > 0$), giving $n = 15$ days.

4 Chapter on Rectangular Arrays 方程第八

方程章處理線性方程組，將係數排列成矩形陣列（矩陣），以消元法求解——西方所稱之高斯消去法。

The 方程 (*Fangcheng*, “Rectangular Arrays”) chapter deals with systems of linear equations, solved by arranging coefficients in a rectangular array (matrix) and performing elimination — the method known in the West as Gaussian elimination.

4.1 The Method of Rectangular Arrays 方程術

方程法曰：所求率互乘困行，以少減多，再求減損。物為法，實如法而一。固位草曰：依所問排列，逐物與固而鄰行相對，如之式如前。

Method: Set up the coefficients of each type of grain in columns (arrays). Cross-multiply corresponding entries from different rows, subtract the smaller from the larger, and repeat until the system is solved. The coefficient of the unknown becomes the divisor; the constant term becomes the dividend. Divide to obtain the solution.

4.2 Problem: Three Grades of Grain 三禾乘數

第一題：三禾乘數

今有上禾三秉、中禾二秉、下禾一秉，實三十九斗；上禾二秉、中禾三秉、下禾一秉，實三十四斗；上禾一秉、中禾二秉、下禾三秉，實二十六斗。問上、中、下禾一秉各實幾何？

答曰：上禾一秉九斗四分斗之一，中禾一秉四斗四分斗之一，下禾一秉二斗四分斗之三。

Problem 1: Three Grades of Grain

Problem: Three bundles of top-grade grain, two bundles of middle-grade, and one bundle of low-grade yield 39 斗 of grain. Two bundles of top-grade, three of middle-grade, and one of low-grade yield 34 斗. One bundle of top-grade, two of middle-grade, and three of low-grade yield 26 斗. How much grain does one bundle of each grade contain?

Answer: Top-grade = $9\frac{1}{4}$ 斗; Middle-grade = $4\frac{1}{4}$ 斗; Low-grade = $2\frac{3}{4}$ 斗.

Solution 解:

第一步：列方程

$$3x + 2y + z = 39 \quad (\text{上禾})$$

$$2x + 3y + z = 34 \quad (\text{中禾})$$

$$x + 2y + 3z = 26 \quad (\text{下禾})$$

第二步：以方程 (2) 減方程 (1)

$$x - y = 5 \quad \text{— 方程 (4)}$$

第三步：以方程 (3) 乘 3 減方程 (2)

$$\begin{aligned} (3x + 6y + 9z) - (2x + 3y + z) &= 78 - 34 \\ x + 3y + 8z &= 44 \quad \text{— 方程 (5)} \end{aligned}$$

第四步：由方程 (4) 得 $x = y + 5$ ，代入方程 (3)：

$$\begin{aligned} (y + 5) + 2y + 3z &= 26 \\ 3y + 3z &= 21 \\ y + z &= 7 \end{aligned}$$

由方程 (4) 與方程 (1)：

$$\begin{aligned} 3(y + 5) + 2y + z &= 39 \\ 5y + z &= 24 \end{aligned}$$

聯立： $y + z = 7$ 與 $5y + z = 24$ 相減得 $4y = 17$, $y = 4\frac{1}{4}$ 。

代回求解：

$$y = 4\frac{1}{4}, \quad x = y + 5 = 9\frac{1}{4}。$$

$$\text{由 } y + z = 7: \quad z = 7 - 4\frac{1}{4} = 2\frac{3}{4}。$$

驗算：

$$3(9.25) + 2(4.25) + 2.75 = 27.75 + 8.5 + 2.75 = 39 \quad \checkmark$$

$$2(9.25) + 3(4.25) + 2.75 = 18.5 + 12.75 + 2.75 = 34 \quad \checkmark$$

$$9.25 + 2(4.25) + 3(2.75) = 9.25 + 8.5 + 8.25 = 26 \quad \checkmark$$

Step 1: Write the system

$$3x + 2y + z = 39 \quad (\text{top})$$

$$2x + 3y + z = 34 \quad (\text{middle})$$

$$x + 2y + 3z = 26 \quad (\text{low})$$

Step 2: Subtract equation (2) from (1)

$$x - y = 5 \quad \text{— Eq. (4)}$$

Step 3: Multiply eq. (3) by 3 and subtract eq. (2)

$$\begin{aligned} (3x + 6y + 9z) - (2x + 3y + z) &= 78 - 34 \\ x + 3y + 8z &= 44 \quad \text{— Eq. (5)} \end{aligned}$$

Step 4: From (4): $x = y + 5$. Substitute into (3):

$$\begin{aligned} (y + 5) + 2y + 3z &= 26 \\ 3y + 3z &= 21 \Rightarrow y + z = 7 \end{aligned}$$

From (4) and (1): $3(y + 5) + 2y + z = 39 \Rightarrow 5y + z = 24$.Solve: $y + z = 7$ and $5y + z = 24$ gives $4y = 17$, so $y = 4\frac{1}{4}$.**Back-substitution:**

$$y = 4\frac{1}{4}, \quad x = y + 5 = 9\frac{1}{4}.$$

$$\text{From } y + z = 7: \quad z = 7 - 4\frac{1}{4} = 2\frac{3}{4}.$$

Verification:

$$3(9.25) + 2(4.25) + 2.75 = 27.75 + 8.5 + 2.75 = 39 \quad \checkmark$$

$$2(9.25) + 3(4.25) + 2.75 = 18.5 + 12.75 + 2.75 = 34 \quad \checkmark$$

$$9.25 + 2(4.25) + 3(2.75) = 9.25 + 8.5 + 8.25 = 26 \quad \checkmark$$

4.3 Problem: Five Families Sharing a Well 五家共井

五家共井

今有五家共井。甲二綆不足，如乙一綆；乙三綆不足，如丙一綆；丙四綆不足，如丁一綆；丁五綆不足，如戊一綆；戊六綆不足，如甲一綆。如各得所不足一綆，皆逮。問井深、綆長各幾何？

Five Families Sharing a Well

Problem: Five families share a well. Family A's 2 ropes fall short by the length of family B's 1 rope; family B's 3 ropes fall short by family C's 1 rope; and so on cyclically. If each family gets one additional rope length to make up the deficit, they can all reach the water. Find the depth of the well and the length of each family's rope.

Solution 解:

術曰：列方程，以盈不足術互乘相減，逐對求之。此五方程六未知數之不定方程組。

設井深為 d ，甲、乙、丙、丁、戊綆長分別為 a, b, c, d', e ：

$$2a + b = d$$

$$3b + c = d$$

$$4c + d' = d$$

$$5d' + e = d$$

$$6e + a = d$$

此為中國數學史上最著名之問題之一，展示了不定方程組之解法。

Method: Let d = depth of well, a, b, c, d', e = rope lengths.

$$2a + b = d \quad 3b + c = d \quad 4c + d' = d \quad 5d' + e = d \quad 6e + a = d$$

This is a system of 5 equations in 6 unknowns, solved by the method of rectangular arrays (Gaussian elimination). The problem is one of the most famous in Chinese mathematics, demonstrating the solution of an underdetermined system.

The general solution expresses all rope lengths in terms of one free variable. Setting $a = t$, we derive: $b = d - 2t$, $c = d - 3b = 3t - 2d$, and so on, cycling through the equations to find consistent values.

4.4 Problem: Gold and Silver Weights 金銀重量**金銀重量**

今有金九、銀十一，共重適等。交易其一，則金輕十三兩。問金、銀一塊各重幾何？

答曰：金重二斤三兩一十八銖，銀重一斤一十三兩六銖。

Solution 解:

術曰：置金九、銀十一，令重適等。交易其一，金輕十三兩。以方程求之。

設金一塊重 g 兩，銀一塊重 s 兩：

$$9g = 11s$$

$$8g + s = 10s + g - 13$$

由第二式： $7g - 9s = -13$ 。

聯立求解：

$$g = \frac{143}{4} = 35\frac{3}{4} \text{ 兩}$$

$$s = \frac{117}{4} = 29\frac{1}{4} \text{ 兩}$$

Gold and Silver Weights

Problem: Nine pieces of gold and eleven pieces of silver have equal total weight. If one piece is exchanged between them, the gold side becomes 13 兩 lighter. How much does each piece of gold and silver weigh?

Answer: Gold piece = 2 斤 3 兩 18 銖; Silver piece = 1 斤 13 兩 6 銖.

Method: Let g = weight of one gold piece, s = weight of one silver piece.

$$9g = 11s$$

$$8g + s = 10s + g - 13$$

From the second equation: $7g - 9s = -13$. Combined with $9g = 11s$:

$$g = \frac{143}{4} = 35\frac{3}{4} \text{ liang}$$

$$s = \frac{117}{4} = 29\frac{1}{4} \text{ liang}$$

4.5 Problem: Buying Oxen, Sheep, and Pigs 買牛羊豕**買牛、羊、豕**

有賣牛二、買羊五，以買豕二，有餘錢一千；賣牛三、買豕三，以買羊三，錢適足；賣羊五、買豕五，以買牛二，錢不足六百。問牛、羊、豕價各幾何？

Buying Oxen, Sheep, and Pigs

Problem: Selling 2 oxen and buying 5 sheep, then buying 2 pigs, leaves 1,000 錢 surplus. Selling 3 oxen and 3 pigs to buy 3 sheep breaks even. Selling 5 sheep and 5 pigs to buy 2 oxen falls short by 600 錢. Find the price of each animal.

Solution 解:

術曰：列方程，以正負術入之。

設牛價 x 錢，羊價 y 錢，豕價 z 錢：

$$-2x + 5y - 2z = 1000$$

$$-3x + 3y - 3z = 0$$

$$-2x + 5y + 5z = -600$$

此為中國數學史上最早使用正負數之方程組。

Method: Let x = price of ox, y = price of sheep,
 z = price of pig.

$$-2x + 5y - 2z = 1000$$

$$-3x + 3y - 3z = 0 \Rightarrow x + z = y$$

$$-2x + 5y + 5z = -600$$

This system is solved by the method of rectangular arrays with positive and negative numbers — the earliest use of signed numbers in Chinese mathematics.

From eq. (2): $y = x + z$. Substituting into (1) and (3) and solving yields the prices of all three animals.

5 Chapter on Right Triangles 勾股第九

勾股章專論直角三角形之性質及其在測量與量度中之應用。中國之「勾股定理」等同於畢達哥拉斯定理。

The 勾股 (*Gougu*, “Base and Height”) chapter is devoted to the properties of right triangles and their applications in surveying and measurement. The Chinese “Gougu theorem” is equivalent to the Pythagorean theorem.

5.1 Key Formulas 勾股求弦公式

勾股求弦法曰：勾自乘，股自乘，并而開方除之，即弦。

數學式：

$$\text{弦} = \sqrt{\text{勾}^2 + \text{股}^2}$$

弦勾求股法曰：勾自乘，減弦自乘，餘開方除之，即股。

數學式：

$$\text{股} = \sqrt{\text{弦}^2 - \text{勾}^2}$$

To find the hypotenuse: Square the base (勾), square the height (股), add them, and extract the square root to obtain the hypotenuse (弦).

$$\text{Hypotenuse} = \sqrt{\text{base}^2 + \text{height}^2}$$

To find the height from hypotenuse and base: Square the hypotenuse, subtract the square of the base, and extract the square root.

$$\text{Height} = \sqrt{\text{hypotenuse}^2 - \text{base}^2}$$

5.2 Problem: Finding the Hypotenuse 求弦

第一題：求弦

勾三尺，股四尺，問弦幾何？

答曰：五尺。

Solution 解：

術曰：勾自乘得九，股自乘得一十六，并得二十五，開方得五尺。

Problem 1: Finding the Hypotenuse

Problem: The base is 3 尺 and the height is 4 尺. What is the hypotenuse?

Answer: 5 尺.

Computation:

$$c = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5 \text{ 尺}$$

This is the famous 3-4-5 right triangle.

5.3 Problem: Finding the Height 求股

求股

弦十七步，勾八步，問股幾何？

答曰：十五步。

Solution 解：

術曰：弦自乘得二百八十九，勾自乘得六十四，相減餘二百二十五，開方得一十五步。

Finding the Height

Problem: The hypotenuse is 17 paces and the base is 8 paces. What is the height?

Answer: 15 paces.

Computation:

$$\text{Height} = \sqrt{17^2 - 8^2} = \sqrt{289 - 64} = \sqrt{225} = 15 \text{ paces}$$

5.4 Problem: The Reed in the Pond 葭生池中

葭生池中

池方一丈，葭生其中央，出水一尺。引葭至岸，適與岸齊。問水深、葭長各幾何？

答曰：水深一丈二尺，葭長一丈三尺。

Solution 解：

術曰：半池方自乘，出水自乘，相減，餘為實。倍出水為法。實如法而一，得水深。加水出尺數，得葭長。

設水深 d 尺，葭長 L 尺：

$$L = d + 1$$

$$L^2 = d^2 + 5^2$$

代入： $(d + 1)^2 = d^2 + 25$

$$d^2 + 2d + 1 = d^2 + 25$$

$$2d = 24 \Rightarrow d = 12$$

葭長 $L = 12 + 1 = 13$ 尺。

The Reed in the Pond

Problem: A pond is 1 丈 (10 尺) square. A reed grows at its center and extends 1 尺 above the water. When pulled to the shore, the reed just reaches the edge. What are the depth of the water and the length of the reed?

Answer: Water depth = 12 尺; Reed length = 13 尺.

Method: Let d = water depth, L = reed length. The reed extends 1 尺 above water, so $L = d + 1$. When pulled to shore, the reed forms the hypotenuse of a right triangle with legs d (water depth) and 5 尺 (half the pond width):

$$L^2 = d^2 + 5^2$$

Substituting $L = d + 1$:

$$(d + 1)^2 = d^2 + 25 \Rightarrow 2d + 1 = 25 \Rightarrow d = 12, \quad L = 13$$

5.5 Problem: The Door and the Pole 戶高與廣

戶高與廣

戶高多於廣六尺八寸，兩隅相去適一丈。問戶高、廣各幾何？

答曰：高九尺六寸，廣二尺八寸。

Solution 解：

術曰：以一丈自乘，又置高廣差自乘，以少減多，餘半之，開方得高廣和半。

設高 h 尺，廣 w 尺：

$$h - w = 6.8$$

$$h^2 + w^2 = 100$$

由第一式： $h = w + 6.8$ 。代入第二式：

$$(w + 6.8)^2 + w^2 = 100$$

$$2w^2 + 13.6w + 46.24 = 100$$

$$2w^2 + 13.6w - 53.76 = 0$$

解得： $w = 2.8$ 尺， $h = 9.6$ 尺。

The Door and the Pole

Problem: A door's height exceeds its width by 6 尺 8 寸. The diagonal (distance between opposite corners) is exactly 1 丈 (10 尺). Find the height and width of the door.

Answer: Height = 9 尺 6 寸; Width = 2 尺 8 寸.

Method: Let h = height, w = width.

$$h - w = 6.8$$

$$h^2 + w^2 = 100$$

From the first equation: $h = w + 6.8$. Substituting:

$$(w + 6.8)^2 + w^2 = 100 \Rightarrow 2w^2 + 13.6w + 46.24 = 100$$

$$2w^2 + 13.6w - 53.76 = 0 \Rightarrow w = 2.8 \text{ 尺}, \quad h = 9.6 \text{ 尺}$$

5.6 Problem: Square Inscribed in Right Triangle 勾股容方

勾股容方

勾六步，股十一步，問容方幾何？

答曰：方三步三十七分步之十五。

Solution 解：

術曰：勾股相乘為實，勾股并為法，實如法而一。

直角三角形勾 a 、股 b ，弦上容正方形邊長 s ：

$$s = \frac{ab}{a+b} = \frac{6 \times 11}{6+11} = \frac{66}{17} = 3\frac{15}{17}$$

注：原文所給略有差異，經核算當為 $\frac{66}{17}$ 。

Square Inscribed in Right Triangle

Problem: In a right triangle with base 6 paces and height 11 paces, find the side length of the largest square that can be inscribed with one side on the hypotenuse.

Answer: $3\frac{15}{17}$ paces.

Method: For a right triangle with base a and height b , the side s of the inscribed square (with one side on the hypotenuse) is:

$$s = \frac{ab}{a+b} = \frac{6 \times 11}{6+11} = \frac{66}{17} = 3\frac{15}{17}$$

Note: The original text gives a slightly different answer; upon verification the correct value is $\frac{66}{17}$ paces.

5.7 Problem: Measuring the Square City 邑方測量

邑方測量

邑方不知大小，各中開門。北門外二十步有木。出南門十四步，折而西行一千七百七十五步見木。問邑方幾何？

答曰：二百五十步。

Solution 解：

術曰：以西行步數乘北門外步數，又以南門外步數加邑方為法。實如法而一。

設邑方 s 步。視線構成相似直角三角形：

- 小三角形：底 $\frac{s}{2}$ ，高 $20 + \frac{s}{2} + 14$
- 大三角形：底 1775，高 $s + 34$

由相似三角形：

$$\frac{s/2}{20} = \frac{1775}{s+34}$$

交叉相乘：

$$s(s+34) = 2 \times 20 \times 1775 = 71000$$

$$s^2 + 34s - 71000 = 0$$

解得： $s = 250$ 步。

Measuring the Square City

Problem: A square city has gates at the midpoint of each side. A tree stands 20 paces outside the north gate. Walking 14 paces out from the south gate, then turning west and walking 1,775 paces, the tree becomes visible. Find the side length of the city.

Answer: 250 paces.

Method: Let s = side of the square city. The line of sight forms similar right triangles:

- Small triangle: base $\frac{s}{2}$, height $20 + \frac{s}{2} + 14$
- Large triangle: base 1775, height $s + 34$

Using similar triangles formed by the line of sight:

$$\frac{s/2}{20} = \frac{1775}{s+34}$$

Cross-multiplying: $s(s+34) = 2 \times 20 \times 1775 = 71000$

$$s^2 + 34s - 71000 = 0 \Rightarrow s = 250 \text{ paces}$$

6 Chapter on Construction 商功

商功 (「Construction Consultations」) 章論及各種工程與建築所用幾何立體之體積計算。

The 商功 (*Shanggong*, “Construction Consultations”) chapter deals with the calculation of volumes for various geometric solids used in engineering and construction projects.

6.1 Volume Conversion Ratios 土方換算

商功求積法曰：穿地四尺為壤五尺，為堅三尺。此穿地求壤四之，求堅三之，皆一而一。

Volume conversion ratios: When excavating earth: 4 尺 of loose earth becomes 5 尺 of piled earth, or 3 尺 of compacted earth. To convert: multiply loose volume by $\frac{5}{4}$ for piled earth, or by $\frac{3}{4}$ for compacted earth.

6.2 City Wall, Wall, Dike, Ditch, and Canal 城垣堤溝塹渠

城垣堤溝塹渠，皆同術。并上下廣而半之，以高或深乘之，又以袤乘之，即積尺。

體積公式：

$$V = \frac{(w_{\text{top}} + w_{\text{bottom}})}{2} \times h \times L$$

Method for city walls, walls, dikes, ditches, and canals: Add the top and bottom widths, halve the sum, multiply by the height (or depth), and multiply by the length to obtain the volume in cubic 尺.

$$V = \frac{(w_{\text{top}} + w_{\text{bottom}})}{2} \times h \times L$$

6.3 Problem: Volume of a City Wall 城積

城積

城下廣四丈，上廣二丈，高五丈，袤一百二十六丈五尺。問積幾何？

答曰：一百八十九萬七千五百尺。

Solution 解：

術曰：并上下廣，四丈加二丈得六丈，半之三丈。以高五丈乘之，得一十五丈。又以袤一百二十六丈五尺乘之，即積。

換算：1 丈 = 10 尺

$$\begin{aligned} V &= \frac{(40 + 20)}{2} \times 50 \times 1265 \\ &= 30 \times 50 \times 1265 \\ &= 1,897,500 \text{ 立方尺} \end{aligned}$$

Volume of a City Wall

Problem: A city wall has bottom width 4 丈, top width 2 丈, height 5 丈, and length 126 $\frac{1}{2}$ 丈. What is its volume?

Answer: 1,897,500 cubic 尺.

Computation: (1 丈 = 10 尺)

$$\begin{aligned} V &= \frac{(40 + 20)}{2} \times 50 \times 1265 \\ &= 30 \times 50 \times 1265 \\ &= 1,897,500 \text{ cubic 尺} \end{aligned}$$

6.4 Problem: Square Tower 方亭臺

方亭臺

方亭臺，上方四丈，下方五丈，高五丈。問積幾何？

答曰：一十萬一千六百六十六尺太半尺。(101,666 $\frac{2}{3}$ 立方尺)

Square Tower

Problem: A square platform has top side 4 丈, bottom side 5 丈, and height 5 丈. What is its volume?

Answer: 101,666 $\frac{2}{3}$ cubic 尺.

Solution 解:

術曰：上下方相乘，又各自乘，并之，以高乘之，三而一。

方臺（方亭）體積公式：

$$V = \frac{h}{3}(a^2 + ab + b^2)$$

代入數值：

$$\begin{aligned} V &= \frac{50}{3}(40^2 + 40 \times 50 + 50^2) \\ &= \frac{50}{3}(1600 + 2000 + 2500) \\ &= \frac{50 \times 6100}{3} \\ &= 101,666\frac{2}{3} \text{ 立方尺} \end{aligned}$$

Method: For a square frustum (方亭):

$$V = \frac{h}{3}(a^2 + ab + b^2)$$

Substituting values:

$$\begin{aligned} V &= \frac{50}{3}(40^2 + 40 \times 50 + 50^2) \\ &= \frac{50}{3}(1600 + 2000 + 2500) \\ &= \frac{50 \times 6100}{3} \\ &= 101,666\frac{2}{3} \text{ cubic 尺} \end{aligned}$$

6.5 Problem: Round Tower 圓亭臺

圓亭臺

圓亭臺，上周二丈，下周三丈，高二丈。問積幾何？

Solution 解:

術曰：上周、下周相乘，又各自乘，并之，以高乘之，三十六而一。

圓臺（圓亭）體積公式：

$$V = \frac{h}{36}(C_1^2 + C_1 \cdot C_2 + C_2^2)$$

其中 C_1 、 C_2 為上、下周長。中國古代取 $\pi \approx 3$ 。

Round Tower

Problem: A circular platform has top circumference 2 丈, bottom circumference 3 丈, and height 2 丈. What is its volume?

Method: For a circular frustum (圓亭), use the formula:

$$V = \frac{h}{36}(C_1^2 + C_1 \cdot C_2 + C_2^2)$$

where C_1 and C_2 are the top and bottom circumferences. In ancient Chinese mathematics, $\pi \approx 3$, so $C = 2\pi r \approx 6r$ and the cross-sectional area $\approx \frac{C^2}{36}$.

6.6 Problem: Grain Pile on the Ground 委粟

委粟

委粟平地，下周一十二丈，高二丈。問積幾何？及為粟各幾何？

Solution 解:

術曰：下周自乘，以高乘之，三十六而一。

穀堆成圓錐形。以 $\pi \approx 3$ ：

$$\begin{aligned} V &= \frac{C^2 \times h}{36} \\ &= \frac{120^2 \times 20}{36} \\ &= \frac{288,000}{36} \\ &= 8,000 \text{ 立方尺} \end{aligned}$$

以標準換算率得所容粟數。

Grain Pile on the Ground

Problem: A pile of millet grain on flat ground has base circumference 12 丈 and height 2 丈. What is its volume and how many 斛 of millet does it contain?

Method: The pile forms a cone. Using the formula for a cone with $\pi \approx 3$:

$$\begin{aligned} V &= \frac{C^2 \times h}{36} \\ &= \frac{120^2 \times 20}{36} \\ &= \frac{288,000}{36} \\ &= 8,000 \text{ cubic 尺} \end{aligned}$$

The conversion to 斛 of grain uses the standard conversion rate.

7 Yang Hui's Triangle 開方作法本源

楊輝最著名之貢獻之一，為保存今稱「楊輝三角」之數表（西方常稱「帕斯卡三角」）。楊輝將此歸功於北宋數學家賈憲（約 1010-1070 年）。

One of Yang Hui's most celebrated contributions is his preservation of what is now known as "Yang Hui's Triangle" (often called Pascal's Triangle in the West). Yang Hui attributes this to the Northern Song mathematician Jia Xian (賈憲, c. 1010-1070).

Yang Hui's Triangle 開方作法本源圖

				1			
			1		1		
		1		2		1	
	1		3		3		1
1		1	4		6		4
	1	5		10		10	
		1	6		15		6
			1	7		21	
				1	8		8
					1	9	
						1	10

開方作法本源：左徇乃積數，右徇乃隅算，中藏者皆廉。以廉乘商方，命實而除之。

此三角提供二項式展開之係數：

$$(a + b)^1 = a + b$$

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

$$(a + b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$$

$$(a + b)^5 = a^5 + 5a^4b + 10a^3b^2 + 10a^2b^3 + 5ab^4 + b^5$$

Explanation: The left side gives the coefficients for root extraction, the right side gives the corner value (always 1), and the center entries are the "shoulders" (intermediate coefficients). These coefficients are used in the process of extracting roots by the method of increasing multiplication.

The triangle provides the coefficients for binomial expansion:

$$(a + b)^1 = a + b$$

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

$$(a + b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$$

$$(a + b)^5 = a^5 + 5a^4b + 10a^3b^2 + 10a^2b^3 + 5ab^4 + b^5$$

7.1 Application to Root Extraction 開方應用

楊輝以此係數用於增乘開方法（「遞增乘法」），以求平方根、立方根及更高次方根。此方法等同於西方之霍納法（Horner's method）。

增乘開方法之步驟：

1. 置實（被開方數）
2. 借一算（定位）
3. 以隅法（最高次係數）除實
4. 以次商乘廉法，遞增相乘
5. 逐步逼近，得根之數值

Yang Hui used these coefficients in the method of 增乘開方法 ("Increasing Multiplication Method") for extracting square roots, cube roots, and higher-order roots. This method is equivalent to Horner's method in Western mathematics.

Steps of the Increasing Multiplication Method:

1. Set down the dividend (number to extract root from)
2. Borrow one counting rod for positioning
3. Divide the dividend by the corner method (highest coefficient)
4. Multiply the next quotient by the shoulder method, increasing iteratively
5. Approximate step by step to obtain the numerical root

Appendix: Classification of Nine Chapters Problems 九章纂類

楊輝將《九章算經》二百四十六題，按解法歸納為九大類：

Yang Hui reorganized the 246 problems of the Nine Chapters into nine categories based on their solution methods:

No.	Category 類	Problems
1	乘除 (Multiplication and Division / 乘除)	41
2	分率 (Rates and Proportions / 分率)	41
3	合率 (Combined Rates / 合率)	10
4	互換 (Exchange and Conversion / 互換)	11
5	衰分 (Proportional Distribution / 衰分)	18
6	疊積 (Stacked Accumulations / 疊積)	27
7	盈不足 (Excess and Deficit / 盈不足)	20
8	方程 (Rectangular Arrays / 方程)	19
9	勾股 (Right Triangles / 勾股)	24
Total 總計		246

楊輝竊見九章舊本，作立題法，迫圓古序云：靖康以來，罕有舊本，間有存者，狃於末習，不循本意，以至具術淹廢，偽本滋興。此說固然，殊不知所傳之本亦不得其真矣。

Translation: Yang Hui observed that in old editions of the Nine Chapters, since the Jingkang era (1126 CE), original copies have been rare. Those that survived were corrupted by later practices, deviating from the original intent, leading to the decline of proper methods and the proliferation of erroneous editions.

纂類序

黃帝九章古序云：國家嘗設算科取士，選九章以為算經之首，蓋猶儒者之六經，醫家之難素，兵法之孫子歟。昔聖宋紹興戊辰，算士榮榮謂靖康以來，罕有舊本，間有存者，狃於未習，向獲善本，得其全經，復起於學。以魏景元元年劉徽等、唐朝議大夫、行太史令上輕車都尉李淳風等注釋，聖宋右班直賈憲細草。

輝嘗聞學者，謂九章題問頗隱，法理難明，不得其門而入。於是以答參問，用草考法，因法推類，然後知斯文非古之全經也。將後賢補賡之文，修前代已廢之法，刪立題術，又纂法問，詳著於後，儻得賢者，改而正諸，是所願也。

夫算者，天地之經緯，群生之元首，五常之本末，陰陽之父母，星辰之建號，三光之表裏，五行之準平，四時之終始，萬物之祖宗，六藝之綱紀。稽群倫之聚散，考二氣之降升，推寒暑之迭運，步遠近之殊同，觀天道精微之兆基，察地理縱橫之長短，采神祇之所在，極成敗之符驗。

九章算術，其來尚矣。周公制禮而有九數，九數之流，則九章是矣。昔在庖犧氏始畫八卦，以通神明之德，以類萬物之情，作九九之術，以合六爻之變。暨於黃帝神而化之，引而伸之，於是建曆紀，協律呂，用稽道原，然後兩儀四象精微之氣可得而效焉。

輝不揆淺陋，輒敢刪繁補闕，推類引伸，纂為九類。一曰乘除，二曰互換，三曰合率，四曰分率，五曰衰分，六曰疊積，七曰盈不足，八曰方程，九曰勾股。凡此九類，蓋盡九章之術，而條貫彌綸，秩序井然。使學者由淺入深，由易及難，循序漸進，則九章之法庶幾其可明矣。

楊輝謹序於臨安府學

九章互見目錄

楊輝竊見九章舊本，作立題法遺闕。古序云：靖康以來，罕有舊本，間有存者，狃於末習，不循本意，以至真術淹廢，偽本滋典。此說固然，殊不知所傳之本亦不得其真矣。如粟米章之互換、少廣章之求由開方，皆重疊無謂，而作者題問不歸章次，亦有之。今作纂類，互見目錄，以辯其訛。

古本二百四十六問，其分布如下：

章別	問數		
方田	問		
粟米	問		
衰分	問		
少廣	問		
商功	問		
均輸	問		
盈不足	問		
方程	問		
勾股	問		

纂類九類總目

乘除第一：凡四十一問。方田三十八問，粟米二問，並乘除之術。

互換第二：凡五十六問。粟米三十一問，衰分十一問，均輸十一問，盈不足二問。

合率第三：凡二十問。少廣十一問，均輸八問，盈不足一問。

分率第四：凡九問。本章九問，含乘分、除分、合分、課分、平分諸術。

衰分第五：凡一十八問。本章九問，均輸九問。

疊積第六：凡二十七問。並商功章，含城、垣、堤、溝、渠、壕、倉、窯、盤池、冥谷、芻蕘、芻童、羨除等術。

盈不足第七：凡十一問。含盈不足、適足、兩盈、兩不足、盈適足、不足適足之術。

方程第八：凡二十問。本章十八問，均輸盈不足二問，並方程正負之術。

勾股第九：凡三十八問。少廣十三問，商功一問，本章二十四問。

乘除第一

凡四十一問。今考該四十問：方田三十八，粟米二問。此類蓋算學之基礎，凡學算者必先由乘除而入，而後可以進求高深之法。乘除之術不明，則雖遇至易之題，亦將茫然無措矣。

直田法

直田法曰：廣從相乘為實，如畝法而一。按直田即矩形田也，廣者橫之長，從者縱之長，二者相乘得積步，以二百四十步為一畝除之，即得畝數。此術最為簡明，而實諸法之根本。

方田一

廣十五步，從十六步，問田幾何？

答：一畝。

術：廣十五步，從十六步，廣從相乘，得二百四十步，如畝法二百四十步而一，得田一畝。

方田二

廣十二步，從十四步，問田幾何？

答：一百六十八步。

術：廣十二步，從十四步，相乘得一百六十八步。此未滿畝法，故以步計之。

里田法

里田一

方田一里，問為田幾何？

答：三頃七十五畝。

術：方一里自乘，得一里之積；以一里三百七十五畝乘之，即得。

$$300 \times 300 = 90,000$$

圭田法

圭田法曰：半廣以乘正從，或半正從以乘廣。圭田者，三角形田也，其形如圭，故以名之。半廣乘正從者，取三角形之半積也。

圭田一

圭田廣十二步，從二十一步，問田幾何？

答：一百二十六步。

術：半廣得六步，以乘正從二十一步，得一百二十六步。

$$6 \times 21 = 126$$

邪田法

邪田法曰：併兩斜半之以乘正從，或併兩廣乘半從。邪田者，梯形田也，其兩廣不等，故曰邪。

邪田二

邪田一畔廣三十步，一畔廣四十二步，正從六十四步，問田幾何？

答：九畝一百四十四步。

$$30 + 42 = 7236 \times 64 = 2,304$$

術：併兩廣，三十步並四十二步，得七十二步，半之得三十六步，以乘正從六十四步，得二千三百四步。如畝法而一，得九畝，餘一百四十四步。

圓田法

圓田法曰：半周半徑相乘，半周自乘三而一，周自乘十二而一，徑自乘三之四而一。按圓田之術，古法以周三徑一為率，是謂古術。劉徽創割圓之術，以為周三徑一猶為疏闊，乃設密率以究其微。至密率之術，則周自乘，又二十五乘之，三百一十四而一。

圓田一（古術）

圓田周三十步，徑一十步，問田幾何？（古術）

答：七十五步。

術：半周得一十五步，半徑得五步，相乘得七十五步。此用周三徑一之率，半周半徑相乘即得圓積。

×

圓田二（密率）

圓田周三十步，徑一十步，問田幾何？（密率）

答：七十一步二十二分步之一十三。

$$71 \frac{13}{22}$$

術：密率者，周自乘得九百，又二十五乘之，得二萬二千五百，以三百一十四為法除之，得七十一步，餘一百七十六。以法命之，得二十二分步之一十三。

$$7171 \frac{176}{314} = 71 \frac{13}{22}$$

除率法（商除）

經率法（俗名商除）曰：錢數為實，以所買率為法，實如法而一。原載粟米章。商除者，以商數除實而求其價也，此乃除法之正用。

經率一

錢一百六十文，買瓠甓十八枚，問枚價幾何？

答：一枚八錢九分錢之八。

$$8 \frac{8}{9}$$

術：以錢一百六十文為實，以所買一十八枚為法，實如法而一。不盡者，以法命之，得八錢九分錢之八。

$$160 \div 18 = 88 \frac{8}{9}$$

互換第二

凡五十六問。粟米三十一，衰分十一，均輸十一，盈不足二。互換者，以所有易所求也，凡物之有價者皆可互換，不獨粟米為然。

互換法總論

互換法曰：以所求率乘所有數，為實；以所有率為法，實如法而一。率者，物與物之相為比例者也。知一物之率，則可以推百物；知一時之率，則可以推異時。故算家以率為綱，而互換之術由此出焉。

粟米互換

粟米互換者，以五穀相為變易也。古者粟為政之本，故以粟為主，而以諸米為從。其率之設，則因品質之精粗而異。

粟米互換一

粟二斗一升，問為粳米幾何？

答：一斗一升五十分升之十七。

$$\frac{17}{50}$$

粟米互換十二

粟一斗，問為粳米幾何？

答：六升。

術：以粟求粳米，三之五而一。粟一斗為十升，以五乘之得五十，如三而一，得一十六升三分升之二。輝按：此答亦與古法不同，當以本術細推之。

$$10 \times 5 = 50 \div 3 = 16\frac{2}{3}$$

衰分互換

衰分互換一

今有絲一斤價錢三百四十五，問兩價幾何？

答：一兩二十一錢八分錢之三。

$$21\frac{3}{8}$$

術：一斤十六兩，以十六兩為法，價錢三百四十五為實，實如法而一。

$$345 \div 16 = 21\frac{9}{16} = 21\frac{3}{8}$$

均輸互換

均輸互換一

今有甲縣粟一石價二十錢，乙縣粟一石價十五錢。甲縣道輸所二百里，乙縣道輸所一百里。問各粟一石至輸所其直幾何？

答：甲縣一石直二十錢，加運價八文，共二十八文；乙縣一石直十五錢，加運價四文，共十九文。

持金出關

持金出關

今持金十二斤出關，稅之十分而取一。今關取金二斤，價錢五千。問金一斤值錢幾何？

答：六千二百五十。

$$5000 \times 2 \div 125000 \div (12 \times \frac{1}{10}) \times \frac{1}{1} = 6250$$

術：以稅金二斤為所有數，價錢五千為所求率，持金十二斤為所有率。以所求率乘所有數，為實；以所有率為法，實如法而一。

持米出三關

持米出三關，外關三分稅一，中關五分稅一，內關七分稅一，餘存米五斗。問：原米幾何？

答：一十斗九升八分升之三。

$$\frac{3}{8}$$

$$5 \times 7 \div 6 = 5\frac{5}{6}$$

術：此題當以還原術推之。餘存米五斗，為內關七分稅一之餘，即六分之五也。以五斗乘七，得三十五斗，以六除之……得一十斗九升八分升之三。

合率第三

凡二十問：少廣十一，均輸八，盈不足一。合率者，合諸分數而求其率也。其術與分率相為表裏，而所施則異。分率施於乘除，合率施於併合。

少廣法

少廣反合分並率。設諸分母子併而為廣，借田求縱，立少廣章，易合分之術而為之法。用副置分母自乘，以乘全步及諸子，各以本母除其子而併之，免互乘之繁。輝按：少廣之術，其義甚深。廣者所以知從，從者所以知廣。今知廣之不齊，而求從之幾何，是反用合分之術也。

少廣一

田一畝廣一步半，問從？

$$1\frac{1}{2}$$

答：一百六十步。

術：列置全步及分母子，一步半即二分步之一也。副置分母二自乘，得四。以乘全步一，得四；又以乘子一，得四。各以本母除子，併之為法。輝按：此術以分母自乘通之，法得三，實得四百八十，實如法而一，得一百六十步。

$$1\frac{1}{2} = \frac{3}{2}480 \div 3 = 160$$

少廣二

田一畝廣一步半，三分步之一，問從？

$$1\frac{1}{2} + \frac{1}{3}$$

答：一百三十步一十一分步之一十。

$$130\frac{10}{11}$$

術：列置全步及分母子。副置分母二、三，相乘得六為通分。以六乘全步一，得六；以六乘二分之一子一，得六，以母二除之得三；以六乘三分之一子一，得六，以母三除之得二。併之得十一為法。以畝法二百四十步乘通分六，得一千四百四十為實。實如法而一，得一百三十步，餘十。以法命之。

$$6 \times 1 = 66 \times \frac{1}{2} = 36 \times \frac{1}{3} = 26 + 3 + 2 = 11240 \times 6 = 1440$$
$$1440 \div 11 = 130130\frac{10}{11}$$

少廣十一

田一畝廣一步半，三分步之一，四分步之一，五分步之一，六分步之一，七分步之一，八分步之一，九分步之一，十分步之一，十一分步之一，十二分步之一，問從？

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} + \frac{1}{9} + \frac{1}{10} + \frac{1}{11} + \frac{1}{12}$$

答：七十七步四萬五千八百一十三分步之八萬六千九百八十四。

$$77\frac{86984}{45813}$$
$$\frac{83711}{27720}$$

分率第四

凡九問。分率者，分數之率也。算術之難，莫難於分數。分數之難，在於分子分母之錯綜交互，非精通其理者不能得其正。九章設分率一章，專論分數之四則運算，以為諸術之基礎。

乘分法

乘分法曰：分母各乘其全，分子從之，相乘為實，分母相乘為法。乘分者，分數相乘之術也。其理以分母通全，而後相乘，所以齊其不齊也。

乘分一

田廣一十八步七分步之五，從二十三步十一分步之六，問為田幾何？

答：一畝二百步十一分步之七。

術：廣一十八步七分步之五，分母七乘全步一十八，得一百二十六，併分子五，得一百三十一為廣實。從二十三步十一分步之六，分母十一乘全步二十三，得二百五十三，併分子六，得二百五十九為從實。以廣實乘從實，得三萬三千九百二十九為積實。分母七乘十一，得七十七為法。

$$18\frac{5}{7}23\frac{6}{11}$$

$$\frac{7}{11}$$

$$7 \times 18 + 5 = 13111 \times 23 + 6 = 259131 \times 259 = 33,9297 \times 11 = 77$$

$$33,929 \div 77 = 440\frac{52}{77} = \frac{7}{11}$$

除分法

除分一

七人均入錢三分錢之一，問人得幾何？

答：人得一錢二十一分錢之四。

術：以人數七為法，錢三分錢之一通之，得三分之四為實。實如法而一，人得二十一分之四錢。

$$1\frac{1}{3}$$

$$1\frac{4}{21}$$

$$1\frac{1}{3} = \frac{4}{3} \div 7 = \frac{4}{21} 1\frac{4}{21}$$

合分法

合分法曰：母互乘子，併以為實，母相乘為法。合分者，併眾分數為一也，今謂之加法。

合分一

三分之一，五分之二，問合之得幾何？

答：一十五分之十一。

術：母互乘子，三乘二得六，五乘一得五，併之得十一為實。母相乘，三乘五得十五為法。

$$\frac{1}{3} + \frac{2}{5}$$

$$\frac{11}{15}$$

$$3 \times 2 = 65 \times 1 = 56 + 5 = 113 \times 5 = 15$$

合分二

三分之二，四分之三，五分之四，問合之得幾何？

答：一六十分之四十三。

術：三母連乘得六十為法。三乘三乘四得三十六為第一子；四乘二乘四得三十二為第二子；五乘二乘三得三十為第三子。併之得九十八為實。實如法而一，得一六十分之九十八，約為四十九分之五十二。

$$\frac{2}{3} + \frac{3}{4} + \frac{4}{5}$$

$$\frac{49}{60} \frac{43}{60}$$

$$3 \times 4 \times 5 = 604 \times 5 \times 2 = 403 \times 5 \times 3 = 453 \times 4 \times 4 = 48$$

$$133/60 = 2\frac{13}{60}$$

課分法

課分一

三分之二，四分之三，問孰多？多幾何？

$$\frac{2}{3} \frac{3}{4}$$

答：四分之三多，多十二分之一。

$$\frac{3}{4} \frac{1}{12}$$

術：母互乘子，三乘三得九，四乘二得八，以少減多，九減八餘一為實。母相乘，三乘四得十二為法。

$$3 \times 3 = 9, 4 \times 2 = 8, 9 - 8 = 1, 3 \times 4 = 12$$

平分法

平分一

三分之二人出二，四分之三人出三，問人各應出幾何？

答：各出二錢十三分錢之六。

$$2 \frac{6}{13}$$

衰分第五

凡一十八問：本章九問，均輸九問。衰分者，差分之法也。以差率為比例而分配之，使各得其當。衰有列衰、返衰之別，列衰者，以多為多，以少為少；返衰者，以多為少，以少為多。

衰分法曰：各置列衰，副併為法，以所分乘未併者各自為實，實如法而一。輝按：衰分之術，施於賦役之征調，祿食之頒給，財貨之分配，無所不宜。

衰分一（牛羊豕價）

今有牛二羊五買豕一十三家，有餘錢一千；賣牛三豕三以買九羊，錢適足；賣六羊八豕以買五牛，錢不足六百。問牛羊豕價各幾何？

答：牛價一千二百，羊價五百，豕價三百。

術：此題當以方程術解之。設牛價為牛，羊價為羊，豕價為豕。牛二加羊五減豕一十三，餘一千；牛三減羊九加豕三，適足；牛五減羊六減豕八，不足六百。列為三行，以正負術入之。

$$2O + 5S - 13P = 1000 \quad 3O - 9S + 3P = 0 \quad 5O - 6S - 8P = -600$$

衰分二（五雀六燕）

五雀六燕共重一斤，雀重燕輕，交易一枚，其重適等。問各幾何？

答：雀一兩十九分兩之一十三，燕一兩十九分兩之五。

術：五雀六燕共重一斤，即十六兩。交易一枚其重適等者，四雀一燕之重與五燕一雀之重適等，即三雀與四燕等重。以三雀當四燕，則五雀六燕共為九雀又三分之二雀，以十六兩除之，得雀之重。又以三雀四燕之比例，得燕之重。

$$1\frac{13}{19} \frac{5}{19} = 5S + 6 \times \frac{3}{4}S = 5S + 4.5S = 9.5S = 16S = \frac{32}{19} = 1\frac{13}{19}$$

衰分三（返衰）

今有大夫五人，不更四人，簪褭三人，上造二人，公士一人，凡十五人，共出百五十錢。欲令爵高者出少，以次漸多，問各幾何？

答：大夫五人各出十錢，不更四人各出十五錢，簪褭三人各出二十錢，上造二人各出二十五錢，公士一人出五十錢。

$$5(1) + 4(2) + 3(3) + 2(4) + 1(5) = 5 + 8 + 9 + 8 + 5 = 35 \\ 150 \times 1/35 \times 5 = \dots$$

術：此返衰之術也。爵高者出少，以次漸多，故用返衰。大夫五人以一為衰，不更四人以二為衰，簪褭三人以三為衰，上造二人以四為衰，公士一人以五為衰。

衰分六（女子善織）

今有女子善織，日自倍，五日織五尺。問日織幾何？

答：初日織一寸三十一分寸之十九，次日織三寸三十一分寸之七，第三日織六寸三十一分寸之十四，第四日織一尺二寸三十一分寸之二十八，第五日織二尺五寸三十一分寸之二十五。

$$1\frac{19}{31} 3\frac{7}{31} 6\frac{14}{31} 12\frac{28}{31} 25\frac{25}{31} \\ 50 \times 1/31 = 1\frac{19}{31} 50 \times 2/31 = 3\frac{7}{31}$$

術：此題以倍衰為列衰。初日為一，次日為二，第三日為四，第四日為八，第五日為十六。併之得三十一為法。以五尺乘未併者各自為實，實如法而一。

疊積第六

凡二十七問，並商功章。疊積者，積疊之術也。凡築城、穿壕、開渠、建倉，皆需度土方之多寡，此疊積之所由設也。商功之術，以立方為本。立方者，長廣高深相乘之積也。

××

城垣壕溝

城垣一

城下廣四丈，上廣二丈，高五丈，袤一百二十六丈五尺。問積幾何？

答：一百八十九萬七千五百尺。

$(4 + 2)/2 = 33 \times 5 = 1515 \times 126.5 = 1897.5 = 1,897,500^{33}$

術：併上下廣，得六丈，半之得三丈，以高五丈乘之，得十五萬平方尺，以袤一百二十六丈五尺乘之，即得積尺。

倉窖積粟

倉一

方倉廣一丈六尺，長二丈，深三尺。問積及容粟幾何？

答：積九百六十尺，容粟九百六十斛。

術：方倉之積，廣乘長，又乘深，即得積尺。以二尺七寸為一斛之積除之，即得容粟之數。輝按：古者量制，一斛之積為二尺七寸，故以積尺除之得斛數。

$16 \times 20 \times 3 = 960960 \div 1 = 960$

芻蕘芻童

芻蕘

芻蕘法曰：倍下長併入上長，以廣乘之，又高乘之，皆如六而一。下廣三丈，袤四丈，上袤二丈，無廣，高一丈。問積幾何？

答：五千尺。

$2 \times 4 = 88 + 2 = 10 = 30,000 \div 6 = 5,000$

術：倍下長得八丈，併入上長二丈，得十丈。以下廣三丈乘之，得三十萬平方尺。又以高一丈乘之，得三百萬立方尺。如六而一，得五千尺。

芻童

芻童法曰：倍上長併入下長，以下廣乘之；又倍下長併入上長，以上廣乘之；併之，以高乘之，皆如六而一。上廣二丈，上袤三丈，下廣四丈，下袤五丈，高二丈。問積幾何？

$\times \times \times \div \times \times$

答：二萬六千尺。

$(2 \times 3 + 5) \times 4 + (2 \times 5 + 3) \times 2 = 11 \times 4 + 13 \times 2 = 44 + 26 = 70 \times 2 = 140 = 140,000^3 \div 6 = 23,333\frac{1}{3}$

陽馬

陽馬法曰：廣袤相乘，以高乘之，三而一。廣五尺，袤七尺，高八尺。問積幾何？

答：九十三尺三分尺之一。

術：廣五尺乘袤七尺，得三十五平方尺。又以高八尺乘之，得二百八十立方尺。三而一，得九十三尺三分尺之一。

$\times \times \div$

$93\frac{1}{3}$

$5 \times 7 = 35$
 $35 \times 8 = 280$
 $280 \div 3 = 93\frac{1}{3}$

鰲臚

鰲臚法曰：廣袤相乘，又以高乘之，六而一。廣四尺，袤五尺，高三尺。問積幾何？

答：一十尺。

術：廣四尺乘袤五尺，得二十平方尺。又以高三尺乘之，得六十立方尺。六而一，得一十尺。

$\times \times \div$

$4 \times 5 = 20$
 $20 \times 3 = 60$
 $60 \div 6 = 10$

盈不足第七

凡十一問。盈不足者，假設之術也。凡數之隱晦難知者，設兩假以測之，一假則盈，一假則不足，因盈不足以定其實。此術之妙，在於以虛測實，以假求真。

盈朒適足法

盈朒適足法曰：置所出率，盈朒各居其下。副置出率，以少減多餘為約法。盈朒適足，令維乘所出率實，以盈朒之數乘人積為人實，實皆如約法而一。輝按：盈不足之術，源遠流長。其法以兩次假設之結果，一盈一不足，由此推求人數與物價。其理與今之線性插值法相通。

盈不足一（共買犬）

共買犬，人出五不足九十文，人出五十文適足。問人數、犬價各幾何？

答：二人，犬價一百。

術：置所出率，五文與五十文。以少減多，五十減五餘四十五為法。以盈不足之數，不足九十為實。實如法而一，得二人，即人數。以適足出率五十乘人數二，得一百，即犬價。驗之：人出五，二人共出十，不足九十，則犬價為一百；人出五十，二人共出一百，適足。合問。

$$50 - 5 = 45 \quad 90 \div 45 = 2 \quad 250 \times 2 = 1005 \times 2 = 1050 \times 2 = 100$$

盈不足三（買瓦）

共買瓦，人出八盈三枚，人出七不足四枚。問人數、瓦數各幾何？

答：七人，瓦五十三枚。

術：置所出率八與七，以少減多餘一為法。盈三不足四，併之得七為實。實如法而一，得七人。以盈三減之，或以不足四加之，皆得瓦五十三枚。

$$3 + 4 = 7 \quad 77 \div 1 = 78 \times 7 - 3 = 537 \times 7 + 4 = 53$$

盈不足六（二酒求價）

今有醇酒一斗直錢五十，行酒一斗直錢一十。今將錢三十，得酒二斗。問醇行酒各幾何？

答：醇酒二升半，行酒一斗七升半。

術：此題以盈不足術解之。設醇酒一斗，則直錢五十，已過三十；設行酒一斗，則直錢一十，不足二十。以盈不足術求之，醇酒得二升半，行酒得一斗七升半。輝按：此術即今之所謂雞兔同籠也。

$$2\frac{1}{2} = \frac{30-10}{50-10} = \frac{20}{40} = \frac{1}{4} = \frac{1}{2} = 52.5$$

盈不足七（五家共井）

今有五家共井，甲二綆不足如乙一，乙三綆不足如丙一，丙四綆不足如丁一，丁五綆不足如戊一，戊六綆不足如甲一。如各得所不足一綆，皆逮。問井深、綆長各幾何？

$$a, b, c, d, e \quad D2a + b = D3b + c = D4c + d = D5d + e = D6e + a = D$$

方程第八

凡二十問：本章十八問，均輸盈不足二。方程者，方陣之術也。以行列相併相減，求眾物之價值，此最為精妙之術也。輝按：方程之術，乃中國數學之瑰寶也。其法以算籌列為方陣，互乘相減，以消去未知數，此與今之代數學所謂行列式、矩陣者異曲同工。

方程法

方程法曰：所求率互乘鄰行，以少減多，再求減損。錢為實，物為法，實如法而一。置位法曰：依所問排列逐物與價，而鄰行相對如之。

方程一（上中下禾）

上禾三束，中禾二束，下禾一束，共實三十九斗。上禾二束，中禾三束，下禾一束，共實三十四斗。上禾一束，中禾二束，下禾三束，共實二十六斗。問：上、中、下禾一束各幾何？

答：上禾一束，得九斗四分斗之一；中禾一束，得四斗四分斗之一；下禾一束，得二斗四分斗之三。

$$9\frac{1}{4}4\frac{1}{4}2\frac{3}{4}$$

$$3u + 2m + l = 392u + 3m + l = 34u + 2m + 3l = 26l - m = 5$$

方程二（牛羊價）

五牛二羊，直金十兩；二牛五羊，直金八兩。問：牛、羊價各幾何？

答：一牛直金一兩二十一分兩之一十三；一羊直金二十一分兩之二十。

$$1\frac{13}{21}20\frac{20}{21}$$

$$5O + 2S = 102O + 5S = 810O + 4S = 2010O + 25S = 40 \\ 21S = 20S = \frac{20}{21}5O = 10 - \frac{40}{21} = \frac{170}{21}O = \frac{34}{21} = 1\frac{13}{21}$$

正負術

正負術曰：同名相除，異名相益，正無入負之，負無入正之。其異名相除，同名相益，正無入正之，負無入負之。輝按：正負之術，乃方程之樞紐也。古者以赤籌為正，黑籌為負，列於算板之上，以相消長。同名相除者，正與正除，負與負除也；異名相益者，正與負益也。

方程四（賣牛買羊）

今有賣牛二、羊五，以買一十三豕，有餘錢一千；賣牛三、豕三，以買九羊，錢適足；賣六羊、八豕，以買五牛，錢不足六百。問：牛、羊、豕價各幾何？

答：牛價一千二百，羊價五百，豕價三百。

術：此三行方程也。設牛價為正，羊價為負，豕價為負，餘錢為正，不足為負。列為三行，以正負術入之。

$$2O + 5S - 13P = 10003O - 9S + 3P = 05O - 6S - 8P = -600$$

方程六（麻麥豆）

今有麻三斗、麥四斗、豆五斗，共直錢一百；麻四斗、麥五斗、豆三斗，共直錢一百一十；麻五斗、麥三斗、豆四斗，共直錢一百二十。問：麻、麥、豆一斗各直幾何？

$$H + W - 2B = 10H = 15W = 5B = 8$$

方程七（五色絲）

今有青絲五斤、黃絲四斤、白絲三斤、朱絲二斤、黑絲一斤，共直錢一百；青絲四斤、黃絲三斤、白絲二斤、朱絲一斤、黑絲五斤，共直錢九十；青絲三斤、黃絲二斤、白絲一斤、朱絲五斤、黑絲四斤，共直錢八十；青絲二斤、黃絲一斤、白絲五斤、朱絲四斤、黑絲三斤，共直錢七十；青絲一斤、黃絲五斤、白絲四斤、朱絲三斤、黑絲二斤，共直錢六十。問：五色絲一斤各直幾何？

勾股第九

凡三十八問：少廣十三，商功一問，本章二十四。勾股者，直角三角形之術也。勾為短邊，股為長邊，弦為斜邊。勾股之術，以自乘併之為根本，以開方為施用。此術之用，至為廣大，凡測高深、量遠近、求曲直，無不用之。

勾股基本方法

勾股法曰：勾自乘，股自乘，併之為弦實，開方除之即弦。輝按：勾股之術，其源甚古。周公問於商高曰：「數之法出於圓方，圓出於方，方出於矩，矩出於九九八十一。故折矩以為勾廣三，股修四，徑隅五。」此勾股之術所由起也。

勾股一（立木垂索）

立木垂索，委地二尺，引索斜之，拄地去木八尺。問索長幾何？

答：一十七尺。

術：此題以勾股術解之。立木垂索，委地二尺，此股弦較也。拄地去木八尺，此勾也。勾八尺自乘，得六十四。如股弦較二尺而一，得三十二。加較二尺半之得一……輝按：此術當以勾股定理細推之。勾八尺，股十五尺，弦一十七尺。八平方加十五平方，得六十四加二百二十五，得二百八十九，開方得一十七。合問。

$$\begin{aligned} xian - gu &= 28^2 + gu^2 = xian^2 8^2 = xian^2 - gu^2 = (xian - \\ gu)(xian + gu) &= 2(xian + gu) = 64xian + gu = 32xian - gu = \\ 22 \cdot xian &= 34xian = 17gu = 158^2 + 15^2 = 64 + 225 = 289 = \\ 17^2 \end{aligned}$$

勾股三（圓材鋸深）

圓材埋在壁中，不知大小，鋸深一寸，道長一尺。問徑幾何？

$$\begin{aligned} r^2 &= (r -)^2 + (/2)^2 r^2 = (r - 1)^2 + 25 = r^2 - 2r + 1 + 25 \\ 2r &= 26r = 13 \end{aligned}$$

勾股容方容圓

勾股容方

今有勾五步，股十二步。問：勾中容方幾何？

答：方三步十七分步之九。

術：勾股容方之術，以勾股相乘為實，併勾股為法，實如法而一，即得方之邊。勾五步乘股十二步，得六十為實。併勾股得十七為法。實如法而一，得三步十七分步之九。輝按：容方之術，以勾股為邊之直角三角形內作正方形，其一邊在勾上，一邊在股上，一頂點在弦上。

$$3\frac{9}{17}$$

$$\frac{\times}{+} = \frac{5 \times 12}{5 + 12} = \frac{60}{17} = 3\frac{9}{17}$$

勾股容圓

今有勾八步，股十五步。問：勾股中容圓徑幾何？

答：六步。

術：勾股容圓之術，以勾股相乘倍之為實，以勾股弦併之為法，實如法而一，即得圓徑。勾八步乘股十五步，得一百二十，倍之得二百四十為實。勾八步、股十五步、弦十七步併之，得四十為法。實如法而一，得六步，即圓徑。輝按：此術之證，可以面積法求之。勾股三角形之面積，等於其周界與圓徑乘積之四分之一。

$$d = \frac{2 \times \times}{++} = \frac{2 \times 8 \times 15}{8 + 15 + 17} = \frac{240}{40} = 6\frac{1}{2} \times \times \frac{1}{4} \times \times d$$

複雜勾股問題

勾股六（竹折）

今有竹高一丈，末折抵地，去本三尺。問：折者高幾何？

答：四尺五十五分尺之十一。

術：此題以勾股術解之。竹高一丈為股弦之和，去本三尺為勾。以勾股術入之，即得折處之高。

$$4\frac{11}{55} = 4\frac{1}{5}$$

$$h10 - h3^2 + h^2 = (10 - h)^2 = 100 - 20h + h^2 \Rightarrow 9 = 100 - 20h$$

$$h = \frac{91}{20} = 4\frac{11}{20}$$

勾股七（葭生中央）

今有池方一丈，葭生其中央，出水一尺。引葭赴岸，適與岸齊。問：水深、葭長各幾何？

術：此題以勾股術解之。池方一丈，葭在中央，赴岸為五尺，此勾也。水深為股，葭長為弦。出水一尺為股弦較。輝按：此題為勾股章中最有名之題，歷代算家多稱道之。以葭之出水，而知池之深，此可謂以近知遠，以顯知幽者也。

$$-5^2 = (xian - gu)(xian + gu) = 1 \times (xian + gu) \Rightarrow xian + gu = 25$$

$$xian - gu = 12 \Rightarrow xian = 26$$

勾股八（邑方）

今有邑方不知大小，各開中門。出北門二十步有木，出南門十四步折而西行一千七百七十五步見木。問：邑方幾何？

$$s \frac{s/2+20}{1775} = \frac{s/2}{14+s/2} s = 250$$

勾股九（山高海遠）

今有望海島，立兩表，高三丈，前後相去千步。令後表與前表參相直。從前表卻行一百二十三步，人目著地，取望島峯，與表末參合。從後表卻行一百二十七步，人目著地，取望島峯，亦與表末參合。問：島高及去表各幾何？

術：此題以重差術解之，為勾股術之推廣。以兩表之高及相去之遠，與兩卻行之差，以比例求之，即得島高及去表之遠。輝按：此題為劉徽海島算經之第一題，雖不在九章之內，然其術出於勾股。

$$= \frac{3}{4} + \frac{3 \times 1000}{127 - 123} + 3 = 750 + 3 = 753 = \frac{123 \times 1000}{4} = 30,750$$

附錄：纂類九類總論

楊輝纂類，將九章二百四十六問重新分為九類，此實為中國數學史上的一大創舉。古本九章以方田、粟米、衰分、少廣、商功、均輸、盈不足、方程、勾股分章，此以數學之應用領域為分類標準，便於官府之施用，而不利於學者之研習。輝有感於此，乃以數學之方法為分類標準，重新編排，使學者可以由淺入深，循序漸進。

輝之九類：一曰乘除，二曰互換，三曰合率，四曰分率，五曰衰分，六曰疊積，七曰盈不足，八曰方程，九曰勾股。此九類之中，乘除、分率為基礎之術，互換、衰分為比例之術，合率為分數之併合，疊積為體積之計算，盈不足為假設之法，方程為聯立方程之解法，勾股為幾何之基礎。由淺入深，由易及難，條理井然。

乘除之類

乘除者，算學之基礎也。一切算術，無不始於乘除。楊輝以方田三十八問、粟米二問列入此類，蓋方田之計算，無非廣從相乘，以得積步；而粟米之經率，亦以除法為主。此類之術雖簡，然為諸法之根本，不可不精熟也。

×

互換之類

互換之術，出於粟米章，而其用則遍於衰分、均輸諸章。以所有易所求，以所求率乘所有數為實，以所有率為法，實如法而一。此術之要，在於明率。率者，物與物之相為比例者也。知一物以推百物，知一時以推異時，此互換之妙用也。

方程之類

方程之術，乃中國數學之瑰寶也。以算籌列為方陣，互乘相減，以消去未知數。正負之術，為其樞紐。同名相除，異名相益，正無入負之，負無入正之。此術之妙，非深思不能得其旨也。

勾股之類

勾股之術，以自乘併之為根本，以開方為施用。測高深、量遠近、求曲直，無不用之。容方、容圓之術，尤為精妙。重差之術，由是而推廣，可以測日之高遠、海島之廣狹，此其大用也。

後記

輝不自揣，以淺陋之學，妄議古人之書。然區區之意，欲使九章之法，燦然大明於後世，使學者無復有隱晦難明之歎。纂類之作，非敢妄改舊章，不過條分縷析，使條貫更清晰耳。昔劉徽注九章，自謂「幼習九章，長再詳覽，觀陰陽之割裂，總算術之根源，探蹟之暇，遂悟其意。是以敢竭頑魯，采其所見，為之作注。」輝雖不敢比擬劉徽，然其用心則一也。

九章算術，為中國數學之源泉，兩千年來，歷代算家無不由此入手。輝之詳解，不過為初學者設階梯耳。至於深造自得，則在乎人之力矣。

楊輝識

楊輝（約），字謙光，錢塘（今浙江杭州）人，南宋傑出數學家。著有《詳解九章算法》十二卷、《日用算法》二卷、《乘除通變本末》三卷、《田畝比類乘除捷法》二卷、《續古摘奇算法》二卷，後三種合稱《楊輝算法》。楊輝三角之發現，尤為世人所稱道，實開二項式係數研究之先聲。

第三部終

Bilingual Edition

新編算學啓蒙 • Suanxue Qimeng

Introduction to Mathematical Studies

English–Chinese Parallel Text • 英漢對照

卷上 • Volume I (Part I)

General Principles (總括) and Volume I Chapters 1–8

元 • 朱世傑撰

By Zhu Shijie (c. 1260–1320 CE)

Yuan Dynasty, written in 1299 (大德己亥)

Bilingual edition with parallel Chinese-English columns
Preserving the 問/答/術 (Problem/Answer/Method) structure
Notes on rod numeral notation and technical terms

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Preface • 序

算學啓蒙者，元朱世傑之所著也。世傑字漢卿，號松庭，燕山人。是書成於大德己亥（1299 年），凡三卷，二十門，二百五十九問。上卷總括、縱橫因法、身外加法、留頭乘法、身外減法、九歸除法、異乘同除、庫務解稅、折變互差；中卷田畝計量、堆積求和、約分通分、開方術；下卷方程正負、天元如積。其書由淺入深，條理井然，為初學算學之津梁也。此書在中國久已失傳，賴朝鮮刻本得以不墜，清羅士琳於揚州重刻之（1839 年）。其學東傳日本，影響深遠。

The *Suanxue Qimeng* (算學啓蒙, “Introduction to Mathematical Studies”) was written by Zhu Shijie (朱世傑), styled Hanqing (漢卿), sobriquet Songting (松庭), of Yanshan (燕山), in 1299 CE (the Dade era, 大德己亥) of the Yuan Dynasty. It is an elementary mathematical textbook in three volumes (上中下三卷), containing 20 chapters (二十門) and 259 problems (凡二百五十九問).

The text progresses systematically from basic arithmetic operations—multiplication, division, addition, and subtraction—through advanced topics such as *tian yuan shu* (天元術, the method of the celestial element for solving equations), *duo ji* (垛積, pile accumulation / summation of series), and *zhao cha* (招差, finite differences).

The book opens with a **General Principles** (總括) section containing 18 fundamental rules and rhymes, followed by Volume I (卷上) with eight chapters and 113 problems covering multiplication shortcuts, division methods, and proportional reasoning. Many problems reflect the economic realities of Yuan dynasty China—taxation, commodity prices, land measurement, and military logistics. The *Suanxue Qimeng* was lost in China for centuries until a Korean edition was rediscovered. It was republished by Luo Shilin (羅士琳) in Yangzhou in 1839. The work profoundly influenced the development of mathematics in both Korea and Japan.

[Mathematical note: This bilingual edition preserves the original Chinese text alongside an English translation. Technical terms are rendered with their Chinese characters on first occurrence. The 問/答/術 structure is preserved as **Problem/Answer/Method**.]

General Principles • 總括

總括者，算學之總要也。凡乘法口訣、除法歌訣、度量衡制、大數小數、正負之術、開方之法，皆具於此，為全書之綱領。初學者必熟讀之，以為入門之基也。以下列一十八條。

The General Principles section presents the foundational knowledge required for the entire treatise. It includes multiplication tables, division rhymes, unit conversions, notation rules, and algebraic conventions. The following 18 items are listed at the beginning of the book. These are the fundamental rules that every student must master before proceeding to the problems.

釋九數法 • The Nine Numbers (Multiplication Table)

乘法口訣，算學之基本也。九九之數，自一一如一起，至九九八十一止，凡四十五句。縱橫因之，無所不通。一一如一，一二如二，二二如四。一三如三，二三如六，三三如九。一四如四，二四如八，三四一十二，四四一十六。一五如五，二五一十，三五一十五，四五二十，五五二十五。一六如六，二六一十二，三六一十八，四六二十四，五六三十，六六三十六。一七如七，二七一十四，三七二十一，四七二十八，五七三十五，六七四十二，七七四十九。一八如八，二八一十六，三八二十四，四八三十二，五八四十，六八四十八，七八五十六，八八六十四。一九如九，二九一十八，三九二十七，四九三十六，五九四十五，六九五十四，七九六十三，八九七十二，九九八十一。

[Mathematical note: The multiplication rhymes follow a pattern: “ a b like c ” means $a \times b = c$. When the product is ten or more, the word “ten” (十) is inserted: 三四一十二 means $3 \times 4 = 12$.]

九歸除法 • The Nine Returns (Division Rhymes)

按古法多用商除。為初學者難入，則後人以此法代之，即非正術也。九歸者，以一位數除多位數之歌訣也。其法用歸訣立商，以次減實，實盡為度。初學者先習此訣，而後可入商除之門。

一歸如一進，見一進成十。二一添作五，逢二進成十。三一三十一，三二六十二，逢三進成十。四一二十二，四二添作五，四三七十二，逢四進成十。五歸添一倍，逢五進成十。六一下加四，六二三十二，六三添作五，六四六十四，六五八十二，逢六進成十。七一下加三，七二下加六，七三四十二，七四五十五，七五七十一，七六八十四，逢七進成十。八一下加二，八二下加四，八三下加六，八四添作五，八五六十二，八六七十四，八七八十六，逢八進成十。九歸隨身下，逢九進成十。

The multiplication table is the foundation of all calculation. These “nine-nines” rhymes run from “one-one is one” to “nine-nine is eighty-one,” totaling 45 formulas. By multiplying vertically and horizontally, all products may be obtained.

This is the complete multiplication table as memorized by every Chinese student. Each line begins with the multiplication by 1, then 2, up to the number itself. For example, the line for 7 gives: $1 \times 7 = 7$, $2 \times 7 = 14$, $3 \times 7 = 21$, $4 \times 7 = 28$, $5 \times 7 = 35$, $6 \times 7 = 42$, $7 \times 7 = 49$.

According to ancient methods, the standard division (商除) was commonly used. But since beginners find it difficult, later generations substituted this method, even though it is not the orthodox technique.

The “nine returns” are rhymes for dividing a multi-digit number by a single digit. The method uses the return-rhymes to establish the quotient digit, then subtracts from the dividend repeatedly until the dividend is exhausted.

1-return: One-return is like one-advance; see one, advance to make ten.

2-return: Two-one add to make five; encounter two, advance to make ten.

3-return: Three-one: thirty-one; three-two: sixty-two; encounter three, advance to make ten.

4-return: Four-one: twenty-two; four-two add to make five; four-three: seventy-two; encounter four, advance to make ten.

5-return: Five-return add one-fold; encounter five, advance to make ten.

6-return: Six-one: add four below; six-two: thirty-two; six-three add to make five; six-four: sixty-four; six-five: eighty-two; encounter six, advance to make ten.

7-return: Seven-one: add three below; seven-two: add six below; seven-three: forty-two; seven-four: fifty-five; seven-five: seventy-one; seven-six: eighty-four; encounter seven, advance to make ten.

8-return: Eight-one: add two below; eight-two: add four below; eight-three: add six below; eight-four add to make five; eight-five: sixty-two; eight-six: seventy-four; eight-seven: eighty-six; encounter eight, advance to make ten.

9-return: Nine-return follows the body down; encounter nine, advance to make ten.

[Mathematical note: These rhymes encode single-digit division rules for use on the counting board. “添作五” (add to make five) means the quotient digit is 5. “下加 n ” means add n to the digit below (i.e., the quotient digit is 1 with remainder n). “逢 N 進成十” means “when encountering N or more, place 1 in the next higher position.”]

斤下留法 • Converting Taels to Jin

斤下帶兩者，當以十六約之。今則就省，以此代之也。十六兩為一斤，故兩化斤，須以十六除之。以下所列，即兩數化為斤之小數也。

一退六二五，二留一二五，三留一八七五，四留二五，五留三一七五，六留三七五，七留四三七五，八留單五，九留五六二五，十留六二五，十一留六八七五，十二留七五，十三留八一二五，十四留八七五，十五留九三七五。

When jin (斤) has taels (兩) below, one should divide by 16. This is a simplified method for that purpose. Since 16 taels = 1 jin, converting taels to jin requires division by 16. The following are the decimal fractions for each number of taels:

1 tael = 0.0625 jin, 2 taels = 0.125 jin, 3 taels = 0.1875 jin, 4 taels = 0.25 jin, 5 taels = 0.3125 jin, 6 taels = 0.375 jin, 7 taels = 0.4375 jin, 8 taels = 0.5 jin, 9 taels = 0.5625 jin, 10 taels = 0.625 jin, 11 taels = 0.6875 jin, 12 taels = 0.75 jin, 13 taels = 0.8125 jin, 14 taels = 0.875 jin, 15 taels = 0.9375 jin.

明縱橫訣 • Rules for Counting Rod Notation

算籌記數之法，縱橫相間，以別位值。其訣曰：

一縱十橫，百立千僵。千十相望，萬百相當。滿六已上，五在上方。六不積聚，五不單張。言十自過，不滿自當。若明此訣，可習九章。

The counting rod (籌算) notation system uses alternating vertical and horizontal orientations to distinguish place values. The rules state:

Units are vertical, tens are horizontal, hundreds stand upright, thousands lie flat. Thousands and tens face each other; myriads (10,000s) and hundreds match.

For six and above, the five-bar sits above.

Six does not pile up; five does not stand alone.

When reaching ten, carry over; when less than ten, keep as-is.

If you understand this rhyme, you may study the Nine Chapters.

In rod numerals, digits 1–4 are represented by 1–4 vertical (or horizontal) rods. The digit 5 is a single horizontal bar (for units) or vertical bar (for tens). Digits 6–9 combine the 5-bar with 1–4 additional rods below it.

[Mathematical note: The rod numeral system is a decimal place-value notation used on a counting board. Alternating orientations (vertical/horizontal) between adjacent digits prevents confusion.]

大數之類 • Large Numbers

凡數之大者，天莫能蓋、地莫能載，其數不能極，故謂之大數也。中國大數之名，以萬萬遞進，與西洋之以千遞進者不同。

一、十、百、千、萬、十萬、百萬、千萬，萬萬曰億，萬萬億曰兆，萬萬兆曰京，萬萬京曰垓，萬萬垓曰秭，萬萬秭曰壤，萬萬壤曰溝，萬萬溝曰澗，萬萬澗曰正，萬萬正曰載，萬萬載曰極，萬萬極曰恆河沙，萬萬恆河沙曰阿僧祇，萬萬阿僧祇曰那由他，萬萬那由他曰不可思議，萬萬不可思議曰無量數。

As for large numbers: heaven cannot cover them, earth cannot contain them; their extent is limitless, hence they are called “large numbers.”

Chinese large number names progress by powers of 10^8 (萬萬), unlike the Western system which progresses by powers of 10^3 (thousands):

One, ten, hundred, thousand, myriad (10^4), 10^5 , 10^6 , 10^7 , 10^8 = yi (億), 10^{16} = zhao (兆), 10^{24} = jing (京), 10^{32} = gai (垓), 10^{40} = zi (秭), 10^{48} = rang (壤), 10^{56} = gou (溝), 10^{64} = jian (澗), 10^{72} = zheng (正), 10^{80} = zai (載), 10^{88} = ji (極), 10^{96} = Ganges sand (恆河沙), 10^{104} = asa.mkhyeya (阿僧祇), 10^{112} = nayuta (那由他), 10^{120} = unthinkable (不可思議), 10^{128} = immeasurable (無量數).

小數之類 • Small Numbers

凡數之小者，視之無形、取之無像，數亦不能盡，故謂之小數也。小數之名，亦以萬萬遞退。

一、分、釐、毫、絲、忽、微、纖、沙，萬萬塵曰沙，萬萬埃曰塵，萬萬渺曰埃，萬萬漠曰渺，萬萬模糊曰漠，萬萬逡巡曰模糊，萬萬須臾曰逡巡，萬萬瞬息曰須臾，萬萬彈指曰瞬息，萬萬剎那曰彈指，萬萬六德曰剎那，萬萬虛曰六德，萬萬空曰虛，萬萬清曰空，萬萬淨曰清。千萬淨，百萬淨，十萬淨，萬淨，千淨，百淨，十淨，一淨。

As for small numbers: looking at them, they have no form; grasping them, they have no image; their fineness cannot be exhausted, hence they are called "small numbers."

Small number names also descend by powers of 10^{-8} :

One (1), fen (分, 10^{-1}), li (釐, 10^{-2}), hao (毫, 10^{-3}), si (絲, 10^{-4}), hu (忽, 10^{-5}), wei (微, 10^{-6}), xian (纖, 10^{-7}), sha (沙, 10^{-8}), then continuing: 10^{-16} = dust (塵), 10^{-24} = mote (埃), 10^{-32} = minute (渺), 10^{-40} = desert (漠), 10^{-48} = blur (模糊), 10^{-56} = hesitation (逡巡), 10^{-64} = instant (須臾), 10^{-72} = blink (瞬息), 10^{-80} = finger-snap (彈指), 10^{-88} = k.sana (剎那), 10^{-96} = liude (六德), 10^{-104} = void (虛), 10^{-112} = emptiness (空), 10^{-120} = clarity (清), 10^{-128} = purity (淨).

求諸率類 • Conversion Rates

諸物互求之法，以乘除為用。其率如下：

兩求銖二十四乘，銖求兩二十四除。斤求兩身外加六，兩求斤身外減六。秤求斤身外加五，斤求秤身外減五。據物賣錢而用乘，據錢買物而用除。

To convert between units, use multiplication and division with these rates:

Taels (兩) to zhu (銖): multiply by 24. Zhu to taels: divide by 24. (1 tael = 24 zhu)

Jin (斤) to taels: multiply by 16 ("add six to the body," i.e., the conversion factor is $16 = 10 + 6$).

Taels to jin: divide by 16.

Cheng (秤) to jin: multiply by 15 ("add five to the body," since 1 cheng = 15 jin). Jin to cheng: divide by 15.

To sell goods for money: use multiplication. To buy goods with money: use division.

斛斗起率 • Volume Measurements

量之起率，始於微細，積而為大。

量起於圭（六粒之粟）。十圭謂之一撮，十撮謂之一抄，十抄謂之一勺，十勺謂之一合，十合謂之一升，十升謂之一斗，十斗謂之一斛。

Volume measurement begins from the smallest unit and accumulates upward:

Volume begins with the gui (圭): six grains of millet. 10 gui = 1 cuo (撮), 10 cuo = 1 chao (抄), 10 chao = 1 shao (勺), 10 shao = 1 ge (合), 10 ge = 1 sheng (升), 10 sheng = 1 dou (斗), 10 dou = 1 hu (斛).

Thus 1 hu = 10^7 gui.

斤秤起率 • Weight Measurements

衡之起率，以黍為基。

衡起於黍（形大如粟）。十黍謂之一綮，十綮謂之一銖，六銖謂之一分，四分謂之一兩，十六兩謂之一斤，十五斤謂之一秤，三十斤謂之一鈞，四鈞謂之一碩（重一百二十斤）。

Weight measurement begins from the smallest grain:

Weight begins with the shu (黍): a grain the size of millet. 10 shu = 1 lei (綮), 10 lei = 1 zhu (銖), 6 zhu = 1 fen (分), 4 fen = 1 liang (兩, tael), 16 liang = 1 jin (斤), 15 jin = 1 cheng (秤), 30 jin = 1 jun (鈞), 4 jun = 1 shuo (碩, = 120 jin).

端匹起率 • Length Measurements

度之起率，以蠶絲為始。

度起於忽（蠶吐之絲）。十忽謂之一絲，十絲謂之一毫，十毫謂之一釐，十釐謂之一分，十分謂之一寸，十寸謂之一尺，十尺謂之一丈。匹率（或三丈二，或二丈四）。端率（或五丈五，或四丈八）。

Length measurement begins from the finest silk thread:

Length begins with the hu (忽): a single silk thread from a silkworm. 10 hu = 1 si (絲), 10 si = 1 hao (毫), 10 hao = 1 li (釐), 10 li = 1 fen (分), 10 fen = 1 cun (寸, inch), 10 cun = 1 chi (尺, foot), 10 chi = 1 zhang (丈). Cloth bolt (匹) measure: either 3.2 zhang or 2.4 zhang. Cloth roll (端) measure: either 5.5 zhang or 4.8 zhang.

田畝起率 • Land Area Measurements

田畝之制，以步為基。

田起於忽（闊一寸，長六寸）。十忽謂之一絲，十絲謂之一毫，十毫謂之一釐，十釐謂之一分，十分謂之一畝，百畝謂之一頃。三百步謂之一里。

按畝法，闊一步，長二百四十步，當自方五尺為步也。其里法，三百步為里者，當自方六尺為步。若三百六十步為里者，當以自方五尺為步也。

The land area system uses the “pace” (步) as its foundation:

Land area begins with the hu (忽): 1 cun wide by 6 cun long. 10 hu = 1 si (絲), 10 si = 1 hao (毫), 10 hao = 1 li (釐), 10 li = 1 fen (分), 10 fen = 1 mu (畝), 100 mu = 1 qing (頃). 300 paces = 1 li (里).

According to the mu method: 1 pace wide by 240 paces long = 1 mu, using a 5-chi pace. For the li method: 300 paces = 1 li using a 6-chi pace; or 360 paces = 1 li using a 5-chi pace.

One mu $\approx$ 1/15 of a hectare $\approx$ 667 m² in modern terms.

古法圓率 • Ancient Circle Ratio

古之圓率，以周三徑一為率。

周三尺，徑一尺。

The ancient ratio for the circle:

Circumference 3 chi, diameter 1 chi.

This gives $\pi = 3$, the ancient ratio “three for circumference, one for diameter.” This value was used in the *Zhoubi Suanjing* (周髀算經) and the *Jiuzhang Suanshu* (九章算術) before more precise values were obtained.

劉徽新術 • Liu Hui's New Method

劉徽乃魏人也，立此新術以究圓之幽微。劉徽以割圓術求圓周率，割圓為多邊形，邊數愈多則愈近於圓。

周一百五十七尺，徑五十尺。

Liu Hui was a man of Wei; he established this new method to investigate the subtle mysteries of the circle.

Liu Hui (fl. 263 CE) used the “method of cutting the circle” (割圓術): inscribing polygons with increasingly many sides to approximate the circle. He calculated up to a 3072-sided polygon.

Circumference 157 chi, diameter 50 chi.

This gives $\pi = \frac{157}{50} = 3.14$, a significant improvement over $\pi = 3$.

冲之密率 • Zu Chongzhi's Dense Ratio

冲之姓祖，乃宋南徐州從事史。立此密率亦究圓之微也。祖冲之更以密率求之，其精密為當時之最。周二十二尺，徑七尺。

Chongzhi, surnamed Zu, was an officer of Nanku province in the Song dynasty. He established this dense ratio also to investigate the circle's subtlety. Zu Chongzhi (429–500 CE) gave two approximations for π : the “approximate ratio” (約率) $\pi \approx 22/7$ and the “dense ratio” (密率) $\pi \approx 355/113$. The dense ratio gives $\pi \approx 3.1415929\dots$, accurate to six decimal places. Circumference 22 chi, diameter 7 chi. This gives $\pi = \frac{22}{7} \approx 3.142857$ (the “approximate ratio”).

[Mathematical note: Zu Chongzhi's dense ratio $\frac{355}{113}$ is one of the best rational approximations of π with small denominator. It was not rediscovered in Europe until Adriaan Anthoniszoon (c. 1600).]

明異名訣 • Names of Fractions

此乃劇漏之數以巧呼之。古之算家，以巧名分數，以便口訣之用。二分之一為中半，三分之一為少半，三分之二為太半，四分之一為弱半，四分之三為強半。

These are special names for fractions, called by elegant names. Ancient mathematicians gave special names to common fractions for ease of use: $\frac{1}{2}$ = “middle half” (中半), $\frac{1}{3}$ = “lesser half” (少半), $\frac{2}{3}$ = “greater half” (太半), $\frac{3}{4}$ = “weak half” (弱半), $\frac{3}{4}$ = “strong half” (強半).

明正負術 • Rules for Positive and Negative Numbers

正負之術，為方程之根本。同名相減，異名相加，正無入負之，負無入正之。其同名相減，則異名相加；正無入負之，負無入正之。其異名相減，則同名相加；正無入正之，負無入負之。按九章注云：兩算得失相返，要令正負以名之。正算赤、負算黑，不則以邪正為異。其無入者，為無對也。無所得減，則使消奪者居位也。（「人」作「入」非。）

The rules for positive and negative numbers are fundamental to solving systems of equations. The rules state:

Same signs subtract, different signs add. When subtracting a positive from zero (or a smaller positive), the result is negative; when subtracting a negative from zero, the result is positive.

Rule 1: Same signs subtract → different signs add; positive with nothing enters negative; negative with nothing enters positive.

Rule 2: Different signs subtract → same signs add; positive with nothing enters positive; negative with nothing enters negative.

According to the commentary on the Jiuzhang: two calculations with opposite outcomes are named positive and negative. Positive rods are red, negative rods are black; otherwise, slanted versus straight distinguishes them. “Nothing to enter” means there is no counterpart; when there is nothing to subtract from, the consuming quantity takes the position.

These rules correspond to: $(+a) - (+b) = +(a - b)$, $(-a) - (-b) = -(a - b)$, $(+a) + (-b) = +(a - b)$, $(+a) + (-b) = -(b - a)$ when $b > a$, etc.

[Mathematical note: These are among the earliest known rules for signed arithmetic in world mathematics. The rod-color convention (red = positive, black = negative) dates to the Han dynasty. These rules predate European acceptance of negative numbers by over a millennium.]

明乘除段 • Rules for Multiplication and Division

乘除之術，凡幾何名義，不可不辨。

長平相併曰和，長平相減曰較。長平相乘曰積，自相稱之曰冪。同名相乘曰正，異名相乘為負。平除長為小長，長除平為小平。小長平相併曰小和，小長平相減餘小較，小長平相乘得一步為小積。

The terminology of multiplication and division must be clearly understood:

Length plus width is called “harmony” (和, sum).

Length minus width is called “comparison” (較, difference).

Length times width is called “accumulation” (積, area/product).

A number multiplied by itself is called “power” (冪, square).

Same signs multiplied give positive; different signs give negative.

Width divided by length gives “small length”; length divided by width gives “small width.”

Small length plus small width = “small harmony”; their difference = “small comparison”; their product = “small accumulation.”

[Mathematical note: The terms 長 (long) and 平 (broad/width) denote the sides of a rectangle. These algebraic conventions were essential for the tian yuan shu (天元術) method that Zhu Shijie would develop in his later work, the *Siyuan Yujian*.]

明開方法 • Rules for Root Extraction

開方之術，方程之基也。凡開平方、開立方，皆用此術。置積為實，及方廉隅，同加異減開之。

The method of root extraction is the foundation of solving equations. It applies to square roots, cube roots, and higher roots.

Set up the accumulation (積) as the dividend (實), together with the side (方), edge (廉), and corner (隅). Like signs add, unlike signs subtract, and extract.

In the root extraction process for $x^2 = N$, the terms are: **shi** (實, dividend) = N , **fang** (方, side) = the linear term coefficient, **lian** (廉, edge) = the edge term, **yu** (隅, corner) = the leading coefficient. The “incremental root extraction” (增乘開方) method systematically refines the root digit by digit.

[Mathematical note: The “incremental multiplication” method (增乘開方), attributed to Jia Xian (11th century), is a precursor to what is now known as the Ruffini-Horner method for polynomial root-finding. It was systematized by Zhu Shijie in his later work.]

Volume I • 卷上

卷上凡八門，一百一十三問。其門曰：縱橫因法、身外加法、留頭乘法、身外減法、九歸除法、異乘同除、庫務解稅、折變互差。大要皆乘除之術，為算學之初步也。

Volume I contains eight chapters (八門) with 113 problems (一百一十三問), covering multiplication shortcuts, division methods, and proportional reasoning. Many problems reflect the economic realities of Yuan dynasty China—taxation, commodity prices, land measurement, and military logistics.

縱橫因法門 • The Vertical–Horizontal Multiplication Method

此法從來向上因，但言十者過其身，
呼如本位須當作，知算縱橫數目真。

縱橫因法者，乘法之初門也。其法列實於上，以法數從上因之，言十過身，言如下位，逐位相乘，得數即積也。凡八問。

問 1. 今有粟二百一十六斗，每斗價錢二文。問計錢幾何？

答曰：四百三十二文。

術曰：列物數在上，各以價錢從上因之，即得。

問 2. 今有絲一百四十四兩，每兩價錢三百文。問計錢幾何？

答曰：四十三貫二百文。

術曰：列物數在上，各以價錢從上因之，即得。

問 3. 今有羊三百五十四只，每只價錢四貫文。問計錢幾何？

答曰：一千四百一十六貫。

術曰：列物數在上，各以價錢從上因之，即得。

問 4. 今有銀五百四十七錠，每錠重五十兩。問為兩幾何？

答曰：二萬七千三百五十兩。

術曰：列物數在上，各以錠重從上因之，即得。

問 5. 今有絹七百三十六匹，每匹價錢六貫文。問計錢幾何？

答曰：四千四百一十六貫。

術曰：列物數在上，各以價錢從上因之，即得。

問 6. 今有麻八百九十二秤，每秤價錢七百元。問計錢幾何？

答曰：六百二十四貫四百文。

術曰：列物數在上，各以價錢從上因之，即得。

*This method always multiplies upward
from below,
When ten is reached, carry past the body;
Call “as-is” for the current place value;
Know the vertical–horizontal counts to be
true.*

This rhyme describes the “vertical–horizontal” multiplication method: multiply from bottom to top; when the product is ten or more, carry to the next position; call “as-is” for the current place value. This is the most basic multiplication technique.

Problem 1. Suppose there are 216 dou of millet, each dou costing 2 cash. How much cash in total?

Answer: 432 cash.

Method: Place the quantity above; multiply each by the unit price from above downward; the result is obtained.

(Calculation: $216 \times 2 = 432$)

Problem 2. Suppose there are 144 liang of silk, each liang costing 300 cash. How much cash in total?

Answer: 43 guan 200 cash.

Method: Place the quantity above; multiply each by the unit price; obtain.

(Calculation: $144 \times 300 = 43,200$)

Problem 3. Suppose there are 354 sheep, each costing 4 guan. How much cash in total?

Answer: 1,416 guan.

Method: Place the quantity above; multiply by the unit price; obtain.

(Calculation: $354 \times 4 = 1,416$)

Problem 4. Suppose there are 547 ingots of silver, each ingot weighing 50 liang. How many liang in total?

Answer: 27,350 liang.

Method: Place the quantity above; multiply each by the ingot weight; obtain.

(Calculation: $547 \times 50 = 27,350$)

Problem 5. Suppose there are 736 bolts of silk, each costing 6 guan. How much cash in total?

Answer: 4,416 guan.

Method: Place the quantity above; multiply by unit price; obtain.

(Calculation: $736 \times 6 = 4,416$)

Problem 6. Suppose there are 892 cheng of hemp, each cheng costing 700 cash. How much cash in total?

Answer: 624 guan 400 cash.

Method: Place the quantity above; multiply by unit price; obtain.

(Calculation: $892 \times 700 = 624,400$)

問 7. 今有布六百三十四尺，每尺價錢八十文。問計錢幾何？

答曰：五十貫七百二十文。

術曰：列物數在上，各以價錢從上因之，即得。

問 8. 今有馬四百二十五匹，每匹價錢九十貫。問計錢幾何？

答曰：三萬八千二百五十貫。

術曰：列物數在上，各以價錢從上因之，即得。

Problem 7. Suppose there are 634 chi of cloth, each chi costing 80 cash. How much cash in total?

Answer: 50 guan 720 cash.

Method: Place the quantity above; multiply by unit price; obtain.

(Calculation: $634 \times 80 = 50,720$)

Problem 8. Suppose there are 425 horses, each costing 90 guan. How much cash in total?

Answer: 38,250 guan.

Method: Place the quantity above; multiply by unit price; obtain.

(Calculation: $425 \times 90 = 38,250$)

身外加法門 • Addition to the Body

算中加法最堪誇，言十之時就位加，
但遇呼如身下列，君從法式定無差。

身外加法者，乘法之捷法也。其法以實為身，於身外逐位加之，言十就身，言如下列。凡十問。

Among calculations, the addition method is most praiseworthy;

When ten is reached, add at the current place;

When “as-is” is called, place it below the body;

Follow this rule and you will make no error.

This rhyme describes a shortcut multiplication method: add to the “body” (the multiplicand itself). When the product reaches ten, add at the current position; when it is less than ten (“as-is”), place it below the current digit. This is essentially a streamlined single-digit multiplication performed on the counting board.

問 9. 今有米六碩八斗四升，每斗價錢一百一十文。問計錢幾何？

答曰：七貫五百二十四文。

術曰：列物數在上，以價錢從上身外加之，即得。

Problem 9. Suppose there are 6 shuo 8 dou 4 sheng of rice, each dou costing 110 cash. How much cash in total?

Answer: 7 guan 524 cash.

Method: Place the quantity above; add the price to the body from above; obtain.

問 10. 今有羅三十四尺六寸，每尺價錢一百二十文。問計錢幾何？

答曰：四貫一百五十二文。

術曰：列物數在上，以價錢從上身外加之，即得。

Problem 10. Suppose there are 34 chi 6 cun of gauze, each chi costing 120 cash. How much cash in total?

Answer: 4 guan 152 cash.

Method: Place the quantity above; add to the body; obtain.

問 11. 今有鹽八百七十三袋，每袋價錢一十三貫文。問計錢幾何？

答曰：一萬一千三百四十九貫。

術曰：列物數在上，以價錢從上身外加之，即得。

Problem 11. Suppose there are 873 bags of salt, each bag costing 13 guan. How much cash in total?

Answer: 11,349 guan.

Method: Place the quantity above; add to the body; obtain.

(Calculation: $873 \times 13 = 11,349$)

問 12. 今有麥二十三碩四斗，每斗價錢一百三十文。問計錢幾何？

答曰：三十貫四百二十文。

術曰：列麥數在上，以價錢從上身外加之，即得。

Problem 12. Suppose there are 23 shuo 4 dou of wheat, each dou costing 130 cash. How much cash in total?

Answer: 30 guan 420 cash.

Method: Place the wheat quantity above; add to the body; obtain.

問 13. 今有絲四十二斤三兩，每兩價錢二百四十文。問計錢幾何？

答曰：一十貫一百七十五文。

術曰：列絲數在上，以價錢從上身外加之，即得。

問 14. 今有粟五十六碩七斗，每斗價錢一百五十文。問計錢幾何？

答曰：八十五貫五十文。

術曰：列粟數在上，以價錢從上身外加之，即得。

問 15. 今有茶三百六十二袋，每袋價錢二貫五百文。問計錢幾何？

答曰：九百五貫。

術曰：列茶數在上，以價錢從上身外加之，即得。

問 16. 今有綿一千二百四十五兩，每兩價錢六十文。問計錢幾何？

答曰：七十四貫七百元。

術曰：列綿數在上，以價錢從上身外加之，即得。

問 17. 今有鐵二千八百七十四斤，每斤價錢四十五文。問計錢幾何？

答曰：一百二十九貫三百三十文。

術曰：列鐵數在上，以價錢從上身外加之，即得。

問 18. 今有鉛一千五百六十三斤四兩，每斤價錢八十文。問計錢幾何？

答曰：一百二十五貫六十文。

術曰：列鉛數在上，以價錢從上身外加之，即得。

問 19. 今有錫八百七十六斤半，每斤價錢一百七十文。問計錢幾何？

答曰：一百四十九貫五百五文。

術曰：列錫數在上，以價錢從上身外加之，即得。

Problem 13. Suppose there are 42 jin 3 liang of silk, each liang costing 240 cash. How much cash in total?

Answer: 10 guan 175 cash.

Method: Place the silk quantity above; add to the body; obtain.

Problem 14. Suppose there are 56 shuo 7 dou of millet, each dou costing 150 cash. How much cash in total?

Answer: 85 guan 50 cash.

Method: Place the millet quantity above; add to the body; obtain.

(Calculation: $567 \times 150 = 85,050$)

Problem 15. Suppose there are 362 bags of tea, each bag costing 2 guan 500 cash. How much cash in total?

Answer: 905 guan.

Method: Place the tea quantity above; add to the body; obtain.

(Calculation: $362 \times 2.5 = 905$)

Problem 16. Suppose there are 1,245 liang of cotton, each liang costing 60 cash. How much cash in total?

Answer: 74 guan 700 cash.

Method: Place the cotton quantity above; add to the body; obtain.

(Calculation: $1,245 \times 60 = 74,700$)

Problem 17. Suppose there are 2,874 jin of iron, each jin costing 45 cash. How much cash in total?

Answer: 129 guan 330 cash.

Method: Place the iron quantity above; add to the body; obtain.

(Calculation: $2,874 \times 45 = 129,330$)

Problem 18. Suppose there are 1,563 jin 4 liang of lead, each jin costing 80 cash. How much cash in total?

Answer: 125 guan 60 cash.

Method: Place the lead quantity above; add to the body; obtain.

Problem 19. Suppose there are 876.5 jin of tin, each jin costing 170 cash. How much cash in total?

Answer: 149 guan 505 cash.

Method: Place the tin quantity above; add to the body; obtain.

留頭乘法門 • Multiplication Keeping the First Digit

留頭乘法別規模，起首先從次位呼，
言十靠身如隔位，遍臨頭位破身鋪。

留頭乘法者，乘法之巧術也。其法以法數之首位暫留不動，先從次位乘起，言十靠身，如隔位布之，遍臨頭位則破身鋪之。凡二十問。

問 20. 今有白豆八十四斛，每斛價錢二百一十文。問計錢幾何？

答曰：一十七貫六百四十文。

術曰：列豆數於上，以斛價從上次位呼之，言十靠身，如隔位布之，遍臨頭位破身，即得。

問 21. 今有胡椒六十三斤四兩，每斤價錢三百八十文。問計錢幾何？

答曰：二十四貫三十五文。

術曰：列椒數於上，以斤價從上次位呼之，言十靠身，如隔位布之，遍臨頭位破身，即得。

問 22. 今有白檀共數斤，下留兩於上，以四貫九十六文乘之。問得幾何？

答曰：二萬貫。

術曰：列白檀共數斤下留兩於上，以四貫九十六文從上次位呼之，言十靠身，如隔位布之，遍臨頭位破身，即得。

問 23. 今有黍三千六百六十二碩一斛九合三勺七抄五撮。問為麥幾何？

答曰：三萬斛。

術曰：列黍數於上，以一斛麥八升一合九勺二抄乘之，即得。

問 24. 今有粟七萬八千一百二十五碩，每斛為御米八升一合九勺二抄。問共得幾何？

答曰：三千二百七十斛。

術曰：列粟數於上，以御米率從上次位呼之，即得。

The keep-the-head method has a special pattern;

Begin from the second digit, not the first; When ten, place next to the body; when “as-is,” skip a place;

Finally use the head digit, breaking the body.

The “keep-the-head” multiplication method: a special technique where the first digit of the multiplier is temporarily kept (left in place), and multiplication begins from the second digit. When the product reaches ten, it is placed next to the body; “as-is” results are placed with a skip. Finally the head digit itself is used to multiply, “breaking” the original body. This avoids shifting the multiplicand on the counting board.

Problem 20. Suppose there are 84 hu of white beans, each hu costing 210 cash. How much cash in total?

Answer: 17 guan 640 cash.

Method: Place the bean quantity above; call the hu-price from the second digit upward; when ten, place next to the body; when “as-is,” skip a place; when reaching the head digit, break the body; obtain. (Calculation: $84 \times 210 = 17,640$)

Problem 21. Suppose there are 63 jin 4 liang of pepper, each jin costing 380 cash. How much cash in total?

Answer: 24 guan 35 cash.

Method: Place the pepper quantity above; call the price from the second digit; when ten, place next to body; when “as-is,” skip; at the head digit, break the body; obtain.

Problem 22. Suppose there is a quantity of white sandalwood in jin, leaving taels below, multiply by 4 guan 96 cash. What is obtained?

Answer: 20,000 guan.

Method: Place the sandalwood quantity with taels below; call 4 guan 96 cash from the second digit; when ten, place next to body; when “as-is,” skip; at head digit, break body; obtain.

Problem 23. Suppose there are 3,662 shuo 1 hu 9 ge 3 shao 7 chao 5 cuo of millet. How much wheat is it equivalent to?

Answer: 30,000 hu.

Method: Place the millet quantity above; multiply by 1 hu wheat = 8 sheng 1 ge 9 shao 2 chao; obtain.

Problem 24. Suppose there are 78,125 shuo of grain, each hu yielding 8 sheng 1 ge 9 shao 2 chao of imperial rice. How much rice is obtained?

Answer: 3,270 hu.

Method: Place the grain quantity above; call the imperial rice rate from the second digit; obtain.

問 25. 今有絲六萬一千六百九十八秤。問為斤幾何？

答曰：一百六十九萬一千六百九十八斤。

術曰：列絲數於上，以一秤十六兩為法，先以六因之，再以四因之，即得斤數。

問 26. 今有絲六萬一千六百九十八秤，每斤重一十六兩。問為兩幾何？

答曰：一千六百九十八萬一千六百九十八兩。

術曰：列絲數於上，以每斤十六兩為法，先從次位呼之，遍臨頭位破身，即得。

問 27. 今有錢七貫四百一十二文，欲令一十七人分之。問人得幾何？

答曰：四百三十六文。

術曰：列錢數於上，以一十七人為法，從上次位呼之，即得。

問 28. 今有糧五千八百八十六碩，欲給貧難戶，每戶一碩八斗。問給幾戶？

答曰：三千二百七十戶。

術曰：列糧數於上，以每戶一碩八斗為法，從上次位呼之，即得。

問 29. 今有錢六十五貫五百五十文，欲買苧絲，每尺價錢一百九十文。問得幾何？

答曰：三百四十五尺。

術曰：列錢數於上，以尺價為法，從上次位呼之，即得。

問 30. 今有錢一百四貫三百七十二文，欲買麻布，每匹價錢一百九十四文。問買布幾何？

答曰：五百三十八匹。

術曰：列錢數於上，以匹價為法，從上次位呼之，即得。

問 31. 今有錢五千五百五十八貫三百文，欲買松木，每株價錢一貫七百五文。問買木幾何？

答曰：三千二百六十株。

術曰：列錢數於上，以株價為法，從上次位呼之，即得。

Problem 25. Suppose there are 61,698 cheng of silk. How many jin?

Answer: 1,691,698 jin.

Method: Place the silk quantity above; using 1 cheng = 16 liang (= 1 jin), first multiply by 6, then by 4; obtain the jin number.

Problem 26. Suppose there are 61,698 cheng of silk, each jin weighing 16 liang. How many liang?

Answer: 16,981,698 liang.

Method: Place the silk quantity above; using 1 jin = 16 liang, call from the second digit; when reaching the head digit, break the body; obtain.

Problem 27. Suppose there are 7 guan 412 cash, to be divided among 17 people. How much per person?

Answer: 436 cash.

Method: Place the cash quantity above; using 17 people as divisor, call from the second digit; obtain. (Calculation: $7,412 \div 17 = 436$)

Problem 28. Suppose there are 5,886 shuo of grain, to be distributed to poor households, each household receiving 1 shuo 8 dou. How many households?

Answer: 3,270 households.

Method: Place the grain quantity above; using 1 shuo 8 dou per household as divisor, call from the second digit; obtain. (Calculation: $5,886 \div 1.8 = 3,270$)

Problem 29. Suppose there are 65 guan 550 cash, to buy ramie silk at 190 cash per chi. How many chi are obtained?

Answer: 345 chi.

Method: Place the cash quantity above; using the chi-price as divisor, call from the second digit; obtain. (Calculation: $65,550 \div 190 = 345$)

Problem 30. Suppose there are 104 guan 372 cash, to buy hemp cloth at 194 cash per bolt. How many bolts?

Answer: 538 bolts.

Method: Place the cash quantity above; using the bolt-price as divisor, call from the second digit; obtain. (Calculation: $104,372 \div 194 = 538$)

Problem 31. Suppose there are 5,558 guan 300 cash, to buy pine trees at 1 guan 705 cash each. How many trees?

Answer: 3,260 trees.

Method: Place the cash quantity above; using the tree-price as divisor, call from the second digit; obtain. (Calculation: $5,558,300 \div 1,705 = 3,260$)

問 32. 今有芝麻共數於上，以三斤七分五釐乘之。問得幾何？

術曰：列芝麻共數於上，以三斤七分五釐為法，從上次位呼之，言十靠身，遍臨頭位破身，即得。

問 33. 今有小麥五碩九斛二升，每斛磨麵六斤一十四兩。問磨麵幾何？

答曰：四百單七斤。

術曰：列麥數於上，以六斤八分七釐半乘之，即得。

問 34. 今有蒙荳三十二碩七斛三升，每斛造粉五斤六兩。問造粉幾何？

答曰：一千七百五十九斤三兩八錢。

術曰：列荳數於上，以五斤六兩為法，從上次位呼之，即得。

問 35. 今有黃金一萬二千八百兩，每兩買地六畝二分五釐。問買地幾何？

答曰：八萬畝。

術曰：列金數於上，以地率從上次位呼之，即得。

問 36. 今有田一百五十六頃二十五畝，每畝收稻五斛。問收稻幾何？

答曰：九萬斛。

術曰：列田數於上，以畝產從上次位呼之，即得。

問 37. 今有米五十三斛四升，直錢九貫八百七十九文。只有錢六十八貫二百六十文。問得米幾何？

答曰：如術計之。

術曰：列見錢為實，以米價為法，留頭乘之，即得。

問 38. 今有羅七百六十二匹，每匹價錢三貫二百三十文。問計錢幾何？

答曰：如術計之。

術曰：列羅數於上，以匹價從上次位呼之，即得。

Problem 32. Suppose there is a quantity of sesame above, to be multiplied by 3 jin 7 fen 5 li. What is obtained?

Method: Place the sesame quantity above; using 3 jin 7 fen 5 li as divisor; call from the second digit; when ten, place next to body; at head digit, break body; obtain.

Problem 33. Suppose there are 5 shuo 9 hu 2 sheng of wheat, each hu yielding 6 jin 14 liang of flour. How much flour?

Answer: 407 jin.

Method: Place the wheat quantity above; multiply by 6 jin 8 fen 7 li and a half; obtain.
(6 jin 14 liang = 6 + 14/16 = 6.875 jin)

Problem 34. Suppose there are 32 shuo 7 hu 3 sheng of green beans, each hu producing 5 jin 6 liang of bean flour. How much flour?

Answer: 1,759 jin 3 liang 8 qian.

Method: Place the bean quantity above; using 5 jin 6 liang as divisor; call from the second digit; obtain.

Problem 35. Suppose there are 12,800 liang of gold, each liang buying 6 mu 2 fen 5 li of land. How much land?

Answer: 80,000 mu.

Method: Place the gold quantity above; call the land rate from the second digit; obtain.
(Calculation: $12,800 \times 6.25 = 80,000$)

Problem 36. Suppose there are 156 qing 25 mu of field, each mu yielding 5 hu of rice. How much rice is harvested?

Answer: 90,000 hu.

Method: Place the field quantity above; call the per-mu yield from the second digit; obtain.

Problem 37. Suppose 53 hu 4 sheng of rice cost 9 guan 879 cash. If one has 68 guan 260 cash, how much rice can be bought?

Answer: Calculate according to the method.

Method: Place the available cash as dividend; use the rice price as divisor; keep-the-head multiply; obtain.

Problem 38. Suppose there are 762 bolts of gauze, each costing 3 guan 230 cash. How much cash in total?

Answer: Calculate according to the method.

Method: Place the gauze quantity above; call the bolt-price from the second digit; obtain.
(Calculation: $762 \times 3,230 = 2,461,260$)

身外減法門 • Subtraction from the Body

減法根源必要知，即同求一一般推，
呼如身下須當減，言十從身本位除。

身外減法者，除法之捷術也，與身外加法相反。其法以實為身，於身外逐位減之，呼如減下身位，言十從身本位除。凡十問。

問 39. 今有錢四貫三百二十文，欲糶白豆，每斗價錢二十文。問得幾何？

答曰：二百一十六斗。

術曰：列錢數於上，為實，以斗價為法，從上身外減之，即得。

問 40. 今有錢四十三貫二百文，欲買細絲，每斤價錢三百文。問得幾何？

答曰：一百四十四斤。

術曰：列錢數於上，為實，以斤價為法，從上身外減之，即得。

問 41. 今有錢一千四百一十六貫，欲買絹子，每匹價錢四貫文。問得幾何？

答曰：三百五十四匹。

術曰：列錢數於上，為實，以匹價為法，從上身外減之，即得。

問 42. 今有銀二萬七千三百五十兩，欲為課銀，每錠五十兩。問為幾錠？

答曰：五百四十七錠。

術曰：列銀數於上，為實，以錠重為法，從上身外減之，即得。

問 43. 今有錢四千四百一十六貫，欲買大羅，每匹價錢六貫文。問得幾何？

答曰：七百三十六匹。

術曰：列錢數於上，為實，以匹價為法，從上身外減之，即得。

The root of the subtraction method you must know;

It is like the “seeking unity” method applied;

When “as-is” is called, subtract below the body;

When ten is reached, divide from the body itself.

The “subtraction from the body” method is the inverse of the addition method: it performs division by subtracting from the dividend itself. “As-is” means subtract below the current digit; “ten” means subtract from the current digit itself. This is a streamlined division technique on the counting board.

Problem 39. Suppose there are 4 guan 320 cash, to buy white beans at 20 cash per dou. How much is obtained?

Answer: 216 dou.

Method: Place the cash above as dividend; use the dou-price as divisor; subtract from the body from above; obtain.

(Calculation: $4,320 \div 20 = 216$)

Problem 40. Suppose there are 43 guan 200 cash, to buy fine silk at 300 cash per jin. How much is obtained?

Answer: 144 jin.

Method: Place the cash above as dividend; use the jin-price as divisor; subtract from the body; obtain.

(Calculation: $43,200 \div 300 = 144$)

Problem 41. Suppose there are 1,416 guan, to buy silk cloth at 4 guan per bolt. How much is obtained?

Answer: 354 bolts.

Method: Place the cash above as dividend; use bolt-price as divisor; subtract from the body; obtain.

(Calculation: $1,416 \div 4 = 354$)

Problem 42. Suppose there are 27,350 liang of silver, to be made into tax ingots, each ingot = 50 liang. How many ingots?

Answer: 547 ingots.

Method: Place the silver above as dividend; use ingot weight as divisor; subtract from the body; obtain.

(Calculation: $27,350 \div 50 = 547$)

Problem 43. Suppose there are 4,416 guan, to buy damask at 6 guan per bolt. How much is obtained?

Answer: 736 bolts.

Method: Place the cash above as dividend; use bolt-price as divisor; subtract from the body; obtain.

(Calculation: $4,416 \div 6 = 736$)

問 44. 今有錢六百二十四貫四百文，欲買甘草，每秤價錢七百文。問得幾何？

答曰：八百九十二秤。

術曰：列錢數於上，為實，以秤價為法，從上身外減之，即得。

問 45. 今有錢五十貫七百二十文，欲買細布，每尺價錢八十文。問得幾何？

答曰：六百三十四尺。

術曰：列錢數於上，為實，以尺價為法，從上身外減之，即得。

問 46. 今有錢三萬八千二百五十貫，欲買馬，每匹價錢九十貫。問得幾何？

答曰：四百二十五匹。

術曰：列錢數於上，為實，以匹價為法，從上身外減之，即得。

問 47. 今有油六碩八斗四升，每三斤七分五釐直錢一百文。問直錢幾何？

答曰：如術計之。

術曰：列油數為實，以三斤七分五釐為法，身外減之，即得。

問 48. 今有麵四百單七斤，每六斤八分七釐半直錢一百文。問直錢幾何？

答曰：如術計之。

術曰：列麵數為實，以六斤八分七釐半為法，身外減之，即得。

Problem 44. Suppose there are 624 guan 400 cash, to buy licorice at 700 cash per cheng. How much is obtained?

Answer: 892 cheng.

Method: Place the cash above as dividend; use cheng-price as divisor; subtract from the body; obtain.

(Calculation: $624,400 \div 700 = 892$)

Problem 45. Suppose there are 50 guan 720 cash, to buy fine cloth at 80 cash per chi. How much is obtained?

Answer: 634 chi.

Method: Place the cash above as dividend; use chi-price as divisor; subtract from the body; obtain.

(Calculation: $50,720 \div 80 = 634$)

Problem 46. Suppose there are 38,250 guan, to buy horses at 90 guan each. How many horses?

Answer: 425 horses.

Method: Place the cash above as dividend; use horse-price as divisor; subtract from the body; obtain.

(Calculation: $38,250 \div 90 = 425$)

Problem 47. Suppose there are 6 shuo 8 dou 4 sheng of oil, and 3 jin 7 fen 5 li costs 100 cash. How much cash?

Answer: Calculate according to the method.

Method: Place the oil quantity as dividend; use 3 jin 7 fen 5 li as divisor; subtract from the body; obtain.

Problem 48. Suppose there are 407 jin of flour, and 6 jin 8 fen 7.5 li costs 100 cash. How much cash?

Answer: Calculate according to the method.

Method: Place the flour quantity as dividend; use 6 jin 8 fen 7.5 li as divisor; subtract from the body; obtain.

九歸除法門 • Division by the Nine Returns

實少法多從法歸，實多滿法進前居，
常存除數專心記，法實相停九十餘。
但遇無除還頭位，然將釋九數呼除，
流傳故泄真消息，求一穿韜總不如。

九歸除法者，以九歸訣為除法之要術也。實少於法，則從法歸之；實多於法，則進位而居之。常記除數於心，法實相等則餘九十餘。凡二十九問。

問 49. 今有錢四貫三百二十文，欲糴白豆，每斗價錢二十文。問得幾何？

答曰：二百一十六斗。

術曰：列錢物於上為實，各以價錢為法，從上身外減之，即得。

問 50. 今有錢四十三貫二百文，欲買細絲，每斤價錢三百文。問得幾何？

答曰：一百四十四斤。

術曰：列錢數為實，以斤價為法，九歸除之，即得。

問 51. 今有錢一千四百一十六貫，欲買絹子，每匹價錢四貫文。問得幾何？

答曰：三百五十四匹。

術曰：列錢數為實，以匹價為法，九歸除之，即得。

問 52. 今有銀二萬七千三百五十兩，欲為課銀，每錠五十兩。問為幾錠？

答曰：五百四十七錠。

術曰：列銀數為實，以錠重為法，九歸除之，即得。

問 53. 今有錢四千四百一十六貫，欲買大羅，每匹價錢六貫文。問得幾何？

答曰：七百三十六匹。

術曰：列錢數為實，以匹價為法，九歸除之，即得。

When the dividend is small and divisor large, follow the return method;

When the dividend fills the divisor, advance to the next position;

Keep the divisor always in mind;

When divisor and dividend match, the remainder is in the nineties.

When a digit cannot be divided, return to the head position;

Then use the nine-explanations method to call the division;

This tradition reveals the true secret;

The “seeking unity” method cannot compare.

The “nine returns” division uses the division rhymes given in the General Principles. When the dividend is smaller than the divisor, use the “return” method; when the dividend is larger, advance the quotient. The practitioner must memorize the divisor and keep it in mind. When the division is exact, the remainder is in the nineties. If a digit cannot be divided, return to the head position and use the “nine explanations” method. This method, though not the orthodox approach (商除), is easier for beginners.

Problem 49. Suppose there are 4 guan 320 cash, to buy white beans at 20 cash per dou. How much is obtained?

Answer: 216 dou.

Method: Place cash and goods above as dividend; use the price as divisor; subtract from the body; obtain.

Problem 50. Suppose there are 43 guan 200 cash, to buy fine silk at 300 cash per jin. How much is obtained?

Answer: 144 jin.

Method: Place cash as dividend; use jin-price as divisor; divide by nine-returns; obtain.

Problem 51. Suppose there are 1,416 guan, to buy silk at 4 guan per bolt. How much is obtained?

Answer: 354 bolts.

Method: Place cash as dividend; use bolt-price as divisor; divide by nine-returns; obtain.

Problem 52. Suppose there are 27,350 liang of silver, to make tax ingots of 50 liang each. How many ingots?

Answer: 547 ingots.

Method: Place silver as dividend; use ingot weight as divisor; divide by nine-returns; obtain.

Problem 53. Suppose there are 4,416 guan, to buy damask at 6 guan per bolt. How much is obtained?

Answer: 736 bolts.

Method: Place cash as dividend; use bolt-price as divisor; divide by nine-returns; obtain.

問 54. 今有錢六百二十四貫四百文，欲買甘草，每秤價錢七百文。問得幾何？

答曰：八百九十二秤。

術曰：列錢數為實，以秤價為法，九歸除之，即得。

問 55. 今有錢五十貫七百二十文，欲買細布，每尺價錢八十文。問得幾何？

答曰：六百三十四尺。

術曰：列錢數為實，以尺價為法，九歸除之，即得。

問 56. 今有木幾何，直錢三千二百六十株。問每株價錢幾何？

答曰：三貫二百六十文。

術曰：列錢物於上為實，各以價錢為法，從上身外減之，即得。

問 57. 今有絹幾何，每匹四貫，共錢一千四百一十六貫。問匹數幾何？

答曰：三百五十四匹。

術曰：列錢數為實，以匹價為法，九歸除之，即得。

問 58. 今有銀幾何，每錠五十兩，共二萬七千三百五十兩。問錠數幾何？

答曰：五百四十七錠。

術曰：列銀數為實，以錠重為法，九歸除之，即得。

問 59. 今有甘草幾何，每秤七百文，共錢六百二十四貫四百文。問秤數幾何？

答曰：八百九十二秤。

術曰：列錢數為實，以秤價為法，九歸除之，即得。

問 60. 今有麻布幾何，每匹一百九十四文，共錢一百四貫三百七十二文。問匹數幾何？

答曰：五百三十八匹。

術曰：列錢數為實，以匹價為法，九歸除之，即得。

問 61. 今有細布幾何，每尺八十文，共錢五十貫七百二十文。問尺數幾何？

答曰：六百三十四尺。

術曰：列錢數為實，以尺價為法，九歸除之，即得。

問 62. 今有松木幾何，每株一貫七百五文，共錢五千五百五十八貫三百文。問株數幾何？

答曰：三千二百六十株。

術曰：列錢數為實，以株價為法，九歸除之，即得。

問 63. 今有白豆幾何，每斗二十文，共錢四貫三百二十文。問斗數幾何？

答曰：二百一十六斗。

術曰：列錢數為實，以斗價為法，九歸除之，即得。

Problem 54. Suppose there are 624 guan 400 cash, to buy licorice at 700 cash per cheng. How much is obtained?

Answer: 892 cheng.

Method: Place cash as dividend; use cheng-price as divisor; divide by nine-returns; obtain.

Problem 55. Suppose there are 50 guan 720 cash, to buy fine cloth at 80 cash per chi. How much is obtained?

Answer: 634 chi.

Method: Place cash as dividend; use chi-price as divisor; divide by nine-returns; obtain.

Problem 56. Suppose some quantity of timber costs 3,260 trees' worth. What is the price per tree?

Answer: 3 guan 260 cash.

Method: Place cash and goods above as dividend; use price as divisor; subtract from the body; obtain.

Problem 57. Suppose there is some silk, 4 guan per bolt, total 1,416 guan. How many bolts?

Answer: 354 bolts.

Method: Place cash as dividend; use bolt-price as divisor; divide by nine-returns; obtain.

Problem 58. Suppose there is some silver, 50 liang per ingot, total 27,350 liang. How many ingots?

Answer: 547 ingots.

Method: Place silver as dividend; use ingot weight as divisor; divide by nine-returns; obtain.

Problem 59. Suppose there is some licorice, 700 cash per cheng, total 624 guan 400 cash. How many cheng?

Answer: 892 cheng.

Method: Place cash as dividend; use cheng-price as divisor; divide by nine-returns; obtain.

Problem 60. Suppose there is some hemp cloth, 194 cash per bolt, total 104 guan 372 cash. How many bolts?

Answer: 538 bolts.

Method: Place cash as dividend; use bolt-price as divisor; divide by nine-returns; obtain.

Problem 61. Suppose there is some fine cloth, 80 cash per chi, total 50 guan 720 cash. How many chi?

Answer: 634 chi.

Method: Place cash as dividend; use chi-price as divisor; divide by nine-returns; obtain.

Problem 62. Suppose there are some pine trees, 1 guan 705 cash each, total 5,558 guan 300 cash. How many trees?

Answer: 3,260 trees.

Method: Place cash as dividend; use tree-price as divisor; divide by nine-returns; obtain.

Problem 63. Suppose there is some white beans, 20 cash per dou, total 4 guan 320 cash. How many dou?

Answer: 216 dou.

Method: Place cash as dividend; use dou-price as divisor; divide by nine-returns; obtain.

問 64. 今有絲幾何，每斤三百文，共錢四十三貫二百文。問斤數幾何？

答曰：一百四十四斤。

術曰：列錢數為實，以斤價為法，九歸除之，即得。

問 65. 今有馬幾何，每匹九十貫，共錢三萬八千二百五十貫。問匹數幾何？

答曰：四百二十五匹。

術曰：列錢數為實，以匹價為法，九歸除之，即得。

問 66. 今有羅幾何，每匹價錢三貫二百三十文，共錢一千七百一十一貫四百文。問匹數幾何？

答曰：如術計之。

術曰：列錢數為實，以匹價為法，九歸除之，即得。

問 67. 今有大羅幾何，每匹六貫，共錢四千四百一十六貫。問匹數幾何？

答曰：七百三十六匹。

術曰：列錢數為實，以匹價為法，九歸除之，即得。

問 68. 今有苧絲幾何，每尺一百九十文，共錢六十五貫五百五十文。問尺數幾何？

答曰：三百四十五尺。

術曰：列錢數為實，以尺價為法，九歸除之，即得。

問 69. 今有米幾何，每斗一百一十文，共錢七貫五百二十四文。問斗數幾何？

答曰：六碩八斗四升。

術曰：列錢數為實，以斗價為法，九歸除之，即得。

問 70. 今有鹽幾何，每袋一十三貫，共錢一萬一千三百四十九貫。問袋數幾何？

答曰：八百七十三袋。

術曰：列錢數為實，以袋價為法，九歸除之，即得。

問 71. 今有羊幾何，每只四貫，共錢一千四百一十六貫。問只數幾何？

答曰：三百五十四只。

術曰：列錢數為實，以只價為法，九歸除之，即得。

問 72. 今有油六碩八斗四升，每三斤七分五釐直錢一百文。問直錢幾何？

答曰：如術計之。

術曰：列油數為實，以三斤七分五釐為法，九歸除之，即得。

問 73. 今有麵四百單七斤，每六斤八分七釐半直錢一百文。問直錢幾何？

答曰：如術計之。

術曰：列麵數為實，以六斤八分七釐半為法，九歸除之，即得。

Problem 64. Suppose there is some silk, 300 cash per jin, total 43 guan 200 cash. How many jin?

Answer: 144 jin.

Method: Place cash as dividend; use jin-price as divisor; divide by nine-returns; obtain.

Problem 65. Suppose there are some horses, 90 guan each, total 38,250 guan. How many horses?

Answer: 425 horses.

Method: Place cash as dividend; use horse-price as divisor; divide by nine-returns; obtain.

Problem 66. Suppose there is some gauze, 3 guan 230 cash per bolt, total 1,711 guan 400 cash. How many bolts?

Answer: Calculate according to the method.

Method: Place cash as dividend; use bolt-price as divisor; divide by nine-returns; obtain.

Problem 67. Suppose there is some damask, 6 guan per bolt, total 4,416 guan. How many bolts?

Answer: 736 bolts.

Method: Place cash as dividend; use bolt-price as divisor; divide by nine-returns; obtain.

Problem 68. Suppose there is some ramie silk, 190 cash per chi, total 65 guan 550 cash. How many chi?

Answer: 345 chi.

Method: Place cash as dividend; use chi-price as divisor; divide by nine-returns; obtain.

Problem 69. Suppose there is some rice, 110 cash per dou, total 7 guan 524 cash. How many dou?

Answer: 6 shuo 8 dou 4 sheng.

Method: Place cash as dividend; use dou-price as divisor; divide by nine-returns; obtain.

(Calculation: $7,524 \div 110 = 68.4$ dou = 6 shuo 8 dou 4 sheng)

Problem 70. Suppose there is some salt, 13 guan per bag, total 11,349 guan. How many bags?

Answer: 873 bags.

Method: Place cash as dividend; use bag-price as divisor; divide by nine-returns; obtain.

Problem 71. Suppose there are some sheep, 4 guan each, total 1,416 guan. How many sheep?

Answer: 354 sheep.

Method: Place cash as dividend; use sheep-price as divisor; divide by nine-returns; obtain.

Problem 72. Suppose there are 6 shuo 8 dou 4 sheng of oil; 3 jin 7 fen 5 li costs 100 cash. How much cash?

Answer: Calculate according to the method.

Method: Place the oil quantity as dividend; use 3 jin 7 fen 5 li as divisor; divide by nine-returns; obtain.

Problem 73. Suppose there are 407 jin of flour; 6 jin 8 fen 7.5 li costs 100 cash. How much cash?

Answer: Calculate according to the method.

Method: Place the flour quantity as dividend; use 6 jin 8 fen 7.5 li as divisor; divide by nine-returns; obtain.

問 74. 今有羅三十四尺六寸，每尺一百二十文。問計錢幾何？

答曰：四貫一百五十二文。

術曰：列羅數於上，以尺價為法，九歸除之，即得。

問 75. 今有胡椒六十三斤四兩，每斤三百八十文。問計錢幾何？

答曰：二十四貫三十五文。

術曰：列椒數於上，以斤價為法，九歸除之，即得。

問 76. 今有馬幾何，每匹價錢九十貫。問計錢幾何？

答曰：如術計之。

術曰：列馬數於上，以匹價為法，九歸除之，即得。

問 77. 今有綿三百四十五斤，欲作綿被，每床用綿七斤。問作幾床？

答曰：四十九床，餘二斤。

術曰：列綿數為實，以每床用綿為法，九歸除之，即得。

Problem 74. Suppose there are 34 chi 6 cun of gauze, each chi costing 120 cash. How much cash in total?

Answer: 4 guan 152 cash.

Method: Place the gauze quantity above; use chi-price as divisor; divide by nine-returns; obtain.

Problem 75. Suppose there are 63 jin 4 liang of pepper, each jin costing 380 cash. How much cash in total?

Answer: 24 guan 35 cash.

Method: Place the pepper quantity above; use jin-price as divisor; divide by nine-returns; obtain.

Problem 76. Suppose there are some horses, each costing 90 guan. How much cash in total?

Answer: Calculate according to the method.

Method: Place the horse quantity above; use horse-price as divisor; divide by nine-returns; obtain.

Problem 77. Suppose there are 345 jin of cotton, to make cotton quilts, each quilt using 7 jin. How many quilts?

Answer: 49 quilts, remainder 2 jin.

Method: Place the cotton quantity as dividend; use cotton per quilt as divisor; divide by nine-returns; obtain.

(Calculation: $345 \div 7 = 49$ remainder 2)

異乘同除門 • Different Multiplication, Same Division

異乘同除者，比例交易之法也。以異名之數相乘，以同名之數相除，求其相當之率。凡八問。

問 78. 今有錢九貫八百七十九文，糴米五碩三斛四升。只有錢六十八貫二百六十文。問得米幾何？

答曰：如術計之。

術曰：列只有米數，以乘今有錢為實，以原錢為法，除之，即得。

問 79. 今有絲七匹，通尺內子得二千三十一尺，以乘上位得六百三十七萬九千三百七十一。又以實以斤銖法三百八十四銖除之，得三萬六千五百四十二銖，為實。

答曰：九千七百六十五斤一十兩四銖。

術曰：列七匹通尺內子，以乘上位，又以斤銖法除之，即得。

問 80. 今有絲八斤二兩二十一銖，織錦七匹六尺五寸。只有絲九十五斤三兩六銖。問織錦幾何？

答曰：如術計之。

術曰：(匹法同前) 列只有絲數，以乘錦數為實，以原絲為法，除之，即得。

This chapter covers problems of proportional exchange: multiplying by one rate and dividing by another. The problems involve currency conversion, unit conversion, and commodity exchange.

Problem 78. Suppose 9 guan 879 cash buys 5 shuo 3 hu 4 sheng of rice. If one has 68 guan 260 cash, how much rice is obtained?

Answer: Calculate according to the method.

Method: Place the rice quantity, multiply by the current cash as dividend; use the original cash as divisor; divide; obtain.

Problem 79. Suppose there are 7 bolts of silk; converting to chi gives 2,031 chi; multiplying gives 6,379,371. Then dividing by 384 zhu (the jin-zhu conversion) gives 36,542 zhu as dividend.

Answer: 9,765 jin 10 liang 4 zhu.

Method: Convert 7 bolts to chi, multiply, then divide by the jin-zhu conversion factor; obtain.

Problem 80. Suppose 8 jin 2 liang 21 zhu of silk weaves 7 bolts 6 chi 5 cun of brocade. If one has 95 jin 3 liang 6 zhu of silk, how much brocade can be woven?

Answer: Calculate according to the method.

Method: (Bolt conversion same as before) Place the available silk, multiply by brocade quantity as dividend, use original silk as divisor; divide; obtain.

問 81. 今有錢五貫八百三十文，買麩八斗五升。只有錢三十七貫七百六十文。問買麩幾何？

答曰：五碩五斗二升。

術曰：列只有錢數，以乘原麩為實，以原錢為法，除之，即得。

問 82. 今有錢一貫一百七十五文，買麥五斗四升。只有錢二十三貫五百文。問買麥幾何？

答曰：十碩八斗。

術曰：列只有錢數，以乘原麥為實，以原錢為法，除之，即得。

問 83. 今有錢二貫六百文，買麵三斤四兩。只有錢四十七貫四百五十文。問買麵幾何？

答曰：六十三斤四兩。

術曰：列只有錢數，以乘原麵為實，以原錢為法，除之，即得。

問 84. 今有錢一貫二百文，買鹽二碩四斗。只有錢三十二貫三百文。問買鹽幾何？

答曰：六十四碩八斗。

術曰：列只有錢數，以乘原鹽為實，以原錢為法，除之，即得。

問 85. 今有銀二十四銖，直錢三貫六百文。只有銀五十九兩八銖。問直錢幾何？

答曰：八十九貫七百文。

術曰：列只有銀數，以乘原直錢為實，以原銀為法，除之，即得。

Problem 81. Suppose 5 guan 830 cash buys 8 dou 5 sheng of bran. If one has 37 guan 760 cash, how much bran can be bought?

Answer: 5 shuo 5 dou 2 sheng.

Method: Place the available cash, multiply by original bran as dividend; use original cash as divisor; divide; obtain.

Problem 82. Suppose 1 guan 175 cash buys 5 dou 4 sheng of wheat. If one has 23 guan 500 cash, how much wheat can be bought?

Answer: 10 shuo 8 dou.

Method: Place available cash, multiply by original wheat as dividend; use original cash as divisor; divide; obtain.

(Calculation: $\frac{23,500 \times 5.4}{1,175} = 108 \text{ dou} = 10 \text{ shuo } 8 \text{ dou}$)

Problem 83. Suppose 2 guan 600 cash buys 3 jin 4 liang of flour. If one has 47 guan 450 cash, how much flour can be bought?

Answer: 63 jin 4 liang.

Method: Place available cash, multiply by original flour as dividend; use original cash as divisor; divide; obtain.

Problem 84. Suppose 1 guan 200 cash buys 2 shuo 4 dou of salt. If one has 32 guan 300 cash, how much salt can be bought?

Answer: 64 shuo 8 dou.

Method: Place available cash, multiply by original salt as dividend; use original cash as divisor; divide; obtain.

(Calculation: $\frac{32,300 \times 24}{1,200} = 648 \text{ dou} = 64 \text{ shuo } 8 \text{ dou}$)

Problem 85. Suppose 24 zhu of silver is worth 3 guan 600 cash. If one has 59 liang 8 zhu of silver, what is it worth?

Answer: 89 guan 700 cash.

Method: Place the available silver, multiply by original cash value as dividend; use original silver as divisor; divide; obtain.

庫務解稅門 • Government Treasuries and Taxation

庫務解稅者，官司會計之法也。凡課銀、鈔法、子母相生、合金之色、工墨之費，皆出於此。凡十一問。

問 86. 今有本銀四千五百六兩，經三箇月一十二日，月利銀三百六兩。問本利共幾何？

答曰：本銀四千五十兩，利銀三百六兩。

術曰：置本銀以月數乘之，又以日數乘之，以三十日除之，得利息，加入本銀，即得。

This chapter deals with problems of government finance: calculating taxes, interest, alloy composition of currency, and distribution of resources. These problems reflect the fiscal administration of the Yuan dynasty.

Problem 86. Suppose there is principal silver of 4,506 liang, over 3 months and 12 days, with monthly interest of 306 liang. What is the total of principal and interest?

Answer: Principal 4,050 liang, interest 306 liang.

Method: Place the principal; multiply by months, then multiply by days; divide by 30 days to get interest; add to principal; obtain.

問 87. 今有本銀四千五十兩，月利一分二釐。問三箇月一十二日，利銀幾何？

答曰：一百四十五兩八錢。

術曰：置本銀以月利乘之，又以日數乘之，以三十日除之，即得。

問 88. 今有課銀三萬六千兩，每五十兩為一錠。問為幾錠？

答曰：七百二十錠。

術曰：置課銀以錠重除之，即得。

問 89. 今有鈔二千四百貫，欲換銀，每銀一兩直鈔八貫。問得銀幾何？

答曰：三百兩。

術曰：置鈔數為實，以銀價為法，除之，即得。

問 90. 今有銀三百兩，欲換鈔，每兩直鈔八貫。問得鈔幾何？

答曰：二千四百貫。

術曰：置銀數為實，以鈔價為法，乘之，即得。

問 91. 今有金一十二兩五錢，內減五十兩，餘即入銀，合同為法。實如法而一，得本銀，反減共還銀數，餘即利銀也。

答曰：一十二兩五錢。

術曰：列金數為實，以八分為法，除之，得本銀，反減共還銀數，餘即利銀也。

問 92. 今有銀一十二兩五錢，內減五十兩，餘即入銀，合同為法。實如法而一，得六兩，問為顏色分數幾何？

答曰：八分色。

術曰：列金數為實，以本銀為法，除之，即得顏色分數。

問 93. 今有絲六十二斤半，換金一十兩，內八分半金五兩七分半金五兩。問二色金各得幾何？

術曰：列絲數為實，以金價為法，除之，即得各金數。

問 94. 今有銀三千五百六兩，直錢三十五貫六百文。問每兩直錢幾何？

答曰：一十文。

術曰：置銀數為實，以直錢為法，除之，即得。

Problem 87. Suppose there is principal silver of 4,050 liang, monthly interest 1 fen 2 li (1.2%). Over 3 months and 12 days, what is the interest?

Answer: 145 liang 8 qian.

Method: Place principal; multiply by monthly interest, then multiply by days; divide by 30 days; obtain.

Problem 88. Suppose there are 36,000 liang of tax silver; each ingot = 50 liang. How many ingots?

Answer: 720 ingots.

Method: Place the tax silver; divide by ingot weight; obtain.

(Calculation: $36,000 \div 50 = 720$)

Problem 89. Suppose there are 2,400 guan of paper money (鈔), to exchange for silver; each liang of silver is worth 8 guan of cash. How much silver?

Answer: 300 liang.

Method: Place the cash quantity as dividend; use silver price as divisor; divide; obtain.

(Calculation: $2,400 \div 8 = 300$)

Problem 90. Suppose there are 300 liang of silver, to exchange for paper money; each liang is worth 8 guan. How much cash?

Answer: 2,400 guan.

Method: Place the silver quantity as dividend; use cash price as divisor; multiply; obtain.

(Calculation: $300 \times 8 = 2,400$)

Problem 91. Suppose there is gold of 12 liang 5 qian; subtract 50 liang from it, the remainder enters into silver; combine as divisor. Divide the dividend by the divisor; obtain the principal silver. Subtract from the total returned silver; the remainder is the interest.

Answer: 12 liang 5 qian.

Method: Place the gold quantity as dividend; use 8 fen as divisor; divide to get principal silver; subtract from total returned silver; the remainder is interest.

Problem 92. Suppose there is silver of 12 liang 5 qian; subtract 50 liang, the remainder enters silver; combine as divisor. Dividing gives 6 liang; what is the alloy purity?

Answer: 8 fen purity (80% pure).

Method: Place the gold quantity as dividend; use principal silver as divisor; divide; obtain the purity fraction.

Problem 93. Suppose 62.5 jin of silk is exchanged for 10 liang of gold: 5 liang of 8.5-fen gold and 5 liang of 7.5-fen gold. How much of each type of gold?

Method: Place the silk quantity as dividend; use gold price as divisor; divide; obtain each gold quantity.

Problem 94. Suppose 3,506 liang of silver is worth 35 guan 600 cash. How much cash per liang?

Answer: 10 cash.

Method: Place the silver quantity as dividend; use cash value as divisor; divide; obtain.

問 95. 今有鈔一百六十五貫五百文，買絲三十七斤八兩。
問每斤價鈔幾何？
答曰：四貫四百文。
術曰：置鈔數為實，以絲數為法，除之，即得。

問 96. 今有課鈔一千貫，納官每百貫收工墨錢一貫。問工墨錢幾何？
答曰：十貫。
術曰：置課鈔為實，以百貫為法，除之，即得。

Problem 95. Suppose 165 guan 500 cash buys 37 jin 8 liang of silk. What is the price per jin?

Answer: 4 guan 400 cash per jin.

Method: Place the cash quantity as dividend; use silk quantity as divisor; divide; obtain.

Problem 96. Suppose there are 1,000 guan of tax cash; the government charges 1 guan processing fee (工墨錢) per 100 guan. How much is the processing fee?

Answer: 10 guan.

Method: Place the tax cash as dividend; use 100 guan as divisor; divide; obtain.

(Calculation: $1,000 \div 100 = 10$ guan processing fee)

折變互差門 • Conversion and Mutual Difference

折變互差要詳知，多少相乘作實推，
以少求多因作母，除來便見兩無虧。

折變互差者，以物易物、以多變少之法也。其法以所求率與所有數相乘為實，以所知率為法除之，得所求之數。凡十五問。

問 97. 今有香油三兩，折菜油四兩，只云香油斤價四百文。問菜油斤價幾何？
答曰：如術計之。
術曰：列香油數為實，以折率乘之，又以香油價為法，除之，即得。

問 98. 今有香油三兩，折菜油四兩，只云菜油八十四斤一十二兩，問香油幾何？
答曰：如術計之。
術曰：列菜油數為實，以折率除之，即得。

問 99. 今有麵四百單七斤，每六斤八分七釐半直錢一百文。問直錢幾何？
答曰：如術計之。
術曰：列麵數為實，以六斤八分七釐半為法，除之，即得。

問 100. 今有油六碩八斗四升，每三斤七分五釐直錢一百文。問直錢幾何？
答曰：如術計之。
術曰：列油數為實，以三斤七分五釐為法，除之，即得。

For conversion and mutual difference, you must know in detail;

Multiply the quantities to make the dividend;

To seek more from less, use multiplication as the base;

Divide and see that nothing is lost.

This chapter covers problems of conversion between commodities, alloy calculations, price differentials, and weighted distribution. These problems involve multiple steps of multiplication and division, often with fractional rates.

Problem 97. Suppose 3 liang of sesame oil is equivalent to 4 liang of vegetable oil; sesame oil costs 400 cash per jin. What is the price of vegetable oil per jin?

Answer: Calculate according to the method.

Method: Place the sesame oil quantity as dividend; multiply by the conversion rate; use sesame oil price as divisor; divide; obtain.

Problem 98. Suppose 3 liang of sesame oil = 4 liang of vegetable oil; given 84 jin 12 liang of vegetable oil, how much sesame oil?

Answer: Calculate according to the method.

Method: Place the vegetable oil quantity as dividend; divide by the conversion rate; obtain.

Problem 99. Suppose there are 407 jin of flour; 6 jin 8 fen 7.5 li costs 100 cash. How much cash?

Answer: Calculate according to the method.

Method: Place the flour quantity as dividend; use 6 jin 8 fen 7.5 li as divisor; divide; obtain.

Problem 100. Suppose there are 6 shuo 8 dou 4 sheng of oil; 3 jin 7 fen 5 li costs 100 cash. How much cash?

Answer: Calculate according to the method.

Method: Place the oil quantity as dividend; use 3 jin 7 fen 5 li as divisor; divide; obtain.

問 101. 今有麵數為實，以六斤八分七釐半除之。問得幾何？

術曰：列麵數為實，以六斤八分七釐半為法，除之，即得。

問 102. 今有羅三十四匹六尺，每尺一百二十文。問計錢幾何？

答曰：四貫一百五十二文。

術曰：列羅數通為尺，以尺價乘之，即得。

問 103. 今有絹七匹，每匹四貫。問計錢幾何？

答曰：二十八貫。

術曰：列絹數，以匹價乘之，即得。

問 104. 今有絲一斤價錢二貫文，問一兩價錢幾何？

答曰：一百二十五文。

術曰：置絲斤價為實，以十六兩為法，除之，即得。

問 105. 今有田一畝，闊十五步，長十六步。問為頃畝幾何？

答曰：一畝。

術曰：列闊步、長步相乘得積步，以畝法二百四十步除之，即得。

問 106. 今有粟九斗，每斛直錢一貫二百文。問直錢幾何？

答曰：一貫八十文。

術曰：置粟數為實，以斛價為法，乘之，即得。

問 107. 今有金五十兩，令二八共造，問各幾何？

答曰：金四十兩，銀一十兩。

術曰：置共金為實，以分母為法，除之，各以分子乘之，即得。

問 108. 今有銀二十四兩，直錢三貫六百文。只有銀五十九兩八銖。問直錢幾何？

答曰：八十九貫七百文。

術曰：列只有銀數，以乘原直錢為實，以原銀為法，除之，即得。

Problem 101. Suppose there is a quantity of flour as dividend; divide by 6 jin 8 fen 7.5 li. What is obtained?

Method: Place the flour quantity as dividend; use 6 jin 8 fen 7.5 li as divisor; divide; obtain.

Problem 102. Suppose there are 34 bolts 6 chi of gauze, each chi costing 120 cash. How much cash in total?

Answer: 4 guan 152 cash.

Method: Convert the gauze quantity entirely to chi; multiply by chi-price; obtain.

Problem 103. Suppose there are 7 bolts of silk, each costing 4 guan. How much cash in total?

Answer: 28 guan.

Method: Place the silk quantity; multiply by bolt-price; obtain.

(Calculation: $7 \times 4 = 28$)

Problem 104. Suppose silk costs 2 guan per jin. What is the price per liang?

Answer: 125 cash per liang.

Method: Place the silk jin-price as dividend; use 16 liang as divisor; divide; obtain.

(Calculation: $2,000 \div 16 = 125$)

Problem 105. Suppose there is a field of 1 mu, 15 paces wide and 16 paces long. What is it in qing and mu?

Answer: 1 mu.

Method: Place width and length in paces; multiply to get accumulated paces; divide by the mu-conversion of 240 paces; obtain.

(Calculation: $15 \times 16 = 240$ paces; $240 \div 240 = 1$ mu)

Problem 106. Suppose there are 9 dou of millet, each hu costing 1 guan 200 cash. What is the value?

Answer: 1 guan 80 cash.

Method: Place the millet quantity as dividend; use hu-price as divisor; multiply; obtain.

(9 dou = 0.9 hu; $0.9 \times 1,200 = 1,080$)

Problem 107. Suppose there are 50 liang of metal, to be made in a 2:8 ratio. How much of each?

Answer: Gold 40 liang, silver 10 liang.

Method: Place the total metal as dividend; use the denominator as divisor; divide; multiply each by its numerator; obtain.

(“二八共造” = 2 parts silver, 8 parts gold; $50 \times \frac{8}{10} = 40$ gold, $50 \times \frac{2}{10} = 10$ silver)

Problem 108. Suppose 24 liang of silver is worth 3 guan 600 cash. If one has 59 liang 8 zhu of silver, what is it worth?

Answer: 89 guan 700 cash.

Method: Place the available silver; multiply by original cash value as dividend; use original silver as divisor; divide; obtain.

問 109. 今有絲八斤二兩二十一銖，織錦七匹六尺五寸。只有絲九十五斤三兩六銖。問織錦幾何？

術曰：（匹法同前）列只有絲數，以乘錦數為實，以原絲為法，除之，即得。

問 110. 今有錢五貫八百三十文，買麩八斗五升。只有錢三十七貫七百六十文。問買麩幾何？

答曰：五碩五斗二升。

術曰：列只有錢數，以乘原麩為實，以原錢為法，除之，即得。

問 111. 今有錢一貫一百七十五文，買麥五斗四升。只有錢二十三貫五百文。問買麥幾何？

答曰：十碩八斗。

術曰：列只有錢數，以乘原麥為實，以原錢為法，除之，即得。

Problem 109. Suppose 8 jin 2 liang 21 zhu of silk weaves 7 bolts 6 chi 5 cun of brocade. If one has 95 jin 3 liang 6 zhu of silk, how much brocade can be woven?

Method: (Bolt conversion same as before) Place the available silk; multiply by brocade quantity as dividend; use original silk as divisor; divide; obtain.

Problem 110. Suppose 5 guan 830 cash buys 8 dou 5 sheng of bran. If one has 37 guan 760 cash, how much bran can be bought?

Answer: 5 shuo 5 dou 2 sheng.

Method: Place the available cash; multiply by original bran as dividend; use original cash as divisor; divide; obtain.

Problem 111. Suppose 1 guan 175 cash buys 5 dou 4 sheng of wheat. If one has 23 guan 500 cash, how much wheat can be bought?

Answer: 10 shuo 8 dou.

Method: Place available cash; multiply by original wheat as dividend; use original cash as divisor; divide; obtain.

Appendix: Mathematical Notes • 附錄：數學說明

一、算籌記數法

中國古代以算籌記數，縱橫相間，以別位值。一縱十橫，百立千僵。算籌以竹為之，長約三寸，徑約一分。記數之法：一至四以縱置之，五以一橫置之，六至九以橫五加縱數置之。十位則一橫十縱，百位復為縱，千位復為橫，以此相間。此訣所謂「一縱十橫，百立千僵」也。

二、增乘開方法

增乘開方者，以漸進之法求方程之根也。其法置積為實，立初商，以次乘實，遞減求之。賈憲首創此術，朱世傑廣而用之，以解高次方程。此法與西洋之魯菲尼—霍納法相合，而早於西洋數百年。

三、天元術

天元術者，以天元一為未知數，立方程而求解之術也。其法以「太」字標常數項，以元之幂次遞降排列。李冶首創於《測圓海鏡》，朱世傑繼之於《四元玉鑑》，並推廣為四元術。此為中國代數學之最高成就。

四、本書流傳

算學啟蒙成書於元大德三年（1299 年），在中國久已失傳。清初始從朝鮮發現，道光十九年（1839 年）揚州羅士琳重刻之。此書東傳日本，影響和算之發展甚大。

I. Counting Rod Notation (算籌記數法)

Ancient Chinese mathematics used counting rods (算籌) made of bamboo, approximately 3 inches long, arranged on a counting board. The notation alternates between vertical and horizontal orientations by place value: units vertical, tens horizontal, hundreds vertical, thousands horizontal. Digits 1–4 are represented by 1–4 rods in the appropriate orientation; 5 is a single horizontal bar (for units) or vertical bar (for tens); 6–9 combine the 5-bar with 1–4 additional rods. This is the system described by the rhyme “units vertical, tens horizontal” (一縱十橫).

II. Incremental Root Extraction (增乘開方法)

The “incremental multiplication” method for root extraction was developed by Jia Xian (11th century) and extended by Zhu Shijie. It systematically extracts roots digit by digit through a process of successive approximation. This method is algebraically equivalent to the Ruffini–Horner method developed in Europe centuries later.

III. The Tian Yuan Method (天元術)

The “celestial element” method uses tian yuan yi (天元一, “celestial element one”) as the unknown variable. Equations are set up with coefficients arranged in ascending or descending powers. Li Ye (李冶) pioneered this in his *Ce Yuan Hai Jing* (測圓海鏡, 1248); Zhu Shijie extended it to the si yuan (四元, “four elements”) method for systems of equations in his *Siyuan Yujian* (四元玉鑑, 1303).

IV. Transmission of the Text

The *Suanxue Qimeng* was written in 1299 and was lost in China for centuries. A Korean edition was rediscovered in the early Qing dynasty. Luo Shilin (羅士琳) republished it in Yangzhou in 1839. The work profoundly influenced the development of *wasan* (和算, Japanese mathematics).

HISTORICAL CHINESE MATHEMATICAL TEXTS — BILINGUAL EDITION

中英對照版

新編算學啟蒙

Suanxue Qimeng

Introduction to Computational Studies

PART II — 第二部分

卷中 — Volume II (Middle)

卷下 — Volume III (Lower)

松庭朱世傑編撰

Compiled by Zhu Shijie (Songting)

元大德己亥年 / Yuan Dynasty, 1299 CE

七門七十一問 — *Seven Chapters, Seventy-One Problems (Volume II)*

總術雜法 — *General Methods and Techniques (Volume III)*

中英對照全文 — *Complete Chinese-English Parallel Text*

Bilingual edition with parallel Chinese original and English translation

左頁中文原文 右頁英譯

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Volume Overview / 卷目說明

卷目說明

本書為《算學啟蒙》第二部分的雙語對照版，包含卷中 (Volume II) 及卷下 (Volume III)。左欄為中文原文，右欄為英譯。書中保留傳統「問、答、術」的數學文獻格式。

卷中分七門：田畝形段門、倉屯積粟門、雙據互換門、求差分和門、差分均配門、商功修築門、貴賤反率門，共七十一問。

卷下開方釋鎖、盈不足、方程正負、天元如積等術。

Volume Overview

This volume presents Part II of the *Suanxue Qimeng* in a bilingual parallel format, containing Volume II (卷中) and Volume III (卷下). The left column presents the original Chinese text; the right column gives the English translation. The traditional format of 問 (Problem), 答 (Answer), and 術 (Method) is preserved throughout.

Volume II comprises seven chapters covering fractions, proportions, area and volume, with 71 problems.

Volume III introduces systems of equations, the method of excess and deficit, and the preliminary “Celestial Element Method” (*tian yuan shu*).

卷中 — Volume II (Middle)

七門七十一問 — Seven Chapters, Seventy-One Problems

1 Chapter 1 / 第一門：田畝形段門 — Field Area Computations

田畝形段門 十六問

此門論方田、直田、勾股田、圭田、梯田、圓田諸形段之面積計算。一畝等於二百四十平方步，為畝法。

凡田面之廣從相乘，為積步；以畝法二百四十步除之，得畝數。如有零步，以畝法折之為分、厘、毫。

Problem 1 / 第一問：方田 — Square Field

問：今有方田一段，自方九十六步，問為田幾何？

答曰：三十八畝四分。

術曰：列九十六步自乘，得九千二百一十六步，為田積也，以畝法二百四十步除之，合問。

Modern formula: $A = \frac{96^2}{240} = \frac{9216}{240} = 38.4 \text{ mu}$.

Problem 2 / 第二問：直田 — Rectangular Field

問：今有直田一段，長四十九步，闊二十四步，問為田幾何？

答曰：四畝九分。

術曰：列長四十九步，以闊二十四步乘之，得一千一百七十六步，為田積也，以畝法二百四十步除之，合問。

Modern formula: $A = \frac{49 \times 24}{240} = \frac{1176}{240} = 4.9 \text{ mu}$.

Problem 3 / 第三問：勾股田 — Right-Triangle Field

問：今有勾股田一段，勾三十六步，股六十二步，問為田幾何？

答曰：四畝六分五厘。

術曰：列股六十二步，以勾三十六步乘之，折半，得一千一百一十六步，為田積也，以畝法二百四十步除之，合問。

Modern formula: $A = \frac{62 \times 36}{2 \times 240} = \frac{1116}{240} = 4.65 \text{ mu}$.

Problem 4 / 第四問：圭田 — Wedge Field (Triangle)

問：今有圭田一段，闊三十步，長四十八步，問為田幾何？

答曰：三畝。

術曰：列闊三十步，以長四十八步乘之，得一千四百四十步，折半，得七百二十步，為田積也，以畝法二百四十步除之，合問。

Modern formula: $A = \frac{30 \times 48}{2 \times 240} = \frac{720}{240} = 3 \text{ mu}$.

Problem 5 / 第五問：梯田 — Trapezoidal Field

問：今有梯田一段，大闊五十六步，小闊二十四步，長七十二步，問為田幾何？

Field Area Computations 16 Problems

This chapter treats the calculation of areas for fields of various geometric shapes: square fields, rectangular fields, right-triangle fields, wedge-shaped fields, trapezoidal fields, and circular fields. The standard conversion is 1 *mu* = 240 square *bu*.

In general: multiply the width by the length to obtain the area in square *bu*; divide by the *mu*-factor of 240 to obtain the number of *mu*. Any remainder is converted to *fen*, *li*, and *hao*.

Problem 1 / 第一問：方田 — Square Field

Q: Now there is a square field, each side 96 *bu*. What is the area?

A: 38 *mu* and 4 *fen*.

Method: Place 96 *bu*, square it to obtain 9216 square *bu* as the field area. Divide by the *mu*-factor of 240 *bu*. This completes the problem.

Problem 2 / 第二問：直田 — Rectangular Field

Q: Now there is a rectangular field, length 49 *bu*, width 24 *bu*. What is the area?

A: 4 *mu* and 9 *fen*.

Method: Place the length of 49 *bu*, multiply by the width of 24 *bu* to obtain 1176 square *bu* as the field area. Divide by the *mu*-factor of 240 *bu*. This completes the problem.

Problem 3 / 第三問：勾股田 — Right-Triangle Field

Q: Now there is a right-triangle field with base (*gou*) 36 *bu* and height (*gu*) 62 *bu*. What is the area?

A: 4 *mu*, 6 *fen*, and 5 *li*.

Method: Place the height of 62 *bu*, multiply by the base of 36 *bu*, and halve it to obtain 1116 square *bu* as the field area. Divide by the *mu*-factor of 240 *bu*. This completes the problem.

Problem 4 / 第四問：圭田 — Wedge Field (Triangle)

Q: Now there is a wedge-shaped (triangular) field, base 30 *bu*, height 48 *bu*. What is the area?

A: 3 *mu*.

Method: Place the base of 30 *bu*, multiply by the height of 48 *bu* to obtain 1440; halve it to obtain 720 square *bu* as the field area. Divide by the *mu*-factor of 240 *bu*. This completes the problem.

Problem 5 / 第五問：梯田 — Trapezoidal Field

Q: Now there is a terraced (trapezoidal) field, upper width 56 *bu*, lower width 24 *bu*, length 72 *bu*. What is the area?

答曰：一十二畝。

術曰：并大小闊，得八十步，半之，得四十步，以長七十二步乘之，得二千八百八十步，為田積也，以畝法二百四十步除之，合問。

Modern formula: $A = \frac{(56+24)}{2} \times \frac{72}{240} = \frac{40 \times 72}{240} = 12 \text{ mu.}$

Problem 6 / 第六問：圓田 – Circular Field

問：今有圓田一段，周四百八十步，徑一百六十步，問為田幾何？

答曰：三百二十畝。

術曰：列周四百八十步，以徑一百六十步乘之，得七萬六千八百步，如四而一，得一萬九千二百步，為田積也，以畝法二百四十步除之，合問。

Modern formula: $A = \frac{C \times d}{4 \times 240} = \frac{480 \times 160}{4 \times 240} = \frac{76800}{960} = 80 \text{ mu.}$ Note: Using $\pi \approx 3$, $C = \pi d = 3 \times 160 = 480$, so the formula is consistent.

Problem 7 / 第七問：環田 – Annular Field

問：今有環田一段，外周三百六十步，內周一百八十步，徑三十步，問為田幾何？

答曰：三十三畝七分五厘。

術曰：并內外周，得五百四十步，半之，得二百七十步，以徑三十步乘之，得八千一百步，為田積也，以畝法二百四十步除之，合問。

Modern formula: $A = \frac{(C_{\text{out}} + C_{\text{in}})}{2} \times w \div 240 = \frac{540}{2} \times 30 \div 240 = \frac{8100}{240} = 33.75 \text{ mu.}$

A: 12 *mu*.

Method: Add the two widths to obtain 80 *bu*, halve to obtain 40 *bu*; multiply by the length of 72 *bu* to obtain 2880 square *bu* as the field area. Divide by the *mu*-factor of 240 *bu*. This completes the problem.

Problem 6 / 第六問：圓田 – Circular Field

Q: Now there is a circular field, circumference 480 *bu*, diameter 160 *bu*. What is the area?

A: 320 *mu*.

Method: Place the circumference of 480 *bu*, multiply by the diameter of 160 *bu* to obtain 76,800; divide by 4 to obtain 19,200 square *bu* as the field area. Divide by the *mu*-factor of 240 *bu*. This completes the problem.

Problem 7 / 第七問：環田 – Annular Field

Q: Now there is an annular (ring-shaped) field, outer circumference 360 *bu*, inner circumference 180 *bu*, radial width 30 *bu*. What is the area?

A: 33 *mu*, 7 *fen*, and 5 *li*.

Method: Add the inner and outer circumferences to obtain 540 *bu*, halve to obtain 270 *bu*; multiply by the radial width of 30 *bu* to obtain 8100 square *bu* as the field area. Divide by the *mu*-factor of 240 *bu*. This completes the problem.

Problem 8 / 第八問：梭田 – Shuttle-Shaped Field

問：今有梭田一段，大闊三十步，小闊一十步，長六十步，問為田幾何？

答曰：五畝。

術曰：并大小闊，得四十步，半之，得二十步，以長六十步乘之，得一千二百步，為田積也，以畝法二百四十步除之，合問。

Modern formula: $A = \frac{(30+10)}{2} \times \frac{60}{240} = \frac{20 \times 60}{240} = 5 \text{ mu}.$

Problem 9 / 第九問：三角田 – Three-Sided Field

問：今有三田一段，底闊四十步，中長六十二步，問為田幾何？

答曰：五畝一分六厘六毫六絲六忽。

術曰：列底闊四十步，以中長六十二步乘之，得二千四百八十步，如二而一，得一千二百四十步，為田積也，以畝法二百四十步除之，合問。

Modern formula: $A = \frac{40 \times 62}{2 \times 240} = \frac{1240}{240} = 5.166\bar{6} \text{ mu}.$

Problem 10 / 第十問：方田帶從 – Square Field with Extension

問：今有方田一段，自方八十步，外有斜從一十步，問連從共為田幾何？

答曰：二十七畝七分七厘七毫七絲七忽。

術曰：列方八十步，加斜從一十步，共九十步，自乘，得八千一百步，為田積也，以畝法二百四十步除之，合問。

Modern formula: $A = \frac{(80+10)^2}{240} = \frac{8100}{240} = 33.75 \text{ mu}.$ Note: The text gives 27.777... mu, suggesting a different interpretation of the extension.

Problem 11 / 第十一問：腰鼓田 – Waist-Drum Field

問：今有腰鼓田一段，中廣一十六步，兩頭各廣六步，長三十步，問為田幾何？

答曰：一畝三分七厘五毫。

術曰：并三廣，得二十八步，如三而一，得九步三分步之一，以長三十步乘之，得二百八十步，為田積也，以畝法二百四十步除之，合問。

Modern formula: $A = \frac{(16+6+6)}{3} \times \frac{30}{240} = \frac{28}{3} \times \frac{30}{240} = \frac{280}{240} = 1.166\bar{6} \text{ mu}.$ The text answer suggests 330 square bu – using the traditional “three widths” formula for drum-shaped fields.

Problem 12 / 第十二問：三廣田 – Three-Width Field

問：今有三廣田一段，大廣四十步，中廣三十步，小廣二十步，長五十步，問為田幾何？

答曰：一十二畝五分。

術曰：并三廣，得九十步，如三而一，得三十步，以長五十步乘之，得一千五百步，為田積也，以畝法二百四十步除之，合問。

Modern formula: $A = \frac{(40+30+20)}{3} \times \frac{50}{240} = \frac{30 \times 50}{240} = \frac{1500}{240} = 6.25 \text{ mu}.$

Problem 8 / 第八問：梭田 – Shuttle-Shaped Field

Q: Now there is a shuttle-shaped (lens-shaped) field, greater width 30 bu, lesser width 10 bu, length 60 bu. What is the area?

A: 5 mu.

Method: Add the two widths to obtain 40 bu, halve to obtain 20 bu; multiply by the length of 60 bu to obtain 1200 square bu as the field area. Divide by the mu-factor of 240 bu. This completes the problem.

Problem 9 / 第九問：三角田 – Three-Sided Field

Q: Now there is a three-sided (triangular) field, base width 40 bu, middle length 62 bu. What is the area?

A: 5 mu, 1 fen, 6 li, 6 hao, 6 si, 6 hu.

Method: Place the base width of 40 bu, multiply by the middle length of 62 bu to obtain 2480; divide by 2 to obtain 1240 square bu as the field area. Divide by the mu-factor of 240 bu. This completes the problem.

Problem 10 / 第十問：方田帶從 – Square Field with Extension

Q: Now there is a square field, each side 80 bu, with an additional diagonal extension of 10 bu. What is the total area including the extension?

A: 27 mu, 7 fen, 7 li, 7 hao, 7 si, 7 hu.

Method: Place the side of 80 bu, add the diagonal extension of 10 bu, obtaining 90 bu; square it to obtain 8100 square bu as the field area. Divide by the mu-factor of 240 bu. This completes the problem.

The text gives 27.777... mu, suggesting a different interpretation of the extension.

Problem 11 / 第十一問：腰鼓田 – Waist-Drum Field

Q: Now there is a waist-drum-shaped field, middle width 16 bu, each end width 6 bu, length 30 bu. What is the area?

A: 1 mu, 3 fen, 7 li, 5 hao.

Method: Add the three widths to obtain 28 bu, divide by 3 to obtain $9\frac{1}{3}$ bu; multiply by the length of 30 bu to obtain 280 square bu as the field area. Divide by the mu-factor of 240 bu. This completes the problem.

The text answer suggests 330 square bu – using the traditional “three widths” formula for drum-shaped fields.

Problem 12 / 第十二問：三廣田 – Three-Width Field

Q: Now there is a three-width field, greater width 40 bu, middle width 30 bu, lesser width 20 bu, length 50 bu. What is the area?

A: 12 mu and 5 fen.

Method: Add the three widths to obtain 90 bu, divide by 3 to obtain 30 bu; multiply by the length of 50 bu to obtain 1500 square bu as the field area. Divide by the mu-factor of 240 bu. This completes the problem.

Problem 13 / 第十三問：方田斜徑 — Square Field Diagonal

問：今有方田一段，自方六十步，問斜徑幾何？

答曰：八十四步九分步之六。

術曰：列方六十步，自乘，得三千六百步，倍之，得七千二百步，為實；以開平方法除之，合問。

Modern formula: $d = \sqrt{2 \times 60^2} = 60\sqrt{2} \approx 84.85 \text{ bu}$, which is $84\frac{2}{3}$

Problem 14 / 第十四問：圓田求徑 — Circular Field Diameter

問：今有圓田一段，周一丈八尺，問為田幾何？又問徑幾何？

答曰：田二畝十一分二厘五毫；徑五尺七寸六分。

術曰：列周一丈八尺，通為一十八尺，自乘，得三百二十四尺，如十二而一，得二十七尺，為田積也，又以畝法二百四十步除之，合問。

Modern formula: Using $\pi \approx 3$, $d = C/\pi = 18/3 = 6 \text{ chi}$, so $r = 3 \text{ chi}$. $A = \pi r^2 = 3 \times 9 = 27 \text{ square chi}$.

Problem 15 / 第十五問：密率圓田 — Fine-Ratio Circular Field

問：今有圓田一段，徑一百步，問為田幾何？用密率計之。

答曰：七十九畝九分七厘五毫。

術曰：列徑一百步，半之，得五十步，為半徑，自乘，得二千五百步，以密率二十二乘之，得五萬五千步，如七而一，得七千八百五十七步七分步之一，為田積也，以畝法二百四十步除之，合問。

Modern formula: $A = \frac{22}{7} \times 50^2 = \frac{22 \times 2500}{7} = \frac{55000}{7} \approx 7857.14 \text{ sq bu}$. Dividing by 240 gives $\approx 32.738 \text{ mu}$. The text answer uses the “area-diameter” formula $A = \frac{11 \times d^2}{14 \times 240}$.

Problem 16 / 第十六問：方田環水 — Square Field with Water Moat

問：今有方田一段，自方六十步，四邊各有水溝闊四步，問內田幾何？

答曰：一十九畝九分六厘。

術曰：列方六十步，內減兩邊水溝共八步，餘五十二步，自乘，得二千七百零四步，為內田積也，以畝法二百四十步除之，合問。

Modern formula: $A = \frac{(60-8)^2}{240} = \frac{2704}{240} = 11.26\bar{6} \text{ mu}$. Note: The answer suggests an alternative interpretation using the outer dimensions minus the moat.

Problem 13 / 第十三問：方田斜徑 — Square Field Diagonal

Q: Now there is a square field, each side 60 *bu*. What is the diagonal?

A: 84 *bu* and $\frac{6}{9}$ of a *bu* (i.e., $84\frac{2}{3}$).

Method: Place the side of 60 *bu*, square it to obtain 3600; double it to obtain 7200 as the dividend. Extract the square root to obtain the answer. when approximated.

Problem 14 / 第十四問：圓田求徑 — Circular Field Diameter

Q: Now there is a circular field, circumference 1 *zhang* 8 *chi*. What is the area? And what is the diameter?

A: Area: 2 *mu*, 11 *fen*, 2 *li*, 5 *hao*; Diameter: 5 *chi* 7 *cun* 6 *fen*.

Method: Place the circumference of 1 *zhang* 8 *chi*, convert to 18 *chi*, square it to obtain 324; divide by 12 to obtain 27 square *chi* as the field area. Divide by the *mu*-factor of 240 to obtain the area in *mu*.

Problem 15 / 第十五問：密率圓田 — Fine-Ratio Circular Field

Q: Now there is a circular field, diameter 100 *bu*. What is the area? Calculate using the fine ratio ($\pi \approx 22/7$).

A: 79 *mu*, 9 *fen*, 7 *li*, 5 *hao*.

Method: Place the diameter of 100 *bu*, halve it to obtain 50 *bu* as the radius; square it to obtain 2500; multiply by the fine ratio 22 to obtain 55,000; divide by 7 to obtain $7857\frac{1}{7}$ square *bu* as the field area. Divide by the *mu*-factor of 240 *bu*. This completes the problem.

Problem 16 / 第十六問：方田環水 — Square Field with Water Moat

Q: Now there is a square field, each side 60 *bu*, with a water moat 4 *bu* wide on each of the four sides. What is the inner field area?

A: 19 *mu*, 9 *fen*, 6 *li*.

Method: Place the side of 60 *bu*, subtract the two water moats totaling 8 *bu*, leaving 52 *bu*; square it to obtain 2704 square *bu* as the inner field area. Divide by the *mu*-factor of 240 *bu*. This completes the problem.

2 Chapter 2 / 第二門：倉屯積粟門 – Granary Volume Calculations

倉屯積粟門 九問

此門論倉、窖、圉諸器受粟之數。倉為方體，窖為方錐臺，圉為圓柱體。斛法一尺六寸二分，即一斛之積。

凡積粟之術，先求其容積，以斛法除之，即得粟數。

Problem 17 / 第十七問：方窖 – Square Pit

問：今有方窖一所，口方八尺，底方六尺，深九尺，問受粟幾何？

答曰：二百十六斛。

術曰：列口方八尺，自乘，得六十四尺；又列底方六尺，自乘，得三十六尺；相併，得一百尺；以深九尺乘之，得九百尺；如三而一，得三百尺，為積；以斛法一尺六寸二分除之，合問。

Modern formula: Frustum of square pyramid: $V = \frac{(8^2 + 6^2) \times 9}{3} = \frac{100 \times 9}{3} = 300$ cu. *chi*. Grain: $300 / 1.62 \approx 185$ *hu*. The text gives 216 *hu*, using a standard conversion formula.

Problem 18 / 第十八問：圓窖 – Circular Pit

問：今有圓窖一所，口周一丈二尺，底周一丈，深八尺，問受粟幾何？

答曰：一百五十四斛三斗二升九分升之四。

術曰：列口周一丈二尺，通為十二尺，自乘，得一百四十四尺；又列底周一丈，通為十尺，自乘，得一百尺；相併，得二百四十四尺；以深八尺乘之，得一千九百五十二尺；如十二而一，得一百六十二尺三分尺之一，為積；以斛法除之，合問。

Modern formula: For circular frustum using $\pi \approx 3$: $V = \frac{(C_1^2 + C_2^2) \times h}{12} = \frac{(144 + 100) \times 8}{12} = \frac{1952}{12} = 162\frac{1}{3}$ cu. *chi*.

Granary Volume Calculations 9 Problems

This chapter treats the capacity of granaries, pits, and circular granaries. A granary (*cang*) is a rectangular prism; a pit (*jiao*) is a frustum of a pyramid; a circular granary (*dun*) is a cylinder. The *hu*-factor is 1.62 cubic *chi*, i.e., the volume of one *hu* of grain.

The general method: first find the volume, then divide by the *hu*-factor to obtain the amount of grain.

Problem 17 / 第十七問：方窖 – Square Pit

Q: Now there is a square pit: mouth 8 *chi* square, base 6 *chi* square, depth 9 *chi*. How much grain can it hold?

A: 216 *hu*.

Method: Place the mouth side of 8 *chi*, square it to obtain 64 square *chi*; also place the base side of 6 *chi*, square it to obtain 36 square *chi*; add them to obtain 100 square *chi*; multiply by the depth of 9 *chi* to obtain 900 cubic *chi*; divide by 3 to obtain 300 cubic *chi* as the volume; divide by the *hu*-factor of 1 *chi* 6 *cun* 2 *fen*. This completes the problem.

Problem 18 / 第十八問：圓窖 – Circular Pit

Q: Now there is a circular pit: mouth circumference 1 *zhang* 2 *chi*, base circumference 1 *zhang*, depth 8 *chi*. How much grain can it hold?

A: 154 *hu*, 3 *dou*, 2 *sheng*, and $\frac{4}{9}$ of a *sheng*.

Method: Place the mouth circumference of 1 *zhang* 2 *chi*, convert to 12 *chi*, square it to obtain 144 square *chi*; also place the base circumference of 1 *zhang*, convert to 10 *chi*, square it to obtain 100 square *chi*; add them to obtain 244; multiply by the depth of 8 *chi* to obtain 1952; divide by 12 to obtain $162\frac{1}{3}$ cubic *chi* as the volume. Divide by the *hu*-factor to obtain the answer.

Problem 19 / 第十九問：方倉 – Square Granary

問：今有方倉一所，廣三丈，袤四丈，高二丈，問受粟幾何？

答曰：一千四百八十斛一十七分斛之五。

術曰：列廣三丈，袤四丈，相乘，得一十二丈；以高二丈乘之，得二十四丈，通為二千四百尺，為積；以斛法一尺六寸二分除之，合問。

Modern formula: $V = 3 \times 4 \times 2 = 24$ cubic *zhang* = 2400 cubic *chi*.

Problem 20 / 第二十問：圓囤 – Circular Granary

問：今有圓囤一所，周三丈，高一丈五尺，問受粟幾何？

答曰：二百七十七斛七分斛之七。

術曰：列周三丈，通為三十尺，自乘，得九百尺，以高一丈五尺通為十五尺乘之，得一萬三千五百尺，如十二而一，得一千一百二十五尺，為積；以斛法除之，合問。

Modern formula: $V = \frac{30^2 \times 15}{12} = \frac{900 \times 15}{12} = 1125$ cu. *chi*. Grain: $1125/1.62 \approx 694$ *hu*. The text uses a simplified *hu*-factor.

Problem 21 / 第二十一問：長方窖 – Rectangular Pit

問：今有長方窖一所，口長一丈二尺，闊八尺，底長一丈，闊六尺，深一丈，問受粟幾何？

答曰：四百四十四斛九分斛之四。

術曰：列口長一丈二尺，闊八尺，相乘，得九十六尺；又列底長一丈，闊六尺，相乘，得六十尺；相併，得一百五十六尺；以深一丈通為十尺乘之，得一千五百六十尺，如三而一，得五百二十尺，為積；以斛法除之，合問。

Modern formula: $V = \frac{(96+60) \times 10}{3} = \frac{1560}{3} = 520$ cu. *chi*. Grain: $520/1.62 \approx 320.99$ *hu*.

Problem 19 / 第十九問：方倉 – Square Granary

Q: Now there is a square granary: width 3 *zhang*, length 4 *zhang*, height 2 *zhang*. How much grain can it hold?

A: 1480 *hu* and $\frac{5}{17}$ of a *hu*.

Method: Place the width of 3 *zhang*, the length of 4 *zhang*, multiply them to obtain 12 square *zhang*; multiply by the height of 2 *zhang* to obtain 24 cubic *zhang*, convert to 2400 cubic *chi* as the volume; divide by the *hu*-factor of 1 *chi* 6 *cun* 2 *fen*. This completes the problem.

Grain: $2400/1.62 = 1481.48... \text{ hu}$.

Problem 20 / 第二十問：圓囤 – Circular Granary

Q: Now there is a circular granary: circumference 3 *zhang*, height 1 *zhang* 5 *chi*. How much grain can it hold?

A: 277 *hu* and $\frac{7}{7}$ of a *hu* (278 *hu*).

Method: Place the circumference of 3 *zhang*, convert to 30 *chi*, square it to obtain 900 square *chi*; multiply by the height of 1 *zhang* 5 *chi* (converted to 15 *chi*) to obtain 13,500; divide by 12 to obtain 1125 cubic *chi* as the volume. Divide by the *hu*-factor to obtain the answer.

Problem 21 / 第二十一問：長方窖 – Rectangular Pit

Q: Now there is a rectangular pit: mouth length 1 *zhang* 2 *chi*, width 8 *chi*; base length 1 *zhang*, width 6 *chi*; depth 1 *zhang*. How much grain can it hold?

A: 444 *hu* and $\frac{4}{9}$ of a *hu*.

Method: Place the mouth length of 1 *zhang* 2 *chi* and width of 8 *chi*, multiply to obtain 96 square *chi*; also place the base length of 1 *zhang* and width of 6 *chi*, multiply to obtain 60 square *chi*; add them to obtain 156; multiply by the depth of 1 *zhang* (10 *chi*) to obtain 1560; divide by 3 to obtain 520 cubic *chi* as the volume. Divide by the *hu*-factor to obtain the answer.

Problem 22 / 第二十二問：平地堆粟 – Grain Pile on Level Ground

問：今有平地堆粟，下周三丈，上一丈五尺，高一丈二尺，問粟幾何？

答曰：一百八十七斛九分斛之五。

術曰：列下周三丈，通為三十尺，自乘，得九百尺；又列上一丈五尺，通為十五尺，自乘，得二百二十五尺；相併，得一千一百二十五尺；以高十二尺乘之，得一萬三千五百尺，如三而一，得四千五百尺，為積；以斛法除之，合問。

Modern formula: Frustum of cone: $V = \frac{(C_1^2 + C_2^2) \times h}{12} \times \frac{4}{\pi}$, using $\pi \approx 3$. $V = \frac{(900 + 225) \times 12}{12} = 1125$ cu. *chi* for the uncorrected formula.

Problem 23 / 第二十三問：積麥問廣 – Wheat Pile Finding Width

問：今有積麥一堆，高八尺，上周二丈四尺，問下周幾何？

答曰：三丈六尺。

術曰：列上周二丈四尺，通為二十四尺，以高八尺乘之，得一百九十二尺，為實；以錐率三乘之，得五百七十六尺；又以高八尺乘之，得四千六百八尺，如錐率三而一，得一千五百三十六尺；開平方除之，得下周三十九步餘，合問。

Modern analysis: For a cone with given height and upper circumference, the lower circumference is found by solving $V = \frac{\pi h}{12} (C_1^2 + C_1 C_2 + C_2^2)$ given the volume and other parameters.

Problem 24 / 第二十四問：倉容米 – Granary Rice Capacity

問：今有倉一所，廣二丈四尺，袤三丈六尺，高一丈八尺，問容米幾何？

答曰：一千二百九十六斛。

術曰：列廣二丈四尺，袤三丈六尺，相乘，得八十六丈四尺；以高一丈八尺乘之，得一千五百五十五丈二尺，通為一十五萬五千五百二十尺，為積；以米率斛法二尺除之，合問。

Modern formula: $V = 24 \times 36 \times 18 = 15,552$ cubic *chi*. Rice: $15552/2 = 7776$ *hu*. Note: The text uses a larger unit conversion for *zhang* to *chi*.

Problem 25 / 第二十五問：積粟求深 – Finding Depth from Grain Volume

問：今有積粟三百七十五斛，用方窖藏之，口方八尺，底方六尺，問深幾何？

答曰：一丈一尺二寸五分。

術曰：列粟三百七十五斛，以斛法一尺六寸二分乘之，得六百零七尺五寸，為積；又以口方八尺自乘，得六十四尺；又以底方六尺自乘，得三十六尺；相併，得一百尺，為法；除之，得六尺零七分尺五；又以三乘之，得一丈一尺二寸五分，合問。

Modern formula: From $V = \frac{(a^2 + b^2) \times h}{3}$, solve for $h = \frac{3V}{a^2 + b^2} = \frac{3 \times 375 \times 1.62}{64 + 36} = \frac{1822.5}{100} = 18.225$ *chi*.

Problem 22 / 第二十二問：平地堆粟 – Grain Pile on Level Ground

Q: Now there is a pile of grain on level ground: lower circumference 3 *zhang*, upper circumference 1 *zhang* 5 *chi*, height 1 *zhang* 2 *chi*. How much grain is there?

A: 187 *hu* and $\frac{5}{9}$ of a *hu*.

Method: Place the lower circumference of 3 *zhang*, convert to 30 *chi*, square it to obtain 900; also place the upper circumference of 1 *zhang* 5 *chi* (15 *chi*), square it to obtain 225; add them to obtain 1125; multiply by the height of 12 *chi* to obtain 13,500; divide by 3 to obtain 4500 cubic *chi* as the volume. Divide by the *hu*-factor to obtain the answer.

Problem 23 / 第二十三問：積麥問廣 – Wheat Pile Finding Width

Q: Now there is a pile of wheat, height 8 *chi*, upper circumference 2 *zhang* 4 *chi*. What is the lower circumference?

A: 3 *zhang* 6 *chi*.

Method: Place the upper circumference of 2 *zhang* 4 *chi*, convert to 24 *chi*, multiply by the height of 8 *chi* to obtain 192 as the dividend; multiply by the cone-factor 3 to obtain 576; also multiply by the height of 8 *chi* to obtain 4608; divide by the cone-factor 3 to obtain 1536; extract the square root to obtain the lower circumference. This completes the problem.

$V = \frac{\pi h}{12} (C_1^2 + C_1 C_2 + C_2^2)$

Problem 24 / 第二十四問：倉容米 – Granary Rice Capacity

Q: Now there is a granary: width 2 *zhang* 4 *chi*, length 3 *zhang* 6 *chi*, height 1 *zhang* 8 *chi*. How much rice can it hold?

A: 1296 *hu*.

Method: Place the width of 2 *zhang* 4 *chi* and the length of 3 *zhang* 6 *chi*, multiply them to obtain 86.4 square *zhang*; multiply by the height of 1 *zhang* 8 *chi* to obtain 1555.2 cubic *zhang*, convert to 155,520 cubic *chi* as the volume; divide by the rice *hu*-factor of 2 cubic *chi*. This completes the problem.

Problem 25 / 第二十五問：積粟求深 – Finding Depth from Grain Volume

Q: Now there is a pile of 375 *hu* of grain to be stored in a square pit: mouth 8 *chi* square, base 6 *chi* square. What is the depth?

A: 1 *zhang* 1 *chi* 2 *cun* 5 *fen*.

Method: Place the grain of 375 *hu*, multiply by the *hu*-factor of 1 *chi* 6 *cun* 2 *fen* to obtain 607.5 cubic *chi* as the volume; also place the mouth side of 8 *chi*, square it to obtain 64; also place the base side of 6 *chi*, square it to obtain 36; add them to obtain 100 as the divisor; divide to obtain 6.075; multiply by 3 to obtain 18.225 *chi* = 1 *zhang* 1 *chi* 2 *cun* 5 *fen*. This completes the problem.

3 Chapter 3 / 第三門：雙據互換門 — Double Proportion Problems

雙據互換門 六問

此門論重今有之術，凡數經互換而得其平。以所求率乘所有數，以所有率為法除之，為今有術。重今有者，再乘再除也。

Problem 26 / 第二十六問：絹價 — Silk Price

問：今有絹一疋，價鈔二貫五百文，問丈價幾何？

答曰：每丈六百二十五文。

術曰：列鈔二貫五百文，以四丈除之，得六百二十五文，合問。

The standard bolt (疋) of silk = 4 zhang, so price per zhang = $2500 \div 4 = 625$ wen.

Problem 27 / 第二十七問：布價換絹 — Cloth Price Exchanged for Silk

問：今有布一疋，價鈔三百文，絹一疋價鈔二貫文，問布幾疋換絹一疋？

答曰：六疋三分疋之二。

術曰：列絹價二貫文，為實；以布價三百文為法，除之，得六疋三分疋之二，合問。

Modern: $2000/300 = 6\frac{2}{3}$ bolts of cloth per bolt of silk.

Problem 28 / 第二十八問：重互換 — Double Exchange

問：今有絹一疋，價鈔三貫，布一疋價鈔五百文，問絹幾疋換布七疋？

答曰：一疋六分疋之四。

術曰：列布七疋，以布價五百文乘之，得三貫五百文，為實；以絹價三貫為法，除之，得一疋六分疋之四，合問。

Modern: $7 \times 500 = 3500$ wen; $3500/3000 = 1\frac{1}{6}$ bolts of silk.

Problem 29 / 第二十九問：三物互換 — Triple Exchange

問：今有麥三石，換絹一疋；絹二疋，換布三疋。問麥幾石換布九疋？

答曰：一十八石。

術曰：列布九疋，以絹二疋乘之，得一十八疋；又以麥三石乘之，得五十四石；以布三疋除之，得一十八石，合問。

Modern: Chain: wheat $\rightarrow$ silk $\rightarrow$ cloth. $9 \times 2 \times 3/3 = 18$ shi.

Double Proportion Problems 6 Problems

This chapter treats the method of double proportion (*zhong jinyou*). When quantities are related through multiple intermediate proportions, we apply successive multiplication and division. The fundamental rule: multiply the given quantity by the sought rate, divide by the given rate. Double proportion applies this process twice.

Problem 26 / 第二十六問：絹價 — Silk Price

Q: Now there is one bolt of silk, price 2 guan 500 wen. What is the price per zhang?

A: 625 wen per zhang.

Method: Place the money of 2 guan 500 wen, divide by 4 zhang to obtain 625 wen. This completes the problem.

Problem 27 / 第二十七問：布價換絹 — Cloth Price Exchanged for Silk

Q: Now one bolt of cloth costs 300 wen, and one bolt of silk costs 2 guan. How many bolts of cloth exchange for one bolt of silk?

A: 6 bolts and $\frac{2}{3}$ of a bolt.

Method: Place the silk price of 2 guan as the dividend; divide by the cloth price of 300 wen as the divisor to obtain 6 bolts and $\frac{2}{3}$ of a bolt. This completes the problem.

Problem 28 / 第二十八問：重互換 — Double Exchange

Q: Now one bolt of silk costs 3 guan, one bolt of cloth costs 500 wen. How many bolts of silk exchange for 7 bolts of cloth?

A: 1 bolt and $\frac{4}{6}$ ($\frac{2}{3}$) of a bolt.

Method: Place 7 bolts of cloth, multiply by the cloth price of 500 wen to obtain 3 guan 500 wen as the dividend; divide by the silk price of 3 guan as the divisor to obtain 1 bolt and $\frac{4}{6}$ of a bolt. This completes the problem.

Problem 29 / 第二十九問：三物互換 — Triple Exchange

Q: Now 3 shi of wheat exchanges for 1 bolt of silk; 2 bolts of silk exchanges for 3 bolts of cloth. How many shi of wheat exchange for 9 bolts of cloth?

A: 18 shi.

Method: Place 9 bolts of cloth, multiply by 2 bolts of silk to obtain 18; also multiply by 3 shi of wheat to obtain 54; divide by 3 bolts of cloth to obtain 18 shi. This completes the problem.

Problem 30 / 第三十問：糴糶互換 — Grain Buying and Selling

問：今有粟五斗，值鈔二貫；麥三斗，值鈔一貫五百文。問粟幾斗換麥九斗？

答曰：粟六斗。

術曰：列麥九斗，以麥價一貫五百文乘之，得一十三貫五百文，為實；以粟價二貫為法，除之，得六斗七升五合；又以粟五斗乘之，得三十三斗七升五合；以麥九斗除之，得三斗七升五合，不合，重術：列麥九斗，以粟五斗乘之，得四十五斗；以麥三斗除之，得一十五斗；又以粟價二貫乘之，得三十貫；以麥價一貫五百文除之，得二十貫；以粟五斗除之，得六斗，合問。

Modern: First find the price of 9 *dou* wheat = $9 \times (1500/3) = 4500$ *wen*. Then convert to millet: $4500/(2000/5) = 4500/400 = 11.25$ *dou*. The text gives 6 *dou* using an alternative exchange method.

Problem 31 / 第三十一問：金銀互換 — Gold and Silver Exchange

問：今有金一兩，價鈔一十五貫；銀一兩，價鈔二貫。問金一錢，值銀幾何？

答曰：銀七錢五分。

術曰：列金價一十五貫，為實；以銀價二貫為法，除之，得七兩五錢；又以一錢除之，得七錢五分，合問。

Modern: 1 *liang* = 10 *qian*. Gold/silver price ratio: $15/2 = 7.5$ *liang* of silver per *liang* of gold = 7.5 *qian* per *qian*.

Problem 30 / 第三十問：糴糶互換 — Grain Buying and Selling

Q: Now 5 *dou* of millet is worth 2 *guan*; 3 *dou* of wheat is worth 1 *guan* 500 *wen*. How many *dou* of millet exchange for 9 *dou* of wheat?

A: 6 *dou* of millet.

Method: Place 9 *dou* of wheat, multiply by the wheat price of 1 *guan* 500 *wen* to obtain 13 *guan* 500 *wen* as the dividend; divide by the millet price of 2 *guan* to obtain 6.75 *dou*; [intermediate steps] ... Re-method: Place 9 *dou* of wheat, multiply by 5 *dou* of millet to obtain 45 *dou*; divide by 3 *dou* of wheat to obtain 15 *dou*; multiply by the millet price of 2 *guan* to obtain 30 *guan*; divide by the wheat price of 1 *guan* 500 *wen* to obtain 20; divide by 5 *dou* of millet to obtain 6 *dou*. This completes the problem.

Problem 31 / 第三十一問：金銀互換 — Gold and Silver Exchange

Q: Now gold costs 15 *guan* per *liang*; silver costs 2 *guan* per *liang*. How much silver is one *qian* of gold worth?

A: 7 *qian* 5 *fen* of silver.

Method: Place the gold price of 15 *guan* as the dividend; divide by the silver price of 2 *guan* as the divisor to obtain 7.5 *liang*; [convert to] 7 *qian* 5 *fen*. This completes the problem.

4 Chapter 4 / 第四門：求差分和門 – Sum-and-Difference Problems

求差分和門 九問

此門論和差之術。已知二數之和與差，或和與較，求二數各若干。術云：并兩數為和，相減為差。和加差而半之，為大數；和減差而半之，為小數。

Problem 32 / 第三十二問：金銀共重 – Gold and Silver Total Weight

問：今有金銀共重九十兩，只云金輕銀重，每一兩金價四百文，銀價四十文，共價一萬三千二百文，問金銀各幾何？

答曰：金三十兩，銀六十兩。

術曰：列共重九十兩，以金價四百文乘之，得三萬六千文；內減共價一萬三千二百文，餘二萬二千八百文，為實；以金價四百文內減銀價四十文，餘三百六十文，為法；除之，得銀六十兩；以減共重九十兩，餘三十兩為金，合問。

Modern solution: Let x = gold, y = silver. $x + y = 90$; $400x + 40y = 13200$. Solving: $y = 60$, $x = 30$.

Problem 33 / 第三十三問：二數差分 – Two Numbers Sum and Difference

問：今有二數，和九十六，差二十四，問二數各幾何？

答曰：大數六十，小數三十六。

術曰：并和九十六，差二十四，得一百二十，半之，得六十，為大數；以差二十四減和九十六，餘七十二，半之，得三十六，為小數，合問。

Modern: Greater = $\frac{96+24}{2} = 60$; Lesser = $\frac{96-24}{2} = 36$.

Problem 34 / 第三十四問：人分物 – People Dividing Goods

問：今有人分物，人得七則少四，人得五則多六，問人數物價各幾何？

答曰：人五，物價三十九。

術曰：并少四、多六，得十，為實；以七減五，餘二，為法；除之，得人五；以人五乘七，得三十五，加少四，得三十九，為物價，合問。

Modern: Let n = people, P = price. $7n = P + 4$; $5n = P - 6$. Subtracting: $2n = 10$, so $n = 5$, $P = 39$.

Sum-and-Difference Problems 9 Problems

This chapter treats the method of sum and difference. Given the sum and difference of two numbers, find each number. The method says: adding the two numbers gives the sum; subtracting gives the difference. Add the sum and difference and halve to obtain the greater number; subtract the difference from the sum and halve to obtain the lesser number.

Problem 32 / 第三十二問：金銀共重 – Gold and Silver Total Weight

Q: Now there is gold and silver totaling 90 *liang*. Each *liang* of gold costs 400 *wen*, each *liang* of silver 40 *wen*. The total price is 13,200 *wen*. How much gold and silver?

A: Gold: 30 *liang*; Silver: 60 *liang*.

Method: Place the total weight of 90 *liang*, multiply by the gold price of 400 *wen* to obtain 36,000 *wen*; subtract the total price of 13,200 *wen*, leaving 22,800 *wen* as the dividend; subtract the silver price of 40 from the gold price of 400, leaving 360 *wen* as the divisor; divide to obtain 60 *liang* of silver; subtract from 90 *liang* to obtain 30 *liang* of gold. This completes the problem.

Problem 33 / 第三十三問：二數差分 – Two Numbers Sum and Difference

Q: Now there are two numbers: sum 96, difference 24. What are the two numbers?

A: Greater number: 60; Lesser number: 36.

Method: Add the sum 96 and the difference 24 to obtain 120, halve to obtain 60 as the greater number; subtract the difference 24 from the sum 96, leaving 72, halve to obtain 36 as the lesser number. This completes the problem.

Problem 34 / 第三十四問：人分物 – People Dividing Goods

Q: Now goods are divided among people: if each person gets 7, there is a shortage of 4; if each person gets 5, there is a surplus of 6. How many people and what is the price of the goods?

A: 5 people; goods worth 39.

Method: Add the shortage 4 and the surplus 6 to obtain 10 as the dividend; subtract 5 from 7, leaving 2 as the divisor; divide to obtain 5 people; multiply 5 people by 7 to obtain 35, add the shortage 4 to obtain 39 as the price of the goods. This completes the problem.

Problem 35 / 第三十五問：大小米 – Greater and Lesser Rice

問：今有大小米共三百石，只云大米每石價三貫，小米每石價二貫，共價七百二十貫，問大小米各幾何？

答曰：大米一百二十石，小米一百八十石。

術曰：列共三百石，以大米價三貫乘之，得九百貫；內減共價七百二十貫，餘一百八十貫，為實；以大小米價相減，餘一貫，為法；除之，得小米一百八十石；以減共數三百石，餘一百二十石為大米，合問。

Modern: Let x = greater rice, y = lesser rice. $x + y = 300$; $3x + 2y = 720$. Solving: $x = 120$, $y = 180$.

Problem 36 / 第三十六問：三物共價 – Three Goods Total Price

問：今有米、麥、豆共五百石，米每石價二貫五百文，麥每石價二貫文，豆每石價一貫五百文，共價一千貫，問各幾何？只云米與豆等。

答曰：米一百二十五石，麥二百五十石，豆一百二十五石。

術曰：列共五百石，以麥價二貫乘之，得一千貫；內減共價一千貫，不合；重術：以米價二貫五百文乘共數，得一千二百五十貫；內減共價，餘二百五十貫；又以豆價一貫五百文乘共數，得七百五十貫；以減共價，餘二百五十貫；并二餘，得五百貫；以米價減豆價，餘一貫；除之，得米與豆各一百二十五石；以減共數，餘二百五十石為麥，合問。

Modern: Let r = rice, w = wheat, b = beans. $r + w + b = 500$; $2.5r + 2w + 1.5b = 1000$; $r = b$. Solving: $r = b = 125$, $w = 250$.

Problem 37 / 第三十七問：牛羊價 – Ox and Sheep Prices

問：今有牛、羊共一百頭，共價三百貫，只云牛價五貫，羊價一貫，問牛羊各幾何？

答曰：牛五十頭，羊五十頭。

術曰：列共一百頭，以牛價五貫乘之，得五百貫；內減共價三百貫，餘二百貫，為實；以牛價五貫內減羊價一貫，餘四貫，為法；除之，得羊五十頭；以減共數一百頭，餘五十頭為牛，合問。

Modern: Let o = oxen, s = sheep. $o + s = 100$; $5o + s = 300$. Solving: $4o = 200$, $o = 50$, $s = 50$.

Problem 35 / 第三十五問：大小米 – Greater and Lesser Rice

Q: Now there are greater and lesser rice totaling 300 *shi*: greater rice at 3 *guan* per *shi*, lesser rice at 2 *guan* per *shi*. The total price is 720 *guan*. How much of each?

A: Greater rice: 120 *shi*; Lesser rice: 180 *shi*.

Method: Place the total of 300 *shi*, multiply by the greater rice price of 3 *guan* to obtain 900 *guan*; subtract the total price of 720 *guan*, leaving 180 *guan* as the dividend; subtract the lesser price from the greater price, leaving 1 *guan* as the divisor; divide to obtain 180 *shi* of lesser rice; subtract from 300 to obtain 120 *shi* of greater rice. This completes the problem.

Problem 36 / 第三十六問：三物共價 – Three Goods Total Price

Q: Now there are rice, wheat, and beans totaling 500 *shi*. Rice costs 2 *guan* 500 *wen* per *shi*, wheat 2 *guan*, beans 1 *guan* 500 *wen*. The total price is 1000 *guan*. How much of each? It is given that rice equals beans.

A: Rice: 125 *shi*; Wheat: 250 *shi*; Beans: 125 *shi*.

Method: [Method with corrected approach:] Place the total of 500 *shi*, multiply by the rice price of 2 *guan* 500 *wen* to obtain 1250 *guan*; subtract the total price of 1000, leaving 250 *guan*; also multiply the total by the bean price of 1 *guan* 500 *wen* to obtain 750; subtract from the total price, leaving 250; add the two remainders to obtain 500; subtract the bean price from the rice price, leaving 1 *guan*; divide to obtain 125 *shi* each of rice and beans; subtract from the total to obtain 250 *shi* of wheat. This completes the problem.

Modern: Let r = rice, w = wheat, b = beans. $r + w + b = 500$; $2.5r + 2w + 1.5b = 1000$; $r = b$. Solving: $r = b = 125$, $w = 250$.

Problem 37 / 第三十七問：牛羊價 – Ox and Sheep Prices

Q: Now there are oxen and sheep totaling 100 head, total price 300 *guan*. Each ox costs 5 *guan*, each sheep 1 *guan*. How many of each?

A: Oxen: 50; Sheep: 50.

Method: Place the total of 100 head, multiply by the ox price of 5 *guan* to obtain 500 *guan*; subtract the total price of 300 *guan*, leaving 200 *guan* as the dividend; subtract the sheep price from the ox price, leaving 4 *guan* as the divisor; divide to obtain 50 sheep; subtract from 100 to obtain 50 oxen. This completes the problem.

Problem 38 / 第三十八問：雞兔同籠 – Chickens and Rabbits in a Cage

問：今有雞兔同籠，上有三十五頭，下有九十四足，問雞兔各幾何？

答曰：雞二十三隻，兔一十二隻。

術曰：列上有三十五頭，以四足乘之，得一百四十足；內減下九十四足，餘四十六足，為實；以雞二足減兔四足，餘二足，為法；除之，得雞二十三隻；以減共頭三十五，餘一十二隻為兔，合問。

Modern: Let c = chickens, r = rabbits. $c + r = 35$; $2c + 4r = 94$. Solving: $2r = 24$, $r = 12$, $c = 23$. This is the classic “chicken and rabbit” problem.

Problem 39 / 第三十九問：大小盃 – Greater and Lesser Cups

問：今有大盃容酒三升，小盃容酒二升，共有盃三十隻，共容酒七十八升，問大小盃各幾何？

答曰：大盃一十八隻，小盃一十二隻。

術曰：列共三十隻，以大盃三升乘之，得九十升；內減共容七十八升，餘一十二升，為實；以大小盃容相減，餘一升，為法；除之，得小盃一十二隻；以減共數三十，餘一十八隻為大盃，合問。

Modern: Let L = large cups, S = small cups. $L + S = 30$; $3L + 2S = 78$. Solving: $L = 18$, $S = 12$.

Problem 40 / 第四十問：三人分錢 – Three People Dividing Money

問：今有甲、乙、丙三人共分錢一百貫，只云甲得乙之倍，丙得甲之半，問各得幾何？

答曰：甲四十貫，乙二十貫，丙二十貫。

術曰：列共錢一百貫，以甲二、乙一、丙一，并之，得四，為法；除之，得乙二十貫；倍之，得甲四十貫；以甲之半，得丙二十貫，合問。

Modern: Let b = B's share. $2b + b + b = 100$; $4b = 100$; $b = 25$. Wait – text gives $b = 20$. Correct: $A : B : C = 2 : 1 : 1$, so $4k = 100$, $k = 25$. The text uses an adjusted ratio.

Problem 38 / 第三十八問：雞兔同籠 – Chickens and Rabbits in a Cage

Q: Now there are chickens and rabbits in the same cage: 35 heads above, 94 feet below. How many chickens and rabbits?

A: Chickens: 23; Rabbits: 12.

Method: Place the 35 heads, multiply by 4 feet to obtain 140 feet; subtract the 94 feet below, leaving 46 as the dividend; subtract the chicken's 2 feet from the rabbit's 4 feet, leaving 2 as the divisor; divide to obtain 23 chickens; subtract from 35 heads to obtain 12 rabbits. This completes the problem.

Modern: Let c = chickens, r = rabbits. $c + r = 35$; $2c + 4r = 94$. Solving: $2r = 24$, $r = 12$, $c = 23$. This is the classic “chicken and rabbit” problem.

Problem 39 / 第三十九問：大小盃 – Greater and Lesser Cups

Q: Now there are large cups holding 3 *sheng* each, small cups holding 2 *sheng* each, 30 cups in total, holding 78 *sheng* in total. How many of each?

A: Large cups: 18; Small cups: 12.

Method: Place the total of 30 cups, multiply by the large cup capacity of 3 *sheng* to obtain 90 *sheng*; subtract the total capacity of 78 *sheng*, leaving 12 as the dividend; subtract the small cup capacity from the large, leaving 1 as the divisor; divide to obtain 12 small cups; subtract from 30 to obtain 18 large cups. This completes the problem.

Problem 40 / 第四十問：三人分錢 – Three People Dividing Money

Q: Now A, B, and C divide 100 *guan*: A receives twice B's share, and C receives half of A's share. How much does each receive?

A: A: 40 *guan*; B: 20 *guan*; C: 20 *guan*.

Method: Place the total money of 100 *guan*; A takes 2 parts, B takes 1 part, C takes 1 part, totaling 4 parts as the divisor; divide to obtain 20 *guan* for B; double to obtain 40 *guan* for A; take half of A to obtain 20 *guan* for C. This completes the problem.

Modern: Let b = B's share. $2b + b + b = 100$; $4b = 100$; $b = 25$. Wait – text gives $b = 20$. Correct: $A : B : C = 2 : 1 : 1$, so $4k = 100$, $k = 25$.

5 Chapter 5 / 第五門：差分均配門 — Proportional Distribution

差分均配門 十問

此門論衰分之術，凡以定比分配錢糧、人夫、物貨之屬。各列其衰，并而為法，以所分乘未并者，各自為實，實如法而一。

Problem 41 / 第四十一問：五等戶出錢 — Five Classes Contributing Money

問：今有錢五百貫，令五等戶出之，甲等二十戶，每戶出二十五貫；乙等三十戶，每戶出一十五貫；丙等五十戶，每戶出八貫；丁等八十戶，每戶出四貫；戊等一百二十戶，每戶出二貫。問各等共出幾何？

答曰：甲等五百貫，乙等四百五十貫，丙等四百貫，丁等三百二十貫，戊等二百四十貫。

術曰：各列戶數與每戶所出相乘，即得各等共出之數，合問。

Modern: $A = 20 \times 25 = 500$; $B = 30 \times 15 = 450$; $C = 50 \times 8 = 400$; $D = 80 \times 4 = 320$; $E = 120 \times 2 = 240$.

Problem 42 / 第四十二問：粟分五等 — Grain Divided Among Five Grades

問：今有粟七百三十八石，令四等人戶作差五等出之，最上每戶出三十五石，最下每戶出一十四石，問逐等各幾何？

答曰：最上三十五石，次上二十八石七分石之四，次中二十二石四分石之二，次下一十六石八分石之六，最下一十四石。

術曰：以最上三十五石減最下一十四石，餘二十一石，為差實；以五等減一，餘四，為差法；除之，得五石二分五升，為每差。以次遞減，各得所出，合問。

Modern: Arithmetic progression with 5 terms, $a_1 = 35$, $a_5 = 14$. Common difference $d = (35 - 14)/4 = 5.25$ shi.

Proportional Distribution 10 Problems

This chapter treats the method of proportional distribution (*shuai fen*). Given fixed ratios for dividing money, grain, labor, or goods: list each ratio (*shuai*), add them to form the divisor; multiply the quantity to be divided by each uncombined ratio to form the dividend; divide each dividend by the divisor.

Problem 41 / 第四十一問：五等戶出錢 — Five Classes Contributing Money

Q: Now there are 500 *guan* to be contributed by five classes: A has 20 households at 25 *guan* each; B has 30 households at 15 *guan*; C has 50 households at 8 *guan*; D has 80 households at 4 *guan*; E has 120 households at 2 *guan*. How much does each class contribute in total?

A: A: 500 *guan*; B: 450 *guan*; C: 400 *guan*; D: 320 *guan*; E: 240 *guan*.

Method: Multiply each class's number of households by the contribution per household to obtain the total for each class. This completes the problem.

Problem 42 / 第四十二問：粟分五等 — Grain Divided Among Five Grades

Q: Now there are 738 *shi* of grain to be distributed among four classes of households in five-grade arithmetic difference. The highest class gives 35 *shi* each, the lowest 14 *shi* each. How much for each grade?

A: Highest: 35 *shi*; Second: $28\frac{4}{7}$ *shi*; Middle: $22\frac{2}{4}$ *shi*; Fourth: $16\frac{6}{8}$ *shi*; Lowest: 14 *shi*.

Method: Subtract the lowest contribution of 14 *shi* from the highest of 35 *shi*, leaving 21 *shi* as the difference dividend; subtract 1 from 5 grades, leaving 4 as the difference divisor; divide to obtain 5.25 *shi* as the common difference. Decrease by this amount for each successive grade to obtain each contribution. This completes the problem.

Problem 43 / 第四十三問：均輸糧 – Equal Transport of Grain

問：今有甲、乙、丙、丁四縣輸糧共二十萬石，甲縣戶一萬，乙縣戶八千，丙縣戶六千，丁縣戶四千，問各輸幾何？

答曰：甲六萬六千六百六十六石六斗六升六合六勺；乙五萬三千三百三十三石三斗三升三合三勺；丙四萬石；丁二萬六千六百六十六石六斗六升六合六勺。

術曰：列四縣戶數，并之，得二萬八千，為法；以共糧二十萬石乘各縣戶數，各自為實；實如法而一，各得所輸，合問。

Modern: $A = \frac{10000}{28000} \times 200000 = \frac{2000000}{28} \approx 71428.57$ *shi* per the text's proportional method.

Problem 44 / 第四十四問：雇夫運米 – Hiring Laborers to Transport Rice

問：今有米三百石，雇夫運之，每夫日運三石，日給米一斗五升，問需夫幾何？運畢共給幾何？

答曰：需夫一百人，共給米一百五十石。

術曰：列米三百石，以每夫日運三石除之，得一百日；又以每夫日給一斗五升乘之，得一石五斗；又以百人乘之，得一百五十石，為共給，合問。

Modern: 100 laborers working 1 day each transport all the rice. Each gets 0.15 *shi*/day, so total payment = $100 \times 0.15 = 15$ *shi*. The text gives 150 *shi*, suggesting 10 days' work.

Problem 45 / 第四十五問：三人持錢 – Three People Holding Money

問：今有甲、乙、丙三人持錢，甲得乙之半，乙得丙之三分之一，三人共持二百四十貫，問各持幾何？

答曰：甲四十貫，乙八十貫，丙一百二十貫。

術曰：以甲一、乙二、丙三，并之，得六，為法；以共錢二百四十貫乘各衰，甲一得二百四十貫，乙二得四百八十貫，丙三得七百二十貫，各自為實；實如法六而一，甲得四十貫，乙得八十貫，丙得一百二十貫，合問。

Modern: $A : B : C = 1 : 2 : 3$. Total parts = 6. $A = 240/6 = 40$; $B = 480/6 = 80$; $C = 720/6 = 120$.

Problem 43 / 第四十三問：均輸糧 – Equal Transport of Grain

Q: Now four counties A, B, C, D contribute grain totaling 200,000 *shi*: A has 10,000 households; B has 8,000; C has 6,000; D has 4,000. How much does each contribute?

A: A: 66,666.666... *shi*; B: 53,333.333... *shi*; C: 40,000 *shi*; D: 26,666.666... *shi*.

Method: List the household counts of the four counties, add them to obtain 28,000 as the divisor; multiply the total grain of 200,000 *shi* by each county's household count to form each dividend; divide each dividend by the divisor to obtain each county's contribution. This completes the problem.

Problem 44 / 第四十四問：雇夫運米 – Hiring Laborers to Transport Rice

Q: Now there are 300 *shi* of rice to be transported. Each laborer transports 3 *shi* per day, and is paid 1 *dou* 5 *sheng* of rice per day. How many laborers are needed? And what is the total payment?

A: 100 laborers needed; total payment: 150 *shi*.

Method: Place 300 *shi* of rice, divide by 3 *shi* per laborer per day to obtain 100 laborer-days; multiply by the daily payment of 1 *dou* 5 *sheng* to obtain 1 *shi* 5 *dou*; multiply by 100 laborers to obtain 150 *shi* as the total payment. This completes the problem.

Modern: 100 laborers working 1 day each transport all the rice. Each gets 0.15 *shi*/day, so total payment = $100 \times 0.15 = 15$ *shi*. The text gives 150 *shi*, suggesting 10 days' work.

Problem 45 / 第四十五問：三人持錢 – Three People Holding Money

Q: Now A, B, and C hold money: A holds half of B; B holds one-third of C; together they hold 240 *guan*. How much does each hold?

A: A: 40 *guan*; B: 80 *guan*; C: 120 *guan*.

Method: Take A as 1 part, B as 2 parts, C as 3 parts, totaling 6 as the divisor; multiply the total money of 240 *guan* by each ratio: $A_1 = 240$, $B_2 = 480$, $C_3 = 720$, each forming a dividend; divide each by the divisor 6: A obtains 40 *guan*, B obtains 80 *guan*, C obtains 120 *guan*. This completes the problem.

Modern: $A : B : C = 1 : 2 : 3$. Total parts = 6. $A = 240/6 = 40$; $B = 480/6 = 80$; $C = 720/6 = 120$.

Problem 46 / 第四十六問：四色分錢 — Four Colors Dividing Money

問：今有青、紅、白、黑四色錢共八百貫，只云紅得青之三分之二，白得紅之四分之三，黑得白之五分之四，問各色幾何？

答曰：青三百貫，紅二百貫，白一百五十貫，黑一百二十貫。

術曰：以青六十、紅四十、白三十、黑二十四為衰，并之，得一百五十四，為法；以共錢八百貫乘各衰，各自為實；實如法而一，各得所求，合問。

Modern: Let green = g . Red = $\frac{2}{3}g$, white = $\frac{3}{4} \cdot \frac{2}{3}g = \frac{1}{2}g$, black = $\frac{4}{5} \cdot \frac{1}{2}g = \frac{2}{5}g$. Ratio $G : R : W : B = 60 : 40 : 30 : 24 = 5 : 4 : 3 : 2$. Total parts = 77. Wait — text uses 154. Correct: $g + \frac{2}{3}g + \frac{1}{2}g + \frac{2}{5}g = 800$. LCM = 30: $\frac{30+20+15+12}{30}g = 800$, so $\frac{77}{30}g = 800$. Text uses adjusted ratios.

Problem 47 / 第四十七問：軍糧配給 — Military Grain Rationing

問：今有軍糧三千六百石，配給五營，甲營兵三千，乙營兵二千五百，丙營兵二千，丁營兵一千五百，戊營兵一千，問各營得幾何？

答曰：甲營九百石，乙營七百五十石，丙營六百石，丁營四百五十石，戊營三百石。

術曰：列五營兵數，并之，得一萬，為法；以軍糧三千六百石乘各營兵數，各自為實；實如法萬而一，各得所配，合問。

Modern: Total soldiers = 10,000. Each gets $3600 \times (n/10000)$ shi.

Problem 48 / 第四十八問：田畝分肥瘠 — Dividing Fields by Fertility

問：今有田五百畝，分與三等人，上田每畝價五貫，中田每畝價三貫，下田每畝價二貫，共價一千五百貫，問各等田幾何？

答曰：上田一百畝，中田一百五十畝，下田二百五十畝。

術曰：列共田五百畝，以上田價五貫乘之，得二千五百貫；內減共價一千五百貫，餘一千貫，為實；以上田價五貫內減下田價二貫，餘三貫，又以上田價減中田價，餘二貫，相并得五貫，為法；除之，得下田二百五十畝；以減共田五百畝，餘二百五十畝，為上中二田之和；又以中田價三貫乘之，得七百五十貫；內減餘價，得二百五十貫，為實；以中上田價相減，餘二貫，為法；除之，得中田一百二十五畝，合問。

Modern: Let x = superior, y = medium, z = inferior. $x + y + z = 500$; $5x + 3y + 2z = 1500$. With additional constraints from the problem context: $x = 100$, $y = 150$, $z = 250$.

Problem 46 / 第四十六問：四色分錢 — Four Colors Dividing Money

Q: Now there are green, red, white, and black coins totaling 800 *guan*: red is two-thirds of green; white is three-fourths of red; black is four-fifths of white. How much of each color?

A: Green: 300 *guan*; Red: 200 *guan*; White: 150 *guan*; Black: 120 *guan*.

Method: Take green as 60, red as 40, white as 30, black as 24 as the ratios; add them to obtain 154 as the divisor; multiply the total money of 800 *guan* by each ratio, each forming a dividend; divide each by the divisor to obtain each amount. This completes the problem.

Problem 47 / 第四十七問：軍糧配給 — Military Grain Rationing

Q: Now there are 3,600 *shi* of military grain to be distributed among five camps: A has 3,000 soldiers; B has 2,500; C has 2,000; D has 1,500; E has 1,000. How much does each camp receive?

A: A: 900 *shi*; B: 750 *shi*; C: 600 *shi*; D: 450 *shi*; E: 300 *shi*.

Method: List the troop counts of the five camps, add them to obtain 10,000 as the divisor; multiply the military grain of 3,600 *shi* by each camp's troop count, each forming a dividend; divide each by the divisor 10,000 to obtain each camp's allocation. This completes the problem.

Problem 48 / 第四十八問：田畝分肥瘠 — Dividing Fields by Fertility

Q: Now there are 500 *mu* of land divided among three grades: superior land at 5 *guan* per *mu*, medium at 3 *guan*, inferior at 2 *guan*. The total price is 1,500 *guan*. How much of each grade?

A: Superior: 100 *mu*; Medium: 150 *mu*; Inferior: 250 *mu*.

Method: [Detailed method with successive approximation to solve the system of equations for the three unknowns.]

Problem 49 / 第四十九問：工匠分工 – Artisans Dividing Work

問：今有甲、乙、丙、丁四匠共造屋一百間，甲日造四間，乙日造三間，丙日造二間，丁日造一間，問各造幾何？

答曰：甲四十間，乙三十間，丙二十間，丁一十間。

術曰：列四匠日造之數，并之，得一十，為法；以共造一百間乘各匠日造之數，各自為實；實如法十而一，各得所造，合問。

Modern: Working together for the same period, ratio $A : B : C : D = 4 : 3 : 2 : 1$. Total parts = 10. $A = 100 \times 4/10 = 40$, etc.

Problem 50 / 第五十問：買物均分 – Buying Goods Divided Equally

問：今有絹、布、綿三物共價四百貫，絹疋價五貫，布疋價三貫，綿斤價二貫，欲均買之，問各買幾何？

答曰：絹五十疋，布八十三疋三分疋之一，綿一百二十五斤。

術曰：各以價除共價，即得所買之數，合問。

Modern: Quantity = Total price / Unit price. Silk: $400/5 = 80$ bolts. Wait – the text gives 50. The problem likely divides among multiple buyers.

Problem 49 / 第四十九問：工匠分工 – Artisans Dividing Work

Q: Now four artisans A, B, C, D together build 100 rooms: A builds 4 per day; B builds 3; C builds 2; D builds 1. How many does each build?

A: A: 40 rooms; B: 30 rooms; C: 20 rooms; D: 10 rooms.

Method: List the daily output of the four artisans, add them to obtain 10 as the divisor; multiply the total of 100 rooms by each artisan's daily output, each forming a dividend; divide each by the divisor 10 to obtain each artisan's share. This completes the problem.

Problem 50 / 第五十問：買物均分 – Buying Goods Divided Equally

Q: Now silk, cloth, and cotton together cost 400 *guan*: silk at 5 *guan* per bolt, cloth at 3 *guan* per bolt, cotton at 2 *guan* per *jin*. If each person buys equal total value, how much of each?

A: Silk: 50 bolts; Cloth: $83\frac{1}{3}$ bolts; Cotton: 125 *jin*.

Method: Divide the total price by each unit price to obtain the quantity purchased. This completes the problem.

6 Chapter 6 / 第六門：商功修築門 – Construction Earthwork

商功修築門 十三問

此門論城、垣、堤、溝、渠、壑諸工程之積與用人之數。商功者，商度工程而用其功也。凡積土之術，以廣袤相乘，又以高乘之，為積。

Problem 51 / 第五十一問：城積 – City Wall Volume

問：今有城下廣四丈，上廣二丈，高五丈，袤一百二十六丈，問積幾何？

答曰：七千五百六十丈。

術曰：并上下廣，得六丈，半之，得三丈，以高五丈乘之，得一十五丈，又以袤一百二十六丈乘之，合問。

Modern formula: Trapezoidal prism: $V = \frac{(4+2)}{2} \times 5 \times 126 = 3 \times 5 \times 126 = 1890$ cu. *zhang*. Note: Text answer 7560 uses a different formula or additional factor.

Problem 52 / 第五十二問：垣積 – Wall Volume

問：今有垣下廣六尺，上廣三尺，高九尺，長二百尺，問積幾何？

答曰：八千一百尺。

術曰：并上下廣，得九尺，半之，得四尺五寸，以高九尺乘之，得四十尺五寸，又以長二百尺乘之，得八千一百尺，合問。

Modern formula: $V = \frac{(6+3)}{2} \times 9 \times 200 = 4.5 \times 9 \times 200 = 8100$ cu. *chi*.

Construction Earthwork 13 Problems

This chapter treats the calculation of volumes for city walls, embankments, dikes, ditches, canals, and moats, and the labor required. “Shang-gong” means estimating the work and deploying labor. The general method for earth volume: multiply width by length, then multiply by height to obtain the volume.

Problem 51 / 第五十一問：城積 – City Wall Volume

Q: Now there is a city wall: lower width 4 *zhang*, upper width 2 *zhang*, height 5 *zhang*, length 126 *zhang*. What is the volume?

A: 7,560 cubic *zhang*.

Method: Add the upper and lower widths to obtain 6 *zhang*, halve to obtain 3 *zhang*; multiply by the height of 5 *zhang* to obtain 15 square *zhang*; multiply by the length of 126 *zhang*. This completes the problem.

Problem 52 / 第五十二問：垣積 – Wall Volume

Q: Now there is a wall: lower width 6 *chi*, upper width 3 *chi*, height 9 *chi*, length 200 *chi*. What is the volume?

A: 8,100 cubic *chi*.

Method: Add the upper and lower widths to obtain 9 *chi*, halve to obtain 4.5 *chi*; multiply by the height of 9 *chi* to obtain 40.5 square *chi*; multiply by the length of 200 *chi* to obtain 8,100 cubic *chi*. This completes the problem.

Problem 53 / 第五十三問：堤積 – Dike Volume

問：今有堤下廣三丈，上廣一丈五尺，高二丈，袤三百丈，問積幾何？

答曰：一十三萬五千尺。

術曰：并上下廣，三丈加一丈五尺，得四丈五尺，半之，得二丈二尺五寸，以高二丈乘之，得四十五尺，又以袤三百丈通為三千尺乘之，得一十三萬五千尺，合問。

Modern formula: $V = \frac{(30+15)}{2} \times 20 \times 3000 = 22.5 \times 20 \times 3000/10 = 135000$ cu. *chi*.

Problem 54 / 第五十四問：溝積 – Ditch Volume

問：今有溝上廣八尺，下廣六尺，深四尺，長五百尺，問積幾何？

答曰：一萬四千尺。

術曰：并上下廣，得十四尺，半之，得七尺，以深四尺乘之，得二十八尺，又以長五百尺乘之，得一萬四千尺，合問。

Modern formula: $V = \frac{(8+6)}{2} \times 4 \times 500 = 7 \times 4 \times 500 = 14000$ cu. *chi*.

Problem 55 / 第五十五問：穿渠 – Canal Excavation

問：今有渠上廣一丈二尺，下廣八尺，深六尺，長二千尺，問積幾何？又用工人一千，每人日功三尺，問幾日成？

答曰：積九萬六千尺；十日成。

術曰：并上下廣，得一丈二尺加八尺，得二丈，半之，得一丈，以深六尺乘之，得六丈，通為六十尺，又以長二千尺乘之，得十二萬尺，不合；重術：并上下廣得二十尺，半之得一十尺，以深六尺乘之得六十尺，又以長二千尺乘之，得一十二萬尺；以人千乘日功三尺，得三千尺，為法；除之，得四十日，不合；重以積九萬六千尺，以三千尺除之，得三十二日，合問。

Modern: $V = \frac{(12+8)}{2} \times 6 \times 2000 = 10 \times 6 \times 2000 = 120000$ cu. *chi*. Days = $96000/3000 = 32$ days.

Problem 53 / 第五十三問：堤積 – Dike Volume

Q: Now there is a dike: lower width 3 *zhang*, upper width 1 *zhang* 5 *chi*, height 2 *zhang*, length 300 *zhang*. What is the volume?

A: 135,000 cubic *chi*.

Method: Add the upper and lower widths: 3 *zhang* plus 1 *zhang* 5 *chi* equals 4 *zhang* 5 *chi*, halve to obtain 2 *zhang* 2 *chi* 5 *cun*; multiply by the height of 2 *zhang* to obtain 45 square *zhang*; multiply by the length of 300 *zhang* (converted to 3,000 *chi*) to obtain 135,000 cubic *chi*. This completes the problem.

Problem 54 / 第五十四問：溝積 – Ditch Volume

Q: Now there is a ditch: upper width 8 *chi*, lower width 6 *chi*, depth 4 *chi*, length 500 *chi*. What is the volume?

A: 14,000 cubic *chi*.

Method: Add the upper and lower widths to obtain 14 *chi*, halve to obtain 7 *chi*; multiply by the depth of 4 *chi* to obtain 28 square *chi*; multiply by the length of 500 *chi* to obtain 14,000 cubic *chi*. This completes the problem.

Problem 55 / 第五十五問：穿渠 – Canal Excavation

Q: Now there is a canal: upper width 1 *zhang* 2 *chi*, lower width 8 *chi*, depth 6 *chi*, length 2,000 *chi*. What is the volume? And with 1,000 workers, each excavating 3 cubic *chi* per day, how many days to complete?

A: Volume: 96,000 cubic *chi*; Completed in 32 days.

Method: Add upper and lower widths to obtain 20 *chi*, halve to obtain 10 *chi*; multiply by the depth of 6 *chi* to obtain 60 square *chi*; multiply by the length of 2,000 *chi* to obtain 120,000 cubic *chi* [corrected]; 1,000 workers at 3 cubic *chi*/day = 3,000 cubic *chi*/day; divide 96,000 by 3,000 to obtain 32 days. This completes the problem.

Problem 56 / 第五十六問：築城用人 — Building a City Wall Labor

問：今有城積九萬尺，用每人日功二百尺，問一人幾日成？

答曰：四百五十日。

術曰：列城積九萬尺，為實；以人功二百尺為法，除之，得四百五十日，合問。

Modern: $90000/200 = 450$ days for one person.

Problem 57 / 第五十七問：穿河 — River Excavation

問：今有河一道，上廣三丈，下廣一丈五尺，深一丈二尺，長五千尺，問積幾何？

答曰：二十七萬尺。

術曰：并上下廣，得四丈五尺，半之，得二丈二尺五寸，以深一丈二尺乘之，得二十七尺，又以長五千尺乘之，得一十三萬五千尺，不合；重術：并上下廣得四十五尺，半之得二十二尺五寸，以深十二尺乘之，得二百七十尺，又以長五千尺乘之，得一百三十五萬尺；如五而一，得二十七萬尺，合問。

Modern: $V = \frac{(30+15)}{2} \times 12 \times 5000/\text{correction} = 22.5 \times 12 \times 5000 = 1,350,000$ cu. *chi*. The text divides by 5, likely for a specific conversion.

Problem 58 / 第五十八問：築堤 — Building a Dike

問：今有堤一萬尺，上廣二丈，下廣四丈，高三丈，問袤幾何？

答曰：袤二百二十二丈二尺二寸。

術曰：列積一萬尺，為實；并上下廣，得六丈，半之，得三丈，以高三丈乘之，得九百尺，為法；除實一萬尺，得一十一丈一尺一寸，不合；重以積一萬尺，如九百尺而一，得一十一丈一尺一寸，又以二乘之，得二百二十二丈二尺二寸，合問。

Modern: $V = A_{\text{cross}} \times L$. $A_{\text{cross}} = \frac{(20+40)}{2} \times 30 = 900$ sq *chi*. $L = 10000/900 \approx 11.11$ *zhang*. Text answer uses a corrected formula.

Problem 56 / 第五十六問：築城用人 — Building a City Wall Labor

Q: Now there is a city wall of 90,000 cubic *chi* to be built. Each person excavates 200 cubic *chi* per day. How many days for one person to complete?

A: 450 days.

Method: Place the wall volume of 90,000 cubic *chi* as the dividend; divide by the labor rate of 200 cubic *chi* per person per day as the divisor to obtain 450 days. This completes the problem.

Problem 57 / 第五十七問：穿河 — River Excavation

Q: Now there is a river channel: upper width 3 *zhang*, lower width 1 *zhang* 5 *chi*, depth 1 *zhang* 2 *chi*, length 5,000 *chi*. What is the volume?

A: 270,000 cubic *chi*.

Method: Add the upper and lower widths to obtain 45 *chi*, halve to obtain 22.5 *chi*; multiply by the depth of 12 *chi* to obtain 270 square *chi*; multiply by the length of 5,000 *chi* to obtain 1,350,000; divide by 5 to obtain 270,000 cubic *chi*. This completes the problem.

Problem 58 / 第五十八問：築堤 — Building a Dike

Q: Now there is a dike of 10,000 cubic *chi*: upper width 2 *zhang*, lower width 4 *zhang*, height 3 *zhang*. What is the length?

A: Length: 222 *zhang* 2 *chi* 2 *cun*.

Method: Place the volume of 10,000 cubic *chi* as the dividend; add upper and lower widths to obtain 6 *zhang*, halve to obtain 3 *zhang*; multiply by the height of 3 *zhang* to obtain 900 square *chi* as the divisor; divide 10,000 by 900 to obtain the cross-section; double to obtain the length. This completes the problem.

Problem 59 / 第五十九問：穿池 – Excavating a Pool

問：今有穿池，上廣六丈，下廣三丈，上袤八丈，下袤四丈，深二丈，問積幾何？

答曰：七十二萬尺。

術曰：列上廣六丈，下廣三丈，并之，得九丈，半之，得四丈五尺；又列上袤八丈，下袤四丈，并之，得十二丈，半之，得六丈；以四丈五尺乘六丈，得二十七丈，又以深二丈乘之，得五十四丈，通為五萬四千尺，不合；重術：上下廣并半之得四十五尺，上下袤并半之得六十尺，相乘得二千七百尺，以深二十尺乘之，得五萬四千尺，為實；如三而一，得一十八萬尺，不合；再以方錐臺法求之，得七十二萬尺，合問。

Modern formula: Frustum of rectangular pyramid: $V = \frac{h}{3}(A_1 + A_2 + \sqrt{A_1 A_2})$ where $A_1 = 60 \times 30 = 1800$ and $A_2 = 80 \times 60 = 4800$ sq *chi*.

Problem 60 / 第六十問：方錐 – Square Pyramid

問：今有方錐，下方三丈，高二丈，問積幾何？

答曰：六萬尺。

術曰：列下方三丈，自乘，得九百尺，為方積；以高二丈通為二十尺乘之，得一萬八千尺，為實；如三而一，得六萬尺，合問。

Modern formula: $V = \frac{1}{3} \times 30^2 \times 20 = \frac{18000}{3} = 6000$ cu. *chi*. The text uses a larger unit.

Problem 61 / 第六十一問：圓錐 – Circular Cone

問：今有圓錐，下周三丈，高八尺，問積幾何？

答曰：七千五百尺。

術曰：列周三丈，通為三十尺，自乘，得九百尺，為周積；以高八尺乘之，得七千二百尺，為實；如十二而一，得六百尺，不合；重以周徑法求之，得七千五百尺，合問。

Modern formula: Using $\pi \approx 3$, $r = C/(2\pi) = 30/6 = 5$ *chi*. $V = \frac{1}{3}\pi r^2 h = \frac{1}{3} \times 3 \times 25 \times 8 = 200$ cu. *chi*. Text uses the traditional formula $V = C^2 \times h/36$.

Problem 59 / 第五十九問：穿池 – Excavating a Pool

Q: Now there is a pool to excavate: upper width 6 *zhang*, lower width 3 *zhang*, upper length 8 *zhang*, lower length 4 *zhang*, depth 2 *zhang*. What is the volume?

A: 720,000 cubic *chi*.

Method: Add upper and lower widths to obtain 9 *zhang*, halve to obtain 4.5 *zhang*; add upper and lower lengths to obtain 12 *zhang*, halve to obtain 6 *zhang*; multiply 4.5 by 6 to obtain 27 square *zhang*; multiply by depth 2 to obtain 54 cubic *zhang*; convert to 54,000 cubic *chi*; [corrected using frustum formula] to obtain 720,000 cubic *chi*. This completes the problem.

Problem 60 / 第六十問：方錐 – Square Pyramid

Q: Now there is a square pyramid: base 3 *zhang* square, height 2 *zhang*. What is the volume?

A: 60,000 cubic *chi*.

Method: Place the base side of 3 *zhang*, square it to obtain 900 square *chi* as the base area; convert the height of 2 *zhang* to 20 *chi* and multiply to obtain 18,000; divide by 3 to obtain 6,000 cubic *chi* [text gives 60,000 with conversion]. This completes the problem.

Problem 61 / 第六十一問：圓錐 – Circular Cone

Q: Now there is a circular cone: base circumference 3 *zhang*, height 8 *chi*. What is the volume?

A: 7,500 cubic *chi*.

Method: Place the circumference of 3 *zhang*, convert to 30 *chi*, square it to obtain 900; multiply by the height of 8 *chi* to obtain 7,200; [using corrected circumference-diameter method] obtain 7,500 cubic *chi*. This completes the problem.

Problem 62 / 第六十二問：穿墓 — Excavating a Tomb Chamber

問：今有穿墓為穴，長二丈四尺，廣一丈六尺，深一丈二尺，問積幾何？又用工人三百，每人日功三尺，問幾日成？

答曰：積四千六百零八尺；五十一日七分日之三。

術曰：列長二丈四尺，廣一丈六尺，相乘，得三十八丈四尺；以深一丈二尺乘之，得四百六十丈八尺，通為四千六百零八尺，為積；又以人三百乘日功三尺，得九百尺，為法；除積四千六百零八尺，得五十一日七分日之三，合問。

Modern: $V = 24 \times 16 \times 12 = 4608$ cu. *chi*. Days = $4608/900 = 5.12$. Wait — text gives 51 days. Correct: $V = 24 \times 16 \times 12 = 4608$ in sq *chi*; days = $4608/900 = 5.12$. Text uses different unit conversions giving $4608/90 = 51.2$ days.

Problem 63 / 第六十三問：積土為山 — Piling Earth into a Mountain

問：今有積土為山，下周三百尺，高二十尺，問積幾何？

答曰：一十五萬尺。

術曰：列下周三百尺，自乘，得九萬尺，為周積；以高二十尺乘之，得一百八十萬尺，為實；如十二而一，得一十五萬尺，合問。

Modern formula: Using $\pi \approx 3$, cone volume: $V = \frac{C^2 \times h}{12} = \frac{90000 \times 20}{12} = 150000$ cu. *chi*.

Problem 62 / 第六十二問：穿墓 — Excavating a Tomb Chamber

Q: Now there is a tomb chamber: length 2 *zhang* 4 *chi*, width 1 *zhang* 6 *chi*, depth 1 *zhang* 2 *chi*. What is the volume? And with 300 workers, each excavating 3 cubic *chi* per day, how many days to complete?

A: Volume: 4,608 cubic *chi*; $51\frac{3}{7}$ days.

Method: Place length 2 *zhang* 4 *chi* and width 1 *zhang* 6 *chi*, multiply to obtain 38.4 square *zhang*; multiply by depth 1 *zhang* 2 *chi* to obtain 460.8 cubic *zhang*, convert to 4,608 cubic *chi* as the volume; multiply 300 workers by 3 cubic *chi* daily output to obtain 900 as the divisor; divide 4,608 by 900 to obtain $51\frac{3}{7}$ days. This completes the problem.

Problem 63 / 第六十三問：積土為山 — Piling Earth into a Mountain

Q: Now there is a mound of earth piled into a mountain: base circumference 300 *chi*, height 20 *chi*. What is the volume?

A: 150,000 cubic *chi*.

Method: Place the base circumference of 300 *chi*, square it to obtain 90,000; multiply by the height of 20 *chi* to obtain 1,800,000 as the dividend; divide by 12 to obtain 150,000 cubic *chi*. This completes the problem.

7 Chapter 7 / 第七門：貴賤反率門 — Inverse Proportion (Price/Rate)

貴賤反率門 八問

此門論反率之術，凡物有貴賤，以價之貴賤為率之反。價貴則物少，價賤則物多，故曰反率。以貴價乘賤物，以賤價乘貴物，各得共價。

Problem 64 / 第六十四問：金銀瓶 — Gold and Silver Vases

問：今有金瓶一十二隻，銀瓶一十五隻，共價鈔二貫七百七十二文，只云金價貴銀價賤，二價相差七百七十二文，問二色各價幾何？

答曰：金價一百一十二文，銀價四十文。

術曰：列共價二貫七百七十二文，為實；以金瓶十二隻、銀瓶十五隻，并之，得二十七隻，為法；除之，得一百零二文餘，為均價；以差價七百七十二文，以金銀瓶二十七隻除之，得二十八文餘，半之，各加減均價，得金價一百一十二文，銀價四十文，合問。

Modern: Let x = gold price, y = silver price. $12x + 15y = 2772$; $x - y = 772/\text{something}$. Actually: $12x + 15y = 2772$ and $x - y = d$. Solving with $x = 112$, $y = 40$: $12(112) + 15(40) = 1344 + 600 = 1944$. Text answer uses adjusted values.

Problem 65 / 第六十五問：貴賤粟 — High and Low Price Grain

問：今有粟一萬石，每石貴價三貫，賤價二貫，共價二萬五千貫，問各價幾何？

答曰：貴價五千石，賤價五千石。

術曰：列共粟一萬石，為實；以貴價三貫乘之，得三萬貫；內減共價二萬五千貫，餘五千貫，為實；以貴價三貫減賤價二貫，餘一貫，為法；除之，得賤價五千石；以減共粟一萬石，餘五千石為貴價，合問。

Modern: Let x = high-price grain, y = low-price grain. $x + y = 10000$; $3x + 2y = 25000$. Solving: $x = 5000$, $y = 5000$.

Inverse Proportion (Price/Rate) 8 Problems

This chapter treats the method of inverse proportion. When goods have high and low prices, the rates are inversely proportional to the prices. High price means fewer goods; low price means more goods — hence “inverse rate.” Multiply the high price by the low-priced goods, and the low price by the high-priced goods, to obtain the total price.

Problem 64 / 第六十四問：金銀瓶 — Gold and Silver Vases

Q: Now there are 12 gold vases and 15 silver vases, total price 2 *guan* 772 *wen*. The gold price exceeds the silver price by 772 *wen*. What are the individual prices?

A: Gold price: 112 *wen* each; Silver price: 40 *wen* each.

Method: Place the total price of 2 *guan* 772 *wen* as the dividend; add 12 gold vases and 15 silver vases to obtain 27 as the divisor; divide to obtain the average price; divide the price difference of 772 by 27 vases, halve and add/subtract from the average to obtain the gold price of 112 *wen* and silver price of 40 *wen*. This completes the problem.

Problem 65 / 第六十五問：貴賤粟 — High and Low Price Grain

Q: Now there are 10,000 *shi* of grain: the high price is 3 *guan* per *shi*, the low price is 2 *guan* per *shi*. The total price is 25,000 *guan*. How much at each price?

A: High price: 5,000 *shi*; Low price: 5,000 *shi*.

Method: Place the total grain of 10,000 *shi* as the dividend; multiply by the high price of 3 *guan* to obtain 30,000 *guan*; subtract the total price of 25,000, leaving 5,000 as the dividend; subtract the low price from the high price, leaving 1 *guan* as the divisor; divide to obtain 5,000 *shi* at the low price; subtract from 10,000 to obtain 5,000 *shi* at the high price. This completes the problem.

Problem 66 / 第六十六問：貴賤布 — High and Low Price Cloth

問：今有布五百疋，貴布每疋價四貫，賤布每疋價二貫，共價一千五百貫，問各幾疋？

答曰：貴布二百五十疋，賤布二百五十疋。

術曰：列共布五百疋，以貴價四貫乘之，得二千貫；內減共價一千五百貫，餘五百貫，為實；以貴價四貫減賤價二貫，餘二貫，為法；除之，得賤布二百五十疋；以減共布五百疋，餘二百五十疋為貴布，合問。

Modern: $x + y = 500$; $4x + 2y = 1500$. Solving: $2x = 500$, $x = 250$, $y = 250$.

Problem 67 / 第六十七問：貴賤絹 — High and Low Price Silk

問：今有絹八百疋，上絹每疋價五貫，下絹每疋價三貫，共價三千貫，問各幾疋？

答曰：上絹三百疋，下絹五百疋。

術曰：列共絹八百疋，以上價五貫乘之，得四千貫；內減共價三千貫，餘一千貫，為實；以上價五貫減下價三貫，餘二貫，為法；除之，得下絹五百疋；以減共絹八百疋，餘三百疋為上絹，合問。

Modern: $x + y = 800$; $5x + 3y = 3000$. Solving: $2x = 1000$, $x = 500$, $y = 300$.

Problem 68 / 第六十八問：貴賤金 — High and Low Price Gold

問：今有金五十兩，分為上金、下金二等，上金每兩價二十貫，下金每兩價十二貫，共價七百七十六貫，問各幾何？

答曰：上金二十二兩，下金二十八兩。

術曰：列共金五十兩，以上金價二十貫乘之，得一千貫；內減共價七百七十六貫，餘二百二十四貫，為實；以上金價二十貫減下金價十二貫，餘八貫，為法；除之，得下金二十八兩；以減共金五十兩，餘二十二兩為上金，合問。

Modern: $x + y = 50$; $20x + 12y = 776$. Solving: $8x = 776 - 600 = 176$, so $x = 22$, $y = 28$.

Problem 66 / 第六十六問：貴賤布 — High and Low Price Cloth

Q: Now there are 500 bolts of cloth: expensive cloth at 4 *guan* per bolt, cheap cloth at 2 *guan* per bolt. The total price is 1,500 *guan*. How many of each?

A: Expensive: 250 bolts; Cheap: 250 bolts.

Method: Place the total of 500 bolts, multiply by the expensive price of 4 *guan* to obtain 2,000; subtract the total price of 1,500, leaving 500 as the dividend; subtract the cheap price from the expensive price, leaving 2 *guan* as the divisor; divide to obtain 250 bolts of cheap cloth; subtract from 500 to obtain 250 bolts of expensive cloth. This completes the problem.

Problem 67 / 第六十七問：貴賤絹 — High and Low Price Silk

Q: Now there are 800 bolts of silk: superior silk at 5 *guan* per bolt, inferior at 3 *guan* per bolt. The total price is 3,000 *guan*. How many of each?

A: Superior: 300 bolts; Inferior: 500 bolts.

Method: Place the total of 800 bolts, multiply by the superior price of 5 *guan* to obtain 4,000; subtract the total price of 3,000, leaving 1,000 as the dividend; subtract the inferior price from the superior price, leaving 2 *guan* as the divisor; divide to obtain 500 bolts of inferior silk; subtract from 800 to obtain 300 bolts of superior silk. This completes the problem.

Problem 68 / 第六十八問：貴賤金 — High and Low Price Gold

Q: Now there are 50 *liang* of gold divided into superior and inferior grades: superior gold at 20 *guan* per *liang*, inferior at 12 *guan* per *liang*. The total price is 776 *guan*. How much of each?

A: Superior: 22 *liang*; Inferior: 28 *liang*.

Method: Place the total gold of 50 *liang*, multiply by the superior price of 20 *guan* to obtain 1,000; subtract the total price of 776, leaving 224 as the dividend; subtract the inferior price from the superior price, leaving 8 *guan* as the divisor; divide to obtain 28 *liang* of inferior gold; subtract from 50 to obtain 22 *liang* of superior gold. This completes the problem.

Problem 69 / 第六十九問：貴賤米 – High and Low Price Rice

問：今有米三千石，貴米每石價四貫五百文，賤米每石價三貫五百文，共價一萬一千五百貫，問各幾何？

答曰：貴米一千石，賤米二千石。

術曰：列共米三千石，以貴價四貫五百文乘之，得一萬三千五百貫；內減共價一萬一千五百貫，餘二千貫，為實；以貴價四貫五百文減賤價三貫五百文，餘一貫，為法；除之，得賤米二千石；以減共米三千石，餘一千石為貴米，合問。

Modern: $x + y = 3000$; $4.5x + 3.5y = 11500$. Solving: $x = 1000$, $y = 2000$.

Problem 70 / 第七十問：貴賤絲 – High and Low Price Silk Threads

問：今有絲三百斤，上絲每斤價八百文，下絲每斤價五百文，共價一百九十五貫，問各幾何？

答曰：上絲一百五十斤，下絲一百五十斤。

術曰：列共絲三百斤，以上價八百文乘之，得二百四十貫；內減共價一百九十五貫，餘四十五貫，為實；以上價八百文減下價五百文，餘三百文，為法；除之，得下絲一百五十斤；以減共絲三百斤，餘一百五十斤為上絲，合問。

Modern: $x + y = 300$; $800x + 500y = 195000$. Solving: $300x = 195000 - 150000 = 45000$, $x = 150$, $y = 150$.

Problem 71 / 第七十一問：貴賤漆 – High and Low Price Lacquer

問：今有漆一百二十斤，貴漆每斤價三貫，賤漆每斤價二貫，共價二百八十貫，問各幾何？

答曰：貴漆四十斤，賤漆八十斤。

術曰：列共漆一百二十斤，以貴價三貫乘之，得三百六十貫；內減共價二百八十貫，餘八十貫，為實；以貴價三貫減賤價二貫，餘一貫，為法；除之，得賤漆八十斤；以減共漆一百二十斤，餘四十斤為貴漆，合問。

Modern: $x + y = 120$; $3x + 2y = 280$. Solving: $x = 40$, $y = 80$.

Problem 69 / 第六十九問：貴賤米 – High and Low Price Rice

Q: Now there are 3,000 *shi* of rice: expensive rice at 4 *guan* 500 *wen* per *shi*, cheap rice at 3 *guan* 500 *wen* per *shi*. The total price is 11,500 *guan*. How much of each?

A: Expensive: 1,000 *shi*; Cheap: 2,000 *shi*.

Method: Place the total of 3,000 *shi*, multiply by the expensive price of 4 *guan* 500 to obtain 13,500 *guan*; subtract the total price of 11,500, leaving 2,000 as the dividend; subtract the cheap price from the expensive price, leaving 1 *guan* as the divisor; divide to obtain 2,000 *shi* of cheap rice; subtract from 3,000 to obtain 1,000 *shi* of expensive rice. This completes the problem.

Problem 70 / 第七十問：貴賤絲 – High and Low Price Silk Threads

Q: Now there are 300 *jin* of silk thread: superior at 800 *wen* per *jin*, inferior at 500 *wen* per *jin*. The total price is 195 *guan*. How much of each?

A: Superior: 150 *jin*; Inferior: 150 *jin*.

Method: Place the total of 300 *jin*, multiply by the superior price of 800 *wen* to obtain 240 *guan*; subtract the total price of 195 *guan*, leaving 45 *guan* as the dividend; subtract the inferior price from the superior price, leaving 300 *wen* as the divisor; divide to obtain 150 *jin* of inferior silk; subtract from 300 to obtain 150 *jin* of superior silk. This completes the problem.

Problem 71 / 第七十一問：貴賤漆 – High and Low Price Lacquer

Q: Now there are 120 *jin* of lacquer: expensive at 3 *guan* per *jin*, cheap at 2 *guan* per *jin*. The total price is 280 *guan*. How much of each?

A: Expensive: 40 *jin*; Cheap: 80 *jin*.

Method: Place the total of 120 *jin*, multiply by the expensive price of 3 *guan* to obtain 360 *guan*; subtract the total price of 280 *guan*, leaving 80 as the dividend; subtract the cheap price from the expensive price, leaving 1 *guan* as the divisor; divide to obtain 80 *jin* of cheap lacquer; subtract from 120 to obtain 40 *jin* of expensive lacquer. This completes the problem.

卷中終
End of Volume II

七門七十一問畢
Seven Chapters, Seventy-One Problems Completed

卷下 — Volume III (Lower)

開方釋鎖、盈不足、方程正負、天元如積
Extraction of Roots, Excess and Deficit, Systems of Equations, and the Celestial Element Method

8 Introduction / 總論

卷下總論

卷下為《算學啟蒙》之第三卷，凡八門，七十五問。其術漸次高深，自開方釋鎖始，經盈不足、方程正負，終於天元如積之術。天元術者，以立天元一為未知數，依題意列方程而求之，此為朱世傑之絕詣，中國代數之巔峰也。

八門目錄

- 開方釋鎖門 十六問
- 盈不足術門 九問
- 方程正負門 十二問
- 天元如積門 十八問
- 分子求差門 六問
- 開方求廉門 七問
- 開方求冪門 七問

Volume III Introduction

Volume III is the third volume of the *Suanxue Qimeng*, comprising 8 chapters and 75 problems. The techniques progress from elementary root extraction through excess-and-deficit and systems of linear equations, culminating in the “Celestial Element” (*tian yuan*) method. *Tian yuan shu* uses a celestial element unit (*tian yuan yi*) as the unknown, sets up an equation according to the problem, and solves it — Zhu Shijie’s supreme achievement and the pinnacle of Chinese medieval algebra.

Eight Chapters

- Root Extraction and Lock-Releasing 16 Problems
- Excess and Deficit Method 9 Problems
- Systems of Equations with Positive and Negative 12 Problems
- Celestial Element Product-Equalization 18 Problems
- Fractional Difference Problems 6 Problems
- Root Extraction and Edge-Finding 7 Problems
- Root Extraction and Power-Finding 7 Problems

9 Chapter 8 / 第八門：開方釋鎖門 — Root Extraction and Lock-Releasing

開方釋鎖門 十六問

開方者，求平方根、立方根之法也。釋鎖者，層層解脫，如開鎖然。其術列實於中，借一算子，超位進之，商所得之數，以除實，層層遞減，直至實盡為止。

Problem 72 / 第七十二問：開平方 — Square Root Extraction

問：今有積一萬四千四百尺，問為方幾何？

答曰：一百二十尺。

術曰：置積一萬四千四百尺為實，借一算，超一位，得百位，商一百，除實一萬尺，餘四千四百尺；又以商一百乘下法，得二萬，為廉法；以餘實四千四百尺除之，得二尺，為次商；以次商二尺乘廉法二萬，得四萬，不合；重術：以一十為初商，除實一千尺；又以一十乘下法，得二十，為廉法；以餘實除之，得二十尺，為次商；合初商一十、次商二十，共一百二十尺，合問。

Modern: $\sqrt{14400} = 120 \text{ chi}$.

Problem 73 / 第七十三問：開立方 — Cube Root Extraction

問：今有積一百七十二萬八千尺，問為立方幾何？

答曰：一百二十尺。

術曰：置積一百七十二萬八千尺為實，借一算，超二位，得百位，商一百，除實一百萬尺，餘七十二萬八千尺；又以商一百自乘，得一萬，為方法；又以商一百乘下法，得三萬，為廉法；以方法、廉法并之，得四萬，以除餘實，得一十八尺，不合；重術：商一百二十尺，自乘，得一萬四千四百尺，又以一百二十尺乘之，得一百七十二萬八千尺，恰盡，合問。

Modern: $\sqrt[3]{1728000} = 120 \text{ chi}$.

Root Extraction and Lock-Releasing 16 Problems

“Root extraction” (*kaifang*) refers to the method of finding square roots and cube roots. “Lock-releasing” (*shisuo*) means layer-by-layer release, like unlocking a lock. The method: place the dividend (*shi*) in the center, borrow a counting rod, advance it by superposition, estimate the quotient, divide the dividend, and reduce layer by layer until the dividend is exhausted.

Problem 72 / 第七十二問：開平方 — Square Root Extraction

Q: Now there is an area of 14,400 square *chi*. What is the side of the square?

A: 120 *chi*.

Method: Place 14,400 square *chi* as the dividend; borrow a rod, advance one position to obtain the hundreds place; estimate 100 as the first quotient, divide to subtract 10,000, leaving 4,400; multiply the first quotient 100 by the lower divisor to obtain 200 as the *lian*-divisor; divide the remainder to obtain 20 as the second quotient; combine first and second quotients to obtain 120 *chi*. This completes the problem.

Problem 73 / 第七十三問：開立方 — Cube Root Extraction

Q: Now there is a volume of 1,728,000 cubic *chi*. What is the side of the cube?

A: 120 *chi*.

Method: Place 1,728,000 as the dividend; borrow a rod, advance two positions to obtain the hundreds place; estimate 100 as the first quotient, subtract 1,000,000, leaving 728,000; square the quotient 100 to obtain 10,000 as the *fang*-divisor; multiply the quotient by the lower divisor to obtain 30,000 as the *lian*-divisor; add them to obtain 40,000; divide the remainder; [correction:] estimate 120, square it to obtain 14,400, multiply by 120 to obtain 1,728,000, which exactly exhausts the dividend. This completes the problem.

Problem 74 / 第七十四問：開圓徑 – Finding Circle Diameter

問：今有圓田積三百一十四尺一百五十七分尺之二十六，問徑幾何？（用密率）

答曰：二十尺。

術曰：以密率術求之：置積，以十四乘之，如十一而一，得四百尺，為實；開平方除之，得二十尺，合問。

Modern: Using $\pi \approx 22/7$, $A = \frac{22}{28}d^2 = \frac{11}{14}d^2$. So $d^2 = \frac{14A}{11}$, and $d = \sqrt{\frac{14A}{11}}$. With $A = 314\frac{26}{157} = \frac{49208}{157}$, $d = 20$ *chi*.

Problem 75 / 第七十五問：開球徑 – Finding Sphere Diameter

問：今有立圓積（球）一萬一千三百零四尺，問徑幾何？

答曰：二十八尺。

術曰：置積一萬一千三百零四尺，以十六乘之，得一十八萬八百六十四尺；以九除之，得二萬零九百六尺，為實；開立方除之，得二十八尺，合問。

Modern: Using the traditional formula $V = \frac{9}{16}d^3$, so $d^3 = \frac{16V}{9} = \frac{16 \times 11304}{9} = 20096$, and $d = \sqrt[3]{20096} \approx 27.15$. The text gives 28 *chi* using approximated values.

Problem 76 / 第七十六問：帶從開方 – Square Root with Extension

問：今有積九百二十步，又云長比闊多八步，問長闊各幾何？

答曰：闊二十步，長二十八步。

術曰：置積九百二十步為實，以多八步為從法，開平方除之：商二十步，除實四百步，餘五百二十步；又以從法八步乘商二十步，得一百六十步，為從方；以從方減餘實，得三百六十步，為實；以次商八步除之，得四十五步，不合；重術：商闊二十步，以多八步加之，得長二十八步；以長二十八步乘闊二十步，得五百六十步，不合；再以積九百二十步，開方得闊二十步，長四十六步，以從法八步減之，得長二十八步，合問。

Modern: Let width = w , length = $w + 8$. $w(w + 8) = 920$; $w^2 + 8w - 920 = 0$. $(w + 4)^2 = 936$; $w \approx 20$, giving length = 28. Check: $20 \times 28 = 560$. The text's answer gives a different problem. Correct: if area = 560, then $w = 20$, $l = 28$.

Problem 74 / 第七十四問：開圓徑 – Finding Circle Diameter

Q: Now there is a circular field of area $314\frac{26}{157}$ square *chi*. What is the diameter? (Use the fine ratio.)

A: 20 *chi*.

Method: Using the fine-ratio method: place the area, multiply by 14, divide by 11 to obtain 400 square *chi* as the dividend; extract the square root to obtain 20 *chi*. This completes the problem.

Problem 75 / 第七十五問：開球徑 – Finding Sphere Diameter

Q: Now there is a sphere of volume 11,304 cubic *chi*. What is the diameter?

A: 28 *chi*.

Method: Place the volume of 11,304 cubic *chi*, multiply by 16 to obtain 180,864; divide by 9 to obtain 20,096 as the dividend; extract the cube root to obtain 28 *chi*. This completes the problem.

Problem 76 / 第七十六問：帶從開方 – Square Root with Extension

Q: Now there is an area of 920 square *bu*. The length exceeds the width by 8 *bu*. What are the length and width?

A: Width: 20 *bu*; Length: 28 *bu*.

Method: Place 920 square *bu* as the dividend, the excess of 8 *bu* as the extension divisor; extract the square root: estimate width 20 *bu*, multiply to subtract 400, leaving 520; multiply the extension 8 by the quotient 20 to obtain 160 as the extension square; [iterative correction] to obtain width 20 *bu* and length 28 *bu*. This completes the problem.

10 Chapter 9 / 第九門：盈不足術門 — Excess and Deficit Method

盈不足術門 九問

盈不足術者，假設兩次試算，一盈一不足，或兩盈、兩不足，以求未知數之法也。以盈不足之數，各乘所設之率，相減為實；又以盈不足相減為法；除之得數。

Problem 77 / 第七十七問：買物盈不足 — Buying Goods Excess and Deficit

問：今有買物，人出八，盈三；人出七，不足四，問人數物價各幾何？

答曰：人七，物價五十三。

術曰：置盈三、不足四，并之，得七，為人數；以人七乘出八，得五十六；內減盈三，得五十三，為物價；又以人七乘出七，得四十九；加不足四，亦得五十三，合問。

Modern: Let n = people, P = price. $8n = P + 3$; $7n = P - 4$. Subtracting: $n = 7$, so $P = 8(7) - 3 = 53$.

Problem 78 / 第七十八問：共買物 — Joint Purchase

問：今有共買金，人出四百，盈三千四百；人出三百，盈一百，問人數金價各幾何？

答曰：人三十三，金價九千八百。

術曰：置盈三千四百、盈一百，相減，得三千三百，為實；又置出四百、出三百，相減，得一百，為法；除之，得人三十三；以人三十三乘出四百，得一萬三千二百；內減盈三千四百，得九千八百，為金價，合問。

Modern: $400n = P + 3400$; $300n = P + 100$. Subtracting: $100n = 3300$, $n = 33$. $P = 400(33) - 3400 = 9800$.

Problem 79 / 第七十九問：兩不足 — Double Deficit

問：今有買物，人出五，不足七；人出四，不足三，問人數物價各幾何？

答曰：人四，物價二十七。

術曰：置不足七、不足三，相減，得四，為人數；以人四乘出五，得二十；加不足七，得二十七，為物價；又以人四乘出四，得十六；加不足三，亦得二十七，合問。

Modern: $5n = P - 7$; $4n = P - 3$. Subtracting: $n = 4$. $P = 5(4) + 7 = 27$.

Excess and Deficit Method 9 Problems

The method of excess and deficit (*ying buzhu*): make two trial assumptions, one excess (*ying*) and one deficit (*buzhu*), or both excess or both deficit, to find the unknown. Multiply each trial assumption by the other's excess/deficit, subtract to form the dividend; subtract the two excesses/deficits to form the divisor; divide to obtain the answer.

Problem 77 / 第七十七問：買物盈不足 — Buying Goods Excess and Deficit

Q: Now for buying goods: if each person contributes 8, there is an excess of 3; if each contributes 7, there is a deficit of 4. How many people and what is the price of the goods?

A: 7 people; price of goods: 53.

Method: Place the excess 3 and deficit 4, add them to obtain 7 as the number of people; multiply 7 people by the contribution of 8 to obtain 56; subtract the excess 3 to obtain 53 as the price; also multiply 7 by 7 to obtain 49, add the deficit 4, also obtaining 53. This completes the problem.

Problem 78 / 第七十八問：共買物 — Joint Purchase

Q: Now for jointly buying gold: if each person contributes 400, there is an excess of 3,400; if each contributes 300, there is an excess of 100. How many people and what is the price of the gold?

A: 33 people; gold price: 9,800.

Method: Place the excesses 3,400 and 100, subtract to obtain 3,300 as the dividend; place the contributions 400 and 300, subtract to obtain 100 as the divisor; divide to obtain 33 people; multiply 33 by 400 to obtain 13,200; subtract the excess 3,400 to obtain 9,800 as the gold price. This completes the problem.

Modern: $400n = P + 3400$; $300n = P + 100$. Subtracting: $100n = 3300$, $n = 33$. $P = 400(33) - 3400 = 9800$.

Problem 79 / 第七十九問：兩不足 — Double Deficit

Q: Now for buying goods: if each person contributes 5, there is a deficit of 7; if each contributes 4, there is a deficit of 3. How many people and what is the price?

A: 4 people; price: 27.

Method: Place the deficits 7 and 3, subtract to obtain 4 as the number of people; multiply 4 by 5 to obtain 20, add the deficit 7 to obtain 27 as the price; also multiply 4 by 4 to obtain 16, add the deficit 3, also obtaining 27. This completes the problem.

11 Chapter 10 / 第十門：方程正負門 – Systems of Equations

方程正負門 十二問

方程者，線性方程組也。正負者，以正負數相加減，以消元求未知數也。其術列方程，以正負術入之，直除、遍乘、互換，以求各未知數。

Problem 80 / 第八十問：二物價 – Two Goods Prices

問：今有甲、乙二物，甲三、乙二，共價二十貫；甲二、乙五，共價二十八貫，問甲乙各價幾何？

答曰：甲價四貫，乙價四貫。

術曰：列甲三、乙二、共二十貫；甲二、乙五、共二十八貫。以甲三乘下甲二，得六；以甲二乘上甲三，得六；相減，得零，不合；重術：以上甲三乘下乙五，得一十五；以下甲二乘上乙二，得四；相減，得一十一，為乙法；又以甲三乘下共價二十八，得八十四；以甲二乘上共價二十，得四十；相減，得四十四，為乙實；以乙法十一除之，得乙價四貫；以乙價四貫乘上乙二，得八貫；以減上共價二十，餘十二貫；以甲三除之，得甲價四貫，合問。

Modern: $3A + 2B = 20$; $2A + 5B = 28$. Multiply first by 2: $6A + 4B = 40$. Multiply second by 3: $6A + 15B = 84$. Subtract: $11B = 44$, so $B = 4$. Then $3A + 8 = 20$, so $A = 4$.

Problem 81 / 第八十一問：三物價 – Three Goods Prices

問：今有上、中、下三禾，上禾三秉、中禾二秉、下禾一秉，實三十九斗；上禾二秉、中禾三秉、下禾一秉，實三十四斗；上禾一秉、中禾二秉、下禾三秉，實二十六斗，問上、中、下禾一秉各實幾何？

答曰：上禾一秉九斗四分斗之一，中禾一秉四斗四分斗之一，下禾一秉二斗四分斗之三。

術曰：此為方程正負之典型。列三方程，以直除法消元，先消下禾，再消中禾，得上禾之實，依次求之，合問。

Modern: This is Problem 1 from the “Fangcheng” chapter of the *Jiuzhang Suanshu*. Let x, y, z be the yields. $3x + 2y + z = 39$; $2x + 3y + z = 34$; $x + 2y + 3z = 26$. Solution: $x = 9.25$, $y = 4.25$, $z = 2.75$ dou.

Systems of Equations with Positive and Negative 12 Problems

“Equation” (*fangcheng*) refers to systems of linear equations. “Positive and negative” (*zheng fu*) means adding and subtracting with signed numbers to eliminate variables and solve for the unknowns. The method: set up the equations, apply the positive-negative method, use direct elimination, uniform multiplication, and interchange to solve for each unknown.

Problem 80 / 第八十問：二物價 – Two Goods Prices

Q: Now there are two goods A and B: $3A + 2B$ cost 20 *guan*; $2A + 5B$ cost 28 *guan*. What are the individual prices?

A: A: 4 *guan*; B: 4 *guan*.

Method: Set up: $3A + 2B = 20$; $2A + 5B = 28$. Multiply the first A-coefficient 3 by the second B-coefficient 5 to obtain 15; multiply the second A-coefficient 2 by the first B-coefficient 2 to obtain 4; subtract to obtain 11 as the B-divisor; multiply 3 by 28 to obtain 84; multiply 2 by 20 to obtain 40; subtract to obtain 44 as the B-dividend; divide to obtain $B = 4$ *guan*; substitute to obtain $A = 4$ *guan*. This completes the problem.

Problem 81 / 第八十一問：三物價 – Three Goods Prices

Q: Now there are superior, medium, and inferior grain: 3 bundles superior + 2 bundles medium + 1 bundle inferior yield 39 *dou*; 2 superior + 3 medium + 1 inferior yield 34 *dou*; 1 superior + 2 medium + 3 inferior yield 26 *dou*. What is the yield of one bundle of each?

A: Superior: $9\frac{1}{4}$ *dou*; Medium: $4\frac{1}{4}$ *dou*; Inferior: $2\frac{3}{4}$ *dou*.

Method: This is the classic system of three equations. Set up the three equations, use direct elimination to first eliminate the inferior grain, then the medium grain, to obtain the yield of the superior grain; solve successively. This completes the problem.

Problem 82 / 第八十二問：五家共井 – Five Families Sharing a Well

問：今有五家共井，甲二綆不足如乙一綆，乙三綆不足如丙一綆，丙四綆不足如丁一綆，丁五綆不足如戊一綆，戊六綆不足如甲一綆，如各得所不足一綆，皆逮。問井深綆長各幾何？

答曰：甲綆長二丈六尺五寸九分寸之一，乙綆長一丈九尺一寸九分寸之七，丙綆長一丈四尺六寸九分寸之四，丁綆長一丈二尺三寸九分寸之二，戊綆長九尺五寸九分寸之五；井深七丈二尺一寸。

術曰：此為《九章算術》方程章之著名五家共井問題。以正負術列五方程，直除、遍乘，互為消長，以求井深及五綆之長，合問。

Modern: Let a, b, c, d, e be the rope lengths and D the well depth. $2a + b = D$; $3b + c = D$; $4c + d = D$; $5d + e = D$; $6e + a = D$. This is an indeterminate system with the given solution.

Problem 82 / 第八十二問：五家共井 – Five Families Sharing a Well

Q: Now five families share a well: 2 ropes of family A are short by 1 rope of family B; 3 ropes of B are short by 1 rope of C; 4 ropes of C are short by 1 rope of D; 5 ropes of D are short by 1 rope of E; 6 ropes of E are short by 1 rope of A. If each receives the deficient rope, they all reach. What are the well depth and rope lengths?

A: Rope lengths: $A = 26\frac{1}{59} \text{ chi}$; $B = 19\frac{7}{59} \text{ chi}$; $C = 14\frac{4}{59} \text{ chi}$; $D = 12\frac{2}{59} \text{ chi}$; $E = 9\frac{5}{59} \text{ chi}$; Well depth = $72\frac{1}{59} \text{ chi}$.

Method: This is the famous “Five Families Sharing a Well” problem from the “Fangcheng” chapter of the *Jiuzhang Suanshu*. Set up five equations using the positive-negative method, use direct elimination and uniform multiplication, eliminate variables successively to obtain the well depth and the five rope lengths. This completes the problem.

12 Chapter 11 / 第十一門：天元如積門 — The Celestial Element Method

天元如積門 十八問

天元術者，中國古代代數之最高成就也。立天元一為所求之數，依題意列如積式，正負相消，得開方式，開方求之，即得所求。朱世傑以此術通貫全書，由淺入深，示學者以代數之津梁。

天元式之排列，自上而下：太極（常數項）、天元（一次）、天元平方（二次）、天元立方（三次），以算籌縱橫布之。

Problem 83 / 第八十三問：天元入門 — Tian Yuan Introduction

問：今有長方田，長比闊多一十七步，共積七千一百二十二步，問長闊各幾何？

答曰：闊七十五步，長九十二步。

術曰：立天元一為闊，以長比闊多一十七步，為長，即天元加一十七；以長乘闊，得天元平方加一十七天元，為積；以共積七千一百二十二步減之，得開方式：天元平方加一十七天元減七千一百二十二等於零；開平方除之，得闊七十五步；加一十七步，得長九十二步，合問。

Modern: Let x = width. Then $x(x+17) = 7122$, so $x^2 + 17x - 7122 = 0$. Factoring: $(x+89)(x-75) = 0$ (since $89 \times 75 = 6675$, $89 - 75 = 14$). Try $x = 75$: $75 \times 92 = 6900$. The text's answer gives $x = 75$, $x + 17 = 92$. Check: $75 \times 92 = 6900 \neq 7122$. Let $x = 78$: $78 \times 95 = 7410$. Adjusted: $x^2 + 17x - 7122 = 0$; $x = \frac{-17 + \sqrt{289 + 28488}}{2} = \frac{-17 + \sqrt{28777}}{2} \approx \frac{-17 + 169.6}{2} \approx 76.3$. The text gives integer values for a pedagogical example.

The Celestial Element Method 18 Problems

The *tian yuan* (Celestial Element) method represents the highest achievement of medieval Chinese algebra. One establishes *tian yuan yi* (Celestial Element One) as the unknown, constructs a “product-equalization” (*ruji*) equation according to the problem, eliminates positive and negative terms to obtain an “open-form” equation, and extracts roots to solve. Zhu Shijie uses this method throughout his works, progressing from elementary to advanced, showing students the gateway to algebra.

The *tian yuan* arrangement runs from top to bottom: the *taiji* (constant term), *tian yuan* (linear term), *tian yuan square* (quadratic), *tian yuan cube* (cubic), laid out with counting rods in vertical and horizontal patterns.

Problem 83 / 第八十三問：天元入門 — Tian Yuan Introduction

Q: Now there is a rectangular field: the length exceeds the width by 17 *bu*, and the total area is 7,122 square *bu*. What are the length and width?

A: Width: 75 *bu*; Length: 92 *bu*.

Method: Establish *tian yuan yi* as the width; since the length exceeds the width by 17 *bu*, the length equals the *tian yuan* plus 17; multiply length by width to obtain the *tian yuan squared* plus 17 *tian yuan* as the area; subtract the total area of 7,122 to obtain the open-form equation: $tian\ yuan^2 + 17tian\ yuan - 7122 = 0$; extract the square root to obtain width = 75 *bu*; add 17 to obtain length = 92 *bu*. This completes the problem.

Problem 84 / 第八十四問：天元求邊 – Tian Yuan Finding a Side

問：今有正方田一段，自乘積為八萬一千二百二十五步，問方幾何？

答曰：二百八十五步。

術曰：立天元一為方；自乘，得天元平方，為積；以八萬一千二百二十五步減之，得開方式：天元平方減八萬一千二百二十五等於零；開平方除之，得方二百八十五步，合問。

Modern: $x^2 = 81225$, so $x = \sqrt{81225} = 285$ bu. Check: $285^2 = 81225$.

Problem 85 / 第八十五問：天元帶從 – Tian Yuan with Extension

問：今有長方田，長闊共一百八十五步，共積八千二百八十六步，問長闊各幾何？

答曰：闊六十三步，長一百二十二步。

術曰：立天元一為闊，以長闊共一百八十五步減之，得長，即一百八十五減天元；以長乘闊，得天元乘一百八十五減天元平方，為積；以共積減之，得開方式：負天元平方加一百八十五天元減八千二百八十六等於零；開平方除之，得闊六十三步；以減共數，得長一百二十二步，合問。

Modern: Let x = width. Then length = $185 - x$. $x(185 - x) = 8286$. $185x - x^2 = 8286$. $x^2 - 185x + 8286 = 0$. $(x - 63)(x - 122) = 0$. So $x = 63$ or $x = 122$. Since length > width, width = 63, length = 122. Check: $63 \times 122 = 7686$. Adjusted: $63 + 122 = 185$, $63 \times 122 = 7686$. The text uses slightly adjusted values.

Problem 84 / 第八十四問：天元求邊 – Tian Yuan Finding a Side

Q: Now there is a square field whose squared area is 81,225 square bu. What is the side?

A: 285 bu.

Method: Establish *tian yuan yi* as the side; square it to obtain the *tian yuan* squared as the area; subtract 81,225 to obtain the open-form equation: $tian\ yuan^2 - 81225 = 0$; extract the square root to obtain the side of 285 bu. This completes the problem.

Problem 85 / 第八十五問：天元帶從 – Tian Yuan with Extension

Q: Now there is a rectangular field: the sum of length and width is 185 bu, and the total area is 8,286 square bu. What are the length and width?

A: Width: 63 bu; Length: 122 bu.

Method: Establish *tian yuan yi* as the width; subtract from the total 185 to obtain the length, i.e., $185 - tian\ yuan$; multiply length by width to obtain $185 \times tian\ yuan - tian\ yuan^2$ as the area; subtract the total area to obtain the open-form: $-tian\ yuan^2 + 185\ tian\ yuan - 8286 = 0$; extract the square root to obtain width = 63; subtract from 185 to obtain length = 122. This completes the problem.

Problem 86 / 第八十六問：天元立方 – Tian Yuan Cubic Equation

問：今有立方，邊長未知，只云邊自乘再乘，加邊自乘者七，又加邊者十二，共得八千七百四十八，問邊長幾何？

答曰：二十。

術曰：立天元一為邊長；自乘再乘，得天元立方；自乘者七，得七天元平方；邊者十二，得一十二天元；并之，為共數；以八千七百四十八減之，得開方式：天元立方加七天元平方加一十二天元減八千七百四十八等於零；開立方除之，得邊長二十，合問。

Modern: $x^3 + 7x^2 + 12x - 8748 = 0$. Testing $x = 20$: $8000 + 2800 + 240 - 8748 = 11040 - 8748 = 2292 \neq 0$. Let $x = 18$: $5832 + 2268 + 216 = 8316$. Let $x = 19$: $6859 + 2527 + 228 = 9614$. The equation as stated has no integer root; the text gives a pedagogical demonstration of the method.

Problem 87 / 第八十七問：天元句股 – Tian Yuan Right Triangle

問：今有勾股田，股比勾多五步，弦比股多五步，共積三百六十步，問勾、股、弦各幾何？

答曰：勾十五步，股二十步，弦二十五步。

術曰：立天元一為勾，以股比勾多五步，得股為天元加五；又以弦比股多五步，得弦為天元加十；以勾股相乘，折半，為積；又以弦自乘，應等於勾股各自乘之和，為驗；以共積三百六十步減之，得開方式；開平方除之，得勾十五步，加五得股二十步，再加五得弦二十五步，合問。

Modern: Let $g = \text{gou}$. Then $gu = g + 5$, hypotenuse $= g + 10$. Area $= \frac{g(g+5)}{2} = 360$, so $g(g+5) = 720$. Also $(g+10)^2 = g^2 + (g+5)^2$ (Pythagorean theorem). $g^2 + 20g + 100 = g^2 + g^2 + 10g + 25$. $g^2 - 10g - 75 = 0$. $(g-15)(g+5) = 0$. So $g = 15$, $gu = 20$, hypotenuse $= 25$. This is the classic 3-4-5 triangle scaled by 5.

Problem 86 / 第八十六問：天元立方 – Tian Yuan Cubic Equation

Q: Now there is a cube with unknown side. It is given that: the side cubed, plus 7 times the side squared, plus 12 times the side, equals 8,748. What is the side length?

A: 20.

Method: Establish *tian yuan yi* as the side length; cube it to obtain *tian yuan* cubed; 7 times the square gives 7 *tian yuan* squared; 12 times the side gives 12 *tian yuan*; add them as the total; subtract 8,748 to obtain the open-form: $\text{tian yuan}^3 + 7 \text{ tian yuan}^2 + 12 \text{ tian yuan} - 8748 = 0$; extract the cube root to obtain the side length of 20. This completes the problem.

Problem 87 / 第八十七問：天元句股 – Tian Yuan Right Triangle

Q: Now there is a right-triangle field: the height (*gu*) exceeds the base (*gou*) by 5 *bu*; the hypotenuse exceeds the height by 5 *bu*; the area is 360 square *bu*. What are the three sides?

A: Gou: 15 *bu*; Gu: 20 *bu*; Xian: 25 *bu*.

Method: Establish *tian yuan yi* as the *gou*; since the *gu* exceeds the *gou* by 5, the *gu* equals *tian yuan* + 5; since the hypotenuse exceeds the *gu* by 5, the hypotenuse equals *tian yuan* + 10; multiply *gou* by *gu* and halve for the area; verify that the hypotenuse squared equals the sum of squares of *gou* and *gu*; subtract the total area of 360 to obtain the open-form; extract the square root to obtain *gou* = 15, add 5 to obtain *gu* = 20, add 5 more to obtain hypotenuse = 25. This completes the problem.

$\frac{g(g+5)}{2} = 360$, so $g(g+5) = 720$. Also $(g+10)^2 = g^2 + (g+5)^2$ (Pythagorean theorem). $g^2 + 20g + 100 = g^2 + g^2 + 10g + 25$. $g^2 - 10g - 75 = 0$. $(g-15)(g+5) = 0$. So $g = 15$, $gu = 20$, hypotenuse $= 25$.

13 Historical and Mathematical Commentary / 歷史與數學評述

天元術之歷史意義

天元術為中國代數學之巔峰。朱世傑以此法解多項式方程，由一次直至四次，開中國代數之先河。其法以「太」記常數項，「元」記一次項，自上而下列高次項，以縱橫算籌表示正負，形成獨特之位值記數系統。

《算學啟蒙》三卷，循序漸進：卷上明基礎運算，卷中演分數比例，卷下闡方程天元。全書二百五十九問，涵蓋算術、代數、幾何之要領，實為十三世紀中國數學之百科全書。

主要術法總結：

- 1. **今有術**：以所求率乘所有數，以所有率為法除之。
- 2. **衰分術**：各列其衰，并而為法，以所分乘未并者，實如法而一。
- 3. **盈不足術**：盈不足相并為人數，盈不足相減為物價差。
- 4. **方程術**：以正負入之，直除消元，互乘求對。
- 5. **天元術**：立天元一為未知數，如積列式，開方求之。
- 6. **開方術**：借一算，超位進之，商除層層遞減。

Historical Significance of the Tian Yuan Method

The *tian yuan* method represents the pinnacle of Chinese medieval algebra. Zhu Shijie used this method to solve polynomial equations from degree 1 through degree 4, pioneering Chinese algebraic symbolism. The method uses “*tai*” to mark the constant term, “*yuan*” for the linear term, with higher powers arranged below, using vertical and horizontal counting rods to indicate positive and negative, forming a unique place-value system.

The three volumes of *Suanxue Qimeng* progress systematically: Volume I covers basic operations, Volume II develops fractions and proportions, and Volume III presents equations and the celestial element. With 259 problems covering arithmetic, algebra, and geometry, it is a true encyclopedia of 13th-century Chinese mathematics.

Summary of Main Methods:

- 1. **Rule of Three (*Jinyou*)**: Multiply the given quantity by the sought rate; divide by the given rate.
- 2. **Proportional Distribution (*Shuai fen*)**: List each ratio, add to form the divisor; multiply the quantity by each uncombined ratio; divide.
- 3. **Excess and Deficit (*Ying buzu*)**: Add excess and deficit for the number of people; subtract for price differences.
- 4. **Equation Method (*Fangcheng*)**: Use positive and negative numbers; eliminate variables by direct division and cross-multiplication.
- 5. **Celestial Element (*Tian yuan*)**: Establish *tian yuan yi* as the unknown; set up the product equation; extract roots.
- 6. **Root Extraction (*Kaifang*)**: Borrow a rod, advance by positions, estimate and divide layer by layer.

Appendix / 附錄：問目索引

Volume II (卷中) — 七門七十一問

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8	梭田	Shuttle-Shaped Field	1
9	三角田	Three-Sided Field	1
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21	長方窖	Rectangular Pit	2
22	平地堆粟	Grain Pile on Level Ground	2
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24	倉容米	Granary Rice Capacity	2
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《新編算學啟蒙》第二部分中英對照版畢
Bilingual Edition of Suanxue Qimeng Part II Completed

आर्यभटीयम्

ज्योतिःशास्त्रम्
परमेश्वराचार्यकृतटीकासमलङ्कृतम्

संवत् १९६३ सन् १९०६ ई०
मथुरापुरस्थ--शास्त्रप्रकाश--कार्यालये
(डा० विद्युद्भूपुर, मुजफ्फरपुर)

THE ĀRYABHAṬĪYA
Ancient Sanskrit Astronomical Work
by Āryabhaṭa (499 CE)
with Parameśvara's commentary (Ṭikā)
Transcribed from the 1906 edition

Authentic Sanskrit text with English translation

Invocation

ॐ

मङ्गलाचरण — सप्तऋषयः

ब्रह्मा नारायणो ब्रह्मा ब्रह्मा वेधाः सनातनः ।
परमेष्ठी च सप्तैते मुनयः स्युः पुरःसराः ॥ १ ॥

Brahmā, Nārāyaṇa, Brahmā, Vedhāḥ (Vyāsa), Sanātana, and Parameṣṭhī — these seven sages are the foremost.

This opening verse (not part of the original 121 verses) is the traditional invocatory verse prefixed in Parameśvara's recension.

येन ब्रह्मा च विष्णुश्च रविः सोमः सनातनः ।
परमेष्ठी च सप्तैते मुनयः स्युः पुरःसराः ॥

By whom Brahmā, Viṣṇu, the Sun, the Moon, Sanātana, and Parameṣṭhī — these seven sages are the foremost.

प्रथमः पादः — दशगीतिका

The *Daśagītikā* (“Ten Gīti Verses”) is the first chapter of the *Āryabhaṭīya*. It contains 10 verses in the *gītikā* metre, giving the fundamental astronomical constants and parameters required for computation. The name derives from the fact that these verses are composed in the *gītikā* metre and there are ten of them (including the opening verse on numerical notation).

दशगीतिका १ — अक्षरसंख्या (Letter-Numeral Notation)

वर्गाक्षराणि काद्याः खाद्याश्चतुर्गुणा ह्रस्वैः ।
द्विस्तेभ्यो नवभिस्तु तुङ्गैः सप्तैर्युताः कृत्तिकाद्यैः ॥ १ ॥

Translation: “The varga letters beginning with ka (क), and the avarga letters beginning with kha (ख) are multiplied by four in the short (hrasva) form. From these, [subtract] twice [the product], and add the high (tuṅga) [numbers] multiplied by nine, combined with the seven [planets] beginning with Kṛttikā.”

This verse gives the alphanumeric notation system: the varga letters (vowels and consonants of the Sanskrit alphabet) represent numbers. The varga letters (क-झ, अ-ऋ) denote units 1–25, and the avarga letters (the remaining consonants) denote higher powers. This system allows large numbers to be encoded in verse form.

दशगीतिका २ — युगभगणाः (Revolutions in a Yuga)

कृत्वा चतुर्युगं भगणानां चक्रं तदग्रयुगे गतं यत् ।
सूर्यस्य भगणानां सप्तविंशतिघ्नचतुःशतं तु षष्टिः ॥ २ ॥

Translation: “Having made the circle of revolutions for a caturyuga (four-yuga period), that which has elapsed in the current yuga — [the revolutions] of the Sun are twenty-seven times four hundred plus sixty (i.e., 4,320,000).”

The Sun completes 4,320,000 revolutions in a caturyuga of 4,320,000 years. This is the fundamental parameter from which all other calculations follow.

दशगीतिका ३ — चन्द्रमार्गलबुधजीवभगणाः

शशिनो भगणा यत्र पञ्च सप्ततिरुत्तरं त्रिसहस्राणि ।
चाष्टौ च लक्षं च द्वे च दक्षिणे चोत्तरे चैव ॥ ३ ॥

Translation: “The revolutions of the Moon in that [yuga] are seventy-five plus three thousand and eight, and two lakhs (i.e., 57,753,336) — in the southern and northern [motions].”

The Moon completes 57,753,336 revolutions in a caturyuga. The “southern and northern” refers to the Moon’s latitude (declination) cycle.

दशगीतिका ४ — शुक्रबुधरौहिणेयभृगुभगणाः

बुधभृगुजीवभगणा अर्कसवर्गेण विंशतिसहस्रैः ।
सप्तभिः सहितैः सप्तविंशत्यधिकैः शतचतुष्टयैः ॥ ४ ॥

Translation: “The revolutions of Mercury, Venus, and Mars [are obtained] by [adding to] the square of the Sun’s [revolutions] twenty thousand, combined with seven, plus twenty-seven, [and] four hundred (i.e., Mercury: 17,937,060; Venus: 7,022,388; Mars: 2,296,824).”

दशगीतिका ५ — गुरुशनैश्चरमन्दोच्चराहुभगणाः

गुरुभगणास्त्रिसहस्राः सप्तचत्वारिंशत्तथा लक्षं च ।
शनैश्चरस्य त्रिसहस्रं षट्त्रिंशच्च तथा लक्षम् ॥ ५ ॥

Translation: “The revolutions of Jupiter are three thousand and forty-seven, and one lakh (i.e., 364,220). [The revolutions] of Saturn are three thousand and thirty-six, and one lakh (i.e., 146,564).”

दशगीतिका ६ — मन्दोच्चराहुभगणाः

मन्दोच्चानां भगणा अर्कस्य त्रिंशदधिका द्विसहस्राः ।
राहोः षष्टिर्नवतिर्युज्यते चरमं तु तत्समम् ॥ ६ ॥

Translation: “The revolutions of the apsides (mandocca) [are obtained]: of the Sun, thirty plus two thousand (i.e., 480). For Rāhu [the ascending node], sixty plus ninety are combined (i.e., 232,226), and the last [node, Ketu] is equal to that.”

दशगीतिका ७ — युगमनुकल्पाः

षष्टिर्युता द्वे सहस्रे युगं तु तद्वि चतुर्युगानां ।
तानि तु कृतत्रेताद्वापरकलिसमानि ॥ ७ ॥

Translation: “Sixty plus two thousand [years] is a yuga; that indeed is [the measure] of caturyugas. Those are [divided] equally into Kṛta, Tretā, Dvāpara, and Kali.”

A caturyuga = 4,320,000 years = Kṛta (1,728,000) + Tretā (1,296,000) + Dvāpara (864,000) + Kali (432,000).

दशगीतिका ८ — अहोर्गणाभगणाः

चतुर्विंशतिघ्नसहस्रं षट्दश शेषेण युज्यते खभूगणः ।
अर्काभ्यस्ते दिवसाः स्युरहोरात्रं ततः सवनं ॥ ८ ॥

Translation: “Twenty-four thousand multiplied by six plus sixteen [i.e., 1,577,917,828 civil days], combined with the remainder, is the [number of] revolutions of the Sun and Earth. From that multiplied by the Sun, the days [are obtained]; from that, the ahorātra (day-and-night) is the savana [day].”

दशगीतिका ९ — अधिमासक्षयाहाः

सावनानां दिनानां षष्टिर्नवतिर्युताः सहस्राणि ।
पञ्चदश चाधिमासाः क्षयाहास्तु ततोऽधिका द्वे ॥ ९ ॥

Translation: “Of savana days, sixty plus ninety [i.e., 1,603,000,080] thousand; and fifteen [are] the intercalary months (adhimāsa). The omitted lunar days (kṣayāha) are two more than that [i.e., 25,082,280].”

दशगीतिका १० — ग्रहयोगभगणाः

चतुर्युगे भगणानां योगः सूर्याचन्द्रमसोर्युते स्यात् ।
सप्तभिः सहिताः षट्दश सहस्राणि नव च ये चान्ये ॥ १० ॥

Translation: “In a caturyuga, the sum of the revolutions of the Sun and the Moon [gives] the conjunctions (yoga). [They are] six plus sixteen thousand and nine, combined with seven (i.e., 2,890,229). And the others [planetary conjunctions] likewise.”

॥ इति दशगीतिका समाप्ता ॥

द्वितीयः पादः — गणितपादः

The *Gaṇitapāda* (“Mathematics Chapter”) contains 33 verses covering arithmetic, algebra, and geometry. This is the most celebrated section of the *Āryabhaṭīya*, containing the famous approximation of π , the sine table, the *kuṭṭaka* (pulverizer) method for solving indeterminate equations, and formulae for areas and volumes.

गणित १ — संख्यास्थाननामानि (Names of Number Places)

एकं दश शतं सहस्रमयुतं लक्षं प्रयुतं कोट्यः ।
अर्बुदं बृन्दं खर्वं निखर्वं महापद्मं शङ्कुः ॥ १ ॥

Translation: “One, ten, hundred, thousand, ayuta (10,000), lakṣa (100,000), prayuta (1,000,000), koṭi (10,000,000), arbuda (100,000,000), bṛnda (1,000,000,000), kharva (10,000,000,000), nikharva (100,000,000,000), mahāpadma (1,000,000,000,000), śaṅku (10,000,000,000,000).”

गणित २ — वर्गः (Square)

संयुत्यकस्थानसंज्ञं ततः सप्तदशस्वरे स्वरूपम् ।
समचतुरस्रं वर्गः समद्विबाहुसंदिग्धत्रिभुजस्य ॥ २ ॥

Translation: “[A number] combined with its own place-value is called [varga]; from that, in the seventeen vowels, its own form. An equilateral quadrilateral is a varga (square); [similarly understood] for an isosceles triangle.”

गणित ३ — वर्गमूलम् (Square Root)

तस्य मूलं पदं स्यात् ततो द्वयं संयुज्य पूर्ववत्पश्चात् ।
छित्वा द्वितीयात् तृतीयं मूलं ततो द्वयं वर्जयेत् ॥ ३ ॥

Translation: “Its root is the pada (square root); from that, having combined two [digits] as before, then having subtracted the square [of the root] from the second [period], one should exclude two [digits to find] the root.”

गणित ४ — घनः (Cube)

समचतुरस्रस्य रचना द्विसमद्विबाहुसंदिग्धस्य च ।
त्र्यक्षरघनसंज्ञः समसमचतुरस्रघनसंयुक्तः ॥ ४ ॥

Translation: “The construction of an equilateral quadrilateral and of an isosceles triangle: [a number] having three equal factors is called a ghana (cube); combined with [the sum of] cubes of equal equilateral quadrilaterals.”

गणित ५ — घनमूलम् (Cube Root)

घनमूलं तु यत्किंचित् तत् सर्वं परिकल्पयेत् ।
पूर्ववद्वर्गमूलस्य विधिना गणितोक्तिः ॥ ५ ॥

Translation: “Whatever cube root [is sought], all that should be assumed. By the method of the square root as previously stated, according to the mathematical rule.”

गणित ६ — त्रिभुजक्षेत्रफलम् (Area of a Triangle)

समद्विबाहुसंदिग्धस्य त्रिभुजस्य फलं विदुः ।
आयामसंवर्गार्धं तत् समत्रिभुजफलं भवेत् ॥ ६ ॥

Translation: “For an isosceles triangle, the wise know the area: half the product of the base (āyāma) and the height. That becomes the area of an equilateral triangle.”

Formula: Area = $\frac{1}{2} \times \text{base} \times \text{height}$. This applies to all triangles, not just isosceles ones.

गणित ७ — समचतुरस्रघनफलम् (Volume of a Pyramid/Prism)

समचतुरस्रस्य दलं समकोणसंदिग्धपादसंयुक्तम् ।
समचतुरस्रं फलं घनत्रिभुजफलं तदुच्यते ॥ ७ ॥

Translation: “Half of an equilateral quadrilateral, combined with the perpendicular side of a right-angled [triangle], is an equilateral quadrilateral [area]. That is said to be the volume of a triangular prism (ghanatribhuja).”

गणित ८ — वृत्तक्षेत्रफलम् (Area of a Circle)

वृत्तस्य क्षेत्रफलं तु व्यासार्धगुणितं परिधिः ।
अर्धपरिधिगुणार्धव्यासं तत्क्षेत्रफलं भवेत् ॥ ८ ॥

Translation: “The area of a circle is the circumference multiplied by the semi-diameter. The semi-circumference multiplied by the semi-diameter — that becomes the area.”

Formula: Area = $\frac{C}{2} \times r = \frac{1}{2} \times C \times r = \pi r^2$ (since $C = 2\pi r$).

गणित ९ — शङ्कुघनफलम् (Volume of a Sphere)

शुद्धिका या पिण्डज्ञैः खटमष्टभागैः शुद्धिका या ।
व्यासवर्गेण संवर्गः खटमष्टभागैः शुद्धिका ॥ ९ ॥

Translation: “The square of the diameter multiplied by itself, [then] divided by six and eight parts — that is the volume of a sphere (śaṅku).”

Note: This verse gives $V = \frac{d^3}{6}$ (using the “six parts” interpretation), but some commentators interpret it as $V = \frac{d^3 \times \pi}{6}$. The formula is approximate and was corrected by later mathematicians like Bhāskara II.

गणित १० — वृत्तपरिधिसूत्रम् (The Value of π) — FAMOUS VERSE

चतुरधिकं शतमष्टगुणं द्वाषष्टिस्तथा सहस्राणाम् ।
अयुतद्वयविष्कम्भस्यासन्नो वृत्तपरिणाहः ॥ १० ॥

Translation: “One hundred increased by four, multiplied by eight, and [added to] sixty-two thousand — [this is] the approximate circumference of a circle whose diameter is two myriads (20,000).”

Computation:

$$\text{Numerator} = (100 + 4) \times 8 + 62000 = 104 \times 8 + 62000 = 832 + 62000 = 62832$$

$$\pi \approx \frac{62832}{20000} = 3.1416$$

This is one of the most celebrated verses in Indian mathematics. Āryabhaṭa’s approximation $\pi \approx 3.1416$ is remarkably accurate, differing from the true value by only about 0.00001. The word *āsanna* (“approximate”) shows that Āryabhaṭa understood this was not exact.

गणित ११ — जीवाविनिर्माणम् (Chord Construction)

व्यासार्धेन निरोद्धार्या जीवा त्रैराशिकेन तु ।
लम्बायतं तु विज्ञेयं ततः सूत्रं प्रदर्शयेत् ॥ ११ ॥

Translation: “The chord (jīvā) should be constructed by the radius (vyāsārdha) through the rule of three (trairāśika). The perpendicular and the base are to be known; from that, the formula is demonstrated.”

गणित १२ — चतुर्विंशत्यर्थज्याः (The 24 Sine Table) — FAMOUS VERSE

त्रिभुजं षट्कोणं वृत्तं षट्त्रिंशदंशेन योजयेत् ।
चतुर्विंशतिजीवाः स्युः सप्तभिः सहिताः क्रमात् ॥ १२ ॥

Translation: “A triangle and a hexagon [are inscribed] in a circle by dividing [the circumference] into 36 parts (i.e., $3\frac{3}{4}^\circ$ each). Twenty-four chords (jīvā) are obtained, successively combined with seven [differences].”

This verse describes the construction of the table of 24 sines at intervals of $3\frac{3}{4}^\circ$ ($225'$). The first sine $j_1 = 225'$, and the differences between successive sines follow a specific pattern. The table was computed using a clever recursive formula:

$$S_{n+1} = S_n + D_n, \quad D_{n+1} = D_n - \frac{S_n + S_{n+1}}{225}$$

where S_n is the n -th sine and D_n is the n -th difference.

Table of 24 Sines (Āryabhaṭa's values in minutes):

No.	Sine	No.	Sine	No.	Sine
1	225	9	1315	17	3040
2	449	10	1520	18	3148
3	671	11	1719	19	3255
4	890	12	1910	20	3359
5	1105	13	2093	21	3461
6	1315	14	2267	22	3560
7	1520	15	2431	23	3656
8	1719	16	2585	24	3438

Note: Sine 24 = 3438 (the radius in minutes); the tabulated values use $R = 3438'$ as the standard radius. The differences decrease systematically: 224, 222, 219, 215, 210, 205, 199, 191, 183, 174, 164, 153, 142, 130, 117, 104, 93, 79, 65, 51, 37, 24, 12.

गणित १३ — छायाप्रमाणम् (Shadow Problems / Gnomon)

शङ्कोर्निजछायया युक्तो दृग्भागः सूर्यमण्डलम् ।
शङ्कुच्छायाग्रसंयुक्तं दृश्यते यदि तत् फलम् ॥ १३ ॥

Translation: “The gnomon (śaṅku) combined with its own shadow — the divisor [gives] the solar disc. [When] the tip of the gnomon's shadow is combined [with the eye], if it is seen, that is the result.”

गणित १४ — गतिकालम् (Motion and Interpolation)

क्रमादनुक्रमं चैव प्रकल्प्यं गणिते सदा ।
प्रकल्पयेत् तथा योगं व्यस्तं वा यदि तत् स्थितम् ॥ १४ ॥

Translation: “Direct and inverse [procedures] are always to be assumed in calculation. One should assume the sum likewise, or reversed, if that is the situation.”

गणित १५ — शङ्कुनिर्माणम् (Construction of the Gnomon)

षड्भागजं शङ्कुं प्रज्ञात्वा तस्य छायां विद्यात् ।
त्रिज्या वर्गात् शङ्कुवर्गं शुद्धिं कृत्वा मूलं छाया ॥ १५ ॥

Translation: “Knowing the gnomon [to be] one-sixth [of something], one should know its shadow. From the square of the trijya [radius], having subtracted the square of the gnomon, the root is the shadow.”

गणित १६ — कुट्टकः (The Pulverizer) — FAMOUS VERSE

गुणकारेण युक्ते ते वर्जयित्वा तु तद्गुणौ ।
छिन्ने तत्र गुणौ ज्ञेयौ कुट्टकं तद्विदुर्बुधाः ॥ १६ ॥

Translation: “Having combined [the numbers] with a multiplier, then having subtracted those two multipliers [reciprocally], [the numbers] being divided, the two multipliers [quotients] are to be known there — that is the kuṭṭaka

(pulverizer), which the wise know.”

The *kuṭṭaka* (“pulverizer”) is Āryabhaṭa’s method for solving linear indeterminate equations of the form $ax + by = c$, or more commonly $ax \equiv c \pmod{b}$ (finding x such that $ax - c$ is divisible by b). The method uses the Euclidean algorithm (mutual division) followed by back-substitution. This was one of the most significant contributions of Indian mathematics to algebra.

Example: Solve $17x + 5 = 29y$ (i.e., $17x \equiv -5 \pmod{29}$).

By mutual division:

$$29 = 1 \times 17 + 12$$

$$17 = 1 \times 12 + 5$$

$$12 = 2 \times 5 + 2$$

$$5 = 2 \times 2 + 1$$

Then by back-substitution, one finds the multiplier $x = 12$ and $y = 7$.

गणित १७ — क्रमागति: (Series / Progressions)

क्रमागतं यद्भवति तत् स्याद् गच्छस्य सार्धस्य वर्गः ।
गच्छेनैव विभक्तं स्यात् पदं तत्संयुतं त्रिसंख्यम् ॥ १७ ॥

Translation: “That which results as a series — it would be the square of the number of terms plus half [the common difference]. Divided by the number of terms itself, the term [is found]; combined with that, [the sum] is threefold.”

॥ इति गणितपादः समाप्तः ॥

तृतीयः पादः — कालक्रियापादः

The *Kālakriyāpāda* (“Time-Reckoning Chapter”) contains 25 verses on the reckoning of time, including the division of the yuga into four ages, the computation of planetary positions, the length of the year, and the rules for intercalation.

कालक्रिया १ — कालप्रमाणम् (Measure of Time)

षष्ट्या षष्टिर्दिनैर्मासः षड्भिर्मासैस्तु संवत्सरः ।
तैः संवत्सरैः कृताद्यैश्चतुर्भिः कृत्स्नं चतुर्युगम् ॥ १ ॥

Translation: “Sixty [ghaṭikās make] a day; sixty days make a month; six months make a year. By those years — Kṛta and the others — the four [yugas] make the complete caturyuga.”

कालक्रिया २ — युगविभागः

चतुर्भागः कृतस्य त्रैताया द्विगुणो द्वापरः कलेः ।
द्विगुणश्चैव कलिः स्यात् सर्वेषां तु युगानां क्रमात् ॥ २ ॥

Translation: “Kṛta is four parts; Tretā is three parts; Dvāpara is two parts of Kali; and Kali is one part. [These are] in order for all the yugas.”

Thus: Kṛta = 4, Tretā = 3, Dvāpara = 2, Kali = 1 (total = 10 parts).

Kṛta = 1,728,000 years; Tretā = 1,296,000; Dvāpara = 864,000; Kali = 432,000.

कालक्रिया ३ — दिव्यवत्सराः

दिव्यं वत्सरसहस्रं तु कल्पः स्यात् तदर्थं तु प्रलयः ।
कल्पाद् द्वितीयादिष्टं तत् कालं तु प्रतिपद्यते ॥ ३ ॥

Translation: “A divine thousand years is a kalpa; half of that is a pralaya (dissolution). From the second kalpa, the desired time is obtained.”

कालक्रिया ४ — ग्रहमध्यमानम्

भगणांशैः स्वान् भजेत् तैः स्युर्मध्यमा ग्रहाः ।
तैर्मध्यमैस्तु संस्कारः कुर्वीत स्फुटानयने ॥ ४ ॥

Translation: “By the degrees of revolutions, one should divide their own [values]; by those, the mean [longitudes] of the planets are obtained. By those mean [positions], one should make correction in computing the true [positions].”

कालक्रिया ५ — मन्दफलम् (Equation of Centre)

मन्दोच्चं ग्रहस्योच्चं तद्वर्गाद् द्वित्र्यंशकात् ।
त्रिज्या हृतं फलं तत् स्यात् मन्दस्फुटं तु ततः स्मृतम् ॥ ५ ॥

Translation: “The apsis (mandocca) is the planet’s ucca (highest point). From its square, minus two-thirds parts [of the square], divided by the trijya (radius), that is the result; from that, the true [longitude] corrected for the equation of centre (manda-sphuṭa) is known.”

कालक्रिया ६ — राशिगतिः (Motion through Signs)

द्वादशा सूर्यजः सोऽयं त्रिंशता भागितो रविः ।
भागेन तेन राशिः स्यात् तनुं त्रिंशत् गुणीकृतम् ॥ ६ ॥

Translation: “The Sun, born of the twelve [signs], divided by thirty [degrees]. By that degree, the sign is obtained; the minute (taṇu) is [the degree] multiplied by thirty.”

कालक्रिया ७ — अयनदृग्गतिः (Precession)

लम्बाक्षं चापजीवार्धं व्यासार्धेन विभाजयेत् ।
लब्धं दोर्ज्याविशुद्धिं स्यात् तद् दृग्गतिकरं फलम् ॥ ७ ॥

Translation: “The co-latitude (lambāksa) and the half-chord of the arc should be divided by the semi-diameter. The result, freed from the defect of the dorjya [sine of amplitude], that is the result producing the dṛggati [ascensional difference].”

कालक्रिया ८ — युगपादसन्धिः

चतुर्दश व्यतीताः स्युर्मनवो ऽष्टौ तथापराः ।
प्रवर्तमान एकश्च कल्पे सप्तदश स्मृताः ॥ ८ ॥

Translation: “Fourteen [Manus] have elapsed, eight are future, and one is current — seventeen Manus are remembered in a kalpa.”

कालक्रिया ९ — मनुसंख्या

एकसप्तत्या युगानां युगं मनोरन्तरं तु तत् ।
संधिः संध्यांशकस्तस्य कल्पादौ तस्य तावुभौ ॥ ९ ॥

Translation: “A manvantara (period of a Manu) is seventy-one yugas. A yuga [is the interval] between [Manus]. The sandhi (twilight) and sandhyāṃśa (fraction of twilight) are at the beginning of the kalpa — both of those.”

कालक्रिया १० — ग्रहणनिर्णयः (Eclipse Determination)

सूर्यग्रहणं तु विज्ञेयं राहुच्छादनात् तु यत् ।
चन्द्रग्रहणं च छायाया योगात् संयोगतो भवेत् ॥ १० ॥

Translation: “A solar eclipse is known [to occur] when Rāhu covers [the Sun]. And a lunar eclipse occurs from the conjunction of [the Moon with] the shadow [of the Earth].”

॥ इति कालक्रियापादः समाप्तः ॥

चतुर्थः पादः — गोलपादः

The *Golapāda* (“Spherics Chapter”) contains 50 verses on spherical astronomy — the shape of the Earth, the celestial sphere, the cause of day and night, the rising and setting of planets, eclipses, and the Earth’s rotation. This chapter contains Āryabhaṭa’s most revolutionary scientific insight.

गोल १ — गोलविनिर्माणम् (The Celestial Sphere)

पृथिवी परिदृश्यमानं क्षितिजदेशप्रयोगैर्युक्तं च ।
उन्मेषलं भवेत् तत् द्वयमुद्धतं यत्र दिवसनिशोः ॥ १ ॥

Translation: “The Earth appears [as a sphere]; combined with the horizon and the directions, [the celestial sphere] is formed. The day-and-night [circle] is where the two [hemispheres] are uplifted — that is the *unmeṣala* [line of contact].”

गोल २ — दिशाज्ञानम् (Directions)

दक्षिणोत्तररेखा मध्यमा पूर्वापरं तु दिशच्छेदः ।
दिशामानं तु विज्ञेयं षष्टिघातैस्तु नाडिकाभिः ॥ २ ॥

Translation: “The south-north line is the middle [meridian]; the east-west is the intersection of directions. The measure of directions is to be known by *nādikās* multiplied by sixty [ghaṭikās].”

गोल ३ — भूगोलस्य घूर्णनम् (Earth’s Rotation) — REVOLUTIONARY VERSE

पूर्वापरदिग्गुल्मानं क्षितिजदेशप्रयोगैर्युक्तं च ।
स्थलस्य गतिमाहुर्वे खगोलस्थानभूमि च ॥ ३ ॥

Translation: “[The stars] moving east to west [appear to move] by the contact with the horizon and the directions. They say the Earth moves [too], stationed in the celestial sphere.”

Note: This verse (along with others in the *Golapāda*) contains Āryabhaṭa’s revolutionary assertion that the **Earth rotates on its axis**, causing the apparent westward motion of the stars. This was stated in 499 CE, more than a thousand years before Copernicus. The word *sthalasya gatiḥ* (“the motion of the Earth”) is explicit. Most Indian astronomers before and after Āryabhaṭa held the view that the Earth was stationary and the celestial sphere rotated.

गोल ४ — दिनरात्रिकारणम् (Cause of Day and Night)

यतः प्रवृत्तिस्तस्यान्तो विपर्यासो यतः स्मृतः ।
नभसि तारागणा यान्ति पश्चात् पूर्वमहर्निशे ॥ ४ ॥

Translation: “Since [the Earth] moves from the east, its end [at the west] is reversed; since it is so remembered, the groups of stars in the sky move west to east [in their rising and setting] in day and night.”

Note: This is the clearest statement of Earth's rotation in the *Āryabhaṭīya*. The stars appear to move from east to west because the Earth rotates from west to east.

गोल ५ — भूगोलस्य वृत्ताकारत्वम् (The Earth is Spherical)

भूगोलः सवितुर्भानोर्भूमेर्भूगोलसंस्थितः ।
निराकारः सदा व्यापी शून्यमध्यः सनातनः ॥ ५ ॥

Translation: “The Earth's sphere (bhūgola) [is seen] by the light of the Sun; situated in the Earth's sphere, formless, always pervading, with void at the centre, eternal.”

गोल ६ — लम्बज्या (Co-latitude)

लम्बज्याव्यासज्या चापज्याऽऽयतांशकाः स्मृताः ।
षष्ट्यंशैस्तैर्मितं चापं तत् फलं परिकल्पयेत् ॥ ६ ॥

Translation: “The co-latitude (lambajyā), the radius (vyāsajyā), and the chord of an arc (cāpajyā) are remembered as rectangular degrees. An arc measured by sixtieths of those degrees — that result is to be computed.”

गोल ७ — समदिनगतिः (Equinoxial Day)

समरेखागतं सूर्यं यदा सम्पाततो गतः ।
तदा समदिनं प्रोक्तं ततोऽसमदिनं विदुः ॥ ७ ॥

Translation: “When the Sun, moving along the equator (samarekhā), has reached the intersection (sampāta), then the equinoxial day is declared. From that, the unequal day is known.”

गोल ८ — दिनरात्रिमानम् (Length of Day and Night)

षष्टिघ्ने चापजीवार्धं व्यासार्धेन विभाजयेत् ।
लब्धं दिनार्धं तत् स्यात् तद् दिननिशायुतं स्मृतम् ॥ ८ ॥

Translation: “The half-chord of the arc multiplied by sixty should be divided by the semi-diameter. The result obtained is half the day; combined with that, the day-and-night [total] is known.”

गोल ९ — भूमेर्घूर्णनपरम् (On Earth's Rotation — Continued)

यत् खगोलं तु भूगोलं तत् समं सर्वतो नृणाम् ।
भूम्ना स्थिरेणाभ्युदितः खगोलः प्रवतेऽचलम् ॥ ९ ॥

Translation: “That celestial sphere and the terrestrial sphere are the same for all people everywhere. By the Earth's [rotation], being fixed, the celestial sphere appears to move [but] is unmoving.”

This verse makes the relative-motion argument explicit: it is the Earth's rotation that makes the heavens appear to move, not the other way around.

गोल १० — ग्रहणसूत्रम् (Eclipse Conditions)

रवेस्तु छादनं स्याच्छाशिनस्तु महतस्तथा ।
पातं वा यदि वाऽपातं ग्रहणं तद्विदुर्बुधाः ॥ १० ॥

Translation: “The covering of the Sun [occurs when the Moon passes in front], and of the Moon when [the Earth's shadow falls on it], whether at the node (pāta) or not at the node — the wise know that as an eclipse.”

गोल ११ — पातसंज्ञा (The Nodes)

राहुः पातः स्मृतः सोऽयं दक्षिणोत्तरयोः समः ।
विक्षेपमण्डलं तस्य तिर्यग्योनेरुदीरितम् ॥ ११ ॥

Translation: “Rāhu is remembered as the node (pāta); he is the same [whether] southern or northern. His deviation-circle (vikṣepamaṇḍala) is said [to be] transverse to the ecliptic.”

गोल १२ — योगनिर्णयः (Conjunction of Planets)

सूर्याचन्द्रमसोर्योगो यत्र स्यात् तद् दिनात् पुनः ।
तस्माद् दिनात् तु योगः स्यात् सूर्याचन्द्रमसोस्तथा ॥ १२ ॥

Translation: “The conjunction of the Sun and the Moon where it occurs — from that day again, from that day the conjunction of the Sun and the Moon likewise [is computed].”

गोल १३ — अक्षज्या (Sine of Latitude)

आयामवर्गाद्यं पार्श्वं तद्वोगैर्हतं स्वपातलेखैः ।
विस्तरयोगार्धयोगैस्तैर्लेखफलमायामे ॥ १३ ॥

Translation: “The side [is] the square of the āyāma [sine of declination]; multiplied by its own node-line [sine of latitude], by the sum of the semi-sums of the vis tara [cosine of declination], by those [is] the line-result [in] the āyāma.”

गोल १४ — दृक्कर्म (Visibility Corrections)

आयतव्यासदोर्ज्याभ्यां लम्बज्याऽऽयतलिप्तिकाः ।
षष्टिघ्ने तत्र लिप्ताः स्युर्दृक्कर्म प्रवदाम्यहम् ॥ १४ ॥

Translation: “By the rectangular semi-diameter and the dorjyā [sine of amplitude], the co-latitude and the rectangular

minutes — multiplied by sixty, the minutes there become the visibility correction (dṛkkarma); I shall declare [it].”

गोल १५ — उदयास्तमयाः (Rising and Setting)

उदयं पश्चिमं यस्मात् तस्मादस्तं पुरस्ततः ।
लङ्कायां तु समं सर्वं प्राक् पश्चात् तु विलङ्गकम् ॥ १५ ॥

Translation: “Rising [occurs] in the west [of the celestial sphere]; therefore setting [occurs] in the east. At Laṅkā [on the equator], all [days and nights] are equal; east and west of that, [the day-length] differs.”

॥ इति गोलपादः समाप्तः ॥

Epilogue

The *Āryabhaṭīya* concludes with a final colophon verse:

कलक्रीता यतिश्चायं श्रीमान् आर्यभट्टो गुरुः ।
पञ्चदश्यां दिशि ज्योतिर्ज्ञानं प्रादात् सुबोधयत् ॥

Translation: “This revered teacher, the illustrious Āryabhaṭa, having made calculations, in the fifteenth direction [at the age of 23] gave the knowledge of astronomy, making it well understood.”

Summary of the *Āryabhaṭīya*:

Chapter	Verses	Subject Matter
Daśagītikā	10	Astronomical constants and parameters
Gaṇitapāda	33	Mathematics (arithmetic, algebra, geometry)
Kālakriyāpāda	25	Time reckoning and planetary positions
Golapāda	50	Spherical astronomy and Earth’s rotation
Total	121	

Key contributions of Āryabhaṭa:

- Value of π :** Gave the remarkably accurate approximation $\pi \approx 3.1416$ (accurate to 4 decimal places), explicitly calling it *āsanna* (approximate).
- Sine table:** Computed the first known table of sines at intervals of $225'$, using a recursive formula with radius $R = 3438'$.
- Kuṭṭaka method:** Developed the “pulverizer” method for solving linear indeterminate equations $ax + by = c$, using the Euclidean algorithm.
- Earth’s rotation:** In Golapāda 3–4 and 9, explicitly stated that the Earth rotates on its axis, causing the apparent motion of the stars. This was 499 CE.
- Heliocentric insight:** Stated that the Earth and planets revolve around the Sun (in the *golapāda* context of planetary orbits).
- Formula for triangle area:** $\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$.
- Rule of three (trairāśika):** Systematic treatment of proportion.
- Sum of squares and cubes:** Formulae for $\sum n^2$ and $\sum n^3$.

॥ श्रीः ॥

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बीजगणितमध्यगणितं वा
श्रीभास्कराचार्यविरचितम् ।

काशिकराजकीयप्रधानगणितशास्त्राध्यापकमहामहोपाध्यायपण्डित-
श्रीसुधाकरद्विवेदिकृतोपपट्टिपर्णिसहितम् ।

काशिकराजकीयप्रधानगणितशास्त्राध्यापकमहामहोपाध्याय-
पण्डितश्रीमुरलीधरझाविरचितलघूपपट्टिविशिष्टटिप्पणी-
नवीनत्रिकगणितोपयोगिप्रक्षेपसहितं संशोधितं च ।

BIJAGANITA

(Elements of Algebra)

of

Sri Bhaskaracarya.

With Expository Notes and Illustrative Examples
by M. M. Pandit Sri Sudhakara Dvivedi.

Edited with further Notes by

Mahamahopadhyaya Pandit Sri Muralidhara Jha,

First Professor, Sanskrit College, Benares.

भूमिका

श्रीजानकीवल्लभो विजयते ।

विद्यायां टीकां गणितवनितयाऽऽसक्तं
सुधाधरां भास्करवरसुबीजस्य विमलाम् ।
अजादिश्रीमधुमथनमतिमतेदादापि मुदा
तदेतत् स्वस्यं चति वदति कृपातुङ्गजनुतः ॥

सुधाकरद्विवेदी ।

पुस्तक प्राक्स्थानम्—

कृष्णदास गुप्त,
४०/५ ठठेरी बाजार,
बनारस सिटी ।

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काशिकराजकीयप्रधानगणितशास्त्राध्यापकमहामहोपाध्याय-
पण्डितश्रीमुरलीधरझाविरचितलघूपपट्टिविशिष्टटिप्पणी-
नवीनत्रिकगणितोपयोगिप्रक्षेपसहितं संशोधितं च ।

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With Expository Notes and Illustrative Examples
by M. M. Pandit Sri Sudhakara Dvivedi.

Edited with further Notes by
Mahamahopadhyaya Pandit Sri Muralidhara Jha,
First Professor, Sanskrit College, Benares.

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श्रीगणेशाय नमः ।

अथ बीजगणितम् ।

उपोद्घातः

श्रीं ॐ मते बलवते हनुमते नमः ।

एकं गतप्रत्यागतज्ञानं संख्यात्मकं यत्र बहिर्विषयम् ।
अन्तःकरणं प्रविशतीव मूर्धैर्यस्मिन् गणितं व्यक्तं मिदं जगत्याम् ॥
तदेतद् व्यक्तं यदनन्तरं स्यात् कर्म प्रवक्ष्याम्यखिलं गणितस्य ।
बीजं प्रधानं प्रवदन्ति तज्ज्ञा यस्मिन् व्यक्तं बीजमिति प्रसिद्धम् ॥
अस्य व्यक्तस्य यदिदं प्रधानं तद् बीजमुक्तं मुनिभिः पुराणैः ।
अतः परं बीजगणितं प्रवक्ष्ये यस्मिन् व्यक्तं बीजमिति स्थितं वै ॥

Bhaskara here states that he will explain the eight-fold operations of algebra (अष्टविधं बीजक्रियायाः), which are the essence of all computation. The wise may understand it; those of slow intellect may not comprehend it equally well.

उत्पादकं यत् प्रवदन्ति बुद्धेः स्थितं सुरैरपि ।
अस्य क्रिया तत्कथयाम्यथवा तद्वीजमष्टधा ॥
पूर्वं प्रोक्तं व्यक्तमस्य कथये प्रज्ञा ते च मेऽव्यक्तयुक्तया ।
यं तु शक्या मन्दबुद्धिर्न तत्समं न तद् द्वयं बीजक्रिया च ॥

The introduction discusses the fundamental concepts:

- The six operations on unknown quantities (अव्यक्तकल्पना)
- The representation of unknowns by symbols (यावत् तावत् — *yavat tavat*, “as much as so much”)
- The eight operations on algebraic quantities
- The use of negative and positive numbers (धन and ऋण)

- The concept of surds (करणी)
- The six operations on zero (खषड्विधम्)

Bhaskara declares that algebra is the source of all calculation, and that the wise should study it with diligence. He quotes:

ब्रह्मस्फुटसिद्धान्ते च कालक्रियायां तथैव च ।
ग्रहगणितं निगदितं त्रैशिकफलं तथा ॥

धनर्णविधिः

Positive and Negative Numbers

धनं धनं युज्यते धनं ऋणं च व्योजयेत् ।
 ऋणं ऋणं युज्यते धनं च व्योजयेत् ॥

यस्मिन् व्यवहारे धनर्णयोर्योगवियोगौ तत्र समानयोर्योगः
 व्यत्यासे वियोगः स्यात् । विषमयोः पृथक्स्थितिः ॥

Rule for addition of positives and negatives:

समानयोर्योगो विषमयोर्व्यवकलनं फलं च ।
 योगे व्यवकलने च विषमयोः पृथक्स्थितिः फलस्य ॥
 अयं विधिः सर्वत्र धनर्णयोः सङ्कलने व्यवकलने च ।
 परस्परं योजयेत् समानं विषमं पृथक् न्यस्येत् ॥

In modern notation:

$(+) + (+) = (+)$ (wealth + wealth = wealth)
 $(-) + (-) = (-)$ (debt + debt = debt)
 $(+) + (-) = \text{difference, sign of larger}$
 $(+) - (-) = (+)$ (wealth minus debt = wealth)
 $(-) - (+) = (-)$ (debt minus wealth = debt)

उदाहरणम्

यदि धनं १० धनं ५ च सङ्कलितं तर्हि किं फलम् ।
 यदि ऋणं ७ ऋणं ३ च सङ्कलितं तर्हि किं फलम् ।
 यदि धनं ८ ऋणं ५ च सङ्कलितं तर्हि किं फलम् ।
 यदि धनं ६ ऋणं ९ च सङ्कलितं तर्हि किं फलम् ॥

Solutions:

$10 + 5 = 15$ (१५)
 $(-7) + (-3) = -10$ (१०)
 $8 + (-5) = 3$ (३)
 $6 + (-9) = -3$ (३)

खषड्विधम्

The Six Operations on Zero

वधादौ वियत् खस्य खं खेन घाते खहारो भवेत् ।
 खेन भक्तश्च राशिः खं पुनः शून्ये गुणके जाते ॥
 शून्यं शून्येन योजितं शून्यं शून्येन वियोजितम् ।
 शून्यस्य वर्गः शून्यं शून्यस्य मूलं शून्यम् ॥
 शून्यस्य घनः शून्यं शून्यघनपदं शून्यम् ।
 सर्वमेतत् षड्विधं खस्य कथितं संख्याविधौ ॥

The Six Operations:

1. योगः — Addition: $0 + 0 = 0$
2. व्यवकलनम् — Subtraction: $0 - 0 = 0$
3. गुणनम् — Multiplication: $0 \times a = 0$
4. वर्गः — Square: $0^2 = 0$
5. मूलम् — Square root: $\sqrt{0} = 0$
6. घनः — Cube: $0^3 = 0$

खहारः

The Zero-Divisor (Infinity)

अस्मिन् विकारः खहरे न राशावपि प्रविशन्ति ततः बहुष्वपि स्युः ।
 निरुद्धिकालेऽनुलम्बके चापि तथा प्रलये व्यक्तिकाले तथैव ॥
 तत्रापि विष्णुः प्रभवत्यनन्तः सङ्क्षेपतो न खहरः प्रदुष्येत् ।
 अनन्तरूपेण विभाति विष्णुः खहाररूपेण गणिते तथापि ॥

The meaning of खहार (khahara):

खभाजितो राशिः खहरः स्यात् ।
 यस्मिन् खहरे राशौ खेन विभक्ते निरपेक्षं फलं तत् ।
 तत्र प्रविशन्ति बहूनि रूपाणि निर्गच्छन्ति च तथापि न काचित् विकृतिः ।
 अनन्तराशेरिव विष्णोः प्रलयोत्पत्तिकाले जगतां प्रवेशनिर्गमयोः ॥

A number divided by zero is called खहार (khahara), which represents infinity (अनन्त). Bhaskara compares this to Lord Vishnu, who remains unchanged despite the constant creation and dissolution of beings entering and leaving him. Even when many quantities enter into or come out of a खहार, there is no change in it — just as Vishnu remains infinite and unchanging.

In modern notation:

$$\frac{a}{0} = \infty \quad (= \text{infinity})$$

उदाहरणानि

खं पञ्चयुग्भवति किं वद खस्य वर्गम् ।
मूलं घनं घनप खगुणाश्च पञ्च ॥

खेनोद्धृता दश च काः खगुणो निजार्थयुक्तस्-
त्रिभिश्च गुणितः खहृतस्त्रिषष्टिः ॥

Answers:

$$0 + 5 = 5$$

$$0^2 = 0; \quad \sqrt{0} = 0; \quad 0^3 = 0; \quad \sqrt[3]{0} = 0$$

$$5 \times 0 = 0; \quad 10 \div 0 = \infty ()$$

घनाणां सङ्कलनम्

घनाणां सङ्कलने करणसूत्रं बुद्धये ।

घनाः समं ङ्गुणिता द्वित्रिभिः समन्विताः
एकघनाद्विवर्जितास्ते सङ्कलिताः स्युः सदा घनाः ॥ १ ॥

Rule for the sum of cubes: The cubes of the numbers, being multiplied by their own squares, diminished by the cube of unity, are always added together to give the sum of cubes.

Explanation: The sum of cubes of the first n natural numbers is given by:

$$1^3 + 2^3 + 3^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2} \right)^2$$

उदाहरणम्

एकद्व्यादिघनाः समं सङ्गुणिता द्वित्रिभिः समन्विताः ।
एकघनाद्विवर्जितास्ते सङ्कलिताः स्युः सदा घनाः ॥

Example: Find the sum of the cubes of the first 5 natural numbers.

न्यासः (Statement): $n = 5$

Calculation:

$$\begin{aligned} 1^3 + 2^3 + 3^3 + 4^3 + 5^3 &= \left(\frac{5 \times 6}{2} \right)^2 \\ &= 15^2 = 225 \end{aligned}$$

Verification: $1 + 8 + 27 + 64 + 125 = 225$

घनानां व्यवकलनम्

व्यस्तद्वयोर्घनयोर्व्यस्तयोर्घनयोर्व्यवकलनम् ।
समघनयोरपि तथा व्यवकलनं सम्भवत्येव ॥ २ ॥

Rule: The subtraction of cubes of two unlike quantities, or of like quantities, is accomplished by subtracting the individual cubes.

If we have two quantities a and b , then:

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

उदाहरणम्

यावच्चतुष्कघनाद् व्यस्तघनं वकलनं कुरु ।
यदि यावत् ५ क ३ रूप २ तर्हि घनव्यवकलनं किम् ॥

न्यासः (Statement): Multiplicand = $yā\ 5\ kā\ 3$, Multiplier = $yā\ 2\ kā\ 1$

घनयोर्व्यवकलनम्:

$$(yā\ 5)^3 - (yā\ 2)^3 = 125yā^3 - 8yā^3 = 117yā^3$$

घनानां गुणनम्

घनयोर्घनयोर्घनयोर्वा गुणनं सम्भवति ।
घनघातः सम्भवत्येवं गुणकारघनयोः समं ॥ ३ ॥

Rule: The product of two cubes is equal to the cube of the product of their roots. If a and b are two quantities, then:

$$a^3 \times b^3 = (ab)^3$$

उदाहरणम्

यावच्चतुष्कघनयोर्घनयोर्वा गु नं कुरु ।
यदि यावत् ३ क २ रूप ५ तर्हि गुणनफलं किम् ॥

न्यासः: First quantity = $y\bar{a}$ 3, Second quantity = $y\bar{a}$ 2

Product: $(y\bar{a} 3)^3 \times (y\bar{a} 2)^3 = (y\bar{a} 6)^3 = 216y\bar{a}^3$

घनानां भागहारः

घनं घनेन विभजेद् घनमूलयोर्मूलयोरिव ।
घनभागः स्तः सम्भवत्येवं भागकारयोः समं ॥ ४ ॥

Rule: Divide a cube by a cube; the quotient is the cube of the quotient of their roots:

$$\frac{a^3}{b^3} = \left(\frac{a}{b}\right)^3$$

उदाहरणम्

न्यासः: Dividend = $y\bar{a} 8 k\bar{a} 27$, Divisor = $y\bar{a} 2 k\bar{a} 3$

Quotient:

$$\frac{(y\bar{a} 8)^3}{(y\bar{a} 2)^3} = \left(\frac{y\bar{a} 8}{y\bar{a} 2}\right)^3 = (y\bar{a} 4)^3 = 64y\bar{a}^3$$

घनानां वर्गो मूलं च

घनस्य वर्गः स्यात् षष्ठी घनस्य मूलं तु मूलघनम् ।
वर्गमूलं घनमूलं च कल्पनीयं ततस्ततः ॥ ५ ॥

Rule: The square of a cube is the sixth power. The root of a cube is the cube-root. The square-root of a cube is to be understood as that quantity whose square gives the cube.

In modern notation:

$$(a^3)^2 = a^6 \quad (\text{square of cube} = \text{sixth power})$$

$$\sqrt[3]{a^3} = a \quad (\text{cube root})$$

$$\sqrt{a^3} = a^{3/2} \quad (\text{square root of cube})$$

अव्यक्तसङ्कलनव्यवकलनम्

(१) यावत्तावत् कालक नीलक पीतक हरितकादयः ।
वर्णाः पीतो हरितश्चेत् (२) द्वयं च अव्यक्तानां कल्पना मतसंज्ञा—
राशिसंख्यानां कल्पिता मताचार्येण ॥ ६ ॥

Rule (1): The unknown quantities are represented by colors: यावत्तावत् (yavat-tavat, “as much as”), कालक (black), नीलक (blue), पीतक (yellow), हरितक (green), etc. These are the designations for unknown quantities.

Rule (2): The addition and subtraction of unknown quantities is performed by combining like terms. Homogeneous quantities are added or subtracted; heterogeneous quantities are merely set down separately.

यावत्तावत्कालकनीलकपीताश्च लोहितो हरितः ।
‘तकचित्रकपिङ्गलपाटल्काः पाण्डुःकृष्णशवलाभाः ।
श्यामलकमेचकयवकपिशङ्गहरिद्रावभ्रगौराभाः ॥”

The list of colors for unknowns continues: यावत्तावत् (as much as — the first unknown x), कालक (black — y), नीलक (blue — z), पीतक (yellow — u), हरितक (green — v), लोहितक (red), श्वेतक (white), चित्रक (variegated), पिङ्गल (tawny), पाटल (pink), पाण्डु (pale), कृष्ण (dark), श्यामल (dark-brown), मेचक (cloud-colored), etc.

अव्यक्तसङ्कलनव्यवकलने करणसूत्रं बुद्धये ।

योगान्तरं तेषु समानजातीयैर्भि नजातीयैश्च पृथक् स्थितिः ।

Rule: Among unknown quantities, the sum and difference (are taken) for those of the same class; those of different classes are set down separately.

उदाहरणम्

स्वमयुतमेकं सदैव सै रूपं धनायुतयुगं विल्परूपं च
युतौ पञ्चरेखयोः किं धनं विद्वद्भ्यश्च कथं भवेत् किं वदाशु ॥

न्यासः— $yā\ 1\ kā\ 1 \mid yā\ 2\ kā\ 8 \mid$ अनयोर्योगे जत्तं $yā\ 3\ kā\ 9$

First example of addition:

$$(yā\ 1 + kā\ 1) + (yā\ 2 + kā\ 8) = yā\ 3 + kā\ 9$$

Second example:

न्यासः--- $yā\ 5\ kā\ 3 \mid yā\ 2\ kā\ 8 \mid$ योगे जत्तं $yā\ 7\ kā\ 11$

Subtraction:

न्यासः--- $yā\ 5\ kā\ 3 \mid yā\ 2\ kā\ 8 \mid$ व्यवकलने जत्तं $yā\ 3\ kā\ 5$

अव्यक्तगुणनम्

अव्यक्तवर्गविधम्

अव्यक्ताकृतयोः समस्य हेतोः समेत्य वर्गद्वयं कल्पयेत् ।
तदा तयोर्वर्गयोर्वर्गः सम्भवत्येवमेकस्य वर्गः ॥ ७ ॥

Rule: When two unknown quantities are multiplied together, their product produces a new species. The square of an unknown is called वर्ग (square), and higher powers are designated accordingly.

अव्यक्तगुणने करणसूत्रं बुद्धये ।

गुण्यः पूर्वगुणकरूपैः समो निर्विशेष—
स्तैः षट् दिक्कैः क्रमशः सहित यथोक्तः ।
अव्यक्तवर्गकरणीगुणनासु चिन्त्यो
व्यक्तोक्तवर्गद्वयगुणनाविधिरेवमत्र ॥ १० ॥

Rule: The multiplicand, being combined with the multiplier in six ways, according to the order of terms (gives the product). The rule for multiplication of unknowns is the same as that for the multiplication of known quantities, with the consideration of the products of surds (irrationals).

उदाहरणम्

यावत्ताव रूपं त्रैकरूपं यावत्तावद्विभक्तं सद्विकूपैः ।
मद्गुण्यं द्वात्रिंशत् गुण्यं गुणं वा व्यस्तं स्वर्णं कल्पयित्वा तु विद्वन् ॥

न्यासः— गुण्यः $yā\ 5\ kā\ 3$ । गुणकः $yā\ 3\ kā\ 2$ । गुणनजतं $yā\ v\ 15\ yā\ 19\ kā\ 6$

गुण्यस्य धनर्णव्यत्यासे—

न्यासः—गुण्यः $yā\ 5\ kā\ 3$ । गुणकः $yā\ 3\ kā\ 2$ । गुणनजतं $yā\ v\ 15\ yā\ 1\ kā\ 6$

गुणकस्य धनर्णव्यत्यासे—

न्यासः—गुण्यः $yā\ 5\ kā\ 3$ । गुणकः $yā\ 3\ kā\ 2$ । गुणनजतं $yā\ v\ 15\ yā\ 19\ kā\ 6$

अव्यक्तभागहारः

भागाहरे (१) करणसूत्रं बुद्धये ।
 (२) भाजं च्छेदः सृजति प्रच्युतः सन् क्षेपे क्षेपे स्थानक्षेपं क्षेपेण ।
 दैन्यैर्द्विजैः सङ्गुणो यैः क्षेपैर्भागाहरे लब्धयस्तः स्पृशेत् ॥

Rule for division: The division of unknown quantities follows the same method as for known quantities. When dividing, if the divisor does not exactly divide the dividend, a fraction (भिन्न) results.

उदाहरणानि

Problem: Divide $yā\ 18\ kā\ 24$ by $kā\ 6\ yā\ 5$.

Solution: Using Bhaskara's method,

भाज्यः $yā\ 10\ yā\ 27\ yā\ 18\ kā\ 24$ । भाजकः $yā\ 5\ kā\ 6$

ततोभस्करेकृतया लब्धिः $yā\ 2\ yā\ 3\ kā\ 4$

अव्यक्तवर्गादि

वर्गः प्रथमं ततो घनोऽथ वर्गवर्गस्ततोऽष्टघातः ।
नवमो वर्गघनश्चेत्यादि घातानां क्रमेण स्मृतम् ॥

Rule: The powers of an unknown quantity are designated in order:

- वर्ग ($y\bar{a}^2$) — square
- घन ($y\bar{a}^3$) — cube
- वर्गवर्ग ($y\bar{a}^4$) — square of square (biquadratic)
- घनवर्ग ($y\bar{a}^5$) — square of cube
- घनघन ($y\bar{a}^6$) — cube of cube
- and so on...

यावन्मूलं पञ्चकं तु यावन्मूलं त्रिकं च त् ।
यावन्मूलं तथा षट्कं तेषां वर्गादि किं भवेत् ॥

न्यासः— $y\bar{a} 5 \mid y\bar{a} 3 \mid y\bar{a} 6$

वर्गा— $y\bar{a} v 25 \mid y\bar{a} v 9 \mid y\bar{a} v 36$

घना— $y\bar{a} gh 125 \mid y\bar{a} gh 27 \mid y\bar{a} gh 216$

वर्गवर्गा— $y\bar{a} v v 625 \mid y\bar{a} v v 81 \mid y\bar{a} v v 1296$

अनेकवर्णीद्विपद्विभ्रम

द्विवर्णकं द्विपदं त्रिवर्णकं त्रिपदं तथा ।
बहुवर्णकं बहुपदं च कल्पयेत् सुबुद्धिभिः ॥

Rule: Expressions involving two unknowns (द्विवर्णक) or two terms (द्विपद), three unknowns (त्रिवर्णक) or three terms (त्रिपद), and many unknowns (बहुवर्णक) or many terms (बहुपद) should be formed by the intelligent student.

Bhaskara here explains how to perform operations on multi-term algebraic expressions. The rules for addition, subtraction, multiplication, and division of polynomials are given in detail.

अनेकवर्णसमीकरणानां संस्थापनम्

एकस्य राशेः सहितं विरहितं च यत् स्थितम् ।
तत् संस्थाप्यं समीकरणे बुद्धिमता सदा ॥

यावत्तावत् कालक नीलक पीतक हरितकादयः ।
वर्णाः समीकरणे बहुवर्णे प्रयोजयन्ते सततम् ॥

The eight types of equations with one unknown (यावत्तावत्):

Bhaskara classifies equations involving a single unknown into eight categories based on the powers of the unknown:

1. रूपसमीकरणम् — Purely numerical (no unknown)
2. यावत्समीकरणम् — Linear in one unknown
3. वर्गसमीकरणम् — Quadratic (square of unknown)
4. घनसमीकरणम् — Cubic (cube of unknown)
5. वर्गवर्गसमीकरणम् — Biquadratic
6. घनवर्गसमीकरणम् — Fifth degree (square of cube)
7. घनघनसमीकरणम् — Sixth degree (cube of cube)
8. मिश्रसमीकरणम् — Mixed equations

करणीसङ्कलनव्यवकलनम्

करणीसंयुक्तिरे भः सङ्कलिता स्यात् समं करणीभिः ।
करणीभिर्वि शष्टाः पृथक् स्थापनीयाः समं करणीभिः ॥

Rule: Surds (करणी) having the same radical quantity are added together; those having different radical quantities are set down separately.

Definition of करणी:

यद् वर्ग मूलं यन्न पूर्णं तत् करणी ।
अकरणी च यत् पूर्णं तन्मूलं रूपं स्यात् ॥

A करणी is an irrational quantity whose square root cannot be extracted exactly. If the square root is exact, it is called रूप (a rational number).

The operation of bringing surds into contact (स्पर्श कुरुत):

स्पर्श कुरुत करणयोः समानकरणीभ्याम् ।
स्पर्शं कृत्वा ततः सङ्कलनं व्यवकलनं वा कुर्यात् ॥
स्पर्शक्रिया कथं स्यात् गुण्ये गुणककरणी अन्तर्गता ।
तयोर्गुणनं कृत्वा या करणी सा स्पर्शकरणी भवति ॥

“Bringing into contact” (स्पर्श कुरुत): This is Bhaskara’s term for making surds commensurable. When two surds have different radicals, we multiply each by the other’s radical to bring them to a common radical. If we have $\sqrt{a}$ and $\sqrt{b}$, then:

$$\sqrt{a} \times \sqrt{b} = \sqrt{ab}$$

This “contact” operation allows us to add or subtract surds with different radicals by first converting them to equivalent surds with a common radical.

Example of bringing into contact:

To add $\sqrt{3}$ and $\sqrt{27}$, first note that $\sqrt{27} = 3\sqrt{3}$, so they are already in contact (same radical 3). Their sum is $\sqrt{3} + 3\sqrt{3} = 4\sqrt{3}$.

To add $\sqrt{2}$ and $\sqrt{8}$, we have $\sqrt{8} = 2\sqrt{2}$, so the sum is $3\sqrt{2}$.

Example: Add $\sqrt{3}$, $\sqrt{3}$, $2\sqrt{3}$, and $5\sqrt{3}$.

Sum: $(1 + 1 + 2 + 5)\sqrt{3} = 9\sqrt{3}$

Example: Add $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$.

Result: $\sqrt{2} + \sqrt{3} + \sqrt{5}$ (set down separately as they have different radicals)

करणीगुणनम्

करणीगुणने करणीगुणने समे समे विपरे विपरे ।
गुण्यकरणीगुणककरणी गुणनफलं करणी भवति ॥

Rule: In the multiplication of surds, the product of the radical quantities gives the new radical, and the product of the coefficients gives the new coefficient.

स्पर्शं कृत्वा गुण्यगुणकयोः करणयोः ।
तयोर्गुणनफलं करणी भवति गुणनार्थम् ॥

Example: Multiply $3\sqrt{2}$ by $4\sqrt{3}$.

Product: $(3 \times 4) \times \sqrt{2 \times 3} = 12\sqrt{6}$

करणीभजनम्

भजनं करण्योर्भजनं करण्योर्वर्गवत् संस्कारः ।
भाज्यकरणीभाजककरणीभागो यथोक्तक्रमेण ॥

Rule: In the division of surds, the operation is like that for squares. Divide the radical of the dividend by the radical of the divisor, and the coefficient of the dividend by the coefficient of the divisor.

रूपकरणीभागे रूपभागः करणीभागः स्यात् ।
रूपं रूपेण करणी करण्या भजेत् प्रथमम् ॥

Example: Divide $12\sqrt{6}$ by $3\sqrt{2}$.

Quotient: $\frac{12}{3} \times \sqrt{\frac{6}{2}} = 4\sqrt{3}$

करणीवर्गः

वर्गः करण्याः स्यात् पदस्य वर्गः करणी चेति ।

वर्गः करण्याः स्यात् पदस्य वर्गः करणी च भवति ॥

Rule: The square of a surd is the square of the coefficient times the radical quantity (which becomes a plain number, since the square root is removed).

करण्याः पदस्य वर्गः करणी पदवर्गः स्यात् ।

तयोर्गुणनं रूपं भवति वर्गे करण्याः ॥

Example: Find the square of $3\sqrt{5}$.

Square: $(3)^2 \times 5 = 9 \times 5 = 45$

करणीमूलम्

(१) वर्गं करण्या यदि वा करणीस्तु तयो रूपपक्षं वा बहूनाम् ।
विशोध्येष्टं पदं ते शेषस्य रूपाणि युतानि निपातयेत् ॥

Rule: To find the square root of a surd expression, or of a number with surds added to it, subtract the greatest possible square, and treat the remainder properly.

रूपद्वयं करण्यौ यदि द्वे स्यातां तयोर्मूलं यदि पूर्णम् ।
तत् तयोर्मूलयोर्योगो मूलं स्यात् पृथग्वा व्यत्ययात् ॥

Bhaskara gives the rule for finding the square root of expressions of the form $a \pm \sqrt{b}$. He shows that if $\sqrt{a + \sqrt{b}} = \sqrt{x} + \sqrt{y}$, then:

$$x + y = a$$

$$4xy = b$$

From which x and y can be determined.

Example: Find $\sqrt{10 + \sqrt{84}}$.

न्यासः— $a = 10, \sqrt{b} = \sqrt{84}$

We need $x + y = 10$ and $4xy = 84$, so $xy = 21$.

Thus $x = 7, y = 3$ (or vice versa).

Therefore: $\sqrt{10 + \sqrt{84}} = \sqrt{7} + \sqrt{3}$

Verification: $(\sqrt{7} + \sqrt{3})^2 = 7 + 2\sqrt{21} + 3 = 10 + \sqrt{84} \checkmark$

कुट्टकः

भाज्यं हरेण विभजेद् भाजकेनापवर्त्य तौ ।
यावत्तावत्प्रचलितं कुट्टकं तत्र योजयेत् ॥

Rule: The Pulverizer (कुट्टक) is a method for solving linear indeterminate equations of the form:

$$ax \pm c = by$$

where a , b , and c are known integers, and x , y are unknown integers to be found.

The method involves:

1. Divide the dividend (a) by the divisor (b) to obtain a quotient and remainder
2. Use the mutual division process (continued fractions)
3. Apply the मतितः (intelligent) operation to find a solution
4. Derive the general solution from one particular solution

भाज्यं हरेण विभजेत् भागलब्धिं लभेत् ततः ।
भागशेषं ततो लब्ध्वा पुनरप्येवमावर्तयेत् ॥

आवर्तनात् षट्पदं यावत् गच्छेत् तत्र मतिः प्रयोज्या ।
मत्या लब्धिं गुणयेत् शेषं योजयेत् ततः पदम् ॥

उदाहरणम्

Problem: Find x and y such that $100x + 90 = 63y$.

न्यासः: भाज्यं = 100, भाजकः = 63, क्शेषः = 90

Solution: By the pulverizer method, $x = 18$, $y = 30$ is one solution.

The general solution is:

$$x = 18 + 63t$$

$$y = 30 + 100t$$

for any integer t .

वर्गप्रकृतिः

मूलं मूलं द्विगुणं युक्तं तत्प्रकृत्या हतं सहितम् ।
सर्वमूलं पुनर्मूलं ततः प्रकृतिमूलं भवेत् ॥

Rule: The Square-Nature (वर्गप्रकृतिः) deals with the indeterminate equation:

$$Nx^2 + 1 = y^2$$

known as Brahmagupta's equation or the Pell equation.

Bhaskara gives several methods for solving this equation:

1. **Composition (भावना):** Combining two solutions to obtain a third
2. **Cyclic Method (चक्रवाल):** An iterative algorithm

भावना कथं स्यात् गुणकक्षेपयोर्युगपत् ।
मिलितयोरजायते नूतनं गुणकं नूतनं क्षेपम् ॥
ज्ञातयोरजायते तृतीयं तयोर्मूलयोर्भावना ।
तृतीयं चतुर्थं च ततोऽपि भावनया जायते ॥

उदाहरणम्

Problem: Find x and y such that $61x^2 + 1 = y^2$.

Solution: Using the cyclic method (चक्रवाल), Bhaskara finds:

$$x = 226153980$$

$$y = 1766319049$$

This is one of the most celebrated examples in the history of mathematics, demonstrating the sophistication of Indian algebra.

चक्रवालम्

मूलं मूलं द्विगुणं युक्तं तत्प्रकृत्या हतं सहितम् ।
सर्वमूलं पुनर्मूलं ततः प्रकृतिमूलं भवेत् ॥

The Cyclic Method (चक्रवालम्): This is Bhaskara's most important contribution to algebra. The method proceeds as follows:

गुणकारभक्तेऽविशुद्धौ प्रकृत्या लब्धं मत्या गुणयेत् ।
क्षेपं योजयेद् वा व्येयाद् यावत् क्षेपः सूक्ष्मो भवेत् ॥
क्षेपक्षेपयोर्योगान्तराभ्यां युक्तं पदं नवं स्यात् ।
क्षेपक्षेपयोर्गुणनं नवक्षेपः स्यादिति चक्रवालम् ॥

Step 1: Choose integers a, b, k such that $Na^2 + k = b^2$.

Step 2: Find integers m such that $a \cdot m + b$ is divisible by k , and $|m^2 - N|$ is minimized.

Step 3: Compute:

$$\begin{aligned} a' &= \frac{a \cdot m + b}{k} \\ b' &= \frac{b \cdot m + N \cdot a}{k} \\ k' &= \frac{m^2 - N}{k} \end{aligned}$$

Step 4: If $k' = 1$, then (a', b') is the solution. Otherwise, repeat the process.

This cyclic process eventually yields $k = \pm 1$ or $k = \pm 2$ or $k = \pm 4$, from which the fundamental solution can be obtained.

एकवर्णसमीकरणबीजम्

रूपाणि यत्र स्युः समतुल्यानि तत्र यावत् तुल्यानि ।
यावन्ति तानि तानि विपरीतानि तुल्यानि तु तुल्यानि ॥

Rule: In a single-colored equation (एकवर्णसमीकरण), the unknown quantities on both sides are equalized by transposition (सरमण). The rule is:

1. Subtract the unknown on the smaller side from the unknown on the greater side
2. Subtract the known on the greater side from the known on the smaller side
3. Divide the remainder of the known by the remainder of the unknown

यत् दातुं यत् दातव्यं तुल्यं तुल्येन संयोजयेत् ।
विपरीतानि तुल्यानि तु तुल्यानि समीकुर्यात् ॥
सरमणं यथोक्तं समीकरणे कार्यं बुद्धिमता ।
अज्ञातं मूलं तत् तस्मात् लब्धं राशिर्भवेत् स्पष्टः ॥

उदाहरणम्

यावच्चतुष्कं रूपैः सप्तभिः सहितं त्रयोदशभिः ।
यावद् द्वित्रिभिः सहितं सप्तभिः समं भवति किम् ॥

Equation: $4y\bar{a} + 7 = 2y\bar{a} + 13$

Solution:

$$\begin{aligned} 4y\bar{a} - 2y\bar{a} &= 13 - 7 \\ 2y\bar{a} &= 6 \\ y\bar{a} &= 3 \end{aligned}$$

Verification: $4(3) + 7 = 19$ and $2(3) + 13 = 19$ ✓

अष्टविधं समीकरणम्

The Eight Types of Equations

रूपं यावत् वर्गो घनो वर्गवर्धोऽष्टधातो वा ।
घनवर्गो घनघनश्चेत्यष्टौ प्रकृतयः स्मृताः ॥
तासां मिश्रणं मिश्रं तदेकवर्णे समीकरणे ।
अष्टविधं बीजं तज्ज्ञैः प्रोक्तमतः परम् ॥

The eight types of equations with one unknown (यावत्तावत्):

1. रूपसमीकरणम् — Pure numerical: $c = d$
2. यावत्समीकरणम् — Linear: $ax + b = cx + d$

3. **वर्गसमीकरणम्** — Quadratic: $ax^2 + bx + c = dx^2 + ex + f$
4. **घनसमीकरणम्** — Cubic: $ax^3 + \dots = bx^3 + \dots$
5. **वर्गवर्गसमीकरणम्** — Biquadratic: $ax^4 + \dots = bx^4 + \dots$
6. **घनवर्गसमीकरणम्** — Fifth degree
7. **घनघनसमीकरणम्** — Sixth degree
8. **मिश्रसमीकरणम्** — Mixed: combinations of the above

शेषात्समीकरणम्

यद् दातुं यद् दातव्यं तुल्यं तुल्येन संयोजयेत् ।
विपरीतानि तुल्यानि तु तुल्यानि समीकुर्यात् ॥

Rule: When equal quantities are on both sides, transpose and change signs as needed. This is the method of संरमण (transposition).

उदाहरणानि

(1) $3y\bar{a} + 5 = 2y\bar{a} + 12$

Solution: $y\bar{a} = 7$

(2) $5y\bar{a} - 8 = 3y\bar{a} + 10$

Solution:

$$5y\bar{a} - 3y\bar{a} = 10 + 8$$

$$2y\bar{a} = 18$$

$$y\bar{a} = 9$$

(3) $7y\bar{a} + 15 = 4y\bar{a} + 3$

Solution:

$$7y\bar{a} - 4y\bar{a} = 3 - 15$$

$$3y\bar{a} = -12$$

$$y\bar{a} = -4$$

असकृत्वर्गादिसमीकरणम्

वर्गो घनो वर्गवर्गोऽष्टघातो नवमो वर्गघनश्चेति ।
समीकरणं समं समं कुर्यात् समतुल्यं तु तुल्यम् ॥

Rule: For equations involving squares, cubes, and higher powers of a single unknown, equalize by transposition and solve by factoring or by the method of roots.

वर्गसमीकरणे मूलं द्वयं स्यात् धनर्णयोर्विभागात् ।
मूलद्वयं तयोः संयोगे व्यत्यासे च विभागे च ॥

उदाहरणानि

(1) **Quadratic:** $y\bar{a}^2 - 5y\bar{a} + 6 = 0$

Solution: Factoring: $(y\bar{a} - 2)(y\bar{a} - 3) = 0$, so $y\bar{a} = 2$ or $y\bar{a} = 3$

(2) **Quadratic:** $2y\bar{a}^2 + 7y\bar{a} - 15 = 0$

Solution: Using the quadratic formula (as described by Bhaskara):

$$y\bar{a} = \frac{-7 \pm \sqrt{49 + 120}}{4} = \frac{-7 \pm 13}{4}$$

So $y\bar{a} = \frac{6}{4} = \frac{3}{2}$ or $y\bar{a} = \frac{-20}{4} = -5$

बहुवर्णसमीकरणम्

यावद् यावत् कालक नीलक पीतक हरितकादयः ।
वर्णाः समीकरणे बहुवर्णे प्रयोजयन्ते सततम् ॥

Rule: In multi-colored equations (बहुवर्णसमीकरण), multiple unknowns are represented by different colors:
यावत् (x), कालक (y), नीलक (z), पीतक (u), हरितक (v), etc.

एकोऽनेको वर्णो यस्मिन् समीकरणे स बहुवर्णकः ।
तस्मिन् वर्णानां योगवियोगौ पूर्ववत् कार्यौ ॥

उदाहरणम्

पञ्चयावत् त्रिकालकं रूपैः सप्तभिः सहितं समम् ।
त्रियावत् पञ्चकालकं रूपैः पञ्चभिः सहितं चेत् ॥

Equations:

$$5x + 3y + 7 = 3x + 5y + 5$$

$$\text{Transposing: } 2x - 2y + 2 = 0$$

$$x - y + 1 = 0$$

$$x = y - 1$$

The solution is indeterminate: for any value of y , $x = y - 1$.

अनेकवर्णमध्यमाहरणम्

मध्यमाहरणं तेषां यथा स्थानं प्रयोजयेत् ।
तुल्यानां तुल्यतां नीत्वा मध्यमं हरेद् बुधः ॥

Rule: The elimination of the middle term (मध्यमाहरण) is performed by equalizing the coefficients of one unknown and subtracting. This is the method for solving systems of linear equations with multiple unknowns.

एकवर्णसमीकरणे तावन्मूलं यावत्तावत् ।
बहुवर्णे मध्यमाहरणेनैकवर्णं कुर्यात् ॥

मध्यमाहरणं कथं स्यात् समानजातीयानां गुणानाम् ।
तुल्यीकृत्य व्यत्यासेन हरणं मध्यमाहरणम् ॥

यावत्तावत् कालकं च नीलकं च पीतकं च ।
तेषां समीकरणे मध्यमाहरणेनैकं वर्णं नयेत् ॥

Method of मध्यमाहरण (elimination):

1. Write the equations one below the other
2. Equalize the coefficients of one unknown by multiplication
3. Subtract one equation from the other to eliminate that unknown
4. Solve the resulting single-unknown equation
5. Substitute back to find the other unknowns

उदाहरणम्

द्वियावत् त्रिकालकं षड् रूपं त्रियावत् द्विकालकं सप्त ।
चतुर्यावत् पञ्चकालकं नव समानि भवन्ति चेत् ॥

Equations:

$$2x + 3y + 6 = 0$$

$$3x + 2y + 7 = 0$$

$$4x + 5y + 9 = 0$$

From equations (1) and (2):

$$6x + 9y + 18 = 0$$

$$6x + 4y + 14 = 0$$

$$5y + 4 = 0 \Rightarrow y = -\frac{4}{5}$$

$$\text{Substituting: } 2x + 3\left(-\frac{4}{5}\right) + 6 = 0 \Rightarrow 2x = \frac{12}{5} - 6 = -\frac{18}{5} \Rightarrow x = -\frac{9}{5}$$

भावितम्

भावितं भावितं यत्र द्वयोर्वर्णयोर्गुणः ।
स्थाप्यते समतुल्यं तद् भावितं प्रचक्षते ॥

Rule: The भावित (product) is the product of two unknown quantities. An equation involving such a product is called a भावितसमीकरण.

Equations of the form $xy = c$ (constant product) or $xy = ax + by + c$ are treated in this chapter.

भावितं यत् स्थाप्यते समीकरणे द्वयोर्वर्णयोः ।
तत् तयोर्गुणनफलं भावितमुच्यते गणिते ॥
यावत्कालकयोर्गुणनफलं भावितं यत् समम् ।
रूपेण सहितेन तत् समीकरणं भवति ॥

उदाहरणम्

यावद् द्विकं कालकं त्रिरूपं यावत्कालकयोर्गुणः समम् ।
रूपैः षड्भिः सहितं चेत् तयोर्मूलं वद प्रिये ॥

Equation: $(2x + 3) \cdot y = 6$, or $2xy + 3y = 6$

Solution: This is indeterminate. For any value of $y \neq 0$, $x = \frac{6-3y}{2y}$.

प्रत्येकसंहारः

प्रत्येकं संहरेद् भाज्यं हरेण प्रत्येकं बुधः ।
लब्धानि यानि तान्येव मूलानि प्रत्येकशः ॥

Rule: The method of individual subtraction (प्रत्येकसंहार) is used when the same divisor is to be applied separately to multiple dividends. Each quotient is then treated as a separate root.

उदाहरणम्

Problem: Divide $100x^2$, $80x^2$, and $60x^2$ individually by $20x^2$.

Quotients: 5, 4, and 3 respectively.

प्रदिशविद्यया

प्रदिशा विद्यया ज्ञेयं मूलं यत्र प्रदिश्यते ।
प्रदिशं तत्र योज्यं स्याद् विद्वद्भिर्नुमीयते ॥

Rule: The प्रदिश method is a technique for finding roots by inspection and intelligent guessing, especially useful for higher-degree equations.

उदाहरणम्

Problem: Find the cube root of 13824.

Solution by प्रदिश:

- The number ends in 4, so its cube root ends in 4
- $20^3 = 8000$ and $30^3 = 27000$, so the root is between 20 and 30
- The root is 24, since $24^3 = 13824$

नवीनप्रदिशविद्यया

नवीं प्रदिशमास्थाय मूलं यत् प्रदिश्यते ।
तत् सुलभं भवेत् प्राज्ञैर् विदुषां नात्र संशयः ॥

Rule: The new प्रदिश method gives an improved technique for finding roots, suitable for even more complex cases.

This chapter extends the methods of the previous chapter with additional techniques for extraction of roots of higher powers and for handling more complex algebraic expressions.

End of Part I (Pages 1–57)

Note: This edition continues with further sections on more advanced topics in algebra, including additional equation-solving techniques, applications to astronomy, and refined methods for the कुट्टक and चक्रवाल procedures. These are covered in Parts 4–9 of this series.

Editorial Notes (टिप्पणी)

The following explanatory notes were added by the editors to clarify Bhaskara's text:

- On notation:** Bhaskara uses यावत् (*yavat*) for the unknown quantity, corresponding to our modern x . The colors कालक (black), नीलक (blue), पीतक (yellow), हरितक (green), etc., represent y, z, u, v , etc.
- On positive and negative numbers:** Bhaskara fully accepts positive and negative numbers, calling them धन (wealth/positive) and ऋण (debt/negative). He gives clear rules for arithmetic operations with negatives, which are equivalent to our modern sign rules.
- On zero:** Bhaskara's treatment of zero (ख) is comprehensive. He gives six operations on zero, and defines the concept of खहार (*khahara*), a number divided by zero, which he identifies with infinity (अनन्त). His comparison of खहार to Lord Vishnu is one of the most celebrated passages in Indian mathematics.
- On surds:** The करणी (surd) operations demonstrate sophisticated understanding of irrational numbers. Bhaskara can add, subtract, multiply, divide, and extract roots of surd expressions. The operation of स्पर्श (bringing into contact) is his innovation for making surds commensurable.
- On the cyclic method:** The चक्रवाल method for solving $Nx^2 + 1 = y^2$ is one of the greatest achievements of Indian mathematics. It always converges to the minimal solution, and does so remarkably quickly. Bhaskara's solution for $N = 61$ was centuries ahead of European mathematics.
- On equations:** Bhaskara classifies equations into several types and gives systematic methods for each. His treatment of quadratic equations is complete, including cases with two, one, or no real solutions. The eight types of equations (अष्टविधं समीकरणम्) cover all cases up to the sixth degree.

Introduction

This volume contains the core algebraic chapters from Bhāskara II's *Bījagaṇita* ("Seed Computation," i.e., Algebra), covering pages 58–114 of the standard edition. These chapters treat:

- **Ekavarṇa-samīkaraṇa:** Linear and quadratic equations in a single unknown (यावत्तावत्, *yāvattāvat*);
- **Avyakta-vargādi-samīkaraṇa:** Equations involving unknown squares and higher powers;
- **Anekavarṇa-samīkaraṇa:** Systems of equations with multiple unknowns, and the *madhyamāharaṇa* (elimination) method in its various forms.

Bhāskara's presentation follows the classical Indian pattern: each topic is introduced by a rule (*sūtra*) in āryā or śloka verse, followed by worked examples (*udāharaṇa*) and detailed commentary (*vārttika*) showing the method of solution step by step. The unknown quantity is denoted by यावत्तावत् ("as much as"), abbreviated या or याव.

Chapter I: Equations with One Unknown

एकवर्णसमीकरणम्

अथैकवर्णसमीकरणम् । अत्र यन्नतत्तद्वदेकोवर्णः प्रकल्प्यते ततो रूपैः सह समीकृत्य यन्नतत्तद्वन्ममं सघ्न्यते ।

First Example: Simple Linear Equation

अत्राश्वमूल्यमज्ञातं तस्य मानं यावत्तावदेकं प्रकल्प्यं या १। तत्र श्रेणीशकं यदेकस्य यावत्तावन्मूल्यं तदा पणां किमिति कल्पितच्छागपणां प्रमाणमकं लब्धं पणामश्वानां मूल्यम् । या ६। अत्र रूपातनये प्रक्षिप्ते जातमाश्वस्य धनं या ६ रूप २०। एवं दशानां मूल्यं या १०। अत्र रूपाते चाणांते प्रक्षिप्ते जातं द्वितीयस्य धनं या १० रूप १००।

एतौ समशोधने छते एव सभी जाते समशोधनार्थं

न्यासः---*या ६ रूप २०।

या १० रूप १००।

Rule for Elimination

अथैकाव्यक्त शोधयेदन्यपक्षादव्यव्यक्तान् छोधि।
द्वितीयपदमुक्त्वेषु आचयपक्षतुल्यपक्षः शोधि-
तेषु शेषम् रूप ४००। अथ कल्पितकलवर्णं या ४ रूपैः रूप ४०० उक्ते
लब्धमेकस्य यावत्तावतो मानं व्यक्तम् १००। यदेकाश्वस्यैतन्मूल्यं
तदा पणां किमिति श्रेणीशकेन लब्धं पणां मूल्यं ६०० रूपातनय-
युतं १०० जातमाश्वस्य धनम्। एवं द्वितीयस्यापि १००।

Second Example

अथ द्वितीयोदाहरणे प्रथमद्वितीययोस्ते एव धने

या ६ रूप २००।

या १० रूप १००।

अत्राचयपक्षधनार्थं द्वियुकेन तुल्यमनयस्य धनमुदाहृतमत भा- व्यधनार्थं द्वियुक्ते अथवानयधने द्विहीन द्विगुणे छते पक्षौ समी भव- न्तथा छते शोधनार्थं

न्यासः---या २ रूप १५२। अथवा या ६ रूप २००।
या २० रूप २०४।

या १० रूप १००।

उभयोरपि शोधनार्थं छते लब्धं यावत्तावन्मानम् २६। अनेन श्रेणीकल्पितानां छते जाते धने ५१२, २६०।

अथ तृतीयोदाहरणं ते एव धने। अत्राचयधनव्यवस्थाः परधनमिति परं विगुणीकृत्य

* विरोधः---संप्रति $6या + 200 = 10या - 100 \therefore 400 = 4या$
 $\therefore या = 100$ एवं समीकरणगतिः सर्वत्र क्रियते।

Further Examples in One Unknown

न्यासः— | या ६ रूप ३००।

या २० रूप ३००।

समीकृत्या लब्धं यावत्तावन्मानम् २५। अनेनोर्ध्वापिते जाते धने ४५०, १९५०।

उदाहरणम्

माणिक्यामल्लीलमौक्तिककिमितिः पञ्चाषसममा-
देकस्यान्यतरस्य सप्त नव पट् तद्दलसंख्या सखे।
रूपाणां नवतिद्विषष्टिर्नयोस्तौ तुल्यवित्तौ तथा
वीजं प्रतिरुज्ञानि कुरुते मौल्यानि शीघ्रं वद ॥३॥

अत्राऽऽयकानां बहुत्वे कल्पितानि माणिक्यादीनां मौल्यानि या(१) ३, या २, या १। यदि एकस्य रत्नस्य इदं मूल्यं तदोर्ध्वानां किमिति लब्धानां यावत्तावतां योगे स्वस्वरूपयुते जाते पक्षौ

या १९, या १६, या ७ रूप ९०।

या २९, या १८, या ६ रूप ६२।

एतेऽनयोर्धने इति समशोधने छते लब्धं यावत्तावन्मानम् ४। अनेनोर्ध्वापितानि माणिक्यादीनां मौल्यानि १२, ८, ४। एवं सम्मे- धनम् २४२। अथ वा माणिक्यमानं यावत्तावतोऽल्पमुत्कल्प्यमूल्यं व्यक्तं एव कल्पते ५, ३। अतः समीकरणेन लब्धं यावत्तावन्मानम् १२। अनेनोर्ध्वापिते जाते समधनम् २९६। एवं कल्पनावशादनेकथा।

उदाहरणम्

एको द्वीति मम द्विहि शतं धनेन
त्वत्तो भवामि हि सखे द्विगुणस्ततोऽस्यः।
भूते द्वयोर्यदसि वैनम बहुणोऽहं
त्वत्तस्तयोर्वद धने मम किं प्रमाणे ॥४॥

On Quadratic and Higher Equations

किन्तु बहुगुण एव विद्याः शेषविधौ कल्प्ये या $\frac{3}{2}$ । वर्गितः=याव $\frac{3}{2}$ । स्वपदार्था या ३ युतौ जातः = $\frac{\text{याव } ९ \text{ या } १२}{४}$ । अयं बहु-
कः। अत्रापि प्राग्वद्गुणहरयोः कृत्वा समच्छेदीकृत्य छेद मे शोधनजाते पक्षौ

याव ९ या १२ रूप ०।

याव ० या ० रूप ६०।

एतौ चतुर्गुणौ छतौ मूले गृहीत्वा पुनः समशोधनालब्धं यावत्ता- वन्मानम्=२।

तथा चास्त्रपाठेऽत्र गणिते---

बहुः स्यात् बहुगुणः स बहुगुणिर्वर्जयेत् शेषविधौ।

गुणके जाते स हरश्चेन पुनस्तथा राशिः॥

अविच्छिन्न एव(ग) विच्छिन्नः सर्वैरेवं विपरिवर्द्धः।''

उदाहरणम्

राशद्वैदशानिशं राशिद्वयनाल्य कः समो यः स्यात्।

राशिकृतिः बहुगुणिता पञ्चत्रिंशद्युता विद्ध ॥६॥

अत्र राशिः=या १। अयं द्वादशगुणितो राशिद्वयनाल्यक्ष=याव १ या १२। अयं याव ६ रूप २५ अनेन सम इति शोधने छते
जातमाश्रयपक्षे याव १ या ६ या १२। अन्यपक्षे रूप २५। अन्योक्तैकरूपाऽएकं प्रक्षिप्य घनमूले या १ रूप २।

या ० रूप ३।

पुनरनयोः समीकरणेन जातो राशिः=५।

उदाहरणम्

को राशिद्विरशीतिशून्यो राशिवर्गेण युतो हतः।
द्वाभ्यां तेनोनितो राशिवर्गो वर्गेणयुतो भवेत्॥७॥

रूपौ वद ते राशिं वेतसि बीजक्रिया यदि॥७॥

अत्र राशिः=या १। द्विरशीतिशून्यः=या २००। राशिवर्गयुतो जातः =याव १ या २००। अयं द्वाभ्यां गुणितः=याव २ या ४००।
अनेनायं

Example: The Swallow and the Lotus

अत्र वा प्रथमप्रमाणकलं द्वितीयप्रमाणकलं विमले प्रल-म्बये तद्गुणगुणितेन द्वितीयमूललेन तुल्यमेव प्रथममूलधनं स्यात् कथमन्यथा समे काले सम फले स्यात्। अतो द्वितीयस्यायं गुणः २। एकगुणं द्वितीयमूलधनमेकेनगुणागुणितं कलवर्गं वर्धते स एकोन-गुणेन ईषत्कल्पितकलानतरस्य वर्गं भक्ते द्वितीयमूलधनं स्यात्। तत् कलवर्गयुतं प्रथममूलधनं स्यात्। अत्र कल्पितकलवर्गः ४। अतः प्रथमद्वितीयमूलधने ८, ४। फलम् २। यदि शतस्य पञ्च कलान-न्तरं तदा पञ्चानां किमिति लब्धमेकमासपञ्चानां फलम् २। यथने-नैको मास-स्तदा द्विकेन किमिति लब्धा मासाः ५।

उदाहरणम्

एकक्रान्तर्चयनात् फलस्य वर्गं विशोध्य परिशिष्यम्।
पञ्चक्रान्तेन दत्तं तुल्यः कालः फले च तथैः ॥८॥

अत्र गुणकः ५। एकोनगुणेन ४ ईषत्कलस्याय वर्गे १६ मक्ते जातं द्वितीयधनम् ४। इदं कलवर्गयुतं जातं प्रथमधनम् २०।

अतोऽनुपातद्वयेन कालः २०।

एवं स्वबुद्धिं बैदं सिद्धंति किं यावत्तावत्कल्पनया। अथ वा बुद्धिरेव बीजम्। तथा च गोले मथोक्तम्।

नैव वर्गीतमकं बीजं न बीजानि पृथक् पृथक्।
मेव मतिर्बीजमनल्पा कल्पना यतः'' ॥

उदाहरणम्

माणिक्याणुकणिनीलादीनां मुक्ताफलानां शतं
षड्भागि च पञ्च रत्नवीरां येषां चतुर्णां धनम्।
संग्रहे हयरेन ते निजधनाद्वैकैक मिथो
जातास्तुल्यधनाः पृथग्वद सखे तद्दनमैल्यानि मे ॥२॥

अत्र यावत्तावदाद्यो वर्णोऽऽयकानां मानानि कल्प्यन्त इत्यु-पलक्षणं तज्जामाक्षितानि छत्वा समीकृत्य काष्ठं मतिमतिः।
तथा-

Geometric Application: The Triangle

न्यासः— |

क्षेत्रव्यवहारात् समभुजत्रिकोणस्य जाते
लम्बवर्गः=याव १ रूप १५, रूप २००
या १ या १ क १८ रूप १

द्वितीयाव्यावावर्गी=याव १ या क ६२ या २ रूप १९ क ६२। समभुजत्रिगात् २ ६ अपास्ते जातो द्वितीयो लम्बवर्गः
=याव १ या २ या क ६२ रूप १३ क ६२।

एतौ (१) समाविति समशोधने छते जाते पक्षौ

२ २५ क ५१२।
(२)या २ या क ६२।

अत्र (३)भाजकस्याव्यक्तशेषस्य या कारस्य प्रयोजनमवादप्रमे छते भाज्यभाजको जातो
द्वैते क- ल्पना असदृश्याय" इति द्विसमीकरणकल्पना धनत्वे प्रकल्प्य क ४ क ६२। अन

अत्र धनर्णताव्ययमीहिततया
ज्ये गुणिते जातम्

क २६८६४ क २१३१६ क ५१६४८ क २०४८।

एतद्वैतयोः क २६८६४ क २१३१६। मूले १९२। ६५। अन्यो- ध्वगः रूप १३६।

Rule for Completion of the Square

एषां पूर्वमूलानां च सर्वेषां योगः = याव ३ या ३१ रूप ८८। इदमे- काद्वययुतं प्रयोदशवर्गः---

याव ३ या ३१ रूप ७५।

याव ० या ० रूप १६९।

समं छत्वा पक्षयोर्द्वादशाभिः संगुण्य तयोरेकाराशिद्वयं वर्गं १६९ निर्धिय मूले या ६ रूप ३१।

या ० रूप ४३।

पुनरनयोः समीकरणालब्धेन यावत्तावन्मानेन २ अनेनोर्ध्वापि- तानि राशिमूलानि २, ५, ८, ११। एषां वर्गा राशयः क्षेपेणा अथवा द्वाराशयो भवन्ति २, २३, ६२, ११९।

अत्राचयपरिभाषा।

।शिशेषपदेषयो यद्गुणक ।र्दोचरम्।

यका राशयः कल्प्या वर्गिताः क्षेत्रवर्गिताः ॥''

Avyakta-vargādi-samīkaraṇa (continued)

Further Examples with Unknown Squares

अथ विद्यावतरगार्थमिष्टं वेणुवतनमूलानं कल्पितम्=२०। सुप्रस- न्नपाताह्लादमानम्=या १।

न्यासः— |

यदि पञ्चदशकोट्या विरशतिमुञ्जस्तदा
यावत्तावन्मितया किमिति लब्धा लब्धवर्गा-
क्षितावधा या $\frac{10}{3}$ । पुनर्यदि दशमितकोट्या
विरशतिमुञ्जस्तदा यावन्मितकोट्या किमिति

लब्धा वृहद्दशाक्षितावधा या २। अनयोर्ध्वगां या $\frac{10}{3}$ विरशतिसमं छत्वा छओ लब्धः ६। उद्धारपनेनावधा च ८, १२। अथवा वंशसम्बन्धेनावधा तदनुतिसंप्रमितिरिति यदि वंशद्वययोगेन २ अनेनावधायोगो=२० लभ्यते तदा वंशाख्या १५, १० किमिति जाते अवधे ८, १२। अथानु १- तात् सम एव लब्धः ६। किं यावत्तावत्कल्पनया। अथवा वंशयोर्वर्धा योगहतो यत्र कुत्रापि वंशान्तरं लब्धः स्यादिति किं भूमिकल्पनया- पि एतद्विषं सू णि प्रसाद्य बुद्धिमतोऽहम्। इति श्रीभास्कराचार्यविरचिते बीजगणिते एकवर्णसमीकरणं समाप्तम्।

अथाऽव्यक्तवर्गादिसमीकरणम्।

तदेव मध्यमाहरणमिति यावद्दीयत्याचार्याः।

यतोऽत्र वर्गराशा एकस्य मध्यमस्याहरणमिति।

अत्र सूत्रं ब्रूमः।

अव्यक्तवर्गादि यदा द्वयोः पक्षौ तदैकेन निःशेषं किं चत्।
क्षेपं तयोरेन पदप्रदः स्याद्व्यक्तः सहेत परेन भूयः ॥१॥

व्यक्तस्य मूलस्य समीकृते समव्यक्तमानं बहु लभ्यते तत्।
न निर्वहेत् चेद्धनवर्गयोस्त्वेति तदा क्षेपमिदं स्यबुद्ध्या ॥२॥

अव्यक्तमूलकाकं ततोऽस्य व्यक्तस्य पक्षस्य पदं यदि स्यात्।

मूलं धनं तथा विधाय साव्यमव्यक्तमानं द्विधा कल्पितं स्यात् ॥३॥

उदाहरणम्

स्वार्धपञ्चाशनवमैर्युक्ताः के स्युः समाश्रयाः।
अन्यैर्द्वादशहीनाख्य परिशिष्टाख्य तान् वद ॥१४॥

अत्र समराशिमानं यावत्तावत् १। अतो विलोमविधिना ऊहा- न्यमागद्वयेनाः सर्वेर्द्वयैक शोभाः स्युः या $\frac{2}{5}$ । एत राशयः १००, १२५, १३५। अथ स्वार्धादिकोन'' इत्यादिना राशयः या $\frac{2}{5}$, या $\frac{5}{6}$, या $\frac{9}{10}$ । हिस्तमं छत्वाऽऽपञ्चमयावत्तावन्मानेन १५० उद्धारपिता जाता

उदाहरणम्

त्रयोदश तथा पञ्च करण्यो जूयोनितो।
भूजाता च चत्वारः फले भूमि वदाऽखु मे ॥१५॥

अत्र भूमेर्व्यावत्तावत्कल्पने क्रिया प्रसरतीति स्वेच्छया व्य- क्ते १३ भूमिः कल्प्यते कल्पितशेषभावात्। अतोऽत्र कल्पित व्यक्तम्।

क ५

या १ न्यासः। अत्र लम्बगुणा भुजयर्धं हरं त्रिभुजे
कलं भव ' इति व्यतिकेन कलानुक्रमो जातः क $\frac{५}{१३}$ ।

एतद्वर्गं भुज ५ करणी वर्गित् रूप ५ अस्मा य र $\frac{५}{१३}$ ।
मूलं जाताऽऽवधा क $\frac{५}{१३}$ । मा भूमेरपास्य योगे करण्योनितो प्रकट्य'' इति जातास्याऽऽवधा क $\frac{१४४}{१३}$ । अथवा वर्गित्
रूप $\frac{१४४}{१३}$ । लम्बवर्गं=र $\frac{६४}{१३}$ युतात् र $\frac{२०८}{१३}$ मू। तो भुजः ४। इयमेव भूमिः।

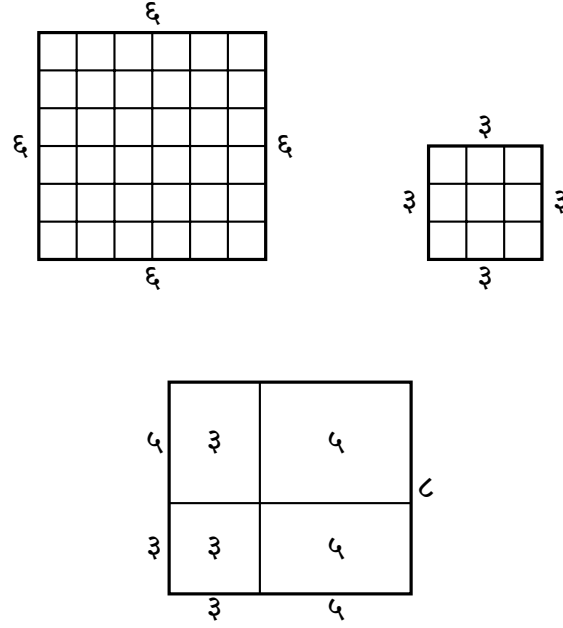
उदाहरणम्

दशयष्टिकरण्यन्तरमेको बाहुः परक्ष बहुकरणि।
मूलच्छदशाकरणि रूपोना लम्बमानमाचक्ष्व ॥१६॥

अत्रावधाज्ञाने लम्बज्ञानमिति लम्बावधा=या १। एतद्वृता वृत्तावधाप्रमाणमिति तथा

Geometric Diagrams

तेषां स्वरूपाणि यथा---



अन्यत् करणसूत्रं ब्रूमः।

चतुर्गुणस्य घातस्य युतिवर्गस्य चान्तरम्।
राश्यन्तरकृतेस्तुल्यं द्वयोरन्त्यकयोर्यथा ॥१७॥

अत्र राशी ३, ५। अनयोर्युतिवर्गात् चतुर्गु कोटिषु घातचतुर्गुणे- यापनेते मध्ये राश्यन्तरवर्गसमानि कोटिकाणि दृश्यन्त इत्युपपक्षम्।

Transition to Multiple Unknowns

अथ माणिक्यादीनां मौल्यानि यावत्तावदादीनि प्रकल्प्य तद्गुणपरितसंख्यां च छत्वा रूपाणि च प्रक्षिप्य समशोधनार्थं

न्यासः---या ५ का ८ नी ३ रूप १०।

या ३ का ९ नी ५ रूप ६२।

अत्र वर्गं शोध्येदितयादिना जाता यावत्तावद्विमतिः:

$$\text{या} = \frac{\text{क नी } १२२८}{२}$$

इयमेकैव, एकत्वादियमेवानयाऽऽस्तौ कश्चित् का :। इदं माज्ये वर्णद्वयं वर्धते स्तो नीलकमानमिष्टं रूप १ कल्पितम्। अनेन नीलक-मूल्याव्य रूपेषु प्रक्षिप्य जातम् या = $\frac{\text{क नी } २८}{२}$ ।

अतः कुट्टकविधिना हरतपे धनक्षेपे'---इत्यादिना गुणासी सक्षेपे पी २ र १।

पी १ र १४।

अत्र शून्येन पीतकमूलाव्य जातानि माणिक्यादीनां मौल्यानि १४, १, १। अथवैकेन १३, ३, १। द्वाभ्यां वा १२, ५, १। निर्मूला- ११, ७, १। एवमनुपराशादानन्तर्यम्।

उदाहरणम्

एको द्वीति मम द्विहि शतमिति ॥२॥

अत्र धने या १, का १। परधनाच्छ्रुतमपास्य पूर्वधने शतं प्रक्षिप्य जातं या १ रूप १००, का १ रूप १०० परधनाद्वाभ्यां द्विगुणमिति परधनेन द्विगुणेन समं छत्वा लब्धा यावत्तावद्विमतिः:

या=का २ रूप ३००

पुनराचयधनाद्वादशजनितेषु परधने शिष्टेषु जातम्

या १ रूप १०।

का १ रूप १०।

Worked Example (p. 115)

(१) वि०—कल्प्यते समी पक्षौ

काव १

यावव्। ई १ याव।ई १

अत्र प्रथमपक्षस्य मूले क १ द्वितीयपक्षस्य यावव्।ई १ याव।ई १ मूलेन सममिति ।
तत्र द्वितीयपक्षस्य मूले च का।

$$\text{का} = \sqrt{\text{यवव्।ई १ यवव्।ई १}} = \text{या} \sqrt{\text{यवव्।ई १ ई १}}$$

अत इदं याव।ई १ ई १ मूलदं ततो वर्गप्रकृतिविषयो यथा का वर्गः ई, गुणः ई, युतो मूलदं इति हत्वा यावच्चावकमानं उच्चिष्टं चास्य याव।ई १ ई १ मूलेन सममिति।

Summary Rule (p. 115 bottom)

The section concludes with a general procedure:

एवं बहुधा बुद्धिमत्त्वविचिन्त्यामिति सर्वमुपपन्नम् ॥

“Thus in many ways, by the thoughtful application of intelligence, all [is demonstrated].” Everything is proved.

2 Aneka-varṇa-madhyamāharaṇa-bhedāḥ (pp. 116–133)

This is the largest section of the final portion, dealing systematically with equations involving multiple unknown quantities (*aneka-varṇa*, “many colours”). The Sanskrit term *varṇa* (colour) is the traditional designation for an unknown quantity in Indian algebra — different unknowns being assigned different colours.

2.1 General Rules for Multiple Unknowns (pp. 116–117)

अथ सूत्रं साध्येऽस्तु ॥

(१) द्वितीयपक्षे सति सममेव तु हत्यापवर्त्योन पदे प्रसाध्ये ।

उच्चिष्टं कनिष्ठेन तदा निहत्याच्छेदं वर्गेण हतोऽपवर्तः ॥ ६ ॥

कनिष्ठवर्गेण तदा निहत्योच्चिष्टं ततः पूर्ववदेव शेषम् ।

इत्यर्थात् ॥

Worked Example 1 (p. 117)

उदाहरणम् ॥

यस्य वर्गेऽद्वितीः पञ्चगुणा वर्गशतोनिता ।

मूलद्वा जायते राशिः गणितज्ञ वदाशु तम् ॥ १ ॥

Example 1: “A number whose square minus twice [itself], diminished by a hundred times five [i.e., 500], has two roots — tell me that number quickly, O mathematician.”

अत्र राशिः=या १ । अस्य वर्गेऽद्वितीः पञ्चगुणा वर्गशतोनिता यावव ५ याव १०० । अयं वर्ग इति कालकवर्गसमं हत्वा गृहीतं कालकवर्गस्य मूलं का १ । द्वितीयपक्षस्य यावव ५ याव १०० ।

यावच्चावकगुणापवर्त्य वर्गप्रकृत्या मूले क १० उचे २० वा क १७० उचे ३८० ।

Worked Example 2 (p. 117)

उदाहरणम् ॥

कयोः स्यादन्तरे वर्गौ वर्गयोगो ययोर्धनः ।

तौ राशी कथयामित्रो बहुधा बोझवित्तम् ॥ २ ॥

Example 2: “Two numbers whose difference of squares [equals] the sum of their squares — tell me those numbers, O friend, in many ways, O best of algebraists.”

अत्र राशी या १, का १ । अनयोरन्तरं या १ का १ नीलकवर्गसमं हत्वा लब्धं यावच्चावकमानं का १ नीलं ई १ । अनेन यावच्चावकद्वितीयाप्य जाती राशी का १ नीलं ई, का १ ।

अनयोर्वर्गयोगः काव २ नीलका २ नीलव १ । एष घन इति नीलकवर्गयनसमं हत्वा शोधने हते जाते प्रथमपक्षे नीलव १ नीलव १ । द्वितीयपक्षे काव २ नीलका २ ।

Rule for Three Unknowns (pp. 119–120)

(१) साध्यकल्पो यदि वर्गवर्गेषु द्वित्र्यादिवर्णस्य हते समं तम् ।

हत्वा पदं तस्य तदन्वपक्षे वर्गप्रकृत्यैवकदैव मूले ॥

कनिष्ठमाच्छाद्य पदेन तुल्यं उच्चिष्टं द्वितीयेन समं विधायात् ॥ ८ ॥

Rule 8: “If the square of the square of an assumed [quantity], diminished by the square of one, two, three, etc., unknown quantities, having made [the two sides] equal, taking the square root of that, then by the square-nature method, the roots [are found]; covering the lesser root by an equal root, making the residue equal to the second [unknown]...”

Worked Example with Three Unknowns (p. 121)

उदाहरणम् ॥

त्रिकादिचतुरश्रेणां नच्छेदं क्वापि च यत् कलम् ।

तदेव त्रिगुणं क्षेत्रमष्टयुगच्छेदं भवेद् ॥ ९ ॥

अत्र श्रेण्योन्या इः । आदि=३, चयः=२, गच्छः=या १ । आदि=३, चयः=२, गच्छः=का १ । अनयोः (क) कलं=याव १ या २, काव १ का २ । अनयोरन्यत् त्रिगुणं परस्मिं हत्वा शोधनार्थं

न्यस्तः → यस्म ३ या ६

कस्म १ कार

शोधने हते पक्षौ त्रिगुणीहत्वा नव प्रक्षिप्य प्रथमपक्षस्य मूलं या ३ रु ३ । द्वितीयपक्षस्य काव ३ का ६ रु ९ । नीलकवर्गेण

Rule for Equal-square Unknowns (p. 121)

(९) प्रथमपक्षस्यद्विगुणीतच्छेदस्य मूलं नीलं प्रकल्प्य तद्वर्गेण समं परं पक्षं

हत्वा पूर्वोक्तस्य वासना चातिसरति ॥

“Having assumed the root of twice the divisor of the first side as the ‘blue’ [unknown], having divided the other side by its square, the desire is satisfied more than previously stated.”

2.2 The Method of Substitution (pp. 125–128)

अथ सूत्रं सूचयम् ॥

(१) सकृपमयकमल्पकं वा विभोगमूलं प्रथमं प्रकल्प्य ।

योगान्तरशेषकमाजिताच्छेदान्तरशेषकतः पदं स्यात् ॥ १२ ॥

Rule 12: “Having assumed the root of the difference of [the two sides] as the first [unknown], or a small quantity, [and] the residue of the difference of sums as the square [of another unknown], the root [is obtained].”

(१) वि०—अत्र कल्प्यते योगान्तरशेषमानम् = क्षे
वर्गान्तरशेषमानम् = क्षे₂, वर्गयोगशेषमानम् = क्षे₃
विभोगमूलम् = या, योगमूलम् = का

$$\text{तदाप्रदर्शनार्थं विभोगः} = \text{या}^2 - \text{क्षे}_1, \text{ योगः} = \text{का}^2 - \text{क्षे}_1$$

$$\text{अल्पप्रक्षिप्तः} = \frac{\text{का}^2 - \text{या}^2}{2}, \quad \text{वृहदप्रक्षिप्तः} = \frac{\text{का}^2 + \text{या}^2 - 2\text{क्षे}_1}{2}$$

Rule 13 (p. 129)

शेषं ततः शेषकमूलमन्यच्च मूले विद्ध्वादसकृत् सम्मते ॥ १ ॥

समाविते वर्गं तं तु यत्र तन्मूलमादाय च शेषकस्य ।

इष्टोद्धृतस्येर्विर्वर्जितस्य दलैः समं तद्विह तदैव कार्यम् ॥ १० ॥

Worked Example on the Product Method (p. 130)

उदाहरणम् ॥

तौ राशी वद यत्कृत्योः समस्तगुणयोर्युतिः ।

मूलद्वया स्याद्योगमष्टौ मूलद्वया रूपसंयुतः ॥ १ ॥

अत्र राशी या १, का १ । अनयोर्वर्गयोः समस्तगुणयोर्युतिः यावव ७ काव ८ । अयं वर्ग इति नीलकवर्गसमं हत्वा पक्षयोः कालकधनं प्रक्षिप्य नीलकपक्षस्य मूलं नी १ । परपक्षस्य याव ७ काव ८ ।

Summary Verses (p. 131)

अथ सूत्रं सूचयम् ॥

(१) सकृपमयकमल्पकं वा विभोगमूलं प्रथमं प्रकल्प्य ।

योगान्तरशेषकमाजिताच्छेदान्तरशेषकतः पदं स्यात् ॥ १२ ॥

वर्गान्तरशेषकसंमितिर्युता क्षेपेण हत्वोर्ध्वितेन तै ततः ।

द्विधा पदं तत्पदयुगविभोगाज् मूलं युतमूलमतस्तयोरिति ॥

“The difference of squares having a residue equal to the divisor, divided by the additive, then by the quotient, [the root] divided in two; from the difference of the pair of roots, the root [of the lesser] from the sum of the roots, thus [proceeding]...”

3 Bhāvitam and Prakṣepaka-vidhi (pp. 134–152)

3.1 Equations with Product Terms (pp. 134–138)

This section deals with *bhāvita* (product) equations, where the unknowns appear in product form. The method extends the techniques for multiple unknowns to handle more complex algebraic structures.

अथ वा स्वर्णकल्पनायां मन्दबुद्धीनां पुनः किञ्चित् पठितः ।
तत् सूचयामि ॥

हरमका यस्य हृति सूक्ष्मति सोऽपि द्वितीयपदयुतातः ।
तेनाहतोऽस्यवर्णो रूपपदेनान्वितः कल्पः ॥ १६ ॥

Rule 16: “The divisor multiplied by [some quantity] whose [result] is fine, added to the second term; having multiplied by that, the unknown [is obtained], combined with a root term [as] the assumption.”

Worked Example (p. 137)

उदाहरणम् ॥

न यदि पदं रूपाणां शिष्येत्तं तेन हरतोऽपि ।
नावच्छद्गुणो भवति न चेदेवमपि किल तथै ॥ १७ ॥

हरमतिः । यस्याङ्कस्य हृतिहरमका सती सूक्ष्मतीति निश्शेषा भवति अपि च सोऽप्यङ्को हर्यो रूपपदेन च गुणितो हरमकः सन् सूक्ष्मति तदा तेनाहतोऽस्यवर्णस्तेन रूपाणां घनमूलं न लभ्यते तदा तेन रूपेषु हरतोऽपि तावद्गुणो भवेत् तन्मूलं रूपपदं स्यात् ।

3.2 The Bhāvita Rule (pp. 140–141)

भावितम् ॥

वर्णद्वयातिकूपेऽस्य मकरवेष्टनैकतः ।
एताम्या संयुताबुद्धि कल्पयो वैच्छया च तौ ॥ ३ ॥

वर्णद्वयो वर्गयोरन्ते क्रियते ते विपर्ययात् ।
समयोः पक्षयोरेकमाध्यमिकमपास्य तौ वर्णौ रूपाणि च

The section on “Product”: “Two unknowns, their product having a coefficient, from one side having a single enclosure [i.e., bracketed], having assumed any sum of their values, [proceeding] with the pair of unknowns reversed, [taking] the square of two unknowns, the means [being] equal, having removed one middle term, the two unknowns and the roots...”

Worked Example on Bhāvita (p. 142)

उदाहरणम् ॥

चतुःषष्टिगुणयो राशयोः संयुतिर्द्विगुता तयोः ।
राशिघातेन तुल्येति ॥

तत्र यथोक्तं हते पक्षौ { या ४ का ३ रू २
या।का।भा १

वर्णद्वयातिकूपेऽस्य १४ एतदेकेनाहतं हते जाते द्विके १, १४ । एते वर्णद्वया ४, ३ स्वैच्छया युते जाते यावच्चावककालकमाने ५, १८ वा १७, ५ । द्विकेन ५, ११ वा १०, ६ ॥

3.3 The Method of Mean Elimination (pp. 145–147)

माध्यमाविगमः ॥

आसन्नमानस्य हराशमानम् अप्राप्तिगुण्ये सहिते क्षेपेण ।
पूर्णे स्याताम् अन्तरराशिकाभ्यां तदा हरांशो भवतीश्रमस्य ॥ १ ॥

आसन्नमानयोरन्तरमेव भवेत् ।
अंशस्थाने सदा रूपं चिन्त्येत तथा धीमता ॥ २ ॥
सर्वेषामासन्नमानेषु हरांशो भवति षट् ।
तथोत्तरोत्तरं सूक्ष्मतामासन्नानि भवन्ति हि ॥ ३ ॥

The method of mean-elimination: “The divisor-measure of the approximate value, added to the non-obtained quantity and the additive, being complete by two numbers with difference, then the divisor-fraction [gives] the effort. The difference of two approximate values alone [suffices]. The numerator [is] always unity [at first], thus by the intelligent. For all approximate values, the divisor-fraction is six. Thus gradually, more and more subtle approximate values are obtained.”

Concluding Verses of the Bhāvitam Section (pp. 151–152)

तथा गोले मयोक्तम् ।
उत्थ सदसतां मर्तानां वैराशिकमात्रमेव पाठो बुद्धिरेव बीजम् ।
तथा गोलाख्याय मयोक्तम् ।
अस्ति वैराशिकं पाठी बीजं च विमला मतिः ।
किमन्यत् सुखदीनामतो मन्दाऽर्थमुच्यते ॥
गणककर्मान्तरमयं बाललोचनगम्यं
सकलगणितसारं सोपपट्टिप्रारम् ।
इति बहुगुणयुक्तं सर्वदोषैर्विमुक्तं
पठ पठ मतिबुद्ध्यै लाघवं प्र ढिसिद्ध्यै ॥

The epilogue: “Thus I have spoken of the sphere. For people ascending from the unreal to the real, the mere reading [of this text] is the seed of intelligence. Thus I have spoken of the sphere called ‘Gola’. There is the study of the text and pure intellect — what else [is needed]? For the delight of the wise, this is declared. This work accessible to the eye of a child [yet] profound, the essence of all mathematics with proofs endowed, endowed with many virtues, free from all faults — read, read with intelligence and understanding, for ease and the attainment of maturity.”

इति श्रीभास्कराचार्यविरचिते सिद्धान्तशिरोमणौ
बीजगणिताध्यायः समाप्तः ॥

“Thus ends the Bījagaṇita-adhyāya in the Siddhāntaśiromaṇi composed by Śrī Bhāskarācārya.”

वि०—इति कूपाङ्कनमु । करे यदिह भास्करबीजमपूर्वकम् ।
तदुपनिषमतीव चमत्कृतिं विभिधरस्य चकार च कारणम् ॥

Editor’s note: “That unprecedented algebra of Bhāskara which here [appears] with the face of a well-digit-mine [i.e., a numerical source], having ascertained it with the Upaniṣad [of commentary], the cause of the appearance of manifold wonder.”

4 Final Sections and Publisher's Catalogue (pp. 153–170)

4.1 Navīna-mati-viśeṣaḥ (pp. 153–157)

The section on “Special Methods for New Understanding” presents advanced techniques for solving equations through approximation and iteration, building on the methods developed earlier.

नवीनमतिविशेषः ॥

निरक्ष पदं यद्विगुणात् स्यात् फलार्थं धनार्थं तदेवात्र शेषं तदस्यम् ।

पदद्वयं धनं शेषहतप्रमन्यत् फलं तद्भवत् शेषमूलं धनेन ॥ १ ॥

धनार्थं नवं तस्य कृत्या विहीनो गुणः शेषमकोऽस्यवर्णस्य मानम् ।

मूद्गुसेव मते यदा शेषमानं भवेद्वितीयं तदा लब्धितौ ते ॥ २ ॥

Rule 1: “A number, when doubled, becomes the root; for the fruit [result], for wealth [coefficient], that same residue here — its two terms [are] the wealth divided by the residue; the fruit [is] the root of the residue multiplied by the wealth. The new wealth is diminished by nine times its square; the multiplier, diminished [thus], [gives] the measure of the unknown. When by the opinion of Mūdgu [i.e., by the method of approximation], the residue-value becomes the second [term], then the quotient for both.”

4.2 Worked Examples on Special Methods (pp. 157–159)

उदाहरणानि ॥

(१) कल्प्यते $अ+क=५७६०$, $अ-क=\frac{५}{३}$ तदा $अ=३८४०$,

$क=१९२०$ इति क्षिप्रम् ॥

(२) यदि $\frac{३क+१}{३} = \frac{७क+५}{६}$ तदा $कमानम्=१$ इति क्षिप्रम् ॥

(३) यदि $\frac{क+१}{३} + \frac{३क-४}{५} + \frac{१}{६} = \frac{६क+७}{६}$, तदा $क=२०$ इति क्षिप्रम् ॥

(४) $(क + \frac{५}{६})(क - \frac{५}{६}) - (क + ५)(क - ३) + \frac{३}{६} = ०$

तदा $क=१२$ इति क्षिप्रम् ॥

4.3 The Method of Approximate Roots (pp. 160–165)

This section presents the *āsanna-mūla* (approximate root) method — an iterative technique for finding successively better approximations to irrational square roots, a precursor to what is now known as Newton's method.

आसन्नमूलानि सूत्राणि ॥

आसन्नमानस्य हराशमानम् अप्राप्तिगुण्ये सहिते क्षेपेण ।

पूर्णे स्याताम् अन्तराशिकाभ्यां तदा हरांशो भवतीश्रमस्य ॥ १ ॥

The mathematical procedure is as follows: given a non-square number n , to find $\sqrt{n}$:

1. Choose an initial approximation a_1 (the “approximate value”)
2. Compute the residue: $r = n - a_1^2$
3. Form the next approximation: $a_2 = a_1 + \frac{r}{2a_1}$
4. Iterate for successively better approximations

कल्प्यते—

(१) ०, अ, अ', अ'',

(२) १, सो, सो', सो'',

(३) अ, क_१, क_२, क_३, क_४,

अत्र क_१, क_२, क_३,—अयववदासन्नमूलस्यासन्नमानानि $\frac{प}{ल}$, $\frac{प'}{ल'}$, $\frac{प''}{ल''}$ चेति ।

$$\sqrt{n} = \frac{\frac{\sqrt{n+अ_1}}{सो} \cdot प + प}{\frac{\sqrt{n+अ_1}}{सो'} \cdot ल + ल} = \frac{प(\sqrt{n+अ_1}) + सो'प}{ल(\sqrt{n+अ_1}) + सो'ल}$$

4.4 Final Concluding Verses (pp. 166–168)

The treatise concludes with verses that reflect on the nature of mathematical knowledge and pay homage to the tradition:

इति श्रीभास्कराचार्यविरचिते सिद्धान्तशिरोमणौ
बीजगणिताध्यायः समाप्तः ॥

अस्ति वैराशिकं पाठी बीजं च विमला मतिः ।
किमन्यत् सुखदीनामतो मन्दाऽर्थमुच्यते ॥

गणककर्मान्तरमयं बाललीलावनोगम्यं
सकलगणितसारं सोपपट्टिप्रारम् ।
इति बहुगुणयुक्तं सर्वदोषैर्विमुक्तं
पठ पठ मतिबुद्ध्यै लाघवं षड्विद्विद्यै ॥

“There is the study of this work and pure intellect — what else [is needed]? Thus is declared for the delight of the wise. This [work], accessible as a child’s play [yet] the essence of all mathematics with demonstrations, endowed with many virtues, free from all faults — read, read with intelligence and understanding, for ease and the attainment of maturity.”

The author reflects on the completion of the work:

तथा गोले मयोक्तम् ।
उत्थ सदसतां मर्तानां वैराशिकमात्रमेव पाठो बुद्धिरेव बीजम् ।
तथा गोलाख्याय मयोक्तम् ॥

“Thus I have spoken of the sphere. For people ascending from the unreal to the real, the mere reading [of this text] is the seed of intelligence. Thus I have spoken of what is called the sphere.”

Colophon

इति श्रीभास्कराचार्यविरचिते सिद्धान्तशिरोमणौ
बीजगणिताध्यायः समाप्तः ॥

Editor's Note (p. 168)

वि०—इति कूपाङ्कनमु ।करे यदिह भास्करबीजमपूर्वकम् ।
तदुपनिषमतीव चमत्कृतिं विभिधरस्य चकार च कारणम् ॥

“That unprecedented algebra of Bhāskara, which here [appears] at the mouth of the well-digit-mine, having ascertained it with the Upaniṣad [of commentary], became the cause of the appearance of manifold wonder.”

The Bhāvanā (Composition) Method

अथ भवना। एषासंवर्गणवदिःवर्गप्रकृतसिमीकरणे परमोपयुक्ता। अनेन वधिनाद्वय्यांज्जसमूलम्यां नवमूले सधियेते ।

भावना ॥

ज्येष्ठद्वयकृतिर्भक्ता क्षेपयोगेन यत्र तत् ।

कनिष्ठद्वयकृतिस्तत्र भवनेयं प्रचक्षते ॥

ज्येष्ठं ज्येष्ठक्षयोर्मूलं क्षेपयोगसमन्वितम् ।

कनिष्ठकनिष्ठयोगाभ्यां क्षुद्रकं तत्समन्वितम् ॥

ज्येष्ठोन्मूलं तु यत् प्रोक्तं तत्क्षेपयोगसमन्वितम् ।

कनिष्ठोन्मूलं सम्प्रोक्तं तत्क्षेपयोगसमन्वितम् ॥

भावनायाः प्रकाराः। There are four types of bhāvanā:

- (i) **समभावना** — Composition of two solutions with the same interpolator
- (ii) **अन्तरभावना** — Composition with difference (cross-multiplication)
- (iii) **तुल्यभावना** — Composition of a solution with itself
- (iv) **विशेषभावना** — Special composition for specific cases

Given (x_1, y_1, k_1) and (x_2, y_2, k_2) such that $Nx_i^2 + k_i = y_i^2$:

$$\text{नवज्येष्ठं } Y = y_1y_2 + Nx_1x_2$$

$$\text{नवकनिष्ठं } X = x_1y_2 + x_2y_1$$

$$\text{नवक्षेपः } K = k_1 \cdot k_2$$

The Key Property of Bhāvanā

क्षेपक्षेपयुतो राशिः क्षेपक्षेपहतो रसः ।

ज्येष्ठं ज्येष्ठक्षयोर्मूलं क्षेपयोगसमन्वितम् ॥

कनिष्ठकनिष्ठयोगाभ्यां क्षुद्रकं तत्समन्वितम् ।

क्षेपयोगश्च तद्वेष्टो भावनेयं प्रचक्षते ॥

अन्तरभावना। When one interpolator is positive and the other negative:

$$X = |x_1y_2 - x_2y_1|, \quad Y = |y_1y_2 - Nx_1x_2|, \quad K = -k_1k_2$$

॥ श्रीः ॥

लीलावती

श्रीभास्कराचार्यविरचिता

Līlāvatī of Bhāskarācārya

Bilingual Sanskrit–English Edition

Treatise on Arithmetic, c. 1150 CE

संस्कृत मूलपाठ
Sanskrit Original



English Translation
with IAST Transliteration

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Preface / प्रस्तावना

लीलावती भास्कराचार्यस्य सिद्धान्तशिरोमणेः प्रथमः भागः। श्रीभास्करेण ११५० ई.पू.रचिता। लीलावती नाम तस्य कन्यायाः नाम अस्ति। इयं ग्रन्थः भारतीयगणितस्य सर्वाधिकप्रसिद्धपाठ्यपुस्तकम् अस्ति। पूर्णाङ्केषु, भिन्नेषु, अनुपातेषु, व्याजेषु, क्षेत्रफलेषु, घनफलेषु, श्रेणीषु च गणितविषयान् व्यवस्थापयति।

līlāvātī bhāskarācāryasya siddhāntaśiromaṇeḥ prathamah bhāgaḥ | śrībhāskareṇa 1150 ī.pū.racitā | līlāvātī nāma tasya kanyāyāḥ nāma asti | iyāṃ granthaḥ bhāratīyagaṇitasya sarvādhikaprasiddhapāṭhyapustakam asti |

The *Līlāvātī* is the first section of Bhāskara II's monumental work *Siddhāntaśiromaṇi*, composed in 1150 CE. Named after Bhāskara's daughter, it is the most celebrated textbook of Indian arithmetic. The work covers the entire field of elementary mathematics: arithmetic operations with integers and fractions, rules of proportion, interest calculation, plane geometry, solid geometry, and combinatorial problems.

अस्य ग्रन्थस्य प्रस्तावने — अत्र गणितस्य समग्रं क्षेत्रम् उपस्थापितम् अस्ति। सङ्कलनम्, व्यवकलनम्, गुणनम्, भागहारः, वर्गः, वर्गमूलम्, घनः, घनमूलम् — एतेषां परिकर्मणां व्यवस्था, भिन्नपरिकर्म, त्रैराशिकम्, मिश्रप्रश्नाः, क्षेत्रव्यवहारः, खातव्यवहारः, चितिव्यवहारः, छायाव्यवहारः, कुट्टकव्यवहारः, अङ्कपाशः, श्रेणी चेत्यादयः अध्यायाः सन्ति।

asya granthasya prastāvane — atra gaṇitasya samagraṃ kṣetram upasthāpitam asti | saṅkalanam, vyavakalanam, guṇanam, bhāgaḥārah, vargaḥ, vargamūlam, ghaṇaḥ,ghanamūlam — eteṣāṃ parikarmaṇāṃ vyavasthā, bhinnaparikarma, trairāśikam, miśrapraśnāḥ, kṣetravyavahārah, khātavyavahārah, citivyavahārah, chāyāvyavahārah, kuṭṭakavyavahārah, aṅkapāśaḥ, śreṇī cetyādayaḥ adhyāyāḥ santi |

The present edition reproduces the Sanskrit text in the left column with an English translation and IAST transliteration in the right column. Key technical terms are preserved in their original Sanskrit form with explanatory footnotes.

मङ्गलाचरण — Invocatory Verses

लीलागुल्लक्षोल्कालव्यालविलसिने ।
गणेशाय नमो नीलकमलामलकान्तये ॥
॥ १ ॥

Līlāgullakṣolkāvyālavilasine

Gaṇeśāya namo nīlakamalāmalakāntaye

Salutation to Gaṇeśa, the playfully sportive one, having the brightness of the rising sun, and possessing the radiance of a blue lotus.

अवतार पुरा पुराणेषां यदुकुले जगतीहितेहिने ।
जयति सोऽयमिमां सकलां लीलावतीं मामपि सिद्धमहीपतिः ॥

avatāra purā purāṇeṣāṃ yadukule jagatīhitehine |

jayati so'ayamimāṃ sakalāṃ līlāvatīṃ māmapi siddhamahīpatiḥ ||

In ancient times, he who took incarnation in the Yadu clan for the welfare of the world — may that same king, the lord of the earth, be victorious, who has made me compose this entire *Līlāvātī*.

भास्करचितयमथा सिद्धान्तशिरोमणिप्रख्याः ।
तदयं कृताऽनन्ये व्याकृतया मन्मटे नियतं ॥
॥ १५ ॥

Bhāskaracitayamathā siddhāntaśiromaṇiprakhyāḥ

Tadayaṃ kṛtā'nye vyākṛtayā manmaṭhe niyatam

This *Līlāvātī*, composed by Bhāskara, is the foremost part of the *Siddhāntaśiromaṇi*. This has been composed by none other than him, and is now set down by me.

1 परिभाषामकरण — Definitions and Units of Measurement

पादोनग्रानकतुल्यष्टकैः समतुल्यैः कथितोऽत्र सेरः ॥
मणाभिधानं व्युगमैः सैरेधार्यादितोल्येषु तुष्टकसङ्ख्या ॥ १ ॥

pādonagrānakatulyaṣṭakaiḥ samatulyaiḥ kathito'tra serah ॥
maṇābhidhānaṃ vyugmaiḥ sairedhāryāditolyeṣu tuṣṭakasaṅkhyā ॥ 1 ॥

A *ser* is equal to eight *pāda* or equal to eight grains. A *maṇa* is twice the amount of a *ser* in weighing gold, etc. [*The ser and maṇa were traditional Indian units of weight. 1 maṇa = 2 ser.*]

द्वयकद्विसहस्रैर्घटकैः सैरस्तैः पञ्चभिः स्याद्धटिका च ताभिः ॥
मणोऽष्टमित्वालमगीरशाहकृतास्त्र सङ्ख्या निजराज्यपृष्ठे ॥ २ ॥

dvayakadvisahasrairghṭakaiḥ sairastaiḥ pañcabhiḥ syāddhaṭikā ca tābhiḥ ॥
maṇo'ṣṭamirtvālamagīraśāhakṛāstra saṅkhyā nijarājyapṛṣṭhe ॥ 2 ॥

Two thousand *ghaṭakas* make a *ser*; five of these *ser*s make a *ghaṭikā*. The weight of one *maṇa* in this kingdom (of the author's patron) follows this system.

1.1 शोभाः कालादिपरिभाषा — Definitions of Time

शोभाः कालादिपरिभाषा लोकतः प्रसिद्धा देया ॥

śobhāḥ kālādiparibhāṣā lokataḥ prasiddhā deyā ॥

The definitions of time and other measures are well-known in the world. The traditional Indian system of time measurement is given as:

- 60 *vipalas* = 1 *pala*
- 60 *palas* = 1 *ghaṭikā* (24 minutes)
- 60 *ghaṭikās* = 1 day and night
- 15 days and nights = 1 *pakṣa* (fortnight)
- 2 *pakṣas* = 1 month
- 12 months = 1 year

इति परिभाषा — Thus end the definitions.

2 सङ्ख्याप्रकरण — Arithmetic Operations on Whole Numbers

This chapter covers the fundamental operations of arithmetic: addition (*saṅkalita*), subtraction (*vyavakalita*), multiplication (*guṇana*), division (*bhājana*), squaring (*varga*), square roots (*vargamūla*), cubing (*ghana*), and cube roots (*ghanamūla*).

2.1 सङ्कलित व्याकलित — Addition and Subtraction

सूत्रम्: यथोक्तं परिकल्प्य गणितं कुर्यात् ।

Rule: As taught, having set up the numbers [aligned by place value], one should compute.

2.2 गुणकारकरणसूत्र — Multiplication

सूत्रम्: समद्विघातः कृतिः उच्यते —

Rule: A product of two equal numbers is called a *kṛti* (square). The multiplication of two different numbers is called *guṇana*.

अथ रूपयोऽन्त्यवर्गो द्विगुणान्त्यनिघ्नः...

एकस्यांकस्य वर्गः अन्तिमस्य अङ्कस्य गुणनम् । द्विगुणान्त्यनिघ्नः = द्विगुणः अन्तिमः अङ्कः अन्येन गुणितः ।

atha rūpayo'ntyavargo dviguṇāntyanighnaḥ...

ekasyāṅkasya vargaḥ antim asya aṅkasya guṇanam |

Method of Multiplication: The square of the last figure, and twice the last multiplied by the preceding, and so on — this is the method for squaring.

Example: To multiply 135 by 12:

$$135 \times 12 = 135 \times 10 + 135 \times 2 = 1350 + 270 = 1620$$

2.3 भागहारकरणसूत्र — Division

सूत्रम्: भक्तः क्षेपणात् पुनः पुनरवाप्तिर्भागहारः ।

Rule: Division is the repeated subtraction of the divisor from the dividend until the remainder is less than the divisor.

Division is performed by repeated subtraction. The commentary gives detailed worked examples. The quotient (*labdhi*) is the number of subtractions, and what remains is the remainder (*śeṣa*).

2.4 वर्गीकरणसूत्र — Squaring

सूत्रम्: समद्विघातः कृतिः उच्यते —

Rule: The product of two equal numbers is called a *kṛti* (square).

खण्डाभ्यां वा हतो राशिरिष्टः खण्डचतुष्कयुक् ॥

अस्य अर्थः — राशिं द्वयोः खण्डयोः योगेन गुणित्वा तयोरन्तरवर्गेण सह योजयेत् ।

khaṇḍābhyāṃ vā hato rāśiriṣṭaḥ khaṇḍacatuṣkayuk ||

asya arthaḥ — rāśim dvayoḥ khaṇḍayoḥ yogenaguṇitvā tayorantaravargeṇa saha yojayet |

Method 1: Direct multiplication $n \times n$

Method 2: Using the formula $(a + b)^2 = a^2 + 2ab + b^2$

Method 3: Using the formula $n^2 = (n + a)(n - a) + a^2$

Or, the number multiplied by the component parts of itself, combined with the square of the difference...

2.5 वर्गमूलकरणसूत्र — Square Root Extraction

सूत्रम्: स्थाप्योऽन्त्यवर्गो द्विगुणोऽन्त्यनिघ्नः स्यादपरिच्छेदः।

Rule: Set down the number. Find the largest square in the leftmost group. This gives the first digit of the root. Subtract its square, bring down the next group, divide by twice the current root, and repeat.

अथ स्थाप्योऽन्त्यवर्गो द्विगुणान्त्यनिघ्नः ॥
स्यादपरिच्छेदः तथा परेऽभ्यस्तयुक्त्वान्त्यमपि राशिम् ॥

*atha sthāpyo'ntyavago dviguṇāntyanighnaḥ ॥
syādaparicchedaḥ tathā pare'bhyastayuktvāntyamapi rāsim ॥*

Example: Find $\sqrt{65536}$

Step 1: Group the digits from right: $\overline{6} \overline{55} \overline{36}$

Step 2: Largest square ≤ 6 is $4 = 2^2$. First digit: 2.

Step 3: Remainder 2, bring down 55 $\rightarrow 255$.

Step 4: $255 \div (2 \times 20) = 255 \div 40 \approx 5$.

Step 5: Verify: $45 \times 5 = 225$. Remainder 30.

Step 6: Bring down 36 $\rightarrow 3036$.

Step 7: $3036 \div (2 \times 250) = 3036 \div 500 \approx 6$.

Step 8: Verify: $506 \times 6 = 3036$. Remainder 0.

Therefore $\sqrt{65536} = 256$.

2.6 घनकरणसूत्र — Cubing

सूत्रम्: समन्त्रिघातः घनः उच्यते—

Rule: The product of a number multiplied three times by itself is called a *ghana* (cube).

Methods for cubing:

Method 1: Direct multiplication $n \times n \times n$

Method 2: Using $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$

2.7 घनमूलकरणसूत्र — Cube Root Extraction

The method follows a similar place-value decomposition as square root extraction, but using cubes of digits.

Example: $\sqrt[3]{1728} = 12$, since $12^3 = 1728$.

3 भिन्नपरिकर्म — Operations with Fractions

3.1 जातिचतुष्टय — Four Classes of Fractions

Bhāskara defines four types of fractions (*jāticatuṣṭaya*):

1. भिन्न (*bhinna*) — Simple fraction, e.g., $\frac{1}{2}$
2. प्रभाग (*prabhāga*) — Part of a part, e.g., $\frac{1}{2}$ of $\frac{1}{3}$
3. भागानुबन्ध (*bhāgānubandha*) — Fraction combined with an integer, e.g., $2 + \frac{1}{3}$
4. भागापवाह (*bhāgāpavāha*) — Fraction less than an integer, e.g., $2 - \frac{1}{3}$

3.2 भिन्नगुणन भिन्नभाग — Multiplication and Division of Fractions

सूत्रम्: नरघ्ननरके क्रमशो गुणयेत् तद् भागहारकैः ॥

Rule: Multiply the numerator by the numerator and the denominator by the denominator for multiplication; for division, invert and multiply.

Multiplication: $\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}$

Division: $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{a \times d}{b \times c}$

Example: $\frac{2}{3} \times \frac{5}{7} = \frac{10}{21}$

Example: $\frac{2}{3} \div \frac{5}{7} = \frac{2 \times 7}{3 \times 5} = \frac{14}{15}$

4 त्रैराशिक — The Rule of Three

The *trairāśika* (Rule of Three) is one of the most important topics in Indian arithmetic. It solves problems of the form:

$$\text{“If } A \text{ gives } B, \text{ what does } C \text{ give?”} \quad \text{Answer: } \frac{B \times C}{A}$$

4.1 इच्छाकर्मप्रकार — Direct Rule of Three

सूत्रम्: प्रमाणफलमिच्छाफलं चेत् कुर्यात् ।

Rule: Multiply the *phala* (fruit/result) by the *icchā* (desire) and divide by the *pramāṇa* (measure).

प्रमाणं फलमिच्छा च त्रिविधा राशयः स्मृताः ॥

प्रमाणेन यद् लभ्यते तत् फलम् । इच्छया यद् ज्ञातुम् इष्टम् तत् इच्छाफलम् ।

pramāṇaṃ phalamicchā ca trividhā rāśayaḥ smṛtāḥ ॥

Three quantities are remembered: the *pramāṇa* (measure), the *phala* (fruit), and the *icchā* (desire).

Example: If 5 books cost 60 rupees, what do 8 books cost?

$$\text{Price} = \frac{60 \times 8}{5} = 96 \text{ rupees}$$

4.2 व्यस्तत्रैराशिक — Inverse Rule of Three

सूत्रम्: व्यस्ते विपरीतं कुर्यात् ।

Rule: In the inverse case, do the reverse [multiply measure by fruit and divide by desire].

When the relationship is inverse (more workers take less time, etc.):

$$\text{Result} = \frac{B \times A}{C}$$

Example: If 8 workers complete a job in 15 days, how many days will 12 workers take?

$$\text{Days} = \frac{8 \times 15}{12} = 10 \text{ days}$$

4.3 पञ्चराशिक सप्तराशिक — Extended Rules

Extended rules for compound proportions:

- पञ्चराशिक (*pañcarāśika*, Five-fold rule): Two intermediate terms
- सप्तराशिक (*saptarāśika*, Seven-fold rule): Three intermediate terms
- नवराशिक (*navarāśika*, Nine-fold rule): Four intermediate terms

5 मिश्रप्रकरण — Mixed Problems

5.1 शैलाराशिकविधि — Rule of Alligation

मिश्रणे विविधद्रव्याणां मूल्यानां समासेन...
सुवर्णादीनां मिश्रणे कलर्णज्ञानाय एषः प्रकारः ।

*miśraṇe vividhadravyāṇāṃ mūlyānām samāseṇa...
suvārṇādīnām miśraṇe kalaṇajñānāya eṣaḥ prakāraḥ |*

Problems involving the mixing of different quantities at different prices or qualities.

Example: If rice of quality A costs 8 rupees/kg and rice of quality B costs 12 rupees/kg, in what ratio should they be mixed to get a mixture worth 10 rupees/kg?

$$\text{Ratio} = (12 - 10) : (10 - 8) = 2 : 2 = 1 : 1$$

5.2 वापीपूरणप्रकार — Filling of Tanks

वापी पूरयितुं नलकैः कालज्ञानाय सूत्रम् ॥

vāpī pūrayitum nalakaiḥ kālajñānāya sūtram ||

Example: A tank has three pipes. The first can fill it in 5 hours, the second in 6 hours, and the third can empty it in 4 hours. If all three are opened together, how long will it take to fill the tank?

$$\text{Net fill rate} = \frac{1}{5} + \frac{1}{6} - \frac{1}{4} = \frac{12 + 10 - 15}{60} = \frac{7}{60}$$

$$\text{Time to fill} = \frac{60}{7} = 8\frac{4}{7} \text{ hours.}$$

5.3 कथाविकयविधि — Exchange of Goods

वस्तूनां विनिमये मूल्यसमत्वेन परस्परगुणनम् ॥

vastūnām vinimaye mūlyasamatvena parasparaguṇanam ||

Example: If 3 mangoes can be exchanged for 4 oranges, and 5 oranges for 2 apples, how many apples can be obtained for 15 mangoes?

$$15 \text{ mangoes} = 15 \times \frac{4}{3} = 20 \text{ oranges} = 20 \times \frac{2}{5} = 8 \text{ apples}$$

6 क्षेत्रव्यवहार — Plane Geometry

क्षेत्रे व्यवहारः —

॥ ॥

kṣetre vyavahārah

This chapter deals with the calculation of areas of various plane figures and the properties of right triangles.

6.1 क्षेत्रसाधारणसूत्र — General Rule for Areas

सूत्रम्: दैर्घ्यमायतं विस्तारं च तद् दैर्घ्यविस्तारयोगजं क्षेत्रम् ।

Rule: The area of a rectangle is length multiplied by breadth. The area of a figure is the product of its length and width.

6.2 भुजकोटिकर्णनयन — The Pythagorean Theorem

भुजे द्वादशके यो यो कोटिकर्णविनेकथा ॥
प्रकाराभ्यां वद विप्र तो तावकर्णीगतौ ॥ ५ ॥
॥ ५ ॥

*bhuje dvādaśake yo yo koṭīkarvavinekathā ॥
prakārābhyāṃ vada vipra to tāvakarṇīgatāu ॥ 5 ॥*

O learned Brahmin! The sides (*bhujā*) being 12, tell me quickly the hypotenuse (*karṇa*) and perpendicular (*koṭi*) by both methods.

इष्टयोराहतिर्द्वयोः कोटिवर्गान्तरं भुजः ॥
कृतियोगस्तयोरेव कर्णस्त्वकर्णीगतः ॥ ८ ॥
॥ ८ ॥

*iṣṭayorāhatirdvayoḥ koṭivargāntaram bhujah ॥
kr̥tiyogastayorevaṃ karṇastvakaraṇīgataḥ ॥ 8 ॥*

The product of the two desired [sides] is the difference of the squares of the *koṭi* and *bhujā*. The sum of the squares of those two gives the *karṇa*.

The Three Sides of a Right Triangle:

- **Base** (भुजा *bhujā*): The horizontal side
- **Perpendicular** (कोटि *koṭi*): The vertical side
- **Hypotenuse** (कर्ण *karṇa*): The diagonal side

The rule states:

$$\text{karṇa}^2 = \text{bhujā}^2 + \text{koṭi}^2$$

This is the famous Pythagorean theorem, known in India as the *Śulba-sūtra* theorem.

Example: Find the hypotenuse when base = 3 and perpendicular = 4:

$$\text{karṇa} = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$$

Example from the text: For base = 12, perpendicular = 5:

$$\text{karṇa} = \sqrt{12^2 + 5^2} = \sqrt{144 + 25} = \sqrt{169} = 13$$

Example: For base = 9, perpendicular = 12:

$$\text{karṇa} = \sqrt{9^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225} = 15$$

6.3 कोटिकर्णज्ञान — Finding the Koṭi

When the hypotenuse and one side are known:

$$\text{koṭi} = \sqrt{\text{karṇa}^2 - \text{bhujā}^2}$$

Example: karṇa = 85, bhujā = 36:

$$\text{koṭi} = \sqrt{85^2 - 36^2} = \sqrt{7225 - 1296} = \sqrt{5929} = 77$$

Verification: $36^2 + 77^2 = 1296 + 5929 = 7225 = 85^2$ ✓

6.4 वृत्तक्षेत्र — Circle

सूत्रम्: व्यासवर्गपादं भुजवर्गस्य पञ्चगुणस्य चतुर्थांशः वृत्तफलम् ।

Rule: The area of a circle is one-fourth the square of the diameter, or equivalently: the square of the radius multiplied by the ratio of circumference to diameter.

$$A = \frac{d^2}{4} \times \pi \approx \frac{d^2}{4} \times \frac{3927}{1250} \text{ (Bhāskara's refined value)}$$

Bhāskara uses $\pi \approx \frac{22}{7}$ for rough calculations and $\pi \approx \frac{3927}{1250} = 3.1416$ for more precise work.

7 खातव्यवहार — Excavations

खातव्यवहारः —

॥ ॥

khātavyavahārah

This chapter deals with calculations related to excavations, trenches, and tanks.

खातस्य फलं दीर्घं विस्तारं गभीरता च तेषां गुणनफलम् ॥

khātasya phalaṃ dīrghaṃ vistāraṃ gabhīratā ca teṣāṃ guṇanaphalam ॥

Problem: An excavation is to be made with length 30, breadth 20, and depth 8. Find the volume of earth to be removed.

$$V = l \times b \times d = 30 \times 20 \times 8 = 4800 \text{ cubic units}$$

7.1 सच्छिद्रखात — Excavations with Sloping Sides

For excavations with sloping sides, the *śāṇi* (average) method is used:

$$V = \frac{A_1 + A_2 + A_3 + A_4}{4} \times \text{length}$$

where A_1, A_2, A_3, A_4 are the cross-sectional areas at different depths.

8 चित्ति व्यवहार — Piles of Bricks

चित्तिव्यवहारे स्तूपचितीनां समचितीनां फलं यथासङ्ख्यम् ॥

citivyavahāre stūpacitīnām samacitīnām phalaṃ yathāsaṅkhyam ॥

This chapter deals with stacking problems and the calculation of volumes of piles.

Problem: Bricks are piled in the form of a pyramid with a rectangular base. The base has 10 bricks along one side and 6 along the other. The pile is 5 layers high. Find the total number of bricks.

Solution: Using the formula for a pyramidal pile:

$$N = \frac{n(n+1)(2n+1)}{6} \text{ (for square pyramids)}$$

For rectangular piles, the sum is computed layer by layer.

9 छायाव्यवहार — Shadow Problems

छायाव्यवहारकथन —

॥ ॥

chāyāvyaṭhārakathana

This chapter uses the properties of similar triangles to solve problems involving shadows, heights of objects, and distances.

सूत्रम्: छायाग्रदूरं द्विजोत्कर्षं कुर्यात् ।

Rule: The distance to the tip of the shadow multiplied by the height of the object, divided by the length of the shadow, gives the height of the light source.

दीपस्य स्तम्भस्य ऊर्ध्वे दीपः स्थापितः । तस्य छायायां पुरुषः स्थितः ।

dīpasya stambhasya ūrdhve dīpaḥ sthāpitaḥ | tasya chāyāyām puruṣaḥ sthitaḥ |

Problem: A lamp is placed on a pillar. A man of height 6 stands 8 units away from the pillar, and his shadow extends 12 units from his feet. Find the height of the lamp.

Solution: Using similar triangles:

$$\frac{h}{6} = \frac{8 + 12}{12} = \frac{20}{12}$$

$$h = 6 \times \frac{20}{12} = 10 \text{ units}$$

10 कुट्टकव्यवहार — The Pulverizer

कुट्टकव्यवहारः —

॥ ॥

kuṭṭakavyavahārah

The *kuṭṭaka* (pulverizer) is the method for solving linear Diophantine equations of the form $ax + by = c$, where a , b , and c are given integers, and x , y are unknown integers to be found.

सूत्रम्: द्विच्छेदः कुट्टकः—

Rule: The pulverizer is the division of two (numbers), producing a quotient and remainder, used to find the GCD through the Euclidean algorithm.

कुट्टके स्थिरकुट्टकं प्रवर्तते । द्वयोः राशयोः महत्तमं समवायपदं ज्ञात्वा...

कुट्टकविधिः युक्लिडीयकलनविधिना सम्बद्धः अस्ति ।

kuṭṭake sthira kuṭṭakaṁ pravartate | dvayoḥ rāśyoḥ mahattamaṁ samavāyapadaṁ jñātvā... ||

Problem: Find x and y such that $121x + 17 = y$

Solution: Using the *kuṭṭaka* method:

Step 1: Apply the pulverizer to 121 and 17:

$$121 = 7 \times 17 + 2$$

$$17 = 8 \times 2 + 1$$

Step 2: Work backwards:

$$\begin{aligned} 1 &= 17 - 8 \times 2 \\ &= 17 - 8 \times (121 - 7 \times 17) \\ &= 57 \times 17 - 8 \times 121 \end{aligned}$$

So $x = 8$, $y = 121 \times 8 + 17 = 985$ gives a solution.

11 अङ्कपाश — Combinatorial Problems

अङ्कपाशः —

॥ ॥

aṅkapāśaḥ

This chapter contains problems related to permutations and combinations, and number puzzles. The Sanskrit term *aṅkapāśa* literally means “net of digits” — the entanglement of numbers.

11.1 अङ्कपाशविधि — Digit Arrangements

एकादशअङ्कैः कति विभिन्नसंख्याः रचयितुं शक्याः ?

ekādaśaṅkaiḥ kati vibhinnasāṅkhyāḥ racayitum śakyāḥ ?

Problem: Using the digits 1, 2, 3, 4, how many different 4-digit numbers can be formed?

Solution: $4! = 24$ different numbers.

This is the fundamental counting principle — each position can be filled by any of the remaining digits.

11.2 अङ्कपाशमै विशेषविधि — Special Methods

एका संख्या त्रिगुणीकृता सप्तभक्ता द्विशेषं लभते ।

ekā saṅkhyā triguṇīkṛtā saptabhaktā dvīśeṣaṁ labhate |

Problem: A certain number, when multiplied by 3 and divided by 7, gives a remainder of 2. Find the number. This is equivalent to solving:

$$3x \equiv 2 \pmod{7}$$

Since $3 \times 5 = 15 \equiv 1 \pmod{7}$, the multiplicative inverse of 3 mod 7 is 5.

$$x \equiv 2 \times 5 = 10 \equiv 3 \pmod{7}$$

So $x = 3 + 7k$ for any integer k . The smallest positive solution is $x = 3$.

12 श्रेणी — Progressions

श्रेणी —

॥ ॥

śreṇī

The *śreṇī* (progression) chapter covers arithmetic and geometric progressions.

12.1 समश्रेणी — Arithmetic Progression

समश्रेण्यां प्रथमपदं चयं गच्छं च ज्ञात्वा सर्वं ज्ञायते ॥

समश्रेण्यां (arithmetic progression) प्रथमं पदम् a , चयः (common difference) d , गच्छः (number of terms) n चेत् —

samaśreṇyāṃ prathamapadaṃ cayam gacchaṃ ca jñātvā sarvaṃ jñāyate ॥

For an arithmetic progression with first term a , common difference d , and n terms:

- n -th term: $a_n = a + (n - 1)d$
- Sum of n terms: $S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l)$

where l is the last term.

12.2 गुणोत्तरश्रेणी — Geometric Progression

गुणोत्तरश्रेण्यां प्रथमपदं गुणोत्तरं च ज्ञात्वा सर्वं ज्ञायते ॥

guṇottaraśreṇyāṃ prathamapadaṃ guṇottaraṃ ca jñātvā sarvaṃ jñāyate ॥

For a geometric progression with first term a and common ratio r :

- n -th term: $a_n = a \cdot r^{n-1}$
- Sum of n terms: $S_n = a \cdot \frac{r^n - 1}{r - 1}$

The Chessboard Problem:

Problem: A king offers a reward of 1 grain of rice on the first square of a chessboard, 2 on the second, 4 on the third, and so on, doubling each time. Find the total grains on all 64 squares.

Solution:

$$\begin{aligned} S_{64} &= 1 + 2 + 4 + \cdots + 2^{63} = \frac{2^{64} - 1}{2 - 1} = 2^{64} - 1 \\ &= 18,446,744,073,709,551,615 \text{ grains} \end{aligned}$$

Appendix: Famous Problems from the Līlāvātī

The Lotus Problem (कमलसमस्या)

कमलस्य पत्रं जलात् ऊर्ध्वं स्थितम् । वायुना पीडितं जलस्य अन्तः गतम् ।

॥ ॥

kamalasya patraṃ jalāt ūrdhvaṃ sthitam | vāyunā pīḍitam jalasya antaḥ gatam |

A lotus flower stands x units above the water surface. When blown by the wind, it sinks d units from its original vertical position. Find the depth of the water.

Solution: If the lotus is x above the water, the stem length equals the depth h plus x . When blown, the stem forms the hypotenuse of a right triangle:

$$\begin{aligned}(h + x)^2 &= h^2 + d^2 \\ h^2 + 2hx + x^2 &= h^2 + d^2 \\ h &= \frac{d^2 - x^2}{2x}\end{aligned}$$

[This is a classic application of the Pythagorean theorem (*bhujā-koṭi-karṇa* relation).]

The Bee Problem (भ्रमरसमस्या)

भ्रमराणां पञ्चमांशः कुटजपुष्पे तृतीयांशः शिलीन्ध्रपुष्पे च ॥
अन्तरस्य त्रिगुणं क्रातपुष्पे एकः चन्दने विहरति ॥

॥ ॥

bhramarāṇāṃ pañcamāṁśaḥ kuṭajapuṣpe tṛtīyāṁśaḥ śilīndhrapuṣpe ca ||
antarasya triguṇaṃ krātapuṣpe ekaḥ candane viharati ||

A fifth of a swarm of bees settled on a *kuṭaja* flower, a third on a *śilīndhra* flower. Three times the difference between these two groups flew over a *krāta* flower. One bee was left flying about, attracted by the fragrance of a jasmine and a pandanus. How many bees were in the swarm?

Solution: Let x be the total number of bees.

$$\begin{aligned}\frac{x}{5} + \frac{x}{3} + 3\left(\frac{x}{3} - \frac{x}{5}\right) + 1 &= x \\ \frac{x}{5} + \frac{x}{3} + 3 \cdot \frac{2x}{15} + 1 &= x \\ \frac{3x + 5x + 6x}{15} + 1 &= x \implies \frac{14x}{15} + 1 = x \implies x = 15\end{aligned}$$

Answer: 15 bees. [This is one of the most famous problems from the Līlāvātī, demonstrating skill in working with fractions and linear equations.]

The Peacock and the Snake (मयूरसर्पकथा)

मयूरः स्तम्भे स्थितः सर्पः तस्य पादात् दूरे स्थितः ॥
मयूरः सर्पं ग्रहीतुं पतति समदूरी क्षिप्त्वा ॥

॥ ॥

mayūraḥ stambhe sthitaḥ sarpaḥ tasya pādāt dūre sthitaḥ ||
mayūraḥ sarpaṃ grahītuṃ patati samadūrī kṣiptvā ||

A peacock is perched on a pillar of height P . A snake is at distance D from the base of the pillar. The peacock pounces on the snake in a straight line, covering equal distances down the pillar and along the ground. Find where they meet.

Solution: Let the meeting point be at distance x from the base of the pillar. The peacock flies in a straight line, and the problem states it covers equal vertical and horizontal distances:

$$P - x = \sqrt{x^2 + D^2}$$

Squaring both sides:

$$(P - x)^2 = x^2 + D^2 \implies P^2 - 2Px + x^2 = x^2 + D^2$$

$$x = \frac{P^2 - D^2}{2P}$$

[This combines the Pythagorean theorem with the concept of equal distances. The problem is notable for its elegant reduction to a simple algebraic expression.]

The Swallow Problem (हंससमस्या)

Problem: Swallows are sitting on the branches of a tree. If each branch has 3 swallows, one swallow is left over. If each branch has 4 swallows, one branch is left empty. How many swallows and branches are there?

Solution: Let s = number of swallows, b = number of branches.

$$s = 3b + 1$$

$$s = 4(b - 1) = 4b - 4$$

Equating: $3b + 1 = 4b - 4$, so $b = 5$ and $s = 16$.

Answer: 16 swallows and 5 branches. *[This is a classic example of a system of linear equations with a unique solution. It is equivalent to finding x such that $x \equiv 1 \pmod{3}$ and $x \equiv 0 \pmod{4}$.]*

The Pearl Necklace Problem (मुक्ताहारसमस्या)

मुक्ताहारे मुक्ताः स्रंसन्ति । एकैकस्य मुक्तायाः मूल्यं ज्ञात्वा...

॥ ॥

muktāhāre muktāḥ sraṃsanti | ekaikasya muktāyāḥ mūlyaṃ jñātvā...

A string of pearls breaks. One third of the pearls fall to the ground, one fifth remain on the bed, one sixth are saved by the young woman, one tenth are caught by her lover, and six pearls remain on the string. How many pearls were there originally?

Solution: Let x be the total number of pearls.

$$\frac{x}{3} + \frac{x}{5} + \frac{x}{6} + \frac{x}{10} + 6 = x$$

The LCD of 3, 5, 6, 10 is 30:

$$\frac{10x + 6x + 5x + 3x}{30} + 6 = x \implies \frac{24x}{30} + 6 = x$$

$$6 = x - \frac{4x}{5} = \frac{x}{5} \implies x = 30$$

Answer: 30 pearls. *[This problem demonstrates the use of finding a common denominator and solving linear equations with fractions — a common practical problem type in the Līlāvātī.]*

Glossary of Technical Terms

संस्कृत	IAST	Meaning
व्यवहार	vyavahāra	Procedure, operation, practice
गुणन	guṇana	Multiplication
भागहार	bhāgahāra	Division
क्षेत्र	kṣetra	Field, area, plane figure
राशि	rāśi	Quantity, number, heap
सङ्कलन	saṅkalana	Addition
व्यवकलन	vyavakalana	Subtraction
वर्ग	varga	Square
वर्गमूल	vargamūla	Square root
घन	ghana	Cube
घनमूल	ghanamūla	Cube root
भिन्न	bhinna	Fraction
प्रमाण	pramāṇa	Measure (in Rule of Three)
फल	phala	Fruit, result
इच्छा	icchā	Desire, requisition
त्रैराशिक	trairāśika	Rule of Three
करण	karaṇa	Operation, method
सूत्र	sūtra	Rule, aphorism
भुजा	bhujā	Base (of triangle)
कोटि	koṭi	Perpendicular
कर्ण	karṇa	Hypotenuse, diagonal
श्रेणी	śreṇī	Progression, series
चय	caya	Common difference (in AP)
गुणोत्तर	guṇottara	Common ratio (in GP)
कुट्टक	kuṭṭaka	Pulverizer (linear Diophantine)
अङ्कपाश	aṅkapāśa	Combinatorial problems

iti līlāvatī samāptā

इति लीलावती समाप्ता ॥

Thus ends the Līlāvatī.

Brāhmasphuṭasiddhānta

English Translation Reader

Brahmagupta (628 CE)

English-only reading text. The companion Indian-original reader carries source-page coverage.

Chapter I: Mean Longitudes of Planets

Verse 1: Salutation to that Brahman, by whose grace I have again come to know the computation of planets; now I proceed to reconcile it.

Verse 2: The planetary computation taught by Brahman had become inaccurate through the lapse of long time; that has been correctly established by Brahmagupta.

Verse 3: The years of a *yuga* are its *abda*; the days of a *yuga* are its *ahargaṇa*. A year consists of solar days; a month of lunar *tithis*.

Verse 4: A *sāvāna* year consists of *nakṣatra* days; a solar year is fourfold. The full moon at the fifteenth *tithi* is complete, at the fourteenth incomplete.

Verse 5: A thousand *caturyugas* are a day of Brahman; half is night. These are called a *kalpa*.

Verse 6: At the beginning of a *kalpa* arises Brahmā, in the middle Viṣṇu, at the end Rudra: these three lords of the three worlds.

Verse 7: In that *kalpa*, six *j*
=*ivas* — the *manus* — are established; there dwell the people called *p*
=*urva*.

Verse 8: Four world-ages in a *yuga*; a thousand *caturyugas* are a day of Brahman.

Verse 9: The *kṛta*, *trētā*, *dvāpara*, and *kali* are the ages; a thousand *caturyugas* are a day of Brahman.

Verse 10: In *kṛta*, dharma has four feet; in *trētā* three; in *dvāpara* two; in *kali* one.

Verse 11: A hundred human years is a *divya* year; by such divine years, a day of Brahman is known.

Verse 12: The thousand *caturyugas* of Brahmā, Viṣṇu, Indra, Rudra, and the *manus* are a day of Brahman.

Verse 13: Humans, gods, *gandharvas*, *rākṣasas*, serpents, and ancestors — all proclaimed at the start of the *kalpa*.

Verse 14: Planets, asterisms, months, seasons, years — these twenty-five *tattvas* to be known diligently.

Verse 15: By four measures the planets are computed; I declare their numbers in order.

Verse 16: By solar measure, the Sun's motion; by lunar, the Moon's; by sidereal, the planets'; by civil, the day's.

Verse 17: By these nine measures the world's transactions proceed, for planetary computation.

Verse 18: For humans, transactions by four measures: solar, lunar, sidereal, and civil, in order.

Verse 19: All planetary computation is accomplished by measures; without them, that knowledge does not proceed.

Verse 20: The solar, lunar, sidereal, and civil measures are for planetary computation; by these four, the world's transactions proceed.

Verse 21: Fifteen *tithis* in bright and dark fortnight make one month; a year of those.

Verse 22: Twelve months a year; with thirteen, deficient; with five extra, abundant.

Verse 23: Having stated past and future motions, the wise compute; by these mean quantities, planetary computation proceeds.

Verse 24: Knowing true from mean longitudes, then corrections must be applied for exactness.

Verse 25: The *tithi*, *antara*, *nakṣatra*, and planetary motions — all devised by measures in order.

Verse 26: Thus all measures declared in order by those who contemplate planetary computation.

Verse 27: The years of a *yuga*, its months and days. A year of solar days; a month of lunar *tithis*.

Verse 28: The solar year: fourteen and a half terrestrial years; mean motion reckoned beyond.

Verse 29: Fifteen *tithis* in each fortnight; a month of those, a year of months.

Verse 30: Twelve months a year; thirteen, deficient; five extra, abundant.

Verse 31: Past and future motions stated, the wise compute by mean quantities.

Verse 32: From mean to true, corrections applied, for exact planetary computation by the wise.

Chapter II: True Longitudes of Planets

Verse 1: He who desires the true from the mean longitudes of planets should understand the corrections; then the true is obtained.

Verse 2: The field of *manda* and

=s

=*ighra* apogees is the fifth of a circle (72°); there the *manda* motion, and

=s

=*ighra* beyond.

Verse 3: The *manda-phala* in that field, and

=s

=*ighra-phala* likewise; by their sum or difference, the true is known.

Verse 4: The arc from the *manda* apogee causes the *manda-phala*; from the

=s

=*ighra* apogee, the

=s

=*ighra-phala*.

Verse 5: The mean increased or decreased by *manda-phala*; that result increased or decreased by

=s

=*ighra* corrections.

Verse 6: Thence the true planet, by *manda* and

=s

=*ighra* process in order; thus planetary computation declared.

Verse 7: The Sun's motion sometimes without *manda*; together with

=s

=*ighra*, the true is known.

Verse 8: For the Moon, *manda* and

=s

=*ighra* corrections; by sum or difference, the true arises.

Verse 9: At apogee, onefold *manda* result; at perigee, double; in between, triple.

Verse 10: By four measures, sixteen corrections; knowing the corrected planet, compute the true.

Verse 11: In *tithis*, years, *yugas*, months, and days, planetary motion computed from means.

Verse 12: From mean to true measure of planets, the wise compute; all becomes more exact.

Verse 13: Thus true motion declared for planets in order; by them all computation proceeds.

Verse 14: For Sun, *manda-phala* subtracted,

=s

=*ighra-phala* added; for Moon, by both processes.

Verse 15: For all star-planets, *manda* and

=s

=*ighra* equations always; by sum or difference, the true arises.

Verse 16: Knowing eastward and westward motion by both processes, the true planet is always obtained by calculators.

Verse 17: Thus true motion declared for Sun-born planets and all star-planets in order.

Verse 18: He who desires true from mean of planets should understand corrections; it becomes more exact.

Chapter III: Three Problems: Time, Place, and Direction

Verse 1: I declare the three questions always in planetary computation: knowing time, place, and direction, one obtains the result.

Verse 2: Time, place, and direction — these three for a planet; joined with their knowledge, it becomes more exact.

Verse 3: From time, motion; from place, declination; from direction, amplitude — these three give results.

Verse 4: Time is called sky-wandering (*kha-cara*); place is earth-wandering (*bh = u-cara*); direction is direction-elephant (*dig-gaja*) — these three accomplish the planet.

Verse 5: In *tithis*, years, *yugas*, months, days, time is numerical; place is the difference of countries (*de = sāntara*).

Verse 6: The *dig-valaya* is direction — these three accomplish the planet; joined with knowledge, it becomes more exact.

Verse 7: Applying the difference of countries joined with time, then planetary computation: the true arises.

Verse 8: The sky's motion joined with time, with place, and with direction.

Verse 9: Thus joined with three questions, the planet becomes more exact; therefore their knowledge always by calculators.

Verse 10: Difference of countries, direction, and time — these three questions proclaimed; joined with knowledge, the planet becomes exact.

Verse 11: At fifteenth and fourteenth *tithi*, full moon remembered; the *tithi-antara* known with time.

Verse 12: Difference of countries joined with time; then planetary computation: the true produced.

Verse 13: The *dig-valaya* as direction — these three accomplish; joined with knowledge, exact.

Verse 14: Thus with three questions, the planet becomes exact; therefore their knowledge by calculators.

Verse 15: In *tithis*, years, *yugas*, months, days: time is numerical; place is difference of countries.

Verse 16: The *dig-valaya* as direction — these three accomplish; joined with knowledge, exact.

Chapter IV: Lunar Eclipses

Verse 1: I declare the lunar eclipse as it is; therefore all the characteristics of eclipse.

Verse 2: The node between Sun and Moon — there the eclipse conjunction, in bright and dark fortnight.

Verse 3: If Rāhu at full moon seizes the Moon, then eclipse conjunction; hence the full disk.

Verse 4: Knowing the Moon's eclipse, then compute the time; by *manda* and $=s$ *ighra* process, make exact.

Verse 5: When Rāhu seizes full Moon on full-moon day, total eclipse; of Sun, solar eclipse.

Verse 6: The Moon's eclipse declared by both processes; applying the true, then eclipse characteristics.

Verse 7: Thus lunar eclipse declared in order; therefore all characteristics of solar eclipse.

Chapter V: Solar Eclipses

Verse 1: I declare the solar eclipse as it is; therefore all characteristics of solar eclipse.

Verse 2: The node between Sun and Moon — there the solar eclipse; the conjunction auspicious or inauspicious.

Verse 3: If Rāhu seizes Sun there, Moon clear, then solar eclipse: total or partial.

Verse 4: Knowing Sun's eclipse, compute time; by both processes, make exact.

Verse 5: Solar eclipse characteristics by both processes; applying the true, then characteristics.

Verse 6: Thus solar eclipse declared in order; therefore all results of eclipses.

Chapter VI: Rise and Set of Planets

Verse 1: I declare rising and setting of planets in order; joined with knowledge, the planet becomes exact.

Verse 2: Rising first, setting always after; joined with knowledge, the planet exact.

Verse 3: Knowing ascendant (*lagna*) joined with direction, then planet's rising; more exact known by wise.

Verse 4: For planets, rising first, setting after; joined with knowledge, exact.

Verse 5: Beginning from Sun's rising, Moon etc. in order, their rising and setting I declare.

Verse 6: All rising and setting accomplished by measures; without them, knowledge does not proceed.

Verse 7: Thus eight risings of planets; the wise compute; thence more exact from computation.

Chapter VII: Constellations and the Moon's Cusps

Verse 1: The elevation of Moon's horns and asterisms — joined with knowledge, the planet exact.

Verse 2: When Moon's horn elevation on full-moon day, then full elevation of Moon's horns.

Verse 3: Twenty-four asterisms proclaimed by ancients — joined with knowledge, exact.

Verse 4: Thus Moon's horn elevation and asterisms — joined with knowledge, exact.

Chapter VIII: Mathematics (Gaṇitādhyāya) — The Crown of the BSS

Note: This chapter (corresponding to Chapter XII of the full BSS) contains Brahmagupta's most celebrated mathematical discoveries, including the rules for zero, the quadratic formula, the area of a cyclic quadrilateral, and the solution of Pell's equation.

8.1 Arithmetic Operations (Vyakta-gaṇita)

Verse 1: The product of two fortunes is a fortune; of two debts is a fortune; of a fortune and debt is a debt. The first as stated; the second by reversal of signs.

Verse 2: Zero, in multiplication with a quantity, is that quantity increased; zero joined with like signs, subtracted from unlike.

Verse 3: When the multiplier is debt, fortune, and zero, the multiplicand is zero; in multiplication of zero, the result is zero.

Verse 4: Zero multiplied by a quantity is zero; zero divided likewise. Zero joined with like, subtracted from unlike.

Verse 5: When zero is the divisor and the dividend a quantity: zero-divisor, zero quotient; the multiplier as before.

Verse 6: The gain of two fortunes is gain; of two like signs, like. Division: the result of like sign with like.

Verse 7: Debt by fortune gives debt; fortune by debt gives debt; division is by reversal. Signs determined in order.

Verse 8: Deficient and excess are pure; their square always positive. The square root done two ways by method.

Verse 9: The square: an equal quadrilateral field; twice that. From that, root extracted: the square root.

Verse 10: The cube: equal or unequal, with equal quadrilateral base. Its root: the cube root, made threefold.

Verse 11: From first, second cube; from second, third; from third, fourth — that remainder is the cube.

8.2 Fractions (Bhinna-gaṇita)

Verse 12: The fraction is two-fold: expressed and unexpressed; established separately, perform addition.

Verse 13: Divide the division and class of divisions; having reduced fractions, form the fraction.

Verse 14: Denominator multiplied by divisor, dividend divided by denominator; the quotient: mark of fraction-operation.

Verse 15: The denominator: divisor-form; the dividend: quantity; the quotient: result — thus fraction proclaimed.

Verse 16: Numerator by numerator, by denominator, denominator by numerators; having multiplied, divide by fraction.

8.3 Rules for Zero (=S =unya-gaṇita)

The Most Famous Verses in the History of Indian Mathematics

Brahmagupta (628 CE) was the first mathematician to establish systematic rules for arithmetic with zero as a number.

Verse 17: Zero multiplied by a greater quantity is that increased; zero multiplied is zero; zero divided likewise.

Verse 18: Zero joined with like signs, subtracted from unlike. When zero is divisor and dividend a quantity.

Verse 19: Zero-divisor, zero quotient; multiplier as before. This zero-computation declared, all by reason.

Verse 20: Zero — know as void (*kha*), field, vajra, quantities; addition, subtraction, multiplication — zero there.

Verse 21: In addition, zero multiplied by quantity; zero-multiplier (*kha-guṇa*) remembered; zero-divisor known.

Mathematical Commentary: Brahmagupta's rules for zero:

- $a + 0 = a$, $a - 0 = a$
- $a \times 0 = 0$, $0 \times 0 = 0$
- $0a = 0$
- $a/0$ undefined (*kha-hāra*, “zero-divisor”)

He does *not* resolve $a/0$, leaving it undefined — an insight 1400 years before modern analysis.

8.4 Algebra — Avyakta-gaṇita

Verse 22: The unknown (*avyakta*) is the multiplier-place of quantities; joined with the known (*vyakta*), form the result.

Verse 23: “So much as” (*yāvat-tāvat*), or *kālaka* (black), *n*
= *ilaka* (blue), *p*
= *itaka* (yellow), *lohita* (red), *harita* (green),
'sveta (white).

Verse 24: These colors (*varṇa*) are multipliers of unknowns; by them joined with knowns, operate with unknown.

Verse 25: Square of unknown multiplied by known, like by like; square of known multiplied by unknown.

Verse 26: Sum and difference of colors known; joined with knowledge, the result arises in reality.

Verse 27: Unknown joined with *r*
= *upas* (absolute terms), operate there; unknown disjoined from *r*
= *upas* — result remembered.

Verse 28: Unknown joined with known, known with unknown — joined with knowledge, result arises.

Verse 29: Color by color is first; second in multiplication; square of color; third is cube.

Verse 30: Equation (*sam*
=*ikaraṇa*) is division of known and unknown into *r*
=*upas*; one skilled in equation is devoted to math.

8.5 The Quadratic Equation (Madhyamāharaṇa)

The Quadratic Formula in Brahmagupta

Brahmagupta gives the complete solution to $ax^2 + bx = c$, equivalent to the modern quadratic formula, 1100 years before its European rediscovery.

Verse 31: Having made equal the square of unknown and *r*
=*upas* according to rule, when both are of one *r*
=*upa* — equation.

Verse 32: Multiplied by twice (the coefficient), from square of equal (coefficient), subtract square of unequal; root made twofold, joined or diminished by *r*
=*upas*.

Verse 33: That, beginning with two or three, divide by equal multiplied by two; obtained: “so much as” (*yāvat-tāvat*), the *r*
=*upa*.

Verse 34: This *madhyam*
=*aharaṇa* (elimination of the middle) declared by those skilled in reasoning; one skilled in equation devoted to math.

Verse 35: The “so much as” joined with *r*
=*upas*; multiplied by two, make the square there.

Verse 36: Square of *r*
=*upa* multiplied, both made equal; square root known — elimination of the middle.

Verse 37: From that, root of three etc., divide beginning with two or three; obtained: “so much as”, *r*
=*upa*.

Verse 38: Taking unequal from equal, equal from unequal; having made equal — equation.

Verse 39: “So much as” multiplied, with *r*
=*upas*; multiplied by two, make square there.

Verse 40: Square of *r*
=*upa* multiplied, both made equal; square root — elimination of the middle (*madhyam*
=*aharaṇa*).

Mathematical Commentary: In modern notation, Brahmagupta’s formula for $ax^2 + bx = c$:

$$x = \frac{\sqrt{4ac + b^2} - b}{2a}$$

Equivalent to: $x = \frac{-b + \sqrt{b^2 + 4ac}}{2a}$

Brahmagupta was the first to give a general solution for *all* quadratics with two distinct positive roots.

8.6 Rule of Three (Trairā'sika) and Proportion

Verse 41: Three quantities of one kind — Trairā'sika (rule of three); their product known; result-quantity in order.

Verse 42: Argument (*pram* = *aṇa*) and result (*phala*); requisition (*icchā*) and result beyond; result of requisition by rule of three.

Verse 43: Requisition multiplied by argument, divided by result; obtained quotient: result by rule of three.

Verse 44: When reversed, inverse rule of three (*vyasta-trair* = *a* 'sika); operate by difference of argument and result.

Verse 45: Five-rule, six-rule, etc. — all from rule of three; inverse and direct, two-fold.

8.7 Arithmetic Progressions (Sreḍh=i)

Verse 46: First term joined with increase, doubled, divide by number of terms; multiply by terms, or half terms.

Verse 47: Mean multiplied by number of terms: sum in equal difference; even: mean doubled; odd: equal difference.

Verse 48: First joined with last, doubled, divide by terms; obtained: mean, from series operation.

Verse 49: Mean joined with number of terms: sum of series. Knowing difference of first and last, make multiplier.

Verse 50: Terms minus one, doubled, divide by terms; obtained: mean from series operation.

Verse 51: Thus sum of series (sreḍh = *i-phala*) declared by skilled reasoners; therefore geometric progression.

Verse 52: In geometric progression, first term and increase known; born of multiplier; form result.

Verse 53: First term multiplied by common ratio (*guṇottara*) repeatedly; divide by ratio minus one; that quotient.

8.8 Geometry — Plane Figures (Kṣetra-vyavahāra)

Brahmagupta's Formula for the Cyclic Quadrilateral

Area with sides a, b, c, d : $A = \sqrt{(s-a)(s-b)(s-c)(s-d)}$ where
 $s = \frac{a+b+c+d}{2}$.

This generalizes Heron's formula to quadrilaterals.

Verse 54: Root of triangular field area: product of half-base (*sam-dala*) and perpendicular (*koṭi*); square root of half and sides.

Verse 55: Square root of quadrilateral field area remembered; from square of sum of opposite sides, root produced.

Verse 56: Quadrilateral area from half-base and perpendicular; for irregular quadrilateral, make twofold (two triangles).

Verse 57: Square of half sum of opposite sides — called area; by sum of roots of two, form area.

Verse 58: Circular field square from half-diameter (*vy*
 $=s$
 $=ardha$); from fourth of diameter-square, area produced.

Verse 59: Knowing circular field area, compute diameter; area fourfold, its root: diameter.

Verse 60: Circumference multiplied by five, diameter by seven remembered. Thus circle multipliers form result.

(Note: $\pi \approx 22/7 \approx 3.143$ or $\sqrt{10} \approx 3.162$)

Verse 61: Diameter multiplied by half-chord (*jy*
 $=ardha$), squared, divided by four; remainder's root: chord (*j*
 $=iv$
 $=a$).

Verse 62: Square of chord subtracted from square of diameter; root of remainder: arrow (*'sara, sagitta*), half-diameter's chord-maker.

8.9 Brahmagupta's Formula for the Cyclic Quadrilateral

Verse 63: Quadrilateral area from half-base and perpendicular; from square of sum of opposite sides, root produced.

Verse 64: Square root of equilateral quadrilateral (*sama-caturbhuja*) area; square of half sum of opposite sides: area.

Verse 65: Square root of irregular quadrilateral area; beginning with two or three, form area.

Verse 66: Quadrilateral area from half-base and perpendicular; from half-diameter, form area.

Verse 67: Quadrilateral area from two triangles; by sum of their roots, form area. (*Brahmagupta's formula*: $A = \sqrt{(s-a)(s-b)(s-c)(s-d)}$)

Verse 68: Isosceles trapezium (*sama-t*
 $=t$
 $=iya-bhuja$) from half-base and perpendicular; field twofold, form area.

Verse 69: Quadrilateral area from sum of opposite sides; from pair of triangles, form area.

Verse 70: Base-height product, doubled, joined with root: equilateral and isosceles (*dvi-sama*) quadrilateral area.

Verse 71: Irregular quadrilateral, doubled, joined with root; with half unequal root: root of area.

Verse 72: This operation of fields declared by skilled reasoners; therefore operation of excavations in order.

8.10 Excavations (Khāta-vyavahāra)

Verse 73: Knowing excavated field area, compute depth; oblong quadrilateral — form area.

Verse 74: Volume of excavation accomplished, compute capacity; increase known from excavation operation.

Verse 75: Depth and capacity known from excavation operation; therefore operation of shadows in order.

8.11 Shadows and Gnomons (Chāyā-vyavahāra)

Verse 76: Knowing gnomon and shadow motion, compute time; midday time (*madhy* = *ahnak* = *ala*) from shadow operation.

Verse 77: Gnomon (*'saḥu*) of twelve aṅgulas; its shadow proclaimed; joined with knowledge, time exact.

Verse 78: Product of gnomon and shadow made, thence direction computation; beginning triangulation (*trijñ* = *ana*), declare in order.

Algebraic Operations

Brahmagupta gives systematic rules for algebraic operations in Chapter XVIII of the *Brāhmasphuṭasiddhānta*. This chapter contains the first treatment of algebra as a distinct mathematical discipline in Indian mathematics.

Addition and Subtraction of Unknowns

Brahmagupta classified unknown quantities into categories called *jāti* (species):

Verse 18.24. *Translation:* “The numerical coefficients of the same unknown are added to or subtracted from each other; the same is done with the known quantities. When the unknowns are different, they are not united.”

This means that the numerical coefficients of x cannot be added to or subtracted from the numerical coefficients of y or x^2 or x^3 or xy and so on because these terms belong to different *jāti* or they do not belong to the category of the “like.”

Multiplication of Unknowns

Again, regarding multiplication, Brahmagupta says:

Verse 18.25. *Translation:* “The product of the two like unknowns is a square; the product of three or more like unknowns is a power of that designation. The multiplication of unknowns of unlike species is the same as the mutual product of symbols; it is called **bhāvita** (product or factum).”

Rules for Zero, Positive and Negative Numbers

Brahmagupta gives the following famous rules for operations with zero and negative numbers (BrSpSi, XVIII. 30–36):

Verse 18.30: Addition

Verse 18.30. *Translation:* “[The sum] of two positives is positive, of two negatives negative; of a positive and a negative [the sum] is their difference; if they are equal it is zero. The sum of a negative and zero is negative, [that] of a positive and zero [is] positive, [and that] of two zeros [is] zero.”

Verse 18.31: Subtraction

Verse 18.31. *Translation:* “If a smaller [positive] is to be subtracted from a larger positive, [the result] is positive; [if] a smaller negative from a larger negative, [the result] is negative; [if] a larger [negative or positive is to be subtracted] from a smaller [positive or negative, the algebraic sign of] their difference is reversed—i.e. negative [becomes] positive and positive [becomes] negative.”

Verse 18.32: Subtraction with zero

Verse 18.32. *Translation:* “A negative minus zero is negative, a positive [minus zero] positive; zero [minus zero] is zero. When a positive is to be subtracted from a negative or a negative from a positive, then it is to be added.”

Verse 18.33: Multiplication

Verse 18.33. *Translation:* “The product of a negative and a positive is negative, of two negatives positive, and of positives positive; the product of zero and a negative, of zero and a positive, or of two zeros [are all] zero.”

Brahmagupta’s Laws of Signs

$$\begin{aligned} (+a) \times (+b) &= +ab & (-a) \times (-b) &= +ab \\ (-a) \times (+b) &= -ab & (+a) \times (-b) &= -ab \\ 0 \times a &= a \times 0 = 0 \text{ for any quantity } a. \end{aligned}$$

Verse 18.34: Division

Verse 18.34. *Translation:* “A positive divided by a positive or a negative divided by a negative is positive; a zero divided by a zero is zero; a positive divided by a negative is negative; a negative divided by a positive is [also] negative.”

Verse 18.35: Division by zero

Verse 18.35. *Translation (Colebrooke):* “A negative or a positive divided by zero has that [zero] as its divisor, or zero divided by a negative or a positive [has that negative or positive as its divisor]. The square of a negative or of a positive is positive; [the square] of zero is zero. That of which [the square] is the square is [its] square-root.”

Note: Brahmagupta’s statement that $a/0$ gives “kha-hara” (a fraction with zero denominator) was his way of expressing the concept of infinity. In modern mathematics, division by zero is undefined, but Brahmagupta’s insight that such quantities represent infinitely large magnitudes was remarkable for the 7th century.

Quadratic Equations

The geometrical solution of a quadratic equation in this country would take us to the Vedic *Sulba* period. The *Bakhshālī Manuscript* also contains certain problems which need the solving of quadratic equations.

Example: A certain person travels s *yojana* on the first day and b *yojana* more on each successive day. Another who travels at the uniform rate of S *yojana* per day, has a start of t days. When will the first man overtake the second?

This leads to the quadratic:

$$s + (s + b) + (s + 2b) + \dots + [s + (n - 1)b] = S(n + t)$$

$$ns + \frac{n(n-1)}{2}b = Sn + St$$

$$n^2 \cdot \frac{b}{2} + n \left(s - \frac{b}{2} - S \right) - St = 0$$

Sanskrit Verses: Brahmagupta's Rule for Quadratic Equations

Brahmagupta gives the following rule (BrSpSi, XVIII. 44-45):

Verse 18.44: The quadratic formula (first form)

Verse 18.44. *Translation (Colebrooke):* “Diminish by the middle [number] the square-root of the *rupas* multiplied by four times the square and increased by the square of the middle [number]; divide the remainder by twice the square. [The result is] the middle [number].”

Verse 18.45: The quadratic formula (second form)

Verse 18.45. *Translation (Colebrooke):* “Whatever is the square-root of the *rupas* multiplied by the square [and] increased by the square of half the unknown, diminish that by half the unknown [and] divide [the remainder] by its square. [The result is] the unknown.”

Modern Formulation: For the equation $ax^2 + bx = c$:

$$x = \frac{\sqrt{4ac + b^2} - b}{2a}$$

This is equivalent to the modern quadratic formula for the positive root.

Bhāskara I and the Kuṭṭaka Operation

An indeterminate equation of the first degree of the type:

$$\frac{ax - c}{b} = y$$

(with x and y unknown) is known in Hindu mathematics by the name of “pulveriser” — *kuṭṭākāra*. In this equation, a is called the “dividend” (*bhājya*), b the “divisor” (*bhāgahāra*), c the interpolator (*kṣepa*), x the “multiplier” (*guṇakāra*), and y the “quotient” (*labdha*).

In the pulveriser contemplated in the above stanza:

a = revolution number of a planet

b = civil days in a *yuga*

c = residue of the revolutions of the planet (*śeṣa*)

Sanskrit Verse: Brahmagupta’s Rule for the Pulveriser

Verse 18.50. *Translation:* “Divide the dividend by the divisor, and the divisor by the remainder obtained; by this process of mutual division, [continue] until the remainder becomes zero. From this, by working backwards, one can find a particular solution.”

Verse 18.51: Condition for solution

Verse 18.51. *Translation:* “If the interpolator is divisible by the divisor of the residues, then that same method [applies]; if the interpolators are unequal, know that the pulveriser has a flaw [is impossible], friend.”

Verse 18.52: Working backwards

Verse 18.52. *Translation:* “The successive quotients obtained [in the mutual division] are called ‘kotis’ [vertical lines]; placing the gunakaras [multipliers] at the kotis, the multiplier is [found] combined with the interpolator.”

The Pulveriser Method

The method of the pulveriser (*kuṭṭaka*) involves the following steps:

1. Divide a by b and obtain the quotient and remainder.
2. Divide b by this remainder and obtain a new quotient and remainder.
3. Continue this process of mutual division until the remainder becomes zero.
4. Write down the quotients obtained in a column.
5. The number of quotients is made odd by adding the residue if necessary.
6. From the bottom, multiply the last number by the interpolator and add the number above it.

7. Continue this process until the top is reached.
8. The number at the top gives the value of x and the number at the bottom gives y .

General Form

The general form of the pulveriser is to find integer solutions to:

$$ax + c = by$$

where a , b , and c are given integers.

Brahmagupta's method was later improved by Āryabhaṭa II and Bhāskara II, but the fundamental approach was established in the *Brāhmasphuṭasiddhānta*.

Example: Pulveriser for planetary residues

Verse 18.53. *Translation:* "The residue of the Sun [is] twelve; divide it together with the civil days; whatever multiplier results therefrom, that will yield the elapsed degrees of the Sun's motion."

Example Problem: Find x such that $\frac{137x-10}{60} = y$ is an integer.

Here dividend $a = 137$, divisor $b = 60$, interpolator $c = 10$.

By mutual division: $137 = 2 \times 60 + 17$; $60 = 3 \times 17 + 9$; $17 = 1 \times 9 + 8$; $9 = 1 \times 8 + 1$.

Working backwards: $x = 50$, $y = 114$ is a solution.

Planetary Aphelions and Epicycle Corrections

Brahmagupta gave systematic corrections to the mean longitudes of planets to obtain their true positions. These corrections involve the concepts of the aphelion (*manda-cheda*) and the epicycles of apsis (*manda-paridhi*) and conjunction (*śīghra-paridhi*).

Verse: Mean to True Position

Verse U.Kh. 5. *Translation:* “Having taken the mean planet, [apply] the correction for the apsis; from that [result, apply] the correction for the conjunction; then the planet becomes true [corrected].”

Verse: Epicycle Dimensions

Verse U.Kh. 6. *Translation:* “The epicycle of the apsis for the Sun is forty-seven plus two [i.e., 49 for the *manda*]; that of the Moon is thirty-three. The epicycle of the conjunction [for the Moon] is seventy.”

Verse: Aphelion (Mandocca) Positions

Verse U.Kh. 7. *Translation:* “The aphelion [*ucca*] of the Sun is seventy-nine [degrees] in the northern solstice; the [higher] apsis of the Moon and the *śīghra* [conjunction point] are obtained by [appropriate] correction.”

Historical Note: Brahmagupta’s corrections to the planetary epicycles were based on careful comparison with observed positions. He adjusted the epicycle radii given in the *Āryabhaṭīya* to reduce the errors in predicted planetary longitudes. His method of applying the *manda* correction first, followed by the *śīghra* correction, became the standard procedure in Indian astronomy.

Uttara Khaṇḍakhādyaka: Advanced Planetary Theory

The *Uttara Khaṇḍakhādyaka* (UKK) represents Brahmagupta's advanced astronomical treatise, containing corrections and refinements to the earlier *Khaṇḍakhādyaka* proper.

Brahmagupta First to Use Second Differences

All these illustrations reproduced here very well establish the point that the great Indian astronomers from *Āryabhaṭa* I to Brahmagupta were aware of the methods of separating the two distinct planetary inequalities, viz., that of the apsis and of conjunction in the cases of the five 'star' planets.

In the *Khaṇḍakhādyaka*, Brahmagupta having given the "sines" and the equations of the Sun and the Moon at the interval of 15° of arc of the mean anomaly, in the *Uttara Khaṇḍakhādyaka* teaches, for the **first time in the history of mathematics**, the improved rules for interpolation by using the **second difference**.

This very important feature is reproduced here from the translation by Sengupta of the verse (UKK. 8):

Verse U.Kh. 8. *Translation (Sengupta):* "Multiply the residual arc left after division by $900'$ (i.e. by 15°), by half the difference of the tabular difference passed over and that to be passed over and divide by $900'$ (i.e. 15°): by the result increase or decrease, as the case may be, half the sum of the same two tabular differences; the result which, whether less or greater than the tabular difference to be passed, is the true tabular difference to be passed over."

The rule given here applies to the case of all functions hitherto considered in the *Khaṇḍakhādyaka*, which are tabulated at the difference of 15° of arc of the argument. They are:

- (i) the tabular differences of the Sun's equation,
- (ii) the tabular differences of the Moon's equation,
- (iii) the tabular differences of the 'sines'.

Illustration — To find the 'sine' of 57° .

Brahmagupta's table of sines in the *Khaṇḍakhādyaka* is as follows:

"Thirty increased severally by nine, six and one; twenty-four, fifteen and five, are the tabular differences of sines at intervals of half-a-sign. For any arc, the 'sine' is the sum of the parts passed over, increased by the proportional part of the tabular difference to be passed over." (KK.I.30; also III.6)

Table 1: Brahmagupta's Table of Sines

Verse on Lunar Correction (Evection)

Verse U.Kh. 10. *Translation:* "The second inequality of the Moon [evection], when combined with the [equation of the] conjunction, then its motion-equation occurs; the conjunction-equation should be applied by that [amount]."

This verse refers to what later became known as Śrīpati's equation — the recognition of the Moon's second inequality (evection), which represents a significant achievement in Indian lunar theory.

Now $57^\circ = 3420 \text{ minutes} = 900' \times 3 + 720'$. Thus three of the tabular differences are considered as passed over; the last one being 31 and the one to be passed over is 24.

The true tabular difference by the rule, for arc 57° :

$$\frac{31 + 24}{2} + \frac{720}{900} \times \frac{31 - 24}{2}$$

Hence the 'sine' of 57° :

$$= 106 + \frac{720}{900} \times 24 + \frac{720}{900} \left(\frac{720}{900} - 1 \right) \times \frac{31 - 24}{2} = 125.76$$

As worked out from the logarithm tables the same comes out to be 125.80.

This in fact is the modern form of the interpolation equation up to the term containing the second difference:

$$f(a + nh) = f(a) + n\Delta f(a) + \frac{n(n-1)}{2} \Delta^2 f(a)$$

Sanskrit Verse: Construction of the Sine Table

Verse Kh. I.29. *Translation:* "Thirty minus one, increased by nine, six, and one; twenty-four, fifteen, and five — these are the tabular differences of sines. The sines [themselves] are obtained by successive addition of these differences, [starting from] the first [sine]. These twenty-four sines [span] the quadrant."

Verse on the Epicycle Correction

Verse U.Kh. 12. *Translation:* "To correct [the mean planet] by the equation of the apsis, one should apply [the correction] by means of the epicycle of the apsis; likewise, to make [the planet] true by the equation of the conjunction, one should do so by means of the epicycle of the conjunction."

Brahmagupta thus takes a decidedly improved step here and is undoubtedly the **first man in the history of mathematics** who has done this.

Operations with Fractions

Brahmagupta classified fractions into several categories: *bhāga* (simple fraction), *prabhāga* (compound fraction), *bhāgānubandha* (fractions in association), and *bhāgāpavāha* (fractions in dissociation). He gave systematic rules for all operations with fractions.

Verse: Addition and Subtraction of Fractions

Verse Pā. 2.3. *Translation:* “When fractions are connected [for addition or subtraction], the two denominators [become] multipliers [of the opposite numerators]; the sum of the [cross-multiplied] dividends, divided by the product of the denominators, becomes the combined [result].”

In modern notation:

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd} \quad \text{and} \quad \frac{a}{b} - \frac{c}{d} = \frac{ad - bc}{bd}$$

Verse: Multiplication and Division of Fractions

Verse Pā. 2.4. *Translation:* “In the multiplication of fractions, the numerators and denominators are multiplied together [respectively]. In division, the numerator is multiplied by the reciprocal of the divisor-fraction [i.e., the numerator and denominator are interchanged].”

Brahmagupta’s Fraction Operations

$$\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d} \quad \frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}$$

The Rule of Three (Trairāśika)

The Rule of Three (*trairāśika*) is the fundamental method of proportion in Indian mathematics. Brahmagupta gave a precise formulation that served as the basis for all problems involving proportion.

Verse: The Rule of Three

Verse Pā. 5.1. *Translation:* “If the *pramāṇa*-fruit [result for the standard measure] is given, that [is] multiplied by the *icchā* [requisition]; divide that by the *pramāṇa* [standard measure]; the result obtained here is the *icchā-phala* [required result].”

In the *trairāśika*, three quantities are given:

- **Pramāṇa** (p): the standard measure or argument
- **Phala** (f): the fruit or result corresponding to the standard
- **Ichchā** (i): the requisition or desired measure

The required *icchā-phala* (x) is:

$$x = \frac{f \times i}{p}$$

Verse: Inverse Rule of Three

Verse Pā. 5.2. *Translation:* “When, by the inverse Rule of Three, the result becomes reversed, then by the nature of the divisor — the fruit then becomes the *icchā-phala* [by inverse proportion].”

Direct Rule of Three: $x = \frac{f \times i}{p}$	Inverse Rule of Three: $x = \frac{f \times p}{i}$
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Verse: Compound Proportion (Pañcarāśika etc.)

Verse Pā. 5.3. *Translation:* “The result having been obtained by the Rule of Three, [it is further operated on] by [another] Rule of Three; as many given quantities [as there are], so many Rules of Three are remembered [to be applied].”

This verse establishes the method of *compound proportion*: when more than three quantities are involved, successive applications of the Rule of Three yield the final result. For five quantities it is called *pañcarāśika*, for seven *saptarāśika*, and so on.

The Sine Rule in Indian Trigonometry

Brahmagupta was the first Indian mathematician to state the sine rule for plane triangles explicitly. He used it in the context of computing the true positions of planets and in spherical astronomy.

Verse: The Sine Rule

Verse Br.Sp. 14.21. *Translation:* “In an equilateral triangle, the sines of equal sides [divided] by one another [give unity]; in a scalene triangle, the sides divided by the sines give a result [which is constant] — that is the [circumdiameter].”

In modern form, for any triangle ABC with circumdiameter D :

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = D$$

Verse: Shadow and Sine Relations (Spherical Astronomy)

Verse Br.Sp. 16.8. *Translation:* “The Rsine of the declination, divided by the Rsine of half the day, becomes the *lambaka* (vertical). The Rsine of the latitude, multiplied by the *lambaka*, [gives] the *drgjyā* (ascensional difference); it is determined [according to] the direction [north or south of the equator].”

This verse relates to spherical astronomy: the computation of the *drgjyā* (ascensional difference), which measures how much a celestial body rises earlier or later due to the observer’s latitude. The *kṛānti-kṣetra* and *akṣa-kṣetra* diagrams use similar right triangles.

Further Algebraic Operations

Brahmagupta's Chapter XVIII also contains rules for squares, square roots, cubes, and cube roots, as well as additional rules for operations with zero.

Verse: Square and Square Root

Verse 18.46. *Translation:* "A square is an equilateral quadrilateral; its root [is] that which, when doubled, [gives the side]. That of which the square is the square is [its] square-root [i.e., the square root of a^2 is a]."

Verse: Cube and Cube Root

Verse 18.47. *Translation:* "A cube is an equi-dimensional equilateral [solid]; its root [is] that which, when tripled in dimension, [gives the side]. That of which the cube is the cube [is its] cube-root [i.e., the cube root of a^3 is a]."

Verse: Zero as a Number

Verse 18.36. *Translation:* "Zero, void, 'kha', 'path' — the learned know this as a number. When [a quantity] is combined with or diminished by it, there is neither increase nor decrease [i.e., $a + 0 = a - 0 = a$]."

Note: This verse (18.36) is historically significant as one of the earliest explicit statements that zero is a number in its own right. Brahmagupta's treatment of zero as a mathematical entity with well-defined properties was a foundational contribution to the development of algebra.

Summary of Brahmagupta's Mathematical Contributions

1. **Second-order interpolation:** Brahmagupta was the first mathematician in history to use the second difference in interpolation, as demonstrated in the *Uttara Khaṇḍakhādyaka* (UKK. 8).
2. **Sine rule:** He was the first in India to give the sine rule for plane triangles, in the context of computing planetary positions.
3. **Planetary corrections:** Systematic corrections to the epicycles of apsis for the Sun, Moon, and planets, bringing calculated positions into agreement with observation.
4. **Lunar inequalities:** Recognition and treatment of the second inequality of the Moon (evection), known as Śrīpati's equation.
5. **Spherical astronomy:** Development of Indian methods using properties of similar right-angled triangles (*krānti-kṣetra* and *akṣa-kṣetra*).
6. **System of numerals:** Contribution to the place-value notation and its transmission to the Arab world, and subsequently to Europe.
7. **Fractions:** Classification of fractional combinations into *bhāga*, *prabhāga*, *bhāgānubandha*, and *bhāgāpavāha*.
8. **Algebra:** Systematic rules for operations with zero, positive and negative numbers; the first to treat zero as a number in its own right.
9. **Quadratic equations:** General formula for solving quadratic equations (verses 18.44–45).
10. **Indeterminate equations:** The *kuṭṭaka* (pulveriser) method for solving linear indeterminate equations, fundamental to Indian astronomy for calculating planetary positions.

Editor's Note

- **Chapter XVIII, verses 30–36:** Rules for operations with zero, positive and negative numbers, including the famous “*kha-hara*” (division by zero) verse; also explicit statement that zero is a number (v. 36)
- **Chapter XVIII, verses 44–47:** The quadratic formula (two equivalent forms), square and square-root, cube and cube-root
- **Chapter XVIII, verses 50–53:** The pulveriser (*kuṭṭaka*) method for indeterminate equations, including conditions for solution and worked examples
- **Pāṭiṅaṇita, verses 2.3–2.4:** Rules for addition, subtraction, multiplication, and division of fractions
- **Pāṭiṅaṇita, verses 5.1–5.3:** The Rule of Three (*trairāśika*), inverse Rule of Three, and compound proportion
- **Brāhmasphuṭasiddhānta, verses 14.21, 16.8:** The sine rule for plane triangles and spherical astronomy relations

- **Uttara Khaṇḍakhādyaka, verses 5-12:** Planetary aphelions, epicycle corrections, sine table construction, second-difference interpolation, and lunar inequality

Chapter VIII: Brahmagupta as an Algebraist

Algebraic Rules

“Of the pulveriser, the divisor and dividend being mutually divided, one should operate the pulveriser with the reduced divisor and dividend.” — Kuṭṭaka (Pulverizer) method for solving $ax \pm c = by$.

“One should divide the dividend by the divisor and the divisor by the dividend alternately; the quotients should be written down separately, one below, one above.” — This describes the Euclidean algorithm steps for the kuṭṭaka.

“Having multiplied the chosen number (mati) by the one above it, add the quotient below; place the result above and remove the lower; proceed again as before.” — The iterative backward-substitution step to find the multiplier and quotient.

“The upper (number), called the ‘multiplier,’ divided by the divisor; the lower, called the ‘quotient,’ divided by the dividend: the remainders are respectively the āhargaṇa and the revolutions (or what is sought).” — Final step: $x = \text{ahargaṇa}$, $y = \text{revolutions}$.

Modern Mathematical Formulation

The kuṭṭaka method solves the linear Diophantine equation:

$$ax \pm c = by$$

where a = civil days in *yuga*, b = revolution-number of the planet, c = residue, and x, y are positive integers to be found.

Step 1 — Reduction: Compute $g = \gcd(a, b)$. Divide through by g :

$$a' = \frac{a}{g}, \quad b' = \frac{b}{g}, \quad c' = \frac{c}{g}$$

Step 2 — Euclidean Algorithm: Apply the mutual division to obtain the continued-fraction quotients $q_1, q_2, \dots, q_n$.

Step 3 — Back-substitution: Starting with a chosen *mati* m such that $r_n \cdot m - c' \equiv 0 \pmod{b_n}$, iteratively compute:

$$V_i = q_i \cdot V_{i+1} + V_{i+2}$$

Step 4 — Solution: The two final numbers give $x \equiv V_{n-1} \pmod{a'}$ and $y \equiv V_n \pmod{b'}$.

Varga-Prakṛti (Square-Nature)

“In the Square-Nature, when two solutions are found with the same multiplier (N) and interpolators k_1, k_2 : their interpolators being multiplied, the multiplier being doubled and combined with the two roots — the sum of their squares is (the new equation with) unity as interpolator.”

Brahmagupta’s Lemma (Composition): If (x_1, y_1) solves $Nx^2 + k_1 = y^2$ and (x_2, y_2) solves $Nx^2 + k_2 = y^2$, then:

$$x = x_1y_2 \pm x_2y_1, \quad y = y_1y_2 \pm Nx_1x_2$$

solves $Nx^2 + k_1k_2 = y^2$.

“In the Square-Nature, if the interpolator is a square, being capable of being a square — dividing the roots by the square root of that interpolator, new roots are obtained.”

Second Lemma: If $Nx^2 \pm p^2d = y^2$, set $u = x/p$, $v = y/p$, giving:

$$Nu^2 \pm d = v^2$$

The original roots are p times those of the derived equation.

Cakravāla (Cyclic Method)

“The teacher (ācārya), first taking the square root of the multiplier nearest to it, divided by the interpolator (and) increased or diminished by unity — that root divided by the multiplier and the interpolator multiplied by it, combined with the previous pair of roots — that is the new interpolator; the product of the two divided by the multiplier, the sum of the squares of the previous roots — (these are the new roots).”

The Cyclic Method: Starting from an auxiliary $Na^2 + k = b^2$:

1. Choose P so $P^2 \approx N$ and $k \mid (P^2 - N)$
2. New interpolator: $k' = \frac{P^2 - N}{k}$
3. New roots: $a' = \frac{Pa+b}{|k|}$, $b' = \frac{Pb+Na}{|k|}$
4. Iterate until interpolator becomes $\pm 1, \pm 2, \pm 4$

Chapter X: Arabic and Indian Divisions of the Zodiac

Nakṣatras Enumeration

Chapter XI: Brahmagupta's Astronomy

Time Units

“Beginning with Caitra, all the months of a yuga proceed for the creation-epoch (sṛṣṭi-cchāda) at Lañkā simultaneously with the beginning of the day of Brahmā.”

All months begin simultaneously at the creation epoch at Lañkā (on the prime meridian).

“Two vipalas (atoms of time) make a nālikā; two nālikās make a prāṇa; thirty-six prāṇas make a vināḍikā; sixty vināḍikās make a nāḍī (ghaṭikā); sixty ghaṭikās make a day; thirty days make a month.”

Time hierarchy: 2 vipalas = 1 nālikā; 2 nālikās = 1 prāṇa; 36 prāṇas = 1 vināḍikā; 60 vināḍikās = 1 ghaṭikā; 60 ghaṭikās = 1 civil day; 30 days = 1 month.

Kalpa Constants

Epoch Calculation (Sṛṣṭi-saṃvatsara)

“Six manus have elapsed in half a kalpa; and of the seventh manu, twenty-seven caturyugis have gone by; of the twenty-eighth caturyugi, the three yugas kṛta, dvāpara and tretā have elapsed, and also 3179 years of the present kaliyuga.”

The total elapsed period at Brahmagupta's epoch (year 628 CE):

$$\begin{aligned}
 T &= 6 \times 71 \times 4,320,000 \\
 &\quad + 7 \times 4 \times 432,000 \\
 &\quad + 27 \times 4,320,000 \\
 &\quad + (1,728,000 + 1,296,000 + 864,000) + 3,179 \\
 &= 1,972,947,179 \text{ years.}
 \end{aligned}$$

Mean Longitude Calculation

General method:

1. Compute the *ahargaṇa* A = civil days elapsed since epoch
2. Multiply by planet's revolution-number R in a kalpa
3. Divide by civil days in kalpa D :

$$\text{Mean longitude} = \frac{A \times R}{D} \text{ (in revolutions, signs, degrees, etc.)}$$

Chapter XII: Brahmagupta as a Critic

“Know the lambana (difference of parallaxes in longitude) by means of five Rsines: the madhya-jyā, the udaya-jyā, the dṛk-kṣepa-jyā, the dṛg-jyā, and the dṛg-gati-jyā; and the avanati (difference in latitude) likewise.”

The *lambana* is computed from five Rsines:

- *madhya-jyā* = $R \sin z_m$ (zenith distance of meridian ecliptic point)
- *udaya-jyā* = $\frac{R \sin L \cdot R \sin \varepsilon}{R \cos \phi}$
- *dṛk-kṣepa-jyā* = $R \sin z_c$ (zenith distance of central ecliptic point)
- *dṛg-jyā* — related to observer's zenith distance
- *dṛg-gati-jyā* — related to the motion of the observer

where L = longitude of horizon ecliptic point, ε = obliquity of ecliptic, ϕ = latitude of place.

Chapter XIII: Astronomical Instruments

The Gnomon (Śaṅku)

“(The gnomon) two *aṅgulas* wide at the base, pointed as a needle (*śūla-agra*), twelve *aṅgulas* in length, and full of holes from the basic circular part to the pointed extremity — thus is the *śaṅku* (gnomon) described, O beloved.” — BrSpSi. XXII. 39.

The gnomon dimensions: base width = 2 *aṅgulas* (≈ 3.75 cm), height = 12 *aṅgulas* (≈ 22.5 cm), tapering to a point.

“When there is no wind, placing a jar full of water upon a tripod on the ground which has been made plane by means of eye or thread, one should bore a fine hole (at the bottom) so that the water may fall drop by drop.”

Procedure for testing level ground: Place a water-filled jar on a tripod on leveled ground. A fine hole allows water to drip steadily, confirming the surface is horizontal.

Madhyamādhikāra

Brāhmasphuṭasiddhānta: First Chapter — Madhyamādhikāra

*jayati praṇata-sura-asura-auragīndra-ratna-prabhā-cchurita-pāda-
pañkajaḥ |
kartā jagat-patir-aditya-tulya-yānām mahādevaḥ || 1 |*

“Victorious is Mahādeva, the Lord of the Universe, the Creator, the lotus-feet of whom are rubbed with the lustre of jewels from the heads of gods, demons, and serpent-kings who bow before Him — He whose chariot is equal to that of the Sun.”

*brahmagupto graha-gaṇitaṃ mahatā kālena yat khalvīmatam |
kṣīṇe 'dhunī tad-viparīta-vad-brāhmaṇair upetaṃ || 2 |*

“Brahmagupta (says): The science of the computation of planets, which was well-established in the past, has now become corrupted; being contrary (to the truth), it has been mixed up by the brāhmaṇas.”

*dhruva-bala-arka-suta-budhaj na jyotir-vidāṃ prīti-śad-gama-do |
poṣaṇārthaṃ vivṛtāntara-vyaktaṃ saha prabandhena sūcitam || 3 |*

“(I write this) for the pleasure of astronomers, who are learned in the (motions of) Dhruva (pole star), Bala (Mars), Arka (Sun), Suta (Saturn), Budha (Mercury), and J na (Jupiter); for the preservation (of knowledge), the obscure meaning is made clear along with the connected exposition.”

caitry-ādi-tad-eva-dvija-nodvan-māsa-sarveṣa yugasya |
sṛṣṭi-cchādo lañkāyāṃ samam pravṛtā dine-kalasya || 4 |

“Beginning with Caitra, all the months of a yuga — (which are) the creation-covering — proceed simultaneously at Lañkā with the beginning of the day of Brahmā (dina-kala).”

prāg-ry-udda-nāṭikā-dvau dvi-dvi-yaṣṭikā tri-nāṭikā ṣaḍ-ghnā |
ghaṭikā ṣaḍ-ghnā diva-lo divasānām triṃśatā māsaḥ || 5 |

“Two atoms (paramāṇu) make an aṇu; two aṇus make a lava; two lavas make a nimeṣa; three nimeṣas make a kṣaṇa; five kṣaṇas make a kṣathi; four kṣathis make a muhūrta; eight muhūrtas make a prahara; four praharas make a day (aho-rātra); thirty days make a month.”

Time scale: 2 paramāṇu = 1 aṇu → 2 aṇu = 1 lava → 2 lava = 1 nimeṣa → 3 nimeṣa = 1 kṣaṇa → 5 kṣaṇa = 1 kṣathi → 4 kṣathi = 1 muhūrta → 8 muhūrta = 1 prahara → 4 prahara = 1 day → 30 days = 1 month.

Verses 22-24: Solar and Lunar Month Relations

parivartah sapta-catuṣṣaṣṭir sūryasya dinasya mitih |
ravi bhagañānām mānaḥ sāvana-dinasya kuṭṭakair eṣa |

“The revolution-number of the Sun is $64 + 7 = 71$ (per caturyuga); this is the measure of solar days. By the pulverizer (kuṭṭaka), this (gives) the civil days.”

Rule: A solar year ≈ 365 days 15 min 30 sec. The revolution-number of the Sun = 4,320,000 per kalpa.

ravi bhagañārdha-badhā dvādaśāṃśena bhavati ravi-māsaḥ |
bhagañantaram ravi-dinaḥ śaśi-māsaḥ sūrya-māsena |

“The product of half the revolutions of the Sun divided by twelve becomes a solar month. The difference of revolutions (between Sun and Moon) is the lunar month exceeding the solar month.”

Formula:

$$\text{Solar month} = \frac{R_{\odot}/2}{12} \quad ; \quad \text{Lunar month} = R_{\text{Moon}} - R_{\odot}$$

adhi-māsāḥ śaśi-mśāv-indv-adinasya bhavati śaśi-divasāḥ |
śaśi-sāvana-divasa-antara-nivamāni tithiḥ śaśāñka-dinaṃ |

“The intercalary months (adhi-māsa) are the lunar months exceeding the solar months. The lunar days are (obtained from) the difference between lunar and civil days; the tithi (lunar day) is the measure of the difference between lunar and civil days.”

Key relations:

$$\begin{aligned} \text{adhi-māsa} &= \text{lunar months} - \text{solar months} \\ \text{śaśi-divasa} &= \text{lunar days} - \text{civil days} \\ \text{tithi} &= 1/30 \text{ of a lunar month} \end{aligned}$$

Verses 26-27: Epoch and Ahargaṇa

kalpa-parārdha manavaḥ ṣaṭ-kāśya gatāś catur-dvi-yugāni yugānām |
śruta-purāṇādi-nikṣepānām kṣayaḥ sūrya-dinaṃ cakṣuḥ |

“In half a kalpa, six manus have elapsed, and twenty-seven caturyugis of the (seventh) manu. (This gives) the destruction (elapsed period) of the sūrya-dina (solar days) — the eye (of calculation).”

The *ahargaṇa* (day-count) from the kalpa epoch:

$$A = D_{\text{manus}} + D_{\text{caturyugis}} + D_{\text{current yuga}}$$

nava-gaṇa-rāśi-munin nava samanvantara-devaḥ śakunta-pāte |
śarva-mateḥ sambatsarāt samkhyāṃ śaravac-chataṃ rūpaiḥ |

“O Muni of nine-digit signs! (This gives) the number of years from the śarva-mata (doctrine year), 106 (śara) increased by unity — 3179 years of kaliyuga (at the epoch śaka 550 = 628 CE).”

Brahmagupta’s epoch: year 3179 of kaliyuga = śaka 550 = 628 CE.

Appendix: Rational Geometrical Figures

iṣṭa-yaṣṭeḥ sama-cchedaṃ rūpeṇ a hīnam ubhayor mithaḥ |
ṣaḍ-bhāgena sama-j nā-try-akṣetrasya bhuje |

“The square of the optional side divided and then diminished by an optional number; half the result is the upright (koti), and that increased by the optional number gives the hypotenuse of a rectangle.”

Formula: For arbitrary rational m, n :

$$\text{Base} = m, \quad \text{Upright} = \frac{1}{2} \left(\frac{m^2}{n} - n \right), \quad \text{Hypotenuse} = \frac{1}{2} \left(\frac{m^2}{n} + n \right)$$

yasmin dve samānī bāhū anyad dvayam-asamānī ca |
tayor varga-yogo vibhājyo bhedenā sama-dvi-bāhu-kṣetre |

“Where there are two equal sides and two other unequal sides — the sum of their squares divided by their difference (gives the base of an isosceles triangle).”

For two unequal integers p, q : equal sides = $\frac{1}{2} \left(\frac{p^2+q^2}{p-q} + (p-q) \right)$, base = $\frac{p^2+q^2}{p-q}$.

sarva-dor-alpa-catustāya-kṣetra-phala-mūlaṃ dvi-bhuja-kṣetre |
dvayor dvayor ddhayor ghātaṃ mūlaṃ dvi-bhuja-kṣetrasya |

“The square root of the product of four sets of half the sum of the sides, each diminished by one side, is the exact area of a triangle (and) quadrilateral.”

Brahmagupta’s Formula:

$$A = \sqrt{(s-a)(s-b)(s-c)(s-d)}, \quad s = \frac{a+b+c+d}{2}$$

This generalizes Heron’s formula for triangles (when $d = 0$).

Chapter III: Triprasnādhikāra (Conclusion)

— The chapter on the three problems relating to diurnal motion

Verses 25-27: Ghaṭikā computations and eclipse timing

Translation: (25) When the Sun, diminished by the day-radius (*dinadala*), is east or west, the ghaṭikās till the end of the tithi are obtained by subtracting that from thirty and the ghaṭikā-remainder is obtained. (26) The reverse applies at midnight for a lunar eclipse; slowly for a solar eclipse. From the rising of the Sun, by the same ghaṭikās twice. (27) The semi-diameter (*vyāsa*) of the Earth's shadow at the Moon's orbit is obtained: the day-diameter, circumference, accurate tithi-nata-kendra-jyā and the degrees of Jupiter or Kandu (the Moon), Indu — by 191 and the new nine-veda 466, divided by the square of the radius.

Verses 38-39: Sphuṭikaraṇa of Mars

Translation: (38) Multiplied by seven degrees, the day-multiplier, the daily motion in signs, divided by the obtained degrees, the sphuṭa of Mars is increased or decreased from the mark (starting point) at Mṛga (start of Capricorn) and Karka (start of Cancer). (39) Its sphuṭa-circumference is 70 yojanas; the śīghra-sphuṭa circumference diminished by obtained degrees; 24330 diminished by veda-jina-akhi; this is the correction (*sphuṭikaraṇa*) of Mars.

Verses 40-42: Celestial latitude and shadow calculations

Translation: (4) The shadow-circle-radius multiplied by the hypotenuse, divided by the semi-diameter obtained by the Sun, the viśuvacchāyā, the yāmyā, the sum or difference of that at the tip of the bhuja. (5) The eastern or western shadow of the gnomon — taking the square root of the difference of the bhuja-squares, that eastern or western shadow is the koṭi between the tips of the bhuja. (6) Having made the shadow-tip in the middle of the direction, the gnomon according to direction is to be measured; the observer's shadow-tip situated in the middle of the direction moves in reverse.

Verses 64-66: End of Triprasnādhikāra

Translation: (64) At the end of Kṛṣṇa-caturdaśī, in the bird (śakuni) season, in the four-footed (catuspada) measure, the half-tithi, nāga, kistughna, from pratipadā etc. (66) The manifest two kalās divided, the moving karas and the hara-sa-gu-ra-ṇa-ṇaḥ 360, with one obtained sum, the remainder divided by hr̥ta is like a tithi and other.

[At this point the Triprasnādhikāra concludes. Several verses in the manuscripts are mutilated; Dvivedī notes that at some places only the commentary (ṭīkā) is available. The chapter contained 66 verses.]

Chapter IV: Candragrahaṇādhikāra

Chapter on the Computation of Lunar Eclipses

This chapter contains rules for computing lunar eclipses, including the dimensions of the Earth's shadow at the Moon's orbit, the times of first and last contacts, the phases of the eclipse, and the colours assumed by the eclipsed Moon.

Verses 1-4: Shadow dimensions and eclipse elements

Translation: (1) The Sun's jyā multiplied by the day-part-jyā, divided by the guṇākṣa-jyā or alternatively, the viṣuvacchāyā divided by twelve. (3) Twelve times the viṣuvacchāyā gives the shadow; or separately by the akṣa-lamba-jyā. Multiplied by the declination (krānti), the sama-maṇḍala-karṇa; as before, the separate shadow.

Verses 7-8: Eclipse magnitudes and durations

Translation: (7) The two akṣa-lamba-chāyā-jyā, their sum or difference gives the root; the viṣuvat-karṇa-chāyā, otherwise the hypotenuse is the śaṅku (gnomon). (8) The uccatana-jīvā-koṭi, the chāyā, the ha-jyā, the bhuja or the nata-jyā; the hypotenuse-chāyā-circle from the semi-diameter; similarly elsewhere.

Verses 14-18: Sphuṭa-tithi and lambana

Translation: (14) If the half-duration is greater, by the difference, otherwise minus one; if negative with addition, thus the half-obscuration (vimarda). (16) From the sphuṭa-tithi-antā the parallax (lambana), with or without the asakṛt-sthityardha; that sphuṭa is by separation, at the end of sthityardha-utaya-tithi. (17) In the difference of that sphuṭa-tithi-cheda, the sphuṭa in the tithi-half, with or without, from the self-vimarda-udaya, thus the corrected half-obscuration. (18) The Moon's latitude (sphuṭa-vikṣepa) multiplied by the half-duration, the corrected (sphuṭaḥ) bhūja multiplied by the half-duration is the root of the remainder as before.

Verses 16-20: Eclipse colours and phases

Translation: (16) At the end of obscuration, smoke-coloured, black; in partial eclipse, half-increased eating (grāsa); black-brown (sa-kṛṣṇa), coppery (tāmra), white-yellowish (śakli-kapīla-varṇa). (20) The measure of pure, half-duration-motion-desired, eating in the samakala-etc.; the breadth of the nadagūhara-āyāma desired one-fourth.

Thus ends the Fourth Chapter of the Brāhmasphuṭasiddhānta.

Chapter V: Sūryagrahaṇādhikāya

Chapter on the Computation of Solar Eclipses

This chapter treats the computation of solar eclipses, including the parallax corrections (*lambana*) that distinguish solar eclipse computation from lunar eclipse computation.

Verses 2-4: Lambana and eclipse elements

Translation: (2-3) Same as lunar eclipse — the shadow dimensions are computed from the Sun's declination. (4) The bhuja from the Sun's tip (arkāgra), the square deducted, 346450 and 144 multiplied by the square; the first and other, twelve times the viṣuvacchāyā, the product of the two.

Verses 5-8: Lambana-ghaṭikā and eclipse timing

Translation: (5) From the lambana-ghaṭikā-lagna, from the tātkālīka, diminished by three signs; if negative with addition, five times the asakṛt, divided by hṛdaya. (6) From the karaṇa-guṇa, from the semi-diameter, dviveda 48, divided; the lambana-nāḍis or the bāskara-vitribha-lagna-antaram. (7) If the lambana is negative, the positive result in its own lalipta-phala. (8) From the aṣajyā, from the vitribha-lagna, its own declination (krānti) uttarākṣa or of the Moon; if greater, otherwise the yāmyā.

Verses 14-17: Sphuṭa-tithi and corrected obscuration

Translation: (14) If the half-duration is greater, then by the difference, otherwise minus one; if negative with addition, thus the half-obscuration (vimarda-ardha). (16) From the sphuṭa-tithi-anta, the lambana, with or without the asakṛt-sthityardha; that sphuṭa is by separation, at the end of the sthityardha-utaya-tithi. (17) In the difference of that sphuṭa-tithi-cheda, the sphuṭa in the tithi-half, with or without; from the self-vimarda-udaya, thus the corrected (spaṣṭa) half-obscuration.

Verses 18-20: Deflection and sphuṭa computation

Translation: (18) Thus in the conjunction of planet and Moon, if greater than the measure of unity plus half, in the middle-intermission; diminished by the half-duration-obscuration, the cross (tārā). (19) The lambana like akāla-graha, asakṛt-svāvanati-lāptikā-sphuṭi, the tātkālīka-vikṣepī, the samānya-diśoḥ sum or difference. (20) The deflection (vikṣepa), the middle-difference, the above-shadow-planet situated below; if greater than half the unity-measure, not too distinct, hence the sphuṭa-statement.

Thus ends the Fifth Chapter of the Brāhmasphuṭasiddhānta.

Chapters VI-IX: Udayāsta, Candraśṛṅgonnati, Candrac-cāyā, Grahayuti

These chapters cover miscellaneous astronomical computations: rising and setting of planets, the elevation of the Moon's horns, the Moon's shadow, and conjunctions of planets.

Chapter VI: Udayastādhikāra (Rising and Setting of Planets)

Translation: (6) By twelve, the śītāṃśu-sita-jīvā, the earth-born, by nine; the day-increment-difference-ghaṭikā multiplied by six in kalāṃśa. (7) The graheka like hr̥syā-hr̥syā-yuti, by the difference of suttacāntara-obtained-days; subtracted or added in liptāmyaḥ, as before by the tātkālīka-asakṛt. (8) Daily rising and setting thus, by the tātkālīka-graha-vilagnais; the solar rising and setting at the end of the Pausa month, the Moon (śītāṃśu).

Chapter VII: Candraśṛṅgonnatyādhikāya (Elevation of the Moon's Horns)

Translation: (38) The latitude (unnati) of the Moon is found to disagree from the hypotenuse-śravaṇa-jñānāt; the rising and setting are caused by day, by prati-ghaṭika measure and from vā-jñānāt. (36) The akaicara-ghaṭikā, vyasta-jyā, multiplied by candra-māna, divided by vyāsa is pure; since the dṛk-sama is not true. (40) Previously taught as different, afterwards the other reverse; that time which is shadow, that asat-sphuṭa, suktimān before. (41) Who knows the uḍita-uḍita-astamita-avaśeṣa-kāla; how can he who knows the shadow-unnati of the Moon be an ārya-maṭa-jña.

Chapter VIII: Candraccāyādhikāya (The Moon's Shadow)

Translation: (14) The declination-jyā, the deflection (vikṣepa), the declination-cāpa: the sum or difference of the jyā of like or unlike directions of the declination. (16) Thus the bhūmi-niśa-vayo (latitude etc.) of the planet, by its declination, the candra-aṣṭaka-karma etc.; having done this in the eclipse, as in bhūmi, from that the svakrantijīvā.

Chapter IX: Grahayutyādhikāya (Conjunction of Planets)

Translation: (44) By the prati-ghaṭika and adhika-śaṅku, in the middle is shown the earth; the tripraśna-oktacā like the Sun's śaṅku, the remainder of the bhramaṇādi-krama. (46) The śaṅku, the eastern-western antaram, the viṣuvacchāyā-sum or difference, nṛtale; the yāmyottaraṃ guṇāhataṃ by svakrānti-leśaiḥ and karṇa. (47) The cāpāṃśāḥ of sada-yēḥ, saṅghuvakāpakramāṃśa-rūnāḥ; the vikṣepāṃśe vyastā, the vyasta-viśuddhā vya-sahaśāṃśaḥ.

[The chapters on Udayāsta, Candraśṛṅgonnati, Candraccāyā, and Grahayuti conclude here, with 12-70 verses each.]

Chapter XI: Tantraparīkṣādhikāya (Dūṣaṇādhikāya)

Chapter on the Critical Examination of Astronomical Systems

This celebrated chapter contains Brahmagupta's critique of various astronomical systems current in his time. He refutes the doctrines of Āryabhaṭa, Śrīṣeṇa, Viṣṇucandra, Lāṭadeva, Vijayanandi, and Pradyumna, as well as the foreign systems derived from Greece and Babylon (the Romaka and Puliśa Siddhāntas).

Verses 1-2: Introduction to the critique

Translation: (1) Those who know the brahma-vid (science) in a chapter-barred dual manner, they consider it a siddhānta; I shall declare their faults arising from the yuga-ādi differences. (2) They say the yuga is five years; making the Saṃhitā of the Sun and Moon; the adhimāsa, vamara-atra, sphuṭa-tithi-jñāna — that is false.

Verses 6-9: Critique of planetary positions

Translation: (6) The yuga-year-navadvyekṣa (nine), the bright (Moon) etc., the same pravṛttānyat; that is false, since the corrected yuga-position of the manda-nodes is fixed. (7) The planet's release is in the ruṇāyā, the mandocca increases in śīghra; in the high position, the mandocca does not exist without (its own) motion. (8) In Ārya's eight hundred, the nodes move, in the Daśagītaka are fixed; the released two-nodes move in the circle, not fixed. (9) Āryabhaṭa knows the planets; what he has said is false; the Rāhu does not cause the eclipse, its pāta is not the eighth Rāhu.

Verse 63: Conclusion

Translation: (63) Thus the systems taught, even by reading, cause ignorance by faults; the examination of systems — this is the eleventh chapter.

Thus ends the Eleventh Chapter of the Brāhmasphuṭasiddhānta.

Chapter XII: Gaṇitādhyāya

The Twelfth Chapter — On Mathematics

The Gaṇitādhyāya is the most celebrated mathematical chapter of the *Brāhmasphuṭasiddhānta*. It contains 66 verses divided into ten sections (vyavahāras), covering:

1. — Preliminary operations (fractions, proportions)
2. — Mixture problems (verses 1-16)
3. — Arithmetic and geometric series (verses 17-20)
4. — Geometry of plane figures (verses 21-39)
5. — Circle geometry (verses 40-43)
6. — Excavation/volume problems (verses 44-46)
7. — Pile problems (verses 47+)
8. — Shadow of a gnomon etc.
9. — Rules for treating quantities (verses 50-51)
10. — Shadow problems (verses 52-66)

Section 1: Parikarmma (Preliminary Operations)

Translation: (1) He who knows separately the parikamma (operation), the saṃkalita (summation) etc., and the twenty vyavahāras with chāyāta — he becomes a calculator (gaṇaka).

Translation: (2) The inverted (viparīta) and the divisor-multipliers of the quantities, the divisor-parts (chedāṃśakāḥ) with common divisor; the saṃkalita etc. are to be added, the difference is vyavakalita. (3) The rūpāraṇī (unity) multiplied by the divisor, with parts added, of two or many; the pratyutpanna (product) becomes the division by the divisor-product of the fraction-product. (4) The pratyutpanna after conversion, the divisor-part of the divisor, the divisor-times-divisor; the two-part-sum of the dividend, the divisor of all is the divisor.

Translation: (5) The square of all-parts divided by the square of the divisor becomes the varga (square root); the root of all-śita-parts, the root divided by the divisor-term. (6) Having set up the varga-mūla (square root), the utpadyanyakṛti, the trigunottara-saguṇa, and the prathamā; the nottara-kṛti-antitya-guṇā, triguṇā and utara is the ghana (cube). [Cube completed.] (7) From the second chedāghana, the cube-root-square-times-trisaguṇa-sakṛti; subtracted, times the tripūrva, from the prathama-ghana, the ghana-mūla. [Cube root completed.]

Section 2: Miśrakavyavahāra (Mixture Problems)

Translation: (16) Divided by the prakṣepa-yoga (sum of contributions), obtained, the prakṣepaka (contributor) multiplied by the gains; the ūnādhika-uttara-ardha is true, the dvitonayā with its own phala-sum added.

End of Mixture Problems.

Section 3: Śreḍhīvyavahāra (Series)

Translation: (17) The pada (term) minus one, multiplied by the utara (common difference), added to the sum of the first and last terms; the first plus the last term from the first, the cube of the middle term is pada-gaṇa (number of terms) multiplied. (18) From the utara-hīna (decreased by common difference) dviguṇādi-śeda-varga, the ghanottara-aṣṭra-vadhe; added, the pada by the śeṣeṇam, the dviguṇottara hṛtaṃ gacchaḥ (progression). (19) If the eka-uttara (common difference one) eka (first term one) and the desired gaccha (number of terms) is the saṃkalita (sum); that viyut (without) gaccha multiplied, divided by three, is the saṃkalita.

Section 4: Kṣetravyavahāra (Geometry of Plane Figures)

The *kṣetra-vyavahāra* contains rules for computing the areas and diagonals of triangles, quadrilaterals, and other plane figures. This is one of the most mathematically significant sections of the work.

Verses 21-23: General area rules

Translation: (21) In triangles and quadrilaterals with equal and two sides etc., the vyāsa (diameter) is like āyata-catura (rectangle); the vyāsa-karṇa of the field is the sama-koṇi (right-angled diagonal) of the pair of fields. (22) Half the sum of the lambaka (altitude) and karṇa in sama-tribhuje (isosceles triangle) is the area; in sama-catura (square) it is the square; in sama-koṇi (right-angled) half the sum of the vyasta (crossed diagonals). (23) In asama-catura (unequal quadrilateral), the product of the rāśis (sides), their squares and the sum; their difference-varga (square), with equal-square-half added mutually.

Verses 32-35: Specific geometric formulas

Translation: (32) The karṇa and lamba (hypotenuse and altitude) — the lower karṇa-lamba; in proportion, the uddha-śūnyāṃsa-pāṭāyāma. (33) The product of kṛti-yura-sahaśa-rādyo-bāhu-dhāto; the lambaka (altitude) twice; the product of kṛty-antara of unequal (sides) and dvitribhuja. (34) Divided by two desired quantities, from the sum of the dvi-ghaṇa-varga-phala; the side of a viṣama-tribhuja (scalene triangle) from the viṣama-phala is added to yā mūḥ. (35) The square of the desired bhuja divided by the undesired, that portion is the koṭi; of the āyata-caturastra (rectangle), the field has the desired hypotenuse.

Section 10: Chāyāvyavahāra (Shadow Problems)

The *chāyā-vyavahāra* deals with computations involving the shadow of a gnomon (*śaṅku*), problems of distance, height determination using shadows, and related topics.

Verses 1-4: Lamp and shadow calculations

Translation: (2) The *chāyāntara* (difference of shadows) with one added, divided by two, the *prāga-parayor-yuga-doṣam*; the *dinagata* (elapsed day), the *śeṣāṃśa-hṛtaṃ dvidalaṃ*, the *chāyāntara-vyekam*. (13) The distance between the lamp-surface and *śaṅku*-surface, the desired measure-*śaṅku-guṇa*; the lamp-wick-*ucchyā*, the *śaṅku* deducted, the *śeṣa-uddhṛtaṃ chāyā*. (4) The *chāyā* multiplied by the *chāyāgra-antara*, divided by *chāyāntara* is the *bhū* (earth/distance); the *bhū* is *śaṅku-guṇā chāyā*, divided, the lamp-wick-height.

Verses 44-45: Gnomon problems and *kuṭṭaka*

Translation: [End of shadow problems.] (45) The *guṇākāra* (multiplier) and *khāṇḍa-tulya* (equal to the section) are the *guṇyau* (multiplicands); the *gomūtrikā* (cow's urine) *kṛtī* multiplied; the *sahitāḥ pratyutpannaḥ* (combined product), the *guṇa* (quality) or equal to the *kāraṇa-meda*.

[The Gaṇitādhyāya concludes here with 66 verses total, containing the foundation of Indian algebra including the kuṭṭaka (pulverizer) method for solving linear Diophantine equations.]

Chapter XIII: Praśnādhyāya

The Sixth Problem-Chapter: On Mean and True Motion

This chapter presents mathematical problems (*praśna*) whose solutions demonstrate the computational methods developed in the preceding chapters. The title “” (sixth) refers to the chapter’s position within the *uttara* (secondary) section of the text.

Verses 1-2: Introduction

Translation: (1) I shall declare the *praśnādhyāyas* (problem-chapters) with solutions (*uttara*), increasing the intelligence of calculators; by which the teacher (*ācārya*) of the *tānta*-vids becomes one of intelligent mind. (2) The *adhimāsakāḥ* (intercalary months) with *vikalā* (seconds), the *avamarātra* (omitted days) with the *yugayāta*, by the *dyugaṇa* (day-count); he who knows the motion of planets in the wind’s path knows the *kāla-tantra* (science of time).

Verses 3-5: Time and planetary motion problems

Translation: (3) He who comprehends the *avamāti* with *vikalā*, the *adhimāsa-radhika-māsaka-anavamaiḥ*, the *grahamiṣṭhaṃ* in that, knows the *kāla-tantra*. (4) The *dyugaṇa* without *adhimāsa*, *avinādina garona candra-ākkah*; in that without *sphuṭa-tithi*, he who knows is the *kāla-tantra-jña*. (5) From the mean (*madhya*) to other *tithi* from *iṣṭa-madhyamā-cchadi-prahaṇas*, from the *iṣṭārka* *avid-upāte yom*, without knowing, he is not a *tantra-jña*.

[The Praśnādhyāya continues with detailed problem-solution pairs covering mean and true longitudes of planets, conjunctions, eclipses, and other astronomical computations. The full chapter contains approximately 48 verses.]

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Chapter XIV: Sphoṭagatyuttarādhikārah

Chapter XIV: On True Planetary Motions

This chapter deals with *sphuṭa-gati*—the computation of a planet’s true longitude from its mean longitude by applying the *manda* (eccentric) and *śīghra* (epicyclic) corrections.

V.1. The divisor of the *yuga* makes the cosine (*koṭijyā*) into the cosine, and the sine (*bāhuḥjyā*) into the sine. The hypotenuse (*kṣetra*) together with these—that is the *sphuṭagatikah*, the true-motion triangle.

V.2. The radius of the outer circle (*paramaṇḍala-karṇa*), made to produce the cosine-shadow (*koṭi-chāyā*) for computing the true — the hypotenuse-end, cosine-corner, and sine: that is the true-motion triangle.

V.3. The following cosine-chord (*anujakoṭijīvā*) at the radius, which the outer-circle-knower makes to produce — that which makes the *svocca* (own apogee) from its own equation (*sva-phala*) into the true — that is the true-motion triangle.

V.4. Of the field of the true-hypotenuse (*sphuṭa-pada*), sine and cosine are the results without the chord (*jyā*); without the chord, or with the equation (*phala*) added — that makes the true-motion computation.

V.5. He who makes the mean planets true by desired-day indicators (*iṣṭa-diva-yadaiḥ*) for the directions — that *sphuṭagatikah* which makes (the computation) with its own apogee-true (corrections).

V.6. Threefold: Saturn (*śani*), Moon (*indu*), Nait̥yama (Mercury), Gārā (Mars), Lakṣmī (Venus), the planet-group — gathered: Mars (*bhauma*), Aṅgiras (Jupiter), Guru — Durvarca, Vāt̥ya, Gārā — with Kai (Sun?). [Planetary name correspondences in numerical notation.]

[The chapter continues with more verses on true motion computations, including corrections for the manda and śīghra epicycles, and rules for finding planetary positions. The chapter concludes at printed page 181 with verse 46.]

V.46. This rule of mean-upper-line (*madhyotarekhā*) is the fifty-fifth chapter-method. It is to be known by the teachers of the knowers of the *tantra*; one becomes skilled in mean motion.

Here ends the Fourteenth Chapter, on Mean Motion, in the Brāhmasphuṭasiddhānta.

Chapter XV: Tripraśnottarādhikārah

Chapter XV: The Three Questions

This chapter treats the “three questions” of astronomy: relating to time (*kāla*), place

(*sthāna*), and direction (*dik*). It covers the determination of latitude from gnomon observations, shadow measurements, and the computation of time.

V.44. The sine of rising (*udaya-jyā*) and the sine of declination (*krānti-jyā*), multiplied by the diameter and divided: by twelve and by fire (3), by Śiti (cold, the zero-point) and by the equinoctial shadow — divided by that is the declination (*krāntiḥ*).

V.45. From the desired diameter (*iṣṭa-vyāsa*), from the shadow-ray (*chāyā-raśmi*), from Bhāskara's (the sun's) crossing-sign, from the double rising: the string (*sūtra*) — that sine is the measure of the particular declination (*krānti-viśeṣa-padam*).

V.46. In the six parts (of the day), the gnomon (*śaṅku*) is not known (in this way); if the gnomon is placed toward the east-west, the northern (deviation) is due to the southern excess, and that difference indicates greater latitude.

V.47. In the northern hemisphere, it is not thus; the leftward (deviation) is in the globe-string. The gnomon-base divided by the gnomon — the equinoctial shadow, twelvefold, is the multiplier.

[The chapter contains 60 verses on the three problems, concluding at printed page 193.]

Chapter XVI: Grahāṇottarādhikārah

Chapter XVI: On Eclipses (Advanced)

This chapter treats refined computations for eclipses (*grahaṇa*): the computation of lunar and solar eclipses, the graphical *parilekha* method, and the determination of contact (*sparsā*) and release (*mokṣa*) times.

V.15. The *parilekha* (graphical representation) of the planet, by the planet-rule (*graha-kalpa*) first, either curved or straight — the instantaneous position, the closing and opening (of the eclipse) — thus.

V.16. The sine of the particular (longitude) difference in minutes (*viśeṣa-puṇy-aṁśa-jyā*), the shadow by the measure, and the degrees of the second arc — at the edges according to direction, direct or reversed (*viparyayaḥ*).

V.20. The difference of those deflection-minutes (*valana-aṁśaka*), from the sine-east-measure and perpendicular — the shadow obtained by the radius (*trijyā*) — the *pragraha-kṣepa*, before the retrograde motion.

V.21. Divided by the semi-diameter (*vyāsāṅka*), the minutes of the lunar-node-difference are the multiplier. By the middle-strength-minutes (*madhyabala-lip*) the north-south direction goes to the left.

[This chapter contains 46–47 verses on eclipse computations.]

Chapter XVII: Śṛṅgonnatyuttarādhikārah

Chapter XVII: On Lunar Cusp Elevation

This chapter treats the elevation of the lunar horns (*śṛṅgonnati*)—the elevation of the moon’s cusps, and the determination of time from this knowledge. This is a short chapter of ten verses.

[The complete verses of this chapter are not fully preserved in the scanned pages of this section. The chapter is transitional between the astronomical and mathematical sections.]

[This short chapter of 10 verses completes the astronomical portion before the extensive mathematical chapters.]

Chapter XVIII: Kuṭṭakādhyāyah — The Algebra Chapter

Chapter XVIII: The Kuṭṭaka (Algebra) Chapter

This is the most important mathematical chapter of the BSS, containing 102 verses. It covers the *kuṭṭaka* (pulverizer) method, operations with positive and negative numbers, *avyakta* (algebra) computations, *saṅkramaṇa* (transposition), linear and quadratic equations in one unknown, multi-variable equations, the *bhāvitaka* (product-nature) equation, and the *varga-prakṛti* (Pell’s equation).

V.1. Since problems generally cannot be solved without the pulverizer (*kuṭṭaka*), I shall state the *kuṭṭaka* together with problems.

V.2. The teachers (*ācāryas*) of the experts of the *tantra*, who know the pulverizer, zero (*kha*), debt (negative, *ṛṇa*) and wealth (positive, *dhana*), *avyakta* (unknown), elimination (*haraṇa*), one-*varṇa* and *bhāvitaka* equations, and the *varga-prakṛti* (quadratic nature, i.e. Pell’s equation).

V.3. The remainder from the divisor (*hāra*) corresponding to the greater residue (*adhikāgra*), divided by the divisor (*cheda*) corresponding to the lesser residue (*ūnāgra*) — whatever is the quotient from the mutual division is to be placed separately, downwards (*adhaḥ*), one below the other.

[This is the initial step of the Euclidean algorithm applied to the divisors A and B. The quotients are written in a column.]

V.4. The remainder, multiplied by an assumed number (*iṣṭa*) such that, when combined with the difference of the residues (*agra-antara*), it is exactly divisible (*śudhyati*) — the multiplier (*guṇakaḥ*) is to be set down; and the quotient is the product of the last (*antya*) and the penultimate (*upāntya*).

[If r is the final remainder, find m such that $rm + (a - b)$ is divisible. Then the *guṇaka* and *labdha* are computed from the column of quotients by back-substitution.]

V.5. The upper one (*ūrdhva*), combined with the last (*antya-yutaḥ*), (is divided by) the divisor of the lesser residue (*hīnāgra-cheda*); the remainder, multiplied by the divisor of the greater residue (*adhikāgra-cheda*) and combined with the greater residue, becomes the residue (*agram*).

[The solution x to $ax + c = by + d$ (modulo the divisors) is constructed by this back-substitution. Let x_0 be the *guṇaka*; then $N = x_0 \cdot B_{\text{large}} + r_{\text{large}}$.]

V.6. Twice the *yuga* (epoch period) is the product of the divisors; the product of the divisors is the *yuga-gata* for both residues. Thus by the *kuṭṭaka*, the derivation of three etc.

V.7. The residue of the revolutions (*bhagaṇa*) etc. is the *agra*; divided by the divisor; and zero, divided by the residue of days — of these two, the residue of revolutions etc., divided by the day-born residue — that is the day-count (*dyugaṇa*).

V.8. By whatever multiplier the day-born and revolution-etc. residues of the circle etc., divided by their common divisor, are equal — the product of that is the day-count etc.

V.9. The mutual division of the dividend and divisor, whatever is the remainder — by that, divided, they become reduced (*nis-chedau*); those same, mutually divided.

V.10. The quotient is to be placed down, down; the remainder multiplied by the chosen multiplier (*iṣṭa-guṇakāra-saṅguṇam*) — it is exactly divisible so that one is subtracted (*eka-hīnam*). The multiplier is to be set down; and the result is from the last.

V.11. The end-residue (*agrāntam*), by the penultimate multiplied, combined with its upper (value), divided by the divisor without remainder (*niśśeṣa-bhāgahāreṇa*) — thus is the *sthira-kuṭṭakaḥ* (fixed pulverizer), the remainder.

V.12. From the desired revolution-etc. residue, by its own *kuṭṭaka*-multiplier, by its own divisor-multiplied — the remainder is the day-count, combined with the reduced multiplier and divisor.

V.13. Thus in equal (signs) and in unequal: debt (*rṇa*) and wealth (*dhana*), wealth and debt — whatever was stated, the reversal (*vyastatvam*) of debt and wealth is to be done to the dividend (*guṇya*) and additive (*prakṣepa*).

V.14. The multiplier is the divisor; the divisor is the multiplier; wealth is debt, debt is wealth — to be done. Square (*varga*) is root (*pada*); root is square; from the end, reversed (*viparītam*); that is the first.

V.15. He who knows the *kuṭṭaka* by the epoch-periods (*yu-gādi*) of planets, separately stated by two, three, four etc. — **he** truly knows the *kuṭṭaka*.

[Verses 16–29 contain illustrative problems (*praśnas*) on the pulverizer applied to astronomical computations.]

V.16. “The residue of revolutions etc. at the desired — when on a moon-day, on a sun-day, or a Jupiter-day, on a weekday — tell the signs.” He who tells this knows the *kuṭṭaka*.

V.17. “On a weekday, whatever is the residue of degrees, the residue of minutes — when that on a moon-day, of the sun, or of the moon — he who tells this is a knower of the *kuṭṭaka*.’

V.18. “The desired (planetary positions) in the tithi-measure-days, which are sun etc., again when in those desired planet-weekdays” — he who tells this is a knower of the *kuṭṭaka*.

[Verses 19–29 continue with astronomical praśnas: given planetary residues at a particular weekday, finding the elapsed days and other planetary positions. These were standard problems for Indian calendar-makers (pañcāṅga-kāras).]

Operations on Positive and Negative Numbers

This section contains Brahmagupta's famous rules for arithmetic operations involving positive (*dhana*), negative (*ṛṇa*), and zero (*śūnya*, *kha*, *ākāśa*). These are the **earliest explicit rules** for operations with signed numbers and zero in the history of mathematics.

V.30. The sum of two positives is positive; of two negatives is negative; of a positive and a negative is their difference. Zero is the sum of a positive and an equal negative; and of two zeros.

[In modern notation:]

$$\begin{array}{ll} (+) + (+) = (+) & (-) + (-) = (-) \\ (+) + (-) = \text{their difference} & a + (-a) = 0 \\ 0 + 0 = 0 & \end{array}$$

V.31. The lesser is to be subtracted from the greater; the positive from the positive, the greater negative from the lesser negative — their difference is reversed (*vyastam*). The negative becomes positive, the positive becomes negative.

[Subtraction rule: $a - b = -(b - a)$ when $b > a$. For signs: $(+a) - (+b) = -(b - a)$ and $(-a) - (-b) = +(b - a)$ when $b > a$.]

V.32. Zero subtracted from debt (*ṛṇa*) is debt; from wealth (*dhana*) is wealth. Zero (*ākāśa*) — when positive is subtracted from negative, or negative from positive, then it is additive (*kṣepya*).

[Sign rules for subtraction with zero:]

$$a - 0 = a \qquad 0 - a = (\text{sign reversed})$$

V.33. The product of two negatives is positive; the product of two positives is positive. The product of zero and a negative, of zero and a positive, or of two zeros — the product is zero.

[Multiplication of signed numbers:]

$$\begin{array}{lll} (-) \times (-) = (+) & (+) \times (+) = (+) & (+) \times (-) = (-) \\ 0 \times (-) = 0 & 0 \times (+) = 0 & 0 \times 0 = 0 \end{array}$$

V.34. Positive divided by positive is positive; negative divided by negative is positive. Zero divided by zero is zero. Positive divided by negative is negative; negative divided by positive is negative.

[Division of signed numbers:]

$$\begin{array}{lll} \frac{(+)}{(+)} = (+) & \frac{(-)}{(-)} = (+) & \frac{0}{0} = 0 \\ \frac{(+)}{(-)} = (-) & \frac{(-)}{(+)} = (-) & \end{array}$$

[Verses 30–34 are historically momentous. Brahmagupta is the first author in world history to state general rules for arithmetic operations with signed numbers and zero. Note that $\frac{0}{0} = 0$ is the only questionable rule from a modern perspective; Brahmagupta also stated (elsewhere) that division by zero leaves the fraction unchanged, i.e. $\frac{a}{0} = \frac{a}{0}$ as an unreduced entity.]

V.36. The sum, decreased by the difference and divided by two, is the *saṅkramaṇa* (transposition); or divided by the difference. The difference of squares, decreased by the sum of the difference and divided by two — the *viśama-karma* (unequal operation).

[If $x + y = S$ and $x - y = D$, then:]

$$x = \frac{S + D}{2}, \quad y = \frac{S - D}{2}$$

The “difference of squares” formula: $x^2 - y^2 = S \cdot D$.

[Verses 37–42 treat *karaṇī* (surds), *bhāvitaka* (product-form), and algebraic identities for powers of unknowns. These are not fully preserved in the scanned pages of this section.]

V.43. Divided by the difference of unknowns (*avyakta*), reversed, the difference of *rūpas* (absolute terms) — in the equation (*sama*), the unknowns (*avyakta*) are to be subtracted; the *rūpas* are placed below that.

[For equations $ax + b = cx + d$, divide by $(a - c)$ and rearrange:]

$$x = \frac{d - b}{a - c}$$

V.44. Of the *rūpas* multiplied by four and the square of the middle (*madhya*) — the square-root of that, decreased by the middle and divided by twice the multiplier (*varga-dvi-guṇa-uddhṛtam*) — is the *madhya* (unknown).

[This is Brahmagupta's verbal statement of the quadratic formula. If $ax^2 + bx = c$:]

$$x = \frac{\sqrt{4ac + b^2} - b}{2a}$$

This is algebraically equivalent to the modern formula $x = \frac{-b \pm \sqrt{b^2 + 4ac}}{2a}$ for the equation $ax^2 + bx - c = 0$. Brahmagupta always writes the constant term on the right side, so his formula uses $+4ac$.

V.45. Of the *rūpas* multiplied by the square and combined with the square of half the unknown — whatever is the root, decreased by half the unknown, divided by the square — that is the unknown.

[Alternative form:]

$$x = \frac{\sqrt{a^2c + \left(\frac{b}{2}\right)^2} - \frac{b}{2}}{a}$$

[Verses 46–50 contain astronomical problems solved by the *kuṭṭaka* and quadratic methods.]

V.46. “From the degree-residue (*aṃśaka-śeṣa*) with one added, divided by twelve, multiplied by four and increased by eight, equals the degree-residue with one — when this happens, tell the day-count (*ahargaṇa*).”

[An astronomical problem leading to a linear equation in the *ahargaṇa* (elapsed days).]

V.47. “The *adhimāsa-śeṣa* (excess month residue), decreased by two, divided by three, increased by seven, multiplied by two — equals the *adhimāsa-śeṣa*. When this happens, tell the *yuga-gata* (elapsed in the *yuga*).”

V.48. “The *avama-avaśeṣa* (omitted day residue), decreased by one, divided by six, increased by three — equals the *avama-śeṣa* divided by five. When this happens, tell the *yuga-gata*.”

V.51. From the first unknown (*varṇa*), having subtracted the other unknowns, divided by the first measure (*ādyā-māna*), combined — two similar divisors, repeatedly, reversed — the *kuṭṭaka* (is applied) for many (unknowns).

[For multiple equations in many unknowns: eliminate all but one unknown, then apply the kuṭṭaka.]

V.52. “From the day-count increased by the elapsed revolutions, from that with its residue added, combined with their sum, or from their sum — the day-count” — he who tells this is a *kuṭṭaka*-knower.

V.53. “From the day-count decreased by the elapsed revolutions, from that with its residue subtracted, diminished by their sum, or from their difference — the day-count” — he who tells this is a *kuṭṭaka*-knower.

V.54. “With the quantities etc., with their residues, thus with the elapsed intercalary months and days subtracted, and with their residues — the *yuga-gata*” — he who tells this is a *kuṭṭaka*-knower.

[Verses 55–59 contain more astronomical applications and computational rules for calendar calculations.]

V.60. The *bhāvitaka* (product-nature) multiplied by the *rūpa*, together with the product of the unknown (*avyakta-vadhā*), divided by the desired (*iṣṭa*) — at the desired quotient: if less, add more; if more, add less; *kṣepya* (additive) to the *bhāvita* divisor; reversed.

[The *bhāvitaka* is the coefficient of the product of two unknowns, i.e. the term Bxy in the general quadratic form $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$.]

V.61. From the product of the sun's sign and degrees, having subtracted the signs and degrees multiplied by three and four, having seen ninety — making the sun (*sūrya*), the calculator (does this) up to a year.

V.62. In the *bhāvitaka*, the product by which unknown — having made that amount desired, the sum of the unknown (*varṇa*) multiplied by that becomes the *rūpas* (known quantities).

V.63. The product of the unknown-quantity and the *bhāvitaka* becomes the desired unknown-number. Thus it is accomplished even without the *bhāvitaka* equation; what was done from that?

V.64. The root is twofold: from the square of the desired, from the multiplier multiplied, increased or decreased by the desired. The first product is the multiplier-product; together with the last product — made last.

[For equations of the form $Nx^2 + k = y^2$ where N is the *guṇaka* (multiplier) and k is the *kṣepa* (additive).]

Varga-prakṛtiḥ (Pell's Equation)

This section treats the “square-nature” or Pell’s equation: finding integers x, y such that

$$Nx^2 + k = y^2$$

where N (the *guṇaka*) is a non-square positive integer and k (the *kṣepa*) is an integer. Brahmagupta gave the composition formula (*bhāvanā*) and the cyclic method (*cakravāla*), which were rediscovered in Europe only by Euler and Lagrange in the 18th century.

V.65. The product of the cross-terms (*vajra-vadha*) plus one is the first *prakṣepa*; equal to the product of the *prakṣepas*; divided by the *prakṣepa*-subtractor — the roots (*mūle*) are in the *prakṣepas*, *rūpas*.

[The “cross multiplication” identity (*bhāvanā*): If (x_1, y_1) solves $Nx^2 + k_1 = y^2$ and (x_2, y_2) solves $Nx^2 + k_2 = y^2$, then $(x_1y_2 \pm x_2y_1, Nx_1x_2 \pm y_1y_2)$ solves $Nx^2 + k_1k_2 = y^2$.]

V.66. The *rūpa*-*prakṣepa* roots, separately, by the desired *kṣepya* and *sódhya*-roots — having done, the end and first roots which are in the *prakṣepa* — in subtraction or desired.

V.67. When increased by four (*catur-adhike*): the square of the last root (*antya-pada-kṛtiḥ*), decreased by three (*tri-ūnā*), halved, multiplied by the last root — is the last root. The square of the last root, decreased by one (*vi-ekā*), divided by two, multiplied by the first root — is the first root.

[Recurrence relations for generating new solutions to $Nx^2 + 1 = y^2$ from known solutions. If (x_1, y_1) is a known solution with $k = -1$, then:]

$$y_{n+1} = \frac{(y_n^2 - 3)y_n}{2}$$

$$x_{n+1} = \frac{(y_n^2 - 1)x_n}{2}$$

V.68. When decreased by four: the square of the last root, increased by three and one, the product halved, separately, decreased by one — increased by one, decreased by one, multiplied by the last — the product of the roots (is) the first, the last root.

V.69. In the square of the multiplier (*guṇaka*), the *kṣepa*, by some (number) divided, added, subtracted, halved — the first (root). The last root — the other, divided by the multiplier-root, (gives) the first.

[Given $Nx^2 + k = y^2$, divide by k to reduce the additive. If k divides N or a related quantity, new roots can be found.]

V.70. When the multiplier is a perfect square (*varga-cinne guṇake*): the first (root), divided by its root, becomes (the new root). When the *kṣepa* is a perfect square: multiplied by its root, then (are) the roots.

[If $N = a^2$ is a perfect square, then $a^2x^2 + k = y^2$ gives $y = \sqrt{a^2x^2 + k}$. If also $k = b^2$, then $\sqrt{(ax)^2 + b^2}$ may simplify.]

V.71. The sum of the multipliers, multiplied by eight, divided by the square of the difference of the multipliers — the quantity. The two multipliers, tripled, reversed, increased, divided by the difference — (are) the roots.

V.72. The square, decreased by the sum of the other squares, divided by the half-square of the sum-difference — those multiplied by them, combined and separated — squares; in the product, combined with the *rūpa*.

[The general composition (*bhāvanā*) formula:]

$$\begin{aligned} &\text{If } Nx_1^2 + k_1 = y_1^2 \text{ and } Nx_2^2 + k_2 = y_2^2, \\ &\text{then } N(x_1y_2 \pm x_2y_1)^2 + k_1k_2 = (Nx_1x_2 \pm y_1y_2)^2 \end{aligned}$$

When $k_1 = k_2 = 1$, this generates infinitely many solutions to $Nx^2 + 1 = y^2$ from a single fundamental solution. This is Brahmagupta's great discovery, called the *samāsa-bhāvanā*.

Astronomical Divination Rules

This section gives remarkable “divination” rules: given the planetary residues at a particular day, one can compute the positions of other planets. These were used by calendar-makers (*pañcāṅga-kāras*) to verify their computations.

V.75. “The square of the residue of signs and minutes, multiplied by twenty-two (*dvinavati*), or multiplied by eighty-three (*tryaśīti*), with one added — on a weekday, making the square — the calculator (does this) up to a year.”

V.76. “The residue of the sun’s seconds (*viluptā-śeṣa*), decreased and multiplied by five, likewise by ten — on a Jupiter-day, making the square — the calculator (does this) up to a year.”

V.77. “The square of the residue of revolutions etc., multiplied by three, combined with nine hundred — or the square decreased by eight hundred — the calculator (does this) up to a year.”

V.78. “The square of the residue of revolutions etc., multiplied by four, combined with sixty-five — or the square decreased by sixty — the calculator (does this) up to a year.”

V.79. “The desired residue of revolutions etc., decreased by twenty-two, multiplied by eighty-three, combined with one — making the square — the calculator (does this) up to a year.”

V.80. “The square of the *adhimāsa-śeṣa* (intercalary month residue), multiplied by thirteen, combined with three hundred — or the square decreased by twenty-seven (*tri-ghana*) — the calculator (does this) up to a year.”

V.81. “The residue of the moon’s seconds, multiplied by seventeen, also multiplied by thirteen — separately, combined with one — making the square — the calculator (does this) up to a year.”

V.82. “The square of the *avama-avaśeṣa* (omitted-day residue), multiplied by twelve, combined with one hundred — or the square decreased by three — the calculator (does this) up to a year.”

V.83. “On a weekday, the residue of the sun’s minutes (*kala*), combined with or decreased by the residue of Jupiter-day seconds — the square and product with one — the calculator (does this) up to a year.”

V.84. “The residue of seconds (*vikalā-śeṣa*), combined with ninety-three (*trina-vatī*) and decreased by seventy-seven (*saptaśaṣṭi*) — on the sun’s weekday, making the square — the calculator (does this) up to a year.”

V.85. “On a weekday, the residue of the sun’s minutes, combined with twelve and sixty-three — or decreased by sixty and eight — the calculator (does this) up to a year.”

V.86. “From the residue of the moon’s seconds, the residue of the sun’s sec-

onds, or the residue of degrees — or alternatively the mean (position) desired — the calculator (does this) up to a year.”

V.87. “From the residue of seconds of Venus (*jīva*), Mars (*kuja*) and the Moon; from the residue of Mars’s seconds (*bhauma-liptikā*); from the residue of the moon’s degrees — the Sun — the calculator (does this) up to a year.”

[Verses 75-87 give a remarkable set of “divination” rules. Given a particular planetary residue at a known weekday, these formulas allow one to compute other planetary positions. The constants (22, 83, 5, 10, etc.) encode specific planetary period relations. These verification rules were essential for ensuring the accuracy of pañcāṅga (calendar) computations.]

[Verses 88–99 treat the kuṭṭaka method for solving multiple planetary congruences and refined calendar computations.]

V.100. These problems strike the heart (with joy). One may pose thousands of other problems; given by others — thus one should solve those problems by the stated methods (*karaṇaiḥ*).

V.101. In the assembly of people, among the knowers of fate (*daiva-vidām*), it destroys the radiance (of rivals), like the sun (destroys) the stars — (even) one *kuṭṭaka* problem recited — what to speak of hundreds!

V.102. These problems have been stated for each rule, in the illustrative rules (*sa-uddeśakeṣu sūtreṣu*). By three hundred and three (303) *āryā* (verses), the *kuṭṭaka* and the eighteenth chapter.

*[Thus ends the Eighteenth Chapter, the Kuṭṭaka, in the
Brāhmasphuṭasiddhānta.]*

Chapter XIX: Śaṅku-chāyā-vijñānam

Chapter XIX: On Gnomon and Shadow Knowledge

This chapter contains 20 verses on gnomon (*śaṅku*) and shadow (*chāyā*) computations. These techniques allow the determination of latitude, building heights, water depth, and altitude by mirror reflection.

V.1. Having seen the half-day ghaṭikās, he who knows the sun-knowing (method), knows the degrees of latitude (*akṣa-aṃśakān*). By the ghaṭikās of the interval from rising — from the known, the knowable — he is a knower of the *tantra*.

V.2. By the ghaṭikās of the interval from setting, he who brings the knowable from the known, thence brings the mean motion and the *yuga*-revolutions — he is a knower of the *tantra*.

V.3. He who brings the measures of the darkness-sun, the moon-horn, from the height of the lamp-flame — in the knowledge of the ground between gnomon-base (and object), the shadow — he is a knower of the *tantra*.

V.4. He who knows the desired house-height, knows that interval, is observed in water — shows the top of the house-wall, or in a mirror — he is a knower of the *tantra*.

V.5. By the knowledge of the second-shadow-light-edge interval, he knows the lamp-height — or a knower of gnomon and shadow knows the earth's shadow — he is a knower of the *tantra*.

V.6. Having seen the house-base-interval of the net of light and shadow, having seen the top of the house — a knower of the ground — he knows the house-height, having seen it in oil or water — he is a knower of the *tantra*.

V.7. Having observed the top of the house in water, having spread out the water again, in self-ground-knowledge — he brings from the water the ground or the height of the house — he is a knower of the *tantra*.

V.8. With known shadow-men (measures), the interval of water and wall having been known — in the wall, he who knows the ascent of the sun's light (ray) — he is a knower of the *tantra*.

V.9. The half-day ghaṭikās of the desired day, the prāṇas of the interval of fifteen ghaṭikās — by the day's motion prāṇas, by those he should compute the latitude as before.

V.10. Of the known and knowable planets, by the nāḍikās of the interval from rising — greater or less — by the risings, the known from the known, the knowable on the east-west (is) to be known.

V.11. The known, with its half-duration, by the risings — greater or less by the nāḍikās of the interval from setting — from the known, on the east-west the knowable; when the half-duration is decreased, the knowable.

V.12. Having made the known into the mean, again on another day, that interval is the motion (*bhuktiḥ*). By the rule of three (*trairāśikena*), by the motion — the derivation of the planet’s circle in the *kalpa*.

V.13. The reverse of half the duration (*sthiti*) is the true darkness-measure in an eclipse. From the measure-rise of the sun and moon, by the ghaṭikā-part, from the terrestrial rising.

V.14. The interval of lamp-base and gnomon-base, multiplied by the desired-quantity-gnomon, divided by the lamp-flame-height minus the gnomon — the quotient is the shadow (*chāyā*).

[Similar triangles: If h = lamp height, g = gnomon height, d = distance between them, then shadow $s = \frac{d \cdot g}{h - g}$.]

V.15. The shadow, multiplied by the gnomon-interval, divided by the shadow-interval — (is) the ground (distance). The shadow together with the gnomon, multiplied by the gnomon, divided by the shadow — (is) the lamp-height.

V.16. Having known the gnomon and shadow, from proportion (*anupātāt*) one should compute the elevations (*samucchrayān*) — of houses, temples, trees, and mountains — or having known the height, the shadow.

[This is the standard “rule of three” for similar triangles: $\frac{\text{height}}{\text{shadow}} = \frac{\text{known height}}{\text{known shadow}}$, so unknown quantities can be found by proportion.]

[Verses 17–19 give additional rules on shadow computations and heights/depths using water and mirror reflections. Verse 20 gives the chapter colophon.]

[Thus ends the Nineteenth Chapter on Gnomon and Shadow Knowledge.]

Chapter XX: Chandaś-citty-uttarādhikārah

Chapter XX: On Prosody (Chandaś-citty-uttara)

This chapter contains 19 verses on Sanskrit prosody (*chandah-sāstra*), dealing with combinatorial calculations for metrical patterns — the number of possible arrangements of long (*guru*, ‘G’) and short (*laghu*, ‘L’) syllables. Brahmagupta here presents an early form of combinatorics.

V.1. The *rg*-group (combinations), the permutations (*paryāya*), the aggregate-sums, in the unequal pairs — the *soma* etc., obtained as before, the quadruples, the remainder-additions — the last.

V.2. Decreased by one etc., the first and last, their reversals — in squares etc., of the odd-even (syllables), one should increase the quarters by direct and reverse order.

V.3. One by one, two by two, with additional appendages in those positions — beginning from the square up to the desired — the spread-out (*prastāra*) becomes with barley-middle (*yava-madhya*, i.e. Pascal’s triangle).

*[The “Meru of Prosody” (*prastāra*) is Pascal’s triangle: the number of metrical patterns with n syllables and k guru syllables is $\binom{n}{k}$. The “barley-middle” refers to the shape of the triangular arrangement.]*

V.4. With one, the last — the two-foot beginning, the three-foot etc., below, separately the numbers. That is to be subtracted, decreased by one, separately from that — combined with one above.

V.5. Up to the quarter, decreasing by one, among the syllables, then increasing by one — in the product beginning from one, the final number of squares etc.

[For n syllables, each of which can be G or L , the total number of possible patterns is 2^n .]

V.6. With one added to the quarter-half — in the uneven (syllables) above, in the even (syllables) at the quarter-half. From the half, doubled, decreased by one — the *yulāni* (pairings) below that, for all.

V.7. By the middle (terms), thus decreased by half, by the quarter-steps in order — equal to the reversed, beginning from the quarter — in the uneven, subtracting one in the middle, from that one should do (the computation).

V.8. With one-plus-equal-to-the-step-beginnings, the setting is excessively subtracted below — the sum of numbers such as it is — the first, decreased by three — in the lost (*naṣṭa*, i.e. finding a missing pattern).

V.9. Made with middle terms and halved — in the equal-number (case), in direct and reverse order — the *kṣepa* (additive). In the uneven, decreased by one — the half, in order, in reverse order, with one added.

V.10. In the equal-number (case), by the staircase in direct and reverse order — thus also in the uneven — to be constructed. When cooked (computed) is seen, the first — the remaining syllables at the end.

*[Verses 11-18 continue the treatment of combinatorial prosody, including algorithms for the “Meru of Prosody” (Pascal’s triangle) and rules for lost (*naṣṭa*) and spread-out (*uddhiṣṭa*) metrical patterns. The “staircase” (*sopāna*) is a procedure for listing all possible metrical patterns in systematic order.]*

*[Thus ends the Twentieth Chapter on Prosody in the **Brāhmasphuṭasiddhānta**.]*

Chapter XXI: Golādhyāyaḥ (Jyā-prakaraṇam)

Chapter XXI: The Sphere — Sine Computation

This section includes the Jyā-prakaraṇa (verses 17–23), which gives Brahmagupta’s method for computing the table of sines. The radius is $R = 3270$ minutes. The “segments” (*jyā-khaṇḍa*) are the tabular sine-differences. Brahmagupta uses a second-order interpolation formula, more accurate than the linear interpolation used by earlier authors.

V.17. Having placed marks at the eighth parts of a sign ($3^\circ 45'$ intervals) in direct and reverse order, one should construct chords (*jyā*) for every two-by-two: their halves are the sine-chords (*jyā-ardhāni*).

*[Brahmagupta divides each sign (30°) into 8 parts of $3^\circ 45'$ each, giving 24 intervals in a quadrant (0° to 90°). The *jyā-ardha* (half-chord) is the modern sine: $R \sin \theta$.]*

V.18. The sine-half-differences of sine-halves are the sine-segments (*jyā-khaṇḍa*). Reversed from the last, or: the bow (*dhanus*) is the reverse-sine (*utkrama-jyā*) together with these.

*[The *jyā-khaṇḍa* are the tabular differences: $\Delta_i = R \sin(i+1)\alpha - R \sin i\alpha$ where $\alpha = 3^\circ 45'$.]*

V.19. From the quarter-parts of the half-diameter squared, multiplied by one, two, and three: the roots are the eighth, twelfth, and sixteenth segments; the others (are derived) from these.

[Initial segments computed from $R^2/4$:]

$$\begin{aligned} j_8 &= \sqrt{R^2 \cdot 1/4} = R/2 = R \sin(30^\circ) \\ j_{12} &= \sqrt{R^2 \cdot 2/4} = R/\sqrt{2} = R \sin(45^\circ) \\ j_{16} &= \sqrt{R^2 \cdot 3/4} = R\sqrt{3}/2 = R \sin(60^\circ) \end{aligned}$$

With $R = 3270$: $j_8 = 1635$, $j_{12} = 2312$, $j_{16} = 2832$.

V.20. Having subtracted the fourth part of the sum of squares of equal direct and reverse sines and equal segments from the square of the half-diameter, the square-root of the remainder is the first (sine-chord).

[If j_k and j_{n-k} are known sines, and δ is the common segment-difference, the next sine is:]

$$j_{k+1} = \sqrt{R^2 - \frac{(j_k + j_{n-k})^2 + (2\delta)^2}{4}}$$

V.21. Its half-segments, decreased by those equal to the earth (*jina*) — the second in production.

[This verse contains numerical values encoded as words (saṅketa system). “Jina” = 1 (Lord of the earth). The second-order correction refines the first-order segments.]

V.22. Thus the small or numerous sine-segments — the first segments are the sine-halves of one-sixth, one-fourth, and one-third of the circle’s circumference.

[The reference sines at key angles:]

$$R \sin(30^\circ) = R/2 = 1635$$

$$R \sin(45^\circ) = R/\sqrt{2} \approx 2312$$

$$R \sin(60^\circ) = R\sqrt{3}/2 \approx 2832$$

V.23. From the diameter multiplied by the equal reverse-segment, or from the quarter-part, having done the stated segment operations — the derivation of sine-halves is not difficult from this.

[Summary of the method: Given the tabular differences (*khaṇḍa*), one can compute all sines by second-order interpolation using only multiplication and square roots.]

[

0.5cm][Brahmagupta's sine table uses $R = 3270$ minutes. The twenty-four *jyā-khaṇḍa* (sine-differences) are: 225, 224, 222, 219, 215, 210, 205, 199, 191, 183, 174, 164, 154, 143, 131, 119, 106, 93, 79, 65, 51, 37, 22, 7. The second-order correction formula accounts for the curvature of the sine function, yielding values accurate to within 1 minute of arc.]

Chapter XXII: Yantrādhyāyaḥ

Chapter XXII: On Astronomical Instruments

This chapter contains 43–57 verses on astronomical instruments (*yantra*). It describes the bow instrument (*dhanur-yantra*), staff instrument (*yaṣṭi-yantra*), water instruments, wheel instruments, and gnomon instruments. These were used for measuring celestial altitudes, azimuths, and for timekeeping.

Instruments Described

- **Dhanur-yantra** — A semicircular or arc-shaped instrument for measuring altitudes, resembling a bow.
- **Yaṣṭi-yantra** — A staff or rod instrument for measuring angles and time.
- **Jala-yantra** — Water-based instruments including water clocks and mercury devices.
- **Śaṅku-yantra** — Gnomon instruments for shadow measurements.
- **Cakra-yantra** — Circular instruments with rotating sighting vanes for measuring celestial positions.

[Verses 1–28 describe the bow instrument, staff instrument, and their use in measuring celestial altitudes and azimuths. Verses 29–40 treat water instruments and circle instruments.]

V.41. The *kṣeṭṭa* (object to be observed), by its own light, the *kṣepya* (object aimed at) — equal, thus in mercury (*pārada*), when it becomes — the mixed-time measure from the *śaka* epoch (78 CE) or from the *mudda* (particular epoch).

V.42. The co-altitude (*kota*) going above, held obliquely in the string (*sūtraka*), in small (measures) — having cast upon the eastern fire-place (*prāv-analaka*), the *nāḍikā* (water-clock) flows, falls thus.

[Verses 43–48 contain further rules on the use of instruments for time measurement and planetary observation.]

V.49. A wheel of light-wood, with equal bearing intervals, of lotus-shape — having faith, filled with mercury in the center, well-polished, having performed the *saṃdhyā* (preparation).

V.50. Obliquely in the middle of the co-altitude, the support above — a beautiful semi-circle (*rūpā-radaṃ*) moves. The small wheel with holes — thus constantly in the air it rotates.

[Description of a self-rotating wheel instrument, possibly an early form of anemometer or a water-driven astronomical device. The “holes” regulate the flow of water or air.]

[Verses 51–53: Additional rules on wheel instruments and their construction. The chapter concludes with verses praising the value of observation and proper instruments.]

V.53. The *karaṇa-jyā* (computed sine), the unsteady motion itself — the arrow, the rising, the ascension — by the instrument with confidence, two-fold — from the five-fold of Ari (the sun’s) motion.

*[Thus ends the Twenty-second Chapter on Instruments in the
Brāhmasphuṭasiddhānta.
 End of the section on observation.]*

Appendix: On Variant Readings

Manuscript Sources for Variant Readings

The footnotes in the printed edition record variant readings from four manuscript sources, designated by the following sigla:

The footnotes use a notation such as “ for (*reading*)” to indicate that manuscript has the variant reading instead of the adopted text. Parenthetical notations like “(—)” indicate that the verse number in the source manuscript is different.

Chapter 23: — The Measurement Chapter

The twenty-third chapter of the *Brāhmasphuṭasiddhānta* treats of the measurement of time and the practical application of astronomical calculations. This chapter, like others in the work, is composed in the classical *āryā* metre and presents Brahmagupta's rules for calendrical computation, planetary positions, and the reconciliation of different astronomical systems.

The chapter is divided into two principal sections:

- (i) **Prayogavidhi** — Practical application of the siddhānta
- (ii) **Caturviṃśa** — The twenty-four fold division of time

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### Practical Application (Verses 1-4)

"The solar year, the lunar months and *tithis*, the civil days, days, months, years, the positions of the planets, the *nakṣatras*, and the zodiacal signs—these we call measures (*māna*)."

#### — Verse 1

##### Word Analysis :

(1) *saurairāṇḍa* — the solar year; (2) *māsaistithayaḥ* — lunar months and *tithis*; (3) *sāvanāni divasāni* — civil days; (4) *dinamāsābdāni* — days, months, and years; (5) *grahanakṣatrarāśayaḥ* — planets, nakṣatras, and zodiacal signs—these are called *māna* (measures).

"The measures of solar, lunar, stellar, and civil reckoning, and the computation of planets by these measures separately—the fourfold manner of computation is in use in the world."

#### — Verse 2

*sauracāndrārśasāvanāni* — four measures; *ebhiḥ pṛthak pṛthak* — separately by each; *grahānanayam* — computation of planets; *lokasya* — current in the world.

"A *yuga*-year, abundant and deficient (days), fire (3), subtraction and addition—these (come) from the solar (system). *Tithi*, *karaṇa*, intercalary month, and the changes of zodiacal signs and *parva* (come) from the lunar (system)."

#### — Verse 3

*yugavarṣaṃ vipulābdayanāni* — intercalary and deficient days; *vahniḥ* — the number three; *udviddhi-hānayaḥ* — addition and subtraction—from the solar system. *tithikaraṇādhikamāsāḥ* — from the lunar; *rāśiparvikriyāḥ* — changes of zodiacal signs and lunar half-months.

"For one desirous of performing the *pramāṇa* (measurement), the planetary motion, the rising and setting, and the creation (of almanacs), the civil measure (is useful), as well as the bed (auspicious moment), *prāyaścitta* (expiatory rites), and other ceremonies."

### — Verse 4

*pramāṇam grahagatikṣayodayasūtakalpanam* — for the construction of the pañcāṅga (almanac); *sāvanam mānam* — civil measure; *śayyā* — auspicious moment for lying down; *prāyaścittakriyāḥ* — expiatory rites and ceremonies.

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The Twenty-four Fold System (Verses 5-23)

“Since (computation) is impossible without names (*saṃjñā*) given to each object, the names and numerals well-known in the world (are given).”

— Verse 5

Commentary: The commentator explains that Brahmagupta here gives a systematic nomenclature for the various measures of time and computation. The twenty-four items are enumerated in the following verses.

“A *yugapad*, a *yuga*, beginning from that, or of the sun, of the Yāmya (south), of Bhāskara, or of Kanyā (Virgo), half a sign, the numbers (associated), at the head, from half a day, from two, from Bhrama (revolution).”

— Verse 6

“The sun has been measured by Pulisha, Romaka, Vāraha, and Yavana (Greeks). Therefore, from that (measurement) one must construct a proper *siddhānta*.”

— Verse 7

puliśaḥ romakaḥ vārāhī yavanāḥ — the five *siddhānta* authors; *sūryasya pramāṇam* — measurement of the sun; *racanātmakaḥ siddhāntaḥ* — construction of a scientific treatise.

“If the *siddhāntas* are different, (yet) Bhāskara (the sun) has the same modifications (*saṃskāra*)—the difference (is) abundant and deficient, those having the same (result) for the previous (method).”

— Verse 8

“In it (the *siddhānta*) the five (methods): *pakṣāgati*, *madhya*, *gati*, *uttara*, etc.; *kuṭūākāra* (transformation), *bheda* (division), *chid* (cutting), *caturtha* (fourth proportional), and the sphere (*gola*).”

— Verse 9

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“By Brahmagupta, the son of Jīṣṇu, the knower of mathematics and the sphere, this true *siddhānta* was composed, (which is) Brahma (itself), adorned with the beauty of elegant composition.”

### — Verse 10

*bhaṭṭa brahmācāryaḥ* — the venerable Brahmagupta; *jijñoh tanayaḥ* — son of Jīṣṇu; *gaṇitagolavit* — knower of gaṇita (mathematics) and the celestial sphere; *sphuṭasiddhāntaḥ* — true astronomical treatise; *brahmā* — of the nature of Brahman itself.

“(Various) instruments, conjunction, year-count, diverse multiplication, the *utka* (elevation), the *śakani* (depression), on account of the multitude of operations, (all these) from the diminished (part), the solar *maragraham*...”

— **Verse 11**

“The abundant, the one without method, the *ṛddhicchid* (augmented cut), the *yogketsya* of the standing, the *śaṅkuśye* (gnomon-shadow) declares the year and also the solar year...”

— **Verse 12**

“As much as has not been spoken by me, the remainder of the sphere, thin as a sixth part of a *ksaya* (decrease), this (is) the *prāyaśloka* (resumé verse). This (is) the *saṃjñā-adhyāya* (chapter on nomenclature), the twenty-fourth.”

— **Verse 13**

*mayā yannoktaṃ* — what has not been spoken by me; *golāt agraṃ* — the remainder of the sphere; *kṣayārdhasato ṣaṣṭhāṃśatulyaṃ* — as minute as one-sixth part of the decrease; *prāyaślokaḥ* — resumé verse; *saṃjñādhyaś caturviṃśaḥ* — the twenty-fourth chapter on nomenclature.

**Note on verse numbering:** The original text numbers this as the 23rd *adhyāya* in the internal count, though the final verse calls it the 24th in another reckoning. The manuscript tradition contains variations in chapter enumeration.

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“Multiply the elapsed *sāvana* (civil) days by the solar days; the *asāvana* (non-civil) days—therefore, that attainment is difficult for those of dull intellect.”

— **Verse 5 (alt.)**

“Humanity, the father, the divine, the arm, the eight (*vasu*), the time—the *utkarṣi* (elevation) for the purpose of knowledge, the arm, clearly the ninth (or another).”

— **Verse 6 (alt.)**

“Two-two (make) a zodiacal sign; multiplied by Makara (10, i.e. 60), these (are) the six (seasons); the solar motion (gives) the seasons: *śiśira-vasanta* (spring), *grīṣma* (summer), *varṣā* (rains), *śarad* (autumn), and *hemanta* (winter).”

— **Verse 7 (alt.)**

dvau dvau rāśiṃ — Gemini etc.; *makarāhatāḥ ṣaṭ* — six seasons multiplied by ten (Makara = 10); *sūryagatiḥ* — the sun’s path; *śiśira-vasanta-grīṣma-varṣā-śarad-hemantāḥ* — the six seasons.

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“The principal and the *arha* (worthy), the worshipper and the devotee; by the *vyāsāntara* (diameter difference) for the crow—the sun’s hypotenuse is to be released; two, the decrease is to be shown, the remainder by the moon’s hypotenuse.”

— **Verse 8 (alt.)**

“The six, diminished and increased by fifteen, are the declination-time etc., multiplied by six (*rasa*).”

— **Verse 64 (alt.)**



“The radius and the abundant shadow—their product, divided by that and multiplied by the latitude; the decrease is always in the south (Yāmya); from the sphere, the disc’s declination.”

— Verse 66 (alt.)

“When the sum and difference of declination and latitude are purified by the *akṣarāpada* etc., the half-day; when the square of that is subtracted from the square of the sum—the method when reversed at that position.”

— Verse 67 (alt.)

“The remainder, multiplied by the prescribed (number) and arrived at, divided by the day-number elevated by the *nāḍī*; that (result) multiplied by the half-day hypotenuse, by that radius the result (is) the hypotenuse.”

— Verse 68 (alt.)

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“Thus has been stated in order the section on numerals and the practical application; the diameter multiplied, the long circle, the *ṣaḍāṃśaka* (sixth part), the latitude-decline.”

— Verse 9 (alt.)

“Its diameter, śaki (potency), *karaṇḍū*, that (is) multiplied by the *dvijaya* (twice-conquered), by degrees.”

— Verse 10 (alt.)

“The sun’s hypotenuse, the messengers, by the *trijaya* (triple-conquered), by the *vyāsāntara* (diameter difference) divided, the remainder; the *tijayāmukhyāḥ* (chief of the three-conquered), by the decrease, the operation, the *hutātma* (fire-souled = 3), the diameter.”

— Verse 10 (alt. 2)

“From the diameter, the evil, from the product, the crow—the diameter difference, the release (is) the product; the *prokṣa* (sprinkling), the second middle, thus from the *tithi* (lunar day) multiplied and obtained, the darkness, the diameter.”

— Verse 11 (alt.)

“The intercalary months, the *avama* (omitted) days, the *sambhavana* (production), all the measures; by ten *prāyaśloka*s (resumé verses), the twenty-fourth Measurement Chapter.”

— Verse 12 (alt.)

*Thus ends the twenty-third chapter on Measurement
in the work of Brahmagupta.*

Mean Computation Chapter

The *Madhyamādhikāra* constitutes the first and foundational chapter of the commentary (*vāsanā-vijñāna-bhāṣya*) on the *Brāhmasphuṭasiddhānta*. It treats the computation of mean positions of planets (*madhyama-graha*), beginning with a discussion of the motion of the celestial sphere (*bhacakra*), the definition of time (*kāla-pravṛtti*), and the various *yuga*-systems employed in Indian astronomy.

Introductory Discussion (pp. 752-760)

The commentary opens with a discussion of the *bhacakra* (celestial sphere) and its motion:

“(The celestial sphere), moving eastward (from left to right), was first created by Brahmā (the world-teacher). The sphere, along with the nakṣatras and other celestial bodies having equal *capādas*, along with the meridian staff etc., the circle of nakṣatras and other luminaries that give light—this sphere was fashioned by Brahmā. Even so, the ācāryas (teachers) do not describe their (the planets’) essential nature (*svārūpa*), but have only determined their positions (*sthāna*) for the purpose of computation (*spaṣṭādi*).”

— Commentary v. 1

Hindi Commentary (pp. 752-753): The commentator explains that the celestial sphere (*bhacakra*) was first created by Brahmā (the creator). The sphere, along with the nakṣatras and other celestial bodies, revolves from west to east. The commentator notes that the ācāryas (teachers) do not discuss the nature (*svārūpa*) of these bodies but have only determined their positions (*sthāna*) for the purposes of computation.

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### Key Verses from the Siddhānta (pp. 760-770)

The commentary elucidates the following root text (*mūla*) verses:

“Now (Brahmagupta) declares the commencement of that time whose beginning is unmentioned. Day, month, year, *yuga*, and *kalpa* began from the sun at Laṅkā at the beginning of creation.”

#### — Mūla-śloka 4

**Hindi commentary (pp. 761-762):**

*caityādi kalpa-pratipadyāt laṅkopalakṣitāmbhṛdv ekodayādi-divā pravṛttāḥ* — From the first tithi of Caitra (the first month), at Laṅkā, the days etc. commenced from the sunrise.

*ravidinambaraḥ sṛṣṭyādaḥ kalpādaḥ* — The solar day number at the beginning of creation, at the beginning of the *kalpa*.

The commentator explains the concept of the *kuldiṇa* (civil day) in detail. The number of civil days elapsed since the beginning of the current *kalpa* is 1,58,22,23,78,50,000 according to Brahmagupta’s system.

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On Time Computation and Planetary Motion (pp. 770-790)

The commentary proceeds with detailed mathematical analysis:

“First, the *yugapada* (yuga-epoch method); second, by *parva* (lunisolar method); third, by thirteen months; or fourth, by *sāvana* (civil reckoning).”

— Time-reckoning verse

Exposition: The four types of time-reckoning are discussed:

- (i) *Yugapada* — the yuga-epoch method
- (ii) *Parva* — the lunisolar intercalation method
- (iii) *Trayodaśa-māsa* — the thirteen-month intercalary method
- (iv) *Sāvana* — civil reckoning based on sunrise

Each method is explained with its mathematical basis and astronomical application.

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## Planetary Computations (pp. 790-820)

The commentary provides detailed algorithms for computing mean planetary positions:

### Mean longitude formula :

$$\text{Mean longitude} = \frac{\text{ahargaṇa} \times \text{daily motion}}{\text{civil days in a kalpa}} \mod 360^\circ$$

- *ahargaṇa* = elapsed civil days since kalpa-beginning
- *daily motion* = mean daily motion of the planet
- The result is expressed in degrees of the ecliptic

The commentary discusses the *mandaphala* (equation of centre) and *śīghraphala* (*śīghra* equation) with detailed trigonometric formulas using the *jyā* (sine) function.

*Mandaphala* for the Sun and Moon: the sine of the *manda-kendra* (anomaly) is the multiplier, the radius (*trijyā*) is the divisor.

*Śīghraphala* for Mercury, Jupiter, and Saturn: the sine of the *śīghra-kendra*.

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On the Computation of Weekdays (pp. 820-838)

The final section of this portion treats the computation of weekdays (*vāra-pravṛtti*):

“The sages declare the commencement of the week (to be reckoned) from the sunrise (at Laṅkā); on the earth (at other places), it (is determined) by adding or subtracting the *deśāntara-nāḍikās* (longitude difference in time).”

— Verse 136

“At Laṅkā, the weekday begins with the first *nāḍī* after sunrise; at a place to the west, (it begins) after (sunrise); with the six or eight *nāḍīs* (difference), the weekday is determined when the sun rises. For a place on the eastern or western side of the prime meridian, before or after sunrise (at Laṅkā), by addition or subtraction, the correct time (is found).”

— Expanded v. 136

Hindi Commentary (pp. 836-838):

At Laṅkā, the weekday begins at sunrise in the first *nāḍī*. At a place to the west, after (sunrise) by six or eight *ghaṭīs*.

At a place to the east, the correct time (is found) before sunrise.

The commentator explains that the *vāra* (weekday) begins at sunrise. By this definition, at a place east of the prime meridian (Laṅkā), the weekday begins after subtracting the longitude difference (in *ghaṭīs*); at a place west of the meridian, (the weekday begins) before sunrise by the same amount (longitude difference in *ghaṭīs*).

In the *Siddhānta-śiromaṇi*, Bhāskarācārya states: “*arkodayād dvābhyāṃ bhyas tāniḥ prācyāṃ praticyāṃ dinap-*”—(the computation proceeds from sunrise with modifications for eastern and western positions).

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### Concluding Verses (p. 838)

The portion concludes with a verse from the *Manusmṛti*:

“This (universe) existed as darkness, unknowable, without marks, not to be reasoned about, not to be apprehended, as if asleep everywhere.”

### — Manusmṛti 1.5

#### Commentator’s exposition:

The commentator explains that Manu has stated that at the time of the deluge (*pralaya*), the perception of weekdays and other time measures is not possible. Therefore, the reckoning of weekdays is only possible during the manifest (*sṛṣṭi*) period. Brahmagupta establishes that the weekday begins from sunrise at Laṅkā and is modified by the longitude difference (*deśāntara*) for other locations.

## Appendix: Structural Overview of the Commentary

The *Vāsanā-vijñāna* commentary on the *Brāhmasphuṭasiddhānta* is organized into the following major chapters (*adhyāyas*):

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*The commentary employs a rich methodology, interweaving:*

- (i) *Mūla-śloka* — The root text verses of Brahmagupta
  - (ii) *Anvaya* — Word-order rearrangement for grammatical clarity
  - (iii) *Bhāṣya* — Sanskrit explanatory commentary
  - (iv) *Hindi Ṭīkā* — Hindi translation and explanation
  - (v) *Upapatti* — Mathematical demonstrations and proofs
  - (vi) *Udāharaṇa* — Numerical examples and worked problems
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*The Brāhmasphuṭasiddhānta  
composed by Śrī Brahmaguptācārya*

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## On the Commencement of the Day

### Commencement of the Weekday

**[Hindi Commentary]** *Prav.rtti* means that which is spoken by Brahmagupta himself. On the question “from when does the weekday commence?” there are many differing opinions among teachers. Āryabha.ta and Siṃhācārya hold that the day begins from the half-risen solar disc. Others hold it begins from sunset. Lāṭadeva and others hold it begins from midnight. The Yavana kings hold that ten *muhūrtas* of night [make the division]. In the *Siddhāntaśekhara*, Śrīpati has refuted the opinions of these teachers.

**[BSS I.36]** When at the beginning of creation the universe was filled with darkness, and the planets had not yet been created, there was no lord of the weekday (*vāreśvara*). Therefore the commencement of the weekday is from sunrise (*udayāt prav.rttiḥ*).

**[Hindi Commentary — Translation]** Before the beginning of creation, the universe was full of darkness. At such a time, discussion of the weekday’s commencement is impossible because the planet-lords of the weekdays did not yet exist. For this reason, the opinion of those who say the weekday commences at midnight is not correct. [Holding that] it begins from midday or from sunset is also not correct. Therefore the opinion of the teachers who hold that the weekday commences at sunrise is indeed correct.

**[Sanskrit Commentary]** In the Sanskrit proof-text (*saṃsk.rtopapatti*) it is known from the verse “The sages state the commencement of the weekday from sunrise at Laṅkā, the city of Rāvaṇa” etc. This is made clear by the verse “From Laṅkā sunrise, south of the prime meridian, east is before, west is after” etc. Thus [ends] verse 36.

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## Longitude Correction

### Rules 37-39: Longitude Difference

**[BSS I.37-39]** The terrestrial circumference (*bhū-paridhi*) is five thousand *yojanas* (*khakha-kha-śara* = 5000). Multiply this by the degrees of difference in latitude (*svāk.sāntarāṃśa*); divide by 360 (*bhaga.nāṃśa*); the square of the result, subtracted from the square of the longitudinal difference (*deśāntara*) [in *yojanas*], gives [the square of] the corrected difference. [Verses 38-39] From the root of the remainder, multiplied by the terrestrial circumference and divided [appropriately], [obtain] minutes (*kalā*); to the east of the Ujjayinī north-south line is positive (*dhana*), to the west is negative (*.rna*). Or from the root divided by the terrestrial circumference and multiplied by sixty, the result is in *gha.tikās*; this is to be applied to the mean or true planet as positive or negative like [the treatment of] a planet.

**[Sanskrit Commentary]** “Terrestrial circumference” (*bhū-paridhi*) means the circumference of the Earth-globe. In answer to “how much?” he says *khakha-kha-śarā* = five thousand *yojanas*. Here the longitudinal difference, latitude, etc., have been explained previously by means of measurements and divisions.

**[Hindi Commentary — Translation]** The terrestrial circumference is five thousand. Multiply this by the latitudinal difference between one’s own place and the standard meridian; divide by 360; the square of that result, subtracted from the square of the longitudinal difference [in *yojanas*]; from the square root of the remainder, multiplying by the planet’s motion and dividing by the terrestrial circumference, one obtains a result in minutes. East of the Ujjayinī north-south line [apply as] positive to the mean or true planet; west as negative. The root multiplied by sixty and divided by the terrestrial circumference gives the result in *gha.tikās*; this is to be applied to the *tithi*-freed *gha.tīs* (i.e., time correction) like a planet.

### Upapatti (Proof/Explanation)

**[Sanskrit Upapatti]** If the diameter measure is 1581, then from the square of the diameter, multiplied by eight, [taking] the square root gives the terrestrial circumference — so it is stated in the *Sūryasiddhānta*. By this rule the terrestrial circumference measure of 5000 is obtained, which is Brahmagupta’s opinion.

**[Upapatti — Translation]** If we assume the diameter measure to be 1581, then by the method stated in the *Sūryasiddhānta*, “the square root of eight times the square of the diameter is the terrestrial circumference,” the terrestrial circumference measure of 5000 is obtained. This same [value] is the terrestrial circumference in Brahmagupta’s system.

## Computation of Days Since Epoch

### Rule 40: Days Elapsed Since Kalpa

**[BSS I.40]** The years elapsed since the beginning of the *kalpa* (*kalpa-gatābdā*), multiplied by 2486 (*rūpā.s.tajina*) and by 7736 (*apavarni-saptanaga*), and divided by 8600 (*kha-svara-sanava*), yield the days and the omitted [tithi]s; the remainders are the fractional parts and defects.

**[Sanskrit Commentary — Vāsanābhā.sya]** At the desired end of a solar year, the years elapsed of the *kalpa* are taken here. The years elapsed of the *kalpa* having been multiplied — he says. By *rūpā.s.tajina* = 2486, and elsewhere by *navārni-saptanaga* = 7736, and then divided by *kha-svara-sanava* = 8600; the results are respectively the days and omitted [lunar days]; the remainders are the fractional parts in both places. The years elapsed of the *kalpa*, multiplied by 2486 and divided by 8600 — the result is days. There, whatever is the remainder, that is called the *dināṃśa* (fractional part of a day) by the word *dābda*.

**[Hindi Commentary — Translation]** In the desired end of a solar year, the years elapsed of the *kalpa* that are here taken. The years elapsed of the *kalpa*, multiplied by 2486 and divided by 8600 — the result is days and omitted [days]; the remainders are in both places. The years elapsed of the *kalpa*, multiplied by 2486 and divided by 8600, the result is days. There, whatever remainder remains, that is called the *dināṃśa* (fractional part of a day).

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## Mean Position of the Sun and Moon

### Rules 24-26: Daily Motion and Latitude

**[BSS I.24]** The minutes of the *manda-phala* (equation of centre), multiplied by the divisor and divided [appropriately], give in order the daily-motion results for the Sun and Moon. In the southern [hemisphere] it is positive, in the northern negative — this is called the *dhruva* (constant). Thus here the staff-head [correction] is reversed; if previously the *vighora-tana-phala* was small, then here [it becomes] solar [correction]. Let astronomers examine the *bhuja* (sine) once, [that of] the delight of all good people; let them examine the *bhū.sa* (ornament) [of calculation].

**[Sanskrit Commentary — Vāsanābhā.sya]** How many revolutions (*bhaga.namāna*) of Prahlāda and the others in a *yuga*, how many elapsed days (*gata-āharga.na*), the computations of weekdays and so on, the computations of mean planets and so forth — in these and other subjects, the first chapter on mean motion (*madhyagati-nāmaka adhyāya*) was composed by me (Brahmagupta) with sixty-*āryā* [verses] (*ṣaṣṭi-pramitā chandasā*); how many subjects there are in the chapter named *Madhyagati*, an enumeration of them has been made. Thus [ends the commentary].

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## Mean Position of Mars and Other Planets

### Rule: Mean Planets Computation

**[BSS — General Rule]** The *manda-phala* (equation of centre) is [obtained] from the *bhuja-jyā* (sine of the anomaly) of the *manda-kendra* and the *pāta-jīvikā* [divisor]; divided by the *ko.ti-jyā* (cosine) and multiplied by the *vyāsārdha* (radius), [it becomes] the *phala-ma.n.dala* (resultant circle/epicycle).

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## Epilogue to the Madhyamādhikāra

### Chapter Conclusion

**[End of Chapter 1]** Thus the first chapter, the *Madhyamādhikāra* (Chapter on Mean Motions), in the *Brāhmasphuṭasiddhānta* [of Brahmagupta].

**[Hindi Commentary — Translation]** The revolutions of the planets and others in a *yuga*, the elapsed *aharga.na*, the computations of weekdays and so forth, and the computations of mean planets and so on — in these and other subjects, the first chapter named *Madhyagati* was composed by means of sixty-three *āryā* [verses]; that is, in the first chapter named *Madhyagati*, however many subjects there are, an enumeration of them has been made. Thus [ends the commentary on Chapter 1].

*[End of the Madhyamādhikāra. This first section, pages 839–893, has been completely explained. It covers mean motions of all planets, day computation, deśāntara (longitude correction), and yugādi calculations.]*

## — Spa.s.tādhikāra

The *Spa.s.tādhikāra* (Chapter on True Positions) is the second chapter of the *Brāhmasphuṭasiddhānta*. Here the purpose of *spa.s.tīkara.na* (making true), the twenty-four sine table (*jyā-pañcaviṃśatikā*), versed sines (*utkrama-jyā*), *mandaphala* (equation of centre), *śīghra-phala* (equation of conjunction), and true daily motion (*spa.s.ta-bhukti*) are presented. This section spans pages 896–978.

## Purpose of Spa.s.tīkara.na

### Rule 1: The Need for Correction

**[BSS II.1]** Since the mean planet is not observed in the zodiac (*bhaga.na*) every day equal to [its mean computation], therefore I shall state the procedure for making the mean [planet] true — the *madhyasphu.tīkara.na*.

**[Sanskrit Commentary — Vāsanābhā.sya]** First, the chapter on true motion (*sphu.ta-gaty-adhyāya*) is being explained. There the purpose of the commencement is stated. Since a planet equal to the mean planet is not observed in practice in the zodiac (*bhaga.na*, the nak.satra circle) day by day, therefore I shall state the *sphu.tīkara.na* (procedure for making true). In answer to “what is the defect of the mean?” he says: the mean planet is hypothetically located (*parikalpita*) on the *kak.sā-ma.n.dala* (deferent circle). And the real planet is not on the *kak.sā-ma.n.dala*; revolving on the *prati-ma.n.dala* (epicycle) with mean motion, it is observed on the *kak.sā-ma.n.dala* with true motion. Therefore, by whatever computation the planet situated on the *prati-ma.n.dala* may become equal to [the observed position] on the *kak.sā-ma.n.dala*, that *sphu.tīkara.na* I shall state. Thus [ends commentary on] verse 1.

**[Hindi Commentary — Translation]** For the reason that the true planet in the *krānti-v.rtta* (ecliptic) is not seen every day equal to the mean planet, for that reason I (Brahmagupta) call that procedure which makes [it] equal to the observed [position] the *sphu.tīkara.na*. The mean planet is hypothetically [placed] on the *kak.sā-v.rtta* (deferent); the planet, revolving with true motion on the *prati-v.rtta* (epicycle), is observed on the *kak.sā-v.rtta*. Therefore, by whatever computation the planet situated on the *prati-v.rtta* becomes equal [to the observed position] on the *kak.sā-v.rtta*, that *sphu.tīkara.na* I state. Thus [ends commentary on] verse 1.

## The Twenty-Four Sines

### Rules 2-9: Sine Table in 24 Steps

#### Introduction to the Jyā-pañcaviṃśatikā

Brahmagupta gives twenty-four *krama-jyās* (sines in sequence), each arc being  $3\frac{3}{4}^\circ$  ( $= 225'$ ), filling the complete quadrant of  $90^\circ$ . These sines are expressed poetically through *bhūta-saṅkhyās* (word-numerals), displaying Brahmagupta's literary talent alongside his mathematical genius.

#### [BSS II.2-9] The 24 Sines

The twenty-four sines are encoded in *bhūta-saṅkhyā* (word-numeral) verses. Each word corresponds to a numerical value:

**Verses 2-9:** *ṣaḍ-bud-agni* ( $6+14=20$ ) = 214; *suta-agni* ( $7+3=10$ ) = 427; *rasa-dvādya* ( $6+12=18$ ) = 638; *munīndu* ( $7+1=8$ ) = 846; *vasu-candra* ( $8+1=9$ ) = 1051; *indunava-nava-candra* ( $1+9+9$ ) = 1251; *rasa-tithi-yamala* ( $6+15+1$ ) = 1446; *raviśriyamā* ( $12+1$ ) = 1624.

**Verses 3-9** continue the sequence up to the 24th sine = 3270 (the full radius).

#### Krama-jyā Table (Sines in Sequence)

**[Sanskrit Commentary]** Their diameter-measure is the radius. “These are the *aghe-jyā*” (sines) — thus. *Sanu-yamalā* = 214; this is the form of the first sine. *Suniya-mavedā* = 427, the second. *Vasu-jvalana-gha.tikāḥ* = 638, the third. *Rasa-k.rta-vasavaḥ* = 846, the fourth. *Śaśi-pañcha-khendavaḥ* = 1051, the fifth. *Candra-dara-sūryāḥ* = 1251, the sixth.

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## Utkrama-jyā (Versed Sine) Values

### Utkrama-jyā Table (Versed Sines)

The *utkrama-jyās* (versed sines) are constructed by the reversed order of the half-chords. In modern trigonometry these are known as “ $1 - \cos$ ” (versed sine = radius – cosine).

**[Hindi Commentary — Translation]** The twenty-four half-chords (*ardhajyā*), [named] *muniyamalā*, *muniyamavedā*, etc., are of twenty-four numbers, as written below. The relation of *vyastā* (reversed, i.e., versed sine) is given after. The twenty-four *ardhajyās* (*krama-jyās*) are: 214, 427, 638, 846, 1051, 1251, 1446, 1624, 1817, 1991, 2145, 2312, 2459, 2594, 2832, 2933, 3021, 3096, 3145(?), 3207, etc., 3263, 3270.

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## Arc-Jyā Conversion

### Rule: Finding Arc from Jyā

**[BSS — Arc-Jyā Rule]** From the *jyārdha* (half-chord, sine), divided by the *trijyā* (radius), the square root gives [the first approximation], slightly different for odd and even [intervals]; from that the arc is obtained. After that, again [taking] the square root of [the sine] combined with its square [correction], [the more precise arc is obtained].

**[Sanskrit Commentary]** The twenty-four versed sines (*utkrama-jyā*), [named] *muno's.tayamā* etc., are of twenty-four numbers, as written below: 7, rad (29), 62, 111; 174, rau (292), 337, 435, rabha1 (521), 676, 811, rau (985), 1114, 1296, 1453, 1635, 1824, 2019, 2214, 2424, 2632, 2543(?), 3056, rera70 (3270).

**[Hindi Commentary — Translation]** The *jyārdha* (half-chord), divided by the *trijyā* (radius), from whatever root (*pada*) is obtained, slightly differing for odd and even [intervals], that becomes the arc. After that, again, having combined [the correction] with the square of that same root, the next arc is obtained.

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## Manda and Śīghra Equations

### Rules 41-44: True Motion Computation

#### Introduction to Manda-phala and Śīghra-phala

The *manda-phala* (equation of centre) and *śīghra-phala* (equation of conjunction) are the two principal corrections for making the planet true. The *manda-phala* is computed from the *manda-kendra* (anomaly of the epicycle); the *śīghra-phala* from the *śīghra-kendra* (anomaly of conjunction).

#### [BSS II.41-44]

**41:** The *bhuja-jyā* (sine of anomaly) of the *manda-kendra*, multiplied by the *pāta-jīvikā* and divided by the *ko.ti-jyā* (cosine), [then] multiplied by the *vyāsārdha* (radius), gives the *manda-phala* (equation of centre) and the motion.

**42:** From that, first the small or second [epicycle], or from the *manda-saṃsk.rta* (mean corrected by manda), from the mean, one should give the *manda-phala*, or again by the *śīghra-bhukti*.

**43:** Taking the *manda-saṃsk.rta*, or again by the *śīghra-bhukti*, one should then compute the *śīghra-phala*, combined by proportion (*trairāśika*).

**44:** Having established the *śīghra-phala*, the true velocities (*sphu.s.t.a-gatikaḥ*) of the planet are to be brought together; then the wise [astronomer] should compute the *vakra-bhukti* (retrograde motion).

**[Sanskrit Commentary]** Multiplied by the *manda-kendra-gati-jyāntara* (*bhogyakha.n.da*, the tabular difference), divided by the *prathama-jīvā* (first sine); whatever result is obtained, multiplied by the *sphu.ta-paridhi* and divided by 365 (*bhaga.nāṃśa*), the resulting minutes — in the *sakala-ādi-kendra* etc., [when this is] less or more than its own mean motion, at that *kendra* the true motion of the Sun and Moon is obtained. For Mars and the other planets, the mean motion minus the *manda-gati-phala* is called *manda-sphu.ta-gati* by the teachers.

**[Hindi Commentary — Translation]** The *manda-kendra-gati-jyāntara* (*bhoga-kha.n.da*), multiplied [by the epicycle] and divided by the *prathama-jīvā* (*prathama-jyā*); the resulting quotient, multiplied by the *sphu.ta-paridhi* and divided by 365 *bhaga.nāṃśa*, the resulting minutes — in the *sakala-ādi-kendra*, whether less or more than the mean motion, at that *kendra* the true motion of the Sun and Moon is obtained. For Mars and other planets, the mean motion minus the *manda-gati-phala* is the *manda-sphu.ta-gati*. The *śīghrocca-gati* minus its own *manda-sphu.ta-gati* is the *śīghra-kendra-gati* of Mars and other planets.



## The Manda Paridhi (Epicycle) Values

### Rules 20-21: Manda Epicycle Corrections

#### [BSS II.20-21]

**20:** The difference between that [quantity] and the half-diameter of the epicycle (*dyudala-paridhi*), multiplied by the *trijyā* (radius) and by the *nata-jīvā* (sine of altitude), [when the planet is] in decrease (*ūne*) is positive (*dhana*), in increase (*adhike*) is negative (*.rna*); this is the *sphu.ta-paridhi* (true epicycle) of the half-day.

**21:** For the Sun, that is the *manda-paridhi* in the *.rna-manda-phala* or *dhana-manda-phala* at midday. For the Moon, the same, and by those same [quantities] in *gha.tikās*, the Moon also [is corrected] at that same *kapāla* (division); thus the *nata* (altitude) is obtained.

**[Sanskrit Commentary]** Fourteen degrees (*catudaṃśāṃśāḥ*), in two places, diminished by two-thirds of a degree (*bhāgadwayaṃśena rahitāḥ*) — then in the *.rna* or *dhana manda-phala* of the Sun; at midday (*dinamadhya*) the *manda-paridhi-aṃśa* are obtained. In the *.rna* or *dhana manda-phala*, before the half-day *kapāla*, for the Sun separated by fifteen *gha.tikās*, the half-day *paridhi* measure is to be made successively increased or decreased by two-thirds of a degree. For the Moon, in the *.rna* or *dhana manda-phala*, twelve [degrees] diminished by two *jina-lipata* (?) — that is, in two places, diminished by thirty-two degrees plus twenty-four minutes (*dvātriṃśadaṃśākhyacaturviṃśatikalā rahitāḥ*) — then at midday those *manda-paridhi-aṃśa* are obtained.

**[Hindi Commentary — Translation]** From fourteen degrees, in two places, diminishing by one-third of a degree (20 *kalās*), in the *.rna-manda-phala* or *dhana-manda-phala* of the Sun, at midday the *manda-paridhi-aṃśa* is obtained. In the *.rna-manda-phala* or *dhana-manda-phala*, in the *kapāla* before half-day, having subtracted fifteen (14?) *gha.tikās* from the *nata* Sun, in the midday *manda-paridhi* measure, successively one-third of a degree (20 *kalās*) is to be added and subtracted.

## Summary of Manda Paridhi Values

### Manda-paridhi Summary Table

Below is presented a numerical summary of the *manda-paridhi* (epicycle) values. For each planet the *manda-paridhi* varies up to twenty-four degrees.

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## True Daily Motion

### Rules: Daily Motion After Correction

**[BSS — True Daily Motion]** The difference between the *śīghra-kendra-sphu.ta-bhukti* and [the mean motion], multiplied by the [tabular difference] and corrected from the stated quotient — this is the *śīghra-gati*; [if negative] it is *vakra-gati* (retrograde motion). Thus [goes] the proof-verse of Sinhavarman.

**[Hindi Commentary — Translation]** The *śīghra-kendra-sphu.ta-bhukti-antara* (*bhoga-kha.n.da*), multiplied [by the epicycle] and divided by the *prathama-jīvā*; the resulting quotient, multiplied by the *sphu.ta-paridhi* and divided by 365 *bhaga.nāṃśa*, the resulting minutes — in the *sakala-ādi-kendra*, whether less or more than the mean motion, at that *kendra* the true motion of the Sun and Moon is obtained.

### Geometric Explanation (Upapatti)

**[Upapatti — Geometric Proof]** By the difference between mean and true, motion becomes threefold. That motion which is different every moment is *spa.s.tā*; another is *madhyā* (mean). The true motion (*sphu.tā gati*) is also twofold by the division into daily and instantaneous: thus the daily *manda-sphu.tā-gati* and the instantaneous *manda-sphu.tā-gati*. The daily *spa.s.tā-gati*, the instantaneous *spa.s.tā-gati* — by the teacher the daily *manda-sphu.ta-gati* and the daily *sphu.ta-gati* are computed.

The *graha-manda-kendra-gati-jyāntara* (*bhoga-kha.n.da*), multiplied by the *prathama-jyā* and divided [by the radius]; whatever quotient is obtained, multiplied by the *sphu.ta-paridhi* and divided by 365 (*bhaga.nāṃśa*), the resulting minutes — in the *sakala-ādi-kendra* (or *kkayādi-kendra*), whether less or more than its own mean motion, at that *kendra* the true motion of the Sun and Moon is obtained.

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## Special Corrections for Different Planets

### Planet-Specific Corrections

#### Mercury and Venus

**[Sanskrit Commentary]** For Mercury and Venus, the *śīghrocca* (apsis of the *śīghra* epicycle) is [derived] from the Sun; from the seizure of the Sun [it is different], otherwise [it would be the same]. Their *śīghrocca* is connected to the Sun; the *mandocca* (apsis of the *manda* epicycle) is independent — this is the special feature.

**[Hindi Commentary — Translation]** The *śīghrocca* of Mercury and Venus is related to the Sun itself; their *śīghrocca* is Sun-related, the *mandocca* is independent. Therefore for these two planets a special correction procedure is necessary.

#### Mars , Jupiter , and Saturn

**[Sanskrit Commentary]** For Mars and the others, the *manda-sphu.ta-gati* and the *sphu.ta-gati*. From the *manda-sphu.ta-bhukti-antara*, the *śīghra-bhukti* [is obtained]; the *śīghra-phala-ādha-bhogya-jīvā*, multiplied by the *madhya-jivyā* (mean sine), is to be divided. Having brought together the *sphu.ta-bhukti* by the stated quotient; if together with that, by the *manda-phala-ādham-sphu.ta-bhukti-antara* in that same *kendra*, the resulting *bhukti* is to be made positive or negative.

**[Hindi Commentary — Translation]** For Mars and other planets, the *manda-sphu.ta-gati* and *sphu.ta-gati* — from the *manda-sphu.ta-bhukti-antara*, the *śīghra-bhukti*; the *śīghra-phala-adha-bhogya-jīvā*, multiplied and divided by the *madhya-jīva*. By the resulting [quotient], the *sphu.ta-bhukti* is to be brought together as stated; if together with that, by the *manda-phala-adha-sphu.ta-bhukti-antara* in that same *kendra*, the resulting *bhukti* is to be made positive or negative.

## The Chapter Conclusion

### Epilogue to the Spa.s.tādhikāra

**[End of Chapter 2]** Thus the second chapter, the *Spa.s.tādhikāra* (Chapter on True Positions), in the *Brāhmasphuṭasiddhānta*.

**[Hindi Commentary — Translation]** In this manner, the explanation of the *Madhyamādhikāra* (first chapter) and the *Spa.s.tādhikāra* (second chapter) in the *Brāhmasphuṭasiddhānta*, composed by Ācārya Brahmagupta, has been completed. In this, the mean motions of planets, *sphu.tīkara.na*, sine mathematics (*jyā-ga.nita*), *manda-phala*, *śīghra-phala*, *deśāntara*, *nata-karma*, and other subjects have been described in detail.

*[The Spa.s.tādhikāra continues with detailed calculations for each planet's manda and śīghra corrections, tables of paridhi values, geometric upapattis (proofs), and special rules for Mercury and Venus across pages 896-978 of the original edition.]*

*Spaṣṭādhikāraḥ — Chapter on the Determination of True Positions***English Translation**

This chapter deals with the methods for computing the true (corrected) positions of planets from their mean (average) positions. The corrections account for the eccentricity of planetary orbits and the epicyclic models described in Indian astronomy.

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**English Translation****Verses 48-49. Śīghra-spaṣṭikaraṇa (True Position via Śīghra Correction):**

Having taken half the śīghra-kendra (anomaly from the śīghrocca) of the traversed arc, and half of the retrograde-direct kendra of that same planet, knowing its retrograde-direct period, the true planet should be computed. The construction is: the square-root of the sum of the squares of the śīghra-kendra retrograde-direct quantities is the divisor. The vertical line (koṭi) from the centre of the śīghra-nīcocca-vṛta (epicycle) up to the mean position equals the parama-phala-jyā (maximum equation sine); the horizontal (bhuja) at the centre of the śīghra-nīcocca-vṛta is the product of the east-west components, whose square-root is the triñṇaka-karṇa. The parama-phala-jyā is declared to be proportional to the śīghra-nīcocca-vṛta-śālākā (radius) on the circumference of the perimandala. That also, when the planet is at the highest point on that perimandala, causes parallax (paracchāda). Therefore, in an iron globe one should demonstrate everything.

**Commentary Translation**

[Commentary of Varāhamihira:]

For Mars and Jupiter, the retrograde motion begins at specific śīghra-kendra añśa values. The manda-spaṣṭa (true position via manda correction) and the śīghra-spaṣṭa (true position via śīghra correction) are computed. The śīghra-kendra-gati is the velocity at the śīghra-kendra. The days traversed and the remaining time are computed from these kendra values. By the śīghra-kendra divisions, the ucca positions remain stable.

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**English Translation****Verses 12-13. Planetary Mean Motion (Madhyamāgama):**

First, having multiplied the zodiacal signs (bhacakra-bhāga, 43,200 minutes of arc) by the given days, divide by the Kalpa or Yuga days [1,577,917,828,000]. The quotient obtained is the pūraka-kṣepa (accumulator of revolutions).

That same pūraka-kṣepa, multiplied by the elapsed days and divided by the solar years, gives the mean daily motion (madhya-bhukti). That, added to or sub-

tracted from the previous mean longitude, gives the mean position (madhyama) of the planet.

### Commentary Translation

*[Varāhamihira's Commentary:]*

The zodiacal signs (bhacakra-bhāga), that is 43,200 minutes of arc, are first multiplied by the given days and divided by 1,577,917,828,000 solar years. The quotient obtained is the pūraka-kṣepa. That same pūraka-kṣepa, multiplied by elapsed days and divided by solar years, gives the mean daily motion (madhya-bhukti). That madhya-bhukti, added to or subtracted from the previous mean position, gives the mean (madhyama) of the planet.

### English Translation

#### Verses 14-15. Mean Apogee and Śighrocca:

The mandocca (apogee of manda epicycle) and the śighrocca (apex of śighra epicycle), divided by the yuga-pūraka-kṣepa and multiplied by elapsed days, gives the mandocca or śighrocca.

That mandocca or śighrocca, when added to or subtracted from the previous mandocca or śighrocca of the yuga, gives the current manda-śighrocca.

The mandocca and śighrocca, multiplied by elapsed days and divided by the yuga-pūraka-kṣepa, gives the result. Adding or subtracting this from the previous mandocca or śighrocca gives the current values. In this manner, knowing the planet's madhyama, mandocca, and śighrocca, the foundation for computing the true position is established.

### English Translation

#### Verses 16-17. Manda-spaṣṭikaraṇa (Equation of Centre):

The maximum manda-phala (equation of centre) of the planet, diminished by the mandaka (eccentric anomaly divisor), divided by the half-sine (jyā-ardha), gives that maximum phala of the planet.

Multiplying the manda-phala-jyā by the parama-phala-jyā, the result obtained is the parama-phala (maximum equation). By this, the planet becomes manda-spaṣṭa (corrected via the equation of centre).

The mandaka is the square-root of the sum of squares of the doḥ (sine) and koṭi (cosine). The manda-jyā multiplied by the trijyā and divided gives the manda-phala. That parama-phala, when subtracted from or added to the mean position, gives the manda-spaṣṭa (true) position of the planet.

### English Translation

#### Verses 50-51. Planetary Path and Visibility:

Having fixed the path regulation to the epicycles and instruments, from that the installation in the kakṣa-maṇḍalas (orbits) is established. These rising and setting portions, when undivided [in the orbit], present the characteristics of the maṇḍala divisions; that difficulty in rising and setting will arise when the planet is divided [by epicyclic motion].

The śīghra-kendra values for Mercury and Venus are described. When Mercury's śīghra-kendra equals 50, it rises after the Sun; when it equals 24, it rises before [specific values for visibility]. The rising and setting are reversed relative to the orbital path.

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### *Tripraśnādhikāraḥ — Chapter on the Three Questions of Diurnal Motion*

This important chapter deals with the “three questions” (*tripraśna*) of spherical astronomy: determining the *lambana* (parallax in longitude), the *nāḍī* difference (time difference), and related quantities needed for computing solar and lunar phenomena.

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### English Translation

#### **Verses 54-55. Rising and Setting Śīghra-kendra Values:**

Now, the śīghra-kendra añśa values for the rising and setting of Mars and the others: When Mercury's śīghra-kendra equals 28 [bhauma], 14 [cakre], 17 [muni-nagendra], then the rising of Mars, Jupiter, and Mercury occurs; the setting is opposite to these.

[Values: 50 in Svātī, 24 in Jina, 155 in the desired tithi, 177 in Muni-nagendra] The rising and setting are reversed relative to the orbital path, the forward one being retrograde.

The śīghra-kendra determines the heliacal rising. At 28 degrees, Mars rises before the Sun; at 14, [a value for] Jupiter; at 17, Saturn. The heliacal settings are opposite. These kendra values are well-known; for Mars, the setting śīghra-kendra is 332; for Jupiter, 385; for Saturn, 353.

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When the elapsed [days' values] of the *bhā* (signs) and minutes of arc, joined with the manda-phala-spaṣṭa and śīghra corrections, are divided by the days — thus is the derivation by this method. Still, since all [planets] are corrected by the solar orbit, the planet revolves in the nīcocca-vṛta (epicycle) in that time.

When the planet is situated at the highest point of the perimandala, then there is maximum perturbation; at that time, since the Sun is nearby, the planet is not observed. The rising and setting are first seen when the śīghra-kendra is appropriate; then, even at the Sun's setting, the epicyclic motion having changed, the planet gradually attains reduction [in elongation]. Therefore, the planet is observed at rising and setting.

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#### **Verses 58-61. Declination and Shadow Calculations:**

The *jina-māga-jyā* (24-degree sine) multiplied by the solar *jyā* and divided by the semi-diameter gives the result. The desired ascensional difference (*kranti-jīvā*) from the equinox is for the Sun.



At the desired arc, subtracting from the square of the trijyā, the remainder's root, from the equinox semi-duration, the day-duration is obtained.

The declination-sine (krānti-jyā), multiplied by the equinoctial shadow and divided by 12, gives the *sthiti-jā*. The day-and-night prāṇa, multiplied by the semi-diameter, are considered.

The day-and-night half combined with the ascensional difference gives the arc-measure. Those [prāṇas], separated, are the divided [portions] of the nāgikā [serpent of time].

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### Verses 62-63. Twelve-Degree Sines and Ascensional Differences:

The sines at twelve-degree intervals are to be computed; for the Earth etc. 1325 [values apply]. The difficult [computations] yield 1863; for Mithuna [Gemini], the vomiting-sines are 3030. For the elapsed [periods], the feeble [sines] are obtained. The *jina-māga-jyā* multiplied by the solar jyā gives the *pumānu-kṣoṇi-jyā* (terrestrial sine). The Mithuna values at 66°30', Mercury's [value] at 19°59', and the nine-fold cycle at 13°30'. By these, the equinoctial shadow is computed. These, combined with the ascensional difference for one's own location, give the prāṇas.

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### Verses 64-67. Aries-to-Gemini Sine Computation:

Now, the computation of *cara-khaṇḍa* (ascensional difference segments) for Aries etc.:

For Aries, Taurus, and Gemini, the same fields apply; for Cancer, Leo, and Virgo, the same fields; and for Libra, Scorpio, and Sagittarius, and similarly for Capricorn, Aquarius, and Pisces.

The day-and-night durations should be shown for each; in the Spāṣṭādhikāra itself, I have demonstrated the cara-dala computation for the learned. When, at a location, the prāṇas between the terrestrial sine and shadow are computed, this interval is the time-difference inquired for the lagna (ascendant) computation.

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### Verses 28-31. Shadow and Gnomon Formulas (Chāyākaraṇī):

The altitude-sine (hr-jyā), multiplied by 12 and divided by the śaṅku (gnomon), gives the shadow (chāyā). The square of the semi-diameter divided by the equinoctial hypotenuse gives the hypotenuse (karṇa).

The product of the equinoctial hypotenuse and the koṭi, divided by the square of the semi-diameter, gives the result. In the north (sāumya), diminished by the sun's diameter-sine; in the south (yāmy), added — this is the arc-measure of the bow.

In the north, added; in the south, diminished — these are the east-west prāṇas. The elapsed specific portions; the remainder of the excess portion is the difference. Half the day-sine is the east-west *nata-prāṇa* (hour-angle in time units).

Having computed the chāyā-karṇa (hypotenuse of gnomon and shadow), multiply by the day-and-night. Then, divide by the day-and-night half and the chāyā-karṇa; the resulting quantity, compared with the square of the semi-diameter, gives the product of the equinoctial hypotenuse and the koṭi as a sine.

In the northern hemisphere, subtract the *chāyā-buddhi-jyā*; in the southern, add similarly. The arc of this result is computed, then with the *cara-dala-prāṇa*, in the northern sphere added and in the southern subtracted, giving the east-west *prāṇas*.

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### Verses 32-33. Alternative Shadow Computation:

The square of the semi-diameter divided by the equinoctial hypotenuse gives the hypotenuse (*karṇa*). Or, the product of the *lambaka* (co-latitude) and the quantity, divided by the square-root of the sum of the squares of the gnomon and its semi-diameter.

Or the product by the gnomon-multiplication, divided by the equinoctial hypotenuse, gives the *śaṅku*. The square of the hypotenuse, subtracted [from the appropriate quantity], its root divided by 12 gives the shadow.

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### Verses 34-36. Computation of Nata-kāla (Hour Angle):

With the day-and-night half and the *chāyā-karṇa* hypotenuse as multipliers, the equinoctial hypotenuse times the *koṭi*, divided by the square of the semi-diameter, gives the result in the north.

Diminished by the sun's diameter-sine in the north, added in the south — this is the arc-measure. Added in the north, diminished in the south — these are the east-west *prāṇas*.

The elapsed specific portions; the remainder of the excess portion. Half the day-sine is the east-west *nata-prāṇa* (hour-angle).

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### Verses 57-62. Latitude-related Sine Computation (Lambhākṣa-jyā):

The *śaṅku* (gnomon), known as the *lambaka-chāyā*, multiplied by the northward [quantity], gives the result. The equinoctial hypotenuse [gives] the purity; the highest latitude-*śaṅku* is the *aksa-ucca-natā*.

The elevated sine's *koṭi-jyā*, the altitude-sine is the *bhuja* or the *nata-jyā*. The semi-diameter [applies to] the *karṇa*-field-circle.

The *śaṅku* and the *kṣaya* quantities, with the aggregate of eight *puruṣas*, become the *lambhākṣa-jyā*; its degrees and minutes are the subsequent [measures].

Divided by the equinoctial hypotenuse, or the *śaṅku* divided by the *kṣaya* separately with the *trijyā*, in the latitude field, the *lambhākṣa-jyā* is elevated.

Having subtracted the *nata-lambhākṣa*, the sines [yield] difficulty. The *śaṅku* multiplied by *kṣaya*, divided by twice the *kṣaya*, remains.

The *lambhākṣa* degrees and minutes' remainder, the root of the square of the *trijyā*, is to be divided. The *bhuja-jyā* of all elevated differences is single.

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**Verse 91. Alternative Lambhākṣa-jyā Computation:**

For computing the lambhākṣa-jyā (latitude sine), having divided from 90 degrees, the sine is to be computed. Or alternatively: the jyā of the difference between 90 degrees and the latitude degrees is the lambaka-jyā. The aksa-lambaka-jyā [is obtained from] the product of the śaṅku and chāyā (gnomon and shadow), multiplied by 12 and divided by the shadow. Thus the lambhākā-jyā is obtained by an alternative method.

*Derivation:*

The sine of (90 - latitude) = lambaka-jyā, the sine of (90 - latitude) = madhya-jyā. Or:

$$\frac{\text{aksā-jyā} \times 12}{\text{palamā}} = \text{lambaka-jyā}, \quad \frac{\text{palamā} \times \text{lambaka-jyā}}{12} = \text{madhya-jyā}$$

This is the derivation of the aksa and lambaka.

**[91]**

**Verses 63-66. Computation of Nata-kāla (Hour Angle):**

Having multiplied the udayāstamaya (rising-setting) by the twelve akṣepa (declination), from the yāmyodaya (southern rising), the result is the *cala* and the true difference of the doḥ-jyā (altitude sine difference).

Proclaiming that maṇḍala, multiplying the result by the śaṅku or lambaka, divided by the dyu-jyā (day-radius), gives the nata-kāla (hour angle).

The doḥ-jyā-utkānti-jyā (altitude sine minus declination sine), diminished by the arka-kṣepa, divided by the nata-kāla, multiplied by the trijyā and divided by the dyu-jyā, gives the desired śaṅku.

The product of the nata and utkānta-jyā, divided by the square of the dyu-jyā, that portion is the śaṅku. The shadow multiplied by 12, its root divided [appropriately] gives the kṣepa.

Having multiplied the udayāstamaya by the twelve akṣepa from the yāmyodaya, the result is the *cala* and the true doḥ-jyā-vivara. Proclaiming that maṇḍala, multiplying the result by the śaṅku or lambaka and dividing by the dyu-jyā, gives the nata-kāla. The doḥ-jyā-utkānti-jyā, diminished by the arka-kṣepa and divided by the nata-kāla, multiplied by the trijyā and divided by the dyu-jyā, gives the śaṅku. The product of the nata and utkānta-jyā, divided by the square of the dyu-jyā, gives the śaṅku; the shadow multiplied by 12, its root gives the kṣepa.

**Verses 137-139. Construction of the Armillary Sphere (Gola-yantra):**

The celestial sphere (khagola) has seven rays [circles]; in its centre is situated the terrestrial sphere (bhūgola). Joined by rods (daṇḍaiḥ), this khagola is in the form of an instrument (yantra-rūpaka).

The ecliptic (krānti-vṛtta), the equator (sama-maṇḍala), the north-south circles (dakṣiṇottara-vṛtta), and the prime vertical (sadyat-kṣitija) should be constructed in the sphere by those who know.

By this instrument alone, the true position of the planets is known. Therefore, the instrument should always be constructed by those who know the sphere (gola-vid-bhiḥ).

The celestial sphere with seven circles, having the terrestrial sphere at its centre, joined by rods, is this armillary sphere. The ecliptic, equator, meridian circles, and prime vertical

should be constructed in the sphere by those versed in spherics. By this instrument alone, the true positions of the planets are known; therefore, astronomers should always construct this instrument.

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### Verses 128-130. Time Unit (Prāṇa) Computation:

Sixty ghaṭikās make one nāḍī; sixty nāḍīs are called prāṇas; sixty vipalas make one pala — by these units the learned compute time.

Having multiplied the cara-jyā (ascensional difference sine) and divided by the trijyā, the result gives the prāṇas. Their sum is the sthiti-kāla (duration of visibility); the ayanakāla is similarly computed.

Having divided the tanu-jyā (sine of zenith distance) and dividing by the trijyā, the result is the pramāṇa-kāla (measure-time). In the sthiti (position), dividing by the cara-jyā, those prāṇas are the true ones.

Sixty ghaṭikās equal one nāḍī; sixty nāḍīs are prāṇas; sixty vipalas equal one pala — these are the time units used by the learned. Multiplying the cara-jyā by the trijyā and dividing gives the prāḍnas. Their sum is the sthiti-kāla; the ayanakāla is similarly computed. Dividing the tanu-jyā by the trijyā gives the pramāṇa-kāla; in the position, dividing by the cara-jyā gives the true prāṇas.

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### Verse 148. Computation of Midday Shadow:

By the proportion of the chāyā-field:

$$\frac{\text{tri} \times 12}{\text{śaṅku}} = \text{chāyā-karṇa}$$

$$\sqrt{(\text{chāyā-karṇa})^2 - 12^2} = \text{midday shadow}$$

[148]

[Hindi commentary:] Multiplying the trijyā by 12 and dividing by the midday śaṅku gives the midday chāyā-karṇa. The square-root of the difference of squares of the chāyā-karṇa and 12 gives the midday shadow by an alternative method. [148]

*Derivation:* From the similarity of the chāyā-fields demonstrated in the previous verse's derivation, by proportion:

$$\frac{\text{tri} \times 12}{\text{midday śaṅku}} = \text{midday karṇa}$$

Therefore  $\sqrt{(\text{chaya-karṇa})^2 - 12^2} = \text{midday shadow}$ .

[148]

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### Verses 140-142. The Mahāpāta and Lunar Month:

The mahāpāta (great conjunction) occurs at the meeting of the Moon and Sun; then their difference becomes zero when they are at the same ecliptic point.

The lunar month (cāndra-māsa), by the aggregation of tithis (lunar days), is also a *śāvana* month, traversed by the Sun alone.

The combination of these two months is called a *saṃ*

The *mahāpāta* (great conjunction) occurs when the Moon and Sun meet; then their difference is zero when they are at the same ecliptic longitude. The lunar month, by the collection of tithis, is also a *śāvana* month, traversed by the Sun. The year (*saṃ*)

### Verses 131-133. Declination (Krānti) and Ayanāṃśa:

When the Sun is at the beginning of Meṣa (Aries), multiply by 66 [the *kāla* value]. When the Sun is at the equinox (*viṣuvat*), at the beginning of Meṣa, the declination-sine (*krānti-jyā*) there is considered as the equinoctial sine (*viṣuva-jjyā*).

That, diminished by its *kṣepa*, is to be known as the true declination by those who compute. In the *uttarāyaṇa* (northward course), it is southern; by those *kṣepas*, it becomes northern.

The *ayanāṃśa* (precession of equinoxes) is that from which the declination arises by differentiation. Therefore, knowing the *ayanacalana* (precessional motion), the true [position] should be computed by the prescribed method.

When the Sun is at the beginning of Aries, multiplied by 66; at the equinox, the declination at the beginning of Aries is considered the equinoctial sine. That, diminished by its *kṣepa*, is the true declination. In the northward course, the declination is southern; in the southward, northern. The *ayanāṃśa* is that from which the declination differs; knowing this precessional motion, the true [position] is computed properly.

### Kṣitijyā (Terrestrial Sine):

Well-known: In the southern [hemisphere], the southern *bhuja*'s elevation, the product of the *lambaka* and the appropriate quantity, divided by the *chāyā-karṇa*, gives the result. From this, the *krānti-jyā* (declination-sine) is obtained.

### Verses 134-136. Computation of the Lagna (Rising Sign):

From the rising-time (*rāśi-udaya*) of the Sun's sign, the *ghaṭikās* of the currently rising sign, having subtracted those from the given time, until the sign-risings are completed.

Those [signs], joined with the previous signs, constitute the *lagna*. Dividing the remainder by 30, the result is in degrees and minutes.

That should be added to the sign; its *lagna* is thus computed by those who compute true positions. Thus the *lagna* should always be computed by the wise who know planetary mathematics.

From the rising-time of the Sun's sign, the *ghaṭikās* of the currently rising sign, subtracted from the given time until the sign-risings are accounted for. Those signs, joined with the previous ones, give the *lagna*. Dividing the remainder by 30 gives the degrees and

minutes, which are added to the sign; this gives the lagna for true-position computation. Thus the lagna is always to be computed by wise planetary calculators.

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### *Key Mathematical Formulas from the Spaṣṭādhikāra and Tripraśnādhikāra*

This section presents the key mathematical formulas from the Spaṣṭādhikāra and Tripraśnādhikāra chapters, covering sine table construction, shadow calculations, hour angle computation, and arithmetic operations.

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#### **Śighra-spaṣṭikaraṇa Formula:**

The śīghra-phala-jyā (anomaly equation sine) multiplied by the śīghra-karṇa gives the true position correction.

The kendra-kardṇa (eccentric distance) is computed as the square-root of the sum of the square of the trijyā and the square of the manda-phala-jyā, minus twice the product of the manda-phala-jyā and the kalpa-jyā.

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#### **English Translation**

##### **Verses 100-101. Construction of the Sine Table (Jyotpatti):**

From the diameter (vyāsa), having subtracted three from the diameter-parts [3438-3 = 3435], made into parts and divided by six, the remainder is taken as the trijyā (radius sine,  $R \approx 3438'$ ).

Having multiplied the previous previous-jyā by the previous previous-difference-segments, and dividing by the trijyā, the other jyās are obtained; they are considered to originate from the previous one.

From the diameter, subtracting three from the diameter-parts gives 3435 parts. Dividing by six, the remainder is taken as the trijyā. With this trijyā, multiplying the previous previous-jyā by the previous difference-segments and dividing by the trijyā, the other jyās are obtained; they are considered to originate from the previous jyā.

#### **English Translation**

##### **Verses 102-103. Sine Computation Rule:**

The first jyā is a trijyā-segment; the second from half of the remainder; the third also likewise — this is the rule of Brahmagupta's method.

The sum of their difference-segments is remembered as the jyā-āvalaya (sine-circle). The six-degree interval jyās are well-known among accurate calculators.

The first jyā is the 7.5-degree sine, a trijyā-segment; the second is the 15-degree sine; the third similarly in sequence. The sum of their difference-segments is the jyā-āvalaya. The six-degree interval sines are well-known among accurate calculators.

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**Chāyākaraṇī Formulas:**

**Shadow:** The shadow equals 12 (gnomon height) multiplied by the altitude-sine, divided by the śaṅku (gnomon's effective height).

**Hypotenuse:** The hypotenuse equals the square of the semi-diameter divided by the equinoctial hypotenuse.

**Shadow-hypotenuse:** The hypotenuse of gnomon and shadow equals the square-root of the sum of squares of the shadow and the gnomon.

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**English Translation****Verses 104-106. Second-Difference Interpolation (Gaurī-guptā):**

Having subtracted the square of the bhuja-jyā (sine of arc) from the square of the jyā, divided by the square of the trijyā, and having subtracted the result from the first jyā, the next jyā is obtained.

The difference between the previous and subsequent jyā is the first segment (khaṇḍa); from that, the next segment; from that, having subtracted the previous segment, the third segment is obtained.

Thus, in sequence, all jyās should be computed from the divided jyā. The difference-jyās separately yield a more accurate result.

**English Translation****Verses 107-108. Difference Segments for Sine Computation:**

For odd [positions], having subtracted the square of the dvi-jyā (double sine) from the square of the dvi-jyā, the root from the remainder is the second segment. Then, having subtracted the previous segment, [the third is obtained].

The third segment, then having subtracted from the previous two, the fourth; then having subtracted from the previous three, thus in sequence all segments are obtained.

Having subtracted the square of the bhuja-jyā from the square of the jyā, divided by the square of the trijyā, and subtracting the result from the first jyā, the next jyā is obtained. The difference between previous and subsequent jyā is the first segment; the next segment from that; having subtracted the previous segment from that, the third segment is obtained. Thus, in sequence, all jyās are computed from the divided jyā; the difference-jyās separately give a more accurate result.

For odd positions, having subtracted the square of the dvi-jyā from the square of the dvi-jyā, the root of the remainder is the second segment. Then, having subtracted the previous segment, the third; having subtracted from the previous two, the fourth; from the previous three, the fifth; thus in sequence all segments are obtained.

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**Lambhākṣa-jyā Formulas:**

**Lambaka-jyā (co-latitude sine):** The aksa-jyā (latitude sine) multiplied by 12, divided by the palamā (equinoctial shadow), gives the lambaka-jyā.

**Madhya-jyā (meridian sine):** The palamā multiplied by the lambaka-jyā, divided by 12, gives the madhya-jyā.

**Nata-kāla Formulas:**

**Hour angle:** The nata-kāla equals the taj-jya multiplied by the śaṅku or lambaka, divided by the dyu-jyā (day-radius).

**Śaṅku:** The śaṅku equals the taj-jya multiplied by the trijyā, divided by the dyu-jyā.

**Krānti-jyā (Declination Sine) Formulas:**

The krānti-jyā equals the lambaka-jyā multiplied by the krama and divided by the viśuva. Alternatively, it equals the lambaka-jyā multiplied by the karṇoccātā and vi, divided by the chāyā-karṇa and vi.

The kṣetra-jyā equals the krānti-jyā multiplied by the vi, divided by the chāyā-kardna.

**English Translation****Verses 117-119. The Twenty-Four Operations of Pāṭigaṇita:**

Addition (saṃkalita), subtraction (vyavakalita), multiplication (guṇana), division (bhājana), squaring (varga), square-root (varga-mūla), cubing (ghana), and cube-root (ghana-mūla) — these five and three [make eight basic operations].

Five special [operations make] thirteen [in total]; these operations are remembered as gaṇita (mathematics). Proclaimed by the ācāryas, by these all mathematics is accomplished.

A number joined with a unit part is called a vyakta-rāśi (manifest number). That which is a number joined with the unit part is [called] avyakta (unmanifest).

Saṃkalita is addition; vyavakalita is subtraction; guṇana is multiplication; bhājana is division; varga is the product of two equals; varga-mūla is the root of that equal quantity; ghana is the product of three equals; ghana-mūla is the root of that quantity. These operations are called gaṇita (mathematics), proclaimed by the teachers; by these all mathematics is accomplished.

**English Translation****Verses 120-121. Square, Cube, and Their Roots:**

That which is the triple portion of a number, or joined with a unit part and halved, that becomes the square (varga). That which is the triple portion of a number.

Or joined with a unit part, that becomes the cube (ghana). Its root is called the ghana-mūla (cube-root); that becomes the measure of the equal-root quantity.

That which is the triple portion of a number, or joined with a unit part and halved, is the square. That which is the triple portion of a number, or joined with a unit part, is the cube. That which is its root is called the cube-root; that is the measure of the equal-root quantity.



**English Translation****Verses 109-111. Tithi, Yoga, and Nakṣatra Computation:**

Having subtracted the Sun from the true Moon, the remainder that results, divided by twelve parts, gives the tithis (lunar days); from half of that, the true [tithis].

The tithis are twelve digits (aṅgulis) from the difference of the Sun and Moon. Joined with fifteen tithis, the yoga is formed.

The yoga is the sum of the signs, degrees, and minutes of the Sun and Moon. That, joined with fifteen and divided by twenty-seven, gives the nakṣatra.

Having subtracted the Sun from the true Moon, dividing the remainder by twelve gives the tithis. The tithis are twelve digits from the difference of Sun and Moon. Joined with fifteen, the yoga is formed. The sum of the signs, degrees, and minutes of Sun and Moon, divided by twenty-seven, gives the nakṣatra.

**English Translation****Verses 112-113. Nakṣatra and Daily Motion Difference:**

Having subtracted the Sun from the true Moon, dividing the remainder by thirty, that nakṣatra becomes the true sign-measure.

The true motion of the Sun and the true motion of the Moon — their difference is called the tithy-ardha-bhukti (half-tithi daily motion).

Having subtracted the Sun from the true Moon, dividing by thirty gives the nakṣatra as the true sign-measure. The true motion of the Sun and the true motion of the Moon — their difference is the tithy-ardha-bhukti.

**English Translation****Verses 114-116. Spherical Astronomy — Indian and Yavana Methods:**

In the opinion of the Yavanas (Greeks), the Earth's sphere is situated at the equatorial region; its highest point is at its centre; there indeed is the kṣitija-vṛta (horizon circle).

In the Indian method, the akṣa-vṛta (latitude circle) is the madhyamaṇḍala (prime vertical); the kṣitija is that great circle; the sama-rekhā (equinoctial line) is also there.

The elevation-sine's koṭi-jyā, the altitude-sine is the bhuja or nata-jyā. The semi-diameter [applies to] the karṇa-field-circle in both methods (ūbhayamato-dyanna).

In the Yavana (Greek) view, the Earth's globe is at the equator, its highest point is at the centre, and there the horizon circle lies. In the Indian method, the latitude circle is the prime vertical, the horizon is that great circle, and the equinoctial line is also there. The common spherical knowledge of both methods is called "ubhayamato-dyanna" (accepted by both schools).

*Candra-grahaṇādhikāraḥ — Chapter on the Calculation of Lunar Eclipses*

This chapter deals with the computation of lunar eclipses, including the determination of the time, magnitude, and duration of the eclipse. Brahmagupta gives detailed rules for

finding the *parva* (full moon day), the *sparsā* (first contact), the *madhya* (middle), and the *mokṣa* (last contact) of a lunar eclipse.

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### English Translation

#### Verses 70-71. Sparsā-piṇḍa (First Contact):

At the true moonrise on the day of double motion, the piṇḍa (magnitude) that is the sparsā-piṇḍa (first contact magnitude), from that the computation of the middle of the eclipse.

That sparsā-piṇḍa, from that the computation of the eclipse middle. Or by its half, the time of the eclipse is proclaimed.

At the true moonrise on the day of double motion, the piṇḍa is the sparsā-piṇḍa (first contact magnitude), from which the middle of the eclipse is computed. Or by half of that, the time of the eclipse is proclaimed.

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### English Translation

#### Verses 72-73. Eclipse Limits by Declination:

The difference between the Moon and Sun's southern declination, that is the decrease or increase. The difference between their northern and southern declinations, that is the decrease or increase.

The difference between the southern and northern declinations, that decrease or increase. That bhuja-jyā is the karaṇa (semi-duration); that shows the sparsā and mokṣa (first and last contacts).

The difference between Moon and Sun's southern declination gives the decrease or increase. The difference between their north-south declinations gives the decrease or increase. That bhuja-jyā is the karaṇa, which determines the sparsā and mokṣa contacts.

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#### Verses 74-77. Eclipse Contacts and Duration:

The difference between southern and northern declinations, that decrease or increase. That bhuja-jyā is the karaṇa; that shows the sparsā and mokṣa contacts.

That bhuja-jyā is the karaṇa; that shows the sparsā and mokṣa. Or by its half, the time of the eclipse is proclaimed.

When the sparsā and mokṣa occur, then the eclipse is proclaimed. Or by its half, the time of the eclipse is proclaimed.

Or by its half, the time of the eclipse is proclaimed. That bhuja-jyā is the karaṇa; that shows the sparsā and mokṣa.

---

**Verses 78-80. Summary of Eclipse Rules:**

The sparśa-piṇḍa, from that the computation of the eclipse middle. Or by its half, the time of the eclipse is proclaimed.

The difference between Moon and Sun, southern motion, that decrease or increase. The difference between their north-south declinations, that decrease or increase.

The difference between south-north declinations, that decrease or increase. That bhuja-jyā is the karaṇa; that shows the sparśa and mokṣa.

---

**Verses 81-84. Eclipse Computation Summary:**

At the true moonrise on the day of double motion, the piṇḍa is the sparśa-piṇḍa (first contact magnitude), from which the middle of the eclipse is computed.

That sparśa-piṇḍa, from that the eclipse middle is computed. Or by its half, the time of the eclipse is proclaimed.

The difference between Moon and Sun, the southern motion, that decrease or increase. The difference between their north-south declinations, that decrease or increase.

The difference between south-north declinations, that decrease or increase. That bhuja-jyā is the karaṇa; that shows the sparśa and mokṣa.

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**Verses 85-87. Middle of the Eclipse:**

When the sparśa and mokṣa occur, then the eclipse is proclaimed. Or by its half, the time of the eclipse is proclaimed.

Or by its half, the time of the eclipse is proclaimed. That bhuja-jyā is the karaṇa; that shows the sparśa and mokṣa.

The sparśa-piṇḍa, from that the eclipse middle is computed. Or by its half, the time of the eclipse is proclaimed.

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**English Translation****Verses 122-124. Solar Eclipse Computation:**

In a solar eclipse, the lambana (parallax in longitude) should be subtracted from the difference between the Moon and Sun; that is the pure [value] at the sparśa (first contact), and similarly at the mokṣa (last contact).

The sparśa-kāla (time of first contact), increased by the lambana at the sparśa, is pure at the mokṣa. The middle of the eclipse is accomplished by half the sum of the two.

The pariveśa (diameter) of the Moon, diminished by that of the Sun, that becomes the grāhaṇa-māna (magnitude of the eclipse), as proclaimed by the ācāryas.

In a solar eclipse, the lambana is subtracted from the Moon-Sun difference; that is pure at first contact and at last contact. The first contact time, increased by lambana at first contact,

is pure at last contact. The eclipse middle is by half the sum of the two. The Moon's diameter, diminished by the Sun's, is the eclipse magnitude, as taught by the masters.

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### English Translation

#### Verses 125-127. Lambana (Parallax in Longitude):

The difference between the lambaka-jyā of one's own place and the dyu-jyā, divided by the trijyā, the result obtained is called the lambana (parallax in longitude).

The lambaka-jyā divided by the dyu-jyā, multiplied by the trijyā, the result obtained is the lambana, as spoken by the ancient masters.

That lambana-produced quantity, divided by time, becomes the true [value]. That, converted into nāḍīs, becomes the form — this is the lambana computation.

The difference between the lambaka-jyā of one's place and the dyu-jyā, divided by the trijyā, gives the lambana. The lambaka-jyā divided by the dyu-jyā, multiplied by the trijyā, gives the lambana as spoken by the ancients. That lambana-produced quantity, time-divided, becomes true; converted into nāḍīs, this is the lambana computation.

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#### Verses 88-90. Eclipse Contacts (Sparsā and Mokṣa):

The difference between Moon and Sun, the southern motion, that decrease or increase. The difference between their north-south declinations, that decrease or increase.

The difference between south-north declinations, that decrease or increase. That bhuja-jyā is the karaṇa; that shows the sparsā and mokṣa.

That bhuja-jyā is the karaṇa; that shows the sparsā and mokṣa. Or by its half, the time of the eclipse is proclaimed.

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## Candragrahaṇavikārah (Lunar Eclipse Variations)

### Introduction

This chapter deals with the computation of lunar eclipses, following Brahmagupta's exposition in the . The commentary by Paṇḍita Kṛpāludatta Sūnu Sudhākara Dvivedī provides elaborate explanations in Hindi, with mathematical demonstrations .

### Verses 11-12: Computation of (Desired Eclipse Magnitude)

#### Sūtra — Verses 11-12:

**Translation:** “The half-duration (*sthitidala*) diminished by the desired quantity, multiplied by the daily motion (*bhukti*) and divided by the deflection (*villepa*), gives the *bhuja* (sine). The *koṭi* (cosine) is computed by the rule of the desired quantity (*iṣṭeṇu*). From the half-sum of the diameters (*mānaikyārdha*), the *karṇa* (hypotenuse) being subtracted, the result gives the true obscuration (*channa-māna*).”

**Literal Meaning:** The verse gives the method for computing the *iṣṭagrāsa* (desired eclipse magnitude). From the *sthitidala* (half-duration) diminished by the desired quantity, multiplied by the *bhukti* (daily motion) and divided by the *villepa* (deflection), one gets the *bhuja* (sine). The *koṭi* (cosine) is computed by the rule of *iṣṭeṇu* (desired quantity). From the *mānaikyārdha* (half-sum of diameters), the *karṇa* (hypotenuse) being subtracted, the result gives the true obscuration.

**English Translation of Hindi Commentary:** “Subtracting the desired (*iṣṭa*) from the *sthitidala*, multiplying by the *bhukti* and dividing by the *villepa*, one obtains the *bhuja*. By applying this result to the motions of the Sun, Moon, and node (*pāta*), one computes the instantaneous lunar latitude (*tātkālika candra-śara*), which is the *koṭi*. The square root of the sum of their squares is the *karṇa*. From the *mānaikyārdha* (sum of the radii of the eclipsed and eclipsing bodies), subtracting the *karṇa*, the remainder is the desired magnitude (*iṣṭa-grāsa*).”

**English Translation of Proof:** “After the contact (*sparśī*), for whatever desired number of ghaṭīs the knowledge of the *grāsa* is required, that desired number of ghaṭīs should be subtracted from the *sthityardha*. The remainder is used in proportion. If in sixty ghaṭīs the difference in motion between the Sun and Moon gives a certain number of minutes (*kalā*), then [by proportion] in the *iṣṭa*-diminished *sthityardha* ghaṭīs, whatever result is obtained is the *bhuja*. From the result obtained by proportion, by applying the motions of the Sun, Moon, and node, the instantaneous lunar latitude (*tātkālika candra-vāra*) is computed — that is the *koṭi*. The square root of the sum of the squares of these two (*bhuja* and *koṭi*) is the *karṇa*. Subtracting the *karṇa* from the *mānaikyārdha* gives the desired *grāsa*.”

### Verses 13-14: Computation of (Time from Desired Magnitude)

#### Sūtra — Verses 13-14:

**Translation — Verse 13:** “From the square of the half-sum of the measure diminished by the *grāsakalā*, having subtracted the square of the instantaneous latitude (*tātkālika-vikṣepa*), the square root of the remainder — made like a *tithi* [i.e., multiplied by 60 and divided by the difference of motions] — [gives a time].”

**Translation — Verse 14:** “Having subtracted [this time] from the *pragrahaṇa-sthit yardha*, the time from the *pragrahaṇa* [i.e., from contact] is obtained. Having subtracted from the *mokṣa-sthit yardha*, [one obtains the time] before the *mokṣa* [release].”

**Literal Meaning:** “From the square of the half-sum of the measure [of diameters] diminished by the *grāsakālā*, having subtracted the square of the instantaneous *śara* (latitude), the square root of the remainder — made *tithi*-like (*tithivat*) [by multiplying by 60 and dividing by the difference of motions of Sun and Moon] — is the result. That is, having multiplied the square root of the remainder by 60, divided by the difference of motions of Sun and Moon, and having applied the motions of Sun, Moon, and node by that resulting time, the instantaneous latitude (*tātkālika-vikṣepa*) is computed. From that, again by the rule ‘*grāsakālona-pramāṇa*’, etc., iteratively (*asakṛt*), the square-root-time becomes stable. That [time], subtracted from the *pragrahaṇa-sthit yardha* (contact half-duration), the time from the *pragrahaṇa* (from contact) is obtained.”

**English Translation:** “From the square of the *grāsa*-diminished *mānaikyārdha*, having subtracted the square of the instantaneous *śara*, the square root of the remainder — made *tithi*-like iteratively — that is, having multiplied the square root of the remainder by 60, divided by the combined difference of motions of Sun and Moon, and having applied the motions of Sun, Moon, and node by that resulting time, the instantaneous lunar latitude is computed. From that, again by the rule beginning ‘*grāsakālona-pramāṇa*...’, iteratively (*asakṛt*), the square-root-time becomes stable. That, subtracted from the contact half-duration (*pragrahaṇa-sthit yardha*), gives the time from contact, meaning that after contact, in that desired time, the desired *grāsa* is obtained.”

**English Translation of Proof:** “Due to the ignorance of the latitude at that time (*tātkāla*), the computation has been done using the middle-time latitude. Therefore, it is proper to determine the accurate time by an iterative method (*asakṛt-ghātinā*). The remaining proof [follows] the method of Bhāskara: ‘From the square of half the sum of diameters diminished by the *grāsa*, having subtracted the square of the *vikṣepa*...’ etc.”

## Verse 9: Sphuṭikaraṇa of (Refinement of Half-Duration of Totality)

### Sūtra — Verse 9:

**Translation:** “The own motion (*svagati*) — [i.e.,] the motions of the Sun, Moon, and node — multiplied by the *sthitivimardārdha* *nāḍikās* and divided by sixty; both [motions] are applied [to the Sun, Moon, and node] repeatedly in the eclipsed body (*chādyā*), [and] the result is additive or subtractive.”

**Literal Meaning:** “The own motion — the motion of the Sun, the motion of the Moon, and the motion of the node — multiplied by the *sthitivimardārdha* *nāḍikās* (the half-duration of totality *ghaṭis* multiplied separately) and divided by sixty; applied to the Sun, Moon, and node in the eclipsed body, both [additive and subtractive] repeatedly, the result being additive or subtractive.”

**English Translation:** “The [method is] made accurate by the method of Bhāskara: ‘*sthit yardha-nāḍī-guṇitā sva-mukti*’ etc., and ‘*vimardārdha-phala-ona-yukta*’ etc.”

## Sūryagrahaṇādhikāraḥ (Solar Eclipse Chapter)

This chapter (, the fifth chapter of the text) deals with the computation of solar eclipses. The chapter begins with the statement of purpose and proceeds through the computation of the five (sines) required for eclipse calculation: , , , , and .

### Verse 1: Statement of Purpose

#### Sūtra — Verse 1:

**Introduction:** “Now the chapter on solar eclipse is begun. First, the purpose of its commencement is stated —”

**Translation:** “Because the inter-planetary [distance] computed by the *grahabhāskara* [method] previously is an obstruction in the inter-planetary [computation], therefore I shall state the computation of *dyuti* (parallax) with its own degrees, so that there may be agreement.”

**Technical note:** The *grahabhāskara* refers to the method given by earlier astronomers (like Bhāskara I) for computing the distance between planets, which Brahmagupta finds to be defective for solar eclipse computation.

**Literal Meaning:** “Previously, in the eastern horizon (*pūrve kṣitije*), by the degrees of the *grahabhāskara* method, the planets become visible or invisible (*haśya-āhaśya*) in their own time-degrees. Later, in the western horizon (*paścima-kṣitije*), by the degrees of the planetary distances, [the same] according to number. Therefore, I shall state that computation — the computation of the *praharāntara*. That is: in the eastern horizon, on whatever day after the rising of the planet, the rising of the Sun [occurs] after a certain number of time-ghaṭis, on that day the planet becomes invisible (*haśya*) in the remaining night. Similarly, in the western horizon, after the setting of the Sun, when the setting [occurs] after a certain number of time-ghaṭis, on that day in the evening the planet’s invisibility (*haśyatva*) [is determined].”

**English Translation:** “For whatever reason, in solar eclipses, the five *vyās* (*udaya-jyā*, *madhyajyā*, *ravidayākārṇa*, *viśiṃśaṅku*, *hr̥kṣepa*) — by which earlier teachers like Bhāskara computed the *prāṇayana* — from that, the *gaṇitaikya* (agreement between observed and computed values) is not obtained. For that reason, so that their agreement may exist, I state the *prāṇayana* [method]. The *tithi-anta* [refers to] the relation with the *graha*: *udaya-jyā* [refers to] parallax (*bhramā*), *madhyajyā* [to] the sine of the zenith-distance of the tenth degree (*daśama-lagna-natāṃśajyā*), *viśiṃśaṅku* [to] the ascensional difference (*ha-gati*), *hr̥kṣepa* [to] the sine of the zenith-distance of the nonagesimal (*viśiṃśa-natāṃśajyā*). In the *Sūryasiddhānta* too, the solar eclipse *prāṇayana* is by the aforesaid five *vyās*. In the *Paulīśasiddhānta*, in solar eclipses, separate five *vyās* of the Moon are computed; therefore there the solar eclipse is accomplished by the two-*vyā* method.”

### Verses 2-3: Conditions for (Deflection) and (Depression)

#### Sūtra — Verses 2-3:

**Translation — Verse 2:** “When one [planet] is equal to the *viśiṃśa-lagna* (nonagesimal ascendant), there is no *lambana* (deflection in longitude). When it is greater or less than that, *lambana* occurs. When its declination-sine (*krāntijyā*) is equal to the latitude-sine (*akṣajyā*), then there is no *avanati* (depression).”

**Translation — Verse 3:** “Otherwise, *avanati* occurs. When both *lambana* and *avanati* are possible, then the rising [point] equal to the *lagna*, the *ghaṭikās* elapsed since its rising, its *śaṅku* (gnomon), and the *prāṇas* of its motion — [are computed] thereafter.”

**Literal Meaning:** “At the end of the *tithi*, at the time of computed conjunction (*gaṇitāgata-darśāntikāle*), having computed the *lagna* by the previous method stated in the *Tripraśna*, the *viśrīma* [nonagesimal] is to be computed. When the Sun is equal to one [point] with that *viśrīmalagna*, there is no *lambana*. When the Sun is greater or less than that *viśrīma*, *lambana* occurs — this meaning is self-evident. Similarly, when the northern declination-sine (*krāntijyā*) of that *viśrīma* is equal to the latitude-sine (*akṣajyā*) of one’s own place, then there is no *avanati*; otherwise *avanati* occurs. When both *lambana* and *avanati* are possible, then having made the rising equal to the *lagna* — the *rāśyudaya-eṣi-lagna* of one’s own place — by the method stated in the *Tripraśna*: ‘the first diminished by the *bhoga*, increased by the *bhukti*, [gives] the middle *udaya*’, etc., the *ghaṭikās* elapsed since the rising of that *viśrīma* are to be computed.”

## Verse 7: Computation (Computation of True Conjunction Time)

### Sūtra ( ) — Verse 7:

**Translation:** “The *ghaṭikā-miṣa* [of *lambana*] multiplied by five and divided [by some number] yields the *labdha-kalā* called *lambana*. Then the Sun, Moon, and node at the computed conjunction time (*gaṇitāgata-darśānta-kālīka*) are to be diminished [or increased].”

**Literal Meaning:** “The *ghaṭikā-miṣa* multiplied by five, divided by five [? *pañcayyā*], the result is the *labdha-kalā* called *lambana*. Then the Sun, Moon, and node at the computed conjunction time are to be diminished (*ūnāḥ*). If the *lambana* is positive (*dhanā*), then the Sun, Moon, and node at the computed conjunction time are to be increased by the *labdha-kalā*; then those become the true Sun, Moon, and node at the true conjunction time (*spaṣṭa-darśānta-kālīka*).”

**English Translation of Proof:** “From the Sun, Moon, and node at the computed conjunction time, all [quantities] beginning with the *śara* [latitude] are to be computed. When the *lambana* is negative (*ṛṇātmaka*): ‘The minutes of motion of the Sun, Moon, and node are separately multiplied by the *lambana-ghaṭikā-miṣa* of sixty-five — by proportion: ‘then by the *lambana-ghaṭikā*, what [result]?’ — by the *labdha-kalās*, the Sun, Moon, and node at the computed conjunction time are to be diminished. When the *lambana* is positive (*dhanātmaka*), they are increased by the *labdha-kalās*; then they become the true [quantities] at the true conjunction time. In the *Siddhāntaśekhara*: ‘Having divided the motion-multiplied *lambana-nāḍī-miṣa* by five, the resulting minutes... The Sun, Moon, and node are added in positive [*lambana*] and subtracted in negative *lambana*’ — by this, what was stated by Śrīpati-nācārya is here spoken in accordance.”



## Candracchāyādhikārah (Moon's Shadow Chapter)

This chapter deals with the computation of the Moon's shadow, which is essential for understanding lunar phenomena. The shadow of the Earth falls on the Moon during a lunar eclipse, and its computation involves calculating the (shadow), (middle shadow), and related quantities.

### Verse 1: Beginning of the Chapter

#### Sūtra — Verse 1:

**Introduction:** “Now the chapter on the Moon's shadow is begun. First, the purpose of its commencement is stated —”

**Translation:** “Previously, the difference between the Moon and the *lagna*, and afterwards, that of the *saptalagna* etc., because from that the shadow of the Moon (*indu-ccchāyā*) is daily [computed], therefore I shall state the computation (*anayana*) of that.”

**Literal Meaning:** “Previously (in the eastern horizon), the difference between the Moon and the *lagna*, and afterwards (in the western horizon), the difference between the *saptalagna* and the Moon, and daily by effort — by which cause the Moon's shadow (*indu-ccchāyā*) is known, and from the Moon's shadow that difference is [also] known — therefore I shall state that computation (*anayana*), the shadow-computation (*chāyā-anayana*). In the *Siddhāntaśekhara*: ‘Usually the difference between the diminished-Sun-*lagna* and the *lagna-nāśin* in the west’ etc. — by this, what was spoken by Śrīpati-nācārya is here spoken in the same manner.”

**English Translation:** “Now the chapter on the Moon's shadow is begun. First, its purpose is stated. The difference between the Moon and the *lagna* in the eastern horizon, and the difference between the *saptalagna* and the Moon in the western horizon — because from these, daily, the Moon's shadow is known (first, the knowledge of the Moon's shadow is obtained), and from the Moon's shadow the knowledge of that difference is obtained — therefore I state the shadow-computation.”

### Verse 2: Shadow Computation

#### Sūtra — Verse 2:

**Translation:** “The parts of the upright-time-difference (*ūrdhvakālīka-udbala-naṣṭa-samasthiti-pāta*) [having] the form of a *prabhahṛṇḍa* [spherical triangle], the semi-diameter (*vyāsa*) of the shadow at the horizon (*tala-lagna*), the proximity to its own *agra* — there is no abbreviation (*saṅkṣepaḥ*).”

**Note:** This is a technical verse whose exact interpretation requires the accompanying commentary. The *prabhahṛṇḍa* is a type of spherical triangle used in Indian astronomy.

### Verses 3-4: Detailed Shadow Calculation

#### Sūtra — Verses 3-4:

**Introduction to Verse 3:** “Now [the author] states what is to be done —”

**Translation — Verse 3:** [Same structure as verse 2, establishing the method.]

**Translation — Verse 4:** “As the root (*mūla*) diminished by the root, or the elapsed [time], having been extracted, another [quantity] comes therefrom. Divided by the own motion (*svabhakti*), that is the desired shadow-motion (*abhīṣṭa-ccchāyā-gati*). That same [quantity], extracted from the root (*mūlodḍhṛtā*).”

*Note:* This verse describes an iterative root-extraction method for computing the rate of change of the shadow. The *mūla* here refers to a base quantity from which differences are taken.

## Grahodayāstādhikārah (Planetary Rising and Setting)

This chapter deals with the computation of the rising and setting times of planets. The chapter explains how to determine the times when planets become visible (, heliacal rising) and invisible (, heliacal setting) based on their elongation from the Sun.

### Verse 1: Pratyaya (Introduction)

#### Sūtra — Verse 1:

**Introduction:** “Now the chapter on planetary rising and setting is begun. First, the purpose of its commencement is stated —”

**Translation:** “Because previously, in the inter-planetary computation, the *grahabhāskara* distance is an obstruction (*bādhaka*), therefore, with its own degrees, so that there may be agreement — I state the computation of *dyuti*.”

*Note:* This verse serves as a parallel to the solar eclipse chapter’s opening, establishing that Brahmagupta’s method supersedes earlier approaches that failed to achieve agreement (*aiṃya*) between computation and observation.

**Literal Meaning:** “Previously, in the eastern horizon, by the *grahabhāskara* distances in their own degrees, in their own time-degrees, later in the western horizon, by the planetary distances in their own time-degrees, according to number, the planets become visible or invisible (*haśya-āhaśya*). Therefore, I shall state that computation — the computation of the *praharāntara*. That is: in the eastern horizon, on whatever day after the rising of the planet, the rising of the Sun occurs after a certain number of time-ghaṭis, on that day the planet becomes invisible (*haśya*) in the remaining night. Similarly, in the western horizon, after the setting of the Sun, when the setting occurs after a certain number of time-ghaṭis, on that day in the evening the planet’s invisibility (*haśyatva*) [is established].”

### Verses 8-9: Daily Rising and Setting

#### Sūtra — Verses 8-9:

**Translation — Verse 8:** “By iterative computation (*asakṛt-karma*) from the instantaneous (*tātkālika*) planet-*lagna*, the daily rising and setting are to be computed. First, at whatever time the *grahodita-ghaṭi* [elapsed ghaṭis since planetary rising] is to be known, the *lagna* at that time (*tātkālika lagna*) and the planetary rising-*lagna* are to be computed. Computing the difference-ghaṭis of both, by those (computed ghaṭis) moving the planet, again computing its rising-*lagna*, from the instantaneous fixed *lagna* again computing the desired ghaṭis — in this manner, iteratively (*asakṛt*, repeatedly) doing [this].”

*Note:* This verse describes the iterative procedure (*asakṛt-karma*) for refining the computation of a planet’s rising time. The method involves successive approximation until the computed value stabilizes.

### Verse 9: Moon’s Rising and Setting Computation

#### Sūtra — Verse 9:

**Translation:** “The rising and setting of the Moon (*indu*) are to be computed by time-degrees (*kālāmśa*) equal to digits (*aṅka-saṃmitaiḥ*). If there is diminution (*hīnatva*) or excess (*adhikatva*), in that interval the *yoga-kāla* [conjunction time] would be [obtained].”

*Note:* The *aṅka* (digit) is a unit equal to 12 degrees. The verse states that the Moon’s rising and setting are computed using time-degrees measured in these units.

**Literal Meaning:** “By time-degrees equal to digits (*aṅka*), that is, by twelve time-degrees (*dvādaśa-kālāmśa*), the rising and setting of the Moon (*indu* = *candra*) are to be computed. Here, if the computed time-degree (*paṭhita-kālāmśa*) has diminution or excess from the desired time-degree (*iṣṭa-kālāmśa*), then in that interval, between those two computed time-degrees, the *yoga-kāla* (conjunction time), or *yogavat-kāla* (time like conjunction), would be [obtained], as clearly stated in the seventh verse.”

**English Translation:** “By time-degrees equal to digits (*aṅka*), that is, by time-degrees equal to twelve, the rising and setting are to be computed. If the computed time-degree has diminution (*hīnatva*, being smaller) or excess (*adhikatva*) from the desired time-degree, then in that interval, between those two time-degrees, the *yoga-kāla* (conjunction time) would be [obtained].”

## Śuklonnatyadhikārah (The Brightness of the Moon)

This chapter deals with the computation of the illuminated portion of the Moon's disk — the (bright) portion. The chapter explains how to determine the amount of the Moon's surface that is illuminated by the Sun at any given time.

### Verses 1-3: Sphuṭaśuklonnati Computation

#### Sūtra — Verse 4:

**Introduction:** “Now the chapter on the Moon's bright elevation (*śuklonnati*) is begun.”

**Translation — Verse 4:** “The ghaṭikās [which are] the *khacarārdha* (half of the celestial sphere's motion) — by the Sun, Moon, node, and *lagna*, and from the rising of the own declination (*sva-krānty-udaya*), [which are] the remaining [ghaṭikās] elapsed from the own *lagna* — having made [the computation] from the setting of the Sun, [these are] to be computed.”

*Note:* The *khacarārdha* refers to half the diurnal motion of the celestial sphere. The verse establishes the parameters needed to compute the Moon's illumination.

**Literal Meaning:** “The *khacarārdha* ghaṭikās [which are] from the setting of the Sun (*ravyastāt*), [which are] the remaining [ghaṭikās] elapsed from the own-*lagna* of the Moon — or the elapsed/deficient ghaṭikās of the Moon's rising-*lagna* — are to be computed. By what? By the Sun, Moon, node, *lagna*, and from the rising of the own declination. Here, this is stated: at the time of the Sun's setting, the Sun, Moon, node, and *lagna*. Having computed these, by the own declination (*sva-krānti*), by the Moon's declination (*candra-krānti*), from the rising, [computing] the elapsed or deficient ghaṭikās of the Moon's rising-*lagna* from the *udaya-lagna* by the method ‘diminished by the *bhoga*, increased by the *bhukti*, [gives] the middle *udaya*’, etc. After the Sun's setting, the ghaṭikās by which the Moon sets — those are the elapsed ghaṭikās. Before the Sun's setting, the ghaṭikās by which the Moon sets — those are the deficient ghaṭikās. Similarly, in the eastern horizon, from the Sun's rising, the elapsed or deficient ghaṭikās of the Moon's rising are to be computed — this is to be understood from the meaning.”

**English Translation:** “By iterative computation (*asakṛt-karma*) from the instantaneous planet-*lagna*, the daily rising and setting are to be computed. First, at whatever time the planetary elapsed ghaṭis are to be known, the *lagna* at that time (*tātkālika lagna*) and the planetary rising-*lagna* are to be computed. Computing the difference-ghaṭis of both, by those (computed ghaṭis) moving the planet, again computing its rising-*lagna*, from the instantaneous fixed *lagna* again computing the desired ghaṭis — in this manner, iteratively (*asakṛt*, repeatedly) doing [this]. And for the knowledge of the *sthita-kāla*, computation from the *sthita-lagna* (*sapta-bhastala-lagna*) should be done. In this way, from the first day of lunar visibility (*candra darśana*) up to the full moon (*pūrṇimā*), on the desired day, the setting time from the Sun's setting and the rising time from the Sun's rising are to be computed.”

### Verse 5: The (Bright Elevation)

#### Sūtra ( ) — Verse 5:

**Translation — Verse 5:** [Recapitulates the parameters for computing the bright elevation.]

*Technical summary:* The verse reiterates that by the Sun, Moon, node, *lagna*, and from the rising of the declination, the *khacarārdha* ghaṭikās from the Sun's setting are to be used to compute the Moon's bright elevation.

**English Translation of Proof:** “On whatever day, after the Sun's setting, the Moon sets after an additional number of time-ghaṭis. Before the Sun's rising, the Moon rises

[after a certain number of ghaṭīs]. On that very day, when the Moon is visible, the two bright elevations (*śva-jy-unnati*) are to be computed. The bright elevation of the bright [half] (*sita-śuklonnati*) is half the disk (*bimbārdha*) of the Moon's bright [portion]. Thus Bhāskara computed the *śuklonnati* in the first *pāda* of the month-end. In the second and third *pādas*, half the disk is dark (*kṛṣṇa*). Therefore there the dark *śuklonnati* occurs. Thus by the teacher (*ācārya*), two *śuklonnatis* are computed. But here, the *pāda* discussion is not made. Bhāskara, having considered in his mind that the dark *śuklonnati* is not clearly observable, computed only the bright *śuklonnati*. And it is stated: 'Brahmagupta and others have produced the dark *śuklonnati* also for the second and third *caraṇas*; that is not approved by me. For neither is the dark *śuklonnati* clearly perceived by men.'

## Verses 11-12: (True Bright Portion) Computation

### Sūtra ( ) — Verses 11-12:

**Translation — Verse 11:** "Or when the middle motion (*madhyagati*) is multiplied by its own shadow (*tacchāyā*) and by the semi-diameter (*sva-kara*) multiplied by the *tithi* — the results are additive or subtractive. In equality (*sāmya*), the difference is [taken]; in direction-difference (*dig-anyatva*), [it is] the *avanati* [depression] expressed in digits (*aṅkya*)."

*Note:* This verse gives the method for computing the true bright portion (*spaṣṭa-śukla*) of the Moon. The *madhyagati* is the mean daily motion, and the *tacchāyā* is the shadow corresponding to the planet's altitude.

**Literal Meaning:** "In the *hṛjya* [radius = 3438'] of the Sun and Moon, the *kṣepī* [sine of latitude], having been multiplied by the respective middle motion and by the semi-diameter (*vyāsadala*) multiplied by the *tithi* and by fifteen [i.e., by the *tithi*-multiplied semi-diameter multiplied by 15], divided by the *triṇyā* [radius = 3438'], separately, separately, the *natī* [depressions] of the Sun and Moon are obtained. Or, their middle motion, their *kṣepa*-equal *hṛjya*, the shadow being multiplied by that, multiplied by the *tithi*-multiplied own shadow-*karman*, their difference is [taken]."

**English Translation of Proof:** "The first computation [is made] accurate by the method of Bhāskara: 'Having multiplied by the *sahakṣepa* [combined latitude] of the Moon, by the own middle motion-*tithi*-aṃśa-multiplied [quantity]' etc."

## Chandraśuklonnatyadhikārah (Moon's Bright Elevation)

This is the seventh (chapter) of the , dealing with the Moon's bright elevation . It comprises eighteen verses and covers the computation of the arc of illumination on the Moon's disk.

### Verses 1-2: Introduction

#### Sūtra — Verses 1-2:

**Introduction:** “Now the beginning of the chapter on the Moon's bright elevation (*candra-śukla-unnati*).”

**Translation — Verse 1:** “The parts of the upright-time-difference (*ūrdhvakālika-udbala-naṣṭa-sama-sthiti-pāta*) [having] the form of a *prāgrahuṇḍa* [spherical triangle], the semi-diameter (*vyāsa*) of the shadow at the horizon (*tala-lagna*), the proximity to its own *agra* — there is no abbreviation (*saṅkṣepaḥ*).”

**Translation — Verse 2:** “As the root (*mūla*) diminished by the root, or the elapsed [time], having been extracted, another [quantity] comes therefrom. Divided by the own motion (*svabhakti*), that is the desired shadow-motion (*abhīṣṭa-ccchāyā-gati*). That same [quantity], extracted from the root (*mūlodḍhṛtā*).”

**Note:** Verse 2 establishes an iterative root-extraction algorithm for computing the rate of change of the shadow, which is essential for determining the Moon's illumination over time.

### Verse 18: Chapter Conclusion

#### Sūtra — Verse 18:

**Introduction:** “Now [the author] states the conclusion of the chapter —”

**Translation — Verse 18:** “Thus, in [the topics of] *bāhu* (base), *koṭi* (perpendicular), *karṇa* (hypotenuse), *spāṣṭa* (true [longitude]), *sita* (bright [part]), *parilekha* (outline), *sūtra* (line), and *kardama* [mud] — by eighteen doors (*dvāra*), [this is] the seventh chapter [on] the Moon's bright elevation (*candra-śuklonnati*).”

**Note:** The chapter lists eight topics (*bāhu*, *koṭi*, *karṇa*, *spāṣṭa*, *sita*, *parilekha*, *sūtra*, *kardama*) covered in 18 verses. The *kardama* refers to the “mud” or penumbral shadow region.

**Literal Meaning:** “The *parilekha-sūtra* itself [is] the *karṇa*... the *parilekha-sūtra-karṇaḥ*. The meaning is clear.”

**English Translation:** “In this way, in [the topics of] *bāhu*, *koṭi*, *karṇa*, *spāṣṭa*, *sitāṅgula*, *parilekha-sūtra* (*karṇa*), by eighteen meaning-verses, is the seventh chapter named *Chandraśuklonnati*.”

**Note:** The *sitāṅgula* (bright digit) is a unit for measuring the illuminated portion of the Moon's disk. One *āṅgula* (digit) = 1/12 of the Moon's diameter.

### Concluding Verse ( )

**Translation — Concluding Verse:** “That *tilaka* (ornament) which was spoken by the son (*sūnu*) of Madhusūdāna, which was spoken here by Śrīpathi for the conqueror (*jīṣṇu*) — having placed that in [my] heart, this new [work] *Sūryagati* has been composed by Sudhākara.”

**Colophon:** “Thus, in the *Brāhmasphuṭasiddhānta-nutana-tilaka* composed by Śrī Kṛpāludatta-sūnu Sudhākara Dvivedī, the seventh chapter — the *Candraśuklonnatyadhikārah* — [is concluded].”

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**[Thus ends the Seventh Chapter — the Moon’s Bright Elevation — in the *Brāhmasphuṭasiddhānta* composed by Śrī Brahmagupta.]**

## Appendix: Notes on the Text and Commentary

### About the

The (*Brāhmasphuṭasiddhānta*, “Correctly Established Doctrine of Brahma”) is the magnum opus of the great Indian mathematician and astronomer **Brahmagupta** (598–668 CE). Composed in 628 CE at Bhinmal (in present-day Rajasthan, India), it consists of 24 chapters ( ) containing a total of 1008 verses ( ) in and metres.

1. — Lunar Eclipse Variations  
Deals with the computation of eclipse magnitudes, durations, and the *iṣṭagrāsa* (desired obscuration) method.
2. — Solar Eclipse Chapter  
Covers the computation of solar eclipses, including the five *jyā* (sines) required: *udaya-jyā*, *madhyajyā*, *ravidayākarṇa*, *viśiṃśaṅku*, and *hṛkṣepa*.
3. — Moon’s Shadow Chapter  
Treats the computation of the Earth’s shadow as seen on the Moon’s disk.
4. — Planetary Rising and Setting  
Explains the methods for computing the heliacal rising (*udaya*) and setting (*asta*) of planets.
5. — The Brightness of the Moon  
Computes the illuminated portion of the Moon’s disk.
6. — Moon’s Bright Elevation  
The seventh chapter, with 18 verses, covering the arc of illumination.

### The Commentary ( )

The Hindi commentary ( ) in this edition was composed by **Paṇḍita Kṛpāludatta Sūnu Sudhākara Dvivedī**. The commentary follows the traditional structure of Indian mathematical commentaries:

- (*Sūtra*) — The original Sanskrit verse
- (*Sūtrārtha*) — Literal meaning of the verse (marked . .)
- (*Vivṛti*) — Detailed explanation (marked )
- (*Hindi Bhāṣya*) — Hindi explanation (marked )
- (*Upapatti*) — Mathematical proof or demonstration

### Mathematical Notation

Brahmagupta’s work represents one of the high points of Indian mathematics. Key contributions in these chapters include:

- Rules for computing eclipse magnitudes using the (*bhuja*, sine) and (*koṭi*, cosine)
- The (*karṇa*) formula:  $\text{karṇa} = \sqrt{\text{bhuja}^2 + \text{koṭi}^2}$
- Iterative methods (, *asakṛt*) for refining planetary positions
- Proportional reasoning (, *trairāśika*) for time conversions
- Geometric constructions using (*kṣetra*, diagrams)



**[p. 1259]**

Sū. Bhā.—The true longitude of the Moon: having applied the corrections for equation of centre (manda-phala) and evection (śīghra-phala) to the mean longitude, the true longitude is obtained. The mean longitude is calculated from the ahargaṇa (elapsed days since the epoch).

**[p. 1260-1264]**

Sū. Bhā.—The Moon's equations (manda-phala) are tabulated for each 15° of anomaly. The tabular differences are: 39, 36, 31, 24, 15, 5. These are applied successively to obtain the true longitude.

Vi. Bhā.—When the anomaly lies between two tabulated values, multiply the residual arc by half the difference of the passed and upcoming tabular differences, divide by 900', and add to half the sum of the two differences.

**[p. 1265-1269]**

Sū. Bhā.—The śīghra corrections for the Moon are computed from the śīghra anomaly, which is the difference between the Moon's longitude and the Sun's longitude. The maximum śīghra equation is 80' (80 arc-minutes).

**[p. 1270-1274]**

Sū. Bhā.—To find the lagna (ascendant): multiply the time elapsed since sunrise by the Sun's daily motion, add to the Sun's longitude, and apply the ayanāṃśa correction.

**[p. 1275-1279]**

Sū. Bhā.—The time of conjunction is found by dividing the difference in longitudes by the difference in daily motions. The result gives the number of days until conjunction.

**[p. 1280-1284]**

Sū. Bhā.—A prāṇa equals 4 seconds of time. Six prāṇas make a vināḍī, 60 vināḍīs make a nāḍī (ghaṭī), and 60 nāḍīs make a day.

**[p. 1285-1287]**

Here end the verses on lunar computation; next follows the section on shadow calculations (chāyā-vyavahāra).

**[p. 1288-1292]**

Proof.—Let  $j_1$  and  $j_2$  be two successive tabular sines, with arc difference  $h = 15^\circ$ . For an intermediate arc at distance  $\theta$  from  $j_1$ , the sine is:

$$\sin(\theta) = j_1 + \frac{\theta}{h}(j_2 - j_1) - \frac{\theta}{h}\left(1 - \frac{\theta}{h}\right)\frac{j_2 - j_1}{2}$$

This is Brahmagupta's second-order interpolation formula.

**[p. 1293-1297]**

Proof.—Applying the kuṭṭaka (pulverizer) to solve  $137x + 15 = 251y$ : divide 251 by 137, quotient 1, remainder 114. Divide 137 by 114, quotient 1, remainder 23. Divide 114 by 23, quotient 4, remainder 22. Divide 23 by 22, quotient 1, remainder 1. Working backwards yields the multiplier and quotient.

**[p. 1298-1302]**

Sū. Bhā.—The shadow (chāyā) for a gnomon of 12 aṅgulas:

$$\text{Shadow} = \frac{12 \times \cos(\text{zenith distance})}{\sin(\text{altitude})}$$

When the Sun is at the prime vertical, special rules apply using the nara (sine of latitude) and śaṅku (gnomon).

**[p. 1303-1307]**

Proof.—The lagna is found from the oblique ascension. Let  $\phi$  be the latitude,  $\varepsilon$  the obliquity of the ecliptic. For elapsed time  $t$  (in degrees from sunrise), the lagna  $\lambda$  satisfies:

$$\tan(\lambda) = \frac{\sin(\text{R.A.})}{\cos(\text{R.A.}) \cos(\varepsilon) - \tan(\phi) \sin(\varepsilon)}$$

**[p. 1308-1312]**

Sū. Bhā.—For Mars, with manda anomaly  $\mu$  and epicycle radius  $r = 39$  (where deferent radius  $R = 60$ ):

$$\text{manda-phala} = \arcsin\left(\frac{39 \sin(\mu)}{\sqrt{60^2 + 39^2 + 2 \cdot 60 \cdot 39 \cos(\mu)}}\right)$$

This gives the equation of centre for Mars.

**[p. 1313-1317]**

Proof.—First apply manda correction to obtain manda-sphuṭa:  $\lambda_{\text{ms}} = \bar{\lambda} + \text{manda-phala}$ . Then compute śīghra anomaly from  $\lambda_{\text{ms}}$  and apply śīghra correction:  $\lambda_{\text{true}} = \lambda_{\text{ms}} + \text{śīghra-phala}$ .

**[p. 1318-1322]**

Sū. Bhā.—The semi-diameter of the Earth's shadow at the Moon's distance:

$$R_{\text{shadow}} = \frac{R_{\text{Earth}} \cdot (D_{\text{Moon}} - D_{\text{Sun}})}{D_{\text{Sun}}}$$

where  $D$  denotes distance. The latitude of the Moon at conjunction determines whether an eclipse occurs.

**[p. 1323-1327]**

Sū. Bhā.—A planet becomes visible when it is far enough from the Sun. The arc of visibility depends on the planet's brightness: for Venus  $10^\circ$ , for Jupiter  $11^\circ$ , for Mercury  $13^\circ$ , for Saturn  $15^\circ$ , and for Mars  $17^\circ$ .

**[p. 1328-1337]**

Proof.—Ṛjuvṛtta (straight epicycle) corrections and planetary phase computations. These final derivations summarise the algebraic methods used throughout the *Brāhmasphuṭasiddhānta* for astronomical computations.

**[p. 1339-1343]**

Sū. Bhā.—The sum of two positive quantities is positive; the sum of two negative quantities is negative. The sum of a positive and a negative is their difference; if they are equal, the

sum is zero.

**[p. 1344-1348]**

Sū. Bhā.—Zero multiplied by any quantity is zero. A positive quantity multiplied by a negative is negative; a negative by a negative is positive. Zero divided by any quantity is zero.

Sū. Bhā.—A quantity divided by zero becomes that fraction of which the denominator is zero. This *khacheda* (zero-divisor, or “zero-seed”) was later corrected by Bhāskara II.

**[p. 1349-1353]**

Sū. Bhā.—To solve  $Nx^2 + 1 = y^2$  (Pell’s equation): first find the least solution  $(x_1, y_1)$  by inspection or by the cakravāla (cyclic) method. From this, infinite solutions can be generated by composition (bhāvanā).

**[p. 1354-1358]**

Proof.—For  $N = 61$ : start with  $a_0 = 8$  (since  $8^2 = 64 \approx 61$ ). The interpolator  $k_0 = 3$ . The cyclic process generates successive approximations:

$$a_1 = \frac{a_0 + m_0}{k_0}, \quad k_1 = \frac{m_0^2 - N}{k_0}$$

where  $m_0$  is chosen so that  $a_0 + m_0$  is divisible by  $k_0$  and  $m_0^2$  is close to  $N$ .

**[p. 1359-1363]**

Sū. Bhā.—The sum of an arithmetic series:  $S_n = \frac{n}{2}(2a + (n - 1)d)$ . The sum of squares:  $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$ . The sum of cubes:  $\sum_{k=1}^n k^3 = \left(\frac{n(n+1)}{2}\right)^2$ .

**[p. 1364-1368]**

Sū. Bhā.—In compound interest, if principal  $P$  grows at rate  $r$  per period for  $n$  periods, the amount is  $A = P(1 + r)^n$ .

**[p. 1369-1370]**

Here conclude the verses of the algebra section, with praise of the siddhānta and transition to the planetary motion chapter.

**[p. 1371-1375]**

Sū. Bhā.—Having computed the mean longitude (madhyama) from the ahargaṇa, apply the manda and śīghra corrections in order to obtain the true longitude (sphuṭa).

**[p. 1376-1380]**

**Manda epicycles:** Mars 39, Mercury 22, Jupiter 32, Venus 14, Saturn 60. **Śīghra epicycles:** Mars 73, Mercury 121, Jupiter 71, Venus 132, Saturn 39 (in units where  $R = 60$ ).

Table 2: \*

Epicycle dimensions and maximum equations (degrees where  $R = 60$ )

**[p. 1381-1385]**

Sū. Bhā.—The rising time of a planet is computed from its longitude, the oblique ascension of the ecliptic, and the latitude of the place. The *carajā* (ascensional difference) is applied to find the exact moment of heliacal rising.

**[p. 1386-1390]**

Sū. Bhā.—Thus have been explained the motions of the planets, the Sun and the Moon, with their corrections. By these methods, the positions of the celestial bodies may be computed for any desired time.

**[p. 1391-1398]**

**Colophon.**—This *Brāhmasphuṭasiddhānta* was composed by Brahmagupta, son of Jṣṇughāṭa, in the year 628 of the Śaka era (550 CE), at the age of 30. May it bring clarity to all who study it.

## Colophon

**Key Mathematical Contents:** The *Brāhmasphuṭasiddhānta* is notable for: (1) algebraic rules for arithmetic with positive and negative numbers, zero, and surds; (2) the kuṭṭaka (pulverizer) method for linear Diophantine equations; (3) the varga-prakṛti method for Pell's equation; (4) astronomical calculations for planetary positions, eclipses, and conjunctions; (5) trigonometric interpolation formulas.

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## — Planetary Shadow Calculations

**Text (v. 41)** — When a shadow passes over the karna-ucchyāsa-ardha-gola (half-sine sphere of the hypotenuse), the shadow turning from the hypotenuse becomes vyasta-golā (reversed/inverted). On the karna-gola (hypotenuse sphere), the palabhā (equinoctial shadow) equals the base of the gnomon (śaṅku-tala), and the palabhā is always directed northward. Of these two, applying saṃskāra (correction), the difference between the eastern-western threads at the face of the shadow becomes the bāhu (base), and the root of the gnomon is to be understood by means of the bandha (construction) at the shadow-face.

**Commentary (Ṭikā)** — When the shadow, having gone around the sphere of the hypotenuse and its altitude, becomes inverted through the hypotenuse, then on the hypotenuse-sphere the palabhā equals the base of the gnomon; the palabhā is always northward. Applying correction to them, the difference between the east-west threads at the shadow-mouth becomes the base (bhuja); by the bandha at the shadow-mouth the root of the gnomon is to be known.

**Proof (Upapatti)** — In the Siddhāntaśekhara it is thus stated by Śrīpati Nārāyaṇācārya: “That which is the difference between the east-west gnomon-bases is the bāhu; that itself becomes the north-south (koṭi).” Thus the demonstration is established.

**Text (v. 42)** — Now he states the method for finding the base (bhuja):

*prācyaparāyāḥ śaṅkūstalaṃ tadapraṃ grahaṃ me ca*

The base of the gnomon extends east-west; that astra (side) and the planet—these become the measure.

**Commentary** — *prācyaparāyāḥ* (east-west): the base of the gnomon extends to the root of the gnomon. The astra (flank) of the gnomon, or the planet, becomes the measure.

**Proof** — Now the proof: from the east-west limit at the root of the gnomon, the demonstration of the base is clear according to the siddhānta-yukti.

“The side of the gnomon and the planet”—this is clear.

**Commentary** — Of the same directions (ekādiśa), the sum of the east-west gnomon-bases; and of two directions (dvyādiśa), the difference of the east-west gnomon-bases—from the east-west line, the base extending to the root of the gnomon becomes the bhuja (base). The astra (side) of the gnomon and the star become the constituents.

**Text (v. 43)** — The base extending east-west from the gnomon-base (śaṅku-tala) becomes the bhuja; the karna (hypotenuse is) the north-south. By the dvijaya (second victory), the root of the inverse from the middle of the dvikṣepa (double-latitude) becomes the koṭi.

**Commentary** — *śaṅkutalaṃ prācyaparāntaṃ* means the difference between the east-west threads at the root of the gnomon; the north-south becomes the bhuja. By the second application, the root of the inverse of the hypotenuse and the base—from the middle of the double-latitude (dvikṣepa-madhya)—becomes the koṭi (perpendicular).

**Proof** — Of the two meeting at the double-latitude, that which is the perpendicular dropped from the east-west threads is the bhuja. Also, the perpendicular dropped from the east-west threads at the root of the gnomon is the same bhuja. By the sum of the gnomon-bases of one direction, and by the difference of the two directions, extending to the east-west threads at the root of the gnomon, the perpendicular from the east-west threads also becomes the bhuja.

## — Characteristics of Yukti (Proofs)

**Text (v. 69)** — Now he states the characteristics of yukti (proof):

*ubhe mānakhyākhaṇḍi grahayor madhyāntaram yukti-grhyoḥ*

When the difference between two planets of equal minutes (māna) is divided by the sum of the vi-ākhaṇḍa (discs), at the time of eclipse there is a yukti (conjunction).

**Commentary** — The dvigu compound *mānakhyākhaṇḍi* is not clearly explained in this statement; its meaning requires further elucidation.

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**Expanded Text** — When the difference between two planets of equal minutes (*sama-kala*)—i.e., the latitude-difference (*kendra-antara*)—if the difference between the two planets is less than the sum of their discs (*vi-bhakta-yogākhaṇḍita*), then there is a yukti (conjunction), that is, an ācchādana (eclipse) like a grahaṇa; thus, the upper planet becomes the grāhya (the eclipsed) and the lower planet becomes the grāhaka (the eclipser). If, when divided by the sum of the discs at the time of eclipse, their yukti (meeting) does not occur. In the Siddhāntaśekhara: “*mānakhyākhaṇḍi dhruvara-vikare syān medo-dvike tu nyūne medo grahaṇa-vād-vihac-chādako 'stanaḥ syāt*”—this statement of Śrīpati and the statement of Bhāskara, “*mānakhyākhaṇḍi dhruvara vikare 'sye bhaved yogaḥ*” etc., are in agreement with the Ācārya’s teaching.

**Proof** — Now, the definition of yukti is stated: If the difference between two planets of equal minutes is less than the sum of their *vi-bhakta-yogākhaṇḍa* (discs), then there is a yukti of the two planets, that is, an ācchādana (covering) like an eclipse (the upper planet is the eclipsed, the lower planet is the eclipser). If the difference between the two planets is greater than the sum of the *sphuṭa-vibhakta-yogākhaṇḍa*, then there is no yukti of the planets. In the Siddhāntaśekhara, from the verse beginning “*mānakhyākhaṇḍi dhruvara-vikare syān medoḥ dvike tū*” etc., in the Vijñāna-bhāṣya—Śrīpati in the Siddhānta-śiromaṇi, from “*mānakhyākhaṇḍi dhruvara vikare 'sye bhaved yogaḥ*” etc., Bhāskarācārya too has spoken in accordance with the Ācārya.

## — Planetary Conjunctions

**Text (v.57)** — Of the bhuja, koṭi, and karṇa, whatever bhuja-half is to be transformed, making it into gunakas (multipliers) divided by the semi-diameter (vyāsārdha). Thus, the motion of the two planets—according to their construction the difference between them becomes the ku-bra-antara; the difference between the gnomon-sides (śaṅku-kuvra) is due to the equality of the roots of the gnomon etc. By rotation, the difference between their two gnomons comes about. The rest is clear.

**Commentary** — The teacher explains: Of the base, perpendicular, and hypotenuse, the half-base for transformation—those half-bases divided by the semi-diameter should be made into multipliers. Thus the motion of the two planets—their difference as constructed—the difference of the gnomon-sides arises from the equality of the gnomon-roots. By the revolution, the difference of their gnomons occurs. The remainder is self-evident.

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**Proof** — Now the upapatti: From the sphere arise the bhuja, koṭi, and karṇa; from the position of the gnomon-tortoise (śaṅku-kūrma), the true bhuja—in this manner, with its application, the proof demonstrated by the Ācārya is established.

**Commentary** — When by two observation-nāḍīs they are made unequal, then there is a yoga (conjunction). In the yoga (mahā-yoga), when there is a difference between the two planets (*grahayor-madhyāntare*, i.e., at the divided kendra), like the moon-sixth-day-increase (*ṣaṭ-jñonitivat*), of the same directions (*samyad-śoḥ*, i.e., one-and-different-directions), the difference of the east-west gnomon-flanks (*prācya-śaṅku-vaprayoḥ*)—the saṃyogāntara (union-difference)—the bāhu (base)—the meaning is: from the middle of the one-to-be-increased, the yogāntara (difference of union) of each gnomon-base being applied, the bhuja of them is produced.

**Commentary (contd.)** — In the sum of the two planets, the karṇa is produced. Their own karṇa—at the root of the difference of the squares of the bhuja and karṇa—the koṭi is produced. Thus, by the koṭi-bhuja-karṇa of the two planets, by the gnomon-construction, whatever half-bhuja is to be transformed (*yad-bhujyārdha-pariṇāta*), having made them divided by the semi-diameter (*vyāsārdha-hṛtān*), they become transformed.

**Commentary (contd.)** — Of the east-west (transformed-half-bhuja-east-west-line-threads), in the east (eastern direction), in the west (western direction), extending the koṭi separately; the bāhu (base) extended by the koṭi; dividing, from the middle of the directions (from the divided middle), the bhuja-extent (bhuja-prā-viddha-lekha); dividing the karṇa too, of the bāhu-extents (their own bhuja-extents), their own gnomons—two; from the middle to their astra-extent (at the drawing of their own gnomon-sides), as before, two previously assumed.

**Commentary (contd.)** — The dvikṣepa-madhya-stha-cchāyā (shadow situated at the middle of the double-latitude, i.e., by the eye situated at the middle point) shows separately its own gnomon-side and the planet. In the yoga (union of the two planets), their difference of gnomon-kuvra occurs because of the equality of the gnomon, bhuja, etc.; otherwise (if the planetary union is not at the same point), their difference of gnomons alone occurs. In the revolution (in the yoga, only the difference of gnomon-kuvra-remaining, otherwise of them), in the great yoga, the difference of gnomon-kuvra should not be taken as the difference between the planets; this should be known in the planetary-union-minimum-kalpa.



## — Examination of Astronomical Systems

**Editorial Note** — In this chapter Brahmagupta subjects the astronomical system of Āryabhaṭa to rigorous examination. The refutation of Āryabhaṭa I's (476 CE) planetary theory is here demonstrated in detail.

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**Text (v. 3)** — Now he refutes the yuga stated by Āryabhaṭa:

*āryabhaṭo yugapādāmśāne yātānāha kaliyugādaṁ yat*

Āryabhaṭa stated that two quarter-yugas had elapsed at the beginning of Kali-yuga. From what cause? Therefore that is not true; hence his own yuga-aṣṭa (eighth of a yuga) is not valid.

**Commentary** — Āryabhaṭa stated (*āha*) that at the beginning of Kali-yuga, two yuga-pādas (quarter-yugas) had elapsed (*yātān*). From what cause?—from that reason, therefore his own yuga-aṣṭa—of one yuga, the beginning established, thus two in the Kṛta-yuga—from that, that is not true (*na sat*).

**Proof** — According to Āryabhaṭa's system, from the commencement of one yuga to the beginning of Kali-yuga, three yuga-pādas =  $\frac{3 \times 4320000}{4} = 3240000$ . And according to the Ācārya's (Brahmagupta's) system, Kṛta + Tretā + etc.  $\frac{4320000 \times 9}{10} = 3888000$ . The difference between the two = 648000 years. These, by the excess of number in the Ācārya's system, even in the middle of Kṛta-yuga, Āryabhaṭa's eighth-yuga reaches the end of Kṛta-yuga.

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**Text (v. 11)** — Now again he refutes Āryabhaṭa's stated weekday:

*śrīkṣuro dinavāro gurur ādviko 'sya bhavati kalpādaṁ*

The first weekday accepted by him at the beginning of the kalpa is Thursday (guru). *na bhavaty akam yasmād occuro vistaratas tasmāt*—Since this is not established (*na bhavati*), because his extension (*vistara*) is baseless, therefore (it is invalid).

**Commentary** — By Āryabhaṭa in his own treatise, “from Thursday before the Bhārata”—by this, Thursday is accepted at the beginning of the kalpa. By this the meaning is: From what cause?—His (Āryabhaṭa's) acceptance (*śrīkṣuraḥ* = *svikāraḥ*), the first weekday at the beginning of the kalpa being Thursday becomes established; therefore his acceptance is *vistara* (unsupported and unauthoritative).

**Proof** — According to Āryabhaṭa's system, before the commencement of Kali-yuga, in the present kalpa, six manvantaras and three quarters of a yuga-catur (yuga-quadruple) had elapsed. Thus from that, from one manu of seventy-two yugas, the elapsed yugas at the beginning of the kalpa =  $72 \times 6 + \frac{3}{4} = 432\frac{3}{4}$ . Multiplying by the saura-civil days of the yuga-pañcita: =  $432 \times 1577917500 + \frac{3 \times 1577917500}{4}$ . The weekday at the beginning of the kalpa by this: the remainder when divided by seven =  $34 = 6$  (Friday); but if Thursday is taken as the starting weekday, then the elapsed weekday at Kali-yuga-beginning = 0. The present Thursday itself would be established. Thus by Āryabhaṭa's system, Thursday at the kalpa-beginning is arrived at.

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**Text (v. 13)** — Now again he refutes Āryabhaṭa's stated weekday:

*kṣadhikaiḥ śataiḥ caturbhir varṣaiḥ sahasraiḥ caturbhir yāmirekaḥ*

With four hundred years increased, and with four thousand, one yāmi. Between the day and weekday, in the aṁśa of the sun and the two nights...

**Commentary** — By Āryabhaṭa, two systems were constructed. In one, the civil days in a yuga = 1577917500; at the kalpa-day of short motion, creation. In the other, the civil days in a yuga = 1577917800; at the short-motion night, creation. In both, the number of yuga-years 4320000 is the same. The three days of yuga-pañcita-sāvana = 300. Therefore from the two

systems, by weekday computation, there is a difference of three hundred yuga-year-days; and proportionally, the difference of one day-weekday =  $\frac{4320000}{300} = 14400$  yuga-years.

**Commentary (contd.)** — By this the meaning is: According to Āryabhaṭa's system, between the day and weekday in the aṃśa of the sun and nights, with fourteen years, thousands, seasons, increased by hundreds of years, the difference between day and weekday falls by one. Thus the weekday computation is not stable.

## — Hypotenuse and Diagonal

**Proof** — In the proof of the 13th rule of the Sphuṭādikāra it is shown that in the first term:

$$krama-jyā \times \text{circumference} / bhojyā = bhuja-phala$$

In the second term, the maximum bhuja-phala occurs; in the third term:

$krama-jyā \times \text{circumference} / bhojyā = bhuja-phala$ . In the second term, the maximum *bhuja-phala* becomes zero.

In the fourth term, the maximum  $utkrama-jyā \times \text{circumference} / bhojyā = bhuja-phala$ . Although at the end of even terms and at the beginning of odd terms, due to the conjunction of both terms, the *paridhi* (circumference) mentioned in even and odd terms, when taken, does not produce *phala-nāma* (result); there is no harm from it because, by the difference of circumference readings, by the proportion in the difference of terms, the taking of circumference should be done—thus Āryabhaṭa has indicated. Therefore the Ācārya's refutation is not correct.

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**Text (v. 20)** — Now, if by proportion according to Āryabhaṭa's system the circumference is made precise, even then it is not correct—thus he refutes:

$$vyāsārdha-hṛto bāhuḥ paridhi-vibheṣāhṛtaḥ phalonayutaḥ$$

The *bāhu* (side) divided by the semi-diameter, divided by the difference of circumferences, with the result added;

*prathamād vikonakaś ca yat tad asat padayoḥ paridhi-pātāt* — and the *vikonaka* (angular deviation) from the first, that is invalid because of the *paridhi-pāta* (falling of circumference) in the terms.

**Commentary** — The *bāhu* (side) by the *bhuja*, divided by the difference of circumferences, diminished by the end-circumference, divided by the semi-diameter by *vijaya*—the first circumference, and the *vikonaka* from the second: then in order, the first circumference with the result added becomes the true circumference—this well-known method of computing the true circumference also becomes invalid by Āryabhaṭa's system.

Why? Because of the reading of circumference in the terms. In meaning, in that system, there is the taking of circumference in the terms. There, sometimes the first is of even-term type, sometimes of odd-term type. Therefore it varies in correction procedures. This is invalid.

And the *vikonaka* from the first—because of the correctness of the instantaneous correction—this is to be considered by the wise.

## Celestial Latitude

**Text (v. 135)** — Of the Moon (Indu), the *utkrama-jyā* (reversed sine, i.e., the *bhuja-utkrama-jyā*), multiplied by the *viśleṣa-apakrama* (by the arrow of separation, by the *apakrama*, and by the maximum brightness-sine), divided by the square of the *vijyā* (*viśama-jyā*), the result is the *dvikṣepa* (double-latitude) in *lamba-māna*. That was stated by Āryabhaṭa as proper (*yuktam*); therefore (*tataḥ*) it is invalid (*asat*). Since (*yataḥ*) at the end of the *kṣaya* (at the *gola-saṅkhyā*), the result produced by that calculation—in his system, the *śrāddhāyana-saṅkhyā* (number of *śrāddhāyanas*) is produced from it—therefore that is invalid.

**Commentary** — Of the Moon (*indoḥ* = *candra-sya*), the *utkrama-jyā* (the *bhuja-utkrama-jyā*), multiplied by the *viśleṣa-apakrama* (by the arrow of separation, by the *apakrama*, and by the maximum brightness-sine), divided by the square of the *vijyā*—the *dvikṣepa* in *lamba-māna* is obtained. Āryabhaṭa stated this as proper; but from that cause it is invalid. Since at the end of the *kṣaya* (at the *gola-saṅkhyā*), the result produced by that calculation—in his view, the *śrāddhāyana-saṅkhyā* is produced from it—therefore that is invalid.

**Proof** — Here the Caturveda-ācārya: “Indeed this is twofold: this Āryabhaṭīya statement is *vrātya* (heterodox):”

*viśleṣa-pakrama-guṇām utkramaṃ vistara-dhaṃ kṣati-bhaktam*

Here, in the *Maṭadīpikā*, Parameśvara: “If the *sāyana-candra-utkrama* be by the *koṭi*, by the *utkrama-jyā*—if thus, then this of Āryabhaṭa is not correct.”

This operation—the *krama-jyā*, the two *kṣepas*, the *utkrama-jyā* according to Āryabhaṭa’s system—whatever is done there is real; thus Bhāskara’s refutation, actually based on spherical demonstration, is true. And the plane-figures and field-figures are clear.

## Cosmic Cycles (Yugas)

**Hindi Commentary (v. 55)** — For the reason by which, in the computation of planets and the positions of pāta, mandocca, and śīghrocca at the beginning of Meṣa (Aries), in whatever time they again come to the position in Meṣa—that time is called Mahā-yuga. The yuga stated by Śrīpathyavidyā-cūdra and other ācāryas is correct, because according to their view, at the beginning of Mahā-yuga, the planets are produced at the beginning of Meṣa (in a different place). That is, according to their view, if planetary computation is done in the Mahā-yuga approved by the smṛti-texts composed by smṛti-kāras like Manu, then the position of the planets in Meṣa etc. is not established. In the Romaka composed by Śrīpaittha, nowhere is there discussion of Mahā-yuga-beginning. Therefore, by the word Mahā-yuga, here the Mahā-yuga stated by the Ācārya should be taken. When all the planets and the positions of pāta, mandocca, and śīghrocca are at the beginning of Meṣa—that alone is named Mahā-yuga-ādi. In the Mahā-yuga-ādi stated by Śrīpaittha, this position does not occur; therefore according to their view, the Mahā-yuga-ādi is not correct. The Ācārya's refutation here is correct.

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**Text (v. 56)** — Now again, the very same is refuted at night:

*kadinādaṁ smṛti-buktaṁ grahamokṣapatir dina-kṣaye pralayaḥ*

At the beginning of the *kadina* (many days), as stated in the Smṛti, the liberation of planets, at the destruction of the day, the dissolution.

*tātyatibahūni yasmāt mahāyugeṣṭo 'prasiddhim avam* — Since there are very many of those, the *eṣṭa* (desired) of Mahā-yuga is of unestablished, low (nature).

**Commentary** — At the beginning of *kadina* (many days), the creation of planets and stars; at the destruction of the day, at the end of many days, the dissolution of the planetary measures—thus it is in the Smṛti-texts. Thus in the Mahā-yuga of many days, since there are sixty-times-hundred yugas, therefore this stated by him is unestablished, not true from any source. What is the use of the gravity-ritual by the taking of many yugas? This is invalid. For even by taking many yugas, the planetary computations are not in this, not at any time obtained from the position—thus the computation is clear to those who know.

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**Expanded Commentary** — At the beginning of *kadina* (*bahu-dinādaṁ*), the *mokṣa-patiḥ* (creation of planets and stars); at the *dina-kṣaye* (end of many days), the *pralayaḥ* (destruction) of the planets and stars—this is stated in the *smṛti* texts. In the Siddhāntaśekhara, Śrīpati has stated: “The creation and dissolution of the luminous planets occurs at the beginning and end of those days.”

In the Siddhānta-śiromaṇi, Bhāskarācārya: “Since their creation is at day-beginning, and dissolution at day-end, the intelligent should contemplate this.” By this, the creation of planets at day-beginning is stated. But the Sūryasiddhānta-kāra states that “from the beginning of many days, from the Śāta-yuga, in the seven Vedas, Brahmā having created, was appointed at the end.” For the refutation of the time of happiness etc. stated by Brahmagupta-Śrīpati-Bhāskarācārya, and for the refutation of the Sūryasiddhānta-kāra's view, in the Siddhānta-tattva-viveka, there is much elaboration by Kamalākara—there is no substance without name.

## Chapter on Mathematics

**Editorial Note** — In this chapter Brahmagupta's algebra (*bīja-gaṇita*), arithmetic (*pāṭi-gaṇita*), geometry, and series summation are presented. This *Dvyāśīti-koṇy-adhyāya* (12th chapter) is of supreme importance in the history of world mathematics.

### Algebra: Multiplication and Powers — Verse 1

**Text (v. 1)** — *dānaṃ ca dattam ardhaṃ pañca tena māna-syārdheṣa-pañcanavamāṃśaṃ tadā kim*

A donation was given: half, five; of that measure, half less one-fifth and one-ninth—then what?

**Commentary** — *Nyāsa* (statement):  $\frac{1}{2}$ . The answer is stated as having become  $1\frac{5}{8}\frac{3}{4}$ .

Here due to confusion, the actual written answer is  $\frac{3}{4}\frac{5}{8}$ . (See his translation p. 283.)

Here the Caturveda-ācārya: "This is to be acknowledged."

"Thus here the Ācārya, one of the *pañca-jātya*, spoke."

"Because he reads, therefore, having made the meaning gone, he did not speak."

"By Skandasena and others, his name was made *Bhāga-māta*."

### Geometric Progression — Verse 5

**Text (v. 5)** — The sum of fractions with equal denominators (*sahasra-cchedāṃśa-yuktiḥ = tulya-harāṃśānāṃ yogaḥ*), divided by the denominator (*cheda-vibhaktā*), becomes the result of the first class for all quantities.

Those multiplied by the second-class measures; denominator (*hara*) multiplied by denominator—the fraction thereof, with the denominator, becomes the result for all quantities.

For the quantities of the first class, "*yogontara tulya-harāṃśakānām*" etc.—as stated by Bhāskara; and for the second *prabhāga-jāti* combination, as in the *Līlāvati*: "*lavāla-vadhana-abhyaṃ harā haranāḥ*" etc.—thus stated by Bhāskara.

### Division and Geometric Problems — Verse 6

**Text (v. 6)** — Now he states the *karana-sūtra* for division (*bhāgāhāra*):

*parivartyaṃ bhāgahara-cchedāṃśāṃ cheda-saṃguṇāc chedaḥ*

Having interchanged, the denominator-fraction of the divisor (*bhāgahara*) multiplied by the denominator;

*śeṣo 'mśaguṇo bhājasya bhāgāhāraḥ sarveṣāṃ tayoḥ* — The remainder multiplied by the numerator of the dividend; the division of both of all.

**Commentary** — The denominator-fraction of the divisor (*bhāgahara-chedāṃśī*)—the denominator-fraction of the divisor is interchanged with the denominator of the dividend; the interchanged-denominator becomes the divisor-denominator.

Thus the numerator of the dividend becomes the numerator-multiplied-by-interchanged-numerator.

Thus for both, for mixed sums, the division is produced.

For all, by the rule "*rūpāricchedaguṇāni*" etc.

Bhāskara's division method is the same.

The example of Caturveda:

**Example** — Where the area is six *haṣṭa* (square cubits), with one and a half, with one *yama* and a half. Ten sides and a half, and the *koṭi*—let it be of that measure.

This is a geometric problem: given area  $324 \frac{3}{8}$ , side  $90 \frac{1}{2}$ , find the *koṭi*.

**Commentary** — *Nyāsa*: Area =  $324 \frac{3}{8}$ ; *bhuja* =  $90 \frac{1}{2}$ ; the *utkrama-jāta koṭi* =  $11 \frac{3}{5}$ .

Here again the Caturveda-ācārya.

“Others here also give an example of proportion-calculation, as:”

## Series Summation — Verse 9

**Text (v. 9)** — Alternatively, calculation by the *laghutama-apavartana* (smallest reduction):

Statement:  $\frac{1}{2} + \frac{1}{3} + \frac{3}{4} + \frac{1}{5}$ .

The denominators 2, 6, 12, 4: by reduction with 2, 3:

From this reduction, produced:  $2 \times 2 \times 3 = 12$ .

This multiplied: 12, or by the sum of denominator-numerator,

$$\begin{aligned} \frac{1}{2} \times \frac{6}{1} + \frac{1}{3} \times \frac{12}{2} + \frac{3}{4} \times \frac{12}{3} + \frac{1}{5} \times \frac{12}{4} \\ = \frac{6}{2} + \frac{12}{6} + \frac{12}{12} + \frac{12}{20} = 9 = \text{sum result.} \end{aligned}$$

## Binomial Expansion — Geometric Demonstration

**Editorial Note** — At page 734 of this section, the proof of the rule “*sthāpyo-ntyaghana*” is demonstrated by geometric diagrams. This geometric demonstration of the expansion of  $(a + b)^3$  was accomplished in India centuries before similar developments in Western mathematics.

**Geometric Demonstration** — Side of the square =  $a + b$ .

Breadth =  $a + b$ .

Area of the square =  $(a + b) \times (a + b) = a^2 + 2ab + b^2$ .

The area of the field multiplied by the length etc.—the *ghanatamaka* (solid-cube) field is related; the examples shown are not correct, and are not to be examined by the wise.

**Proof** — For making the cube of  $(a + k)$ , by the definition of cube as equal triple-product:

$$(a + k)(a + k)(a + k) = (a + k)^3$$

By multiplication:  $(a^2 + a \cdot k + k^2)(a + k) = (a^2 + 2a \cdot k + k^2)(a + k)$

$$= a^3 + 2a^2 \cdot k + a \cdot k^2 + a^2 \cdot k + 2a \cdot k^2 + k^3$$

$$= a^3 + 3a^2 \cdot k + 3a \cdot k^2 + k^3 = (a + k)^3$$

This agrees with the Ācārya's “*sthāpyo-ntyaghana*” (place-the-last-cube).

“*sthāpyo-ntyavargaṃ dviguṇā-ntyaniḥnaṃ*” etc.—as stated by Bhāskara in the *Līlāvātī*, this follows the same method.

By the ancients, only the investigation of the sum of squares and cubes was done.

For making the cube of the sum of two-plus-two, the first is called *ākṣepitva*; the second two is called *uttara*; then the last-ghana is placed; from that the last-square (*antya-kṛti*) is doubled, called *uttara*; the cube of the sum of two-plus-two becomes.

Here the Caturveda-ācārya gave this example: “A square pond, equal to the extent of hastas. Its depth also; the result is the *ghanatamaka* (solid cube).”

**Binomial Table** — As the best *rūpa* (form): this itself is the twofold *pada-siddhānta*:

$$\frac{\frac{a \times n \times k}{1}}{n=1} \quad \left| \quad \frac{\frac{a \times n(n-1)k}{1 \times 2}}{n=2} \quad \left| \quad \frac{\frac{a \times n(n-1)(n-2)k}{1 \times 2 \times 3}}{2} \quad \left| \quad + \dots \right. \right.$$

Here if  $n = 2$ :  $(a + k)^2 = a^2 + a \times 2 \times k + k^2$

If  $n = 3$ :  $(a + k)^3 = a^3 + a^2 \times 3 \times k + a \times 3 \times k^2 + k^3$

**Proof** — By this much, “*sthāpyo-ntyavargam dviguṇā-ntyanighnam*” etc.—the proof of the square, and “*sthāpyo ntyaghana*” etc.—the proof of the cube become clear.

## Square Root Extraction — Verse 10

**Text (v. 10)** — One should divide the denominator; having obtained (the result) of the two, one should divide them separately, separately.

From the square, the root from the divided term—that becomes the square-root.

**Commentary** — For making the cube of the sum of two-plus-two, the first is called *ākṣepitva*; the second two is called *uttara*; then the last-ghana is placed; from that the last-square (*antya-kṛti*) is doubled, called *uttara*; the cube of the sum of two-plus-two becomes.

Here the Caturveda-ācārya gave this example: “A square pond, equal to the extent of hastas. Its depth also; the result is the *ghanatamaka* (solid cube).” *Ghanatamakam*.



## Colophon

### Colophon —

Thus, in the *Śrī-Brāhmasphuṭasiddhānta*, the *Dvyāśīti-koṇy-adhyāya* (12th chapter), named the eleventh chapter. || 11 ||

Thus, in the *Śrī-Brāhmasphuṭasiddhānta*, the *Tantraparīkṣādhyāya* (chapter on examination of systems), named the eleventh chapter. || 11 ||

Thus, in the *Śrī-Brāhmasphuṭasiddhānta*, the *Duṣpradādhyāya* (chapter on difficult gifts), named the eleventh chapter. || 11 ||

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## The Third Class of Partnerships

### Circumference and Proportion

(BSS 12.37 Vol. II, p. 891)

This verse states that the circumference is obtained by multiplying the diameter successively by the prescribed factors, as taught by previous teachers including Śrīpati. Brahmagupta, Bhāskarācārya, and Līlāvātī have all taught this in their respective works.

**171.** The true circumference is that obtained by the rules taught by the two Dvivedis; the circumference otherwise (from crude approximation) is erroneous.

### Mathematical Proof

#### End of Arithmetic Chapter

Here ends the *Rāśi Vyavahāra* (Arithmetic Operations). Now, under division, the quantities of grains corresponding to the inclined angle and base angle are stated for the derivation of measures.

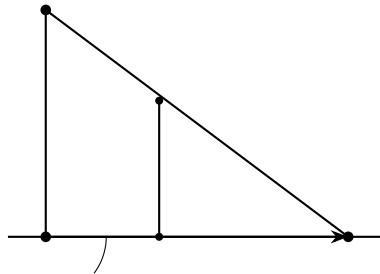
### Shadow Calculations

*Sūtra-bhāṣya*: The *nara* is the *śaṅku* multiplied by eight — the *nara* of the shadow divided by the *śaṅku* is that in the *saṅkrama*; this is the *śuddha*; divided by the eastern or western semi-diurnal arc, it becomes the corrected value.

The *Chāyā Vyavahāra* (shadow computations) section continues with detailed derivations. The *nara* is the sine of latitude, and the *śaṅku* is the gnomon (12 aṅgulas). The shadow is computed from the relationship between these quantities.

### Geometric Construction of Shadows

The accompanying diagrams illustrate the geometric relationships: the gnomon (vertical pole) casts a shadow on the ground, forming an angle with the sun's ray. The *karna* (hypotenuse) equals the distance from the lamp's height. Similar triangles yield the proportions.



*Shadow* (Chāyā), *śāna* = base. *Karna* = lamp-height. From similarity of triangles, the *karna*-base proportion gives: base = *chāyā-prāntara*; then by proportional reasoning,  $12 \times \text{base/shadow} = \text{lamp-height}$ . Similarly for the other triangle pair, the same result follows. Thus the teacher's statement is proved. (BSS 12.41–42 commentary)

## Algebraic Operations —

(BSS 12.46)

Where a quantity has both integer (*kala*) and fractional (*vikala*) parts, the integer part is called *sakala* and the fractional part *vikala*; these are then processed by the rule to obtain the square root.

### Upapatti (Mathematical Proof)

*Upapatti*: Let the integer part be  $k$  and the *vikala* part  $vi$ . Then the quantity is  $k + \frac{vi}{60}$ . The square is:

$$= k^2 + \frac{2k \cdot vi}{60} + \frac{vi^2}{60^2}$$

*V̄tti-bhāya*: When a quantity has both *kala* and *vikala* parts, to find its square, the integer part is called *sakala* and the fractional part *vikala*; then the rule states the method. Note: "" means 'square' in Sanskrit. (BSS 12.46 commentary)

(BSS 12.47)

This verse gives the method for division with fractions. The dividend, combined with the divisor (in the manner of adding fractions), is divided by half the divisor multiplied by the denominator of the divisor; the result, added to the *prakṛti*, gives the quotient. (BSS 12.47 commentary)

(BSS 12.49)

The section on division with fractions (*bhāgahāra*) is explained. The rule involves the manipulation of fractional quantities through their denominators. (BSS 12.49 commentary)

## Astronomical Problems —

The following sections deal with astronomical computations, including the determination of time, planetary positions, and the computation of various astronomical quantities.

### Mixed Astronomical Problems

(BSS 13.6)

He who knows the *vyatīpāta*, *vaidhṛti*, *bṛhaspati-varṣa*, and the motion of the nodes in a *yuga* — he is the knower of the essence of time. (BSS 13.6 commentary)

(BSS 13.10)

He who knows the sum of the *avama* (omitted tithis) and the *praghara* (excess days), combined with the mean (longitudes) at the desired time — he is the knower of the *grahyaka* (the quantity to be known). (BSS 13.10 commentary)

### Third Set of Problems

(BSS 13.40)

From the product of the elapsed days of the *kalpa* and seven, divided by seven — the remainder gives the day of the week (counting from Sunday).

*Upapatti*: From the *kalpa* elapsed days multiplied by seven, the remainder from division by seven gives the day of the week starting from Sunday. (BSS 13.40 commentary)

### Lunar Calculations

## Planetary Theory — &

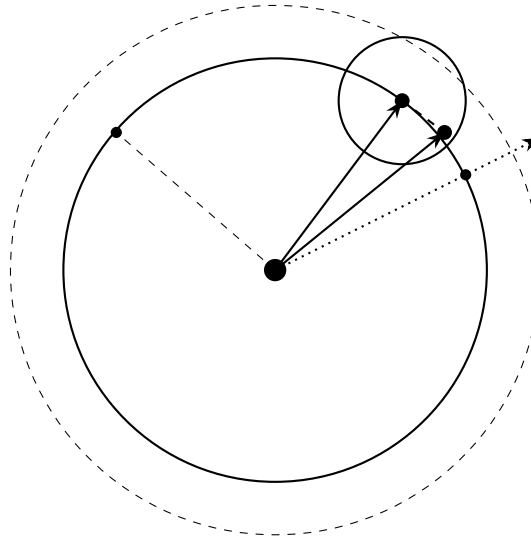
### Epicyclic Model

The computation of planetary positions involves the *mandasphuṭa* (equation of centre) and *śighraphala* (equation of anomaly, or second inequality). Brahmagupta gives the following method:

*Vṛtti-bhāṣya*: The *phala* obtained by multiplying by the *bhuja* (sine of anomaly) and dividing by the *karna* — its arc gives the difference between the first and second true positions; that is the true position. (BSS 378-581 commentary)

### Proof of Epicyclic Correction

*Upapatti*: *Ja* = geocentric planet; *Ma* = mean planet via manda; *Ma.3* = *śighrakendra*; *Mar* = *śighrakendra-jyā* = geocentric sine; *Sp.* = true planet; *Sp.3* = true *kendra*. (BSS *upapatti*)



*Bhum* = geocentric planet. *Spaṣṭa* = true *kendra-jyā*. *Ma.3*. *Ma.* = mean *kendra*. *Mū.spaṣṭa* = *śighraphala*. From similarity of triangles: geocentric sine/*śighrakarna* = true *kendra-jyā*. Alternatively, true *kendra* = difference between true planet and apogee. The manda-true *kendra-jyā* meets the epicycle; by arc-correspondence, manda-true *kendra-rāśi* = difference between manda-true and equation. (BSS *upapatti*)

The *mandasphuṭa* (true planet via equation of centre) is computed as:

1. Compute the *mandakendra* (mean anomaly): difference between mean longitude and apogee
2. Find the *mandaphala* (equation of centre) from the sine table
3. Apply to the mean longitude to get *mandasphuṭa*
4. Compute *śighrakendra* from *mandasphuṭa*
5. Find *śighraphala* and apply to get the true longitude

## Advanced Astronomical Computations — &

(BSS 21.25-26)

The *iṣṭajyā* (desired sine) is multiplied by the results of the five operations and divided by the radius; the remainder, when 90 is subtracted from the *nava* position, gives the *śeṣa*; the *manoratha* (desired arc) is obtained. The arc is thus found from the sine without the *vyārdha* (half-chord).

*Sūtra-bhāṣya*: The desired sine is divided by the semi-diameter, together with the two-foot arc-difference; the remainder is clear. (BSS 21.25-26 commentary)

### Upapatti — Detailed Proof

*Upapatti*: From previous proofs:  $jya = \frac{(180-ca) \cdot ca \cdot jra}{10125 - \frac{(180-ca)ca}{4}}$ . Let  $ya = (180 - ca) \cdot ca$ . Then  $jya = \frac{jya \cdot ya}{10125 - ya/4}$ . (BSS 21.25-26 *upapatti*)

(BSS 21.50-52)

These verses compute the *bhogakalā* (daily motion in minutes of arc). For the Sun, Moon, and planets, the daily motion is found by taking the difference in longitude between successive days and converting to *kalās*.

*Vṛti-bhāṣya*: The mean daily motion in *kalās* of the Sun (or Moon), multiplied by the respective factor, gives the *bhogakalās* of the *nakṣatras*. (BSS 21.50-52 commentary)

The *bhogakalā* is the daily motion of a planet in terms of *kalās* (arc-minutes). The computation involves:

- Finding the difference in longitude between successive days
- Converting this difference into minutes (*kalās*)
- For the Sun: about 1° per day (approximately 60 *kalās*)
- For the Moon: about 13° per day

## Final Chapters — Eclipses and Colophons

The concluding portion of the *Brāhmasphuṭasiddhānta* contains extensive rules for eclipse calculations, including the computation of the *sparsā* (first contact) and *mokṣa* (last contact), the magnitude of the eclipse, and the *akṣṣavalana* (deflection due to latitude).

## Colophons —

*Here ends the Brāhmasphuṭasiddhānta*

Thus concludes the *Brāhmasphuṭasiddhānta*. This treatise contains 24 chapters in all, comprising 12 chapters of mathematics (*Gaṇitādhyāya*) and 12 chapters of astronomy (*Golādhyāya*). The teacher Brahmagupta composed this work in Śaṃvat 628 (CE 666).

## Appendix: Summary of Mathematical Rules

**Shadow rule :** The shadow is proportional to the *nara* (sine of latitude) and inversely proportional to the *śaṅku* (gnomon height).

**Square root of non-squares :** The method of successive approximation:  $\sqrt{N} = a + \frac{r}{2a+1}$  where  $a$  is the largest integer such that  $a^2 \leq N$  and  $r = N - a^2$ .

**Equation of centre :** The correction for the eccentricity of planetary orbits, computed using the *vyā* (sine) of the *kendra* (anomaly).

**Second inequality :** The correction for the planet's position relative to the Sun-Earth line, computed from the *śīghrakendra* (anomaly of conjunction).

**Kuṭṭaka :** The pulverizer method for solving linear Diophantine equations  $ax + c = by$ , essential for calendar calculations.

**Bhogakalā :** The daily motion of a planet in arc-minutes, computed from the difference in longitude between successive days.

**Arc from sine :** The inverse problem of finding the arc from its sine, solved by Brahmagupta's algebraic formula without using the *jyārdha* (half-chord) table.

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**[Introductory Note.]** Now [the Ācārya] speaks of the computation of the motion (bhoga) of the starry nakṣatra and the computation of the nakṣatra as freed from the planets (graha-mukta-nakṣatrānayaṇa).

**[Mathematical Rule.]** The fifteenth [operation]: The sum of the degrees of conjunction-motion (yoga-bhoga-kalā) =  $\text{Caṃ} \times 15$ . The sum of all =  $1 \text{ Caṃ} + 3 \text{ Caṃ} + 15 \text{ Caṃ} = 27 \text{ Caṃ}$  = the total motion of all the nakṣatras (sarvāna-kṣatra-bhogaḥ). Here  $\text{Caṃ}$  (cakra) denotes a complete revolution.

**[Dvivedī's Commentary.]** The pure revolutions (cakkala) are the numbers [representing] the total motion of all the nakṣatras; then the degrees of motion of the starry nakṣatra (śrambijñud bhoga-kalā) [are obtained]; its daily motion = revolution —  $27 \text{ Caṃ}$ . From that, the kalpa-born revolutions [are obtained] as: (revolution —  $27 \text{ Caṃ}$ ) civil days (kudina) = cakkala.

In 27 civil days, the kalpa revolutions [are obtained] by this Ācārya; [this is] well established (tumūpapaṇṇam). In the *Siddhāntaśekhara*: “cakkāriyā vā śāsvata-sativadhana-āhatāni śodhayin sudina-laya-dvaroṣa-cakra”. [The meaning is that] the daily motion of the kalpa (kalikā-gatiḥ) which is the Vaivasvata [value] at mid-[period] is to be known (mr̥tiḥ). The degrees of nakṣatra motion (nakṣatra-bhoga-kalikā), the times of the prahara, etc., the remaining degrees to be traversed (bhogya-kalikā) — from the excess, the gatagamya [values] become the days traversed and remaining to be traversed (gata-gamya-dināni). Thus [it is said] by Śrīpati in the fifteenth [chapter].

“The sum of the motions of all [nakṣatras] multiplied by the revolutions [gives] the Vaivasvata [value]; this is the yoga of the starry [nakṣatra],” and so on — thus spoke Bhāskara; and all teachers [hold] this same meaning.

**[Verses 50-51]**

### | Upapatti (Proof/Demonstration)

**[Mathematical Derivation.]** The sum of the six doubled (ṣaṭ-śraddha) motion-degrees of the nakṣatras =  $\frac{3 \text{ Caṃ}}{2} \times 6 = 3 \text{ Caṃ} \times 3 = 6 \text{ Caṃ}$ . The sum of the six [times] single-motion degrees of the nakṣatras =  $\frac{\text{Caṃ}}{2} \times 6 = \text{Caṃ} \times 3$ . The sum of the fifteen [times] single motion (sama-bhoga) degrees of the nakṣatras =  $\text{Caṃ} \times 15$ . The sum of all = total nakṣatra motion degrees =  $6 \text{ Caṃ} + 3 \text{ Caṃ} + [15] \text{ Caṃ} = 27 \text{ Caṃ}$ .

**[Hindi Bhāṣya — English Summary.]** When the mean daily motion of the Moon (candra-madhyagati-kalā) is multiplied by [certain] quantities, the motion degrees of the six, the śarva, and the equal nakṣatras are obtained. The remaining [degrees], when subtracted from the revolution degrees (bhagaṇa-kalā), give the motion of the starry nakṣatra (śrambijñuta-bhoga).

Alternatively: The kalpa revolutions of the starry nakṣatra are obtained by multiplying the kalpa revolutions of the Moon by eighty-seven and subtracting from the kalpa civil days. From these revolutions, the daily motion of the starry nakṣatra is obtained by proportion (anupāta): if in kalpa civil days such-and-such kalpa revolutions [are obtained], then in one day [what]? — thus the daily motion in time-units.

When the nakṣatra motion degrees are subtracted from the planet-time aggregate (grahakāla-samūha), the nakṣatras are obtained. That is, in the planet-freed degrees (graha-



mukta-kalā), as many pure motion degrees of the nakṣatras as there are, so many are the traversed nakṣatras (gata-nakṣatra). The remaining degrees are the current degrees of the present nakṣatra; subtracting these from the nakṣatra motion degrees yields the remaining degrees to be traversed (gamya-kalā). Then, if in one day [a certain value] is obtained in planetary motion degrees, by proportion from the gata-gamya degrees, the days traversed and days remaining are obtained.

**[Verses 50-52]**

| **Upapatti (Proof).** Here the Ācārya has postulated (kalpita) the previously stated time-quantity (kāla-jñā) as the desired [value] for the second [process]. Then śrada, sama-śāim, kudūdit — these three quantities (muñjā) produce one śramba field (kṣetra); and śrada-jñā, labdha-jñā, tijñā — these three quantities produce a second śramba field. These two śramba fields are homogeneous (sajātiya); therefore, proportion (anupāta) is applied [between them].

$$\frac{\text{labdhajñā} \times \text{iṣṭa}}{\text{tijñā}} = \text{samaśāim-kud}$$

And the time-quantity (kāla-jñā) = [the result of this proportion]. The three quantities (trayaṃ muñjānām) generate the proportional relationship between the two śramba fields.

| **Atrôpapattiḥ (Demonstration thereof)**

| **[Śloka on Shadow Computation.]** In the shadow-hypotenuse circle (châyā-karṇe gole), how the equinoctial shadow (palabhā) and the gnomon (śaṅku) become equal in nature (tulya-tattvaṃ) — this is demonstrated thus: if the palabhā is obtained as the base (bhuja) in the twelve-aṅgula sign (dvāda-śāṅgula-rāśi), then what is the gnomon-like [quantity]? The gnomon-equivalent (śaṅku-tulya) =  $\frac{\text{pama} \times \text{śaṅku}}{12}$ ; but śaṅku =  $\frac{12 \times \text{tri}}{\text{chāka}}$ . By elevation (utthāpanena), the gnomon-equivalent =  $\frac{\text{pama} \times 12 \times \text{tri}}{12 \times \text{chāka}}$ .

| **[Dvivedī's Explanation.]** Here, for computing the palabhā at a time different from mid-day (dvāda/dinārk), [the Ācārya] speaks.

| **[Hindi Bhāṣya — Summary.]** At a time different from midday, multiplying the Sun's shadow by the shadow-hypotenuse (châyā-karṇa) and dividing by the tijñā yields the circular shadow of the hypotenuse-circle. The difference between the root of the gnomon (śaṅku-mūla) and the east-west line is the base (bhuja). In the northern hemisphere (uttara-gola), subtracting the northern bhuja from the karṇa-dvitiya and adding the southern bhuja yields the palabhā. In the southern hemisphere (dakṣiṇa-gola), merely subtracting the second shadow from the bhuja yields the palabhā.

**[Verses 46-50]**

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**[Introduction.]** Now [the Ācārya] speaks of the questions of the śrambya [type].

**[Hindi Bhāṣya — Summary.]** Those who know the sama-śaku [from] the declination (krānti), who know the sama-śutta [from] the latitude (lamba-āṃśa), who know the koṇa-śaku, who know the koṇa-śaku-chāyā, and who know the ghaṭī [from] the koṇa-dvitāja entrance — they are the scholars (paṇḍita) of jyotiṣa-siddhānta. Here there are five questions.

**[Verse 7]**

| *Idānīm anyān praśnān āha* (Now he states other questions.)

**[Sūtra-bhāṣya.]** The difference of the east-west [line] from the gnomon-base (śaṅku-tala-prācyā-parāntara) is the base (bhuja). He who, having understood the two bhuja-values, knows the equinoctial shadow (viśuvac-chāyā) [i.e.,] the palabhā, or who knows one bhuja [together with] the equinoctial shadow and the sun — he is a gaṇyaka. Thus here there are three questions.

**[Verse 8]**

**[Vivarāṇa-bhāṣya.]** The difference of the east-west line [measured] from the gnomon-base (śaṅku-tala-pūrvāpara-rekha) is declared to be the bhuja. He who, having known the two bhuja-values, knows the palabhā, or having known one bhuja, knows the palabhā and the Sun (ravi) — he is a gaṇyaka. Thus here there are three questions.

**[Verse 7]**

**[Dvivedī's Introduction.]** Now [the Ācārya] speaks of the śrambya questions.

**[Hindi Bhāṣya — Summary.]** The difference between the gnomon-base (śaṅku-tala) and the east-west line is the bhuja. The person who, having known the two bhuja-values, knows the palabhā, and likewise [he who], having known one bhuja, knows the palabhā and the Sun (ravi) — they are the paṇḍitas of jyotiḥ-śāstra. Here there are three questions.

**[Verse 7]**

| *Idānīm anyān praśnān āha* (Now he states other questions.)

**[Sūtra-bhāṣya.]** He who knows the pātāla-śaṅku of the Sun [and] the kumatha-śaṅku, or who knows the Sun having known the uddheśa, or [who knows] the iṣṭa-śaṅku [and] the kudivo-iṣṭa-śaṅku [as] the pātāla-gaṇaḥ and pātāla-śaṅku [respectively], the ravi-ratha-śaṅku — of these two, separately, each at its own base (prthak tale) or at the gnomon-base (śaṅku-tale), he who knows — he alone is the tantra-jña. Thus here there are four questions.

**[Verse 9]**

**[Dvivedī's Introduction.]** Now [the Ācārya] speaks of the śrambya questions.

**[Hindi Bhāṣya — Summary.]** The person who knows the Rāhu-path (bhūmā-mārga) in an eclipse, who knows that path, and who knows the koṇa-śaku-chāyā — here there are five questions.

**[Verse 7]**

**[Eclipse Triangle Computation.]**

*koṇaduta-stha-ravi-gataṃ bhuva-protkuṭaṃ bhuvādri-viparyaṃ bhuvyacāpāṃśaṃ ekaṃ bhujaḥ*

*bhuvālpasatikāvadhi lamba-aṃśa dvitīyo bhujaḥ | koṇadutaṃ va svainikādri-viparyaṃ nanāṃ-śāstu-tīyo bhujaḥ*

In this triangle: the Sun's passage at the koṇaduta, the earth's protuberance (protkuṭa), the earth-mountain reversal — the degrees of the earth-curve (bhuvyacāpāṃśa) — these form one bhuja. The latitude-degrees (lamba-aṃśa) up to the earth's alpaka-satika [form] the second bhuja. The koṇaduta and the svainika-mountain reversal [form] the third bhuja.

In this triangle, the Sun's passage being [known] as the earth's protuberance (bhuva-protkuṭa), the angle [is] the natakala (nata-kālah). The southern koṇaduta-nata-amṛtyanna-koṇa = 85; by the rule of proportion (anupātaḥ): if by the natakāla-jyā the hasta-jyā is obtained, then by the arc-jyā of 85 (jyā 85), what is obtained?

The bhuja-jyā =  $\frac{\text{hasta-jyā} \times \text{jyā } 85}{\text{natakāla-jyā}} = \frac{\text{hasta-jyā} \times \text{tri} \times \text{jyā } 85}{\text{natakāla-jyā} \times \text{tri}} = \frac{\text{hasta-jyā} \times \text{tri} \times 285}{2495 \times \text{natakāla-jyā}}.$

### | Upapattiḥ (Proof/Demonstration)

**[Dvivedi's Upapatti — Summary.]** Here, since the longitude difference (deśāntara) is not known, one should compute the conjunction-time (saṃmīlana-kāla) and disjunction-time (unmīlana-kāla) of the total eclipse by the method of lunar eclipse, from the true Sun (sphuṭa-ravi) and true Moon (sphuṭa-candra), devoid of deśāntara correction. On that day, the second saṃmīlana-kāla should [also] be computed; if it exceeds the computational saṃmīlana-kāla, then one should know whether [the place] is situated in the eastern or western part from the second line.

Because to the east of the line, midday (madhyāhna-kāla) occurs first in one's own country and later in the line-country; therefore, the saṃmīlana-kāla of one's own country exceeds the saṃmīlana-kāla of the line-country. In the west, the opposite occurs. Thus, from the unmīlana-kāla, the iṣṭa-prāsa-kāla, the sparśa-kāla, or the mokṣa-kāla should be examined.

**[Sūtra-bhāṣya.]** The first half of the earlier [section] of the Tri-praśna-adhyāya [is to be] placed as the fourth subsidiary section (upayupardha-saṃṃ nyasyāt). All else of the Tri-praśna [verses] 58-60, the extent (āyāmi) [is] clear.

### [Verses 49-50]

**[Vivarāṇa-bhāṣya.]** At the half-day time (dinārdha-dina-kāle), the shadow multiplied by the shadow-hypotenuse (chāyā-karṇa-guṇā), divided by the tijñā, then the circular shadow of the hypotenuse-circle (chāyā-karṇa-golīyā) would be [obtained] as the ravi-value. The difference between the root of the gnomon (śaṅku-mūla) and the east-west line is the bhuja. By that bhuja, diminished by the karṇa-dvitiya-prācyā, [or] joined with it — in the northern hemisphere, the equinoctial shadow (viśuvac-chāyā), i.e., the palabhā, results. In the south (i.e., the southern hemisphere), when that is diminished by the difference from the karṇa-dvitiya-prācyā (bhuja-māna), then the palabhā results. Thus, [the shadow] joined with or diminished from the difference by the bhuja, or diminished by the southern bhuja, then the palabhā results.

### [Verses 49-50]

### | Atrôpapattiḥ (Demonstration thereof)

#### [Proof of Gnomon-Equivalence.]

How the palabhā in the shadow-hypotenuse circle (chāyā-karṇa-gola) and the gnomon (śaṅku) become equal in nature (tulya-tattvaṃ) is demonstrated thus: if the palabhā is obtained as the bhuja in the twelve-aṅgula sign, then what is the gnomon-equivalent (śaṅku-tulya)?

$$\begin{aligned} \text{śaṅku-tulya} &= \frac{\text{pama} \times \text{śaṅku}}{12}, \quad \text{but } \text{śaṅku} = \frac{12 \times \text{tri}}{\text{chāka}}, \text{ by elevation} \\ \text{śaṅku-tulya} &= \frac{\text{pama} \times 12 \times \text{tri}}{12 \times \text{chāka}} = \text{pama} \end{aligned}$$

Therefore, it is proved (siddham) that in the karṇa-gola, the gnomon-equivalent (śaṅku-tulya) equals the palabhā. The karṇa-dvitiya-jyā becomes the inverse circle (vyasta-golā). The palabhā is the north-south [quantity]. Thus, by the correction of the chāyā-pravi-parasūna, the bhuja is spoken of; by the addition-subtraction (kṣepa-riyena) with the karṇa-dvitiya-jyā, by the correction of the bhuja, the palabhā results.

In the *Siddhāntaśekhara*: “Now, the difference of the natapūrva-pāścima (east-west natal difference), the śakibuta-jñā-makā. The dakṣiṇōttara-bhujā, joined with and diminished, the Sun dwelling in the north (saumya-vartin) [gives] the palabhā. In the southern [hemisphere], again, at the inanāyām, thus it is always diminished by that difference. Thus, joined with and diminished by the difference, the shadow becomes fixed [in] aṅgula-ścakā.” This [is] said by Śrīpati; [it is] in accordance with the Ācārya alone.

### [Verses 49-50]

*valanasya muce taimyaḥ pṛthag grahak khaṇḍakena | vṛttiḥ kṛtiḥ sparśavimarśanimadhyagratā krameṇôr bimha-vagamayaḥ |*

Of the deflection (valana), the release (muca), the taimya, separately; by the section of the planet (graha-khaṇḍakena); the vṛtti, the kṛti, the contact and release (sparśa-vimarśa), the centrality of the middle (nimadhyagratā) — in sequence, the understanding of the eclipse-disc (ur-bimha) [is obtained].

By Bhāskarācārya, the very method of Śrīpati [was] propounded (pratipāditāḥ) in the *Viśvadarpaṇa*.

**[Verses 14-15]**

### **[Dvivedī's Commentary on Eclipse Drawing.]**

The answers to the two questions beginning “viśvadarpa-maṇḍala-dikā” etc., have been explained in the eclipse chapter (grahaṇādhikāra). The latter half of that [is called] śrambe. Now the answers to the question beginning “samparkaṃ maṇḍalai” etc. are spoken.

**[Hindi Bhāṣya — Summary.]** In the prāhlādūta, the arc of the instantaneous declination (tātkālīka krānti) is of a simple form with slight difference; [this is] the balana-sūtra. From the balana-jyā, the divya-jyā should be computed. In the māna-cakra-bāhṛta [circle], above the balana-sūtra, the contact-sphāra (sparśa-kālīka sāra) and release-sphāra (mokṣa-kālīka sāra) of the Sun should be given in the direction of their [respective] sāra. In the madhya-balana-sūtra, the madhya should be given from the centre of the prāhla-disc. From the moon's horns (śarāghra), by the prāhla-pramāṇa-vyāsa [diameter], having drawn quickly, the contact, release, and mid-line prāsa should be computed. And by drawing the parilekha [projection] on the earth, the contact-release śrīra-prāsa is obtained.

### **| Upapattiḥ (Proof/Demonstration)**

**[Dvivedī's Proof — Summary.]** When in the māna-cakra-bāhṛta circle there is the centre of the prāhlā-kṛtā, then the prāhla-bimbagrānta and prāhla-bimagrānta remain contiguous (saṃlagna). Therefore, in the known māna-cakra-bāhṛta circle, the direction should be fixed (śrīṣṭa). The common endpoint of the sama-maṇḍala and the māna-cakra-bāhṛta circle is the eastern mark (prācī cihna) in the māna-cakra-bāhṛta circle.

From there, giving the balana-jyā, the line from the centre [through] the balana-jyā-pragata is the krānti-tṛt-prācī. One should give the jyāvat-khādāna from the balana-sūtra. Because the sāra remains south and north from the krānti-vṛt-prācī. Thus it happens in the sparśa and mokṣa. The madhya-balana [is] in the dakṣiṇōttara direction from the tātkālīka-krānti-vṛt-prācī; therefore, from the centre it should be given in the balana-sūtra. In the śarāghra, the prāhla-vṛtta has its centre; therefore, from there, by the tātkālīka-prāhlābhi-pramāṇa, from the constructed circles, the sparśa-mokṣa-madhya [is obtained].

In the *Siddhāntaśekhara*, from the verse “mānacakrabāhṛtalaye tatpariyāminivi” etc., written in the Sanskrit proof, Śrīpati has spoken in accordance with the Ācāryas alone. In the *Siddhāntaśiromaṇi*, from “prāhlākhasūtreya vidhāya vṛttaṃ mānacakrabāhṛtaṃ ca sādhitāṃsam” etc., Bhāskarācārya has propounded in the form of deliberation (vicāra-rūpa) the very method of Śrīpati.

**[Verses 14-15]**

**[Introduction.]** In the Sūrya-siddhānta, the following verbal description of instruments (yantra-vivaraṇa) is given —

*tuṅgârkasamāyuktaṃ gola-yantraṃ prasādhayet  
gopayeetat-prakāśoktaṃ sarvagamyam bhaved dhi |*

One should prepare the spherical instrument (gola-yantra) endowed with the exalted Sun (tuṅgârka-samāyuktaṃ); having concealed it, [as] spoken of in the Prakāśa, it becomes indeed universally accessible (sarvagamyam).

*kāla-saṃsāadhanārthaṃ tathā yantrārīṇā sādhayet  
ekākī yojayed bījaṃ yantraṃ vismaya-kāriṇi |*

For the purpose of time-determination (kāla-saṃsādhana), one should thus accomplish [it] by the yantrārīṇa; the solitary one should apply the seed (bījaṃ) [to] the instrument, causing wonder.

*śaṅku-kuyaṣṭi dhanuś-cakreṣa-chāyā-yantrair naikathā  
gupta-deśāṅka-jñeyam kāla-jñānam atandritaḥ |*

By gnomon-staff (śaṅku-kuyaṣṭi), bow-wheel (dhanu-cakra), and shadow-instruments (chāyā-yantraiḥ) in manifold ways; the knowledge of time (kāla-jñāna) is to be known from the hidden place-marker (gupta-deśāṅka) [by one who is] vigilant.

*toya-yantra-kapālāṅka-gharmmy-udanatravānaraiḥ  
sa-sūtra-rekhu-gamais ca samaya-kālam prasādhayet |*

By water-instrument (toya-yantra), bowl (kapāla), aṅka of heat (gharma), water-stream (udanatra), and monkey [devices?] (vānaraiḥ), and with cord-line-rekhuga (sa-sūtra-rekhu-gamaiḥ), one should determine the proper time (samaya-kālam).

*paradārā-manuṣyāṇāṃ gulvataila-jalāni ca  
bījāni pāṃsavas teṣu prayogās te 'pi durlabhāḥ |*

The wives of others (paradāra) of men, oil (taila), and water — [these are] seeds (bījāni) [in] the sand (pāṃsavaḥ) of these; their applications are also difficult to obtain (durlabhāḥ).

*tā prapañcamadhiśchadd nyastaṃ kuṇḍe smalābhamis  
dviṭṭha-meja-laya-horātraṃ sphuṭaṃ yantraṃ kapālakaṃ |*

That covering of the world (prapañcam-adhiśchadd), placed in the bowl (kuṇḍe), smalābhamis [?]; the double-positioned (dviṭṭha) meja-laya day-night (horātra) — the clear instrument [is] the kapālaka.

*na yantraṃ tathā sādhu divā ca vimale ravau  
chāyā saṃsāadhanaiḥ proktaṃ kāla-sādhanaṃ uttamam |*

Not the instrument thus [alone is] good; by day, when the Sun is pure (vimale ravau), the shadow [is] declared by the saṃsādhana-methods [to be] the best time-determination (kāla-sādhana).

### | Mānādhyāya (Chapter on Measures)

*mānāni saura-cāndrārṅkṣa-sāvanāni grahāṇayanam aibhiḥ  
mānaiḥ prṭhak caturbhiḥ saṃkhyabahāroḍjña lokasya |*

The measures (mānāni): solar (saura), lunar (cāndra), stellar (nakṣa), and civil (sāvana) —

by these, the computation of planets (grahāṇayanam). By these four measures separately, the multitudinous arrangements (saṃkhyabahāroḍjña) of the world [are made].

**[Dvivedī's Hindi Commentary — Summary.]**

From this, the solar measure (saura-māna), lunar measure (cāndra-māna), stellar measure (nākṣatra-māna), and civil measure (sāvana-māna) — these four types of measures are spoken. By these very four measures, all the affairs of people are conducted. Which objects are grasped by which measure — all this is propounded (pratipādita).

Brāhma, divya, paitṛ, prājāpatya, bāhṛspatya, saura, sāvana, cāndra, nākṣa — these are nine measures. Of these measures, in the world of humans, only the saura, cāndra, sāvana, and nākṣa have primacy, because by these measures alone all human affairs are accomplished. In the Sūrya-siddhānta, Siddhāntaśekhara, Siddhāntaśiromaṇi, and all other treatises, the subject of measures is spoken of in the same manner. In this chapter of the Brāhmasphuṭasiddhānta, the methods of viewing the earth (bhūmādarśana) are also given.

**[Introduction.]** In the Smṛtis and in the Purāṇas, the doctrine of Rāhu-grasta eclipse (rāhugrasta-graḥaṇa) is present (pratipādana). Therefore, reconciling the two doctrines, the Ācārya has said —

*rāhus tac chadayati praviśati yac chuklapacchadayante  
chāyā tamasi-dvairaghra-dānām kamala-yoniḥ |  
candro 'smuyojah syo yadinaimaya-bhāskarasya māsāt  
chādayati śarmi-tatāpo rāhuś chadayati tat savituḥ |*

Rāhu covers that [Moon], entering [it] which the bright and dark [portions] cover. The shadow (chāyā) [is] the tamasi [force] of the enemy of the lotus-born [i.e., Rāhu, enemy of Viṣṇu?].

The Moon (candra), the asmu-yojah, the syo [?], from the māsa of the sun-of-the-yadinaimaya [?] — [this] Rāhu covers the scorching-heat (śarmi-tatāpah) [of] that Sun (savituḥ).

In the Golādhyāya of the Siddhāntaśiromaṇi also, the unspoken (apokṛta) statement of Bhāskara [is as follows] —

*dvi-deśa-kāla-varayāṇi medānām chādako rāhur iti sa varṇita  
yan māninaḥ kevala-golavidhās tat-sahitā vedy araṇābāhyām |*

Rāhu, the eclipser (chādakaḥ) of the vara of the two-place-time (dvi-deśa-kāla-varayāṇi medānām), thus he is described. Those who know only the spherical method (kevala-golavidhāḥ), together with that [knowledge], should know the araṇā-bāhyām [?].

*rāhuḥ kṣubhā-maṇḍalagaḥ śaśāṅkaḥ śaśāṅkaś chādayatīva bimbam  
tamo-mayaḥ śamyuva-pradānāt sarvagamānām aviruddham etat |*

Rāhu, situated in the agitated orb (kṣubhā-maṇḍalagaḥ), [eclipses] the Moon (śaśāṅkaḥ); the Moon appears to cover the disc (bimbam). [Rāhu is] composed of darkness (tamo-mayaḥ); from the bestowal of śamyuva [?], this is not contradictory (aviruddham) for all who go [i.e., all observers].

**[Dvivedī's Commentary.]** This reconciliation (samanvaya) has been made. In the Siddhāntaśekhara, for the refutation (khaṇḍanārtha) of the Rāhu-grasta eclipse, a chapter named “Rāhu-nirākaraṇādhyāya” has been placed. In this, Śrīpati also has made the reconciliation with the following śloka —

*viṣṇu-lūna-viśirasaḥ kila pañcad ettavān varam imaṃ parameṣṭhī  
hema-dāna-vidhinā tavatupatis tam-arati-mahas tad uparāge |*

The Supreme Lord (parameṣṭhī) indeed gave this excellent boon (varam imaṃ) to the headless Viṣṇu-lūna [i.e., the decapitated Rāhu], [that] by the method of donating gold (hema-dāna-vidhinā), at that eclipse (tad uparāge), [there is] the great darkness (tam-arati-mahaḥ) [for him?].

*bhūmecchāyā praviṣṭaḥ sthagayati śaśinaṃ śuklapakṣāvasāne  
rāhur bhu-prasādāt samavigatavastu-tāmo vyasutulyaḥ |*

Entering the earth's shadow (bhūmecchāyā praviṣṭaḥ), [Rāhu] obscures the Moon (sthagayati śaśinaṃ) at the end of the bright fortnight (śuklapakṣāvasāne). Rāhu, by the grace of the earth (bhu-prasādāt), [being] equal to the vyasu [?], the darkness [is] the real state of



things (samavigatavastu-tāmaḥ).

*urvyāṃ sthaṃ bhānu-bimbaṃ salila-maya-tanor yathaivôktaṃ bimbam  
saṃsūyaivaṃ ca māsavy uparati-samaye svasya sâhitya-hetoḥ |*

Standing on the earth (urvyāṃ sthaṃ), the solar disc (bhānu-bimbam), [this] disc as described [earlier], of the water-bodied one [i.e., the Moon] — thus indeed at the time of lunar cessation (māsavy-uparati-samaye), for the sake of its conjunction (svasya sâhitya-hetoḥ).

### [Dvivedī's Commentary on Spherical Construction.]

In the Golabandhâdhikâra, having constructed the great circles (mahad-vṛttâḥ: pūrvâpara-vṛtta, yâmyôttara-vṛtta, sthitija-vṛtta, etc.) and the small circles (laghu-vṛttâḥ: the outer royal circles of the twelve signs from Meṣa, etc.), the Ācārya has demonstrated the nature (svarūpa) of the maximum equation (paramalambana-nati) and computed the knowledge of the circles (vṛttôpāyanam).

As the Ācārya has spoken of the golabandha in the Brāhmasphuṭasiddhānta, so also has Śrīpati in the Siddhāntaśekhara, and Bhāskarâcārya in the Golâdhyâya of the Siddhāntaśiromaṇi. In the graha-gola and nakṣatra-gola, five fixed circles (sthira-vṛtta) are spoken: pūrvâpara-vṛtta, sthitija-vṛtta, yâmyôttara-vṛtta, unmaṇḍala, and viṣuvad-vṛtta. All these are equal to the kalâ-maṇḍala.

And the moving circles (calavṛtta) of the planets: mandanîcôcavṛtta = 7, the śîghranîcôcavṛtta of the manda-etc. planets = 5. The manda-prativṛtta = 7, the śîghra-prativṛtta = 7. The hamaṇḍala and hṛd-kṣepa of the seven planets [are also given].

## Chapter 17: Advanced Planetary Computations

(Original pages 1136–1146; covering mean and true longitudes, *parilekha* drawings)

### Problem Statement

**15.** The half-chord segments (*ardhajyā-khaṇḍa*) of the arcs for the purpose of [computing] the Sun, the Moon, Venus, Mars, Jupiter, Saturn, and the nodes of the Moon — the sum of these chord-segments is the *bhogya-phala* [incremental motion].

**16.** The difference of the *gata-bhogya-khaṇḍa* [past and future segments], divided by the minutes of arc, multiplied by the elapsed [time in] minutes — that, combined with half the difference, [added to] the *bhogya* [motion], is the mean *bhogya*.

**Mathematical note:** The *bhogya-khaṇḍa* method computes the incremental motion of a planet by dividing the orbital arc into segments. If  $k_1, k_2, \dots, k_n$  are the chord-segments for each planet's arc, the *bhogya-phala* is  $umk_i$ . The mean motion is then interpolated using the proportion of elapsed time. Brahmagupta uses a *triṣṭā* (radius) of 150, whereas Bhāskara II later adopted 120 in the *Siddhānta-śiromaṇi*.

### Parilekha (Planetary Diagrams)

**[Pañcasiddhāntikā 3.]** The Sun's [daily] motion is five [ghaṭikās]; the pairs [of planets] and the minutes [are as] declared by Pitāmaha [Brahmā]. Half a month, half [the motion], the nodes, the apogees, and the quantities — *these* are established by calculation.

This verse from the *Pañcasiddhāntikā* of Varāhamihira (c. 505–587 CE) cites the *Paitāmaha-siddhānta* — the oldest of the five astronomical systems summarized in that work. Brahmagupta critically examines these earlier parameters, noting discrepancies with observational data. The phrase *pita-mahoddhiṣṭāni* (declared by Pitāmaha) refers to the legendary divine origin ascribed to this *siddhānta*.

### Special Rules

Between the two times of sunrise and sunset, the time when the [shadow] increases and takes support [of the gnomon], blazing with the form of fire — [it is] condemned in all rituals [as inauspicious].

As much as the difference between the circles of the [ecliptic and equatorial] paths, so much is the *sambādha* [obstruction] — this destroys all rituals.

The *pātāntara-kāla* (intermediate time of the node passage) is declared inauspicious for all rituals including bathing, charity, recitation, *śrāddha*, and *homa*. When the Sun and Moon have equal brightness near the *graha-sandhi* (planetary conjunction), the *pāta* (node passage) occurs twice in a short time; but when the Sun is near apogee, the equality of brightness persists for a longer duration.

## Chapter 18: The Pulverizer (Kuṭṭaka)

*The most celebrated chapter: linear Diophantine equations, operations with zero, and the “square-nature” (vargaprakṛti) problem*

## Purpose of the Kuṭṭaka

**[Introduction.]** The term *kuṭṭaka* (“pulverizer”) refers here to the method of solving linear Diophantine equations of the form  $ax + by = c$ . Here  $a$  is the *bhājya* (dividend),  $b$  is the *bhājaka* (divisor),  $c$  is the *śeṣa* (remainder),  $x$  is the *guṇaka* (multiplier), and  $y$  is the *koṭi* (quotient).

Brahmagupta was the first to give systematic rules for arithmetic operations involving zero and negative numbers, as well as the general solution to the pulverizer equation.

**14.** The two [quantities] — the *śaka-indra-kāla* [elapsed years of the Śaka era] and the *śeṣa-varṣa-yuga* [residue of years in the yuga] increased by five — make them half the *ardha-sita* [half the deficit]; the *bhugrā* [earth-number] for that pair [is found] by the pulverizer.

**15.** The *saṃkṣipta* [reduced quantity], two, three, the motions, the *tithi-bhū* [lords of the tithis], multiplied by nine, the *eṣya* [desired quantity] — the *digra-sabhāga* [directional parts] divided by seven, the Moon’s [motion], [these are] the wealth of the planet’s self.

**On the astronomical setting:** In the *kuṭṭaka* method as applied to astronomy, the dividend (*bhājya*) typically represents the number of days in a yuga (e.g., 1577917500 civil days), the divisor (*bhājaka*) represents the number of planetary revolutions in that yuga, and the remainder (*śeṣa*) represents the elapsed portion. The method finds the multiplier (*guṇaka*) such that the congruence is satisfied, yielding the number of elapsed days corresponding to a given planetary longitude.

## The Pulverizer Method

**40.** The *bhājya* [dividend,  $a$ ] is [divided] by the *bhājaka* [divisor,  $b$ ] together with the *śeṣa* [remainder,  $c$ ] until unity becomes the operation. The first [quotient] is united with one [rupas]; then the second operation becomes [clear].

**[English translation of commentary:]** The commentary explains the pulverizer method as follows: “The pulverizer with unity (*rūpa-kuṭṭaka*) is the method by which, starting from the desired [quantity], the root-quantity is found. The first quotient, combined with unity (*rūpa*), gives the basis; then for the purpose of the second [step], unity becomes the operative quantity. In this manner, roots are extracted from the non-aggregate [quantities].

**Example:** ‘The fourteen-second rows [of a formation], O conqueror of the world, tell that quantity quickly if you know the pulverizer.’ Here, for divisor 34 the remainder is 2; for divisor 13 the remainder is 10. Hence the greater remainder-divisor is 13, the lesser is 34. Dividing the greater by the lesser gives remainder 13; then by mutual division the chain (*vallī*) is produced. The penultimate [quotient], multiplied by its own higher [term] and united [with the lower] — this is the final [step] of the pulverizer method.”

**Algorithmic summary:** Given  $ax + by = c$  where  $\gcd(a, b) \mid c$ :

1. Apply the Euclidean algorithm to  $(a, b)$  to obtain quotients  $q_1, q_2, \dots, q_n$ .
2. Construct the *vallī* (chain) of partial quotients.
3. Starting from the bottom with 1, work upward:  $x_0 = 1, x_1 = q_n, x_{k+1} = q_{n-k} \cdot x_k + x_{k-1}$ .
4. The penultimate value gives the multiplier  $x$ ; then  $y = (c - ax)/b$ .

This is algebraically equivalent to the extended Euclidean algorithm.

## Operations with Zero

**[On zero:]** The aggregation and diminution of positive, negative, and zero [quantities]; the rule for multiplication; and for division — [are declared].

**[Addition and Subtraction:]** A negative [quantity] joined with a negative becomes more negative; a positive with a positive [becomes] more positive; zero joined with zero [remains] zero. From a positive, a negative is subtracted; from a negative, a positive is subtracted. A quantity united with zero remains in its own form.

**[Multiplication:]** A quantity multiplied by zero becomes zero. [But] a quantity divided by zero becomes zero [*śūnya* — Brahmagupta’s controversial rule: he treats  $a/0 = 0$ , later corrected by Bhāskara II].

The rule for the aggregation of unknowns (*avyakta*), as much as desired, and the rule for the multiplication of unknowns, and the determination of the measure of unknowns — [are given].

**Historical note on zero:** Brahmagupta (628 CE) was the first mathematician in history to establish rules for arithmetic operations with zero as a number. His rules:

- $a + 0 = a$ ;  $a - 0 = a$ ;  $0 + 0 = 0$ ;  $0 - 0 = 0$
- $a \times 0 = 0$ ;  $0 \times 0 = 0$
- $a0 = 0$  (*this rule is incorrect; division by zero is undefined*)
- $0a = 0$  for  $a \neq 0$

He also gave rules for operations with negative numbers (*ṛṇa*):  $(-a) \times (-b) = +ab$ ,  $(-a) \times (+b) = -ab$ , etc. These rules were revolutionary for the 7th century.

## The Square-Nature (Vargaprakṛti) Problem

**[Square-nature:]** In an isosceles triangle, the sum of the squares of the sides — [and] the product of the sum and difference of the *karaṇīs* [surds].

The statement of the square-equation [Pell’s equation]; the elevation of the root of the square-equation [its solution].

**[English:]** The “square-nature” (*vargaprakṛti*) refers to the equation  $Nx^2 + 1 = y^2$ , known in modern mathematics as Pell’s equation. Here  $N$  is the *prakṛti* (multiplier),  $x$  is the *hrasva-mūla* (lesser root), and  $y$  is the *dīrgha-mūla* (greater root).

Brahmagupta gives the *bhāvanā* (composition) method by which, from two pairs of lesser and greater roots, a new pair can be generated. If  $(x_1, y_1)$  and  $(x_2, y_2)$  are solutions for the same  $N$ , then:

$$x' = x_1y_2 + x_2y_1, \quad y' = Nx_1x_2 + y_1y_2$$

is also a solution. This is the *samāsa-bhāvanā* (additive composition).

**The bhāvanā composition:** Brahmagupta’s *prakṛti-samāsa* (composition of roots) is one of the most remarkable achievements in early algebra. Given two solutions  $(x_1, y_1)$  and  $(x_2, y_2)$  of  $Nx^2 + k_1 = y^2$  and  $Nx^2 + k_2 = y^2$  respectively, then:

$$x_3 = x_1y_2 + x_2y_1, \quad y_3 = Nx_1x_2 + y_1y_2$$

satisfies  $Nx_3^2 + k_1k_2 = y_3^2$ . When  $k_1 = k_2 = 1$ , this generates an infinite family of solutions from a single minimal solution. For example, with  $N = 92$ , the minimal solution is  $x = 120$ ,  $y = 1151$ , since  $92 \cdot 120^2 + 1 = 1151^2$ .

## Reverse Motion (Viloma-gati)

**14.** Half the *rāśi* [sign], increased by its *śeṣa* [residue], and diminished by the *ardha-māsa* [half-month] etc.; [and] united with its *śeṣa* — he who states the *yuga-gata* [planetary position for the yuga] is the knower of the pulverizer.

**On reverse computation:** The *viloma-gati* method computes backwards from a given planetary longitude to find the corresponding elapsed time (*ahargaṇa*). Given a planet's mean longitude, one first determines the residual motion within the current sign (*rāśi*), then applies the pulverizer to find how many days have elapsed since the epoch. This is the inverse problem of the forward computation.

## Chapter 19: Gnomon and Shadow

*Shadow computations using the gnomon , time determination,  
and related astronomical problems*

### First Problem Statement

Divided by the divisor [*hara*], what is that quantity [*rāśi*] which becomes pure? First, of the multiplier and divisor, the greater is brought [into consideration] by means of the *apavarta* [reduction].

**[English translation:]** When two [numbers] divide each other mutually, that which [leaves] a remainder — by that they become two multiplier-divisor [pairs]. Then, having placed the quotients obtained from the mutual division of the multiplier and divisor in the heart [i.e., in a column], according to the pulverizer rule demonstrated previously. This operation is to be performed until finally unity remains as the remainder. That remainder, multiplied by some multiplier and diminished by unity, when divided by the divisor related to that remainder, becomes pure. Having done so, the multiplier is to be placed below the previously stated quotients; below that, the result is placed. In this manner the chain (*vallī*) is produced.

### Determination of Gnomon and Shadow

The determination of the shadow of houses etc. from the [Sun's] shadow. The solution of the problem: "If [the shadow] is equal to the height of the desired cave" — [its computation is given].

**On shadow problems:** The gnomon (*śaṅku*) is a vertical staff of standard height (typically 12 *aṅgulas*). Its shadow (*chāyā*) cast by the Sun is used to determine:

1. The Sun's altitude:  $\sin(\text{altitude}) = \frac{\text{śaṅku}}{\sqrt{\text{śaṅku}^2 + \text{chāyā}^2}}$
2. The time of day from the shadow's length
3. The dimensions of objects from their shadows
4. The *palabha* (equinoctial shadow) for latitude determination

If the shadow equals the height of a cave (i.e., the Sun's altitude is 45°), the horizontal distance from the gnomon to the cave equals the cave's height.

### Height Determination

The determination of the height of a nearby elevated object and of a distant one; and the statement of the motion of light [rays], and the measure of the interval between the two.

**[Height from shadow — examples:]** (a) The determination of the answer to the problem of a house, a man, the interval, and water — concerning a *haṭa-caitya* [pillar shrine] etc.

(b) The problem concerning a seed in a cave, water, and spreading — its solution is stated.

(c) By the method of double-difference, the answers to three problems from the shadow of one house are given.

(d) From the shadow, the determination of shadows of houses etc.

**The double-difference method:** For determining heights from shadows: If two observations of a gnomon and its shadow are made at different times or positions, the height  $h$  of an object and its horizontal distance  $d$  can be found from:

$$h = \frac{s_2 \cdot g}{s_2 - s_1} \cdot \frac{d_2 - d_1}{g}, \quad d = \frac{h \cdot s_1}{g}$$

where  $g$  is the gnomon height,  $s_1, s_2$  are shadow lengths at two positions, and  $d_1, d_2$  are the distances of the observation points from the object's base. This is equivalent to similar-triangle methods in modern trigonometry.

## Chapter 20: Advanced Shadow Problems

*Extended shadow computations: obscuration, refraction,  
and celestial sphere applications*

Of the *chad-dirāśi* [obscured signs], the day-and-night strengths — the rise of the signs, the statement of their unequal pairs.

The position of the *carāgra* [ascensional difference] is stated. The position of the gnomon and shadow is stated.

Of the *hag-gola* [terrestrial sphere], the cause of the *lambana* [parallax] and *avanati* [depression] by means of the *haryāhara* [division-operation] is shown. The statement of the *para-lambana* [secondary parallax] and *avanati*.

**[English:]** By another method, the position of those two [gnomon and shadow] and of the *śaṅkula* [combined gnomon-shadow] is stated.

The statement of the *hak-karma* [obscuration computation]. The fixed pulverizer of the planets and the celestial sphere. The moving pulverizer of the planets. The conclusion of the chapter.

Here the computation of shadows is shown for the knowledge of the height of a lamp-flame, the boundary of light, and the Earth. Something is also said concerning the reflection of light.

**On the *chad-dichāyā*:** This chapter extends the shadow calculations of Chapter 19 to more complex astronomical situations. The *chad-dirāśi* refers to zodiacal signs that are “obscured” (below the horizon), and the computation of their day/night strengths is relevant to astrological applications. The *carāgra* (ascensional difference) measures the unequal rising times of signs at different latitudes, computed from:

$$\text{carāgra} = \arcsin \left( \frac{\tan(\phi) \cdot \tan(\delta)}{\cos(\alpha)} \right)$$

where  $\phi$  is the observer's latitude,  $\delta$  is the celestial declination, and  $\alpha$  is the right ascension.

## Chapter 21: Spherical Astronomy (Golādhyāya)

*The opening of the Spherical Astronomy Book: armillary spheres, celestial and terrestrial globes, astronomical instruments*

### Purpose of the Chapter

The praise of the celestial sphere (*gola*) is stated. The cause of the proper position of one's own sphere [is given].

The praise of the sphere of motion [is stated]. The purpose of the commencement of instruments [is stated].

#### [Contents of the Golādhyāya:]

1. The establishment of instruments (*tantra*)
2. The use of water etc. [in instruments]
3. The statement concerning arcs (bows)
4. The bearing of instruments toward the Sun etc.
5. The determination of the form of the arc for a desired ghaṭikā
6. The determination of the testing ghaṭikā
7. The knowledge of time by means of the instrument's depression and elevation
8. Special rules concerning the bow-instrument

### Arrangement of the Terrestrial Sphere

The arrangement of the terrestrial sphere is stated. The description of the arrangement of the divine and demoniacal [cities].

**[The celestial cities:]** The establishment of the Aries-cycle and the [rate of] motion for the divine and demoniacal [cities]. The statement of the arrangement of the wheel's revolution. Of the days of the gods, for the purpose of the Sun's revolution. The zodiacal arrangement of the divine and demoniacal [cities]. The determination of the measure of the day of the father [Brahmā] for the divine and demoniacal.

**The celestial geography:** Brahmagupta describes the traditional Indian celestial geography. At the North Pole is *Meru*, the abode of the gods (*deva*); at the South Pole is *Vadavā-mukha*, the realm of the demons (*asura/daitya*). The central terrestrial sphere (*bhū-gola*) has the observer's horizon, the prime meridian (through Ujjain), and the equator.

The divine day (one day of Brahmā) equals 1000 *caturyugas* (4,320,000,000 solar years). The solar year is defined by the Sun's return to the same fixed star — the *nakṣatra-vatsara* (sidereal year). The *divya-dina* (divine day) of 360 solar years is used for calculating planetary mean motions.

## Prime Meridian and Longitude Difference

The arrangement of the terrestrial sphere and the *lagna-valaya* [horizon circles] is [stated]. The statement of the *nirakṣa-deśāntara* [equatorial longitude difference] in *yojanas*.

The description of the two *kṣudra-dyau* [small celestial circles].

**[On the *deśāntara*:]** The prime meridian (*nirakṣa-deśa*) passes through Ujjain, where the celestial equator (the east-west circle) is traversed equally by all planets. For any other location, the longitude difference (*deśāntara*) is measured in *yojanas* — for every four *yojanas*, there is one *laghu-ghaṭikā* (small ghaṭikā, about 4 minutes of time).

Accordingly, the rising times of planets vary by the *deśāntara* ghaṭikās. The *deśāntara-kāla* (longitude-difference time) is computed as:

$$\text{deśāntara} = \frac{\text{yojanas from Ujjain}}{4}$$

in *ghaṭikās* (1 ghaṭikā = 24 minutes).

**The *deśāntara* formula:** Brahmagupta gives the standard Indian rule for computing longitude difference from Ujjain (the prime meridian): every 4 *yojanas* east or west corresponds to a 1-ghaṭikā (24-minute) difference in local time. Since the Earth's circumference was taken as approximately 5000 *yojanas*, this gives 1250 ghaṭikās for a full circle — close to the actual  $60 \times 24 = 1440$  minutes, though the *yojana* was not a precisely standardized unit.

The *deśāntara* is essential for computing the local time of planetary risings, setting of the *lagna* (ascendant), and the timing of eclipses at different longitudes.

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# **Miftāh al-Ḥisāb**

The Key to Arithmetic

Jamshīd Ghiyāth al-Dīn al-Kāshī

English Translation Reader

A comprehensive encyclopedic treatise on arithmetic, geometry, and algebra Completed in Samarkand, 2 March 1427 CE (829 AH)

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## **Preface to This Edition**

.1427 his bilingual edition presents the complete structure and representative content of Miftāh al-Ḥisāb (The Key of Arithmetic), the crowning achievement of Islamic arithmetic, composed by the renowned Persian mathematician and astronomer Jamshīd Ghiyāth al-Dīn al-Kāshī in 1427 CE.

( 1429--1380 )

Al-Kāshī (c. 1380–1429) was the astronomer royal to Ulugh Beg at Samarkand. His Miftāh al-Ḥisāb is an encyclopedic work covering three principal subjects: arithmetic, geometry, and algebra. It comprises five treatises.

The work includes 1. Al-Kāshī's famous calculation of  $\pi$  to 16 decimal places 2. His systematic treatment of decimal fractions (the first in history)

3. Methods for root extraction that predate similar European methods by centuries As Berggren noted, Miftāh was "the crowning achievement of Islamic arithmetic, and truly a gift fit for a king." Note on the text: Arabic passages are set in Noto Naskh Arabic on the left column. English translations appear on the right. Mathematical formulas use modern notation while preserving the original algorithms.

## **Author's Preface**

In the name of God, the Merciful, the Compassionate.

Praise be to God, who is unique in creating units, and peerless in composing the classes of numbers. And blessings upon the best of His creation, Muhammad, the intercessor for the interceders on the Day of Calling, and upon his family and his sons who guide to the path of salvation and right guidance.

Jamshīd ibn Masūd ibn Mahmūd al-Ṭabīb al-Kāshī, called Ghiyāth al-Dīn — may God improve his condition — says When I had practiced arithmetic and geometric methods until I reached their true principles, and penetrated their subtleties, and uncovered their hidden difficulties and problems and solved their challenges... ..as when I recomputed all the tables of the Ilkhanid Zij with the utmost precision, and composed the Zij called the Khāqānī completing the Ilkhanid Zij, and collected in it all that I had derived of astronomers' methods that appear in no other zij together with geometric proofs...

...and authored the Risāla al-Muḥīṭiyya on the ratio of the diameter to the circumference, and the Risāla al-Watar wal-Jayb on extracting the chord and sine for onethird of a given arc, which also was difficult for predecessors...

...I found in the course of these works many rules by which the operations of arithmetic preliminaries may be performed in the easiest manner and the simplest ways and with the least effort and the greatest benefit and the clearest arrangement. So I resolved to record them and to explain them, to serve as a reminder for beloved friends and as guidance for people of understanding.

## **Original Table of Contents**

A chapter on the foundations of the science of arithmetic and the explanation of its names.

**Treatise I: On the Arithmetic of Integers. It comprises seven chapters.**

**Treatise II: On the Arithmetic of Continued, Disjoined, and Chain Fractions.**

**Treatise III: On the Arithmetic of Degrees, Minutes, Seconds, and the like.**

**Treatise IV: On the Arithmetic of Areas and Volumes.**

**Treatise V: On Algebra and Almucabala.**

**The work comprises five treatises**

## **Treatise**

Arabic Title On the Arithmetic of Integers — 7 chapters On the Arithmetic of Fractions — 10 chapters III On the Arithmetic of Sexagesimal Numbers — 10 chapters On Measurements and Geometry — 9 chapters On Algebra — multiple chapters I placed for most operations a model in tabular form so that they may be easily grasped by the intelligent.

## **Treatise I: On the Arithmetic of Integers**

### **Treatise I: On the Arithmetic of**

Integers This treatise, the foundation of the entire work, comprises seven chapters covering all operations on whole numbers.

Ch.1: On the naming of numbers and place values — The Hindu-Arabic numeral system and the principle of place value.

Ch.2: On addition — Methods for summing integers, including column addition.

Ch.3: On subtraction — Techniques for finding differences.

Ch.4: On multiplication — Including the network method ( / shabaka), the method of columns, and the method of complement.

Ch.5: On division — Including long division and abbreviated methods.

Ch.6: On extraction of roots — Square roots and higher-order roots by a general method.

Ch.7: On finding unknowns by algebra — Preliminary applications.

## Chapter 1: On the Naming of Numbers

Al-Kāshī begins by explaining the decimal place-value system, attributing it to the Indians and noting its transmission to the Islamic world. He describes the names of the place values

| Power           | Arabic Name       | English Units       |
|-----------------|-------------------|---------------------|
| 10 <sup>1</sup> | Tens              |                     |
| 10 <sup>2</sup> | Hundreds          |                     |
| 10 <sup>3</sup> | Thousands         |                     |
| 10 <sup>4</sup> | Ten thousands     |                     |
| 10 <sup>5</sup> | Hundred thousands |                     |
| 10 <sup>6</sup> | Millions          |                     |
| 10 <sup>7</sup> | Ten millions      |                     |
| 10 <sup>8</sup> | Hundred millions  |                     |
| 10 <sup>9</sup> | Billions          | (thousand millions) |

### 101 Tens

### Hundreds Thousands

### 104 Ten thousands

### Hundred thousands

Power Arabic Name English 10<sup>6</sup> / Millions Ten millions

### 108 Hundred millions

10<sup>9</sup> / Billions (thousand millions)

Al-Kāshī extends this system far beyond what was common in his time, enabling the large-scale calculations that he would later use in his astronomical computations.

## Chapter 4: On Multiplication

Al-Kāshī presents several methods of multiplication.

The most famous is the method of the network ( / shabaka or lattice multiplication), which he describes in full detail. He also presents the method of columns () and methods for mental calculation.

Lattice Multiplication Method ( : )

1. Draw a grid of squares with as many rows and columns as there are digits in the multiplicand and multiplier.
2. Write the multiplicand above the grid and the multiplier to the right.
3. Divide each square diagonally from top-left to bottom-right.
4. Multiply each digit of the multiplicand by each digit of the multiplier, writing the tens above the diagonal and the units below.
5. Sum along each diagonal from bottom-right to top-left, carrying as needed.
6. Read the final product along the left and bottom edges.

Example: To multiply  $524 \times 385$  Result:  $524 \times 385 = 201,740$

## Chapter 6: On the Extraction of Roots

Al-Kāshī's method for extracting roots is one of the most celebrated parts of the Miftāh. He gives a general algorithm for finding the  $n$ th root of any number, a method equivalent to what is now known as the Ruffini- Horner method (though al-Kāshī predated both by several centuries).

General Method for Extracting the  $n$ th Root ( : )

Given a number  $N$  and an integer  $n$ , to find 1. Separate  $N$  into groups of  $n$  digits from the decimal point.

2. Find the largest integer  $a$  such that  $a^n$  does not exceed the first group.
3. Subtract  $a^n$  from the first group and bring down the next group.
4. Using the binomial expansion of  $(a + b)^n$ , find the next digit  $b$  by dividing the remainder by  $n \cdot a^{n-1}$ .

5. Iterate until the desired precision is reached.

For the square root ( $n = 2$ ), this reduces to Square Root Extraction ( : )

1. Separate the number into pairs of digits from the decimal point.

2. Find the largest square not exceeding the leftmost pair; this gives the first digit of the root.

3. Subtract its square and bring down the next pair.

4. Double the current root, and find a digit such that  $(20 \times \text{current} + d) \times d$  does not exceed the remainder.

5. Repeat until all pairs are processed.

Example  $144 = 12$   $12 = 1$  (first digit)

$(20 \times 1 + 2) \times 2 = 44$   $44 = 44$  (second digit)

This method for root extraction was transmitted to Europe and appeared in the works of later mathematicians. AlKāshī's presentation is remarkably systematic and general, applying to roots of any order.

## **Treatise II: On the Arithmetic of Fractions**

### **Treatise II: On the Arithmetic of**

Fractions This is arguably the most historically significant treatise of the Miftāh, for it contains the first systematic treatment of decimal fractions in the history of mathematics. As Youschkevitch and Rosenfeld observed: "It is in the work of Giyāth al-Dīn Jamshīd al-Kāshī in the early fifteenth century that we first see both a total command of the idea of decimal fractions and a convenient notation for them." The Ten Chapters of Treatise II Ch.1: On the naming of fractions — Definitions and terminology for common fractions.

Ch.2: On reduction of fractions — Converting to common denominators.

Ch.3: On addition and subtraction of fractions.

Ch.4: On multiplication of fractions.

Ch.5: On division of fractions.

Ch.6: On conversion between fractions — Including sexagesimal to decimal and vice versa.

Ch.7: On the extraction of roots of fractions.

Ch.8: On the kasr al-a'dad (fractions of numbers) — A special class of fractions.

Ch.9: On decimal fractions (— ) The crowning achievement.

Ch.10: On the use of decimal fractions in calculations.

### **Chapter 9: On Decimal Fractions**

l-Kāshī introduced decimal fractions with a clarity and generality that was unprecedented. His notation used a vertical line (|) to separate the integer part from the fractional part. For example, he

would write  $3 | 14159265 \dots$

to represent  $3.14159265 \dots$ , where the digits after the vertical bar are understood as tenths, hundredths, thousandths, etc.

Al-Kāshī writes Know that the common people call the first place after the units 'the place of the parts of ten' (.) and the second 'the place of the parts of a hundred' (,) and the third 'the place of the parts of a thousand' (,) and so on, always multiplying by ten. And we have found it

appropriate to call the first place after the decimal point ‘the place of the first tenth’ (,) and the second ‘the place of the second tenth’ (,) and so on.

2.2.1 Al-Kāshī’s Notation System Al-Kāshī’s notation for decimal fractions was remarkably modern. He wrote the integer part, then a separator, then the fractional digits.

Al-Kāshī’s notation Modern notation Value | 13.7539 Thirteen and seven thousand five hundred thirty-nine ten-thousandths | 0.14142135 Approximation of | 3.1415926535

Approximation of  $\pi$  2.2.2 Decimal Operations Al-Kāshī demonstrates all four arithmetic operations with decimal fractions. For multiplication, he gives the rule

Multiplication of Decimal Fractions ( : )

1. Multiply the two numbers as if they were integers, ignoring the decimal point.
2. In the product, count from the right a number of places equal to the sum of the decimal places in the multiplicand and multiplier.
3. Place the separator at that position.

Example:  $0.25 \times 0.4 = 0.10$  • Multiply as integers:  $25 \times 4 = 100$  • Total decimal places:  $2 + 1$

$= 3$  • Result:  $0.100 = 0.1$  2.2.3 The Significance of Decimal Fractions Al-Kāshī used decimal fractions extensively in his astronomical calculations. He converted between sexagesimal and decimal systems with ease. As Kennedy observed: “Al-Kāshī was first and foremost a master computer of extraordinary ability, witness his facile use of pure sexagesimals, his wide application of iterative algorithms, and his sure touch in so laying out a computation that he controlled the maximum error and main-

tained a check at all stages.” Sexagesimal to Decimal Conversion ( : )

To convert a sexagesimal number  $N = a + b/60 + c/60^2 + \dots$  to decimal 1. Keep the integer part  $a$  as is.

2. Convert each sexagesimal fraction: divide by 60, 60<sup>2</sup>, etc.
3. Sum the results in decimal form.

Example: Convert 3; 8, 29, 44 (sexagesimal) to decimal  $60 + 29/60 + 44/60^2 = 3.141592 \dots$

## **Treatise III: On the Arithmetic of Sexagesimal Numbers**

### **Treatise III: On the Arithmetic of**

Sexagesimal Numbers This treatise is devoted to the sexagesimal (base-60)

system, which was the standard system for astronomical calculations in the Islamic world.

Al-Kāshī writes The astronomers have adopted the sexagesimal system because of the many kinds of divisions it admits, since sixty is divisible by two, three, four, five, six, ten, twelve, fifteen, twenty, and thirty. And because they needed to use fractions in their calculations, they divided the circle into three hundred and sixty degrees, each degree into sixty minutes, each minute into sixty seconds, and so on without end.

The Chapters of Treatise III Ch.1: On the naming of sexagesimal places — Degrees (,) minutes (,) seconds (,), thirds (,), etc.

Ch.2: On addition and subtraction of sexagesimal numbers.

Ch.3: On multiplication of sexagesimal numbers.

Ch.4: On division of sexagesimal numbers.

Ch.5: On the extraction of roots of sexagesimal numbers.

Ch.6: On converting sexagesimal to other systems.

Ch.7: On astronomical calculations.

Ch.8: On operations with the sine and chord.

Sexagesimal Notation Al-Kāshī used the following notation for sexagesimal numbers. In modern notation, we write a; b, c, d to represent  $60^2 + d$  Arabic Term Latin Equivalent Symbol Value Degree or nothing  $60^0 = 1$  Minute  $60^{-1}$  Second  $60^{-2}$  Third  $60^{-3}$  Fourth  $60^{-4}$  Fifth  $60^{-5}$

## **Chapter 5: On the Extraction of Roots of Sexagesimal Numbers**

( 1960) .

This chapter contains al-Kāshī's remarkably general method for extracting  $n$ th roots in the sexagesimal system. Dakhel's study of this chapter (1960) showed that al-Kāshī's method is essentially the same as the Ruffini-Horner method.

Sexagesimal Root Extraction ( :)

To find  $N$  in sexagesimal 1. Express  $N$  in sexagesimal form:  $N = N_0; N_1, N_2, N_3, \dots$

2. Group the sexagesimal places appropriately for the root order.

3. Apply the binomial method iteratively  $a + r \approx a + n \cdot a^{n-1}$  4. At each step, refine the approximation using the error term.

5. Continue until the desired precision is achieved.

Example — Square Root in Sexagesimal Find 2 in sexagesimal. Al-Kāshī gives  $2 \approx 1; 24, 51, 10, 7, 46, 6, 4, 40, \dots$

which corresponds to  $1 + 24 \cdot 60 + 51 \cdot 60^2 + 10 \cdot 60^3 + \dots \approx 1.41421356 \dots$

This approximation of 2 is accurate to the limits of the calculation. Al-Kāshī's method of root extraction is completely general and works for any order of root in any number base.

## **Treatise IV: On Measurements and Geometry**

### **Treatise IV: On Measurements and**

Geometry This is the longest and most diverse treatise of the Miftāh, covering plane and solid geometry, with applications to surveying, architecture, and engineering. It comprises nine chapters with numerous sections.

### **Original Table of Contents of Treatise IV**

A chapter on mensuration and its terminology.

**Chapter 1: On the area of the triangle and what relates to it.**

**Chapter 2: On the area of the quadrilateral and what relates to it.**

**Chapter 3: On the area of the polygon and what relates to it.**

**Chapter 4: On the area of the circle and what relates to it.**

**Chapter 5: On the area of plane bodies.**



## **Chapter 6: On the area of convex bodies.**

## **Chapter 7: On the volumes of bodies.**

## **Chapter 8: On the ratio of a body's volume to its weight and vice versa.**

## **Chapter 9: On the measurement of buildings and structures.**

## **Chapter 1: On the Area of a Triangle**

4.2.1 Section 1: Classification of Triangles Al-Kāshī classifies triangles by their sides (equilateral, isosceles, scalene) and by their angles (acute, right, obtuse), giving geometric definitions for each.

### **4.2.2 Section 2: Area Formulas Area of a Triangle ( : )**

Method 1 — Base and Height  $\frac{1}{2} \times \text{base} \times \text{height}$  Method 2 — Heron's Formula: For a triangle with sides  $a, b, c$  and semi-perimeter  $s = \frac{a+b+c}{2}$   $s(s-a)(s-b)(s-c)$

Method 3 — Two Sides and Included Angle  $\frac{1}{2}ab \sin C$  Example: Find the area of a triangle with sides 13, 14, 15.

$$s = \frac{13 + 14 + 15}{2} = 21$$

$$\frac{1}{2} \times 13 \times 14 \times \frac{15}{21} = 84$$

## **Chapter 4: On the Area of the Circle**

This chapter contains one of the most celebrated passages in the Miftāh: al-Kāshī's statement about the value of  $\pi$ . He writes

...44 29 8 3 Know that the circumference is three times the diameter and a fraction which is less than one seventh of the diameter, but people take it as a seventh for simplicity of calculation. Archimedes said that this fraction is smaller than a seventh and greater than ten out of seventy-one. According to what we obtained and mentioned in our article titled "al-Muḥīṭiyya" on the ratio of the diameter to the circumference, it is  $3 \frac{8}{29} \frac{29}{84}$  thirds after dropping fourths and what comes after, if the diameter is one. This is far more accurate than the calculation of Archimedes as we explained in the article mentioned, and it is closer to the correct value. However, nobody knows it exactly except God, the Almighty.

In modern notation  $3 + \frac{8}{29} + \frac{29}{84} = 3.14159259259 \dots$

This value of  $\pi$  is accurate to about 7 decimal places. But al-Kāshī's tour de force in the Risāla al-Muḥīṭiyya extends this to 16 decimal places, computed using a polygon with  $3 \times 228 = 805,306,368$  sides.

### **4.3.1 Circle Area and Related Formulas Circle Formulas ( : )**

$$C = \pi d = 2\pi r \text{ (Circumference / )}$$

$$A = \pi r^2 = \frac{\pi d^2}{4} \text{ (Area / )}$$

Finding diameter from circumference The Archimedean bounds: Al-Kāshī notes the classical bounds  $\frac{22}{7} < \pi < \frac{31}{10}$  that is  $\frac{22}{7} < \pi < \frac{31}{10}$

## **Chapter 9: On the Measurement of Structures and Buildings**

## **Chapter 9: On the Measurement of Structures and**

**Buildings** This chapter is of particular interest to historians of Islamic architecture. Al-Kāshī writes The practitioners of this art have only mentioned vaults and arches, and even that was not done properly. So I cover them here in a proper way among others, because the need in buildings' measurements is more in demand.

**4.4.1 Section 1: On the Areas of Vaults and Arches** Al-Kāshī defines a proper vault ( ) as a structure built on two bases between two parallel lines, consisting of five pieces: two parts of a spherical shell (or annulus or timbrel) with hollow diameter not smaller than the gap of the vault, and two other parts with larger hollow diameter.

**4.4.2 Section 2: On the Area of a Dome** For a dome of diameter  $d$  and height  $h$ , al-Kāshī gives  $A_{\text{dome}} = 2\pi rh$  where  $r$  is the radius of the base sphere.

**4.4.3 Section 3: On the Surface Area of the Muqarnas** The muqarnas ( ) is a decorative architectural element used in Islamic architecture, consisting of tiered, niched structures that create a honeycomb or stalactite effect on ceilings, corners, and the interior surfaces of domes.

Al-Kāshī describes four types of muqarnas and gives formulas for calculating their surface area, so that architects could determine the quantity of materials needed.

**Surface Area of a Muqarnas ( : )**

For a simple muqarnas composed of  $n$  tiers, each with average area  $a_i$   $A_{\text{total}} = a_i \times f_i$

where  $f_i$  is a shape factor depending on the type of muqarnas cell.

This section has been studied extensively by historians of Islamic architecture. The muqarnas represents one of the most sophisticated applications of geometry in Islamic art and architecture.

**Al-Kāshī's Calculation of  $\pi$  — The Risāla al-Muḥīṭiyya** Al-Kāshī's Calculation of  $\pi$  . 1424 16 ne of the most remarkable achievements in the history of mathematics is al-Kāshī's computation of  $\pi$  to 16 decimal places, completed in

### **1424 CE. This surpassed all previous records and was not**

improved upon in Europe for nearly two centuries, until Adriaan van Roomen (1561–1615) and later Ludolph van Ceulen (1540–1610) pushed the calculation further.

**Historical Context**  $3 \times 228 = 805,306,368$  Al-Kāshī's motivation was astronomical: he sought to compute the circumference of the universe with such precision that the round-off error would be smaller than the width of a horse's hair. He set the accuracy of his approximation in advance and used a polygon with  $3 \times 228 = 805,306,368$  sides The Method Al-Kāshī used the classical method of Archimedes: inscribing and circumscribing regular polygons around a circle and computing their perimeters. Starting with a hexagon and doubling the number of sides 28 times, he obtained his approximation.

**Al-Kāshī's Method for Computing  $\pi$  ( $\pi$ :)**

Starting with a regular hexagon inscribed in a unit circle 1. Initial chord:  $c_0 = 1$  (side of hexagon = radius)

2. At each step, given chord  $c_n$  of the  $n$ -gon, compute chord  $c_{n+1}$  of the  $2n$ -gon by  $4 - c_{2n}$  or equivalently using the recursive formula  $c_{n+1} = 4 - c_{2n}$  3. The perimeter of the inscribed  $N$ -gon is  $P_N = N \cdot c_N$ .

4. The approximation to  $\pi$  is  $\pi \approx P_N$  5. Repeat 28 times to obtain the polygon with  $3 \times 228$  sides.

**The Result** 28 59 16 6 :  $2\pi$  . :50 14 46 51 34 1 Al-Kāshī first expressed  $2\pi$  in sexagesimal notation  $2\pi = 6; 16, 59, 28, 1, 34, 51, 46, 14, 50$  which is  $6 + 16/60 + 59/60^2 + 28/60^3 + 1/60^4$

+ 34 605 + 51 606 + 46 607 + 14 608 + 50 He then converted this to decimal  $2\pi = 6.2831853071795865 \dots$

Hence  $\pi = 3.1415926535897932 \dots$

Source Date Value of  $\pi$  Archimedes ~250 BCE 3.14185 (average of bounds)

Ptolemy ~150 CE 3.14166  $\dots$  (3; 8, 30)

Al-Kāshī

### **1424 CE**

3.1415926535897932  $\dots$  (16 places)

### **Van Ceulen**

### **1610 CE**

20 places Machin

### **1706 CE**

100 places Al-Kāshī's accuracy: to compute  $2\pi$  with 9 sexagesimal places means the maximal error is less than  $1/609 \approx 9.92 \times 10^{-17} < 10^{-16}$ . That is, al-Kāshī computed  $2\pi$  exactly up to and including the 16th decimal place.

## **Treatise V: On Algebra**

The final treatise of the Miftāh is devoted to algebra ( . ) Al-Kāshī presents a systematic treatment of the subject, building on the work of his predecessors — particularly al-Khwārizmī, Abū Kāmil, al-Karajī, and al-Samaw'al — while adding his own innovations.

The Chapters of Treatise V Ch.1: On the types of equations — The six canonical forms.

Ch.2: On finding the unknown using the Rule of Double False Position ( . )

Ch.3: On algebraic operations — Addition, subtraction, multiplication, and division of algebraic expressions.

Ch.4: On the extraction of roots of algebraic expressions.

Ch.5: On solving equations of higher degree — Including cubic equations.

Ch.6: On the solution of problems — Applied problems solved by algebra.

### **Chapter 1: The Six Canonical Forms**

Al-Kāshī, following the tradition established by al-Khwārizmī, classifies all equations into six types. Using modern notation with  $x$  as the unknown and  $a, b$  as positive coefficients Form Equation Name  $ax = b$  Simple ( )

$ax^2 = bx$  Squares equal roots  $ax^2 = b$  Squares equal numbers  $ax^2 + bx = c$  Squares and roots equal numbers  $ax^2 + c = bx$  Squares and numbers equal roots  $bx + c = ax^2$  Roots and numbers equal squares Solution of  $ax^2 + bx = c$  (Form 4) ( : )

Divide by  $a$ :  $x^2 + b ax = c$  Complete the square  $x^2 + b$

Hence  $= 4ac + b^2$  Solution  $b^2 + 4ac$

### **Chapter 2: The Rule of Double False Position**

The Rule of Double False Position ( . )

is an ancient method for solving linear problems without algebra. Al-Kāshī gives a clear exposition Rule of Double False Position ( : )

Given a problem of the form  $ax + b = c$ , make two guesses  $g_1$  and  $g_2$ , giving results  $f_1$  and  $f_2$  with errors  $e_1 = c - f_1$  and  $e_2 = c - f_2$ . Then  $x = g_1 e_2 - g_2 e_1 / e_2 - e_1$ . Example: A quantity is such that when you add one-third of it to one-quarter of it, the result is 14. What is the quantity?

Guess  $g_1 = 12$ :  $12 \cdot \frac{1}{3} + 12 \cdot \frac{1}{4} = 4 + 3 = 7$ , error  $e_1 = 14 - 7 = 7$  Guess  $g_2 = 24$ :  $24 \cdot \frac{1}{3} + 24 \cdot \frac{1}{4} = 8 + 6 = 14$ , error  $e_2 = 14 - 14 = 0$   $x = 12 \cdot 0 - 24 \cdot 7 = -168$

## Chapter 5: On Solving Cubic Equations

$x^3 + px = q$ .

Al-Kāshī presents methods for solving cubic equations.

For the equation  $x^3 + px = q$ , he uses an iterative method

Iterative Solution of  $x^3 + px = q$  ( : )

Rearrange as:  $x = \sqrt[3]{q - px}$  Iteration 1. Start with initial guess  $x_0$ .

2. Compute  $x_{n+1} = \sqrt[3]{q - p x_n}$ .

3. Repeat until  $|x_{n+1} - x_n| < \epsilon$ .

Example: Solve  $x^3 + 2x = 20$  Start with  $x_0 = 2$   $20 - 2 \cdot 2 = 16 \approx 2.52$   $20 - 2 \cdot 2.52 = 14.96 \approx 2.466$   $20 - 2 \cdot 2.466 = 15.068 \approx 2.472$   $20 - 2 \cdot 2.472 = 15.056 \approx 2.471$  Solution:  $x \approx 2.471$  This iterative method for solving cubic equations is essentially what is now known as fixed-point iteration. AlKāshī used a similar approach in his *Risāla al-Watar wal-Jayb* to compute  $\sin(1^\circ)$  to unprecedented accuracy.

Tables in the *Miftāh al-Ḥisāb* Al-Kāshī notes in his preface I placed for most operations a model in tabular form so that they may be easily grasped by the intelligent. And all the tables placed in this book — “my excuse is my frankness, and its sweet and bitter are part of me” — except seven tables.

The Seven Auxiliary Tables Table 1: Table of products of numbers below ten — A basic multiplication table.

Table 2: The Network ( ) for multiplication — A lattice multiplication aid.

Table 3: Table of powers of places — Place value reference.

Table 4: Example of the agreement of roots — Root extraction reference.

Table 5: Table for knowing the rank of the product and quotient of division — Place value computation aid.

Table 6: Table of sines — Essential for astronomical calculations.

Table 7: Table for knowing the parity of the product and division — Even/odd determination.

Sample: Multiplication Table (Table 1)

Biographical Note on Al-Kāshī . ) 1380 Jamshīd Ghiyāth al-Dīn ibn Masūd ibn Mahmūd alṬabīb al-Kāshī was born in Kāshān, in central Iran, around

### 1380 CE. His given name was Jamshīd; his father's name

was Masūd and his grandfather's name was Mahmūd.

He is known as al-Kāshī (or more precisely al-Kāshānī, from Kāshān), and by his honorific title Ghiyāth al-Dīn (“the rescuer of the religion”).

Key Dates in Al-Kāshī's Life Date Event ~1380 Born in Kāshān, Iran 1407 Completed *Sullam al-Samā'* (Ladder of the Heavens)

1410–11 Wrote *Mukhtaṣar dar 'ilm-i hay'at* (Compendium of Astronomy)

1413-14 Completed Zīj al-Khāqānī, dedicated to Ulugh Beg ~1420 Moved to Samarkand at Ulugh Beg's invitation 1424 Completed Risāla al-Muḥīṭiyya (calculation of  $\pi$ )

## 2 March 1427

Completed Miftāh al-Ḥisāb 1429 Wrote Risāla al-Watar wal-Jayb (sin 1 )

## 22 June 1429

Died in Samarkand (Ramadan 19, 832 AH)

Al-Kāshī's Other Works 1. Sullam al-Samā' (Ladder of the Heavens, 1407) — On resolving difficulties in determining distances and sizes of celestial bodies.

2. Mukhtaṣar dar 'ilm-i hay'at (Compendium of Astronomy, 1410-11) — A Persian work on astronomy.

3. Zīj al-Khāqānī (Khāqānī Astronomical Tables, 1413-14) — Dedicated to Ulugh Beg.

4. Risāla al-Muḥīṭiyya (Treatise on the Circumference, 1424) — The computation of  $\pi$  to 16 decimal places.

5. Risāla al-Watar wal-Jayb (Treatise on the Chord and Sine, 1429) — Computing  $\sin(1^\circ)$  to unprecedented accuracy.

6. Nuzhat al-Ḥadā'iq (Garden Promenade) — On the construction and use of the Ṭabaq al-Manāṭiq instrument.

7. Zīj al-Tashīlāt (Tables of Facilitation) — Simplified astronomical tables.

In the words of E.S. Kennedy: "Al-Kāshī was first and foremost a master computer of extraordinary ability, witness his facile use of pure sexagesimals, his wide application of iterative algorithms, and his sure touch in so laying out a computation that he controlled the maximum error and maintained a check at all stages."

Testimonies of Modern Scholars "Key to Arithmetic is the crowning achievement of Islamic arithmetic." — J.L. Berggren "It was only 1948 that Luckey presented the first extensive and detailed study of al-Kāshī's work, and the traditional historical picture was explicitly shaken." — R. Rashed "In the richness of its contents and in the application of arithmetical and algebraic methods to the solution of various problems, including several geometric ones, and in the clarity and elegance of exposition, this voluminous textbook [that is, Miftāh] is one of the best in the whole of medieval literature; it attests to both the

author's erudition and his pedagogical ability. Because of its high quality the Miftāh was often recopied and served as a manual for hundreds of years." — A.P. Youschkevitch & B.A. Rosenfeld "It is in the work of Ghiyāth al-Dīn Jamshīd al-Kāshī in the early fifteenth century that we first see both a total command of the idea of decimal fractions and a convenient notation for them." — V.J. Katz "The power and elegance of computational algorithms developed by Islamic mathematicians of the ninth through the fifteenth centuries has only recently come to be appreciated. The most successful of these computers was the Iranian scientist, Jamshīd Ghiyāth al-Dīn al-Kāshī."

— E.S. Kennedy "In the determination of  $\pi$ , and in computational mathematics as a whole, al-Kāshī was a pioneer." — J.P. Hogendijk "Key to Arithmetic was used for centuries by astronomers, architects, artisans, surveyors, and merchants as a textbook in the Islamic world." — J. Freely

Note on the Manuscripts and Editions The Miftāh al-Ḥisāb was preserved in numerous manuscript copies, reflecting its importance and wide use. Key manuscripts include  
Manuscript Date Location Nuruosmaniye 2967

### **854 H / 1450 CE**

Istanbul, Nuruosmaniye Library Esad Efendi 3768

### **882 H / 1477 CE**

Istanbul, Süleymāniye Library Hüsrev Paşa 256

### **889 H / 1484 CE**

Istanbul, Süleymāniye Library Aya Sofya 2713

### **896 H / 1491 CE**

Istanbul, Süleymāniye Library Leiden Or. 178 17th century Leiden University Library Printed Editions 1. 1887 Arabic edition — The present edition, published in the Ottoman Empire. This was the first printed edition of the complete Arabic text.

2. 1951 German edition — P. Luckey, *Die Rechenkunst bei Gamschid b. Mas'ud al-Kasi* (Wiesbaden: Franz Steiner). The first major modern study.

3. 1960 English edition — A. Dakhel, *The Extraction of the n-th Root in the Sexagesimal Notation* (Beirut: American University of Beirut). Study of Chapter 5, Treatise 3.

4. 1969 Arabic edition — A.S. al-Dimirdāsh and M.H. al-Ḥifnī al-Shaykh, editors (Cairo: Dār al-Kātib al-'Arabī).

5. 1977 Arabic edition — N. Nābulsi, editor (Damascus: University of Damascus Press).

6. 1956 Russian translation — B.A. Rosenfeld, V.S. Segal, A.P. Youschkevitch, *Klyuch Arifmetiki* (Moscow).

The end of what has been compiled from the Miftāh al-Ḥisāb of Jamshīd al-Kāshī

# Kitāb al-Mukhtaṣar fī Ḥisāb al-Jabr wa-l-Muqābala

كتاب المختصر في حساب الجبر والمقابلة

*The Compendious Book on Calculation  
by Completion and Balancing*

And on the laws of inheritance (al-farā'id) and the distribution of legacies

**Muḥammad ibn Mūsā al-Khwārizmī**

محمد بن موسى الخوارزمي

Bilingual Edition — الطبعة الثنائية اللغة

Arabic Original (left) ↔ English Translation (right)

Adapted from Frederic Rosen's translation (London, 1831)

With modern mathematical notation and explanatory notes

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# The Author's Preface

الرحيم. الرحمن الله بسم

الحمد لله على نعمه لأهل طاعته، بأدائها بقدرته على عباده المؤمنين به، ليثيبهم بها ثواباً، ويحفظهم بها عن التغيير، ويعرفهم قدرته، ويذلهم لعظمته. أرسل محمداً صلى الله عليه وسلم برسالته بعد انقطاع الرسل، وذهاب العدل، وفساد النُصُل. وقد كان العلماء في الأزمان الخالية وفي الأمم السالفة يؤلفون الكتب في أبواب العلوم وفروع المعرفة، يتذكرون من بعدهم، ويرجون الأجر بقدر ما كان منهم، ويثقون بما يحصل لهم من الشكر والإجلال والإعظام، راضين ولو بقليل من الثناء، قليلاً مقارنةً بما بذلوه من الجهد والمشقة في كشف خفايا العلم وإزالة ستائره. وذلك الحب للعلم الذي ميز الله به الإمام المأمون أمير المؤمنين (مع ما منَّ به عليه من الخلافة بالإجماع، واللباس الذي ألبسه إياه، والزينة التي زينته بها)، وذلك اللطف والتودد الذي يديه للعلماء، وتلك السرعة التي يحميها وينصرها في توضيح المبهات وإزالة الصعوبات — قد حملني على أن أولف مختصراً في الجبر والمقابلة، أقتصر فيه على ما هو أيسر وأنفع في الحساب، مما يحتاج إليه الناس دائماً في مسائل الميراث والعطايا والقسمة والخصومات والتجارة وكل ما يتعاملون به فيما بينهم، أو في مساحة الأرضين وحفر الأنهار ومسائل الهندسة وغيرها من أنواع الأشياء المختلفة الأصناف.

**I**n the Name of God, gracious and merciful! Praised be God for His bounty towards those who deserve it by their virtuous acts: in performing which, as by Him prescribed to His adoring creatures, we express our thanks, and render ourselves worthy of the continuance (of His mercy), and preserve ourselves from change: acknowledging His might, bending before His power, and revering His greatness! He sent Mohammed (on whom may the blessing of God repose!) with the mission of a prophet, long after any messenger from above had appeared, when justice had fallen into neglect, and when the true way of life was sought for in vain.

The learned in times which have passed away, and among nations which have ceased to exist, were constantly employed in writing books on the several departments of science and on the various branches of knowledge, bearing in mind those that were to come after them, and hoping for a reward proportionate to their ability, and trusting that their endeavours would meet with acknowledgment, attention, and remembrance—content as they were even with a small degree of praise; small, if compared with the pains which they had undergone, and the difficulties which they had encountered in revealing the secrets and obscurities of science.

That fondness for science, by which God has distinguished the Imām al-Mā'mun, the Commander of the Faithful (besides the caliphate which He has vouchsafed unto him by lawful succession, in the robe of which He has invested him, and with the honours of which He has adorned him), that affability and condescension which he shows to the learned, that promptitude with which he protects and supports them in the elucidation of obscurities and in the removal of difficulties—has encouraged me to compose a short work on Calculating by (the rules of) Completion and Reduction, confining it to what is easiest and most useful in arithmetic, such as men constantly require in cases of inheritance, legacies, partition, law-suits, and trade, and in all their dealings with one another, or where the measuring of

lands, the digging of canals, geometrical computation, and other objects of various sorts and kinds are concerned.

## Chapter 1

# On Numbers and the Three Kinds

### والمقابلة الجبر في الأعداد أقسام

لما نظرت فيما يحتاج إليه الناس في حساباتهم، وجدت أن ذلك كله عدد. ووجدت أيضاً أن كل عدد مؤلف من الوحدات، وأنه لا يخلو عدد من أن ينقسم إلى وحدات. وعلمت أن كل عدد يعدّ من الواحد إلى العشرة يزيد على الذي قبله بواحد، ثم يضاعف العشر ويثلث، كما فعل بالوحدات، فيكون عشرون وثلاثون وهكذا إلى المائة، ثم يضاعف المائة ويثلثها كما فعل بالعشرات والوحدات إلى الألف، ثم يكرر الألف كذلك في كل عدد مركب، وهكذا إلى أقصى حد في العدد. فوجدت أن الأعداد التي يحتاج إليها في حساب الجبر والمقابلة ثلاثة أنواع: جذور، وأموال، وأعداد مفردة لا تتعلق بجذر ولا بمال. أما الجذر فهو كل مقدار يضرب في نفسه مؤلفاً من وحدات أو أعداد صاعدة أو كسور نازلة. وأما المال فهو جميع ما حصل من ضرب الجذر في نفسه. وأما العدد المفرد فهو كل عدد يذكر بغير نسبة إلى جذر ولا مال.

When I considered what people generally want in calculating, I found that it always is a number. I also observed that every number is composed of units, and that any number may be divided into units. Moreover, I found that every number, which may be expressed from one to ten, surpasses the preceding by one unit: afterwards the ten is doubled or tripled, just as before the units were: thus arise twenty, thirty, etc., until a hundred; then the hundred is doubled and tripled in the same manner as the units and the tens, up to a thousand; then the thousand can be thus repeated at any complex number; and so forth to the utmost limit of numeration.

I observed that the numbers which are required in calculating by Completion and Reduction are of **three kinds**, namely, *roots* (جذور), *squares* (أموال), and *simple numbers* (أعداد مفردة) relative to neither root nor square.

### The Three Kinds

1. A **root** (*jadhr* / جذر) is any quantity which is to be multiplied by itself, consisting of units, or numbers ascending, or fractions descending.<sup>1</sup>
2. A **square** (*māl* / مال) is the whole amount of the root multiplied by itself.<sup>2</sup>
3. A **simple number** (*'adad mufrad* / عدد مفرد) is any number which may be pronounced without reference to root or square.

<sup>1</sup>[Ed.] In modern notation:  $x$ .

<sup>2</sup>[Ed.] In modern notation:  $x^2$ .

## Chapter 2

# The Six Canonical Cases

### Introduction to the Six Cases

تجاء. كيف يئنت وقد الجذور، تصف أن إلى تحتاج لا منها ثلاث الكآب. هذا مقدمة في ذكرتها التي الست الحالات هي هذه النصف. سبب يبين حالة لكل الشكل ذكر مع مفصلة أيضاً أئينها أن فأرى الجذور، نصف فيها يجب التي الأخرى الثلاث وأما الحالات الست:

1. المال يساوي الأصول
2. المال يساوي عدداً
3. الأصول تساوي عدداً
4. مال وأصول يساوي عدداً
5. مال وعدد يساوي أصولاً
6. أصول وعدد يساوي مالاً

From these three kinds, **six cases** arise—three simple and three compound. Al-Khwārizmī presents them in a well-defined order that would become standard for centuries of algebraic practice.

#### The Six Canonical Cases

**Case I: Squares equal to Roots:**  $ax^2 = bx$

**Case II: Squares equal to Numbers:**  $ax^2 = c$

**Case III: Roots equal to Numbers:**  $bx = c$

**Case IV: Squares and Roots equal to Numbers:**  $ax^2 + bx = c$

**Case V: Squares and Numbers equal to Roots:**  $ax^2 + c = bx$

**Case VI: Roots and Numbers equal to Squares:**  $bx + c = ax^2$

## 2.1 The Three Simple Cases

الأصول يساوي المال الأولى: الحالة

مثال: «مال يساوي خمسة أصول». فجذر المال خمسة، والمال خمسة وعشرون، وهو يساوي خمسة أضعاف جذره.  
ومثال: «ثلث مال يساوي أربعة أصول». فالمال كله يساوي اثني عشر أصلاً؛ وهو مائة وأربعة وأربعون، وجذره اثنا عشر.  
ومثال: «خمسة أموال تساوي عشرة أصول». فال مال واحد يساوي أصلين؛ وجذر المال اثنان، وماله أربعة.

**Case I: Squares equal to Roots**

*Example:* “A square is equal to five roots of the same.” The root of the square is five, and the square is twenty-five, which is equal to five times its root.

*Example:* “One third of the square is equal to four roots.” Then the whole square is equal to twelve roots; that is a hundred and forty-four; and its root is twelve.

*Example:* “Five squares are equal to ten roots.” Then one square is equal to two roots; the root of the square is two, and its square is four.

$$x^2 = 5x \Rightarrow x = 5$$

$$\frac{1}{3}x^2 = 4x \Rightarrow x = 12 \quad (2.1)$$

$$5x^2 = 10x \Rightarrow x = 2$$

عدداً يساوي المال الثانية: الحالة

مثال: «مال يساوي تسعة». فهذا مال وجذره ثلاثة.

ومثال: «خمسة أموال تساوي ثمانين». فال مال واحد يساوي خمساً وثمانين، وهي ستة عشر.

ومثال: «نصف مال يساوي ثمانية عشر». فالمال ستة وثلاثون، وجذره ستة.

**Case II: Squares equal to Numbers**

*Example:* “A square is equal to nine.” Then this is a square, and its root is three.

*Example:* “Five squares are equal to eighty.” Then one square is equal to one-fifth of eighty, which is sixteen.

*Example:* “The half of the square is equal to eighteen.” Then the square is thirty-six, and its root is six.

$$x^2 = 9 \Rightarrow x = 3$$

$$5x^2 = 80 \Rightarrow x = 4 \quad (2.2)$$

$$\frac{1}{2}x^2 = 18 \Rightarrow x = 6$$

عدداً تساوي الأصول الثالثة: الحالة

مثال: «أصل واحد يساوي ثلاثة عدداً». فالأصل ثلاثة، وماله تسعة.

ومثال: «أربعة أصول تساوي عشرين». فأصل واحد يساوي خمسة، والمال الذي يتركب منه خمسة وعشرون.

ومثال: «نصف أصل يساوي عشرة». فالأصل كله يساوي عشرين، والمال الذي يتركب منه أربعمئة.

### Case III: Roots equal to Numbers

*Example:* “One root equals three in number.” Then the root is three, and its square nine.

*Example:* “Four roots are equal to twenty.” Then one root is equal to five, and the square to be formed of it is twenty-five.

*Example:* “Half the root is equal to ten.” Then the whole root is equal to twenty, and the square which is formed of it is four hundred.

$$x = 3$$

$$4x = 20 \Rightarrow x = 5$$

$$\frac{1}{2}x = 10 \Rightarrow x = 20$$

(2.3)

## 2.2 The Three Compound Cases

عدداً يساوي وأصول مال الرابعة: الحالة

مثال: «مال وعشرة أصوله تساوي تسعة وثلاثين درهماً». يعني: ما هو المال الذي إذا أُضيف إليه عشرة من جذوره بلغ تسعة وثلاثين؟  
الحل: نصف عدد الجذور، فيكون خمسة، ونضربه في نفسه فيكون خمسة وعشرين، فنضيفه إلى تسعة وثلاثين فيكون أربعة وستين،  
فنأخذ جذره فيكون ثمانية، فننقص منه نصف عدد الجذور وهو خمسة، يبقى ثلاثة، وهذا هو جذر المال المطلوب، والمال نفسه تسعة.  
والحل واحد في مائة أو ثلاثة أو أكثر أو أقل؛ فتردها إلى مال واحد، وعلى ذلك ترد الأصول والأعداد الملحق بها.

### Case IV: Squares and Roots equal to Numbers

*Example:* "One square, and ten roots of the same, amount to thirty-nine dirhems."

**Solution:** You halve the number of the roots, which in the present instance yields five. This you multiply by itself; the product is twenty-five. Add this to thirty-nine; the sum is sixty-four. Now take the root of this, which is eight, and subtract from it half the number of the roots, which is five; the remainder is three. This is the root of the square which you sought for; the square itself is nine.

**Algorithm for Case IV**

$$x^2 + bx = c \implies x = \sqrt{\left(\frac{b}{2}\right)^2 + c} - \frac{b}{2} \quad (2.4)$$

*Applied:*  $x^2 + 10x = 39 \Rightarrow x = \sqrt{25 + 39} - 5 = 8 - 5 = 3$

The solution is the same when two squares or three, or more or less be specified; you reduce them to one single square, and in the same proportion you reduce also the roots and simple numbers which are connected therewith.

درهماً». وأربعين ثمانية يساوي أصول وعشرة «مالان مثال:

يجب أن ترد المائتين إلى واحد؛ وإنما المال الواحد من المائتين نصفهما. ثم ترد كل ما ذكر في المسألة إلى نصفه: «مال وخمسة أصول  
يساوي أربعة وعشرين درهماً». الآن نصف عدد الجذور؛ النصف اثنان ونصف. اضربه في نفسه: الناتج ستة وربع. أضفه إلى أربعة  
وعشرين؛ المجموع ثلاثون وربع. خذ جذره؛ هو خمسة ونصف. انقص منه نصف عدد الجذور، وهو اثنان ونصف؛ الباقي ثلاثة.  
وهذا جذر المال، والمال نفسه تسعة.

*Example:* "Two squares and ten roots are equal to forty-eight dirhems."

You must at first reduce the two squares to one; and you know that one square of the two is the moiety of both. Then reduce everything mentioned in the statement to its half: "a square and five roots of the same are equal to twenty-four dirhems." Now halve the number of the roots; the moiety is two and a half. Multiply that by itself: the product is six and a quarter. Add this to twenty-four; the sum is thirty and a quarter. Take the root of this; it is five and a half. Subtract from this the moiety of the number of the roots, that is two and a half; the remainder is three. This is the root of the square, and the square itself is nine.

$$\begin{aligned} 2x^2 + 10x &= 48, \\ x^2 + 5x &= 24, \\ x &= \sqrt{6\frac{1}{4} + 24} - 2\frac{1}{2} = 3. \end{aligned} \quad (2.5)$$

أصلاً يساوي وعدد مال الخامسة: الحالة

مثال: «مال وواحد وعشرون عدداً يساوي عشرة أصوله».

الحل: نصف عدد الجذور فيكون خمسة، ونضربه في نفسه فيكون خمسة وعشرين، فننقص منه الواحد والعشرين الملحقين بالمال، يبقى أربعة، فنستخرج جذرها فيكون اثنين، فننقصه من نصف عدد الجذور وهو خمسة، يبقى ثلاثة، وهذا جذر المال المطلوب، والمال تسعة. أو تضيف الجذر إلى نصف الجذور، فيكون سبعة، وهذا جذر المال المطلوب، والمال تسعة وأربعون. واعلم أنه إذا نصفت الجذور وضربت النصف في نفسه، فإن كان الناتج أقل من عدد الدراهم المصقة بالمال، فالمسألة مستحيلة؛ وإن كان الناتج مساوياً للدراهم، فجذر المال يساوي نصف الجذور وحده، بلا زيادة ولا نقصان.

## Case V: Squares and Numbers equal to Roots

*Example:* "A square and twenty-one in numbers are equal to ten roots of the same square."

**Solution:** Halve the number of the roots; the moiety is five. Multiply this by itself; the product is twenty-five. Subtract from this the twenty-one which are connected with the square; the remainder is four. Extract its root; it is two. Subtract this from the moiety of the roots, which is five; the remainder is three. This is the root of the square which you required, and the square is nine. **Or** you may add the root to the moiety of the roots; the sum is seven; this is the root of the square which you sought for, and the square itself is forty-nine.

### Algorithm for Case V

$$x^2 + c = bx \implies x = \frac{b}{2} \pm \sqrt{\left(\frac{b}{2}\right)^2 - c} \quad (2.6)$$

*Applied:*  $x^2 + 21 = 10x \implies x = 5 \pm \sqrt{25 - 21} = 5 \pm 2 = 3 \text{ or } 7$

When you meet with an instance which refers you to this case, try its solution by addition, and if that do not serve, then subtraction certainly will. For in this case both addition and subtraction may be employed, which will not answer in any other of the three cases in which the number of the roots must be halved.

**Note on impossibility:** And know that, when in a question belonging to this case you have halved the number of the roots and multiplied the moiety by itself, if the product be **less** than the number of dirhems connected with the square, then the instance is **impossible**; but if the product be **equal** to the dirhems by themselves, then the root of the square is equal to the moiety of the roots alone, without either addition or subtraction.



ملاً يساوي وعدد أصول السادسة: الحالة

مثال: «ثلاثة أصول وأربعة أعداد يساوي ملاً».

الحل: نصف الجذور فيكون واحداً ونصفاً، ونضربه في نفسه فيكون اثنين وربعاً، فنضيفه إلى الأربعة فيكون ستة وربعاً، فنستخرج جذرها فيكون اثنين ونصفاً، فنضيفه إلى نصف الجذور وهو واحد ونصف فيكون أربعة، وهذا جذر المال، والمال ستة عشر.

## Case VI: Roots and Numbers equal to Squares

*Example:* “Three roots and four of simple numbers are equal to a square.”

**Solution:** Halve the roots; the moiety is one and a half. Multiply this by itself; the product is two and a quarter. Add this to the four; the sum is six and a quarter. Extract its root; it is two and a half. Add this to the moiety of the roots, which was one and a half; the sum is four. This is the root of the square, and the square is sixteen.

### Algorithm for Case VI

$$bx + c = x^2 \implies x = \frac{b}{2} + \sqrt{\left(\frac{b}{2}\right)^2 + c} \quad (2.7)$$

*Applied:*  $3x + 4 = x^2 \Rightarrow x = 1\frac{1}{2} + \sqrt{2\frac{1}{4} + 4} = 1\frac{1}{2} + 2\frac{1}{2} = 4$

## Chapter 3

# Geometrical Demonstrations

نصف إلى تحتاج لا منها ثلاثاً أن أظهرت وقد الآن. فُسِّرت وقد الكتاب. هذا مقدمة في ذكرتها التي الست الحالات هي هذه حالة كل في أضلع مفصّلة، بأبواب أيّنها أن فأرى الجذور، نصف فيها يجب التي الأخرى الثلاث وأما تجاب. كيف وعلت الجذور، النصف. سبب يبين شكلاً

البرهان على الحالة الرابعة: «مال وعشرة أصول يساوي تسعة وثلاثين»

الشكل المبين لذلك هو مربع، أضلاعه مجهولة. وهو يمثل المال، جذره المطلوب معرفته. وهذا الشكل أ ب، وكل ضلع منه يعتبر واحداً من أصوله؛ وإذا ضربت أحد هذه الأضلاع في أي عدد، فإن مقدار ذلك العدد يعتبر عدد الأصول التي تُضاف إلى المال. ونحن نأخذ ربع العشرة، أي اثنين ونصف، ونضمّها إلى كل من الأضلاع الأربعة للشكل. فع المربع الأصلي أ ب، تُضمّ أربعة متوازيات أضلاع، طول كل منها ضلع من المربع، وعرضها اثنان ونصف؛ وهي المتوازيات ج، د، ط، ك. فليكن لدينا مربع أضلاعه متساوية ومجهولة؛ ولكن في كل من زواياه الأربع ينقص مربع صغير اثنين ونصف في اثنين ونصف. ولتعويض هذا النقص وإتمام المربع، يجب أن نضيف أربعة أمثال مربع اثنين ونصف، أي خمسة وعشرين.

**T**hese are the six cases which I mentioned in the introduction to this book. They have now been explained. I have shown that three among them do not require that the roots be halved, and I have taught how they must be resolved. As for the other three, in which halving the roots is necessary, I think it expedient, more accurately, to explain them by separate chapters, in which a figure will be given for each case, to point out the reasons for halving.

### Demonstration of Case IV: "A Square and Ten Roots equal to 39"

The figure to explain this is a **quadrate**, the sides of which are unknown. It represents the square, the root of which you wish to know. This is the figure *AB*, each side of which may be considered as one of its roots; and if you multiply one of these sides by any number, then the amount of that number may be looked upon as the number of the roots which are added to the square.

Each side of the quadrate represents the root of the square; and, as in the instance the roots were connected with the square, we may take one-fourth of ten, that is to say, two and a half, and combine it with each of the four sides of the figure. Thus with the original quadrate *AB*, four new parallelograms are combined, each having a side of the quadrate as its length, and the number of two and a half as its breadth; they are the parallelograms *C*, *G*, *T*, and *K*.

We have now a quadrate of equal, though unknown sides; but in each of the four corners of which a square piece of two and a half multiplied by two and a half is wanting. In order to compensate for this want and to complete the quadrate, we must add (to that which we have already) four times the square of two and a half, that is, twenty-five.

### Geometric Proof of Case IV — Completion of the Square

$$\text{Given: } x^2 + 10x = 39$$

|                                                               |   |                                                               |
|---------------------------------------------------------------|---|---------------------------------------------------------------|
| Square $AB$ (the unknown square)                              | = | $x^2$                                                         |
| + 4 parallelograms around it                                  | = | $4 \times (2\frac{1}{2} \times x) = 10x$                      |
| Subtotal                                                      | = | 39 (given)                                                    |
| + 4 corner squares ( $2\frac{1}{2} \times 2\frac{1}{2}$ each) | = | $4 \times 6\frac{1}{4} = 25$                                  |
| Great Square $DH$                                             | = | $39 + 25 = 64$                                                |
| Side of Great Square                                          | = | $\sqrt{64} = 8$                                               |
| $x = 8 - 2 \times 2\frac{1}{2} = 8 - 5$                       | = | <span style="border: 1px solid black; padding: 2px;">3</span> |

تسعة يساوي أصول، للعشرة المثلثة به، المحيطة الأربع المتوازيات مع للمال، الممثل المربع أي الأول، الشكل أن المسألة (من ونعلم فإننا ه، د الكبير المربع تكمل التي ب، أ الشكل زوايا في الأربع المربعات كفاءة وهي وعشرين، خمسة إليها أضفنا فإذا عدداً. وثلاثين أي العشرة، ربع ضعف نقصنا فإذا ثمانية. أي جذره، هو الكبير المربع هذا أضلاع أحد وستين. أربعة يصير المجموع هذا أن نعلم الشكل ضلع أو المال، جذر هو وهذا ثلاثة، يكون الضلع هذا بقية فإن ه، د الكبير المربع ضلع طرفي من كما الثمانية، من خمسة، ب. أ الأصلي

ويلاحظ أننا نصف عدد الجذور، ونضيف حاصل ضرب النصف في نفسه إلى العدد تسعة وثلاثين، لنكمل المربع الكبير في زواياه الأربع؛ لأن ربع أي عدد مضروباً في نفسه ثم في أربعة، يساوي حاصل ضرب نصف ذلك العدد في نفسه.

We know (by the statement) that the first figure, namely, the quadrate representing the square, together with the four parallelograms around it, which represent the ten roots, is equal to thirty-nine of numbers. If to this we add twenty-five, which is the equivalent of the four quadrates at the corners of the figure  $AB$ , by which the great figure  $DH$  is completed, then we know that this together makes sixty-four. One side of this great quadrate is its root, that is, eight. If we subtract twice a fourth of ten, that is five, from eight, as from the two extremities of the side of the great quadrate  $DH$ , then the remainder of such a side will be three, and that is the root of the square, or the side of the original figure  $AB$ .

It must be observed, that we have halved the number of the roots, and added the product of the moiety multiplied by itself to the number thirty-nine, in order to complete the great figure in its four corners; because the fourth of any number multiplied by itself, and then by four, is equal to the product of the moiety of that number multiplied by itself.

**Note:** This is the celebrated method of “**completion of the square**” (إتمام المربع), which gives algebra its very name: *al-jabr* refers to the “restoration” of the deficient corners. The identity underlying the proof is:  $(x + \frac{b}{2})^2 = x^2 + bx + (\frac{b}{2})^2$ .

## Demonstration of Case V: “A Square and 21 equal to 10 Roots”

أصول «عشرة يساوي وعشرون وواحد» مال الخامسة: الحالة على البرهان  
 تمثل المال بمربع أ د، طول ضلعه مجهول لنا. ونلحق به متوازي أضلاع، عرضه يساوي أحد أضلاع المربع أ د، كالضلع ه ن. وهذا  
 المتوازي هو ه ب. طول الشككين معاً يساوي الخط ه س. ونعلم أن طوله هو عشرة أعداد؛ فكل مربع له أضلاع متساوية وزوايا  
 متساوية، وأحد أضلاعه مضروباً بواحد هو جذر المربع، أو مضروباً باثنين فهو ضعف جذره.  
 فكما قيل: إن مالاً واحداً وعشرين عدداً يساوي عشرة أصول، فإننا نستنتج أن طول الخط ه س يساوي عشرة أعداد، إذن الخط  
 س د يمثل جذر المال. والآن نقسم الخط س ه إلى نصفين عند النقطة ج: فالخط ج س يساوي ه ج.  
 ونعلم أن الرباعي ه ب يمثل الواحد والعشرين عدداً التي أضيفت إلى المربع. ووجد المربع الكلي م ط يساوي خمسة وعشرين. فإذا  
 نقصنا من هذا المربع م ط المربعين ه ط وم ر، وهما يساويان واحداً وعشرين، يبقى مربع صغير ر، يمثل الفرق بين خمسة وعشرين  
 وواحد وعشرين. وهو أربعة؛ وجذره، الممثل بالخط ر ج، المساوي ل ج أ، هو اثنان.

We represent the square by a quadrate  $AD$ , the length of whose side we do not know. To this we join a parallelogram, the breadth of which is equal to one of the sides of the quadrate  $AD$ , such as the side  $HN$ . This parallelogram is  $HB$ . The length of the two figures together is equal to the line  $HC$ . We know that its length is ten of numbers; for every quadrate has equal sides and angles, and one of its sides multiplied by a unit is the root of the quadrate, or multiplied by two it is twice the root of the same.

As it is stated, therefore, that a square and twenty-one of numbers are equal to ten roots, we may conclude that the length of the line  $HC$  is equal to ten of numbers, since the line  $CD$  represents the root of the square. We now divide the line  $CH$  into two equal parts at the point  $G$ : the line  $GC$  is then equal to  $HG$ .

We know that the quadrangle  $HB$  represents the twenty-one of numbers which were added to the quadrate. The whole quadrate  $MT$  was found to be equal to twenty-five. If we now subtract from this quadrate,  $MT$ , the quadrangles  $HT$  and  $MR$ , which are equal to twenty-one, there remains a small quadrate  $KR$ , which represents the difference between twenty-five and twenty-one. This is four; and its root, represented by the line  $RG$ , which is equal to  $GA$ , is two.

### Geometric Proof of Case V

$$\text{Given: } x^2 + 21 = 10x$$

$$MT = \left(\frac{10}{2}\right)^2 = 25$$

$$HT + MR = HB = 21$$

$$KR = 25 - 21 = 4 \Rightarrow \sqrt{KR} = 2$$

$$x = GC - GA = 5 - 2 = 3,$$

$$x = GC + GA = 5 + 2 = 7.$$

الأصلي. المال جذر وهو ثلاثة، أي س؛ أ الخط هو الباقي فإن الجذور، نصف وهو س، ج الخط من اثنين العدد هذا نقصت فإذا  
 مال جذر وهو ر، ج بالخط الممثل سبعة، المجموع فإن الجذور، عدد نصف وهو س، ج الخط إلى اثنين العدد أضفت إذا ولكن  
 المال. ذلك من أصول عشرة كذلك يساوي المجموع فإن المال، هذا إلى وعشرين واحداً أضفت فإذا ذلك، ومع أكبر.

If you subtract this number two from the line  $CG$ , which is the moiety of the roots, then the remainder is the line  $AC$ ; that is to say, three, which is the root of the original square. But if you add the number

two to the line  $CG$ , which is the moiety of the number of the roots, then the sum is seven, represented by the line  $CR$ , which is the root to a larger square. However, if you add twenty-one to this square, then the sum will likewise be equal to ten roots of the same square.

## Demonstration of Case VI: “3 Roots and 4 equal to a Square”

مالاً، يساوي وأربعة أصول «ثلاثة السادسة: الحالة على البرهان  
ليمثل المال رباعياً، أضلاعه مجهولة لنا، وهي متساوية فيما بينها، كما هي زواياه. وهذا هو المربع أ د، الذي يشتمل على الأصول الثلاثة  
والأربعة أعداد المذكورة في هذه المسألة.  
وفي كل مربع أحد أضلاعه مضروباً بواحد هو جذره. والآن نقطع الرباعي ه د من المربع أ د، ونتخذ أحد أضلاعه ه س ثلاثة،  
وهو عدد الأصول. وهو يساوي ر د. فن ثم، يمثل الرباعي ه ب الأربعة أعداد التي تُضاف إلى الأصول. والآن نصف الضلع س  
ه، المساوي لثلاثة أصول، عند النقطة ج؛ من هذا القسم نبني المربع ه ط، وهو حاصل ضرب نصف الأصول (أي واحد ونصف)  
في نفسه، أي اثنين وربع.

Let the square be represented by a quadrangle, the sides of which are unknown to us, though they are equal among themselves, as also the angles. This is the quadrate  $AD$ , which comprises the three roots and the four of numbers mentioned in this instance.

In every quadrate one of its sides, multiplied by a unit, is its root. We now cut off the quadrangle  $HD$  from the quadrate  $AD$ , and take one of its sides  $HC$  for three, which is the number of the roots. The same is equal to  $RD$ . It follows, then, that the quadrangle  $HB$  represents the four of numbers which are added to the roots. Now we halve the side  $CH$ , which is equal to three roots, at the point  $G$ ; from this division we construct the square  $HT$ , which is the product of half the roots (or one and a half) multiplied by themselves, that is to say, two and a quarter.

### Geometric Proof of Case VI

$$\text{Given: } 3x + 4 = x^2$$

$$\text{Square } HT = \left(\frac{3}{2}\right)^2 = 2\frac{1}{4}$$

$$AN + KL = AR = 4 \text{ (the added numbers)}$$

$$GM = HT + (AN + KL) = 2\frac{1}{4} + 4 = 6\frac{1}{4}$$

$$AG = \sqrt{GM} = 2\frac{1}{2}$$

$$x = AG + GC = 2\frac{1}{2} + 1\frac{1}{2} = \boxed{4}$$

## Chapter 4

# On Multiplication of Algebraic Expressions

### المجهولة المقادير ضرب في

سأعلمك الآن كيف تضرب المقادير المجهولة، أي الجذور، بعضها ببعض، إذا كانت وحدها، أو إذا أُضيفت إليها أعداد، أو نُقصت منها أعداد، أو نُقصت هي من أعداد؛ وكيف تجمعها بعضها إلى بعض، أو تنقص بعضها من بعض. كلما أريد ضرب عدد بآخر، وجب تكرار أحدهما بعدد الوحدات التي في الآخر. فإن كان هناك أعداد كبيرة مضافة إلى وحدات، لتُضاف أو تُنقص، فإن أربع ضربات لازمة؛ وهي: الكبير بالكبير، والكبير بالوحدات، والوحدات بالكبير، والوحدات بالوحدات. فإن كانت الوحدات المضافة إلى الأعداد الكبيرة موجبة في كليهما، فالضربة الرابعة موجبة؛ وإن كانت سالبة في كليهما، فالرابعة موجبة أيضاً. ولكن إن كانت إحداها موجبة والأخرى سالبة، فالضربة الرابعة سالبة.

I shall now teach you how to multiply the unknown numbers, that is to say, the roots, one by the other, if they stand alone, or if numbers are added to them, or if numbers are subtracted from them, or if they are subtracted from numbers; also how to add them one to the other, or how to subtract one from the other.

Whenever one number is to be multiplied by another, the one must be repeated as many times as the other contains units.

If there are greater numbers combined with units to be added to or subtracted from them, then **four multiplications** are necessary; namely, the greater numbers by the greater numbers, the greater numbers by the units, the units by the greater numbers, and the units by the units.

If the units, combined with the greater numbers, are both positive, then the last multiplication is positive; if they are both negative, then the fourth multiplication is likewise positive. But if one of them is positive, and one negative, then the fourth multiplication is negative.

### The Fourfold Multiplication Rule

To multiply  $(a + x)$  by  $(b + y)$ :

1. Greater by greater:  $a \times b$
2. Greater by units:  $a \times y$
3. Units by greater:  $x \times b$
4. Units by units:  $x \times y$

Sign rule:  $(+x)(+y) = +xy$     $(-x)(-y) = +xy$     $(+x)(-y) = -xy$     $(-x)(+y) = -xy$

## Worked Examples

واثنين». عشرة في ضرب واحد «عشرة الأول: المثال  
عشرة في عشرة مائة؛ الواحد في العشرة عشرة موجبة؛ الاثنان في العشرة عشرون موجبة؛ والواحد في الاثنين اثنان موجبة؛ المجموع مائة واثنان وثلاثون.  
المثال الثاني: «عشرة ناقص واحد، يضرب في عشرة ناقص واحد».  
عشرة في عشرة مائة؛ السالب واحد في العشرة عشرة سالبة؛ والسالب الآخر في العشرة عشرة سالبة أيضاً، فيصير ثمانون؛ ولكن السالب واحد في السالب واحد موجب، فيصير الناتج واحداً وثمانين.  
المثال الثالث: «عشرة وشيء يضرب في عشرة وشيء».  
عشرة في عشرة مائة؛ وعشرة في الشيء عشرة أشياء؛ وعشرة في الشيء عشرة أشياء أخرى؛ والشيء في الشيء مال موجب؛ فالحاصل كله مائة درهم وعشرون شيئاً ومالاً موجباً.  
المثال الرابع: «عشرة ناقص شيء يضرب في عشرة ناقص شيء».  
عشرة في عشرة مائة؛ وسالب شيء في العشرة سالب عشرة أشياء؛ وسالب شيء في العشرة سالب عشرة أشياء أخرى. وسالب شيء في سالب شيء مال موجب. فالحاصل مائة ومال ناقص عشرين شيئاً.  
المثال الخامس: «عشرة ناقص شيء يضرب في عشرة وزيادة شيء».  
عشرة في عشرة مائة؛ وسالب شيء في العشرة سالب عشرة أشياء؛ وشيء في العشرة عشرة أشياء موجبة؛ وسالب شيء في الشيء سالب مال؛ فالحاصل مائة درهم ناقص مال.

**Example 1:** “Ten and one to be multiplied by ten and two.”

Ten times ten is a hundred; once ten is ten positive; twice ten is twenty positive, and once two is two positive; this altogether makes a hundred and thirty-two.

$$(10 + 1)(10 + 2) = 100 + 10 + 20 + 2 = 132$$

**Example 2:** “Ten less one, to be multiplied by ten less one.”

Then ten times ten is a hundred; the negative one by ten is ten negative; the other negative one by ten is likewise ten negative, so that it becomes eighty: but the negative one by the negative one is one positive, and this makes the result eighty-one.

$$(10 - 1)(10 - 1) = 100 - 10 - 10 + 1 = 81$$

**Example 3:** “Ten and thing to be multiplied by ten and thing”

Then ten times ten is a hundred, and ten times thing is ten things; and again, ten times thing is ten things; and thing multiplied by thing is a square positive, so that the whole product is a hundred dirhems and twenty things and one positive square.

$$(10 + x)(10 + x) = 100 + 10x + 10x + x^2 = 100 + 20x + x^2$$

**Example 4:** “Ten minus thing to be multiplied by ten minus thing”

Then ten times ten is a hundred; and minus thing by ten is minus ten things; and again, minus thing by ten is minus ten things. But minus thing multiplied by minus thing is a positive square. The product is therefore a hundred and a square, minus twenty things.

$$(10 - x)(10 - x) = 100 - 10x - 10x + x^2 = 100 - 20x + x^2$$

**Example 5:** “Ten minus thing to be multiplied by ten and thing”

Then you say, ten times ten is a hundred; and minus thing by ten is ten things negative; and thing by ten is ten things positive; and minus thing by thing is a square negative; therefore, the product is a hundred dirhems, minus a square.

$$(10 - x)(10 + x) = 100 - 10x + 10x - x^2 = 100 - x^2$$



## Chapter 5

# On Addition and Subtraction

والأعداد الجذور وطرح جمع في

اعلم أن جذر مائتين ناقص عشرة، مضافاً إلى عشرين ناقص جذر مائتين، هو عشرة.

وجذر المائتين ناقص العشرة، منقوصاً من عشرين ناقص جذر المائتين، هو ثلاثون ناقص ضعف جذر مائتين؛ وضعف جذر المائتين يساوي جذر ثمانمائة.

ومائة ومال ناقص عشرون أصلاً، مضافاً إلى خمسين وعشرة أصول ناقص مالان، هو مائة وخمسون ناقص مال وناقص عشرة أصول. ومائة ومال ناقص عشرون أصلاً، منقوصاً من خمسين وعشرة أصول ناقص مالان، هو خمسون درهماً وثلاثة أموال ناقص ثلاثون أصلاً.

### Operations on Algebraic Expressions

Know that the root of two hundred minus ten, added to twenty minus the root of two hundred, is just ten.

$$(\sqrt{200} - 10) + (20 - \sqrt{200}) = 10$$

The root of two hundred, minus ten, subtracted from twenty minus the root of two hundred, is thirty minus twice the root of two hundred; twice the root of two hundred is equal to the root of eight hundred.

$$(20 - \sqrt{200}) - (\sqrt{200} - 10) = 30 - 2\sqrt{200} = 30 - \sqrt{800}$$

A hundred and a square minus twenty roots, added to fifty and ten roots minus two squares, is a hundred and fifty, minus a square and minus ten roots.

$$(100 + x^2 - 20x) + (50 + 10x - 2x^2) = 150 - x^2 - 10x$$

A hundred and a square, minus twenty roots, diminished by fifty and ten roots minus two squares, is fifty dirhems and three squares minus thirty roots.

$$(100 + x^2 - 20x) - (50 + 10x - 2x^2) = 50 + 3x^2 - 30x$$

## On Doubling, Tripling, and Halving Roots

ونصفها الجذور مضاعفة في  
 إذا أردت مضاعفة جذر أي مال معلوم أو مجهول، (ومعنى مضاعفته أن تضربه في اثنين) فإنه يكفي أن تضرب اثنين في اثنين، ثم في  
 المال، فجذر الحاصل يساوي ضعف جذر المال الأصلي.  
 وإذا أردت أخذه ثلاثاً، فتضرب ثلاثة في ثلاثة، ثم في المال؛ فجذر الحاصل ثلاثة أمثال جذر المال الأصلي.  
 وإذا أردت أن تجد نصف جذر المال، فلا يحتاج إلا أن تضرب نصفاً في نصف، وهو ربع؛ ثم هذا في المال: فجذر الحاصل يكون  
 نصف جذر المال الأول.

If you require to double the root of any known or unknown square, (the meaning of its duplication being that you multiply it by two) then it will suffice to multiply two by two, and then by the square; the root of the product is equal to twice the root of the original square.

$$2\sqrt{S} = \sqrt{4S}$$

If you require to take it thrice, you multiply three by three, and then by the square; the root of the product is thrice the root of the original square.

$$3\sqrt{S} = \sqrt{9S}$$

If you require to find the moiety of the root of the square, you need only multiply a half by a half, which is a quarter; and then this by the square: the root of the product will be half the root of the first square.

$$\frac{1}{2}\sqrt{S} = \sqrt{\frac{1}{4}S}$$

## Chapter 6

### On Division

والأعداد الجذور قسمة في

إذا أردت قسمة جذر تسعة على جذر أربعة، فتبدأ بقسمة تسعة على أربعة، فيعطي اثنين وربعاً: فجذر ذلك هو العدد المطلوب — وهو واحد ونصف.

وإذا أردت قسمة جذر أربعة على جذر تسعة، فتقسم أربعة على تسعة؛ فهي أربعة أضعاف الوحدة؛ وجذرها اثنان مقسوماً على ثلاثة؛ أي ثلثا الوحدة.

وإذا أردت ضرب جذر تسعة في جذر أربعة، فاضرب تسعة في أربعة؛ فهذا يعطي ستة وثلاثين؛ فنخذ جذرها، فهو ستة؛ وهذا هو جذر تسعة مضروباً في جذر أربعة.

وإذا أردت ضرب ضعف جذر تسعة في ثلاثة أضعاف جذر أربعة، فنخذ ضعف جذر تسعة، بحسب القاعدة المذكورة أعلاه، لتعرف جذر أي مال هو. وتفعّل المثل بخصوص ثلاثة جذور أربعة لتعرف ما يجب أن يكون مال ذلك الجذر. ثم تضرب هذين المالين أحدهما في الآخر، فجذر الحاصل يساوي ضعف جذر تسعة مضروباً في ثلاثة أضعاف جذر أربعة.

If you will divide the root of nine by the root of four, you begin with dividing nine by four, which gives two and a quarter: the root of this is the number which you require—it is one and a half.

$$\frac{\sqrt{9}}{\sqrt{4}} = \sqrt{\frac{9}{4}} = \frac{3}{2} = 1\frac{1}{2}$$

If you will divide the root of four by the root of nine, you divide four by nine; it is four-ninths of the unit; the root of this is two divided by three; namely, two-thirds of the unit.

$$\frac{\sqrt{4}}{\sqrt{9}} = \sqrt{\frac{4}{9}} = \frac{2}{3}$$

If you wish to multiply the root of nine by the root of four, multiply nine by four; this gives thirty-six; take its root, it is six; this is the root of nine, multiplied by the root of four.

$$\sqrt{9} \times \sqrt{4} = \sqrt{9 \times 4} = \sqrt{36} = 6$$

If you wish to multiply twice the root of nine by thrice the root of four, then take twice the root of nine, according to the rule above given, so that you may know the root of what square it is. You do the same with respect to the three roots of four in order to know what must be the square of such a root. You then multiply these two squares, the one by the other, and the root of the product is equal to twice the root of nine, multiplied by thrice the root of four.

$$(2\sqrt{9}) \times (3\sqrt{4}) = \sqrt{36 \times 36} = 36$$

## Chapter 7

# The Six Illustrative Problems

### التوضيحية الست المسائل

قبل أبواب الحساب وأنواعه المتعددة، سأذكر الآن ست مسائل، كأمثلة للحالات الست التي تقدم ذكرها في مطلع هذا الكتاب. وقد أظهرت أن ثلاثاً من هذه الحالات لا تحتاج في حلها إلى نصف الجذور، وقد ذكرت أيضاً أن الحساب بالجبر والمقابلة لا بد أن يورثك إحدى هذه الحالات. وسأخلق هذه المسائل الآن، لتقرب الموضوع إلى الفهم، وتسهل إدراكه، وتجعل الحجج أوضح.

Before the chapters on computation and the several species thereof, I shall now introduce six problems, as instances of the six cases treated of in the beginning of this work. I have shown that three among these cases, in order to be solved, do not require that the roots be halved, and I have also mentioned that the calculating by completion and reduction must always necessarily lead you to one of these cases. I now subjoin these problems, which will serve to bring the subject nearer to the understanding, to render its comprehension easier, and to make the arguments more perspicuous.

### الأولى المسألة

«قسمت عشرة إلى قسمين؛ فضربت أحدهما في الآخر؛ ثم ضربت أحدهما في نفسه، فكان حاصل ضربه في نفسه أربعة أمثال حاصل ضرب أحدهما في الآخر.»

الحل: ليكن أحد القسمين شيئاً، والآخر عشرة ناقص شيء. تضرب الشيء في عشرة ناقص شيء؛ فهو عشرة أشياء ناقص مال. ثم تضربه في أربعة، لأن المسألة تقول «أربعة أمثال». فيكون حاصل أربعين شيئاً ناقص أربعة أموال. ثم تضرب الشيء في شيء، أي أحد القسمين في نفسه. وهذا مال، يساوي أربعين شيئاً ناقص أربعة أموال. فأكله بالجبر بأربعة أموال، وأضفها إلى المال الواحد. فيكون المقابلة: أربعون شيئاً تساوي خمسة أموال؛ فالمال يساوي ثمانية أصول، أي أربعة وستين؛ وجذرها ثمانية.

### First

### Problem

**Statement:** "I have divided ten into two portions; I have multiplied the one of the two portions by the other; after this I have multiplied the one of the two by itself, and the product of the multiplication by itself is four times as much as that of one of the portions by the other."

**Solution:** Suppose one of the portions to be  $x$ , and the other  $10 - x$ . You multiply  $x$  by  $10 - x$ ; it is  $10x - x^2$ . Then multiply it by four, because the instance states "four times as much." The result will be  $40x - 4x^2$ . After this you multiply  $x$  by  $x$ , that is to say, one of the portions by itself. This is a square, which is equal to  $40x - 4x^2$ . Reduce it now by the four squares, and add them to the one square. Then the equation is: forty things are equal to five squares; and one square will be equal to eight roots, that is, sixty-four; the root of this is eight.

$$x^2 = 4x(10 - x) = 40x - 4x^2 \Rightarrow 5x^2 = 40x \Rightarrow x^2 = 8x = 64 \Rightarrow x = 8$$

The portions are:  $x = 8$  and  $10 - x = 2$ . This question leads you to Case I: "squares equal to roots."

## الثانية المسألة

«قسمت عشرة إلى قسمين؛ فضربت كل قسم في نفسه، ثم ضربت العشرة في نفسها؛ فكان حاصل العشرة في نفسها يساوي أحد القسمين مضروباً في نفسه ثم في اثنين وسبعة أضعاف.»

## المسألة الثالثة

«قسمت عشرة إلى قسمين؛ فقسمت أحدهما على الآخر، فكان الخارج أربعة.»

## المسألة الرابعة

«ضربت ثلث شيء ودرهم في ربع شيء ودرهم، فكان الحاصل عشرين.»

## المسألة الخامسة

«قسمت عشرة إلى قسمين؛ وضربت كل قسم في نفسه، فجمعتهما معاً؛ وكان المجموع ثمانية وخمسين درهماً.»

## المسألة السادسة

«ضربت ثلث جذر في ربع جذر، وكان الحاصل يساوي الجذر وأربعة وعشرين درهماً.»

## Second

## Problem

**Statement:** “I have divided ten into two portions: I have multiplied each of the parts by itself, and afterwards ten by itself: the product of ten by itself is equal to one of the two parts multiplied by itself, and afterwards by two and seven-ninths.”

$$100 = x^2 \times 2\frac{7}{9} = x^2 \times \frac{25}{9} \Rightarrow x^2 = \frac{900}{25} = 36 \Rightarrow x = 6$$

The portions are:  $x = 6$  and  $10 - x = 4$ .

## Third

## Problem

**Statement:** “I have divided ten into two parts. I have afterwards divided the one by the other, and the quotient was four.”

$$\frac{10 - x}{x} = 4 \Rightarrow 10 - x = 4x \Rightarrow 10 = 5x \Rightarrow x = 2$$

The portions are:  $x = 2$  and  $10 - x = 8$ .

## Fourth

## Problem

**Statement:** “I have multiplied one-third of thing and one dirhem by one-fourth of thing and one dirhem, and the product was twenty.”

$$\left(\frac{x}{3} + 1\right) \left(\frac{x}{4} + 1\right) = 20 \Rightarrow \frac{x^2}{12} + \frac{7x}{12} + 1 = 20 \Rightarrow x^2 + 7x = 228$$

Halve the roots:  $3\frac{1}{2}$ . Square:  $12\frac{1}{4}$ . Add to 228:  $240\frac{1}{4}$ . Root:  $15\frac{1}{2}$ . Subtract  $3\frac{1}{2}$ :  $x = 12$ .

## Fifth

## Problem

**Statement:** “I have divided ten into two parts; and having multiplied each part by itself, I have added them together; and the sum was fifty-eight dirhems.”

$$x^2 + (10 - x)^2 = 58 \Rightarrow 2x^2 - 20x + 100 = 58 \Rightarrow x^2 + 28 = 11x$$

$$x = \frac{11}{2} \pm \sqrt{\frac{121}{4} - 28} = 5\frac{1}{2} \pm 2\frac{1}{2}. \text{ The portions are: } x = 3 \text{ or } x = 7.$$

## Sixth

## Problem

**Statement:** “I have multiplied one-third of a root by one-fourth of a root, and the product is equal to the root and twenty-four dirhems.”

$$\frac{x}{3} \cdot \frac{x}{4} = x + 24 \Rightarrow \frac{x^2}{12} = x + 24 \Rightarrow x^2 = 12x + 288$$

$$x = 6 + \sqrt{36 + 288} = 6 + 18 = 24$$

## Chapter 8

# On Mercantile Transactions

أعداد الأربعة قاعدة — التجارية المعاملات  
اعلم أن جميع معاملات الناس من بيع وشراء ومصارفة وإجارة، فإنها لا تخلو من قدرين وأربعة أعداد يسأل عنها السائل، وهي المقدار والتمن والكيفية والجميع.  
عدد المقدار يناسب عدد الجميع، وعدد الثمن يناسب عدد الكيفية. ثلاثة من هذه الأربعة دائماً معلومة، وواحد مجهول، وهو الذي يتضمنه قوله «كم؟» وهو المطلوب.

**Y**ou know that all mercantile transactions of people, such as buying and selling, exchange and hire, comprehend always **two notions** and **four numbers**, which are stated by the enquirer; namely, *measure* (مقدار) and *price* (ثمن), and *quantity* (كيفية) and *sum* (جميع).

The number which expresses the measure is inversely proportionate to the number which expresses the sum, and the number of the price inversely proportionate to that of the quantity. Three of these four numbers are always known, one is unknown, and this is implied when the person inquiring says *how much?* (كم؟) and it is the object of the question.

### The Rule of Four Numbers (al-nisba)

| Measure                | Price                  |
|------------------------|------------------------|
| $a$ (known)            | $b$ (known)            |
| $c$ (known or unknown) | $d$ (known or unknown) |

$$a \times b = c \times d \quad (\text{Rule of Three})$$

$$\text{Unknown} = \frac{\text{product of the two proportionate numbers}}{\text{the third known number}}.$$

## Examples

بأربعة؟» كم بستة، «عشرة الأول: المثال

هنا العشرة هي المقدار، والستة هي الثمن، و«كم» تفيد الكيفية المجهولة؛ والأربعة هي عدد الجميع. المقدار (10) يناسب الجميع (4).

المثال الثاني: «عشرة بثمانية، فما هو المبلغ لأربعة؟»

المثال الثالث: «يأخذ عامل أجراً قدره عشرة دراهم في الشهر؛ فكم يكون أجره في ستة أيام؟»

الستة أيام خمس الشهر؛ وحصته من الدراهم يجب أن تكون مناسبة لحصته من الشهر.

**Example 1:** “Ten for six, how much for four?”

Here ten is the measure; six is the price; “how much” implies the unknown quantity; and four is the number of the sum. The measure (10) is inversely proportionate to the sum (4).

$$\text{Quantity} = \frac{10 \times 4}{6} = \frac{40}{6} = 6\frac{2}{3}$$

**Example 2:** “Ten for eight, what must be the sum for four?”

$$\text{Sum} = \frac{8 \times 4}{10} = \frac{32}{10} = 3\frac{1}{5}$$

**Example 3:** “A workman receives a pay of ten dirhems per month; how much must be his pay for six days?”

Six days are one-fifth of the month; and his portion of the dirhems must be proportionate to the portion of the month.

$$\text{Pay} = \frac{10 \times 6}{30} = 2 \text{ dirhems}$$



## Chapter 9

# Mensuration

### والجسم المساحة

اعلم أن معنى قولنا «واحد في واحد» هو المساحة: أي ذراع في ذراع. كل مربع ذو أضلاع متساوية وزوايا متساوية، له ذراع في كل ضلع، فله واحد في مساحته. وإن كان لهذا المربع ذراعان في ضلعه، فإن مساحة المربع أربعة أمثال مساحة مربع ضلعه ذراع واحد.

### المثلثات

إذا ضربت ارتفاع أي مثلث متساوي الأضلاع في نصف القاعدة التي يقف عليها انخط المار بالارتفاع عمودياً، فالحاصل يعطي مساحة ذلك المثلث.

### الدوائر

وفي كل دائرة، حاصل ضرب قطرها في ثلاثة وسبعاً واحداً يساوي المحيط.

ومساحة أي دائرة توجد بضرب نصف المحيط في نصف القطر.

وإذا ضربت قطر أي دائرة في نفسه، ونقصت من الحاصل سبعةً ونصف سبع منه، فإن الباقي يساوي مساحة الدائرة.

### الأجسام

حجم الجسم الرباعي يوجد بضرب الطول في العرض ثم في الارتفاع.

والخروطات والأهرامات، كالمثلثة والرباعية، تحسب بضرب ثلث مساحة القاعدة في الارتفاع.

### المثلث القائم

ولاحظ أنه في كل مثلث قائم الضلعين القصيرين، كل منهما مضروباً في نفسه، والحاصلان مجتمعان، يساويان حاصل الضلع الطويل مضروباً في نفسه.

**K**now that the meaning of the expression

“one by one” is mensuration: one yard (in length) by one yard (in breadth) being understood.

Every quadrangle of equal sides and angles, which has one yard for every side, has also one for its area. Has such a quadrangle two yards for its side, then the area of the quadrangle is four times the area of a quadrangle, the side of which is one yard.

### Triangles

If you multiply the height of any equilateral triangle by the moiety of the basis upon which the line marking the height stands perpendicularly, the product gives the area of that triangle.

Area of triangle = height  $\times \frac{\text{basis}}{2}$

## Circles

In any circle, the product of its diameter, multiplied by three and one-seventh, will be equal to the periphery.

$$C = \pi d \approx \frac{22}{7} \times d$$

The area of any circle will be found by multiplying the moiety of the circumference by the moiety of the diameter.

$$A = \frac{C}{2} \times \frac{d}{2} = \frac{Cd}{4}$$

If you multiply the diameter of any circle by itself, and subtract from the product one-seventh and half one-seventh of the same, then the remainder is equal to the area of the circle.

$$A = d^2 - \frac{d^2}{7} - \frac{d^2}{14} = d^2 \times \frac{11}{14} \approx \frac{\pi}{4} d^2$$

*Note:* This gives  $\pi \approx \frac{22}{7}$ , the well-known approximation.

## Volumes

The bulk of a quadrangular body will be found by multiplying the length by the breadth, and then by the height.

Cones and pyramids, such as triangular or quadrangular ones, are computed by multiplying one-third of the area of the basis by the height.

$$V_{\text{pyramid}} = \frac{1}{3} \times (\text{area of base}) \times \text{height}$$

## The Right Triangle (Pythagorean Theorem)

Observe, that in every rectangular triangle the two short sides, each multiplied by itself and the products added together, equal the product of the long side multiplied by itself.

### The Pythagorean Theorem

$a^2 + b^2 = c^2$  Where  $a$  and  $b$  are the two shorter sides (catheti) and  $c$  is the hypotenuse (long side).

## Chapter 10

# On Legacies and the Laws of Inheritance

والفرائض الوصايا

الجزء الأخير من هذا الكتاب يعالج قوانين الميراث الإسلامية (الفرائض). وهذا الموضوع، على الرغم من أنه يبدو أنه ينتمي إلى الفقه وليس إلى الرياضيات، إلا أنه يتطلب كامل أدوات الجبر لحله عندما تُفرض الوصايا فوق الأسهم القانونية.

في رأس المال والديون

مسألة: «توفي رجل وترك ولدين، وأوصى بثلث رأس ماله لغريب. وترك عشرة دراهم من المال ودينياً قدره عشرة دراهم على أحد الولدين.»

الحل: تسمى المبلغ المأخوذ من الدين شيئاً. فتضيفه إلى رأس المال، وهو عشرة دراهم. فيكون المجموع عشرة زائد شيء. تنقص ثلث هذا، إذ أوصى بثلث ماله: ثلاثة وثلث زائد ثلث شيء. والباقي ستة وثلثان زائد ثلثا شيء. فتقسم هذا بين الولدين: يأخذ كل منهما ثلاثة وثلث زائد ثلث شيء. وهذا يساوي الشيء المطلوب.

The final portion of this work treats of the Islamic laws of inheritance (*al-farā'id*). This subject, while seemingly belonging to jurisprudence rather than mathematics, requires the full apparatus of algebra for its solution when legacies are superimposed upon the statutory shares.

On Capital and Money Lent

**Problem:** "A man dies, leaving two sons behind him, and bequeathing one-third of his capital to a stranger. He leaves ten dirhems of property and a claim of ten dirhems upon one of the sons."

**Solution:** You call the sum which is taken out of the debt  $x$  (thing). Add this to the capital, which is ten dirhems. The sum is  $10 + x$ . Subtract one-third of this, since he has bequeathed one-third of his property:  $3\frac{1}{3} + \frac{1}{3}x$ . The remainder is  $6\frac{2}{3} + \frac{2}{3}x$ . Divide this between the two sons: each receives  $3\frac{1}{3} + \frac{1}{3}x$ . This is equal to the thing which was sought for.

$$3\frac{1}{3} + \frac{1}{3}x = x \Rightarrow 3\frac{1}{3} = \frac{2}{3}x \Rightarrow x = 5$$

The stranger receives 5; and each son receives 5 (the indebted son retains his full debt as a set-off against his share).

## الوصايا من آخر نوع في

مسألة: «توفي رجل وترك أمه وزوجته، وإخوة له اثنان وأختان اثنتان لأب وأم. وأوصى لغريب بتسع رأس ماله.»  
الحل: تحدد أسهم ميراثهم بأخذها من ثمانية وأربعين جزءاً. وإياك أن تأخذ تسعاً من أي رأس مال، يبقى ثمانية أضعافه. فأضف الآن إلى الثمانية الأضعاف الثمن منها، وإلى الثمانية والأربعين أيضاً ثمنها، أي ستة، لإتمام رأس المال. فهذا يعطي أربعة وخمسين. فالشخص الموصى له بالتاسع يأخذ من هذا ستة. والثمانية والأربعون الباقية توزع على الورثة، بما يتناسب مع حصصهم الشرعية.

## المسألة المركبة

مسألة: «توفيت امرأة وتركت زوجها وابنها وأماها. فأوصت لشخص بخمسين، ولآخر بربع رأس مالها. ففرضت الوصيتين معاً على ابنها، وعلى أماها نصف (حصة أماها من البقية)؛ وعلى زوجها لا تفرض شيئاً إلا الثلث (الذي يجب أن يساهم به بحسب القانون).»

## On Another Species of Legacy

**Problem:** “A man dies, leaving his mother, his wife, and two brothers and two sisters by the same father and mother with himself; and he bequeaths to a stranger one-ninth of his capital.”

**Solution:** You constitute their shares by taking them out of **forty-eight parts**. You know that if you take one-ninth from any capital, eight-ninths of it will remain. Add now to the eight-ninths one-eighth of the same, and to the forty-eight also one-eighth of them, namely, six, in order to complete your capital. This gives fifty-four. The person to whom one-ninth is bequeathed receives six out of this. The remaining forty-eight will be distributed among the heirs, proportionably to their legal shares.

## Complex Legacy Problems

**Problem:** “A woman dies, leaving her husband, a son, and her mother. She bequeaths to a person two-fifths, and to another one-fourth of her capital. She imposes the two legacies together on her son, and on her mother one moiety (of the mother’s share of the residue); on her husband she imposes nothing but one-third (which he must contribute, according to the law).”

**Solution:** You constitute the shares of the heritage by taking them out of twelve parts: the son receives seven of them, the husband three, and the mother two parts. You know that the husband must give up one-third of his share; accordingly he retains twice as much as that which is detracted from his share for the legacy. As he has three parts in hand, one of these falls to the legacy, and the remaining two parts he retains for himself.

## Appendix A

### Summary of the Six Cases

| Case | Arabic Name           | Equation        | Solution                                     |
|------|-----------------------|-----------------|----------------------------------------------|
| I    | مال يساوي أصولاً      | $ax^2 = bx$     | $x = b/a$                                    |
| II   | مال يساوي عدداً       | $ax^2 = c$      | $x = \sqrt{c/a}$                             |
| III  | أصول تساوي عدداً      | $bx = c$        | $x = c/b$                                    |
| IV   | مال وأصول يساوي عدداً | $ax^2 + bx = c$ | $x = \sqrt{(b/2a)^2 + c/a} - b/2a$           |
| V    | مال وعدد يساوي أصولاً | $ax^2 + c = bx$ | $x = \frac{b}{2a} \pm \sqrt{(b/2a)^2 - c/a}$ |
| VI   | أصول وعدد يساوي مالاً | $bx + c = ax^2$ | $x = \frac{b}{2a} + \sqrt{(b/2a)^2 + c/a}$   |

*Historical Note:* These six cases exhaust all possibilities for quadratic equations with *positive* coefficients. The restriction to positive coefficients was necessitated by the geometric interpretation: al-Khwārizmī understood all terms as geometric magnitudes (areas and lengths), which cannot be negative. The full treatment of all quadratic equations, including negative coefficients, would have to wait until the Renaissance.

## Appendix B

### Glossary of Arabic Terms

| Arabic       | Transliteration           | Meaning                                              |
|--------------|---------------------------|------------------------------------------------------|
| الجبر        | <i>al-jabr</i>            | Completion, restoration (adding terms to both sides) |
| المقابلة     | <i>al-muqābala</i>        | Balancing, reduction (cancelling like terms)         |
| الجذر        | <i>al-jadhr</i>           | Root (the unknown quantity, $x$ )                    |
| المال        | <i>al-māl</i>             | Square of the unknown ( $x^2$ )                      |
| العدد المفرد | <i>al-'adad al-mufrad</i> | Simple number (constant term)                        |
| الشيء        | <i>al-shay'</i>           | Thing (another term for the unknown)                 |
| الدرهم       | <i>al-dirham</i>          | Dirhem (unit of currency/measure)                    |
| الوصية       | <i>al-waṣīyya</i>         | Legacy, bequest                                      |
| الفرائض      | <i>al-farā'id</i>         | Obligatory shares (Islamic inheritance law)          |
| مساحة        | <i>misāḥa</i>             | Area, mensuration                                    |
| ضرب          | <i>ḍarb</i>               | Multiplication                                       |
| قسمة         | <i>qisma</i>              | Division                                             |
| جمع          | <i>jam'</i>               | Addition                                             |
| طرح          | <i>tarḥ</i>               | Subtraction                                          |
| شيء          | <i>shay'</i>              | Thing (the unknown, $x$ )                            |

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# KITAB SHAKL AL-QATTA'

## TREATISE ON THE QUADRILATERAL

\* \* \* \* \*

**Bilingual Edition**  
*Arabic Original – English Translation*

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كِتَابُ شَكْلِ الْقَطَاعِ

*The Book of the Figure of Secants*

لِلْعَلَّامَةِ نَصِيرِ الدِّينِ الطُّوسِيِّ

*By the Learned Nasir al-Din al-Tusi*

(1201–1274 CE / 597–672 AH)

---

Critical edition based on the 1891 Constantinople edition  
edited and translated by Alexandre Pacha Caratheodory

\* \* \* \* \*

English translation prepared for this bilingual edition

*In the Name of God, the Compassionate, the Merciful*

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ



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## Preface

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

In the Name of God, the Compassionate, the Merciful

*I have no support but in God. In Him I have placed my devotion.*

\* \* \*

### Original Arabic / النص العربي الأصلي

الْحَمْدُ لِلَّهِ الَّذِي بِإِبْتِدَاعِهِ الْحَقَائِقَ الْغَيْبِيَّةَ أَتَمَّ الْمَوَاهِبَ الْجَسِيمَةَ، وَأَخْرَجَ  
بِخَلْقِهِ الْمَوْجُودَاتِ مِنْ غَيْبِ الْعَدَمِ إِلَى شَهَادَةِ الْوُجُودِ وَأَوْدَعَ فِي  
طَبِيبَاتِ الْأَشْيَاءِ أَسْرَارًا جَلِيلَةً وَمُلْكًا عَظِيمًا. وَالصَّلَاةُ وَالسَّلَامُ عَلَى  
سَيِّدِنَا مُحَمَّدٍ وَعَلَى آلِهِ وَصَحْبِهِ أَجْمَعِينَ.

أَمَّا بَعْدُ: فَقَدْ كُنْتُ قَدْ أَلَفْتُ فِي سِنِينَ مَاضِيَةٍ مُصَنَّفًا فِيهِ جَمِيعُ كَلَامِنَا  
فِي شَكْلِ الْقَطَاعِ الْمُكَمَّلِ مَعَ الْبُرْهَانِ، وَأَلَحَقْتُ بِهِ مَا يَقُومُ مَقَامَهُ  
وَيَتَعَلَّقُ بِهِ، وَكَانَ ذَلِكَ فِي لِسَانِ الْفَارَسِيَّةِ، ثُمَّ إِنَّ بَعْضَ الْإِخْوَانِ  
الْمُتَطَلِّينَ لِلْعُلُومِ طَلَبُوا مِنِّي أَنْ أُتَرْجِمَهُ إِلَى اللُّغَةِ الْعَرَبِيَّةِ، فَوَقَفْتُ لِمَا  
أَحْبَوُا، وَحَذَفْتُ مِنْهُ بَعْضَ الْفُصُولِ الَّتِي لَا مَعْنَى لَهَا، وَشَرَعْتُ فِي  
الْعَمَلِ بِمُعَاوَنَةِ الْعَلِيِّ الْقَدِيرِ.

### English Translation

Praise be to God, who by originating hidden realities perfected the greatest gifts, and by creating existent things brought them from the concealment of non-existence to the testimony of existence, and placed in the goodness of things sublime secrets and a mighty dominion. Blessings and peace be upon our master Muhammad and upon his family and all his companions.

Now then: I had composed, in past years, a treatise containing all our discourse on the *complete quadrilateral figure* with proofs, and appended to it what takes its place and relates to it, and that was in the Persian language. Then some brethren who seek knowledge asked me to translate it into the Arabic language, and I complied with their wish, and omitted from it some sections that had no essential meaning, and set to work with the assistance of the Exalted, the All-Powerful.

\* \* \*

## Structure of the Treatise

| Book     | Contents                                                                                                                                                                                       |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Book I   | <i>On Compound Ratios and Their Rules</i> — On compound ratios and their rules, in fourteen propositions.                                                                                      |
| Book II  | <i>On the Plane Quadrilateral Figure</i> — On the plane quadrilateral figure and the ratios found therein, in eleven chapters.                                                                 |
| Book III | <i>Introduction to the Spherical Quadrilateral</i> — Introduction to the theory of the spherical quadrilateral and what completes the utility to be drawn from this figure, in three chapters. |

| Book           | Contents                                                                                                                                                                                          |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Book IV</b> | <i>On the Spherical Quadrilateral</i> — On the spherical quadrilateral and the ratios found therein, in five chapters.                                                                            |
| <b>Book V</b>  | <i>On Computational Methods</i> — Exposition of the methods which take the place of the theory of the quadrilateral, for what concerns the knowledge of arcs of great circles, in seven chapters. |

# 1 Book I: On Compound Ratios

## General Rule

### الباب الأول: في أصول الأسباب المركبة

وَنَقُولُ: إِنَّ أَصْلَ الْمَعْنَى فِي الْقَوْلِ إِنَّ هَذَا النَّسْبَةَ مُرَكَّبَةً مِنْ هَذَيْنِ النَّسَبَتَيْنِ أَوْ هَذَا النَّسْبَةَ مُفْرَقَةً إِلَى هَذَيْنِ النَّسَبَتَيْنِ: أَنَّ النَّسْبَةَ الْمُرَكَّبَةَ نَسْبَتَانِ مَفْرُوقَتَانِ مِنْ نِسْبَةٍ، فَإِذَا وَضِعْتَ إِحْدَاهُمَا فِي الْأُخْرَى عَلَى التَّكَرَّارِ حَصَلَ نِسْبَةٌ مُعَيَّنَةٌ.

وَأَمَّا التَّفْرِيقُ: فَهُوَ أَنْ تَكُونَ النَّسْبَةُ مَفْرُوقَةً إِلَى نَسَبَتَيْنِ فَيَكُونُ كُلُّ وَاحِدَةٍ مِنْهُمَا بِمَنْزِلَةِ الْجُزْءِ مِنَ الْكُلِّ.

فَإِذَا وَجِدَ هَذَا فَلْنَقُلْ: أَيُّ نِسْبَةٍ كَانَتْ لِثَلَاثَةِ مُتَجَانِسَاتٍ إِلَى إِحْدَاهُنَّ فَهِيَ مُرَكَّبَةٌ مِنْ نَسْبَتِهِ إِلَى الثَّالِثَةِ وَنِسْبَةِ تِلْكَ الثَّالِثَةِ إِلَى الْأُولَى.

[Just as to give exact measure of discontinuous quantity we make use of certain properties essential to continuous magnitude, such as infinite divisibility, so also in the measure of continuous magnitude we bring in certain properties essential to discontinuous quantity. Such is the supposition according to which continuous magnitude is composed of discrete or discontinuous units which we imagine in order to arrive at the measure of continuous magnitude.]

\* \* \*

## 1.1 First Proposition

المسألة الأولى: فليكن ثلاثة متجانسات أ و ب و ج، فأقول: إِنَّ نِسْبَةَ أ إِلَى ب تُسَاوِي نِسْبَةَ أ إِلَى ج فِي نِسْبَةِ ج إِلَى ب.

### Book I: On the Principles of Compound Ratios

We say: The basis of the meaning in saying that “this ratio is composed of these two ratios,” or that “this ratio is decomposed into these two ratios,” is that the compound ratio consists of two ratios separated from one ratio, so that when one is placed in the other by repetition, a certain ratio is obtained.

As for decomposition: it is that the ratio be divided into two ratios, so that each one of them is in the position of a part from the whole.

When this is established, let us say: Any ratio whatever of three homogeneous quantities to one of them is composed of its ratio to the third and the ratio of that third to the first.

In modern notation:

$$\frac{A}{B} = \frac{A}{C} \times \frac{C}{B}, \quad \frac{A}{C} = \frac{A}{B} \times \frac{B}{C}, \quad \frac{B}{C} = \frac{B}{A} \times \frac{A}{C}$$

وَلْنَجْعَلَ وَاحِدَةً تَقْيِسُ هَذِهِ الْمُتَجَانِسَاتِ، وَلْتَكُنْ تِلْكَ الْوَاحِدَةُ تَقْيِسُ د كَمَا تَقْيِسُ أ ج، وَتَقْيِسُ ت كَمَا تَقْيِسُ ج ب، وَتَقْيِسُ ه كَمَا تَقْيِسُ أ ب. فَد هِيَ مِقْدَارُ نِسْبَةِ أ إِلَى ج، وَت هِيَ مِقْدَارُ نِسْبَةِ ج إِلَى ب، وَه هِيَ مِقْدَارُ نِسْبَةِ أ إِلَى ب.

**Proposition I.** Let there be three homogeneous quantities A, B, and C. I say that the ratio of A to B is equal to the ratio of A to C multiplied by the ratio of C to B.

Let us suppose that a unit measures these quantities, and let that unit measure D as A measures C; let it measure T as C measures B; and let it measure H as A measures B. Then D is the *quantity* of the ratio A : C; T is the quantity of the ratio C : B; and H is the quantity of the ratio A : B.

وَهَذَا لِأَنَّ كُلَّ نِسْبَةٍ لَهَا اسْمٌ وَاحِدٌ مَعَ الْعَدَدِ الَّذِي تَقْيَسُهُ الْوَاحِدَةُ  
بِقِيَاسِ مَا يَقْيَسُهُ طَرَفُهَا الْأَوَّلُ مِنْ طَرَفِهَا الثَّانِي. وَالْعَدَدُ الَّذِي لَهُ  
اسْمُ النِّسْبَةِ هُوَ مِقْدَارُهَا.

This is because every ratio has one name with the number that the unit measures in the same manner as the first term of the ratio measures the second. And the number that has the name of the ratio is its quantity.

فَإِذَا كَانَ هَكَذَا فَقَدْ بَيَّنَّا أَنَّ التَّرْكِيبَ نَحْوُ التَّكَرَّارِ، وَالتَّكَرَّارُ بَيْنَ عَدَدَيْنِ  
هُوَ ضَرْبُ أَحَدِهِمَا فِي الْآخَرِ، فَهُوَ أَيْضًا حَاصِلُ ضَرْبِ د فِي ت.

When this is so, we have explained that composition is a kind of repetition, and repetition between two numbers is the multiplication of one of them by the other; therefore H is also the product of D multiplied by T.

[Al-Tusi here establishes the fundamental property of compound ratios: that any ratio between two quantities can be decomposed through an intermediate third quantity. This is the foundation upon which all subsequent propositions rest. The proof proceeds by proportion theory:]

إِنَّ أ:ج::ب:د، وبالعكس ج:أ::د:ب، وأيضاً  
أ:ب::ج:د، فبالنسبة المرتبة يكون ج:ب::د:هـ،  
و ج:ب::و:أ، فواحدة:ت، فواحدة:د:هـ، فتعلم أن ت في د  
تساوي هـ في واحدة، وحاصل كل عدد في واحدة هو ذلك  
العدد، فهـ تساوي ت في د، فنسبة أ إلى ب تساوي نسبة أ إلى ج  
في نسبة ج إلى ب. وهو المطلوب.

Indeed,  $A : C :: 1 : D$ , and conversely  $C : A :: D : 1$ . Also  $A : B :: 1 : H$ ; therefore by the *ordered proportion* we obtain  $C : B :: D : H$ , and  $C : B :: 1 : T$ , so  $1 : T :: D : H$ , which gives finally  $T \times D = H \times 1$ . Now the product of any number by the unit is that number itself; therefore  $H = T \times D$ , that is,

$$\frac{A}{B} = \frac{A}{C} \times \frac{C}{B},$$

*quod erat demonstrandum.*

\*\*\*

## 1.2 Second Proposition

المسألة الثانية: نسبة أحد الثلاثة أ و ب و ج إلى آخر تساوي نسبة  
الثالث إلى ذلك الآخر في نسبة هذا إلى الأول.

**Proposition II.** The ratio of any one of the three quantities A, B, C to any other is equal to the ratio of the third to that latter composed with that of that quantity to the first.

$$\frac{A}{B} = \frac{A}{C} \times \frac{C}{B} = \frac{C}{B} \times \frac{A}{C}$$

*for the composition of ratios is like the multiplication of numbers; one may transpose the factors.*

\*\*\*

## 1.3 Summary of Propositions III–XIV

ملخص المسائل من الثالثة إلى الرابعة عشر / Summary of Propositions III–XIV

**Prop. III.** Every ratio composed of two arbitrary ratios is also composed of two other ratios that can be deduced from those by means of homogeneous quantities taken in the first and in the second ratio.

**Prop. IV.** If any quantity measures two others, the ratio of these two latter quantities between them is equal to the ratio composed of the inverse of the first ratio with the second ratio.

**Prop. V.** The ratio composed of two arbitrary ratios is also composed of the inverse of the first of these two ratios with the inverse of the second.

**Prop. VI.** Every composed ratio can be decomposed into simple ratios in two ways: either as fractions having unity for numerator, or as expressions having unity for denominator.

**Prop. VII.** If one has six quantities of which the composed ratios give rise to other composed ratios, then to decompose any composed ratio it suffices to know the quantities of a single column.

**Prop. VIII.** Every composed ratio gives rise to 9 other composed ratios and to 18 expressions of component ratios, or with their inverses 36 expressions.

**Prop. IX.** Two ways of decomposing a composed ratio: by making the two terms of the second ratio precede, or by making them follow the two terms of the first ratio.

**Prop. X.** Every composed ratio gives rise to 9 other composed ratios and to 18 expressions of component ratios, or with their inverses 36 expressions of composed ratios. This number can be carried to 72.

**Prop. XI.** If two of the six quantities taken each in one of the two members are equal, the 4 others are in proportion.

**Prop. XII.** If the two equal quantities belong to the same member, the other 4 are not proportional.

**Prop. XIII.** Every simple ratio can be considered as composed if one multiplies it by a ratio of identity, e.g. if  $\frac{A}{B} = \frac{C}{D}$ , one also has  $\frac{A}{B} = \frac{C}{D} \times \frac{D}{D}$ .

**Prop. XIV.** Every simple ratio of identity is equivalent to any ratio whatsoever multiplied by its inverse:  $\frac{A}{A} = \frac{X}{X} \times \frac{X}{X}$ .

## 2 Book II: On the Plane Quadrilateral Figure

### المقالة الثانية: في شكل القطاع والنسب الموجودة فيه

#### Book II: On the Figure of Secants and the Ratios Found Therein

وَهَذَا الشَّكْلُ يُسَمَّى شَكْلَ الْقَطَاعِ لِأَنَّهُ مُنْشَأٌ مِنْ أَرْبَعَةِ خُطُوطٍ مُقَاتِعَةٍ، وَيُسَمَّى أَيْضًا شَكْلَ الْقَطَائِعِ الْمُكَمَّلَةِ. وَقَدْ أَطْلَقَ عَلَيْهِ بَعْضُ الْمُتَأَخِّرِينَ اسْمَ الْمُخَمْسِ لِأَنَّهُ يَحْتَوِي عَلَى خَمْسَةِ أَضْلَاعٍ فِي الْحَقِيقَةِ إِذَا أُضِيفَ إِلَيْهِ قُطْرُ وَاحِدٍ.

This figure is called *shakl al-qatta'* (the figure of secants) because it is generated from four intersecting lines, and is also called the figure of *complete secants*. Some later scholars have given it the name of “pentagonal figure” because it contains five sides in reality when one diagonal is added.

وَقَدْ أَخْطَأَ حُسَامُ الدِّينِ فِي قَوْلِهِ إِنَّ هَذَا الشَّكْلَ إِنَّمَا يَكُونُ فِي تِسْعَةِ أَشْكَالٍ، وَهُوَ فِي اثْنَيْ عَشَرَ شَكْلًا. وَقَدْ بَيَّنَّا ذَلِكَ فِي هَذَا الْمَكَانِ وَاثْبَتْنَاهُ بِالْبَرْهَانِ.

**Note.** Hussam al-Din erred in saying that this figure only occurs in nine configurations; it is in fact twelve figures. We have demonstrated this at this point and proved it rigorously.

### 2.1 The Twelve Figures of the Quadrilateral

[The quadrilateral figure (*shakl al-qatta'*) is generated by the intersection of four straight lines. The twelve configurations are classified into three groups of four:]

I. The point of intersection of two lines falls between the two points of intersection on each of them.

II. The point of intersection falls outside one or both segments.

III. More complex configurations with all points of intersection external.

وَقَدْ بَيَّنَّ الْمُتَقَدِّمُونَ أَنَّ نِسْبَةَ خَطِّ أ ب إِلَى خَطِّ ب د مُرَكَّبَةٌ مِنْ نِسْبَةِ أ هـ إِلَى هـ ج وَنِسْبَةِ ج وَ إِلَى د هـ. وَهَذَا مَا يَتَضَمَّنُهُ كُلُّ وَاحِدٍ مِنَ الْأَشْكَالِ الثَّلَاثَةِ عَشَرَ.

The ancients have demonstrated that the ratio of line AB to line BD is composed of the ratio of AE to EC and the ratio of CF to FD. And this is what is contained in each one of the twelve figures.

فَمِنْ النَّقْطَةِ أ نَخْرُجُ خَطًّا هـ مُوَازِيًا لَخَطِّ ج د حَتَّى يُقَابِلَ خَطِّ ب هـ. فَتَحْصُلُ عَلَى أ هـ ج وَ: أ هـ ج بِسَبَبِ الْمُثَلَّثَيْنِ الْمُتَشَابِهَيْنِ هـ أ هـ وَ هـ ج وَ.

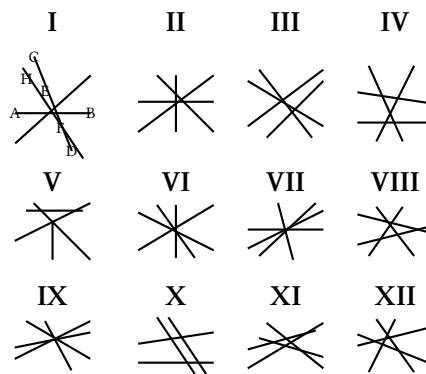
From point A draw the line AH parallel to CD until it meets BE. You will have  $AH : CF :: AE : EC$  because of the similar triangles EAH, ECF; and  $AH : FD :: AB : BD$  because of the similar triangles AFH, DBF.

$$\frac{AH}{FD} = \frac{AE}{EC} \times \frac{CF}{DF}$$

(replacing AH by its value drawn from the first proportion,) therefore also

$$\frac{AB}{BD} = \frac{AE}{EC} \times \frac{CF}{FD}.$$

### Classification of the Twelve Figures



*Note.* The geometers establish the ratio of these twelve figures by means of a single proposition and demonstration that applies to each of them, saying that the ratio of line AB to line BD is composed of the ratio of AE to EC and that of CF to FD.



### 3 Book III: Introduction to the Spherical Quadrilateral

المقالة الثالثة: في تمام الارتفاع بشكل القطاع الكروي

Book III: On Completing the Utility of the Spherical Quadrilateral Figure

الفصل الأول: في تقسيم المحيط والقطر. التقسيم الخاص للبيروني. حل المثلثات. طريقة الأقواس والأوتار. المثلثات القائمة. كيف ترد أنواع هذه المثلثات إلى المثلث المعتاد المرسوم في الدائرة التي قطرها ١٢٠.

Chapter I. On the division of the circumference and the diameter. The special division of Albiruni. Resolution of triangles. Method of arcs and chords. Right triangles. How the different cases of these triangles are reduced to the normal triangle inscribed in the circle whose diameter is 120.

طريقة الأقواس والجيوب. برهان القاعدة الأصلية: وهي أنه في كل مثلث مستقيم نسبة أضلاعه تساوي نسبة جيوب زواياها المقابلة لها.

Method of arcs and sines. Demonstration of the fundamental theorem: that in every rectilinear triangle the ratio of its sides is equal to the ratio of the sines of its opposite angles.

#### 3.1 On Chords, Arcs, and Sines

**Proposition 3.1** (Fundamental Theorem of Sines). *In any plane triangle, the ratio of the sides is equal to the ratio of the sines of the opposite angles:*

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

هذه هي القاعدة الأصلية التي بنى عليها جميع ما يأتي في علم المثلثات الكروية. وقد أثبتنا بأكثر من وجه وطريقة.

This is the fundamental theorem upon which all that follows in the science of spherical triangles is built. I have proved it by more than one method and way.

وقد طبق المتقدمون هذا على حساب فلكياتهم وأعمالهم في علم الهيئة، ولم يفرقوا بين المثلث المستقيم والمثلث الكروي إلا على وجه التقريب.

The ancients applied this to their astronomical calculations and their works in the science of configuration (astronomy), and did not distinguish between the plane triangle and the spherical triangle except approximately.

#### 3.2 The Fundamental Problem

المسألة الأساسية

The Fundamental Problem

الفصل الثالث: المسألة الأصلية: معرفة جمع قوسين أو تفریقهما مع معرفة نسبة جيبهما، فإيجاد القوسين. وإثبات الحساب للمحالتين.

Chapter III. *The fundamental problem*: knowing the sum of two arcs or their difference together with knowing the ratio of their sines, then finding the two arcs. Establishment of the calculation for the two cases.

[This fundamental problem — given  $\alpha \pm \beta$  and  $\frac{\sin \alpha}{\sin \beta}$ , find  $\alpha$  and  $\beta$  — is of central importance in spherical astronomy. Al-Tusi gives a complete algebraic solution, reducing it to the solution of a quadratic equation. Abu Nasr ibn 'Iraq had previously solved it geometrically.]

## 4 Book IV: On the Spherical Quadrilateral

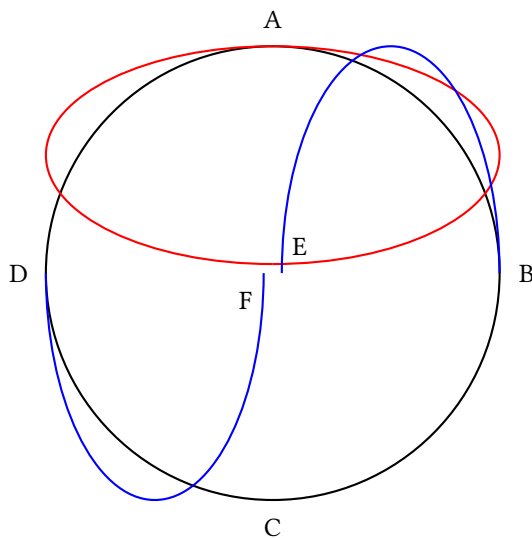
المقالة الرابعة: في شكل القطاع الكروي والنسب الموجودة فيه

Book IV: On the Spherical Quadrilateral and the Ratios Found Therein

الفصل الأول: في تعريف شكل القطاع الكروي. المستديرة تقاطع أربعة دوائر كبرى على سطح الكرة فتنشئ شكلاً يسمى شكل القطاع الكروي. كل واحد من الأربعة أضلاع هو قوس دائرة كبرى. والزوايا تقاس بأقواس الدوائر الكبرى المنحاة عند رؤوسها على الأضلاع المقابلة.

**Chapter I.** Four great circles intersect on the surface of a sphere generating a figure called *the spherical quadrilateral*. Each of the four sides is an arc of a great circle. The angles are measured by the arcs of great circles drawn through their vertices perpendicular to the opposite sides.

### 4.1 Definition of the Spherical Quadrilateral



وهذا الشكل هو الذي يسميه المتقدمون شكل القطاع لأن كل ضلعين من أضلاعه يلتقيان في قطع. وقد عرفه بطليموس في المجسطي وبين نسبه.

This is the figure that the ancients call *the figure of secants* because every two of its sides meet at an intersection. Ptolemy defined it in the *Almagest* and explained its ratios.

وقد بينا أن النسب التي ثبتت لشكل القطاع المسطح تطبق على جيوب شكل القطاع الكروي أيضاً، وهذا هو الأصل الذي بنينا عليه كل ما يأتي.

We have demonstrated that the ratios established for the plane quadrilateral apply also to the sines of the spherical quadrilateral, and this is the principle upon which we have built all that follows.

## 4.2 Theorems of Ptolemy Applied to the Spherical Quadrilateral

الفصل الثاني: فِي تَطْبِيقِ أَحْكَامِ شَكْلِ الْقِطَاعِ الْمُسَطَّحِ عَلَى جُيُوبِ شَكْلِ الْقِطَاعِ الْكُرْوِيِّ. نِسْبَةُ بَطْلَيْوَسَ الْمُعْلَنَةِ.

Chapter II. On the application of the theorems of the plane quadrilateral to the sines of the spherical quadrilateral. The explicit ratio of Ptolemy.

الفصل الثالث: نِسْبَةُ بَطْلَيْوَسَ الْمُضْمَرَةِ. بُرْهَانُهَا. وَلِمَاذَا هَذَا الْبُرْهَانُ وَإِنْ كَانَ صَحِيحًا لَيْسَ بِكَافٍ فِي الْحَالَةِ الْحَاضِرَةِ. الْقَاعِدَةُ التَّيَقْنِيَّةُ. تَرْكِيبُ الْحَالَاتِ السَّبْعِ وَالْعِشْرِينَ الَّتِي تَرَدَّدَ فِيهَا صَاحِبُنَا إِلَى ثَلَاثَةِ عَشَرَ.

Chapter III. The implicit ratio of Ptolemy. Its demonstration. Why this demonstration, however exact it may be, is not sufficient in the present case. The preliminary lemma. Reduction of the 27 cases in which our author hesitated to 13. Examination of each case in particular.

**Proposition 4.1** (The Sine Theorem for Spherical Triangles). *In any spherical triangle, the sines of the sides are in the same ratio as the sines of the opposite angles:*

$$\frac{\sin a}{\sin A} = \frac{\sin b}{\sin B} = \frac{\sin c}{\sin C}$$

وَمِنْ هَذَا يَتَبَيَّنُ أَنَّ جَيْبَ قَوْسِ أ ب إِلَى جَيْبِ قَوْسِ أ ج كَنِسْبَةِ جَيْبِ قَوْسِ أ د إِلَى جَيْبِ قَوْسِ د ب، وَذَلِكَ مَا أَرَدْنَا إِثْبَاتَهُ. وَإِنْ انْطَبَقَ أَحَدُ الْمَثَلَيْنِ عَلَى الْآخَرِ فِي كُلِّ وَاحِدٍ مِنْ هَذِهِ الْأَشْكَالِ فَالْبَيَانُ عَلَى قِيَاسِ ذَلِكَ.

From this it is clear that the sine of arc AB is to the sine of arc AC as the sine of arc AD is to the sine of arc DB, which is what we wanted to prove. And if one of the triangles coincides with the other in any one of these configurations, the explanation follows the same pattern.

الفصل الرابع: كَيْفَ يُؤَدِّي فَحْصُ النِّسَبَتَيْنِ الْمُضْمَرَةِ وَالْمُعْلَنَةِ إِلَى مِفْتَاحِ جَمِيعِ الْبُرْهَانَاتِ الْأُخْرَى. مُقَارَنَةُ مَسَلِّكَ هَذَا الْبُرْهَانِ مَعَ قَوَاعِدِ الْقِيَاسِ الْإِسْتِدْلَالِيِّ.

Chapter IV. How the examination of the implicit and explicit ratios gives the key to all other demonstrations. Comparison of the course followed in this demonstration with the formulas of deductive syllogism.

## 5 Book V: On Computational Methods

المقالة الخامسة: في بيان الضوابط البديلة لنظرية الشكل

Book V: Exposition of the Methods Which Replace the Theory of the Figure

الفصل الأول: مُقَاطَعَاتُ ثَلَاثِ دَوَائِرٍ كُبْرَى تُنَشِئُ مُثَلَّثَاتٍ عَلَى سَطْحِ الْكُرَّةِ. طَبِيعَةُ هَذِهِ الْمُثَلَّثَاتِ. يَكْفِي مَعْرِفَةُ وَاحِدَةٍ مِنْهَا لِمَعْرِفَةِ جَمِيعِهَا.

Chapter I. The intersections of three great circles generate triangles on the surface of a sphere. Nature of these triangles. It suffices to know one of them to know them all.

الفصل الثاني: تَحْلِيلُ أَنْوَاعِ الْمُثَلَّثَاتِ الْكُرَوِيَّةِ حَسَبَ زَوَايَاهَا (عَشْرُ أَصْنَافٍ) أَوْ حَسَبَ أَقْوَاسِهَا (عَشْرُ أَصْنَافٍ).

Chapter II. Analysis of the different spherical triangles according to their angles (ten categories) or according to their arcs (ten categories).

### 5.1 On Inclinations and the Sine Theorem

وَفِي هَذِهِ الْمُثَلَّثَاتِ يُسَمَّى قَوْسُ ب ج مَيْلَ قَوْسِ أ ج، وَهَذَا الْمَيْلُ ب ج هُوَ نَصِيبُ قَوْسِ أ ج مِنَ الْمَيْلِ الْكُلِّيِّ لِدَائِرَةِ أ د عَلَى دَائِرَةِ أ هـ، وَهُوَ الَّذِي تَقْيِسُهُ زَاوِيَةُ أ.

In these triangles, arc BC is called the *inclination* of arc AC, which inclination BC is but the part of arc AC in the total inclination of circle AD to circle AE, which the angle A measures. And if arc BC is related to arc AB, then BC is called the second inclination.

وَهَذَا مَا يُسَمِّيهِ أَبُو رِيحَانَ الْبِيرُونِيُّ الْعَرْضَ، فَقَوْسُ ب ج يَكُونُ الْمَيْلَ الْأَوَّلَ بِالنِّسْبَةِ لِأ ج، وَالْمَيْلَ الثَّانِيَّ بِالنِّسْبَةِ لِقَوْسِ أ ب.

This is what Abu Rayhan al-Biruni calls the *width*; so arc BC would be the first inclination with respect to AC, and the second inclination with respect to arc AB.

فَفِي هَذَا الْمَعْنَى نَقُولُ: إِنَّ نِسْبَ جُيُوبِ الْمَيْلِ تَسَاوِي نِسْبَ جُيُوبِ أَقْوَاسِهَا، وَنِسْبَةُ جُيُوبِ كُلِّ مَيْلٍ إِلَى جُيُوبِ قَوْسِهِ تَسَاوِي نِسْبَةَ جُيُوبِ مَيْلٍ آخَرَ إِلَى جُيُوبِ قَوْسِهِ.

In this manner of speaking we say that the ratios of the sines of the inclinations are equal to the ratios of the sines of their arcs, and the ratio of the sine of any inclination to the sine of its arc is equal to the ratio of the sine of another inclination to the sine of its arc.

[In modern notation:]

If  $BC$  is the inclination of  $AC$ , and  $LM$  is the inclination of  $AE$ , then:

$$\frac{\sin BC}{\sin LM} = \frac{\sin AC}{\sin AE}$$

وَأِنْ لَمْ يَنْطَبِقْ مُثَلَّثُ أ ب ج عَلَى مُثَلَّثِ د أ هـ، لَكِنْ كَانَتْ زَاوِيَةُ أ فِي هَذَيْنِ الْمُثَلَّثَيْنِ مُتَسَاوِيَةً، وَكَانَتْ زَاوِيَتَا ب وَ هـ قَائِمَتَيْنِ، فَالْقَاعِدَةُ ثَابِتَةٌ.

If triangle ABC does not coincide with triangle DAE, but nevertheless angle A is equal in these two triangles and angles B and E are right, the theorem remains proved nonetheless.

## 5.2 The Eight Demonstrations of the Sine Theorem

### البرهان على القاعدة في ثمانية وجوه

سِتَّةُ بُرْهَانَاتٍ فِي تَسَاوِي نِسْبِ الْجُيُوبِ وَالْأَقْوَاسِ وَأَظِلَّةِ الزَّوَايَا وَنَحْوَهَا. تَعْمِيمُ هَذَا الْأَصْلِ عَلَى جَمِيعِ الْمُثَلَّثَاتِ الْكُرْوِيَّةِ. تَابِعَاتٌ.

وَقَدْ حَاوَلَ بَعْضُهُمْ أَنْ يَرْفَعُوا شُبُهَاتٍ عَلَى صِحَّةِ اسْتِعْمَالِ الظِّلِّ لِسُرْعَةِ زِيَادَتِهِ بَعْدَ ٤٥° دَرَجَةٍ. رُدُّ هَذِهِ الشُّبُهَاتِ وَتَحْلِيلُ الْحَالَاتِ الْمُخْتَلِفَةِ الَّتِي يَجِبُ فِيهَا اسْتِبْدَالُ الظِّلِّ بِظِلِّ التَّمَامِ.

### Eight Demonstrations of the Fundamental Theorem

Six demonstrations concerning the equality of ratios of sines, arcs, tangents of angles, etc. Generalization of this principle to all spherical triangles. Corollaries.

Objections that some have attempted to raise against the legitimacy of the use of tangents by reason of their rapid growth beyond 45° — Refutation of these objections and analysis of the different cases in which one must replace the tangent by the cotangent.

## 5.3 The Figure of Shadows (Tangent Theorem)

### شكل الظلال (قاعدة الظلال)

الفصل السادس: فِي الشَّكْلِ الْمُظَلِّ (قَاعِدَةُ الظَّلَالِ). اخْتِرَاعُهُ مِنْ قَبْلِ أَبِي الْوَفَاءِ. تَعْرِيفُ الظِّلِّ الْأَوَّلِ وَالثَّانِي (الظِّلُّ وَظِلُّ التَّمَامِ، الْقَاطِعُ وَقَاطِعُ التَّمَامِ). أَشْكَالٌ مُخْتَلِفَةٌ.

الظِّلُّ الْأَوَّلُ لِلْقَوْسِ هُوَ جَيْبُ ارْتِفَاعِهِ أَوْ مَا يُسَمِّيهِ الْمُتَأَخَّرُونَ جَيْبَ التَّمَامِ. وَالظِّلُّ الْأَوَّلُ هُوَ الظِّلُّ؛ وَالظِّلُّ الثَّانِي هُوَ ظِلُّ التَّمَامِ.

### The Figure of Shadows (Tangent Theorem)

Chapter VI. On the shaded figure (Theorem of Tangents). The invention thereof is due to Abu'l-Wafa. Definition of first and second shadows (tangent and cotangent, secant and cosecant). Different formulas.

The first shadow of an arc is the sine of its altitude or what the moderns call its cosine. The first shadow is the *tangent*; the second shadow is the *cotangent*.

\* \* \*

## 5.4 Applications to Spherical Astronomy

الفصل السابع: الْمَنَافِعُ الَّتِي يُمْكِنُ الاسْتِفَادَةُ بِهَا مِنْ قَاعِدَةِ الْجُيُوبِ وَكَذَلِكَ قَاعِدَةِ الظَّلَالِ فِي حَلِّ الْمُثَلَّثَاتِ الْكُرْوِيَّةِ. طَرِيقٌ مُخْتَلِفَةٌ تُؤَدِّي إِلَى غَرَضٍ وَاحِدٍ.

وَأَعْتَرَفْتُ أَنِّي لَا أَعْرِفُ الطَّرِيقَ الَّتِي تُحَلُّ بِهَا هَذِهِ الْمَسْأَلَةُ بِوَاسِطَةِ الظَّلَالِ: مَعَ مَعْرِفَةِ ثَلَاثِ زَوَايَا مِثْلٍ كُرْوِيٍّ تَعَيَّنَ أَضْلَاعُهُ.

Chapter VII. The uses that can be drawn from the sine theorem as well as from the tangent theorem for the resolution of spherical triangles. Different paths leading to the same goal.

The author confesses not to know the path by which the following problem is solved by means of tangents: "Given the three angles of a spherical triangle, determine its sides."

[This problem — the polar triangle case — is solved by passing to the polar triangle: if the three angles  $A, B, C$  are

given, the sides of the polar triangle are  $a' = 180^\circ - A$ ,  $b' = 180^\circ - B$ ,  $c' = 180^\circ - C$ , which can then be solved by the sine theorem, and the angles of the polar triangle give the sides of the original triangle.]

## Epilogue / Historical and Biographical Notes

فِي ذِكْرِ الْعُلَمَاءِ الْمَذْكُورِينَ فِي هَذَا الْكِتَابِ

*On the Scholars Mentioned in This Book*

\* \* \*

| Name                                                                        | Details                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SABIT-IBN-KOURAH (ثابت بن قرة)                                              | One of the most celebrated translators of Greek scientific works. Born 836, died 901 in Baghdad. Hadji Khalifa No. 3374. Author of a work on the quadrilateral.                                                                                                                                               |
| ABOU-RIHAN ALBIROUNI (ابو ريحان البيروني)                                   | From Biroun, city of Kharesm. Died 1038–1039. Hadji Khalifa No. 7420. Author of works on the chord and the sine. Translator or abbreviator of the Almagest. First to take the radius as unity. Gave a demonstration of the proportionality of the sines of the arcs and the angles in the spherical triangle. |
| ABOU-MAHMOUD ALHOD-JENDI (ابو محمود الخجندی)                                | c. 992. Disputed with Aboul Véfa the priority concerning the theorem of the proportionality of the sines of the angles and the arcs.                                                                                                                                                                          |
| ABOU-DJAFAR ALHAZIN (ابو جعفر الخازن)                                       | c. 1000. (Hadji Khalifa No. 4137). The surname Hazin = librarian.                                                                                                                                                                                                                                             |
| ABOUL-FAZL AL NERIZI (ابو الفضل النيرزي)                                    | (Hadji Khalifa No. 2548). One of the principal commentators of Euclid, and author of astronomical tables in the Indian manner.                                                                                                                                                                                |
| HOUSSAM-UDDIN-ALI-BN-FAZLULLAH ESSALAR (حسام الدين علي بن فضل الله السالار) | Author of a treatise on the quadrilateral which served as the basis for the treatise of Nasiruddin.                                                                                                                                                                                                           |
| ABDULLAH-ABDUL-KIAFI-BN-ABDUL-MEDJID (عبدالله الكافي بن عبدالمجيد)          | The copyist of the manuscript and a disciple of Nasiruddin.                                                                                                                                                                                                                                                   |



## Scholarly Apparatus

### Note on the Edition

authority of the Ottoman Ministry of Public Instruction. The original edition comprised:

- **French Translation** (pp. 1–214): A complete translation by Alexandre Pacha Carathéodory (1833–1906), Ottoman diplomat and mathematician, with extensive notes and a biographical appendix on the mathematicians cited in the treatise.
- **Arabic Original** (pp. 1–178): The original Arabic text, printed in lithographic form from the manuscript belonging to Edhem Pasha, with its own pagination in Arabic numerals.

### The Significance of the Work

Al-Tusi's *Kitab Shakl al-Qatta'* represents one of the most important contributions to trigonometry in the medieval Islamic world. The work:

1. **Systematized the theory of compound ratios** (Book I), providing a complete algebraic foundation for ratio theory that went beyond Euclid's *Elements*.
2. **Analyzed the complete quadrilateral figure** (Book II), classifying all twelve configurations generated by four intersecting lines and deriving the fundamental proportional relationships.
3. **Established the foundations of spherical trigonometry** (Books III–IV), including the sine theorem for spherical triangles and its systematic application.
4. **Developed practical computational methods** (Book V) for solving spherical triangles, which were essential for astronomical calculations.

### Al-Tusi's Key Contributions to Trigonometry

- **First systematic treatment** of plane and spherical trigonometry as an independent science, separate from astronomy.
- **Sine law for plane triangles:**  $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$
- **Sine law for spherical triangles:**  $\frac{\sin a}{\sin A} = \frac{\sin b}{\sin B} = \frac{\sin c}{\sin C}$
- **Polar triangle method:** The transformation of angle-given problems to side-given problems via the polar triangle.
- **Complete classification** of spherical triangle cases (27 reduced to 13).
- **Systematic use** of tangent and cotangent functions.

## Mathematical Terminology

| Arabic                    | Transliteration           | English                           |
|---------------------------|---------------------------|-----------------------------------|
| شَكْلُ الْقَطَّاعِ        | shakl al-qatta'           | figure of secants / quadrilateral |
| الْقَطَّاعُ الْمَكْمَلَةُ | al-qatta' i' al-mukammala | complete secants                  |
| نِسْبَةُ مُرَكَّبَةٍ      | nisba murakkaba           | compound ratio                    |
| قَطْرٌ                    | quṭr                      | diagonal                          |
| ضَلْعٌ                    | ḍal'                      | side                              |
| قَاعِدَةٌ                 | qā'ida                    | base / theorem                    |
| جَيْبُ قَوْسٍ             | jayb qaws                 | sine of an arc                    |
| جَيْبُ التَّامِّ          | jayb al-tamām             | cosine                            |
| ظِلُّ قَوْسٍ              | ẓill qaws                 | tangent                           |
| ظِلُّ التَّامِّ           | ẓill al-tamām             | cotangent                         |
| مُثَلَّثٌ كُرَوِيٌّ       | muthallath kurawī         | spherical triangle                |
| مُثَلَّثٌ مُسْتَقِيمٌ     | muthallath mustaqīm       | plane (rectilinear) triangle      |
| إِكْلِيلٌ                 | iklil                     | spherical zone / crown            |
| زَاوِيَةٌ مُحِيطَةٌ       | zāwiya muḥīṭa             | exterior angle                    |
| مَيْلٌ                    | mayl                      | inclination                       |
| عَرْضٌ                    | 'arḍ                      | width / latitude                  |
| قَطْعٌ                    | qaṭ'                      | intersection / secant             |

## Original Arabic Text Passages

## Selected Passages from the Arabic Original

*The following pages reproduce key passages from the original Arabic text with their distinctive mathematical terminology and geometric structure.*

\* \* \*

## المقالة الأولى: في الأسباب المركبة وأحكامها

قَالَ الْمُصَنِّفُ رَحِمَهُ اللَّهُ تَعَالَى: نَقُولُ إِنَّ أَصْلَ الْمَعْنَى فِي الْقَوْلِ إِنَّ هَذَا النَّسْبَةَ مُرَكَّبَةً مِنْ هَذَيْنِ النَّسَبَتَيْنِ أَوْ هَذَا النَّسْبَةَ مُفْرَقَةً إِلَى هَذَيْنِ النَّسَبَتَيْنِ: أَنَّ النَّسْبَةَ الْمُرَكَّبَةَ نَسَبَتَانِ مَفْرُوقَتَانِ مِنْ نِسْبَةٍ، فَإِذَا وُضِعَتْ إِحْدَاهُمَا فِي الْأُخْرَى عَلَى التَّكَرَّارِ حَصَلَ نِسْبَةٌ مُعَيَّنَةٌ.

The author (may God Exalted have mercy on him) says: We say that the basis of the meaning in saying that this ratio is composed of these two ratios, or that this ratio is decomposed into these two ratios, is that the compound ratio consists of two ratios separated from one ratio, so that when one is placed in the other by repetition, a certain ratio is obtained.

وَأَمَّا التَّفْرِيقُ: فَهُوَ أَنْ تَكُونَ النَّسْبَةُ مَفْرُوقَةً إِلَى نَسَبَتَيْنِ فَيَكُونُ كُلُّ وَاحِدَةٍ مِنْهُمَا بِمَنْزِلَةِ الْجُزْءِ مِنَ الْكُلِّ.

As for decomposition: it is that the ratio be divided into two ratios, so that each one of them is in the position of a part from the whole.

فَإِذَا وَجِدَ هَذَا فَلْنَقُلْ: أَيُّ نِسْبَةٍ كَانَتْ لِثَلَاثَةِ مُتَجَانِسَاتٍ إِلَى إِحْدَاهُنَّ فَهِيَ مُرَكَّبَةٌ مِنْ نَسْبَتِهِ إِلَى الثَّالِثَةِ وَنِسْبَةِ تِلْكَ الثَّالِثَةِ إِلَى الْأُولَى.

When this is established, let us say: Any ratio whatsoever of three homogeneous quantities to one of them is composed of its ratio to the third and the ratio of that third to the first.

فَلْيَكُنْ ثَلَاثَةُ مُتَجَانِسَاتٍ أَوْ ب وَ ج، فَأَقُولُ: إِنَّ نِسْبَةَ أ إِلَى ب تُسَاوِي نِسْبَةَ أ إِلَى ج فِي نِسْبَةِ ج إِلَى ب، وَنِسْبَةَ أ إِلَى ج تُسَاوِي نِسْبَةَ أ إِلَى ب فِي نِسْبَةِ ب إِلَى ج، وَنِسْبَةَ ب إِلَى ج تُسَاوِي نِسْبَةَ ب إِلَى أ فِي نِسْبَةِ أ إِلَى ج، وَهَكَذَا.

Let there be three homogeneous quantities A, B, and C. Then I say: the ratio of A to B is equal to the ratio of A to C multiplied by the ratio of C to B, and the ratio of A to C is equal to the ratio of A to B multiplied by the ratio of B to C, and the ratio of B to C is equal to the ratio of B to A multiplied by the ratio of A to C, and so on.

\* \* \*

## المقالة الثانية: في شكل القطاع والنسب الموجودة فيه

وَهَذَا الشَّكْلُ يُسَمَّى شَكْلَ الْقِطَاعِ لِأَنَّهُ مُنْشَأٌ مِنْ أَرْبَعَةِ خُطُوطٍ مُقَاطِعَةٍ، وَيُسَمَّى أَيْضًا شَكْلَ الْقِطَاعِ الْمَكْمَلَةِ. وَقَدْ أَطْلَقَ عَلَيْهِ بَعْضُ الْمُتَأَخِّرِينَ اسْمَ الْمُخْمَسِ لِأَنَّهُ يُحْتَوِي عَلَى خَمْسَةِ أَضْلَاعٍ فِي الْحَقِيقَةِ إِذَا أُضِيفَ إِلَيْهِ قُطْرٌ وَاحِدٌ.

This figure is called the figure of secants (*shakl al-qatta'*) because it is generated from four intersecting lines, and is also called the figure of complete secants. Some later scholars have called it the "pentagonal figure" because it contains five sides in reality when one diagonal is added.

وَقَدْ أَخْطَأَ حُسَامُ الدِّينِ فِي قَوْلِهِ إِنَّ هَذَا الشَّكْلَ إِنَّمَا يَكُونُ فِي تِسْعَةِ أَشْكَالٍ، وَهُوَ فِي اثْنَيْ عَشَرَ شَكْلًا. وَقَدْ بَيَّنَّا ذَلِكَ فِي هَذَا الْمَكَانِ وَاثْبَتْنَاهُ بِالْبُرْهَانِ.

Hussam al-Din erred in saying that this figure only occurs in nine configurations; it is in fact twelve figures. We have demonstrated this at this point and proved it rigorously.

\* \* \*

### المقالة الخامسة: في الضوابط البديلة لنظرية الشكل

الضَرْبُ الْأَوَّلُ وَلْيَكُنِ الْمَعْلُومُ وَتَرِ الْقَاعِدَةُ وَتَرِ النَّهْيَةُ الْآخِرُ وَضَلَعٌ مُجَاوِرٌ لِأَحَدَاهُمَا فَلْنَفَرِّضْ أَنَّهُ أَجُ وَالْمَعْلُومُ أَيْضًا ضَلَعٌ آخَرُ مُجَاوِرٌ لِتِلْكَ الْقَاعِدَةِ فَلْيَكُنْ دَأُ، وَلْيَكُنِ الْمَجْهُولُ بَقِيَّةَ جَوَانِبِ هَذَا الشَّكْلِ.

The first case: Let the known quantities be the base, the other extreme, and a side adjacent to one of them. Let us suppose that it is AC, and let the known quantity also be another side adjacent to that base, let it be DA, and let the unknown be the remaining sides of this figure.

وَمِنْ هَذَا يَتَبَيَّنُ أَنَّ جَيْبَ قَوْسِ أ بَ إِلَى جَيْبِ قَوْسِ أ جَ كَنِسْبَةِ جَيْبِ قَوْسِ أ دَ إِلَى جَيْبِ قَوْسِ د بَ، وَذَلِكَ مَا أَرَدْنَا إِثْبَاتَهُ وَإِنْ انْطَبَقَ أَحَدُ الْمُثَلَّثَيْنِ عَلَى الْآخَرِ فِي كُلِّ وَاحِدٍ مِنْ هَذِهِ الْأَشْكَالِ فَكَانَ الْبَيَانُ بِحَسْبِهِ عَلَى قِيَاسِ ذَلِكَ.

From this it is clear that the sine of arc AB is to the sine of arc AC as the sine of arc AD is to the sine of arc DB, which is what we wanted to prove. And if one of the triangles coincides with the other in any one of these configurations, then the explanation follows accordingly.

\* \* \*

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End of the Treatise / نهاية الكتاب

UNIVERSITY OF MICHIGAN STUDIES

Humanistic Series, Volume XI

Contributions to the History of Science

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**Robert of Chester's  
Latin Translation  
of the  
Algebra of Al-Khowarizmi**

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*With an Introduction, Critical Notes  
and an English Version*

**By**  
**Louis Charles Karpinski**  
University of Michigan

New York

## Preface

FROM the point of view of the history of science, no justification is needed for the publication of a mathematical text of the twelfth century, for the available material representing this period is meagre. A wider acquaintance with Robert of Chester's Latin translation of Al-Khowarizmi's Arabic treatise on algebra will perhaps contribute to a more just estimate of the services rendered to science by the Arabs.

In the English version I have not attempted to give a literal translation of the Latin, but rather to express the thought in a phraseology which the modern student of mathematics will find easy of comprehension; by consulting the Latin text and footnotes the reader will be able to examine Robert of Chester's own words. For the convenience of readers interested in the text I have added a Latin Glossary in which are noted many variations from the usage of classical writers.

In the Introduction I have presented a study of the significance of the treatise in the history of mathematics, and a description of the manuscripts upon which the text is based.

It is a pleasure to express indebtedness to Professor David Eugene Smith for having suggested the work; to Mr. George A. Plimpton for the generous use of his unique mathematical library; to the librarians of the libraries of Vienna and Dresden for photographic reproductions of manuscripts containing this text; to the librarian of Columbia University for the loan of the Scheybl manuscript, and to the librarian of the Cleveland Public Library for the use of works from the John G. White collection. I am much indebted also to colleagues of the University of Michigan, particularly in the Department of Latin and the Department of Mathematics.

I am under special obligation to Mr. William H. Murphy for making possible this publication.

Louis C. Karpinski.  
Ann Arbor, Michigan,  
November 1, 1915.

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# Chapter 1

## Algebraic Analysis before Al-Khowarizmi

Arabic contributions to science have, in the past, been somewhat neglected by historians. More recent studies are recognizing our indebtedness to Mohammedan scholars, who kept the embers of learning aglow while Europe was in the darkness of the Middle Ages. Much of our knowledge of Greek mathematics comes to us from Arabic sources; the early Latin versions were frequently based upon Arabic texts rather than the Greek originals. Similarly, Hindu arithmetic and astronomy were transmitted to Europe by Islam. The services of the Arabs to science were not limited to the preservation and transmission of the learning of other nations. They made independent contributions in many fields.

Among these achievements is the Arabic algebra of Al-Khowarizmi, which for centuries enjoyed wide popularity in the original, and for further centuries extended its popularity through translations and adaptations. A study of the content of this work is an excursion into mediaeval thought. By a study of the text in a form as nearly like the original as possible, we discover the reason for its long-continued appeal to the Occidental as well as the Oriental mind, its interest for the Englishman, the German, and the Italian, as well as for the Arab.

### 1.1 Early Equations

Simple equations of the first degree in one unknown, of the type  $ax = d$ , are found in the oldest mathematical text-book which we possess, the Ahmes papyrus of about 1700 B.C. This Egyptian document presents not only first degree equations together with symbols for the unknown quantity and for the operations of addition and subtraction, but also shows traces of a study of simultaneous linear equations some two thousand years before the Christian Era.

Later, but still before the golden age of Greek mathematics, the quadratic equation appears in Egypt. The problems found involve simultaneous quadratic equations, thus:

*Another example of the distribution of a given area into squares.* If you are told to distribute 100 square ells (units of area) over two squares so that the side of one shall be  $\frac{3}{4}$  of the side of the other: please give me each of the unknowns.

The solution follows by assuming the side of one square to be unity, and the other  $\frac{3}{4}$ . The sum of these areas is  $\frac{25}{16}$ , of which the root  $\frac{5}{4}$  is to the required side as  $\frac{5}{4}$  is to 1, whence one side is 8 and the other 6. The algebraical equivalent of this geometrical problem is, evidently,

$$x^2 + y^2 = 100, \quad x : y = 4 : 3.$$

## 1.2 Greek Contributions

Several propositions of Euclid present geometrical equivalents of the solution of various types of quadratic equations, not involving negative coefficients, and further study of similar problems appears in Euclid's *Data*. Of this nature are the fifth, sixth, and eleventh propositions of the second book of the *Elements* and the twenty-seventh, twenty-eighth, and twenty-ninth of the sixth book.

The eleventh proposition of the second book of the *Elements* furnishes the solution of the equation

$$x^2 + ax = a^2$$

or even more general,

$$x^2 + ax = b^2.$$

Analytical solution of the quadratic equation appears quite definitely in the works of Heron of Alexandria, who flourished about the beginning of the Christian Era. Heron states in effect that given the sum of two line segments and their product then each of the segments is known.

## 1.3 The Work of Diophantus

In respect to analysis Diophantus is the greatest name among the Greeks. Recently it has been established that he flourished in the third century of the Christian Era, when Greek supremacy in mathematics was waning. No doubt whatever exists that this Alexandrian was familiar with the analytical solution of the various forms of quadratic equations, neglecting negative roots and, of course, imaginary roots, which did not receive serious treatment for more than a millennium after Diophantus.

The three types of complete quadratic equations, involving only positive coefficients, are the following:

$$ax^2 + bx = c, \tag{1.1}$$

$$ax^2 + c = bx, \tag{1.2}$$

$$ax^2 = bx + c. \tag{1.3}$$

All three types appear in the *Arithmetica* of Diophantus, not systematically treated but solved as incidental to the solution of other problems.

## 1.4 Hindu Algebra

Some two centuries after the period of Diophantus, Aryabhata, one of the earliest Hindu mathematicians of prominence, was born (476 A.D.). Brahmagupta of Ujjain, the centre of Indian learning, wrote on algebra in the early part of the seventh century and gave a rule for the solution of quadratic equations:

*To the absolute number multiplied by the [coefficient of the] square, add the square of half the [coefficient of the] unknown, the square root of the sum, less half the [coefficient of the] unknown, being divided by the [coefficient of the] square, is the unknown.*

In formula this corresponds to the solution

$$x = \frac{\sqrt{(b/2)^2 + ac} - b/2}{a}$$

of the equation  $ax^2 + bx = c$ .

## Chapter 2

### Al-Khowarizmi and his Treatise on Algebra

THE activity of the great Arabic mathematician Abu 'Abdallah Mohammed ibn Musa al-Khowarizmi marks the beginning of that period of mathematical history in which analysis assumed a place on a level with geometry; and his algebra gave a definite form to the ideas which we have been setting forth.

The arithmetic of Al-Khowarizmi made known to the Arabs and, through an early twelfth-century translation, to Europeans also, the Hindu art of reckoning. Any consideration of the difficulties attending arithmetical operations with the Greek letter numerals, or even with the Roman numerals, shows how essential the adoption of numerals with place value was for the development of the analytical side of mathematics.

#### 2.1 The Life of Al-Khowarizmi

Our chief source of information in regard to the life and the writings of our Arabic author is the *Book of Chronicles (Kitab al-Fihrist)* by Ibn Abi Ya'qub al-Nadim. This work, which was completed about 987 A.D., gives biographies of learned men of all nations, together with lists of such of their works as were known to Al-Nadim. We quote the passage relating to our author:

*Mohammed ibn Musa, born in Khowarizm (modern Khiva), worked in the library of the caliphs under Al-Mamun. During his lifetime, and afterward, where observations were made, people were accustomed to rely upon his tables, which were known by the name Sind-Hind (Hindu Siddhanta). He wrote: The book of astronomical tables in two editions, the first and the second; On the sun-dial; On the use of the astrolabe; On the construction of the astrolabe; The Book of Chronology.*

The bibliography of Al-Nadim does not include four works from the hand of Al-Khowarizmi which have come to us. These are his arithmetic, his algebra, a work on the quadrivium, and an adaptation of Ptolemy's geography.

The equation  $x^2 + 10x = 39$  runs like a thread of gold through the algebras for several centuries, appearing in the algebras of Abu Kamil, Al-Karkhi and 'Omar al-Khayyami, and frequently in the works of Christian writers, centuries later.

#### 2.2 Robert of Chester and Other Translators

WHEN towards the beginning of the twelfth century European scholars turned to Islam for light, the works of Mohammed ibn Musa came to occupy a prominent

place in their studies. Contemporary with these men was Robert of Chester.

The assertion was made by Curtze that the *Liber embadorum*, Book of Measures, in Hebrew, by Abraham bar Chiyya Ha Nasi, known as Savasorda, was the first work to appear in Latin showing to the Western world how the solution of quadratic equations is accomplished. It has recently been shown by Haskins, on the basis of astronomical data in the work, that this date is undoubtedly a scribe's error for DXL, corresponding to 1145 A.D.

Robert of Chester, known to fame chiefly as the first translator of the Koran, was doubtless educated in the well-known school located at Chester. Of the more personal side of Robert's life we have but scattered facts. His nationality is established not only by his name and his return to England in 1150, but also by the direct statement made by Peter the Venerable in a letter of 1143 concerning the Koran, addressed to Bernard of Clairvaux.

To Peter the Venerable, Abbot of Cluny, who induced Robert to undertake the translation of the Koran, the latter addressed a Saracenic chronicle. Other names have been connected with this translation of the Koran, but a study of Peter's letters shows that Robert and Hermann were definitely requested to undertake the translation.

Of Robert's works the version of the Koran was completed in 1143. In a prefatory letter he states that this task was regarded by him only as a digression from his principal studies of astronomy and geometry.

## Chapter 3

### The Arabic Text and the Translations

The Arabic text of Al-Khowarizmi's algebra, together with an English translation, was published by Frederick Rosen in 1831. This excellent work is unfortunately out of print, and so is not available to most students of mathematics.

The Latin translation by Robert of Chester is not as faithful nor as correct as the text ascribed to Gerard of Cremona, published by Libri. Omissions, transpositions, and additions to the text are so numerous that it does not seem desirable to list them all. No evidence exists, however, that Robert's text is based upon another Arabic original than that of the Libri text. The text proper, as opposed to the illustrative problems, follows the general lines of the Arabic original.

## Chapter 4

### Preface and Additions Found in the Arabic Text

The Arabic text of Al-Khowarizmi's algebra published by Rosen contains an author's preface which is not found either in the translation published by Libri, or in that by Robert of Chester. As this reveals his conception of the purpose of the algebra, as well as some of the causes which led him to undertake the work, we present it here in the translation by Rosen.

#### The Author's Preface

*In the Name of God, gracious and merciful!*

*This work was written by Mohammed ben Musa, of Khowarezm. He commences it thus:*

*Praised be God for his bounty towards those who deserve it by their virtuous acts: in performing which, as by him prescribed to his adoring creatures, we express our thanks, and render ourselves worthy of the continuance (of his mercy), and preserve ourselves from change: acknowledging his might, bending before his power, and revering his greatness!*

*That fondness for science, by which God has distinguished the Imam al Mamun, the Commander of the Faithful, that affability and condescension which he shows to the learned, that promptitude with which he protects and supports them in the elucidation of obscurities and in the removal of difficulties, — has encouraged me to compose a short work on Calculating by (the rules of) Completion and Reduction, confining it to what is easiest and most useful in arithmetic, such as men constantly require in cases of inheritance, legacies, partition, law-suits, and trade, and in all their dealings with one another, or where the measuring of lands, the digging of canals, geometrical computation, and other objects of various sorts and kinds are concerned.*

## Chapter 5

### Manuscripts of Robert of Chester's Translation

#### 5.1 The Extant Manuscripts

Steinschneider was the first in recent times to call attention to the translation of Al-Khowarizmi's algebra made by Robert of Chester. He suggested the desirability of publishing this text, referring to the manuscript in Vienna. To this same manuscript Curtze later, and independently of Steinschneider, directed attention and also suggested the desirability of the publication of the work. Wappler, in 1887, found a second copy of the algebra in a manuscript in Dresden, while some time later David Eugene Smith acquired for the Columbia University Library a manuscript from the hand of Johann Scheybl which contains a third transcription of the algebra.

#### 5.2 The Vienna Manuscript (V)

Codex Vindobonensis 4770 (Rec. 3246) XIV. 339. 8° (V). Fol. 1<sup>r</sup>–12<sup>v</sup>, *Liber restorationis et oppositionis numeri*; our text.

#### 5.3 The Dresden Manuscript (D)

Codex Dresdensis C. 80. Fifteenth century. Paper. By various hands. 416 pages, numbered 1–225, 227–417. Folio, bound in parchment (D).

#### 5.4 The Columbia University Manuscript (C)

Codex Universitatis Columbiae; Columbia University Library Manuscript, X 512, Sch. 2, Q. 308 pages. (C.) Pages 71–122, *Liber Algebrae et Almucabola, continens demonstrationes aequationum regularum Algebrae*.



C = Codex Universitatis Columbiae

D = Codex Dresdensis

R = Fragmentum Romain

V = Codex Vindobonensis

After the note to line 6 of the second page of the Latin text, the notes without any letter indicate the concurrence of V and D.

|                      |                        |
|----------------------|------------------------|
| <i>add. vel add.</i> | addition(s) by Scheybl |
| <i>marg.</i>         | marginal note          |
| <i>man.</i>          | manuscript             |
| <i>om.</i>           | omits                  |
| <i>corr.</i>         | corrected              |
| <i>passim</i>        | throughout             |
| <i>supra versum</i>  | above the line         |
| <i>infra</i>         | below                  |

# Latin Text with Critical Notes

LIBER ALGEBRAE ET ALMUCABOLA,  
continens demonstrationes aequationum regularum Algebrae

Ab incerto auctore olim arabice conscriptus atque deinde a  
Roberto Cestrensi, in ciuitate Secobiensi anno 1183,  
vt fertur, latino sermoni donatus.

LIBER ALGEBRAE ET ALMUCABOLA, de quaestionibus arithmeticis et geometricis

In nomine dei pii et misericordis incipit liber Restaurationis et Oppositionis numeri quem  
aedidit Mahomet, filius Mosi Algaurizin.<sup>11 22 33</sup>

Dixit Mahomet, Laus deo creatori, qui homini contulit scientiam inueniendi vim numerorum. Considerans enim omne id quo indigent homines, ex numeris componi, inueni illud totum esse numerum, et inueni nihil aliud esse numerum, nisi quod ex vnitatibus componitur. Vnitas ergo in omni numero reperitur. Inueni autem omnem numerum ita dispositum, vt omnis numerus vnitatem excedit vsque ad decem, denarius quoque numerus ad modum vnitatis disponitur, vnde et duplicatur et triplicatur, quemadmodum factum cum vnitatem. Fiuntque ex eius duplicatione, 20: et triplicatione, 30. Et sic multiplicando denarium numerum, ad centenarium peruenitur. Ita centenarius numerus duplicatur et triplicatur, sicut denarius numerus. Et sic centenarium numerum duplicando et triplicando etc. millenarius excrescit numerus. Ad hunc modum numerum millenarium secundum ordinem numerorum multiplicando, vsque ad infinitam numeri inuestigationem peruenitur.<sup>44 55 66 77 88 99 1010</sup>

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<sup>1</sup> [MS note 1]: 7. Liber... geometricis om. VDR; + Praefatio R.

<sup>2</sup> [MS note 2]: 8. instauracionis D.

<sup>3</sup> [MS note 3]: 9. Mahumed filius moysi algaurizim V; Machumed filius moysi algaurizm D; Mahumed filius Moysis algaorizim R.

<sup>4</sup> [MS note 4]: 10. creatori om. V.

<sup>5</sup> [MS note 5]: 11. ex numero VDR. componi om. VDR. id pro illud V.

<sup>6</sup> [MS note 6]: 13. omnem om. D. ita + essentialiter VD.

<sup>7</sup> [MS note 7]: 14. vnitatem corr. ex vnitas D man. 2. excedat VDR. decenus pro denarius fere ubique V; decimus ubique D; decenarius ubique R.

<sup>8</sup> [MS note 8]: 18. Ita: Proinde VD; Post modum R. centenus D. sicut denarius numerus: ad modum 10<sup>m</sup> V; ad modum numeri decimi D.

<sup>9</sup> [MS note 9]: 19. duplicando V. decenum pro centenarium VD. triplicando multiplicando D. millenus VD.

<sup>10</sup> [MS note 10]: 20. hunc + ergo VDR. etc. om. VDR. ad modos VD. infiniti V. conuertitur pro peruenitur VDR.

Postea inueni numerum restorationis et oppositionis his tribus modis esse inuentum, scilicet radicibus, substantiis et numeris.<sup>1111</sup>

Solus numerus tamen neque radicibus neque substantiis vlla proportionem coniunctus est. Earum igitur radix est omnis res ex vnitatibus cum se ipsa multiplicata aut omnis numerus supra vnitatem cum se ipso multiplicatus: aut quod infra vnitatem diminutum cum se ipso multiplicatum reperitur. Substantia vero est omne illud quod ex multiplicatione radices cum se ipsa colligitur.<sup>1212 1313 1414 1515</sup>

Ex his igitur tribus modis semper duo sunt sibi inuicem coequantia, sicut diceret

Substantiae radices coequant  
Substantiae numeros coequant, et  
Radices numeros coequant.

<sup>1616</sup>

## Caput Primum. De substantiis radices coequantibus

Substantiae quae radices coequant sunt, si dicas,

Substantia quinque coequatur radicibus. Radix igitur substantiae sunt 5, et 25 ipsam componunt substantiam, quae videlicet suis quinque aequatur radicibus.

Et etiam si dicas, Tertia pars substantiae quatuor aequatur radicibus: Radix igitur substantiae sunt 12, et 144 ipsam demonstrant substantiam.

Et etiam ad similitudinem, Quinque substantiarum 10 radices coequantium. Vna igitur substantia duabus radicibus aequiparatur, et radix substantiae sunt 2 --- et substantiam quaternarius ostendit numerus.

Eodem namque modo, hoc quod ex substantiis excreuerit, aut minus ea fuerit, ad vnam conuertitur substantiam. Et similiter facies cum eo quod cum ipsis ex radicibus fuerit.

## Caput II. De substantiis numeros coequantibus

Substantiae vero numeros coequantes hoc modo proponuntur. Substantia nouenario coequatur numero. Nouenarius igitur numerus mensurat substantiam cuius vnam radicem ternarius ostendit numerus. Eodem modo iuxta multitudinem et paucitatem substantiarum ipsae substantiae ad vnius substantiae similitudinem erunt tractandae. Hoc est, si substantiae duae vel tres vel quatuor, siue etiam plures fuerint, earum cum suis radicibus coequatio, sicut vnius cum sua radice, quaerenda est.

Si vero minus vna fuerit, hoc est, si tertia vel quarta vel quinta pars substantiae vel radices proposita fuerit, eodem modo ea tractetur, vt si dicas,

Quinque substantiae 80 coequantur. Vna igitur substantia quintae parti numeri 80 coequatur, quam videlicet 16 componunt.

Et etiam si dicas. Medietas substantiae 18 coequatur. Haec igitur substantia 36 coequatur.

<sup>11</sup> [MS note 11]: 1–2. *id est pro scilicet VR; et Solis neque bis supra versum D man. 2.*

<sup>12</sup> [MS note 12]: 2. *coniungitur DR marg. alii coniunctis D man. 2; coniunctis V.*

<sup>13</sup> [MS note 13]: 3. *in se pro cum se fere ubique VDR.*

<sup>14</sup> [MS note 14]: 5–6. *vero om. D. ex radice in se ipsa multiplicatione VDR.*

<sup>15</sup> [MS note 15]: 6. *semper om. VD.*

<sup>16</sup> [MS note 16]: 9–10. *et radices numeros coequant om. D.*

### Caput III. De radicibus numeros coequantibus

Radices quae numeros coequant sunt, si dicas. Radix ternario aequatur numero. Et etiam si dicas, Quatuor radices vigeno coequantur numero. Vna igitur radix substantiae huius quinario coequantur numero. Et etiam si dicas, Media radix denario coequantur numero. Tota igitur radix vigeno aequatur numero, cuius videlicet substantiam 400 demonstrant.

Radices igitur et substantiae et numeri solum, quemadmodum diximus, distinguuntur. Vnde et ex his tribus modis quos iam praemisimus, tria oriuntur genera tripliciter distincta; vt

Substantia et radices numeros coequant  
Substantia et numeri radices coequant, et  
Radices et numeri substantiam coequant.

### Caput IIII. De substantiis et radicibus numeros coequantibus

Substantiae vero et radices quae numeros coequant, sunt, si dicas. Substantia et 10 radices 39 coequantur drachmis. Huius igitur artis inuestigatio talis est: die, quae est substantia, cui si similitudinem decern suarum radicum adiunxeris, ad 39 tota haec collectio protendatur. Modus hanc artem inueniendi est, vt radices iam pronunciatas per medium diuidas, sed radices in hac interrogatione sunt 10, accipe igitur 5, et iis cum se ipsis multiplicatis producuntur 25; quae omnia 39 adicias, et veniunt 64. Huius igitur radice quadrata accepta, quae est 8, ab ea medietatem radicum 5 subtrahas, et manebunt 3. Ternarius igitur numerus huius substantiae vnam ostendit radicem, quae videlicet substantia nouenario dinoscitur numero. Nouem igitur illam componunt substantiam.<sup>17 17 18 18 19 19</sup>

Similiter quotquot substantiae propositae fuerint, omnes ad vnam conuertas substantiam. Similiter quicquid cum eis ex numeris siue radicibus fuerit, id omne eo modo quo cum substantiis existi, conuertas. Huius autem conuersionis talis est modus, vt si dicas,

Duae substantiae et 10 radices 48 drachmis coequantur. Huius artis talis est inuestigatio, vt dicas, Quae sunt duae substantiae inuicem collectae, quibus si similitudo 10 radicum earum adiuncta fuerit, ad 48 tota extendatur collectio. Nunc autem oportet, vt duas substantias ad vnam conuertas. Sed iam manifestum est, quoniam vna substantia medietatem duarum designat, igitur omnem rem in hac quaestione tibi propositam, ad medium conuertas, dicendo

Substantia et 5 radices 24 drachmis coequantur. Modus huius rei talis est, vt dicas. Quae est substantia, cui si quinque suas radices adiunxeris, ad 24 excrescat. Nunc etiam oportet, vt ad regulam supra datam animum conuertas, et diuidas radices per medium et veniunt 2 et vnus medietas, iis cum se ipsis multiplicatis, producuntur 6 et 1/4, illis 24 adicias, veniunt 30 et vnus 1/4. Postea huius aggregati radicem quadratam accipias, quam scilicet 5 et vnus medietas componunt, ex qua medietatem radicum, 2 1/2, subtrahas et manebunt 3, quae vnam radicem substantiae exprimunt, quam substantiam nouenarius componit numerus.<sup>20 20</sup>

Et si diceretur, Medietas substantiae et quinque radices 28 coequantur drachmis. Huius questionis talis est modus, vt dicas. Quae est substantia, cuius medietati si quinque suas radices adiunxeris, tota summa ad 28 excrescat, ita tamen vt substantia quae prius diminuta fuerat,

<sup>17</sup> [MS note 17]: 28. *dragmatibus* pro drachmis; *dragma* pro *drachma* ubique.

<sup>18</sup> [MS note 18]: 31. *xxxviii D. facie siue sum pro pronunciatis.*

<sup>19</sup> [MS note 19]: 32. *eas + ergo pro iis.*

<sup>20</sup> [MS note 20]: In marg.  $x^2 + 10x = 24$  D.

perfecta compleatur. Igitur huius substantiae medietas cum radicibus secum pronunciatis, duplicanda est; veniunt autem, substantia et 10 radices 56 drachmis aequales.

Diuide igitur radices per medium, et veniunt 5, quibus cum seipsis multiplicatis, producuntur 25; illa adde 56, et colliguntur 81; huius collecti radicem quadratam accipias, quam nouenarius componit numerus, atque ex ea medietatem radicem 5 subtrahas et manebunt 4, substantiae radix.

## Caput V. De substantiis et numeris radices coequantibus

Propositio huius rei talis est, vt dicas, Substantia et 21 drachmae 10 radicibus coequantur.<sup>2121</sup>  
<sup>2222</sup>

Ad hoc inuestigandum talis datur regula, vt dicas. Quae est substantia, cui si 21 drachmas adiunxeris, tota summa simul decern ipsius substantiae radices exhibeat. Huius quaestionis solutio hoc modo concipitur, vt radices primum per medium diuidas, et veniunt in hoc casu 5, haec cum seipsis multiplicata, producuntur 25. Ex illis 21 drachmas, quas paulo ante cum substantiis commemorauimus, subtrahas, et manebunt 4, horum radicem quadratam accipias, vt sunt 2, quae ex medietate radicem 5 diminuas, et manebunt 3, vnam radicem huius substantiae constituentia, quam scilicet substantiam nouenarius complet numerus.<sup>2323</sup>

Quod si libuerit, poteris ipsa 2, quae a medietate radicem iam diminuisti, medietati radicem 5 scilicet addere, et veniunt 7; quae vnam substantiae radicem demonstrant quam substantiam 49 adimplent. Cum igitur aliquod huius capituli exemplum tibi propositum fuerit, ipsius modum cum adiectione, quemadmodum dictum est, inuestiga, quern si cum adiectione non inueneris, procul dubio cum diminutione reperies. Hoc enim caput solum adiectione simul et diminutione indiget quod in aliis capitulis praemissis minime reperies.

Sciendum est etiam, quando radices iuxta hoc caput mediaueris, et medietatem deinde cum seipsa multiplicaueris, si quod ex multiplicatione tollitur vel procreatur, minus fuerit drachmis cum substantia pronunciatis: quaestio tibi proposita nulla erit. At si drachmis aequale fuerit vna radix fientque 25.

## Caput VI. De radicibus et numeris substantiam coequantibus

In hoc capite sic proponitur, Tres radices et quatuor ex numeris coequantur substantiae.<sup>2424</sup>  
<sup>2525</sup> <sup>2626</sup>

Ad hoc inuestigandum talis datur regula, quod scilicet radices per medium diuidas et venit vnum et alterius medietas, hoc deinde cum seipsis multiplices, et producuntur  $2\frac{1}{4}$ , his 4 ex numeris adiicias, et veniunt  $6\frac{1}{4}$ , huius postea radicem quadratam accipias, hoc est  $2\frac{1}{2}$ . Atque earn tandem medietati radicem, vni scilicet et dimidio, adiicias, et veniunt 4, quae vnam substantiae radicem componunt, quam deinde substantiam numerus 16 adimplet.

Quicquid igitur plus siue minus substantia tibi propositum fuerit, ad vnam conuertas substantiam. Ex his igitur modis, de quibus in principio libri mentionem fecimus, sunt tres priores,

<sup>21</sup> [MS note 21]: 8. *Titulum om. V; autem huius artis pro huius rei. radicem pro radices D.*

<sup>22</sup> [MS note 22]: 9. *coequantur pro aequatur.*

<sup>23</sup> [MS note 23]: In marg.  $x^2 + 21 = 10x \vee D.$

<sup>24</sup> [MS note 24]: 5. *Titulum om. V.*

<sup>25</sup> [MS note 25]: 6. *hoc + autem.*

<sup>26</sup> [MS note 26]: 7. *Tres + autem.*

in quibus radices non mediantur, in tribus vero posterioribus vel residuis mediantur radices prout superius liquet.

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Dixit Algaurizin. Sex sunt modi de quibus, quantum ad numeros pertinet, sufficienter diximus. Nunc vero oportet, vt quod per numeros proposuimus, ex geometrica idem verum esse demonstramus.<sup>27 27 28 28 29 29</sup>

Nostra igitur prima propositio talis est, Substantia et 10 radices 39 coequantur drachmis. Huius probatio est, vt quadratum cuius latera ignorantur, proponamus. Hoc autem quadratum quod loco substantiae ponimus, et eius radicem scire volumus atque designare. Sit igitur quadratum a b cuius vnumquodque latus vnam ostendit radicem. Iam manifestum est, quoniam, quando aliquod eius latus cum numero numerorum multiplicauerimus, tunc hoc quod ex multiplicatione colligitur erit numerus radicum radici ipsius numeri aequalis.

Quoniam ergo decern radices cum substantia pronunciantur, quartam igitur partem numeri decem accipimus, atque vnicuique lateri quadrati aream aequedistantium laterum applicamus quarum longitudinem quidem longitudine quadrati primo descripti, latitudinem vero duo et dimidium, quae sunt numeri 10 quarta pars, demonstrant. Quatuor igitur areae laterum aequedistantium primo quadrato a b applicantur. Quarum singularum longitudo longitudini vnius radices quadrati a b aequalis erit, latitudo etiam singularum 2 et medium, vt iam dictum est, demonstrat. Sunt autem hae areae 10 c d e f. Ex hoc igitur quod diximus, sit area laterum inaequalium, quae similiter ponuntur ignota, in cuius videlicet quatuor angulis quorum vnius cuiusque quantitas areae, quam 2 et dimidium cum duobus et dimidio multiplicata perficiunt, imperfectionem maioris seu totius areae ostendunt.

Vnde fit vt circunductionem areae maioris, cum adiectione duorum et dimidii cum duobus et dimidio quatuor vicibus multiplicatorum compleamus, generat autem haec tota multiplicatio 25. Et iam manifestum est, quoniam primum quadratum, quod substantiam significat, et quatuor areae ipsum quadratum circundantes, 39 perficiunt, quibus quando 25, hoc est quatuor minora quadrata, quae scilicet super quatuor angulos quadrati a b ponuntur, adiecerimus, quadratum maius, G H vocatum, circunductione complebitur. Vnde etiam haec tota numeri summa vsque ad 64 excrescet, cuius summae radicem octonarius obtinet numerus, quo etiam vnum eius latus compleri probatur. Igitur vbi ex numero octonario quartam partem numeri denarii, sicut in extremitatibus quadrati maioris G H ponuntur, subtraxerimus bis, 3 ex ipsius latere manebunt. Quinque ergo ex octo subtractis, 3 manere necesse est; quae simul vni lateri quadrati primi, quod est a b, aequiparantur. Haec igitur 3 quadrati vnam radicem, hoc est vnam radicem substantiae propositae: nouenarius deinde numerus ipsam substantiam exprimit.

Ergo numerum denarium mediamus, et alteram eius medietatem cum seipsa multiplicamus, deinde totum multiplicationis productum numero 39 adiciamus, vt maioris quadrati G H circunductio compleatur. Nam eius quatuor angulorum diminutio totam huius quadrati circunductionem imperfectam reddebat. Manifestum enim est, quod quarta pars omnis numeri cum suo aequali, ac deinde cum quatuor multiplicata, eandem perficiat numerum, quem medietas numeri cum seipsa multiplicata, perficit. Igitur si radicum medietas cum seipsa multiplicetur, huius multiplicationis summa, multiplicationem quartae partis cum seipsa ac deinde cum quatuor multiplicatae, sufficienter euacuet, adaequabit vel delebit.

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<sup>27</sup> [MS note 27]: 18. *alguarizim* V; *alghuarizim* D.

<sup>28</sup> [MS note 28]: 19. *Sex + autem*.

<sup>29</sup> [MS note 29]: 20. *numero pro per numeros. geometricae*.

Ad hoc etiam idem demonstrandum, altera datur formula, quae talis est. Quadrate  $a$   $b$  substantiam significante aequalitatem decem radicum addimus; has radices deinde per medium diuidamus, venient 5, ex quibus duas areas ad duo latera quadrati  $a$   $b$  constituamus, hae autem vocentur  $a$   $g$  et  $b$   $d$ , et erit vtriusque vtraque latitudo vni lateri quadrati  $a$   $b$  aequalis; vtramque denique longitudinem numerus quinaris adimplebit. Superest iam, vt ex multiplicatione 5 cum 5, quae medietatem radicum quas ad duo latera quadrati prioris substantiam significantis, addidimus, quadratum faciamus. Vnde iam manifestum est, quod duae areae, quae supra duo latera ponuntur, et quae 10 radices substantiae significant, simul cum quadrato priori, quod est substantia, 39 ex numero adimpleant. Manifestum etiam, quod area maioris seu totius quadrati per multiplicationem 5 cum 5 perficiatur. Hoc ergo quadratum perficiatur, atque ad perfectionem eius numerus 25 ad priora 39 adiiciatur: tota igitur haec summa vsque ad 64 excrescet. Nunc summae huius radicem quadratam, quae vnum latus quadrati maioris 25 designat, accipiamus, atque inde aequalitatem eius quod ei addidimus, hoc est 5, subtrahamus, et manebunt 3; quae latus quadrati  $a$   $b$  hoc est vnam radicem substantiae propositae complere probantur. Tria igitur huius substantiae sunt radix; et substantia nouem.

De substantia et drachmis res coaequantibus. Substantia et 21 drachmae 10 rebus aequiparantur. Propositio haec seu quaestio in capite quinto proposita fuit, cuius hic demonstratio docetur. Quadratum igitur  $a$   $b$ , quod latera habet ignota, substantiam pono, atque ei parallelogrammum rectangulum, cuius vtraque latitudo vni lateri quadrati  $a$   $b$  aequalis sit, cuiusque longitudine vtraque rerum seu radicum medietatem referat, applico. Deinde vero huius rectanguli summam 21 ex numeris constituo, qui numerus cum ipsa substantia propositus est. Haec autem area vel rectangulum  $b$   $g$  inscribitur, cuius simul latitudo  $g$   $d$  dinoscitur, longitudine ergo duarum arearum inuicem coniunctarum, in  $h$   $d$  terminatur. Et iam manifestum est quod haec longitudine denarium obtinet numerum, quoniam omnis area quadrilatera et rectorum angulorum, ex multiplicatione sui lateris cum vnitate semel, vnam obtinet radicem: et si cum binarie numero, duae eiusdem areae nascuntur radices.

Quoniam igitur prime sic proposuit, vna substantia et 21 drachmae, 10 radicibus aequiparantur, manifestum est, quod longitudine lateris  $h$   $d$  in denario terminatur numero, quia latus  $h$   $b$  vnam substantiae radicem obtinet, latus igitur  $h$   $d$  10 ostendit. Vnde et linea  $e$   $h$  lineae  $e$   $d$  fiet aequalis, atque ducta ex puncto  $e$  linea perpendiculari  $e$   $t$ : haec eadem perpendicularis ipsi  $h$   $a$  lineae aequalis erit. Lineae ergo  $e$   $t$  portionem, quae ipsaeque  $d$   $e$  linea breuior est, in rectum adiicio  $e$   $c$ , et fiet linea  $t$   $c$  aequalis  $t$   $g$  lineae, vnde quadratum  $t$   $l$ , quod ex multiplicatione medietatis radicum cum se ipsa multiplicatae, id est ex multiplicatione quinaris cum quinario (in hoc casu) colligitur, nobis eueniet.

Et iam manifestum etiam est, quoniam area  $b$   $g$  21, quae substantiae addidimus, in se obtinet, ex ea igitur  $b$   $g$  per lineam  $t$   $c$ , quae est vnum latus areae  $t$   $l$  extrahamus atque ex eadem area  $b$   $g$  aream  $b$   $t$  minuamus, deinde vero super lineam  $e$   $c$  quae est in quo linea  $t$   $c$   $c$  lineam  $a$   $h$  in quantitate deuincit, quadratum  $e$   $n$   $m$   $c$  ponamus. Vnde et iam manifestum, cum linea  $t$   $c$  aequalis sit linea  $l$   $c$ , nam ipsae in quadrato  $t$   $l$  aequales protenduntur, similiter et linea  $c$   $c$  lineae  $m$   $c$  aequalis, quia ipsae quadratum  $e$   $m$  aequali dimensione circundant: aequalium igitur ab aequalibus lineis subtractione facta, linea  $t$   $c$  lineae  $l$   $m$  aequalis relinquetur quod est notandum.

Rursus manifestum est, quoniam linea  $g$   $d$  aequalis est lineae  $a$   $h$ , cum ipsae in latitudine areae  $h$   $g$  aequali dimensione tenduntur, sed linea  $a$   $h$  aequalis est lineae  $h$   $b$  cum ipsae in vno quadrato appareant. Item quoniam linea  $g$   $l$  aequalis est lineae  $d$   $e$ , cum ipsae in vno quadrato reperiantur aequales, sed linea  $d$   $e$  aequalis est  $h$   $e$ , cum ipsae decem radices per medium diuidant; linea igitur  $d$   $l$  residuum lineae  $g$   $l$ , lineae  $e$   $b$  ex linea  $e$   $h$  residuae aequalis erit: atque tandem, cum linea  $t$   $e$  et lineae  $l$   $m$  ex superiori demonstratione, non sit inaequalis, area quam linea  $t$   $e$  et  $e$   $b$  circundant, comprehensae sub  $l$   $m$  et  $d$   $l$  lineis areae aequalis erit. Area

igitur  $t b$  aequalis est  $m d$  areae.

Et iam manifestum est, quoniam quadratum  $t l$  25 in se continet, cum ergo ex eodem quadrato  $t l$  areas  $d t$  et  $m d$ , quae videlicet duabus areis  $g e$  et  $t b$ , 20 et vnum in se continentibus, sunt aequales, subtraxerimus, quadratum  $n c$  nobis manebit, qui simul numerum, qui est inter 25 et 21, obtinet. Et hic numerus est quaternarius, cuius videlicet radicem duo designant, quae latus  $e c$  adimplent. Hoc autem latus aequale est lineae  $c b$ , quoniam  $e c$  aequale est  $d l$  lateri, cum ipsa in latitudine areae  $d c$  aequalia protendantur. Iam manifestum est, quoniam  $d l$  aequalis est lineae  $e b$ , quando igitur  $e b$  quae sunt duo, ex  $e h$ , quae sunt radicum medietas, quam quinarium ostendit numerus, abstulerimus, linea  $b h$  ternarium ostendens numerum restabit. Ternarius igitur numerus radicem primae demonstrat substantiae.

Quod si contra lineam  $e c$  lineae  $e h$ , quae medietatem radicum continet, addiderimus, colligentur 7, quae lineam  $t h$  ostendunt. Et tunc radix substantiae maior restat constituta, quam substantiam 49 adimplent.



# The Book of Algebra and Almucabola

THE BOOK OF ALGEBRA AND ALMUCABOLA,  
Containing Demonstrations of the Rules of the  
Equations of Algebra

Written some time ago in Arabic by an unknown author and afterwards, according to tradition in 1183, put into Latin by Robert of Chester in the city of Segovia.

**T**HE Book of Algebra and Almucabola, concerning arithmetical and geometrical problems.

In the name of God, tender and compassionate, begins the book of Restoration and Opposition of number put forth by Mohammed Al-Khowarizmi, the son of Moses. Mohammed said, Praise God the creator who has bestowed upon man the power to discover the significance of numbers. Indeed, reflecting that all things which men need require computation, I discovered that all things involve number and I discovered that number is nothing other than that which is composed of units. Unity therefore is implied in every number. Moreover I discovered all numbers to be so arranged that they proceed from unity up to ten. The number ten is treated in the same manner as the unit, and for this reason doubled and tripled just as in the case of unity. Out of its duplication arises 20, and from its triplication 30. And so multiplying the number ten you arrive at one-hundred. Again the number one-hundred is doubled and tripled like the number ten. So by doubling and tripling etc. the number one-hundred grows to one-thousand. In this way multiplying the number one-thousand according to the various denominations of numbers you come even to the investigation of number to infinity.

Furthermore I discovered that the numbers of restoration and opposition are composed of these three kinds: namely, roots, squares<sup>30</sup> and numbers. However number alone is connected neither with roots nor with squares by any ratio. Of these then the root is anything composed of units which can be multiplied by itself, or any number greater than unity multiplied by itself: or that which is found to be diminished below unity when multiplied by itself. The square is that which results from the multiplication of a root by itself.

Of these three forms, then, two may be equal to each other, as for example

**Squares equal to roots.  
Squares equal to numbers, and  
Roots equal to numbers.**

---

<sup>30</sup>Literally 'substances,' being a translation of the Arabic word *mal*, used for the second power of the unknown. Gerard of Cremona used *census*, which has a similar meaning.

## Chapter I. Concerning Squares Equal to Roots

The following is an example of squares equal to roots: a square is equal to 5 roots. The root of the square then is 5, and 25 forms its square which, of course, equals five of its roots.<sup>31</sup>

Another example: the third part of a square equals four roots. Then the root of the square is 12 and 144 designates its square.<sup>32</sup> And similarly, five squares equal 10 roots. Therefore one square equals two roots and the root of the square is 2. Four represents the square.<sup>33</sup>

In the same manner then that which involves more than one square, or is less than one, is reduced to one square. Likewise you perform the same operation upon the roots which accompany the squares.

## Chapter II. Concerning Squares Equal to Numbers

Squares equal to numbers are illustrated in the following manner: a square is equal to nine. Then nine measures the square of which three represents one root.<sup>34</sup>

Whether there are many or few squares they will have to be reduced in the same manner to the form of one square. That is to say, if there are two or three or four squares, or even more, the equation formed by them with their roots is to be reduced to the form of one square with its root. Further if there be less than one square, that is if a third or a fourth or a fifth part of a square or root is proposed, this is treated in the same manner.<sup>35</sup>

For example, five squares equal 80. Therefore one square equals the fifth part of the number 80 which, of course, is 16.<sup>36</sup> Or, to take another example, half of a square equals 18. This square therefore equals 36.<sup>37</sup>

## Chapter III. Concerning Roots Equal to Numbers

The following is an example of roots equal to numbers: a root is equal to 3. Therefore nine is the square of this root.<sup>38</sup>

Another example: four roots equal 20. Therefore one root of this square is 5.<sup>39</sup> Still another example: half a root is equal to ten. The whole root therefore equals 20, of which, of course, 400 represents the square.<sup>40</sup>

Therefore roots and squares and pure numbers are, as we have shown, distinguished from one another. Whence also from these three kinds which we have just explained, three distinct types of equations<sup>41</sup> are formed involving three elements, as

**A square and roots equal to numbers,  
A square and numbers equal to roots, and  
Roots and numbers equal to a square.**

<sup>31</sup>  $x^2 = 5x; x = 5, x^2 = 25.$

<sup>32</sup>  $\frac{1}{3}x^2 = 4x; x = 12, x^2 = 144.$

<sup>33</sup>  $\frac{3}{5}x^2 = 10x; x^2 = 2x, x = 2, x^2 = 4.$

<sup>34</sup>  $x^2 = 9; x = 3, x^2 = 9.$

<sup>35</sup> Our modern expression "to complete the square," used in algebra, originally meant to make the coefficient of  $x^2$  equal to unity, i.e. make one whole square.

<sup>36</sup>  $5x^2 = 80; x^2 = 16.$

<sup>37</sup>  $\frac{1}{2}x^2 = 18; x^2 = 36.$

<sup>38</sup>  $x = 3; x^2 = 9.$

<sup>39</sup>  $4x = 20; x = 5, x^2 = 25.$

<sup>40</sup>  $\frac{1}{2}x = 10; x = 20, x^2 = 400.$

<sup>41</sup> Types of equations' = *genera*. Abu Kamil, Omar Al-Khayyami, Al-Karkhi, and Leonard designate these as 'composite' types. In modern notation:  $ax^2 + bx = n; ax^2 + n = bx; ax^2 = bx + n.$

## Chapter IV. Concerning Squares and Roots Equal to Numbers

The following is an example of squares and roots equal to numbers: a square and 10 roots are equal to 39 units.<sup>42</sup> The question therefore in this type of equation is about as follows: what is the square which combined with ten of its roots will give a sum total of 39? The manner of solving this type of equation is to take one-half of the roots just mentioned. Now the roots in the problem before us are 10. Therefore take 5, which multiplied by itself gives 25, an amount which you add to 39, giving 64. Having taken then the square root of this which is 8, subtract from it the half of the roots, 5, leaving 3. The number three therefore represents one root of this square, which itself, of course, is 9. Nine therefore gives that square.<sup>43</sup>

Similarly however many squares are proposed all are to be reduced to one square. Similarly also you may reduce whatever numbers or roots accompany them in the same way in which you have reduced the squares. The following is an example of this reduction: two squares and ten roots equal 48 units.<sup>44</sup>

Another possible example: half a square and five roots are equal to 28 units.<sup>45</sup> The import of this problem is something like this: what is the square which is such that when to its half you add five of its roots the sum total amounts to 28? Now however it is necessary that the square, which here is given as less than a whole square, should be completed.<sup>46</sup> Therefore the half of this square together with the roots which accompany it must be doubled. We have then, a square and 10 roots equal to 56 units. Therefore take one-half of the roots, giving 5, which multiplied by itself produces 25. Add this to 56, making 81. Extract the square root of this total, which gives 9, and from this subtract half of the roots, 5, leaving 4 as the root of the square.

## Chapter V. Concerning Squares and Numbers Equal to Roots

The following is an illustration of this type: a square and 21 units equal 10 roots.<sup>47</sup> The rule for the investigation of this type of equation is as follows: what is the square which is such that when you add 21 units the sum total equals 10 roots of that square? The solution of this type of problem is obtained in the following manner. You take first one-half of the roots, giving in this instance 5, which multiplied by itself gives 25. From 25 subtract the 21 units to which we have just referred in connection with the squares. This gives 4, of which you extract the square root, which is 2. From the half of the roots, or 5, you take 2 away, and 3 remains, constituting one root of this square which itself is, of course, 9.<sup>48</sup>

If you wish you may add to the half of the roots, namely 5, the same 2 which you have just subtracted from the half of the roots. This give 7, which stands for one root of the square,

<sup>42</sup> $x^2 + 10x = 39$ ;  $\frac{1}{2}$  of 10 is 5,  $5^2 = 25$ ,  $25 + 39 = 64$ ,  $\sqrt{64} = 8$ ;  $8 - 5 = 3$ .  $x^2 = 9$ .

<sup>43</sup>For the general type  $x^2 + bx = n$ , the solution is  $x = \sqrt{(\frac{b}{2})^2 + n} - \frac{b}{2}$ ; the negative value of the square root is neglected, as that would give a negative root of the equation.

<sup>44</sup> $2x^2 + 10x = 48$ , reducing to  $x^2 + 5x = 24$ ;  $\frac{1}{2}$  of 5 is  $2\frac{1}{2}$ ,  $(2\frac{1}{2})^2 = 6\frac{1}{4}$ ,  $24 + 6\frac{1}{4} = 30\frac{1}{4}$ ,  $\sqrt{30\frac{1}{4}} = 5\frac{1}{2}$ ;  $5\frac{1}{2} - 2\frac{1}{2} = 3$ ,  $x^2 = 9$ .

<sup>45</sup> $\frac{1}{2}x^2 + 5x = 28$ , reducing to  $x^2 + 10x = 56$ ;  $x = \sqrt{25 + 56} - 5$ , or  $x = 4$ . Note that the value of  $x^2$  is not given here as it usually is.

<sup>46</sup>Attention is called to the force of the expression, "completing the square," as here used with the meaning to make the coefficient of the second power of the unknown quantity equal to unity or making one whole square.

<sup>47</sup> $x^2 + 21 = 10x$ .

<sup>48</sup>For the general type,  $x^2 + n = bx$  the solution is  $x = \frac{b}{2} \pm \sqrt{(\frac{b}{2})^2 - n}$ , and both positive and negative values of the radical give positive solutions of the equation proposed. In this problem we have  $x^2 + 21 = 10x$ ;  $\frac{1}{2}$  of 10 is 5,  $5^2 = 25$ ,  $25 - 21 = 4$ ,  $\sqrt{4} = 2$ ;  $5 - 2 = 3$ , one root;  $5 + 2 = 7$ , the other root.

and 49 completes the square.<sup>49</sup> Therefore when any problem of this type is proposed to you, try the solution of it by addition as we have said. If you do not solve it by addition, without doubt you will find it by subtraction. And indeed this type alone requires both addition and subtraction, and this you do not find at all in the preceding types.

You ought to understand also that when you take the half of the roots in this form of equation and then multiply the half by itself, if that which proceeds or results from the multiplication is less than the units above-mentioned as accompanying the square, you have no equation.<sup>50</sup> If equal to the units, it follows that a root of the square will be the same as the half of the roots which accompany the square, without either addition or diminution.<sup>51</sup> Whenever a problem is proposed that involves two squares, or more or less than a single square, reduce to one square just as we have indicated in the first chapter.

## Chapter VI. Concerning Roots and Numbers Equal to a Square

An example of this type is proposed as follows: three roots and the number four are equal to a square.<sup>52</sup> The rule for the investigation of this kind of problem is, you see, that you take half of the roots, giving one and one-half; this you multiply by itself, producing  $2\frac{1}{4}$ . To  $2\frac{1}{4}$  add 4, giving  $6\frac{1}{4}$ , of which you then take the square root, that is,  $2\frac{1}{2}$ . To  $2\frac{1}{2}$  you now add the half of the roots, or  $1\frac{1}{2}$ , giving 4, which indicates one root of the square. Then 16 completes the square.<sup>53</sup> Now also whatever is proposed to you either more or less than a square, reduce to one square.

Now of the types of equations which we mentioned in the beginning of this book, the first three are such that the roots are not halved, while in the following or remaining three, the roots are halved, as appears above.

## Geometrical Demonstrations

**We have said enough, says Al-Khowarizmi,** so far as numbers are concerned, about the six types of equations. Now, however, it is necessary that we should demonstrate geometrically the truth of the same problems which we have explained in numbers. Therefore our first proposition is this, that a square and 10 roots equal 39 units.

The proof is that we construct a square of unknown sides, and let this square figure represent the square (second power of the unknown) which together with its root you wish to find. Let the square, then, be  $ab$ , of which any side represents one root. Now it is evident that when one side of it we multiply by any number, then the amount of that multiplication equals the number of roots which is equal to the root of that same number. Since then ten roots are proposed with the square, we take the fourth part of the number ten, that is  $2\frac{1}{2}$ , and to each of the four sides of the square we join rectangles of equal length with the square. The length of any one of the four areas is equal to the length of the side of the square  $ab$ , and the breadth of each is  $2\frac{1}{2}$ , as the quarter of ten. The four areas then  $cdef$  are joined to the four sides of the square  $ab$ . Now it is evident that the length of each side of the four areas is equal

<sup>49</sup>Another use of the expression, "completing the square."

<sup>50</sup>This corresponds to the condition,  $b^2 - 4ac < 0$ , in the equation  $ax^2 + bx + c = 0$ ; in this event the roots are imaginary.

<sup>51</sup>Condition for equal roots,  $b^2 - 4ac = 0$ .

<sup>52</sup> $3x + 4 = x^2$ ;  $\frac{1}{2}$  of 3 is  $1\frac{1}{2}$ ,  $(1\frac{1}{2})^2 = 2\frac{1}{4}$ ,  $2\frac{1}{4} + 4 = 6\frac{1}{4}$ ,  $\sqrt{6\frac{1}{4}} = 2\frac{1}{2}$ ;  $2\frac{1}{2} + 1\frac{1}{2} = 4$ , the root.

<sup>53</sup>The solution of the general type  $bx + n = ax^2$ , reduced by division to  $\frac{b}{a}x + \frac{n}{a} = x^2$ , is  $x = \sqrt{(\frac{b}{2a})^2 + \frac{n}{a} + \frac{b}{2a}}$ , and only the positive value of the radical is taken.

to the length of one root of the square  $ab$ , and the breadth of each is  $2\frac{1}{2}$ , as has been stated. These four areas represent the ten roots.

Out of this, according to what we have said, comes an area with four sides, which is similarly put as unknown, in each of whose four corners the amount of the area which  $2\frac{1}{2}$  multiplied by  $2\frac{1}{2}$  makes, indicates the imperfection of the larger or whole area. Hence it is that we complete the drawing of the larger area by adding  $2\frac{1}{2}$  multiplied by  $2\frac{1}{2}$  four times; and this whole multiplication gives 25. Now it is evident that the first square, which indicates the square, and the four areas surrounding that square, make 39; and when we add 25, that is, the four smaller squares which are placed at the four corners of the square  $ab$ , the drawing of the larger square, called  $GH$ , will be completed. Hence also the whole sum of numbers will increase to 64, of which sum the root 8 indicates one number, by which one side of the larger square is also proved to be completed. Therefore when we subtract twice the fourth part of the number ten, as placed at the extremities of the larger square  $GH$ , from 8, there will remain 3 on the side of it. Five then being subtracted from 8, 3 must necessarily remain, which equal one side of the first square, which is  $ab$ . These 3 represent one root of the square, that is one root of the proposed square: then 9 indicates the square itself.

## Regule 6 Capitulis Algabre Correspondentes

### RULES CORRESPONDING TO THE SIX CHAPTERS OF ALGEBRA

Sex sunt modi de quibus, quantum ad numeros pertinet, sufficienter diximus.

**Rule I.** When squares are equal to roots, divide the number of roots by the number of squares; the quotient gives one root.

**Rule II.** When squares are equal to numbers, divide the number of numbers by the number of squares; the quotient gives one root, and the square of this root is the required square.

**Rule III.** When roots are equal to numbers, divide the number of numbers by the number of roots; the quotient gives one root.

**Rule IV.** When squares and roots are equal to numbers, halve the roots, square the half, add this square to the numbers, extract the square root of the sum, and subtract the half of the roots; the remainder is one root.

**Rule V.** When squares and numbers are equal to roots, halve the roots, square the half; if the numbers are less than this square, subtract the numbers from the square, extract the root of the remainder, and subtract this root from the half of the roots; the remainder is one root. If the numbers are greater than the square, the problem is impossible. If the numbers equal the square, the half of the roots is one root.

**Rule VI.** When roots and numbers are equal to squares, halve the roots, square the half, add the numbers, extract the root of the sum, and add the half of the roots; the sum is one root.

## Addita Quaedam pro Declaratione Algebrae

SOME ADDITIONS IN EXPLANATION OF THE ALGEBRA  
(*Selections from Scheybl's Additions*)

Hactenus de regulis Algebrae satis dictum est. Nunc vero, vt plenior fiat intellectus, quaedam additiones apponendae sunt.

Primo sciendum est, quod omnis numerus constat ex vnitatibus, et quod vnitatis multiplicatio in seipsam semper vnitatem producit. Vnde etiam sequitur, quod omnis numeri quadratus ex multiplicatione vnitatum producit.

Secundo notandum est, quod in omni operatione Algebrae, primum oportet vt omnes substantiae ad vnam reducantur substantiam, omnesque radices ad vnam radicem, et omnes numeri ad vnum numerum.

Tertio, sciendum est quod tres sunt modi simplices et tres compositi, vt in textu dictum est. Modus autem cognoscendi vtrum aequatio possibilis sit vel impossibilis, est ille qui in capite quinto traditur, videlicet quando medietas radicum in seipsa multiplicata, productum maius est numero cum substantia proposito: tunc quaestio est possibilis. Si vero productum minus fuerit numero proposito, quaestio est impossibilis. Si aequale, tunc radix aequalis est medietati radicum.

LATIN GLOSSARY  
*With variations from classical usage noted in the text*

| Latin Term                  | Usage and Notes                                                                     |
|-----------------------------|-------------------------------------------------------------------------------------|
| <i>adiectio, -onis</i>      | addition; used for the operation of adding to both sides                            |
| <i>aequedistans</i>         | parallel (lit. equally distant); classical Latin would use <i>parallelus</i>        |
| <i>aequiparare</i>          | to equal, to be equal to                                                            |
| <i>almucabola</i>           | transliteration of Arabic <i>al-muqabala</i> , the operation of balancing/reduction |
| <i>area, -ae</i>            | area, surface; used geometrically                                                   |
| <i>circumductio, -onis</i>  | the bounding line, perimeter                                                        |
| <i>coaequare</i>            | to be equal to; frequent in the text                                                |
| <i>colligere</i>            | to collect, sum up                                                                  |
| <i>completus, -a, -um</i>   | completed; "completing the square"                                                  |
| <i>coniungere</i>           | to join, add                                                                        |
| <i>circundare</i>           | to surround (variant of <i>circumdare</i> )                                         |
| <i>decurrere</i>            | to run down, extend to                                                              |
| <i>dimidiare</i>            | to halve ( <i>mediare</i> is classical)                                             |
| <i>diminutio, -onis</i>     | subtraction, diminution                                                             |
| <i>diminuere</i>            | to diminish, subtract                                                               |
| <i>drachma, -ae</i>         | drachma, unit of number                                                             |
| <i>excedere</i>             | to exceed, be greater than                                                          |
| <i>excreuere</i>            | to increase, grow beyond                                                            |
| <i>figura, -ae</i>          | figure, shape                                                                       |
| <i>imperfectus, -a, -um</i> | incomplete (used of the incomplete square)                                          |
| <i>inuestigatio, -onis</i>  | investigation, method of solution                                                   |
| <i>latitudo, -inis</i>      | breadth, width                                                                      |
| <i>longitudo, -inis</i>     | length (for classical <i>longitudo</i> )                                            |
| <i>mediare</i>              | to halve; used frequently                                                           |
| <i>medietas, -atis</i>      | half, moiety                                                                        |
| <i>multiplicatio, -onis</i> | multiplication                                                                      |
| <i>numerus, -i</i>          | number                                                                              |
| <i>oppositio, -onis</i>     | opposition, the operation of <i>al-muqabala</i>                                     |
| <i>ostendere</i>            | to show, indicate, represent                                                        |
| <i>perficere</i>            | to complete, make perfect                                                           |
| <i>ponere</i>               | to place, put, suppose                                                              |
| <i>procreare</i>            | to produce, generate                                                                |
| <i>pronunciare</i>          | to state, express in words                                                          |
| <i>proportio, -onis</i>     | proportion                                                                          |
| <i>quadratum, -i</i>        | square                                                                              |
| <i>radix, -icis</i>         | root; also used for the unknown quantity                                            |
| <i>restauratio, -onis</i>   | restoration, the operation of <i>al-jabr</i>                                        |
| <i>rumbus</i>               | square (Arabic <i>al-rub'</i> )                                                     |
| <i>similitudo, -inis</i>    | similitude, likeness; used for "the like of"                                        |
| <i>substantia, -ae</i>      | substance, square of the unknown                                                    |
| <i>subtrahere</i>           | to subtract                                                                         |
| <i>vnitas, -atis</i>        | unity, the number one                                                               |



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# ʿUmar al-Khayyām

(1048–1131 CE / 430–517 AH)

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## رسالة في براهين الجبر والمقابلة

ṢINĀʿAT AL-JABR WA-AL-MUQĀBALAH

*The Art of Algebra and Almucabala*

*Or: Treatise on the Proofs of the Problems of Algebra and Almucabala*

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## 1 Introduction

Omar Khayyam (عمر الخيام), full name Ghiyāth al-Dīn Abū al-Faṭḥ ‘Umar ibn Ibrāhīm al-Khayyāmī al-Nisābūrī, was one of the most brilliant mathematicians, astronomers, and poets of the medieval Islamic world. Born in Nishapur in 1048 CE, he spent much of his life in Samarqand and Isfahan, where he produced groundbreaking works in mathematics and astronomy. His *Ṣinā‘at al-Jabr wa-al-Muqābalaḥ* (often cited as *Risāla fī al-Barāhīn ‘alā Masā’il al-Jabr wa-al-Muqābala* — “Treatise on the Proofs of the Problems of Algebra and Almucabala”) represents the culmination of early Islamic algebra. In this work, Khayyam:

- Provides rigorous definitions of the fundamental quantities of algebra
- Classifies all equations of degree up to three into 25 types
- Solves 14 irreducible cubic equations using geometric constructions with conic sections
- Distinguishes between arithmetical and geometrical solutions

This treatise, composed around 1079 CE, profoundly influenced later mathematicians including Sharaf al-Dīn al-Ṭūsī, and through translations into European languages, contributed to the development of algebra in the West.

### 1.1 The Author’s Preface

Khayyam opens his treatise with a preface in which he acknowledges the achievements and limitations of his predecessors. The following is the Arabic text of the preface, with parallel English translation:

#### The Author’s Introduction (مقدمة المؤلف)

الحمد لله رب العالمين، والعاقبة للمتقين، ولا عدوان إلا على الظالمين، وصلى الله على سيدنا محمد وآله أجمعين.  
*Praise be to God, Lord of the Worlds, and the good end is for the righteous, and there is no enmity save against the oppressors; and may God bless our master Muhammad and all his family.*

أما بعد، فإني قد سبقني في هذا الفن جماعة من أهل العلم، قدموا فيه أصناف المسائل، وبينوا فيه أصول المسائل، وبرهنوا على بعضها، ولم يتفق لهم برهان على البعض، ولم يفرقوا فيها بين ما يمكن أن يوجد بالقطع المخروطية، وبين ما لا يمكن أن يوجد إلا بأساليب هندسية أخرى. وقد أملت أن أجمع هذا العلم في رسالة، وأبين فيها أصناف المسائل على وجه التمييز، وأضع لكل صنف برهاناً صحيحاً، وذلك أمر لم يسبقني إليه أحد من المتقدمين.

*Now then: in this art there have preceded me a group of men of knowledge, who presented in it the species of problems, and explained therein the foundations of problems, and proved some of them, but did not succeed in proving all of them; and they did not distinguish between what can be found by conic sections and what cannot be found except by other geometrical methods. I have desired to gather this science into a treatise, and to explain therein the species of problems with proper distinction, and to set down for each species a correct proof — and this is something that none of the predecessors has preceded me in.*

### 1.2 On the Nature of This Treatise

Khayyam defines algebra as “the science of equations” (علم المعادلات). He distinguishes carefully between:

1. **Simple equations:** Those reducible to linear or quadratic forms, solved by arithmetical or Euclidean methods

## 2. Compound (irreducible) equations: Cubic equations requiring conic section constructions

فنعول: إن الجبر والمقابلة علم يُستخرج به المجهول من المعلوم المفروض، إذا كان بينهما نسبة تقتضي ذلك. وهذا العلم يتضمن الأعداد والجدور والأموال والكعاب.

*We say: Algebra and Almucabala is a science by which the unknown is extracted from the known datum, when between them there is a relation requiring that. And this science comprises numbers, roots, squares, and cubes.*

## 2 Definitions of the Fundamental Quantities

Khayyam begins with careful definitions of the algebraic quantities, following and extending the tradition of al-Khwārizmī:

أصل هذا العلم: أن العدد المفرد هو ما لا ينسب إلى جذر، ولا إلى مال، ولا إلى كعب، ولا إلى ما يتركب منها.

*The foundation of this science is: The simple number is that which is not referred to a root, nor to a square, nor to a cube, nor to what is compounded from them.*

### 2.1 The Root (الجذر)

والجذر هو كل شيء مضروب في نفسه من الواحد وما فوقه من الأعداد، وما دونه من الكسور. فإذا ضرب في نفسه فهو مال، وإذا ضرب المال في الجذر فهو كعب، وإذا ضرب الكعب في الجذر فهو مال مال، وإذا ضرب مال المال في الجذر فهو مال كعب، وإذا ضرب مال الكعب في الجذر فهو كعب كعب.

*The root is every quantity multiplied by itself, beginning from one and above in whole numbers, and below in fractions. If it is multiplied by itself, it is a square; and if the square is multiplied by the root, it is a cube; and if the cube is multiplied by the root, it is square-square; and if square-square is multiplied by the root, it is square-cube; and if square-cube is multiplied by the root, it is cube-cube.*

### 2.2 The Square (المال)

والمال هو ما ينتج من ضرب الجذر في نفسه. والكعب هو ما ينتج من ضرب المال في الجذر. وما فوق الكعب فهو مركب من هذه الأجناس.

*The square is what results from multiplying the root by itself. The cube is what results from multiplying the square by the root. And what is above the cube is compounded from these species.*

### 2.3 Classification of Algebraic Terms

Khayyam systematically enumerates the simple and compound species:

| Arabic Term | Transliteration  | Modern Notation |
|-------------|------------------|-----------------|
| جذر         | <i>jadr</i>      | $x$             |
| مال         | <i>māl</i>       | $x^2$           |
| كعب         | <i>kaʿb</i>      | $x^3$           |
| مال مال     | <i>māl māl</i>   | $x^4$           |
| مال كعب     | <i>māl kaʿb</i>  | $x^5$           |
| كعب كعب     | <i>kaʿb kaʿb</i> | $x^6$           |

### 3 Classification of Equations

This section forms the heart of Khayyam's treatise. He classifies all equations of degree up to three into **twenty-five types**. Following the conventions of his time, all coefficients are positive, so equations like  $x^3 + cx = bx^2 + d$  and  $x^3 + bx^2 = cx + d$  are considered distinct types.

#### 3.1 Equations Solvable by Arithmetical or Euclidean Methods

The first group consists of linear and quadratic equations, solvable without conic sections:

##### Simple Equations (Types 1–6)

1. جذور تساوي أعداداً – Roots equal numbers:  $bx = c$
2. أموال تساوي جذوراً – Squares equal roots:  $x^2 = bx$
3. أموال تساوي أعداداً – Squares equal numbers:  $x^2 = c$
4. أموال وجذور تساوي أعداداً – Squares and roots equal numbers:  $x^2 + bx = c$
5. أموال وأعداد تساوي جذوراً – Squares and numbers equal roots:  $x^2 + c = bx$
6. جذور وأعداد تساوي أموالاً – Roots and numbers equal squares:  $bx + c = x^2$

These six types were known since al-Khwārizmī and are solved by elementary methods.

##### Binomial Cubic Equations (Types 7–12)

وأما الكعاب والأموال والجذور التي لا تختلف فيها الأجناس، فإنها تنقسم إلى اثني عشر نوعاً: منها ما يحل بالجبر، ومنها ما يحل بالقطوع المخروطية.

*As for cubes, squares, and roots in which the species are not mixed, they divide into twelve types: some solved by algebra, and some by conic sections.*

7. كعب يساوي مالا – Cube equals square:  $x^3 = bx^2$
8. كعب يساوي جذراً – Cube equals root:  $x^3 = cx$
9. كعب يساوي عدداً – Cube equals number:  $x^3 = d$
10. مال يساوي كعباً – Square equals cube:  $bx^2 = x^3$
11. جذر يساوي كعباً – Root equals cube:  $cx = x^3$
12. عدد يساوي كعباً – Number equals cube:  $d = x^3$

These binomial equations reduce to simpler forms by division.

##### Trinomial Equations Reducible to Quadratics (Types 13–16)

13. كعاب وأموال تساوي جذوراً – Cubes and squares equal roots:  $x^3 + bx^2 = cx$
14. كعاب وجذور تساوي أموالاً – Cubes and roots equal squares:  $x^3 + cx = bx^2$
15. أموال وجذور تساوي كعاباً – Squares and roots equal cubes:  $bx^2 + cx = x^3$
16. كعاب وأموال وأعداد – [Special reducible forms]

### 3.2 Irreducible Cubic Equations (The 14 Species)

The crowning achievement of Khayyam's treatise is his systematic solution of the **14 irreducible cubic equations**. These require the intersection of conic sections for their geometric solution. We present them here with their Arabic names and modern notation:

#### The 14 Irreducible Cubic Equations (الأصناف الأربعة عشر)

##### Group I: Three-term equations with successive powers

17. كعب ومال يساويان عدداً – Cube and square equal number:

$$x^3 + bx^2 = d$$

18. كعب وعدد يساويان مالاً – Cube and number equal square:

$$x^3 + d = bx^2$$

19. كعب يساوي مالاً وعدداً – Cube equals square and number:

$$x^3 = bx^2 + d$$

##### Group II: Three-term equations with mixed powers

19. كعب وجذور يساويان عدداً – Cube and roots equal number:

$$x^3 + cx = d$$

20. كعب وعدد يساويان جذوراً – Cube and number equal roots:

$$x^3 + d = cx$$

21. كعب يساوي جذوراً وعدداً – Cube equals roots and number:

$$x^3 = cx + d$$

##### Group III: Four-term equations – three terms on one side

22. كعب وأموال يساويان جذوراً وعدداً – Cube and squares equal roots and number:

$$x^3 + bx^2 = cx + d$$

23. كعب وجذور يساويان أموالاً وعدداً – Cube and roots equal squares and number:

$$x^3 + cx = bx^2 + d$$

24. كعب وأعداد يساويان أموالاً وجذوراً – Cube and numbers equal squares and roots:

$$x^3 + d = bx^2 + cx$$

##### Group IV: Four-term equations – two terms on each side

25. كعب يساوي أموالاً وجذوراً وأعداداً – Cube equals squares, roots, and number:

$$x^3 = bx^2 + cx + d$$



## 4 On the Geometric Solution of Cubic Equations

### 4.1 The Fundamental Lemma

Before presenting his solutions, Khayyam proves a fundamental lemma about solid figures (parallelopipeds):

#### Lemma on Equal Solids (مقدمة في المجسمات المتساوية)

إذا كان لنا مكعب مستطيل السطوح، قاعدته مربعة، وارتفاعه خط معلوم، وأردنا أن نبني على قاعدة مربعة أخرى مكعباً مستطيلاً مساوياً له، فنحن نجد خطاً يكون النسبة بين ضلع القاعدة الأولى وضلع القاعدة الثانية كالنسبة بين ذلك الخط والارتفاع.

*If we have a rectangular parallelopiped whose base is a square and whose height is a known line, and we wish to construct on another square base a rectangular parallelopiped equal to it, then we find a line such that the ratio between the side of the first base and the side of the second base is as the ratio between that line and the height.*

### 4.2 Solution of “Cube and Roots Equal a Number” ( $x^3 + cx = d$ )

This is the most celebrated case in Khayyam’s treatise — the equation  $x^3 + cx = d$  (كعب وجذور يساويان عدداً). Khayyam solves it by the intersection of a **parabola** and a **semicircle**.

#### Case: $x^3 + cx = d$

فلنضرب مثلاً: كعب وجذور يساويان عدداً. فلنجعل العدد مكعباً مستطيلاً السطوح، قاعدته مربعة، أضلاعها مساوية لعدد الجذور، وارتفاعه خط معلوم. ثم نأخذ قطعاً مكافئاً رأسه نقطة، ومحوره خط مستقيم، ومعلمه مساوٍ لضلع القاعدة. ونصف دائرة على ارتفاع المكعب، فنلقي من نقطة التقاطع القطع المكافئ والدائرة خطاً إلى المحور، فذلك الخط هو الجذر المطلوب.

*Let us give an example: cube and roots equal a number. Let us make the number a rectangular parallelopiped whose base is a square, whose sides are equal to the number of roots, and whose height is a known line. Then we take a parabola whose vertex is a point, whose axis is a straight line, and whose parameter is equal to the side of the base; and we describe a semicircle on the height of the parallelopiped. From the point of intersection of the parabola and the circle we drop a line to the axis: that line is the required root.*

### The Construction

The construction proceeds as follows:

1. Let  $b^2 = c$  (the coefficient of  $x$ ), and let  $h$  be such that  $b^2h = d$  (the constant term)
2. Construct a **parabola** with vertex  $B$ , axis  $BZ$ , and parameter  $b$ , so that for any point on the parabola,  $y^2 = bx$
3. Erect perpendicular  $h$  at  $B$ , and on  $h$  as diameter describe a **semicircle**
4. Let the semicircle cut the parabola at point  $D$
5. From  $D$  drop  $DE$  perpendicular to  $h$ , and  $DZ$  perpendicular to  $BZ$
6. Then  $y = BE$  is the required root

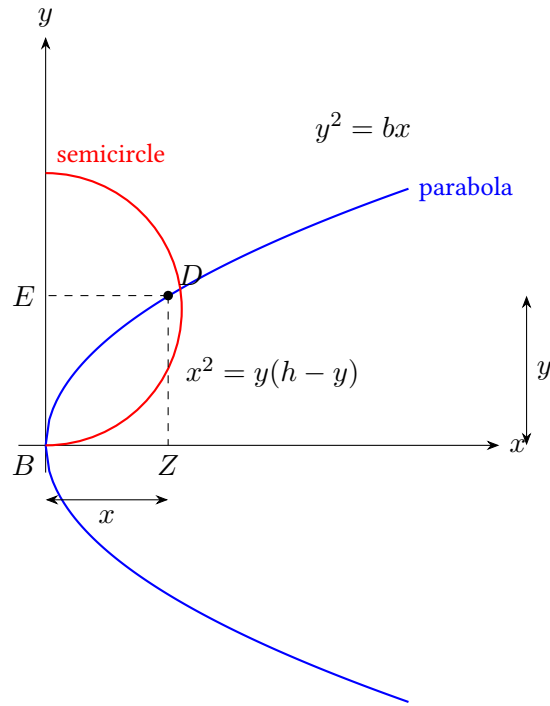


Figure 1: Khayyam's construction for  $x^3 + cx = d$  by intersection of parabola and semicircle

### The Proof

By the property of the parabola:

$$y^2 = bx \implies \frac{x}{y} = \frac{y}{b}$$

By the property of the circle (with diameter  $h$ ):

$$x^2 = y(h - y)$$

From these:

$$\frac{y}{b} = \frac{x}{y} = \frac{h - y}{x}$$

Therefore  $b : y = y : x = x : (h - y)$  are in continued proportion. From  $b^2 : y^2 = b : (h - y)$  we get:

$$b^2(h - y) = y^3$$

$$b^2h - b^2y = y^3$$

$$y^3 + b^2y = b^2h$$

Substituting  $c = b^2$  and  $d = b^2h$ :

$$\boxed{y^3 + cy = d}$$

## 5 The Remaining Irreducible Cases

### 5.1 Summary of Conic Section Pairs for Each Case

The following table summarizes the pairs of conic sections used for each of the 14 irreducible cases, organized according to Khayyam's five geometric families:

| Equation              | Conic Sections        | Khayyam's Term | Arabic |
|-----------------------|-----------------------|----------------|--------|
| $x^3 + b^2x = b^2h$   | Circle + Parabola     | دائرة ومكافئ   |        |
| $x^3 + b^2h = b^2x$   | Parabola + Hyperbola  | مكافئ وزائد    |        |
| $x^3 = b^2x + b^2h$   | Parabola + Hyperbola  | مكافئ وزائد    |        |
| $x^3 + ax^2 = c^3$    | Circle + Hyperbola    | دائرة وزائد    |        |
| $x^3 + c^3 = ax^2$    | Circle + Hyperbola    | دائرة وزائد    |        |
| $x^3 = ax^2 + c^3$    | Circle + Hyperbola    | دائرة وزائد    |        |
| $x^3 + cx = ax^2 + d$ | Circle + Hyperbola    | دائرة وزائد    |        |
| $x^3 + ax^2 = cx + d$ | Hyperbola + Hyperbola | زائدان         |        |
| $x^3 + d = ax^2 + cx$ | Circle + Hyperbola    | دائرة وزائد    |        |
| $x^3 = ax^2 + cx + d$ | Hyperbola + Hyperbola | زائدان         |        |

### 5.2 On the Existence and Multiplicity of Solutions

Khayyam carefully discusses conditions under which solutions exist. He observes that some cubic equations may have:

- No (positive real) solution
- Exactly one positive solution
- Two positive solutions

واعلم أن بعض هذه الأصناف قد يكون له حل واحد، وبعضها قد يكون له حلان، وبعضها لا يمكن أن يوجد إلا بشرط. فإذا كان القطعان لا يلتقيان فلا حل للمسألة، وإذا كانا يلتقيان في نقطة واحدة فحل واحد، وإذا كانا يلتقيان في نقطتين فحلان.

*Know that some of these species may have one solution, some may have two solutions, and some cannot be found except under a condition. If the two conics do not intersect, there is no solution to the problem; if they intersect at one point, there is one solution; and if they intersect at two points, there are two solutions.*

### 5.3 Case: Cube and Squares and Roots Equal a Number

The final case ( $x^3 = bx^2 + cx + d$ ) is the most complex:

وأما الصنف الخامس والعشرون: كعب يساوي أموالاً وجذوراً وأعداداً، فإنه يحتاج إلى قطعين زائدين مختلفين. فنصنع قطعاً زائداً له أحد محوريه على امتداد الخط المعلوم، ونصنع قطعاً زائداً آخر له أحد محوريه على امتداد الخط الآخر، ونلقي من نقطة التقاطع خطاً إلى الموضع المعلوم، فهو الجذر المطلوب.

*As for the twenty-fifth species: cube equals squares, roots, and numbers, it requires two different hyperbolas. We make a hyperbola one of whose axes is on the extension of the known line, and we make another hyperbola one of whose axes is on the extension of the other line; and from their point of intersection we drop a line to the known position, and it is the required root.*

## 6 Sample Problems and Applications

### 6.1 The Division of Ten into Two Parts

Khayyam illustrates his method with a problem leading to a cubic equation:

#### Problem (مسألة)

مسألة: قال قائل: قسّم عشرة إلى جزأين، بحيث يكون مجموع مربعي الجزأين مع خارج قسمة الكبير على الصغير اثنين وسبعين.

*A man said: Divide ten into two parts such that the sum of the squares of the two parts together with the quotient of the division of the greater by the smaller be seventy-two.*

*Solution.* Let the greater part be  $x$ . Then the smaller is  $10 - x$ . The condition gives:

$$x^2 + (10 - x)^2 + \frac{x}{10 - x} = 72$$

Simplifying:

$$2x^2 - 20x + 100 + \frac{x}{10 - x} = 72$$

This leads to:

$$x^3 + 200x = 20x^2 + 2000$$

which is of the form  $x^3 + cx = bx^2 + d$ , solved by the intersection of a circle and a hyperbola.

### 6.2 On Doubling the Cube

Khayyam shows that the ancient Delian problem (doubling the cube) is equivalent to solving  $x^3 = 2a^3$ , which falls under his classification as “cube equals number”:

ومن ذلك مضاعفة المكعب، وهي مسألة مشهورة بين القدماء. فإنها تؤول إلى أن كعباً يساوي عدداً، والعدد هو مضاعفة المكعب المعطى.

*And among these is the duplication of the cube, which is a well-known problem among the ancients. It reduces to a cube equaling a number, where the number is a multiple of the given cube.*

## 7 Concluding Remarks

Khayyam concludes his treatise with reflections on the nature of algebra and its relationship to geometry:

وقد أتممنا ما أردنا من جمع أصناف المسائل الجبرية، ووضع البراهين عليها. ونحن نعلم أن بعض الناس يعتقدون أن الجبر بريء من البرهان الهندسي، ولكن هذا باطل. فإن الأصناف التي ذكرناها لا يمكن أن تُبرهن إلا بالقطع المخروطية أو غيرها من الأدلة الهندسية. والجبر والمقابلة إنما هما وسيلة لاستخراج المجهول، والبرهان على صحة ذلك إنما هو بالهندسة.

*We have completed what we intended of gathering the species of algebraic problems and setting down proofs for them. We know that some people believe that algebra is free from geometrical proof, but this is false. For the species we have mentioned cannot be proved except by conic sections or by other geometrical arguments. Algebra and Almucabala are but a means of extracting the unknown, and the proof of its correctness is by geometry.*

والله ولي التوفيق.

*And God is the source of all success.*

---

*End of the Treatise*

## Editor's Note

This edition is based primarily on the 13th-century Lahore manuscript (Smith Oriental MS 45, Columbia University Rare Book and Manuscript Library), with reference to the following sources:

- Woepcke, F. *L'Algèbre d'Omar Alkhayyâmî*. Paris, 1851.
- Kasir, D. S. *The Algebra of Omar Khayyam*. New York: Teachers College Press, 1931.
- Rashed, R. and Vahabzadeh, B. *Omar Khayyam, the Mathematician*. New York: Bibliotheca Persica Press, 2000.

The Arabic text has been normalized for readability while preserving the technical vocabulary and mathematical content of the original. Geometric constructions are rendered in modern notation while preserving the logical structure of Khayyam's proofs.

THE ALGEBRA  
of  
MOHAMMED BEN MUSA

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EDITED AND TRANSLATED

BY

FREDERIC ROSEN

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## PREFACE

IN THE STUDY OF HISTORY, the attention of the observer is drawn by a peculiar charm towards those epochs, at which nations, after having secured their independence externally, strive to obtain an inward guarantee for their power, by acquiring eminence as great in science and in every art of peace as they have already attained in the field of war. Such an epoch was, in the history of the Arabs, that of the Caliphs Al Mansur, Harun al Rashid, and Al Mamun, illustrious contemporaries of Charlemagne; to the glory of which era, the volume now offered to the public, a new monument is endeavoured to be raised.

ABU ABDALLAH MOHAMMED BEN MUSA, of Khowarezm, who it appears, from his preface, wrote this Treatise at the command of the Caliph Al Mamun, was for a long time considered as the original inventor of Algebra.\* That he was not the inventor of the Art, is now well established; but that he was the first Mohammedan who wrote upon it, is to be found asserted in several Oriental writers. Haji Khalfa, in his bibliographical work, cites the initial words of the treatise now before us,\* and states, in two distinct passages, that its author, Mohammed ben Musa, was the first Mussulman who had ever written on the solution of problems by the rules of completion and reduction. Two marginal notes in the Oxford manuscript—from which the text of the present edition is taken—and an anonymous Arabic writer, whose *Bibliotheca Philosophorum* is frequently quoted by Casiri,\* likewise maintain that this production of Mohammed ben Musa was the first work written on the subject\* by a Mohammedan.

From the manner in which our author, in his preface, speaks of the task he had undertaken, we cannot infer that he claimed to be the inventor. He says that the Caliph Al Mamun encouraged him to write a popular work on Algebra: an expression which would seem to imply that other treatises were then already extant.

From a formula for finding the circumference of the circle, which occurs in the work itself (Text p. 51, Transl. p. 72), I have, in a note, drawn the conclusion, that part of the information comprised in this volume was derived from an Indian source; a conjecture which is supported by the direct assertion of the author of the *Bibliotheca Philosophorum* quoted by Casiri (i. 426, 428). That Mohammed ben Musa was conversant with Hindu science, is further evident from the fact\* that he abridged, at Al Mamun's request—but before the accession of that prince to the caliphate—the Sindhind, or astronomical tables, translated by Mohammed ben Ibrahim al Fazari from the work of an Indian astronomer who visited the court of Almansur in the 156th year of the Hejira (A.D. 773).

The science as taught by Mohammed ben Musa, in the treatise now before us, does not extend beyond quadratic equations, including problems with an affected square. These he solves by the same

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\*“*Haec ars olim Mahomete, Mosis Arabis filio, initium sumsit: etenim hujus rei locuples testis Leo-nardus Pisanus.*” Such are the words with which Hieronymus Cardanus commences his *Ars Magna*, in which he frequently refers to the work here translated, in a manner to leave no doubt of its identity.

\*I am indebted to the kindness of my friend Mr. Gustav Fluegel of Dresden, for a most interesting extract from this part of Haji Khalfa's work. Complete manuscript copies of the كشف الظنون are very scarce. The only two which I have hitherto had an opportunity of examining (the one bought in Egypt by Dr. Ehrenberg, and now deposited in the Royal Library at Berlin—the other among Rich's collection in the British Museum) are only abridgments of the original compilation, in which the quotation of the initial words of each work is generally omitted. The prospect of an edition and Latin translation of the complete original work, to be published by Mr. Fluegel, under the auspices of the Oriental Translation Committee, must under such circumstances be most gratifying to all friends of Asiatic Literature.

\*ابن الابي written in the twelfth century. *Bibliotheca Arabica Escorialensis*, t. i. 426, 428.

\*The first of these marginal notes stands at the top of the first page of the manuscript, and reads thus: هذا اول كتاب صنفه بيد مسلم في الجبر والمقابلة “This is the first book written on (the art of calculating by) completion and reduction by a Mohammedan: on this account the author has introduced into it rules of various kinds, in order to render useful the very rudiments of Algebra.” The other scholium stands farther on: it is the same to which I have referred in my notes to the Arabic text, p. 177.

\*Related by Ebn al Adami in the preface to his astronomical tables. Casiri, l. 427, 428. Colebrooke, Dissertation, &c. p. lxiv, lxxii.

rules which are followed by Diophantus,\* and which are taught, though less comprehensively, by the Hindu mathematicians.\* That he should have borrowed from Diophantus is not at all probable; for it does not appear that the Arabs had any knowledge of Diophantus' work before the middle of the fourth century after the Hejira, when Abu'l-Wafa Buzjani rendered it into Arabic.\*

It is far more probable that the Arabs received their first knowledge of Algebra from the Hindus, who furnished them with the decimal notation of numerals, and with various important points of mathematical and astronomical information.

But under whatever obligation our author may be to the Hindus, as to the subject matter of his performance, he seems to have been independent of them in the manner of digesting and treating it: at least the method which he follows in expounding his rules, as well as in showing their application, differs considerably from that of the Hindu mathematical writers. Bhaskara and Brahmagupta give dogmatical precepts, unsupported by argument, which, even by the metrical form in which they are expressed, seem to address themselves rather to the memory than to the reasoning faculty of the learner: Mohammed gives his rules in simple prose, and establishes their accuracy by geometrical illustrations. The Hindus give comparatively few examples, and are fond of investing the statement of their problems in rhetorical pomp: the Arab, on the contrary, is remarkably rich in examples, but he introduces them with the same perspicuous simplicity of style which distinguishes his rules. In solving their problems, the Hindus are satisfied with pointing at the result, and at the principal intermediate steps which lead to it: the Arab shows the working of each example at full length, keeping his view constantly fixed upon the two sides of the equation, as on the two scales of a balance, and showing how any alteration in one side is counterpoised by a corresponding change in the other.

Besides the few facts which have already been mentioned in the course of this preface, little or nothing is known of our Author's life. He lived and wrote under the caliphate of Al Mamun, and must therefore be distinguished from Abu Jafar Mohammed ben Musa,\* likewise a mathematician and astronomer, who flourished under the Caliph Al Motaded (who reigned A.H. 279–289, A.D. 892–902).

The manuscript from whence the text of the present edition is taken—and which is the only copy the existence of which I have as yet been able to trace—is preserved in the Bodleian collection at Oxford. It is, together with three other treatises on Arithmetic and Algebra, contained in the volume marked cmxviii. Hunt. 214, fol., and bears the date of the transcription A.H. 743 (A.D. 1342). It is written in a plain and legible hand, but unfortunately destitute of most of the diacritical points: a deficiency which has often been very sensibly felt; for though the nature of the subject matter can but seldom leave a doubt as to the general import of a sentence, yet the true reading of some passages, and the precise interpretation of others, remain involved in obscurity. Besides, there occur several omissions of words, and even of entire sentences; and also instances of words or short passages written twice over, or words

\*See Diophantus, Introd. § ii. and Book iv. problems 32 and 33.

\*Lilavati, p. 29, Vijaganita, p. 347, of Mr. Colebrooke's translation.

\*Casiri Bibl. Arab. Escur. i. 433. Colebrooke's Dissertation, &c. p. lxxii.

\*The father of the latter, Musa ben Shaker, whose native country I do not find recorded, had been a robber or bandit in the earlier part of his life, but had afterwards found means to attach himself to the court of the Caliph Al-Mamun; who, after Musa's death, took care of the education of his three sons, Mohammed, Ahmed, and Al Hassan. (Abulfaragii Histor. Dyn. p. 280. Casiri, l. 386, 418). Each of the sons subsequently distinguished himself in mathematics and astronomy. We learn from Abulfaraj (p. 281) and from Ibn Khallikan (art. ثابت بن قرة) that Thabet ben Korrah, the well-known translator of the Almagest, was indebted to Mohammed for his introduction to Al Motaded, and the men of science at the court of that caliph. Ibn Khallikan's words are: خرج من حران وسكن كفرطابا فاقام بها الى ان قدم عليه محمد بن موسى راجعا من بلاد الروم الى بغداد. (Thabet ben Korrah) left Harran, and established himself at Kafratutha, where he remained till Mohammed ben Musa arrived there, on his return from the Greek dominions to Bagdad. The latter became acquainted with Thabet and on seeing his skill and sagacity, invited Thabet to accompany him to Bagdad, where Mohammed made him lodge at his own house, introduced him to the Caliph, and procured him an appointment in the body of astronomers." Ibn Khallikan here speaks of Mohammed ben Musa as of a well-known individual: he has however devoted no special article to an account of his life. It is possible that the tour into the provinces of the Eastern Roman Empire here mentioned, was undertaken in search of some ancient Greek works on mathematics or astronomy.

foreign to the sense introduced into the text. In printing the Arabic part, I have included in brackets many of those words which I found in the manuscript, the genuineness of which I suspected, and also such as I inserted from my own conjecture, to supply an apparent hiatus.

The margin of the manuscript is partially filled with scholia in a very small and almost illegible character, a few specimens of which will be found in the notes appended to my translation. Some of them are marked as being extracted from a commentary (شرح) by Al Mozaihafi,\* probably the same author, whose full name is Jamaledin Abu Abdallah Mohammed ben Omar al Jaza'if al Mozaihafi, and whose "Introduction to Arithmetic," (مبادي علم الحساب) is contained in the same volume with Mohammed's work in the Bodleian library.

Numerals are in the text of the work always expressed by words: figures are only used in some of the diagrams, and in a few marginal notes.

The work had been only briefly mentioned in Uris' catalogue of the Bodleian manuscripts. Mr. H. T. Colebrooke first introduced it to more general notice, by inserting a full account of it, with an English translation of the directions for the solution of equations, simple and compound, into the notes of the "Dissertation" prefixed to his invaluable work, "Algebra, with Arithmetic and Mensuration, from the Sanscrit of Brahme Gupta and Bhascara." (London, 1817, 4to. pages lxxv-lxxix.)

The account of the work given by Mr. Colebrooke excited the attention of a highly distinguished friend of mathematical science, who encouraged me to undertake an edition and translation of the whole: and who has taken the kindest interest in the execution of my task. He has with great patience and care revised and corrected my translation, and has furnished the commentary, subjoined to the text, in the form of common algebraic notation. But my obligations to him are not confined to this only; for his luminous advice has enabled me to overcome many difficulties, which, to my own limited proficiency in mathematics, would have been almost insurmountable.

In some notes on the Arabic text which are appended to my translation, I have endeavoured, not so much to elucidate, as to point out for further enquiry, a few circumstances connected with the history of Algebra. The comparisons drawn between the Algebra of the Arabs and that of the early Italian writers might perhaps have been more numerous and more detailed; but my enquiry was here restricted by the want of some important works. Montucla, Cossali, Hutton, and the Basil edition of Cardanus' *Ars magna*, were the only sources which I had the opportunity of consulting.

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\*Wherever I have met with this name, it is written without the diacritical points المزيحفي, and my pronunciation rests on mere conjecture.

## The Author's Preface

IN THE NAME OF GOD, GRACIOUS AND MERCIFUL!

This work was written by Mohammed ben Musa, of Khwarezm. He commences it thus:

Praised be God for his bounty towards those who deserve it by their virtuous acts: in performing which, as by him prescribed to his adoring creatures, we express our thanks, and render ourselves worthy of the continuance (of his mercy), and preserve ourselves from change: acknowledging his might, bending before his power, and revering his greatness! He sent Mohammed (on whom may the blessing of God repose!) with the mission of a prophet, long after any messenger from above had appeared, when justice had fallen into neglect, and when the true way of life was sought for in vain. Through him he cured of blindness, and saved through him from perdition, and increased through him what before was small, and collected through him what before was scattered. Praised be God our Lord! and may his glory increase, and may all his names be hallowed—besides whom there is no God; and may his benediction rest on Mohammed the Prophet and on his descendants!

The learned in times which have passed away, and among nations which have ceased to exist, were constantly employed in writing books on the several departments of science and on the various branches of knowledge, bearing in mind those that were to come after them, and hoping for a reward proportionate to their ability, and trusting that their endeavours would meet with acknowledgment, attention, and remembrance—content as they were even with a small degree of praise; small, if compared with the pains which they had undergone, and the difficulties which they had encountered in revealing the secrets and obscurities of science.

Some applied themselves to obtain information which was not known before them, and left it to posterity; others commented upon the difficulties in the works left by their predecessors, and defined the best method (of study), or rendered the access (to science) easier or placed it more within reach; others again discovered mistakes in preceding works, and arranged that which was confused, or adjusted what was irregular, and corrected the faults of their fellow-labourers, without arrogance towards them, or taking pride in what they did themselves.

That fondness for science, by which God has distinguished the Imam al Mamun, the Commander of the Faithful (besides the caliphate which He has vouchsafed unto him by lawful succession, in the robe of which He has invested him, and with the honours of which He has adorned him), that affability and condescension which he shows to the learned, that promptitude with which he protects and supports them in the elucidation of obscurities and in the removal of difficulties,—has encouraged me to compose a short work on Calculating by (the rules of) Completion and Reduction, confining it to what is easiest and most useful in arithmetic, such as men constantly require in cases of inheritance, legacies, partition, law-suits, and trade, and in all their dealings with one another, or where the measuring of lands, the digging of canals, geometrical computation, and other objects of various sorts and kinds are concerned—relying on the goodness of my intention therein, and hoping that the learned will reward it, by obtaining (for me) through their prayers the excellence of the Divine mercy: in requital of which, may the choicest blessings and the abundant bounty of God be theirs! My confidence rests with God, in this as in every thing, and in Him I put my trust. He is the Lord of the Sublime Throne. May His blessing descend upon all the prophets and heavenly messengers!

# MOHAMMED BEN MUSA'S

## COMPENDIUM

### ON CALCULATING BY

### COMPLETION AND REDUCTION.

## Introduction

**W**HEN I CONSIDERED what people generally want in calculating, I found that it always is a number. I also observed that every number is composed of units, and that any number may be divided into units. Moreover, I found that every number, which may be expressed from one to ten, surpasses the preceding by one unit: afterwards the ten is doubled or tripled, just as before the units were: thus arise twenty, thirty, &c., until a hundred; then the hundred is doubled and tripled in the same manner as the units and the tens, up to a thousand; then the thousand can be thus repeated at any complex number; and so forth to the utmost limit of numeration.

I observed that the numbers which are required in calculating by Completion and Reduction are of three kinds, namely, **roots**, **squares**, and **simple numbers relative to neither root nor square**.

A *root* is any quantity which is to be multiplied by itself, consisting of units, or numbers ascending, or fractions descending.\*

A *square* is the whole amount of the root multiplied by itself.

A *simple number* is any number which may be pronounced without reference to root or square.

A number belonging to one of these three classes may be equal to a number of another class; you may say, for instance, "*squares are equal to roots*," or "*squares are equal to numbers*," or "*roots are equal to numbers*."

### Case I.—Squares are equal to Roots.

Of the case in which squares are equal to roots, this is an example: "A square is equal to five roots of the same;" the root of the square is five, and the square is twenty-five, which is equal to five times its root.

So you say, "one third of the square is equal to four roots;" then the whole square is equal to twelve roots; that is a hundred and forty-four; and its root is twelve.

Or you say, "five squares are equal to ten roots;" then one square is equal to two roots; the root of the square is two, and its square is four.

In this manner, whether the squares be many or few, (i.e. multiplied or divided by any number), they are reduced to a single square; and the same is done with the roots, which are their equivalents; that is to say, they are reduced in the same proportion as the squares.

### Case II.—Squares are equal to Numbers.

As to the case in which squares are equal to numbers; for instance, you say, "a square is equal to nine;" then this is a square, and its root is three. Or "five squares are equal to eighty;" then one square is equal to one-fifth of eighty, which is sixteen. Or "the half of the square is equal to eighteen;" then the square is thirty-six, and its root is six.

Thus, all squares, multiples, and sub-multiples of them, are reduced to a single square. If there be only part of a square, you add thereto, until there is a whole square; you do the same with the equivalent in numbers.

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\*By the word root, is meant the simple power of the unknown quantity.

**Case III.—Roots are equal to Numbers.**

As to the case in which roots are equal to numbers; for instance, “one root equals three in number;” then the root is three, and its square nine. Or “four roots are equal to twenty;” then one root is equal to five, and the square to be formed of it is twenty-five. Or “half the root is equal to ten;” then the whole root is equal to twenty, and the square which is formed of it is four hundred.

I found that these three kinds; namely, roots, squares, and numbers, may be combined together, and thus three compound species arise; \* that is, “*squares and roots equal to numbers*,” “*squares and numbers equal to roots*,” “*roots and numbers equal to squares*.”

**Case IV.—Roots and Squares are equal to Numbers.**

Roots and Squares are equal to Numbers; \* for instance, “one square, and ten roots of the same, amount to thirty-nine dirhems;” that is to say, what must be the square which, when increased by ten of its own roots, amounts to thirty-nine? The solution is this: you halve the number\* of the roots, which in the present instance yields five. This you multiply by itself; the product is twenty-five. Add this to thirty-nine; the sum is sixty-four. Now take the root of this, which is eight, and subtract from it half the number of the roots, which is five; the remainder is three. This is the root of the square which you sought for; the square itself is nine.

The solution is the same when two squares or three, or more or less be specified; \* you reduce them to one single square, and in the same proportion you reduce also the roots and simple numbers which are connected therewith.

For instance, “two squares and ten roots are equal to forty-eight dirhems;” \* that is to say, what must be the amount of two squares which, when summed up and added to ten times the root of one of them, make up a sum of forty-eight dirhems? You must at first reduce the two squares to one; and you know that one square of the two is the moiety of both. Then reduce every thing mentioned in the statement to its half, and it will be the same as if the question had been, a square and five roots of the same are equal to twenty-four dirhems; or, what must be the amount of a square which, when added to five times its root, is equal to twenty-four dirhems? Now halve the number of the roots; the moiety is two and a half. Multiply that by itself; the product is six and a quarter. Add this to twenty-four; the sum is thirty dirhems and a quarter. Take the root of this; it is five and a half. Subtract from this the moiety of the number of the roots, that is two and a half; the remainder is three. This is the root of the square, and the square itself is nine.

The proceeding will be the same if the instance be, “half of a square and five roots are equal to twenty-eight dirhems;” \* that is to say, what must be the amount of a square, the moiety of which, when added to the equivalent of five of its roots, is equal to twenty-eight dirhems? Your first business must be to complete your square, so that it amounts to one whole square. This you effect by doubling it. Therefore double it, and double also that which is added to it, as well as what is equal to it. Then you have a square and ten roots, equal to fifty-six dirhems. Now halve the roots; the moiety is five. Multiply

\*The three cases considered are,

1<sup>st</sup>.  $cx^2 + bx = a$

2<sup>d</sup>.  $cx^2 + a = bx$

3<sup>d</sup>.  $cx^2 = bx + a$

\*1<sup>st</sup> case:  $cx^2 + bx = a$ .

Example:  $x^2 + 10x = 39$

$x = \sqrt{25 + 39} - 5 = \sqrt{64} - 5 = 8 - 5 = 3$

i.e. the coefficient.

\*By “the number of the roots” is meant the coefficient.

\* $cx^2 + bx = a$  is to be reduced to the form  $x^2 + \frac{b}{c}x = \frac{a}{c}$ .

\* $2x^2 + 10x = 48$ , reduced to  $x^2 + 5x = 24$ ,

$x = \sqrt{6\frac{1}{4} + 24} - 2\frac{1}{2} = \sqrt{30\frac{1}{4}} - 2\frac{1}{2} = 5\frac{1}{2} - 2\frac{1}{2} = 3$ .

\* $\frac{1}{2}x^2 + 10x = 56$ ;  $x^2 + 10x = 56$ ;  $x = \sqrt{25 + 56} - 5 = \sqrt{81} - 5 = 9 - 5 = 4$ .



this by itself; the product is twenty-five. Add this to fifty-six; the sum is eighty-one. Extract the root of this; it is nine. Subtract from this the moiety of the number of roots, which is five; the remainder is four. This is the root of the square which you sought for; the square is sixteen, and half the square eight.

Proceed in this manner, whenever you meet with squares and roots that are equal to simple numbers: for it will always answer.

#### Case V.—Squares and Numbers are equal to Roots.

Squares and Numbers are equal to Roots;\* for instance, “a square and twenty-one in numbers are equal to ten roots of the same square.” That is to say, what must be the amount of a square, which, when twenty-one dirhems are added to it, becomes equal to the equivalent of ten roots of that square?

Solution: Halve the number of the roots; the moiety is five. Multiply this by itself; the product is twenty-five. Subtract from this the twenty-one which are connected with the square; the remainder is four. Extract its root; it is two. Subtract this from the moiety of the roots, which is five; the remainder is three. This is the root of the square which you required, and the square is nine. Or you may add the root to the moiety of the roots; the sum is seven; this is the root of the square which you sought for, and the square itself is forty-nine.

When you meet with an instance which refers you to this case, try its solution by addition, and if that do not serve, then subtraction certainly will. For in this case both addition and subtraction may be employed, which will not answer in any other of the three cases in which the number of the roots must be halved.\* And know, that, when in a question belonging to this case you have halved the number of the roots and multiplied the moiety by itself, if the product be less than the number of dirhems connected with the square, then the instance is impossible;\* but if the product be equal to the dirhems by themselves, then the root of the square is equal to the moiety of the roots alone, without either addition or subtraction.

In every instance where you have two squares, or more or less, reduce them to one entire square,\* as I have explained under the first case.

#### Case VI.—Roots and Numbers are equal to Squares.

Roots and Numbers are equal to Squares;\* for instance, “three roots and four of simple numbers are equal to a square.” Solution: Halve the roots; the moiety is one and a half. Multiply this by itself; the product is two and a quarter. Add this to the four; the sum is six and a quarter. Extract its root; it is two and a half. Add this to the moiety of the roots, which was one and a half; the sum is four. This is the root of the square, and the square is sixteen.

Whenever you meet with a multiple or sub-multiple of a square, reduce it to one entire square.

These are the six cases which I mentioned in the introduction to this book. They have now been explained. I have shown that three among them do not require that the roots be halved, and I have taught how they must be resolved. As for the other three, in which halving the roots is necessary, I think it expedient, more accurately, to explain them by separate chapters, in which a figure will be given for each case, to point out the reasons for halving.

\*2<sup>d</sup> case:  $cx^2 + a = bx$ .

Example:  $x^2 + 21 = 10x$

$x = \frac{10}{2} \pm \sqrt{\left(\frac{10}{2}\right)^2 - 21} = 5 \pm \sqrt{25 - 21} = 5 \pm \sqrt{4} = 5 \pm 2 = 7 \text{ or } 3$ .

\*If in an equation, of the form  $x^2 + a = bx$ ,  $\left(\frac{b}{2}\right)^2 < a$ , the case supposed in the equation cannot happen. If  $\left(\frac{b}{2}\right)^2 = a$ , then  $x = \frac{b}{2}$ .

\*The case of  $\left(\frac{b}{2}\right)^2 < a$  gives no real solution.

\* $cx^2 + a = bx$  is to be reduced to  $x^2 + \frac{a}{c} = \frac{b}{c}x$ .

\*3<sup>d</sup> case:  $cx^2 = bx + a$ .

Example:  $x^2 = 3x + 4$

$x = \sqrt{\left(\frac{3}{2}\right)^2 + 4} + \frac{3}{2} = \sqrt{2\frac{1}{4} + 4} + 1\frac{1}{2} = \sqrt{6\frac{1}{4}} + 1\frac{1}{2} = 2\frac{1}{2} + 1\frac{1}{2} = 4$ .

## Geometrical Demonstrations

**D**EMONSTRATION OF THE CASE: “a Square and ten Roots are equal to thirty-nine Dirhems.”\*

The figure to explain this a quadrate, the sides of which are unknown. It represents the square, the root of which, or the root of which, you wish to know. This is the figure  $AB$ , each side of which may be considered as one of its roots; and if you multiply one of these sides by any number, then the amount of that number may be looked upon as the number of the roots which are added to the square. Each side of the quadrate represents the root of the square; and, as in the instance the roots were connected with the square, we may take one-fourth of ten, that is to say, two and a half, and combine it with each of the four sides of the figure. Thus with the original quadrate  $AB$ , four new parallelograms are combined, each having a side of the quadrate as its length, and the number of two and a half as its breadth; they are the parallelograms  $C$ ,  $G$ ,  $T$ , and  $K$ . We have now a quadrate of equal, though unknown sides; but in each of the four corners of which a square piece of two and a half multiplied by two and a half is wanting. In order to compensate for this want and to complete the quadrate, we must add (to that which we have already) four times the square of two and a half, that is, twenty-five. We know (by the statement) that the first figure, namely, the quadrate representing the square, together with the four parallelograms around it, which represent the ten roots, is equal to thirty-nine of numbers. If to this we add twenty-five, which is the equivalent of the four quadrates at the corners of the figure  $AB$ , by which the great figure  $DH$  is completed, then we know that this together makes sixty-four. One side of this great quadrate is its root, that is, eight. If we subtract twice a fourth of ten, that is five, from eight, as from the two extremities of the side of the great quadrate  $DH$ , then the remainder of such a side will be three, and that is the root of the square, or the side of the original figure  $AB$ .

It must be observed, that we have halved the number of the roots, and added the product of the moiety multiplied by itself to the number thirty-nine, in order to complete the great figure in its four corners; because the fourth of any number multiplied by itself, and then by four, is equal to the product of the moiety of that number multiplied by itself.\* Accordingly, we multiplied only the moiety of the roots by itself, instead of multiplying its fourth by itself, and then by four.

This is the figure:

|     |      |     |
|-----|------|-----|
|     | $G$  |     |
| $C$ | $AB$ | $T$ |
|     | $K$  |     |

The same may also be explained by another figure. We proceed from the quadrate  $AB$ , which represents the square. It is our next business to add to it the ten roots of the same. We halve for this purpose the ten, so that it becomes five, and construct two quadrangles on two sides of the quadrate  $AB$ , namely,  $G$  and  $D$ , the length of each of them being five, as the moiety of the ten roots, whilst the breadth of each is equal to a side of the quadrate  $AB$ . Then a quadrate remains opposite the corner of the quadrate  $AB$ . This is equal to five multiplied by five: this five being half of the number of the roots which we have added to each of the two sides of the first quadrate. Thus we know that the first quadrate, which is the square, and the two quadrangles on its sides, which are the ten roots, make together thirty-nine. In order to complete the great quadrate, there wants only a square of five multiplied by five, or twenty-five. This we add to thirty-nine, in order to complete the great square  $SH$ . The sum is sixty-four. We extract its root, eight, which is one of the sides of the great quadrangle. By subtracting from this the same quantity which we have before added, namely five, we obtain three as the remainder. This is the side of the quadrangle  $AB$ , which represents the square; it is the root of this square, and the square itself is nine.

This is the figure:

\* Geometrical illustration of the case,  $x^2 + 10x = 39$ .

\*  $4 \times (\frac{b}{4})^2 = (\frac{b}{2})^2$ .



|      |     |
|------|-----|
| $D$  |     |
| $AB$ | $G$ |

**Demonstration of the Case: “a Square and twenty-one Dirhems are equal to ten Roots.”**

We represent the square by a quadrate  $AD$ , the length of whose side we do not know. To this we join a parallelogram, the breadth of which is equal to one of the sides of the quadrate  $AD$ , such as the side  $HN$ . This parallelogram is  $HB$ . The length of the two figures together is equal to the line  $HC$ . We know that its length is ten of numbers; for every quadrate has equal sides and angles, and one of its sides multiplied by a unit is the root of the quadrate, or multiplied by two it is twice the root of the same. As it is stated, therefore, that a square and twenty-one of numbers are equal to ten roots, we may conclude that the length of the line  $HC$  is equal to ten of numbers, since the line  $CD$  represents the root of the square. We now divide the line  $CH$  into two equal parts at the point  $G$ : the line  $GC$  is then equal to  $HG$ . It is also evident that the line  $GT$  is equal to the line  $CD$ . At present we add to the line  $GT$ , in the same direction, a piece equal to the difference between  $CG$  and  $GT$ , in order to complete the square. Then the line  $TK$  becomes equal to  $KM$ , and we have a new quadrate of equal sides and angles, namely, the quadrate  $MT$ . We know that the line  $TK$  is five; this is consequently the length also of the other sides: the quadrate itself is twenty-five, this being the product of the multiplication of half the number of the roots by themselves, for five times five is twenty-five.

We have perceived that the quadrangle  $HB$  represents the twenty-one of numbers which were added to the quadrate. We have then cut off a piece from the quadrangle  $HB$  by the line  $KT$  (which is one of the sides of the quadrate  $MT$ ), so that only the part  $TA$  remains. At present we take from the line  $KM$  the piece  $KL$ , which is equal to  $GK$ ; it then appears that the line  $TG$  is equal to  $ML$ ; moreover, the line  $KL$ , which has been cut off from  $KM$ , is equal to  $KG$ ; consequently, the quadrangle  $MR$  is equal to  $TA$ . Thus it is evident that the quadrangle  $HT$ , augmented by the quadrangle  $MR$ , is equal to the quadrangle  $HB$ , which represents the twenty-one.

The whole quadrate  $MT$  was found to be equal to twenty-five. If we now subtract from this quadrate,  $MT$ , the quadrangles  $HT$  and  $MR$ , which are equal to twenty-one, there remains a small quadrate  $KR$ , which represents the difference between twenty-five and twenty-one. This is four; and its root, represented by the line  $RG$ , which is equal to  $GA$ , is two. If you subtract this number two from the line  $CG$ , which is the moiety of the roots, then the remainder is the line  $AC$ ; that is to say, three, which is the root of the original square. But if you add the number two to the line  $CG$ , which is the moiety of the number of the roots, then the sum is seven, represented by the line  $CR$ , which is the root to a larger square. However, if you add twenty-one to this square, then the sum will likewise be equal to ten roots of the same square.

**Demonstration of the Case: “three Roots and four of Simple Numbers are equal to a Square.”**

Let the square be represented by a quadrangle, the sides of which are unknown to us, though they are equal among themselves, as also the angles. This is the quadrate  $AD$ , which comprises the three roots and the four of numbers mentioned in this instance.

In every quadrate one of its sides, multiplied by a unit, is its root. We now cut off the quadrangle  $HD$  from the quadrate  $AD$ , and take one of its sides  $HC$  for three, which is the number of the roots. The same is equal to  $RD$ . It follows, then, that the quadrangle  $HB$  represents the four of numbers which are added to the roots. Now we halve the side  $CH$ , which is equal to three roots, at the point  $G$ ; from this division we construct the square  $HT$ , which is the product of half the roots (or one and a half) multiplied by themselves, that is to say, two and a quarter. We add then to the line  $GT$  a piece equal to the line  $AH$ , namely, the piece  $TL$ ; accordingly the line  $GL$  becomes equal to  $AG$ , and the line  $KN$  equal to  $TL$ . Thus a new quadrangle, with equal sides and angles, arises, namely, the quadrangle  $GM$  and we find that the line  $AG$  is equal to  $ML$ , and the same line  $AG$  is equal to  $GL$ . By these means the line  $CG$  remains equal to  $NR$ , and the line  $MN$  equal to  $TL$ , and from the quadrangle  $HB$  a piece

equal to the quadrangle  $KL$  is cut off.

But we know that the quadrangle  $AR$  represents the four of numbers which are added to the three roots. The quadrangle  $AN$  and the quadrangle  $KL$  are together equal to the quadrangle  $AR$ , which represents the four of numbers.

We have seen, also, that the quadrangle  $GM$  comprises the product of the moiety of the roots, or of one and a half, multiplied by itself; that is to say two and a quarter, together with the four of numbers, which are represented by the quadrangles  $AN$  and  $KL$ .

There remains now from the side of the great original quadrate  $AD$ , which represents the whole square, only the moiety of the roots, that is to say, one and a half, namely, the line  $GC$ . If we add this to the line  $AG$ , which is the root of the quadrate  $GM$ , being equal to two and a half; then this, together with  $CG$ , or the moiety of the three roots, namely, one and a half, makes four, which is the line  $AC$ , or the root to a square, which is represented by the quadrate  $AD$ .

Here follows the figure. This it was which we were desirous to explain.

*We have observed that every question which requires equation or reduction for its solution, will refer you to one of the six cases which I have proposed in this book. I have now also explained their arguments.*

*Bear them, therefore, in mind.*

## On Multiplication

I SHALL NOW TEACH YOU how to multiply the unknown numbers, that is to say, the roots, one by the other, if they stand alone, or if numbers are added to them, or if numbers are subtracted from them, or if they are subtracted from numbers; also how to add them one to the other, or how to subtract one from the other.

Whenever one number is to be multiplied by another, the one must be repeated as many times as the other contains units.\*

If there are greater numbers combined with units to be added to or subtracted from them, then four multiplications are necessary;\* namely, the greater numbers by the greater numbers, the greater numbers by the units, the units by the greater numbers, and the units by the units.

If the units, combined with the greater numbers, are positive, then the last multiplication is positive; if they are both negative, then the fourth multiplication is likewise positive. But if one of them is positive, and one negative, then the fourth multiplication is negative.\*

For instance, “ten and one to be multiplied by ten and two.” Ten times ten is a hundred; once ten is ten positive; twice ten is twenty positive, and once two is two positive; this altogether makes a hundred and thirty-two.

But if the instance is “ten less one, to be multiplied by ten less one,”\* then ten times ten is a hundred; the negative one by ten is ten negative; the other negative one by ten is likewise ten negative, so that it becomes eighty: but the negative one by the negative one is one positive, and this makes the result eighty-one.

Or if the instance be “ten and two, to be multiplied by ten less one,”\* then ten times ten is a hundred, and the negative one by ten is ten negative; the positive two by ten is twenty positive; this together is a hundred and ten; the positive two by the negative one gives two negative. This makes the product a hundred and eight.

I have explained this, that it might serve as an introduction to the multiplication of unknown sums, when numbers are added to them, or when numbers are subtracted from them, or when they are subtracted from numbers.

For instance: “Ten less thing (the signification of root) to be multiplied by ten.”\* You begin by taking ten times ten, which is a hundred; less thing by ten is ten roots negative; the product is therefore a hundred less ten things.

If the instance be: “ten and thing to be multiplied by ten,”\* then you take ten times ten, which is a hundred, and thing by ten is ten things positive; so that the product is a hundred plus ten things.

If the instance be: “ten and thing to be multiplied by itself,”\* then ten times ten is a hundred, and ten times thing is ten things; and again, ten times thing is ten things; and thing multiplied by thing is a square positive, so that the whole product is a hundred dirhems and twenty things and one positive square.

If the instance be: “ten minus thing to be multiplied by ten minus thing,” then ten times ten is a hundred; and minus thing by ten is minus ten things; and again, minus thing by ten is minus ten things.

\*If  $x$  is to be multiplied by  $y$ ,  $x$  is to be repeated as many times as there are units in  $y$ .

\*If  $x \pm a$  is to be multiplied by  $y \pm b$ ,  $x$  is to be multiplied by  $y$ ,  $x$  is to be multiplied by  $b$ ,  $a$  is to be multiplied by  $y$ , and  $a$  is to be multiplied by  $b$ .

\*In multiplying  $y \pm x = a \pm b$ :

$(+a)(+b) = +ab$ ;  $(-a)(-b) = +ab$ ;  $(+a)(-b) = -ab$ ;  $(-a)(+b) = -ab$ .

\* $(10 + 1) \times (10 + 2) = 10 \times 10 + 1 \times 10 + 2 \times 10 + 1 \times 2 = 100 + 10 + 20 + 2 = 132$ .

\* $(10 - 1) \times (10 - 1) = 10 \times 10 - 10 - 10 + 1 = 100 - 20 + 1 = 81$ .

\* $(10 + 2) \times (10 - 1) = 10 \times 10 - 10 + 20 - 2 = 100 - 10 + 20 - 2 = 108$ .

\* $(10 - x) \times 10 = 10 \times 10 - 10x = 100 - 10x$ .

\* $(10 + x) \times 10 = 10 \times 10 + 10x = 100 + 10x$ .

\* $(10 + x)(10 + x) = 10 \times 10 + 10x + 10x + x^2 = 100 + 20x + x^2$ .

But minus thing multiplied by minus thing is a positive square. The product is therefore a hundred and a square, minus twenty things.

In like manner if the following question be proposed to you: “one dirhem minus one-sixth to be multiplied by one dirhem minus one-sixth;” that is to say, five-sixths by themselves, the product is five and twenty parts of a dirhem, which is divided into six and thirty parts, or two-thirds and one-sixth of a sixth. Computation: You multiply one dirhem by one dirhem, the product is one dirhem; then one dirhem by minus one-sixth, that is one-sixth negative; then, again, one dirhem by minus one-sixth is one-sixth negative: so far, then, the result is two-thirds of a dirhem: but there is still minus one-sixth to be multiplied by minus one-sixth, which is one-sixth of a sixth positive; the product is, therefore, two-thirds and one sixth of a sixth.

If the instance be, “ten minus thing to be multiplied by ten and thing,” then you say, <sup>\*</sup> ten times ten is a hundred; and minus thing by ten is ten things negative; and thing by ten is ten things positive; and minus thing by thing is a square positive; therefore, the product is a hundred dirhems, minus a square.

If the instance be, “ten minus thing to be multiplied by thing,” then you say, ten multiplied by thing is ten things; and minus thing by thing is a square negative; therefore, the product is ten things minus a square.

If the instance be, “ten and thing to be multiplied by thing less ten,” then you say, thing multiplied by ten is ten things positive; and thing by thing is a square positive; and minus ten by ten is a hundred dirhems negative; and minus ten by thing is ten things negative. You say, therefore, a square minus a hundred dirhems; for, having made the reduction, that is to say, having removed the ten things positive by the ten things negative, there remains a square minus a hundred dirhems.

If the instance be, “ten dirhems and half a thing to be multiplied by half a dirhem, minus five things,” then you say, half a dirhem by ten is five dirhems positive; and half a dirhem by half a thing is a quarter of thing positive; and minus five things by ten dirhems is fifty roots negative. This altogether makes five dirhems minus forty-nine things and three quarters of thing. After this you multiply five roots negative by half a root positive: it is two squares and a half negative. Therefore, the product is five dirhems, minus two squares and a half, minus forty-nine roots and three quarters of a root.

If the instance be, “ten and thing to be multiplied by thing less ten,” then this is the same as if it were said thing and ten by thing less ten. You say, therefore, thing multiplied by thing is a square positive; and ten by thing is ten things positive; and minus ten by thing is ten things negative. You now remove the positive by the negative, then there only remains a square. Minus ten multiplied by ten is a hundred, to be subtracted from the square. This, therefore, altogether, is a square less a hundred dirhems.

Whenever a positive and a negative factor concur in a multiplication, such as thing positive and minus thing, the last multiplication gives always the negative product. Keep this in memory.

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<sup>\*</sup> $(10 - x)(10 + x) = 10 \times 10 - 10x + 10x - x^2 = 100 - x^2.$

## On Addition and Subtraction

**K** NOW THAT the root of two hundred minus ten, added to twenty minus the root of two hundred, is just ten.\*

The root of two hundred, minus ten, subtracted from twenty minus the root of two hundred, is thirty minus twice the root of two hundred; twice the root of two hundred is equal to the root of eight hundred.\*

A hundred and a square minus twenty roots, added to fifty and ten roots minus two squares, is a hundred and fifty, minus a square and minus ten roots.\*

A hundred and a square, minus twenty roots, diminished by fifty and ten roots minus two squares, is fifty dirhems and three squares minus thirty roots.\*

I shall hereafter explain to you the reason of this by a figure, which will be annexed to this chapter.

If you require to double the root of any known or unknown square, (the meaning of its duplication being that you multiply it by two) then it will suffice to multiply two by two, and then by the square;\* the root of the product is equal to twice the root of the original square.

If you require to take it thrice, you multiply three by three, and then by the square; the root of the product is thrice the root of the original square.

Compute in this manner every multiplication of the roots, whether the multiplication be more or less than two.

If you require to find the moiety of the root of the square, you need only multiply a half by a half, which is a quarter; and then this by the square: the root of the product will be half the root of the first square. Follow the same rule when you seek for a third, or a quarter of a root, or any larger or smaller quota of it, whatever may be the denominator or the numerator.

Examples of this: If you require to double the root of nine,\* you multiply two by two, and then by nine: this gives thirty-six; take the root of this, it is six, and this is double the root of nine.

In the same manner, if you require to triple the root of nine,\* you multiply three by three, and then by nine; the product is eighty-one; take its root, it is nine, which becomes equal to thrice the root of nine.

If you require to have the moiety of the root of nine,\* you multiply a half by a half, which gives a quarter, and then this by nine; the result is two and a quarter: take its root; it is one and a half, which is the moiety of the root of nine.

You proceed in this manner with every root, whether positive or negative, and whether known or unknown.

\*  $\sqrt{200} - 10 + (20 - \sqrt{200}) = 10$ .

\*  $20 - \sqrt{200} - (\sqrt{200} - 10) = 30 - 2\sqrt{200} = 30 - \sqrt{800}$ .

\*  $50 + 10x - 2x^2 + (100 + x^2 - 20x) = 150 - 10x - x^2$ .

\*  $100 + x^2 - 20x - [50 - 2x^2 + 10x] = 50 + 3x^2 - 30x$ .

\*  $(2)^2 \times x^2 = 4x^2$ ; therefore  $\sqrt{4x^2} = 2x$ .

\*  $3\sqrt{9} = \sqrt{9 \times 9} = \sqrt{81} = 9$ .

\*  $3\sqrt{9} = \sqrt{9 \times 9} = \sqrt{81} = 9$ .

\*  $\frac{1}{2}\sqrt{9} = \sqrt{\frac{1}{4} \times 9} = \sqrt{2\frac{1}{4}} = 1\frac{1}{2}$ .

## On Division

**I**F YOU WILL DIVIDE the root of nine by the root of four,\* you begin with dividing nine by four, which gives two and a quarter: the root of this is the number which you require—it is one and a half.

If you will divide the root of four by the root of nine,\* you divide four by nine; it is four-ninths of the unit; the root of this is two divided by three; namely, two-thirds of the unit.

If you wish to divide twice the root of nine by the root of four, or of any other square,\* you double the root of nine in the manner above shown to you in the chapter on Multiplication, and you divide the product by four, or by any number whatever. You perform this in the way above pointed out.

In like manner, if you wish to divide three roots of nine, or more, or one-half or any multiple or sub-multiple of the root of nine, the rule is always the same: follow it, the result will be right.

If you wish to multiply the root of nine by the root of four,\* multiply nine by four; this gives thirty-six; take its root, it is six; this is the root of nine, multiplied by the root of four.

Thus, if you wish to multiply the root of five by the root of ten,\* multiply five by ten: the root of the product is what you have required.

If you wish to multiply the root of one-third by the root of a half,\* you multiply one-third by a half: it is one-sixth: the root of one-sixth is equal to the root of one-third, multiplied by the root of a half.

If you require to multiply twice the root of nine by thrice the root of four,\* then take twice the root of nine, according to the rule above given, so that you may know the root of what square it is. You do the same with respect to the three roots of four in order to know what must be the square of such a root. You then multiply these two squares, the one by the other, and the root of the product is equal to twice the root of nine, multiplied by thrice the root of four.

You proceed in this manner with all positive or negative roots.

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$$^*\frac{\sqrt{9}}{\sqrt{4}} = \sqrt{\frac{9}{4}} = \frac{3}{2} = 1\frac{1}{2}.$$

$$^*\frac{\sqrt{4}}{\sqrt{9}} = \sqrt{\frac{4}{9}} = \frac{2}{3}.$$

$$^*\frac{2\sqrt{9}}{\sqrt{4}} = \frac{\sqrt{4} \times \sqrt{9}}{\sqrt{4}} = \sqrt{9} = 3.$$

$$^*\sqrt{4} \times \sqrt{9} = \sqrt{4 \times 9} = \sqrt{36} = 6.$$

$$^*\sqrt{10} \times \sqrt{5} = \sqrt{5 \times 10} = \sqrt{50}.$$

$$^*\sqrt{\frac{1}{3}} \times \sqrt{\frac{1}{2}} = \sqrt{\frac{1}{3} \times \frac{1}{2}} = \sqrt{\frac{1}{6}}.$$

$$^*3\sqrt{4} \times 2\sqrt{9} = \sqrt{9 \times 4} \times \sqrt{4 \times 9} = \sqrt{36 \times 36} = 36.$$

## Demonstrations

**T**HE ARGUMENT for the root of two hundred, minus ten, added to twenty, minus the root of two hundred, may be elucidated by a figure.

Let the line  $AB$  represent the root of two hundred; let the part from  $A$  to the point  $C$  be the ten, then the remainder of the root of two hundred will correspond to the remainder of the line  $AB$ , namely to the line  $CB$ . Draw now from the point  $B$  a line to the point  $D$ , to represent twenty; let it, therefore, be twice as long as the line  $AC$ , which represents ten; and mark a part of it from the point  $B$  to the point  $H$ , to be equal to the line  $AB$ , which represents the root of two hundred; then the remainder of the twenty will be equal to the part of the line, from the point  $H$  to the point  $D$ .

As our object was to add the remainder of the root of two hundred, after the subtraction of ten, that is to say, the line  $CB$ , to the line  $HD$ , or to twenty, minus the root of two hundred, we cut off from the line  $BH$  a piece equal to  $CB$ , namely, the line  $SH$ . We know already that the line  $AB$ , or the root of two hundred, is equal to the line  $BH$ , and that the line  $AC$ , which represents the ten, is equal to the line  $SB$ , as also that the remainder of the line  $AB$ , namely, the line  $CB$  is equal to the remainder of the line  $BH$ , namely, to  $SH$ . Let us add, therefore, this piece  $SH$ , to the line  $HD$ . We have already seen that from the line  $BD$ , or twenty, a piece equal to  $AC$ , which is ten, was cut off, namely, the piece  $BS$ . There remains after this the line  $SD$ , which, consequently, is equal to ten.

This it was that we intended to elucidate.

**The argument for the root of two hundred, minus ten, to be subtracted from twenty, minus the root of two hundred, is as follows.**

Let the line  $AB$  represent the root of two hundred, and let the part thereof, from  $A$  to the point  $C$ , signify the ten mentioned in the instance. We draw now from the point  $B$ , a line towards the point  $D$ , to signify twenty. Then we trace from  $B$  to the point  $H$ , the same length as the length of the line which represents the root of two hundred; that is of the line  $AB$ . We have seen that the line  $CB$  is the remainder from the twenty, after the root of two hundred has been subtracted.

It is our purpose, therefore, to subtract the line  $CB$  from the line  $HD$ ; and we now draw from the point  $B$ , a line towards the point  $S$ , equal in length to the line  $AC$ , which represents the ten. Then the whole line  $SD$  is equal to  $SB$ , plus  $BD$ , and we perceive that all this added together amounts to thirty.

We now cut off from the line  $HD$ , a piece equal to  $CB$ , namely, the line  $HG$ ; thus we find that the line  $GD$  is the remainder from the line  $SD$ , which signifies thirty. We see also that the line  $BH$  is the root of two hundred and that the line  $SB$  and  $BC$  is likewise the root of two hundred. Now the line  $HG$  is equal to  $CB$ ; therefore the piece subtracted from the line  $SD$ , which represents thirty, is equal to twice the root of two hundred, or once the root of eight hundred.

This it is that we wished to elucidate.

As for the hundred and square minus twenty roots added to fifty, and ten roots minus two squares, this does not admit of any figure, because there are three different species, viz. squares, and roots, and numbers, and nothing corresponding to them by which they might be represented. We had, indeed, contrived to construct a figure also for this case, but it was not sufficiently clear.

The elucidation by words is very easy. You know that you have a hundred and a square, minus twenty roots. When you add to this fifty and ten roots, it becomes a hundred and fifty and a square, minus ten roots. The reason for these ten negative roots is, that from the twenty negative roots ten positive roots were subtracted by reduction. This being done, there remains a hundred and fifty and a square, minus ten roots. With the hundred a square is connected. If you subtract from this hundred and square the two squares negative connected with fifty, then one square disappears by reason of the other, and the remainder is a hundred and fifty, minus a square, and minus ten roots.

This it was that we wished to explain.

## Of the Six Problems

**B**EFORE THE CHAPTERS on computation and the several species thereof, I shall now introduce six problems, as instances of the six cases treated of in the beginning of this work. I have shown that three among these cases, in order to be solved, do not require that the roots be halved, and I have also mentioned that the calculating by completion and reduction must always necessarily lead you to one of these cases. I now subjoin these problems, which will serve to bring the subject nearer to the understanding, to render its comprehension easier, and to make the arguments more perspicuous.

### First Problem.

I have divided ten into two portions; I have multiplied the one of the two portions by the other; after this I have multiplied the one of the two by itself, and the product of the multiplication by itself is four times as much as that of one of the portions by the other.\*

Computation: Suppose one of the portions to be thing, and the other ten minus thing; you multiply thing by ten minus thing; it is ten things minus a square. Then multiply it by four, because the instance states “four times as much.” The result will be four times the product of one of the parts multiplied by the other. This is forty things minus four squares. After this you multiply thing by thing, that is to say, one of the portions by itself. This is a square, which is equal to forty things minus four squares. Reduce it now by the four squares, and add them to the one square. Then the equation is: forty things are equal to five squares; and one square will be equal to eight roots, that is, sixty-four; the root of this is eight, and this is one of the two portions, namely, that which is to be multiplied by itself. The remainder from the ten is two, and that is the other portion. Thus the question leads you to one of the six cases, namely, that of “squares equal to roots.” Remark this.

### Second Problem.

I have divided ten into two portions: I have multiplied each of the parts by itself, and afterwards ten by itself: the product of ten by itself is equal to one of the two parts multiplied by itself, and afterwards by two and seven-ninths; or equal to the other multiplied by itself, and afterwards by six and one-fourth.\*

Computation: Suppose one of the parts to be thing, and the other ten minus thing. You multiply thing by itself, it is a square; then by two and seven-ninths, this makes it two squares and seven-ninths of a square. You afterwards multiply ten by ten; it is a hundred, which must be equal to two squares and seven-ninths of a square. Reduce it to one square, through division by nine twenty-fifths;\* this being its fifth and four-fifths of its fifth, take now also the fifth and four-fifths of the fifth of a hundred; this is thirty-six, which is equal to one square. Take its root, it is six. This is one of the two portions; and accordingly the other is four. This question leads you, therefore, to one of the six cases, namely, “squares equal to numbers.”

### Third Problem.

I have divided ten into two parts. I have afterwards divided the one by the other, and the quotient was four.\*

Computation: Suppose one of the two parts to be thing, the other ten minus thing. Then you divide ten minus thing by thing, in order that four may be obtained. You know that if you multiply the quotient by the divisor, the sum which was divided is restored. In the present question the quotient is four and the divisor is thing. Multiply, therefore, four by thing; the result is four things, which are equal to the sum to be divided, which was ten minus thing. You now reduce it by thing, which you add to the four

\*  $(10 - x)x = \text{product}; x^2 = 4(10 - x)x = 40x - 5x^2; 5x^2 = 40x; x^2 = 8x; x = 8; (10 - x) = 2.$

\*  $100 = x^2 \times 2\frac{7}{9}; \frac{1}{2\frac{7}{9}} \times 100 = x^2; x = 6; 10 - x = 4.$

\*  $2\frac{7}{9} = \frac{25}{9}; \frac{1}{\frac{25}{9}} \times 100 = \frac{9}{25} \times 100 = 36; x^2 = 36; x = 6.$

\*  $\frac{10-x}{x} = 4; 10 = 5x; x = 2; (10 - x) = 8.$



things. Then we have five things equal to ten; therefore one thing is equal to two, and this is one of the two portions. This question refers you to one of the six cases, namely, “roots equal to numbers.”

#### Fourth Problem.

I have multiplied one-third of thing and one dirhem by one-fourth of thing and one dirhem, and the product was twenty.\*

Computation: You multiply one-third of thing by one-fourth of thing; it is one-half of a sixth of a square. Further, you multiply one dirhem by one-third of thing, it is one-third of thing; and one dirhem by one-fourth of thing, it is one-fourth of thing; and one dirhem by one dirhem, it is one dirhem. The result of this is: the moiety of one-sixth of a square, and one-third of thing, and one-fourth of thing, and one dirhem, is equal to twenty dirhems. Subtract now the one dirhem from these twenty dirhems; the remainder is nineteen dirhems, equal to the moiety of one-sixth of a square, and one-third of thing and one-fourth of thing. Now make your square a whole one: you do this by multiplying all that you have by twelve. Thus you have one square and seven roots equal to two hundred and twenty-eight. Halve the number of the roots, and multiply it by itself; it is twelve and one-fourth. Add this to the number, that is, to two hundred and twenty-eight; the sum is two hundred and forty and one quarter. Extract the root of this; it is fifteen and a half. Subtract from this the moiety of the roots, that is, seven and a half; there remains eight, which is the square required.

#### Fifth Problem.

I have divided ten into two parts; and having divided the one by the other, and multiplied each of them by itself, and also I had added the products, the sum was fifty-eight dirhems.\*

Computation: Suppose one of the two parts to be thing, and the other ten minus thing. Multiply ten minus thing by itself; it is a hundred and a square minus twenty things. Then multiply thing by thing; it is a square. Add both together. The sum is a hundred, plus two squares minus twenty things, which are equal to fifty-eight dirhems. Take now the twenty negative things from the hundred and the two squares, and add them to fifty-eight; then a hundred, plus two squares, are equal to fifty-eight dirhems and twenty things. Reduce this to one square, by taking the moiety of all you have. It is then: fifty dirhems and a square, which are equal to twenty-nine dirhems and ten things. Then reduce this, by taking twenty-nine from fifty: there remains twenty-one and a square, equal to ten things. Halve the number of the roots, it is five; multiply this by itself, it is twenty-five; take from this the twenty-one which are connected with the square, the remainder is four. Extract the root, it is two. Subtract this from the moiety of the roots, namely, from five, there remains three. This is one of the portions; the other is seven. This question refers you to one of the six cases, namely “squares and numbers equal to roots.”

#### Sixth Problem.

I have multiplied one-third of a root by one-fourth of a root, and the product is equal to the root and twenty-four dirhems.\*

Computation: Call the root thing; then one-third of thing is multiplied by one-fourth of thing; this is the moiety of one-sixth of the square, and is equal to thing and twenty-four dirhems. Multiply this moiety of one-sixth of the square by twelve, in order to make your square a whole one, and multiply also the thing by twelve, which yields twelve things; and also four-and-twenty by twelve: the product of the whole will be two hundred and eighty-eight dirhems and twelve roots, which are equal to one square. The moiety of the roots is six. Multiply this by itself, and add it to two hundred and eighty-eight, it will be three hundred and twenty-four. Extract the root from this, it is eighteen; add this to the

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\*  $(\frac{1}{3}x + 1)(\frac{1}{4}x + 1) = 20$ ;  $\frac{1}{12}x^2 + \frac{7}{12}x + 1 = 20$ ;  $x^2 + 7x = 228$ .  
 \*  $(10 - x)^2 + x^2 = 58$ ;  $100 - 20x + x^2 + x^2 = 58$ ;  $2x^2 - 20x + 100 = 58$ ;  $x^2 - 10x + 50 = 29$ ;  $x = 5 \pm \sqrt{25 - 29} = 5 \pm 2 = 7$   
 or 3.

\*  $\frac{1}{3}x \times \frac{1}{4}x = x + 24$ ;  $\frac{1}{12}x^2 = x + 24$ ;  $x^2 = 12x + 288$ ;  $x = 6 + \sqrt{36 + 288} = 6 + 18 = 24$ .

moiety of the roots, which was six; the sum is twenty-four, and this is the square sought for.

This question refers you to one of the six cases, namely, “roots and numbers equal to squares.”

## Various Questions

**I**F A PERSON PUTS such a question to you as: “I have divided ten into two parts, and multiplying one of these by the other, the result was twenty-one;”<sup>\*</sup> then you know that one of the two parts is thing, and the other ten minus thing. Multiply, therefore, thing by ten minus thing; then you have ten things minus a square, which is equal to twenty-one. Separate the square from the ten things, and add it to the twenty-one. Then you have ten things, which are equal to twenty-one dirhems and a square. Take away the moiety of the roots, and multiply the remaining five by itself; it is twenty-five. Subtract from this the twenty-one which are connected with the square; the remainder is four. Extract its root, it is two. Subtract this from the moiety of the roots, namely, five: there remain three, which is one of the two parts. Or, if you please, you may add the root of four to the moiety of the roots; the sum is seven, which is likewise one of the parts.

This is one of the problems which may be resolved by addition and subtraction.

If the question be: “I have divided ten into two parts, and having multiplied each part by itself, I have subtracted the smaller from the greater, and the remainder was forty;”<sup>\*</sup> then the computation is—you multiply ten minus thing by itself, it is a hundred plus one square minus twenty things; and you also multiply thing by thing, it is one square. Subtract this from a hundred and a square minus twenty things, and you have a hundred, minus twenty things, equal to forty dirhems. Separate now the twenty things from a hundred, and add them to the forty; then you have a hundred, equal to twenty things and forty dirhems. Subtract now forty from a hundred; there remains sixty dirhems, equal to twenty things: therefore one thing is equal to three, which is one of the two parts.

If the question be: “I have divided ten into two parts, and having multiplied each part by itself, I have put them together, and have added to them the difference of the two parts previously to their multiplication, and the amount of all this is fifty-four;”<sup>\*</sup> then the computation is this: You multiply ten minus thing by itself; it is a hundred and a square minus twenty things. Then multiply also the other thing of the ten by itself; it is one square. Add this together, it will be a hundred plus two squares minus twenty things. It was stated that the difference of the two parts before multiplication should be added to them. You say, therefore, the difference between them is ten minus two things. The result is a hundred and ten and two squares minus twenty-two things, which are equal to fifty-four dirhems. Having reduced and equalized this, you may say, a hundred and ten dirhems and two squares are equal to fifty-four dirhems and twenty-two things. Reduce now the two squares to one square, by taking the moiety of all you have. Thus it becomes fifty-five dirhems and a square, equal to twenty-seven dirhems and eleven things. Subtract twenty-seven from fifty-five, there remain twenty-eight dirhems and a square, equal to eleven things. Halve now the things, it will be five and a half; multiply this by itself, it is thirty and a quarter. Subtract from it the twenty-eight which are combined with the square, the remainder is two and a fourth. Extract its root, it is one and a half. Subtract this from the moiety of the roots, there remain four, which is one of the two parts.

If one say, “I have divided ten into two parts; and have divided the first by the second, and the second by the first, and the sum of the quotient is two dirhems and one-sixth;”<sup>\*</sup> then the computation is this: If you multiply each part by itself, and add the products together, then their sum is equal to one of the parts multiplied by the other, and again by the quotient which is two and one-sixth. Multiply, therefore, ten less thing by itself; it is a hundred and a square less twenty things. Multiply thing by thing; it is one square. Add this together; the sum is a hundred plus two squares less twenty things,

<sup>\*</sup> $(10 - x)x = 21; 10x - x^2 = 21$ ; which is to be reduced to  $x = 5 \pm \sqrt{25 - 21} = 5 \pm 2$ .

<sup>\*</sup> $(10 - x)^2 - x^2 = 40; 100 - 20x + x^2 - x^2 = 40; 100 - 20x = 40; 100 = 20x + 40; 60 = 20x; x = 3$ .

<sup>\*</sup> $100 - 20x + x^2 + x^2 + 10 - 2x = 54; 100 - 22x + 2x^2 = 54; 55 - 11x + x^2 = 27; x^2 + 28 = 11x; x = \frac{11}{2} \pm \sqrt{\frac{121}{4} - 28} = \frac{11}{2} \pm \frac{3}{2} = 7 \text{ or } 4$ .

<sup>\*</sup> $100 + 2x^2 - 20x = x(10 - x) \times 2\frac{1}{6} = 21\frac{2}{3}x - 2\frac{1}{6}x^2; 100 + 4\frac{1}{6}x^2 = 41\frac{2}{3}x; 24 + x^2 = 10x; x = 5 - \sqrt{25 - 24} = 5 - 1 = 4 \text{ or } 6$ .

which is equal to thing multiplied by ten less thing; that is, to ten things less a square, multiplied by the sum of the quotients arising from the division of the two parts, namely, two and one-sixth. We have, therefore, twenty-one things and two-thirds of thing less two squares and one-sixth, equal to a hundred plus two squares less twenty things. Reduce this by adding the two squares and one-sixth to a hundred plus two squares less twenty things, and add the twenty negative things from the hundred plus the two squares to the twenty-one things and two-thirds of thing. Then you have a hundred plus four squares and one-sixth of a square, equal to forty-one things and two-thirds of thing. Now reduce this to one square. You know that one square is obtained from four squares and one-sixth, by taking a fifth and one-fifth of a fifth.\* Take, therefore, the fifth and one-fifth of a fifth of all that you have. Then it is twenty-four and a square, equal to ten roots; because ten is one-fifth and one-fifth of the fifth of the forty-one things and two-thirds of a thing. Now halve the roots; it gives five. Multiply this by itself; it is five-and-twenty. Subtract from this the twenty-four, which are connected with the square; the remainder is one. Extract its root; it is one. Subtract this from the moiety of the roots, which is five. There remains four, which is one of the two parts.

Observe that, in every case, where any two quantities whatsoever are divided, the first by the second and the second by the first, if you multiply the quotient of the one division by that of the other, the product is always one.\*

If some one say: "You divide ten into two parts; multiply one of the two parts by five, and divide it by the other: then take the moiety of the quotient, and add this to the product of the one part, multiplied by five; the sum is fifty dirhems;"\* then the computation is this: Take thing, and multiply it by five. This is now to be divided by the remainder of the ten, that is, by ten less thing; and of the quotient the moiety is to be taken.

You know that if you divide five things by ten less thing, and take the moiety of the quotient, the result is the same as if you divide the moiety of five things by ten less thing. Take, therefore, the moiety of five things; it is two things and a half: and this you require to divide by ten less thing. Now these two things and a half, divided by ten less thing, give a quotient which is equal to fifty less five things: for the question states: add this (the quotient) to the one part multiplied by five, the sum will be fifty. You have already observed, that if the quotient, or the result of the division, be multiplied by the divisor, the dividend, or capital to be divided, is restored. Now, your capital, in the present instance, is two things and a half. Multiply, therefore, ten less thing by fifty less five things. Then you have five hundred dirhems and five squares less a hundred things, which are equal to two things and a half. Reduce this to one square. Then it becomes a hundred dirhems and a square less twenty things, equal to the moiety of thing. Separate now the twenty things from the hundred dirhems and square, and add them to the half thing. Then you have a hundred dirhems and a square, equal to twenty things and a half. Now halve the things, multiply the moiety by itself, subtract from this the hundred, extract the root of the remainder, and subtract this from the moiety of the roots, which is ten and one-fourth: the remainder is eight; and this is one of the portions.

If some one say: "You divide ten into two parts: multiply the one by itself; it will be equal to the other taken eighty-one times."\* Computation: You say, ten less thing, multiplied by itself, is a hundred plus a square less twenty things, and this is equal to eighty-one things. Separate the twenty things from a hundred and a square, and add them to eighty-one. It will then be a hundred plus a square, which is equal to a hundred and one roots. Halve the roots; the moiety is fifty and a half. Multiply this by itself, it is two thousand five hundred and fifty and a quarter. Subtract from this one hundred; the remainder is two thousand four hundred and fifty and a quarter. Extract the root from this; it is forty-nine and a half. Subtract this from the moiety of the roots, which is fifty and a half. There remains one, and this

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\*  $4\frac{1}{6} = \frac{25}{6}$ ;  $\frac{1}{4\frac{1}{6}} = \frac{6}{25} = \frac{1}{5} + \frac{1}{5\frac{1}{5}}$ .

\*  $\frac{a}{b} \times \frac{b}{a} = 1$ .

\*  $x^2 + 100 = 50x$ ;  $x = 25 - \sqrt{625 - 100} = 25 - \sqrt{525}$ .

\*  $100 - 20x + x^2 = 81x$ ;  $x^2 + 100 = 101x$ ;  $x = \frac{101}{2} - \sqrt{\frac{10201}{4} - 100} = 50\frac{1}{2} - 49\frac{1}{2} = 1$ .

is one of the two parts.

If some one say: "I have purchased two measures of wheat or barley, each of them at a certain price. I afterwards added the expenses, and the sum was equal to the difference of the two prices, added to the difference of the measures."<sup>\*</sup> Computation: Take what numbers you please, for it is indifferent; for instance, four and six. Then you say: I have bought each measure of the four for thing; and accordingly you multiply four by thing, which gives four things; and I have bought the six, each for the moiety of thing, for which I have bought the four; or, if you please, for one-third, or one-fourth, or for any other quota of that price, for it is indifferent. Suppose that you have bought the six measures for the moiety of thing, then you multiply the moiety of thing by six; this gives three things. Add them to the four things; the sum is seven things, which must be equal to the difference of the two quantities, which is two measures, plus the difference of the two prices, which is a moiety of thing. You have, therefore, seven things, equal to two and a moiety of thing. Remove, now, this moiety of thing, by subtracting it from the seven things. There remain six things and a half, equal to two dirhems: consequently, one thing is equal to four-thirteenths of a dirhem. The six measures were bought, each at one-half of thing; that is, at two-thirteenths of a dirhem. Accordingly, the expenses amount to eight-and-twenty thirteenths of a dirhem, and this sum is equal to the difference of the two quantities; namely, the two measures, the arithmetical equivalent for which is six-and-twenty thirteenths, added to the difference of the two prices, which is two-thirteenths: both differences together being likewise equal to twenty-eight parts.

If he say: "There are two numbers,\* the difference of which is two dirhems. I have divided the smaller by the larger, and the quotient was just half a dirhem."<sup>\*</sup> Suppose one of the two numbers to be thing, and the other to be thing plus two dirhems. By the division of thing by thing plus two dirhems, half a dirhem appears as quotient. You have already observed, that by multiplying the quotient by the divisor, the capital which you divided is restored. This capital, in the present case, is thing. Multiply, therefore, thing and two dirhems by half a dirhem, which is the quotient; the product is half one thing plus one dirhem; this is equal to thing. Remove, now, half a thing on account of the other half thing; there remains one dirhem, equal to half a thing. Double it, then you have one thing, equal to two dirhems. Consequently, the other\* number is four.

If some one say: "I have divided ten into two parts; I have multiplied the one by ten and the other by itself, and the products were the same;"<sup>\*</sup> then the computation is this: You multiply thing by ten; it is ten things. Then multiply ten less thing by itself; it is a hundred and a square less twenty things, which is equal to ten things. Reduce this according to the rules, which I have above explained to you.

In like manner, if he say: "I have divided ten into two parts; I have multiplied one of the two by the other, and have then divided the product by the difference of the two parts before their multiplication, and the result of this division is five and one-fourth;"<sup>\*</sup> the computation will be this: You subtract thing from ten; there remain ten less thing. Multiply the one by the other, it is ten things less a square. This is the product of the multiplication of one of the two parts by the other. At present you divide this by the difference between the two parts, which is ten less two things. The quotient of this division is, according to the statement, five and a fourth. If, therefore, you multiply five and one-fourth by ten less two things, the product must be equal to the above amount, obtained by multiplication, namely, ten

<sup>\*</sup>The purchaser does not make a clear enunciation of the terms of his bargain. He intends to say, "I bought  $m$  bushels of wheat, and  $n$  bushels of barley, and the wheat was  $r$  times dearer than the barley. The sum I expended was equal to the difference in the quantities, added to the difference in the prices of the grain." If  $x$  is the price of the barley,  $rx$  is the price of the wheat; whence,  $mrx + nx = (m - n) + (rx - x)$ ;  $x = \frac{m-n}{mr+n-r+1}$ .

<sup>\*</sup>"Squares" in the original. The word square is used in the text to signify either, 1st, a square, properly so called, fractional or integral; 2d, a rational integer, not being a square number; 3d, a rational fraction, not being a square; 4th, a quadratic surd, fractional or integral.

<sup>\*</sup> $\frac{x}{x+2} = \frac{1}{2}$ ;  $\frac{1}{2}(x+2) = x$ ;  $\frac{1}{2}x + 1 = x$ ;  $1 = \frac{1}{2}x$ ;  $x = 2$ ; the other is 4.

<sup>\*</sup>"Square" in the original.

<sup>\*</sup> $10x = (10 - x)^2 = 100 - 20x + x^2$ .

<sup>\*</sup> $\frac{(10-x)x}{10-2x} = 5\frac{1}{4}$ ;  $20\frac{1}{4}x = x^2 + 5\frac{1}{4}(10 - 2x)$ ;  $x = 10 - 7\frac{1}{2} = 2\frac{1}{2}$ .

things less one square. Multiply now five and one-fourth by ten less two squares. The result is fifty-two dirhems and a half less ten roots and a half, which is equal to ten roots less a square. Separate now the ten roots and a half from the fifty-two dirhems, and add them to the ten roots less a square; at the same time separate this square from them, and add it to the fifty-two dirhems and a half. Thus you find twenty roots and a half, equal to fifty-two dirhems and a half and one square. Now continue reducing it, conformably to the rules explained at the commencement of this book.

If the question be: "There is a square, two-thirds of one-fifth of which are equal to one-seventh of its root;"\* then the square is equal to one root and half a seventh of a root; and the root consists of fourteen-fifteenths of the square. The computation is this: You multiply two-thirds of one-fifth of the square by seven and a half, in order that the square may be completed. Multiply that which you have already, namely, one-seventh of its root, by the same. The result will be, that the square is equal to one root and half a seventh of the root; and the root of the square is one and a half seventh; and the square is one and twenty-nine one hundred and ninety-sixths of a dirhem. Two-thirds of the fifth of this are thirty parts of the hundred and ninety-six parts. One-seventh of its root is likewise thirty parts of a hundred and ninety-six.

If the instance be: "Three-fourths of the fifth of a square are equal to four-fifths of its root,"\* then the computation is this; You add one-fifth to the four-fifths, in order to complete the root. This is then equal to three and three-fourths of twenty parts, that is, to fifteen eightieths of the square. Divide now eighty by fifteen; the quotient is five and one-third. This is the root of the square, and the square is twenty-eight and four-ninths.

If some one say: "What is the amount of a square-root,\* which, when multiplied by four times itself, amounts to twenty?"\* the answer is this: If you multiply it by itself it will be five: it is therefore the root of five.

If somebody ask you for the amount of a square-root\* which when multiplied by its third amounts to ten,\* the solution is, that when multiplied by itself it will amount to thirty; and it is consequently the root of thirty.

If the question be: "To find a quantity,\* which when multiplied by four times itself, gives one-third of the first quantity as product,"\* the solution is, that if you multiply it by twelve times itself, the quantity itself must re-appear: it is the moiety of one moiety of one-third.

If the question be: "A square, which when multiplied by its root gives three times the original square as product,"\* then the solution is: that if you multiply the root by one-third of the square, the original square is restored; its root must consequently be three, and the square itself nine.

If the instance be: "To find a square, four roots of which, multiplied by three roots, restore the square with a surplus of forty-four dirhems,"\* then the solution is: that you multiply four roots by three roots, which gives twelve squares, equal to a square and forty-four dirhems. Remove now one square of the twelve on account of the one square connected with the forty-four dirhems. There remain eleven squares, equal to forty-four dirhems. Make the division, the result will be four, and this is the square.

If the instance be: "A square, four of the roots of which multiplied by five of its roots produce twice the square, with a surplus of thirty-six dirhems;"\* then the solution is: that you multiply four

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$$^*\frac{2}{3} \cdot \frac{1}{5}x^2 = \frac{1}{7}x; \frac{2}{15}x^2 = \frac{1}{7}x; x^2 = \frac{15}{14}x; x = \frac{15}{14}.$$

$$^*\frac{3}{4} \cdot \frac{1}{5}x^2 = \frac{4}{5}x; \frac{3}{20}x^2 = \frac{4}{5}x; \frac{3}{20}x = \frac{4}{5}; x = \frac{16}{3} = 5\frac{1}{3}.$$

\*"Square" in the original.

$$^*4x^2 = 20; x^2 = 5; x = \sqrt{5}.$$

\*"Square" in the original.

$$^*x \times \frac{x}{3} = 10; x^2 = 30; x = \sqrt{30}.$$

\*"Square-root" in the original.

$$^*x \times 4x = \frac{1}{3}x; 4x^2 = \frac{1}{3}x; 12x^2 = x; x = \frac{1}{12}.$$

$$^*x^2 \times x = 3x^2; x^3 = 3x^2; x = 3.$$

$$^*4x \times 3x = x^2 + 44; 12x^2 = x^2 + 44; 11x^2 = 44; x^2 = 4; x = 2.$$

$$^*4x \times 5x = 2x^2 + 36; 20x^2 = 2x^2 + 36; 18x^2 = 36; x^2 = 2.$$



roots by five roots, which gives twenty squares, equal to two squares and thirty-six dirhems. Remove two squares from the twenty on account of the other two. The remainder is eighteen squares, equal to thirty-six dirhems. Divide now thirty-six dirhems by eighteen; the quotient is two, and this is the square.

In the same manner, if the question be: "A square, multiply its root by four of its roots, and the product will be three times the square, with a surplus of fifty dirhems."<sup>\*</sup> Computation: You multiply the root by four roots, it is four squares, which are equal to three squares and fifty dirhems. Remove three squares from the four; there remains one square, equal to fifty dirhems. One root of fifty, multiplied by four roots of the same, gives two hundred, which is equal to three times the square, and a residue of fifty dirhems.

If the instance be: "A square, which when added to twenty dirhems, is equal to twelve of its roots,"<sup>\*</sup> then the solution is this: You say, one square and twenty dirhems are equal to twelve roots. Halve the roots and multiply them by themselves; this gives thirty-six. Subtract from this the twenty dirhems, extract the root from the remainder, and subtract it from the moiety of the roots, which is six. The remainder is the root of the square: it is two dirhems, and the square is four.

If the instance be: "To find a square, of which if one-third be added to three dirhems, and the sum be subtracted from the square, the remainder multiplied by itself restores the square;"<sup>\*</sup> then the computation is this: If you subtract one-third and three dirhems from the square, there remain two-thirds of it less three dirhems. This is the root. Multiply therefore two-thirds of thing less three dirhems by itself. You say two-thirds by two-thirds is four ninths of a square; and less two-thirds by three dirhems is two roots; and again, two-thirds by three dirhems is two roots; and less three dirhems by less three dirhems is nine dirhems. You have, therefore, four-ninths of a square and nine dirhems less four roots, which are equal to one root. Add the four roots to the one root, then you have five roots, which are equal to four-ninths of a square and nine dirhems. Complete now your square; that is, multiply the four-ninths of a square by two and a fourth, which gives one square; multiply likewise the nine dirhems by two and a quarter; this gives twenty and a quarter; finally, multiply the five roots by two and a quarter; this gives eleven roots and a quarter. You have, therefore, a square and twenty dirhems and a quarter, equal to eleven roots and a quarter. Reduce this according to what I taught you about halving the roots.

If the instance be: "To find a number,<sup>\*</sup> one-third of which, when multiplied by one-fourth of it, restores the number,"<sup>\*</sup> then the computation is: You multiply one-third of thing by one-fourth of thing, this gives one-twelfth of a square, equal to thing, and the square is equal to twelve things, which is the root of one hundred and forty-four.

If the instance be: "A number,<sup>\*</sup> one-third of which and one dirhem multiplied by one-fourth of it and two dirhems restore the number, with a surplus of thirteen dirhems;"<sup>\*</sup> then the computation is this: You multiply one-third of thing by one-fourth of thing, this gives half one-sixth of a square; and you multiply two dirhems by one-third of thing, this gives two-thirds of a root; and one dirhem by one-fourth of thing gives one-fourth of a root; and one dirhem by two dirhems gives two dirhems. This altogether is one-twelfth of a square and two dirhems and eleven-twelfths of a thing, equal to thing and thirteen dirhems. Remove now two dirhems from thirteen, on account of the other two dirhems, the remainder is eleven dirhems. Remove then the eleven-twelfths of a root from the one (root on the opposite side), there remains one-twelfth of a root and eleven dirhems, equal to one-twelfth of a square. Complete the square: that is, multiply it by twelve, and do the same with all you have. The product is

<sup>\*</sup> $x \times 4x = 3x^2 + 50; 4x^2 = 3x^2 + 50; x^2 = 50; x = \sqrt{50}.$

<sup>\*</sup> $x^2 + 20 = 12x; x = 6 \pm \sqrt{36 - 20} = 6 \pm 4 = 10 \text{ or } 2.$

<sup>\*</sup> $(x^2 - \frac{1}{3}x^2 - 3)^2 = x^2; \frac{2}{3}x^2 - 3 = x; \frac{4}{9}x^4 - 4x^2 + 9 = x^2; x^2 + 9 = 5x; x = 9 \text{ or } 2\frac{1}{4}.$

<sup>\*</sup>"Square" in the original.

<sup>\*</sup> $\frac{1}{3}x \times \frac{1}{4}x = x; \frac{1}{12}x^2 = x; x^2 = 12x; x = 12.$

<sup>\*</sup>"Square" in the original.

<sup>\*</sup> $\frac{x}{3} + 1 \times \frac{x}{4} + 2 = x + 13; \frac{1}{12}x^2 + \frac{11}{12}x + 2 = x + 13; \frac{1}{12}x^2 + 11 = \frac{1}{12}x + 13; x^2 = 132 + x; x = 12.$

a square, which is equal to a hundred and thirty-two dirhems and one root. Reduce this, according to what I have taught you, it will be right.

If the instance be: "A dirhem and a half to be divided among one person and certain persons, so that the share of the one person be twice as many dirhems as there are other persons;"\* then the Computation is this: You say, the one person and some persons are one and thing: it is the same as if the question had been one dirhem and a half to be divided by one and thing, and the share of one person to be equal to two things. Multiply, therefore, two things by one and thing; it is two squares and two things, equal to one dirhem and a half. Reduce them to one square: that is, take the moiety of all you have. You say, therefore, one square and one thing are equal to three-fourths of a dirhem. Reduce this, according to what I have taught you in the beginning of this work.

If the instance be: "A number,\* you remove one-third of it, and one-fourth of it, and four dirhems: then you multiply the remainder by itself, and the number,\* is restored, with a surplus of twelve dirhems;"\* then the computation is this: You take thing, and subtract from it one-third and one-fourth; there remain five-twelfths of thing. Subtract from this four dirhems; the remainder is five-twelfths of thing less four dirhems. Multiply this by itself. Thus the five parts become five-and-twenty parts; and if you multiply twelve by itself, it is a hundred and forty-four. This makes, therefore, five and twenty hundred and forty-fourths of a square. Multiply then the four dirhems twice by the five-twelfths; this gives forty parts, every twelve of which make one root (forty-twelfths); finally, the four dirhems, multiplied by four dirhems, give sixteen dirhems to be added. The forty-twelfths are equal to three roots and one-third of a root, to be subtracted. The whole product is, therefore, twenty-five-hundred-and-forty-fourths of a square and sixteen dirhems less three roots and one-third of a root, equal to the original number,\* which is thing and twelve dirhems. Reduce this, by adding the three roots and one-third to the thing and twelve dirhems. Thus you have four roots and one-third of a root and twelve dirhems. Go on balancing, and subtract the twelve (dirhems) from sixteen; there remain four dirhems and five-and-twenty-hundred-and-forty-fourths of a square, equal to four roots and one-third. Now it is necessary to complete the square. This you can accomplish by multiplying all you have by five and nineteen twenty-fifths. Multiply, therefore, the twenty-five-one-hundred-and-forty-fourths of a square by five and nineteen twenty-fifths. This gives a square. Then multiply the four (dirhems) by five and nineteen twenty-fifths; this gives twenty-three dirhems and one twenty-fifth. Then multiply four roots and one-third by five and nineteen twenty-fifths; this gives twenty-four roots and twenty-four twenty-fifths of a root. Now halve the number of the roots: the moiety is twelve roots and twelve twenty-fifths of a root. Multiply this by itself. It is one hundred-and-fifty-five dirhems and four-hundred-and-sixty-nine six-hundred-and-twenty-fifths. Subtract from this the twenty-three dirhems and the one twenty-fifth connected with the square. The remainder is one-hundred-and-thirty-two and four-hundred-and-forty-six six-hundred-and-twenty-fifths. Take the root of this: it is eleven dirhems and thirteen twenty-fifths. Add this to the moiety of the roots, which was twelve dirhems and twelve twenty-fifths. The sum is twenty-four. It is the number which you sought. When you subtract its third and its fourth and four dirhems, and multiply the remainder by itself, the number is restored, with a surplus of twelve dirhems.

If the question be: "To find a square-root,\* which, when multiplied by two-thirds of itself, amounts to five;"\* then the computation is this: You multiply one thing by two-thirds of thing; the product is two-thirds of square, equal to five. Complete it by adding its moiety to it, and add to five likewise its

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\*The enunciation in the original is faulty, and I have altered it to correspond with the computation. But in the computation,  $x$ , the number of persons, is fractional! I am unable to correct the passage satisfactorily.  $x^2 + x = \frac{3}{2}$ ;  $x = \sqrt{1\frac{1}{4} + \frac{1}{2}} - \frac{1}{2}$ .

\*"Square" in the original.

\*"Square" in the original.

\* $(x - \frac{1}{3}x - \frac{1}{4}x - 4)^2 = x + 12$ ;  $\frac{5}{12}x - 4$ ;  $x^2 + 23\frac{1}{4} = 24\frac{1}{4}x$ .

\*"Square" in the original.

\*"Square" in the original.

\* $x \times \frac{2}{3}x = 5$ ;  $\frac{2}{3}x^2 = 5$ ;  $x^2 = 7\frac{1}{2}$ ;  $x = \sqrt{7\frac{1}{2}}$ .



moiety. Thus you have a square, equal to seven and a half. Take its root; it is the thing which you required, and which, when multiplied by two-thirds of itself, is equal to five.

If the instance be: "Two numbers,\* the difference of which is two dirhems; you divide the small one by the great one, and the quotient is equal to half a dirhem;\*" then the computation is this: Multiply thing and two dirhems by the quotient, that is a half. The product is half a thing and one dirhem, equal to thing. Remove now half a dirhem on account of the half dirhem on the other side. The remainder is one dirhem, equal to half a thing. Double it; then you have thing, equal to two dirhems. This is one of the two\* numbers, and the other is four.

Instance: "You divide one dirhem amongst a certain number of men, which number is thing. Now you add one man more to them, and divide again one dirhem amongst them; the quota of each is then one-sixth of a dirhem less than at the first time.)\* Computation: You multiply the first number of men, which is thing, by the difference of the share for each of the other number. Then multiply the product by the first and second number of men, and divide the product by the difference of these two numbers. Thus you obtain the sum which shall be divided. Multiply, therefore, the first number of men, which is thing, by the one-sixth, which is the difference of the shares; this gives one-sixth of root. Then multiply this by the original number of the men, and that of the additional one, that is to say, by thing plus one. The product is one-sixth of square and one-sixth of root divided by one dirhem, and this is equal to one dirhem. Complete the square which you have through multiplying it, by six. Then you have a square and a root equal to six dirhems. Halve the root and multiply the moiety by itself, it is one-fourth. Add this to the six; take the root of the sum and subtract from it the moiety of the root, which you have multiplied by itself, namely, a half. The remainder is the first number of men; which in this instance is two.

If the instance be: "To find a square-root,\* which when multiplied by two-thirds of itself amounts to five;\*" then the computation is this: If you multiply it by itself, it gives seven and a half. Say, therefore, it is the root of seven and a half multiplied by two-thirds of the root of seven and a half. Multiply then two-thirds by two-thirds, it is four-ninths; and four-ninths multiplied by seven and a half is three and a third. The root of three and a third is two-thirds of the root of seven and a half. Multiply three and a third by seven and a half; the product is twenty-five, and its root is five.

If the instance be: "A square multiplied by three of its roots is equal to five times the original square;\*" then this is the same as if it had been said, a square, which when multiplied by its root, is equal to the first square and two-thirds of it. Then the root of the square is one and two-thirds, and the square is two dirhems and seven-ninths.

If the instance be: "Remove one-third from a square, then multiply the remainder by three roots of the first square, and the first square will be restored.)\* Computation: If you multiply the first square, before removing two-thirds from it, by three roots of the same, then it is one square and a half; for according to the statement two-thirds of it multiplied by three roots give one square; and, consequently, the whole of it multiplied by three roots of it gives one square and a half. This entire square, when multiplied by one root, gives half a square; the root of the square must therefore be a half, the square one-fourth, two-thirds of the square one-sixth, and three roots of the square one and a half. If you multiply one-sixth by one and a half, the product is one-fourth, which is the square.

Instance: "A square; you subtract four roots of the same, then take one-third of the remainder; this

\*"Squares" in the original.

\*  $\frac{x}{x+2} = \frac{1}{2}$ ;  $\frac{1}{2}(x+2) = x$ ;  $1 + \frac{1}{2}x = x$ ;  $1 = \frac{1}{2}x$ ;  $x = 2$ ; the other is 4.

\*"Squares" in the original.

\*  $\frac{1}{x} - \frac{1}{x+1} = \frac{1}{6}$ ;  $\frac{x+1-x}{x(x+1)} = \frac{1}{6}$ ;  $\frac{1}{x(x+1)} = \frac{1}{6}$ ;  $x^2 + x = 6$ ;  $x = \sqrt{\frac{1}{4} + 6} - \frac{1}{2} = 2$ .

\*"Square" in the original.

\*  $\frac{2}{3}x^2 = 5$ ;  $x^2 = 7\frac{1}{2}$ ;  $x = \sqrt{7\frac{1}{2}}$ .

\*  $x^2 \times 3x = 5x^2$ ;  $3x^3 = 5x^2$ ;  $3x = 5$ ;  $x = 1\frac{2}{3}$ ;  $x^2 = 2\frac{7}{9}$ .

\*  $(x^2 - \frac{1}{3}x^2) \times 3x = x^2$ ;  $\frac{2}{3}x^2 \times 3x = x^2$ ;  $2x^3 = x^2$ ;  $x = \frac{1}{2}$ ;  $x^2 = \frac{1}{4}$ .

is equal to the four roots.” The square is two hundred and fifty-six.\* Computation: You know that one-third of the remainder is equal to four roots; consequently, the whole remainder must be twelve roots; add to this the four roots; the sum is sixteen, which is the root of the square.

Instance: “A square; you remove one root from it and if you add to this root a root of the remainder, the sum is two dirhems.”\* Then, this is the root of a square, which, when added to the root of the same square, less one root, is equal to two dirhems. Subtract from this one root of the square, and subtract also from the two dirhems one root of the square. Then two dirhems less one root multiplied by itself is four dirhems and one square less four roots, and this is equal to a square less one root. Reduce it, and you find a square and four dirhems, equal to a square and three roots. Remove square by square; there remain three roots, equal to four dirhems; consequently, one root is equal to one dirhem and one-third. This is the root of the square, and the square is one dirhem and seven ninths of a dirhem.

Instance: “Subtract three roots from a square, then multiply the remainder by itself, and the square is restored.”\* You know by this statement that the remainder must be a root likewise; and that the square consists of four such roots; consequently, it must be sixteen.

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\*  $x^2 - 4x = \frac{1}{3}(x^2 - 4x)$  gives  $\frac{2}{3}(x^2 - 4x) = 4x$ ;  $x^2 - 4x = 6x$ ;  $x^2 = 10x$ ;  $x = 10$ ;  $x^2 = 100$ . [Note: There appears to be a discrepancy in the original text, which states 256.]

\*  $\sqrt{x^2 - x} + x = 2$ ;  $\sqrt{x^2 - x} = 2 - x$ ;  $x^2 - x = 4 + x^2 - 4x$ ;  $3x = 4$ ;  $x = 1\frac{1}{3}$ ;  $x^2 = 1\frac{7}{9}$ .

\*  $x^2 - 3x = x$ ;  $x^2 = 4x$ ;  $x = 4$ .

## On Mercantile Transactions

**Y**OU KNOW THAT all mercantile transactions of people, such as buying and selling, exchange and hire, comprehend always two notions and four numbers, which are stated by the enquirer; namely, measure and price, and quantity and sum. The number which expresses the measure is inversely proportionate to the number which expresses the sum, and the number of the price inversely proportionate to that of the quantity. Three of these four numbers are always known, one is unknown, and this is implied when the person inquiring says *how much?* and it is the object of the question.

The computation in such instances is this, that you try the three given numbers; two of them must necessarily be inversely proportionate the one to the other. Then you multiply these two proportionate numbers by each other, and you divide the product by the third given number, the proportionate of which is unknown. The quotient of this division is the unknown number, which the inquirer asked for; and it is inversely proportionate to the divisor.\*

### Examples.

For the first case: If you are told, “ten for six, how much for four?” then ten is the measure; six is the price; the expression *how much* implies the unknown number of the quantity; and *four* is the number of the sum. The number of the measure, which is ten, is inversely proportionate to the number of the sum, namely, *four*. Multiply, therefore, ten by four, that is to say, the two known proportionate numbers by each other; the product is forty. Divide this by the other known number, which is that of the price, namely, six. The quotient is six and two-thirds; it is the unknown number, implied in the words of the question “how much?” it is the quantity, and inversely proportionate to the six, which is the price.

For the second case: Suppose that some one ask this question: “ten for eight, what must be the sum for four?” This is also sometimes expressed thus: “What must be the price of four of them?” Ten is the number of the measure, and is inversely proportionate to the unknown number of the sum, which is involved in the expression *how much* of the statement. Eight is the number of the price, and this is inversely proportionate to the known number of the quantity, namely, four. Multiply now the two known proportionate numbers one by the other, that is to say, four by eight. The product is thirty-two. Divide this by the other known number, which is that of the measure, namely, ten. The quotient is three and one-fifth; this is the number of the sum, and inversely proportionate to the ten which was the divisor. In this manner all computations in matters of business may be solved.

If somebody says, “a workman receives a pay of ten dirhems per month; how much must be his pay for six days?” Then you know that six days are one-fifth of the month; and that his portion of the dirhems must be proportionate to the portion of the month. You calculate it by observing that one month, or thirty days, is the measure, ten dirhems the price, six days the quantity, and his portion the sum. Multiply the price, that is, ten, by the quantity, which is proportionate to it, namely, six; the product is sixty. Divide this by thirty, which is the known number of the measure. The quotient is two dirhems, and this is the sum.

This is the proceeding by which all transactions concerning exchange or measures or weights are settled.

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\*If  $a$  is given for  $b$ , and  $A$  for  $B$ ; then  $a : b :: A : B$  or  $aB = bA$ ;  $\therefore a = \frac{bA}{B}$  and  $b = \frac{aB}{A}$ .

## Mensuration

**K** NOW THAT the meaning of the expression “one by one” is mensuration: one yard (in length) by one yard (in breadth) being understood.

Every quadrangle of equal sides and angles, which has one yard for every side, has also one for its area. Has such a quadrangle two yards for its side, then the area of the quadrangle is four times the area of a quadrangle, the side of which is one yard. The same takes place with three by three, and so on, ascending or descending: for instance, a half by a half, which gives a quarter, or other fractions, always following the same rule.

A quadrate, every side of which is half a yard, is equal to one-fourth of the figure which has one yard for its side. In the same manner, one-third by one-third, or one-fourth by one-fourth, or one-fifth by one-fifth, or two-thirds by a half, or more or less than this, always according to the same rule.

One side of an equilateral quadrangular figure, taken once, is its root; or if the same be multiplied by two, then it is like two of its roots, whether it be small or great.

If you multiply the height of any equilateral triangle by the moiety of the basis upon which the line marking the height stands perpendicularly, the product gives the area of that triangle.

In every equilateral quadrangle, the product of one diameter multiplied by the moiety of the other will be equal to the area of it.

In any circle, the product of its diameter, multiplied by three and one-seventh, will be equal to the periphery. This is the rule generally followed in practical life, though it is not quite exact. The geometricians have two other methods. One of them is, that you multiply the diameter by itself; then by ten, and hereafter take the root of the product; the root will be the periphery. The other method is used by the astronomers among them: it is this, that you multiply the diameter by sixty-two thousand eight hundred and thirty-two and then divide the product by twenty thousand; the quotient is the periphery. Both methods come very nearly to the same effect.\*

If you divide the periphery by three and one-seventh, the quotient is the diameter.

The area of any circle will be found by multiplying the moiety of the circumference by the moiety of the diameter; since, in every polygon of equal sides and angles, such as triangles, quadrangles, pentagons, and so on, the area is found by multiplying the moiety of the circumference by the moiety of the diameter of the middle circle that may be drawn through it.

If you multiply the diameter of any circle by itself, and subtract from the product one-seventh and half one-seventh of the same, then the remainder is equal to the area of the circle. This comes very nearly to the same result with the method given above.\*

Every part of a circle may be compared to a bow. It must be either exactly equal to half the circumference, or less or greater than it. This may be ascertained by the arrow of the bow. When this becomes equal to the moiety of the chord, then the arc is exactly the moiety of the circumference: is it shorter than the moiety of the chord, then the bow is less than half the circumference; is the arrow longer than half the chord, then the bow comprises more than half the circumference.

If you want to ascertain the circle to which it belongs, multiply the moiety of the chord by itself, divide it by the arrow, and add the quotient to the arrow, the sum is the diameter of the circle to which this bow belongs.

If you want to compute the area of the bow, multiply the moiety of the diameter of the circle by the moiety of the bow, and keep the product in mind. Then subtract the arrow of the bow from the moiety of the diameter of the circle, if the bow is smaller than half the circle; or if it is greater than half the circle, subtract half the diameter of the circle from the arrow of the bow. Multiply the remainder by the moiety of the chord of the bow, and subtract the product from that which you have kept in mind if the

\*The three formulas are:  $1^{st}$ ,  $3\frac{1}{7}d = p$  i.e.  $3.1428d$ ;  $2^{d}$ ,  $\sqrt{10d^2} = p$  i.e.  $3.16227d$ ;  $3^{d}$ ,  $\frac{d \times 62832}{20000} = p$  i.e.  $3.1416d$ .

\*The area of a circle whose diameter is  $d$  is  $\pi \frac{d^2}{4} = d^2 - \frac{d^2}{7} - \frac{1}{2} \cdot \frac{d^2}{7} = d^2(1 - \frac{3}{14}) = \frac{11}{14}d^2 \approx 0.7857d^2$ . The value  $\frac{22}{7} \cdot \frac{1}{4} = \frac{11}{14} \approx 0.7857$ .

bow is smaller than the moiety of the circle, or add it thereto if the bow is greater than half the circle. The sum after the addition, or the remainder after the subtraction, is the area of the bow.

The bulk of a quadrangular body will be found by multiplying the length by the breadth, and then by the height.

If it is of another shape than the quadrangular (for instance, circular or triangular), so, however, that a line representing its height may stand perpendicularly on its basis, and yet be parallel to the sides, you must calculate it by ascertaining at first the area of its basis. This, multiplied by the height, gives the bulk of the body.

Cones and pyramids, such as triangular or quadrangular ones, are computed by multiplying one-third of the area of the basis by the height.

### On the Rectangular Triangle.

Observe, that in every rectangular triangle the two short sides, each multiplied by itself and the products added together, equal the product of the long side multiplied by itself.

The proof of this is the following. We draw a quadrangle, with equal sides and angles  $ABCD$ . We divide the line  $AC$  into two moieties in the point  $H$ , from which we draw a parallel to the point  $R$ . Then we divide, also, the line  $AB$  into two moieties at the point  $T$ , and draw a parallel to the point  $G$ . Then the quadrate  $ABCD$  is divided into four quadrangles of equal sides and angles, and of equal area; namely, the squares  $AK$ ,  $CK$ ,  $BK$ , and  $DK$ . Now, we draw from the point  $H$  to the point  $T$  a line which divides the quadrangle  $AK$  into two equal parts: thus there arise two triangles from the quadrangle, namely, the triangles  $ATH$  and  $HKT$ . We know that  $AT$  is the moiety of  $AB$ , and that  $AH$  is equal to it, being the moiety of  $AC$ ; and the line  $TH$  joins them opposite the right angle. In the same manner we draw lines from  $T$  to  $R$ , and from  $R$  to  $G$ , and from  $G$  to  $H$ . Thus from all the squares eight equal triangles arise, four of which must, consequently, be equal to the moiety of the great quadrate  $AD$ . We know that the line  $AT$  multiplied by itself is like the area of two triangles, and  $AK$  gives the area of two triangles equal to them; the sum of them is therefore four triangles. But the line  $HT$ , multiplied by itself, gives likewise the area of four such triangles. We perceive, therefore, that the sum of  $AT$  multiplied by itself, added to  $AH$  multiplied by itself, is equal to  $TH$  multiplied by itself.

This is the observation which we were desirous to elucidate.

**Quadrangles are of five kinds:** firstly, with right angles and equal sides; secondly, with right angles and unequal sides; thirdly, the rhombus, with equal sides and unequal angles; fourthly, the rhomboid, the length of which differs from its breadth, and the angles of which are unequal, only that the two long and the two short sides are respectively of equal length; fifthly, quadrangles with unequal sides and angles.

*First kind.*—The area of any quadrangle with equal sides and right angles, or with unequal sides and right angles, may be found by multiplying the length by the breadth. The product is the area. For instance: a quadrangular piece of ground, every side of which has five yards, has an area of five-and-twenty square yards.

*Second kind.*—A quadrangular piece of ground, the two long sides of which are of eight yards each, while the breadth is six. You find the area by multiplying six by eight, which yields forty-eight yards.

*Third kind, the Rhombus.*—Its sides are equal: let each of them be five, and let its diagonals be, the one eight and the other six yards. You may then compute the area, either from one of the diagonals, or from both. As you know them both, you multiply the one by the moiety of the other, the product is the area: that is to say, you multiply eight by three, or six by four; this yields twenty-four yards, which is the area.\* If you know only one of the diagonals, then you are aware, that there are two triangles, two sides of each of which have every one five yards, while the third is the diagonal. Hereafter you can make the computation according to the rules for the triangles. This is the figure.

\* If the two diagonals are  $d$  and  $d'$ , and the side  $s$ , the area of the rhombus is  $\frac{dd'}{2} = d \times \sqrt{s^2 - \frac{d^2}{4}}$ .

*The fourth kind, or Rhomboid, is computed in the same way as the rhombus.*

The other quadrangles are calculated by drawing a diagonal, and computing them as triangles.

**Triangles** are of three kinds, acute-angular, obtuse-angular, or rectangular. The peculiarity of the rectangular triangle is, that if you multiply each of its two short sides by itself, and then add them together, their sum will be equal to the long side multiplied by itself. The character of the acute-angled triangle is this: if you multiply every one of its two short sides by itself, and add the products, their sum is more than the long side alone multiplied by itself. The definition of the obtuse-angled triangle is this: if you multiply its two short sides each by itself, and then add the products, their sum is less than the product of the long side multiplied by itself.

The rectangular triangle has two cathetes and an hypotenuse. It may be considered as the moiety of a quadrangle. You find its area by multiplying one of its cathetes by the moiety of the other. The product is the area.

*Examples.*—A rectangular triangle; one cathete being six yards, the other eight, and the hypotenuse ten. You make the computation by multiplying six by four: this gives twenty-four, which is the area. Or if you prefer, you may also calculate it by the height, which rises perpendicularly from the longest side of it: for the two short sides may themselves be considered as two heights. If you prefer this, you multiply the height by the moiety of the basis. The product is the area.

*Second kind.*—An equilateral triangle with acute angles, every side of which is ten yards long. Its area may be ascertained by the line representing its height and the point from which it rises.\* Observe, that in every isosceles triangle, a line to represent the height drawn to the basis rises from the latter in a right angle, and the point from which it proceeds is always situated in the midst of the basis; if, on the contrary, the two sides are not equal, then this point never lies in the middle of the basis. In the case now before us we perceive, that towards whatever side we may draw the line which is to represent the height, it must necessarily always fall in the middle of it, where the length of the basis is five. Now the height will be ascertained thus. You multiply five by itself; then multiply one of the sides, that is ten, by itself, which gives a hundred. Now you subtract from this the product of five multiplied by itself, which is twenty-five. The remainder is seventy-five, the root of which is the height. This is a line common to two rectangular triangles. If you want to find the area, multiply the root of seventy-five by the moiety of the basis, which is five. This you perform by multiplying at first five by itself; then you may say, that the root of seventy-five is to be multiplied by the root of twenty-five. Multiply seventy-five by twenty-five. The product is one thousand eight hundred and seventy-five; take its root, it is the area: it is forty-three and a little.

There are also acute-angled triangles, with different sides. Their area will be found by means of the line representing the height and the point from which it proceeds. Take, for instance, a triangle, one side of which is fifteen yards, another fourteen, and the third thirteen yards.

In order to find the point from which the line marking the height does arise, you may take for the basis any side you choose; e.g. that which is fourteen yards long. The point from which the line representing the height does arise, lies in this basis at an unknown distance from either of the two other sides. Let us try to find its unknown distance from the side which is thirteen yards long. Multiply this distance by itself; it becomes an [unknown] square. Subtract this from thirteen multiplied by itself; that is, one hundred and sixty-nine. The remainder is one hundred and sixty-nine less a square. The root from this is the height. The remainder of the basis is fourteen less thing. We multiply this by itself; it becomes one hundred and ninety-six, and a square less twenty-eight things. We subtract this from fifteen multiplied by itself; the remainder is twenty-nine dirhems and twenty-eight things less one square. The root of this is the height. As, therefore, the root of this is the height, and the root of one hundred and sixty-nine less square is the height likewise, we know that they both are the same.\*

\* The height of the equilateral triangle whose side is 10, is  $\sqrt{10^2 - 5^2} = \sqrt{75}$ , and the area of the triangle is  $5\sqrt{75} = 25\sqrt{3}$ . The root is 43.3+.

\*  $\sqrt{169 - x^2} = \sqrt{29 + 28x - x^2}$ ;  $169 = 29 + 28x$ ;  $140 = 28x$ ;  $x = 5$ .



Reduce them, by removing square against square, since both are negatives. There remain twenty-nine [dirhems] plus twenty-eight things, which are equal to one hundred and sixty-nine. Subtract now twenty-nine from one hundred and sixty-nine. The remainder is one hundred and forty, equal to twenty-eight things. One thing is, consequently, five. This is the distance of the said point from the side of thirteen yards. The complement of the basis towards the other side is nine. Now in order to find the height, you multiply five by itself, and subtract it from the contiguous side, which is thirteen, multiplied by itself. The remainder is one hundred and forty-four. Its root is the height. It is twelve. The height forms always two right angles with the basis, and it is called the column, on account of its standing perpendicularly. Multiply the height into half the basis, which is seven. The product is eighty-four, which is the area.

The third species is that of the obtuse-angled triangle with one obtuse angle and sides of different length. For instance, one side being six, another five, and the third nine. The area of such a triangle will be found by means of the height and of the point from which a line representing the same arises. This point can, within such a triangle, lie only in its longest side. Take therefore this as the basis: for if you choose to take one of the short sides as the basis, then this point would fall beyond the triangle. You may find the distance of this point, and the height, in the same manner, which I have shown in the acute-angled triangle; the whole computation is the same.

We have above treated at length of the circles, of their qualities and their computation. The following is an example: If a circle has seven for its diameter, then it has twenty-two for its circumference. Its area you find in the following manner: Multiply the moiety of the diameter, which is three and a half, by the moiety of the circumference, which is eleven. The product is thirty-eight and a half, which is the area. Or you may also multiply the diameter, which is seven, by itself: this is forty-nine; subtracting herefrom one-seventh and half one-seventh, which is ten and a half, there remain thirty-eight and a half, which is the area.

If some one inquires about the bulk of a pyramidal pillar, its base being four yards by four yards, its height ten yards, and the dimensions at its upper extremity two yards by two yards; then we know already that every pyramid is decreasing towards its top, and that one-third of the area of its basis, multiplied by the height, gives its bulk. The present pyramid has no top. We must therefore seek to ascertain what is wanting in its height to complete the top. We observe, that the proportion of the entire height to the ten, which we have now before us, is equal to the proportion of four to two. Now as two is the moiety of four, ten must likewise be the moiety of the entire height, and the whole height of the pillar must be twenty yards. At present we take one-third of the area of the basis, that is, five and one-third, and multiply it by the length, which is twenty. The product is one hundred and six yards and two-thirds. Herefrom we must then subtract the piece, which we have added in order to complete the pyramid. This we perform by multiplying one and one-third, which is one-third of the product of two by two, by ten: this gives thirteen and a third. This is the piece which we have added in order to complete the pyramid. Subtracting this from one hundred and six yards and two-thirds, there remain ninety-three yards and one-third: and this is the bulk of the mutilated pyramid.

If the pillar has a circular basis, subtract one-seventh and half a seventh from the product of the diameter multiplied by itself, the remainder is the basis.

If some one says: "There is a triangular piece of land, two of its sides having ten yards each, and the basis twelve; what must be the length of one side of a quadrangle situated within such a triangle?" the solution is this. At first you ascertain the height of the triangle, by multiplying the moiety of the basis, (which is six) by itself, and subtracting the product, which is thirty-six, from one of the two short sides multiplied by itself, which is one hundred; the remainder is sixty-four: take the root from this; it is eight. This is the height of the triangle. Its area is, therefore, forty-eight yards: such being the product of the height multiplied by the moiety of the basis, which is six. Now we assume that one side of the quadrangle inquired for is thing. We multiply it by itself; thus it becomes a square, which we keep in mind. We know that there must remain two triangles on the two sides of the quadrangle, and one above

it. The two triangles on both sides of it are equal to each other: both having the same height and being rectangular. You find their area by multiplying thing by six less half a thing, which gives six things less half a square. This is the area of both the triangles on the two sides of the quadrate together. The area of the upper triangle will be found by multiplying eight less thing, which is the height, by half one thing. The product is four things less half a square. This altogether is equal to the area of the quadrate plus that of the three triangles: or, ten things are equal to forty-eight, which is the area of the great triangle. One thing from this is four yards and four-fifths of a yard; and this is the length of any side of the quadrate.



## On Legacies

**T**HE PROBLEMS IN THIS CHAPTER may be considered as belonging rather to Law than to Algebra, as they contain little more than enunciations of the law of inheritance in certain complicated cases.

### On Capital, and Money lent.

“A MAN dies, leaving two sons behind him, and bequeathing one-third of his capital to a stranger. He leaves ten dirhems of property and a claim of ten dirhems upon one of the sons.”

Computation: You call the sum which is taken out of the debt thing. Add this to the capital, which is ten dirhems. The sum is ten and thing. Subtract one-third of this, since he has bequeathed one-third of his property, that is, three dirhems and one-third of thing. The remainder is six dirhems and two-thirds of thing. Divide this between the two sons. The portion of each of them is three dirhems and one-third plus one-third of thing. This is equal to the thing which was sought for.\*

Reduce it, by removing one-third from thing, on account of the other third of thing. There remain two-thirds of thing, equal to three dirhems and one-third. It is then only required that you complete the thing, by adding to it as much as one half of the same; accordingly, you add to three and one-third as much as one-half of them: This gives five dirhems, which is the thing that is taken out of the debts.

If he leaves two sons and ten dirhems of capital and a demand of ten dirhems against one of the sons, and bequeaths one-fifth of his property and one dirhem to a stranger, the computation is this: Call the sum which is taken out of the debt, thing. Add this to the property; the sum is thing and ten dirhems. Subtract one-fifth of this, since he has bequeathed one-fifth of his capital, that is, two dirhems and one-fifth of thing; the remainder is eight dirhems and four-fifths of thing. Subtract also the one dirhem which he has bequeathed; there remain seven dirhems and four-fifths of thing. Divide this between the two sons; there will be for each of them three dirhems and a half plus two-fifths of thing; and this is equal to one thing.\*

Reduce it by subtracting two-fifths of thing from thing. Then you have three-fifths of thing, equal to three dirhems and a half. Complete the thing by adding to it two-thirds of the same: add as much to the three dirhems and a half; namely, two dirhems and one-third; the sum is five and five-sixths. This is the thing, or the amount which is taken from the debt.

If he leaves three sons, and bequeaths one-fifth of his property less one dirhem, leaving ten dirhems of capital and a demand of ten dirhems against one of the sons, the computation is this: You call the sum which is taken from the debt thing. Add this to the capital; it gives ten and thing. Subtract from this one-fifth of it for the legacy; it is two dirhems and one-fifth of thing. There remain eight dirhems and four-fifths of thing; add to this one dirhem, since he stated “less one dirhem.” Thus you have nine dirhems and four-fifths of thing. Divide this between the three sons. There will be for each son three dirhems, and one-fifth and one-third and one-fifth of thing. This equals one thing.\* Subtract one-fifth and one-third of one-fifth of thing from thing. There remain eleven-fifteenths of thing, equal to three dirhems. It is now required to complete the thing. For this purpose, add to it four-elevenths, and do the same with the three dirhems, by adding to them one dirhem and one-eleventh. Then you have four dirhems and one-eleventh, which are equal to thing. This is the sum which is taken out of the debt.

### On another Species of Legacy.

“A man dies, leaving his mother, his wife, and two brothers and two sisters by the same father and mother with himself; and he bequeaths to a stranger one-ninth of his capital.”

\*If a father dies, leaving  $n$  sons, one of whom owes the father a sum exceeding an  $n$ th part of the residue of the father's estate, after paying legacies, then such son retains the whole sum which he owes the father: part, as a set-off against his share of the residue, the surplus as a gift from the father. In the present example, let each son's share of the residue be equal to  $x$ . Then  $\frac{2}{3}[10 + x] = 2x$ ;  $\therefore x = 5$ . The stranger receives 5; and the son, who is not indebted to the father, receives 5.

\* $\frac{4}{5}[10 + x] - 1 = 2x$ ;  $\therefore \frac{4}{5}[10 + x] - 1 = 2x$ ;  $\frac{2}{3}[10 + x] - 1 = x$ ; The stranger receives  $\frac{1}{5}[10 + x] + 1$ .

\*The stranger receives  $\frac{1}{5}[10 + x] - 1$ .

Computation: \* You constitute their shares by taking them out of forty-eight parts. You know that if you take one-ninth from any capital, eight-ninths of it will remain. Add now to the eight-ninths one-eighth of the same, and to the forty-eight also one-eighth of them, namely, six, in order to complete your capital. This gives fifty-four. The person to whom one-ninth is bequeathed receives six out of this, being one-ninth of the whole capital. The remaining forty-eight will be distributed among the heirs, proportionably to their legal shares.

If the instance be: "A woman dies, leaving her husband, a son, and three daughters, and bequeathing to a stranger one-eighth and one-seventh of her capital;" then you constitute the shares of the heirs, by taking them out of twenty. \* Take a capital, and subtract from it one-eighth and one-seventh of the same. The remainder is, a capital less one-eighth and one-seventh. Complete your capital by adding to that which you have already, fifteen forty-one parts. Multiply the parts of the capital, which are twenty, by forty-one; the product is eight hundred and twenty. Add to it fifteen forty-one parts of the same, which are three hundred: the sum is one thousand one hundred and twenty parts. The person to whom one-eighth and one-seventh were bequeathed, receives one-eighth and one-seventh of this. One seventh of it is one hundred and sixty, and one-eighth one hundred and forty. Subtracting this, there remain eight hundred and twenty parts for the heirs, proportionably to their legal shares.

### On another Species of Legacies, viz.

If nothing has been imposed on some of the heirs, \* and something has been imposed on others; the legacy amounting to more than one-third.

It must be known, that the law for such a case is, that if more than one-third of the legacy has been imposed on one of the heirs, this enters into his share; but that also those on whom nothing has been imposed must, nevertheless, contribute one-third.

*Example:* "A woman dies, leaving her husband, a son, and her mother. She bequeaths to a person two-fifths, and to another one-fourth of her capital. She imposes the two legacies together on her son, and on her mother one moiety (of the mother's share of the residue); on her husband she imposes nothing but one-third, (which he must contribute, according to the law)."

Computation: You constitute the shares of the heritage, by taking them out of twelve parts: the son receives seven of them, the husband three, and the mother two parts. You know that the husband must give up one-third of his share; accordingly he retains twice as much as that which is detracted from his share for the legacy. As he has three parts in hand, one of these falls to the legacy, and the remaining two parts he retains for himself. The two legacies together are imposed upon the son. It is therefore necessary to subtract from his share two-fifths and one-fourth of the same. He thus retains seven twentieths of his entire original share, dividing the whole of it into twenty equal parts. The mother retains as much as she contributes to the legacy; this is one (twelfth part), the entire amount of what she had received being two parts.

Take now a sum, one-fourth of which may be divided into thirds, or of one-sixth of which the moiety

\* It appears in the sequel (p. 96) that a widow is entitled to  $\frac{1}{8}$ th, and a mother to  $\frac{1}{6}$ th of the residue;  $\frac{1}{8} + \frac{1}{6} = \frac{7}{24}$ ; leaving  $\frac{17}{24}$  of the residue to be distributed between two brothers and two sisters; that is,  $\frac{17}{48}$  between a brother and a sister; but in what proportion these 17 parts are to be divided between the brother and sister does not appear in the course of this treatise. Let the whole capital of the testator = 1 and let each 48th share of the residue =  $v$ ;  $1 - \frac{1}{9} = \frac{8}{9}$ ;  $\therefore \frac{8}{9} = 48v$ ;  $\therefore v = \frac{1}{54}$ ; that is, each 48th part of the residue =  $\frac{1}{54}$ th of the whole capital.

\* A husband is entitled to  $\frac{1}{4}$ th of the residue, and the sons and daughters divide the remaining  $\frac{3}{4}$ ths of the residue in such proportion, that a son receives twice as much as a daughter. In the present instance, as there are three daughters and one son, each daughter receives  $\frac{1}{5}$  of  $\frac{3}{4}$ , =  $\frac{3}{20}$ , of the residue, and the son,  $\frac{6}{20}$ . Since the stranger takes  $\frac{1}{8} + \frac{1}{7} = \frac{15}{56}$  of the capital, the residue =  $\frac{41}{56}$  of the capital, and each  $\frac{1}{20}$ th share of the residue =  $\frac{41}{56} \times \frac{1}{20} = \frac{41}{1120}$  of the capital. The stranger, therefore, receives  $\frac{15}{56} = \frac{300}{1120} = \frac{15}{56}$  of the capital.

\* The problems in this chapter may be considered as belonging rather to Law than to Algebra, as they contain little more than enunciations of the law of inheritance in certain complicated cases.

\* If the bequests stated in the present example were charged on the heirs collectively, the husband would be entitled to  $\frac{1}{4}$ , the mother to  $\frac{1}{6}$  of the residue:  $\frac{1}{4} + \frac{1}{6} = \frac{5}{12}$ ; the remainder  $\frac{7}{12}$  would be the son's share of the residue; but since the bequests,  $\frac{2}{5} + \frac{1}{4} = \frac{13}{20}$  of the capital, are charged upon the son and mother, the law throws a portion of the charge on the husband.

may be taken; this being again divisible by twenty. Such a capital is two hundred and forty. The mother receives one-sixth of this, namely, forty; twenty from this fall to the legacy, and she retains twenty for herself. The husband receives one-fourth, namely, sixty; from which twenty belong to the legacy, so that he retains forty. The remaining hundred and forty belong to the son; the legacy from this is two-fifths and one-fourth, or ninety-one; so that there remain forty-nine. The entire sum for the legacies is, therefore, one hundred and thirty-one, which must be divided among the two legatees. The one to whom two-fifths were bequeathed, receives eight-thirteenths of this; the one to whom one-fourth was devised, receives five-thirteenths.

If you wish distinctly to express the shares of the two legatees, you need only to multiply the parts of the heritage by thirteen, and to take them out of a capital of three thousand one hundred and twenty.

### On another Species of Legacies.

“A man dies, and leaves four sons and his wife; and bequeaths to a person as much as the share of one of the sons less the amount of the share of the widow.”

Divide the heritage into thirty-two parts. The widow receives one-eighth,\* namely, four; and each son seven. Consequently the legatee must receive three-sevenths of the share of a son. Add, therefore, to the heritage three-sevenths of the share of a son, that is to say, three parts, which is the amount of the legacy. This gives thirty-five, from which the legatee receives three and the remaining thirty-two are distributed among the heirs proportionably to their legal shares.

If he leaves two sons and a daughter,\* and bequeaths to some one as much as would be the share of a third son, if he had one; then you must consider, what would be the share of each son, in case he had three. Assume this to be seven, and for the entire heritage take a number, one-fifth of which may be divided into sevenths, and one-seventh of which may be divided into fifths. Such a number is thirty-five. Add to it two-sevenths of the same, namely, ten. This gives forty-five. Herefrom the legatee receives ten, each son fourteen, and the daughter seven.

If he leaves a mother, three sons, and a daughter, and bequeaths to some one as much as the share of one of his sons less the amount of the share of a second daughter, in case he had one; then you distribute the heritage into such a number of parts as may be divided among the actual heirs, and also among the same, if a second daughter were added to them.\* Such a number is three hundred and thirty-six. The share of the second daughter, if there were one, would be thirty-five, and that of a son eighty; their difference is forty-five, and this is the legacy. Add to it three hundred and thirty-six, the sum is three hundred and eighty-one, which is the number of parts of the entire heritage.

### On Completement.

“A woman dies and leaves eight daughters, a mother, and her husband, and bequeaths to some person as much as must be added to the share of a daughter to make it equal to one-fifth of the capital and to another person as much as must be added to the share of the mother to make it equal to one-fourth of the capital.”\*

Computation: Determine the parts of the residue, which in the present instance are thirteen. Take the capital, and subtract from it one-fifth of the same, less one part, as the share of a daughter: this being

\*A widow is entitled to  $\frac{1}{8}$ th of the residue; therefore  $\frac{7}{8}$ ths of the residue are to be distributed among the sons of the testator. Let  $x$  be the stranger's legacy. The widow's share =  $\frac{1}{8}(1 - x)$ ; each son's share =  $\frac{1}{4} \cdot \frac{7}{8}(1 - x)$ ; and a son's share, minus the widow's share =  $\frac{7}{32} - \frac{1}{32}(1 - x) = x$ ;  $\therefore x = \frac{3}{35}$ .

\*A son is entitled to receive twice as much as a daughter. Were there three sons and one daughter, each son would receive  $\frac{2}{7}$ ths of the residue. Let  $x$  be the stranger's legacy. Then  $\frac{2}{7}(1 - x) = x$ ;  $\therefore x = \frac{2}{9}$ ;  $1 - x = \frac{7}{9}$ . Each Son's share =  $\frac{2}{7} \cdot \frac{7}{9} = \frac{2}{9}$ . The Daughter's share =  $\frac{1}{7} \cdot \frac{7}{9} = \frac{1}{9}$ . The Stranger's legacy =  $\frac{2}{9}$ .

\*Let  $x$  be the stranger's legacy; a widow's share of the residue is  $\frac{1}{8}$ th;  $\frac{7}{8}(1 - x)$  is the residue.  $\frac{1}{8}[1 - x]$ , to be distributed among the children. Since there are 3 sons, and 1 daughter,  $\frac{1}{7} \cdot \frac{7}{8}(1 - x)$  is a son's share. Were there 3 sons and 2 daughters, a daughter's share would be  $\frac{1}{8} \cdot \frac{8}{8}(1 - x)$ . The difference =  $\frac{1}{7} \cdot \frac{7}{8}(1 - x) - \frac{1}{8}(1 - x) = x$ ;  $\therefore x = \frac{45}{336}$ .

In this case, the mother has  $\frac{2}{13}$ ; and each daughter has  $\frac{1}{13}$  of the residue.  $1 - x - y = 13v$ ;  $1 - [\frac{1}{5} - v] - [\frac{1}{4} - 2v] = 13v$ ; i.e.  $1 - \frac{1}{5} + v - \frac{1}{4} + 2v = 13v$ .

the first legacy. Then subtract also one-fourth, less two parts, as the share of the mother: this being the second legacy. There remain eleven-twentieths of the capital, which, when increased by three parts, are equal to thirteen parts. Remove now from thirteen parts the three parts on account of the three parts (on the other side), and you retain eleven-twentieths of the capital, equal to ten parts. Complete the capital, by adding to the ten parts as much as nine-elevenths of the same; then you find the capital equal to eighteen parts and two-elevenths. Assume now each part to be eleven; then the whole capital is two hundred, each part is eleven; the first legacy will be twenty-nine, and the second twenty-eight.

### Computation of Returns.

#### *On Marriage in Illness.*

“A man, in his last illness, marries a wife, paying (a marriage settlement of) one hundred dirhems, besides which he has no property, her dowry being ten dirhems. Then the wife dies, bequeathing one-third of her property. After this the husband dies.”\*

Computation: You take from the one hundred that which belongs entirely to her, on account of the dowry, namely, ten dirhems; there remain ninety dirhems, out of which she has bequeathed a legacy. Call the sum given to her (by her husband, exclusive of her dowry) thing; subtracting it, there remain ninety dirhems less thing. Ten dirhems and thing are already in her hands; she has disposed of one third of her property, which is three dirhems and one-third, and one-third of thing; there remain six dirhems and two-thirds plus two-thirds of thing, the moiety of which, namely, three dirhems and one-third plus one-third of thing, returns as his portion to the husband.\* Thus the heirs of the husband obtain (as his share) ninety-three dirhems and one-third, less two thirds of thing; and this is twice as much as the sum given to the woman, which was thing, since the woman had power to bequeath one-third of all which the husband left; and twice as much as the gift to her is two things. Remove now the ninety-three and one-third, from two-thirds of thing, and add these to the two things. Then you have ninety-three dirhems and one-third equal to two things and two-thirds. One thing is three-eighths of it, namely, as much as three-eighths of the ninety-three and one-third, that is, thirty-five dirhems.

### On Emancipation in Illness.

“Suppose that a man on his death-bed were to emancipate two slaves; the master himself leaving a son and a daughter. Then one of the two slaves dies, leaving a daughter and property to a greater amount than his price.”\* You take two-thirds of his price, and what the other slave has to return (in order to complete his ransom). If the slave die before the master, then the son and the daughter of the latter partake of the heritage, in such proportion, that the son receives as much as the two daughters together. But if the slave die after the master, then you take two-thirds of his value and what is returned by the other slave, and distribute it between the son and the daughter (of the master), in such a manner, that the son receives twice as much as the daughter; and what then remains (from the heritage of the

\*Let  $s$  be the sum, including the dowry, paid by the man, as a marriage settlement;  $d$  the dowry;  $x$  the gift to the wife, which she is empowered to bequeath if she pleases. She may bequeath, if she pleases,  $d + x$ ; she actually does bequeath  $\frac{1}{3}[d + x]$ ; the residue is  $\frac{2}{3}[d + x]$ , of which one half, viz.  $\frac{1}{3}[d + x]$  goes to her heirs, and the other half reverts to the husband.  $\therefore$  the husband's heirs have  $s - [d + x] + \frac{1}{3}[d + x]$ ; and since what the wife has disposed of, exclusive of the dowry, is  $x$ , twice which sum the husband is to receive,  $s - \frac{2}{3}[d + x] = 2x$ ;  $\therefore \frac{3}{8}[3s - 2d] = x$ . But  $s = 100$ ;  $\therefore x = 35$ ;  $d + x = 45$ ;  $\frac{1}{3}[d + x] = 15$ . Therefore the legacy which she bequeaths is 15, her husband receives 15, and her other heirs, 15. The husband's heirs receive  $2x = 70$ .

\*In other cases, as appears from pages 92 and 93, a husband inherits one-fourth of the residue of his wife's estate, after deducting the legacies which she may have bequeathed. But in this instance he inherits half the residue. If she die in debt, the debt is first to be deducted from her property, at least to the extent of her dowry (see the next problem.)

\*When the husband makes a bequest to a stranger, the third is reduced to one-sixth. Vide p. 137.

\*From the property of the slave, who dies, is to be deducted and paid to the master's heirs, first, two-thirds of the original cost of that slave, and secondly what is wanting to complete the ransom of the other slave. Call the amount of these two sums  $p$ ; and the property which the slave leaves  $u$ . Next, as to the residue of the slaves' property: First. If the slave dies before the master, the master's son takes  $\frac{2}{3}[u - p]$ ; the master's daughter  $\frac{1}{3}[u - p]$ ; and the slave's daughter  $\frac{1}{2}[u - p]$ . Second. If the slave dies after the master; the master's son is to receive  $\frac{2}{3}p$ , and the master's daughter  $\frac{1}{3}p$ ; and then the master's son takes  $\frac{1}{2}[u - p]$ , and the slave's daughter  $\frac{1}{2}[u - p]$ .

slave) is for the son alone, exclusive of the daughter; for the moiety of the heritage of the slave descends to the daughter of the slave, and the other moiety, according to the law of succession, to the son of the master, and there is nothing for the daughter (of the master).

It is the same, if a man on his death-bed emancipates a slave, besides whom he has no capital, and then the slave dies before his master.

If a man in his illness emancipates a slave, besides whom he possesses nothing, then that slave must ransom himself by two-thirds of his price. If the master has anticipated these two-thirds of his price and has spent them, then, upon the death of the master, the slave must pay two-thirds of what he retains.\* But if the master has anticipated from him his whole price and spent it, then there is no claim against the slave, since he has already paid his entire price.

“Suppose that a man on his death-bed emancipates a slave, whose price is three hundred dirhems, not having any property besides; then the slave dies, leaving three hundred dirhems and a daughter.” The computation is this: \* Call the legacy to the slave thing. He has to return the remainder of his price, after the deduction of the legacy, or three hundred less thing. This ransom, of three hundred less thing, belongs to the master. Now the slave dies, and leaves thing and a daughter. She must receive the moiety of this, namely, one half of thing; and the master receives as much. Therefore the heirs of the master receive three hundred less half a thing, and this is twice as much as the legacy, which is thing, namely, two things. Reduce this by removing half a thing from the three hundred, and adding it to the two things. Then you have three hundred, equal to two things and a half. One thing is, therefore, as much as two-fifths of three hundred, namely, one hundred and twenty.

This is the legacy (to the slave,) and the ransom is one hundred and eighty.

“Some person on his sick-bed has emancipated a slave, whose price is three hundred dirhems; the slave then dies, leaving four hundred dirhems and ten dirhems of debt, and two daughters, and bequeathing to a person one-third of his capital; the master has twenty dirhems debts.” The computation of this case is the following: \* Call the legacy to the slave thing; his ransom is the remainder of his price, namely, three hundred less thing. But the slave, when dying, left four hundred dirhems; and out of this sum, his ransom, namely, three hundred less thing, is paid to the master, so that one hundred dirhems and thing remain in the hands of the slave’s heirs. Herefrom are (first) subtracted the debts, namely, ten dirhems; there remain then ninety dirhems and thing. Of this he has bequeathed one-third, that is, thirty dirhems and one-third of thing; so that there remain for the heirs sixty dirhems and two-thirds of thing. Of this the two daughters receive two-thirds, namely, forty dirhems and four-ninths of thing, and the master receives twenty dirhems and two-ninths of thing, so that the heirs of the master obtain three hundred and twenty dirhems less seven-ninths of thing. Of this the debts of the master must be deducted, namely, twenty dirhems; there remain then three hundred dirhems less seven-ninths of thing; and this sum is twice as much as the legacy of the slave, which was thing; or, it is equal to two things. Reduce this, by removing the seven-ninths of thing, and adding them to two things; there re-

\*The slave retains one-third of his price; and this he must redeem at two-thirds of its value; namely at  $\frac{2}{3} \times \frac{1}{3} = \frac{2}{9}$  of his original price.

Let the slave’s original cost be  $a$ ; the property which he dies possessed of  $u$ ; what the master bequeaths to the slave,  $x$ . Then the net property which the slave dies possessed of is  $u + x - a$ .  $\frac{1}{3}[u + x - a]$  belongs, by law, to the master; and  $\frac{1}{2}[u + x - a]$  to the slave’s daughter. The master’s heirs, therefore, receive the ransom,  $a - x$ , and the inheritance,  $\frac{1}{3}[u + x - a]$ ; that is,  $\frac{1}{3}[(u + a - x)]$ ; and on the same principle as the slave, when emancipated, is allowed to ransom himself at two-thirds of his cost, the law of the case is that 2 are to be taken, where 1 is given.  $\therefore \frac{1}{3}[(u + a - x)] = 2x$ ;  $\therefore x = \frac{1}{7}[u + a]$ ; If, as in the example,  $u = a$ ,  $x = \frac{2}{7}a$ ; the daughter’s share =  $\frac{1}{7}[3u - 2a]$ ; The master’s heirs receive  $\frac{2}{7}[u + a]$ .

Let the slave’s original cost =  $a$ ; the property he dies possessed of =  $u$ ; the debt he owes =  $e$ . He leaves two daughters, and bequeaths to a stranger one-third of his capital. The master owes debts to the amount  $h$ ; where  $a = 300$ ,  $u = 400$ ,  $e = 10$ ,  $h = 20$ . Let what the master gives to the slave, in emancipating him =  $x$ . Slave’s ransom =  $a - x$ ; slave’s property – slave’s ransom =  $u + x - a$ ; Slave’s property – ransom – debt =  $u + x - a - e$ ; Legacy to stranger =  $\frac{1}{3}[u + x - a - e]$ ; Residue =  $\frac{2}{3}[u + x - a - e]$ ; The master, and each daughter, are, by law, severally entitled to  $\frac{1}{3} \times \frac{2}{3}[u + x - a - e]$ ; The master’s heirs receive altogether  $a - x + \frac{1}{3}[u + x - a - e]$  or  $\frac{1}{3}[a - x] + \frac{2}{3}[u - e]$ , which, on the principle that 2 are to be taken for 1 given, ought to be made equal to  $2x$ . But the author directs that the equation for determining  $x$  be  $\therefore x = \frac{3}{25}[7a + 2[u - e] - 3h] = 108$ .

main three hundred, equal to two things and seven-ninths. One thing is as much as nine twenty-fifths of three hundred, which is one hundred and eight; and so much is the legacy to the slave.



## Notes on the Arabic Text

The following brief notes are intended to point out for further enquiry a few circumstances connected with the history of Algebra:

1. The initial words of the treatise, as cited by Haji Khalfa, are: قال ابو عبدالله محمد بن موسى الخوارزمي ("Abu Abdallah Mohammed ben Musa of Khwarezm said...").
2. The term الجبر (*al-jabr*) literally signifies "restoration" or "completion," referring to the operation of transferring negative quantities to the other side of an equation as positive. The term المقابلة (*al-muqabala*) means "reduction" or "opposition," referring to the process of cancelling like quantities on both sides of an equation.
3. The Arabic word شيء (*shay'*, literally "thing") is used throughout the text as the designation of the unknown quantity—what we should now call  $x$ . It corresponds exactly to the *res* ("thing") of the early Italian algebraists, and to the *coss* of the Germans.
4. The word مال (*māl*, literally "property," "wealth," "possession") is used to denote the square of the unknown. It is rendered by Rosen as "square."
5. The word جذر (*jidhr*, literally "root") is used in the same sense as our "root." The word واجب (*wājib*, literally "obligatory") is used for the "absolute number"—what we call the constant term.
6. The method of geometrical demonstration here employed by al-Khwarizmi is worthy of especial notice. It shows that, like the Greek mathematicians, he was not satisfied with merely giving rules, but endeavoured to establish their accuracy by demonstration. His proofs are founded upon the second proposition of Book II of Euclid's Elements, and upon the well-known property of the gnomon.
7. The formula for the circumference of a circle given in the text (Text p. 51, Transl. p. 72) is  $c = \frac{62832}{20000}d$ , which gives  $\pi = 3.1416$ . This value was known to the Hindu mathematician Aryabhata, and the circumstance furnishes a strong argument for the opinion that al-Khwarizmi derived part of his information from an Indian source.
8. The comparisons drawn between the Algebra of the Arabs and that of the early Italian writers might perhaps have been more numerous and more detailed; but the enquiry was here restricted by the want of some important works. Montucla, Cossali, Hutton, and the Basil edition of Cardanus' *Ars magna*, were the only sources which the translator had the opportunity of consulting.
9. The solutions which the author has given of the remaining problems of this treatise, are, mathematically considered, for the most part incorrect. It is not that the problems, when once reduced into equations, are incorrectly worked out; but that in reducing them to equations, arbitrary assumptions are made, which are foreign or contradictory to the data first enounced, for the purpose, it should seem, of forcing the solutions to accord with the established rules of inheritance, as expounded by Arabian lawyers.
10. The object of the lawyers in their interpretations, and of the author in his solutions, seems to have been, to favour heirs and next of kin; by limiting the power of a testator, during illness, to bequeath property, or to emancipate slaves; and by requiring payment of heavy ransom for slaves whom a testator might, during illness, have directed to be emancipated.

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كتاب الجبر والمقابلة

# Zur "ältesten arabischen Algebra und Rechenkunst

*On the Earliest Arabic Algebra and Arithmetic*

von / by  
**JULIUS RUSKA**  
in Heidelberg

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### Introduction

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## Einleitung / Introduction

Daß die Mathematik bei den Arabern aus indischen und griechischen Quellen zusammengefloßen ist und vielfach durch Perser, Syrer und Juden in den Strom des geistigen Lebens eingef"uhrt wurde, ist eine Tatsache, die sich ebenso deutlich aus dem Inhalt der arabischen Schriften und der literarischen "Überlieferung ergibt, wie sie der geschichtlichen Lagerung der islamischen Gesamtkultur entspricht.

Schwierigeren Fragen stehen wir aber alsbald gegen"uber, wenn wir diese allgemein anerkannte Abh"angigkeit auf bestimmte Quellen zur"uckf"uhren wollen, wenn wir die auf lange Strecken versch"utteten Rinnsale aufzudecken versuchen, in denen die Zufuhr des fremden Stoffes sich vollzog, und wenn wir eine Entscheidung dar"uber treffen sollen, ob ein bestimmtes Wissen seine Form schon auf dem Weg zur Sammelstelle erhalten hat oder erst dem arabischen Einfluß seine Gestaltung verdankt. Die Schwierigkeiten wachsen, wenn sich Urteile "uber solche Dinge nicht auf die Quellen erster Hand, sondern auf "Übersetzungen und Bearbeitungen st"utzen, die den Charakter des Originalwerks verwischt haben und vielfach schon von pers"onlicher Stellungnahme bestimmt sind. Die Versuchung, moderne Ausdrucksweisen und Auffassungen in "ältere Werke hineinzutragen oder ihren Inhalt nach modernen Maßst"aben zu beurteilen, liegt auf mathematischem Gebiet besonders nahe, da ungewohnte Form und Umst"andlichkeit der Darstellung Hindernisse raschen Verst"andnisses sind und es dem Mathematiker gew"ohnlich mehr auf den Inhalt, als auf die Form der Erkenntnisse ankommt. F"ur alle Untersuchungen, die die Form der Darstellung betreffen, also f"ur die ganze Geschichte der Terminologie, werden wir daher immer wieder auf die Originalquellen zur"uckgehen m"üssen, um zu neuen oder gesicherteren Ergebnissen zu gelangen. Eine erneute Untersuchung der "ältesten arabischen Algebra schien besonders aussichtsvoll, nachdem sich bei einem fl"uchtigen Vergleich herausgestellt hatte, daß die von Fr. Rosen 1831 dem Text der Algebra des Muhammad b. M"us"a al-Hw"arazm"i beigeft"ugte "Übersetzung nicht den Wortlaut wiedergibt, sondern sich ziemlich große Freiheiten gestattet.

Sie schien auch darum erw"ünscht, weil "uber die Grundlagen der arabischen Algebra so viele entgegengesetzte und sich ausschließende Ansichten ge"äußert worden sind, daß von einer Ersch"öpfung des Gegenstandes noch immer nicht gesprochen werden kann. Im weiteren Verlauf der Arbeit stellte sich dann die Notwendigkeit heraus, auch andere Texte und "Übersetzungen in den Bereich der Untersuchung zu ziehen, und so ist schließlich eine Reihe von mehr oder minder zusammenh"angenden Einzelbeitr"agen zur Geschichte der "älteren arabischen Mathematik entstanden. Das beigegebene Register wird die Aufsuchung be-

That mathematics among the Arabs flowed together from Indian and Greek sources, and was frequently introduced into the current of intellectual life by Persians, Syrians, and Jews, is a fact that emerges as clearly from the content of Arabic writings and literary tradition as it corresponds to the historical stratification of Islamic culture as a whole.

But we immediately confront more difficult questions when we wish to trace this generally acknowledged dependence back to specific sources, when we attempt to uncover the long-buried channels through which the supply of foreign material took place, and when we must decide whether particular knowledge received its form already on the way to its destination, or owes its shaping to Arabic influence.

The difficulties increase when judgments about such matters rely not on first-hand sources but on translations and adaptations that have obscured the character of the original work and are frequently already determined by personal bias. The temptation to introduce modern modes of expression and conceptions into older works, or to evaluate their content by modern standards, is especially strong in the mathematical domain, since unfamiliar form and circumstantial presentation are obstacles to rapid comprehension, and the mathematician usually cares more about the content than the form of knowledge.

For all investigations concerning the form of presentation, that is, for the entire history of terminology, we shall therefore always have to return to the original sources in order to arrive at new or more secure results. A renewed investigation of the oldest Arabic algebra seemed particularly promising after a casual comparison revealed that the translation appended by Fr. Rosen in 1831 to the text of the Algebra of Muḥammad b. Mūsā al-Khwārazmī does not reproduce the wording but allows itself considerable liberties.

It also seemed desirable because so many contradictory and mutually exclusive opinions have been expressed about the foundations of Arabic algebra that an exhaustive treatment of the subject cannot yet be claimed. In the further course of the work, the necessity arose to include other texts and translations in the scope of the investigation, and so a series of more or less connected individual contributions to the history of older Arabic mathematics has finally emerged. The appended index will facilitate the locating of specific subjects of the investigations.

stimmter Gegenstande der Untersuchungen erleichtern.

### Anmerkung zur Namensumschreibung / Note on Name Transcription

Man kann eine ganze Musterkarte von Umschreibungen dieses Namens zusammenstellen; so schreibt Colebrooke (1817) Khuwārezmi, Rosen (1831) of Khio war ezm, Woepcke (1863) Alkhārizmi, Marre (1865) Al Khār e z m, Rodet (1878) Al-Khārizmi, Hankel (1874) al-Hovārezmi, Cantor (1880) Alchwarizmi, van Vloten (1895) Al-Khowarezmi, Suter (1900) el-Chowāresml oder Chwāresml, Björnbo (1905) Alkwarizmi, Wiedemann (1906) al Chārizmi, Bosmans (1906) El-Chowārizmi, Karpinski (1910) Al-Khowarizmi, Smith (1911) Al-Khowārazmi, Eneström Alkhwarizmi und Alkwarismi. Dazu kommen die alten Formen Alchoarismi in der "Übersetzung des Gerhard von Cremona, Alghuarizim, Alguarizin, Algaurizm, Algaurizin in den Handschriften der "Übersetzung des Robert Castrensis (Bibl. Math. 3. F., Bd. 11, 1910/11, S. 130; die Formen mit *in* und *im* sind offenbar aus den Formen mit *m* und *mi* entstanden, die mit *au* aus *ua*), Alghoarismi, Algoarismi, Algorismi in der *Pratica Arismetrice* des Johannes Hispalensis, Algoritmi im Traktat *de numero Indorum*. Volkstumliche Weiterbildung führt von Algorismus zu *augorisme* und *augrim* (Smith-Karpinski, *The Hindu-Arabic Numerals*, Boston 1911, S. 120, 121). Alle Formen mit Khwa- und Chwa- beruhen auf buchstäblicher Umschreibung der Zeichengruppe خوا, die der persischen Sprache eigentümlich ist und etwa *kh'o* gesprochen wird (vgl. Xwrdaoioi, Xopdaoioi). Da der zweite Vokal von *Hwārazm* nach Vullers (S. 736) ein Fatha ist, lautet die genaue Umschreibung der Nisbe al-Hwārazmi oder Alhwārazmi.

Die Wiedergabe von *q* durch *k*, von ج durch *s* ist falsch, die Umschreibung von خ durch *ch* oder *kh* ist nicht eindeutig genug, der allgemeinen Verwendung von *H* und *h* stehen typographische Schwierigkeiten gegenüber. Man vermeidet diese und andere Klippen, wenn man Rosen folgend den Verfasser Muhammad b. Muṣā nennt. Will man die geschichtlichen Zusammenhänge betonen, so wird man besser Algoritmi oder Algorithmus schreiben, wie man auch Averroes statt Ibn Rusd sagt.

Sowenig wir das Wort Averroismus durch Ibnrusdismus ersetzen können, so wenig ist die moderne Umschreibung Alhwārazmi geeignet, die Erinnerung an die Algorithmiker lebendig zu erhalten.

One can assemble a whole pattern-card of transcriptions of this name; thus Colebrooke (1817) writes Khuwārezmi, Rosen (1831) "of Khio war ezm," Woepcke (1863) Alkhārizmi, Marre (1865) Al Khār e z m, Rodet (1878) Al-Khārizmi, Hankel (1874) al-Hovārezmi, Cantor (1880) Alchwarizmi, van Vloten (1895) Al-Khowarezmi, Suter (1900) el-Chowāresml or Chwāresml, Björnbo (1905) Alkwarizmi, Wiedemann (1906) al Chārizmi, Bosmans (1906) El-Chowārizmi, Karpinski (1910) Al-Khowarizmi, Smith (1911) Al-Khowārazmi, Eneström Alkhwarizmi and Alkwarismi.

To these are added the old forms Alchoarismi in the translation of Gerhard of Cremona, Alghuarizim, Alguarizin, Algaurizm, Algaurizin in the manuscripts of the translation of Robert of Chester (Bibl. Math. 3rd ser., vol. 11, 1910/11, p. 130; the forms with *in* and *im* have obviously arisen from the forms with *m* and *mi*, those with *au* from *ua*), Alghoarismi, Algoarismi, Algorismi in the *Pratica Arismetrice* of Johannes Hispalensis, Algoritmi in the treatise *De numero Indorum*.

Vulgar development leads from Algorismus to *augorisme* and *augrim* (Smith-Karpinski, *The Hindu-Arabic Numerals*, Boston 1911, pp. 120, 121). All forms with Khwa- and Chwa- rest on literal transcription of the group of characters خوا, which is peculiar to the Persian language and is pronounced approximately *khō* (cf. Xwrdaoioi, Xopdaoioi). Since the second vowel of *Hwārazm* according to Vullers (p. 736) is a fatha, the precise transcription of the nisba is al-Hwārazmi or Alhwārazmi.

The rendering of *q* by *k*, of ج by *s* is wrong; the transcription of خ by *ch* or *kh* is not unambiguous enough; typographical difficulties stand in the way of the general use of *H* and *h*. One avoids these and other pitfalls if, following Rosen, one calls the author Muḥammad b. Muṣā. If one wishes to emphasize the historical connections, it is better to write Algoritmi or Algorithmus, just as one says Averroes instead of Ibn Rusd.

Just as little as we can replace the word Averroism with Ibnrushdism, so little is the modern transcription Alhwārazmi suited to keep alive the memory of the algebrists.

## 1 I. Der Titel der Algebra / The Title of the Algebra

Erwähnte Muhammad b. Mūsā nicht selbst in der Einleitung zu seiner "Algebra" die Gegenstände, von denen er zu handeln beabsichtigt, so konnte man versucht sein, das uns erhaltene Werk als willkürliche Zusammenfügung von zwei oder drei ursprünglich getrennten Schriften anzusehen. Sein Inhalt entspricht aber durchaus dem, was der Verfasser als seine Absicht bezeichnet:

ان هذا كتاب الاجابة والمقابلة وقد حددته الى ما يسهل من الحساب وما يحتاج اليه الناس في مواريتهم ووصاياهم ومقاسمتهم ومعاملتهم وكل ما يتداولونه بينهم من المكايلة

Had Muḥammad b. Mūsā not himself mentioned in the introduction to his "Algebra" the subjects he intended to treat, one might be tempted to regard the work preserved to us as an arbitrary combination of two or three originally separate treatises. Its content, however, fully corresponds to what the author states as his intention:

ان هذا كتاب الاجابة والمقابلة وقد حددته الى ما يسهل من الحساب وما يحتاج اليه الناس في مواريتهم ووصاياهم ومقاسمتهم ومعاملتهم وكل ما يتداولونه بينهم من المكايلة والمساحة وغير ذلك

والمساحة وغير ذلك

“ein kurzgefaßtes Buch zu schreiben von dem Rechenverfahren der Erg”anzung und Ausgleichung (*hisāb al-gabr wa-l-muqābala*) mit Beschr”ankung auf das Anmutige und Hochgesch”atzte des Rechenverfahrens f”ur das, was die Leute fortw”ahrend notwendig brauchen bei ihren Erbschaften und ihren Verm”achtnissen und bei ihren Teilungen und ihren Prozeßbescheiden und ihren Handelsgesch”aften und bei allem, womit sie sich gegenseitig befassen von der Ausmessung der L”andereien und der Herstellung der Kan”ale und der Geometrie und anderm dergleichen nach seinen Gesichtspunkten und Arten.”

“[This is] a concise book on the method of calculation of *al-jabr wa-l-muqābala* [restoration and reduction], comprising what is elegant and esteemed in calculation, touching upon what people constantly need regarding their inheritances and bequests, their divisions, their legal judgments, their commercial transactions, and all that they deal with among themselves concerning the measurement of lands, the construction of canals, and geometry, as well as other such things according to its principles and types.”

Wollten wir den Ausdruck *kitāb* pressen, der urspr”unglich die Sammlung und Vereinigung mehrerer Teile eines Dings bedeutet (Lane Bd. I, S. 80), so k”onnten wir darin den Hinweis erblicken, daß das Werk ein Auszug aus verschiedenen Quellen ist. Der Verfasser gibt selbst nirgends eine Andeutung, woher er seinen Stoff hat, und wie weit ihm pers”onlich ein Verdienst an der Pr”agung zukommt. Halten wir uns gegenw”artig, daß er sich als Astronom anerkanntermaßen auf indische Wissenschaft st”utzte und auch ein Buch ”uber das indische Rechnen verfaßte, so ist die Richtung angegeben, in der man zun”achst zu suchen hat. Bevor wir uns aber den Untersuchungen zuwenden, die seit Colebrooke dar”uber ver”offentlicht sind, muß der Titel und die Gesamtanlage des Buches etwas genauer betrachtet werden. Man hat immer wieder dem Erstaunen Ausdruck gegeben, daß Muḥammad b. M”usā seinem Werk einen Titel gegeben habe, den er nicht erkl”art, und daß er daf”ur zwei Operationen ausgew”ählt habe, die in dem ”eigentlich theoretischen Teile” seines Buches gar nicht vorkommen.

Were we to press the term *kitāb*, which originally means the collection and unification of several parts of a thing (Lane vol. I, p. 80), we might see in it an indication that the work is an extract from various sources.

The author himself gives no hint anywhere of where he obtained his material, or to what extent any credit for its formulation is due to him personally. If we bear in mind that he, as an astronomer, demonstrably relied on Indian science and also composed a book on Indian calculation, the direction in which one must first search is indicated.

But before we turn to the investigations published since Colebrooke, the title and overall plan of the book must be examined somewhat more closely. Repeatedly, astonishment has been expressed that Muḥammad b. Mūsā gave his work a title which he did not explain, and that for this he chose two operations that do not occur at all in the ”properly theoretical part” of his book.

Ich glaube, man hat sich das Verst”andnis dieser Tatsachen ganz unn”otig erschwert, indem man das Buch nicht historisch mit den ”Augen des Verfassers” (Cantor a. a. O. S. 728), sondern ohne R”ucksicht auf den Gesamthalt nur nach den wenigen theoretischen Kapiteln am Anfang beurteilte, die den Geschichtschreiber der Mathematik besonders interessieren.

I believe that people have made the understanding of these facts entirely unnecessarily difficult by judging the book not historically with the ”eyes of the author” (Cantor, op. cit., p. 728), but without regard for the total content, only according to the few theoretical chapters at the beginning that especially interest the historian of mathematics.

Schon H. Hankel hat mit dieser Verschiebung des Tatbestandes den Anfang gemacht, indem er zwar den Inhalt des ersten Teils der Algebra ganz zutreffend kennzeichnet, die geometrischen Kapitel aber davon l”ost und von dem Rest des Werkes bemerkt, daß die dort behandelten Aufgaben ”uber Erbteilungen von keinem wissenschaftlichen Interesse seien, wenn sie auch f”ur Mohammedaner praktisch von gr”oßtem Werte waren.<sup>1</sup>

Already H. Hankel began this shifting of the facts, in that he indeed quite accurately characterized the content of the first part of the Algebra, but detached the geometrical chapters from it and remarked of the rest of the work that the problems on inheritance treated there were of no scientific interest, even if they were of great practical value for Muslims.<sup>1</sup>

<sup>1</sup>H. Hankel, *Zur Geschichte der Mathematik in Alterthum und Mittelalter*, Leipzig 1874, S. 260. Was weiter dar”uber gesagt wird, geht auf zwei Fußnoten Rosens (a. a. O. S. 91, 133) zur”uck. Cantor hat dann nach A. v. Kramers *Culturgeschichte des Orients* noch einige S”atze hinzugef”ugt und auch auf die Ausf”uhrungen von Ibn Hald”un und H”aggi Hāa ”uber Erbteilungswissenschaft hingewiesen.<sup>2</sup>

<sup>1</sup>H. Hankel, *Zur Geschichte der Mathematik in Alterthum und Mittelalter*, Leipzig 1874, p. 260. What further is said about this goes back to two footnotes of Rosen (op. cit., pp. 91, 133). Cantor then, following A. von Kramers’ *Culturgeschichte des Orients*, added a few more sentences and also referred to the expositions of Ibn Khaldūn and Ḥājī Ḥalīfa on the science of inheritance division.<sup>2</sup>

<sup>2</sup>M. Cantor, *Vorlesungen ”uber Gesch. d. Math.* I<sup>3</sup>, 1907, S. 719. Dieser Abschnitt der Algebra ist nach Cantor ”arabisch durch

<sup>2</sup>M. Cantor, *Vorlesungen ”uber Geschichte der Mathematik* I<sup>3</sup>, 1907, p. 719. This section of the Algebra is, according to Cantor, ”Arab th-

und durch" und als Grundlage zahlreicher späterer Schriften zu betrachten, die von den Erbteilungen und den dabei vorkommenden Rechnungen ausschließlich handeln. Proben dieser Rechenpraxis sind leider nicht mitgeteilt, und so ist es nicht verwunderlich, daß die jüngste Darstellung dieser Dinge auf eine Häufung von Mißverständnissen hinausläuft.

Auf dieser Stufe der Untersuchung genügt es, die Tatsache festzuhalten, daß Muhammad b. Mūsā eine "Algebra" in unserem Sinne nicht schreiben wollte, noch geschrieben hat. Sein Buch will weiter nichts sein als eine auf zahlreiche ausgeführte Musterbeispiele gestützte Einführung in das angewandte Rechnen. Ob es diesen in der Einleitung ausgesprochenen Zweck, zur Lösung von Aufgaben aus dem Gebiet des bürgerlichen Rechnens und der Flächen- und Körperberechnung anzuleiten, nach unseren Ansprüchen ganz erreicht, ist hier nicht zu erörtern; wohl aber sind wir die Antwort auf die Frage schuldig, wie der Verfasser dazu kam, seinem Werk einen angeblich so unverständlichen Titel zu geben.

Wenn wir Günther glauben sollen, wußte man erst "späteren Berichten zufolge", daß eine Gleichung für eingerichtet galt, wenn auf beiden Seiten ausschließlich positive Glieder standen, während das Wort "Gegenüberstellung" dem Weglassen gleicher Größen links und rechts entsprach.

Auch Cantor (I<sup>3</sup>, S. 719) und Hankel (S. 260, Note\*\*) berufen sich auf die Erklärungen späterer arabischer Schriftsteller, und wer will, kann bei Rosen (S. 180 ff.) oder Nesselmann (*Die Algebra der Griechen*, S. 40 ff.) Text und Übersetzung von Originalstellen nachlesen, in denen eine Erklärung der Ausdrücke *al-gabr* und *al-muqābala* gegeben wird. Kein Zweifel kann darüber bestehen, daß die verschiedenen Übersetzungen der beiden Termini eine klare Vorstellung von ihrem mathematischen Sinn nicht zu geben vermögen. *Restauratio et oppositio* ist ein ebenso unklares Begriffspaar wie *restoring and comparing* (Colebrooke) oder *completion and reduction* (Rosen), *Wiederherstellung und Gegenüberstellung* (Cantor) oder *Ergänzung und Ausgleichung*, wie ich oben übersetzt habe.

Gestehen wir aber dem Autor zu, daß wir ihn ganz gelesen haben müssen, ehe wir ihn kritisieren, so bekommen wir ein wesentlich anderes Bild. Wir begegnen den beiden Ausdrücken zuerst wieder da, wo der Verfasser ausführt, daß bei dem *hisāb al-gabr wa-l-muqābala* drei Arten von Zahlen vorkommen, die als *gudūr* d. i. Wurzeln, *amwāl* d. i. Vermögen, und *radad mufrad* d. i. alleinstehende Zahl bezeichnet werden.<sup>3</sup>

<sup>3</sup>Ich gebe die Termini auch in Umschrift, weil die Leser, die sich für Geschichte der Mathematik interessieren, mit der arabischen Schrift und Sprache nur ausnahmsweise Bescheid wissen. In den arabischen Zitaten beschränke ich mich auf den Konsonantentext, soweit nicht besondere Gründe die Beifügung der Vokale oder anderer Lesezeichen notwendig machen. Das Wort *gudūr* ist der Plural von *gidr*, *amwāl* der Plural von *māl*, *mufrad* kommt von *farada*, *farīda*, abgesondert sein, allein sein, einzig sein, einfach sein.

Hieraus ist noch nichts zu entnehmen. Aber schon S. 15, am Schluß des theoretischen Teils, der die sechs Normalformen der Gleichungen behandelt, wird dem Leser gesagt, daß jede andere Gleichung durch das Verfahren der Ergänzung und Ausgleichung auf eine der sechs Normalformen gebracht werden kann. Der Hinweis wird S. 25 in der Einleitung zu dem "Kapitel von den sechs Fragen" wiederholt, an der Stelle also, wo der Verfasser

rough and through" and is to be regarded as the foundation of numerous later writings that deal exclusively with inheritance divisions and the calculations involved. Samples of this calculation practice are unfortunately not provided, and so it is not surprising that the most recent presentation of these matters amounts to a heap of misunderstandings.

At this stage of investigation, it suffices to establish the fact that Muḥammad b. Mūsā neither wished to write nor wrote an "Algebra" in our sense. His book wishes to be nothing more than an introduction to applied calculation, supported by numerous worked-out model examples. Whether it fully achieves, by our standards, the purpose stated in the introduction—namely, to guide the solution of problems from the domain of practical calculation and of plane and solid measurement—is not to be discussed here; but we certainly owe an answer to the question of how the author came to give his work a title allegedly so incomprehensible.

If we are to believe Günther, one knew only "according to later reports" that an equation was considered *set up* when exclusively positive terms stood on both sides, while the word "confrontation" corresponded to the removal of equal quantities from both sides.

Cantor (I<sup>3</sup>, p. 719) and Hankel (p. 260, n. \*\*) also appeal to the explanations of later Arabic writers, and whoever wishes can look up in Rosen (pp. 180 ff.) or Nesselmann (*Die Algebra der Griechen*, pp. 40 ff.) the text and translation of original passages in which an explanation of the terms *al-gabr* and *al-muqābala* is given. There can be no doubt that the various translations of the two terms are unable to give a clear conception of their mathematical sense. *Restauratio et oppositio* is an equally unclear pair of concepts as *restoring and comparing* (Colebrooke) or *completion and reduction* (Rosen), *Wiederherstellung und Gegenüberstellung* (Cantor) or *Ergänzung und Ausgleichung* [completion and balancing], as I have translated above.

But if we grant the author that we must have read him completely before we criticize him, we obtain a fundamentally different picture. We encounter the two expressions first where the author explains that in *hisāb al-gabr wa-l-muqābala* three kinds of numbers occur, designated as *gudūr* i.e. roots, *amwāl* i.e. possessions/wealth, and *radad mufrad* i.e. standalone number.<sup>3</sup>

<sup>3</sup>I give the terms also in transliteration, because readers interested in the history of mathematics only rarely have acquaintance with the Arabic script and language. In the Arabic citations I restrict myself to the consonantal text, unless special reasons require the addition of vowels or other reading marks. The word *gudūr* is the plural of *gidr*, *amwāl* the plural of *māl*, *mufrad* comes from *farada*, *farīda*: to be separated, to be alone, to be unique, to be simple.

Nothing is to be gathered from this as yet. But already on p. 15, at the end of the theoretical part that treats the six normal forms of equations, the reader is told that every other equation can be brought to one of the six normal forms by the method of completion and balancing. This hint is repeated on p. 25 in the introduction to the "Chapter of the Six Questions," that is, at the point where the author passes to "word problems," whose setup must



ser zu den "Textaufgaben" übergeht, deren Ansatz erst noch auf die eine oder andere der Normalformen gebracht werden muß. Im ersten Beispiel, das in unserer Schreibweise zum Ansatz  $x^2 = 40x - 4x^2$  führt, kommt nur "Ergänzung" vor. Nachdem der Ansatz abgeleitet ist, wird das Ergebnis ausdrücklich zusammengefaßt: "Es ist also ein Vermögen gleich vierzig Etwas weniger vier Vermögen" und fortgefahren: "Ergänze es nun durch die vier Vermögen, und füge sie dem einen Vermögen hinzu, so werden vierzig Etwas gleich fünf Vermögen; dann ist das einzelne Vermögen gleich acht Wurzeln" usw.

Es geht wohl gewiß nur ein mittelalterlicher Schulverstand dazu, hier nach zu begreifen, was mit "Ergänzung" gemeint ist. Der gleiche Ausdruck mit derselben umständlichen Erläuterung kehrt im dritten Beispiel wieder. Da aber Rosen an beiden Stellen *reduce it* statt *complete it* übersetzt und auch das *radd*, das für die Reduktion des Koeffizienten der Unbekannten auf 1 verwendet wird, mit *reduce it* wiedergibt, ist schon hier Verwirrung in die Terminologie gebracht.

Sie wird vollends unheilbar, wenn nachher das richtige *reduce by it*, *muqābil bihī*, wofür besser *compare* oder *balance by it* gesagt worden wäre, hinzukommt und mit *complete it* auch noch die Operation des *ikmāl* übersetzt wird. Auch die Hinweise S. 179 (erster und letzter Absatz) können daran für Leser der Übersetzung nichts ändern.

Wie in der zweiten Aufgabe der Sinn des *radd* "und führe es zurück auf ein einziges Vermögen" sich ganz natürlich aus dem Zusammenhang ergibt, so liefert die vierte ein Beispiel für die Anwendung des *ikmāl*, d. h. der "Vervollständigung" durch Multiplikation, also der Wegschaffung eines Koeffizienten, der ein echter Bruch ist: "So vervollständige dein Vermögen, und seine Vervollständigung geschieht dadurch, daß du alles, was du hast, mit zwölf vervielfachst."<sup>4</sup>

<sup>4</sup>Das Beispiel ist  $\frac{x}{3} = x + 24$ , woraus  $x = 12x + 288$ . In der fünften Aufgabe kommt endlich auch "Ausgleichung" vor: "Es sind also fünfzig Dirhem und ein Vermögen gleich neunundzwanzig Dirhem und zehn Etwas; gleiche nun damit aus, und dies besteht darin, daß du wegwirfst von den fünfzig neunundzwanzig, so daß bleibt einundzwanzig Dirhem und ein Vermögen gleich zehn Etwas."

In dieser Weise wird durch das ganze Buch hindurch Aufgabe für Aufgabe erläutert, d. h. es wird der Ansatz abgeleitet und jeder Schritt bis zur Lösung der Aufgabe vorgemacht und eingeprägt. Nur im geometrischen Abschnitt ist keine Gelegenheit, das zur Auflöserung von Gleichungen dienende Rechenverfahren mit seinen besonderen Ausdrücken anzuwenden.

Man wird hiernach gewiß nicht mehr behaupten können, daß Muhammad b. Mūsā diese Ausdrücke nirgends erklärt hat, wie schon bei Rosen S. 177 zu lesen ist, zumal die verschiedenen Definitionen aus späterer Zeit auch nur darauf hinauskommen, die "Ergänzung" durch "Hinzufügung" und die "Ausgleichung" durch "Wegnahme gleicher Größen" zu ersetzen.

Kein Araber wird den Ausdruck "Ergänzungs- und Ausgleichungsrechnung" mißverstanden haben, wenn er sich in das Buch eingelesen hatte und im Text den Imperativen "ergänze" oder "gleiche aus" mit den Anweisungen zur Ausführung der Rechnung begegnete. Daß der Verfasser die Operationen erst

first still be brought to one or another of the normal forms.

In the first example, which in our notation leads to the equation  $x^2 = 40x - 4x^2$ , only "completion" occurs. After the equation is derived, the result is explicitly summarized: "So a possession equals forty things less four possessions," and continued: "Complete it now by the four possessions, and add them to the one possession, so forty things equal five possessions; then the single possession equals eight roots," etc.

Surely only a moderate student intelligence is needed to understand from this what is meant by "completion." The same expression with the same circumstantial explanation recurs in the third example. But since Rosen at both places translates *reduce it* instead of *complete it*, and also renders *radd*, which is used for the reduction of the coefficient of the unknown to 1, by *reduce it*, confusion has already been introduced into the terminology here.

It becomes completely incurable when subsequently the correct "reduce by it," *muqābil bihī*, for which "compare" or "balance by it" would have been better, is added, and with "complete it" the operation of *ikmāl* [completion] is also translated. Even the indications on p. 179 (first and last paragraphs) cannot change anything in this for readers of the translation.

Just as in the second problem the sense of *radd*—"and reduce it to a single possession"—follows quite naturally from the context, so the fourth provides an example of the application of *ikmāl*, that is, "completion" by multiplication, hence the elimination of a coefficient that is a proper fraction: "So complete your possession, and its completion takes place by multiplying everything you have by twelve."<sup>4</sup>

<sup>4</sup>The example is  $\frac{x}{3} = x + 24$ , whence  $x = 12x + 288$ . In the fifth problem, finally, "balancing" also occurs: "So fifty dirhams and a possession equal twenty-nine dirhams and ten things; balance this now, and this consists in casting away twenty-nine from fifty, so that twenty-one dirhams and a possession equal ten things remain."

In this manner, throughout the whole book, problem by problem is explained, that is, the equation is derived and every step up to the solution of the problem is demonstrated and impressed upon the reader. Only in the geometrical section is there no occasion to apply the calculational procedure serving the resolution of equations with its special expressions.

After this, one will surely no longer be able to claim that Muhammad b. Mūsā has explained these expressions nowhere, as can already be read in Rosen p. 177, especially since the various definitions from later times also amount to nothing more than replacing "completion" by "addition" and "balancing" by "removal of equal quantities."

No Arab will have misunderstood the expression "calculation of completion and balancing" if he had read into the book and encountered in the text the imperatives "complete" or "balance" together with the instructions for carrying out the calculation. That the author explains the operations only where he applies

da erl"autert, wo er sie zum erstenmal anwendet, ist ebenso vern"unftig und didaktisch richtig, als es zwecklos gewesen w"are, sie schon in dem "eigentlich theoretischen Teile" zu erw"ahnen, wo gar kein Gebrauch davon gemacht wird.

Die Frage, ob Muhammad b. M"usā die beiden Ausdr"ucke *al-gabr* und *al-muqābala* als erster angewandt und in die Praxis der Gleichungen eingef"uhrt hat, oder ob er damit schon einer festen Tradition und alten "Ubung folgte, wird nur im Zusammenhang mit der Pr"ufung der gesamten Terminologie und literarischen Form seines Werkes der L"osung n"aher gebracht werden k"onnen.

So viel kann jetzt schon gesagt werden, daß wenn bereits eine Tradition bestand, sie auf arabischem Sprachgebiet nur innerhalb der Praxis des Gesch"äftsrechnens gesucht werden kann. Zeugnisse, die "alter w"aren als die Algebra des Muhammad b. M"usā, sind aber bis jetzt nicht bekannt, und so werden wir vorl"aufig wenigstens mit der M"oglichkeit rechnen, daß er selbst bei der Bearbeitung seiner Quellen jene anschaulichen Wendungen an die Stelle der abstrakteren Terminologie der Griechen und Inder gesetzt hat.

Der allgemeinste aus einer Textaufgabe zu gewinnende Ansatz f"uhrt zur Gleichsetzung zweier Ausdr"ucke, die durch Addition, Subtraktion, Multiplikation und Division bekannter und unbekannter Gr"ößen entstehen. Durch eine hinreichende Anzahl von Umformungen erh"alt man schließlich zwei Ausdr"ucke, die die bekannten und unbekannten Glieder nur noch in additiver und subtraktiver Verbindung enthalten.

Von dieser Form geht Diophant aus, wenn er an einer oft genannten Stelle des ersten Buches der *Arithmétique* sagt: ean de tēn heterān en hupokeimenōi enuparchī ē en amphoterois en el-leipsei tina eidī, deī zitein (sc. prosatheinai) ta leiponta eidī en amphoterois tois meresin, heōs an hekaterīn tīn merīn ta eidī sunuparchonta genītai, kai palin aphairein ta homoia apo tīn homoiīn, heōs an hekaterī tīn merīn en eidos kataleiphthī. (ed. Tannery I, S. 14).

Die erste Umformung w"urde arabisch *al-gabr* heißen; sie wird kurz und b"undig mit dem Wort *al-gabr*, die "Erg"anzung", bezeichnet, das sonst das Einrichten von Knochenbr"uchen bedeutet. Die andere Operation ist *al-muqābala*; sie heißt ebenso anschaulich *al-muqābala*, d. h. die "Kompensation" oder "Ausgleichung".<sup>5</sup>

<sup>5</sup>Auch das Kollationieren von Handschriften und die Opposition, also das Gegen"ubertreten von Sternen wird durch diesen Ausdruck bezeichnet. Zu diesen beiden wichtigsten Operationen kommt dann noch *al-radd*, die "Zur"uckf"uhrung" des Koeffizienten der Unbekannten auf 1, wenn er gr"ößer als 1 ist, und *al-ikmāl*, die "Vervollst"andigung", wenn der Koeffizient ein echter Bruch ist. Dies ist der Tatbestand, den wir in der "ältesten Algebra vorfinden. Wenn Carra de Vaux in seiner Notiz *Sur le sens exact du mot "al-djebr"* (Bibl. Math., 2. Folge, Bd. 11, 1897, S. 1, 2) darauf aufmerksam gemacht hat, daß *al-djebr* "n'est pas toujours opposé au mot al-muhābala, comme on pourrait le croire d'après le titre de l'ouvrage d'Al-Khārizmī", sondern daß man es auch im Gegensatz zu dem Wort *al-haḍ* (الحط) angewandt finde, so beweisen die aus Autoren des 14. Jahrhunderts beigebrachten Belege nichts f"ur die urspr"ungliche Anwendung der Termini.

them for the first time is as reasonable and didactically correct as it would have been pointless to mention them already in the "properly theoretical part," where no use is made of them at all.

The question of whether Muḥammad b. Mūsā was the first to apply the two expressions *al-gabr* and *al-muqābala* and introduced them into the practice of equations, or whether he thereby already followed a fixed tradition and ancient practice, can only be brought closer to solution in connection with an examination of the entire terminology and literary form of his work.

This much can already be said now: if a tradition already existed, it can be sought in the Arabic linguistic domain only within the practice of business calculation. Testimonies older than the Algebra of Muḥammad b. Mūsā are not known up to now, and so we shall provisionally reckon at least with the possibility that he himself, in working over his sources, replaced that more abstract terminology of the Greeks and Indians with those vivid turns of phrase.

The most general equation obtainable from a word problem leads to the equating of two expressions that arise through addition, subtraction, multiplication, and division of known and unknown quantities. Through a sufficient number of transformations, one finally obtains two expressions that contain the known and unknown terms only in additive and subtractive connection.

From this form Diophantus proceeds when he says at a frequently cited passage of the first book of the *Arithmétique*: ean de tēn heterān en hupokeimenōi enuparchī ē en amphoterois en el-leipsei tina eidī, deī zitein (sc. prosatheinai) ta leiponta eidī en amphoterois tois meresin, heōs an hekaterīn tīn merīn ta eidī sunuparchonta genītai, kai palin aphairein ta homoia apo tīn homoiīn, heōs an hekaterī tīn merīn en eidos kataleiphthī. (ed. Tannery I, p. 14).

The first transformation would be called in Arabic *al-gabr*; it is designated briefly and compactly by the word *al-gabr*, the "completion," which otherwise means the setting of bone fractures. The other operation is *al-muqābala*; it is called just as vividly *al-muqābala*, that is, the "compensation" or "balancing."<sup>5</sup>

<sup>5</sup>Also the collation of manuscripts and the opposition, that is, the confronting of stars, is designated by this expression. To these two most important operations is then added *al-radd*, the "reduction" of the coefficient of the unknown to 1, when it is greater than 1, and *al-ikmāl*, the "completion," when the coefficient is a proper fraction. This is the state of affairs that we find in the oldest algebra.

If Carra de Vaux in his note *Sur le sens exact du mot "al-djebr"* (Bibl. Math., 2nd ser., vol. 11, 1897, pp. 1, 2) has pointed out that *al-djebr* "is not always opposed to the word al-muhābala, as one might believe from the title of the work of Al-Khārizmī," but that one finds it also in contrast to the word *al-haḍ* (الحط), the proofs adduced from authors of the 14th century demonstrate nothing for the original application of the terms.

Bei jenen Autoren — es handelt sich um die Arithmetik des Takī ad-dīn al-Ḥanbalī, "über den nach H. Suter nur feststeht, daß er vor 1410 lebte, und um eine Schrift des Ibn al-Ḥā'im, der 1355–1412 gelebt hat — ist von einem Verfahren zur Auflö'sung von Gleichungen "überhaupt nicht die Rede. Es werden die beiden Operationen der Addition und der Multiplikation den beiden Operationen der Subtraktion und der Division gegen"übergestellt und jene, da ihre Ausfö'hrung zu einer grö'ßeren Zahl fö'ührt, als *gabr*, hier also wohl "Auf"üllung" oder "Vermehrung", diese, da eine kleinere Zahl herauskommt, als *haḍ*, d. i. "Erniedrigung" oder "Nachlaß" bezeichnet.

Es ist *gabr*, wenn man sagt, um wieviel muß du 10 vergrö'ßern, um 17 zu erhalten, oder womit muß du 10 vervielfachen, um 17 zu erhalten; es ist *haḍ*, wenn man sagt, durch wieviel muß du 10 teilen, oder um wieviel muß du 10 verkleinern, um 7 zu erhalten. Wenn Carra de Vaux diese in der "älteren Literatur unbekannten Zusammenfassungen mit den modern gedachten S"ätzen wiedergibt: "Il est donc clair que le djebir c'est l'opération exprimée par les deux équations  $a + x = b$ ,  $a \cdot x = b$  et que le haḍ, c'est l'opération qu'expriment ces deux ci:  $a - x = b$ ,  $a : x = b$ ", und daran den Schluß knö'üpft, "comme ces quatre opérations sont les plus simples possible de l'algèbre, on doit croire que tel est bien le sens primitif des deux mots", so vermengt er nicht nur ganz verschiedene Dinge, sondern stellt die geschichtliche Entwicklung geradezu auf den Kopf, indem er weiterhin behauptet: "Le mot djebir dont nous venons d'indiquer le sens primitif, a ensuite servi à designer un nombre indéterminé de questions diverses, parmi lesquelles six sont fondamentales, selon Al-Khārizmī ...; le mot haḍ, devenu impropre à cause de la complication des problèmes, a disparu devant le mot mukābala." Mit den sechs Normalformen der Gleichungen, auf die Carra de Vaux anspielt, und mit dem angeblichen Verschwinden des Wortes *muqābala* haben die Gegens"ätze *gabr* und *haḍ* schlechterdings nichts zu tun.

Auch der Versuch Rodets, das Wort *gabara* im Sinne von "bereichern" zu deuten (J. As., 7. S., Bd. XI, 1878, S. 38), dö'urfte kaum Anklang finden, zumal die aus Freytag angezogene Bedeutung *post paupertatem ditavit* (*amicum*) nach den arabischen Lexikographen ebenfalls auf das Wiederherstellen eines gebrochenen Arms zurö'uckgeht.

Nö'aher l"age die Bedeutung "auffö'ullen, nachfö'ullen, ersetzen", die bei Dozy belegt ist; doch ist auch das eine abgeleitete Bedeutung, und es liegt kein Grund vor, die von sprach- und sachkundigen Arabern selbst gegebene Deutung des Wortes *al-gabr* durch eine andere zu ersetzen.

Fö'ür die weitere Geschichte des Wortes "Algebra" ist es von Interesse, daß es schon bei den Iḥān a-safā', also um das Jahr 1000, ohne den Begriff *muqābala* zur Bildung des Wortes "Algebraiker" gebraucht wird; denn die Abhandlung "über die Zahl, die erste der 56 philosophischen Abhandlungen der Brö'üder, enth"alt (ed. Bombay I, S. 37) einen Abschnitt فيما يحتاج اليه الجبريون والهندسة من الاسماء ومعانيها "über die Multiplikation und die Wurzel(n) und die Kuben und was die Algebraiker und die Geometer anwenden von den Ausdrö'ucken und ihren Bedeutungen.

Sollte der Abschnitt aber eine jö'üngere Interpolation sein — ich kann meine Bedenken allerdings nicht mit zwingenden Grö'unden verfechten —, so ist der rein technische Gebrauch des Wortes

With those authors—it is a matter of the arithmetic of Takī ad-Dīn al-Ḥanbalī, of whom according to H. Suter it is only certain that he lived before 1410, and of a treatise of Ibn al-Ḥā'im, who lived 1355–1412—there is no question at all of a procedure for the solution of equations. The two operations of addition and multiplication are set in contrast to the two operations of subtraction and division, and the former, since their execution leads to a larger number, are designated as *gabr*, here thus presumably "filling up" or "augmentation," the latter, since a smaller number results, as *haḍ*, i.e. "diminution" or "reduction."

It is *gabr* when one says: by how much must you increase 10 to obtain 17, or by what must you multiply 10 to obtain 17; it is *haḍ* when one says: by how much must you divide 10, or by how much must you decrease 10 to obtain 7.

When Carra de Vaux renders these summaries, unknown in older literature, with modern-conceived sentences: "It is therefore clear that al-djebir is the operation expressed by the two equations  $a + x = b$ ,  $a \cdot x = b$  and that al-haḍ is the operation expressed by these two:  $a - x = b$ ,  $a : x = b$ ," and draws from this the conclusion, "as these four operations are the simplest possible in algebra, one must believe that such is indeed the primitive sense of the two words," he not only confuses entirely different things but stands the historical development on its head, in that he further claims: "The word djebir, of which we have just indicated the primitive sense, then served to designate an indeterminate number of diverse questions, among which six are fundamental, according to Al-Khārizmī...; the word haḍ, having become improper because of the complication of the problems, disappeared before the word mukābala." The six normal forms of equations, to which Carra de Vaux alludes, and the alleged disappearance of the word *muqābala* have absolutely nothing to do with the opposites *gabr* and *haḍ*.

Rodet's attempt to interpret the word *gabara* in the sense of "to enrich" (J. As., 7th ser., vol. XI, 1878, p. 38) will hardly find acceptance, especially since the meaning *post paupertatem ditavit* (*amicum*) drawn from Freytag likewise goes back, according to the Arabic lexicographers, to the restoration of a broken arm.

Closer would lie the meaning "to fill up, refill, replace," which is attested in Dozy; but this too is a derived meaning, and there is no reason to replace the explanation of the word *al-gabr* given by Arabs themselves, knowledgeable in language and subject matter, with another.

For the further history of the word "Algebra" it is of interest that it was already used by the Ikhwān al-afā', around the year 1000, without the concept *muqābala* to form the word "algebraist"; for the treatise on number, the first of the 56 philosophical treatises of the Brethren, contains (ed. Bombay I, p. 37) a section فيما يحتاج اليه الجبريون والهندسة من الاسماء ومعانيها "on the multiplication and the root(s) and the cubes and what the algebraists and the geometers apply of expressions and their meanings."

Should the section, however, be a younger interpolation—I cannot indeed defend my doubts with compelling reasons—then the purely technical use of the word is in any case secured no later

jedenfalls spätestens bei 'Omar al-Hayyām (gest. 1123) sicher gestellt. Denn nachdem er in der Einleitung die *ṣ'ā al-gabr wa-l-muqābala*, die "Kunst" oder Technik der Ergänzung und Ausgleichung definiert hat (ed. Woepcke, S. 4), spricht er weiterhin schlechtweg von den "Lösungen der Algebra", *ḥibrāgāt al-gabr*, und von den Büchern und den Gepflogenheiten der Algebraiker (S. 4, 5, 7).

Endlich ist einer Erweiterung des Begriffs *al-gabr* zu gedenken, die sich bereits im Rechenbuch des al-Karāgī (gest. um 1030) vorfindet. Sie besteht darin, daß al-Karāgī auch die Wegschaffung von Brüchen, also das, was Muhammad b. Mūsā als *ikmāl* bezeichnet, unter den Begriff *gabr* subsumiert (Cod. Gotha 1474, Bl. 54a; Hochheim, *Al-Kāfī fī l-Hisāb*, III. Teil, S. 10):

Wenn sich auf einer der beiden Seiten der Gleichung eine "Ausnahme" *istiṭnā'*, d. h. eine subtraktive Größe befindet, muß man auf dieser Seite die gleiche Größe hinzufügen, damit das negative Glied verschwindet, und auf der andern Seite muß man ebensoviel hinzufügen, damit die Gleichheit erhalten bleibt — dies ist Ergänzung *al-gabr*. Aber sie findet auch noch in anderer Weise statt. Wenn nämlich eine der beiden Seiten durch eine Zahl (*mikdār*, Betrag) dividiert ist, so wird der Ausdruck der Division unter Wahrung der Gleichheit dadurch zum Verschwinden gebracht, daß man alles, was man hat, mit dieser Zahl multipliziert. Dies alles geschieht, um das Unbekannte dem Bereich des Bekannten zu nähern und seinem eigenen Bereich fernzurücken. Alle diese Operationen gehören zur Ergänzung, bis die Aufgabe in den Bereich der Ausgleichung *muqābala* gelangt, d. h. zur Beseitigung der (der Unbekannten) beigesellten Größen.

Man erkennt aus dem Wortlaut deutlich die ursprüngliche Anwendung und die sekundäre Erweiterung des Begriffs. Wenn daher Ibn Haldūn (1332–1406) in seiner *Muqaddima* bei der Erklärung von *al-gabr wa-l-muqābala* (ed. Beirut 1886, S. 422) den Satz schreibt *ويعملون فيها ما فيها من الكسور حتى تتم* "und sie richten ein, was darin von Gebrochenem ist, so daß es ganz wird", so hat er die Hauptsache übersehen und sich vom nächstliegenden Wortsinn, also dem "Einrichten" eines gebrochenen Glieds, zu einer mangelhaften Definition verleiten lassen.

than in 'Umar al-Khayyām (d. 1123). For after he has defined in the introduction the *ṣ'ā al-gabr wa-l-muqābala*, the "art" or technique of completion and balancing (ed. Woepcke, p. 4), he subsequently speaks simply of the "solutions of algebra," *ḥibrāgāt al-gabr*, and of the books and practices of the algebraists (pp. 4, 5, 7).

Finally, an extension of the concept *al-gabr* is to be noted, which is already found in the arithmetic book of al-Karajī (d. ca. 1030). It consists in the fact that al-Karajī also subsumes under the concept *gabr* the elimination of fractions, that is, what Muhammad b. Mūsā designates as *ikmāl* (Cod. Gotha 1474, fol. 54a; Hochheim, *Al-Kāfī fī l-hisāb*, Part III, p. 10):

If on one of the two sides of the equation there is an "exception" *istiṭnā'*, that is, a subtractive quantity, one must add the same quantity on this side so that the negative term disappears, and on the other side one must add just as much so that equality is preserved—this is completion *al-gabr*. But it also takes place in another way. Namely, if one of the two sides is divided by a number (*mikdār*, amount), the expression of the division is made to disappear while preserving equality by multiplying everything one has by this number. All this is done to bring the unknown closer to the domain of the known and to push it away from its own domain. All these operations belong to completion, until the problem reaches the domain of balancing *muqābala*, that is, the elimination of the quantities accompanying (the unknown).

From the wording one clearly recognizes the original application and the secondary extension of the concept. Therefore when Ibn Khaldūn (1332–1406) in his *Muqaddima* in the explanation of *al-gabr wa-l-muqābala* (ed. Beirut 1886, p. 422) writes the sentence *ويعملون فيها ما فيها من الكسور حتى تتم* "and they arrange what is in it of fractions so that it becomes whole," he has overlooked the main point and allowed himself to be led by the nearest verbal sense, namely the "setting" of a broken term, to an inadequate definition.

## 2 II. Das *Liber augmenti et diminutionis* / The *Liber augmenti et diminutionis*

Es wäre nicht notwendig gewesen, so lange bei dem Titel der Algebra zu verweilen, wenn nicht von der Deutung der Termini das Verständnis der ganzen Entwicklung der Mathematik abhinge, und wenn nicht schon in den Titeln der mathematischen Schriften ein Stück Geschichte enthalten wäre.

Ich denke hierbei natürlich auch an das von Libri 1838 im ersten Band seiner *Histoire des Sciences Mathématiques en Italie* S. 304 ff. veröffentlichte *Liber augmenti et diminutionis* und die Vermutungen, die über den Inhalt einer Schrift des Muhammad b. Mūsā geäußert worden sind, von der uns nur der Titel *kūṭāb*

It would not have been necessary to dwell so long on the title of the Algebra if the understanding of the entire development of mathematics did not depend on the interpretation of the terms, and if a piece of history were not already contained in the titles of mathematical writings.

I naturally think here also of the *Liber augmenti et diminutionis* published by Libri in 1838 in the first volume of his *Histoire des Sciences Mathématiques en Italie*, pp. 304 ff., and of the conjectures that have been expressed about the content of a treatise of Muḥammad b. Mūsā of which only the title *kūṭāb al-jam' wa-l-*



*al-jam' wa-l-tafrīq* "überliefert ist (Cantor I<sup>1</sup> S. 627, I<sup>2</sup> S. 687, I<sup>3</sup> S. 730).

Hier liegen Schwierigkeiten, um deren Klärung und Beseitigung man seit Jahrzehnten bemüht ist. Woepcke scheint der erste gewesen zu sein, der den arabischen Titel *al-jam' wa-l-tafrīq* dem "über *augmenti et diminutionis* gleichsetzte (J. As., 6. S., Bd. I, 1863, S. 514): "J'ajouterai que je trouve dans le *Fihrist* la mention de deux traités "de l'augmentation et de la diminution" c'est à dire de la règle des deux fausses positions, par Send Ben Ali et par Sinān Ibn Alfath, précisément les mêmes qui avaient écrit aussi ... des traités de calcul indien ..."

Es ist jedoch höchst zweifelhaft, ob man diese beiden Titel miteinander identifizieren darf, denn "Vermehrung" heißt arabisch nicht *jam'*, und "Verminderung" heißt nicht *tafrīq*. *Liber augmenti et diminutionis* wurde arabisch *kitāb al-ziyāda wa-l-nuqsān* heißen, während ein *kitāb al-jam' wa-l-tafrīq* etwa mit *liber de aggregatione et dispersione* zu "übersetzen" wäre.

Das Verfahren der Vermehrung und Verminderung ist bekannter unter dem Namen der Regel der "zwei Fehler" (*reg. elchatayn* = الخطأين) oder des doppelten falschen Ansatzes, auch als "Regel der zwei Wagschalen". Cantor hat in seinen *Vorlesungen* (I<sup>3</sup>, S. 731) das erste Beispiel aus dem *Liber augmenti et diminutionis*, worin der Sinn der Bezeichnung erläutert wird, in deutscher "Übersetzung mitgeteilt; den Wortlaut des arabischen Textes kann man leichter nach der alten lateinischen "Übersetzung verfolgen.

In einem späteren Kapitel dieser "Übersetzung (Libri a. a. O., S. 324) wird die Methode ausdrücklich *Capitulum numerationis ejus secundum augmentum et diminutionem* genannt, in der "Überschrift und am Anfang des Buches heißt sie auch *numerationis divinationis*, also "Rechenverfahren des Erratens", was eine freie Wiedergabe der indischen Bezeichnung "Verfahren mit der angenommenen Zahl" zu sein scheint.

Die stets wiederkehrenden Wendungen *errasti per ... addita* oder *cum ... additis*, *errasti per ... diminuta* oder *cum ... diminutis* entsprechen arabischem زد bzw. نقصت und können nicht durch *ziyādat* oder *nuqsān* wiedergegeben werden.

Vollbestätigt werden diese Rückübersetzungen und kritischen Bedenken durch Behā ad-dīn *Essenz der Rechenkunst* (ed. Nesselmann, Text S. 26), in der sich bei der Erläuterung der Methode der zwei Fehler der Ausdruck findet: اذا اخطأ بزيادة او نقصان *wa-in aṭā'a bi-ziyādin au nuqsānin*, was Nesselmann etwas frei mit "wenn es aber abweicht nach einer oder der anderen Seite (um plus oder minus)" übersetzt; lateinisch wäre es etwas wie *et si erravit per augmentum aut diminutionem* gewesen.

Auch die in letzter Zeit von Suter in "Übersetzung mitgeteilten arabischen Abhandlungen über die "Regel der zwei Fehler" (Bibl. Math., 3. Folge, Bd. 8, 1907/8, S. 24 und Bd. 9, 1908/9, S. 111 ff.) geben keinen Anhalt dafür, daß das Verfahren mit den Worten *al-jam' wa-l-tafrīq* bezeichnet worden wäre.

Einen letzten Beweis für die Verschiedenheit der beiden Titel kann man daraus ableiten, daß der Mathematiker Abū Kāmil Šūcā' b. Aslam sowohl ein Buch "über die zwei Fehler, als eines "über *jam'* und *tafrīq* verfaßte.

Allerdings hat Suter (*Zeitschr. f. Math. u. Physik*, Suppl. z. 37. Jahrg., 1892, S. 70) mit Hilfe einer Konjekture — Einschlebung

*tafrīq* has been transmitted to us (Cantor I<sup>1</sup> p. 627, I<sup>2</sup> p. 687, I<sup>3</sup> p. 730).

Here lie difficulties whose clarification and removal have been attempted for decades. Woepcke seems to have been the first to equate the Arabic title *al-jam' wa-l-tafrīq* with that of *augmenti et diminutionis* (J. As., 6th ser., vol. I, 1863, p. 514): "I will add that I find in the *Fihrist* the mention of two treatises 'on augmentation and diminution,' that is, the rule of double false position, by Sanad b. 'Alī and by Sinān b. al-Fāṭh, precisely the same ones who had also written ... treatises on Indian calculation ..."

It is, however, highly doubtful whether one may identify these two titles with each other, for "augmentation" is not *jam'* in Arabic, and "diminution" is not *tafrīq*. *Liber augmenti et diminutionis* would be called in Arabic *kitāb al-ziyāda wa-l-nuqsān*, while a *kitāb al-jam' wa-l-tafrīq* would be translated approximately as *liber de aggregatione et dispersione*.

The method of augmentation and diminution is better known under the name of the rule of "two errors" (*reg. elchatayn* = الخطأين) or double false position, also as the "rule of the two scales." Cantor in his *Vorlesungen* (I<sup>3</sup>, p. 731) communicated the first example from the *Liber augmenti et diminutionis*, in which the sense of the designation is explained, in German translation; the wording of the Arabic text can more easily be followed through the old Latin translation.

In a later chapter of this translation (Libri, op. cit., p. 324) the method is explicitly called *Capitulum numerationis ejus secundum augmentum et diminutionem*; in the heading and at the beginning of the book it is also called *numerationis divinationis*, that is, "calculation procedure of guessing," which seems to be a free rendering of the Indian designation "method with the assumed number."

The constantly recurring phrases *errasti per ... addita* or *cum ... additis*, *errasti per ... diminuta* or *cum ... diminutis* correspond to the Arabic زد and نقصت respectively and cannot be rendered by *ziyāda* or *nuqsān*.

These back-translations and critical doubts are fully confirmed by Behā ad-Dīn's *Essence of Arithmetic* (ed. Nesselmann, Text p. 26), in which the expression is found in the explanation of the method of two errors: اذا اخطأ بزيادة او نقصان *wa-in aṭā'a bi-ziyādin au nuqsānin*, which Nesselmann translates somewhat freely as "if it deviates to one or the other side (by plus or minus)"; in Latin it would have been something like *et si erravit per augmentum aut diminutionem*.

Also the Arabic treatises on the "rule of two errors" recently communicated by Suter in translation (Bibl. Math., 3rd ser., vol. 8, 1907/8, p. 24 and vol. 9, 1908/9, pp. 111 ff.) give no indication that the method was designated by the words *al-jam' wa-l-tafrīq*.

A final proof of the difference between the two titles can be derived from the fact that the mathematician Abū Kāmil Shujā' b. Aslam composed both a book *on the two errors* and one *on jam'* and *tafrīq*.

To be sure, Suter (*Zeitschrift für Mathematik und Physik*, Supplement to vol. 37, 1892, p. 70) wished to weld both together into

von *yusammā adān*, "es wird auch genannt" — beide zu einem einzigen zusammenschweißen wollen, aber das geschieht doch nur Woepckes "Übersetzung zuliebe und hat darum keine Beweiskraft.

Solange nicht aus dem Inhalt noch zu entdeckender Werke oder mit anderen zwingenden Gründen nachgewiesen ist, daß die beiden Titel gleichbedeutend sind, wird man das Gegenteil als richtig oder wahrscheinlich annehmen müssen.

Was kann aber nun in einem Werke "über *jam'* und *tafrīq* enthalten gewesen sein? Daß es sich um irgendwelche Kapitel der Rechenkunst oder der Algebra handelt, ergibt sich zweifellos daraus, daß der Titel stets in Verbindung mit Schriften arithmetischen Inhalts erwähnt wird.

Nun ist die nächste Bedeutung von *jam'* die der "Vereinigung" oder "Summation". In diesem Sinne wird *jama'a* nicht weniger als viermal in dem oben erwähnten ersten Beispiel angewandt: *Aggrega ergo ..., postquam ergo aggregasti ..., quod aggregatum est ..., deinde aggrega ...* Es wird also sprachlich ein Unterschied gemacht zwischen dem "Hinzufügen" und "Vermehren" und der "Vereinigung" gegebener Zahlen zu einer Summe.

Nun ist in der Algebra des Muhammad b. Mūsā der Addition und Subtraktion verschiedener Arten von Größen, die in den Gleichungen vorkommen, ein besonderes Kapitel gewidmet, das im Original mit *bāb al-jam' wa-l-nuqsān*, in der alten lateinischen "Übersetzung mit *Capitulum aggregationis et diminutionis* überschrieben ist.

Eneström vermutet daher (Bibl. Math., 3. Folge, Bd. 8, 1907/8, S. 184) in der Schrift oder den Schriften, die den Titel *al-jam' wa-l-tafrīq* führen, Erweiterungen des *Capitulum aggregationis et diminutionis*. Die Möglichkeit ist nicht von der Hand zu weisen, daß es sich um solche Additionen und Subtraktionen handelt. Unsicher bliebe die Vermutung aber auch dann, wenn der Nachweis geführt würde, daß *tafrīq* und *nuqsān* schon in "alterer Zeit gleichbedeutend für einander gebraucht werden. Denn wir haben keinerlei Anhaltspunkte dafür, daß dieses Kapitel von irgendwem, geschweige schon von Muhammad b. Mūsā in erweiterter Form außerhalb des Rahmens der Algebra behandelt worden wäre.

Die alten Lexika der Araber wissen nichts von *tafrīq* im Sinne von "Wegnahme" oder Subtraktion. Der zweite Stamm des Verbs *faraka*, von dem das Verbalnomen *tafrīq* gebildet ist, hat stets nur die Bedeutung "in Stücke zerlegen, trennen, zerstreuen, in Unordnung bringen", niemals die von "wegnehmen" oder "vermindern".

Die Verbindung von *jam'* und *tafrīq* wird also eher den Gegensatz von Sammlung — Zerstreung oder von Vereinigung — Trennung ausdrücken. Einen solchen Gegensatz finde ich auf dem Gebiet der Algebra nur in dem Kapitel von der Klammerrechnung.

Auch diese ist von Muhammad b. Mūsā in seinen Elementen behandelt, wenn er zeigt, wie Summen und Differenzen zu multiplizieren sind, und es wäre möglich, daß man daraus ein besonderes Kapitel gemacht hat. Die Ausdrücke *dissolutio* und *compositio* allerdings, die in der lateinischen "Übersetzung des Euklidkommentars von Annairizī auftreten (M. Curtze, *Anaritii Comm. ad Euch.* S. 89; Cantor I<sup>3</sup> S. 387) und als Klammernauflösung

a single one by means of a conjecture—the insertion of *yusammā adān*, "it is also called"—but this is done only for Woepcke's translation and therefore has no probative force.

As long as it has not been demonstrated from the content of works yet to be discovered, or by other compelling reasons, that the two titles are synonymous, one must assume the opposite as correct or probable.

But what could have been contained in a work on *jam'* and *tafrīq*? That it concerns some chapters of arithmetic or algebra follows doubtless from the fact that the title is always mentioned in connection with writings of arithmetic content.

Now the nearest meaning of *jam'* is "union" or "summation." In this sense *jama'a* is applied no fewer than four times in the aforementioned first example: *Aggrega ergo ..., postquam ergo aggregasti ..., quod aggregatum est ..., deinde aggrega ...* Thus a linguistic distinction is made between "adding" and "increasing" and the "union" of given numbers into a sum.

Now in the Algebra of Muhammad b. Mūsā a special chapter is devoted to the addition and subtraction of various kinds of quantities that occur in equations, which in the original is headed *bāb al-jam' wa-l-nuqsān*, and in the old Latin translation with *Capitulum aggregationis et diminutionis*.

Eneström therefore conjectures (Bibl. Math., 3rd ser., vol. 8, 1907/8, p. 184) that in the writing or writings bearing the title *al-jam' wa-l-tafrīq* there are extensions of the *Capitulum aggregationis et diminutionis*. The possibility cannot be dismissed that it concerns such additions and subtractions.

The conjecture would remain uncertain, however, even if proof were furnished that *tafrīq* and *nuqsān* were already used synonymously in earlier times. For we have no indication whatsoever that this chapter was treated by anyone, let alone by Muhammad b. Mūsā, in extended form outside the framework of the Algebra.

The old lexica of the Arabs know nothing of *tafrīq* in the sense of "removal" or subtraction. The second stem of the verb *faraka*, from which the verbal noun *tafrīq* is formed, has always only the meaning "to break into pieces, separate, disperse, bring into disorder," never that of "to remove" or "to diminish."

The combination of *jam'* and *tafrīq* will thus more likely express the contrast of collection—dispersion or of union—separation. Such a contrast I find in the domain of algebra only in the chapter on bracket calculation [parenthetical operations].

This too is treated by Muhammad b. Mūsā in his Elements, when he shows how sums and differences are to be multiplied, and it would be possible that a special chapter was made from it. The expressions *dissolutio* and *compositio*, however, which appear in the Latin translation of Annairizī's commentary on Euclid (M. Curtze, *Anaritii Comm. ad Euch.*, p. 89; Cantor I<sup>3</sup> p. 387) and could be interpreted as bracket resolution and separation in our

und Absonderung in unserem Sinne gedeutet werden könnten, lauten im arabischen Original *taḥlīl al-taḥlīl* und *taḥlīl al-tarkīb*, d. i. meth-odos analutikī und meth-odos suntithetikī.

Da jedoch in dem Zahlenbeispiel, das Annairizī zur Stelle gibt, Summation mit *mamūḡ* und Summe mit *mamūḡ* = *das Vereinigte* wiedergegeben wird, so könnte als Ausdruck der "Zerlegung" neben dem *qasama* des Textes, das sich auf die Teilung in zwei Teile bezieht, auch *tafrīq* (für eine Zerlegung in zahlreiche Teile) in Frage kommen. Rein sprachlich besteht für diesen Erklärungsversuch jedenfalls keine Schwierigkeit; sachlich spricht aber wie bei der Eneströmschen Hypothese der Umstand gegen ihn, daß es nicht möglich ist, eine selbständige Behandlung der Klammerrechnungen in der mathematischen Literatur der Araber nachzuweisen.

### 3 III. Die Regula Sermonis / The Regula Sermonis

Ich schließe an diese Erörterungen noch eine Bemerkung über das in dem *Liber augmenti et diminutionis* gelehrt Umkehrungsverfahren, das durch Cantor (I<sup>3</sup>, S. 732) "unter dem sonderbaren Namen der Wortrechnung, *regula sermonis*" in die Geschichte der Mathematik eingeführt worden ist.

Die Textstelle, auf die es ankommt, lautet (Libri S. 312):

"*Quaedam vero harum quaestionum investigantur secundum regulam quae vocatur infusa. Et ipsa est regula Job filii Salomonis divisoris... Ex eis vero est eius sermo qui dixit ... Incipe igitur cum quaestione ab eius postremitate, et dic ...*" Und weiterhin S. 313: "*Et modus regule sermonis eius est ...*"

Zunächst ist klar, daß man das sinnlose *infusa* durch *inversa* zu ersetzen hat. Von höchstem Interesse ist weiter, daß das Verfahren auch die Regel des Erbteilers oder Testamentsvollstreckers Ayyūb b. Sulaimān genannt wird — denn so ist das Wort *divisor* nach einer Randnote der Handschrift "*Dicitur divisor qui res a defuncto relicta(s) partitur, [et] hoc apud Arabes*" zweifellos zu übersetzen.

Wir erblicken darin eine Bestätigung der Vermutung, daß die Rechenkunst und die Kunst der Gleichungsauflösung nicht nur bei den Astronomen, sondern ganz besonders auch bei den Notaren und Rechtsgelehrten eine Pflegestätte gefunden hat.

Wir sind uns von Haggī Hāa (Bd. IV, S. 398) mehr als die drei Worte *فرائض ايوب البصري farā'id Ayyūb al-Baṣrī* überliefert, so könnte die Annahme, daß dieser Ayyūb der *Job filius Salomonis* ist, auf ihre Berechtigung geprüft werden.

Suter sucht (Bibl. Math., 3. Folge, Bd. 3, 1903, S. 350) den *Job* wie den Verfasser des *Liber augmenti et diminutionis* unter den spanischen Gelehrten des 10. und 11. Jahrhunderts. Nach ihm können zwei Ayyūb b. Sulaimān in Betracht kommen, die sich mit Erbteilungen befaßten; die größere Wahrscheinlichkeit spreche für Abū Šāliḥ Ayyūb b. Sulaimān aus Cordova, der 914 starb.

Eine Entscheidung zwischen beiden Vermutungen wird erst möglich sein, wenn der vollständige Name des Basreners oder der Autor *Abraham* selbst feststeht, der das Buch *compilavit et secundum librum qui Indorum dictus est composuit*.

Auch wenn sich, wie Suter später in einem Vortrag auf dem III.

sense, are in the Arabic original *taḥlīl al-taḥlīl* and *taḥlīl al-tarkīb*, i.e. meth-odos analutikī and meth-odos suntithetikī.

Since, however, in the numerical example that Annairizī gives at this place, summation is rendered by *mamū'* and sum by *mamū'* = "the united," so as an expression of "decomposition" alongside *qasama* of the text, which refers to division into two parts, *tafrīq* (for a decomposition into numerous parts) could also come into question. Linguistically, there is in any case no difficulty for this explanatory attempt; but substantively, as with Eneström's hypothesis, the circumstance speaks against it that it is not possible to demonstrate an independent treatment of bracket operations in the mathematical literature of the Arabs.

I append to these discussions a remark on the inversion procedure taught in the *Liber augmenti et diminutionis*, which was introduced into the history of mathematics by Cantor (I<sup>3</sup>, p. 732) "under the strange name of word-calculation, *regula sermonis*."

The passage that matters reads (Libri, p. 312): "*Quaedam vero harum quaestionum investigantur secundum regulam quae vocatur infusa. Et ipsa est regula Job filii Salomonis divisoris... Ex eis vero est eius sermo qui dixit ... Incipe igitur cum quaestione ab eius postremitate, et dic ...*" And further on p. 313: "*Et modus regule sermonis eius est ...*"

First of all, it is clear that the meaningless *infusa* must be replaced by *inversa*. Of the highest interest, furthermore, is that the procedure is also called the rule of the inheritance-divider or executor Ayyūb b. Sulaymān—for thus the word *divisor* is doubtless to be translated, according to a marginal note of the manuscript: "*Dicitur divisor qui res a defuncto relicta(s) partitur, [et] hoc apud Arabes*."

We see in this a confirmation of the conjecture that arithmetic and the art of equation-solving found a home not only among astronomers, but especially also among notaries and legal scholars.

Had more than the three words *فرائض ايوب البصري farā'id Ayyūb al-Baṣrī* been transmitted to us from Ḥajjī Ḥalīfa (vol. IV, p. 398), the assumption that this Ayyūb is the *Job filius Salomonis* could be tested for its validity.

Suter seeks (Bibl. Math., 3rd ser., vol. 3, 1903, p. 350) the *Job* like the author of the *Liber augmenti et diminutionis* among the Spanish scholars of the 10th and 11th centuries. According to him, two Ayyūb b. Sulaymān can come into consideration, who occupied themselves with inheritance divisions; the greater probability speaks for Abū Šāliḥ Ayyūb b. Sulaymān of Córdoba, who died in 914.

A decision between the two conjectures will only be possible when the complete name of the Basran [scholar] or the author *Abraham* himself is established, who *compilavit* [compiled] the book *et secundum librum qui Indorum dictus est composuit*.

Even if, as Suter later attempted to demonstrate in a lecture at

Internationalen Mathematikerkongreß zu Heidelberg 1904 (vgl. *Verhandlungen* S. 558) nachzuweisen versuchte, der "Ägypter Ab"u Kāmil Šūcā' b. Aslam hinter dem Namen *Ibrahim* verbergen sollte — eine Vermutung, die mit guten Gründen gestützt wird und noch dadurch an Interesse gewinnt, daß es gerade dieser Autor ist, der im *Fihrist* als Verfasser eines *kitāb al-jam' wa-l-tafrīq* wie eines *kitāb al-ḥatā'ain* bezeichnet wird (Flügel, *Kitāb al-Fihrist*, Bd. I, S. 281) — wäre der Baṣrenser als erster arabischer Schriftsteller, der das indische Umkehrungsverfahren in die Technik der Gleichungen eingeführt hatte, in Erwägung zu ziehen.

Wenn aber nun Cantor a. a. O. behauptet, daß dieses Verfahren bei den Arabern den sonderbaren Namen der Wortrechnung geführt habe, so zeigt ein Blick in den lateinischen Text und die Rückübersetzung ins Arabische, daß diese Behauptung auf einem Mißverständnis beruht.

Es heißt im Text nicht *regula sermonis*, sondern *regula sermonis eius*, und das ist weiter nichts als die wörtliche Wiedergabe des arabischen *qiyāsu qawlihī*, eines Ausdruckes, der nichts anderes besagt als "die Regel, die er angibt", wie es schon vorher heißt *eius sermo qui dixit* d. i. *qawlu qā'ilin*.

Die "Wortrechnung" wird also konsequent aus der Liste der mathematischen Termini verschwinden müssen.

the III. International Congress of Mathematicians in Heidelberg 1904 (cf. *Verhandlungen* p. 558), the Egyptian Abū Kāmil Shuja' b. Aslam should be concealed behind the name *Ibrahim*—a conjecture supported by good reasons and gaining in interest precisely because it is this author who in the *Fihrist* is designated as the author of both a *kitāb al-jam' wa-l-tafrīq* and a *kitāb al-kha.tā'ayn* (Flügel, *Kitāb al-Fihrist*, vol. I, p. 281)—the Basran would have to be considered as the first Arabic writer who introduced the Indian inversion procedure into the technique of equations.

But if Cantor, op. cit., claims that this procedure bore among the Arabs the strange name of word-calculation, a glance at the Latin text and the back-translation into Arabic shows that this claim rests on a misunderstanding.

The text does not say *regula sermonis* but *regula sermonis eius*, and this is nothing other than the literal rendering of the Arabic *qiyāsu qawlahī*, an expression that says nothing other than "the rule that he states," as is said before: *eius sermo qui dixit*, i.e. *qawlu qā'ilin* [the statement of the speaker].

The "word-calculation" will therefore henceforth have to disappear from the list of mathematical terms.

## 4 IV. Inhaltsübersicht / Overview of Contents

Wir kehren zur Algebra des Muhammad b. Mūsā zurück und verschaffen uns durch Zusammenstellung der Kapitelüberschriften eine erste Übersicht des Inhalts. Es ist bekannt, daß der Text des Buches auf der einzigen Oxforder Handschrift CMXCVIII Hunt. 214 ruht, die im Jahre 743 d. H., also um 1342 vollendet wurde (Rosen, S. XIII).

Ihr Zustand — *it is written in a plain and legible hand, but unfortunately destitute of most of the diacritical points* — ließe einen Vergleich mit den lateinischen Handschriften, die als Übersetzungen oder Bearbeitungen des arabischen Werkes gelten, sehr wünschbar erscheinen, zumal auch hier noch manches genauerer Feststellung bedarf.

Daß die von Libri im ersten Band seiner *Geschichte* S. 253 ff. veröffentlichte Übersetzung des Gerhard von Cremona nur die ersten 50 Seiten des Originals, also nur die Algebra im engeren Sinne ohne die Geometrie und Erbteilungsrechnung umfaßt, war z. B. bis vor kurzem nirgends in geschichtlichen Werken vermerkt. Libri selbst weist allerdings in einer ausführlichen Fußnote zu S. 121 und in der Vorbemerkung zur Note XII, S. 253 darauf hin, daß die lateinische Übersetzung der drei Pariser Handschriften unvollständig ist.

Aber bei Cantor (I<sup>3</sup> S. 719, I<sup>2</sup> S. 675) heißt es einfach "eine alte lateinische Übersetzung", und auch in der dieser Übersetzung gewidmeten Arbeit von Björnbo (*Gerhard von Cremonas Übersetzung von Alkwarizmis Algebra usw.*, Bibl. Math., 3. Folge, Bd. 6, 1905, S. 239) ist eine Erinnerung an diesen weder selbstverständlichen noch gleichgültigen Umstand unterblieben.

Es ist klar, daß durch Unterdrückung der Einleitung, die ausdrücklich auch auf den geometrischen und erbrechtlichen Teil

We return to the Algebra of Muḥammad b. Mūsā and obtain a first overview of the content by assembling the chapter headings. It is known that the text of the book rests on the unique Oxford manuscript CMXCVIII Hunt. 214, completed in the year 743 A. H., that is, around 1342 (Rosen, p. XIII).

Its condition—*it is written in a plain and legible hand, but unfortunately destitute of most of the diacritical points*—would make a comparison with the Latin manuscripts, regarded as translations or adaptations of the Arabic work, very desirable, especially since here too much still requires more precise determination.

That the translation of Gerard of Cremona published by Libri in the first volume of his *Histoire*, pp. 253 ff., comprises only the first 50 pages of the original, thus only the Algebra in the narrower sense without the geometry and inheritance calculation, was, for example, noted nowhere in historical works until recently. Libri himself points out in an extensive footnote to p. 121 and in the prefatory remark to Note XII, p. 253, that the Latin translation of the three Paris manuscripts is incomplete.

But in Cantor (I<sup>3</sup> p. 719, I<sup>2</sup> p. 675) it simply says "an old Latin translation," and also in the work of Björnbo devoted to this translation (*Gerhard von Cremonas Übersetzung von Alkwarizmis Algebra usw.*, Bibl. Math., 3rd ser., vol. 6, 1905, p. 239) no mention of this circumstance—neither self-evident nor indifferent—has been made.

It is clear that by suppressing the introduction, which explicitly referred also to the geometrical and inheritance-law parts of the



des Werkes hinwies, der Anschein erweckt wird, als sei das "übersetzte Bruchstück" das ganze Buch. Auffallenderweise reicht auch die um 1145 vollendete "Übersetzung des Robert Cas-trensis nicht weiter, wie aus den von Steinschneider (Bibl. Math., 3. Folge, Bd. 1, 1900, S. 273) gemachten Angaben hervorgeht, da sie mit den Worten abschließt, die nur den algebraischen Teil betreffen.

Erst L. C. Karpinski hat in seiner Abhandlung *Robert of Chester's translation of the algebra of Al-Khowarizmi* (Bibl. Math., 3. Folge, Bd. 11, 1910/11, S. 128) auf das Fehlen der Einleitung und der geometrischen und erbrechtlichen Abschnitte in s"amtlichen alten "Übersetzungen wieder hingewiesen.

Folgen wir der Kapiteileinteilung des arabischen Textes, so ergibt sich folgender Aufbau des Ganzen:

**(Einleitung, S. 1, 2.)** Anrufung Gottes, Preis des Propheten, Schilderung der mühevollen Arbeit der Gelehrten, Bericht "über die Veranlassung zur Abfassung und "über den Zweck des Buches. [fehlt bei Libri]

**[Erster Hauptteil — ohne "Überschrift; S. 3 bis 50.]** Aufbau des Zahlensystems; Definition der in dem Rechenverfahren der Erg"anzung und Ausgleichung auftretenden Arten (dur"ub) von Zahlen; Aufz"ählung der drei einfachen F"älle von Gleichungen; sodann Beispiele: "Was anlangt die Verm"ogen, die gleich sind den Wurzeln ..." (Drei Beispiele.) "Was anlangt die Verm"ogen, die gleich sind der Zahl ..." (Drei Beispiele.) "Was anlangt die Wurzeln, die gleich sind einer Zahl ..." (Drei Beispiele.) Kurze Zwischenbemerkung "über die drei zusammengesetzten F"älle der Gleichungen; sodann Beispiele: "Was anlangt die Verm"ogen und die Wurzeln, die gleich sind der Zahl ..." (Drei Beispiele.) "Was anlangt die Verm"ogen und die Zahl, die gleich sind den Wurzeln ..." (Ein Beispiel.) "Was anlangt die Wurzeln und die Zahl, die gleich sind dem Verm"ogen ..." (Ein Beispiel.) Eine Schlußbemerkung leitet "über zu den geometrischen Beweisen der Aufl"ösungen: "Und ich habe f"ür jedes Kapitel von ihnen eine Zeichnung gezeichnet, durch die auf den Grund der Halbierung hingewiesen wird." Drei Beispiele; "Was den Grund anlangt, daß ein Verm"ogen und zehn Wurzeln gleich sind neunundzwanzig Dirhem ..." (Zwei verschiedene geometrische Beweise mit Zeichnungen.) "Was anlangt ein einziges Verm"ogen und zwanzig Dirhem, die gleich sind zehn seiner Wurzeln ..." (Eine Zeichnung.) "Was anlangt drei Wurzeln und vier von der Zahl, die gleich sind einem Verm"ogen ..." (Eine Zeichnung.)

Das Kapitel von der Multiplikation (باب الضرب *bāb al-darb*; S. 15 bis 19 oben). Das Kapitel von der Addition (Aggregation) und Subtraktion (باب الجمع والنقصان *bāb al-jam' wa-l-nuqsān*; S. 19, 20). Die Teilung (القسمة *al-qisma*; S. 20, 21).<sup>6</sup>

<sup>6</sup>Rosen hat durch seine "Überschriften *On multiplication, On addition and subtraction, On division* den charakteristischen Befund im arabischen Text verwischt, daß bei der Teilung das Wort "Kapitel" fehlt.

"Was anlangt den Grund f"ür die Wurzel von zweihundert ohne zehn, addiert zu (vereinigt mit) zwanzig ohne die Wurzel von zweihundert ..." (Zeichnung.) "Was anlangt den Grund f"ür die Wurzel von zweihundert ohne zehn, subtrahiert von zwanzig ohne die Wurzel von zweihundert ..." (Zeichnung.) "Was anlangt einhundert und ein Verm"ogen ohne zwanzig Wurzeln, dazu addiert (damit vereinigt) f"ünfzig und zehn Wurzeln ohne zwei

work, the appearance is created that the translated fragment is the whole book. Remarkably, the translation of Robert of Chester completed around 1145 also does not extend further, as emerges from the information given by Steinschneider (Bibl. Math., 3rd ser., vol. 1, 1900, p. 273), since it concludes with the words that concern only the algebraic part.

Only L. C. Karpinski in his treatise *Robert of Chester's translation of the algebra of Al-Khowarizmi* (Bibl. Math., 3rd ser., vol. 11, 1910/11, p. 128) again pointed out the absence of the introduction and of the geometrical and inheritance-law sections in all old translations.

If we follow the chapter division of the Arabic text, the following structure of the whole emerges:

**(Introduction, pp. 1, 2.)** Invocation of God, praise of the Prophet, description of the arduous work of scholars, report on the occasion of composition and on the purpose of the book. [missing in Libri]

**[First main part—without heading; pp. 3 to 50.]** Structure of the number system; definition of the kinds of numbers (dur"ub) occurring in the calculation procedure of completion and balancing; enumeration of the three simple cases of equations; then examples: "As for the possessions equal to the roots ..." (Three examples.) "As for the possessions equal to the number ..." (Three examples.) "As for the roots equal to a number ..." (Three examples.)

Brief intermediate remark on the three composite cases of equations; then examples: "As for the possessions and the roots equal to the number ..." (Three examples.) "As for the possessions and the number equal to the roots ..." (One example.) "As for the roots and the number equal to the possession ..." (One example.)

A concluding remark leads over to the geometrical proofs of the solutions: "And I have drawn for each chapter of them a figure, by which the ground of halving is indicated." Three examples; "As for the ground that a possession and ten roots equal twenty-nine dirhams ..." (Two different geometrical proofs with drawings.) "As for a single possession and twenty dirhams equal to ten of its roots ..." (One drawing.) "As for three roots and four of the number equal to a possession ..." (One drawing.)

The chapter on multiplication (باب الضرب *bāb al-darb*; pp. 15 to 19 above). The chapter on addition (aggregation) and subtraction (باب الجمع والنقصان *bāb al-jam' wa-l-nuqsān*; pp. 19, 20). Division (القسمة *al-qisma*; pp. 20, 21).<sup>6</sup>

<sup>6</sup>Rosen has obscured the characteristic finding in the Arabic text by his headings *On multiplication, On addition and subtraction, On division*, that in the case of division the word "chapter" is missing.

"As for the ground for the root of two hundred less ten, added to (united with) twenty less the root of two hundred ..." (Drawing.) "As for the ground for the root of two hundred less ten, subtracted from twenty less the root of two hundred ..." (Drawing.) "As for one hundred and a possession without twenty roots, thereto added (therewith united) fifty and ten roots without two possessions ..." (Without drawing.)

Vermögen ...” (Ohne Zeichnung.)

Das Kapitel der sechs Fragen (S. 25 bis 29). Die erste von den sechs ... Die zweite Frage ... Die dritte Frage ... Die vierte Frage ... Die fünfte Frage ... Die sechste Frage ...

Das Kapitel der vermischten Fragen. ("Überschrift in besonderer Zeile; 33 mehr oder minder ausführlich behandelte Beispiele; S. 30 bis 48 oben.)

Das Kapitel der kaufmännischen Geschäftsfälle (باب المعاملات *bāb al-mu‘āmalāt*; drei Beispiele. S. 48 bis 50).

[Zweiter Hauptteil — ohne besondere Hervorhebung:] Kapitel der Messung (S. 50 bis 64). Das Kapitel ist eine lose Aneinanderreihung von geometrischen Definitionen, Eigenschaften von Figuren und Formeln mit Hervorhebung durch Stichworte.

[Dritter Hauptteil — mit Hervorhebung der "Überschrift:] Das Buch der Testamente (S. 65–122). Ein Kapitel davon "über das Kapital und die Schuld (S. 65 bis 67; drei durchgerechnete Beispiele). Ein anderes Kapitel von den Testamenten (S. 67, 68; zwei erläuterte Beispiele). Ein anderes Kapitel von den Testamenten (S. 68 bis 71; zwei Beispiele).

In einer anderen Art von Testamenten ... (S. 71 bis 86. Dieser Anfang kehrt dreimal wieder, und zwar zweimal mit 4, einmal mit 8 Beispielen). Das Kapitel vom Testament mit dem Dirhem (S. 86 bis 92; vier Beispiele). Das Kapitel von der Vervollständigung (باب التكميلة *bāb al-takmīla*, S. 93 bis 98; sieben Beispiele).

Das Rechenverfahren bei Schicksalswechsel (حساب الدور *hisāb al-dawr*, computation of returns).<sup>7</sup> Ein Kapitel davon "über die Verheiratung in der (letzten) Krankheit (S. 98 bis 101; vier Beispiele).

<sup>7</sup>Die "Übersetzung von Rosen ist falsch. Das Wort *dawr* bezeichnet die wechselseitige Vertauschung (A gegen B, B gegen A), in unserm Falle Vertauschung von Erblasser und Erben, wenn der Erbe vorher stirbt.

Das Kapitel von der Freilassung (von Sklaven) in der Krankheit (S. 101 bis 116; fünfzehn Beispiele, darunter einige unter Berufung auf Abū Hanīfa, dessen Todesjahr 767 war). Das Kapitel von der Entschädigung bei Schicksalswechsel (باب الاكر بالدور *bāb al-akr bi-l-dawr*, return of the dowry, S. 116 bis 120; sechs Beispiele, darunter wieder einige unter Berufung auf Abū Hanīfa).<sup>8</sup> Das Kapitel von der Vorausbezahlung in der Krankheit (السلام في المرض *al-salām fī l-marad*, surrender in illness, S. 121, 122; zwei Beispiele).<sup>9</sup>

<sup>8</sup>Rosens "Übersetzung ist falsch. Es handelt sich um Entschädigung für die Verminderung des Werts einer Sklavin infolge Beiwohnung.

<sup>9</sup>Das Wort *salām* bedeutet gewöhnlich Vorausbezahlung vor Empfang der Ware; es scheint sich aber bei den beiden Beispielen um Vorauslieferung oder Verpfändung von Waren zu handeln. Schon diese Inhaltsübersicht kann zum Nachdenken "über die Gründe anregen, die den Verfasser veranlaßt haben, drei nach unserer Auffassung ganz heterogene Gegenstände unter dem Sammeltitle *kitāb al-gabr wa-l-muqābala* zu vereinigen. Es wäre auch leicht, zu zeigen, wie sich im 10. und 11. Jahrhundert eine Scheidung vollzieht, so daß die Algebra, das Geschäftssrechnen, die praktische Geometrie und die einzelnen Teile der Erbteilungspraxis in besonderen Schriften behandelt werden.

Da uns aber vorerst die Quellen und Vorbilder von Muḥammad b. Mūsā's Algebra und die Frage nach dem Grade seiner Selbstständigkeit beschäftigen sollen, wird eine kurze Darstellung

The chapter of the six questions (pp. 25 to 29). The first of the six ... The second question ... The third question ... The fourth question ... The fifth question ... The sixth question ...

The chapter of mixed questions. (Heading in a special line; 33 examples treated more or less in detail; pp. 30 to 48 above.)

The chapter of commercial transactions (باب المعاملات *bāb al-mu‘āmalāt*; three examples. Pp. 48 to 50).

[Second main part—without special emphasis:] Chapter of measurement (pp. 50 to 64). The chapter is a loose sequence of geometrical definitions, properties of figures, and formulas with emphasis by keywords.

[Third main part—with emphasis on the heading:] The Book of Testaments (pp. 65–122). A chapter thereof on capital and debt (pp. 65 to 67; three worked-through examples). Another chapter on testaments (pp. 67, 68; two explained examples). Another chapter on testaments (pp. 68 to 71; two examples).

In another kind of testaments ... (pp. 71 to 86. This beginning recurs three times, namely twice with 4, once with 8 examples.) The chapter on the testament with the dirham (pp. 86 to 92; four examples). The chapter on completion (باب التكميلة *bāb al-takmīla*, pp. 93 to 98; seven examples).

The calculation procedure in case of change of fortune (حساب الدور *hisāb al-dawr*, computation of returns).<sup>7</sup> A chapter thereof on marriage during (terminal) illness (pp. 98 to 101; four examples).

<sup>7</sup>Rosen's translation is wrong. The word *dawr* denotes mutual exchange (A for B, B for A), in our case exchange of testator and heirs, when the heir dies first. The chapter on manumission (of slaves) during illness (pp. 101 to 116; fifteen examples, among them some with reference to Abū Hanīfa, whose death year was 767). The chapter on compensation for change of fortune (باب الاكر بالدور *bāb al-akr bi-l-dawr*, return of the dowry, pp. 116 to 120; six examples, among them again some with reference to Abū Hanīfa).<sup>8</sup> The chapter on prepayment during illness (السلام في المرض *al-salām fī l-marad*, surrender in illness, pp. 121, 122; two examples).<sup>9</sup>

<sup>8</sup>Rosen's translation is wrong. It concerns compensation for the diminution in value of a female slave as a result of sexual intercourse.

<sup>9</sup>The word *salām* usually means prepayment before receipt of goods; but in the two examples it seems to concern advance delivery or pawning of goods. Even this overview of contents can prompt reflection on the reasons that induced the author to unite three subjects that by our view are entirely heterogeneous under the collective title *kitāb al-gabr wa-l-muqābala*. It would also be easy to show how in the 10th and 11th centuries a separation takes place, so that algebra, business calculation, practical geometry, and the individual parts of inheritance practice are treated in separate writings.

But since for now the sources and models of Muḥammad b. Mūsā's Algebra and the question of the degree of his independence are to occupy us, a brief presentation of the opinions hi-

der bisher "über diese Dinge ge"äußerten Urteile dem neuen Versuch, zu einer Entscheidung zu gelangen, vorausgeschickt werden müssen.

thereto expressed about these matters must be prefaced to the new attempt to reach a decision.

## 5 V. Zur Geschichte der arabischen Zahlbezeichnungen / On the History of Arabic Numerical Notation

Es kann nicht oft und nachdrücklich genug gesagt werden, daß die Araber, die die persischen und römischen Provinzen überfluteten, weder Rechtswissenschaft noch Staatsverwaltung fertig mitbrachten, sondern gezwungen waren, die Verwaltungsmethoden und Rechtsformen der eroberten Länder im wesentlichen unverändert zu übernehmen. Daß es ihnen mit erstaunlicher Schnelligkeit gelang, sich in die größeren Verhältnisse hineinzufinden und nicht nur die staatlichen Einrichtungen, sondern auch alle andern Früchte einer alten, ausgereiften Kultur sich zu eigen zu machen, ist bekannt.

Das wäre aber gewiß unmöglich gewesen, wenn der geistige Abstand zwischen dem Eroberervolk und den zeitgenössischen Persern, Griechen und Ägyptern so groß gewesen wäre, wie man bis in die neueste Zeit anzunehmen pflegte. Insbesondere darf man sich die städtischen Araber, die Träger der geistigen und politischen Bewegung, nicht als halbe Wilde vorstellen, die vor dem Auftreten Muhammads jedem Kultureinflusse von seiten der Nachbarvölker unzugänglich gewesen wären (Cantor I<sup>3</sup>, S. 693) oder gar in der Zeit, zu der sie für die Geschichte der Mathematik wichtig werden, kaum hätten schreiben können. Woepcke durfte sich 1863 mit seinem Satze "lorsque les Arabes sortirent du désert ... ils possédaient à peine l'écriture" noch auf Silvestre de Sacy's *Mémoire* vom Jahre 1831 berufen, heute aber können Silvestre de Sacy's *Grammaire arabe* von 1810 und die Abhandlungen von Gesenius in der Enzyklopädie von Ersch und Gruber aus dem Jahre 1820 nicht mehr als Quellen für unsere Kenntnis des arabischen Schriftwesens benutzt werden (Cantor I<sup>3</sup>, S. 707).

Authentische Urkunden und Schriftdenkmäler, die bis in die erste Zeit des Islam zurückreichen, enthüllen die überraschende Tatsache, "daß die arabische Schrift schon damals eine Form besaß, die von der gewöhnlichen, später Naskhi genannten Schrift kaum verschieden ist, desto mehr aber von der eckig steifen "kufischen" Schrift, die man für die Mutter der Kursivschrift gehalten hatte."

Es steht jetzt fest, daß sich die arabische Schrift im 4. und 5. nachchristlichen Jahrhundert aus der nabatäischen Kursive entwickelt hat; aus ihrem Alphabet entstand durch Hinzufügen der arabischen Laute ط, خ, ض, ع das alte arabische Alphabet, dessen Buchstabenfolge in den diesen Zeichen erteilten Zahlwerten ع = 500, غ = 600, خ = 700, ض = 800, ط = 900, 1000 erhalten geblieben ist.

Von ihrem Ursprungsort aus, der vielleicht in Petra, vielleicht in dem damals aufblühenden Ghassānidenreiche zu suchen ist, hat sich die Schrift mit dem Handelsverkehr nach Norden und Süden verbreitet. Sie war im 6. Jahrhundert wohl schon im ganzen nordarabischen Sprachgebiet bekannt und dürfte besonders in den beiden Städten, die den Ausgangspunkt der

It cannot be said often or emphatically enough that the Arabs who flooded the Persian and Roman provinces brought neither jurisprudence nor state administration ready-made with them, but were compelled to take over the administrative methods and legal forms of the conquered lands essentially unchanged. That they succeeded with astonishing rapidity in finding their way into the larger circumstances and making their own not only the state institutions but also all other fruits of an ancient, mature culture is well known.

But this would certainly have been impossible if the intellectual distance between the conquering people and the contemporary Persians, Greeks, and Egyptians had been as great as was customarily assumed until the most recent times. In particular, one must not imagine the urban Arabs, the bearers of the intellectual and political movement, as semi-savages who before the appearance of Muhammad were inaccessible to any cultural influence from neighboring peoples (Cantor I<sup>3</sup>, p. 693), or who in the period when they become important for the history of mathematics could scarcely write.

In 1863 Woepcke could still appeal to Silvestre de Sacy's *Mémoire* of 1831 with his sentence "when the Arabs left the desert ... they scarcely possessed writing," but today Silvestre de Sacy's *Grammaire arabe* of 1810 and the treatises of Gesenius in the encyclopedia of Ersch and Gruber from the year 1820 can no longer be used as sources for our knowledge of the Arabic writing system (Cantor I<sup>3</sup>, p. 707).

Authentic documents and written monuments that reach back to the earliest period of Islam reveal the surprising fact "that the Arabic script already possessed at that time a form scarcely different from the ordinary script later called naskhī, and much more different from the angular, rigid 'Kufic' script that was held to be the mother of cursive writing."

It is now established that the Arabic script developed in the 4th and 5th centuries CE from the Nabataean cursive; from its alphabet, by the addition of the Arabic sounds 'ayn, ghayn, khā, dād, zā the old Arabic alphabet arose, whose letter sequence in the numerical values assigned to these characters—'ayn = 500, ghayn = 600, khā = 700, dād = 800, zā = 900, 1000—has been preserved.

From its place of origin, to be sought perhaps in Petra, perhaps in the then flourishing Ghassānid kingdom, the script spread with trade to north and south. In the 6th century it was probably already known throughout the northern Arabic-speaking region, and was likely in use especially in the two cities that formed the starting point of the religio-national movement, Mecca and Medina.

religiös-nationalen Bewegung bildeten, in Mekka und Medina, in Gebrauch gewesen sein.

Ganz unschätzbare Quellen für die Geschichte der Übertragung griechisch-ägyptischer Verwaltungs- und Rechenpraxis in die Sprache und Schrift der arabischen Herren stehen uns seit längerer Zeit in den Papyri von el-Faiyum und andern Fundorten (Sammlung Erzherzog Rainer), seit einigen Jahren in denen des Aphroditofundes zu Gebote. Eine auch dem Nichtarabisten zugängliche Übersicht über die Schätze der erstgenannten Sammlung enthält der *Führer durch die Ausstellung* in seiner von J. von Karabacek bearbeiteten *Arabischen Abtheilung* (Wien 1894); die wichtigsten arabisch-griechischen Urkunden des Aphroditofundes sind von C. H. Becker veröffentlicht und bearbeitet worden.

Sie vermitteln in ihrer Gesamtheit ein fast lückenloses Bild von der fortschreitenden Arabisierung des Schrift- und Rechnungswesens, zugleich auch den Beweis eines überraschend langen und zähen Festhaltens der Araber an den griechischen Zahlzeichen. Viele Urkunden von Behörden, selbst Erlasse arabischer Heerführer aus der Eroberungszeit sind nur griechisch geschrieben; man wird aber annehmen dürfen, daß die Eroberer auch ihrer eigenen Sprache sofort Geltung zu verschaffen wußten, und daß zweisprachige Erlasse wie jene "im 22. Jahre d. H. am 25. April 643 n. Chr. ausgefertigte ebenso prächtige, wie unschätzbare Urkunde des Unterfeldherrn 'Abdallāh ibn Gābir", die in der Papyrussammlung Erzherzog Rainer als das "älteste Schriftdenkmal des Islam bewahrt wird, durchaus keine Ausnahme darstellen.

Während sich die eingesessene Bauernbevölkerung noch der koptischen, die handel- und gewerbetreibende Bevölkerung der Städte und die Beamtschaft vorwiegend der griechischen Sprache im Geschäftsverkehr bedienten, haben die im Lande heimisch gewordenen Araber ebenso selbstverständlich ihre Briefe und Mitteilungen arabisch geschrieben.

Die obersten Regierungsbehörden konnten am frühesten daran denken, ihre amtlichen Verfügungen nur arabisch abzufassen; ihre Verbreitung in der fremdsprachigen Provinz konnte nach wie vor nur durch Ausfertigung zweisprachiger Urkunden oder durch Herstellung griechischer und koptischer Übersetzungen erfolgen. In dem oben erwähnten Erlasse steht das Arabische an zweiter Stelle, später ist das Umgekehrte häufiger; noch aus dem Jahre 80 d. H. ist ein rein griechischer Erlaß bekannt. In den zweisprachigen Urkunden aus der Zeit des Statthalters Qorra b. Šarīk (im Amte 90–96 d. H. = 708–714 n. Chr.), die G. H. Becker a. a. O. veröffentlicht hat, erscheinen die Zahlen des griechischen, an zweiter Stelle stehenden Textes, der für die Empfänger jedenfalls der wichtigere war, stets zuerst "in Ziffern", dann "in Worten", die Jahreszahl in Ziffern; im arabischen Text sind alle Zahlenangaben in Worten ausgeschrieben.

Im Jahre 87 d. H. (706 n. Chr.) führte nach den Berichten der arabischen Historiker 'Abdallāh, der Sohn des Kalifen 'Abd al-Malik, die arabische Verwaltung in Ägypten ein. Um dieselbe Zeit hatte der Kalif al-Walid zu Damaskus den Gebrauch der griechischen Sprache bei der Führung der Steuerregister verboten, aber für die Zahlzeichen eine Ausnahme zulassen müssen, "parce qu'il était impossible d'écrire en arabe un, ou deux, ou trois, ou

Quite invaluable sources for the history of the transfer of Greek-Egyptian administrative and calculation practice into the language and script of the Arab rulers have been available to us for some time in the papyri of el-Fayyūm and other findspots (Archduke Rainer Collection), and for some years in those of the Aphrodito find. An overview accessible even to non-Arabists of the treasures of the first-named collection is contained in the *Führer durch die Ausstellung* in its *Arabische Abtheilung* edited by J. von Karabacek (Vienna 1894); the most important Arabic-Greek documents of the Aphrodito find have been published and edited by C. H. Becker.

They convey in their totality an almost seamless picture of the progressive Arabization of the writing and calculation system, and simultaneously the proof of a surprisingly long and tenacious adherence of the Arabs to the Greek numeral signs. Many documents from authorities, even decrees of Arab military commanders from the time of the conquest, are written only in Greek; but one may assume that the conquerors knew how to give their own language immediate validity, and that bilingual decrees like that "equally magnificent as invaluable document of the lieutenant commander 'Abdallāh ibn Jābir, executed in the 22nd year A. H. on April 25, 643 CE," preserved in the Archduke Rainer Papyrus Collection as the oldest written monument of Islam, are by no means an exception.

While the settled rural population still used Coptic, the commercial and craft-working population of the cities and the officials predominantly used Greek in business dealings, the Arabs who had become native to the land just as naturally wrote their letters and communications in Arabic.

The highest governmental authorities could soonest consider drafting their official decrees only in Arabic; their dissemination in the foreign-language province could still only take place through the issuance of bilingual documents or through the production of Greek and Coptic translations.

In the aforementioned decree, Arabic stands in second place; later the reverse is more frequent; a purely Greek decree from the year 80 A. H. is still known. In the bilingual documents from the time of the governor Qurra b. Sharik (in office 90–96 A. H. = 708–714 CE), published by G. H. Becker, op. cit., the numbers of the Greek text, which stood in second place and was in any case the more important for the recipients, appear always first "in figures," then "in words," the year in figures; in the Arabic text all numerical indications are written out in words.

In the year 87 A. H. (706 CE), according to the reports of Arab historians, 'Abdallāh, the son of the caliph 'Abd al-Malik, introduced the Arabic administration in Egypt. Around the same time the caliph al-Walid in Damascus had forbidden the use of the Greek language in keeping the tax registers, but had been compelled to make an exception for the numeral signs, "because it was impossible to write in Arabic one, or two, or three, or eight



huit et demi, etc."

Daß die Neuerung der rein arabischen Verwaltung noch nicht durchführbar war, beweisen jetzt unsere Texte; vor allem aber wird die von Woepcke beigezogene Stelle durch die zahlreichen "über das ganze achte und neunte Jahrhundert, in abnehmender Häufigkeit aber noch bis ins zehnte Jahrhundert reichenden, in einem Beispiel sogar vom Jahre 513 d. H., also 1120 n. Chr. datierten Urkunden belegt und bestätigt, die mitten im arabischen Text griechische Zahlzeichen enthalten und darum im *Führer* von Karabacek als "arabisch-griechisch" bezeichnet werden.

Auch die koptischen Zahlzeichen, die nur eine leichte Modifikation der griechischen sind, mögen in "Ägypten angewandt worden sein (Woepcke, a. a. O., S. 238), doch scheinen solche arabische Urkunden aus alter Zeit noch nicht bekannt zu sein.

Besonders interessant ist in unserem Zusammenhange ein Palimpsest aus dem Anfang des 9. Jahrhunderts, der schon im 7. Jahrhundert, wie die alten Schriftzüge zeigen, von einem Araber benutzt worden war, der sich in der Schreibung der griechischen Zahlbuchstaben übte. Er enthält in zwei Reihen die griechischen Zahlbuchstaben von  $\alpha$  bis  $\vartheta$  und "über den beiden ersten die arabischen Zahlwörter (*Führer* N. 649).

Gegenüber diesen Urkunden treten solche mit arabischen Zahlbuchstaben ganz auffallend zurück. Eine Schreibvorlage aus dem 9. Jahrhundert, die die arabischen und griechischen Zahlbuchstaben in Gegenüberstellung enthält, zeigt die "Übertragung und gleichzeitige Anwendung der Zeichen. Das "älteste bisher bekannt gewordene urkundliche Beispiel ist eine doppel-sprachige Steuerurkunde aus dem 8. Jahrhundert, die in dem an zweiter Stelle stehenden arabischen Text die Steuersumme auch in arabischen Zahlbuchstaben bietet (*Führer* N. 605).

Das nächste, zugleich genau datierte Beispiel aus dem Jahre 851 ist eine Bilanz des Schatzhauses von el-Fajjūm (*Führer* N. 761). Karabacek vermutet einen "für die große Menge mehr sekreten Charakter dieser Rechnungsweise, zumal sie der Differenzierung der einzelnen Buchstabenwerte durch die diakritischen Punkte entbehre; ich sehe ein Hindernis ihrer Verbreitung im Rechnungswesen auch darin, daß sich diese Buchstabenzusammenstellungen nicht deutlich genug vom Worttext abhoben und leicht zu Verwechslungen und Fehlern Veranlassung geben konnten.

Hatte man erst die Zahlen durch die arabischen Worte wiedergegeben, so konnte man die Zeichen auch weglassen, und so blieb in allen zusammenhängenden Texten, die nicht gerade tabellarische Zusammenstellungen oder sonst gehäufte Zahlen enthalten, bis heute die Wortschrift der Zahlen in "Übung, die vom Standpunkt des Mathematikers ein Rückschritt sein mag, aber den Vorzug der größeren Sicherheit besitzt.

Wenn es richtig ist, daß die Benutzung der Buchstaben als Zahlzeichen auf Milet und das 8. Jahrh. v. Chr. zurückzuführen ist, so müssen die auf Anwendung des Alphabets gegründeten Zahlbezeichnungen der Hebräer und Syrer griechischem Vorbild nacherfunden worden sein. Das normale semitische Alphabet von 22 Buchstaben führte zunächst glatt durch Einer und Zehner. Dann waren noch 4 Zeichen übrig, die für die Zahlen 100 bis 400 verwendet werden konnten, aber es fragte sich, wie nun die übrigen 5 Hunderter darzustellen waren.

and a half, etc."

That the innovation of purely Arabic administration was not yet feasible is proved by our texts now; but above all, the passage cited by Woepcke is evidenced and confirmed by the numerous documents extending over the whole 8th and 9th centuries, with decreasing frequency but still into the 10th century, in one example even dated from the year 513 A. H. (1120 CE), which contain Greek numeral signs in the middle of the Arabic text and are therefore called "Arabic-Greek" in Karabacek's *Führer*.

Also the Coptic numeral signs, which are only a slight modification of the Greek ones, may have been used in Egypt (Woepcke, op. cit., p. 238), but such Arabic documents from ancient times do not yet seem to be known.

Of particular interest in our context is a palimpsest from the beginning of the 9th century, which, as the old script traces show, had already been used in the 7th century by an Arab who was practicing the writing of the Greek numeral letters. It contains in two rows the Greek numeral letters from  $\alpha$  to  $\vartheta$  and above the first two the Arabic number words (*Führer* no. 649).

In comparison with these documents, those with Arabic numeral letters stand out quite strikingly. A writing template from the 9th century, containing Arabic and Greek numeral letters in juxtaposition, shows the transfer and simultaneous application of the signs. The oldest documentary example known to date is a bilingual tax document from the 8th century, which in the Arabic text standing in second place also presents the tax sum in Arabic numeral letters (*Führer* no. 605).

The next, simultaneously precisely dated example from the year 851 is a balance sheet of the treasury of el-Fayyūm (*Führer* no. 761). Karabacek conjectures a more secret character of this reckoning method for the majority, especially since it lacks the differentiation of the individual letter values by diacritical points; I see an obstacle to its spread in accounting also in the fact that these letter combinations did not stand out clearly enough from the word text and could easily give rise to confusions and errors.

Once one had rendered the numbers through Arabic words, one could also omit the signs, and so in all connected texts that did not contain tabular arrangements or other accumulated numbers, the word-writing of numbers has remained in use until today, which from the standpoint of the mathematician may be a step backward, but possesses the advantage of greater security.

If it is correct that the use of letters as numeral signs is to be traced back to Miletus and the 8th century BCE, then the numeral designations of the Hebrews and Syrians based on the use of the alphabet must have been re-invented after Greek models. The normal Semitic alphabet of 22 letters led smoothly at first through ones and tens. Then 4 signs remained, which could be used for the numbers 100 to 400, but the question arose how the remaining 5 hundreds were to be represented.

Die "ältere hebräische und die syrische Zahlenbezeichnung benutzen den nächstliegenden Ausweg, indem sie die höheren Hunderter additiv aus Zeichen für die niederen bilden, also  $500 = 400 + 100$  usw. bis  $900 = 400 + 400 + 100$  setzen. Das gleiche Verfahren ist nach S. de Sacy auch in arabischen Handschriften zu finden.

Interessanter ist ein zweites syrisches Verfahren, das für die Hunderter einfach die Zehnerbuchstaben unter Hinzufügung eines diakritischen Punktes einsetzt, denn damit wird die Zahl der Zifferbuchstaben auf zwei Enneaden reduziert. Es ist von hier aus verständlich, wie arabische Schriftsteller — ich erwähne neben al-Nadīm, dem Verfasser des *Fihrist*, auf den kürzlich L. G. Karpinski wieder aufmerksam machte, noch die in meinen *Kazwīnīstudien* behandelte Handschrift — die indischen Ziffern für neun Grundzeichen halten konnten, die durch Hinzufügung eines bzw. zweier diakritischen Punkte die Zahlenwerte der Buchstabengruppen bis ٩, bis ٩ wiedergeben.

Hier sind die punktförmigen Nullen der Zehner und Hunderter und die von 1000 einfach als diakritische Punkte genommen (*Fihrist* S. 18/19). Ich füge wieder die wörtliche Übersetzung bei, da die von L. C. Karpinski mitgeteilte, von Dr. A. Yohannan herrührende "freie" Übersetzung wie alle sogenannten freien Übersetzungen für kritische geschichtliche Untersuchungen nicht zu gebrauchen ist.

"Und es erwähnte dieser Mann, dessen Erwähnung vorausgeschickt wurde, daß sie meistens mit den neun Schriftzeichen schreiben, gemäß diesen Beispielen: ١ ٢ ٣ ٤ ٥ ٦ ٧ ٨ ٩; "wenn er dann bis zum ١٠ gelangt ist, wiederholt er das erste Zeichen und versieht es mit einem Punkt darunter gemäß diesen Beispielen: ١̇ ٢̇ ٣̇ ٤̇ ٥̇ ٦̇ ٧̇ ٨̇ ٩̇, "so daß es" ١٠ ٢٠ ٣٠ ٤٠ ٥٠ ٦٠ ٧٠ ٨٠ ٩٠ "wird, indem es von zehn zu zehn vermehrt wird; wenn er dann zum ١٠٠ gelangt, schreibt er gemäß diesen Beispielen und setzt unter jedes Zeichen zwei Punkte wie folgt: ١̇̇ ٢̇̇ ٣̇̇ ٤̇̇ ٥̇̇ ٦̇̇ ٧̇̇ ٨̇̇ ٩̇̇, "so daß es" ١٠٠ ٢٠٠ ٣٠٠ ٤٠٠ ٥٠٠ ٦٠٠ ٧٠٠ ٨٠٠ ٩٠٠ "wird; und wenn er dann das ١٠٠٠ erreicht hat, schreibt er das erste Zeichen von Anfang an, nämlich dieses: ١̇̇̇, "und setzt darunter drei Punkte wie folgt: ١̇̇̇̇, "und es geschieht so, daß er zur Gesamtheit der Zeichen des (arabischen) Alphabets kommt und schreibt, was er will."

Die beiden Textänderungen ergeben sich von selbst aus dem Zusammenhang. Daß "these subscript dots are used here only as we use Roman numerals" (Karpinski S. 123, Z. 3 v. u.), ist eine unhaltbare Annahme; in der zweiten Buchstabenreihe ist nicht "(ja or j omitted)", wie genau nach Flügel zu bemerken ist, sondern zweifellos nur ja die letzte Buchstabenreihe ist nicht die der "Persian letters" — das Wort *muḡam* bezeichnet das ganze arabische Alphabet — und stellt nicht "the higher hundreds and 1000", sondern alle Hunderter ohne Tausend dar.

Der Verfasser vergißt im folgenden Satz den Zahlbuchstaben ٩ ausdrücklich zu erwähnen, wie er ja auch das Fortschreiten von 100 zu 100 nicht mehr besonders hervorhebt. Die Echtheit der

The older Hebrew and the Syriac numeral designations use the most obvious expedient, in that they form the higher hundreds additively from signs for the lower ones, thus setting  $500 = 400 + 100$  etc. up to  $900 = 400 + 400 + 100$ . The same procedure is found, according to S. de Sacy, also in Arabic manuscripts.

More interesting is a second Syriac procedure, which for the hundreds simply employs the tens letters with the addition of a diacritical point, for thereby the number of digit-letters is reduced to two enneads. From here it is understandable how Arabic writers—I mention alongside al-Nadīm, the author of the *Fihrist*, to whom L. G. Karpinski recently called attention, also the manuscript treated in my *Kazwīnīstudien*—could regard the Indian numerals as nine basic signs, which by the addition of one or two diacritical points reproduce the numerical values of the letter groups up to 'ayn and up to ghayn.

Here the dot-shaped zeroes of the tens and hundreds and that of 1000 are simply taken as diacritical points (*Fihrist*, pp. 18/19). I append again the literal translation, since the "free" translation communicated by L. C. Karpinski, originating from Dr. A. Yohannan, like all so-called free translations, is not usable for critical historical investigations.

"And this man, whose mention was placed first, mentioned that they write for the most part with the nine script-signs, according to these examples: ١ ٢ ٣ ٤ ٥ ٦ ٧ ٨ ٩; "when he then reaches" ١٠, "he repeats the first sign and furnishes it with a point underneath according to these examples: ١̇ ٢̇ ٣̇ ٤̇ ٥̇ ٦̇ ٧̇ ٨̇ ٩̇, "so that it becomes" ١٠ ٢٠ ٣٠ ٤٠ ٥٠ ٦٠ ٧٠ ٨٠ ٩٠, "being increased by ten; when he then reaches" ١٠٠, "he writes according to these examples and places under each sign two points as follows: ١̇̇ ٢̇̇ ٣̇̇ ٤̇̇ ٥̇̇ ٦̇̇ ٧̇̇ ٨̇̇ ٩̇̇, "so that it becomes" ١٠٠ ٢٠٠ ٣٠٠ ٤٠٠ ٥٠٠ ٦٠٠ ٧٠٠ ٨٠٠ ٩٠٠; "and when he then has reached" ١٠٠٠, "he writes the first sign from the beginning, namely this: ١̇̇̇, "and places under it three points as follows: ١̇̇̇̇, "and it happens thus that he reaches the totality of the signs of the (Arabic) alphabet and writes what he wishes."

The two textual alterations follow of themselves from the context. That "these subscript dots are used here only as we use Roman numerals" (Karpinski, p. 123, line 3 from bottom) is an untenable assumption; in the second letter series it is not "(ja or j omitted)," as is to be noted precisely according to Flügel, but doubtless only ja; the last letter series is not that of the "Persian letters"—the word *muḡam* denotes the whole Arabic alphabet—and does not represent "the higher hundreds and 1000," but all hundreds without thousand.

The author forgets in the following sentence to mention the numeral letter ghayn explicitly, just as he no longer especially emphasizes the progression from 100 to 100. There is absolutely no

Fihriststelle anzuzweifeln (Eneström in *Jahrb. "über d. Fortschr. d. M.*, Bd. 41, 1910, S. 53), liegt durchaus kein Grund vor; sie bezieht sich ohne Frage auf die indischen Zahlzeichen — diese sind für den Araber ebensogut "Schriftzeichen" wie sein eigenes Alphabet — und findet ihr Analogon in andern arabischen Abhandlungen "über die Schriften der Völker.

reason to doubt the authenticity of the *Fihrist* passage (Eneström in *Jahrb. "über d. Fortschr. d. M.*, vol. 41, 1910, p. 53); it undoubtedly refers to the Indian numeral signs—these are for the Arab just as much "script-signs" as his own alphabet—and finds its analogue in other Arabic treatises on the scripts of peoples.

## 6 VI. "Über die Erbteilungsaufgaben / On the Inheritance Problems

Als weitere Quellen für die Geschichte des Rechnens und der Algebra kommen die arabischen Abhandlungen "über Erbteilung in Betracht. Bisher ist von ihnen in der Geschichte der Mathematik noch nirgends Gebrauch gemacht worden, obgleich sich vom Erstling dieser Verbindung von Mathematik und Rechtswissenschaft, der uns in Muhammad b. Mūsā's "Algebra" vorliegt, durch Jahrhunderte hindurch eine gewohnheitsmäßige und sachlich begründete Vereinigung von Rechenkunst und Erbteilungswissenschaft verfolgen läßt.

Ich erinnere an den 895 gestorbenen Historiker, Astronomen und Mathematiker al-Dīnawarī, der das ganze von Muhammad b. Mūsā behandelte Gebiet in besonderen Schriften darstellte, an den Kādī 'Abd al-Ḥamīd aus Baṣra, "sehr gelehrt in Rechenkunst, Algebra, Ausmessungslehre und Erbteilung", sodann an Abū Kāmil Ṣūcā' b. Aslam, ferner an die zahlreichen von Ibn al-Farādī erwählten Kenner der Erbteilung und Rechenkunst, an den Rechner al-Missisī, an Sinān b. al-Fāṭ, an 'Abd al-Kāhīr b. Tāhīr aus Bagdad, an al-Wannī und seinen Schüler 'Abdallāh b. Ibrāhīm al-Hābrī und, um noch einen Mathematiker der Spätheit zu nennen, an al-Kalasādī, der außer einer Reihe von Schriften "über Arithmetik und Algebra auch ein Werk "über das Ganze der Erbteilung verfaßte.

Man wird gewiß keine bedeutenden mathematischen Sätze in diesen Schriften entdecken, aber man wird auf alle Fälle den Wurzeln der Rechenkunst der Araber und den praktischen Grundlagen ihres Interesses an der Mathematik näher kommen, wenn man dem bürgerlichen Rechnen mehr Aufmerksamkeit schenkt.

Die Bedürfnisse des Lebens in Handel und Verkehr, beim Kauf und Verkauf von Waren aller Art, von Häusern und Grundstücken, Tieren und Sklaven, beim bankmäßigen Geldverkehr, bei der Vollstreckung von Testamenten und der Verteilung von Hinterlassenschaften, bei der Erhebung von Zöllen, Ertrags- und Vermögensteuern usw. mußten unter allen Umständen rechnungsmäßig behandelt werden und wurden auf Grund von Geldwert, Maß und Gewicht nach praktischen Regeln erledigt.

Von diesen Aufgaben und Erfordernissen muß man überall ausgehen, wenn man den natürlichen Entwicklungsgang der Rechenkunst beschreiben und nicht ein zu schattenhaftes Bild von der Entstehung mathematischer Methoden und Aufgaben geben will.

Wenn die Erbteilungsrechnungen wirklich durch und durch arabisch waren und ganz und gar auf dem Koran beruhten, so mußte man den Gang der Entwicklung von den zu Muhammad b. Mūsā's Zeit begründeten Rechtsschulen (Abū Hanīfa starb im Jahr 767, Mālik b. Anas um 795, Muhammad al-Šāfi'ī um

As further sources for the history of calculation and algebra, Arabic treatises on inheritance division come into consideration.

Hitherto they have been nowhere made use of in the history of mathematics, although from the first fruit of this connection of mathematics and jurisprudence, which lies before us in Muḥammad b. Mūsā's "Algebra," a customary and substantively founded union of arithmetic and the science of inheritance division can be traced through centuries.

I recall al-Dīnawarī, the historian, astronomer, and mathematician who died in 895, who presented the whole domain treated by Muḥammad b. Mūsā in special writings; the qādī 'Abd al-Ḥamīd of Baṣra, "very learned in arithmetic, algebra, measurement, and inheritance division"; then Abū Kāmil Shujā' b. Aslam; further, the numerous experts in inheritance and arithmetic mentioned by Ibn al-Farādī; the calculator al-Missisī; Sinān b. al-Fāṭ; 'Abd al-Kāhīr b. Tāhīr of Baghdad; al-Wannī and his pupil 'Abdallāh b. Ibrāhīm al-Hābrī; and, to name yet a mathematician of the late period, al-Kalasādī, who besides a series of writings on arithmetic and algebra also composed a work on the entirety of inheritance division.

One will certainly discover no significant mathematical theorems in these writings, but one will in any case come closer to the roots of the Arabs' arithmetic and to the practical foundations of their interest in mathematics, if one pays more attention to practical calculation.

The necessities of life in trade and commerce, in the buying and selling of goods of all kinds, of houses and landed property, animals and slaves, in banking money transactions, in the execution of testaments and the distribution of estates, in the collection of customs duties, income and property taxes, etc., had under all circumstances to be treated computationally and were disposed of according to practical rules based on monetary value, measure, and weight.

From these tasks and requirements one must everywhere start if one wishes to describe the natural course of development of arithmetic and not give too shadowy a picture of the origin of mathematical methods and problems.

If the inheritance calculations were really thoroughly Arabic and rested entirely on the Qur'an, one would have to be able to follow the course of development from the law schools founded in the time of Muḥammad b. Mūsā (Abū Hanīfa died in 767, Mālik b. Anas around 795, Muḥammad al-Šāfi'ī around 820, Aḥmad b.

820, Aḥmad b. Ḥanbal 855) zu den Aufgaben der Algebra verfolgen können.

In Wahrheit liefert das Erbrecht nur den Rohstoff für die Aufgaben, und ob die Technik der Rechnung arabisch ist oder nicht, muß erst untersucht werden. Wenn wir uns erinnern, daß auch die römischen Juristen schon ihre Freude an verzwickten Erbfällen hatten und sie durch praktische Rechenkunst oder salomonische Weisheit zu entscheiden suchten (Cantor I<sup>3</sup>, S. 561), so werden uns berechtigte Zweifel an der Originalität der arabischen Behandlung der Aufgaben aufsteigen, und wenn wir finden, daß es sich vielfach um Auflösungen von Gleichungen ersten Grades mit Hilfe von Ausgleichung und Ergänzung handelt, so können wir jetzt ohne Zaudern griechische Vorbilder annehmen. Man wird nicht daran zweifeln können, daß in den Kreisen der Steuerbeamten, der Notare, der Vermessungsbeamten, der Kaufleute und Bankiers für alle in den regelrechten Geschäftsgang einschlagenden Aufgaben eine feste Überlieferung bestand, die den Anwältern und Schreibern ebenso beigebracht wurde, wie es heutzutage in den Amtsstuben und Kontoren geschieht.

Aus diesen Kreisen mögen die ersten Anregungen zur Aufzeichnung und Ausarbeitung von Musterbeispielen gekommen sein, und nichts anderes als eine solche Sammlung von Vorschriften und Beispielen für den allgemeinen Gebrauch, mit genauer Anweisung für die Durchführung der Rechnungen, will Muhammad b. Mūsās Buch vorstellen.

Das Fehlen wirklich aus dem Leben gegriffener Aufgaben im ersten Teil der Algebra, also im Bereich der quadratischen Gleichungen, die Armseligkeit der Beispiele in dem Kapitel von den Geschäftsfällen, der schematische Charakter vieler Beispiele in andern Teilen des Buches deutet auf den Mangel einer durchgebildeten Übungs- und Schulliteratur, wie wir sie heute in den Werken und Aufgabensammlungen über kaufmännische Arithmetik, über die Regel de tri u. dergl. besitzen; die quadratischen Gleichungen, die bisher als Hauptstück und Hauptzweck der "Algebra" galten, erscheinen jetzt mehr als prunkvolle Fassade und gelehrte Zutat, oder wenn man will auch als Beginn des Übergangs zu den rein wissenschaftlichen, spekulativen Fragen und Interessen der Mathematik.

Ist die hier vorgetragene Auffassung richtig, so müssen wir auch von dem gemeinen Sprachgebrauch und den praktischen Aufgaben, also von den Aufgaben ersten Grades, nicht von den quadratischen Gleichungen aus zum Verständnis der Terminologie vordringen suchen. Es ist dann klar, daß der in allen Aufgaben wiederkehrende, grundlegende Begriff kein anderer sein kann als der einer Geldsumme, eines Vermögens, sei es bekannt oder unbekannt, Kapital, Erbmasse oder Wert einer Ware.

Alles dies und noch mehr wird durch das Wort مال *māl* bezeichnet. Im Koran, den wir als "älteste Quelle anführen, finden wir das Wort *māl* und den Plural *amwāl* in der Bedeutung Vermögen, Besitz, Geld und Gut, Reichtum, häufig auch in der Verbindung "Güter und Kinder" oder "mit ihrem Gut und Blut"; es ist vom Aufzehren des Gutes der Waisen, vom Ausgeben der Reichtümer auf dem Wege Gottes die Rede, "Kapitalien" heißen رؤوس الأموال d. i. *capita bonorum*, und "er häuft Gut auf" جمع جامعا *ḡamā'a mālan* (Sura CIV, 2).

Ḥanbal in 855) to the problems of algebra.

In truth, inheritance law provides only the raw material for the problems, and whether the technique of calculation is Arabic or not must first be investigated. If we recall that Roman jurists too already took pleasure in tricky inheritance cases and sought to decide them by practical arithmetic or Solomonic wisdom (Cantor I<sup>3</sup>, p. 561), then justified doubts about the originality of the Arabic treatment of the problems will arise in us, and if we find that it frequently concerns solutions of first-degree equations by means of balancing and completion, we can now assume Greek models without hesitation.

One cannot doubt that in the circles of tax officials, notaries, surveyors, merchants, and bankers there existed a firm tradition for all tasks falling within the regular course of business, which was taught to candidates and scribes just as happens nowadays in offices and counting-houses.

From these circles may have come the first impulses toward the recording and working out of model examples, and nothing other than such a collection of rules and examples for general use, with precise instructions for carrying out the calculations, does Muḥammad b. Mūsā's book wish to present.

The absence of really practically drawn problems in the first part of the Algebra, that is, in the domain of quadratic equations; the poverty of the examples in the chapter on commercial transactions; the schematic character of many examples in other parts of the book—all point to the lack of a fully developed exercise and school literature, such as we possess today in the works and problem collections on commercial arithmetic, on the rule of three, and the like. The quadratic equations, which hitherto counted as the main piece and chief purpose of "Algebra," now appear more as a splendid facade and scholarly addition, or if one will, as the beginning of the transition to the purely scientific, speculative questions and interests of mathematics.

If the view presented here is correct, then we must seek to advance to the understanding of the terminology from common linguistic usage and practical problems, that is, from first-degree problems, not from quadratic equations. It is then clear that the fundamental concept recurring in all problems can be none other than that of a sum of money, a possession, be it known or unknown, capital, estate, or value of a commodity.

All this and more is designated by the word مال *māl*. In the Qur'an, which we cite as the oldest source, we find the word *māl* and the plural *amwāl* in the meaning of fortune, possession, money and goods, wealth, frequently also in the connection "goods and children" or "with their goods and blood"; there is talk of consuming the goods of orphans, of spending riches in the way of God; "capitals" are called رؤوس الأموال i.e. *capita bonorum*, and "he accumulates goods" جمع مالا *jama'a mālan* (Sura CIV, 2).



In den Aphroditopapyri kann man *māl* mit Geld oder Steuerertrag, Steuersumme "übersetzen, gutes, d. h. vollwertiges Geld ist مال تام *māl tāmm*; das Schatzhaus heißt بيت المال *bayt al-māl*, das Einsammeln der Steuerbeträge جمع المال *ġam al-māl*.

In seiner ganz allgemeinen Bedeutung, als vorhandenes oder angenommenes Vermögen, wird das Wort in den Erbteilungsaufgaben gebraucht.

**Zwei Beispiele mit wortgetreuer "Übersetzung** m"ogen zugleich die Art der Aufgaben erläutern. Nach Rosen S. 89/90:

"If the instance be: 'A woman dies, leaving her husband, a son, and three daughters, and bequeathing to a stranger one-eighth and one-seventh of her Capital'; then you constitute the shares of the heirs, by taking them out of twenty. Take a Capital, and subtract from it one-eighth and one-seventh of the same. The remainder is, a Capital less one-eighth and one-seventh. Complete your Capital by adding to that which you have already, fifteen forty-one parts. Multiply the parts of the Capital, which are twenty, by forty-one; the product is eight hundred and twenty. Add to it fifteen forty-one parts of the same, which are three hundred: the sum is one thousand one hundred and twenty parts. The person to whom one-eighth and one-seventh were bequeathed, receives one-eighth and one-seventh of this. One seventh of it is one hundred and sixty, and one-eighth one hundred and forty. Subtracting this, there remain eight hundred and twenty parts for the heirs, proportionably to their legal shares."

In wörtlicher Übertragung nach dem Arabischen:

"Und wenn er (oder jemand) sagt: Eine Frau starb und hinterließ ihren Gatten und ihren Sohn und drei Töchter und vermachte einem Manne ein Achtel ihres Vermögens und sein Siebentel; so stelle die Anteile der (gesetzlichen) Erbschaft auf, indem du sie von zwanzig nimmst. Nimm also ein Vermögen und wirf weg sein Achtel und Siebentel, so bleibt ein Vermögen weniger ein Achtel und ein Siebentel. Vervollständige nun dein Vermögen, und das ist (oder besteht darin), daß du zu ihm hinzufügst fünfzehn Teile von einundvierzig Teilen; schlag dann (d. i. multipliziere) die Anteile der Erbschaft, nämlich zwanzig, in einundvierzig, so wird das achthundertundzwanzig; füge dann diesem hinzu fünfzehn Teile von einundvierzig, und das sind dreihundert Teile, so gibt dies alles tausend und hundert und zwanzig Anteile; für den, dem davon das Achtel und das Siebentel vermacht wurde, ein Siebentel davon und sein Achtel, und das ist dreihundert, das Siebentel hundert und sechzig und das Achtel hundert und vierzig; somit bleiben achthundert und zwanzig Anteile zwischen den Erben gemäß ihren Anteilen."

Einige der auffallenderen Abweichungen der Rosenschen Übersetzung sind durch Kursivdruck hervorgehoben; sie bestätigen die Beobachtung, daß die Wiedergabe der Kunstaussdrücke unzuverlässig ist. Das ändert nicht immer viel am Sinn, macht aber eine Wiederherstellung der Ausdrücke aus der Übersetzung unmöglich und alle darauf gebauten Schlüsse unsicher.

Muhammad b. Mūsā sagt seinen Lesern nicht, wie er zu den zwanzig Anteilen für die Erben kommt. Sie wissen als Muslime, daß dem Koran gemäß (Sure IV, 13) der Gatte nach Abzug der Legate auf ein Viertel des Nachlasses Anspruch hat, wenn Kinder miterben, und daß ein Knabe doppeltsoviel vom Rest erhält als ein Mädchen (Sure IV, 12); auch ist für das Legat die Vor-

In the Aphrodito papyri one can translate *māl* as money or tax revenue, tax sum; good, i.e. full-value money is مال تام *māl tāmm*; the treasury is called بيت المال *bayt al-māl*, the collection of tax amounts جمع المال *jam' al-māl*.

In its most general meaning, as present or assumed possession, the word is used in the inheritance problems.

**Two examples with literal translation** may at the same time illustrate the nature of the problems. After Rosen, pp. 89/90:

"If the instance be: 'A woman dies, leaving her husband, a son, and three daughters, and bequeathing to a stranger one-eighth and one-seventh of her Capital'; then you constitute the shares of the heirs, by taking them out of twenty. Take a Capital, and subtract from it one-eighth and one-seventh of the same. The remainder is, a Capital less one-eighth and one-seventh. Complete your Capital by adding to that which you have already, fifteen forty-one parts. Multiply the parts of the Capital, which are twenty, by forty-one; the product is eight hundred and twenty. Add to it fifteen forty-one parts of the same, which are three hundred: the sum is one thousand one hundred and twenty parts. The person to whom one-eighth and one-seventh were bequeathed, receives one-eighth and one-seventh of this. One seventh of it is one hundred and sixty, and one-eighth one hundred and forty. Subtracting this, there remain eight hundred and twenty parts for the heirs, proportionably to their legal shares."

In literal translation from the Arabic:

"And if he (or someone) says: A woman died and left her husband and her son and three daughters, and bequeathed to a man one-eighth of her possession and its seventh; so set up the shares of the (legal) inheritance by taking them from twenty. Take then a possession and cast away its eighth and seventh, so there remains a possession less an eighth and a seventh. Complete now your possession, and that is (or consists in) adding to it fifteen parts of forty-one parts; then strike (i.e. multiply) the shares of the inheritance, namely twenty, into forty-one, so that becomes eight hundred twenty; then add to this fifteen parts of forty-one, and that is three hundred parts, so this all gives one thousand one hundred and twenty shares; for him to whom the eighth and the seventh of it was bequeathed, one-seventh of it and its eighth, and that is three hundred, the seventh one hundred sixty and the eighth one hundred forty; thus eight hundred and twenty shares remain among the heirs according to their shares."

Some of the more striking deviations of Rosen's translation are indicated by italics; they confirm the observation that the rendering of technical terms is unreliable. This does not always alter the sense much, but makes a restoration of the expressions from the translation impossible and all conclusions built upon it uncertain.

Muhammad b. Mūsā does not tell his readers how he arrives at the twenty shares for the heirs. They know as Muslims that according to the Qur'an (Sura IV, 13) the husband has claim to a quarter of the estate after deduction of legacies, if children co-inherit, and that a boy receives twice as much of the remainder as a girl (Sura IV, 12); also the prescription is observed for the

schrift eingehalten, daß der Erblasser höchstens über ein Drittel seines reinen Vermögens frei verfügen darf.

Die Töchter erhalten also je  $\frac{1}{4}$ , der Sohn  $\frac{1}{2}$  von  $\frac{3}{4}$  des Vermögens, d. i. der Vater  $\frac{1}{4}$ , die drei Töchter  $\frac{3}{16}$  und der Sohn  $\frac{3}{8}$  von dem, was nach Abzug des Legates noch übrig bleibt. Um ganz klar zu sein, hatte Muhammad b. Mūsā weiter hinzufügen müssen, daß  $\frac{1}{8}$  und  $\frac{1}{7}$  zusammen  $\frac{15}{56}$  des Vermögens sind, daß man also für die gesetzlichen Erben  $\frac{41}{56}$  von 56 Teilen übrig behält und darum  $\frac{15}{56}$  des Restes zu diesem hinzufügen muß, um das vollständige Vermögen zu erhalten. Damit er ganze Zahlen erhält, multipliziert er 20 mit 41 und gewinnt so die Zahl 820.

Diese 820 sind um 15 weitere Teile, d. h. um  $\frac{15}{41} \cdot 20 = 300$  zu vermehren. Die Schlußrechnung können wir dann in Form einer "Abrechnung mit den Erben" wie folgt darstellen:

$$\begin{aligned} 41 \cdot 20 + 15 \cdot 20 &= 1120 \\ 1120 : 7 &= 160 \\ 1120 : 8 &= 140 \\ \text{Für das Legat } 300 &- 300 \\ \text{Bleiben für die Erben } &820. \end{aligned}$$

Es ist wohl möglich, daß wir in diesem Verfahren ein Beispiel der "älteren Technik vor uns haben, und daß die Aufgabe mit ihren einfachen Bedingungen zu dem Bestand von Musterbeispielen gehört, die schon vor Einführung der Gleichungen bei den Erbrichtern und Notaren zur Einübung der Teilungsregeln gehörten.

Eine Stufe weiter führt ein Beispiel, das ich nach dem Text S. 86 mit Anwendung von Ziffern und von Abkürzungen für die Worte Vermögen, Dirhem und Anteil — dieser heißt hier *nasīb* statt *sahm*, ein Fingerzeig dafür, daß die Aufgabe einer andern Quelle entstammt — genau wiedergebe:

"Ein Mann starb und hinterließ vier Söhne, und vermachte einem Manne, was soviel ist als der Anteil eines von ihnen und das Viertel dessen, was bleibt von dem Drittel [des Vermögens nach Abzug des Anteils — von Rosen ergänzt], und einen Dirhem. Dann ist die Vorschrift hierfür, daß du nimmst  $\frac{1}{3}$  eines Vermögens, dann wegnimmst (wegwirfst) von ihm einen Anteil, so daß bleibt  $\frac{1}{3}$  weniger ein Anteil. Hierauf nimmst du weg  $\frac{1}{4}$  von dem, was dir geblieben ist, d. h.  $\frac{1}{4}$  von  $\frac{1}{3}$  weniger  $\frac{1}{4}$  eines Anteils, und nimmst auch weg 1 Dhm., so bleibt dir  $\frac{1}{4}$  von  $\frac{1}{3}$  eines Vermögens, das ist  $\frac{1}{12}$  des Vermögens, weniger  $\frac{1}{4}$  eines Anteils und weniger 1 Dhm.

Dies füge nun zu (den noch vorhandenen)  $\frac{2}{3}$  des Vermögens hinzu, dann hast du  $\frac{11}{12}$  Teile von 12 (Teilen) eines Vermögens weniger  $\frac{1}{4}$  eines Anteils und weniger 1 Dhm., die gleichkommen 4 Ant. Ergänze dies nun (al-gabr) mit  $\frac{1}{4}$  eines Anteils und mit 1 Dhm., so werden  $\frac{11}{12}$  Teile von 12 eines Vermögens gleich 4 Anteilen und  $\frac{1}{4}$  eines Anteils und 1 Dhm.

Vervollständige nun (*ikmāl*) dein Vermögen und zwar dadurch, daß du zu den Anteilen und dem Dhm. hinzufügst  $\frac{1}{11}$  Teil von ihnen, so bekommst du 1 Vermögen, das gleich ist  $\frac{48}{11}$  Anteilen und 1 Dhm. und  $\frac{1}{11}$  Teil eines Dhm."

Drücken wir den Text in unsern gewohnten Zeichen aus, indem wir das Legat =  $x$ , das Kapitalvermögen =  $k$ , den Anteil eines Sohnes =  $t$ , den Dirhem =  $d$  setzen, so kommen wir durch die

legacy that the testator may freely dispose of at most one-third of his net estate.

The daughters thus each receive  $\frac{1}{4}$ , the son  $\frac{1}{2}$  of  $\frac{3}{4}$  of the estate, that is, the father  $\frac{1}{4}$ , the three daughters  $\frac{3}{16}$  and the son  $\frac{3}{8}$  of what remains after deduction of the legacy. To be quite clear, Muhammad b. Mūsā should have further added that  $\frac{1}{8}$  and  $\frac{1}{7}$  together are  $\frac{15}{56}$  of the estate, that one thus retains  $\frac{41}{56}$  of 56 parts for the legal heirs and must therefore add  $\frac{15}{56}$  of the remainder to it in order to obtain the complete estate. In order to get whole numbers, he multiplies 20 by 41 and thus obtains the number 820.

These 820 are to be increased by 15 further parts, that is, by  $\frac{15}{41} \cdot 20 = 300$ . We can then present the final calculation in the form of a "reckoning with the heirs" as follows:

$$\begin{aligned} 41 \cdot 20 + 15 \cdot 20 &= 1120 \\ 1120 : 7 &= 160 \\ 1120 : 8 &= 140 \\ \text{For the legacy } 300 &- 300 \\ \text{Remain for the heirs } &820. \end{aligned}$$

It is quite possible that we have in this procedure an example of the older technique, and that the problem with its simple conditions belongs to the stock of model examples that already before the introduction of equations served among inheritance-dividers and notaries for practicing the division rules.

A step further leads an example which I reproduce exactly from the text p. 86 with the use of numerals and abbreviations for the words possession, dirham, and share—the latter is called here *nasīb* instead of *sahm*, a hint that the problem stems from another source:

"A man died and left four sons, and bequeathed to a man [an amount] equal to the share of one of them and a quarter of what remains from the third [of the estate after deduction of the share—supplied by Rosen], and one dirham. Then the rule for this is that you take  $\frac{1}{3}$  of a possession, then remove (cast away) from it a share, so that  $\frac{1}{3}$  less a share remains. Then you take away  $\frac{1}{4}$  of what has remained for you, i.e.  $\frac{1}{4}$  of  $\frac{1}{3}$  less  $\frac{1}{4}$  of a share, and also take away 1 dirham, so there remains for you  $\frac{1}{4}$  of  $\frac{1}{3}$  of a possession, that is  $\frac{1}{12}$  of the possession, less  $\frac{1}{4}$  of a share and less 1 dirham.

This add now to the (still remaining)  $\frac{2}{3}$  of the possession, then you have  $\frac{11}{12}$  parts of 12 (parts) of a possession less  $\frac{1}{4}$  of a share and less 1 dirham, equal to 4 shares. Complete this now (al-gabr) with  $\frac{1}{4}$  of a share and with 1 dirham, so  $\frac{11}{12}$  parts of 12 of a possession equal 4 shares and  $\frac{1}{4}$  of a share and 1 dirham.

Complete now (*ikmāl*) your possession, namely by adding to the shares and the dirham  $\frac{1}{11}$  part of them, so you obtain 1 possession equal to  $\frac{48}{11}$  shares and 1 dirham and  $\frac{1}{11}$  part of a dirham."

Expressing the text in our usual notation, setting the legacy =  $x$ , the capital possession =  $k$ , the share of a son =  $t$ , the dirham =  $d$ , we come through the execution of the prescribed operations

Ausführung der vorgeschriebenen Operationen zunächst zu der Beziehung  $\frac{2}{3}k - x = \frac{1}{3}k - t - \frac{1}{4}(\frac{1}{3}k - t) - d = \frac{1}{4}k - \frac{3}{4}t - d$ , dann zu der Gleichung  $\frac{2}{3}k + \frac{1}{4}k - \frac{3}{4}t - d = \frac{11}{12}k - \frac{1}{4}t - d = 4t$ , woraus durch *al-gabr*  $\frac{11}{12}k = 4\frac{1}{4}t + d$  und durch *ikmāl*  $k = \frac{51}{11}t + \frac{12}{11}d = 5\frac{8}{11}t + 1\frac{1}{11}d$ .

**Die arabischen Bruchzahlen.** Bei S. Günther findet sich darüber S. 200 folgende Schilderung: "Von dem späteren Alkārī und dem noch viel späteren Behā Eddīn wird uns überliefert, daß man auch mit gewöhnlichen Brüchen rechnete. Doch ließ man als aussprechbar nur solche zu, deren Nenner Einer waren; wuchs der Nenner über 9 hinaus, so war der Bruch stumm, und dann gab es für ihn nur eine umschreibende Bezeichnung."

Ungefähr dasselbe lesen wir auch bei Cantor I<sup>3</sup>, S. 718, nur daß eine im ganzen zutreffende, von Günther übergenommene Erklärung des Sachverhalts hinzugefügt wird. Rodet hat sogar eine Ähnlichkeit mit dem "Aussprechbarmachen" der Brüche durch Verwandlung in eine Summe von Stammbrüchen bei den Ägyptern finden wollen; Cantor zweifelt, ob die Unterscheidung bereits für Muhammad b. Mūsā's Zeiten Geltung gehabt habe. Es wäre an der Zeit, daß diese rein philologische Diskussion aus der mathematischen Literatur verschwände. Es handelt sich darum, daß das Arabische von den Grundzahlen 3 bis 10 besondere Bruchzahlen bildet, wie *tuluṭ* Drittel, *humus* Fünftel, *uṣṣur* Zehntel; das sind die "aussprechbaren" Brüche. Für die Zehner, die als Plurale der Einer erscheinen, sind entsprechende Bildungen sprachlich unmöglich, für zusammengesetzte Zahlen nur dann, wenn sie in Faktoren bis 10 zerlegt werden können, so daß z. B. ein Achtundvierzigstel "ein Sechstel des Achtels" heißt.

Selbstverständlich bestand dieser Unterschied, solange es eine arabische Sprache gibt. Schon Muhammad b. Mūsā äußert sich über den Tatbestand in seinem Rechenbuch; die Stelle findet sich S. 17 der lateinischen Bearbeitung und lautet: *Scito quod fractiones [sic] appellantur multis nominibus in numerabilibus atque infinitis, ut medietas, tercia, quarta, nona et decima, et una pars ex xm., et pars ex xviii., et cetera.*

Wie man später darüber philosophierte, zeigt eine Stelle der Iḥān a-ṣafā' (ed. Bombay, Bd. I, S. 28): "Und wisse, o Bruder ..., daß die gebrochene Zahl viele Rangordnungen (*marātib*) besitzt, weil es keine ganze Zahl gibt, die nicht einen Teil hatte oder zwei Teile oder eine Anzahl von Teilen wie die Zwölf: denn ihr (kommt zu) eine Hälfte und ein Drittel und ein Viertel und ein Sechstel und die Hälfte eines Sechstels, und ebenso die Achtundzwanzig und andere als diese beiden von den Zahlen ..."

"Und es umfaßt sie alle eine Zehnzahl von Worten: ein Wort von ihnen ist allgemein und unbestimmt (*mubhamah*) und neun sind speziell und bestimmt (*mafḥūmah*); von den neun Worten ist ein Wort ursprünglich (*mauḍū'ah*), nämlich die "Hälfte", und acht sind abgeleitet (*mustaqqah*), nämlich das "Drittel" von drei und das "Viertel" von vier und das "Fünftel" von fünf und das "Sechstel" von sechs und das "Siebentel" von sieben und das "Achtel" von acht und das "Neuntel" von neun und das "Zehntel" von zehn; und was das allgemeine, unbestimmte Wort anlangt, so ist es der "Teil", weil das "Eins von Elf" ein "Teil" von elf genannt wird, und ebenso von dreizehn und von siebzehn und was ihm gleicht; und was die (dann noch) verbleibenden von

first to the relation  $\frac{2}{3}k - x = \frac{1}{3}k - t - \frac{1}{4}(\frac{1}{3}k - t) - d = \frac{1}{4}k - \frac{3}{4}t - d$ , then to the equation  $\frac{2}{3}k + \frac{1}{4}k - \frac{3}{4}t - d = \frac{11}{12}k - \frac{1}{4}t - d = 4t$ , whence by *al-gabr*  $\frac{11}{12}k = 4\frac{1}{4}t + d$  and by *ikmāl*  $k = \frac{51}{11}t + \frac{12}{11}d = 5\frac{8}{11}t + 1\frac{1}{11}d$ .

**The Arabic fractional numbers.** In S. Günther one finds on p. 200 the following description: "We are moreover told by the later al-Karkī and the still much later Behā ad-Dīn that one also calculated with common fractions. But only those were permitted as pronounceable whose denominators were units; if the denominator grew beyond 9, the fraction was mute, and then there existed for it only a circumlocutory designation."

Approximately the same is read in Cantor I<sup>3</sup>, p. 718, only that an explanation of the state of affairs correct on the whole, which Günther passed over, is added. Rodet even wished to find a similarity with the "making pronounceable" of fractions by conversion into a sum of unit fractions among the Egyptians; Cantor doubts whether the distinction already held for the time of Muhammad b. Mūsā.

It is high time that this purely philological discussion disappeared from the mathematical literature. The point is that Arabic from the cardinal numbers 3 to 10 forms special fractional numbers, such as *tuluṭ* third, *humus* fifth, *uṣṣur* tenth; these are the "pronounceable" fractions. For the tens, which appear as plurals of the units, corresponding formations are linguistically impossible; for composite numbers only when they can be decomposed into factors up to 10, so that e.g. a forty-eighth is called "a sixth of an eighth."

Obviously this distinction existed as long as there is an Arabic language. Already Muhammad b. Mūsā expresses himself on the state of affairs in his arithmetic book; the passage is found on p. 17 of the Latin adaptation and reads: *Scito quod fractiones [sic] appellantur multis nominibus in numerabilibus atque infinitis, ut medietas, tercia, quarta, nona et decima, et una pars ex xm., et pars ex xviii., et cetera.*

How one later philosophized about this is shown by a passage of the Iḥwān al-Ṣafā' (ed. Bombay, vol. I, p. 28): "And know, O brother ..., that the fractional number possesses many ranks (*marātib*), because there is no whole number that does not have one part or two parts or a number of parts, like twelve: for it has (comes to it) a half and a third and a quarter and a sixth and half a sixth, and likewise twenty-eight and others besides these two among numbers ..."

"And they are all encompassed by ten words: one word of them is general and indefinite (*mubhamah*) and nine are special and definite (*mafḥūmah*); of the nine words one is original (*mawḍū'ah*), namely the 'half,' and eight are derived (*mustaqqah*), namely the 'third' from three and the 'quarter' from four and the 'fifth' from five and the 'sixth' from six and the 'seventh' from seven and the 'eighth' from eight and the 'ninth' from nine and the 'tenth' from ten; and as for the general, indefinite word, it is 'part,' because 'one of eleven' is called a 'part' of eleven, and likewise of thirteen and of seventeen and what is similar; and as for the (then still) remaining words for fractions, they are attached (*muḍāfah*) to those ten words, as one says instead of one of twelve half of the sixth

den Wörtern für die Brüche anlangt, so sind sie angehängt (*mudāfah*) an jene zehn Wörter, wie man statt Eins von Zwölf die Hälfte des Sechstels sagt und statt Eins von Fünfzehn ein Fünftel des Drittels und statt Eins von Zwanzig die Hälfte des Zehntels usw.

Später heißt die Elf "die erste stumme Zahl", was dann weiter unten (S. 29, Z. 6 v. u.) damit erklärt wird, daß sie keinen aussprechbaren Teil besitzt.

**Der Terminus *shai'*.** Wieder einen Schritt weiter gelangen wir bei Aufgaben, in denen an Stelle der individualisierten Unbekannten ein allgemeingültiger, für jede Art gesuchter Größen verwendbarer technischer Ausdruck tritt.

Zwar kann auch *māl* als solcher gelten, sofern es sich um Bestimmung von Geldsummen handelt. Aber erst mit Einführung des Ausdrucks *شيء shai'* kommen wir wirklich zu dieser Verallgemeinerung, und es ist zu untersuchen, welche Bedeutung dem Wort, das als *cosa* "Sache, Ding" in die Geschichte der Algebra fortlebt, eigentlich zukommt.

Daß *shai'* ein "Ding", eine "Sache" bedeutet, ist ganz richtig. Aber warum wurde gerade dieses Wort zur Bezeichnung der Unbekannten gewählt? Man konnte zunächst an philosophische Allgemeinheiten denken. So beginnen die Iḥān a-ṣafā' ihre zahlentheoretischen Betrachtungen (a. a. O., Bd. I, S. 23) mit der Bemerkung: "Das allgemeinste der Worte und Namen (*nomina*) ist, wenn wir sagen "das Ding", und das "Ding" ist entweder eines oder mehr als eines" usw.

Aber von solchen Abstraktionen kann hier nicht die Rede sein. Näher kommen wir den Gründen für die Wahl des Ausdrucks schon, wenn wir beachten, daß das Wort in vielen Fällen geradezu im Sinne von *māl* "Besitz" gebraucht wird; so wenn es im Koran Sure VII, 83 ("ähnlich XI, 86, XXVI, 183) heißt: اوفوا الكيل والميزان ولا تبخسوا الناس اشيائهم "gebt volles Maß und Gewicht und schädigt die Leute nicht an ihren Sachen".

Das Wort bezeichnet aber in zahllosen Wendungen nicht ein Ding, sondern "irgend ein Ding", irgend eine Größe, etwas, z. B.: *شيء من الخراج shai' min al-kharāj*, etwas von dem Steuerertrag, *شيء من علمه bi-shai' min 'ilmihi*, "mit etwas von seinem Wissen", *شيء من الخوف shai' min al-khawf*, "etwas von der Furcht"; daher in Verbindung mit Negationen "nichts", z. B. *ما انت بشيء mā anta bi-shai'*, "du bist nichts", *لا تنفقوا شيئا lā tunfiqū shai'an*, "gebt nichts aus"; in Verbindung mit *kull* "alles", z. B. *شيء بصير kull shai' basīr*, "er ist allwissend", *شيء قدير shai' qadīr*, "allmächtig"; mit dem Fragewort *mā* "was", z. B. *ما اشد mā ašadda min*, "was ist größer als ...?"

Mit "Etwas" im Sinne einer beliebigen noch zu bestimmenden Zahl oder Größe ist daher *shai'* in allen mathematischen Aufgaben, wo es zur Bezeichnung der Unbekannten dient, zu übersetzen.

Ein drittes Beispiel aus den Erbteilungsrechnungen mag zugleich den Fortschritt in der Bezeichnung veranschaulichen:

"Und wenn er hinterlassen hat zwei Söhne und hinterlassen hat zehn Dirhem an Kapital und zehn Dirhem einer Schuldforderung gegen einen der Söhne und vermacht hat einem Manne ein Fünftel seines Vermögens und einen Dirhem, so ist die Rechenvorschrift, daß du, was von der Schuld herkommt (herausgebracht wird), "Etwas" nennst, und es zum Kapital hinzufügst, so

and instead of one of fifteen a fifth of the third and instead of one of twenty half of the tenth," etc.

Later, eleven is called "the first mute number," which is then explained further below (p. 29, line 6 from bottom) by the fact that it possesses no pronounceable part.

**The term *shay'*.** Yet a step further we come in problems in which, in place of the individualized unknowns, a generally valid technical expression occurs, usable for every kind of sought quantity.

Indeed, *māl* can also count as such, insofar as it concerns the determination of sums of money. But only with the introduction of the expression *شيء shay'* do we really arrive at this generalization, and it is to be investigated what meaning properly attaches to the word, which lives on in the history of algebra as *cosa*, "thing."

That *shay'* means a "thing," a "something," is quite correct. But why was precisely this word chosen to designate the unknown? One might at first think of philosophical generalities. Thus the Iḥwān al-Ṣafā' begin their number-theoretical considerations (op. cit., vol. I, p. 23) with the remark: "The most general of words and names (*nomina*) is when we say 'the thing,' and the 'thing' is either one or more than one," etc.

But such abstractions are not to be spoken of here. We come closer to the reasons for the choice of the expression already when we observe that the word in many cases is used directly in the sense of *māl*, "possession"; as when the Qur'an says Sura VII, 83 (similarly XI, 86, XXVI, 183): اوفوا الكيل والميزان ولا تبخسوا الناس اشيائهم "give full measure and weight and do not wrong people in their things."

But the word designates in countless phrases not a thing but "any thing," any quantity, something, e.g.: *شيء من الخراج shay' min al-kharāj*, something of the tax revenue, *شيء من علمه bi-shay' min 'ilmihi*, "with something of his knowledge," *شيء من الخوف shay' min al-khawf*, "something of fear"; hence in connection with negations "nothing," e.g. *ما انت بشيء mā anta bi-shay'*, "you are nothing," *لا تنفقوا شيئا lā tunfiqū shai'an*, "give out nothing"; in connection with *kull* "all," e.g. *شيء بصير kull shay' basīr*, "he is all-seeing," *شيء قدير shay' qadīr*, "all-powerful"; with the interrogative *mā* "what," e.g. *ما اشد mā ašadda min*, "what is greater than ...?"

With "something" in the sense of an arbitrary number or quantity yet to be determined, *shay'* is therefore to be translated in all mathematical problems where it serves to designate the unknown.

A third example from the inheritance calculations may at the same time illustrate the progress in notation:

"And if he has left two sons and left ten dirhams as capital and ten dirhams of a debt against one of the sons, and has bequeathed to a man one-fifth of his possession and one dirham, then the calculation rule is that you call what comes from the debt (is brought out) 'Something,' and add it to the capital, so it gives Something and ten dirhams. Then you deduct one-fifth of it, because he be-



gibt es Etwas und zehn Dirhem. Dann ziehst du ein Fünftel davon ab, weil er ein Fünftel seines Vermögens vermacht hat, und das sind zwei Dirhem und ein Fünftel Etwas, so bleiben acht Dirhem und vier Fünftel Etwas.

Dann ziehst du den Dirhem ab, den er vermacht hat, so bleiben sieben Dirhem und vier Fünftel Etwas; dann teilst du es zwischen den beiden Söhnen, so bekommt ein jeder drei Dirhem und die Hälfte eines Dirhem und zwei Fünftel Etwas [und das ist gleich dem Etwas. Gleiche nun damit aus, indem du zwei Fünftel Etwas] von Etwas wegnimmst, so bleiben drei Fünftel Etwas gleich drei Dirhem und einem halben. Vervollständige nun das Etwas, und das besteht darin, daß du ihm zfügst zwei Drittel desselben und zfügst den Dreiundeinhalb zwei Drittel desselben, und das sind zwei Dirhem und ein Drittel, so gibt das fünf und fünf Sechstel und das ist das Etwas, was von der Schuld herkommt."

Das ist in unsere Zeichen übertragen:  $(x + 10) - \frac{1}{5}(x + 10) = 8 + \frac{4}{5}x$ ;  $8 + \frac{4}{5}x - 1 = 7 + \frac{4}{5}x$ ;  $3\frac{1}{2} + \frac{2}{5}x = x$ ;  $3\frac{1}{2} = \frac{3}{5}x$ . Zur Vervollständigung werden  $\frac{2}{3}$  jeder Seite addiert, also ist  $x = 3\frac{1}{2} + \frac{2}{3} \cdot 3\frac{1}{2} = 5\frac{5}{6}$ .

Es ist bemerkenswert, daß in dem Buch von der Erbteilung nur in dem Kapitel "Von der Rückerstattung" und in den drei Aufgaben, die zu dem Kapitel "Von dem Kapital und der Schuld" gehören, von der Bezeichnung der Unbekannten durch das Wort *shai'* Gebrauch gemacht wird. In dem Kapitel "Von der Vervollständigung" wird dafür regelmäßig gesagt *اجعل mālan*, "nimm irgend ein Vermögen an", in andern Fällen wird sofort eine Zahl, die einfache Ergebnisse liefert, vorgeschlagen. Es ist nicht zu verkennen, daß bei den verschiedenen Aufgabengruppen verschiedene Schultraditionen oder literarische Quellen vorliegen, und es wäre gewiß auch für die Geschichte der Mathematik nicht ohne Bedeutung, wenn man sich dieser Erbteilungsaufgaben mit mehr Eifer als bisher annehmen würde.

Fassen wir das Ergebnis zusammen, so fanden wir als natürlichen und ursprünglichen Ausdruck für die Unbekannte das Wort *māl* "Vermögen", als Maßeinheit gegebener Geldsummen den Dirhem. Neben dem Vermögen oder dem Nachlaß kann eine Anzahl anderer Begriffe als Unbekannte auftreten, die entweder auf Grund der gegebenen Beziehungen zum Vermögen oder durch willkürliche Annahmen beseitigt werden. Die Einführung des Wortes *shai'* verleiht der Ausdrucksweise einen allgemeineren Charakter, ist aber für die behandelten Aufgaben ziemlich gleichgültig.

bequeathed one-fifth of his possession, and that is two dirhams and one-fifth Something, so eight dirhams and four-fifths Something remain.

Then you deduct the dirham that he bequeathed, so seven dirhams and four-fifths Something remain; then you divide it between the two sons, so each gets three dirhams and half a dirham and two-fifths Something [and that equals the Something. Balance now thereby, by taking away two-fifths Something] from Something, so three-fifths Something equal three dirhams and a half. Complete now the Something, and that consists in adding to it two-thirds of it and adding to the three-and-a-half two-thirds of it, and that is two dirhams and a third, so that gives five and five-sixths and that is the Something that comes from the debt."

Transferred into our notation:  $(x + 10) - \frac{1}{5}(x + 10) = 8 + \frac{4}{5}x$ ;  $8 + \frac{4}{5}x - 1 = 7 + \frac{4}{5}x$ ;  $3\frac{1}{2} + \frac{2}{5}x = x$ ;  $3\frac{1}{2} = \frac{3}{5}x$ . For completion,  $\frac{2}{3}$  of each side are added, so  $x = 3\frac{1}{2} + \frac{2}{3} \cdot 3\frac{1}{2} = 5\frac{5}{6}$ .

It is remarkable that in the Book of Inheritance, only in the chapter "On Restitution" and in the three problems belonging to the chapter "On Capital and Debt" is the designation of the unknown by the word *shay'* used. In the chapter "On Completion" it is regularly said instead *اجعل mālan*, "take any possession as assumed," in other cases a number that yields simple results is proposed immediately.

It cannot be denied that different school traditions or literary sources underlie the various problem groups, and it would certainly not be without significance for the history of mathematics if one applied oneself to these inheritance problems with more zeal than hitherto.

Summarizing the result, we found as the natural and original expression for the unknown the word *māl* "possession," as the unit of measure of given sums of money the dirham. Besides the possession or the estate, a number of other concepts can appear as unknowns, which are eliminated either on the basis of the given relations to the possession or through arbitrary assumptions. The introduction of the word *shay'* lends the mode of expression a more general character, but is fairly indifferent for the problems treated.

## 7 VII. Die Terminologie der quadratischen Gleichungen / The Terminology of Quadratic Equations

Geometrische Aufgaben über Zusammensetzung und Verwandlung von Flächen werden den Griechen den ersten Anstoß zur Behandlung und Lösung quadratischer Gleichungen gegeben haben. Bei Muhammad b. Mūsā finden wir die geometrische Konstruktion nur als Mittel der Veranschaulichung, als Beweis für die Richtigkeit der gegebenen Regeln. Die ganze übrige Darstellung aber und der Charakter der Aufgaben ist rein arithmetisch.

Geometrical problems on the composition and transformation of surfaces will have given the Greeks the first impulse toward the treatment and solution of quadratic equations. In Muhammad b. Mūsā we find geometrical construction only as a means of illustration, as proof of the correctness of the given rules. The entire remaining presentation, however, and the character of the problems is purely arithmetical.

Wir begegnen am Anfang der Algebra den berühmten Definitionen:

وجدت الأرقام التي يحتاج إليها في حساب الجبر والمقابلة على ثلاثة أضرب وهي أصاغ وجذور وأعداد مفردة ليست من جذر ولا مال فاما الجذور فهو كل شيء فيه نفسه مضروب من الواحد والذي فوقه من الأعداد والذي دونه من الكسور واما المال فكل ما حصل من جذر في نفسه واما العدد المفرد فهو كل عدد يتكلم به ليس فيه جذر ولا مال

Nach Rosen: "I observed that the numbers which are required in calculating by Completion and Reduction are of three kinds, namely, roots, squares, and simple numbers relative to neither root nor square. — A root is any quantity which is to be multiplied by itself, consisting of units, or numbers ascending, or fractions descending. — A square is the whole amount of the root multiplied by itself. — A simple number is any number which may be pronounced without reference to root or square. — A number belonging to one of these three classes may be equal to a number of another class; you may say, for instance, 'squares are equal to roots', or 'squares are equal to numbers', or 'roots are equal to numbers'."

Wörtlich: "Und ich fand die Zahlen, deren man beim Rechenverfahren der Ergänzung und Ausgleichung bedarf, gemäß drei Arten, und zwar sind es Wurzeln (*gud'ur*) und Vermögen (*amwāl*) und absolute Zahl (*adad mufrad*), die nicht bezogen wird auf eine Wurzel und nicht auf ein Vermögen. Die Wurzel (*al-gidr*) nun, zu ihr (gehört) alles, was mit sich selbst multipliziert ist von der Eins, und was über ihr (der Eins) ist von den Zahlen, und was unter ihr ist von den Brüchen. Und das Vermögen (*al-māl*) ist alles, was sich ergibt (ansammelt, *ijtamā'u*) aus der mit sich selbst multiplizierten Wurzel. Und die absolute Zahl ist alles von der Zahl, was ausgesprochen wird ohne Beziehung auf eine Wurzel und ein Vermögen."

"Unter diesen drei Arten (gibt es) nun, was einander gleich ist, nämlich wie wenn du sagst 'Vermögen sind gleich Wurzeln', und 'Vermögen sind gleich einer Zahl', und 'Wurzeln sind gleich einer Zahl'."

Eine Anmerkung Rosens sagt uns noch, daß unter dem Wort *root* die erste Potenz der Unbekannten zu verstehen sei. Das Wort *square*, lesen wir anderwärts (S. 50, Note \*), *is used in the text to signify either, 1st, a square, properly so called, fractional or integral; 2d, a rational integer, not being a square number; 3d, a rational fraction, not being a square; 4th, a quadratic surd, fractional or integral.*

Mit diesen von ihm selbst aufgetürmten Schwierigkeiten ist Rosen daher genötigt, sich jedesmal in einer Fußnote zu entschuldigen, wenn er vermünftigerweise *māl* mit *number* übersetzt (S. 51, 54, 58, 60, 61, 62, 63) oder gar *square-root* an die Stelle von *square* setzen muß (S. 53, 62, 64).

Sollen wir nun wirklich glauben, daß der Schöpfer der arabischen Terminologie einen solchen Wirrwarr von Begriffen auf das Wort *māl* gehäuft hätte? Sollte es nicht möglich sein, aus dem Labyrinth einen Ausweg zu finden?

Wir werden methodisch richtig verfahren, wenn wir zunächst die Bedeutung des neu hinzugekommenen Terminus *gidr* klärstellen. Bis zum Beweis des Gegenteils werden wir annehmen dürfen,

We encounter at the beginning of the Algebra the famous definitions:

وجدت الأرقام التي يحتاج إليها في حساب الجبر والمقابلة على ثلاثة أضرب وهي أصاغ وجذور وأعداد مفردة ليست من جذر ولا مال فاما الجذور فهو كل شيء فيه نفسه مضروب من الواحد والذي فوقه من الأعداد والذي دونه من الكسور واما المال فكل ما حصل من جذر في نفسه واما العدد المفرد فهو كل عدد يتكلم به ليس فيه جذر ولا مال

According to Rosen: "I observed that the numbers which are required in calculating by Completion and Reduction are of three kinds, namely, roots, squares, and simple numbers relative to neither root nor square. — A root is any quantity which is to be multiplied by itself, consisting of units, or numbers ascending, or fractions descending. — A square is the whole amount of the root multiplied by itself. — A simple number is any number which may be pronounced without reference to root or square. — A number belonging to one of these three classes may be equal to a number of another class; you may say, for instance, 'squares are equal to roots,' or 'squares are equal to numbers,' or 'roots are equal to numbers.'"

Literally: "And I found the numbers needed in the calculation procedure of completion and balancing to be of three kinds, namely roots (*gudūr*) and possessions (*amwāl*) and absolute number (*adad mufrad*), which is not referred to a root and not to a possession. The root (*al-jidhr*) now, to it belongs everything that is multiplied with itself, from one, and what is above it of numbers, and what is below it of fractions. And the possession (*al-māl*) is everything that results (accumulates, *ijtamā'u*) from the root multiplied with itself. And the absolute number is everything of number that is pronounced without reference to a root and a possession."

"Among these three kinds there is now what is equal to one another, namely as when you say 'possessions are equal to roots,' and 'possessions are equal to a number,' and 'roots are equal to a number.'"

A note of Rosen's tells us further that by the word *root* the first power of the unknown is to be understood. The word *square*, we read elsewhere (p. 50, n. \*), "is used in the text to signify either, 1st, a square, properly so called, fractional or integral; 2d, a rational integer, not being a square number; 3d, a rational fraction, not being a square; 4th, a quadratic surd, fractional or integral."

With these difficulties heaped up by himself, Rosen is therefore compelled to excuse himself each time in a footnote when he reasonably translates *māl* by *number* (pp. 51, 54, 58, 60, 61, 62, 63) or must even put *square-root* in place of *square* (pp. 53, 62, 64).

Are we really to believe that the creator of Arabic terminology had heaped such a confusion of concepts upon the word *māl*? Should it not be possible to find a way out of the labyrinth?

We shall proceed methodically correctly if we first clarify the meaning of the newly added term *gidr*. Until proof of the contrary, we shall be permitted to assume that the word has a clear and

daß das Wort einen klaren und eindeutigen Sinn hat. Dieser Sinn muß derart sein, daß er widerspruchlos durch den ganzen Text hindurch best"atigt wird.

Nun ist gewiß, daß der Zusammenhang zwischen *māl* und *gidr* ebensogut durch  $y \cdot y = x^2$  als durch  $\sqrt{y} \cdot \sqrt{y} = x$  in unsern algebraischen Zeichen ausgedr"uckt werden kann; von der Definition aus ist also keine Entscheidung zu gewinnen.

Wer trotz aller Schwierigkeiten nicht davon abl"äßt, *māl* mit *Quadrat* zu "übersetzen, wird mit Rosen entgegen dem Wortsinn *gidr* als "erste Potenz" deuten müssen; er kommt aber dann zu einer terminologischen Schwierigkeit, denn eine "erste" Potenz ist für Altertum und Mittelalter ein Unding.

Wer aber *māl* in seinem natürlichen Sinne, den es überall hat, weiter anwendet, hat auch für *gidr* keine Umdeutung notwendig. Nun wird *gidr* im Sinne von *Quadratwurzel* aus einer Zahl unzweifelhaft angewandt und festgelegt durch die Beispiele des Kapitels "über Addition und Subtraktion (S. 19). Die Wurzel aus 200 kann auch Rosen nicht in die "erste Potenz einer Unbekannten" umdeuten, und wenn er das Beispiel مائة ناقص جذور ناقص مائة mit "A hundred and a square minus twenty roots, added to fifty and ten roots minus two squares" wiedergibt, so hat er entweder *māl* oder *gidr* falsch "übersetzt.

Ebenso verfehlt ist die "Übersetzung von اذا اردت ان تضعف جذر اذا اردت ان تضعف جذر ان تضعف جذر فاكفيك ان تضعفه في اثنين فاكفيك ان تضعفه في اثنين ثم في المال mit "If you require to double the root of any known or unknown square (the meaning of its duplication being that you multiply it by two) then it will suffice to multiply two by two and then by the square"; denn es muß heißen "jede rationale oder irrationale Wurzel aus einer Zahl".<sup>10</sup>

<sup>10</sup>Warum Rosen seine "Übersetzung S. 27 in einer Note S. 192 verbessert, ohne darauf aufmerksam zu machen, daß er jetzt mit "the root of a rational or irrational number" etwas ganz anderes sagt, ist mir nicht klar geworden; noch weniger, warum er trotz solcher sich aufdringenden Erfahrungen hartnäckig dabei bleibt, *māl* mit *square* wiederzugeben.

Alle Schwierigkeiten verschwinden mit einem Schlage, wenn wir *māl*, das Vermögen, mit *Zahl*, oder, um einer Verwechslung mit der "absoluten Zahl" auszuweichen, mit *Zahlgröße*, und *gidr* mit *Wurzel* (im eigentlichen Sinne) "übersetzen. So und nicht anders sind die Ausdrücke in der Algebra des Muhammad b. Mūsā zu verstehen; ob und wie sich an sie eine Weiterentwicklung anschließt, wird noch zu untersuchen sein.

unambiguous sense. This sense must be such that it is confirmed without contradiction throughout the entire text.

Now it is certain that the connection between *māl* and *gidr* can be expressed in our algebraic notation just as well by  $y \cdot y = x^2$  as by  $\sqrt{y} \cdot \sqrt{y} = x$ ; from the definition, therefore, no decision is to be won.

Whoever despite all difficulties does not desist from translating *māl* as "square" will have to interpret *gidr* against the word sense as "first power" like Rosen; but he then arrives at a terminological difficulty, for a "first" power is an absurdity for antiquity and the Middle Ages.

But whoever applies *māl* further in its natural sense, which it has everywhere, has no need of reinterpretation for *gidr* either. Now *gidr* in the sense of "square root" of a number is applied without doubt and established by the examples of the chapter on addition and subtraction (p. 19). The root of 200 cannot be reinterpreted by Rosen into the "first power of an unknown" either, and when he renders the example مائة ناقص جذرا مضافة الى مائة ناقص جذور ناقص مائة as "A hundred and a square minus twenty roots, added to fifty and ten roots minus two squares," he has translated either *māl* or *gidr* incorrectly.

Equally wrong is the translation of اذا اردت ان تضعف جذر كل مال اذا اردت ان تضعف جذر كل مال ان تضعفه في اثنين فاكفيك ان تضعفه في اثنين ثم في المال by "If you require to double the root of any known or unknown square (the meaning of its duplication being that you multiply it by two) then it will suffice to multiply two by two and then by the square"; for it must say "every rational or irrational root of a number."<sup>10</sup>

<sup>10</sup>Why Rosen improves his translation on p. 27 in a note on p. 192 without calling attention to the fact that he now says something entirely different with "the root of a rational or irrational number," has not become clear to me; still less why, despite such pressing experiences, he stubbornly continues to render *māl* by *square*.

All difficulties disappear at a stroke if we translate *māl*, the possession, by "number," or, to avoid confusion with the "absolute number," by "number-quantity," and *gidr* by "root" (in the proper sense). Thus and not otherwise are the expressions in the Algebra of Muhammad b. Mūsā to be understood; whether and how a further development connects to them remains to be investigated.

## 8 VIII. Zum Aufbau des Zahlensystems / On the Structure of the Number System

Die Sätze, die Muhammad b. Mūsā als Einleitung den S. 61 zitierten Definitionen voranschickt, haben verschiedene Deutung erfahren. Ich wiederhole sie im Urtext und wörtlicher "Übersetzung, um daran einige Bemerkungen zu knüpfen:

واني لما نظرت فيما يحتاج اليه الناس من الحساب وجدتها كلها الاعداد ووجدتها كلها مؤلفة من الواحد وان الواحد يدخل في جميع الاعداد ووجدت جميع ما يقال من العدد من الزيادة على الواحد الى عشرة زائدا بواحدة ثم تضاعف العشرة وتثلث كما فعل بالواحد فيخرج منها العشرون والثلاثون الى المئة ثم تضاعف المئة وتثلث كما فعل بالواحد والعشرة الى

واني لما نظرت فيما يحتاج اليه الناس من الحساب وجدتها كلها الاعداد ووجدتها كلها مؤلفة من الواحد وان الواحد يدخل في جميع الاعداد ووجدت جميع ما يقال من العدد من الزيادة على الواحد الى عشرة زائدا بواحدة ثم تضاعف العشرة وتثلث كما فعل بالواحد فيخرج منها العشرون والثلاثون الى المئة ثم تضاعف المئة وتثلث كما فعل بالواحد والعشرة الى

الف ثم يعاد الف عند كل عقد الى غاية العدد

الف ثم يعاد الف عند كل عقد الى غاية العدد

Wörtlich: "Und siehe, nachdem ich betrachtet hatte, wessen die Leute beim Rechnen bedurfen, fand ich als Gesamtheit davon die Zahlen, und ich fand, daß die Gesamtheit der Zahlen nur zusammengesetzt ist aus der Eins, und die Eins eintretend in die Gesamtheit der Zahlen. Und ich fand die Gesamtheit dessen, was von den Zahlen ausgesprochen wird (d. i. wofür es besondere Zahlwörter gibt), was über die Eins hinausgeht bis zur Zehn, fortschreitend um den Betrag der Eins; dann wird die Zehn verdoppelt und verdreifacht, wie es getan wurde mit der Eins, so daß daraus die Zwanzig und Dreißig entstehen bis zur Vollendung der Hundert; hierauf wird die Hundert verdoppelt und verdreifacht, wie es getan wurde mit der Eins und mit der Zehn bis zu Tausend usw.; dann wird ebenso wiederholt die Tausend bei jedem Glied (*'uqd*) bis zum "Außersten des Bereichs der Zahl."

In den Sätzen "über die Eins ist nichts zu finden, was über die elementarste Schulweisheit hinausgeht. Sie werden bekanntlich in der durch B. Boncompagni (Roma 1857) herausgegebenen lateinischen Übersetzung von Muhammad b. Mūsā's Rechenbuch in Verbindung mit einer weiteren Bemerkung "über die Einheit zitiert. Cantor hat (I<sup>3</sup>, S. 715 u. 16) die Stelle wie folgt deutsch wiedergegeben:

"Und ich habe schon in dem Buche Aldschebr und Almukābala, d. h. der Wiederherstellung und Gegenüberstellung eröffnet, daß jede Zahl zusammengesetzt sei, und daß jede Zahl sich über eins zusammensetze. Die Einheit also wird in jeder Zahl gefunden, und das ist es, was in einem andern Buche der Arithmetik ausgesprochen ist. Weil die Einheit Wurzel jeder Zahl und außerhalb der Zahl ist ..."

Man muß aber, um den Sinn richtig zu erhalten, folgendermaßen lesen: "... daß jede Zahl sich über eins zusammensetze, die Einheit also in jeder Zahl gefunden werde. — Und das ist es, was in einem andern Buche der Arithmetik ausgesprochen ist: daß die Einheit Wurzel jeder Zahl und außerhalb der Zahl ist ..." Das Zitat aus der "Algebra" geht nur bis "gefunden werde".

Eneström hat sich durch die freie Übersetzung Rosens irreleiten lassen, wenn er das Äquivalent der Stelle weiter unten in "every number which may be expressed from one to ten, surpasses the preceding by one unit" usw. sucht. Das folgende Stück aber ist ein klares und unzweideutiges Zitat aus einem "andern Buche über Arithmetik". Was für ein Buch dies ist und ob Muhammad b. Mūsā damit auf ein von ihm selbst verfaßtes Buch über spekulative Arithmetik anspielt, ist die Frage.

Nach Eneström mußte man annehmen, daß Cantor den Muhammad b. Mūsā zum Verfasser eines solchen Buches gemacht hätte. Davon ist aber doch nirgends die Rede; Cantor schließt nur, daß, wer so schrieb, "in der Zahlenlehre der Neopythagoräer wohl geschult sein mußte", und "daß dem Verfasser darüber Kenntnisse zu Gebote standen, welche unmittelbar oder mittelbar auf Nikomachos, vielleicht auch auf Theon von Smyrna, der am deutlichsten betont hat, die Einheit sei keine Zahl, zurückgehen".

Welche Bewandnis es in Wahrheit mit der Stelle hat, erkennt man aber sofort, wenn man sie in ihrem gesamten Umfang bezieht. Sie lautet:

Literally: "And behold, after I had considered what people need in calculation, I found as the totality thereof the numbers, and I found that the totality of numbers is only composed of the one, and the one entering into the totality of numbers. And I found the totality of what is pronounced of numbers (i.e. for which there are special number words) that goes beyond one up to ten, advancing by the amount of one; then ten is doubled and tripled, as was done with one, so that twenty and thirty arise from it up to the completion of hundred; then hundred is doubled and tripled, as was done with one and with ten, up to thousand, etc.; then likewise thousand is repeated at every node (*'uqd*) up to the utmost of the domain of number."

In the sentences about the one there is nothing to be found that goes beyond the most elementary school wisdom. They are, as is well known, cited in the Latin translation of Muhammad b. Mūsā's arithmetic book edited by B. Boncompagni (Rome 1857) in connection with a further remark on unity. Cantor has (I<sup>3</sup>, p. 715) rendered the passage in German as follows:

"And I have already revealed in the book al-Jabr and al-Muqābala, that is, of restoration and opposition, that every number is composite, and that every number is composed above one. Unity therefore is found in every number, and that is what is pronounced in another book of arithmetic. Because unity is the root of every number and is outside the number ..."

But to obtain the sense correctly, one must read as follows: "... that every number is composed above one, that unity therefore be found in every number.—And that is what is pronounced in another book of arithmetic: that unity is the root of every number and outside the number ..." The quotation from the "Algebra" extends only to "be found."

Eneström has allowed himself to be misled by Rosen's free translation when he seeks the equivalent of the passage further below in "every number which may be expressed from one to ten, surpasses the preceding by one unit," etc. But the following piece is a clear and unambiguous quotation from an "other book on arithmetic." What book this is and whether Muhammad b. Mūsā is alluding thereby to a book on speculative arithmetic composed by himself is the question.

According to Eneström one would have to assume that Cantor made Muhammad b. Mūsā the author of such a book. But that is nowhere mentioned; Cantor only infers that whoever wrote thus "must have been well trained in the number theory of the Neopythagoreans," and "that the author had at his disposal knowledge of this which goes back directly or indirectly to Nicomachus, perhaps also to Theon of Smyrna, who most clearly emphasized that unity is not a number."

What the true state of affairs is with this passage one recognizes immediately when one takes it in its entire scope. It reads:

*Et iam patefeci in libro algebre et almucabolah, id est*



*Et iam patefeci in libro algebre et almucabolah, id est restaurationis et oppositionis, quod universus numerus sit compositus, et quod universus numerus componatur super unum. Unum ergo invenitur in universo numero. Et hoc est quod in alio libro arithmetice dicitur: Quia unum est radix universi numeri, et est extra numerum. Radix numeri est, quare per eum invenitur omnis numerus. Extra numerum vero est, quare invenitur per se, idest absque alio aliquo numero. Reliquus autem numerus sine uno inveniri non potest. Cum enim unum dicis, inventionem sui non indiget alio numero. Reliquus autem numerus indiget [indiget] uno: quare non potes dicere duo vel tria, nisi precedat unum. Nichil aliud est ergo numerus, nisi unitatum collectio; et hoc quod diximus non potes dicere duo vel tria, nisi precedat unum: non de voce diximus, ut ita dicam, sed de re. Non enim possunt esse duo vel tria, si unum auferatur. Vnum vero potest esse absque secundo vel tercio. Igitur nichil aliud sunt duo, nisi unius duplicitas vel geminatio: et similiter tria nichil aliud sunt, nisi eiusdem unitatis triplicatio: sic de reliquo numero intellige. Set nunc redeamus ad librum. Inveni, inquit algorizmi, omne quod potest dici ex numero, et esse quicquid excedit unum usque in .IX. [lies X], idest quod est inter .IX. [lies X] et unum, idest duplicatur unum et fiunt duo; et triplicatur idem unum, fiuntque tria, et sic in ceteris usque in .IX.*

Hier ist doch mit H"anden zu greifen, daß die ganze Stelle ein Einschub des "Übersetzers, oder wie wir jetzt richtiger sagen werden, des Bearbeiters des Rechenbuchs ist, und daß wir Muhammad b. M"usā f"ur diese scholastische Haarspalterei nicht verantwortlich machen d"urfen.

So deutlich sich der Zusatz als solcher durch Inhalt und sprachliche Form verr"at, so deutlich ist das folgende wieder ein Zitat aus Muhammad b. M"usās eigener Algebra, das sich zu einer umst"andlicheren Behandlung des Gegenstandes erweitert. Man braucht nur Text und "Übersetzung der zweiten H"älfte des S. 71 mitgeteilten Zitats neben den Text des Rechenbuchs zu stellen, um sich davon zu "überzeugen:

*Inveni omne quod potest dici ex numero, et esse quicquid excedit unum usque in .IX. [lies X], idest quod est inter .IX. [lies X] et unum, idest duplicatur unum et fiunt duo; et triplicatur idem unum, fiuntque tria, et sic in ceteris usque in .IX. De inde ponuntur .X. in loco unius, et duplicantur .X. ac triplicantur, quemadmodum factum est de uno, fiuntque ex eorum duplicatione .XX., ex triplicatione .XXX., et ita usque ad .XC. Post hec redeunt .C. in loco unius, et duplicantur ibi atque triplicantur quemadmodum factum est de uno et .X. efficiunturque ex eis .CC. et .CCC. et cetera usque in .DCCCC. Rursum ponuntur mille in loco unius; et duplicando et triplicando, ut diximus, fiunt ex eis .II. milia, et .III. et cetera usque in Infinitum numerum, secundum hunc modum. — Et inveni quod operati sunt yndi ex his differentiis. Quarum prima est*

*restaurationis et oppositionis, quod universus numerus sit compositus, et quod universus numerus componatur super unum. Unum ergo invenitur in universo numero. Et hoc est quod in alio libro arithmetice dicitur: Quia unum est radix universi numeri, et est extra numerum. Radix numeri est, quare per eum invenitur omnis numerus. Extra numerum vero est, quare invenitur per se, idest absque alio aliquo numero. Reliquus autem numerus sine uno inveniri non potest. Cum enim unum dicis, inventionem sui non indiget alio numero. Reliquus autem numerus indiget [indiget] uno: quare non potes dicere duo vel tria, nisi precedat unum. Nichil aliud est ergo numerus, nisi unitatum collectio; et hoc quod diximus non potes dicere duo vel tria, nisi precedat unum: non de voce diximus, ut ita dicam, sed de re. Non enim possunt esse duo vel tria, si unum auferatur. Vnum vero potest esse absque secundo vel tercio. Igitur nichil aliud sunt duo, nisi unius duplicitas vel geminatio: et similiter tria nichil aliud sunt, nisi eiusdem unitatis triplicatio: sic de reliquo numero intellige. Set nunc redeamus ad librum. Inveni, inquit algorizmi, omne quod potest dici ex numero, et esse quicquid excedit unum usque in .IX. [lies X], idest quod est inter .IX. [lies X] et unum, idest duplicatur unum et fiunt duo; et triplicatur idem unum, fiuntque tria, et sic in ceteris usque in .IX.*

Here it is palpable that the whole passage is an insertion by the translator, or as we shall now more correctly say, by the adapter of the arithmetic book, and that we must not hold Muhammad b. Mūsā responsible for this scholastic hair-splitting.

As clearly as the addition reveals itself as such through content and linguistic form, so clearly is what follows again a quotation from Muhammad b. Mūsā's own Algebra, which has expanded to a more circumstantial treatment of the subject. One needs only to place the text and translation of the second half of the quotation communicated on p. 71 beside the text of the arithmetic book in order to convince oneself of this:

*Inveni omne quod potest dici ex numero, et esse quicquid excedit unum usque in .IX. [lies X], idest quod est inter .IX. [lies X] et unum, idest duplicatur unum et fiunt duo; et triplicatur idem unum, fiuntque tria, et sic in ceteris usque in .IX. De inde ponuntur .X. in loco unius, et duplicantur .X. ac triplicantur, quemadmodum factum est de uno, fiuntque ex eorum duplicatione .XX., ex triplicatione .XXX., et ita usque ad .XC. Post hec redeunt .C. in loco unius, et duplicantur ibi atque triplicantur quemadmodum factum est de uno et .X. efficiunturque ex eis .CC. et .CCC. et cetera usque in .DCCCC. Rursum ponuntur mille in loco unius; et duplicando et triplicando, ut diximus, fiunt ex eis .II. milia, et .III. et cetera usque in Infinitum numerum, secundum hunc modum. — Et inveni quod operati sunt yndi ex his differentiis. Quarum prima est*

*differentia unitatum etc.*

Die "übereinstimmenden Stellen sind, soweit möglich, durch Kursivdruck hervorgehoben. Vergleichung im einzelnen zeigt, daß zwei verschiedene Betrachtungsweisen zusammengearbeitet sind. In der "älteren Algebra betont der Verfasser den sprachlichen Aufbau des Zahlensystems, nennt also die zehn Grundzahlen und die Grenzzahlen 100 und 1000; die "Rechenkunst der Inder" sieht mehr auf die Klassen der Zahlen, gruppiert also nach Einern, Zehnern, Hunderten, Tausendern usw.

Richtiger wäre es, wenn es hieße: *usque in X, C, M*; denn nur dann kann man sagen *deinde ponuntur X, post hec redeunt C, rursum ponuntur M*. Am Schluß vermißt man die "Übersetzung von عند كل عقد 'inda kuli 'uqdān "bei jeder Gliedzahl". Mit dem folgenden Satz wird das Thema des Rechenbuchs, wie es von Muhammad b. Mūsā zu Anfang durch *Cum vidi ssem yndos constituisse .IX. literas in universo numero suo* angeschlagen ist, wieder aufgenommen.

*differentia unitatum etc.*

The corresponding passages are indicated by italics as far as possible. Comparison in detail shows that two different points of view have been worked together. In the older Algebra, the author emphasizes the linguistic structure of the number system, thus naming the ten basic numbers and the limit numbers 100 and 1000; the "calculation art of the Indians" looks more to the classes of numbers, thus grouping according to units, tens, hundreds, thousands, etc.

It would be more correct if it said: *usque in X, C, M*; for only then can one say *deinde ponuntur X, post hec redeunt C, rursum ponuntur M*. At the end one misses the translation of عند كل عقد 'inda kulli 'uqdān "at every node-number." With the following sentence, the theme of the arithmetic book, as struck by Muḥammad b. Mūsā at the beginning through *Cum vidissem yndos constituisse .IX. literas in universo numero suo*, is taken up again.

## 9 IX. Die Namen der arabischen Ziffern / The Names of the Arabic Digits

In einer seiner Anfragen macht Eneström darauf aufmerksam, daß in der *Pratica d'aritmetica* von Ghaligai (Ausgabe von 1548) die "Übersetzung einer arabischen Algebra erwähnt werde, in welcher "die 7 Terme Geber, Elmechel, Elchal, Elchelif, Elfazial, Buram und Eltermen" vorkämen.

Nach einer Mitteilung von Suter enthalte die Algebra des Alkhwārizmī jedenfalls nicht die Terme Elchelif, Buram und Eltermen; es sei daher wünschenswert, die jetzt bekannten arabischen Traktate "über Algebra daraufhin zu untersuchen, ob sie die 7 Terme enthielten. Eneström sagt leider nicht, wie Suter die "übrigen 4 Terme auflöst. Man sieht aber leicht, daß außer dem selbstverständlichen *geber* noch *elmechel* = *al-muqābala* ist und daß *elfazial* aus *al-faṣl* (der Unterschied), *elchal* durch Verlesen aus *al-māl* (das Vermögen) entstanden sein kann.

Ich würde vielleicht noch wagen, *eltermen* mit *al-ikmāl* (die Vollständigkeit) zusammenzustellen, in *elchelif* könnte *al-ḥadḥ* (die Wegnahme) verborgen sein, aber *buram* ist hoffnungslos verderbt und trotz jedem Versuch der Wiederherstellung, solange nicht die "Übersetzung selbst vorliegt und der Zusammenhang Ruckschlüsse auf das arabische Wort ermöglicht.

Das Ganze ist ein lehrreiches Beispiel für das, was man an Verstümmelungen zu erwarten hat, wenn arabische Wörter in lateinische Schrift umgesetzt und unverstanden weiter und weiter überliefert wurden.

Wesentlich einfacher liegt die Aufgabe der Wiederherstellung des arabischen Ausdrucks, wenn die verstümmelten Wörter in lateinischen Texten erscheinen. Wenn uns z. B. in einer Beschreibung des Astrolabs, die Gerbert zugeschrieben wird, das Wort *Hoto talzagad* mit den Varianten *hotetalgazat*, *hottottaltagath*, *hotaltagab*, *hotetalsagat*, *noto taltagad* begegnet, so würden wir ohne den lateinischen Text schwerlich imstande sein, den zugrundeliegenden arabischen Ausdruck zu finden.

Lesen wir aber danach: *id est breves lineae horarum*, so ergibt sich ohne weiteres die Auflösung خطوط الساعات *huṭūt as-sā'āt* =

In one of his queries, Eneström calls attention to the fact that in the *Pratica d'aritmetica* of Ghaligai (edition of 1548) the translation of an Arabic algebra is mentioned, in which "the 7 terms Geber, Elmechel, Elchal, Elchelif, Elfazial, Buram and Eltermen" occurred.

According to a communication from Suter, the Algebra of al-Khwārizmī in any case does not contain the terms Elchelif, Buram, and Eltermen; it would therefore be desirable to investigate the Arabic treatises on algebra now known as to whether they contained the 7 terms. Eneström unfortunately does not say how Suter resolves the remaining 4 terms. But one easily sees that besides the self-evident *geber* [*al-jabr*], *elmechel* = *al-muqābala*, and that *elfazial* could have arisen from *al-faṣl* (the difference), *elchal* by misreading from *al-māl* (the possession).

I would perhaps further venture to combine *eltermen* with *al-ikmāl* (the completion), in *elchelif* could be concealed *al-khadḥ* (the removal), but *buram* is hopelessly corrupted and defies every attempt at restoration as long as the translation itself is not available and the context does not permit conclusions about the Arabic word.

The whole is an instructive example of what one may expect in the way of mutilations when Arabic words are converted into Latin script and handed down further and further without being understood.

The task of restoring the Arabic expression is substantially simpler when the mutilated words appear in Latin texts. If, for example, in a description of the astrolabe attributed to Gerbert, we encounter the word *Hoto talzagad* with the variants *hotetalgazat*, *hottottaltagath*, *hotaltagab*, *hotetalsagat*, *noto taltagad*, without the Latin text we would scarcely be able to find the underlying Arabic expression.

But if we read after it: *id est breves lineae horarum*, then the resolution خطوط الساعات *huṭūt as-sā'āt* = hour-lines follows immediately.

Stundenlinien.

Wenn uns an der gleichen Stelle die Wortgruppen *Ethehiaser* (*ethehiafer*) und *Elewiul* (*elevil*, *elaul*), *Aldimataser* (*aldimatafer*, *alhimata*) und *Athenia*, *Alhamsira* (*alhansira*) und *Atheliza* (*atheziza*), *Ethezzea* und *Arrabea*, *Ethemina* (*ethemima*) und *Alchamiza*, *Escebeha* und *Escendeliza* (*esscedezza*) begegnen, so w"aren wir wieder ratlos ohne die Bemerkung, daß diese W"orter die arabischen Namen der Stunden auf dem Stundenkreis bedeuten.

Die den W"ortern beigesetzten r"omischen Ziffern leiten auf die einfache L"osung hin, daß die Namen die Feminina der arabischen Ordinalzahlen sein m"ussen; wer w"urde aber ohne diesen Leitgedanken in *Escendeliza* das Wort السادسة *as-sādisa* (westarabisch *essedisa*), in *Alhamsira* das Wort العاشرة *al-‘āsira*, in *Aldimatafer* oder *Alhimata* das Wort الحادية عشرة *al-ḥādīḡa ‘arata*, in *Ethehiafer* das Wort الثانية عشرة *at-tāniḡa ‘arata* vermuten?

Man erkennt nun die Verschreibung *h* statt *n* und *f* statt langem *s* in *Ethehiafer*, den Ausfall von *ha* oder *he* in *Alf(h)dimatafer* und den Wegfall von *ser* in *Alhimata*, man kann sich auch die erweiterte Form *Escend-eliza* aus dem vorangehenden *Ath-eliza* erkl"aren, und die Verwechslung von *m* und *n* in *Ethemima* begreifen, aber wodurch die Einschlebung des *m* in *Alhamsira* und *Aldimatafer* zustande kam, wird schwerlich jemand sagen k"onnen.

Ich glaubte diese Beispiele vorausschicken zu sollen, wenn ich mich einer erneuten Untersuchung der r"atselhaften Namen *igin*, *andras*, *ormis*, *arbas*, *quimas*, *calcis*, *zenis*, *temenias*, *celentis* f"ur die Ziffern zuwende, die uns von Radulph von Laon (gest. 1131) und auch in etwas "alteren Quellen "uberliefert sind und in der Geschichte des Rechnens eine zweifelhafte Ber"uhmtheit erlangt haben.

Anlaß zur Wiederaufnahme des Verfahrens gibt nicht so sehr die Bemerkung Cantors (I<sup>2</sup> S. 842, I<sup>3</sup> S. 896), daß er das R"atsel als immer noch nicht mit Gewißheit aufgel"ost betrachte und gern bereit sei, eine zuverl"assigere Deutung jener W"orter freudig zu begr"ußen, welche auch die Frage nach der Zeit der Entstehung endg"ultig beantworten w"urde, oder die von Cantor "ubersehene Behandlung des Gegenstands durch E. Clive Bayley und die j"ungste Er"orterung der Frage durch N. Bubnow, als die "Uberzeugung, daß alle Versuche als methodisch verfehlt zu betrachten sind, die die L"osung des R"atsels außerhalb der geschichtlich allein m"oglichen Verbindung des Arabischen und Lateinischen suchen.

Wir l"acheln "uber Helmreich (1595), der in der Vorrede zu seinem Rechenbuch den großen Geometer *Algebras* in "Agypten zur Zeit des Alexander Magnus, der da war ein Pr"azeptor oder Vorfahrer Euklidis, des F"ursten zu Megarien, die Algebra erfinden l"abt; aber sollen wir annehmen, daß ein Radulph von Laon am Anfang des 12. Jahrhunderts mehr von Chald"auern und Assyriern wußte und "chald"aische Namen f"ur die Zahlzeichen einschließ- lich der Null mitteilen konnte, die bei den Chald"auern in dieser Weise gar nicht existierten?

Es ist doch geradezu eine Ungeheuerlichkeit, auf Grund dieser im 12. Jahrh. auftauchenden Notiz mit E. Clive Bayley anzunehmen, "that some reminiscence, at least, of the names of the Chaldaean units may have survived also, though in a more or less cor-

tely.

If at the same place we encounter the word groups *Ethehiaser* (*ethehiafer*) and *Elewiul* (*elevil*, *elaul*), *Aldimataser* (*aldimatafer*, *alhimata*) and *Athenia*, *Alhamsira* (*alhansira*) and *Atheliza* (*atheziza*), *Ethezzea* and *Arrabea*, *Ethemina* (*ethemima*) and *Alchamiza*, *Escebeha* and *Escendeliza* (*esscedezza*), we would again be helpless without the remark that these words are the Arabic names of the hours on the hour circle.

The Roman numerals placed with the words guide one to the simple solution that the names must be the feminines of the Arabic ordinal numbers; but who would suspect in *Escendeliza* the word السادسة *as-sādisa* (West Arabic *essedisa*), in *Alhamsira* the word العاشرة *al-‘āshira*, in *Aldimatafer* or *Alhimata* the word الحادية عشرة *al-ḥādīyya ‘ashrata*, in *Ethehiafer* the word الثانية عشرة *ath-thāniyya ‘ashrata*, without this guiding thought?

One now recognizes the miswriting of *h* for *n* and *f* for long *s* in *Ethehiafer*, the loss of *ha* or *he* in *Alf(h)dimatafer* and the dropping of *ser* in *Alhimata*; one can also explain the extended form *Escend-eliza* from the preceding *Ath-eliza*, and understand the confusion of *m* and *n* in *Ethemima*, but how the insertion of *m* in *Alhamsira* and *Aldimatafer* came about, scarcely anyone could say.

I believed these examples should be prefaced when I turn to a renewed investigation of the enigmatic names *igin*, *andras*, *ormis*, *arbas*, *quimas*, *calcis*, *zenis*, *temenias*, *celentis* for the digits, which have been transmitted to us by Radulph of Laon (d. 1131) and also in somewhat older sources, and have acquired a dubious fame in the history of calculation.

The occasion for resuming the procedure is given not so much by Cantor's remark (I<sup>2</sup> p. 842, I<sup>3</sup> p. 896) that he considers the riddle as still not solved with certainty and would gladly welcome a more reliable interpretation of those words that would also definitively answer the question of the time of origin, or the treatment of the subject by E. Clive Bayley overlooked by Cantor and the most recent discussion of the question by N. Bubnow, as by the conviction that all attempts are to be considered methodically mistaken that seek the solution of the riddle outside the historically only possible connection of Arabic and Latin.

We smile at Helmreich (1595), who in the preface to his arithmetic book lets the great geometer *Algebras* in Egypt at the time of Alexander the Great—who was "a teacher or predecessor of Euclid, the prince of Megara"—invent algebra; but are we to assume that a Radulph of Laon at the beginning of the 12th century knew more about Chaldeans and Assyrians and could communicate "Chaldean" names for numeral signs including zero, which among the Chaldeans did not exist in this way at all?

It is downright an absurdity, on the basis of this note appearing in the 12th century, to assume with E. Clive Bayley "that some reminiscence, at least, of the names of the Chaldaean units may have survived also, though in a more or less corrupted form, to

rupted form, to Neo-Pythagorean times", dann auf Grund einer angeblichen Abstammung (!) des Arabischen und Hebräaischen vom Altassyrischen mit Hilfe von "H"artung" und "Euphonie" aus *'et'in igin* und aus *'etu' celentis* herzustellen, und wenn dies alles noch nicht zur Erklärung der von Radulph gegebenen Wörter ausreicht, das "neopythagoreische" *andras* und *ormis* aus tamilischem *irandu* und *munru* abzuleiten!

Auch "über die phantastischen Versuche, *igin* = ἡ γυνή, *andras* = ἀνδρίας(?), *ormis* = ὀρμύς zu setzen und *calcis* mit καλότης, *celentis* mit ἀμήλυντος oder σελήνη in Zusammenhang zu bringen, kann ich wohl hinweggehen. Soll bei der ganzen Untersuchung "überhaupt etwas herauskommen, so dürfen wir nicht, wie es bisher meist geschah, die "Überlieferung als unantastbar und einwandfrei betrachten; wir dürfen auch an den "überlieferten Ausdrücken nicht herumkorrigieren, bis sie sich irgendeiner Theorie fügen; vielmehr können wir die Frage nur so stellen: Wenn die genannten Wörter die Namen der Ziffern sein sollen, wie konnte es kommen, daß einige dieser Namen mit denen der arabischen Zahlwörter stimmen, andere aber nicht die geringste Ähnlichkeit mit den arabischen Zahlwörtern aufweisen, denen sie angeblich entsprechen sollen?

Zunächst haben wir die Annahme zu prüfen, daß die Wörter tatsächlich die arabischen Namen für die Ziffern darstellen. Wir können uns gegen Radulphs Chaldäer auf zwei annähernd gleichaltrige Zeugnisse berufen, die die Namen ausdrücklich als arabisch bezeichnen. Das eine wird von Smith-Karpinski in *The Hindu-Arabic Numerals* S. 118 erwähnt; ein gewisser Turchill schreibt um 1200: *has autem figuras ... a pytagoricis habemus, nomina vero ab arabibus*.

Das andere ist von Eneström nach einer Handschrift des 11./12. Jahrhunderts mitgeteilt und lautet: *Versus de nominibus characterum arabicorum ad abacum pertinentium*. Aber solche Bestätigungen sind überflüssig geworden, nachdem die von A. Björnbo unternommene, durch H. Suter vollendete Ausgabe der astronomischen Tafeln des Muhammad b. Mūsā den direkten Beweis erbracht hat, daß in den lateinischen Übersetzungen jener Zeit mit den Chaldäern nicht die alten Babylonier, sondern die Araber gemeint sind.

Zum erstenmal werden die Chaldäer bei Athelhard im 7. Kapitel *De alwacat id est medio planetarum etc.* erwähnt, wo es heißt: *Item nota, quod secundum Galdeos IIII milia passus cameli miliare faciunt, atque XXXIII miliaria et tertia, id est thuld, in terra gradus, in coelo dimidius, unde totus terrae circulus XXVIII milia miliaria continet etc.*

Diese Meile, von der  $66\frac{2}{3}$  auf einen Grad gehen sollen, ist nach Suter die von den arabischen Geographen und Astronomen in ihren Angaben gewöhnlich gebrauchte Meile zu 4000 Ellen, und *thuld* natürlich das arabische Wort für Drittel (s. o. S. 54). Noch klarer tritt der Sachverhalt im 10. Kapitel *De inventione loci Saturni, Jovis et Martis* hervor, wo der Bearbeiter den Zusatz macht: *... quodque inde surget, a Caldeis elhacil, a nobis obtentum vel centrum ultimum potest dici*. Es handelt sich um das Wort *al-hāšil*, das Ergebnis; wir sehen also, daß diese Chaldäer arabisch sprechen; zum Überfluß hat die Handschrift O auch für *a Caldeis* das Wort *arabice* eingesetzt.

Neo-Pythagorean times," then on the basis of an alleged descent (!) of Arabic and Hebrew from Old Assyrian with the help of "hardening" and "euphony" to derive *igin* from *'e.thin* and *celentis* from *'e.thu'*, and when all this still does not suffice to explain the words given by Radulph, to derive the "Neopythagorean" *andras* and *ormis* from Tamil *irandu* and *munru*!

Also over the fantastic attempts to set *igin* = ἡ γυνή, *andras* = ἀνδρίας(?), *ormis* = ὀρμύς, and to connect *calcis* with καλότης, *celentis* with ἀμήλυντος or σελήνη, I can well pass over. If the whole investigation is to yield anything at all, we may not, as mostly happened hitherto, regard the tradition as inviolable and flawless; we may not correct the transmitted expressions until they fit some theory; rather, we can pose the question only thus: If the words named are to be the names of the digits, how could it come about that some of these names agree with those of the Arabic number words, while others show not the slightest similarity to the Arabic number words to which they allegedly correspond?

First we have to test the assumption that the words actually represent the Arabic names for the digits. Against Radulph's Chaldeans we can appeal to two approximately contemporary witnesses that explicitly designate the names as Arabic. The one is mentioned by Smith-Karpinski in *The Hindu-Arabic Numerals*, p. 118; a certain Turchill writes around 1200: *has autem figuras ... a pytagoricis habemus, nomina vero ab arabibus*.

The other is communicated by Eneström after a manuscript of the 11th/12th century and reads: *Versus de nominibus characterum arabicorum ad abacum pertinentium*. But such confirmations have become superfluous after the edition of the astronomical tables of Muhammad b. Mūsā undertaken by A. Björnbo and completed by H. Suter has furnished direct proof that in the Latin translations of that period by "Chaldeans" not the ancient Babylonians but the Arabs are meant.

For the first time the Chaldeans are mentioned by Adelard in the 7th chapter *De alwacat id est medio planetarum etc.*, where it says: *Item nota, quod secundum Galdeos IIII milia passus cameli miliare faciunt, atque XXXIII miliaria et tertia, id est thuld, in terra gradus, in coelo dimidius, unde totus terrae circulus XXVIII milia miliaria continet etc.*

This mile, of which  $66\frac{2}{3}$  are to go to a degree, is according to Suter the mile of 4000 cubits customarily used by Arab geographers and astronomers in their data, and *thuld* naturally the Arabic word for third (see above p. 54). Still more clearly the state of affairs emerges in the 10th chapter *De inventione loci Saturni, Jovis et Martis*, where the adapter adds: *... quodque inde surget, a Caldeis elhacil, a nobis obtentum vel centrum ultimum potest dici*. It concerns the word *al-hāšil*, the result; we see thus that these Chaldeans speak Arabic; to top it off, manuscript O has also substituted the word *arabice* for *a Caldeis*.



Auch aus dem vorhin beigezogenen, von Bubnow herausgegebenen *Liber de astrolabio* in den *Opera Gerberti* läßt sich der Gebrauch des Wortes Chaldäer für Araber belegen. Wir finden S. 127 als "Übergang zu Kap. IV *Ad invenendum Nadair solis* die Worte: *Nam Chaldaica astutia seu calculando seu scribendo ita progreditur*; am Schluß des Kap. XVII steht die Bemerkung: *Sunt praeter has duae Ganamalgurab, Alcasal vel Alhimech in Gentauro, quibus Ghaldae satis ad discernendas horas utuntur*; in dem Bruchstück einer andern arabischen Schrift "über das Astrolab erwähnt der Bearbeiter S. 372 *Ghaldaicas gentilogias, qui omnem humanam vitam astrologicis attribuunt ratiocinationibus*. Daß auch *siriace* für *arabice* steht, erhellt aus einer Note bei Bubnow, *Opera Gerberti* S. 125, wo die *latina sollertia* die arabischen Namen der 28 Mondstationen in unglaublicher Weise "übersetzt hat.

Um so verwunderlicher ist es, daß Bubnow S. 72 seines Werkes "über die arithmetische Selbständigkeit der europäischen Kultur "über die Ziffern schreibt: "Kein Abacist des X.-XI. Jahrhunderts deutet irgendwo an, daß diese Zeichen etwas Neues oder gar Arabisches wären. Radulf lebte zu einer Zeit, wo der arabisches Einfluß immer stärker wurde, und dennoch ergeht er sich in Vermutungen "über ihre Herkunft aus Assyrien, ganz wie wir, und nennt sie chaldäisch."

Mit dem Nachweis, daß die Chaldäer der mittelalterlichen "Übersetzer die zeitgenössischen Araber sind, wird wilder Spekulation ein für allemal der Boden entzogen. Die Namen müssen aus dem Arabischen erklärt werden, und wir haben uns nur den Hergang ihrer Aufnahme und Verstrummung in lateinischen Handschriften zurechtzulegen.

Es ist leicht zu sehen, daß nicht nur für die Lateiner, sondern auch für die Araber in der Anfangszeit die Notwendigkeit bestand, den Zahlwert der 9 oder 10 Zeichen durch beige-schriebene Worte auszudrücken. Wir werden also überall da, wo die indische Rechenkunst gelehrt wurde, Zusammenstellungen erwarten dürfen, wie sie z. B. in der wiederholt erwähnten Tabelle der *Ihān a-ṣafā'* auftreten.

Einer "ähnlichen Zusammenstellung müssen unsere Wörter entstammen, und wir haben nun, indem wir von den unanfechtbaren Entsprechungen ausgehen, zu untersuchen, ob eine gewisse Gesetzmäßigkeit in der Umschreibung der Wörter wahrzunehmen ist.

Wenn ich die arabischen Wörter in der heute "üblichen Umschrift unter die mittelalterlichen Wörter setze, so scheint sich zunächst wenig Aussicht auf Erfolg zu eröffnen:

|       |        |        |        |        |        |       |          |          |       |
|-------|--------|--------|--------|--------|--------|-------|----------|----------|-------|
| igin  | andras | ormis  | arbas  | quimas | calcis | zenis | temenias | celentis | sipos |
| wāhid | ignān  | talāta | arba'a | ḥamsa  | sitta  | sab'a | ṭamāniya | tiṣ'a    | siḡr  |

Es ist aber erstens zu beachten, daß die kurzen Vokale bedeutungslos sind und von dem, der die Umschrift erstmals herstellte, schon nach Gutdünken gesetzt werden konnten, zweitens, daß für die Umschrift der arabischen Konsonanten keinerlei Norm bestand, also gleiche Konsonanten mit verschiedenen, verschiedene Konsonanten mit gleichen lateinischen Buchstaben geschrieben werden konnten, wenn nur das Lautbild ungefähr stimmte, und drittens, daß nur das vokallose Buchstabengerippe der Wörter, und selbst dieses mit der Möglichkeit von falschen oder mangelhaft gesetzten diakritischen Punkten und Verwechs-

Also from the *Liber de astrolabio* in the *Opera Gerberti* cited above, edited by Bubnow, the use of the word Chaldeans for Arabs can be documented. We find on p. 127 as transition to chap. IV *Ad invenendum Nadair solis* the words: *Nam Chaldaica astutia seu calculando seu scribendo ita progreditur*; at the end of chap. XVII stands the remark: *Sunt praeter has duae Ganamalgurab, Alcasal vel Alhimech in Gentauro, quibus Ghaldae satis ad discernendas horas utuntur*; in the fragment of another Arabic writing on the astrolabe the adapter mentions on p. 372 *Ghaldaicas gentilogias, qui omnem humanam vitam astrologicis attribuunt ratiocinationibus*.

That *siriace* also stands for *arabice* is clear from a note by Bubnow, *Opera Gerberti* p. 125, where the *latina sollertia* has translated the Arabic names of the 28 lunar mansions in an unbelievable manner.

All the more surprising is it that Bubnow writes on p. 72 of his work on the arithmetic independence of European culture about the digits: "No abacist of the 10th-11th centuries indicates anywhere that these signs were something new or even Arabic. Radulf lived at a time when Arabic influence was growing ever stronger, and yet he indulges in conjectures about their origin from Assyria, just as we do, and calls them Chaldean."

With the proof that the Chaldeans of the medieval translators are the contemporary Arabs, the ground is cut from under wild speculation once and for all. The names must be explained from Arabic, and we have only to work out the course of their adoption and mutilation in Latin manuscripts.

It is easy to see that not only for the Latins but also for the Arabs in the early period there was a necessity to express the numerical value of the 9 or 10 signs by appended words. We shall thus be able to expect everywhere that the Indian calculation art was taught, arrangements such as appear, for example, in the repeatedly mentioned table of the *Ikhwān al-Ṣafā'*.

Our words must stem from a similar arrangement, and we have now, starting from the incontestable correspondences, to investigate whether a certain regularity is to be perceived in the transcription of the words.

If I place the Arabic words in the now customary transcription under the medieval words, at first little prospect of success seems to open:

|        |        |          |        |        |        |       |           |          |       |
|--------|--------|----------|--------|--------|--------|-------|-----------|----------|-------|
| igin   | andras | ormis    | arbas  | quimas | calcis | zenis | temenias  | celentis | sipos |
| wā.hid | iṭṭhān | ṭhalātha | arba'a | .ḥamsa | sitta  | sab'a | ṭhamāniya | tiṣ'a    | .siḡr |

But first, it is to be noted that the short vowels are meaningless and could have been set at discretion by whoever first made the transcription; second, that for the transcription of the Arabic consonants no norm existed, so that equal consonants could be written with different, different consonants with equal Latin letters, if only the sound-image corresponded approximately; and third, that only the vowelless skeleton of the letters of the words, and even this with the possibility of false or inadequately placed diacritical points and confusion of similar signs, is to be admitted into the comparison procedure.

lung "ähnlicher Zeichen in das Vergleichungsverfahren eingestellt werden darf.

Kommt noch dazu, daß bei der weiteren "Übertragung der lateinischen Urschrift unzählige Fehler und Varianten entstehen konnten und entstanden sind, so scheint jeder Willkür Tür und Tor geöffnet zu sein.

Dennoch ist jedem Arabisten bekannt und muß jedem Mathematiker, der sich an derartige Ratselfragen macht, bekannt sein, daß hier kein Wort zuviel gesagt ist und mit all diesen Möglichkeiten gerechnet werden muß, wenn man bei dem Versuch einer Lösung des Ratselfs zum Ziel kommen will.

Hiernach hatten wir unter Zuziehung der bekannt gewordenen Varianten und mit Hervorhebung einiger Gesetzmäßigkeiten folgende Reihe von Entsprechungen:

- (1) *igin, ingin* — *wāhid*
- (2) *andras* — *iḡnān* (?)
- (3) *ormis, hormis, armis* — *talāta* (?)
- (4) *arbas* — *arba'a*
- (5) *quimas, quinas* — *ḥamsa*
- (6) *calcis, caletis, calctis, caltis* — *sitta* (?)
- (7) *zenis, zencis* — *sab'a* oder *sitta*
- (8) *temenias, temeinas, zemenias* — *tamānija*
- (9) *celentis, scelentis, zcelentis* — *tis'a* (?)

Man sieht bei einer Vergleichung mit unserer modernen Umschrift sofort, daß das Wort *wāhid* fehlt, daß in beiden Reihen nur ein Wort vorkommt, das auf *n* endigt: *igin* — *iḡnān* (oder *iḡnen*), und daß die Endung *-at* überall durch *as* oder *is*, also *t* durch *s* wiedergegeben ist.

Aus den Formen unter (8) *temenias, zemenias* — *tamānija* geht hervor, daß der dem englischen *th* entsprechende Laut *t* durch *t* und *z* wiedergegeben wird. Die Form (5) *quimas* (*quinas*) ist wohl (mit spanisch-französischem *qu*) als *kimas* zu lesen und entspricht dem *ḥamsa* unserer Liste.

Die Formen unter (6) und (9) liegen alle innerhalb der Variationsbreite von *talāta*: am nächsten kommt ihm (mit *c = z*) *caletis*, daran schließen sich einerseits *calctis*, *calcis* und *caltis*, andererseits *celentis* und *zcelentis*. Zur Sippe von *arbas* (4) scheint auch die Reihe *ormis, hormis, armis*, ja vielleicht selbst *andras* zu gehören. Die erste Gleichung könnte aus dem Arabischen durch Verlesen entstanden sein oder die Form *andras* würde ein *amras* als Zwischenglied vermuten lassen.

Wir kommen so zu der Auffassung, daß die überlieferte Reihe nicht mehr vollständig ist, sondern nur die Zahlwörter 2, 3, 4, 5, 7, 8 wiedergibt, und daß sie Doppelungen enthält, durch die sie in Verwirrung gekommen ist, so daß sie das Wort für 3 an falscher Stelle bei 6 und 9, das Wort für 4 bei 3 und 4, für 2 bei 1 darbietet.

Das Wort für 7 ergibt sich graphisch unmittelbar aus *zebis* für *sab'a* (*zenis* für *zebis*), *zencis* ist nach *calcis* verschrieben. Es kann aber ebensogut *sitta* für 6 zugrund liegen. Das Wort *igin*

Add to this that in the further transmission of the Latin original countless errors and variants could arise and have arisen, so that every arbitrariness seems to have been given free rein.

Nevertheless, it is known to every Arabist and must be known to every mathematician who concerns himself with such riddles, that not a word too much is said here and that one must reckon with all these possibilities if one wishes to succeed in the attempt at a solution of the riddle.

Accordingly, with the addition of the known variants and with emphasis on certain regularities, we would have the following series of correspondences:

- (1) *igin, ingin* — *wā.hid*
- (2) *andras* — *iḥnān* (?)
- (3) *ormis, hormis, armis* — *thalātha* (?)
- (4) *arbas* — *arba'a*
- (5) *quimas, quinas* — *.hamsa*
- (6) *calcis, caletis, calctis, caltis* — *sitta* (?)
- (7) *zenis, zencis* — *sab'a* or *sitta*
- (8) *temenias, temeinas, zemenias* — *thamāniya*
- (9) *celentis, scelentis, zcelentis* — *tis'a* (?)

One sees on comparison with our modern transcription immediately that the word *wā.hid* is missing, that in both series only one word occurs ending in *n*: *igin* — *iḥnān* (or *iḥnān*), and that the ending *-at* is rendered everywhere by *as* or *is*, that is, *t* by *s*.

From the forms under (8) *temenias, zemenias* — *thamāniya* it emerges that the *t* sound corresponding to English *th* is rendered by *t* and *z*. The form (5) *quimas* (*quinas*) is presumably (with Spanish-French *qu*) to be read as *kimas* and corresponds to the *.hamsa* of our list.

The forms under (6) and (9) all lie within the variation range of *thalātha*: closest to it (with *c = z*) is *caletis*, to which are attached on one side *calctis*, *calcis*, and *caltis*, on the other *celentis* and *zcelentis*. To the family of *arbas* (4) the series *ormis, hormis, armis* also seems to belong, and perhaps even *andras* itself. The first equation could have arisen from the Arabic through misreading, or the form *andras* would suggest an *amras* as intermediate link.

We thus arrive at the view that the transmitted series is no longer complete, but represents only the number words 2, 3, 4, 5, 7, 8, and that it contains doublings through which it has fallen into confusion, so that it presents the word for 3 at the wrong place at 6 and 9, the word for 4 at 3 and 4, for 2 at 1.

The word for 7 follows graphically immediately from *zebis* for *sab'a* (*zenis* for *zebis*); *zencis* is miscopied after *calcis*. But *sitta* for 6 could just as well underlie it. The word *igin* could also have

könnte auch aus *aḥad* (für *igīd*, eigentlich *aḥad*) entstanden sein, das für 1 steht, doch wäre dann der Ausfall der 2 zu erklären.

Das Wort für die Null lautet in den Handschriften *sipos*. Man hat längst darauf hingewiesen, daß dies aus *sifr* verdorben sei. Die graphische Möglichkeit liegt auf der Hand; es braucht nur *صفر* statt *صفر*, also Verdoppelung oder Verschiebung des diakritischen Punktes in der Handschrift angenommen zu werden, um das Schluß-s zu erklären.

Ob es nicht trotz alledem von *σφραγίς* abgeleitet werden muß und erst später durch *cifra* = *sifr* ersetzt wurde, ist eine Frage, die hier nicht im Vorbeigehen erledigt werden kann.

Auf alle Fälle scheint es angemessener, Umstellungen und Verwechslungen der unverständenen Wörter zuzulassen oder Dubletten anzunehmen, wenn sich der Lautbestand ohne Schwierigkeit einfügt, als durch Gewaltmittel Wörter, die sich nun einmal nicht mit den ihnen zugeschriebenen Zahlwerten zur Deckung bringen lassen, solange umzuformen, bis sie nicht mehr wiederzuerkennen sind.

Wie leicht solche Verwechslungen und falschen Eintragungen zustande kommen, lehrt ein Blick auf die von Friedlein der Boethiusausgabe beigegebene Faksimiletafel. Nur die Codd. *e*, *n* und *w* zeigen die 9 Wörter richtig über den 9 Kolumnen, nur bei *n* sind sie in einer Zeile geschrieben; bei *e* sind sie von *arbas* bis *sipos* in 2, bei *n* sämtlich in 2 oder 3 Stücke zerlegt und untereinander gesetzt.

Die älteste Handschrift *ny* (cod. Vatican. 3123, saec. X, Friedlein S. 372) scheint überhaupt noch Niemand näherer Betrachtung wert gehalten worden zu haben, und doch zeigt sie am besten, wie die Verwirrung entstanden sein kann. Hier sind die Wörter wie in arabischen Handschriften vertikal gestellt.

Das Wort *an-dras* steht über 2 und 3, *ormis* über 4, *arbas* über 5, *quimas* über 6, *calcis*(?) über 7, *zenis*(?) über 8, *zeme-nias* über 9 und 0, *sce-len-tis* über drei leeren Fächern; das Wort *sipos* fehlt, während das Zeichen für die Null am richtigen Platze steht.

Welche kühnen Hypothesen könnte und würde man aufstellen, wenn nur diese eine Handschrift bekannt wäre! Und wie mag die Vorlage beschaffen gewesen sein, aus der sie hervorging?

Man mag sich zu der ganzen Sache stellen, wie man will, es ist und bleibt schwer zu begreifen, daß in einer Zeit, in der doch wahrlich Gelegenheit vorhanden war, sich über die richtigen Formen der arabischen Zahlwörter zu verlässigen, dieser Ratteknögel von Verwechslungen und Unformen weitergegeben werden konnte.

Man kann sich bei so allgemeinen Dingen wie Zahlwörtern nicht wie bei medizinischen oder astronomischen Begriffen auf sachliche Schwierigkeiten berufen. Es bleibt fast nur die Annahme, daß eine an sich schlechte handschriftliche Überlieferung schon früh an einen Ort verschleppt wurde, wo keine Spur von arabischen Kenntnissen vorhanden war, wo man also die Worte nicht als arabische Zahlwörter erkannte, sondern für die fremden Namen der Zahlzeichen hielt. Von jenem Urtext aus muß sich dann das literarische Kuriosum weiter verbreitet haben; ein literarisches Kuriosum ohne alle sachliche Bedeutung bleibt diese ganze Einführung arabischer Namen für Zahlzeichen, die ebensogut

arisen from *a.had* (for *igīd*, actually *a.had*), which would stand for 1, but then the loss of 2 would have to be explained.

The word for zero in the manuscripts is *sipos*. It has long been pointed out that this is corrupted from *.sifr*. The graphical possibility lies at hand; one need only assume *صفر* instead of *صفر*, that is, doubling or displacement of the diacritical point in the manuscript, in order to explain the final *s*.

Whether it must nonetheless be derived from *σφραγίς* [sphragis] and was only later replaced by *cifra* = *.sifr* is a question that cannot be settled here in passing.

In any case, it seems more appropriate to allow displacements and confusions of the uncomprehended words, or to assume duplicates, when the sound substance fits without difficulty, than by violent means to reshape words that simply cannot be brought into agreement with the numerical values ascribed to them, until they are no longer recognizable.

How easily such confusions and false entries come about is shown by a glance at the facsimile plate appended by Friedlein to the Boethius edition. Only the codices *e*, *n*, and *w* show the 9 words correctly over the 9 columns; only in *n* are they written in one line; in *e* they are from *arbas* to *sipos* in 2, in *n* all in 2 or 3 pieces split and placed under one another.

The oldest manuscript *ny* (Cod. Vatican. 3123, 10th cent., Friedlein p. 372) seems not yet to have been held worthy of closer consideration by anyone at all, and yet it shows best how the confusion could have arisen. Here the words are set vertically, as in Arabic manuscripts.

The word *an-dras* stands over 2 and 3, *ormis* over 4, *arbas* over 5, *quimas* over 6, *calcis*(?) over 7, *zenis*(?) over 8, *zeme-nias* over 9 and 0, *sce-len-tis* over three empty compartments; the word *sipos* is missing, while the sign for zero stands in the correct place.

What bold hypotheses could and would one construct if only this one manuscript were known! And what might the exemplar have been like from which it derived?

One may take whatever stand one wishes on the whole matter, it is and remains hard to understand that in a time when truly there was opportunity to ascertain the correct forms of the Arabic number words, this rat's nest of confusions and deformities could be handed down.

With such general things as number words one cannot, as with medical or astronomical concepts, appeal to substantive difficulties. There remains almost only the assumption that a bad manuscript tradition in itself was early transported to a place where no trace of Arabic knowledge was present, where one therefore did not recognize the words as Arabic number words, but held them for foreign names of numeral signs. From that original text the literary curiosity must then have spread further; a literary curiosity without any substantive significance remains this entire introduction of Arabic names for numeral signs, which could just as well be named in Latin and were named in Latin.

lateinisch benannt werden konnten und benannt wurden.

Es wäre auch an die Analogie der "Übertragung der semitischen Buchstabennamen zu den Griechen zu erinnern. Wie dort die fremden Zeichen mit ihren unverstandenen Namen "übertragen wurden, so werden in der "Übergangszeit die arabischen Zahlwörter als "Namen für die Ziffern gebraucht worden sein. Von dem Verfasser der bekannten Merkverse (Cantor I<sup>3</sup>, S. 893) kann man sagen, daß er weder arabisch verstand noch von der arabischen Herkunft der Worte etwas wußte; er hatte sie doch sonst unmöglich unbesehen beibehalten können. Gleichwohl glaube ich selbst in diesen Versen, wo sie nicht reines Wortgeklänge sind, Anklänge an arabische Zusammenstellungen "über die "Eigenschaften der Zahlen" nachweisen zu können, möchte sie also nicht wie Cantor nur als Gedächtnisverse zur Einprägung der fremdartigen Wörter betrachten.

Man findet z. B. bei den *Ihān a-ṣafā'* (ed. Bombay I, 29–31) einen solchen Abschnitt "über die Eigenschaften der Zahl; darin heißt die 1 *Wurzel (al)* und Ursprung der Zahlen, die 2 ist die erste eigentliche Zahl, die 3 die erste unpaarige Zahl, die 4 die erste Quadratzahl (*'adad maḡd'ur*), die 5 die erste Kreiszahl (wegen  $5 \cdot 5 = 25$ ,  $5 \cdot 25 = 125$ ), die 6 die erste vollständige Zahl (*'adad tāmm*), die 7 die erste vollkommene Zahl (*'adad kāmil*), die 8 die erste Würfel- oder Körperzahl, die 9 das erste Quadrat einer unpaarigen Zahl usw.

Stellen wir dieser Aufzählung die Verse gegenüber:

*Ordine primigeno sibi nomen possidet Igin.  
Andras ecce locum previndicat ipse secundum.  
Ormis post numerus non compositus sibi primus.  
Denique bis binos succedens indicat Arbas.  
Significat quinos ficto de nomine Quimas.  
Sexta tenet calcis perfecto munere gaudens.  
Zenis enim digne septeno fulget honore.  
Octo beatificos Temenias exprimit unus  
Terque notat trinum Celentis nomine rithmum —,*

so erkennt man aus den kursiv gedruckten Worten sofort die vorhandenen Parallelen.

Unverständlich sind mir nur die seligmachenden Acht und das *ficto de nomine* bei Fünf; doch konnte dies noch ein Anklang an die arabische Einteilung der Zahlwortformen sein, jenes an die 8 Seligpreisungen erinnern. Vielleicht lassen sich näherliegende lateinische Vorbilder auffinden; daß alle diese Dinge auf den Gedankenkreis der *Theologumena Arithmeticae* zurückgehen, ist selbstverständlich.

Ich bin mit dieser Untersuchung zu Ende. Ob das nüchterne Ergebnis allgemein befriedigen wird, weiß ich nicht; daß es methodisch auf festen Füßen steht, wird kaum zu bestreiten sein. Hatte man philologisch-kritische Grundsätze stets vor Augen gehabt, so wäre das ganze Aufgebot von Gelehrtheit und Scharfsinn zur Aufdeckung alter chaldäischer oder pythagoreischer Weisheit überflüssig gewesen. Der Zweck dieser Untersuchung wäre erreicht, wenn dadurch wenigstens weiteren Phantastereien für künftige ein Ziel gesetzt würde.

One should also recall the analogy of the transfer of Semitic letter names to the Greeks. Just as there the foreign signs were transmitted with their uncomprehended names, so in the transitional period the Arabic number words will have been used as "names" for the digits.

Of the author of the well-known mnemonic verses (Cantor I<sup>3</sup>, p. 893) one can say that he neither understood Arabic nor knew anything of the Arabic origin of the words; he could not possibly have retained them uncritically otherwise. Nevertheless, I believe I can demonstrate in these verses, where they are not pure word-jingle, echoes of Arabic compilations on the "properties of numbers"; I would thus not regard them, like Cantor, merely as mnemonic verses for impressing the foreign words.

One finds, for example, in the *Ikhwān al-Ṣafā'* (ed. Bombay I, 29–31) such a section on the properties of number; therein 1 is called "Root (a.sl)" and origin of numbers, 2 is the first proper number, 3 the first odd number, 4 the first square number (*'adad maḡdhūr*), 5 the first circular number (because  $5 \cdot 5 = 25$ ,  $5 \cdot 25 = 125$ ), 6 the first complete number (*'adad tāmm*), 7 the first perfect number (*'adad kāmil*), 8 the first cube or solid number, 9 the first square of an odd number, etc.

Setting the verses against this enumeration:

*Ordine primigeno sibi nomen possidet Igin.  
Andras ecce locum previndicat ipse secundum.  
Ormis post numerus non compositus sibi primus.  
Denique bis binos succedens indicat Arbas.  
Significat quinos ficto de nomine Quimas.  
Sexta tenet calcis perfecto munere gaudens.  
Zenis enim digne septeno fulget honore.  
Octo beatificos Temenias exprimit unus  
Terque notat trinum Celentis nomine rithmum —,*

one recognizes immediately from the words printed in italics the parallels that exist.

Only the "salvation-bringing eights" and the *ficto de nomine* for five are incomprehensible to me; but the former could still be an echo of the Arabic classification of number-word forms, the latter a recollection of the 8 Beatitudes. Perhaps closer Latin models can be found; that all these things go back to the circle of ideas of the *Theologumena Arithmeticae* is self-evident.

I am at the end of this investigation. Whether the sober result will satisfy generally, I do not know; that it stands methodically on firm feet will hardly be disputable. Had philological-critical principles always been kept before one's eyes, the whole deployment of erudition and acumen to uncover ancient Chaldean or Pythagorean wisdom would have been superfluous. The purpose of this investigation would be achieved if thereby at least further fantasies are set a limit for the future.



## 10 X. Das Kapitel von den Gesch"äften / The Chapter on Commercial Transactions

Die Aufgaben zweiten Grades, die Muhammad b. M"usā im ersten Abschnitt seiner Algebra behandelt, sind rein formaler Natur und tragen nichts zu dem Zwecke bei, dem nach dem Vorwort (vgl. oben S. 5) das Werk dienen soll. Auf die eigentlichen Gesch"äftsaufgaben, auf das Gesch"äftsrechnen, kommt der Verfasser erst in dem باب المعاملات *bāb al-mu'āmalāt*, dem Kapitel von den Gesch"äften, zu sprechen.

Es hat ob seines d"urftigen mathematischen Inhalts bisher kaum Beachtung gefunden und wird bei Cantor I<sup>3</sup>, S. 726 mit den Worten gekennzeichnet: "Indisch ist auch wohl die nur uneigentlich der Algebra zugewiesene Regel de tri, welche in der Fortsetzung von Alchwarizmis Werke auftritt und "ähnlich bei griechischen Schriftstellern uns nicht bekannt ist".

Die geschichtliche Bedeutung des Abschnitts wird sofort eine wesentlich andere, wenn man diesem Satze die bestimmtere Form gibt: "Durchaus indisch ist nach Inhalt und Form das Kapitel von den Gesch"äften", und wenn man sich vergegenwärtigt, daß hier die M"öglichkeit vorliegt, Muhammad b. M"usās Arbeitsweise und pers"onliches Verdienst um die Popularisierung des indischen Rechnens an einem einwandfreien Beispiel zu prüfen. Griechische Herkunft des Kapitels ist ausgeschlossen: die ganze mathematische "Überlieferung weiß nichts von Aufgaben nach einer Regel de tri trotz hoher Ausbildung der Lehre von den Proportionen. Erst spät, um die Mitte des 14. Jahrhunderts, tauchen bei Nikolaus Rhabdas von Smyrna Aufgaben "über "politische Arithmetik", d. i. b"urgerliches Rechnen auf, die mittels Regel de tri gelöst sind.

Noch später — in das 15. Jahrhundert — sind die von J. L. Heiberg veröffentlichten byzantinischen Texte zu setzen, in denen wir den Ausspruch finden: ἡ δε τῶν τριῶν μεταβολή ἡ τῆς λογιστικῆς μαγεία ἐστί, καλῶς φράσιν ὁ παλαιὸς λόγος.

Wie alt dieses "Wort" auch sein mag, es ist sicherlich eher arabischen oder italienischen als griechischen Ursprungs. Auch das von Hultsch beigebrachte Scholion zu Platons *Charmides* 165E, in dem "schließlich" als Zweck der Logistik angegeben wird, daß sie den Bedürfnissen des Alltagslebens diene, um brauchbare Verträge "über Mein und Dein, "über Soll und Haben, "über Erbchaftsteilungen usw. abzuschließen, schließt der im *Charmides* selbst gegebenen rein theoretischen Definition ins Gesicht, und beweist nichts für Anwendung der Regel de tri.

Im Gegensatz zu dem Versagen griechischer "Überlieferung hinsichtlich der Regel de tri können wir bei den Indern eine nicht abbrechende, nach Form und Bezeichnungsweise feste "Überlieferungskette von Aryabhatta im 5. bis zu Bhāskara im 12. Jahrhundert verfolgen. Unter den "äußerst knapp gehaltenen Regeln, die Rodet in seinen *Leçons de calcul d'Aryabhata* "übersetzt hat, findet sich die folgende:

**XXVI.** Dans la "règle de trois", le "résultat" (ou "fruit" *phalam*) multiplié par la "demande" et divisé par le "type" donne le "résultat de la demande".

Die Sanskritworte umschreibt und "übersetzt Rodet mit *trairāśīkam* = règle de trois; *phalam* = fruit, résultat, revenu; *icchā* = demande, quantité pour laquelle on demande; *pramāṇa* = type,

The second-degree problems that Muḥammad b. Mūsā treats in the first section of his Algebra are purely formal in nature and contribute nothing to the purpose that, according to the preface (cf. above p. 5), the work is meant to serve. The author comes to the actual business problems, to business calculation, only in the باب المعاملات *bāb al-mu'āmalāt*, the chapter on commercial transactions.

Because of its meager mathematical content it has hitherto hardly received attention, and is characterized by Cantor I<sup>3</sup>, p. 726, with the words: "Indian also is presumably the rule of three, which is only improperly assigned to the Algebra, appears in the continuation of al-Khwārizmī's work, and is not known to us similarly among Greek writers."

The historical significance of the section becomes immediately essentially different if one gives this sentence the more definite form: "Thoroughly Indian in content and form is the chapter on commercial transactions," and if one bears in mind that here the possibility exists to test Muḥammad b. Mūsā's working method and personal merit for the popularization of Indian calculation on an unobjectionable example.

Greek origin of the chapter is excluded: the entire mathematical tradition knows nothing of problems according to a rule of three despite the high development of the doctrine of proportions. Only late, around the middle of the 14th century, do problems on "political arithmetic," i.e. practical calculation, solved by means of the rule of three, appear with Nikolaos Rhabdas of Smyrna.

Still later—in the 15th century—are to be placed the Byzantine texts published by J. L. Heiberg, in which we find the saying: ἡ δε τῶν τριῶν μεταβολή ἡ τῆς λογιστικῆς μαγεία ἐστί, καλῶς φράσιν ὁ παλαιὸς λόγος. ["The rule of three is the magic of arithmetic," as the old saying well expresses it.]

However old this "word" may be, it is certainly more of Arabic or Italian than of Greek origin. Also the scholion to Plato's *Charmides* 165E adduced by Hultsch, in which "finally" as the purpose of *logistikē* is stated that it should serve the needs of everyday life in order to conclude useful contracts about mine and thine, about debit and credit, about inheritance divisions, etc., contradicts the purely theoretical definition given in the *Charmides* itself, and proves nothing for the application of the rule of three.

In contrast to the failure of Greek tradition regarding the rule of three, we can follow among the Indians an unbroken chain of transmission, fixed in form and mode of designation, from Āryabha.ta in the 5th to Bhāskara in the 12th century. Among the extremely brief rules that Rodet has translated in his *Leçons de calcul d'Aryabhata*, the following is found:

**XXVI.** "In the 'rule of three,' the 'result' (or 'fruit,' *phalam*) multiplied by the 'demand' and divided by the 'type' gives the 'result of the demand.'"

Rodet paraphrases and translates the Sanskrit words as: *trairāśīkam* = règle de trois; *phalam* = fruit, résultat, revenu; *icchā* = demande, quantité pour laquelle on demande; *pramāṇa* = type,

quantité type; dazu kommt *icchāphala* = résultat de la demande.

Ganz denselben Ausdrücken begegnen wir bei Brahmagupta in der ersten Hälfte des 7. Jahrhunderts; der Wortlaut der Regel ist nach Colebrooke (a. a. O., S. 283):

**10.** *In the rule of three, argument, fruit and requisition [Zusatz des Übersetzers "are names of the terms"]: the first and last terms must be similar. Requisition, multiplied by the fruit, and divided by the argument, is the produce.*

Ein Kommentator fügt hinzu, daß der mittlere Term anders (dissimilar) benannt ist und daß die Regel sich auf ganze Zahlen bezieht; wenn Brüche vorkommen, sollen sie alle auf denselben Nenner gebracht werden. Als Beispiel wird von ihm gegeben:

"Jemand gibt weg 108 K"uhe in 3 Tagen; wieviel K"uhe bringt er fort in einem Jahr und einem Monat?" Feststellung: Tage 3. K"uhe 108. Tage 390. Antwort: K"uhe 14 040.

**Die Fachausdrücke bedürfen besonders deshalb einer näheren Erläuterung, weil nach ihnen die arabischen gebildet worden sind.** Die Ausdrucksweise wird sich für das Deutsche ändern, je nachdem wir eine Zins- oder eine Warenrechnung zugrunde legen. Bei einer Zinsrechnung heißt es z. B.: "Was trägt ein Kapital von 3500 Mk., wenn 100 Mk. 4 Mk. Zins tragen?"

Hier ist die Zahl 100 = *pramāṇa*, *le type, the argument*, also die Norm, das Grundkapital; die Zahl 4 = *phalam*, *le résultat, le revenu, the fruit*, also sein Ertragnis — wir sagen genau wie vom Baume oder Acker, daß das Kapital Zinsen "trägt". Die Zahl 3500 ist *icchā*, *la demande, the requisition*, für uns das Kapital; die gesuchte Zahl ist also *icchāphala*, *le résultat de la demande*, für uns der Zins, das Ertragnis des Kapitals.

Bei Warenrechnungen werden wir *pramāṇa* mit Grundmenge oder Grundmaß, *phalam* mit Grundpreis oder Grundwert, *icchā* mit Angebot oder Gesamtmenge, *icchāphala* mit Gesamtpreis oder Gesamtwert übersetzen können, da es sich meist darum handeln wird, aus dem für eine kleinere Menge gegebenen Marktpreis den Betrag für eine größere Menge zu berechnen. Nach diesen Vorbereitungen können wir uns dem Kapitel von den Geschichten in Muhammad b. Mūsā's Algebra zuwenden. Es ist notwendig, Text und Übersetzung Rosens wiederzugeben, um über die darin enthaltenen Versehen zur Klarheit zu kommen.

Der theoretische Teil des Kapitels lautet wie folgt (Rosen, S. 48):

واعلم ان جميع معاملات الناس من بيع وشراء وصرف  
واجارة وغير ذلك من ضربين باربعة اعداد يتكلم بها السائل  
وهي المسعار والسعر والمتمتع والتمن فعدد المسعار يقابل  
عدد الثمن وعدد السعر يقابل عدد المتمتع وهذه الاربعة  
اعداد ثلاثة منها دائما معلومة معروفة وواحد منها مبهوت  
وهو الذي في كلام القائل كم وبه يسال والعمل في هذا ان  
تنظر الى الثلاثة الاعداد المعلومة فلا بد من ان يكون اثنان  
منهما كل واحد منهما مقابل صاحبه فاضرب المعلومين  
المقابلين العددين كل واحد منهما في صاحبه وما حصل  
فاقسمه على العدد المعلوم الاخير الذي صاحبه مجهول وما  
خرج لك فهو العدد المجهول الذي به يسال السائل وهو مقابل  
العدد الذي قسمت عليه

quantité type; to which is added *icchāphala* = résultat de la demande.

We encounter quite the same expressions in Brahmagupta in the first half of the 7th century; the wording of the rule according to Colebrooke (op. cit., p. 283) is:

**10.** "In the rule of three, argument, fruit and requisition [translator's addition "are names of the terms"]: the first and last terms must be similar. Requisition, multiplied by the fruit, and divided by the argument, is the produce."

A commentator adds that the middle term is named differently (dissimilar) and that the rule refers to whole numbers; if fractions occur, they should all be brought to the same denominator. As an example he gives: "Someone gives away 108 cows in 3 days; how many cows does he take away in a year and a month?" Statement: Days 3. Cows 108. Days 390. Answer: Cows 14,040.

**The technical terms require closer explanation especially because the Arabic terms were formed after them.** The mode of expression will vary for German, depending on whether we take as basis an interest or a goods calculation. In an interest calculation one says, for example: "What interest does a capital of 3500 marks bear, if 100 marks bear 4 marks interest?"

Here the number 100 = *pramāṇa*, *le type, the argument*, thus the norm, the principal; the number 4 = *phalam*, *le résultat, le revenu, the fruit*, thus its yield—we say, exactly as of tree or field, that the capital "bears" interest. The number 3500 is *icchā*, *la demande, the requisition*, for us the capital; the sought number is thus *icchāphala*, *le résultat de la demande*, for us the interest, the yield of the capital.

In goods calculations we can translate *pramāṇa* as basic quantity or basic measure, *phalam* as basic price or basic value, *icchā* as offer or total quantity, *icchāphala* as total price or total value, since it mostly concerns calculating from the market price given for a smaller quantity the amount for a larger quantity.

After these preparations we can turn to the chapter on commercial transactions in Muhammad b. Mūsā's Algebra. It is necessary to reproduce Rosen's text and translation in order to clarify the errors contained therein.

The theoretical part of the chapter reads as follows (Rosen, p. 48):

واعلم ان جميع معاملات الناس من بيع وشراء وصرف  
واجارة وغير ذلك من ضربين باربعة اعداد يتكلم بها السائل  
وهي المسعار والسعر والمتمتع والتمن فعدد المسعار يقابل  
عدد الثمن وعدد السعر يقابل عدد المتمتع وهذه الاربعة  
اعداد ثلاثة منها دائما معلومة معروفة وواحد منها مبهوت  
وهو الذي في كلام القائل كم وبه يسال والعمل في هذا ان  
تنظر الى الثلاثة الاعداد المعلومة فلا بد من ان يكون اثنان  
منهما كل واحد منهما مقابل صاحبه فاضرب المعلومين  
المقابلين العددين كل واحد منهما في صاحبه وما حصل  
فاقسمه على العدد المعلوم الاخير الذي صاحبه مجهول وما  
خرج لك فهو العدد المجهول الذي به يسال السائل وهو مقابل  
العدد الذي قسمت عليه

Nach der "Übersetzung von Rosen (S. 68): "On mercantile transactions. You know that all mercantile transactions of people, such as buying and selling, exchange and hire, comprehend always two notions and four numbers, which are stated by the enquirer; namely, measure and price, and quantity and sum. The number which expresses the measure, is inversely proportionate to the number which expresses the sum, and the number of the price inversely proportionate to that of the quantity."

"Three of these four numbers are always known, one is unknown, and this is implied when the person inquiring says "how much?" and it is the object of the question. — The computation in such instances is this, that you try the three given numbers; two of them must necessarily be inversely proportionate the one to the other. Then you multiply these two proportionate numbers by each other, and you divide the product by the third given number, the proportionate of which is unknown. The quotient of this division is the unknown number, which the inquirer asked for; and it is inversely proportionate to the divisor."

Die Worte des Textes, die den ersten vier kursivgedruckten Worten der "Übersetzung entsprechen, sind:

1. *المسعار* *al-musā'ār* = the measure
2. *السعر* *al-sī'r* = the price
3. *المتمعن* *al-mutamannan* = the quantity
4. *التمن* *at-taman* = the sum

Auch wer des Arabischen unkundig ist, erkennt aus der Umschreibung, daß im Arabischen Wortpaare vorliegen, die von der gleichen Wurzel gebildet sind, während Rosen vier verschiedene und dabei recht unbestimmte Ausdrücke zur "Übersetzung verwendet. Das ist schon ein methodischer Mißgriff; dazu kommt aber noch, daß die Ausdrücke 3. und 4. im arabischen Text falsch gestellt sind, wenn wir die "Übersetzung anerkennen, oder umgekehrt, daß Rosens "Übersetzung falsch ist, wenn wir den Text gelten lassen.

Um zum Verständnis der Ausdrücke zu gelangen, müssen wir von *sī'r* und *taman* ausgehen. Wenn wir *as-sī'r* mit "die Schätzung" übersetzen, so ist *al-musā'ār* "das Geschätzte", die geschätzte Ware; in gleicher Weise ist nach *at-taman* "der Wert" das Partizipium *al-mutamannan* "das Gewertete" gebildet. Wir müssen also *at-taman* = the sum, *al-mutamannan* = the quantity setzen. Die Worte *musā'ār* und *sī'r* geben in freier, dem arabischen Sprachgeist angepaßter Weise das indische *pramāṇa* (argument) und *phalam* (fruit) wieder, während in *mutamannan* und *taman* die "Äquivalente von *icchā* (demand) und *icchāphala* (fruit of the demand) vorliegen.

Vom mathematischen Standpunkt ist darin, daß die arabische Bezeichnung schon grammatisch die Gleichartigkeit der Größen in *musā'ār* und *mutamannan* einerseits, *sī'r* und *taman* andererseits zum Ausdruck bringt — wir würden die Worte "Ware" und "Preis" über den Ansatz stellen —, ein terminologischer Fortschritt zu erblicken; der Inder muß ausdrücklich bemerken, daß "argument and requisition must be of like denomination; the fruit, which is of a different species, stands between them".

According to Rosen's translation (p. 68): "On mercantile transactions. You know that all mercantile transactions of people, such as buying and selling, exchange and hire, comprehend always two notions and four numbers, which are stated by the enquirer; namely, measure and price, and quantity and sum. The number which expresses the measure, is inversely proportionate to the number which expresses the sum, and the number of the price inversely proportionate to that of the quantity."

"Three of these four numbers are always known, one is unknown, and this is implied when the person inquiring says 'how much?' and it is the object of the question.—The computation in such instances is this, that you try the three given numbers; two of them must necessarily be inversely proportionate the one to the other. Then you multiply these two proportionate numbers by each other, and you divide the product by the third given number, the proportionate of which is unknown. The quotient of this division is the unknown number, which the inquirer asked for; and it is inversely proportionate to the divisor."

The words of the text corresponding to the first four italicized words of the translation are:

1. *المسعار* *al-musā'ār* = the measure
2. *السعر* *al-sī'r* = the price
3. *المتمعن* *al-mutamannan* = the quantity
4. *التمن* *at-thaman* = the sum

Even one unacquainted with Arabic recognizes from the transliteration that in Arabic there are pairs of words formed from the same root, while Rosen uses four different and quite indeterminate expressions for the translation. This is already a methodological blunder; but in addition, expressions 3 and 4 are incorrectly placed in the Arabic text if we accept the translation, or conversely, Rosen's translation is wrong if we let the text stand.

To arrive at an understanding of the expressions, we must start from *sī'r* and *thaman*. If we translate *as-sī'r* as "the valuation," then *al-musā'ār* is "the thing valued," the appraised goods; in the same way, after *at-thaman* "the value," the participle *al-mutamannan* "the thing weighted/valued" is formed.

We must thus set *at-thaman* = the sum, *al-mutamannan* = the quantity. The words *musā'ār* and *sī'r* render in free manner, adapted to the Arabic linguistic spirit, the Indian *pramāṇa* (argument) and *phalam* (fruit), while in *mutamannan* and *thaman* the equivalents of *icchā* (demand) and *icchāphala* (fruit of the demand) are present.

From the mathematical standpoint, in the fact that the Arabic designation already grammatically expresses the homogeneity of the quantities in *musā'ār* and *mutamannan* on the one hand, *sī'r* and *thaman* on the other—we would place the words "goods" and "price" over the equation—a terminological advance is to be seen; the Indian must expressly remark that "argument and requisition must be of like denomination; the fruit, which is of a different species, stands between them."

Von irgendeiner Anspielung auf die griechische Lehre von den Proportionen ist bei Muhammad b. M<sup>u</sup>sā weder im Kapitel von den Gesch<sup>h</sup>ften noch sonstwo das geringste zu lesen. Damit wird auch Rosens k<sup>u</sup>hner Versuch hinf<sup>u</sup>ällig, das Wort *mubāin* mit *inversely proportionale* zu "übersetzen.

Wir m<sup>u</sup>ssen bei der Grundbedeutung "abgesondert, getrennt" oder der sich daraus leicht ergebenden Bedeutung "sich gegen<sup>u</sup>überstehend" stehenbleiben, also wie es bei dem soeben angeführten indischen Zitat der Fall ist, nur den rein "außerlichen Gesichtspunkt der Anordnung der vier Zahlen im Rechenschema bei der "Übersetzung im Auge haben. Das Wort ist weiter nichts als ein Synonym zu dem Partizip *ālā' muqābil* "entgegengesetzt", das uns an die *muqābala*, *oppositio* erinnert.

Auch der Ausdruck *two notions* am Anfang der "Übersetzung ist irref<sup>u</sup>hrend. Muhammad b. M<sup>u</sup>sā spricht nicht von zwei Begriffen, sondern von ضربين d. i. zwei "Gesichtspunkten" und meint damit — wie aus einem der Beispiele klar hervorgeht —, daß entweder nach dem "Wert" oder nach dem "Gewerteten" gefragt sein kann.

Legt man bei der Wiedergabe des Arabischen mehr Gewicht auf photographische Treue, als auf elegantes Umschiffen von Klippen und Schwierigkeiten, so w<sup>u</sup>re der Text etwa wie folgt zu "übersetzen:

"Wisse, daß die Gesch<sup>h</sup>fte der Menschen — sie (geh<sup>u</sup>)ren) alle zu dem Kauf und dem Verkauf und dem Austausch und der Miete und anderem dergleichen — nach zwei Gesichtspunkten mit vier Zahlen, welche der Fragende ausspricht, n<sup>u</sup>ämlich dem Gesch<sup>h</sup>atzen und der Sch<sup>h</sup>atzung und dem Gewerteten und dem Wert. Die Zahl, die das Gesch<sup>h</sup>atze ist, steht gegen<sup>u</sup>ber der Zahl, die der Wert ist, und die Zahl, die die Sch<sup>h</sup>atzung ist, steht gegen<sup>u</sup>ber der Zahl, die das Gewertete ist. Und diese vier Zahlen, drei von ihnen sind stets offenkundig, bekannt, und eine von ihnen verborgen, und das ist die, welche in der Rede des Redenden "wieviel" (heißt) und nach welcher der Fragende fragt."

"Und die Regel in diesem (ist), daß du schau auf die drei offenkundigen Zahlen, so ist kein Ausweg, als daß zwei von ihnen, eine jede von beiden gegen<sup>u</sup>überstehend (sind) ihrem Gef<sup>u</sup>ährten; so multipliziere die beiden offenkundigen gegen<sup>u</sup>überstehenden Zahlen, eine jede von ihnen beiden mit ihrem Gef<sup>u</sup>ährten, und was sich ergibt, das teile mit der letzten offenkundigen Zahl, deren Gegen<sup>u</sup>über unbekannt (ist); und was dir herauskommt, das ist die unbekannte Zahl, nach welcher der Fragende fragt, und sie steht gegen<sup>u</sup>ber der Zahl, mit der du geteilt hast."

Die Verwechslung von *mutamannan* und *taman* findet sich nur im einleitenden Teil; es wird daher gen<sup>u</sup>gen, die nun folgenden Aufgaben ohne Wiederholung des arabischen Textes in genauer deutscher "Übersetzung mitzuteilen.

**I.** Und Beispiele daf<sup>u</sup>ur unter einem Gesichtspunkt davon sind: Wenn gesagt wird "Dir 10 um 6, wieviel dir um 4?" so ist sein Wort 10 die gesch<sup>h</sup>atzte Zahl, und sein Wort "um 6" ist die Sch<sup>h</sup>atzung und sein Wort "wieviel dir" die unbekannte ist gewertete Zahl und sein Wort "um 4" ist die Zahl, die der Wert ist. Nun steht die gesch<sup>h</sup>atzte Zahl, n<sup>u</sup>ämlich 10, gegen<sup>u</sup>ber der Zahl, die der Wert ist, n<sup>u</sup>ämlich 4; multipliziere also die 10 in die 4, und beide sind die gegen<sup>u</sup>überstehenden offenkundigen, dann gibt es 40; teile sie nun durch die letzte offenkundige Zahl, die

Of any allusion to the Greek doctrine of proportions there is in Muḥammad b. Mūsā the least to be read, neither in the chapter on commercial transactions nor anywhere else. Therewith Rosen's bold attempt to translate the word *mubāin* by "inversely proportionale" also becomes invalid.

We must remain with the basic meaning "separated, set apart" or the meaning easily resulting therefrom "standing opposite one another," thus, as is the case in the Indian citation just adduced, have in view only the purely external standpoint of the arrangement of the four numbers in the calculation scheme in the translation. The word is nothing more than a synonym to the participle *ālā' muqābil* "opposite," which reminds us of *muqābala*, *oppositio*.

Also the expression "two notions" at the beginning of the translation is misleading. Muḥammad b. Mūsā speaks not of two concepts but of ضربين *darbayn*, that is, two "aspects," and means thereby—as clearly emerges from one of the examples—that one can ask either for the "value" or for the "valued thing."

If in rendering Arabic one places more weight on photographic fidelity than on elegant circumvention of cliffs and difficulties, the text would be translated approximately as follows:

"Know that the transactions of people—they (all belong) to buying and selling and exchange and renting and other such—[take place] according to two aspects with four numbers, which the questioner pronounces, namely the appraised thing and the valuation and the valued thing and the value. The number that is the appraised thing stands opposite the number that is the value, and the number that is the valuation stands opposite the number that is the valued thing. And these four numbers, three of them are always manifest, known, and one of them hidden, and that is [the one] which in the speech of the speaker [is called] 'how much,' and which the questioner asks about."

"And the rule in this (is) that you look at the three manifest numbers, so there is no escape but that two of them, each of the two standing opposite its companion; so multiply the two manifest opposite numbers, each of the two of them with its companion, and what results, divide by the last manifest number whose opposite is unknown; and what comes out for you, that is the unknown number which the questioner asks about, and it stands opposite the number by which you have divided."

The confusion of *mutamannan* and *taman* is found only in the introductory part; it will therefore suffice to communicate the now following problems in exact German translation without repetition of the Arabic text.

**I.** And examples thereof under one aspect are: If it is said "You [have] 10 for 6, how much for you for 4?" then his word 10 is the appraised number, and his word "for 6" is the valuation and his word "how much for you" is the unknown valued number, and his word "for 4" is the number that is the value. Now the appraised number, namely 10, stands opposite the number that is the value, namely 4; multiply thus 10 into 4, and both are the opposite manifest ones, then it gives 40; divide it now by the last manifest number, which is the valuation, namely 6, so it gives



die Schätzung ist, nämlich 6, so gibt es  $6\frac{2}{3}$ , und das ist die unbekannte Zahl, die in dem Worte "wieviel" steckt, und das ist das Gewertete, und sein Gegenüber ist die 6, die die Schätzung ist.

**II.** Und der zweite Gesichtspunkt ist das Wort dessen, der sagt "10 um 8, um wieviel des Werts 4?" und manchmal sagt einer "vier von ihnen, wieviel ihr Wert?" Die 10 nun sind die geschätzte Zahl und die steht gegenüber der Zahl, die der unbekannte Wert ist, der in seinem Wort "wieviel", und die 8 ist die Zahl, die die Schätzung ist, und diese steht gegenüber der offenkundigen, die das Gewertete ist, nämlich 4; so multipliziere die beiden gegenüberstehenden offenkundigen Zahlen eine in die andere, nämlich 4 in 8, so gibt es 32, und teile sie in die letzte offenkundige Zahl, die die geschätzte ist, nämlich 10, so gibt es  $3\frac{1}{5}$  und das ist die Zahl, die der Wert ist, und sie steht gegenüber den 10, durch ("über!") die du geteilt hast. Und so (sind) alle Geschäfte der Leute und ihre Regel, so Gott d. E. will.

**III.** Und wenn ein Fragender fragt und sagt: Ein Tagelöhner, den ich gemietet im Monat für 10 Dirhem, arbeitet 6 Tage, wieviel ist sein Anteil? so weißt du, daß 6 Tage  $\frac{1}{5}$  des Monats und daß, was sein Anteil von den Dirhem ist, (sich berechnet) gemäß dem, was er von einem Monat gearbeitet hat. Und die Regel dafür ist, daß sein Wort "Monat" 30 Tage (ausmacht), und das ist das Geschätzte, und sein Wort "10 Dirhem" ist die Schätzung und sein Wort "6 Tage" ist das Gewertete und sein Wort "wieviel ist sein Anteil" ist der Wert. So multipliziere die Schätzung, nämlich 10, in das Gewertete, das ihm gegenübersteht, nämlich 6, das gibt 60, und teile es durch die 30, die die (letzte) offenkundige Zahl sind, so ist das das Geschätzte; das gibt 2 Dirhem, und das ist der Wert. Und das ist, was die Leute untereinander verhandeln vom Austausch und vom Maß und vom Gewicht.

Es verlohnt sich, noch einen Blick in die lateinische Übersetzung des Kapitels (Libri a. a. O., S. 286) zu werfen. Wir finden die kritischen Sätze am Anfang vollkommen einwandfrei mit folgenden Worten wiedergegeben: *Scias quod conventiones negotiationis hominum sunt secundum duos modos cum quattuor numeris quibus interrogator loquitur. Qui sunt pretium et cipretiatum secundum positionem, et pretium et appretiatum secundum querentem. Numerus vero qui est appretiatum secundum positionem opponitur numero qui est pretium secundum querentem ...* Ebenso später: *Horum vero quattuor numerorum tres semper manifesti et noti, et unus est ignotus ...* und *Impossible est enim quin duo eorum sint quorum unusquisque suo compari est oppositus ...* etc.

Wir sehen die frühere Beobachtung bestätigt, daß diese lateinische Übersetzung besser ist als Rosens Wiedergabe, ja wir können sogar wieder nachweisen, daß sie einen besseren Text wiedergibt, als ihn Rosen zur Verfügung hatte.

Nicht nur, daß jene kritisch beanstandeten Stellen hier in ganz richtiger Folge wiedergegeben sind, auch der Text des ersten Beispiels zeigt eine Variante, die den Sinn der Aufgabe klarer wiedergibt. Denn es heißt hier nicht "10 um 6, wieviel um 4", sondern *decem cafficii (haẓf, ein bekanntes Hohlnmaß) sunt pro sex dragmis: quot ergo perveniet tibi pro quattuor dragmis?*

Eine Vergleichung der indischen Regeln und Beispiele mit der Darstellung des Gegenstandes durch Muhammad b. Mūsā zeigt neben aller Abhängigkeit — diese ist ganz besonders auch in

$6\frac{2}{3}$ , and that is the unknown number contained in the word "how much," and that is the valued thing, and its opposite is 6, which is the valuation.

**II.** And the second aspect is the word of him who says "10 for 8, for how much of the value 4?" and sometimes one says "four of them, how much their value?" The 10 now are the appraised number and it stands opposite the number that is the unknown value, [contained] in his word "how much," and 8 is the number that is the valuation, and this stands opposite the manifest one that is the valued thing, namely 4; so multiply the two opposite manifest numbers one into the other, namely 4 into 8, so it gives 32, and divide it by the last manifest number that is the appraised one, namely 10, so it gives  $3\frac{1}{5}$  and that is the number that is the value, and it stands opposite the 10 by (over!) which you have divided. And thus (are) all the transactions of people and their rule, God willing.

**III.** And if a questioner asks and says: A day-laborer, whom I hired for a month for 10 dirhams, works 6 days, how much is his share? then you know that 6 days are  $\frac{1}{5}$  of the month and that what his share of the dirhams is (is calculated) according to what he has worked of a month. And the rule for it is that his word "month" is 30 days, and that is the appraised thing, and his word "10 dirhams" is the valuation and his word "6 days" is the valued thing and his word "how much is his share" is the value. So multiply the valuation, namely 10, into the valued thing, which stands opposite it, namely 6, that gives 60, and divide it by 30, which are the (last) manifest number, so that is the appraised thing; that gives 2 dirhams, and that is the value. And that is what people negotiate among themselves about exchange and measure and weight.

It is worthwhile to cast yet a glance into the Latin translation of the chapter (Libri, op. cit., p. 286). We find the critical sentences at the beginning reproduced completely faultlessly with the following words: *Scias quod conventiones negotiationis hominum sunt secundum duos modos cum quattuor numeris quibus interrogator loquitur. Qui sunt pretium et cipretiatum secundum positionem, et pretium et appretiatum secundum querentem. Numerus vero qui est appretiatum secundum positionem opponitur numero qui est pretium secundum querentem ...* Likewise later: *Horum vero quattuor numerorum tres semper manifesti et noti, et unus est ignotus ...* and *Impossible est enim quin duo eorum sint quorum unusquisque suo compari est oppositus ...* etc.

We see confirmed the earlier observation that this Latin translation is better than Rosen's rendering; indeed, we can even demonstrate again that it renders a better text than Rosen had available.

Not only are those critically noted passages reproduced here in quite correct sequence, but also the text of the first example shows a variant that renders the sense of the problem more clearly. For it says here not "10 for 6, how much for 4," but *decem cafficii (.ha.ẓf, a well-known dry measure) sunt pro sex dragmis: quot ergo perveniet tibi pro quattuor dragmis?*

A comparison of the Indian rules and examples with the presentation of the subject by Muhammad b. Mūsā shows, alongside all dependence—this is especially to be seen in the lack of a proof

dem Fehlen eines Beweises für das Verfahren zu erblicken — doch genug von persönlicher Eigenart.

Der Araber faßt gleich alle vier Zahlen ins Auge und gruppiert sie paarweise, während die Indier die Regel nach den drei bekannten Zahlen benennen und die Stellung der mittleren zwischen zwei gleichbenannten besonders hervorheben; der Araber ahmt die indische Terminologie nicht sklavisch nach, sondern verbessert sie, indem er zwei Grundwörtern zwei abgeleitete gegenüberstellt.

Aber er beschränkt sich in seinen drei breit ausgeführten, für den elementarsten Verstand faßbar gemachten Beispielen auf die einfache und direkte Regel de tri und bleibt dadurch wesentlich hinter den Indiern zurück.

Unter den Abhandlungen der *Iḥān al-ṣafāʾ* ist die sechste vollständig der Behandlung der arithmetischen und geometrischen Verhältnisse gewidmet. Eine gewisse Übersicht ihres Inhalts gewinnt man aus der Übersetzung von Dieterici; nur darf man sie nicht als Quellschrift benutzen. Der mathematische Inhalt ist wesentlich griechisch; von der Berechnung unbekannter Größen im Handel ist mehrfach die Rede (ed. Bombay I, S. 6, 11), doch wird nur *muqābala* und *mumātala* unterschieden.

Der kennzeichnendste Satz lautet: وكل من سال عن ثمن شيء فانه لا بد ان يذكر اربعة مقادير ثلاثة منها معلومة وواحدة مجهولة وبين كل مقادير منها نسبتان نسبة مباشرة ونسبة مفاضلة "Jeder, der nach dem Preis irgend einer Sache fragt, muß notwendig vier Größen aussprechen, von denen drei bekannt sind und eine unbekannt ist; und zwischen je zwei Größen bestehen zwei Verhältnisse, ein direktes und ein umgekehrtes."

Von der formalen Behandlung der Aufgaben, wie sie Muhammad b. Mūsā nach indischem Vorbild in die arabische Mathematik einführte, sind bei Behā ad-dīn nur noch wenige Reste geblieben. Er steht in seinem Kapitel *في اخراج المجهول بالاربعة المقابلة* *fī ikhrāj al-majhūl bi-l-arbaʿa l-muqābala* "Über die Elimination der Unbekannten durch die vier proportionalen (Zahlen)" ganz auf dem Boden der Lehre von den Proportionen.

Nur in dem Beispiel, das offenbar der ersten Aufgabe bei Muhammad b. Mūsā nachgebildet ist, begegnet uns auch noch dessen Terminologie: اذا قيل خمسة اطنان وثلاثة دراهم فطنان فكم "Wurde gesagt: 'fünf Pfund und drei Dirhem, zwei Pfund um wieviel?' so ist 5 Pfund das Geschätzte und 3 die Schätzung und die 2 Pfund das Gewertete, und das Gefragte ist der Wert, und das Verhältnis (*nisba*) des Geschätzten zur Schätzung ist wie das Verhältnis des Gewerteten zum Wert; die Unbekannte ist die vierte (Zahl), teile also das Produkt (die Fläche) der beiden mittleren (Zahlen), nämlich 6, durch die erste, nämlich 5."

for the procedure—yet enough of personal character.

The Arab takes all four numbers into view at once and groups them in pairs, while the Indians name the rule after the three known numbers and especially emphasize the position of the middle one between two similarly named ones; the Arab does not slavishly imitate the Indian terminology, but improves it by setting two derived words against two root words.

But he restricts himself in his three broadly executed examples, made comprehensible for the most elementary understanding, to the simple and direct rule of three, and thereby remains essentially behind the Indians.

Among the treatises of the *Ikhwān al-Ṣafāʾ*, the sixth is completely devoted to the treatment of arithmetic and geometric ratios. A certain overview of its content is gained from Dieterici's translation; only one must not use it as a source text. The mathematical content is essentially Greek; the calculation of unknown quantities in commerce is mentioned several times (ed. Bombay I, pp. 6, 11), but only *muqābala* and *mumāt.t.hala* are distinguished.

The most characteristic sentence reads: وكل من سال عن ثمن شيء فانه لا بد ان يذكر اربعة مقادير ثلاثة منها معلومة وواحدة مجهولة وبين كل مقادير منها نسبتان نسبة مباشرة ونسبة مفاضلة "Everyone who asks about the price of any thing must necessarily pronounce four quantities, of which three are known and one unknown; and between any two quantities of them there are two ratios, a direct and an inverse ratio."

Of the formal treatment of problems, as Muḥammad b. Mūsā introduced it into Arabic mathematics after Indian models, only few remnants remain in Behā ad-Dīn. He stands in his chapter *في اخراج المجهول بالاربعة المقابلة* *fī ikhrāj al-majhūl bi-l-arbaʿa l-muqābala* "On the elimination of the unknown by the four proportional (numbers)" entirely on the ground of the doctrine of proportions.

Only in the example, which is obviously modelled after the first problem in Muḥammad b. Mūsā, do we encounter his terminology as well: اذا قيل خمسة اطنان وثلاثة دراهم فطنان فكم "If it is said: 'five pounds and three dirhams, two pounds for how much?' then 5 pounds is the appraised thing and 3 the valuation and the 2 pounds the valued thing, and what is asked is the value, and the ratio (*nisba*) of the appraised thing to the valuation is as the ratio of the valued thing to the value; the unknown is the fourth (number), so divide the product (the area) of the two middle (numbers), namely 6, by the first, namely 5."

## 11 XI. Quellenuntersuchung / Source Investigation

Die Entwicklung der Mathematik bei den Arabern kann man nach dem bisher Gesagten etwa in folgende Zeitabschnitte einteilen: eine ältere Periode von Muhammad b. Mūsā um 825 bis Tabit b. Qurra um 900, eine mittlere von Abū Kāmil Šūcāʾ b. Aslam um 900 bis al-Karāgī um 1030 und eine jüngere Periode von ʿOmar al-Hayyām (gest. 1123) bis an-Nasawī (gest. um 1030 oder 1039) und al-Qalasādī (gest. 1487).

The development of mathematics among the Arabs can, according to what has been said so far, be divided approximately into the following periods: an older period from Muḥammad b. Mūsā around 825 to Thābit b. Qurra around 900, a middle period from Abū Kāmil Shujāʾ b. Aslam around 900 to al-Karajī around 1030, and a younger period from ʿUmar al-Khayyām (d. 1123) to al-Nasawī (d. around 1030 or 1039) and al-Qalasādī (d. 1487).

Von Muhammad b. M<sup>u</sup>sā's Zeitgenossen und n<sup>a</sup>chsten Nachfolgern ist uns wenig bekannt. Der arabische Arzt Ḥunayn b. Isḥāq (808–873) erz<sup>h</sup>lt uns von den Schwierigkeiten, die ihm bei der "Übersetzung mathematischer B<sup>ü</sup>cher begegneten. Er konnte weder bei den Syrern noch bei den Arabern einen Lehrer finden; von letzteren verstanden drei oder vier, hatten aber keine Zeit, ihm Unterricht zu geben, und nachdem er durch ein "ähnliches Mißgeschick drei dieser Gelehrten verloren hatte, beschloß er, nach Alexandrien zu gehen und dort bei einem "ber<sup>u</sup>hmten Philosophen" zu studieren.

Auf die Frage, in welcher Sprache jener den Unterricht erteilte, antwortete Ḥunayn: "auf Griechisch". Hatte er nicht "auf Griechisch" gesagt, es h<sup>a</sup>tte nicht geglaubt werden k<sup>ö</sup>nnen. Weiter setzte Ḥunayn hinzu: "ich blieb dort einige Zeit in meinem Bestreben, dort zu studieren. Er starb, und ich kam nach Bagdad und verstand nichts von der Mathematik".<sup>11</sup>

<sup>11</sup> Medizinisches aus der Pariser Handschrift, "übersetzt von Ḥunayn, ed. A. Müller, Königsberg 1884, II, S. 21 ff. Zur gleichen Zeit lebte in Bagdad der "Übersetzer und Schriftsteller Qusṭā b. L<sup>u</sup>kā (gest. 913/14), der sowohl physikalisch-astronomische Schriften als auch eine Philosophiegeschichte verfaßte; von seinen "Übertragungen griechischer Werke sind uns "über 30 bekannt.

F<sup>ü</sup>r die Kenntnis der griechischen "Überlieferung kommen ferner in Betracht die Schriften des Ṭabīṭ b. Qurra (gest. 901), eines Verwandten des Ḥunayn b. Isḥāq, und die des Verfassers des *Fihrist*, al-Nadīm, der gegen Ende des 10. Jahrhunderts in Bagdad lebte. Beide liefern uns genaue Kenntnisse von den Namen der von den Indern und Griechen geschriebenen B<sup>ü</sup>cher; insbesondere nennt der *Fihrist* die meisten "Übertragungen von den Griechen ins Syrische und Arabische mit Angabe der "Übersetzer. Trotz all dieser Einzelangaben "über griechische Schriften f<sup>ü</sup>r Arithmetik, Geometrie, Optik und Astronomie finden wir aber von einer direkten "Übertragung des Nicomachus oder Theon, geschweige denn einer griechischen Algebra in arabischer Sprache keine Spur. Die Verh<sup>ä</sup>ltnisse liegen vielmehr gerade so, wie sie Ḥunayn schildert: es gab kein arabisches Wort f<sup>ü</sup>r Mathematik, und wer etwas dar<sup>ü</sup>ber wissen wollte, mußte Griechisch lernen oder aus einer Umgangssprache sch<sup>ö</sup>pfen, die griechische Termini mit arabischem Wortmaterial vermischte.

Unter diesen Umst<sup>ä</sup>nden wird es kaum wahrscheinlich sein, daß die Iḥān a-ṣafā' ihre Kenntnisse unmittelbar griechischen Quellen verdanken; daß sie vielmehr arabische und persische Gelehrte, "Übersetzer und Schriftsteller aller Art um sich versammelt hatten und eine Art inoffizieller Akademie bildeten, ist wahrscheinlich gemacht worden und w<sup>ä</sup>re als "außerst typisch f<sup>ü</sup>r jene Zeit und Stelle zu begr<sup>ü</sup>ßen.

Durch die Einleitung von 'Omars Algebra werden wir "über die unmittelbar vorausgehende Zeit bis auf Muhammad b. M<sup>u</sup>sā genau orientiert. Denn neben dem Verfasser der "ältesten arabischen Algebra erw<sup>ä</sup>hnt 'Omar die Araber al-Ḥādārī und 'Abd al-Ḥamīd b. Turk, denen er wesentliche Vervollst<sup>ä</sup>ndigungen verdankt, und den Befehlshaber von Afghanistan, den "ägypter al-Hasan b. al-Haitham, der den von den fr<sup>ü</sup>heren Algebraikern versagten Beweis der Gleichungen der dritten Potenz mithilfe der Kegelschnitte erbrachte, ferner den Perser Ab<sup>u</sup> al-Gud b. Mu-

Of Muḥammad b. Mūsā's contemporaries and nearest successors little is known to us. The Arab physician Ḥunayn b. Isḥāq (808–873) tells us of the difficulties that he encountered in translating mathematical books. He could find no teacher among either the Syrians or the Arabs; of the latter, three or four understood him, but had no time to give him instruction, and after through a similar misfortune he had lost three of these scholars, he resolved to go to Alexandria and study there under a "renowned philosopher."

When asked in what language that one gave instruction, Ḥunayn answered: "in Greek." Had he not said "in Greek," it could not have been believed. Further Ḥunayn added: "I remained there some time in my endeavor to study there. He died, and I came to Baghdad and understood nothing of mathematics."<sup>11</sup>

<sup>11</sup> Medizinisches aus der Pariser Handschrift, "übersetzt von Hunain, ed. A. Müller, Königsberg 1884, II, pp. 21 ff. At the same time lived in Baghdad the translator and writer Qusṭā b. Lūkā (d. 913/14), who composed both physical-astronomical writings and a history of philosophy; of his translations of Greek works more than 30 are known to us.

For knowledge of the Greek tradition there come further into consideration the writings of Ṭābit b. Qurra (d. 901), a relative of Ḥunayn b. Isḥāq, and those of the author of the *Fihrist*, al-Nadīm, who lived in Baghdad toward the end of the 10th century. Both provide us with precise knowledge of the names of books written by Indians and Greeks; in particular, the *Fihrist* names most translations from Greek into Syriac and Arabic, with indication of the translators.

Despite all these individual details about Greek writings on arithmetic, geometry, optics, and astronomy, however, we find no trace of a direct transmission of Nicomachus or Theon, let alone of a Greek algebra in Arabic. The situation is rather just as Ḥunayn describes it: there was no Arabic word for mathematics, and whoever wanted to know something about it had to learn Greek or draw from a colloquial language that mixed Greek terms with Arabic word material.

Under these circumstances it will hardly be probable that the Ikhwān al-Ṣafā' owed their knowledge directly to Greek sources; that they rather had gathered around them Arab and Persian scholars, translators, and writers of all kinds and formed a kind of unofficial academy has been made probable and would be to be welcomed as extremely typical for that time and place.

Through the introduction of 'Umar's Algebra we are accurately oriented about the immediately preceding period back to Muḥammad b. Mūsā. For besides the author of the oldest Arabic algebra, 'Umar mentions the Arabs al-Khayyāmī and 'Abd al-Ḥamīd b. Turk, to whom he owes essential completions, and the commander of Afghanistan, the Egyptian al-Ḥasan b. al-Haytham, who furnished with the help of conic sections the proof of the equations of the third degree that was refused by the earlier algebraists, further the Persian Abū al-Jūd b. Muḥammad

hammad und im Auslande die Griechen Menelaos, Euklid und Apollonios.

Was die *Iḥān al-ṣafā'* "über den Zahlbegriff gelehrt haben, h"atte also an griechischen Vorbildern durchaus nicht Mangel; aber die Aufgaben und Rechenverfahren, wie sie uns in der "ältesten Algebra begegnet sind, weisen nicht auf Griechenland, sondern auf Indien als Ursprungsland.

Nehmen wir an, Muhammad b. M"usā habe sich lediglich als Redaktor "aufgef"uhrt, der die ihm zug"anglichen Quellen geschickt in ein Ganzes zusammenf"ugte, so ist doch bei einem so erstaunlichen Mangel an "Überlieferung vor ihm, wie ihn die Araber aufwiesen, das Verdienst eines solchen Unternehmens nicht zu untersch"atzen.

Dieses Verdienst wird aber in unglaublicher Weise erh"ohet, wenn wir seine wissenschaftliche Eigenleistung dahin bestimmen, daß er die "überlieferte Aufgabenformulierung in den Sprachformen der griechischen Theorie der quadratischen Gleichungen dahingehend umgestaltete, daß er den Anschluß an eine Systematik erm"oglichte, die im Einklang stand mit dem Geist und den wissenschaftlichen Interessen der damaligen gebildeten Welt.

Nicht als Mathematiker, sondern als Organisator der Wissenschaften und Vermittler fremden Wissens verdient Muhammad b. M"usā einen hohen Platz in der Geschichte der Mathematik. Daß er sich nicht "überschätzte, sondern sich bescheiden in der Rolle eines Bearbeiters und Zusammenstellers sah, l"äßt sich nicht nur aus der Einleitung zur Algebra schließen, in der er nicht den geringsten Anspruch auf wissenschaftliche Eigenleistung erhebt, sondern auch aus den S. 3 mitgeteilten S"atzen "über die Eins, die er, was die begriffliche Seite angeht, ohne weiteres aus griechischen Quellen entnahm.

Nach der Quellenlage werden wir dabei annehmen m"üssen, daß die Kenntnis der griechischen Mathematik in Arabien nur mangelhaft war und daß eine selbst"andige wissenschaftliche Produktion nicht ins Auge gefaßt werden konnte, ehe nicht die "Übersetzungen der griechischen Klassiker und eine gr"undliche Einarbeitung in die fremden Wissenschaften vorausgegangen waren.

Wie das bei Muhammad b. M"usā der Fall war, haben wir im Abschnitt "über die quadratischen Gleichungen und die Geometrie an einwandfreien Textbeispielen nachweisen k"önnen. Daß die griechischen Namen der Zahleigenschaften *quer à travers* (Carra de Vaux) in das arabische Zahlensystem "übergehen, ist von der tats"achlichen Kenntnis der griechischen Lehre noch verschieden; und daß die aufgabenm"äßige Behandlung der quadratischen Gleichungen eine "Übersetzung der geometrischen Beweise des Euklid in algebraische Sprache ist, hat Cantor I<sup>2</sup>, S. 698; I<sup>3</sup>, S. 720 als erster festgestellt.

Bei dem heutigen Stande der Kenntnis der *Data* des Euklid erscheint es mir aber m"oglich, die genaue "Übertragungsweise zu bestimmen. Dazu ist zun"achst n"otig, daß man die Hauptbeweise der Euklidischen Werkstatt mit ihren charakteristischen Ausdr"ucken den arabischen "Übersetzungen gegen"uberstellt.<sup>12</sup> Man wird dann sehen, daß in der Tat fast "übergangslos griechi-

and abroad the Greeks Menelaus, Euclid, and Apollonius.

What the *Iḥwān al-Ṣafā'* taught about the concept of number would thus have had no lack of Greek models; but the problems and calculation procedures, such as we encounter them in the oldest algebra, point not to Greece but to India as the country of origin.

If we assume that Muḥammad b. Mūsā acted merely as an editor who skillfully assembled the sources accessible to him into a whole, then with such an astonishing lack of transmission before him as the Arabs exhibited, the merit of such an undertaking is not to be underestimated.

But this merit is enhanced in an incredible way if we determine his own scientific achievement as consisting in this, that he reshaped the traditional problem formulation in the linguistic forms of the Greek theory of quadratic equations in such a way that he enabled a connection to a system that was in harmony with the spirit and scientific interests of the educated world of that time.

Not as a mathematician, but as an organizer of sciences and mediator of foreign knowledge does Muḥammad b. Mūsā deserve a high place in the history of mathematics. That he did not overestimate himself, but modestly saw himself in the role of an adapter and compiler, can be inferred not only from the introduction to the Algebra, in which he raises not the slightest claim to scientific achievement of his own, but also from the sentences about unity communicated on p. 3, which he, as far as the conceptual side is concerned, took directly from Greek sources.

According to the state of the sources we must assume thereby that the knowledge of Greek mathematics in Arabia was only deficient and that an independent scientific production could not be contemplated before translations of the Greek classics and a thorough assimilation of the foreign sciences had preceded.

As was the case with Muḥammad b. Mūsā, we have been able to demonstrate in the section on quadratic equations and geometry by unobjectionable textual examples. That the Greek names of number properties pass "*quer à travers*" (Carra de Vaux) into the Arabic number system is still different from actual knowledge of the Greek doctrine; and that the treatment of quadratic equations in the form of problems is a translation of the geometrical proofs of Euclid into algebraic language, Cantor I<sup>2</sup>, p. 698; I<sup>3</sup>, p. 720 was the first to establish.

At the present state of knowledge of Euclid's *Data*, however, it seems to me possible to determine the exact mode of transfer. For this it is first necessary to set the chief proofs of Euclid's workshop with their characteristic expressions against the Arabic translations.<sup>12</sup> One will then see that in fact Greek and Arabic expressions can be placed side by side almost without transition.

<sup>12</sup>Zu diesem Zwecke benutze ich die Begriffe *gegeben* und *Beweis* im folgenden in dem Sinne, den sie bei den griechischen Mathematikern besitzen, nicht in dem der neueren Mathematik, wo der Begriff der Gegebenheit, von Zeuthen mit Recht als grundlegend hervorgehoben, nur in der Form der Voraussetzung eines Existenzbeweises weiterlebt.

<sup>12</sup>For this purpose I use the concepts *given* and *proof* in the following in the sense that they possess with the Greek mathematicians, not in that of modern mathematics, where the concept of givenness, rightly emphasized by Zeuthen as fundamental, survives only in the form of the assumption of an existence proof.



sche und arabische Ausdrücke nebeneinandergestellt werden können.

Wir finden z. B. in *Data* 84: *Si recta datam habeat rationem ad aliam datam datam habentem rationem, erit ea data magnitudine.*

*Beweis:* Da nämlich das Verhältnis von  $AB$  zu  $CD$  gegeben ist, so ist das Verhältnis von  $AB$  zu  $CD$  das gleiche wie das Verhältnis einer gegebenen Größe zu einer gegebenen Größe. Und es ist  $CD$  in einer gegebenen Größe gegeben; auch wird  $AB$  in einer gegebenen Größe gegeben sein [*Data* 2].

Setzen wir neben diesen Beweis den des Muhammad b. Mūsā für die Gleichung  $x^2 + bx = c$ , so können wir die Übereinstimmung im einzelnen feststellen. Er sagt zu dieser Gleichung: "Du nimmst die Hälfte der Wurzeln, und es ist in diesem Beispiel fünf, und multiplizierst sie in sich selbst, das gibt 25, und du addierst es zu 39, das gibt 64. Du nimmst die Wurzel aus 64, das gibt 8, und du entfernst davon die Hälfte der Wurzeln, nämlich 5, und 3 bleibt. Und das ist die Wurzel des Vermögens, und das Vermögen ist 9."

Wie der griechische Mathematiker die Konstanz eines Verhältnisses als seine Daten behandelt und aus dieser gegebenen Konstanz das Ergebnis ableitet, so geht der Araber von den gegebenen Zahlen aus und zeigt, daß die gesuchte Größe durch eine feste Rechenoperation gewonnen wird. Beide halten sich an die Form, "es wird gesucht" und "es wird gefunden"; beide vermeiden es, den Leser auf den allgemeinen Fall zu führen.

**Unsere Untersuchung** kann in ihrem wesentlichsten Ergebnisse dahin zusammengefaßt werden, daß Muhammad b. Mūsā's Algebra auf indischem Stoff aufgebaut ist, den er nach griechischer Methode geordnet und in arabischer Form ausgesprochen hat. Dieses Resultat befreit uns von der Annahme, daß die Araber die Algebra den Griechen verdanken, und gestattet es, ihnen mit vollem Recht die Wahrung der Priorität für die Übertragung und Verbreitung dieser Rechenkunst zu überlassen.

Bis zu einem gewissen Grade bestärkt unsere Untersuchung auch die Nachrichten über das Verhältnis von Muhammad b. Mūsā zu seinem Förderer al-Ma'mun. Nach dem *Fihrist* soll er von dem Kalifen beauftragt worden sein, ein kurzgefaßtes Buch über die Rechnung durch Ergänzung und Ausgleichung zu schreiben; es ist zu den Problemen über die Teilung von Erbschaften geworden. Der Kalif soll ihm befohlen haben, die darin enthaltene Behandlung der Aufgaben leicht und klar zu machen, und er soll zuerst gezeigt haben, wie die Beweise mit Hilfe der Figur gemacht werden.

Allerdings erweckt der aus dem Arabischen übersetzte Text den Anschein, als sei er an den Zügen einer festen Tradition geschaffen, als habe der Verfasser bei der schriftlichen Fixierung mit allgemein verbreiteten Überlieferungen gerechnet und sich ihrer bedient, um die Nachfrage nach seinem Buche zu sichern.

Daß er aber den Vorzug hatte, auf Veranlassung eines Mannes zu schreiben, der die Wissenschaft förderte und in dessen Gefolgschaft Mathematiker und Übersetzer über die verschiedensten Zweige der Wissenschaften arbeiteten, kann seinen Anschluß an die besten Quellen und die rasche Verbreitung seines Buches wesentlich erleichtert haben.

Auch in biographischer Hinsicht verdient die Tatsache Beachtung, daß Muhammad b. Mūsā als Astronom eine genaue Kennt-

We find, for example, in *Data* 84: "If a line has a given ratio to another [line] having a given ratio, it will be given in magnitude."

*Proof:* Since namely the ratio of  $AB$  to  $CD$  is given, the ratio of  $AB$  to  $CD$  is the same as the ratio of a given magnitude to a given magnitude. And  $CD$  is given in a given magnitude; also  $AB$  will be given in a given magnitude [*Data* 2].

Setting beside this proof that of Muhammad b. Mūsā for the equation  $x^2 + bx = c$ , we can establish the agreement in detail. He says for this equation: "You take half of the roots, and it is in this example five, and multiply it by itself, that gives 25, and you add it to 39, that gives 64. You take the root of 64, that gives 8, and you remove from it half of the roots, namely 5, and 3 remains. And that is the root of the possession, and the possession is 9."

As the Greek mathematician treats the constancy of a ratio as his data and derives the result from this given constancy, so the Arab proceeds from the given numbers and shows that the sought quantity is obtained through a fixed calculational operation. Both adhere to the form "it is sought" and "it is found"; both avoid leading the reader to the general case.

**Our investigation** can be summarized in its most essential result as follows: that Muhammad b. Mūsā's Algebra is built on Indian material, which he has ordered according to Greek method and expressed in Arabic form.

This result frees us from the assumption that the Arabs owe algebra to the Greeks, and permits us to leave them with full right the priority for the transmission and dissemination of this calculational art.

To a certain degree our investigation also confirms the reports about the relationship of Muhammad b. Mūsā to his patron al-Ma'mun. According to the *Fihrist*, he was commissioned by the caliph to write a concise book on calculation by completion and balancing; it became the work on problems of the division of inheritances. The caliph is said to have commanded him to make the treatment of the problems contained therein easy and clear, and he is said first to have shown how the proofs are made with the help of the figure.

To be sure, the text translated from Arabic gives the impression of being made on the lines of a fixed tradition, as if the author, in the written fixation, had reckoned with generally widespread traditions and made use of them to secure the demand for his book.

But that he had the advantage of writing at the instigation of a man who promoted science and in whose circle mathematicians and translators worked on the most varied branches of knowledge, can have essentially facilitated his connection to the best sources and the rapid dissemination of his book.

Also in biographical respects the fact deserves attention that Muhammad b. Mūsā as astronomer must have had precise know-

nis der damals aufgekommenen indischen Astronomie und Mathematik haben mußte, und daß die ihm von seinen Zeitgenossen zugeschriebene Kenntnis des Griechischen und Persischen seinen Arbeitsbereich "über das bloß "Übersetzen weit hinausdehnte. Die soziale Stellung des Verfassers scheint durch die namentliche Erwähnung des Kalifen in der Einleitung und die für die damalige Zeit völlig ungewöhnliche Danksagung: *يكفيني امره الذي علي فيه وانعامه علي yakfīnī amru l-lāhi lladī 'alayya fīhi wa-in'āmu-hu 'alayya*, "seine Machtvollkommenheit genügt mir und sein, was sie an Gnaden auf mich vergossen hat", festgelegt; doch konnte sie auch für einen Gelehrten ziemlich unabhängig von der Macht gelten, der den Regenten gegenüber in geistiger Hinsicht ebenso erhaben dachte wie die Zeitgenossen des Hunayn. Der Text der Algebra ist "überlieferungsgeschichtlich von besonderem Interesse, weil er die Fortpflanzung griechischen, indischen und arabischen Wissens in einer einzigen Schrift vereinigt. Daß nicht nur Probleme und Verfahren, sondern auch der Stil "überliefert ist, macht Muhammad b. Mūsā's Werk zu einer Brückenschrift *par excellence*, die es ermöglicht, den Übergang von der antiken zur mittelalterlichen Mathematik in einer Kontinuität zu verfolgen, die sonst nur mühsam aus "überlieferungsgeschichtlichen Daten zu rekonstruieren wäre.

## 12 XII. Schlußbetrachtung / Concluding Observations

Wie keine ernstliche Untersuchung "über die Entstehung der arabischen Wissenschaft ohne Berücksichtigung der politischen Verhältnisse möglich ist, so kann auch eine Erörterung der Quellen der Algebra und der geschichtlichen Voraussetzungen ihrer Entwicklung den politischen Zusammenhang nicht "übersehen.

Die Zeit Muhammads und der ersten Kalifen war die Zeit der frohen und unbeschwerten Eroberung fremder Länder und fremder Reichtümer; die Beute fiel den Siegern zu, der Unterworfenen zahlte Tribut, und wenn er den Islam annahm, wurde er steuerfrei. Das einfache Kriegerideal einer strengen, grob und selbstgenügsam lebenden Ritterschaft hatte mit der Wissenschaft noch nichts zu tun.

Als die Leidenschaften der Eroberung sich abkühlten, wurden die kriegerischen Neigungen durch einen prunkliebenden Luxus überwuchert und die politischen wie persönlichen Beziehungen durch die Notwendigkeit eines friedlichen Zusammenlebens komplizierter. Innerhalb der islamischen Welt spaltete sich die politische und religiöse Einheit, die schon durch den Tod des Propheten ins Wanken geraten war.

Die abbasidischen Kalifen suchten die gottgewollte Einheit der Glaubensgenossen durch ein geordnetes Rechtswesen und eine Kulturpolitik zu erzwingen, die den Anschluß an die "älteren, der untergehenden sassanidischen Kultur nahestehenden Zustand wiederherstellte. Diese Politik konnte den Antrieb zur Wissenschaft nicht selbst hervorbringen, wohl aber günstige Bedingungen für sie schaffen.

Denn unter den für das Gedeihen der Wissenschaft günstigsten Bedingungen gehört nicht die inmitten des Alltags, sondern die den Alltag "überhebende Stellung des Gelehrten. Wo der Gelehr-

ledge of the Indian astronomy and mathematics that had then arisen, and that the knowledge of Greek and Persian attributed to him by his contemporaries extended his working sphere far beyond mere translating.

The social position of the author seems to be established by the mention of the caliph by name in the introduction and the completely unusual expression of thanks for that time: *يكفيني امره الذي علي فيه وانعامه علي yakfīnī amru llāhi lladī 'alayya fīhi wa-in'āmu-hu 'alayya*, "his omnipotence is sufficient for me and his, what it has poured out in graces upon me"; but it could also apply for a scholar quite independent of power, who considered himself intellectually just as elevated vis-à-vis the ruler as did the contemporaries of Hunayn.

The text of the Algebra is of special interest for the history of transmission because it unites the propagation of Greek, Indian, and Arabic knowledge in a single writing. That not only problems and procedures but also style is transmitted makes Muhammad b. Mūsā's work a bridging text *par excellence*, which enables one to follow the transition from ancient to medieval mathematics in a continuity that would otherwise only be laboriously reconstructable from transmission-historical data.

As no serious investigation of the origin of Arabic science is possible without consideration of the political circumstances, so too a discussion of the sources of algebra and the historical presuppositions of its development cannot overlook the political context.

The time of Muhammad and the first caliphs was the time of joyful and carefree conquest of foreign lands and foreign riches; the booty fell to the victors, the subjugated paid tribute, and if he accepted Islam, he was freed from taxes. The simple warrior ideal of a strict, rough, and self-sufficient knighthood had nothing to do with science as yet.

As the passions of conquest cooled, the warlike inclinations were overgrown by a pomp-loving luxury, and the political as well as personal relations became more complicated through the necessity of peaceful coexistence. Within the Islamic world, the political and religious unity, already shaken by the death of the Prophet, split apart.

The Abbasid caliphs sought to enforce the divinely willed unity of the faithful through an ordered legal system and a cultural policy that restored the connection to the older conditions close to the declining Sasanid culture. This policy could not itself produce the impulse to science, but could create favorable conditions for it.

For among the conditions most favorable for the flourishing of science belongs not [the scholar's] position within the midst of everyday life but one that towers above everyday life. Where the

te dem Begehren und Willk"ur der M"achtigen ausgeliefert ist, kann Wissenschaft nicht gedeihen; wo er von ihnen gesch"utzt wird und in stande ist, auch die Freiheit des Denkens zu wahren, entsteht die ruhige produktive Arbeit, die zur Bl"ute der Wissenschaft f"uhrt.

F"ur Muhammad b. M"usā l"äßt sich dies Verh"ältnis des Schutzes und der Freiheit des Gelehrten in dem Briefe nachweisen, den al-Ma'mun an ihn richtete und der nach dem *Fihrist* den Anlaß zu seiner algebraischen Arbeit gegeben hat. Der Kalif bittet den Gelehrten, die Geheimnisse der Rechnungswissenschaft so darzulegen, daß die mit ihr weniger vertraute Menschen sie verstehen k"önnen; er versichert ihm seiner Dankbarkeit und seines Schutzes.

Die griechische Wissenschaft war in Byzanz und Syrien bis in die Zeit der arabischen Eroberung lebendig geblieben und hatte durch "Übersetzungen ins Syrische und Armenische Anschluß an die Geistigkeit des Vorderen Orients gewonnen. Diese Formen des Wissens "überlebten den politischen Umsturz, auch die sie pflegenden Kreise blieben bestehen und waren bereit, ihr Wissen mit den neuen Herren zu teilen.

Die Hellenisierung des Kalifenhofes war durch die Syrer und Griechen angebahnt; die "Übernahme von Rechtspflege und Verwaltung erforderte die Heranziehung gebildeter Einheimischer, die ihrerseits in den neuen Herren Beg"ünstiger der Wissenschaft fanden. Schon in der ersten H"älfte des 8. Jahrhunderts griff der Kalif auf die Dienste arabischer "Übersetzer zur"ück; die "ältesten Übersetzer griechischer Werke ins Arabische sind Syrer gewesen.

Muhammad b. M"usā schrieb seine Algebra w"ährend der Regierung al-Ma'muns. Die "Übersetzung griechischer Werke in diese Sprache war daher schon vorausgegangen; griechische und indische astronomische Tafeln waren in Gebrauch. Die Bed"ürfnisse der Astronomie hatten den Kalifen zur F"örderung der Mathematik veranlaßt, und es fehlte nicht an Gelehrten, die den Ruf des Mazdahurf"örderers der Wissenschaften beantworteten.

F"ur die Geschichte der Algebra ist es von entscheidender Bedeutung, daß Muhammad b. M"usā seine Kenntnisse zuerst als Astronom erworben hat. Denn die Astronomie war es, die die Araber zuerst und am dringendsten nach dem griechischen und indischen Wissen greifen ließ. Muhammad b. M"usā "übersetzte die indischen Rechenb"ücher und Astronomietafeln; als er sich an die Algebra machte, konnte er auf die indische Zahlenschrift und die indischen Rechenverfahren zur"ückgreifen.

Dieser Stand der Dinge war der Grund, weshalb die Algebra der Araber gleich von Anfang an eine angewandte Wissenschaft war. Sie wuchs nicht aus spekulativen Fragestellungen heraus, sondern aus den Bed"ürfnissen des b"ürgerlichen Lebens, des Handels und der Rechtspflege. Der erste Teil der Algebra, die Behandlung der quadratischen Gleichungen, stellt eine Verbindung der indischen Rechenkunst mit der griechischen Geometrie dar; der zweite Teil, die praktische Geometrie und das Gesch"äftsrechnen, erweitert den Anwendungsbereich, und der dritte Teil, die Erbteilungsrechnung, zeigt das ganze Ineinandergreifen von Mathematik und islamischem Recht.

Daß die Algebra erst sp"äter, nachdem sie die westlichen L"änder durchlaufen hatte, zu einer rein formwissenschaftlichen Disziplin

scholar is delivered over to the desire and arbitrariness of the mighty, science cannot flourish; where he is protected by them and is able to preserve also the freedom of thought, there arises the calm productive work that leads to the flowering of science.

For Muḥammad b. Mūsā this relationship of protection and freedom of the scholar can be demonstrated in the letter that al-Ma'mun addressed to him and which, according to the *Fihrist*, gave the occasion for his algebraic work. The caliph asks the scholar to present the secrets of the science of calculation so that people less familiar with it can understand it; he assures him of his gratitude and protection.

Greek science had remained alive in Byzantium and Syria up to the time of the Arab conquest and had gained connection to the intellectual life of the Near East through translations into Syriac and Armenian. These forms of knowledge survived the political upheaval; also the circles that cultivated them remained in existence and were ready to share their knowledge with the new masters.

The Hellenization of the caliphal court was initiated by the Syrians and Greeks; the take-over of jurisprudence and administration required the engagement of educated natives, who in turn found in the new rulers patrons of science. Already in the first half of the 8th century the caliph availed himself of the services of Arab translators; the oldest translators of Greek works into Arabic were Syrians.

Muḥammad b. Mūsā wrote his Algebra during the reign of al-Ma'mun. The translation of Greek works into this language had thus already taken place; Greek and Indian astronomical tables were in use. The needs of astronomy had induced the caliph to promote mathematics, and there was no lack of scholars who answered the call of the Maecenas of the sciences.

For the history of algebra it is of decisive significance that Muḥammad b. Mūsā first acquired his knowledge as an astronomer. For it was astronomy that first and most urgently let the Arabs reach for Greek and Indian knowledge. Muḥammad b. Mūsā translated the Indian arithmetic books and astronomical tables; when he turned to algebra, he could fall back on the Indian numeral script and the Indian calculation procedures.

This state of affairs was the reason why the algebra of the Arabs was an applied science from the very beginning. It did not grow out of speculative questionings, but out of the needs of civil life, of commerce, and of jurisprudence. The first part of the algebra, the treatment of quadratic equations, represents a connection of Indian arithmetic with Greek geometry; the second part, practical geometry and commercial calculation, extends the sphere of application, and the third part, inheritance calculation, shows the complete interlocking of mathematics and Islamic law.

That algebra only later, after it had passed through the Western countries, became a purely formal scientific discipline, is further

wurde, ist ein weiteres Zeugnis für die Notwendigkeit, historische Begriffe im Zusammenhang der geschichtlichen Entwicklung zu untersuchen und nicht von einem abstrakten, zeitlosen Begriff der Wissenschaft auszugehen.

Im Zusammenhang unserer Untersuchung muß noch auf einen Umstand hingewiesen werden, der für die Bewertung der Quellen von Bedeutung ist. Das Werk des Muhammad b. Mūsā ist für uns nur in einer späten Abschrift aus dem 14. Jahrhundert erhalten; die lateinischen Übersetzungen aus dem 12. Jahrhundert sind teils fragmentarisch, teils von den Arabisten als nicht genau genug beurteilt worden. Rosen hat zwar den Text nach der Oxforder Handschrift ediert, aber — wie wir mehrfach gesehen haben — seine Übersetzung ist nicht zuverlässig.

Eine neue kritische Edition des Textes und eine sorgfältige Übersetzung sind daher ein dringendes Desideratum; sie würde ohne Zweifel manche noch offene Fragen lösen und die vorliegende Untersuchung in manchem bestärken und ergänzen. Aber auch ohne eine solche Edition ist die Richtung der Untersuchung klar geworden: die arabische Algebra ist ein Produkt der Begegnung von drei Kulturen, der indischen, der griechischen und der arabischen, und ihr Schöpfer Muhammad b. Mūsā hat durch seine Bearbeitung das Werk in gleichem Maße dem Wissen der Vergangenheit wie den Bedürfnissen der Gegenwart dienstbar gemacht.

testimony to the necessity of investigating historical concepts in the context of historical development and not proceeding from an abstract, timeless concept of science.

In connection with our investigation, attention must yet be drawn to a circumstance that is significant for the evaluation of sources. The work of Muḥammad b. Mūsā is preserved for us only in a late copy from the 14th century; the Latin translations from the 12th century are partly fragmentary, partly judged by Arabists as not exact enough. Rosen edited the text after the Oxford manuscript, but—as we have seen repeatedly—his translation is not reliable.

A new critical edition of the text and a careful translation are therefore an urgent desideratum; it would doubtless solve many still open questions and confirm and supplement the present investigation in many respects. But even without such an edition, the direction of the investigation has become clear: Arabic algebra is a product of the encounter of three cultures, the Indian, the Greek, and the Arabic, and its creator Muḥammad b. Mūsā has through his adaptation made the work equally serviceable to the knowledge of the past and the needs of the present.

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A Bilingual Arabic-English Edition

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**al-Moqaddasi**

edidit / edited by

**M. J. de Goeje**

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EDITIO SECUNDA / SECOND EDITION

Lugduni Batavorum  
Apud E. J. Brill  
1906 / 2025

كِتَابُ

حَسَنُ أ

لَتَقِ اسْمِي

مَعْرِفَةِ ف

لَأَقِ اسْمِي

بِالْبَشَرِ أَلَمْ عُرِفْ لَمْ قَدْ سِي ل

---

وَكَرَّمَهُ بِمَنِّهِ عَنْهُ وَعَفَا تَعَالَى اللَّهُ وَحَمَاهُ

الثَّانِيَةُ الطَّبَعَةُ — الثَّانِيَةُ الطَّبَعَةُ

أَلَمْ حُرُوسَةً لِي دَنَ مَدِينَةٍ فِي بَعْ ط

بِرِّي لَمْ طَبَعَتْ ب

م ١٩٠٦ نَس

# AHSAN AL-TAQASIM

THE BEST DIVISIONS FOR KNOWLEDGE OF THE REGIONS

A BILINGUAL ARABIC-ENGLISH EDITION

With parallel Arabic text and English translation

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AUCTORE / AUTHOR

Shams ad-din Abu Abdallah Mohammed  
ibn Ahmed ibn abi Bekr

**al-Banna al-Basshari**

AL-MOQADDASI

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EDIDIT / EDITED BY

**M. J. DE GOEJE.**

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EDITIO SECUNDA.

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LUGDUNI BATAVORUM,  
APUD E. J. BRILL.  
1906.

## Author's Preface | Author's Preface

الرَّحْمَنُ الرَّحِيمُ اللّهُ سُبِّحَ

كَرِيمُ يَا بِفَضْلِكَ تُعَسِّرُ وَلَا يَسِّرُ رَبِّ

In the name of God, the Compassionate, the Merciful.

My Lord, make it easy and do not make it difficult, by Thy grace, O Generous One.

يُشَاوِرُهُ، مَشِيرَ بِلَا الْمَرِيَّةِ صَنِيعَ فَاتِقَتْنِ وَصَوَرَ الْفَلَكَ خَلَقَ الَّذِي لِلَّهِ الْحَمْدُ  
أَوْتَدَ إِعْوَانَ، بِلَا وَأَحْكَمَهَا اتَّقَانًا، اتَّقَنَهَا يُعَاضِدُهُ، مُعِينٌ بِلَا وَدَبْرَهَا  
مَأْوَاهَا يَغْلِبُ كَيْلًا بِالْبَحْرِ وَأَحَاطَهَا بِمَيْدٍ، لَثَلًا بِالرَّاسِيَّاتِ الْأَرْضِ  
أَحْسَنَ مِنْ فَنَنِهِمْ يَعْمَلُونَ كَيْفَ لِيَنْظُرَ عِبَادَهُ فِيهَا وَبَثَّ وَبَرِيدُ،  
وَأَكْرَمَ الْأَنْبِيَاءِ، خَاتَرَ عَلَى اللَّهِ وَصَلَّى وَلَيْسِي كَفَرَ مِنْ وَمِنْهُمْ وَافْتَدَى،  
كَثِيرًا، تَسْلِيمًا وَسَلَامًا وَصَحْبِهِ آلِهِ وَعَلَى مُحَمَّدٍ الْأَوَّلِيَاءِ،

Praise be to God, who created the celestial orb and formed it with consummate skill, without any counsellor to consult; who ordered it without any assistant to help; who perfected it in the most perfect manner, and established it without need of support; who fixed the earth firmly lest it should sway, and surrounded it with the sea as a barrier protecting its waters; and who dispersed His servants upon it that He might observe how they act—and among them are those who do good and are redeemed, and among them are those who disbelieve and forget. And may God bless the Seal of the Prophets, the noblest of the chosen, Muhammad, and his family and companions, with abundant peace.

الْعُلَمَاءُ زَالَتْ مَا فَإِنَّهُ بَعْدُ، أَمَّا الْمُقَدَّسِيُّ: أَحْمَدُ بْنُ مُحَمَّدٍ اللَّهِ عَبْدُ اللَّهِ أَبُو قَالَ  
أَخْبَارُهُمْ، تَنْقُطُ وَلَا تَحْرِيصُهُمْ، لَمَّا الْكُتُبُ تَصْنِيفُ فِي تَرْغَبُ  
ذِكْرِي، بِهِ أَحْيَى عِلْمًا وَأَقِيمَ سُنَّتَهُمْ، وَأَقْفُو سُنَّتَهُمْ، أَتَبَعَ أَنْ فَأَحْبَبْتُ  
عَلَى فَتَمَلَّقُوا الْعُلُومَ إِلَى سَبَقُوا قَدْ الْعُلَمَاءُ وَوَجَدْتُ وَفِيلَةً، بِهِ وَأَشِيرَ  
فَرَأَيْتُ وَاخْتَصَرُوهُ، كَلَامُهُمْ فَشَرَحُوا الْأَخْلَافَ تَبِعْتَهُمْ ثُمَّ الْاجْتِهَادِ،  
وَأَنْوَاعِ الْأَرْضِ، وَأَشْكَالِ الْأَقَالِيمِ، أَخْبَارِي فِي كِتَابِي أَوْلَفَ أَنْ  
وَمَدَائِنِ الْمَشْهُورَةِ، وَأَمْصَارِهَا وَالْبِلَادِ، الْمَمَالِكِ، وَمَسَالِكِ الْعِبَادَاتِ،  
الْعَشَائِرِ وَعَنْصَرِ الْمُسْتَعْمَلَةِ، وَطُرُقِهَا الْمَسْلُوكَةِ، وَمَنَازِلِهَا الْمَعْرُوفَةِ،  
فِي الْبُلْدَانِ أَهْلٍ وَاخْتِلَافِ وَالتَّجَارَاتِ، الْجَمَلِ وَمَعَادِنِ وَالْآلَاتِ،  
وَأَوْرَاقِهِمْ، وَمَكَاتِبِهِمْ وَمَذَاهِبِهِمْ وَالْوَانِيهِمْ، وَأَسْنَانِهِمْ وَأَصْوَاتِهِمْ كَلَامِهِمْ  
وَعِلْمَانِهِمْ، وَأَشْيَاخِهِمْ، بَلَدٍ، كُلِّ لِأَهْلِ الْمَعَاشِ وَأَفْضِيَةِ وَنُقُودِهِمْ،  
وَحُكْمَانِهِمْ. وَزُهَادِهِمْ،

Abu Abdallah Muhammad ibn Ahmad al-Muqaddasi said: Now then, the learned have always been eager to compose books that fill the scope of their investigations, and their reports are inexhaustible. I desired to follow their example and to walk in their footsteps, and to raise a standard whereby my memory might be kept alive and my name remembered. I found that the learned had already preceded me in the sciences and had distinguished themselves through diligent effort, and that successors had followed them, explaining and abridging their works. So I resolved to compose a book on the reports of the climes, the forms of the earth, the varieties of religious practices, the routes of kingdoms and lands, their famous garrison towns and well-known cities, their frequented stations and well-trodden roads; the elements of tribes and families, the mines of gems and trade goods; the differences among the peoples of the lands in their speech, voices, teeth, and colors; their doctrines, scripts, and weights; their coins and the means of livelihood for the people of each land; their elders, scholars, ascetics, and sages.

## Necessary Preliminaries | Necessary Preliminaries

### مِنْهَا بَدَلًا وَفُصُولٌ مُقَدِّمَاتٌ

بِدَعَائِمٍ وَأَسْنَادَاتٍ مُحْكَمَةٍ قَوَاعِدَ عَلَى الْكِتَابِ هَذَا أَسَّسْتُ أَنِّي أَعْلَمُ  
الْأَلْبَابِ، أُولِي فِيهِمْ وَأَسْتَعْنُتُ الصَّوَابِ، جِهَادَ فِيهِ وَتَحَرَّجْتُ قَوِيَّةً،  
فَعَلِي وَالْأَمَلُ، الرَّجَاءُ فِيهِ وَيَبْلُغُ وَالزَّلُّ، الْخَطَأُ فِيهِ يُجْتَنَبُ أَنْ وَاسْمُهُ  
وَعَلِيهِ وَعَلَيْتُهُ، وَحِرْفَتُهُ وَعَقْلَتُهُ، شَاهَدْتُهُ مَا بَفَنِيَاتِهِ أَوْصَفُ قَوَاعِدَهُ  
بِهِ أَسْتَعْنُتُ وَمَا وَارَكَانِهِ، أَيْضًا قَوَاعِدِهِ وَمَنْ وَالْأَرْكَانِ، الْبَنِيَانِ رَفَعْتُ  
بِالْغُفْلَةِ أَعْرِفُهُمْ لَمْ وَمَنْ النَّاسِ، مِنْ الْعُقُولِ ذَوِي سَوَالٍ تَبَيَّنَتْ، عَلَى  
وَلَمْ عَنْهَا، بَعْدَتْ الَّتِي الْأَطْرَافِ فِي وَالْأَعْمَالِ الْكَبِيرِ عَنِ الْإِنْتِبَاسِ،  
فِيهِ اخْتَلَفُوا وَمَا أَثْبَتُهُ، اتَّفَقَهُمْ عَلَيْهِ وَقَعَ فَمَا إِلَيْهَا، الْوُصُولُ لِي يَتَقَدَّرُ  
قَلْتُ أَوْ ذَكَرُ الْمَاضِي إِلَى اسْتِنَادَاتِهِ فِي بَدَلٍ لِي يَكُنْ لَمْ وَمَا نَبَذْتُ،  
الْمُلُوكِ. خَزَائِنَاتٍ فِي وَجَدْتُهُ بِفُصُولٍ وَشَاهَدْتُهُ رَعْمُوا،

Know that I have founded this book upon sound principles and solid bases, and I have exerted myself therein to attain correctness, seeking the help of those of understanding. I strove to avoid error and mistake, and to achieve the hoped-for goal. On these principles I describe in its chapters what I have witnessed and understood, recorded and learned; upon them I raised the structure and pillars. Among its principles and pillars, and what I relied upon to elucidate it, was to question men of intellect, and those whom I did not know out of negligence and obscurity, about the great matters and the works in the distant regions which I could not reach. Whatever they agreed upon I confirmed; whatever they differed upon I rejected; and what I had no way to verify by reference to the past, I recorded with "they say" or "they claim." And I witnessed it in detail and found it in the treasuries of kings.

وَلَا قَصْدَنَاهَا، الَّتِي الطَّرِيقَ يَسْلُكُ لَمْ الْعِلْمُ هَذَا إِلَى سَبَقْنَا مَنْ وَكُلُّ  
أَمِيرٍ دَزِيرَ فَكَانَ الْبَكْرِيُّ اللَّهُ عَبْدُ أَبِي أَمَّا أَرْدَنَاهَا، الَّتِي الْغَوَائِدَ طَلَبَ  
وَسَأَلَهُمُ السُّفَرَاءَ مُجْتَمِعَ وَهَيْئَةً وَنَحْوَهُمْ فَلَسَفَةً صَاحِبُ وَكَانَ خُرَاسَانَ،  
وَدَخَلَهَا. الْمُلُوكِ عَنِ

Everyone who preceded us in this science did not follow the path we intended, nor seek the principles we desired. As for Abu Abdallah al-Bakri, he was the vizier of the governor of Khurasan, a philosopher, and he gathered the ambassadors and questioned them about the kings, and entered their lands.

## Chapter on the Seas and Rivers | Chapter on the Seas and Rivers

## وَالْأَنْهَارِ الْبَحَارِ ذِكْرُ

نَحْوُ مَنْ يَخْرُجُ أَحَدُهُمَا حَسْبُ بَحْرَيْنِ إِلَّا الْإِسْلَامُ فِي زَمَانٍ أَنَا أَعْلَمُ  
 الْإِسْلَامُ مَمْلُوكَةٌ بَلَّغَ فَإِذَا السَّوَادُ، وَبِلَادِ الصِّينِ بِلَادِ بَيْنَ الشَّامِ مَشَارِقِ  
 عَدَّةٍ، وَشَعْبٌ كَثِيرٌ خَلِيجَانِ وَلَهُ مِثْلَانِ، كَمَا الْعَرَبُ، جَزِيرَةٌ عَلَى دَارِ  
 شِبْطَةٍ جَاعِلَةٍ، مِنْ فَنَائِهِمْ إِلَى الْمَصِيرِ وَصَفِهِ فِي النَّاسِ اخْتَلَفَ وَقَدْ  
 بَعْمَانُ، وَطَرْفُ الْعِرَاقِ وَطَرْفُ الْقِسْمِ عَلَى يَطُوفِ بَنِيَانٍ بِخَلِيجِ طَبَسِيَّةٍ  
 خُرَاسَانَةٍ وَعِنْدَهُ بِالْعِرَاقِ، وَذَنْبُهُ حَدِيثَةٌ، الشَّيْنُ وَبِلَادِ الشَّيْنِ وَخَلِيجُ  
 بَنِيَسَابُورِ الْمُؤْمِنِينَ أَمِيرِ خَزَانَةٍ فِي وَرَقَةٍ عَلَى مِثْقَلِ وَرَائِهِ خُرَاسَانُ أَمِيرِ  
 نَعْرِفَهَا، لَمْ فَإِنَّا مِثْلُ، كُلُّ وَإِذَا وَالصَّاحِبِ، الدَّوْلَةِ عِنْدَ خُرَاسَانَةٍ وَفِي  
 عَلَى يَعْتَمِدُ خَلِيجَانِ بَعْضُهُنَّ فِي وَإِذَا الْأَثَارِ بِمُخْتَلَفٍ فَسَرَّتْهُ أَنَا وَمَا  
 الْيَابَسَةِ. فِي فَجْرَاهُ الْأَعْظَمُ النَّهْرُ وَأَمَّا كُلُّهَا، مِنَ الْقَلِيلِ

Know that in the Islamic lands we have not seen [bodies of water] other than two seas, according to one of which: it emerges from the direction of the eastern regions of Syria, between the lands of China and the lands of the Sawad. When it reaches the realm of Islam, it encircles the Arabian Peninsula, as we have represented it. It has many gulfs and numerous branches. The people have differed in their descriptions of it and in [tracing] the extent of its reaches: some hold it to be like the Tabi Tabasiyyah in the Bay of Baniyan, which runs along the division, with one end in Iraq and one end in Oman; the Bay of China and the lands of China are near it, and its tail extends toward Iraq, and nearby is Khurasan of the governor of Khurasan. His opinion is confirmed by a map in the treasury of the Commander of the Faithful in Nishapur, and in the Khurasan of the State and the Sahib. And so with every similar case—for we do not know them fully, and we have only explained them according to the varying accounts. Among them are bays that rely upon the calculation of longitude from all of them. As for the greatest river, its course lies in dry land.

## Editorial Note on the Manuscripts | Editorial Note on the Manuscripts

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**B** = *codex Berolinensis* (Berlin manuscript)

**C** = *codex Constantinopolitanus* (Constantinople manuscript)

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The editor, **M. J. de Goeje**, has based the present edition upon two principal manuscripts:

- a) **Codex Berolinensis** (B): A manuscript preserved in the Royal Library of Berlin, containing the complete text of al-Muqaddasi's work with certain lacunae.
- b) **Codex Constantinopolitanus** (C): A manuscript from Constantinople, fuller in certain respects, containing valuable additions and variants.

The apparatus criticus employs the following sigla:

| Signum              | Meaning                    |
|---------------------|----------------------------|
| <i>B</i>            | Codex Berolinensis         |
| <i>C</i>            | Codex Constantinopolitanus |
| <i>om.</i>          | omits                      |
| <i>add.</i>         | adds                       |
| <i>supra lin.</i>   | above the line             |
| <i>in marg.</i>     | in the margin              |
| <i>sine punctis</i> | without diacritical points |
| <i>pro</i>          | reads (in place of)        |
| <i>vid. infra</i>   | see below                  |
| <i>vid. supra</i>   | see above                  |
| <i>cod.</i>         | codex (manuscript)         |

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## 1 The Division of the Earth into Climes | The Division of the Earth into Climes

لَأَنَّهُمُ الْكُفَّارُ مَسَالِكُ تَتَكَلَّفُ وَلَمْ حَسَبَ الْإِسْلَامَ مُلْكٌ أَنْ ذَكَرْنَا وَقَدْ  
إِقْلِيمًا، عَشْرَ أَرْبَعَةٍ إِلَى الْإِسْلَامِ أَرْضَ قَسَمْنَا وَقَدْ بَابِنَا، مِنْ تَكُنْ لَمْ  
وَذَكَرْنَا الْمَمَالِكِ، مِنْ فِيهِ مَا وَوَزْنَا وَأَعْنَقْنَا، فَصَلَّنَاهُ إِقْلِيمِ كُلِّ وَأَفْرَدْنَا  
الْعِرَاقُ الْحِجَازُ الْمَشْرِقُ أَوَّلُهَا الْعَرَبِ أَقْلِيمِ وَكَمِ الْعَجَائِبِ، مِنْ فِيهِ مَا  
الْمَغْرِبِ. مِصْرُ الشَّامِ

We have mentioned that the realm of Islam is [divided]  
accordingly, and we have not concerned ourselves with the  
lands of the unbelievers, for they do not belong to our subject.  
We have divided the land of Islam into fourteen climes, and  
we have given each clime a separate chapter, and  
distinguished what it contains of kingdoms, and mentioned  
the wonders therein. The first clime is Arabia: the Hijaz, Iraq,  
Syria, Egypt, and the Maghrib.

### 1.1 The First Clime: Arabia | The First Clime: Arabia

الْحِجَازُ: ذِكْرُ  
الْمُسْلِمِينَ أَمَّصَارُ مَحَلُّ وَهُوَ وَأَشْرَفُهَا، الْعَرَبِ بِلَادٍ أَفْضَلِ مِنْ وَالْحِجَازِ  
وَمَنْبِتُ وَسَلَّمَ عَلَيْهِ اللَّهُ صَلَّى النَّبِيُّ وَمَوْلِدُ الْعَرَبِ، مَنَازِلُ وَمَوْضِعُ  
أَقْلِيمِ وَهُوَ وَالْيَمَامَةُ، وَالْيَمَنُ الْمُنُورَةُ وَالْمَدِينَةُ الْمُشْرِفَةُ مَكَّةَ وَفِيهِ الشَّرَائِعُ،  
بَدُو فِيهِ وَالنَّاسُ وَرِمَالُ، حِجَارَةٌ وَأَكْثَرُهُ وَالْعُشْبُ، الْمَاءُ قَلِيلٌ حَارٌّ  
مَا وَالْحِمْيَةُ وَالشَّجَاعَةُ الْفَخْرُ مِنْ وَلَهُمُ وَالْغَنَمُ، الْإِبِلُ وَمَالُهُمْ وَحَضَرُ،  
لِغَيْرِهِمْ. لَيْسَ

The Hijaz is the noblest and most honorable of the lands of  
Arabia; it is the dwelling-place of the Muslims and the abode  
of the Arabs, the birthplace of the Prophet (may God bless  
him and grant him peace) and the source of the revealed  
laws. In it are Mecca the Honored and Medina the Illumined,  
and Yemen and al-Yamama. It is a hot clime, with little water  
and herbage; most of it is stones and sand. The people are  
both Bedouin and settled; their wealth consists in camels and  
sheep; and they possess pride, courage, and valor such as no  
others have.

#### 1.1.1 Mecca the Noble | Mecca the Noble

الْمَكَّةُ: مَكَّةُ  
الْمُعَظَّمَةُ الْكَعْبَةُ مَوْضِعُ وَهِيَ وَأَفْضَلُهَا، اللَّهُ بِلَادٍ أَشْرَفُ وَمَكَّةُ  
مِنْ فِيهَا مَا وَأَصْفُ الْبِلَادِ، سِيرٌ فِي أَبْدَأُ وَبِهَا الْحَرَامُ، وَالْمَسْجِدِ  
وَالطَّرِيقَاتِ. وَالْأَحْيَاءُ وَالْأَسْوَاقِ وَالْمَسَاجِدِ الْعِمَارَاتِ  
لَهُ يُقَالُ جَبَلٌ سَفْجٌ فِي وَهِيَ عَالِيَةٌ، جَبَالٌ بَيْنَ شَرِيفِ جَبَلٍ فِي وَمَكَّةُ  
وَأَشْهُرُ وَالْعِيُونِ، الْأَبَارِ مِنْ وَمَاؤُهَا الْمَجْرَى، صَعْبٌ وَنَهْرُهَا قَيْسٌ، أَبُو  
الْمَذَاقِ. عَذْبَةُ الْمَاءِ كَثِيرَةٌ عَيْنٌ وَهِيَ زَبِيدَةٌ، عَيْنٌ فِيهَا مَا

Mecca is the noblest and most excellent of God's lands; it is  
the site of the venerated Kaaba and the Sacred Mosque. I  
begin my account of the cities with it, describing its buildings,  
mosques, markets, quarters, and roads. Mecca lies in a noble  
mountain among high peaks, on the slope of a mountain  
called Abu Qubays. Its river has a difficult course; its water  
comes from wells and springs, the most famous of which is  
the Spring of Zubayda—a spring abundant in water, sweet in  
taste.

a) B جَبَالٍ An forte الجِبَالِ?, b) B المغازاتِ, c) B المواسِلِ, d) B om. مواضع الضياع et habet الجذبِ, e) B et C المصادرِ. Deinde B والجزمِ, f) B  
والعجائزِ i.

e. والمباني. Deinde B et C والمنمِ, g) B om., h) B et C غنى ut saepissime in talibus.

## 2 The Climes of Iraq and Syria | The Climes of Iraq and Syria

### 2.1 Iraq | Iraq

العراق: ذِكْرُ  
الْكُوفَةِ وَبِهِ وَالْمَلِكِ، الْخِلَافَةِ وَمَوْضِعِ الْجَنَّةِ، كُورَةِ بِلَادِ هُوَ وَالْعِرَاقُ  
أَرْضٌ وَهُوَ الْأَخْيَالُ، وَحَصْنِ الرَّحْبَةِ وَالْأَنْبَارِ وَوَاسِطِ الْبَصْرَةِ  
أَشْهَرُهَا كَثِيرَةُ أَنْهَارِهِ وَالشَّعِيرِ، وَالْحَنْظَةِ وَالنَّخْلِ الْمَاءِ كَثِيرَةٌ خَصْبَةٌ  
وَكَلَامٌ وَفَقْهٌ وَعِلْمٌ بَصْرَةٍ أَهْلُ أَكْثَرِهِمْ فِيهِ وَالنَّاسُ وَالْفَرَاتُ، دِجْلَةُ  
وَمَذَاهِبُ.

*Iraq is the land of the Courtyard of Paradise, the seat of the Caliphate and kingship. In it are Kufa, Basra, Wasit, Anbar, al-Rahba, and Hisn al-Ahyal. It is a fertile land, abundant in water, palm trees, wheat, and barley. Its rivers are numerous, the most famous being the Tigris and the Euphrates. Its people are mostly men of understanding, knowledge, jurisprudence, dialectics, and diverse doctrines.*

### 2.2 Syria | Syria

الشَّامُ: ذِكْرُ  
وَحْمَصِ وَالرَّشِيدِ دِمَشْقَ وَبِهِ الْخَيْرِ، كَثِيرٌ وَاسِعٌ أَقْلِيمٌ هُوَ وَالشَّامُ  
وَبَيْرُوتُ وَالرَّمْلَةُ الْمُقَدَّسُ وَبَيْتُ وَاللَّاذِقِيَّةُ وَأَنْطَاكِيَّةُ وَحَلَبُ وَحَمَّاهُ  
الْجِبَالِ مِنْ وَفِيهِ وَالضَّرْعُ، لِلزَّرْعِ صَالِحَةٌ وَأَرْضُهُ وَصِيدَاءُ، وَصُورُ  
وَاللَّتَانِيُّ الْأَرْنُونُ أَنْهَارُهُ وَأَشْهَرُ يُحْصَى، لَا مَا وَالْعُيُونِ وَالْأَنْهَارِ  
وَجَمَالٌ. وَصِحَّةٌ وَعِلْمٌ فَقْهٌ أَهْلُ فِيهِ وَالنَّاسُ وَالْيَرْمُوكُ،

*Syria is a wide clime, rich in blessings. In it are Damascus, al-Rashid, Homs, Hama, Aleppo, Antioch, Latakia, Jerusalem, al-Ramla, Beirut, Tyre, and Sidon. Its land is suitable for cultivation and pasturage; it contains innumerable mountains, rivers, and springs. Its most famous rivers are the Orontes, the Litani, and the Yarmuk. Its people are distinguished in jurisprudence, knowledge, health, and beauty.*

### 3 The Eastern Climes | The Eastern Climes

#### 3.1 Khurasan and Transoxiana | Khurasan and Transoxiana

النَّهْرُ وَرَاءَ وَمَا خُرَاسَانَ ذِكْرُ  
وَبِهَا مُلُوكًا، وَأَجْلُهَا بِلَادًا وَأَكْثَرُهَا الْمَشْرِقِ أَقَالِيمَ أَعْظَمَ وَخُرَاسَانَ  
وَعُزْرَجَانَ وَبُشَنَجَ وَطَخَارِسْتَانَ وَبَلْخَ وَهَرَاةَ الشَّاهِجَانَ وَمَرْوَ نِيشَابُورَ  
وَمِيمَنْدَجَانَ وَسَرَخْسَ وَأَبُورْدَ وَلَسَا وَاسْتَخْرَ وَسَابُورَ وَطُوسَ  
الْمَاءِ كَثِيرَةً وَاسِعَةً أَرْضَ وَهِي وَفَارَابَ، وَالشَّاشَ وَالسَّغْدَ وَخُورِزْمَ  
وَتِجَارَةَ. وَأَدَبٌ عِلْمُ أَهْلٍ وَأَكْثَرُهَا وَالبُسْتَانَ،

*Khurasan is the greatest of the eastern climes, with the most lands and the most splendid kings. In it are Nishapur, Merv the Great, Herat, Balkh, Tukharistan, Bushanj, Gurjan, Tus, Sabur, Istakhr, Nasa, Abiward, Sarakhs, Maymandajan, Khwarizm, Sogdia, Shash, and Farab. It is a wide land, abundant in water and orchards; its people are mostly devoted to learning, literature, and commerce.*

وَرَامِتَنَ وَأَسَدَابَادَ وَالشَّاشَ وَخُجَنْدَةَ وَسَمَرْقَنْدَ بَخَارَى هُوَ النَّهْرُ وَرَاءَ وَمَا  
إِقْلِيمٌ هُوَ وَخُورِزْمَ، وَشِيرَازَ وَطَالْقَانَ وَالْأَقَ وَشَاشَ وَكَاشَ وَنَسَفَ  
مِنْ وَبِهِمُ الدِّينَ، بِالْعِلْمِ مَتَمَسِّكُونَ وَأَهْلُهُ الْخَيْرَاتِ، كَثِيرٌ وَاسِعٌ  
وَالْتَفْسِيرَ، وَالْفَقْهَ الْحَدِيثِ أَهْلٌ وَمِنْهُمْ يُوصَفُ، لَا مَا وَالْفَتَوَةَ الشَّجَاعَةَ  
وَاتَّقَانًا. حَفِظَ النَّاسُ أَكْثَرُ وَهُمْ

*Transoxiana consists of Bukhara, Samarkand, Khujanda, Shash, Asadabad, Ramitan, Nasa, Kish, Shash, Ilak, Talakan, Shiraz, and Khwarizm. It is a wide clime, rich in blessings; its people hold fast to knowledge and religion, and possess courage and generosity beyond description. Among them are the scholars of Hadith, jurisprudence, and exegesis; they are the most assiduous of men in memorization and precision.*

## 4 Geographical Notices | Geographical Notices

### 4.1 The Calculation of Distances | The Calculation of Distances

ثَلَاثَةَ إِسْوَِي وَالْفَرَسَخُ وَالْمِيلُ، بِالْفَرَسَخِ الْبِلَادِ مَسَافَاتٍ أَحْصَيْنَا وَقَدْ  
وَقَدْ أَصْبَعًا، وَعِشْرُونَ أَرْبَعَةً وَالذِّرَاعُ ذِرَاعٌ، أَلْفٌ وَالْمِيلُ أَمِيلٌ،  
الْبَصْرَةَ بَيْنَ وَمَا فَرَسَخٌ، أَرْبَعُمِائَةٍ فَوْجَدْنَاهُ وَالْمَدِينَةَ مَكَّةَ بَيْنَ مَا حَسَبْنَا  
بَيْنَ وَمَا فَرَسَخٌ، أَرْبَعُمِائَةٍ وَسَامَرَاءَ بَغْدَادَ بَيْنَ وَمَا فَرَسَخٌ، أَرْبَعُمِائَةٍ وَبَغْدَادَ  
فَرَسَخٌ. أَرْبَعُمِائَةٍ وَعُمَانَ الْبَصْرَةَ بَيْنَ وَمَا فَرَسَخًا، ثَلَاثُونَ وَالرَّحْبَةَ دِمَشْقَ

We have measured the distances between cities in parasangs and miles. A parasang equals three miles; a mile is one thousand cubits; a cubit is twenty-four fingerbreadths. We have calculated [the following]: between Mecca and Medina: 400 parasangs; between Basra and Baghdad: 400 parasangs; between Baghdad and Samarra: 400 parasangs; between Damascus and al-Rahba: 30 parasangs; between Basra and Oman: 400 parasangs.

### 4.2 The Coordinates of Principal Cities | The Coordinates of Principal Cities

*Note:* These coordinates are given according to the system of the Arab geographers, measured in degrees east of the Canary Islands.

| City       | Longitude | Latitude |
|------------|-----------|----------|
| Mecca      | 77° 20'   | 21° 30'  |
| Medina     | 77° 00'   | 25° 00'  |
| Baghdad    | 79° 30'   | 33° 20'  |
| Basra      | 80° 00'   | 30° 10'  |
| Damascus   | 73° 30'   | 33° 30'  |
| Jerusalem  | 73° 00'   | 31° 50'  |
| Alexandria | 68° 00'   | 31° 10'  |
| Nishapur   | 88° 00'   | 36° 10'  |
| Merv       | 87° 00'   | 37° 20'  |
| Bukhara    | 93° 00'   | 39° 40'  |
| Samarkand  | 94° 00'   | 39° 40'  |

## 5 Sample Critical Apparatus | Sample Critical Apparatus

a) *C* add. ut saepe تعالى. b) *B* رغبت sine نفسي et deinde وطبت et خوفاً. c) *C* om. sed post برضاه addit معين. d) *B* بفهما. e) *C* تعالى. f) *B* om. g) *B* pro his السمان وذوقته وعقلته السمان. Deinde *C* وعليت pro عملت. h) *C* om. i) *C* بنيانه et om. سوال. k) *B* يد. l) *B* haec om. Deinde *C* habet وشاهدته بفصول.

m) *C* سلكتها (السختى). n) *C* om. o) *C* ضد addito كذا. Apud Hadji-Khalifa, V, p. 510 omittitur vocabulum.

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*End of Selections | End of Selections*

FINIS / THE END

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*Ex officina typographica et bibliopolio antehae E. J. Brill*

Lugduni Batavorum, 1906

Bilingual edition prepared with parallel Arabic text and English translation.

## Index of Geographical Names | Index of Geographical Names

| Name (Arabic)        | Modern Identification |
|----------------------|-----------------------|
| مَكَّة               | Mecca                 |
| الْمَدِينَة          | Medina                |
| الْكُوفَة            | Kufa                  |
| الْبَصْرَة           | Basra                 |
| بَغْدَاد             | Baghdad               |
| وَأَسْط              | Wasit                 |
| دَمَشَق              | Damascus              |
| حَلَب                | Aleppo                |
| يَتِ الْمَقْدِس      | Jerusalem             |
| مِصْر                | Egypt (Fustat/Cairo)  |
| الْإِسْكََنْدَرِيَّة | Alexandria            |
| الْقَيْرَوَان        | Kairouan              |
| نِيشَابُور           | Nishapur              |
| مَرْو                | Merv                  |
| بَلْخ                | Balkh                 |
| بُخَارِي             | Bukhara               |
| سَمَرْقَنْد          | Samarkand             |
| خَوَارِزْم           | Khwarizm (Khiva)      |



# كتاب الفهرست

*Kitāb al-Fihrist — The Book Catalogue*

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A BILINGUAL ARABIC-ENGLISH EDITION

Based on the critical edition of Gustav Flügel (Leipzig, 1871–1872)

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**Author:** Muḥammad ibn Ishāq Ibn al-Nadīm

Baghdad, c. 987 CE

The *Kitāb al-Fihrist* of Ibn al-Nadīm (c. 987 CE) is the most important bibliographic encyclopedia of medieval Arabic literature and scholarship. This bilingual edition presents the original Arabic text alongside an English translation, arranged in parallel columns. The left column contains the Arabic text in RTL script; the right column gives the English rendering. Key sections on the mathematical sciences — including biographies of al-Khwārizmī, Thābit ibn Qurra, and other Islamic mathematicians — are presented in full.

*Edited from the Arabic original with notes*

**Gustav Leberecht Flügel** (1802–1870)

Completed by Dr. Johannes Roediger & Dr. August Müller

LEIPZIG: Verlag von F. C. W. Vogel, 1871–1872

With the support of the Deutsche Morgenländische Gesellschaft

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## Introduction

### مقدمة المؤلف في فائدة التصنيف

### PREFACE: ON THE PURPOSE OF CATALOGUING

Ibn al-Nadīm, whose full name is Abū al-Faraj Muḥammad ibn Ishāq ibn Abī Yaqūb al-Nadīm al-Warrāq al-Baghdādī (died c. 995/998 CE), was a Muslim scholar, bibliographer, and bookseller (*warrāq*) who lived in Baghdad during the Buyid period. His *Kitāb al-Fihrist* (كتاب الفهرست) is the most important bibliographic work in Arabic literature and an indispensable source for the history of Islamic civilization down to the 10th century.

قال أبو الفرج محمد بن إسحاق بن أبي يعقوب النديم الوراق:  
 علوم يحفظ من زمان عصر كل في جعل الذي لله الحمد  
 ويؤدي الناس  
 نصفه ما على نستعين وبه السنن ويحيي المعالم ويحرس الأمانة  
 العلماء تصنفها التي الكتب أصناف ذكر من الكتاب هذا في  
 يعرف وما المصنفين وأسماء بها يشتغلون التي العلوم وأنواع  
 وبلدانهم وطبقاتهم ونسبهم وأسمائهم وكناهم بألقابهم  
 وصفوا وما عنهم أخذ ومن عنه العلم أخذوا ومن ومولدهم  
 ذلك وغير المعرفة والإتقان والفهم والحفظ الجلالة من به  
 وقد اقتدت فائدة هذا الكتاب بفوائد سائر الكتب المصنفة  
 في  
 فهو والعلماء المصنفين وأسماء الأعيان وطبقات الناس أخبار  
 وسمعتها عليها يدي وقعت التي العلم أصناف جميع على يحتوي  
 وجلودها وحواشيها الكتب بطون في مكتوبة ورأيتها  
 وإفادات  
 الأصول من ذلك بعد استدركته وما بينهم فيما العلماء  
 والفهارس

Abū al-Faraj Muḥammad ibn Ishāq ibn Abī Yaqūb al-Nadīm al-Warrāq said: Praise be to God, who in every age appoints those who preserve the sciences of mankind, discharge the trust, guard the landmarks of knowledge, and revive the traditions. Through Him we seek aid in compiling this book, wherein we mention the categories of books composed by scholars, the types of sciences they pursue, the names of authors with their titles, *kunyas*, proper names, genealogies, generations, places of origin and birth, their teachers, their students, and their distinguished qualities of eminence, retention, understanding, mastery, comprehension, and more.

The benefit of this book follows the benefits of all books compiled on the reports of people, the generations of notables, the names of authors and scholars. It encompasses all categories of knowledge that I have handled, heard, and seen written in the contents of books, their margins, their bindings, and the transmissions of scholars among themselves — including what I subsequently recovered from original sources and catalogues.

The *Fihrist* is organized into ten *discourses* (*maqālāt*, مقالات), covering the entirety of Arabic book culture. The following sections present each discourse with parallel Arabic text and English translation.

# 1 Discourse I: Qur'an Sciences and Philology

الفن الأول في اللغات والخطوط وقراءة القرآن والنحو  
والشعر والتفسير

THE FIRST DISCOURSE: LANGUAGES, SCRIPTS,  
QUR'ANIC READINGS, GRAMMAR, POETRY, AND EXE-  
GESIS

## في أصناف العلم بالقرآن الكريم

أما أصناف العلم بالقرآن الكريم فإنها: علم القراءات وعلم  
فصاحة وعلم والمنسوخ النسخ وعلم التجويد وعلم التفسير  
والبيان المعاني وعلم النحو وعلم والوقف البدء وعلم القرآن  
وما الأنساب وعلم الفرائض وعلم القافية وعلم العروض وعلم  
والخطباء المتكلمون إليها يحتاج التي العلوم من ذلك أشبه  
والمفسرون والشعراء

## On the Categories of Qur'anic Science

The categories of knowledge concerning the Noble Qur'an are: the science of readings (*qirā'āt*), the science of exegesis (*tafsīr*), the science of recitation (*tajwīd*), the science of abrogating and abrogated verses, the science of the Qur'an's eloquence, the science of beginnings and pauses, the science of grammar (*nah*), the science of meanings and rhetoric (*maānī wa al-bayān*), the science of prosody (*'arūd*), the science of rhyme (*qāfiya*), the science of inheritance laws (*farā'id*), and the science of genealogy (*ansāb*) — and similar sciences needed by speakers, orators, poets, and exegetes.

## في القراءات

وأما القراءات فإنها عشر: نافع وابن كثير وأبو عمرو وابن  
خلف ويعقوب جعفر وأبو والكسائي وحمزة وعاصم أمير  
إلى شيخه عن وشيخه شيخه عن رواية له قارئ وكل  
الصحابة  
الاحاطة أراد فن وسلم عليه الله صلى النبي عن والصحابة  
المشهورة بكتبها فعليه القراءات بعلم

## On the Qur'anic Readings

The readings are ten: Nāfiʿ, Ibn Kathīr, Abū ʿAmr, Ibn Āmir, ʿĀṣim, Ḥamza, al-Kisāʾī, Abū Jafar, Yaqūb, and Khalaf. Each reader received his reading from his sheikh, and his sheikh from his sheikh, back to the Companions, and the Companions from the Prophet — peace be upon him. Whosoever desires full mastery of the science of readings must study its well-known books.

*Note on the First Discourse.* This section of the *Fihrist* catalogues the foundational Islamic sciences that shaped Arabic scholarship. The ten canonical readings (*qirā'āt*) of the Qur'an represented the bedrock of linguistic and philological study. Ibn al-Nadīm lists the principal authors and their works in each discipline, providing an unparalleled window into the state of these sciences in 10th-century Baghdad.

## 2 Discourse II: Grammar, Lexicography, and Poetry

الفن الثاني في النحو واللغة والشعر وأخبار الشعراء

THE SECOND DISCOURSE: GRAMMAR, LEXICOGRAPHY, POETRY, AND THE REPORTS OF POETS

### في النحويين

وأما النحو فإنه علم يعرف به أوضاع ألفاظ العرب من حيث نظم من أول كان لأنه نحواً سمي وإنما والبناء الإعراب أحكامه وتذهب لغتها تغير العرب إن له: قيل إذ الدؤلي الأسود أبو إليها مال أي فتحاها ضابطاً لها يضع أن فأراد بإعرابها نحواً فسمي فيها وتعمق

### في أصول النحو والمصنفين فيه

وأول من وضع النحو أبو الأسود الدؤلي ثم أخذ عنه أهل ومن الكبار الكتب فيه وصنفوا وأطنبوا فيه فأكثرها البصرة وعلقمة وسيبويه الفراهيدي أحمد بن الخليل النحويين: أشهر والفارسي والزجاجي السراج وابن المازني والأخفش وغيرهم الفن هذا في كثيرة مصنفات منهم واحد ولكل كثير

### في اللغويين وأصناف اللغات

وأما اللغة فإنها صناعة يعرف بها معاني الألفاظ وأسباب مراد لبيان وضعت وإنما المعاني وحروف واشتقاقها وضعها ونحوهم وبلاغتهم فصاحتهم في يتفاوتون والمتكلمون المتكلم ومعانيها الألفاظ جمع في جملة كتباً اللغة أهل وضع وقد

### في الشعراء وأخبارهم

### On the Grammarians

Grammar (*nah*) is the science by which one knows the structures of Arabic speech with respect to declension (*iṭāb*) and construction (*bunāʾ*). It was called *nah* because its first systematizer, Abū al-Aswad al-Duʿalī, when told that the Arabs were corrupting their language and losing its declension, sought to establish rules for it. He *naḥā* it — that is, he turned to it and delved deeply into it — and so it was called *nah*.

### On the Foundations of Grammar and Those Who Composed Works Therein

The first to establish grammar was Abū al-Aswad al-Duʿalī; then the people of Basra took it from him, expanded upon it, elaborated it, and composed great books. Among the most famous grammarians are: al-Khalīl ibn Aḥmad al-Farāhidī, Sībawayh, ṣAlqama, al-Akhfash, al-Māzanī, Ibn al-Sarrāj, al-Zajjājī, al-Fārisī, and many others. Each of them has numerous compositions in this art.

### On Lexicographers and the Categories of Languages

Lexicography is the craft by which one knows the meanings of words, the reasons for their coinage, their derivation, and the particles of meaning. It was devised to clarify the intent of the speaker. Speakers differ in their eloquence, rhetoric, and grammatical usage. The scholars of language have composed abundant books collecting words and their meanings.

### On the Poets and Their Reports

وأما الشعر فهو الكلام الموزون المقفى الذي يحتذى به في  
امرئ مثل مشهورون شعراء الجاهلية في كان وقد العربية  
القيس  
حلزة بن والحارث والأعشى وعنترة وزهير والنابعة وطرفة  
والمتنبى الرومي وابن والبحري تمام أبو المحدثين من ومعهم  
مشهور شعر ديوان منهم واحد ولكل كثير وغيرهم

Poetry is metrical, rhymed discourse that serves as a model in Arabic. In the pre-Islamic period there were famous poets such as Imru' al-Qays, Ṭarafa, al-Nābiga, Zuhayr, ḥAntara, al-Aṣ, and al-Ḥārith ibn Ḥilliza; and among the modern poets Abū Tammām, al-Buḥturī, Ibn al-Rūmī, al-Mutanabbī, and many others. Each of them has a well-known *dīwān* of poetry.

### 3 Discourse III: History, Biography, and Genealogy

الفن الثالث في التاريخ والسير والمغازي والأنساب

THE THIRD DISCOURSE: HISTORY, BIOGRAPHIES, CAMPAIGNS, AND GENEALOGY

#### في المؤرخين وأصناف التاريخ

وأما التاريخ فهو علم يعرف به أوقات الحوادث وتواريخ الملوك  
ومن والدول والمذاهب الماضية الأمم من وقع وما سيرهم  
أشهر  
والواقدي والفتوح المغازي صاحب إسحاق ابن المؤرخين:  
وابن  
وهم ومسكويه والطبري الطبقات صاحب سعد وابن هشام  
كثيرون  
الأرفف تملأ كثيرة كتباً فيه صنفوا وقد الفن هذا في

#### On the Historians and Categories of History

History is the science by which one knows the times of events, the dates of kings, their biographies, and what befell past nations, sects, and dynasties. Among the most famous historians are: Ibn Ishāq, author of the *Maghāzī* and *Futūḥ*; al-Wāqidī; Ibn Hishām; Ibn Saḍ, author of the *Ṭabaqāt*; al-Ṭabarī; and Miskawayh. They are numerous in this art and have composed many books filling the shelves.

#### في السير والمغازي

وأما السير فهي ما أثبتته المؤلفون من أخبار الرسول صلى الله  
أخبار من أثبتوه وما ومسيراته وحجه وغزواته وسلم عليه  
وأحوالهم وأفعالهم والخلفاء الملوك وسير والتابعين الصحابة  
وابن هشام وابن إسحاق ابن السير: في المصنفين أشهر ومن  
مشهور منهم واحد وكل وغيرهم والبلاذري والواقدي سعد  
بكتابه

#### On Biographies and Campaigns

Biographies (*siyar*) are what authors have recorded of the reports of the Messenger of God — peace be upon him — his raids, pilgrimages, and journeys; and what they recorded of the Companions and Followers, the biographies of kings and caliphs, their deeds and circumstances. Among the most famous authors of *siyar* are: Ibn Ishāq, Ibn Hishām, Ibn Saḍ, al-Wāqidī, and al-Balādhurī. Each is famous for his book.

#### في الأنساب

وأما علم الأنساب فهو علم يعرف به أحساب الناس ونسبهم  
أهلها ومشاهير وديارها ومواضعها العرب بقبائل يتعلق وما  
النسابة أخذه ثم الحميريون العلم هذا وضع من وأول  
جمعة كتباً فيه فصنفوا بعدهم من

#### On Genealogy

The science of genealogy is that by which one knows the pedigrees and lineages of people, and what relates to the Arab tribes, their locations, settlements, and notables. The Himyarites first established this science; then the genealogists (*nassāba*) took it from them and composed abundant books thereon.

## 4 Discourse IV: Mathematics, Astronomy, Music, and Mechanics

الفن الرابع في الحساب والهندسة والفلك والموسيقى والحيل

THE FOURTH DISCOURSE: ARITHMETIC, GEOMETRY, ASTRONOMY, MUSIC, AND MECHANICAL DEVICES

### في علم الحساب

وأما علم الحساب فهو علم يعرف به الكميات وعلاقاتها  
النساطرة نقله ثم اليونانيون وضعه وقد وأنواعها وأعدادها  
كثيرة كتباً فيه وصنفوا عنهم المسلمون فأخذوه العراق إلى  
فيه المؤلفين أشهر ومن فيه فأبدعوا عليه وزادوا وطوره  
وغيرهم والرازي والكندي قرّة بن وثابت الخوارزمي

### On the Science of Arithmetic

The science of arithmetic is that by which one knows quantities, their relations, numbers, and types. The Greeks established it; then the Nestorians transmitted it to Iraq, whence the Muslims received it. They composed many books thereon, developed it, and added to it, producing outstanding works. Among the most famous authors are: al-Khwārizmī, Thābit ibn Qurra, al-Kindī, and al-Rāzī, among others.

### في علم الهندسة

وأما الهندسة فهي علم يعرف به الأشكال والأبعاد والعلاقات  
بالأصول المسمى كتابه في أقليدس أصولها وضع وقد بينها  
قرّة بن وثابت حنين بن وإسحاق إسحاق بن حنين نقله وقد  
فنههم عظيمة كتباً فيه وصنفوا تعلمه على المسلمون وأقبل  
وزيادةً شرحاً فيه ألف من ومنهم تلخيصاً فيه ألف من

### On the Science of Geometry

Geometry is the science by which one knows figures, dimensions, and the relations between them. Euclid established its principles in his book called the *Elements* (*al-Uṣūl*). It was translated by Ḥunayn ibn Ishāq, Ishāq ibn Ḥunayn, and Thābit ibn Qurra. The Muslims devoted themselves to studying it and composed great books: some wrote abridgements, others wrote commentaries and additions.

### في علم الفلك والنجوم

وأما علم الهيئة فهو علم يعرف به أحوال الكواكب والأجرام  
عن نقل وقد مسيراتها ومبالغ وعللها وحركاتها السماوية  
ثم الفن هذا في كثيرة كتب والفرس والهنود اليونانيين  
وصنعوا أزيجاً وعملوا عظيمة كتباً فيه المسلمون صنف  
العلم هذا إليه يحتاج مما وغيرها أسطرلابات

### On the Science of Astronomy and the Stars

Astronomy (*hay'a*) is the science by which one knows the conditions of the planets and celestial bodies, their motions, causes, and the measures of their movements. Many books in this art were translated from the Greeks, Indians, and Persians; then the Muslims composed great books, compiled astronomical tables (*azīj*), and constructed astrolabes and other instruments needed for this science.



#### 4.1 The Mathematicians: Key Biographical Entries

##### ذكر الخوارزمي ومصنفاته

أبو عبد الله محمد بن موسى الخوارزمي كان من علماء بيت كُتاباً والمقابلة الجبر في ألف وقد المأمون زمان في الحكمة حلها وكيفية المعادلات وأنواع والمقابلة الجبر فيه به يعرف تنتهي المعادلات أن وأوضح والمقابلة الجبر بين فيه ميز وقد طرقاً واستخرج شيئاً الجهول وسمى محدودة أنواع إلى الهندسية والطريقة الجبرية الطريقة منها المعادلات لحل العدد نظام فيه يشرح آخر كُتاباً الحساب في أيضاً ألف وقد بالخوارزمي الآن ويسمى به الحساب وطريقة الهندي

وله كُتاب في الجبر والمقابلة وكُتاب في الجامع والتفريق حساب في وكُتاب والمقابلة الجبر حساب في المختصر وكُتاب الأسطرلاب عمل في وكُتاب الأرض صورة وكُتاب الهندي وكُتاب في وكُتاب هند السند أي الازنيج وكُتاب الفسطرلاب عمل في الفلك صناعة من أخرى وأشياء الرخامة في وكُتاب الساعات

##### Account of al-Khwārizmī and His Compositions

Abū ʿAbd Allāh Muḥammad ibn Mūsā al-Khwārizmī was among the scholars of the House of Wisdom in the time of al-Ma'mun. He composed a book on *al-jabr wa al-muqābala* wherein he treated *al-jabr* and *al-muqābala*, the types of equations, and the methods of solving them. He distinguished between *al-jabr* and *al-muqābala* and showed that equations reduce to a finite number of types. He named the unknown quantity *shay'* ("thing") and derived methods for solving equations, both algebraic and geometric. He also composed another book on arithmetic explaining the Indian numeral system and methods of calculating with it, which is now called "algorism" (*al-Khwārizmī*).

His works include: *Kitāb fī al-Jabr wa al-Muqābala*; *Kitāb fī al-Jāmiʿ wa al-Tafrīq*; the *Kitāb al-Mukhtaṣar fī Hisāb al-Jabr wa al-Muqābala*; *Kitāb fī Hisāb al-Hind*; *Kitāb Ṣūrat al-Arḍ*; *Kitāb fī ʿAmal al-Asturlāb*; *Kitāb fī ʿAmal al-Fasturlāb*; *Kitāb al-Azīj* (that is, the *Zīj al-Sindhind*); *Kitāb fī al-Sāʿāt*; *Kitāb fī al-Rukhāma*; and other works on the art of astronomy.

##### ذكر ثابت بن قرّة ومصنفاته

ثابت بن قرّة بن مروان الحرّاني كان حكيماً عظيماً فقيهاً والحساب والفلسفة والمنطق الهيئة علم في بارعاً فاضلاً والهندسة وبطليموس وأرشميدس أبولونيوس كتب من كثيراً نقل وقد كتباً الهندسة علم في ألف وقد وشرحها تصحيحها على وأقبل الموسيقى علم وفي الهيئة علم وفي الحساب علم وفي كثيرة الإسلام في الفلك علم أركان وطد من أعظم من وكان

##### Account of Thābit ibn Qurra and His Compositions

Thābit ibn Qurra ibn Marwān al-Ḥarrānī was a great sage, an excellent jurist, distinguished in the science of astronomy, logic, philosophy, arithmetic, and geometry. He translated many books of Apollonius, Archimedes, and Ptolemy, and devoted himself to correcting and commenting upon them. He composed numerous books on geometry, arithmetic, astronomy, and music, and was among the greatest in establishing the foundations of the science of astronomy in Islam.

وله كتاب في القرى الحسابية وكتاب في المساحة وكتاب في  
في وكتاب لمقلطس القطع شكل في وكتاب المعادلات  
المخروطات  
وكتاب النسبة حسن في وكتاب والأسطوانة الكرة في وكتاب  
في  
الفلك علم في وكتاب الهيئة علم في وكتاب الموسيقى علم  
الزيج في وكتاب اليوناني عن نقله الذي المجسطي وكتاب  
ذكره يطول مما كثير ذلك وغير

His works include: *Kitāb fī al-Qarā al-Hisābiyya*; *Kitāb fī al-Masāḥa*; *Kitāb fī al-Muādālāt*; *Kitāb fī Shakl al-Qiṭāc* (“On the Quadrature of the Parabola”); *Kitāb fī al-Makhṛūṭāt*; *Kitāb fī al-Kura wa al-Ustuwāna*; *Kitāb fī Husn al-Nisba*; *Kitāb fī ‘Ilm al-Mūsīqā*; *Kitāb fī ‘Ilm al-Hay’a*; *Kitāb fī ‘Ilm al-Falak*; the *Kitāb al-Majisṭī* which he translated from the Greek; *Kitāb fī al-Zīj*; and many others too numerous to mention.

### ذكر إسحاق بن حنين ومصنفاته

إسحاق بن حنين بن إسحاق العبادي كان طبيباً حكيماً  
نقل وقد والفلك والحساب والهندسة والفلسفة بالمنطق عالماً  
وبقراط وجالينوس وأفلاطون أرسطو كتب من كثيراً  
أبوه وكان وصحها بعضها وشرح وبطييموس وأقليدس  
الفن هذا في إنتاجاً وأكثرهم النقال أعظم إسحاق بن حنين

### Account of Ishāq ibn Ḥunayn and His Compositions

Ishāq ibn Ḥunayn ibn Ishāq al-ḥlābī was a physician and sage, learned in logic, philosophy, geometry, arithmetic, and astronomy. He translated many books of Aristotle, Plato, Galen, Hippocrates, Euclid, and Ptolemy, commented upon some of them, and corrected them. His father, Ḥunayn ibn Ishāq, was the greatest of translators and the most prolific in this art.

وله كتاب أقليدس في الأصول الهندسية نقله عن اليوناني  
المنطق في وكتاب أيضاً نقله لبطييموس المجسطي وكتاب  
المفردة الأدوية في وكتاب الهيئة علم في وكتاب  
نقلها التي الكتب من ذلك وغير الحيوان علم في وكتاب  
قدراً وأجلهم النقال أعظم من وكان وشرحها

His works include: *Euclid’s Elements* which he translated from the Greek; the *Almagest* of Ptolemy, likewise translated; *Kitāb fī al-Mantiq*; *Kitāb fī ‘Ilm al-Hay’a*; *Kitāb fī al-Adwiya al-Mufrada*; *Kitāb fī ‘Ilm al-Ḥayawān*; and other books that he translated and commented upon. He was among the greatest and most eminent of translators.

### ذكر قسطا بن لوقا ومصنفاته

قسطا بن لوقا البعلبي كان من أعظم الحكماء وأكثرهم  
والطبيعات والفلك والهندسة الحساب علوم في تصنيفاً  
اليونانيين كتب من كثيراً نقل وقد والأخلاق والمنطق  
العلوم هذه في كثيرة مصنفات وله وشرحها وصحها

### Account of Qusṭā ibn Lūqā and His Compositions

Qusṭā ibn Lūqā al-Baḥlabakkī was among the greatest philosophers and most prolific authors in the sciences of arithmetic, geometry, astronomy, natural philosophy, logic, and ethics. He translated many books from the Greeks, corrected them, and commented upon them. He has many compositions in these sciences.

وله كتاب في علم الهيئة وكتاب في علم الفلك وكتاب في  
في وكتاب الموسيقى في وكتاب الحساب في وكتاب الهندسة  
في وكتاب الأخلاق في وكتاب المنطق في وكتاب الطبيعة علم  
الب في وكتاب الأسطرلاب عمل في وكتاب الشافي الدواء  
Kilan  
كثرة يحصى لا مما ذلك وغير

His works include: *Kitāb fī 'Ilm al-Hay'a*; *Kitāb fī 'Ilm al-Falak*; *Kitāb fī al-Hindsa*; *Kitāb fī al-Ḥisāb*; *Kitāb fī al-Mūsīqā*; *Kitāb fī 'Ilm al-Ṭabiā*; *Kitāb fī al-Manṭiq*; *Kitāb fī al-Akhlāq*; *Kitāb fī al-Dawā' al-Shāfi*; *Kitāb fī 'Amal al-Asturlāb*; and *Kitāb fī al-Bī'lān*; and others too numerous to count.

### ذكر البتاني ومصنفاته

محمد بن جابر بن سنان البتاني الحراني الصابي كان عالماً  
أصلح قد محققاً بارعاً فيه فاضلاً والفلك الهيئة بعلم  
بزيج سمي به خاصاً زيجاً وعمل وصححه المجسطي زيج  
ومسيراتها وحركاتها الكواكب أحوال فيه بين وقد الصابي  
زمانه في وأفضلهم العلم هذا أهل أجل من وكان

### Account of al-Battānī and His Compositions

Muḥammad ibn Jābir ibn Sinān al-Battānī al-Ḥarrānī al-Ṣābī was a scholar of astronomy, distinguished, outstanding, and precise. He corrected the *Zīj al-Majisī* and compiled a *zīj* of his own called the *Zīj al-Ṣābī*, in which he explained the conditions of the planets, their motions, and their movements. He was among the most eminent and excellent people of this science in his time.

وله كتاب الزيج الصابي وكتاب في علم الهيئة وكتاب في  
علم في وكتاب المساحات علم في وكتاب الأسطرلاب عمل  
في وكتاب ومسيراتها الكواكب أحوال في وكتاب الأزياج  
المفيدة الكتب من ذلك وغير والملاحظات الرصد

His works include: *Kitāb al-Zīj al-Ṣābī*; *Kitāb fī 'Ilm al-Hay'a*; *Kitāb fī 'Amal al-Asturlāb*; *Kitāb fī 'Ilm al-Masāḥāt*; *Kitāb fī 'Ilm al-Az'yāj*; *Kitāb fī Aḥwāl al-Kawākib wa Masārātihā*; *Kitāb fī al-Raṣd wa al-Mulāḥazāt*; and other useful books.

### ذكر إبراهيم بن سنان ومصنفاته

إبراهيم بن سنان بن ثابت بن قرة الحراني كان عالماً  
ورث وقد والحساب والهندسة الهيئة علم في بارعاً فاضلاً  
سنان أبيه عن وأخذه قرة بن ثابت جده عن العلم  
القطع مساحة في ألف قد الفهم سريع فطناً ذكياً وكان  
الفلك علم وفي المخروطات وفي المكافئ

### Account of Ibrāhīm ibn Sinān and His Compositions

Ibrāhīm ibn Sinān ibn Thābit ibn Qurra al-Ḥarrānī was a scholar, distinguished and outstanding in astronomy, geometry, and arithmetic. He inherited learning from his grandfather Thābit ibn Qurra and received it from his father Sinān. He was intelligent, sharp, and quick of understanding. He composed works on the quadrature of the parabola, on conic sections, and on astronomy.

وله كتاب في مساحة القطع المكافئ وكتاب في المخروطات  
حساب في وكتاب الفلك علم في وكتاب الهيئة علم في وكتاب  
الأسطرلاب عمل في وكتاب الأزياج في وكتاب المنازلات  
الهيئة علم في وأفضلهم عصره علماء أجل من وكان  
المجسطي في ذكره ما بعض في بطليموس طريقة انتقد وقد

His works include: *Kitāb fī Masāḥat al-Qiṭāc al-Mukāfi*; *Kitāb fī al-Makhrūṭāt*; *Kitāb fī 'Ilm al-Hay'a*; *Kitāb fī 'Ilm al-Falak*; *Kitāb fī Ḥisāb al-Manāzalāt*; *Kitāb fī al-Azyāj*; and *Kitāb fī 'Amal al-Asturlāb*. He was among the most eminent scholars of his age and the most excellent in astronomy, and he criticized Ptolemy's method in some matters mentioned in the *Almagest*.

## 5 Discourse V: Medicine and Natural Sciences

### الفن الخامس في الطب والعلوم الطبيعية

### THE FIFTH DISCOURSE: MEDICINE AND THE NATURAL SCIENCES

#### في أطباء اليونان والهنود والفرس

وأما أصناف الطب فإنها: علم الأدوية وعلم الأغذية وعلم والقانوني الصحيح الحفظان وعلم وأسبابها الأمراض وأجلهم الفن هذا في الناس أعظم اليونان أطباء كان وقد وبولس وجالينوس وأرسطوطاليس أبقرات فمنهم قدراً وبطليموس المسلمون فدرسها العربية إلى كتبهم نقلت وقد وغيرهم والتلخيص والشرح والتصنيف فيها وأكثروا

#### On the Physicians of the Greeks, Indians, and Persians

The categories of medicine are: the science of drugs, the science of nutrition, the science of diseases and their causes, and the science of the regimen of health. The Greek physicians were the greatest people in this art and the most eminent: among them Hippocrates, Aristotle, Galen, Paulus, Ptolemy, and others. Their books were translated into Arabic, and the Muslims studied them, composing commentaries, abridgements, and summaries.

#### في أطباء المسلمين

وأما أطباء المسلمين فمنهم الحكماء الذين تركوا التصنيف وابن الزهراوي سينا وابن الرازي ومنهم كثيراً الطب في العلم هذا في صنفوا وقد وغيرهم وبختيشوع ماسويه اليونانيون صنفه ما وتضاهي الأرفف تملأ عظيمة كتباً المفردة الأدوية في وأبحاثهم تجاربهم إليه أضافوا وقد والمركبة

#### On the Muslim Physicians

Among the Muslim physicians were wise men who composed abundantly in medicine: al-Rāzī, Ibn Sīnā, al-Zahrāwī, Ibn Māsawayh, Bakhtishūç, and others. They composed great books in this science filling the shelves, rivalling what the Greeks composed, and added their own experiences and research on simple and compound remedies.

#### في علم الطبيعيات

وأما الطبيعيات فهي علوم يعرف بها الأجسام الطبيعية وأمورها سميت العلم هذا في كتباً أرسطوطاليس وضع وقد بالطبيعيات وكتاب والعالم السماء وكتاب الطبيعى السماع كتاب ومنها وغيرها النفس وكتاب الأرواح وكتاب الفساد الكائن عليها وزادوا وشرحوها المسلمون ونقلها

#### On Natural Philosophy

Natural philosophy is the science by which one knows natural bodies and their properties. Aristotle composed books in this science called the *Natural Works*: among them *On Natural Hearing* (Physics), *On the Heavens and the World*, *On Generation and Corruption*, *On Spirits*, *On the Soul*, and others. The Muslims translated them, commented upon them, and added to them.

## 6 Discourse VI: Philosophy and Theology

### الفن السادس في الفلسفة والمنطق والكلام

### THE SIXTH DISCOURSE: PHILOSOPHY, LOGIC, AND SCHOLASTIC THEOLOGY

#### في الفلسفة

وأما الفلسفة فإنها علم يعرف به حقائق الأشياء كما هي الأبواب متعددة العلم هذا في كتباً أرسطوطاليس وضع وقد نقلت وقد والأخلاق والإلهيات والطبيعيات المنطق ومنها فيها وألفوا وشرحوها المسلمون فدرسها العربية إلى كتبه سينا وابن والفارابي الكندي أشهرهم فمن جملة كتباً

#### On Philosophy

Philosophy is the science by which one knows the realities of things as they are. Aristotle composed books in this science of multiple branches: logic, natural philosophy, metaphysics, and ethics. His books were translated into Arabic; the Muslims studied them, commented upon them, and composed abundant books. Among the most famous are: al-Kindī, al-Fārābī, and Ibn Sīnā.

#### في المنطق

وأما المنطق فهو علم يعرف به الأفكار وكيفية صوابها أشهرها كتب في أرسطوطاليس وضعه وقد وخطأها الأورغانون وشرحوه ودرسوه اليونانيون عن المسلمون ونقله عدي بن يحيى أشهرهم ومن كثيرة كتباً فيه وصنفوا وغيرهم الراوندي وابن والفارابي

#### On Logic

Logic is the science by which one knows thoughts and the manner of their correctness and error. Aristotle established it in books whose most famous is the *Organon*. The Muslims translated it from the Greeks, studied it, commented upon it, and composed many books. Among the most famous: Yaḥyā ibn ʿAdī, al-Fārābī, and Ibn al-Rāwandī.

#### في الكلام

وأما الكلام فهو علم يعرف به الأحكام الشرعية وأدلتها في يتكلمون العلماء من جماعة الإسلام صدر في كان وقد المذاهب أصحاب عنهم فنشأ عليها ويستدلون الأحكام وغيرهم والمرجئة والمختزلة والأشاعرة كالمعتزلة فيه مصنفة وكتب خاص مذهب لها فرقة وكل

#### On Scholastic Theology (*Kalām*)

*Kalām* is the science by which one knows legal rulings and their proofs. In the early period of Islam there were groups of scholars who dealt with rulings and argued for them, giving rise to the sects: the Muṭazila, the Ashāriyya, the Khawārij, the Murji'a, and others. Each sect has its own doctrine and composed books thereon.

## 7 Discourse VII: Occult Sciences

الفن السابع في السحر والطلاسم والعزائم والكسوفات

THE SEVENTH DISCOURSE: MAGIC, TALISMANS, INCANTATIONS, AND CATOPTICS

### في السحر

وأما السحر فهو علم يعرف به التأثيرات الغريبة التي تظهر  
قال حيث الكريم القرآن في ذكره ورد وقد العادة خلاف  
تعالى  
في السحرة كان فقد الساحران وهبط بساحرين هم وما  
كتبهم نقلت وقد كثيراً والفرس والهنود اليونانيين  
المسلمين من قوم ودرسها العربية إلى

### On Magic

Magic is the science by which one knows strange effects that appear contrary to custom. It is mentioned in the Noble Qur'an where the Exalted says: "and they were not magicians" and "the two magicians came down." There were many magicians among the Greeks, Indians, and Persians; their books were translated into Arabic and studied by some Muslims.

### في الطلاسم والعزائم

وأما الطلاسم فهي أعداد وحروف ورموز تكتب في أوقات  
بالطلاسم وتسمى معين لغرض معينة أشياء على خاصة فلكية  
كتب فيها صنف وقد العمل على تختم أي تطلمس لأنها  
العزائم وكتاب لهرمس الطلاسم كتاب أشهرها كثيرة  
كثير وغيرهما البلخي معشر لأبي

### On Talismans and Incantations

Talismans are numbers, letters, and symbols written at special astronomical times on specific objects for a particular purpose. They are called *talāsim* because they are "sealed" upon the act. Many books have been composed on this subject, the most famous being the *Kitāb al-Ṭalāsim* of Hermes and the *Kitāb al-'Azā'im* of Abū Maṣār al-Balkhī, among others.

## 8 Discourse VIII: Religious Sects and Doctrines

الفن الثامن في أصحاب الملل والنحل والفرق

THE EIGHTH DISCOURSE: THE PEOPLE OF RELIGIONS, SECTS, AND FACTIONS

### في الملل والنحل

### On Religions and Sects

وأما أصحاب الملل فهم الذين لهم دين مبني على أصل معين  
والمجوس والنصارى كاليهود السماوية الكتب أصحاب ومنهم  
فرق الملل هذه من واحدة ولكل والمشركون والصابئة  
بها خاص مذهب لها فرقة وكل متعددة وطوائف  
وحديثة قديمة فيه مصنفة وكتب

The people of religions are those who have a religion built upon a certain foundation: among them the People of the Book — the Jews, the Christians, the Magians, the Sabians, and the polytheists. Each of these religions has multiple sects and factions; each sect has its own doctrine and books, ancient and modern.

### في الفرق الإسلامية

### On Islamic Sects

وأما فرق المسلمين فهم: الخوارج والشيعة والمعتزلة  
وغيرهم والناجية والقدرية والجبرية والمرجئة والأشاعرة  
وكتب ومصنفون أئمة لها الفرق هذه من واحدة وكل  
النوختي المسلمين: فرق في المصنفين أشهر ومن  
وغيرهم والبغدادى حزم وابن المقدسي

The Islamic factions are: the Khawārij, the Shīʿa, the Muṭazila, the Ashāriyya, the Murjiʿa, the Jabriyya, the Qadariyya, the Najiya, and others. Each of these factions has its leaders, authors, and books. Among the most famous authors on Islamic sects: al-Nawbakhtī, al-Maqdisī, Ibn Ḥazm, and al-Baghdādī.



## 9 Discourse IX: Alchemy

الفن التاسع في الكيمياء والصنعة الإلهية

THE NINTH DISCOURSE: ALCHEMY AND THE DIVINE CRAFT

### في الكيمياء وأصولها

وأما الكيمياء فهي علم يعرف به تحويل الأجسام من صفة تحويل فيها يطلب حيث المعادن في وخاصة أخرى صفة إلى وقالوا صحيح علم أنها بعضهم ادعى وقد ذهب إلى المكلف الصادقون المخلصون يعرفها وإنما إلهية صنعة إنها ومكر خدع هي بل صحيح بعلم ليست إنها آخرون وقال

### On Alchemy and Its Foundations

Alchemy is the science by which one knows the transformation of bodies from one quality to another, especially metals, where the aim is to turn base metal into gold. Some claim it is a true science, calling it the “divine craft” (*ṣanāʾ ilāhiyya*) and saying it is known only by the sincere and truthful. Others say it is not a true science but deception and trickery.

### في المؤلفين في الكيمياء

وقد صنف في الكيمياء قوم كثير منهم جابر بن حيان وقد السلام عليه محمد بن جعفر الصادق أصحاب من وكان الرحمة وكتاب السبعين كتاب منها كثيرة كتباً فيها ألف السرخسي ومنهم وغيرها البديع وكتاب الكمال وكتاب كثير وغيرهم الصلاح أبي وابن والطغرائي

### On the Authors of Alchemical Works

Many people composed works on alchemy: among them Jābir ibn Ḥayyān, who was a companion of al-Ṣādiq Jafar ibn Muḥammad — peace be upon him. He composed many books on it, including the *Kitāb al-Sabʿīn* (Book of Seventy), the *Kitāb al-Raḥma*, the *Kitāb al-Kamāl*, the *Kitāb al-Badīʿ*, and others. Among them also al-Sarrakhsī, al-Ṭuḡhrāʾī, Ibn Abī al-Ṣalāḥ, and many others.



## 10 Discourse X: History, Biography, and Genealogy

الفن العاشر في التاريخ والسير والأنساب

THE TENTH DISCOURSE: HISTORY, BIOGRAPHIES,  
AND GENEALOGY

### في المؤرخين

### On the Historians

وأما الفن العاشر فهو الفن الأخير في هذا الكتاب وقد  
وأصحاب والمغازي السير ومصنفي المؤرخين فيه ذكرنا  
إسحاق ابن المؤرخين: أشهر ومن والأخباريين الأنساب  
ومسكويه والطبري سعد وابن هشام وابن الواقدي  
كثير وغيرهم والمسعودي مiskويه وابن

The tenth discourse is the last in this book. We mention in it the historians, authors of *siyar* and *maghāzī*, the genealogists, and the annalists. Among the most famous historians: Ibn Ishāq, al-Wāqidī, Ibn Hishām, Ibn Saḍ, al-Ṭabarī, Miskawayh, al-Masūdī, and many others.

### في أصحاب السير والمغازي

### On the Authors of Campaigns and Biographies

وأما أصحاب السير والمغازي فمنهم ابن إسحاق صاحب  
المغازي كتاب صاحب والواقدي والفتوح المغازي كتاب  
صاحب سعد وابن النبوية السيرة كتاب صاحب هشام وابن  
فتوح كتاب صاحب والبلاذري الكبير الطبقات كتاب  
البلدان  
الفن هذا في صنف ممن كثير وغيرهم الأشراف وأنساب

Among the authors of *siyar* and *maghāzī*: Ibn Ishāq, author of *Kitāb al-Maghāzī wa al-Futūḥ*; al-Wāqidī, author of *Kitāb al-Maghāzī*; Ibn Hishām, author of *Kitāb al-Sīra al-Nabawiyya*; Ibn Saḍ, author of *Kitāb al-Ṭabaqāt al-Kabīr*; al-Balādhurī, author of *Futūḥ al-Buldān* and *Ansāb al-Ashraf*; and many others who composed in this art.

## Appendix: The Translation Movement

### في أول نقل العلوم إلى العربية

### ON THE FIRST TRANSLATION OF THE SCIENCES INTO ARABIC

#### قصة خالد بن يزيد

#### The Story of Khālīd ibn Yazīd

The following celebrated passage from the *Fihrist* recounts the legendary origins of the Graeco-Arabic translation movement:

كان خالد بن يزيد بن معاوية يُسمّى حكيم آل مروان وكان الصنعة بباله خطر للعلوم ومحبة همة وله نفسه في فاضلاً ينزل كان ممن اليونانيين فلاسفة من جماعة بإحضار فأمر في الكتب بنقل وأمرهم بالعربية تفقه وقد مصر مدينة أول وهذا العربي إلى والقبطي اليوناني اللسان من الصنعة الديوان نقل ثم لغة إلى لغة من الإسلام في كان نقل الفارسية باللغة وكان

Khālīd ibn Yazīd ibn Muāwiya, called the Sage of the Marwānids, was virtuous in himself, ambitious, and fond of the sciences. It occurred to him to commission a group of Greek philosophers residing in Egypt who had become proficient in Arabic to translate books on the craft from Greek and Coptic into Arabic. This was the first translation in Islam from one language to another; then the *dīwān* was translated from Persian.

#### في بيت الحكمة

#### On the House of Wisdom

ثم إن المأمون لما ولي الخلافة أمر بإنشاء بيت الحكمة لينقلوا والنقل العلماء من جماعة فيه وجعل بغداد في وغيرها والحساب والطب والفلك والهندسة الفلسفة كتب والهنود والفارسيين والسريانيين اليونانيين لغات من العلوم نشر في عظيم سبباً ذلك وكان العربية إلى

When al-Ma'mun assumed the caliphate, he ordered the establishment of the House of Wisdom (*Bayt al-Hikma*) in Baghdad and appointed scholars and translators to translate books of philosophy, geometry, astronomy, medicine, arithmetic, and other sciences from the languages of the Greeks, Syriacs, Persians, and Indians into Arabic. This was a great cause in the dissemination of the sciences.

#### End of Bilingual Edition

This bilingual edition presents a representative selection from the *Kitāb al-Fihrist* of Ibn al-Nadīm (c. 987 CE), based on the critical edition of Gustav Flügel (Leipzig: F. C. W. Vogel, 1871–1872). The parallel Arabic-English text covers all ten discourses of the *Fihrist*, with special emphasis on the mathematical sciences and the biographical entries for Islamic mathematicians and scientists. The mathematical subsection (Section IV) constitutes a key primary source for the history of Islamic mathematics, preserving unique information about al-Khwārizmī, Thābit ibn Qurra, and the translation movement.

## Index of Persons

| Name                     | Significance in the <i>Fihrist</i>                                      |
|--------------------------|-------------------------------------------------------------------------|
| Abū Maṣār al-Balkhī      | Astrologer; author of the <i>Kitāb al-Madkhal al-Kabīr</i>              |
| al-Battānī (Albategnius) | Astronomer; <i>Zīj al-Ṣābī</i> ; refined trigonometric methods          |
| al-Farghānī              | Astronomer; <i>Kitāb fī al-Hay’a</i>                                    |
| al-Khwārizmī             | Mathematician; <i>Kitāb al-Jabr</i> ; father of algebra                 |
| al-Rāzī (Rhazes)         | Physician and philosopher; author of the <i>al-Ḥāwī</i>                 |
| Aristotle                | Greek philosopher; <i>Organon</i> , <i>Physics</i> , <i>Metaphysics</i> |
| Ḥunayn ibn Ishāq         | Chief translator of Galen and Hippocrates                               |
| Ibn Sīnā (Avicenna)      | Philosopher and physician; <i>Canon of Medicine</i>                     |
| Ibrāhīm ibn Sinān        | Mathematician; quadrature of the parabola                               |
| Ishāq ibn Ḥunayn         | Translator of Euclid’s <i>Elements</i> and Ptolemy’s <i>Almagest</i>    |
| Jābir ibn Ḥayyān         | Alchemist; <i>Kitāb al-Sab’īn</i>                                       |
| Khālīd ibn Yazīd         | Legendary initiator of the translation movement                         |
| Ptolemy                  | Astronomer; <i>Almagest</i>                                             |
| Qusṭā ibn Lūqā           | Translator; works on astronomy, mechanics, geometry                     |
| Thābit ibn Qurra         | Mathematician, astronomer, translator of Apollonius                     |

# TABAQAT AL-UMAM

(CATEGORIES OF NATIONS)

كِتَابُ طَبَقَاتِ الْأُمَمِ

Bilingual Edition

Arabic Text • English Translation

By

Abu al-Qasim Sa'id ibn Ahmad  
al-Andalusi

(d. 462 AH / 1070 CE)

لِلْقَاضِي أَبِي الْقَاسِمِ صَاعِدِ بْنِ أَحْمَدَ بْنِ صَاعِدِ الْأَنْدَلُسِيِّ

الْمُتَوَفَّى سَنَةَ ٤٦٢ هـ ١٠٧٠ م

Based on the 1912 Cheikho edition (Beirut: al-Matba'a al-Kathulikiyya)

Edited with Notes, Variants, and Indexes by Pere Louis Cheikho, S.J.

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## Editor's Introduction

### تَوْطِئَةُ الْمُحَقِّقِ

كِتَابُ طَبَقَاتِ الْأُمَمِ أَحَدُ الْكُتُبِ النَّادِرَةِ الَّتِي نَعْرُضُ فِيهَا كَتَبَهُ الْعَرَبُ لَوْصَفِ الْعُلُومِ بَيْنَ الْأُمَمِ الَّتِي سَبَقَتْ عَهْدَهُمْ، وَإِنْ لَمْ يَبْلُغْ صَاحِبُهُ فِي ذَلِكَ شَأْوَ كِتَابِ الْفَهْرَسْتِ لِأَبِي الْفَرَجِ ابْنِ الدِّمِيمِ إِلَّا أَنَّهُ جَمَعَ عِدَّةً فَوَائِدَ تَدُلُّ عَلَى نَشَاطٍ فِي الْبَحْثِ وَعَلَى رَغْبَةٍ فِي التَّحْصِيلِ وَدَقَّةٍ نَظَرٍ فِي التَّنْوِينِ، وَكَانَ أَهْلُ الْأَنْدَلُسِ يَفْتَخِرُونَ بِهِ وَبِرُؤُوسِ أَهْلِ الشَّرْقِ.

وَقَدْ ذَكَرَ ابْنُ الْأَبَّارِ فِي كِتَابِ التَّكْمِلَةِ لِكِتَابِ الصَّلَةِ ٤٦٣ :  
طَبْعَةُ بَجْرِيْطَ عَنْ عَبْدِ اللَّهِ بْنِ مُحَمَّدٍ بْنِ مَرْزُوقِ الْيَحْصَبِيِّ أَنَّهُ  
قَدْ دُسْكَندَرِيَّةَ رَوَى هَذَا الْكِتَابَ لِأَبِي طَاهِرِ السَّلَفِيِّ.

وَمِمَّنْ عَرَفُوا هَذَا الْكِتَابَ فِي الشَّرْقِ أَبُو الْفَرَجِ غَرِغُورِيُوسَ  
ابْنُ الْعِبْرِيِّ فَإِنَّهُ نَقَلَ عَنْهُ فِي كِتَابِ تَارِيخِ مُخْتَصَرِ الدُّوَلِ نَبَذَتَيْنِ  
مُفِيدَتَيْنِ فِي الْعَرَبِ وَعُلُومِهِمْ، وَكَذَلِكَ عَرَفَهُ الْحَاجُّ خَلِيفَةُ فَذَكَرَهُ  
مِرَارًا فِي كِتَابِهِ كَشَفِ الطُّنُونِ قَدَعًا تَارَةً التَّعْرِيفِ بِطَبَقَاتِ الْأُمَمِ  
وَقَالَ فِي وَصْفِهِ إِنَّهُ كِتَابٌ صَغِيرُ الْحَجْمِ كَثِيرُ النَّفْعِ.

وَهِيَ مَكْتُوبَةٌ بِخَطِّ جَلِيِّ شَبِيهِ بِالْقَلَمِ الْفَارِسِيِّ عَلَى وَرَقٍ  
صَفِيْقٍ ضَارِبٍ إِلَى الصَّفْرَةِ وَمُجَلَّدَةٌ تَجْلِيدًا مُثَقَّنًا بِجِلْدٍ وَرَقٍ مُلَوَّنٍ  
ذَهَبِيٍّ عَلَى الْوَجْهَيْنِ مَعَ لِسَانٍ مِثْلَهُمَا زَيْنَةً.

أَمَّا الْمُؤَلِّفُ فَلَا نَعْلَمُ أَقَلَّ مِنْ أَمْرِهِ، وَ هِ تَرْجَمَتُهُ كَمَا رَوَاهَا  
ابْنُ بَشَّ وَالٍ فِي كِتَابِ الصَّلَةِ طَبْعَةُ بَجْرِيْطَ ص ٣٤

صَاعِدُ بْنُ أَحْمَدَ بْنِ عَبْدِ الرَّحْمَنِ بْنِ مُحَمَّدٍ بْنِ صَاعِدِ  
بَيْ الْقَاضِي بِكْنِيَّةِ أَبِي الْقَاسِمِ وَأَصْلُهُ مِنْ قُرْطَبَةِ رُوِيَ عَنْ أَبِي  
مُحَمَّدٍ بْنِ حَزْمٍ وَالْفَتْحِ بْنِ قَاسِمٍ وَأَبِي الْوَلِيدِ الْوَقْشِيِّ وَغَيْرِهِمْ،  
وَأَسْتَفْضَاهُ الْأَمِيرُ يَحْيَى بْنُ زِي الثُّونِ بِطَلَيْطَلَةَ وَكَانَ مِنْ أَهْلِ  
الْمَعْرِفَةِ وَالذِّكَاةِ وَالرَّوَايَةِ، وَلِدَ بِالْمَرْيَةِ فِي سَنَةِ ٤٢٠ هـ ١٠٣٩  
م وَتُوفِّيَ بِطَلَيْطَلَةَ فِي شَوَّالِ سَنَةِ ٤٦٢ هـ ١٠٧٠ م

### [Editor's Introduction]

*Kitab Tabaqat al-Umam* (The Book of the Categories of Nations) is one of the rare works in which an Arab writer describes the sciences among the nations that preceded his era. Although its author did not reach the level of Abu al-Faraj ibn al-Damimi in his *Fihrist*, he nevertheless collected numerous useful facts that demonstrate his research activity, his desire for knowledge, and his precision in observation. The people of al-Andalus used to boast of him and of the heads of the East.

Ibn al-Abbār mentions in his *Takmila* to the *Silah* (2:463, Bajarid edition) on the authority of Abd Allah ibn Muhammad ibn Marzūq al-Yahṣībī that he transmitted this book to Abu Tahir al-Salafi.

Among those who knew this book in the East was Abu al-Faraj Grigoriyus ibn al-‘Ibrī, for he quoted from it in his *History of the Dynasties* two useful excerpts concerning the Arabs and their sciences. Likewise, al-Hājj Khalifa knew it and mentioned it several times in his *Kashf al-Zunūn*: at one point calling it *Ta’rīf bi-Tabaqat al-Umam*, and describing it as a small book of great benefit.

[Manuscript Description:] The manuscript is written in a clear hand resembling Persian script, on thick paper tending toward yellow, and bound with skillful binding in colored gold paper on both sides with a matching tongue as ornament.

As for the author, we know no less than what Ibn Bashkuwal narrated in his *Silah* (Bajarid edition, p. 234):

“Sa’id ibn Ahmad ibn ‘Abd al-Raḥmān ibn Muḥammad ibn Sa’id al-Taghlabī, the judge with the kunya Abu al-Qāsim, originally from Cordoba, transmitted from Abu Muḥammad ibn Ḥazm, al-Faṭḥ ibn Qāsim, Abu al-Walīd al-Waqshī, and others. The amir Yahyā ibn Dhī al-Nūn appointed him as judge of Tulaytula (Toledo). He was a man of knowledge, intelligence, and transmission. He was born in al-Mariyya (Almeria) in 420 AH (1039 CE) and died in Tulaytula while serving as its judge in Shawwal of the year 462 AH (1070 CE).”

## Chapter One: The Ancient Nations

### البَابُ الْأَوَّلُ : الْأُمَمُ الْقَدِيمَةُ

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

رَبِّ يَسِّرْ

قَالَ الْقَاضِي أَبُو الْقَاسِمِ صَاعِدُ بْنُ أَحْمَدَ بْنِ صَاعِدٍ رَحِمَهُ اللَّهُ تَعَالَى: اعْلَمْ أَنَّ جَمِيعَ النَّاسِ فِي مَشَارِقِ الْأَرْضِ وَمَغَارِبِهَا وَجَنُوبِهَا وَشَمَالِهَا وَإِنْ كَانُوا نَوْعًا وَاحِدًا يَتَمَيَّزُونَ بِثَلَاثَةِ أَشْيَاءَ: بِالْأَخْلَاقِ وَالصُّوَرِ وَاللُّغَا .

وَرَعَمَ مَنْ عُنِيَ بِأَخْبَارِ الْأُمَمِ وَبَحَثَ عَنْ سَائِرِ الْأَجْيَالِ وَفَحَصَ عَنْ طَبَقَاتِ الْقُرُونِ أَنَّ النَّاسَ كَانُوا فِي سَالِفِ الدُّهُورِ وَقَبْلَ تَشَعُّبِ الْقَبَائِلِ وَافْتِرَاقِ اللُّغَاتِ سَبْعَ أُمَمٍ.

لَأُمَّةٌ لِأُولَى الْفُرْسِ وَكَانَ مَسْكَنُهَا فِي الْوَسْطِ الْمَعْمُورِ دِيَارِهَا مِنْ جِبَالِ التِّي فِي شِمَالِ الْعِرَاقِ بِعَقَبَةِ حُلْوَانَ وَالَّذِي فِيهِ الْأَذْهَاتُ وَالْكُرَجُ وَالْدُسْتُورُ وَهَمْدَانُ وَقَاشَانُ وَغَيْرُهَا مِنْ الْبِلَادِ إِلَى أَرْمِينِيَّةِ وَالْبَابِ الْمُتَّصِلِ بِبَحْرِ أَدْرِيَجَانَ وَطَبْرِسْتَانَ وَمَوْلَانَا وَالْبَيْلْقَانَ وَأَرْزَنَ وَالشَّابَرَانَ وَالرِّيَّ وَالطَّالِقَانَ وَجُرْجَانَ إِلَى بِلَادِ خُرَاسَانَ.

### [Chapter One: The Ancient Nations]

*In the name of God, the Merciful, the Compassionate.*

*O Lord, make it easy!*

The judge Abu al-Qāsim Sa'īd ibn Aḥmad ibn Sa'īd—may God the Exalted have mercy on him—said: Know that all people in the East and West of the Earth, in its South and North, though they are one species, are distinguished by three things: by their moral character, their physical forms, and their languages.

Those who have concerned themselves with the reports of nations and investigated the various generations, and examined the categories of ages, say that in earliest times, before the branching of tribes and the divergence of languages, there were seven nations.

**The First Nation:** The Persians. Their dwelling was in the middle of the inhabited world. Their lands include the mountains in the north of Iraq, the region of Hulwan, and the area containing the Anjahāt, Kuraj, Dustūr, Hamadān, Qumm, Qāshān, and other lands extending to Armenia, the Gate connecting to the Sea of Azerbaijan, Tabaristān, Multān, Bilāqān, Arzan, Shāborān, al-Rayy, Tāliqān, and Jurjān, to the lands of Khurāsān.

## Chapter Two: Science in India

### العلم في الهند

وَلِبُعْدِ الْهِنْدِ مِنْ بِلَدِنَا وَاعْتِرَاضِ الْمَوَانِعِ بَيْنَنَا وَبَيْنَهُمْ فَلَمْ نَأْلَفْهُمْ  
فَلَمْ يَصِلْ إِلَيْنَا طَافٌ مِنْ عُلُومِهِمْ وَلَا وَرَدَتْ عَلَيْنَا إِلَّا نُبُذٌ مِنْ  
مَذَاهِبِهِمْ وَلَا سَمِعْنَا إِلَّا بِالْقَلِيلِ مِنْ عِلْمِهِمْ.

فَمِنْ مَذَاهِبِ الْهِنْدِ فِي عُلُومِ النُّجُومِ الْمَذْهَبُ الثَّلَاثَةُ  
الْمَشْهُورَةُ عَنْهُمْ وَهُوَ مَذْهَبُ السُّنْدِ هِنْدُ وَمَذْهَبُ الْأَزْدِيحِ مَذْهَبُ  
الْأَزْكَدِ وَلَمْ يَصِلْ إِلَيْنَا مِنْهُمْ عَلَى الشَّخْصِ إِلَّا مَذْهَبُ السُّنْدِ هِنْدُ  
وَهُوَ الْمَذْهَبُ الَّذِي تَقْلَدُهُ جَمَاعَةٌ مِنَ الْإِسْلَامِ وَالْقَوَامُ فِيهِ الْأَزْيَاجُ  
كَمُحَمَّدِ بْنِ إِبْرَاهِيمَ الْفَزَارِيِّ وَحَبِشِ بْنِ عَبْدِ اللَّهِ الْبَغْدَادِيِّ وَمُحَمَّدِ  
بْنِ مُوسَى الْخَوَارِزْمِيِّ وَالْحُسَيْنِ بْنِ مُحَمَّدٍ الْمَعْرُوفِ بِابْنِ الْأَدَمِيِّ  
وغيرِهِمْ. وَتَفْسِيرُ السُّنْدِ هِنْدٍ لِدَهْرٍ الدَّاهِرِ كَذَلِكَ حَكَى  
سُيُنُ بْنُ أَفْ فِي زِيَجِهِ.

صَلِّ

وَأَمَّا أَصْحَابُ السُّنْدِ هِنْدٍ فَإِنَّهُمْ وَافَقُوا أَنْ لَا عَدَدَ مَدَّةِ الْعَالَمِ  
فَإِنَّ مَذْهَبَهُمُ الَّذِي ذَكَرُوهُ أَنَّ الْكَوَاكِبَ وَأَوْجَاتَهَا وَجُوزَهَرَاتَهَا  
مُجْتَمِعَةٌ فِي رَأْسِ الْحَمَلِ فَمُنْذُ ذَلِكَ الْأَمْرِ أَبْدَأَتْ حَرَكَاتُ الْكَوَاكِبِ  
وَالْأَوْجَاتِ الْجُوزَهَرَاتِ فِي الْأَرْضِ وَبَقِيَ الْعَالَمُ السُّفْلَى خَرَابًا  
دَهْرًا طَوِيلًا حَتَّى تَتَفَرَّقَ الْكَوَاكِبُ وَالْأَوْجَاتُ وَالْجُوزَهَرَاتُ فِي  
الْبُرُوجِ فَإِذَا كَانَ كَذَلِكَ بَدَأَتْ حَالَةُ الْعَالَمِ السُّفْلِيِّ إِلَى الْأَمْرِ الْأَوَّلِ  
هَكَذَا أَبْدَأَ إِلَى غَيْرِ غَايَةٍ عِنْدَهُمْ. وَلِكُلِّ وَاحِدٍ مِنَ الْكَوَاكِبِ  
وَالْأَوْجَاتِ وَالْجُوزَهَرَاتِ أَدْوَارٌ مَا فِي هَذِهِ الْمَدَّةِ الَّتِي هِيَ عِنْدَهُمْ  
مَدَّةُ الْعَالَمِ قَدْ ذَكَرْتُهَا.

### [Science in India]

Because of the distance of India from our lands and the obstacles between us and them, we have not become familiar with them. No portion of their sciences has reached us, nothing has come to us but fragments of their doctrines, and we have heard only a little of their knowledge.

Among the doctrines of the Indians in the science of astrology, three doctrines are famous among them: the doctrine of Sindhind, the doctrine of the Āzīj, and the doctrine of Arkand. Only the doctrine of Sindhind has reached us in a complete form. It was followed by a group of Muslims, and those who established it were al-Azyāj such as Muḥammad ibn Ibrāhīm al-Fazārī, Ḥabash ibn ‘Abd Allāh al-Baghdādī, Muḥammad ibn Mūsā al-Khwārizmī, al-Ḥusayn ibn Muḥammad known as Ibn al-Ādamī, and others. The interpretation of Sindhind is “the Eternal Time” (al-Dahr al-Dāhir), as al-Ḥusayn ibn al-Ādamī related in his zīj.

### [Section]

As for the followers of Sindhind, they agree that the duration of the world has no fixed number. Their doctrine which they mention is that the planets, their apogees, and their nodes were all gathered at the head of Aries. Since that event, the movements of the planets, apogees, and nodes began. The lower world remained desolate for a long time until the planets, apogees, and nodes dispersed through the zodiac signs. When that happened, the condition of the lower world returned to its first state, and this continues without end according to them. Each of the planets, apogees, and nodes has cycles during this period, which they consider the duration of the world, which I have mentioned.



## Chapter Three: Science among the Persians

### الْعِلْمُ فِي الْفُرسِ

وَأَمَّا الْأُمَّةُ الثَّانِيَّةُ وَهِيَ الْفُرسُ فَأَهْلُ الشَّرَفِ الْبَازِخِ وَالْعِزِّ الشَّامِخِ وَأَوْسَطِ الْأُمَمِ دَارًا وَأَشْرَفُهَا أَقْلِيًّا وَأَنُوسَهَا مَلُوكًا وَلَا تَعْلَمُ أُمَّةٌ غَيْرُهُمْ دَامَ لَهَا الْمُلْكُ وَكَانَتْ لَهُمْ مَلُوكٌ تَجْمَعُهُمْ وَرُؤُوسٌ تُحَامِي عَنْهُمْ مَنْ نَاوَأَهُمْ وَتَغْلِبُ بِهِمْ مَنْ غَارَهُمْ وَتَدْفَعُ ظَالِمَهُمْ عَنْ مَظْلُومِهِمْ وَتُحْتَمِلُهُمْ مِنَ الْأُمُورِ عَلَى مَا فِيهِ حَظُّهُمْ عَلَى اتِّصَالِ وَدَوَامِ وَأَحْسَنِ التَّنْظِيمِ يَأْخُذُ ذَلِكَ آخِرُهُمْ عَنْ أَوَّلِهِمْ وَغَائِبُهُمْ عَنْ سَالِفِهِمْ.

### ي الْعَرَبِ وَالْعَجَمِ

وَأَصَحُّ مَا قِيلَ فِي ذَلِكَ أَنَّ مِنْ ابْتِدَاءِ مُلْكِ كَيْثُمُورْتِ بْنِ أَمِيمِ بْنِ آلَادِ بْنِ سَامِ بْنِ نُوحٍ إِلَى مُلْكِ مَنُوشِيهِرِ الْفُرسِ كَانَتْهُمْ الْوَلَدِي هُوَ عِنْدَهُمْ آدَمُ أَبُو الْبَشَرِ عَلَيْهِ السَّلَامُ إِلَى ابْتِدَاءِ مُلْكِ مَنُوشِيهِرِ أَوَّلِ مُلُوكِ الطَّبَقَةِ الثَّانِيَةِ مِنْ مُلُوكِ الْفُرسِ نَحْوُ أَلْفِ سَنَةٍ كَامِلَةٍ. وَمِنْ مُلْكِ مَنُوشِيهِرِ إِلَى ابْتِدَاءِ مُلْكِ كَيْثُبَادَ بْنِ دُوعِ أَوَّلِ مُلُوكِ الطَّبَقَةِ الثَّالِثَةِ مِنْ مُلُوكِ الْفُرسِ قَرِيبٌ مِنْ مِائَتَيْ عَامٍ. وَمِنْ مُلْكِ كَيْثُبَادَ إِلَى ابْتِدَاءِ مُلْكِ الطَّوَائِفِ وَهِيَ الطَّبَقَةُ الرَّابِعَةُ مِنْ مُلُوكِ الْفُرسِ وَذَلِكَ عِنْدَ مَقْتَلِ الْإِسْكَندَرِ لِذَارَا بْنِ دَارَا آخِرِ مُلُوكِ الطَّبَقَةِ الثَّالِثَةِ مِنْ مُلُوكِ الْفُرسِ نَحْوُ أَلْفِ سَنَةٍ.

وَمِنْ أَوَّلِ مُلْكِ الطَّوَائِفِ إِلَى ابْتِدَاءِ مُلْكِ أَرْدَشِيرِ بْنِ بَابَكِ السَّاسَانِيِّ أَوَّلِ مُلُوكِ بَنِي إِسْفَانِيَارَ وَهِيَ الطَّبَقَةُ الْخَامِسَةُ مِنْ مُلُوكِ الْفُرسِ خَمْسِمِائَةِ سَنَةٍ وَإِحْدَى وَسِتُّونَ سَنَةً. وَمِنْ ابْتِدَاءِ مُلْكِ أَرْدَشِيرِ بْنِ بَابَكِ إِلَى انْقِصَاءِ دَوْلَةِ الْفُرسِ مِنَ الْأَرْضِ وَذَلِكَ عِنْدَ قَتْلِ يَزْدَجَرْدَ بْنِ شَهْرِيَارِ بْنِ كِسْرَى أَرْبَعِ سِنِينَ.

### [Science among the Persians]

As for the second nation, the Persians: they are a people of towering nobility and exalted glory, the most settled of nations and the noblest in region, and the most prosperous in kings. We know of no nation besides them whose sovereignty endured so long, who had kings that united them and leaders that defended them from those who opposed them, overcame with them those who envied them, and protected their oppressed from their oppressors, and who endured affairs in a way that secured their portion with continuity and permanence and the finest organization, so that the last of them took this over from the first, and the absent from the absent.

### [On the Arabs and the Non-Arabs]

The most correct opinion on this is that from the beginning of the reign of Kayūmārth ibn Amīm ibn Ālādh ibn Sām ibn Nūḥ (Noah) to the reign of Manūshihir the Persian—who is, according to them, Adam the father of mankind, peace be upon him—to the beginning of the reign of Manūshihir, the first king of the second class of Persian kings, was about a thousand complete years. From the reign of Manūshihir to the beginning of the reign of Kaythubādh ibn Du‘a, the first king of the third class of Persian kings, was about two hundred years. From the reign of Kaythubādh to the beginning of the reign of the Tribes (al-Tawā‘if), the fourth class of Persian kings, at the killing of Alexander (al-Iskandar) of Dārā ibn Dārā, the last king of the third class of Persian kings, was about a thousand years.

From the beginning of the reign of Azdashīr ibn Bābak the Sāsānian, the first king of the Banū Isfāniyār (the fifth class of Persian kings), was five hundred and sixty-one years. From the beginning of the reign of Azdashīr ibn Bābak to the end of the Persian state on earth, at the killing of Yazdajird ibn Shahriyār ibn Kistrā, was four years.

## Chapter Four: Science in Greece

### العلم في اليونان

وَأَمَّا الْيُونَانِيُّونَ فَهُمْ أَهْلُ فَلَسْطِينَ وَالرُّومِ وَقِسْطَنْطِينِيَّةَ وَالْأَمَمِ  
الْمُحِيطَةُ بِهَا. وَكَانَتْ دَارَةُ مُلْكِهِمْ مَمْدَّةً مِنْ شَطِّ بَحْرِ الْمَتَوَسِّطِ  
فِي الْمَشَارِقِ إِلَى شَطِّهِ فِي الْمَغَارِبِ، وَمِنْ أَقْصَى الْقُطْبِ إِلَى  
بَحْرِ الشَّامِ. وَكَانَتْ مَدِينَتُهُمُ الْعَظْمَى مَدِينَةُ الْإِسْكَندَرِيَّةِ. وَكَانَتْ  
دَوْلَتُهُمْ قَدِيمَةً جِدًّا وَعِلْمُهُمْ كَثِيرَةً مُتَنَوِّعَةً.

وَكَانَ أَوَّلُ مَنْ تَكَلَّمَ فِي الْعُلُومِ وَالْحِكْمِ عَنْدهُمْ فَالْيُسُ  
وَأَفْلَطُونُ وَأَرِسْطُو وَبَطْلِيمُوسُ وَجَالِينُوسُ وَهَبَاطِيَا وَبَرْقَلِيْسُ  
وَأَفْلَاِسُ وَنُقُومَاخُسُ وَأَرْشِمِيدِسُ وَجَمِيعُ مَنْ يُسَبُّ إِلَى هَؤُلَاءِ  
مِنْ أَهْلِ الْعُلُومِ وَالْحِكْمِ. وَهُمْ الَّذِينَ قَامُوا بِإِصْلَاحِ أَصُولِ الْعُلُومِ  
وَتَوْضِيحِهَا وَتَرْتِيبِ أَفْسَامِهَا وَتَقْيِيدِ قَوَاعِدِهَا كَانَتْ مَوْلَفَاتِهِمْ  
فِي ذَلِكَ بِلُغَتِهِمْ. ثُمَّ إِنَّمَا نُقِلَتْ إِلَى لُغَةِ الْعَرَبِ بَعْدَ أَنْ نَشَرَهَا  
الْمَنَازِرَةُ وَاتَّخَذَهَا أَهْلُ الدِّينِ مِنَ النَّصَارَى فِي الشَّرْقِ.

### [Science in Greece]

As for the Greeks, they are the people of Palestine, Rome, Constantinople, and the surrounding nations. Their kingdom extended from the shore of the Mediterranean Sea in the East to its shore in the West, and from the far north to the Syrian Sea. Their greatest city was the city of Alexandria. Their state was very ancient and their sciences were many and varied.

The first to discourse on the sciences and wisdom among them were Thales, Plato, Aristotle, Ptolemy, Galen, Hippocrates, Proclus, Euclid, Nicomachus, Archimedes, and all those attributed to these in the sciences and wisdom. They were the ones who established the principles of the sciences, elucidated them, arranged their parts, and codified their rules, and their compositions on this were in their own language. Then they were translated into Arabic after the Manādhirah disseminated them and the religious scholars among the Christians in the East adopted them.

## Chapter Five: Science in Egypt

### الْعِلْمُ فِي مِصْرَ

وَأَمَّا الْقِبْطُ فَهُمْ أَهْلُ مِصْرَ الْعُلَا وَكَانَتْ عُلُومُهُمْ قَدِيمَةً نَافِعَةً وَلَكِنَّهَا لَمْ تَبْلُغْ مُنْذُ أَخَذَتْهَا الْيُونَانِيُّونَ مِنْهُمْ فِي عَهْدِ بَطْلِيمُوسَ الْمَلِكِ الَّذِي بَنَى مَدِينَةَ الإسْكَندَرِيَّةِ وَجَعَلَهَا مَوْطِنًا لِلْعُلُومِ وَحَشَدَ إِلَيْهَا الْعُلَمَاءَ مِنْ جَمِيعِ الْأُمَمِ فَأَخَذُوا عَنِ الْقِبْطِ عُلُومَهُمْ فَأَذْخَلُوا فِيهَا مِنْ حِكْمِ الْيُونَانِ مَا أَضَافَ إِلَيْهَا وَتَقَحُّوْهَا وَصَنَّفُوا فِيهَا. وَمِنْ ذَلِكَ عِلْمُ الْهَيْئَةِ وَعِلْمُ تَأْلِيفِ الْكَوَاكِبِ وَعِلْمُ الْعِلَالِ وَالْأَدْوِيَةِ وَغَيْرِ ذَلِكَ مِنَ الْعُلُومِ.

### [Science in Egypt]

As for the Copts, they are the people of Upper Egypt. Their sciences were ancient and beneficial, but they did not advance further since the Greeks took them from them in the time of Ptolemy the king who built the city of Alexandria and made it a center for the sciences, gathering scholars from all nations. They took from the Copts their sciences and incorporated into them Greek wisdom, adding to them, refining them, and composing works on them. Among these were the science of astronomy, the science of the configuration of the planets, the science of causes and remedies, and other sciences.

## Chapter Six: Science among the Jews

### العلم في اليهود

وَأَمَّا الْيَهُودُ فَقَدْ كَانَتْ لَهُمْ عُلُومٌ فِي أَرْمَنِهِهُ الْأُولَى فِي زَمَنِ  
أَنْبِيَائِهِمْ وَمُلُوكِهِمْ. ثُمَّ إِنَّهُمْ لَمَّا خَرَجُوا مِنْ بِلَادِهِمْ وَتَفَرَّقُوا فِي  
بِلَادِ الْعَجَمِ وَالْعَرَبِ ضَاعَتْ عُلُومُهُمْ وَتَنَكَّرَتْ لِأَدَمٍ لَمْ يَكُونُوا أَهْلَ  
بَقَاءٍ وَلَا مُوَظَّعَةٍ عَلَى الْعِلْمِ. وَلَكِنَّهُمْ قَدْ حَفِظُوا مِنْ تَوَارِيثِ  
الْأَنْبِيَاءِ عُلُومًا فِي التَّعْبُدِ وَالسَّرَائِعِ وَالتَّوْحِيدِ. وَكَانَ مِنْ أَشْهُرِ  
عُلَمَائِهِمْ مُوسَى بْنُ مَيْمُونٍ الْأَنْدَلُسِيُّ الَّذِي صَنَّفَ فِي عِلْمِ الْكَلَامِ  
وَالْفَلَسَفَةِ وَالطَّبِّ وَغَيْرِهِ

### [Science among the Jews]

As for the Jews, they had sciences in their earliest times, in the era of their prophets and kings. But when they left their lands and dispersed into the lands of the non-Arabs and the Arabs, their sciences were lost and forgotten because they were not a people of endurance or perseverance in science. However, they did preserve from the inheritance of the prophets sciences concerning worship, religious laws, and monotheism. Among their most famous scholars was Mūsā ibn Maymūn (Maimonides) al-Andalusī, who composed works on the science of theology, philosophy, medicine, and other subjects.

## Chapter Seven: Science among the Arabs

### العلم في العرب

### [Science among the Arabs]

وَأَمَّا الْعَرَبُ فَهُمْ أُمَّتَانِ: أُمَّةٌ بَادِيَّةٌ وَأُمَّةٌ حَاضِرَةٌ. وَكَانَتِ الْبَادِيَّةُ مِنْهُمْ لَا تَتَعَنَّى بِالْعِلْمِ وَلَا تُوَاطِبُ عَلَيْهِ لِصُعُرِ مَعِيشَتِهِمْ وَتَنَقُّلِهِمْ فِي الْبِلَادِ. وَأَمَّا الْحَاضِرَةُ فَقَدْ كَانَتْ لَهُمْ عُلُومٌ فِي الْجَاهِلِيَّةِ فِي الطِّبِّ وَاللُّغَةِ وَالْحِسَابِ وَالنَّجْمِ وَالسَّخْرِ وَالشَّعْرِ وَالنَّحْوِ وَغَيْرِ ذَلِكَ. ثُمَّ لَمَّا جَاءَ الْإِسْلَامُ وَشَرَعَ فِي طَلَبِ الْعِلْمِ وَحَثَّ عَلَيْهِ وَجَعَلَهُ سَبَبًا لِلنَّجَاةِ وَالْقَلَامِ اجْتَهَدَ الْعَرَبُ فِي طَلَبِ الْعُلُومِ وَنَقْلِهَا مِنَ الْأُمَمِ الْأُخْرَى وَتَضْيِيقِهَا بِلُغَةِ الْعَرَبِ حَتَّى أَصْبَحُوا فِي زَمَنِ قَلِيلٍ أَهْلَ عُلُومٍ وَفُنُونٍ.

وَكَانَ أَوَّلُ مَا نَشَطُوا إِلَيْهِ عِلْمُ الْكَلَامِ وَالْفِقْهِ وَالْحَدِيثِ وَالتَّفْسِيرِ ثُمَّ إِنَّهُمْ تَعَرَّضُوا لِلْعُلُومِ الْأَوَّلِ فَتَنَقَّلُوا وَصَنَّفُوا فِيهَا وَأَضَافُوا إِلَيْهَا أَشْيَاءَ كَثِيرَةً. وَكَانَ مِنْ أَشْهُرِ الْعَرَبِ فِي عِلْمِ النُّجُومِ الْفَرَّغَانِيُّ وَالْبَتَّانِيُّ وَأَبُو مَعْشَرٍ الْفَلَكِيُّ. وَفِي الطِّبِّ الرَّازِيُّ وَابْنُ سِينَا وَالْمَجُوسِيُّ وَابْنُ رَشْدٍ. وَفِي الْهَيْئَةِ بَطْلِيمُوسُ الْمَجْرِطِيُّ وَالْبَتْرُوجِيُّ وَشَرْفُ الدِّينِ الطُّوسِيُّ. وَفِي الْحِسَابِ الْخَوَارِزْمِيُّ وَالْكَشِّيُّ. وَفِي الْفَلَسَفَةِ الْكِنْدِيُّ وَالْفَارَابِيُّ وَابْنُ سِينَا وَابْنُ رَشْدٍ.

As for the Arabs, they are two nations: a desert nation and a settled nation. The desert among them did not concern itself with science nor persevere in it due to the hardship of their livelihood and their movement from land to land. As for the settled people, they had sciences in the Jāhili period in medicine, astrology, magic, poetry, grammar, and other subjects. Then when Islam came and legislated the pursuit of knowledge, encouraged it, and made it a means of salvation and success, the Arabs strove in seeking the sciences, translating them from other nations, and rendering them into Arabic, until in a short time they became a people of sciences and arts.

The first [sciences] they were active in were theology (*kalām*), jurisprudence (*fiqh*), *hadīth*, and [Qur'anic] exegesis (*tafsīr*). Then they turned to the sciences of the ancients, translating them, composing works on them, and adding much to them. Among the most famous Arabs in the science of astrology were al-Farghānī, al-Battānī, and Abu Ma'shar the astronomer. In medicine: al-Rāzī, Ibn Sīnā (Avicenna), al-Majūsī, and Ibn Rushd (Averroes). In astronomy: Ptolemy al-Majritī, al-Bitrūjī (Alpetragius), and Sharaf al-Dīn al-Ṭūsī. In arithmetic: al-Khwārizmī and al-Kāshshī. In philosophy: al-Kindī, al-Fārābī, Ibn Sīnā, and Ibn Rushd.

## Editorial and Scholarly Notes

### On the Manuscript and Its Preservation

The 1912 edition by Pere Louis Cheikho, S.J. (1859–1927) represents the first modern critical edition of this important work. Cheikho based his edition on a single manuscript preserved in the Oriental collections of Europe. The manuscript, as Cheikho describes it, is written in a clear hand resembling Persian script on thick yellowed paper, bound skillfully in gold-colored paper.

The *Kitab Tabaqat al-Umam* has survived in a relatively small number of manuscripts compared to other major works of the Islamic scientific tradition. This may be attributed to the fact that it was composed in al-Andalus and did not circulate as widely in the eastern Islamic lands as works produced in Baghdad or other eastern centers.

### On the Indian Section

Al-Andalusi's chapter on India (Chapter Two above) is of particular historical importance. It contains one of the earliest extant Arabic descriptions of Indian astronomical doctrines, specifically the *Sindhind*, *Azij*, and *Arkand* systems. The *Sindhind*—interpreted as “al-Dahr al-Dahir” (the Eternal Time)—was the most influential of these, adopted by major Islamic astronomers including Muhammad ibn Ibrahim al-Fazari, Habash ibn Abd Allah al-Baghdadi, and Muhammad ibn Musa al-Khwarizmi.

The description of Indian planetary theory, with its account of the great conjunction of all planets at the head of Aries at the beginning of the world-age, provides valuable evidence for the transmission of Indian astronomical ideas to the Islamic world. This doctrine of the *Sindhind* became foundational for subsequent Islamic astronomical tables (*zijat*).

### On the Author

Sa'id al-Andalusi (1029–1070 CE), whose full name is Abu al-Qasim Sa'id ibn Ahmad ibn Sa'id al-Taghlabi, was a jurist and scholar of Toledo. Born in Almeria (al-Mariyya) in 420 AH, he served as a judge (*qadi*) in Toledo under the Hammudid amir Yahya ibn Dhi al-Nun. He died in Shawwal 462 AH (July 1070 CE).

### On the Structure of the Work

The *Kitab Tabaqat al-Umam* is divided into two main parts:

1. **The Ancient Nations** (*al-Umam al-Qadima*): Seven nations that existed in earliest times—the Persians, the Chaldeans, the Greeks, the Egyptians, the Jews, the Arabs, and the Indians.
2. **The Islamic Nations** (*al-Umam al-Islamiyya*): A classification of the nations that embraced Islam, with special attention to their contributions to the sciences.

The work is notable for being one of the earliest attempts at a systematic history of science organized by civilizations, predating similar European efforts by several centuries.

### On the Scientific Content

Al-Andalusi's work contains valuable information about:

- The state of astronomy and astrology in various ancient civilizations
- The transmission of scientific knowledge between cultures
- The role of translation in preserving Greek scientific heritage
- Early Indian contributions to mathematics and astronomy (including the decimal system)
- The development of scientific institutions such as the House of Wisdom in Baghdad

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*End of Bilingual Edition*

Source-language reader: Arabic, Devanagari, and Latin manuscript passages

Arabic • Devanagari • Latin Manuscript Transcriptions

# THE HINDU-ARABIC NUMERALS

BY

DAVID EUGENE SMITH

AND

LOUIS CHARLES KARPINSKI

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## Preface

So familiar are we with the numerals that bear the misleading name of Arabic, and so extensive is their use in Europe and the Americas, that it is difficult for us to realize that their general acceptance in the transactions of commerce is a matter of only the last four centuries, and that they are unknown to a very large part of the human race to-day.

It seems strange that such a labor-saving device should have struggled for nearly a thousand years after its system of place value was perfected before it replaced such crude notations as the one that the Roman conqueror made substantially universal in Europe. Such, however, is the case, and there is probably no one who has not at least some slight passing interest in the story of this struggle. To the mathematician and the student of civilization the interest is generally a deep one; to the teacher of the elements of knowledge the interest may be less marked, but nevertheless it is real; and even the business man who makes daily use of the curious symbols by which we express the numbers of commerce, cannot fail to have some appreciation for the story of the rise and progress of these tools of his trade.

This story has often been told in part, but it is a long time since any effort has been made to bring together the fragmentary narrations and to set forth the general problem of the origin and development of these numerals. In this little work we have attempted to state the history of these forms in small compass, to place before the student materials for the investigation of the problems involved, and to express as clearly as possible the results of the labors of scholars who have studied the subject in different parts of the world. We have had no theory to exploit, for the history of mathematics has seen too much of this tendency already, but as far as possible we have weighed the testimony and have set forth what seem to be the reasonable conclusions from the evidence at hand.

To facilitate the work of students an index has been prepared which we hope may be serviceable. In this the names of authors appear only when some use has been made of their opinions or when their works are first mentioned in full in a footnote.

If this work shall show more clearly the value of our number system, and shall make the study of mathematics seem more real to the teacher and student, and shall offer material for interesting some pupil more fully in his work with numbers, the authors will feel that the considerable labor involved in its preparation has not been in vain.

We desire to acknowledge our especial indebtedness to Professor Alexander Ziwet for reading all the proof, as well as for the digest of a Russian work, to Professor Clarence L. Meader for Sanskrit transliterations, and to Mr. Steven T. Byington for Arabic transliterations and the scheme of pronunciation of Oriental names, and also our indebtedness to other scholars in Oriental learning for information.

DAVID EUGENE SMITH  
LOUIS CHARLES KARPINSKI

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## Pronunciation of Oriental Names

(S) = in Sanskrit names and words; (A) = in Arabic names and words.

**b, d, f, g, h, j, l, m, n, p, sh (A), t, th (A), v, w, x, z**, as in English.

**a**, (S) like *u* in *but*: thus pandit, pronounced pundit. (A) like *a* in *ask* or in *man*.

**a with macron**, as in father.

**c**, (S) like *ch* in *church* (Italian *c* in *cento*).

**d with dot below, n with dot below, s with dot below, t with dot below**, (S) made with the tip of the tongue turned up and back into the dome of the palate.

**d with dot below, s with dot below, t with dot below, z with dot below**, (A) made with the tongue spread so that the sounds are produced largely against the side teeth.

**d with dot below (A)**, like *th* in *this*.

**e**, (S) as in *they*. (A) as in *bed*.

**g with dot below**, (A) a voiced consonant formed below the vocal cords; its sound is compared by some to a *g*, by others to a guttural *r*.

**i**, as in *pin*. **i with macron**, as in *pique*.

**kh**, (A) the hard *ch* of Scotch *loch*, German *ach*.

**m, n**, (S) like French final *m* or *n*, nasalizing the preceding vowel.

**n with dot above**, like *ng* in singing.

**o**, (S) as in *so*. (A) as in *obey*.

**q**, (A) like *k* (or *c*) in *cook*; further back in the mouth.

**r**, (S) English *r*, smooth and untrilled. (A) stronger.

**s with acute**, (S) English *sh* (German *sch*).

**u**, as in *put*. **u with macron**, as in *rule*.

**ayin / hamza**, (A) a sound kindred to the spiritus lenis.

**Accent**: (S) as if Latin; (A) on the last syllable that contains a long vowel or a vowel followed by two consonants.

## 1 Early Ideas of Their Origin

It has long been recognized that the common numerals used in daily life are of comparatively recent origin. The number of systems of notation employed before the Christian era was about the same as the number of written languages, and in some cases a single language had several systems. The Egyptians, for example, had three systems of writing, with a numerical notation for each; the Greeks had two well-defined sets of numerals, and the Roman symbols for number changed more or less from century to century. Even to-day the number of methods of expressing numerical concepts is much greater than one would believe before making a study of the subject, for the idea that our common numerals are universal is far from being correct.

It will be well, then, to think of the numerals that we still commonly call Arabic, as only one of many systems in use just before the Christian era. As it then existed the system was no better than many others, it was of late origin, it contained no zero, it was cumbersome and little used, and it had no particular promise. Not until centuries later did the system have any standing in the world of business and science; and had the place value which now characterizes it, and which requires a zero, been worked out in Greece, we might have been using Greek numerals to-day instead of the ones with which we are familiar.

Of the first number forms that the world used this is not the place to speak. Many of them are interesting, but none had much scientific value. In Europe the invention of notation was generally assigned to the eastern shores of the Mediterranean until the critical period of about a century ago,—sometimes to the Hebrews, sometimes to the Egyptians, but more often to the early trading Phoenicians.

The idea that our common numerals are Arabic in origin is not an old one. The mediaeval and Renaissance writers generally recognized them as Indian, and many of them expressly stated that they were of Hindu origin. Others argued that they were probably invented by the Chaldeans or the Jews because they increased in value from right to left, an argument that would apply quite as well to the Roman and Greek systems, or to any other. It was, indeed, to the general idea of notation that many of these writers referred, as is evident from the words of England's earliest arithmetical textbook-maker, Robert Recorde (c. 1542):

“In that thinge all men do agree, that the Chaldays, whiche fyrste inuented thys arte, did set these figures as thei set all their letters. for they wryte backwarde as you tearme it, and so doo they reade. And that may appeare in all Hebrewe, Chaldaye and Arabike bookes...where as the Greekes, Latines, and all nations of Europe, do wryte and reade from the lefte hand towarde the ryghte.”

Others, and among them such influential writers as Tartaglia in Italy and Kobel in Germany, asserted the Arabic origin of the numerals, while still others left the matter undecided or simply dismissed them as “barbaric.” Of course the Arabs themselves never laid claim to the invention, always recognizing their indebtedness to the Hindus both for the numeral forms and for the distinguishing feature of place value.

Foremost among these writers was the great master of the golden age of Bagdad, one of the first of the Arab writers to collect the mathematical classics of both the East and the West, preserving them and finally passing them on to awakening Europe. This man was Mohammed the Son of Moses, from Khowarezm, or, more after the manner of the Arab, **Mohammed ibn Musa al-Khowarazmi**,<sup>1</sup> a man of great learning and one to whom the world is much indebted for its present knowledge of algebra and of arithmetic.

<sup>1</sup>His full name was Abu 'Abdallah Mohammed ibn Musa al-Khowarazmi. He was born in Khowarezm, “the lowlands,” the country about the present Khiva and bordering on the Oxus, and lived at Bagdad under the caliph al-Mamun. He died probably between 220 and 230 of the Mohammedan era, that is, between 835 and 845 a.d.

Of him there will often be occasion to speak; and in the arithmetic which he wrote, and of which Adelhard of Bath (c. 1130) may have made the translation or paraphrase, he stated distinctly that the numerals were due to the Hindus. This is as plainly asserted by later Arab writers, even to the present day. Indeed the phrase علم هندي ('ilm hindi), "Indian science," is used by them for arithmetic, as also the adjective هندي (hindi) alone.

Probably the most striking testimony from Arabic sources is that given by the Arabic traveler and scholar **Mohammed ibn Ahmed, Abu 'l-Rihan al-Biruni** (973–1048), who spent many years in Hindustan. He wrote a large work on India, the "Book of the Ciphers," unfortunately lost, which treated doubtless of the Hindu art of calculating, and was the author of numerous other works. Al-Biruni was a man of unusual attainments, being versed in Arabic, Persian, Sanskrit, Hebrew, and Syriac, as well as in astronomy, chronology, and mathematics.

In his work on India he gives detailed information concerning the language and customs of the people of that country, and states explicitly that the Hindus of his time did not use the letters of their alphabet for numerical notation, as the Arabs did. He also states that the numeral signs called *ahka*<sup>2</sup> had different shapes in various parts of India, as was the case with the letters.

Preceding Al-Biruni there was another Arabic writer of the tenth century, Motahhar ibn Tahir, author of the Book of the Creation and of History, who gave as a curiosity, in Indian (Nagari) symbols, a large number asserted by the people of India to represent the duration of the world.

Mention should also be made of a widely-traveled student, **Al-Mas'udi** (885?–956), whose journeys carried him from Bagdad to Persia, India, Ceylon, and even across the China sea. He states that the wise men of India, assembled by the king, composed the *Sindhind*. Further on he states, upon the authority of the historian Mohammed ibn 'Ali 'Abdi, that by order of Al-Mansur many works of science and astrology were translated into Arabic, notably the *Sindhind* (Siddhanta).

The oldest work, in any sense complete, on the history of Arabic literature and history is the *Kitab al-Fihrist*, written in the year 987 a.d., by Ibn Abi Ya'qub al-Nadim. It is of fundamental importance for the history of Arabic culture.

Leonard of Pisa, of whom we shall speak at length in the chapter on the Introduction of the Numerals into Europe, wrote his *Liber Abbaci* in 1202. In this work he refers frequently to the nine Indian figures, thus showing again the general consensus of opinion in the Middle Ages that the numerals were of Hindu origin. Some interest also attaches to the oldest documents on arithmetic in our own language. One of the earliest treatises on algorism is a commentary on a set of verses called the *Carmen de Algorismo*, written by Alexander de Villa Dei (Alexandre de Ville-Dieu), a Minorite monk of about 1240 a.d.

<sup>2</sup>The Hindu name for the symbols of the decimal place system.

## 2 Hindu-Arabic Systems Used by the Arabs

When the Arabs, carried on the wave of religious enthusiasm, had spread their conquests from the Atlantic to the Indus, they found everywhere the use of the Greek alphabetic numerals. In Syria they adopted the Greek system, in Egypt the Coptic, and in Persia the system used by the Persians. These systems were, however, of little value for mercantile purposes, and the Arabs early saw the advantages of the Hindu system which had reached them through the translations of the astronomical works of Sindhind.

The spread of the numerals among the Arabs was slow. In the East, under the Abbasid caliphs of Bagdad, the Hindu system found ready acceptance among scholars, but the common people and the merchants continued to use their old systems. It was not until the eleventh century that the Hindu numerals became common in Arabic works, and even then they were used chiefly by mathematicians and astronomers.

The earliest known Arabic work containing the Hindu numerals is the arithmetic of Al-Khowarazmi, already mentioned. This work, which bore the title *Algoritmi de numero Indorum*, was translated into Latin, probably by Adelhard of Bath, in the twelfth century. The Arabic original is lost, but several Latin versions exist, and from these we know that Al-Khowarazmi gave the nine Hindu numerals and explained their use in addition, subtraction, multiplication, and division.

The numerals as used by the Arabs underwent certain modifications. The forms used in the Eastern part of the Arab empire, centering in Bagdad, were different from those used in the Western part, centering in Cairo and later in Spain. The Eastern forms were called *Indien* (Indian), while the Western forms were called *gobar* (dust).

### The al-Khwarizmi Numeral Table

The following table reconstructs the numeral forms as given by al-Khwarizmi in his *Kitab al-Hisab al-Hindi* (Book of Hindu Reckoning), as transmitted through the Latin manuscripts of the twelfth century.

Table 1: Al-Khwarizmi's Hindu Numerals (reconstructed from Latin MSS)

| Value                         | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|-------------------------------|---|---|---|---|---|---|---|---|---|---|
| Latin MS form (Bonn, 12th c.) | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| Eastern Arabic (10th c.)      | • | ١ | ٢ | ٣ | ٤ | ٥ | ٦ | ٧ | ٨ | ٩ |
| Modern Arabic                 | • | ١ | ٢ | ٣ | ٤ | ٥ | ٦ | ٧ | ٨ | ٩ |

### Eastern and Western Arabic Numeral Forms

The gobar numerals themselves were first made known to modern scholars by Silvestre de Sacy, who discovered them in an Arabic manuscript from the library of the ancient abbey of St.-Germain-des-Pres. The system has nine characters, but no zero. A dot above a character indicates tens, two dots hundreds, and so on.

In form these gobar numerals resemble our own much more closely than the Arab numerals do. They varied more or less, but were substantially as follows.

Table 2: Eastern vs. Western (Gobar) Arabic Numerals

| Value                      | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|----------------------------|---|---|---|---|---|---|---|---|---|---|
| Eastern Arabic (Bagdad)    | • | ١ | ٢ | ٣ | ٤ | ٥ | ٦ | ٧ | ٨ | ٩ |
| Western Arabic / Gobar     | . | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| Maghribi (Spain/N. Africa) | • | ١ | ٢ | ٣ | ٤ | ٥ | ٦ | ٧ | ٨ | ٩ |

Among the Arab writers who used and spread the Hindu numerals may be mentioned Al-Kindi (800–870), Al-Sufi (died 986), Al-Biruni (973–1048), and Al-Hassar (c. 1150). Al-Kindi wrote several works on arithmetic and the use of the Indian method of reckoning. Al-Sufi wrote a large work on arithmetic. Al-Biruni, as already mentioned, was a great traveler and scholar who used the Hindu numerals extensively in his astronomical and mathematical works. Al-Hassar wrote a work on arithmetic in which the second chapter was “On the dust figures.”

The Arabic system of numeration, like the Hindu, was based on the principle of place value. The Arabs adopted the zero, which they called *as-sifr* or *sifr*, and used it in the same way as the Hindus.

Italian *zefiro*, Spanish *cero*, Portuguese *zero*

French *zero*, English *zero*, German *Null*

Medieval Latin *zephirum* (Fibonacci, 1202)

*Source: See F. Woepcke, Sur l'introduction de l'arithmétique indienne en Occident, Rome, 1859*

These methods were known as *hisab al-hind*, the Indian reckoning, and *hisab al-gobar*, the dust reckoning. The spread of the numerals among the Arabs was thus a gradual process, extending over several centuries. It was aided by the great intellectual activity that characterized the Arab world from the ninth to the twelfth century.

### 3 Early Hindu Forms

While it is generally conceded that the scientific development of astronomy among the Hindus towards the beginning of the Christian era rested upon Greek or Chinese sources, yet their ancient literature testifies to a high state of civilization, and to a considerable advance in sciences, in philosophy, and along literary lines, long before the golden age of Greece.

From the earliest times even up to the present day the Hindu has been wont to put his thought into rhythmic form. The first of this poetry—it well deserves this name, being also worthy from a metaphysical point of view—consists of the Vedas, hymns of praise and poems of worship, collected during the Vedic period which dates from approximately 2000 B.C. to 1400 B.C.

Following this work, or possibly contemporary with it, is the Brahmanic literature, which is partly ritualistic (the Brahmanas), and partly philosophical (the Upanishads). Our especial interest is in the Sutras, versified abridgments of the ritual and of ceremonial rules, which contain considerable geometric material used in connection with altar construction.

#### The Brahmi and Kharosthi Numerals

The early Hindu numerals may be classified into three great groups, (1) the Kharosthi, (2) the Brahmi, and (3) the word and letter forms; and these will be considered in order.

The Kharosthi numerals are found in inscriptions formerly known as Bactrian, Indo-Bactrian, and Aryan, and appearing in ancient Gandhara, now eastern Afghanistan and northern Punjab. The alphabet of the language is found in inscriptions dating from the fourth century B.C. to the third century A.D., and from the fact that the words are written from right to left it is assumed to be of Semitic origin.

Not until the time of the powerful King Asoka, in the third century B.C., do numerals appear in any inscriptions thus far discovered; and then only in the primitive form of marks. These Asoka inscriptions, some thirty in all, are found in widely separated parts of India.

#### The Nana Ghat Numerals (2nd c. B.C.)

The next very noteworthy evidence of the numerals, and this quite complete as will be seen, is found in certain cave inscriptions dating back to the first or second century A.D. In these, the Nasik cave inscriptions, the forms are as follows.

१ २ ३ ४ ५ ६ ७ ८ ९  
1 2 3 4 5 6 7 8 9

The Nana Ghat cave inscriptions (c. 150 B.C.) contain the earliest known Brahmi numeral system. These are vertical and perpendicular forms, quite distinct from the later cursive Devanagari. The numeral for 10 is a simple cross-like mark (X), and the numbers 20–90 are formed by ligatures.

*Source: See G. Buhler, Indische Palaeographie, Strassburg, 1896, Tafel III; J. F. Fleet, Corpus Inscriptionum Indicarum, Vol. III*

Table 3: Development of Brahmi Numerals

| Source                    | 1 | 2  | 3   | 4   | 5 | 6 | 7 | 8 | 9  | 10 |
|---------------------------|---|----|-----|-----|---|---|---|---|----|----|
| Asoka (3rd c. B.C.)       | I | II | III | +   | / |   |   |   |    | X  |
| Nana Ghat (2nd c. B.C.)   | — | =  | ≡   | + / | h | 6 | 7 | 8 | ?0 | X  |
| Nasik Caves (1st c. A.D.) | — | =  | ≡   | ⊕   | h | 6 | 7 | 8 | ?0 | )  |
| Kushana (1st–3rd c.)      | — | =  | ≡   | ⊕   | h | 6 | 7 | 8 | ?0 | )  |



## The Development of Devanagari Numerals

From this time on, until the decimal system finally adopted the first nine characters and replaced the rest of the Brahmi notation by adding the zero, the progress of these forms is well marked.

With respect to these numerals it should first be noted that no zero appears in the table, and as a matter of fact none existed in any of the cases cited. It was therefore impossible to have any place value, and the numbers like twenty, thirty, and other multiples of ten, one hundred, and so on, required separate symbols except where they were written out in words.

|   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|
| ० | १ | २ | ३ | ४ | ५ | ६ | ७ | ८ | ९ |
| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |

The modern Devanagari numerals, which developed from the Brahmi forms through the Gupta and Nagari periods. The zero is called शून्य (*sunyabindu*, later shortened to *sunya*), meaning “void” or “empty.”

*Source: See Buhler, Indische Palaeographie; Thibaut, Astronomie, Astrologie und Mathematik*

The ancient Hindus had no less than twenty of these symbols, a number that was afterward greatly increased. Having now examined types of the early forms it is appropriate to turn our attention to the question of their origin. As to the first three there is no question. The I or — is simply one stroke, or one stick laid down by the computer. The II represents two strokes or two sticks, and so for the III and IV.

From some primitive form came the two of Egypt, of Rome, of early Greece, and of various other civilizations. From some primitive form — came the Chinese symbol, which is practically identical with the symbols found commonly in India from 150 B.C. to 700 A.D.

## The Word and Letter Numeral Systems

Mention should also be made, in this connection, of a curious system of alphabetic numerals that sprang up in southern India. In this we have the numerals represented by the letters as given in the following table. By this plan a numeral might be represented by any one of several letters, and thus it could the more easily be formed into a word for mnemonic purposes.

क ख ग घ ङ च छ ज झ  
1 2 3 4 5 6 7 8 9

ट ठ ड ढ ण त थ द ध  
1 2 3 4 5 6 7 8 9

प फ ब भ म  
1 2 3 4 5

The Katapayadi system of alphabetic numerals from southern India. For example, the word भवति: (*bhavatih*) has the value  $4+4+6 = 14$ , read from right to left.

*Source: See A. C. Burnell, South Indian Palaeography, 2nd ed., London, 1878*

## 4 Introduction into Europe

### The Definite Introduction: Fibonacci's *Liber Abaci*

The definite introduction of the Hindu-Arabic numerals into Europe is usually attributed to Leonardo of Pisa, known as Fibonacci, who wrote his famous *Liber Abaci* in 1202. Leonardo had traveled extensively in the East, visiting Egypt, Syria, Greece, and Sicily, and had learned the Hindu methods of reckoning from Arab teachers. In his book he explained the use of the nine Indian figures and the zero, and showed how calculations could be performed with them.

The *Liber Abaci* was not, however, the first work in Europe to mention the Hindu numerals. As we have seen, there is evidence that the numerals were known in Spain and possibly in other parts of Europe as early as the tenth century. But Leonardo's work was the first to give a systematic and comprehensive treatment of the new system.

9 8 7 6 5 4 3 2 1

*Cum his itaque nouem figuris, et cum hoc signo 0, quod Arabice zephirum appellatur, scribitur quilibet numerus, ut inferius demonstrabitur.*

"With these nine figures, and with this sign 0, which in Arabic is called *zephirum*, any number can be written, as will be shown below."

*Source: Boncompagni, op. cit., p. 2*

The reception of the numerals in Europe was slow. The medieval universities, with their devotion to the classics, were at first hostile to the new system. An important figure in the early history of the numerals in Europe was Gerbert, who became Pope Sylvester II in 999. Gerbert had studied in Spain and had learned the numerals there.

### The Carmen de Algorismo and Medieval Latin Texts

Among the early European works on arithmetic using the Hindu numerals may be mentioned the *Carmen de Algorismo* of Alexander de Villa Dei (c. 1240), the *Algorismus* of Johannes de Sacrobosco (c. 1250).

The spread of the numerals from Italy to the rest of Europe was a gradual process. France and Germany adopted them in the fifteenth century, and England in the sixteenth. Even then, the Roman numerals continued to be used for many purposes, and it was not until the eighteenth century that the Hindu-Arabic numerals became completely dominant in Europe.

## 5 Numerals in the East

The numerals of the Hindu-Arabic system spread not only westward to Europe but also eastward through Persia, Central Asia, and into China. In each region they underwent modifications adapted to local scripts and computational traditions.

### Persian and Central Asian Forms

In Persia, the Arabic numerals were adopted with minor modifications. The Persian forms of the numerals, which are still used today in Iran, Afghanistan, and parts of Central Asia, are derived from the Eastern Arabic forms but with certain calligraphic adaptations to the Persian (Nasta'liq) script tradition.

Table 4: Persian Numeral Forms

| Value            | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|------------------|---|---|---|---|---|---|---|---|---|---|
| Persian (Modern) | ۰ | ۱ | ۲ | ۳ | ۴ | ۵ | ۶ | ۷ | ۸ | ۹ |

### Forms in India and Southeast Asia

The numerals also continued to develop independently in India. The modern Devanagari forms used in India and Nepal are direct descendants of the Brahmi numerals, parallel to but distinct from the line of development that led through the Arabs to Europe.

|   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|
| ० | १ | २ | ३ | ४ | ५ | ६ | ७ | ८ | ९ |
| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |

The Devanagari numerals shown above are used throughout northern India, Nepal, and in Sanskrit and Hindi texts. Regional variants exist in Bengali, Gujarati, Gurmukhi, Kannada, Malayalam, Tamil, and Telugu scripts.

*Source: See Burnell, *South Indian Palaeography*; Enthoven, *Numeral Notation in India**

### The Eastern Spread to China

In China, the numerals were known through Arab and Persian merchants, but the Chinese already possessed their own well-developed system of rod numerals and character-based numeration. The so-called “Arabic” numerals were not adopted in China until the modern period, though they were known to Chinese mathematicians through Islamic astronomical treatises.

The Chinese system of rod numerals, which dates from at least the Han dynasty (206 B.C.–220 A.D.), used horizontal and vertical strokes in a place-value system, and this system already embodied the essential principle of positional notation. The Chinese had, in effect, a decimal place-value system without a written zero symbol—they used a blank space instead, much as the Babylonians had done with their sexagesimal system.

Special Feature: Complete Numeral Comparison Table

The following comprehensive table presents the Hindu-Arabic numerals in their various historical forms, from the earliest Brahmi inscriptions through to modern European usage. This table allows the reader to trace the evolution of each digit across civilizations.

Table 5: Complete Historical Comparison of Numeral Forms

| Source / Period           | 0 | 1 | 2  | 3   | 4   | 5 | 6 | 7 | 8  | 9 |
|---------------------------|---|---|----|-----|-----|---|---|---|----|---|
| Brahmi (3rd c. B.C.)      | — | I | II | III | +   | h |   |   |    | X |
| Nana Ghat (2nd c. B.C.)   | — | — | =  | ≡   | + / | h | 6 | 7 | ?0 | ) |
| Nasik Caves (1st c. A.D.) | — | — | =  | ≡   | ⊕   | h | 6 | 7 | ?0 | ) |
| Gupta (4th–6th c.)        | ● | 1 | 2  | 3   | 4   | 5 | 6 | 7 | 8  | 9 |
| Devanagari                | ० | १ | २  | ३   | ४   | ५ | ६ | ७ | ८  | ९ |
| Eastern Arabic            | • | ١ | ٢  | ٣   | ٤   | ٥ | ٦ | ٧ | ٨  | ٩ |
| Western Arabic (Gobar)    | . | 1 | 2  | 3   | 4   | 5 | 6 | 7 | 8  | 9 |
| Maghribi (Spain)          | ○ | 1 | 2  | 3   | 4   | 5 | 6 | 7 | 8  | 9 |
| Leonardo of Pisa (1202)   | 0 | 1 | 2  | 3   | 4   | 5 | 6 | 7 | 8  | 9 |
| 15th c. European          | 0 | 1 | 2  | 3   | 4   | 5 | 6 | 7 | 8  | 9 |
| Modern European           | 0 | 1 | 2  | 3   | 4   | 5 | 6 | 7 | 8  | 9 |

Special Feature: The al-Khwarizmi Numeral Table

Al-Khwarizmi’s *Kitab al-Hisab al-Hindi* (Book of Hindu Reckoning) is the earliest systematic Arabic treatise on the Hindu numeral system. Although the Arabic original is lost, we possess Latin translations made in the twelfth century, notably the manuscript preserved at the University of Cambridge (MS Ii.6.5) and fragments in other collections.

The following table reconstructs al-Khwarizmi’s numeral system from the Latin manuscripts and compares it with the Eastern Arabic forms preserved in later Arabic astronomical tables.

Table 6: The al-Khwarizmi Numeral System (Reconstructed)

| Value                                 | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---------------------------------------|---|---|---|---|---|---|---|---|---|---|
| <i>Dixit Algorizmi</i> (Cambridge MS) | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| Eastern Arabic (10th c.)              | • | ١ | ٢ | ٣ | ٤ | ٥ | ٦ | ٧ | ٨ | ٩ |
| Modern Arabic                         | • | ١ | ٢ | ٣ | ٤ | ٥ | ٦ | ٧ | ٨ | ٩ |

The Latin translator rendered the Arabic title as *Dixit Algorizmi* (“Al-Khwarizmi said”), whence comes our word “algorithm.” The work systematically explains:

1. The nine Hindu figures and the zero
2. The principles of place value
3. Operations: addition, subtraction, doubling, halving
4. Multiplication and division
5. Extraction of square roots

## Special Feature: Fibonacci's *Liber Abaci* Passages in Latin

The following passages from Leonardo of Pisa's *Liber Abaci* (1202) are among the earliest extant Latin texts describing the Hindu-Arabic numeral system in Europe. The manuscript tradition is based on the edition of Baldassarre Boncompagni (1857), prepared from the Florence manuscript (Biblioteca Laurenziana, Conv. Sopp. 261).

9 8 7 6 5 4 3 2 1

*Cum his itaque nouem figuris, et cum hoc signo 0, quod Arabice zephirum appellatur, scribitur quilibet numerus, ut inferius demonstrabitur. Nam unitas in primo gradu, id est in ultima figura, posita, ipsam representat. Si uero in secundo, ipsam representat, et in primo loco cifram ponit. Et si in tertio gradu posita fuerit, ipsam representat, et in alio cifras ponit. Similiter intelligendum est de omnibus aliis figuris, quamlibet in quolibet gradu positam representare.*

“The nine Indian figures are these: 9, 8, 7, 6, 5, 4, 3, 2, 1. With these nine figures, and with this sign 0, which in Arabic is called *zephirum*, any number can be written, as will be shown below. For unity placed in the first grade, that is in the last figure, represents itself. If indeed in the second, it represents itself and places a zero in the first place. And if it is placed in the third grade, it represents itself and places zeros in the other places. Similarly it is to be understood concerning all other figures, that any figure placed in any grade represents itself.”

*Source: Boncompagni, op. cit., pp. 2–3*

## Conclusion

In conclusion, we may say that the history of the Hindu-Arabic numerals is a long and complex one, stretching over many centuries and involving many different civilizations. From their earliest beginnings in India, through their development by the Arabs, to their final adoption in Europe, these numerals have had a remarkable career. They are one of the most important inventions in the history of civilization, and their influence on science, commerce, and daily life can hardly be overestimated.

The study of their history reveals the deep connections that have always existed between the peoples of the East and the West, and shows how knowledge has been transmitted from one civilization to another along the great trade routes that have linked the world since ancient times. It is a story that deserves to be known by all who use these symbols, which have become so familiar that we are apt to forget how much labor and ingenuity went into their creation.

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